ORDERED

Architecting Commerce Platforms

Željko Obrenović

ORDERED: Architecting Commerce Platforms

Željko Obrenović

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ORDERED: Architecting Commerce Platforms

Željko Obrenović

AI-assisted: researched and written with Claude Code, images generated with the Gemini API. How this works

How commerce systems fit together, why the architectural decisions behind them matter, and how those decisions evolve as a business grows — from the catalog, the cart and the payment to the warehouse, the stores, the marketplaces and the back office, following one order from discovery to delivery and back. Written for readers new to commerce architecture, platform-neutral, with a fictional running example and dated public evidence. This is a living journal and a work in progress: I revise it as I learn more through my work.

Preface

Introduction: Why Commerce Architecture Matters

IN THIS SECTION, YOU WILL: Learn what this book argues, who it is for, how its eleven parts follow one order, how to find the chapter for your decision, and how to read its evidence.

KEY POINTS:

- **A commerce platform is a promise-keeping machine.** It shows a product and promises a price, "in stock", a delivery date and a return; then a dozen systems owned by different teams and vendors have to keep those promises. It is not one product you buy but the whole set of systems that make and keep promises to a customer.
- **Most commerce failures are a promise one system made and another could not keep.** A storefront sells a jacket the warehouse no longer has; two systems refund the same return; three systems report three statuses for one order. The failures live in the joins between systems, rarely inside one of them.

- **Architecture decides who may make which promise, how the others learn of it, and what happens when they disagree.** The right answer changes with the size and shape of the business, so the book follows one fictional merchant as it grows, compares every decision across four scale stages, and says in each chapter what is established, what depends on context and what is the author's view.

At 21:14 on 13 April 2027, Eva Janssen in Nijmegen buys a €199 Ridgewalker jacket and two €18 pairs of socks from **Kestrel & Pine, a fictional outdoor brand**. She pays €235 through iDEAL, the Dutch bank-transfer method. The shop confirms order #1072455. For Eva, the next step is waiting for delivery. For the merchant, several systems must now turn that confirmation into the right parcel, delivered when promised.

The shop has promised the described product, an agreed price, available stock, a delivery date and a way to return it. Pricing supplies the amount, the warehouse supplies stock, the carrier helps determine delivery and finance records payment and any refund. The page taking the order depends on those systems' answers. If one is wrong or out of date, checkout can succeed while committing the merchant to a promise it cannot keep.

On Black Friday 2026, the same shop oversold **1,340 units**. The jacket sold out at 19:40, but channels continued selling for up to two hours from old stock copies. The component systems stayed up. Their handoffs failed the customer.

This book follows Eva's order from discovery to delivery, exchange and accounting. It explains who should own each fact, how other systems learn about changes and what to do when they disagree. The choices evolve as Kestrel & Pine adds channels, locations and complexity.

Who This Book Is For

The book is for readers **new to commerce architecture** who need to understand how selling systems fit together and judge changes to them. You may have just joined:

- **A retailer, a brand or a wholesaler**, as an engineer, product manager, analyst or engineering lead, and found a dozen systems that nobody drew.
- **A commerce software vendor**, building one of those systems (a warehouse system, a returns portal, a marketplace connector), and need to understand the systems on either side of yours.
- **An agency or a consultancy**, asked to recommend a platform, a migration or an integration.
- **A leadership role**, as a CTO, head of ecommerce or finance or operations director, and need to know which technology questions are really business questions.

Starting knowledge. You should recognize an API, database, message queue and web page. You do not need commerce experience.

Vocabulary stays local. Chapters explain the terms their decisions need, such as **SKU**, a code for a sellable variant, or **system of record**, the authoritative home of a fact. They also explain mechanisms such as **idempotency**, making a repeated request have the same effect as one request. The Glossary: Plain-Language Definitions collects these definitions and links to fuller treatments.

The book speaks from the position of **an architect who advises growing merchants**. When I give my judgement, I say so in the first person, with my reasons and the conditions under which I would decide differently.

The Argument: A Promise-Keeping Machine

A commerce platform is usually described as a product you subscribe to, license or assemble. This book uses the word differently: **a commerce platform is the whole set of systems that make and keep promises to a customer**: the storefront, the catalog, pricing, promotions, search, inventory, checkout, payments, the order record, tax and invoicing, the warehouse, the carriers, the stores, the returns process, customer service, the marketplaces, the wholesale portal, the EDI links, the ERP (enterprise resource planning, the system that runs finance, purchasing and the books) and the data warehouse.

Seen that way, the platform's job is simple to state and hard to do. **It makes promises in one place and keeps them in another**. The storefront promises; the warehouse, the carrier, the payment provider and the finance system deliver. Between the promise and its keeping sits a handoff, a message from one system to another, and every handoff can be late, lost, duplicated or misread.

That is why the book's central claim is about joins, not components: **most commerce failures are a promise one system made and another could not keep**. The Black Friday oversell was a storefront promising from a two-hour-old copy. Kestrel & Pine's Three-Status Order, which showed *Delivered* on the platform, *In picking* in the warehouse and *Not invoiced* in the ERP for six days, was three honest systems describing three different objects. Its Double Refund, 212 returns refunded twice, was two systems each entitled to refund the same return.

Architecture, in this book, is the discipline of answering three questions for every promise:

Question	What it means in practice	Where the book answers it first
Who may make this promise?	Which system owns the fact the promise rests on (the stock count, the price, the order's status), so its copy wins when copies disagree	Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record
How do the others learn of it?	Whether they call it, receive an event, read a nightly file or copy it by hand, and how fresh their copy must be	APIs, Events, Files and Middleware: How Commerce Systems Talk
What happens when they disagree?	Which messages can arrive twice or not at all, what a repeat does, and which job compares the copies and repairs them	Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation

A one-store brand and a group of brands in six countries need different answers, which is why the book follows one merchant as it grows.

Figure 1: *A commerce platform makes promises in one place and keeps them in another,*

so most failures happen at a join between systems, not inside one of them.

The Order's Journey as the Spine

The book is organized around **the order's journey**: the path one order takes from a shopper's search to the doorstep, back again as a return, and into the books. Eva's jacket is that order. Its canonical facts never change: ordered Tuesday 13 April 2027 at 21:14, picked and packed at Kestrel & Pine's Duiven warehouse on Wednesday, delivered by PostNL on Thursday 15 April, and exchanged for a size L through the returns portal on Monday 19 April.

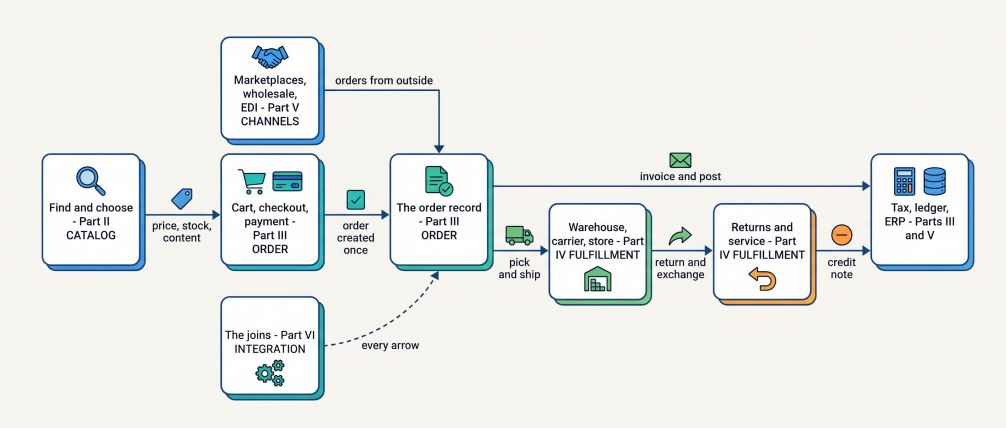


Figure 2: *Eva's order crosses Parts II to V in the order it happens; Part VI covers every arrow between them, and Parts VII to XI step back to look at the platform, the risks, the whole journey and how the decisions change with scale.*

Parts II to V follow the order in the sequence it happens; Part VI is about the arrows between systems. Each chapter has a short paragraph near its top headed "**Where we are in the order's journey**", so you always know which step of Eva's order you are looking at, or how a chapter off the order's path bears on it.

Four Scale Stages, and Why Every Chapter Compares Them

The right architecture for a merchant with one shop and a back room is the wrong one for a group of brands in six countries, and the reverse is just as true. The book therefore compares its decisions across **four scale stages**:

Stage	Rough size	Typical shape	Kestrel & Pine
1 — Starting	under about €2 million revenue	One channel, a hosted platform, the founders running operations, a third-party logistics provider (3PL) or a back room	2016–2018
2 — Growing	about €2–20 million	Two or three channels, a first store or wholesale line, first integrations (accounting or ERP, 3PL, a marketplace), a small team and an agency	2019–2022
3 — Omnichannel	about €20–200 million	Stores, wholesale, marketplaces and several countries, an own warehouse, a dozen or more systems; the joins become the problem	2023 to now
4 — Group	over about €200 million, or several brands or business units	Several brands, countries or business units; platform teams; shared services against autonomy; acquisitions	not yet

The revenue bands are illustrative markers, not thresholds. **The shape of the business decides the stage**: a €15 million brand with stores, wholesale and two countries already has Omnichannel problems.

Every chapter in Parts II to VIII has a section titled "**Across Scale Stages: When It Fits and When It Limits**", with a table that says, for each stage, what is typical, what becomes necessary and which warning signs show you have outgrown an approach.

The most common mistake in commerce architecture is not a bad design but one that was right two stages ago, or will be right two stages from now. Kestrel & Pine's 2026 troubles are mostly Growing-stage designs, such as an hourly stock sync built for fewer channels, still running under Omnichannel load. What Changes at Each Scale: From First Store to Group gathers all the stage tables in one place.

How the Book Is Organized

Three preface pages, eleven parts and four reference pages. Each part asks one question:

Section	The question it answers	Chapters
Preface	What is the book for, how do commerce platforms go wrong, and which merchant do the examples follow?	This page, Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong, Meet Kestrel & Pine
Part I — COMMERCE: What a Commerce Platform Is	What is a commerce platform, who sells what to whom, and which system owns which fact?	1–5: the order's journey, business models, the capability map, systems of record, the 2026 platform landscape
Part II — CATALOG: Products, Prices and Promises	What does the shop sell, at what price and discount, how do shoppers find it, and is it really in stock?	6–10: product information, pricing, promotions, search, inventory and availability
Part III — ORDER: Cart, Checkout, Payment and the Order Record	How does a cart become a paid order with one record, and how does that order reach the books?	11–14: checkout, payments, order management, tax and invoicing
Part IV — FULFILLMENT: From Warehouse to Doorstep and Back	How is the promise kept in the physical world, and what is the customer told along the way?	15–19: warehouse, shipping, stores, returns, service and notifications

Section	The question it answers	Chapters
Part V — CHANNELS: Selling Everywhere	How do orders from marketplaces, business customers and trading partners join the same order record?	20–23: marketplaces, B2B, EDI and e-invoicing, the ERP
Part VI — INTEGRATION: Making Separate Systems Behave Like One	How do separate systems talk, recover, keep their contracts, mean the same customer and count the same order?	24–28: integration styles, events and reconciliation, contracts and versioning, identity and master data, analytics
Part VII — PLATFORM: How Commerce Platforms Are Built	How are the platforms a merchant rents built, shared, extended, rendered and kept up on the peak day?	29–33: architecture styles, multi-tenancy, extensibility, storefronts, peak load
Part VIII — TRUST: Security, Privacy and Regulation	What do attackers take, whose data is it, and which laws turn into data requirements with a date?	34–36: security, privacy, regulation
Part IX — JOURNEYS: One Order, End to End	What does the whole book look like for one order, one return, one EDI reorder and one Black Friday?	37–40: four journeys, each told on two architectures
Part X — EVOLUTION: Decisions That Change With Scale	Which decisions change with scale: platform choice, migration, stage, ownership and AI?	41–45: build or buy, replatforming, scaling stages, organizing engineering, AI and agents
Part XI — PRACTICE: How Others Built It	What did Shopify, Amazon, Zalando and the composable vendors actually build, and what transfers?	46–49: four dated cases

Section	The question it answers	Chapters
Reference	Where do the research maps land, which templates help, what does a term mean, and where is the source?	Domain Coverage: From Three Model Families to Forty-Nine Chapters, Toolkit: Cards and Templates for Commerce Architecture Decisions, Glossary: Plain-Language Definitions, Bibliography: Every Source the Book Cites

Part I is the one part not to skip. It introduces the vocabulary, the capability map and the entity–owner table that every later chapter points back to.

Every chapter in Parts II to VIII answers the same five questions: **what problem this solves and why it matters** (business first), **how the components work together**, **which choices exist**, **what trade-offs they introduce**, and **when the approach fits and when it limits**. It opens with a scene at Kestrel & Pine and ends with "What to Take Away", questions to ask in a design review, and its sources.

The four journeys of Part IX deserve a word. Each one takes a Kestrel & Pine story with fixed dates and amounts, Eva's order (Journey 1: Eva's Jacket, From Search to Doorstep), her exchange (Journey 2: The Jacket Comes Back), the Bergzicht Outdoor chain's spring reorder by EDI (Journey 3: A Chain Reorders by EDI, From Order to Invoice) and Black Friday (Journey 4: Black Friday, Hour by Hour), and runs it twice: through Kestrel & Pine's late-2027 architecture and through its January 2027 one. **The dates do not change; what changes is whether the promises matched them**. If you learn best from a whole example, read a journey early and come back to the chapters it names.

Start With the Decision in Front of You

The parts are a learning path for a first reading. If a decision is already on your desk, find the row that matches it, read the chapter in **Start with**, and follow **Then** if you need more. Return to Part I whenever a term or an owner is unclear.

If you need to decide…	Start with	Then
Whether to replatform, stay, or add systems beside the current platform, and how to move without a big-bang cutover	Build, Buy or Assemble: Choosing and Changing Platforms	Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026, Moving Without Burning the Bridge: Replatforming and Migration, Case: commercetools, Shopware and the Composable Movement
Why channels keep overselling, and which system should own stock	Is It in Stock? Inventory, Availability and the Promise You Can Keep	Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record, Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation, Journey 4: Black Friday, Hour by Hour
Whether you need an order management system, and what it should own	The Order Record: Lifecycle, State and the Order Management System	Follow the Order: What Happens Between "Buy" and "Delivered", Where Is My Order? Notifications, Service and the Customer Record, Journey 1: Eva's Jacket, From Search to Doorstep
Whether you need a PIM, and who should own product content	What Are We Selling? Products, Variants and the Catalog	One Customer, One Product: Identity and Master Data Across Systems, Selling on Someone Else's Platform: Marketplaces and Channel Integration, Finding the Product: Search, Navigation and Merchandising

If you need to decide…	Start with	Then
Why the same product shows different prices in different channels	What Does It Cost? Price Lists, Contract Prices, Currencies and Tax	Discounts, Coupons and Loyalty: Promotions Without Chaos, Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation, Promises Between Systems: API Contracts, Webhooks and Versioning
How to stop refunds being paid twice or reconciled by hand	The Way Back: Returns, Exchanges, Refunds and Reverse Logistics	Taking the Money: Payment Providers, Authorization, Capture and Refunds, Tax, Invoices and the Books: From Order to Ledger, Journey 2: The Jacket Comes Back
How to connect marketplaces without a second order system	Selling on Someone Else's Platform: Marketplaces and Channel Integration	The Order Record: Lifecycle, State and the Order Management System, Is It in Stock? Inventory, Availability and the Promise You Can Keep, Promises Between Systems: API Contracts, Webhooks and Versioning
How to serve wholesale customers and EDI trading partners, and what role the ERP should play	Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals	Documents Between Companies: EDI, Punchout and E-Invoicing, The Back Office: ERP Integration and Who Is the Hub, Journey 3: A Chain Reorders by EDI, From Order to Invoice

If you need to decide…	Start with	Then
How to tame dozens of hand-built integrations with no owners	APIs, Events, Files and Middleware: How Commerce Systems Talk	Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation, Promises Between Systems: API Contracts, Webhooks and Versioning, Who Builds and Who Owns: Teams, Vendors, Agencies and Governance
Whether to go microservices, composable or headless	Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says	Delivering the Storefront: Themes, Headless, Rendering and Performance, Case: Shopify, the Modular Monolith That Serves Millions of Merchants, Case: Zalando, Microservices, Platform Strategy and the Return to Discipline
How to survive the next Black Friday	Black Friday Every Year: Peak Load, Resilience and Recovery	Journey 4: Black Friday, Hour by Hour, Is It in Stock? Inventory, Availability and the Promise You Can Keep, Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation

If you need to decide…	Start with	Then
Which apps, scripts or AI agents may write stock, prices or orders	Apps, Plugins and Functions: Extensibility and Ecosystem Governance	Who Builds and Who Owns: Teams, Vendors, Agencies and Governance, Assistants and Agents: AI in Commerce and What Is Real in 2026, Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers

One row starts somewhere you might not expect: overselling starts with inventory, not with the storefront or the marketplace connector, because the fix is deciding which system owns stock, not making the copies faster.

How to Read the Evidence

Architecture writing becomes unreliable when facts, habits, guesses, opinions, examples and estimates blur into one voice. Every chapter keeps **six kinds of statement** apart:

Kind of statement	What it is	How to recognize it
Documented fact	A company's public statement, a standard, a regulation, a date, a number from a filing or documentation	Linked inline, with the **year in the sentence** ("Zalando's March 2025 engineering post…"), and listed under Sources and Evidence. Regulations cite the official text and the dates they apply from.
Industry practice	What merchants and vendors commonly do	Phrased as "commonly" or "a typical pattern is", with no invented percentages

Kind of statement	What it is	How to recognize it
Reference-architecture inference	How a system is plausibly built, reasoned from engineering principles	When it concerns a real company, it is labeled **INFERENCE, NOT DOCUMENTED**
Recommendation	My judgement as an architect advising growing merchants	First person ("I would…"), with the reasoning and the conditions under which it does not hold
Illustration	An example set at Kestrel & Pine	Labeled fictional in each chapter; its numbers agree across the book
Assumption	An illustrative volume, latency, cost or percentage that is neither documented nor a Kestrel & Pine fact	Marked "assume…" in the text or *(assumption)* in a table; follow the reasoning, do not use it as a benchmark

Two distinctions are easy to lose. A Kestrel & Pine figure, such as the 24% item return rate, is an illustration, not an industry benchmark. And a recommendation that sounds like a fact, such as "the warehouse is the system of record for physical stock", is still a recommendation, with reasons and exceptions.

Why claims carry dates. An undated claim quietly goes stale. In **March 2025** Zalando's engineering blog described replacing a price-and-stock event stream, which could fall up to 30 minutes behind during Cyber Week, with a synchronous read API (Zalando Engineering, 2025); a chapter describing the earlier design without its date would teach the opposite lesson (Case: Zalando, Microservices, Platform Strategy and the Return to Discipline).

On **3 November 2026** bol announced that its Offers API v10 stops on 1 February 2027 (bol partner platform, 2026), and Shopify releases a new API version every quarter (Shopify API versioning, accessed November 2026).

Laws run on a timetable too: the EU rule that a price reduction must be shown against the lowest price of the previous 30 days has applied since 28 May 2022 (Directive (EU) 2019/2161). So every company and regulatory claim carries its date, evidence older than about three years is flagged, and announced features are kept apart from generally available ones.

Three tiers of take-away

Every chapter ends its argument with "**What to Take Away**", split into three tiers, so you can tell how much weight each conclusion carries:

Tier	What it holds	How to use it
Established	What the field agrees on, with the evidence or standard behind it (for example, that a message between systems may arrive more than once, so the receiver must handle repeats)	Treat as a starting point; argue with it only with evidence
Depends on context	Recommendations that hold under stated conditions: a scale stage, a business model, a team size, a vendor	Check the condition against your own business before you adopt it
The author's synthesis	My own position, marked as such, with the reasoning in one clause	Weigh it, disagree with it, and write down why

What the Book Draws On

The book uses three kinds of material, and no others.

- **Public, dated sources.** Filings, engineering posts, developer documentation, standards such as EDIFACT and GS1, legal texts and published research, cited with their year and collected in the Bibliography: Every Source the Book Cites.
- **Verified research briefs.** For each subject area a brief collected the public evidence and tagged each claim by how well it is verified; chapters use only verified or widely reported claims, or claims they checked themselves.
- **Domain models, used as maps only.** Productscape domain models in my sibling repository, structured descriptions of a merchant platform of the Shopify kind, a retail marketplace of the Amazon kind and a set of specialist commerce products, served as research maps of capabilities, roles and data. **They are models inferred from public product behavior, not descriptions of any company**, and nothing in them is used as a fact about Shopify, Amazon or any vendor. Domain Coverage:

From Three Model Families to Forty-Nine Chapters maps every capability in them to the chapter that covers it, and lists what the book adds that the models lack, from integration and reconciliation to scale stages and failure patterns.

Vendor-neutral by design. The book recommends architectures, not products. Kestrel & Pine's systems are named by role ("the platform", "the ERP", "the WMS", "the marketplace connector"), never by vendor. Real companies appear as dated cases with their sources: **Shopify** (Case: Shopify, the Modular Monolith That Serves Millions of Merchants), **Amazon** (Case: Amazon, From Monolith to Services to the Everything Store), **Zalando** (Case: Zalando, Microservices, Platform Strategy and the Return to Discipline), and **commercetools and Shopware** with the composable movement around them (Case: commercetools, Shopware and the Composable Movement). They show what was decided, why and at what size; none of them is presented as the answer.

No internal material. Nothing here comes from confidential documents or inside knowledge of any company. Where a chapter needs a situation the public record cannot supply, it uses Kestrel & Pine.

Kestrel & Pine and the Toolkit

Kestrel & Pine is fictional. Founded in Arnhem in 2016, it earns **€48 million in 2026**: €27 million online across the Netherlands, Belgium and Germany, €12 million wholesale, €6 million in stores and €3 million through marketplaces. It serves 1,100 wholesale accounts, six stores and three marketplaces: bol, Amazon.de and Zalando.

Its Duiven warehouse connects to a landscape of thirteen systems and 34 integrations. Orders and product content each live in four places, without shared order or product management. The recurring incidents expose the cost: about €96,000 from overselling, €19,400 refunded twice and €14,200 deducted for pallet labels that did not match shipment notices.

In January 2027 **Noor Visser** joins as Head of Engineering. She is the reader's stand-in: experienced in software, new to this merchant's commerce landscape. In March 2027 she launches **Waypoint**, a €1.1 million program that decides ownership first and buys the missing hubs second: the warehouse owns physical stock, an order management system becomes the one order record, a product information system (PIM) owns product content, integrations get owners, contracts and reconciliation, and the company **stays on its hosted platform** rather than replatforming. Its eight decisions, W1 to W8, are canonical; chapters explain them and never change them.

Kestrel & Pine is a composite of patterns common among growing omnichannel brands, not a disguise for any company; Meet Kestrel & Pine introduces it in full. When Noor decides, ask what the equivalent would be in your business.

The **Toolkit: Cards and Templates for Commerce Architecture Decisions** collects the book's working templates, sixteen in all, from a capability map and an entity–owner table to an integration contract card, a reconciliation card, a peak plan and an architecture decision record. Start with the one that changes the decision you face this week. The Glossary: Plain-Language Definitions defines the book's terms and names the chapter that owns each one.

What the Book Leaves to Others

Three neighboring subjects appear only where they touch commerce architecture, and each has a book of its own in this collection:

- **Payments and ledgers in depth**. This book covers taking the money as a merchant does: authorization, capture, refunds, webhooks and the match of orders against payouts (Taking the Money: Payment Providers, Authorization, Capture and Refunds, Tax, Invoices and the Books: From Order to Ledger). How card networks, acquirers and double-entry ledgers work inside is the subject of *RECONCILED*, starting with Payments for Finance Architects: Eleven Reasons the Numbers Disagree.
- **Running a marketplace**. This book covers selling on marketplaces as a merchant (Selling on Someone Else's Platform: Marketplaces and Channel Integration). Operating one, with its two-sided economics, liquidity, ranking and trust, is the subject of *LISTED*, starting with Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics.
- **The internals of AI agents**. This book covers what AI assistants and shopping agents do in commerce as of 2026 and what a merchant must have in place before one touches stock, prices or orders (Assistants and Agents: AI in Commerce and What Is Real in 2026). How agents are built, given tools and kept within bounds is the subject of *Architecting Agentic Systems*, starting with Tool Ecosystems, MCP and the Tool Broker.

The Failure Patterns

Commerce platforms break promises through recurring patterns, each of which looked reasonable when chosen. The next page, Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong, names **sixteen** of them in four families:

- **Truth**, a fact with two writers or no owner: **The Phantom Stock**, **Schrödinger's Order**, **The Familiar Stranger**, **The Boomerang** and **The Price Nobody Can Explain**.

- **Integration**, a message with no key, no contract or no owner: **The Spaghetti Junction**, **The Message in a Bottle** and **The Pulled Rug**.
- **Platform and change**, a decision copied from a bigger company or made on one date: **The First Snowfall**, **The Giant's Boots**, **The Burned Bridge**, **The Forked Platform** and **One Size Fits None**.
- **Governance**, a dependency, a deadline or an actor nobody owns: **The Lynchpin**, **The Long Fuse** and **The Secret Agent**.

The patterns travel in chains that usually start at the joins: a Spaghetti Junction of unowned connectors breeds Messages in a Bottle, which breed Phantom Stock and Boomerangs. Kestrel & Pine's Black Friday 2026 met two at once: **The Phantom Stock** in the two-hour-old copy and **The First Snowfall** in a stock chain that met real load for the first time on the day itself. Every chapter opens by naming the patterns it works on, so you can use the list as an index.

A Living Journal

This book is a living journal and a work in progress, which I revise as I learn more through my work. Its **research cut-off is 7 November 2026**: facts were checked against public sources up to that date unless a chapter says otherwise. Time-sensitive facts, such as API versions, program closures, regulatory dates and market positions, carry their date in the sentence; check the source before relying on one. The story at Kestrel & Pine is set in 2027, a year after the cut-off, so where it touches real dates (an API retirement, a law's date of application) it uses the dates published by November 2026.

Every chapter has a "View spec" link in its byline that opens the specification it was written against, including its open questions. Read the book as current thinking with its limits stated, not as a finished text.

I developed the book through AI-mediated authoring: I set the concept, the structure, the chapter contracts and the arguments; AI tools supported research against public sources, drafting from those contracts, consistency checks across chapters and editing; I reviewed and decided. The specs and the scripts that build this site are in the book's GitHub repository. The content, its judgements and its errors are my responsibility.

Where to Go Next

If you read in order, turn next to Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong for the sixteen patterns, then to Meet Kestrel & Pine, so that every chapter's opening scene lands in a company you know, and begin Part I with Follow the Order: What Happens Between "Buy" and "Delivered". If a decision is waiting, use the routing table above and come back to Part I when a term is unclear. Either way, keep

Eva's five promises in mind: this jacket, at this price, in stock, on Thursday, returnable. Every chapter is about one of them, and about which system keeps it.

Sources and Evidence

Company, product and investor sources

- From event-driven to API: Zalando's price and stock pipeline, Zalando Engineering, 2025: event delays of up to 30 minutes during Cyber Week and the move to a synchronous read API.
- bol partner platform, bol, 2026: the announcement on 3 November 2026 that Offers API v10 stops on 1 February 2027.
- API versioning, Shopify developer documentation, n.d., accessed 2026-11: quarterly API versions and their support period.

Standards, regulation and official documentation

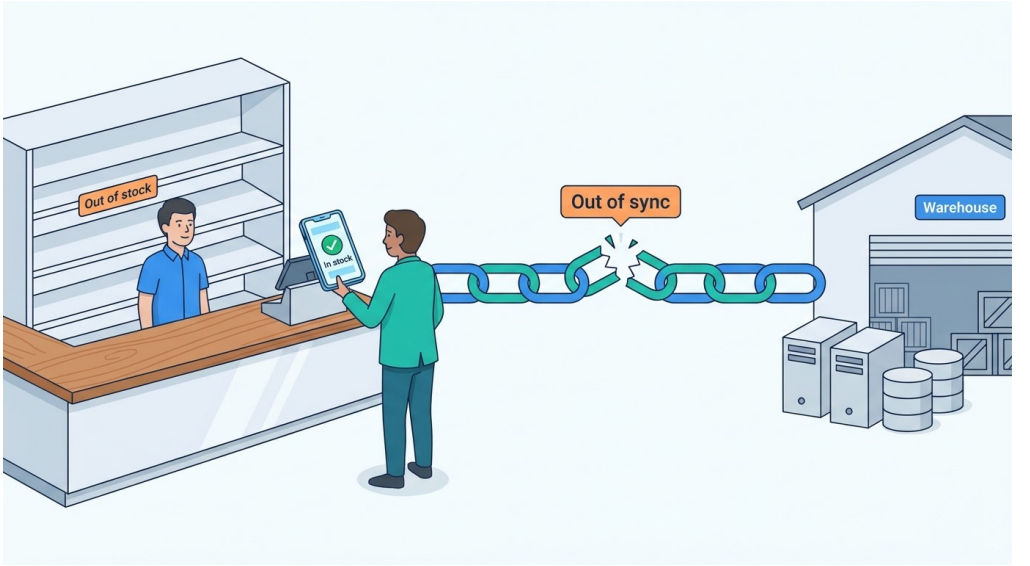
- Directive (EU) 2019/2161 (Omnibus), European Parliament and Council, 2019: the 30-day prior-price rule for price reductions, applied from 28 May 2022.

Research and practitioner sources

- Productscape domain models (author's sibling repository, 2026): research maps of capabilities, roles and data, inferred from public product behavior and mapped in Domain Coverage: From Three Model Families to Forty-Nine Chapters. Not a description of any company's systems.

Company, regulatory and research sources for each topic are listed in the chapter that owns the topic and collected in the Bibliography: Every Source the Book Cites. Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong



IN THIS SECTION, YOU WILL: Learn sixteen named ways commerce platforms break promises, recognize each by its warning sign, see what it costs in public cases and at the book's fictional merchant, and find the chapters that prevent or repair it.

KEY POINTS:

- **Nearly every pattern is a promise one system made and another could not keep.** A storefront promises a unit the warehouse no longer has, a portal promises a refund the payment provider already sent, a page promises a delivery the warehouse's cut-off cannot meet. The sixteen patterns are the recurring ways the promise and its keeper come apart.
- **The patterns fall into four families, and the family tells you where to look.** Truth patterns are a fact with two writers; integration patterns

are a message with no key, no contract or no owner; platform-and-change patterns are a decision copied from a bigger company or made on one date; governance patterns are a dependency, a deadline or an actor nobody owns.

- **The patterns travel in chains, and the chains start at the joins**. A Spaghetti Junction breeds Messages in a Bottle, which breed Phantom Stock, Schrödinger's Orders and Boomerangs; a Lynchpin sets up a Pulled Rug; a Giant's Boots ends as a Forked Platform or a Burned Bridge. Fixing the downstream symptom rarely holds; naming the origin usually does.

The storefront accepts a sale, but the warehouse never receives the order. Both systems can remain available while the customer waits for a parcel nobody is preparing. Other failures follow the same shape: prices, stock or status cross a handoff incorrectly. This guide names sixteen recurring patterns so you can recognize the warning signs, trace the mechanism and choose a remedy. The Introduction: Why Commerce Architecture Matters gives the wider argument.

Many patterns accumulate through reasonable local decisions: an app for a campaign, a script for a new channel or a shortcut for support. The risk becomes visible when someone examines their combined effect. Use the patterns to find that effect and assign responsibility.

Each entry gives a warning sign, mechanism, cost, remedy and reading links. Examples from **Kestrel & Pine, a fictional outdoor brand**, are invented, as are the opening quotes (Meet Kestrel & Pine). Public examples retain dates and sources; assigning one to a pattern is the author's interpretation.

Sixteen Patterns to Recognize

Four families organize the patterns: **truth**, disagreements over stock, orders, customers, returns and prices; **integration**, failures to exchange data reliably; **platform and change**, choices about capacity, structure and migration; and **governance**, responsibility for the systems and their changes.

Pattern	Warning sign	Family
The Phantom Stock	Two channels sell the same last unit; "out of stock" cancellations rise in busy weeks.	Truth
Schrödinger's Order	Support checks three systems to say where an order is, and they disagree.	Truth
The Familiar Stranger	Nobody can say whether the store customer and the online customer are the same person, because each system counts customers its own way.	Truth
The Boomerang	Refunds, restocking and credit notes are reconciled by hand after the return arrives, and sometimes twice.	Truth
The Price Nobody Can Explain	The same product carries different prices in different channels, and nobody can say which system set which price or why.	Truth
The Spaghetti Junction	Every pair of systems, sometimes every customer, has its own hand-built connector with its own owner, or none.	Integration
The Message in a Bottle	Messages between systems are sent once and trusted; customers find the duplicates and the gaps.	Integration
The Pulled Rug	A vendor's API or a partner's document format changes and an integration breaks before anyone is told.	Integration

Pattern	Warning sign	Family
The First Snowfall	The busiest day of the year is the first time the system, and its integrations, meet real load.	Platform and change
The Giant's Boots	An architecture is copied from a company a hundred times larger, or from a trend, without its reasons.	Platform and change
The Burned Bridge	Everything moves to a new platform on one date, with no way back.	Platform and change
The Forked Platform	The platform's core or theme has been customized so deeply that it can no longer be upgraded, and every vendor release is a project.	Platform and change
One Size Fits None	Several brands or business units are forced onto one stack before anyone has measured the benefit, or kept apart when they should share.	Platform and change
The Lynchpin	A capability customers depend on runs through one vendor, one connector, one agency or one person, with no exit plan.	Governance
The Long Fuse	A legal deadline (e-invoicing, accessibility, consumer rules) is discovered when it is months away.	Governance

Pattern	Warning sign	Family
The Secret Agent	A script, app, bot or AI agent changes orders, stock, prices or money without a named owner, its own identity or a person's approval.	Governance

A warning sign is a prompt to look, not a verdict: **ask for the number that would confirm it**.

The Phantom Stock

"The site said we had twelve. The warehouse had none since seven o'clock."

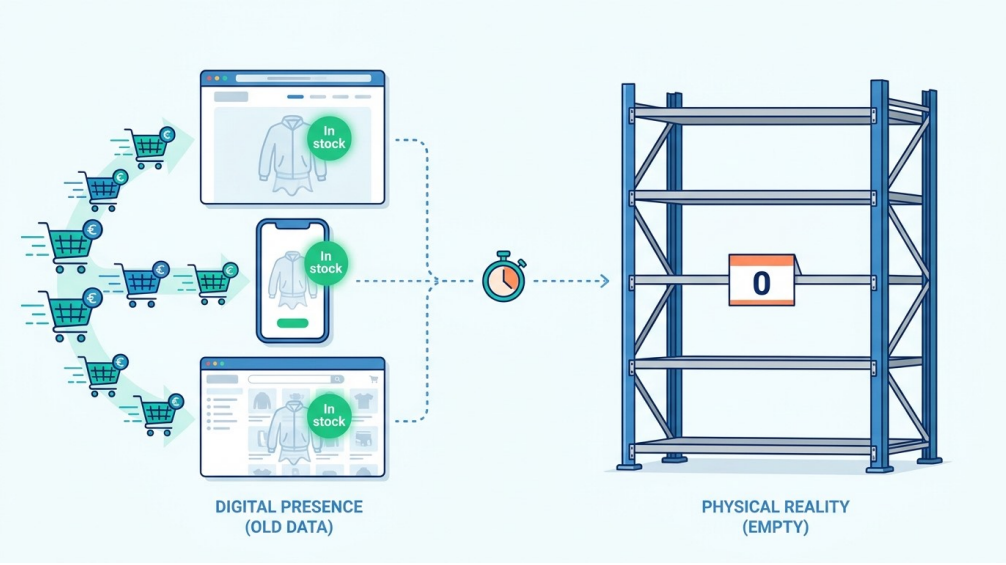


Figure 1: *The Phantom Stock: every channel sells from an aging copy of a stock number, so the shelf can be empty while the screens still say "in stock".*

What it looks like. The storefront's stock number is a copy of unknown age; click-and-collect promises what the store sold that afternoon.

Why it happens. The warehouse knows physical stock, while sales channels use copies. Scheduled updates introduce delay; extra apps and imports can change the same number. Define the authoritative source and permitted writers before trying to synchronize faster.

What it costs. Canceled orders, vouchers, marketplace penalties and seller ratings. Stock records are less accurate than assumed: the classic field study found 65% of about

370,000 inventory records inaccurate at one US retailer's stores (DeHoratius and Raman, *Management Science*, 2008; dated, still the field's reference). Even commercetools documents that its available quantity is eventually consistent "with a delay of up to 10 seconds" (n.d., accessed 2026-11).

At Kestrel & Pine (fictional). The Ridgewalker sells out at Duiven at 19:40 on 27 November 2026. Stock updates pass from the warehouse management system (**WMS**) to the enterprise resource planning system (**ERP**) every 15 minutes, then hourly to the platform and hourly to marketplaces. Sales continue for up to two hours: **1,340 units oversold**, **1,120 orders canceled or delayed**, about **€96,000 in costs**.

December's store-pickup pilot repeats the problem: nightly store-stock updates leave **37 promised items** already sold.

In public. Shopify's August 2026 developer changelog moved draft-order and transfer holds from its `reserved` to its `committed` inventory state (5 August 2026): the stock vocabulary changes under the apps that read it.

Remedy direction. One writer per location's stock, changes published as events, available-to-promise computed in one place, copies reconciled so drift is measured (Kestrel & Pine's W1).

Where this book helps: Is It in Stock? Inventory, Availability and the Promise You Can Keep owns available-to-promise, reservations and buffers; Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record names the owner per fact; Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation measures the drift.

Schrödinger's Order

"The customer sees Delivered. The warehouse says In picking. Finance says it was never invoiced."

Figure 2: *Schrödinger's Order: each system keeps its own status for the same order, and each is honest and wrong at once.*

What it looks like. The platform's status is set by the first message it received; the ERP waits for one that never came; a split shipment has no status at all.

Why it happens. The money, each shipment and each return have their own state. Without one record that keeps them apart, every system's status is honest and wrong.

What it costs. Service load first: "where is my order" (WISMO) is the largest single category of e-commerce support contacts, at 18% of incoming requests in one helpdesk vendor's data (Gorgias, updated 13 November 2023). Then money: invoices that never post and duplicate shipments.

At Kestrel & Pine (fictional). Order **#1052877** in December 2026, a fleece and a pair of socks: the fleece shipped on 7 December and the socks on 14 December after replenishment. The platform showed *Delivered* (the agency's flow had marked the whole order fulfilled on the first warehouse confirmation), the WMS showed *In picking* and the ERP *Not invoiced*, for six days, while one customer opened **11 support tickets**.

In public. Shopify's help documentation keeps separate order, payment and fulfillment status families on one order, and states that without a tracking number "no shipping updates are received" (n.d., accessed 2026-11).

Remedy direction. One order record for every channel, with separate states for money, each shipment and returns, and a summary computed from the least-advanced shipment (Kestrel & Pine's W2).

Where this book helps: The Order Record: Lifecycle, State and the Order Management System owns the order record and its state model; Where Is My Order? Notifications,

Service and the Customer Record builds the service view; The Back Office: ERP Integration and Who Is the Hub stops the ERP from being a second order system.

The Familiar Stranger

"Is the Eva who bought a jacket online the Eva on the store loyalty list? Probably."

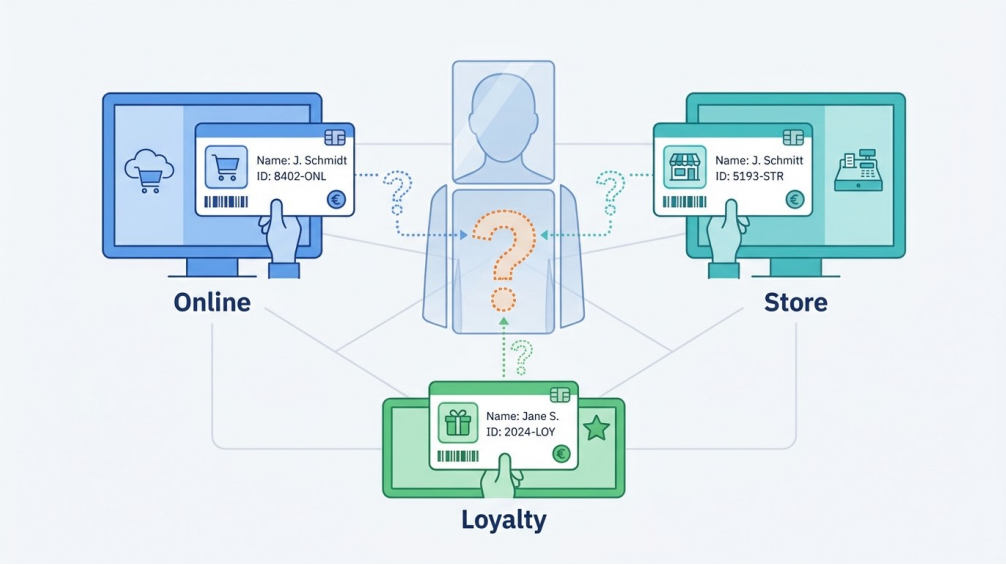


Figure 3: *The Familiar Stranger: each channel keeps its own customer record, so one person becomes several strangers nobody can match.*

What it looks like. A store cannot see an online order; a data-subject request becomes a search across nine systems.

Why it happens. Each channel's system has its own customer table, keyed on what it collects. Matching them is a project nobody owns, because every system works without it.

What it costs. Duplicate marketing, blind service, and loyalty built on a wrong count. Under the GDPR a data-subject request must be answered within one month, extendable by two (Regulation (EU) 2016/679, Article 12(3), 2016); a merchant that cannot find the person in all its systems cannot meet it.

At Kestrel & Pine (fictional). Eva Janssen, the recurring order's customer, exists as an online account, a store customer in the POS loyalty list, a newsletter subscriber and a returns-portal contact, with no shared identifier. When she returns her jacket to the Nijmegen store, the store cannot see that she is the online Eva. In 2027 the ecommerce manager and the head of engineering agreed not to launch an online loyalty program until the customer record is unified.

In public. Shopify's developer documentation marks its legacy customer accounts deprecated as of 26 February 2026 (accessed 2026-11): identity is a model to rebuild, not a table to keep.

Remedy direction. A customer master with one identifier per person, written matching rules, channel records linked to it, privacy requests served from it. Loyalty comes after the match.

Where this book helps: One Customer, One Product: Identity and Master Data Across Systems owns the customer master and matching; Whose Data Is It? Privacy, Consent and Retention serves data-subject requests across systems; Discounts, Coupons and Loyalty: Promotions Without Chaos explains why loyalty waits for it.

The Boomerang

"We refunded it. Then the portal refunded it. The customer kept both, and who can blame her."

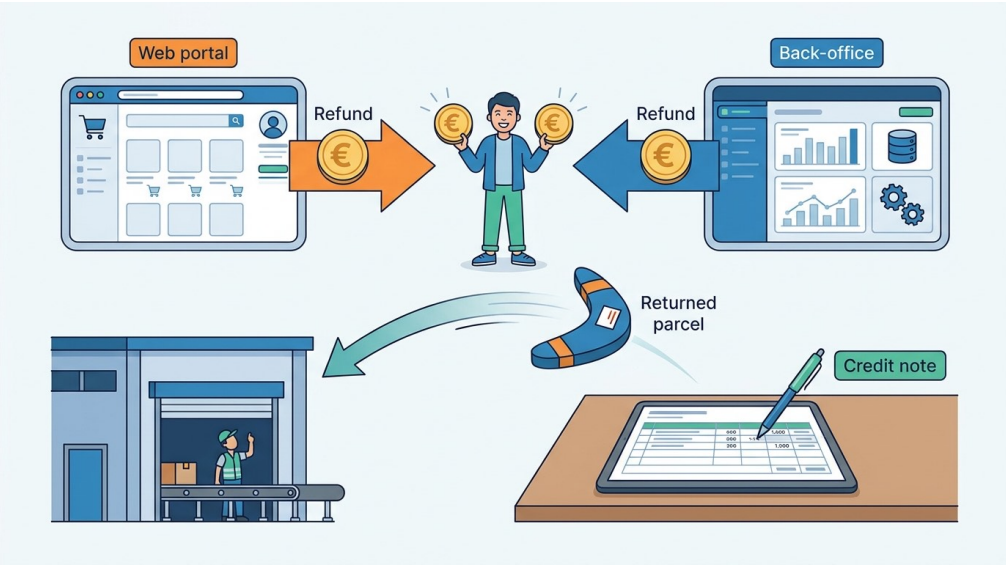


Figure 4: *The Boomerang: a return touches goods, money and books with three owners, so the refund can be paid twice and the credit note made by hand.*

What it looks like. Two systems can issue a refund; stock rises at the dock scan; credit notes come from a spreadsheet.

Why it happens. A return touches goods, money and books, with three owners. The way back was assembled from parts, and nothing says which event releases the refund.

What it costs. The NRF put the 2025 US online return rate at 19.3% of online sales, with 9% of all returns fraudulent (October 2025). Stripe notes a "risk of double refund" for

bank-debit methods, and Mollie "blocks duplicate refunds made in a short time frame" (both n.d., accessed 2026-11).

At Kestrel & Pine (fictional). Over three weeks in January 2027, **212 returns** were refunded twice: once by the returns portal through the platform's API and once by hand by support in the payment provider's console; **€19,400** went out twice, most of it recovered by asking customers to transfer it back. In the Black Friday week before, **190** returned jackets were refunded at €159.20 instead of the €143.28 actually paid, about €3,000, because the order-level discount had never been allocated to lines in the order export.

In public. Shopify's returns documentation separates the *return* (the buyer's intent and the money) from the *reverse fulfillment order* (the warehouse's work) and names four dispositions, restocked, processing required, not restocked and missing (n.d., accessed 2026-11): money and stock are two records.

Remedy direction. One return record; the warehouse's disposition as the single trigger for restock, refund and credit note; one system that refunds, keyed on the return id; a daily refund match. Kestrel & Pine's February 2027 fix.

Where this book helps: The Way Back: Returns, Exchanges, Refunds and Reverse Logistics owns the return record, dispositions and refund reconciliation; Taking the Money: Payment Providers, Authorization, Capture and Refunds owns refunds and keys; Tax, Invoices and the Books: From Order to Ledger books credit notes.

The Price Nobody Can Explain

"Germany is at 179. Everyone else is at 199. Which one did we mean?"

Figure 5: *The Price Nobody Can Explain: prices live in several systems with their own calendars and imports, so the same product carries several prices and nobody can say why.*

What it looks like. Prices live in five systems with five campaign calendars; discounts stack unconfigured; a refund uses a price nobody paid.

Why it happens. Several roles set prices under several rules, each selling system keeps its own number, imports fail silently, and promotion settings outlive their campaign.

What it costs. Margin, parity penalties and regulatory exposure: since 28 May 2022 the EU's Omnibus rules require any price-reduction announcement to show the "prior price", the lowest price in at least the 30 days before the reduction (Directive (EU) 2019/2161, 2019), and the Dutch regulator ACM announced supervision of those rules from 1 January 2023 (6 December 2022).

At Kestrel & Pine (fictional). In October 2026 the German storefront sold the Ridgewalker jacket at **€179** for **nine days** while the marketplace listing and the stores charged €199: the restore row of a campaign import had failed in an agency script that logged the error and reported success. **410 orders**, about **€8,200** of margin lost, a price-parity warning from a marketplace.

In Black Friday week a 20% outerwear discount stacked with a 10% welcome code on **2,860** orders because a combination setting from an October campaign had never been reverted: about **€52,000**.

In public. The Omnibus rule (2022) and ACM's supervision (2023) make a 30-day price history per SKU and channel a legal requirement.

Remedy direction. One owner per price kind, price events with effective dates, campaigns as data, a 30-day price history, promotions with an evaluation order and

stop rule, daily reconciliation, prices frozen on the order.

Where this book helps: What Does It Cost? Price Lists, Contract Prices, Currencies and Tax owns price ownership, price lists and the price history; Discounts, Coupons and Loyalty: Promotions Without Chaos owns stacking rules and line-level allocation; Selling on Someone Else's Platform: Marketplaces and Channel Integration handles parity and channel price rules.

The Spaghetti Junction

"Thirty-four integrations. Thirteen have no owner. Six of them write stock."

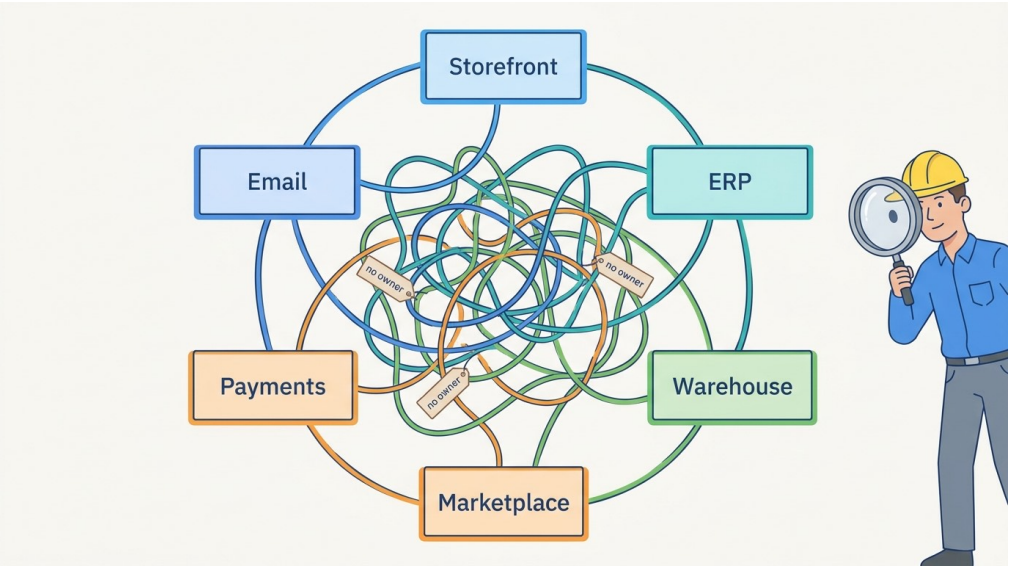


Figure 6: *The Spaghetti Junction: each connector was the fastest way to meet one need, until nobody can list them or say who owns them.*

What it looks like. Flows on the integration platform, agency scripts and vendor plugins move the same fact along several paths. Nobody can produce the list.

Why it happens. Each connector was the fastest way to meet one need. Point-to-point is cheap the first ten times.

What it costs. Undiagnosable incidents, unplannable upgrades, and money: the MACH Alliance's own February 2026 survey of 600 enterprise technology decision-makers found 37% citing integration complexity as a primary concern (18 February 2026).

At Kestrel & Pine (fictional). The February 2027 integration inventory counted **34 integrations**: 14 on the iPaaS, 9 agency scripts, 11 vendor-built connectors or plugins; **13 had no named owner**; only 3 had a reconciliation job, none for stock or orders;

four flows wrote stock into the platform, and the head of engineering's capability map found **six writers** of Duiven stock in all. The pre-order app that wrote inventory beside the WMS feed (March 2026) is the pattern at its smallest.

In public. Zalando's engineering blog described how its developer platform, built on Backstage, replaced "more than 100 disconnected interfaces" with one portal holding over 40,000 registered entities (3 August 2023).

Remedy direction. An integration inventory with an owner and a contract per flow, shared facts moving as events through one place, a reconciliation for every copy (Kestrel & Pine's W5).

Where this book helps: APIs, Events, Files and Middleware: How Commerce Systems Talk owns the styles and the inventory; The Back Office: ERP Integration and Who Is the Hub untangles the ERP's flows; Who Builds and Who Owns: Teams, Vendors, Agencies and Governance assigns the owners.

The Message in a Bottle

"We sent the order to the warehouse. If it didn't arrive, that's on them."

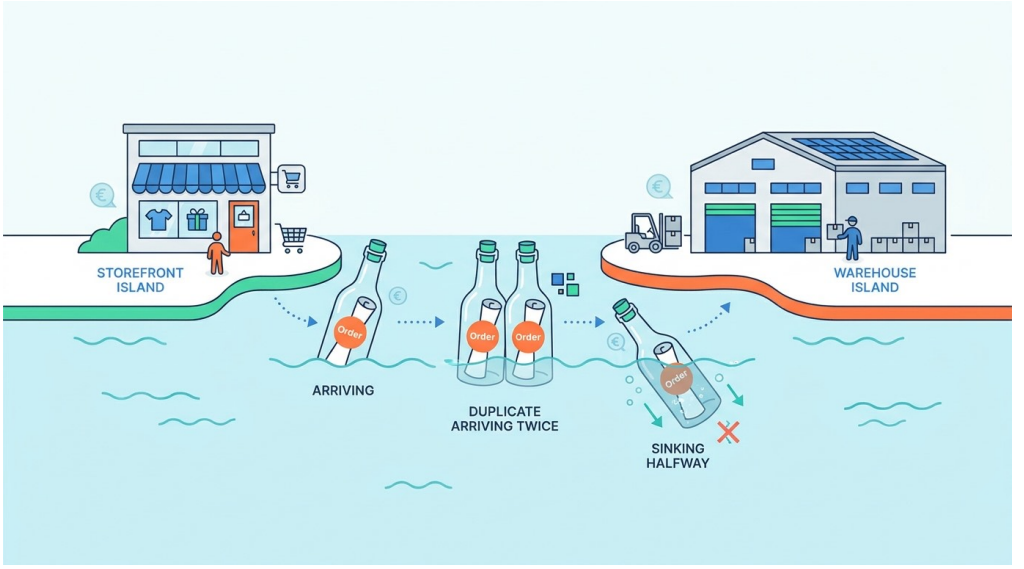


Figure 7: *The Message in a Bottle: messages between systems arrive once, twice, late or never, and a design that trusts them to arrive once finds the duplicates and gaps through customers.*

What it looks like. A retry creates a second order or refund; a dropped message leaves an order paid and never picked.

Why it happens. Delivery is at least once, sometimes twice, late or never, and vendors say so. Shopify's webhook documentation says an app "might receive the same webhook more than once", and retries an unanswered webhook 8 times over 4 hours before deleting the subscription; Stripe retries for up to three days; Mollie calls a webhook ten times over 26 hours and then stops (all n.d., accessed 2026-11). Integrations assume exactly once, because on a quiet day it holds.

What it costs. Duplicates, lost orders, drifting copies and peak backlogs. Zalando's engineering blog reported that its event-driven price and stock pipeline, reprocessing payloads of which "over 90%" were unchanged, delayed events "by up to 30 minutes" during Cyber Week and left consumers with "divergent versions of product data", before it was replaced by a synchronous read API for the critical path (7 March 2025).

At Kestrel & Pine (fictional). On Black Friday 2026 the iPaaS flow that created orders in the WMS one at a time was throttled by the WMS vendor's rate limit; the queue backed up for **four hours**, and roughly **8,000 orders** missed the next-day delivery the storefront had promised.

In public. Stripe's fulfillment guide: "You can't rely on triggering fulfillment only from your checkout landing page, because it's not guaranteed customers visit that page" (n.d., accessed 2026-11).

Remedy direction. An idempotency key from the business event on every message, a record of handled events, state fetched from its owner, a dead-letter queue, a daily reconciliation.

Where this book helps: Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns idempotency, outbox, dead letters and reconciliation; APIs, Events, Files and Middleware: How Commerce Systems Talk chooses the style; Taking the Money: Payment Providers, Authorization, Capture and Refunds applies the rules to money.

The Pulled Rug

"They announced it in November. The e-mail went to someone who left in 2025."

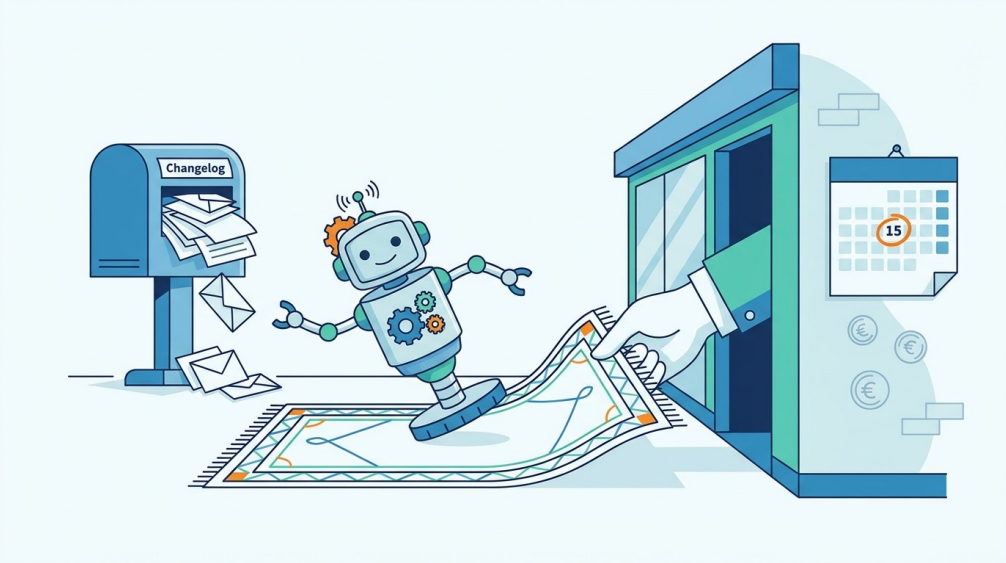


Figure 8: *The Pulled Rug: vendors announce their changes, but without a record of the dependency and someone reading the changelog, the integration breaks first and the news arrives later.*

What it looks like. A script keeps calling a retired version; chargebacks arrive before the new mapping; changelogs reach a mailbox nobody reads.

Why it happens. Vendors announce changes, but the merchant has no record of the dependency, no reader and no test. The contract exists on one side only.

What it costs. Stale stock, seller-account strikes, chargebacks, emergency rewrites. The calendars are public: Amazon's Marketplace Web Service was "no longer available" from 1 April 2024, with all private developers told to migrate to the Selling Partner API by 31 March 2024 (SP-API changelog, 2023–2024); bol's partner platform announced on 3 November 2026 that version 10 of its Offers API would be switched off on 1 February 2027.

At Kestrel & Pine (fictional). On 8 February 2027 the marketplace connector retired the API version a 2021 agency script used to push stock for the bol listings, announced in November 2026 to a contact who had left the agency in 2025. The script logged errors and kept running; stock to bol was stale for **two days**. In Q3 2026 Bergzicht Outdoor, the wholesale chain, deducted **€14,200** for despatch advices whose pallet labels did not match what arrived.

In public. Shopify's API versioning policy, quarterly releases with each version supported for at least 12 months and a nine-month overlap (n.d., accessed 2026-11), is one calendar among many.

Remedy direction. Pinned versions and sunset dates per integration, a role mailbox for changelogs, partner cards with owners, contract tests, despatch advices generated from

what was packed.

Where this book helps: Promises Between Systems: API Contracts, Webhooks and Versioning owns versions, sunsets and contract tests; Documents Between Companies: EDI, Punchout and E-Invoicing owns the partner card and the despatch advice.

The First Snowfall

"It worked fine all year. Then Friday happened."

Figure 9: *The First Snowfall: the busiest day is the first time the system and its integrations meet real load, and it is usually the joins, not the storefront, that give way.*

What it looks like. The storefront holds; the stock sync and the order queue do not.

Why it happens. Peak is 20 to 40 times an ordinary hour, and every design decision was made against the ordinary hour. Load tests, when they exist, test the storefront and not the joins.

What it costs. Oversells and missed promises on the day the brand is measured. Target's new site, launched in 2011 after it left Amazon's platform, crashed on 13 September 2011 under the Missoni launch and was "down most of the day", with a second 2.5-hour outage on 25 October (MPR News, 2011). Shopify's 2025 readiness post describes tests at 150% of the previous year's peak, reaching 200 million requests a minute (20 November 2025).

At Kestrel & Pine (fictional). Black Friday week 2026, 23–29 November: **31,000** online orders, **10.6%** of the year's; a peak hour on Friday of **2,300 orders**, 38 a minute, with storefront traffic about **41×** an ordinary hour. The platform held. The stock sync chain

and the WMS connector queue did not: the oversell above, a four-hour backlog, 8,000 late orders, a stacking bug worth about €52,000. Kestrel & Pine had no kill switch to disable guest card payments without a deploy.

In public. Shopify's 2025 Black Friday weekend peaked at $5.1 million a minute (2 December 2025); the merchant's own connectors are the merchant's to size.

Remedy direction. Test the joins at a multiple of last year's peak; switches to degrade without a deploy; a dated change freeze; a named runbook.

Where this book helps: Black Friday Every Year: Peak Load, Resilience and Recovery owns the peak plan, degradation and switches; Journey 4: Black Friday, Hour by Hour walks the 2026 Friday hour by hour and the 2027 plan; Pick, Pack, Ship: Warehouse Management and Fulfillment sizes the order queue.

The Giant's Boots

"Amazon does it with microservices. Zalando does it with 4,000 applications. So should we."

Figure 10: *The Giant's Boots: copying the architecture of a company a hundred times larger brings its costs without its reasons.*

What it looks like. A 14-person team runs a dozen composable vendors; a €48 million merchant builds a headless storefront. The giant's reasons are absent; the costs are not.

Why it happens. The giants publish good work, and vendors sell the shape that produces the most work. "Do we have that problem" is easy to skip.

What it costs. Cost and slower change without the benefit. The reversals are public: Segment described in 2018 growing to over 140 services, three engineers spending most of their time keeping them running, and consolidating back into one (10 July 2018); an Amazon Prime Video team reported in 2023 that a serverless monitoring design supported about 5% of expected load and that one process cut cost by 90% (InfoQ, 3 May 2023; the original post has been removed).

At Kestrel & Pine (fictional). In February 2027 the head of engineering received two pitches citing the Black Friday Oversell: an outside consultancy's composable rebuild replacing the platform (about **€3.2 million** over three years) and the agency's headless rebuild of the three storefronts on the current platform (about **€240,000** and nine months, plus a second runtime to operate). She declined both after measuring. Waypoint's W4, stay on the SaaS platform and surround it, is the decision not to wear the boots.

In public. Zalando's 2025 replacement of an event stream with a synchronous read API on its critical path (7 March 2025) is a giant narrowing a choice others copied from it.

Remedy direction. Count channels, warehouses, countries, teams and facts with two writers; add a capability when a measured problem demands it.

Where this book helps: Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says compares the styles with their reasons; Build, Buy or Assemble: Choosing and Changing Platforms works the stay-and-surround decision; What Changes at Each Scale: From First Store to Group gives the stage test.

The Burned Bridge

"Go-live is 19 November. The old platform is switched off the same day."

Figure 11: *The Burned Bridge: moving everything on one date with the old platform switched off leaves no way back when something goes wrong.*

What it looks like. Everything moves at once; the old platform is switched off to save a month's subscription; a contract sets the date, weeks before peak.

Why it happens. A big bang looks cheaper than running two systems, and each risk was fine in the demo.

What it costs. Lost sales, sometimes a reversal. Finish Line switched to new e-commerce site technology on 19 November 2012 and reverted to the former platform, "which it had kept operable", 17 days later, at a cost of about $3 million in quarterly sales (Retail Dive, 8 January 2013; dated).

At Kestrel & Pine (fictional). Waypoint replaced nothing on one date. The OMS came in seam by seam in 2027: a shadow copy of all orders in August, a parallel run on 1–14 September, marketplace orders on 15 September, D2C orders on 13 October, a change freeze around Black Friday, wholesale orders on 8 December, the old paths retired in January 2028. The PIM went live one storefront at a time.

In public. Target's 2011 launch weeks before its holiday season, and its two outages (MPR News, 2011), is a date chosen against the calendar.

Remedy direction. Migrate by seams, keep the old path operable until the new one has run a peak, run in parallel, rehearse rollback, design exit in at purchase.

Where this book helps: Moving Without Burning the Bridge: Replatforming and Migration owns seams, parallel runs and rollback; Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says explains the strangler pattern; Case: commercetools, Shopware and the Composable Movement reads the public replatforming record.

The Forked Platform

"We can't take the upgrade. The theme and the checkout changes would all have to be redone."

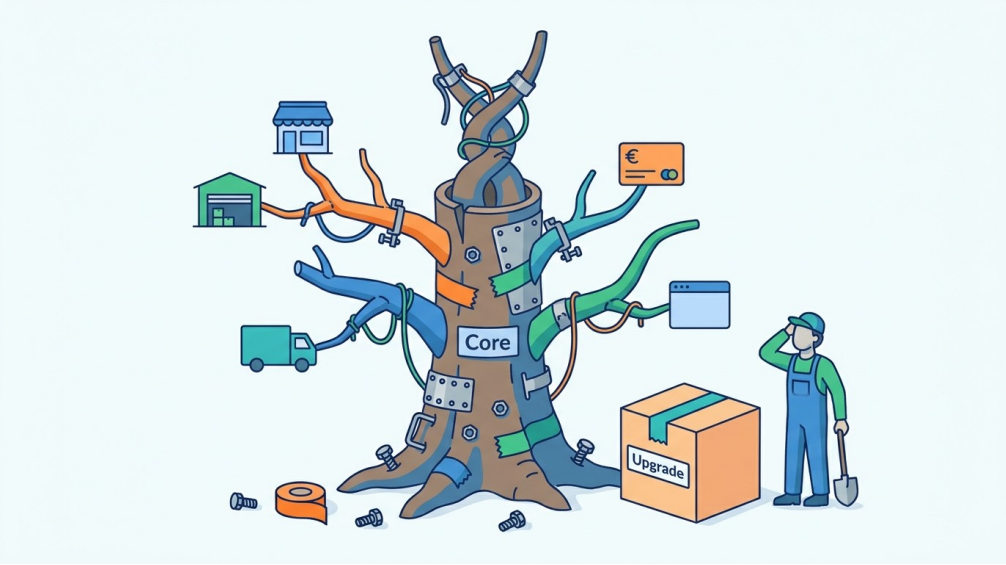


Figure 12: *The Forked Platform: customize the core deeply enough and every vendor upgrade becomes a project.*

What it looks like. Agency edits to files the vendor has since replaced; scripts inside the checkout; apps nobody uses.

Why it happens. Each customization was the fastest route, and nobody owns upgradeability.

What it costs. Lost features, security exposure, a forced migration on the vendor's date. Adobe Commerce gives each release line three years of standard support, and from 1 June 2027 cloud environments on unsupported versions "may be taken offline" (lifecycle policy, accessed 2026-11). Shopify's `checkout.liquid` customizations stopped working on 13 August 2024 (announced 13 February 2023), and Shopify Scripts stopped executing on 30 June 2026 (changelog, 9 April 2026).

At Kestrel & Pine (fictional). The storefront theme has been customized by the agency since 2017; the checkout has not, which is one reason the decision to stay on the platform (W4) is possible. The **41 installed apps**, of which 22 were installed by the agency, 6 are unused and 3 wrote inventory, are the smaller fork: the April 2027 app governance rule gives every app a named owner and takes away the right to write stock.

In public. The sunset calendars above.

Remedy direction. Upgradeability with an owner; supported extension points only; a register of customizations and apps; upgrades budgeted as routine.

Where this book helps: Apps, Plugins and Functions: Extensibility and Ecosystem Governance owns extension points, app governance and the register; Delivering the Storefront: Themes, Headless, Rendering and Performance separates theme from core; From Cart to Confirmation: Checkout Orchestration keeps the checkout unforked.

One Size Fits None

"All three brands go on the group platform next year. The business case is 'synergies'."

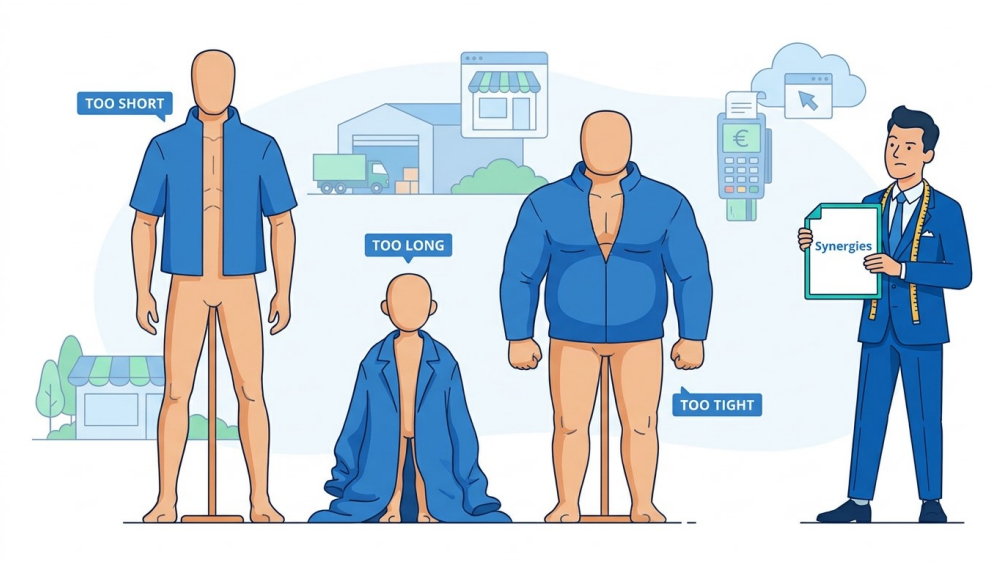


Figure 13: *One Size Fits None: forcing different brands onto one stack before anyone measures the benefit, or keeping them apart when they should share, decides on cost and misses the operational differences.*

What it looks like. One tenant serves brands with different catalogs, rules and peaks, and every change is a negotiation; or two brands run two of everything because nobody measured what sharing would save.

Why it happens. The decision is made on cost, and the operational differences are invisible in the business case.

What it costs. Sharing: slower change, one brand's peak degrading another. Separation: duplicated contracts and integrations.

At Kestrel & Pine (fictional). Three regional storefronts share **one tenant**, a merchant account on the hosted platform. They share a catalog, warehouse, season and €199 jacket price, so the arrangement fits.

A second brand, fourth country or revenue near €200 million would trigger a review. None is planned for 2027–2028. The company defers the group-sharing decision to 2029 rather than choosing a structure before knowing the requirements.

In public. Zalando completed its acquisition of ABOUT YOU on 11 July 2025 as a dual-brand consumer business "with shared logistics and payment infrastructure", with group synergies targeted at around €100 million a year (corporate.zalando.com, 11 July 2025).

Remedy direction. Decide sharing per capability: share what has scale economics and the same rules (payments, logistics), keep apart what differs by brand, and write the benefit as a number first.

Where this book helps: Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation owns tenancy and isolation; What Changes at Each Scale: From First Store to Group names the Group-stage signals; Case: Zalando, Microservices, Platform Strategy and the Return to Discipline reads a group's sharing decisions.

The Lynchpin

"Only the EDI consultant understands the mappings. He's on holiday."

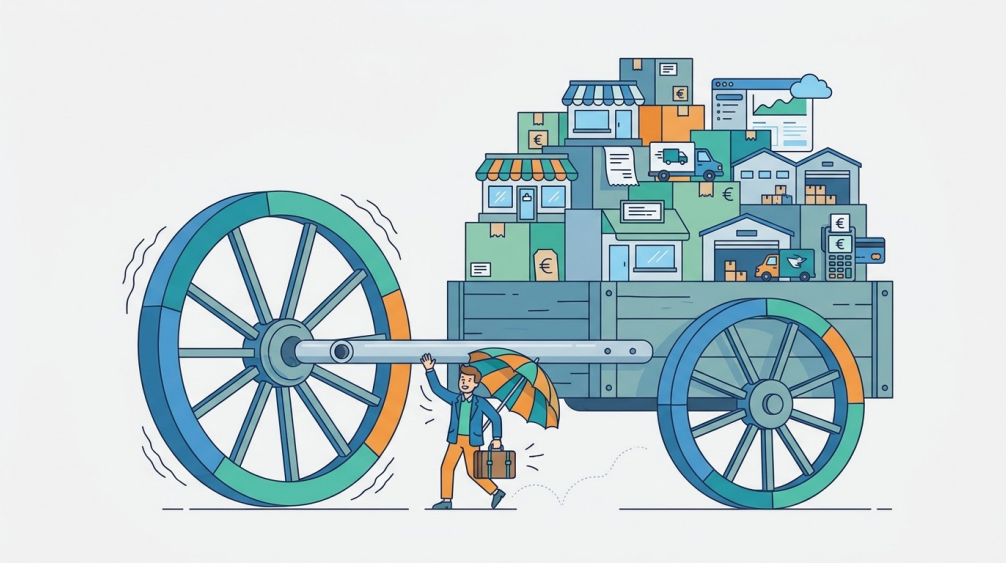


Figure 14: *The Lynchpin: when a capability customers rely on runs through one vendor, connector, agency or person, the whole cart depends on one pin.*

What it looks like. A connector documented nowhere; mappings in one consultant's head.

Why it happens. Specialists are efficient, and an exit plan has no visible return until the day it is needed.

What it costs. Outages as long as one person's absence, and programs that end on the vendor's date: Shopify sold its logistics business to Flexport in a deal that closed on 6 June 2023, and Zalando's Connected Retail program, through which physical shops sold branch stock on its platform, ends by the end of June 2027 (TextilWirtschaft, 24 April 2026).

At Kestrel & Pine (fictional). The EDI mappings for Bergzicht Outdoor, worth €2.1 million a year, were built in 2025 by one external consultant; when despatch advices and pallet labels disagreed, the fix waited on him. The ERP partner's platform-to-ERP connector was undocumented and could be retired only after the lead integration engineer wrote its contract in Q3 2027. The February 2027 ownership map found **5 integrations with no named owner**, among them the scripts behind the Deprecated Endpoint and the Price Drift.

In public. Connected Retail's end (announced April 2026, effective June 2027) is a channel retired by the platform that owned it.

Remedy direction. An ownership map for every system, integration and app; contracts in the merchant's hands; a second person per mapping; exit cost assessed at purchase.

Where this book helps: Who Builds and Who Owns: Teams, Vendors, Agencies and Governance owns the ownership map; Build, Buy or Assemble: Choosing and Changing Platforms prices exit at purchase; Documents Between Companies: EDI, Punchout and E-Invoicing puts the partner knowledge on a card.

The Long Fuse

"Belgium wants structured invoices from January. It's October."

Figure 15: *The Long Fuse: a legal deadline found only months before it lands leaves too little time to meet it.*

What it looks like. Finance learns of a mandate from a newsletter; the storefront has never had an accessibility audit.

Why it happens. Rules land per topic and per country, nobody in a 14-person team reads the Official Journal, and each rule arrives without an owner.

What it costs. Fines, blocked invoices, emergency prices. The dates are public: Belgium's structured B2B e-invoicing obligation applied from 1 January 2026, announced 18 September 2024, with fines of €1,500, €3,000 and €5,000 for lacking the technical means (einvoice.belgium.be); the European Accessibility Act applied to e-commerce services from 28 June 2025 (Directive (EU) 2019/882); PCI DSS requirements 6.4.3 and 11.6.1 on payment-page scripts became mandatory on 31 March 2025; Germany has required businesses to receive e-invoices since 1 January 2025.

At Kestrel & Pine (fictional). The fuse was caught once: Peppol e-invoices have gone to Belgian business customers since January 2026, issued and numbered in the ERP through the EDI provider as access point. The 2027 regulatory calendar, kept by the head of engineering with the finance director, turns each remaining rule (accessibility of the three storefronts and the checkout widgets, the Dutch e-invoicing plan for 2030, packaging rules, the product passport) into a dated requirement with an owner.

In public. The Belgian mandate, announced September 2024 for January 2026: fifteen months of notice, then fines.

Remedy direction. A regulatory calendar with a named owner, listing each rule's countries, date and systems; vendor conformance as a procurement item.

Where this book helps: Rules Turned Into Requirements: Consumer, Payment,

Invoicing and Platform Regulation owns the calendar; Documents Between Companies: EDI, Punchout and E-Invoicing and Tax, Invoices and the Books: From Order to Ledger implement e-invoicing; Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers covers the PCI requirements.

The Secret Agent

"Something changed 600 stock levels on Tuesday night. We think it was the pre-order app."

Figure 16: *The Secret Agent: a script, app or AI agent that writes orders, stock, prices or money with no owner, no identity of its own and no approval changes the business in the dark.*

What it looks like. Scripts and apps share one admin's API key and write beside the official feeds.

Why it happens. Apps are easy to add, hard to see and broadly permitted by default; AI agents arrive through the same door with more autonomy.

What it costs. Phantom stock, price drift and, at worst, a compromised checkout: Sansec reported on 25 June 2024 that the `cdn.polyfill.io` script, embedded by more than 100,000 sites, had been sold in February 2024 and was injecting malware (Sansec, 2024). Visa reported 31 issuers and 4 merchants live with agent-initiated transactions in Europe (2 July 2026), and the agent protocols (Agentic Commerce Protocol releases through April 2026; Universal Commerce Protocol, launched 11 January 2026) carry an identity and a mandate for the agent.

At Kestrel & Pine (fictional). In March 2026 a pre-order app adjusted inventory in the platform while the WMS feed also wrote it: two weeks of phantom reservations on **600 variants**. In October 2026 the agency's price-import script failed on the row that restored the German price, logged the error and reported success. Waypoint's answer is W5 (every writer named) and W8 (agent readiness, not agents: accurate product data, availability and order status by API first, no shopper-facing AI agent in 2027).

In public. Polyfill.io (June 2024) is checkout code changing hands unseen.

Remedy direction. Every writing actor with an owner, its own credentials and stated rights; least privilege; human approval over a threshold; a person at the gate for AI agents.

Where this book helps: Apps, Plugins and Functions: Extensibility and Ecosystem Governance owns app governance and permissions; Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers covers API keys and script monitoring; Assistants and Agents: AI in Commerce and What Is Real in 2026 covers agents, mandates and W8.

Where the Patterns Strike

Across the four scale stages of What Changes at Each Scale: From First Store to Group, most patterns are born at Growing and paid for at Omnichannel (the author's judgement, not a survey).

Pattern	Starting	Growing	Omnichannel	Group
The Phantom Stock	Rare: one channel, one stock	**Born:** second channel, first sync	**Hurts:** stores, marketplaces, own warehouse	Per brand and shared pool
Schrödinger's Order	Rare: the platform is the record	**Born:** the first 3PL or ERP copy	**Hurts:** orders in four systems	Group reporting disagrees by brand
The Familiar Stranger	Rare: one customer table	**Born:** the first store or marketplace	**Hurts:** service and loyalty blocked	Cross-brand identity
The Boomerang	Rare: a refund button	**Born:** the first returns portal or 3PL	**Hurts:** returns from portal, stores, marketplaces	Shared reverse logistics

Pattern	Starting	Growing	Omnichannel	Group
The Price Nobody Can Explain	Rare	**Born**: wholesale price list, first campaign import	**Hurts**: five price copies, parity rules	Group price governance
The Spaghetti Junction	Absent	**Born**: connectors one at a time	**Hurts**: 30-plus flows, unknown writers	Platform team or chaos
The Message in a Bottle	Hidden by the platform	**Born**: the first custom integration	**Hurts**: at peak and in refunds	Event platform discipline
The Pulled Rug	Vendor handles it	**Born**: first marketplace or partner API	**Hurts**: many calendars, one mailbox	Contract management function
The First Snowfall	Platform absorbs it	**Born**: first big campaign with integrations	**Hurts**: joins fail at 40× load	Shared peak across brands
The Giant's Boots	**Born**: the wrong platform category	Agency-driven over-architecture	**Hurts**: composable estate for a small team	Appropriate at last, with reasons
The Burned Bridge	Low cost to migrate	**Born**: first replatforming	**Hurts**: migration touches a dozen systems	Brand-by-brand migrations
The Forked Platform	Rare	**Born**: theme and checkout edits, apps	**Hurts**: upgrades become projects	Per-brand forks
One Size Fits None	Absent	Rare	**Born**: second country or line in one tenant	**Hurts**: brands forced together or apart

Pattern	Starting	Growing	Omnichannel	Group
The Lynchpin	The founder	**Born**: the agency, the one consultant	**Hurts**: EDI, ERP connector, unowned flows	Vendor concentration
The Long Fuse	Platform defaults, per country	**Born**: second country, first B2B mandate	**Hurts**: many rules, many countries	Compliance function
The Secret Agent	Apps with wide permissions	**Born**: scripts and apps that write	**Hurts**: six writers of stock	Agents at scale

Kestrel & Pine's April 2027 stage map attributes eight of its ten incidents to Growing-stage designs still running in 2026. The remedy is rarely a new platform; it is ownership, contracts and reconciliation.

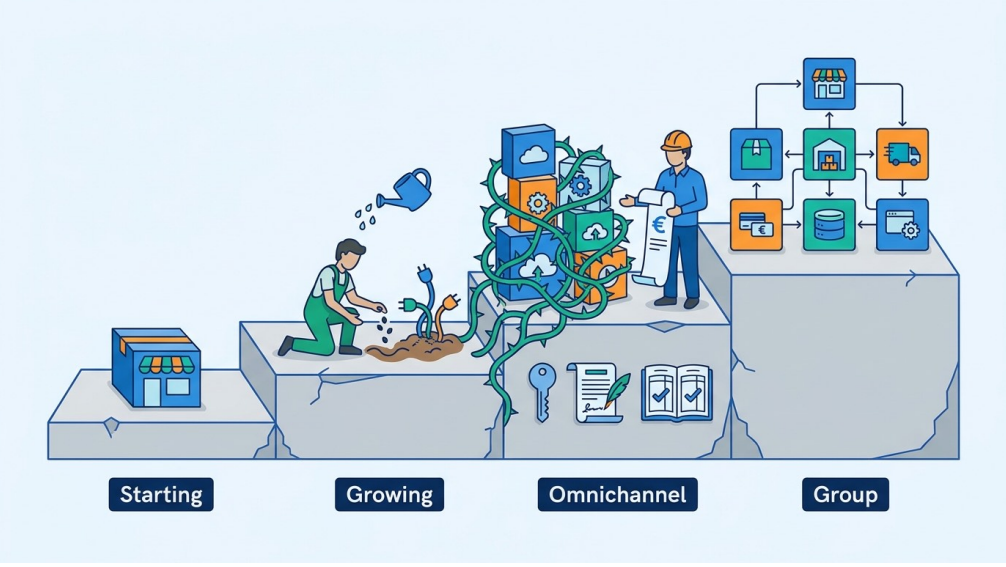


Figure 17: *Most failure patterns are born at the Growing stage and paid for at the Omnichannel stage, and the remedy is rarely a new platform; it is ownership, contracts and reconciliation.*

How the Patterns Reinforce Each Other

The integration chain. A Spaghetti Junction produces Messages in a Bottle, and those produce the Phantom Stock, Schrödinger's Order, the Boomerang and, through a price import, the Price Nobody Can Explain. The First Snowfall makes all of them visible in one evening.

The governance chain. A Lynchpin sets up the Pulled Rug: nobody reads the calendar for a flow nobody owns. The Secret Agent is a Lynchpin with write access; the Long Fuse is a deadline with no owner.

The platform chain. The Giant's Boots lead to the Forked Platform or the Burned Bridge. **The identity thread**, the Familiar Stranger, runs under all of them.

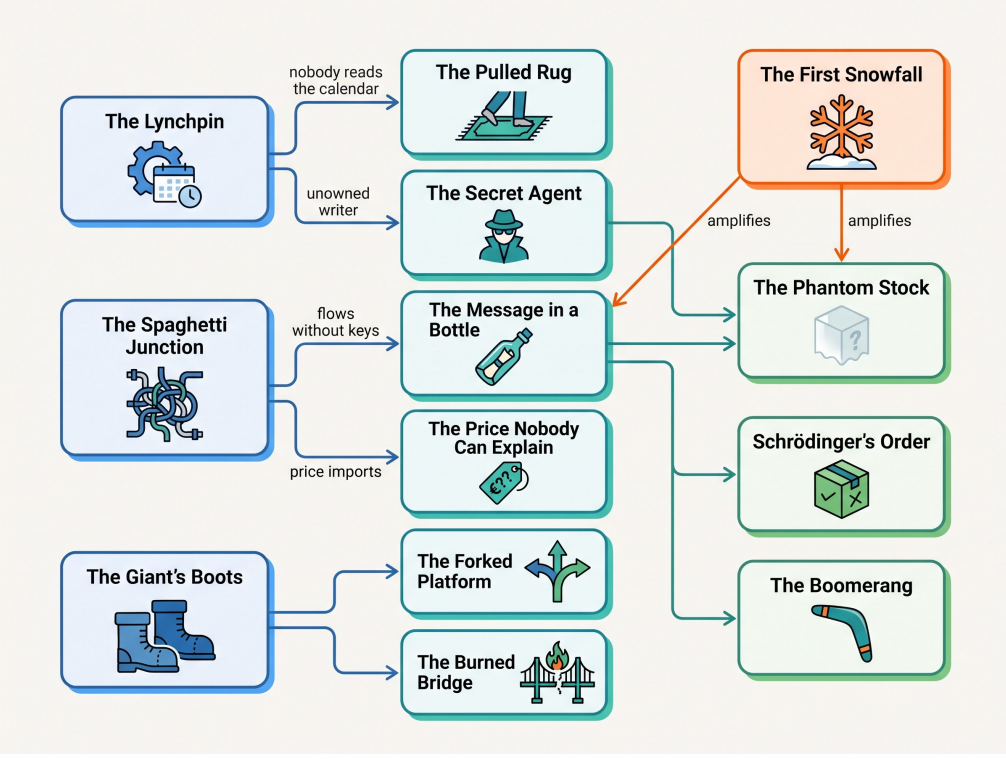


Figure 18: *The patterns travel in chains: the integration chain runs from the Spaghetti Junction through the Message in a Bottle to phantom stock, undecidable orders and double refunds; the governance chain runs from the Lynchpin to the Pulled Rug and the Secret Agent; the platform chain runs from the Giant's Boots to the fork or the burned bridge; the First Snowfall amplifies whatever is already there.*

Of Kestrel & Pine's ten incidents, eight are a fact with two writers; two are what the outside world does to a landscape nobody watches. So Waypoint decides ownership

first, then adds the hubs, then makes the copies trustworthy, and keeps the platform.

Using the Patterns

In design reviews. Walk one order through the design with the step table of Journey 1: Eva's Jacket, From Search to Doorstep and ask at every handoff which pattern could occur and what prevents it.

In vendor demos. Ask to see the pattern, not the feature: a split shipment's status, a refund sent twice, what an app may write.

In due diligence. Ask for the integration inventory, the writers of stock, prices and money, the version register, the regulatory calendar and the last reconciliation reports. Each missing document is a pattern with no owner.

The next page introduces the merchant this book follows: its business lines, systems, people, and the ten incidents above as they were lived. Continue with Meet Kestrel & Pine.

Sources and Evidence

Company, product and investor sources

- Draft order and transfer shipment inventory is moving from reserved to committed — Shopify developer changelog, 2026-08-05; a stock state's meaning changing under its readers.
- Order status and Customer notifications — Shopify Help Center, n.d., accessed 2026-11; separate status families; no tracking number, no updates.
- Customer accounts — Shopify developer documentation, n.d., accessed 2026-11; legacy customer accounts deprecated as of 2026-02-26.
- Returns apps — Shopify developer documentation, n.d., accessed 2026-11; return versus reverse fulfillment order; dispositions.
- Webhooks and HTTPS delivery — Shopify developer documentation, n.d., accessed 2026-11; duplicate and unordered deliveries; 8 retries over 4 hours.
- API versioning — Shopify developer documentation, n.d., accessed 2026-11; quarterly versions, 12 months of support, nine months of overlap.
- The checkout.liquid theme file is being deprecated — Shopify changelog, 2023-02-13; the 2024-08-13 sunset.
- Shopify Scripts deprecation — Shopify developer changelog, 2026-04-09; script editing ends 2026-04-15, execution 2026-06-30.
- BFCM readiness 2025 — Shopify Engineering, 2025-11-20; load tests at 150% of the previous peak, 200 million requests per minute, 80,000+ checkouts per minute.

- Black Friday Cyber Monday 2025 data — Shopify, 2025-12-02; $14.6 billion and $5.1 million per minute.
- Shopify completes sale of Shopify Logistics to Flexport — Shopify, 2023-06-06; a vendor exiting a capability.
- Inventory — commercetools API documentation, n.d., accessed 2026-11; the 10-second availability consistency window.
- Refunds, Webhooks and Fulfill orders with Checkout — Stripe documentation, n.d., accessed 2026-11; double-refund risk, three-day retries, fulfillment from webhooks.
- Refunds and Webhooks — Mollie documentation, n.d., accessed 2026-11; duplicate blocking, "do not retry automatically", ten retries over 26 hours.
- Important Amazon MWS removal dates — Amazon SP-API changelog, 2023–2024; MWS unavailable from 2024-04-01.
- bol partner platform — bol, 2026-11-03; Offers API v10 switched off 2027-02-01.
- From event-driven to API: Zalando's price and stock pipeline — Zalando Engineering, 2025-03-07; 30-minute event delays at Cyber Week, divergent copies, the synchronous read API.
- Sunrise: Zalando's developer platform based on Backstage — Zalando Engineering, 2023-08-03; "more than 100 disconnected interfaces" replaced.
- Zalando and ABOUT YOU successfully complete transaction — Zalando, 2025-07-11; shared logistics and payment infrastructure, about €100 million of synergies.
- Zalando stellt Connected Retail ein — TextilWirtschaft, 2026-04-24; the program ends by the end of June 2027.
- Goodbye Microservices: from 100s of problem children to 1 superstar — Twilio Segment, 2018-07-10; 140+ services consolidated into one (dated).
- Prime Video switched from serverless to EC2 and ECS to save costs — InfoQ, 2023-05-03; 90% cost reduction; the original post has been removed.
- MACH Alliance research: 6x more organizations achieve AI ROI with a composable foundation — MACH Alliance, 2026-02-18; 37% cite integration complexity as a primary concern (600 respondents).
- Adobe Commerce lifecycle policy — Adobe, accessed 2026-11; three years of standard support; cloud enforcement from 2027-06-01.
- Target's website crashes again — MPR News (AP), 2011-10-25; the Missoni crash of 2011-09-13 and the second outage (dated).
- Finish Line pulls a U-turn online after losing $3 million — Retail Dive, 2013-01-08; the 17-day reversal (dated).
- Visa press release on agentic transactions in Europe — Visa UK, 2026-07-02; 31 issuers and 4 merchants live.
- Universal Commerce Protocol announcement — Google, 2026-01-11; launch of the agent checkout protocol.
- Agentic Commerce Protocol — OpenAI and Stripe, specification releases through 2026-04-17.

- Polyfill supply chain attack hits 100K+ sites — Sansec, 2024-06-25; the script domain sold in February 2024.
- How to automate WISMO requests — Gorgias, updated 2023-11-13; WISMO as 18% of requests.

Standards, regulation and official documentation

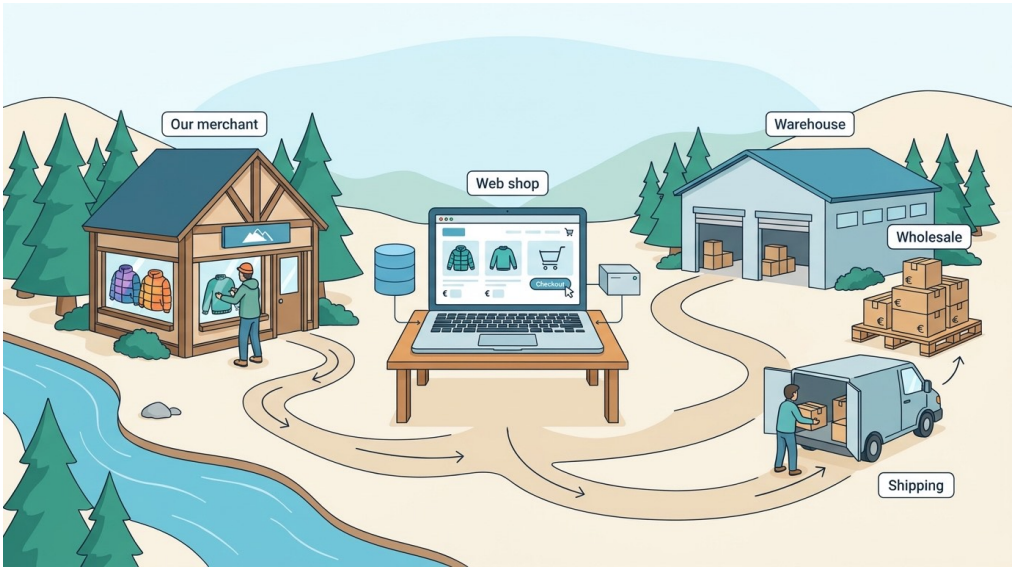
- Directive (EU) 2019/2161 (Omnibus) — Official Journal, 2019-12-18; the 30-day prior-price rule, applied from 2022-05-28.
- ACM wil einde aan nepkortingen na komst strengere regels — Authority for Consumers and Markets, 2022-12-06; supervision from 2023-01-01.
- Regulation (EU) 2016/679 (GDPR) — Official Journal, 2016-05-04; Article 12(3), one month to answer a data-subject request.
- Directive (EU) 2019/882 (European Accessibility Act) — Official Journal, 2019-06-07; application from 2025-06-28.
- Structured electronic invoices between companies are compulsory from 2026 and FAQ — FPS Finance Belgium, 2024-09-18 and n.d., accessed 2026-11; the mandate and the fines.
- E-Rechnung FAQ — German Federal Ministry of Finance, accessed 2026-11; receiving since 2025-01-01, issuing transition to end 2027.
- Guidance for PCI DSS e-commerce requirements effective after 31 March 2025 — PCI Security Standards Council, 2025; mandatory from 2025-03-31.

Research and practitioner sources

- Are inventory records inaccurate? Evidence from retail — DeHoratius and Raman, *Management Science* 54(4), 2008; 65% of records inaccurate (dated).
- 2025 Retail Returns Landscape — NRF with Happy Returns, 2025-10-15; 19.3% online return rate, 9% fraudulent.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Meet Kestrel & Pine



IN THIS SECTION, YOU WILL: Meet Kestrel & Pine, the fictional outdoor brand this book follows, its systems, its people, its 2026 incidents, the Waypoint program and the recurring order every chapter returns to.

KEY POINTS:

- **Kestrel & Pine is a fictional composite, not a disguised company**. It borrows the shape of many growing omnichannel brands so the book can show real tensions with consistent numbers. Every name, price, figure and incident is invented and recorded in one canonical ledger.
- **Four channels, thirteen systems, no order hub and no product hub**. Orders live in four systems, product data in four, stock in four, and the chain that copies stock between them takes up to two hours. Nearly every incident on this page is that gap, seen from a different side.
- **Waypoint decides ownership first and buys the hubs second**. The 2027 program names a system of record for stock, orders and product content, adds an availability service, an order management system and a product information system beside the existing platform, and sets integration standards, without replatforming.

The worked examples follow Kestrel & Pine, a fictional outdoor brand that grows from an online shop into a business with stores, wholesale customers and marketplaces. Keeping its catalog, people, systems and incidents consistent lets you see how one architectural decision affects the next. This page supplies the full reference; individual chapters repeat the facts their examples need. Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong names the recurring failure patterns.

The company combines common growth patterns: an online shop adds stores, wholesale, marketplaces and its own warehouse, connecting each new system as the need arises. It does not portray a real retailer. Systems are named by role to keep the discussion platform-neutral. Named marketplaces and carriers are real; their dealings with Kestrel & Pine are invented.

Use the tables to look up a figure, person, incident or planned change. Terms are explained locally, with links to the chapters that examine them.

Kestrel & Pine at a Glance

Kestrel & Pine B.V. is an outdoor apparel and equipment brand founded in **Arnhem in 2016** by **Mara Lindqvist and Teun Baas**. They retain majority ownership; fictional investor **Veluwe Capital** has held a minority since **2022**. Own-brand clothing and equipment produce about **75%** of revenue; third-party equipment accounts for **25%**. The company sells in euros in the Netherlands, Belgium since 2020 and Germany since 2022.

In **2026**, revenue is **€48.0 million**, gross margin **54%**, and earnings before interest, tax, depreciation and amortization (**EBITDA**) about **€3.4 million**. Of **310 employees**, 96 work in stores, 88 in the warehouse and 126 at head office; **14** work in technology. Annual technology spend is about **€2.6 million**, or **5.4% of revenue**.

Four business lines share the company but place different demands on its systems. The table is the baseline for the examples; Who Sells What to Whom: Business Models and the Economics That Shape Architecture explains the differences.

Line	Revenue 2026	Orders or transac-tions	Average value	Notes
Online D2C storefronts (NL, BE, DE)	**€27.0M** (NL €18.5M, DE €5.5M, BE €3.0M)	**293,000 orders**	**€92**	Free shipping over €75; "ordered before 22:00, delivered tomorrow" in NL
Wholesale (B2B)	**€12.0M**	**9,400 orders**	**€1,277**	**1,100 active accounts** (NL 640, BE 210, DE 250); 30-day payment terms
Stores	**€6.0M**	**118,000 transac-tions**	**€51**	Six stores
Marketplaces	**€3.0M**	**43,000 orders**	**€70**	bol, Amazon.de, Zalando through a channel connector

Direct-to-consumer (D2C) means selling through the brand's own storefronts. **Business-to-business (B2B)** means selling to other companies, here shops that resell the goods. A €51 store receipt, €92 web order and €1,277 wholesale order can contain the same jacket but need different prices, payment terms and fulfillment.

How Kestrel & Pine Grew, 2016–2027

The history matters because every stage left a system behind, chosen for the stage it was bought in. The book describes four **scale stages** (Starting, Growing, Omnichannel, Group; What Changes at Each Scale: From First Store to Group owns them), and Kestrel & Pine has passed through three.

Years	What happened	Stage
2016–2018	Founded in Arnhem; one Dutch web shop on a hosted SaaS commerce platform (software as a service: rented by the month and run by the vendor); one payment provider; orders shipped by a third-party logistics company (3PL, a warehouse that stores and ships on a merchant's behalf) in Tiel; an agency retained for the theme and apps from 2017.	1 — Starting
2019–2022	First store in Arnhem (2019) on a cloud point-of-sale system (POS, the till software that records store sales) from a different vendor; Belgium (2020); the Utrecht store, the marketplace connector, the integration platform and GTINs on every own-brand item (2021); Germany, Veluwe Capital's investment and Ruben Alves as lead integration engineer (2022).	2 — Growing
2023–2026	The Duiven warehouse on its own WMS, the ERP and the platform's enterprise tier with B2B features (2023); Amsterdam, Nijmegen and Eindhoven stores and the data warehouse (2023–24); Antwerp, EDI with Bergzicht Outdoor and the returns portal (2025); Peppol e-invoicing, the Utrecht click-and-collect pilot and the Black Friday Oversell (2026).	3 — Omnichannel
2027	Noor Visser joins as Head of Engineering (January); Waypoint is approved (March).	3 — Omnichannel

The revenue bands behind the stages are rough markers (Starting under about €2 million, Growing to about €20 million, Omnichannel to about €200 million, Group beyond that or several brands). The *shape* decides the stage: Kestrel & Pine became Omnichannel in 2023, when it had stores, wholesale, marketplaces, three countries and its own warehouse at the same time, and a dozen systems to join. The joins are where its problems now live.

The Catalog and the Products

Kestrel & Pine sells **3,400 products** in **26,000 sellable variants**. A product is "the Ridgewalker shell jacket, women's"; a variant is one size in one color of it. The book calls a sellable variant a **SKU** (stock-keeping unit, the thing you can count on a shelf and put in a parcel). About **17,000 variants are active** at any time; two main seasons a year plus carry-over lines bring about **1,200 new variants** per season. Every own-brand SKU has carried a **GTIN** (Global Trade Item Number, the number behind the EAN-13 barcode) since 2021; third-party SKUs use their brand's GTIN.

The hero product is the **Ridgewalker** shell jacket, in women's and men's cuts, six sizes and four colors including "moss". A few prices recur through the book, all fictional:

Item	Consumer price (incl. VAT)	Wholesale price (excl. VAT)	Landed cost
Ridgewalker shell jacket	**€199**	**€95**	**€58**
Trailhead fleece	€119	€57	
Merino hiking socks	**€18**	**€8.50**	

Where the product data lives is the first sign of trouble. In January 2027 it lives in **three places**, the platform (storefront content and variants), the ERP item master (purchasing and costs) and shared spreadsheets (seasonal enrichment and translations), with a **fourth copy** in the marketplace connector for listings. There is **no PIM** (product information management system: one place where product content and attributes are kept and published to every channel). What Are We Selling? Products, Variants and the Catalog works this out.

Channels and Customers

Online. Three localized storefronts (the Netherlands; Belgium in Dutch and French; Germany) run in **one tenant** of the platform, meaning one account on the vendor's shared service with one configuration, theme and set of apps. The Dutch storefront promises next-day delivery for orders before **22:00**, a promise written into the page rather than computed from the warehouse's state. Shoppers pay by iDEAL, card, PayPal, Klarna, Bancontact or Apple Pay.

Stores. Six stores, each holding about **1,800 variants**: Arnhem (2019, the flagship next to head office), Utrecht (2021; click-and-collect pilot since 2026, meaning order online and pick up in store), Amsterdam and Nijmegen (2023), Eindhoven (2024) and Antwerp

(2025, where Belgian fiscal receipt rules apply). Store stock is tracked in the POS and copied to the ERP **nightly**. There is **no link between the POS and the platform**, so the click-and-collect pilot is checked by phone and e-mail. Store card payments go through a separate terminal provider (Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment).

Wholesale. **1,100 active accounts**, mostly independent outdoor shops. **62%** of their orders arrive through the platform's B2B portal (company accounts, price lists, payment terms); **38%** arrive by e-mail or phone and are keyed into the ERP by the wholesale team (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals).

The largest account is **Bergzicht Outdoor**, a fictional Dutch outdoor chain with **38 shops** and a distribution center in **Zwolle**, worth **€2.1 million** a year. Since 2025 Bergzicht orders by **EDI** (electronic data interchange, the long-established standard for exchanging business documents such as orders and invoices between companies' systems without anyone typing them).

Belgian business customers have received **Peppol** e-invoices (a European network for structured electronic invoices) since January 2026 (Documents Between Companies: EDI, Punchout and E-Invoicing).

Marketplaces. **bol** (NL and BE), **Amazon.de** and **Zalando** (DE and NL), all through **the marketplace connector**, a channel integration service that publishes listings, pulls orders and pushes stock levels. About **40%** of marketplace units for Germany are fulfilled by the marketplace's own fulfillment program from stock Kestrel & Pine sends there in advance, a location outside its control (Selling on Someone Else's Platform: Marketplaces and Channel Integration).

The Volumes Every Design Is Sized Against

Chapters quote these 2026 figures when they size a queue, a sync interval or a peak plan. All are illustrative.

Measure	2026 value
Online D2C orders	**293,000** (about 800 a day on average)
Marketplace orders	**43,000**
Wholesale orders	**9,400**
Store transactions	**118,000**
Active variants	about **17,000** of 26,000
Shipped from Duiven	**336,000** parcels plus **9,400** wholesale shipments; **1.9 million units**

Measure	2026 value
Returns	**72,000** parcels; item return rate **24%**, order return rate **19%** on D2C; **28%** of returns are exchanges; **61%** of returned items resold as new
Black Friday week (23–29 November)	**31,000** D2C orders, **10.6%** of the year's
Peak hour (Friday 27 November, 20:00–21:00)	**2,300** orders, **38 per minute**; storefront traffic about **41×** an average hour
Stock sync delay at peak	up to **two hours** from a sale to the stock shown elsewhere; the WMS connector queue backed up for **four hours**
Installed platform apps	**41**, of which **6** unused and **3** write to inventory
Integrations	**34** (point-to-point and iPaaS flows)

Two numbers deserve a pause. A 24% item return rate means one item comes back for every four shipped, so the return path is a main road, not a side street (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics). And the peak hour is 41 times an ordinary one: a design that works at 800 orders a day meets 2,300 in one hour, once a year, which is what the pattern **The First Snowfall** warns about (Black Friday Every Year: Peak Load, Resilience and Recovery).

The Systems Landscape in January 2027

When Noor Visser arrives in January 2027, this is what she finds. Each system is named by its role.

Area	System	Since	Notes
Storefront, cart, checkout, B2B portal	**the platform**, a hosted SaaS commerce platform, enterprise tier since 2023	2016	Three storefronts in one tenant; theme customized by the agency; 41 apps

Area	System	Since	Notes
Finance and back office	**the ERP** (enterprise resource planning system: the books, purchasing, inventory valuation, invoicing, item master, manual wholesale order entry)	2023	The platform-to-ERP connector was built by the ERP partner
Warehouse	**the WMS** (warehouse management system: receiving, picking, packing and shipping at Duiven)	2023	Orders in from the platform, stock out to the ERP, via the iPaaS
Order management	**none**	—	Orders live in the platform, the connector, the ERP and the POS
Product information	**none**	—	Product data in the platform, the ERP, spreadsheets and the connector
Marketplaces	**the marketplace connector**	2021	Listings, orders, hourly stock sync
Stores	**the POS**	2019	Store stock to the ERP nightly; no link to the platform
Returns	**the returns portal** (SaaS)	2025	Labels, reasons, exchanges; refunds through the platform's API
Shipping	**the shipping service** (multi-carrier labels and tracking)	2023	Inside the WMS flow

Area	System	Since	Notes
Payments	**the PSP** (payment service provider: the company that takes the shopper's payment and settles the money to the merchant)	2016	A separate terminal provider for stores since 2019
EDI	**the EDI provider** (a value-added network that relays EDI documents between trading partners)	2025	The order, despatch advice and invoice messages (ORDERS, DESADV, INVOIC) with Bergzicht; Peppol e-invoices
Integration	**the iPaaS** (integration platform as a service: a hosted tool for building and running the flows that move data between systems) plus agency scripts and vendor connectors	2021	34 integrations; owned by Ruben Alves
Data	**the data warehouse** and a BI tool (business intelligence: reports and dashboards)	2024	Three definitions of "order" in use

Two missing systems explain much of the difficulty. An **order management system (OMS)** would coordinate orders from every channel, with shared payment, fulfillment, delivery and return states. A **product information management system (PIM)** would own descriptions and attributes.

Instead, orders begin in the storefront, marketplace connector, ERP or store till, and product descriptions are maintained four times. Buying a useful system for each new need has left the company without shared ownership (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record, The Order Record: Lifecycle, State and the Order Management System).

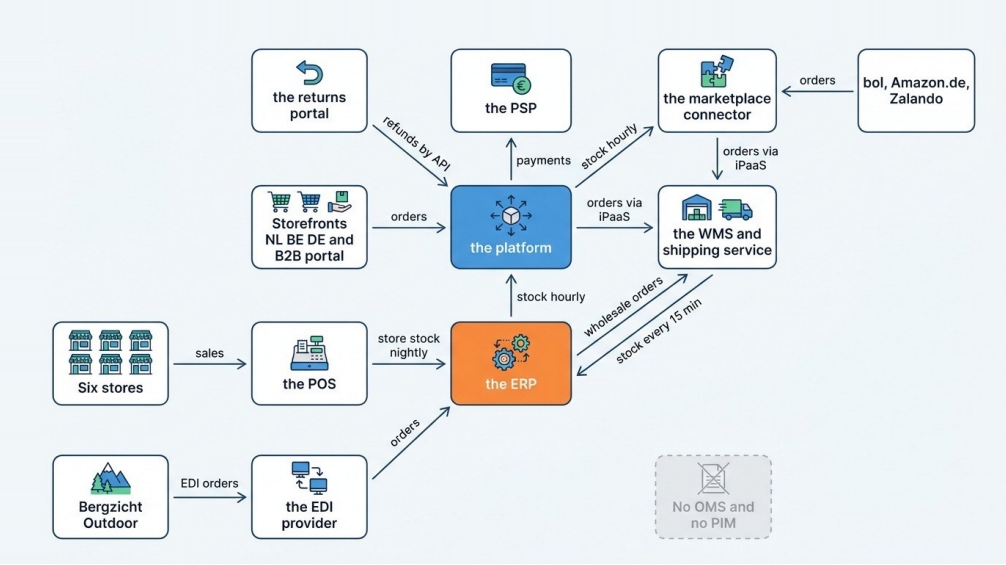


Figure 1: *Kestrel & Pine's January 2027 landscape: four channels create orders in four different systems, stock travels from the WMS to the ERP to the platform to the marketplace connector in quarter-hourly and hourly hops, and no system holds all orders or all product data.*

The **stock sync chain** in the middle of the figure is the mechanism behind the Black Friday Oversell. The WMS knows what is physically on the shelf. It tells the ERP every 15 minutes; the ERP tells the platform every hour; the platform tells the connector every hour; the connector tells the marketplaces. A sale at Duiven can take up to **two hours** to reduce the number a shopper sees on Amazon.de, and in those two hours every channel believes it can still sell the unit. Is It in Stock? Inventory, Availability and the Promise You Can Keep owns the fix.

The People

Each person owns something the architecture depends on, and most of the book's tensions are between their legitimate goals. All are fictional.

Person	Role	Pronouns	Owns
Mara Lindqvist	Co-founder and CEO	she/her	The Waypoint budget
Teun Baas	Co-founder, Chief Product and Brand	he/him	The catalog and the seasons
Noor Visser	Head of Engineering since January 2027	she/her	**The reader's stand-in.** Previously lead engineer at a logistics software company; proposed Waypoint
Daniel Okafor	Head of Operations	he/him	The warehouse, the stores and the carriers
Sanne de Wit	Finance Director	she/her	The ERP, invoicing and month-end
Milan Petrov	Ecommerce and Marketplaces Manager	he/him	The storefronts, the apps and the marketplace connector
Lea Moreau	Head of Wholesale	she/her	The B2B portal and the EDI relationship
Ruben Alves	Lead Integration Engineer since 2022	he/him	The iPaaS and the integration inventory
Hanna Berg	Data Lead	she/her	The data warehouse and metric definitions
Eva Janssen	A customer in Nijmegen	she/her	The recurring order
Pieter Smit	Buyer at Bergzicht Outdoor	he/him	The spring reorder
the agency	External storefront and apps agency since 2017	—	The theme, the apps and several scripts
the ERP partner	External ERP implementation partner	—	The platform-to-ERP connector

Noor leads a technology team of **14**: herself, nine engineers (four integration and back-end, three storefront and apps, two data), two product managers and two application

support specialists. When a chapter says "you", it usually means Noor: new to this company, not new to engineering, and asked to decide about systems she did not choose.

The Problems Kestrel & Pine Starts 2027 With

When Noor starts, every system works and the company is profitable. The problems are in the joins. Ten incidents define her starting position; each is an instance of a named pattern from Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong, and each is worked in full by the chapter that owns it.

1. **The Black Friday Oversell** (Friday 27 November 2026). The Ridgewalker sold out at Duiven at 19:40; the storefronts and marketplaces kept selling it for up to two hours. **1,340 units** oversold, **1,120 orders** canceled or delayed, about **€96,000** in cost including goodwill vouchers and marketplace penalties. *The Phantom Stock* and *The First Snowfall*; Is It in Stock? Inventory, Availability and the Promise You Can Keep, Journey 4: Black Friday, Hour by Hour.
2. **The Three-Status Order** (December 2026). Order **#1052877** showed *Delivered* in the platform, *In picking* in the WMS and *Not invoiced* in the ERP for six days; one customer opened **11 support tickets**. *Schrödinger's Order*; The Order Record: Lifecycle, State and the Order Management System, Where Is My Order? Notifications, Service and the Customer Record.
3. **The Double Refund** (January 2027). Over three weeks, **212 returns** were refunded twice: once by the returns portal through the platform's API and once by hand in the PSP console. **€19,400** went out twice, most of it recovered. *The Boomerang* and *The Message in a Bottle*; The Way Back: Returns, Exchanges, Refunds and Reverse Logistics, Taking the Money: Payment Providers, Authorization, Capture and Refunds.
4. **The Price Drift** (October 2026). The German storefront sold the Ridgewalker at **€179** for **nine days** while the marketplace listing and the stores charged **€199**; a price-list sync had failed silently. **410 orders**, about **€8,200** of margin lost, and a price-parity warning from a marketplace. *The Price Nobody Can Explain*; What Does It Cost? Price Lists, Contract Prices, Currencies and Tax.
5. **The ASN Chargeback** (Q3 2026). Bergzicht deducted **€14,200** for despatch advices whose pallet labels did not match what arrived. A despatch advice (**DESADV** in EDIFACT, the UN message standard most European EDI uses; also called an ASN, advance shipping notice) tells the buyer what is on its way and on which pallets, each identified by an **SSCC** (serial shipping container code, the 18-digit number on a pallet label); Kestrel & Pine's DESADV came from the ERP's view of the order, not from what the warehouse packed. *The Pulled Rug* (the contract) and *The Lynchpin* (one EDI consultant); Documents Between Companies: EDI, Punchout and E-Invoicing.

6. **The Store Stock Mirage** (December 2026). Click-and-collect at Utrecht promised **37** items the store had already sold, because store stock reaches the ERP only nightly. *The Phantom Stock*, from the store side; Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment.
7. **The App That Wrote Stock** (March 2026). A pre-order app adjusted inventory in the platform while the WMS feed also wrote it: two weeks of phantom reservations on **600 variants**. *The Secret Agent* (an unowned writer) and *The Spaghetti Junction*; Apps, Plugins and Functions: Extensibility and Ecosystem Governance.
8. **The Card-Testing Night** (August 2026). **48,000** bot attempts with stolen card numbers hit the checkout in one night; **€2,900** in PSP fees for declined authorizations, and the PSP paused the account for 40 minutes. A trust failure rather than one of the sixteen patterns; Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers.
9. **The Familiar Stranger**. Eva Janssen exists as an online account, a POS loyalty customer, a newsletter subscriber and a returns-portal contact, with no shared identifier, so nobody can say she is one person. *The Familiar Stranger*; One Customer, One Product: Identity and Master Data Across Systems.
10. **The Deprecated Endpoint** (February 2027). The marketplace connector retired an API version on a published date; the agency script that read it had not been updated, and stock to bol was stale for **two days**. *The Pulled Rug*; Promises Between Systems: API Contracts, Webhooks and Versioning.

Eight of the ten are the same problem: a fact (stock, status, price, identity) that lives in more than one system, with no agreed owner and no reliable way for the copies to learn of a change. The other two are what the outside world does to a landscape nobody watches as a whole.

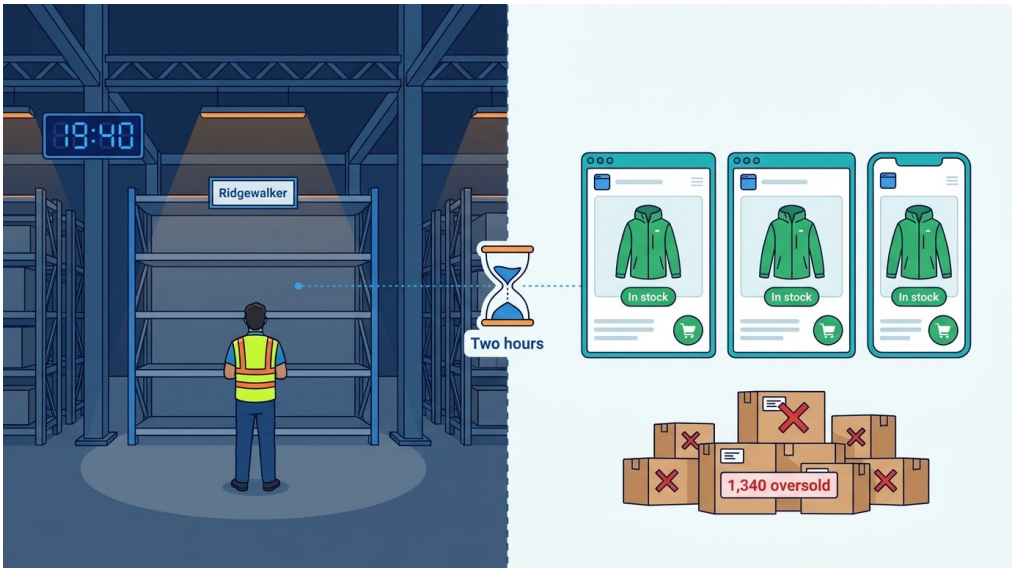


Figure 2: *The Black Friday Oversell: the Ridgewalker sold out at Duiven at 19:40, but the storefronts and marketplaces kept selling it from a copy up to two hours old, and 1,340 units were oversold.*

The Waypoint Program

In **March 2027** Mara Lindqvist approved **Waypoint**, on Noor's proposal: **€1.1 million** over 2027–2028 on top of the running technology spend. The decisions are canonical; chapters explain them and never change them.

Decision	What changes	When
W1 — Stock ownership	The WMS becomes the **system of record** for physical stock at Duiven (the one system whose value is the truth when copies disagree); the POS for store stock; the marketplace fulfillment stock becomes a tracked external location. An **availability service**, bought as a module of the OMS below, computes **available-to-promise** per channel (what a channel may still sell: physical stock minus what is already promised or reserved) from stock events. The sync chain is retired.	Q4 2027
W2 — One order record	A bought **OMS** becomes the order hub: every channel's order (platform, marketplaces, EDI, manual wholesale, later stores) is created there, with one state model and one order number family. The platform keeps the checkout and the storefront order history.	Q4 2027
W3 — Product information	A **PIM** becomes the system of record for product content and attributes; the ERP keeps costs and purchasing; the platform and the connector become consumers.	Q3 2027

Decision	What changes	When
W4 — Stay and surround	Kestrel & Pine **stays on its SaaS platform**; no replatforming. Capabilities the platform does not own well (OMS, PIM, availability) are added beside it. The platform's own POS is evaluated in 2028.	2027; review 2029
W5 — Integration standards	Every integration gets a named owner, a contract (fields, direction, frequency, system of record), **idempotent** handling of repeated messages (receiving a message twice has the same effect as once) and a reconciliation job; the inventory is kept current. Stock, orders and prices move by **events** (a message sent the moment something changes, instead of a periodic copy); products by scheduled publication; finance by daily batch.	From 2027
W6 — EDI compliance	SSCC labels generated by the WMS at pack time; DESADV sent from the WMS pack confirmation, not from the ERP; one EDI partner card per trading partner.	2027
W7 — One definition of an order	The data warehouse adopts one metric definition for order, revenue and return, owned by the Data Lead.	2027

Decision	What changes	When
W8 — Agent readiness, not agents	Accurate product data (via the PIM), availability and delivery promise by API, and order status by API are prerequisites; no shopper-facing AI agent in 2027.	2027

The order is the argument. Waypoint decides **who owns each fact** first (W1, W3, W7), then adds the hubs the landscape lacks (the availability service, the OMS, the PIM), then makes the copies trustworthy (W5, W6). It does not replace the platform (W4), because the platform is not what failed; the joins failed. Build, Buy or Assemble: Choosing and Changing Platforms works the stay-and-surround decision, Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation the integration standards, Assistants and Agents: AI in Commerce and What Is Real in 2026 W8.

The Recurring Worked Examples

Five stories recur through the book. Their facts never change, so you can follow one order across many chapters and the numbers will always agree.

Eva's jacket, the recurring order. On **Tuesday 13 April 2027 at 21:14**, Eva Janssen in Nijmegen orders on the Dutch storefront: one **Ridgewalker shell jacket, women's, size M, color moss (€199)** and **two pairs of merino hiking socks (€18 each)**. Total **€235**, free shipping, paid by **iDEAL** (the Dutch bank-transfer method most online shoppers there use). Order number **#1072455**; because she ordered before 22:00, delivery is promised for tomorrow, Wednesday.

The order is picked and packed at Duiven on **Wednesday 14 April**, shipped with PostNL and delivered **Thursday 15 April**, a day later than promised, because Duiven's last pick wave leaves at 21:00 (Pick, Pack, Ship: Warehouse Management and Fulfillment).

On **Monday 19 April** she requests an **exchange of the jacket for size L** through the returns portal (too small). The return is received and inspected on **22 April**, graded resellable, and the exchange order ships the same day.

If a chapter needs the refund variant, it is a refund of **€199** by iDEAL reversal, booked as a credit note.

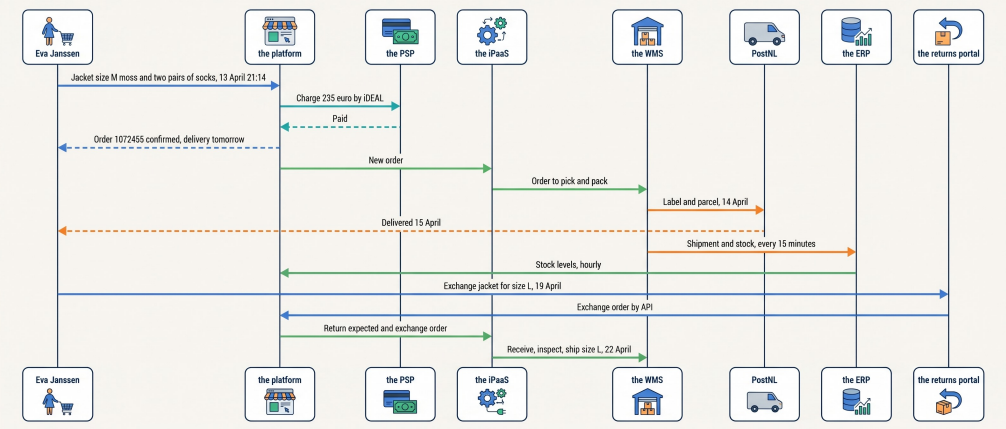


Figure 3: *Eva's jacket crosses seven systems in nine days: the platform sells and takes the money, the iPaaS carries the order to the WMS, PostNL delivers, stock trickles back through the ERP, and the exchange re-enters through the returns portal; every arrow is a handoff that can be late, lost or duplicated.*

This is the order Follow the Order: What Happens Between "Buy" and "Delivered" walks step by step and Journey 1: Eva's Jacket, From Search to Doorstep and Journey 2: The Jacket Comes Back work in full. Every chapter says near its top where in this journey it sits.

Bergzicht's spring reorder. An EDI ORDERS message arrives on **Tuesday 6 April 2027** for **240 units** across **18 SKUs** (Ridgewalker jackets and Trailhead fleeces), worth **€19,860** excluding VAT. It is picked to **three pallets** with SSCC labels; the DESADV is sent at pack confirmation; the pallets reach the Zwolle distribution center on **9 April**; the INVOIC (the EDI invoice) is sent the same day; payment is due **9 May**, on 30-day terms. Journey 3: A Chain Reorders by EDI, From Order to Invoice and Documents Between Companies: EDI, Punchout and E-Invoicing.

The marketplace variant. On 13 April 2027, the day of Eva's order, one Amazon.de customer orders a Ridgewalker that Kestrel & Pine ships itself from Duiven (seller-fulfilled), and another orders one that the marketplace's fulfillment program ships from the stock sent there in advance. The two orders look alike to the customers and nothing alike to the systems. Selling on Someone Else's Platform: Marketplaces and Channel Integration.

Black Friday 2026. The week of 23–29 November and its Friday evening: 31,000 orders, a peak hour of 2,300, the sell-out at 19:40, the two-hour stock lag and the four-hour WMS queue, with the numbers in the volumes table and in problem 1. Journey 4: Black Friday, Hour by Hour, Black Friday Every Year: Peak Load, Resilience and Recovery and Pick, Pack, Ship: Warehouse Management and Fulfillment.

The stock reconciliation. In **May 2027**, the first weekly reconciliation between the WMS and the platform (a job that compares the two systems' quantities and lists the

differences) found **2.3%** of active variants with a different quantity, most left behind by the three apps that wrote stock. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation.

The 2026 incidents are the evidence behind Waypoint. The 2027 examples happen while Waypoint runs and test its priorities: ownership before hubs, contracts before connectors, reconciliation for every copy.

How the Chapters Use Kestrel & Pine

Each chapter opens with a 2027 scene at Kestrel & Pine, then answers its question with concepts, dated public evidence from real companies and standards, and a Kestrel & Pine worked example. Chapters in Parts II to VIII end with a table across the four scale stages, using Kestrel & Pine's own path as one column. The journeys of Part IX — JOURNEYS: One Order, End to End follow the recurring orders end to end; the case studies of Part XI — PRACTICE: How Others Built It leave Kestrel & Pine and test the book against the public record.

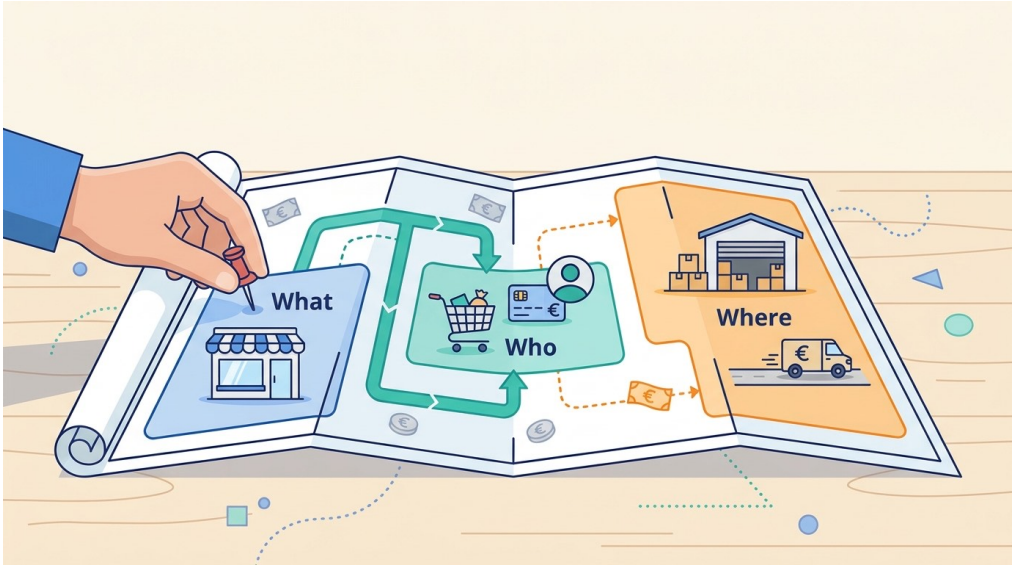
Three conventions hold. **Kestrel & Pine is labeled fictional** the first time it appears in a chapter. **Its systems are named by role**, never by vendor, so nothing in the book reads as a recommendation for or against a product. **Its facts come from one canonical record**, so the 1,340 units, the €19,400 and order #1072455 mean the same on every page; numbers that are neither Kestrel & Pine's nor documented are labeled as assumptions.

With the company, its problems and its program introduced, Part I begins where Noor began in her first week: by following one order through every system it touches, and asking at each handoff which system made the promise and which one has to keep it. Continue with Part I — COMMERCE: What a Commerce Platform Is.

Sources and Evidence

Kestrel & Pine, Bergzicht Outdoor, Veluwe Capital and every person named here are fictional; all figures are illustrative.

Part I — COMMERCE: What a Commerce Platform Is



KEY POINTS:

- **A commerce platform is a promise-keeping machine, not one product.** The storefront makes the promises; the warehouse, carrier, payment provider and finance system keep them, and every handoff between them can fail.
- **Who sells what to whom decides the systems.** Consumer, wholesale, marketplace and store selling make different promises, so they need different systems around a shared core.
- **Draw the map in capabilities, then name one record per fact.** Every other value is a copy; choose a platform by the integration work you can own, because every kind moves that work rather than removing it.

A shopper places one order, but the warehouse, carrier, payment provider and finance team each have work to do before it is complete. The storefront's promises about product, price, stock, delivery and returns depend on those other systems. Understanding who supplies each answer makes it possible to locate a failure and decide what must change. This part introduces the business jobs, their owners and the software supporting them.

Figure 1: *The storefront writes the promises in seconds; the warehouse, the carrier, the payment provider and the finance system have to honour them over days.*

The first chapter traces one order end to end. The second compares consumer, wholesale, marketplace and store sales. The remaining chapters map capabilities, assign authority over facts and explain platform categories. Start with the decision you need to understand; each chapter supplies its own context.

The example is **Kestrel & Pine**, **a fictional outdoor brand** with €48 million in 2026 revenue, thirteen systems, 34 integrations and 41 apps. Orders and product data each live in four places; six writers affect warehouse stock. Its **Waypoint** improvement program begins in March 2027 by deciding who owns what and how other systems learn of changes.

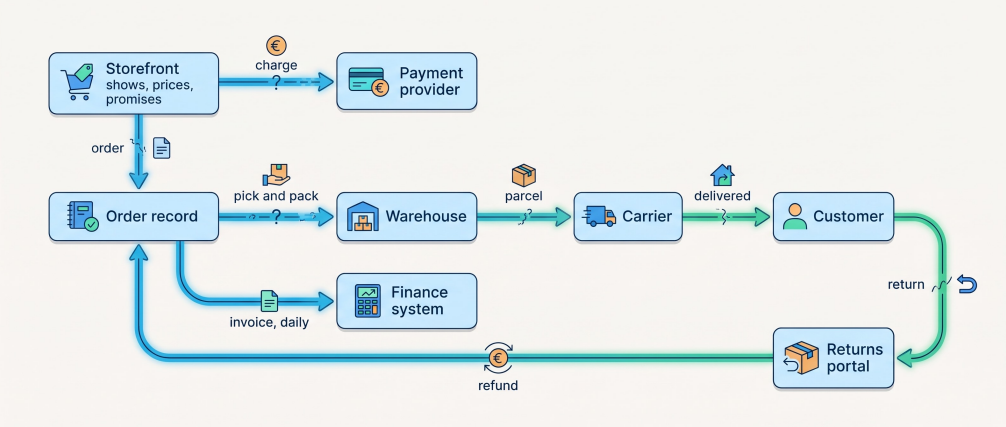


Figure 2: *The order's journey as a chain of promises: the storefront makes them early, the other systems keep them late, and every arrow is a handoff that can be late, lost, duplicated or misread.*

Chapters in this part

Follow the Order: What Happens Between "Buy" and "Delivered" follows **one purchase through thirteen steps**, including payment, delivery, return and accounting. Eva's jacket order shows the normal path. Another order, simultaneously marked delivered, in picking and uninvoiced, shows why one status word cannot explain everything.

Who Sells What to Whom: Business Models and the Economics That Shape Architecture explains **how the buyer and sales channel change the system requirements**. One jacket sold to a consumer, shop, marketplace customer or store visitor needs different pricing, payment and fulfillment. The chapter also introduces four growth stages as business shapes, rather than purchase thresholds.

The Map of a Commerce Platform: Capabilities and Layers maps **the jobs a merchant must perform before choosing products**. Head of Engineering Noor Visser places Kestrel & Pine's systems against those jobs and exposes missing capabilities and overlapping writers.

Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record assigns **an authoritative owner to each fact**. It distinguishes records from copies and explains why systems can mean different things by "order". The worked example resolves six competing answers about stock.

Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026 compares **seven platform categories and their operating costs**, using dated evidence through 2026. Two rebuild proposals test whether a new product would solve Kestrel & Pine's actual problems. The company chooses to keep its platform and add missing capabilities.

Begin with Follow the Order: What Happens Between “Buy” and “Delivered”.

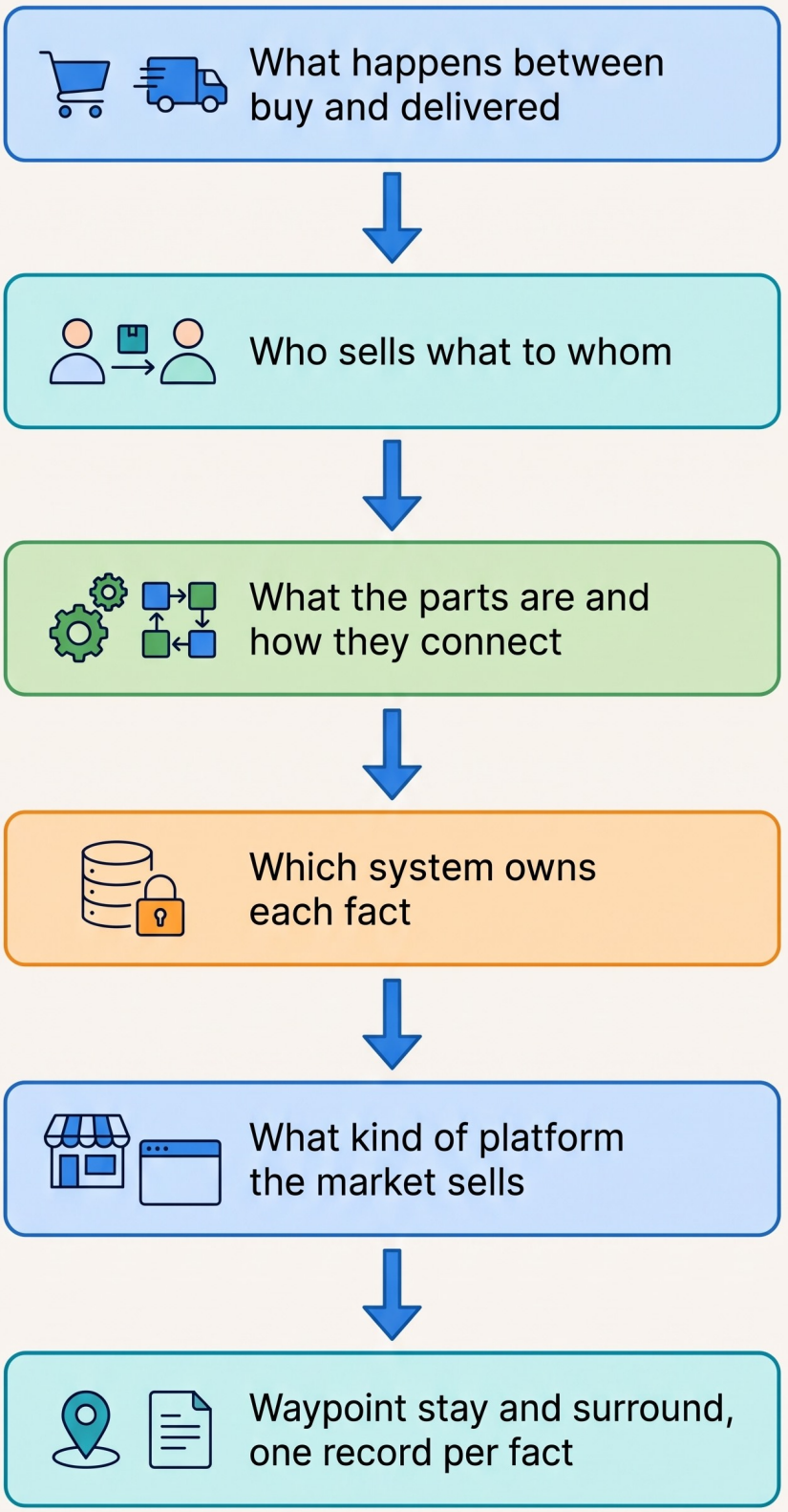
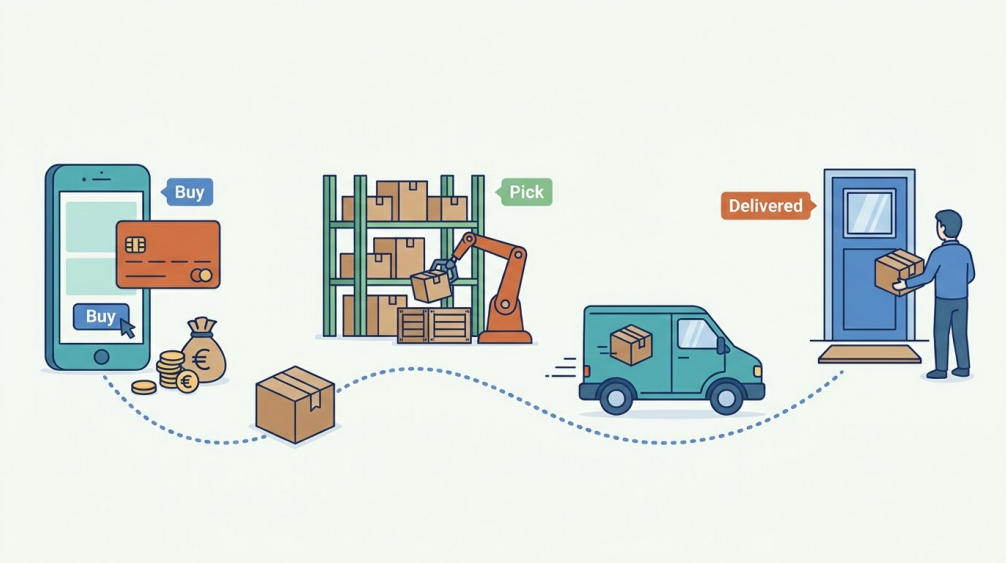

What happens between buy and delivered
Who sells what to whom
What the parts are and how they connect
Which system owns each fact
What kind of platform the market sells
Waypoint stay and surround, one record per fact

Figure 3: *The five questions of Part I in reading order: each answer is the vocabulary the next question needs, and together they are the reasoning behind Kestrel & Pine's 2027 decisions.*

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine is fictional; recommendations and labeled assumptions are the author's judgement.

1. Follow the Order: What Happens Between “Buy” and “Delivered”



IN THIS CHAPTER, YOU WILL: Follow one online order through every step and every system it touches, and learn the words the rest of the book uses for the things, the promises and the handoffs along the way.

KEY POINTS:

- **An order is a chain of promises, not a database row.** Between “buy” and “delivered” the shop promises a product, a price, an availability, a delivery date, a return right and a refund. Each promise is made by one system and has to be kept by another, often owned by a different team or vendor.
- **Every step hands the order to a different system, and a handoff is a message.** Storefront, payment provider, warehouse, carrier, returns portal and finance system each hold their own copy of the order. Messages between them can be late, lost, duplicated or misread, and nearly every commerce failure is one of those four.

- **The order has three status lines, not one.** Money, goods and returns each move through their own states. Reducing them to one "status" string is how a support agent ends up with three systems that disagree.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Schrödinger's Order** — the order's three status lines live in three systems, and an integration that flattens them into one word makes them disagree.
- **The Message in a Bottle** — every handoff in the journey is a message sent from one system to another, and a message sent once and trusted is the cheapest way to lose or duplicate an order.
- **The Phantom Stock** — the first promise in the journey, "in stock", is made by the storefront from a number the warehouse produced up to two hours earlier.

After a shopper pays for an order, the warehouse needs to pick it, finance needs to account for it and support may need to explain where it is. Each team sees a different record: a pick list, an invoice or a service ticket. When those records disagree, the shopper can be left waiting while every team believes its part is complete.

An online order passes through about a dozen steps, from a search for a jacket to the accountant's month-end close. It moves between the storefront, payment provider, warehouse and carrier. A return sends it back through the warehouse and finance. **Each handoff must carry a promise another system can keep**: the price agreed, the stock committed, the parcel expected or the refund owed.

Tracing those handoffs shows where responsibility or information goes missing. This chapter follows the records an order needs, the systems that hold them, and the checks that let a merchant explain what happened.

Where we are in the order's journey. Eva Janssen buys a jacket and socks from Kestrel & Pine, the fictional outdoor brand used throughout this book. We follow her order from search to delivery and return. Meet Kestrel & Pine introduces the company; Part IX — JOURNEYS: One Order, End to End revisits the journey in detail.

Setting the Scene at Kestrel & Pine: Four Drawings of One Order

In her second week as Head of Engineering, in January 2027, Noor Visser asked four colleagues at Kestrel & Pine to draw what happens to an online order. She gave each a whiteboard and twenty minutes.

Milan Petrov, who runs the storefronts and marketplaces, put the hosted commerce platform at the center, with arrows to payments and the warehouse. Daniel Okafor, Head of Operations, put Duiven’s warehouse system at the center, with orders arriving and PostNL parcels leaving.

Finance Director Sanne de Wit centered her drawing on the **ERP**, the enterprise resource planning system that keeps the books. Lead integration engineer Ruben Alves drew thirty-four boxes and ran out of time. Each had drawn the part they could see.

Four drawings, four centers, and not one of them wrong. Each person drew the part of the journey their system owns, which is the part they can see. What nobody had drawn, because nobody owned it, was the order’s path *between* the boxes: which message carries it from one to the next, what each message promises, and what happens when a message is late.

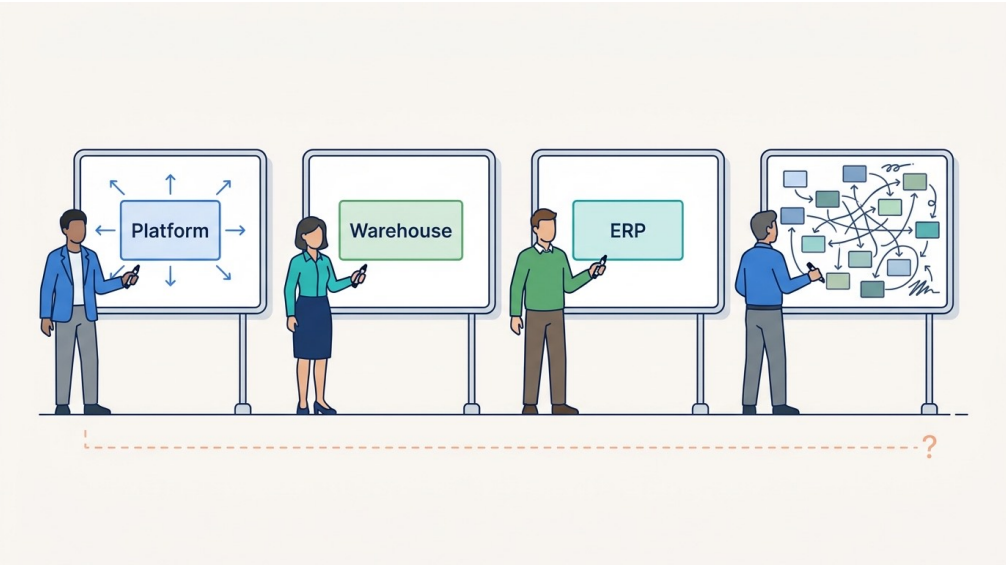


Figure 1: *Four people drew the same order around four different centers; none of them drew the path between the boxes, because nobody owned it.*

Noor’s response was to pick one order and follow it, with a notebook, through every system it touched. The order the book uses for that walk is Eva’s jacket, placed in April 2027. Before following it, the chapter needs the thirteen steps, the things an order is made of, and the two words that describe how it moves.

The Journey in Thirteen Steps

Every consumer order, on every platform, passes through roughly the same steps. The names vary by company; the shape does not. The table lists each step, what happens,

the **promise** the merchant makes to the customer at that step, and which system usually has to keep it.

#	Step	What happens	The promise made	Who must keep it
1	**Discovery**	The shopper finds the product through search, a category page, an advertisement or a marketplace listing	"We sell this, and it is what the page says"	Product information, search (Finding the Product: Search, Navigation and Merchandising)
2	**Product page**	Price, sizes, colors, images, stock and delivery promise are shown	"This price, this variant, in stock, delivered by then"	Pricing, inventory, delivery promise
3	**Cart**	The shopper collects items; the cart is a draft, not a commitment	"We will hold this draft for you"	The platform
4	**Checkout**	Identity, address, delivery option, final price with tax and shipping, payment method	"You pay exactly this, delivered there, by then"	Pricing, tax, shipping rates, availability
5	**Payment**	The payment service provider (PSP) takes the money or reserves it	"Your money is safe and matches the order"	The PSP, the platform

#	Step	What happens	The promise made	Who must keep it
6	**Order creation**	The order record is created, numbered and confirmed; stock is committed to it	"The order exists and is being kept"	The platform or an order management system
7	**Fulfillment**	The warehouse picks, packs and labels the parcel	"What you ordered is in the box"	The warehouse system
8	**Shipping**	The carrier collects the parcel and scans it along the way	"It is on its way, here is where"	The carrier, the shipping service
9	**Delivery**	The parcel is handed over, left at a pickup point, or delivery fails	"Delivered when we said"	The carrier, the notification layer
10	**Service**	The customer asks "where is my order", changes an address or cancels	"We know where your order is"	Service tools over all of the above
11	**Return**	The customer sends an item back; the warehouse receives and inspects it	"You can send it back within 14 days"	Returns portal, warehouse
12	**Refund or exchange**	Money goes back or a replacement ships	"Your money is back within 14 days"	The PSP, the platform, the warehouse
13	**Accounting**	Invoice, credit note, payout reconciliation, month-end	"The books agree with what happened"	The ERP, finance

Three things about the table matter before any technology is named. **The promises are made early and kept late**: steps 1 to 6 take minutes, steps 7 to 13 take days or weeks. **The system that makes a promise is rarely the one that keeps it**: the product page promises “in stock” from a number it was given, and the warehouse has to find the unit. And **steps 10 to 13 exist for every order**, not only the problem ones: at Kestrel & Pine, 24% of items sold online came back in 2026, and every order became an invoice. A journey that ends at “delivered” is half a journey.

The Things an Order Is Made Of

Before following an order through systems, it helps to name the things the systems hold. These are the **entities** of commerce: the nouns that every platform, in its own words, keeps records of.

Thing	In plain words	Where the book explains it
Product	What the shop sells, as the shopper thinks of it: “the Ridgewalker jacket”	What Are We Selling? Products, Variants and the Catalog
Variant or **SKU**	One sellable version of a product: the Ridgewalker jacket, women’s, size M, color moss. SKU (stock keeping unit) is the identifier of that one thing. Kestrel & Pine has 3,400 products and 26,000 SKUs	What Are We Selling? Products, Variants and the Catalog
Price	What a variant costs in one channel, country and currency, including or excluding tax	What Does It Cost? Price Lists, Contract Prices, Currencies and Tax
Stock or **inventory**	How many units of a SKU exist at a location, and how many of those may still be promised	Is It in Stock? Inventory, Availability and the Promise You Can Keep

Thing	In plain words	Where the book explains it
Cart	The shopper’s draft: a list of variants and quantities with no commitment on either side	From Cart to Confirmation: Checkout Orchestration
Order	The record of what was bought, by whom, for how much, to be delivered where. It gets a number, and it is the thing every later system refers to	The Order Record: Lifecycle, State and the Order Management System
Payment	The money side of the order: what was reserved, taken, refunded, and the references the payment provider gave for each	Taking the Money: Payment Providers, Authorization, Capture and Refunds
Fulfillment order	The instruction to one warehouse: ship these lines of this order to this address	Pick, Pack, Ship: Warehouse Management and Fulfillment
Shipment	A physical parcel with a carrier label and a tracking number, and the stream of carrier scans that follow it	Getting It There: Carriers, Labels, Tracking and the Delivery Promise
Return	The customer’s request to send items back, and the warehouse’s record of what arrived and what was done with it	The Way Back: Returns, Exchanges, Refunds and Reverse Logistics
Invoice and **credit note**	The legal documents: what was sold, with how much tax, under which sequential number; and the mirror document that reduces it after a return	Tax, Invoices and the Books: From Order to Ledger

Thing	In plain words	Where the book explains it
Customer	The person or company behind the order, as each system knows them	One Customer, One Product: Identity and Master Data Across Systems

Two of these are often merged by newcomers. **A cart is not an order**. A cart is a draft that commits nobody, and most are abandoned: the Baymard Institute's running average across fifty studies published between 2006 and 2025 is a 70.22% cart-abandonment rate (page updated 28 October 2026). An order is a commitment on both sides. **An order is not a shipment**. One order can become two parcels from two locations, and the fulfillment order is the piece in between: the warehouse's unit of work.

Promises and Handoffs

Two words carry the argument of this book, and this chapter owns their definitions.

A **promise** is a statement the merchant makes to the customer that some system will later have to make true. "In stock" is a promise. "€199" is a promise. "Ordered before 22:00, delivered tomorrow" is a promise. "Free returns within 14 days" is a promise. Promises are made by the systems closest to the customer (the storefront, the checkout, the returns portal) and kept by the systems closest to the goods and the money (the warehouse, the carrier, the payment provider, the finance system). **A commerce platform, in the sense this book uses, is the whole set of systems that make and keep those promises**, not one product with that name on the invoice.

A **handoff** is the moment responsibility for the order passes from one system to another. A handoff is always carried by a message: an API call, a file, a webhook, a queue entry, an EDI document (electronic data interchange: a structured document exchanged between two companies' systems), sometimes an e-mail. The storefront hands the order to the payment provider ("charge €235"); the payment provider hands the result back ("paid"); the storefront hands the order to the warehouse ("pick these three lines"). Each handoff has a sender, a receiver, a message format, a timing and, if things are done well, an owner.

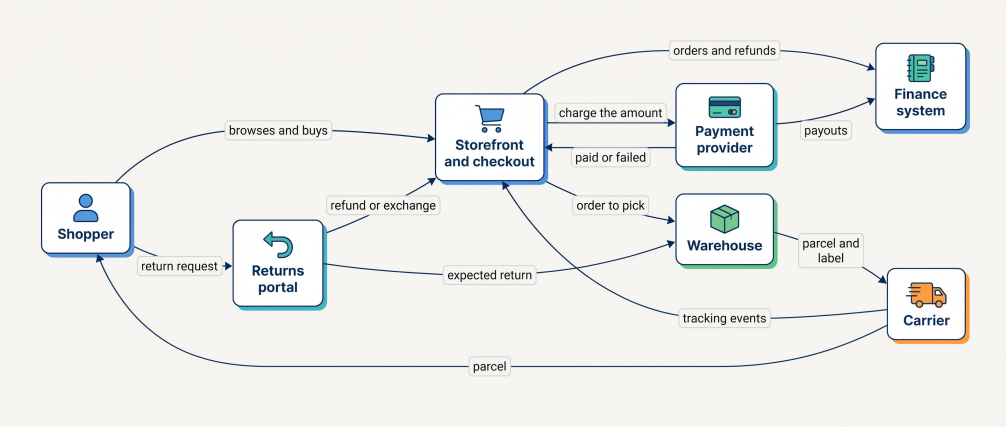


Figure 2: *The order’s journey as handoffs: every arrow is a message from one system to another, and every message can be late, lost, duplicated or misread.*

Each handoff carries requirements the business sets, usually without saying so: how fresh the information must be, how consistent with the sender’s copy, how fast it must arrive, and what happens if the receiver is down. The table states them for the main handoffs of a consumer order; the targets are illustrative, and a real merchant sets them from its own promises.

Handoff	Freshness the promise needs	Consistency	Speed	If the receiver is down
Warehouse stock → storefront “in stock”	Seconds to minutes during a sale; hours on a quiet Tuesday *(assumption)*	The storefront’s number may lag but must never exceed the warehouse’s	Urgent in aggregate, not per message	The storefront keeps selling from a stale number: the Phantom Stock
Checkout → payment provider → checkout	Immediate	Exactly one charge per order	Seconds; the shopper is waiting	Checkout cannot complete; the shopper leaves

Handoff	Freshness the promise needs	Consistency	Speed	If the receiver is down
Storefront → warehouse (order to pick)	Before the cut-off the delivery promise depends on	Every paid order, exactly once, with the right lines	Minutes; it decides whether "tomorrow" holds	Orders queue; the promise slips silently
Warehouse → storefront (shipped, tracking number)	At label print	One shipment record per parcel	Minutes; it triggers the customer's e-mail	The customer sees "being prepared" after the parcel has left
Returns portal → warehouse → payment provider (refund)	On inspection, not on request	Exactly one refund per return	Within 14 days by EU law	A refund issued by hand in the PSP console: the Double Refund
Storefront → finance system (invoice, payment, refund)	Daily is usually enough	Every order, invoice and credit note exactly once, with stable identifiers	Hours to a day	Month-end does not reconcile

Every merchant has all of these rows; what differs is whether anyone has written them down. That is why Part VI — INTEGRATION: Making Separate Systems Behave Like One is about integration.

One Order, Three Status Lines

A newcomer's instinct is to give an order one status: *new*, *paid*, *shipped*, *delivered*, *returned*. Platforms learned long ago that this does not work, because the money, the goods and the return move independently.

Shopify's documentation (n.d., accessed November 2026) keeps three status families on one order: an *order status* (open, archived, canceled), a *payment status* (from authorized through paid to refunded, eight values) and a *fulfillment status* (from unfulfilled

through in progress to fulfilled, six values). commercetools’ order documentation (n.d., accessed November 2026) does the same with an `OrderState`, a `PaymentState` and a `ShipmentState`, plus separate states for each returned item. Neither vendor’s “fulfilled” means “delivered”; delivery is known only if the carrier’s events are fed back.

In plain words: an order can be *paid and not shipped*, *shipped and not paid* (an invoice customer), *delivered and partly refunded*, or *returned and not yet refunded*, and each of those is a normal state, not an error. The goods line alone, for a simple order, looks like this:

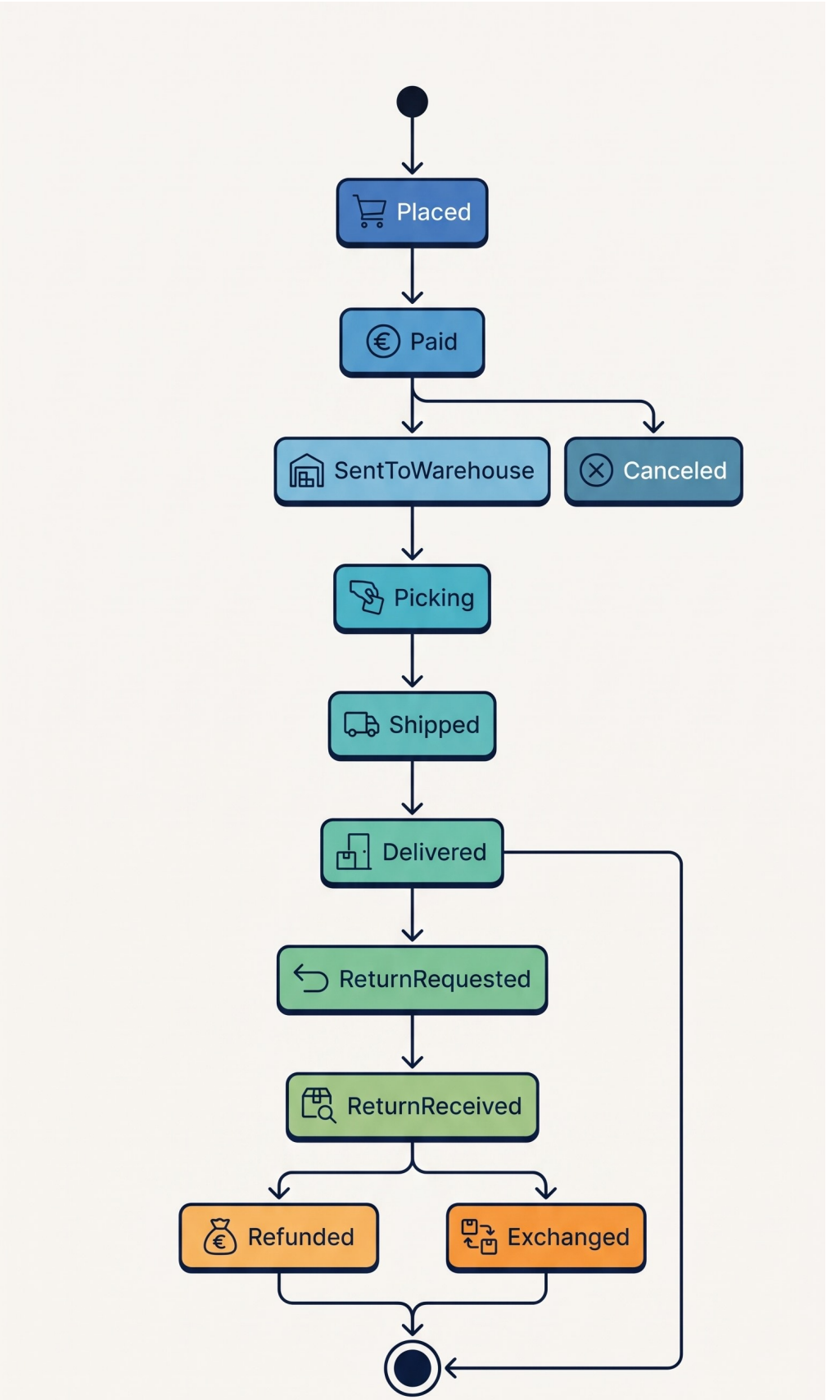

Placed
Paid
SentToWarehouse
Canceled
Picking
Shipped
Delivered
ReturnRequested
ReturnReceived
Refunded
Exchanged

Figure 3: *The goods line of a simple consumer order; money and returns have their own lines, and a support agent needs all three to answer “where is my order”.*

Kestrel & Pine’s **Three-Status Order**, #1052877, shows the cost of that gap. The fleece shipped on 7 December 2026; the socks waited a week for replenishment. The integration could not represent a partial shipment, so it marked the whole order fulfilled when the fleece left. The platform then showed *Delivered* after the carrier’s scan.

The warehouse management system (WMS) kept the order *In picking* until the socks shipped. The ERP waited for the WMS’s “order complete” message before invoicing. Each status described part of the order, but nobody could give the customer the useful answer: **your order ships in two parcels**. An order management system can hold that combined view (The Order Record: Lifecycle, State and the Order Management System). The eleven support tickets are followed in Where Is My Order? Notifications, Service and the Customer Record.

Eva’s Jacket, System by System

Now the walk. Kestrel & Pine is fictional, and so is Eva Janssen, a customer in Nijmegen; every number below is fixed for the whole book, so it will match every later chapter.

On **Tuesday 13 April 2027 at 21:14**, Eva orders on the Dutch storefront: one Ridgewalker shell jacket, women’s, size M, color moss, at **€199**, and two pairs of merino hiking socks at **€18** each. The total is **€235**; shipping is free because the order is over €75. She pays by **iDEAL**, the Dutch bank-transfer method. The order number is **#1072455**. Because she ordered before the 22:00 cut-off on the storefront, the confirmation promises delivery tomorrow, Wednesday. The iPaaS in the figure is the integration tool that carries messages between Kestrel & Pine’s systems; step 6 explains it.

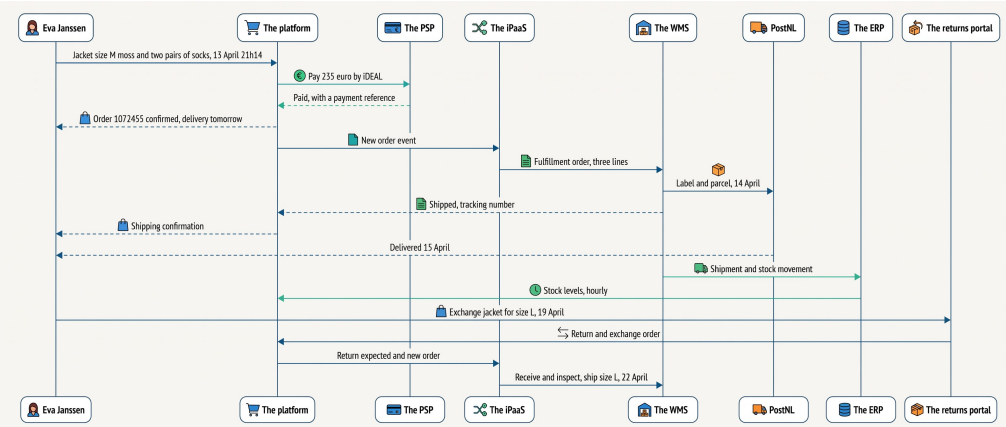


Figure 4: *Eva's jacket crosses seven systems in nine days; every arrow is a handoff, and each one is a place where a later chapter's problems live.*

Steps 1 and 2, discovery and product page (the platform). Eva searched for "waterproof jacket" on the Dutch storefront. The product page showed the moss color in size M as in stock, at €199, deliverable tomorrow. The price came from the platform's own price list. The stock number came from the platform's inventory field, which the ERP updated up to an hour earlier from a number the WMS gave it up to fifteen minutes before that. The delivery promise came from a static rule in the theme (the storefront's page templates): before 22:00 means tomorrow. Three promises, three sources, and only the price is the platform's own.

Steps 3 and 4, cart and checkout (the platform). The cart held three lines. At checkout the platform asked for Eva's address, offered home delivery by PostNL at €0, computed the total of €235 including Dutch VAT (value-added tax) at the standard rate of 21%, and offered iDEAL, cards, PayPal, Klarna, Bancontact and Apple Pay. This is the step the book calls **checkout orchestration**: in a few seconds the platform has called pricing, promotions, tax, availability, shipping rates and the payment provider, and turned a draft into a set of promises.

Step 5, payment (the PSP). The platform asked the payment provider (the **PSP**, payment service provider: the company that takes the shopper's money and later pays it out to the merchant) to collect €235 by iDEAL. iDEAL is different from a card: Mollie's documentation (n.d., accessed November 2026) describes it as an instant charge, captured automatically, with no chargeback and a 15-minute session timeout. So within a minute the PSP told the platform, by a **webhook** (a message the PSP pushes to a web address the merchant registered), that the payment was complete, with a reference. For a card payment this step has two halves, an **authorization** that reserves the money and a **capture** that takes it, which Taking the Money: Payment Providers, Authorization, Capture and Refunds explains.

Step 6, order creation (the platform). Strictly, the platform created order #1072455 a moment before the iDEAL result arrived, with the payment *pending*, and the webhook then marked it paid; steps 5 and 6 overlap. It committed one moss M jacket and two pairs of socks from its own available quantity, sent Eva a confirmation e-mail and published an order event to the **iPaaS** (integration platform as a service: the hosted tool Kestrel & Pine uses to move data between systems). The order now exists in one system. Within an hour it will exist in three.

Step 7, fulfillment (the WMS). The iPaaS created a **fulfillment order** in the warehouse management system (the **WMS**, the software that runs the physical work at Duiven: where stock sits, who picks what, into which box). On Wednesday 14 April the jacket and the socks were picked in a morning wave, packed in one box and labeled. The WMS reported the shipment and the tracking number back through the iPaaS to the platform, which sent Eva a shipping confirmation, and to the ERP, which recorded the stock movement.

Steps 8 and 9, shipping and delivery (PostNL). PostNL collected the parcel on Wednesday evening and delivered it on **Thursday 15 April**. Its scans (accepted, sorted, out for delivery, delivered) reached the platform through the shipping service inside the WMS flow, and the order status page moved to *Delivered*. The promise made on Tuesday evening was broken by a day. The “before 22:00” rule dates from the years when a third-party warehouse in Tiel shipped the orders; at Duiven the last pick wave starts at 21:00, so an order placed at 21:14 is picked the next morning and leaves with the evening collection. Nobody updated the sentence on the page, and Eva did not complain, which is why nobody noticed. Getting It There: Carriers, Labels, Tracking and the Delivery Promise replaces the rule with a computed promise.

Step 10, service. Eva did not call. Had she asked “where is my order” on Wednesday evening, the agent would have worked from what the support team calls “the five tabs”: the platform’s order screen, the WMS, the shipping service’s tracking page, the PSP console and the returns portal, with the ERP visible only to finance.

Step 11, return (the returns portal and the WMS). On **Monday 19 April** Eva used the returns portal to request an **exchange** of the jacket for size L: too small. The portal, a separate SaaS (software as a service) product, issued a PostNL return label and told the platform, through its API, that a return was expected and a new order for size L should be created. The portal has made two promises: that Eva may return the jacket, which EU law backs (Directive 2011/83/EU gives a consumer 14 days from receiving goods to withdraw from a distance purchase without giving a reason), and that size L will ship, which depends on size L being in stock.

Step 12, exchange (the WMS, the platform). The return was received at Duiven on **22 April**, inspected and graded resellable; the WMS reported the receipt, the platform closed the return, and the held exchange order for size L shipped the same day. Had Eva chosen a refund, the PSP would have returned €199 by iDEAL reversal and the ERP would have issued a credit note.

Step 13, accounting (the ERP). On dispatch, the ERP generated an invoice for order #1072455: €235 gross, of which €194.21 net revenue and €40.79 VAT at 21%. When the PSP paid out Tuesday’s sales a few days later, finance matched the payout batch to the orders in it, fees deducted. The exchange produced a credit note for the size M jacket and a new invoice for size L, netting to zero. Under the accounting standard IFRS 15, revenue is recognized when the goods are delivered, not when Eva paid, which is why finance cares about step 9 and not only step 5. Tax, Invoices and the Books: From Order to Ledger owns this.

The table says what each system knew at each moment. Read it down a column and you see one system’s world; across a row, why Noor’s colleagues drew four pictures.

Moment	The platform	The PSP	The WMS	The ERP	The returns portal
Tue 21:14, paid	Order #1072455, paid, unfulfilled	Payment, €235, complete	Nothing yet	Nothing yet	Nothing
Wed morning, picked	Paid, unfulfilled	Complete	Fulfillment order in picking	Order copy, not invoiced	Nothing
Wed afternoon, shipped	Paid, fulfilled, tracking number	Complete	Shipped	Invoice issued, stock reduced	Nothing
Mon 19 April	Delivered, return open, exchange order held	Complete	Return expected	Invoiced	Return requested, label issued
22 April	Return closed, exchange shipped	Complete	Return received, resellable; size L shipped	Credit note and new invoice	Return complete

Six systems, six copies of the order, each correct about its own part. That is normal. The question the book keeps asking is which copy is the one to believe when two disagree, and how the others learn when it changes. Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record names that copy the **system of record**.

Why the Joins Are Where Things Break

Each arrow in Figure 2 is a message. Messages can go wrong in only four ways, and all four have happened at Kestrel & Pine.

Late. The stock number on the product page was up to two hours old because it had traveled WMS → ERP → platform → marketplace connector in fifteen-minute and hourly hops. On the evening of Black Friday 2026 the Ridgewalker jacket sold out in the warehouse at 19:40 and the storefronts and marketplaces kept selling it for up to two hours: 1,340 units oversold, 1,120 orders canceled or delayed, about €96,000 in cost. The same evening the queue carrying orders to the WMS backed up for four hours.

Lost. In February 2027 the marketplace connector (the software that sends listings and stock to marketplaces such as bol, the Dutch and Belgian marketplace, and brings their orders back) retired an API version on a published date. The agency script that pushed the platform’s stock figures into the connector still used the retired version, so the connector kept sending bol its last figures for two days. A message that is not sent produces no error.

Duplicated. Over three weeks in January 2027, 212 returns were refunded twice: once by the returns portal through the platform’s API, and once by a support agent in the PSP’s console who could not see that the first refund had happened. €19,400 went out twice; most was recovered. Two senders, one receiver, no shared record of what had already been done.

Misread. The Three-Status Order was three systems reading the same two-parcel order differently, each consistently with the messages it had received: “shipped” meant “all shipped” to one and “first parcel shipped” to another.

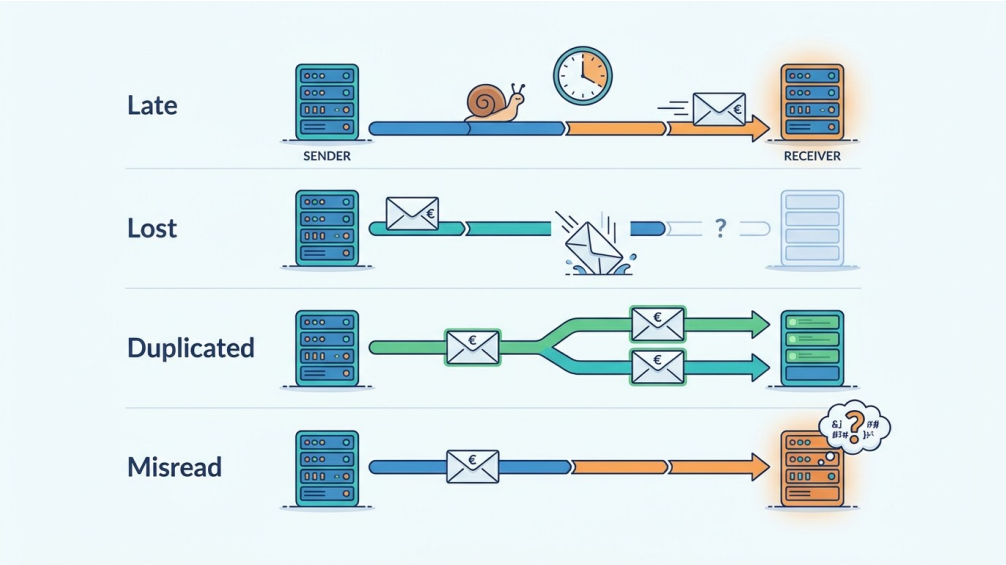


Figure 5: *Every handoff is a message, and a message can fail in only four ways: late, lost, duplicated or misread.*

Two ideas, introduced here and worked in Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation, are the field’s answer. **Idempotency** means that receiving the same message twice has the same effect as receiving it once: a second “refund order #1072455” request must not produce a second refund. Stripe’s documentation (n.d., accessed November 2026) describes the mechanism on the payment side: a client sends an idempotency key with each request, and a retry with the same key returns the saved result instead of acting again. **Reconciliation** means regularly

comparing two systems' copies and listing the differences: the platform's orders against the WMS's shipments, the returns against the refunds, the PSP's payouts against the invoices. The first weekly reconciliation Kestrel & Pine ran, in May 2027, found 2.3% of active variants with different stock quantities in the WMS and the platform. Neither idea is advanced; both are routinely skipped, because each integration was built to make the happy path work.

In Practice: What the Public Record Says

Vendor documentation and regulation describe the journey's steps with surprising precision.

Payment success is learned by message, not by page. Stripe's checkout fulfillment guide (n.d., accessed November 2026) says a merchant "can't rely on triggering fulfillment only from your checkout landing page, because it's not guaranteed customers visit that page", and that the fulfillment function must "correctly handle being called multiple times with the same Checkout Session ID". commercetools' checkout documentation (n.d., accessed November 2026) creates the order *before* the payment provider has answered, with the payment pending, and warns to start fulfilling "only if the Payment contains a Transaction" with a successful state. In both designs the handoff at step 5 is asynchronous: the order waits for a message.

Messages arrive more than once and out of order. Stripe's webhook documentation (n.d., accessed November 2026) retries delivery "for up to three days", "doesn't guarantee the delivery of events in the order that they're generated", and warns that endpoints "might occasionally receive the same event more than once", so receivers should track event identifiers. Mollie's webhook (n.d., accessed November 2026) carries only a payment identifier and no status, so that a forged call can never mark an order paid; the receiver must fetch the status, and Mollie retries ten times over about 26 hours. These are the vendors' own descriptions of the Message in a Bottle, and of its remedy.

The warehouse handoff is a handshake. Shopify's fulfillment-service documentation (n.d., accessed November 2026) sends a fulfillment request that the warehouse's service must accept or reject; fulfillments "can't be created for a fulfillment order before the fulfillment service accepts", and once accepted, a cancellation becomes a *request* the warehouse may refuse. That handshake is what closes the order's edit window.

Tracking is the customer's view of the journey. Shopify's customer-notification page (n.d., accessed November 2026) lists the default messages: order confirmation (which cannot be switched off), shipping confirmation at fulfillment, and "out for delivery" and "delivered" when a carrier reports them. Its order-status page warns that "if you fulfill an order without a tracking number, then no shipping updates are received". Gorgias, a helpdesk vendor, reported from its own customer base that "where is my order" requests were about 18% of incoming support contacts (article updated November 2023; one vendor's data, now approaching three years old).

The law fixes the outer bounds of three promises. Directive 2011/83/EU on consumer rights (2011, in force) requires delivery "not later than 30 days from the conclusion of the contract" unless agreed otherwise (Article 18); gives the consumer 14 days from receiving the goods to withdraw without reason (Article 9); and requires the trader to reimburse "not later than 14 days" after being informed, by the same means of payment, while allowing the trader to wait until the goods or proof of sending arrive (Article 13). Each clause is a date the order record must be able to compute.

The books are the last handoff. Odoo's accounting documentation (version 18, accessed November 2026), representative of how ERPs behave, assigns an invoice its number from a sequence on confirmation, after which "an invoice can no longer be updated"; a credit note "is the only legal method for canceling, refunding, or modifying a validated invoice". Eva's exchange therefore produced two new documents, not an edit.

Across Scale Stages: When It Fits and When It Limits

The thirteen steps stay much the same as a merchant grows. The number of systems, channels and stock locations changes:

- **Starting**: one hosted platform handles most of the order, with an outside payment provider, warehouse and carrier. This was Kestrel & Pine in 2016–2018.
- **Growing**: another channel creates orders elsewhere. An ERP or third-party logistics provider (3PL) holds copies of orders and stock. People start asking which copy to trust.
- **Omnichannel**: orders arrive through several channels and stock sits in warehouses, stores and marketplace fulfillment centers. An order hub, an availability service and written integration contracts become necessary.
- **Group**: several brands or countries share services while retaining local differences.

Kestrel & Pine reached Omnichannel in 2023. Its 2027 problems concern the handoffs between systems. Who Sells What to Whom: Business Models and the Economics That Shape Architecture defines the stages; What Changes at Each Scale: From First Store to Group examines the transitions.

When It Goes Wrong at Each Step

None of these needs a disaster; all happen on normal weeks.

Step	Ordinary failure	How it is detected	How it is recovered
Product page	Stock shown is stale; price differs from another channel	Oversell cancellations; a marketplace’s price-parity warning	Stock events instead of periodic sync; one price owner (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax)
Checkout	Payment succeeded but no order was created (the shopper closed the tab)	Payments in the PSP with no matching order	A job that finds paid-without-order and creates or refunds; the webhook as the trigger
Payment	Webhook arrives twice or never; the PSP pauses the account (the Card-Testing Night, 40 minutes)	Duplicate order numbers; orders stuck in *pending*	Idempotent handling by event identifier; a daily PSP-to-orders reconciliation
Order to warehouse	Order arrives late, never, or twice	Orders paid but not picked past the cut-off; two parcels for one order	Queue monitoring on age, not only on errors; the request-accept handshake
Fulfillment and shipping	Wrong item picked; tracking number not passed back; parcel lost	Returns marked “wrong item”; “where is my order” tickets; carrier exceptions	Scan-to-verify at pack; tracking number required on the shipped message

Step	Ordinary failure	How it is detected	How it is recovered
Return and refund	Refund issued twice; refund without receipt; restock without inspection	Returns-to-refunds reconciliation; PSP refunds not linked to a return	One refund path, triggered by the inspection event; one refund per return
Accounting	Invoice skipped or duplicated; payout does not match orders	Month-end three-way match of orders, PSP payouts and bank	Idempotent order-to-invoice posting; credit notes, never edits

The common thread: **detection is a comparison** between two systems' copies, and **recovery is a message sent again safely**. A merchant that can compare and can safely resend is most of the way to a reliable journey; one that can do neither will meet every pattern in Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong in turn.

What to Take Away

Established.

- An order's money, goods and return move through separate states; the major platforms model them as separate status families on one record (Shopify's order, payment and fulfillment statuses; commercetools' order, payment and shipment states), and "fulfilled" never means "delivered" without carrier events.
- Payment outcomes arrive as messages that can be delayed, duplicated and re-ordered; payment providers document this and tell receivers to deduplicate by event identifier and to treat the message, not the landing page, as the trigger for fulfillment.
- EU consumer law fixes three promises for every distance sale: delivery within 30 days unless agreed, a 14-day right of withdrawal from receipt, and a refund within 14 days of notice (Directive 2011/83/EU).
- Invoices are numbered sequentially at posting and never edited; a return produces a credit note.

Depends on context.

- Whether the platform, an ERP or an order management system should hold the order record depends on how many channels create orders and how many locations fulfill them. One channel and one warehouse: the platform’s order is enough. Several of each: an order hub becomes necessary (The Order Record: Lifecycle, State and the Order Management System).
- Whether a static delivery promise is acceptable depends on how often it is broken. “Before 22:00, tomorrow” served Kestrel & Pine for years; it fails for every order placed after Duiven’s last pick wave, Eva’s included, and worst when the warehouse queue backs up, which is when the promise matters most (Getting It There: Carriers, Labels, Tracking and the Delivery Promise).
- How fresh stock must be on the product page depends on how fast the merchant sells its last units. Hourly is fine for a slow-moving catalog and ruinous on a peak evening (Is It in Stock? Inventory, Availability and the Promise You Can Keep).

The author’s synthesis.

- I treat a commerce platform as the whole set of systems that make and keep promises to a customer, and I judge an architecture by its handoffs, not by its boxes: ten good systems with no owner for the messages between them make a worse platform than five ordinary systems with written contracts for every arrow.
- Before any ownership decision, I follow one real order through every system with a notebook, as Noor did. The table of what each system knows at each moment is the cheapest architecture document a merchant can produce, and the one most merchants have never produced.

Questions to Ask

Take these into your first week, a design review or a vendor demo.

1. Can someone draw the order’s path through every system, with the message that carries each handoff and the person who owns it?
2. For a given order number, which systems hold a copy, and which copy is believed when they disagree?
3. How does the storefront learn that the warehouse sold the last unit, and how long does that take on the busiest evening of the year?
4. What triggers fulfillment: a page, a webhook or a scheduled job? What happens when the trigger arrives twice, or never?
5. When the warehouse starts picking, what stops an order edit, a cancellation or a second fulfillment order?
6. How many ways can a refund be issued, and do they share a record of what has already been refunded?

7. Which of the merchant's promises (stock, price, delivery date, return right) are computed from data, and which are rules someone typed in years ago?
8. When was the last time anyone compared two systems' copies of the same orders, and what did they find?

The journey is the same for every merchant, but the systems that run it and the promises they make differ sharply by who is selling what to whom. A brand selling its own jackets to consumers, a wholesaler shipping pallets to a retail chain, a marketplace seller and the marketplace itself run different journeys over the same thirteen steps, with different economics pulling on every one of them. Who Sells What to Whom: Business Models and the Economics That Shape Architecture takes that up next.

Sources and Evidence

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

2. Who Sells What to Whom: Business Models and the Economics That Shape Architecture

IN THIS CHAPTER, YOU WILL: Learn the commerce business models, the unit economics of an order and the four scale stages, and see how each one pulls on the architecture behind it.

KEY POINTS:

- **A business model is a decision about who makes which promise**. Selling to consumers, to retailers, through a marketplace or as a marketplace changes who captures the order, who sets the price, who is the legal seller, who holds the stock and who ships. The systems follow from that, not from the technology.
- **Each number in the unit economics of an order pulls on a different part of the platform**. Conversion pulls on checkout, basket size on promotions, margin on price control, returns on the reverse flow, payment fees on the provider choice, and the peak day on capacity.

A merchant that knows its numbers knows where its architecture must be strong.

- **Scale stages are defined by shape, not revenue.** A merchant moves from Starting to Growing to Omnichannel to Group by adding channels, locations and brands, and vendors' own price tiers confirm it. Each transition turns the joins between systems into the problem, which is why the order hub, the product hub and integration standards arrive together.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **One Size Fits None** — business lines with different promises are forced onto one stack, or kept apart, before anyone has worked out what they share.
- **The Giant's Boots** — an architecture is copied from a marketplace operator or a hundred-times-larger retailer whose business model, and therefore whose problems, the merchant does not have.
- **The First Snowfall** — seasonality is a property of the business model, and the merchant that has not measured its peak meets it in production.

A brand's web shop sells a jacket to a shopper; its wholesale team sells the same jacket to a retail chain. The shopper expects an agreed price, a delivery date and a return right. The chain may require a contract price, correctly labeled pallets and an invoice its systems can read. Software built for the first sale can accept the second order yet fail to meet its conditions.

Marketplace sellers must also meet the marketplace's rules or face penalties. The marketplace operator has a different job: coordinate the shopper's experience across sellers and pay those sellers. Similar-looking orders can create very different obligations.

This chapter explains the business models through those obligations, then follows the money an order earns and costs. Its practical question comes from Kestrel & Pine, the fictional outdoor brand: **what should four business lines share, and what must stay separate?** The four scale stages introduced here help answer that question as the company grows.

Where we are in the order's journey. Eva's Ridgewalker jacket is a consumer purchase. The same jacket also leaves Kestrel & Pine through a wholesale reorder from retail chain Bergzicht Outdoor, an Amazon.de order, or a sale in the Antwerp store. Each needs a different route from order to payment and delivery. Follow the Order: What Happens Between "Buy" and "Delivered" traces the consumer route.

Setting the Scene at Kestrel & Pine: One Jacket, Four Ways to Sell It

In February 2027, Head of Engineering Noor Visser reviewed the accounts with Finance Director Sanne de Wit before proposing **Waypoint**, the company's program to improve its commerce systems. Kestrel & Pine's 2026 revenue was **€48.0 million**, with a **54%** gross margin: €27.0 million online, €12.0 million wholesale, €6.0 million through six stores and €3.0 million through three marketplaces. All four lines shared one warehouse in Duiven and a catalog of 3,400 products.

Then Noor asked each head of a business line what they needed from technology in 2027.

Milan Petrov, responsible for storefronts and marketplaces, wanted more shoppers to complete checkout, credible delivery dates and fresher marketplace stock. Wholesale lead Lea Moreau needed correct contract prices and fewer orders typed from email into the enterprise resource planning system (ERP), which keeps the books and wholesale orders. Manual entry still accounted for 38% of wholesale orders. She also wanted Bergzicht's electronic data interchange (EDI), the exchange of structured business documents, to stop producing chargebacks.

Operations lead Daniel Okafor needed pallets and parcels to stop competing for the same pickers, and the web shop to see store stock. Sanne needed one definition of an order so finance could close the month faster.

Each request exposed a different business obligation. The systems served individual needs but did not carry each order reliably from sale to completion. Noor's proposal had to identify the facts the four lines could share and the rules each needed to keep.

The Business Models, Defined by the People Who Report Them

Marketing uses the model names loosely. Statistics agencies and annual reports use them precisely, because they have to count something. The table lifts their definitions.

Model	Working definition	Who reports it that way
B2C (business to consumer)	Sales to private consumers over websites or apps	Eurostat counts "web sales" to private consumers separately from sales to enterprises and public authorities (data extracted February 2026)
D2C (direct to consumer)	A brand selling through channels it operates itself: its own web shop, its own stores	VF Corporation's fiscal 2026 annual report (May 2026): direct-to-consumer "includes VF-operated retail stores, brand e-commerce sites, concession retail locations and other digital platforms", 44% of its revenue
Wholesale (B2B for a brand)	Selling to retailers and distributors who resell	Columbia Sportswear's 2025 annual report (February 2026) separates its wholesale channel (department store chains, internet retailers, international distributors) from its direct-to-consumer stores and sites
B2B e-commerce (business to business)	Business buyers ordering electronically, by web shop or by structured messages between systems	Eurostat splits "web sales" from "EDI-type sales": in 2024, EDI-type sales were 11.07% of EU enterprise turnover and web sales 8.39%, so most business ordering is still by structured message, not by web shop

Model	Working definition	Who reports it that way
B2B2C (business to business to consumer)	A brand sells through a partner's consumer channel while keeping control of its product, content and price	Zalando's 2025 annual report (March 2026) describes a partner business that lets brands sell on Zalando "while maintaining full control over their offer, content and pricing"
Marketplace selling	Selling on someone else's marketplace	Eurostat: web sales through marketplaces were 1.30% of EU enterprise turnover in 2024, against 7.08% through own websites or apps
Marketplace operating	Running the marketplace and taking a fee from sellers	Amazon's 2025 annual report (February 2026): "We offer programs that enable sellers to sell their products in our stores, and fulfill orders using our services. We are not the seller of record in these transactions"; third-party seller services were $172.2 billion of net sales
Omnichannel	Stores and digital channels sharing stock, orders and customers	Target's fiscal 2025 annual report (March 2026): stores fulfilled 97.6% of its merchandise sales, including shipment from stores, order pickup and same-day delivery

Model	Working definition	Who reports it that way
Subscription	Recurring orders under a standing agreement, or a paid membership that changes the promises	Amazon reports "subscription services" of $49.6 billion in 2025 (Prime and digital subscriptions); Target's paid Target Circle 360 membership gives "unlimited same-day delivery on eligible orders over $35"

Figure 1: *The nine business models at a glance: each is a different answer to who sells, to whom, and through whose channel, and a merchant usually runs several at once.*

Three terms in the table need plain words. The **seller of record** is the party that legally sells to the customer: the one whose name is on the invoice, who owes the VAT (value-added tax) and who answers for the consumer's rights. A marketplace operator usually is not, which is why Amazon says so in its annual report. **GMV** (gross merchandise value) is the value of orders passing through a platform, including what the platform does not keep: Shopify's 2025 annual report (February 2026) defines it as "the total dollar value of orders facilitated through our platform" net of refunds. **EDI-type sales** are orders received as structured messages between systems, which Documents Between Companies: EDI, Punchout and E-Invoicing explains; the point here is that for business

customers they are the majority.

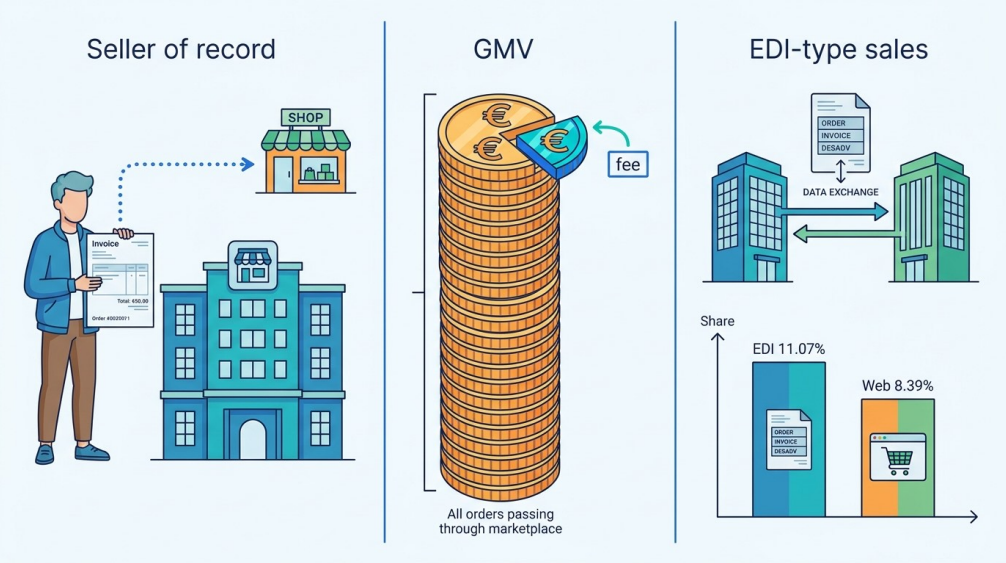

Figure 2: *Three terms in pictures: the seller of record is the name on the invoice, GMV counts every order but the marketplace keeps only its fee, and in the EU more business turnover arrives by structured message (11.07%) than through web shops (8.39%) (Eurostat, 2024 data).*

The models are not exclusive. Kestrel & Pine runs four at once, which is what "omnichannel" means in practice, and most merchants above the Starting stage run at least two.

What Each Model Changes in the Promise Chain

Follow the Order: What Happens Between "Buy" and "Delivered" listed the promises an order carries: a product, a price, an availability, a delivery, a return right, a refund, an invoice. A business model is a decision about who makes and keeps each of them. The table says who, per model; the architectural consequence follows in the last column.

Promise	D2C brand	Wholesale	Marketplace seller	Marketplace operator	Store (omnichannel)
Who captures the order	The brand's storefront	The buyer's system (EDI, punchout) or the brand's B2B portal	The marketplace; the seller imports it	The marketplace, for thousands of sellers	The till (the POS, point of sale)
Who sets the price	The brand, per market	Negotiated per account; price lists and terms	The seller, under parity rules and repricing pressure	The sellers; the operator sets fees	The brand, but often a separate price list
Who is seller of record	The brand	The brand (to the retailer)	The seller	Usually not the operator	The brand
Who holds the stock	The brand's warehouse or a 3PL (a third-party logistics warehouse)	The brand's warehouse; pallets, not parcels	The seller, or the marketplace's fulfillment program	The sellers; the operator in its fulfillment program	The store, as a location of its own
Who ships	The brand via carriers	The brand to a distribution center, with labels the buyer requires	The seller or the marketplace	The sellers or the operator	The customer carries it; or the store ships

Promise	D2C brand	Wholesale	Marketplace seller	Marketplace operator	Store (omnichannel)
Who handles returns	The brand's portal and warehouse	Rare; defects and chargebacks instead	The marketplace's policy, debited to the seller	The operator's policy	The store, often for online orders too
Who invoices	The brand, per order	The brand, per shipment, often electronically	The seller (or the marketplace on its behalf)	The operator invoices fees to sellers	The till, with fiscal rules per country

Promise	D2C brand	Wholesale	Marketplace seller	Marketplace operator	Store (omnichannel)
Architecture consequence	The brand owns the whole chain and every system in it	Order capture moves into the buyer's systems; prices and terms per account are the hard part	Someone else's catalog format and order format become the contract	Tenancy, payouts and seller-of-record questions; a different book (Selling on Someone Else's Platform: Marketplaces and Channel Integration and LISTED's Rules of the Marketplace: DSA, DAC7 and Vertical Regulation)	Store stock becomes sellable online; the till becomes a system

A few words in the table need plain definitions. **Punchout** lets a business buyer open the supplier's catalog from inside the buyer's own purchasing system and send the filled cart back there for approval. **Parity rules** are a marketplace's terms that a seller may not offer the same product cheaper elsewhere; **repricing** is the automatic adjustment of marketplace prices against competing sellers. **Tenancy** is the vendor's job of running many sellers or merchants on one shared system while keeping their data apart.

Two rows deserve emphasis. **Who captures the order** decides where the order record is born; a merchant with orders born in four systems needs a fifth to hold them (The Order Record: Lifecycle, State and the Order Management System). **Who holds the stock**

decides how many locations the availability calculation must cover; a marketplace's fulfillment program is a location the merchant can see but not count (Is It in Stock? Inventory, Availability and the Promise You Can Keep).

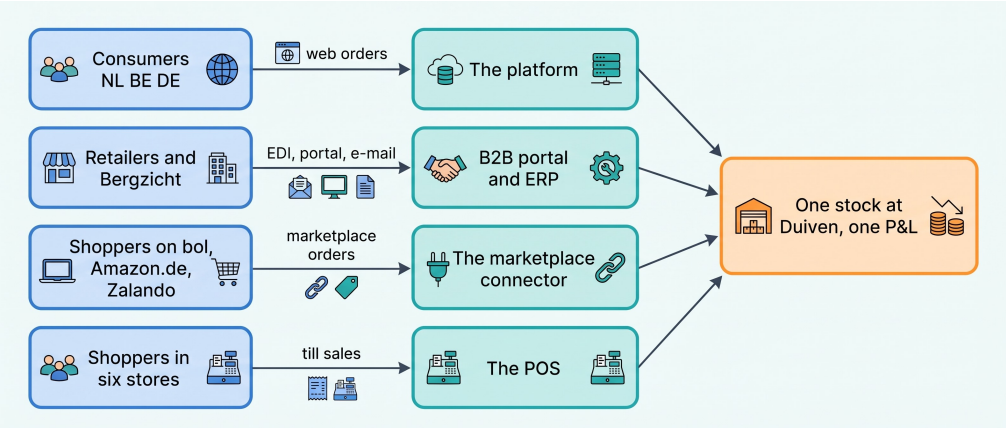

Figure 3: *Kestrel & Pine's four business lines capture orders in four systems and draw on one stock and one profit-and-loss statement; what the lines share is the stock and the money, and what they do not share is the promise.*

The **merchant's view and the vendor's view** of the same platform differ in the same way. Shopify's 2025 annual report (February 2026) notes that while most of its merchants subscribe to its Basic and Grow plans, the majority of its GMV comes from merchants on Shopify Plus and enterprise offerings. For the vendor, the product is a tenancy and GMV is the metric. For the merchant, the platform is one system among a dozen and the order is the metric. A merchant reading a vendor's roadmap should remember which metric it was written for.

The Unit Economics of an Order

How much money does one order make? The answer decides which promises a merchant can afford, and each number in it pulls on a different part of the architecture. The table gives the metrics, a dated public benchmark for each, Kestrel & Pine's own figure, and the part of the platform the metric pulls on.

Metric	In plain words	Public benchmark (dated)	Kestrel & Pine 2026	What it pulls on
Conversion rate	Purchases divided by visits	Median 1.3%, mean 1.9% across 438 Shopify stores in the 90 days to 31 October 2026 (Littledata, November 2026)	Not recorded *(assume about 1.8%)*	Storefront speed, checkout steps, visible total cost (From Cart to Confirmation: Checkout Orchestration, Delivering the Storefront: Themes, Headless, Rendering and Performance)
Cart abandonment	Carts started and not bought	70.22% average of 50 studies, 2006–2025 (Baymard, October 2026); top reasons: extra costs 40%, slow delivery 20%, forced account 18%, site errors 17%	—	The same: the top reasons are architecture failures, not design failures

Metric	In plain words	Public benchmark (dated)	Kestrel & Pine 2026	What it pulls on
Average order value (AOV)	Revenue per order	Shopify merchants' average cart $114.70 over the 2025 Black Friday weekend (December 2025); Zalando's average basket €62.8 in 2025 (March 2026)	€92 online, €70 market-places, €51 stores, €1,277 wholesale	Promotions, bundles, free-shipping thresholds (Discounts, Coupons and Loyalty: Promotions Without Chaos); a wholesale order is a different object, not a bigger basket
Gross margin	Revenue minus cost of goods, as a share of revenue	VF 54.8% (FY2026), Columbia 50.5% (2025), Nike 42.9% (FY2026), Zalando 39.9% (2025), Target 27.9% (2025)	54%	Markdown and price control: one price owner (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax)

Metric	In plain words	Public benchmark (dated)	Kestrel & Pine 2026	What it pulls on
Return rate	Share of items or orders that come back; say which	19.3% of US online sales projected for 2025 (the National Retail Federation, NRF, with Happy Returns, October 2025); 24.4% of online apparel orders in a 2023 survey (Coresight; evidence age)	24% of items, 19% of orders online; 72,000 return parcels; 61% resold as new	The reverse flow: restock, refund, credit note across systems (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics)
Shipping cost and threshold	What a parcel costs and when it is free	Extra costs are the top abandon-ment reason (Baymard); Zalando charges a “minimum order fee” on small orders (2025 annual report)	Free over €75; 336,000 parcels	The delivery promise and carrier choice (Getting It There: Carriers, Labels, Tracking and the Delivery Promise)

Metric	In plain words	Public benchmark (dated)	Kestrel & Pine 2026	What it pulls on
Payment fees	What the PSP (payment service provider) keeps per transaction	EU interchange capped at 0.2% debit and 0.3% credit for consumer cards since 9 December 2015 (Regulation 2015/751); Dutch list prices, n.d., accessed November 2026: Stripe 1.5% + €0.25, Mollie 1.80% + €0.25, Shopify Payments 1.9% to 1.4% + €0.25, iDEAL about €0.29–0.32 per payment	Not recorded	PSP choice and platform tier (Taking the Money: Payment Providers, Authorization, Capture and Refunds)

Metric	In plain words	Public benchmark (dated)	Kestrel & Pine 2026	What it pulls on
Customer acquisition cost	Marketing spend per new customer	Warby Parker spent $110.2 million on marketing against $871.9 million revenue in 2025 (12.6%), up from 11.7% in 2023 (annual report, February 2026)	Not recorded	Analytics and identity: knowing who is new (Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics)

Two observations follow for the newcomer. First, **the abandonment reasons are an architecture list**. Unknown shipping costs until the last step, slow delivery promises, forced account creation and site errors are each a failure of a system to make a promise early and keep it. Second, **payment fees are an architecture decision disguised as a finance one**. For a merchant at €27 million online revenue, the gap between 1.9% and 1.4% plus a fixed fee is six figures a year, and whether the platform forces its own payment gateway, and at what fee, is one of the platform's pricing levers.

The Same Jacket in Three Lines: A Worked Example

Kestrel & Pine's Ridgewalker jacket sells at **€199** to a consumer (including 21% Dutch VAT), at **€95** excluding VAT to a wholesale account, and at €199 on Amazon.de; its landed cost is **€58**. The table works one unit through three lines. Unmarked figures are Kestrel & Pine's own (fictional); everything marked *(assumption)* is illustrative and not a benchmark.

Line	Online D2C (Netherlands)	Wholesale (Bergzicht)	Marketplace (Amazon.de, seller-fulfilled)
Price the customer pays	€199 incl. VAT	€95 excl. VAT	€199 incl. VAT
Net revenue after VAT	€164.46	€95.00	€167.23 at German VAT of 19%
Landed cost	€58.00	€58.00	€58.00
Gross profit per unit	€106.46 (65% of net)	€37.00 (39%)	€109.23
Payment or channel fee	iDEAL, about €0.30 *(from list prices)*	None; invoice on 30-day terms	Referral fee: Amazon's US pricing page lists 15% for most categories (n.d., accessed November 2026); *(assume 15% of €199, €29.85)*
Outbound shipping	One parcel, *(assume €5.50)*	240 units on three pallets, *(assume €0.60 per unit)*	One parcel, *(assume €7.50 cross-border)*
Returns	24% of items; *(assume €7.50 per return in carriage and handling, €1.80 per unit sold)*	Rare; chargebacks instead (€14,200 in Q3 2026)	Marketplace policy; *(assume as D2C)*
Marketing	*(assume 12% of net revenue, in the range Warby Parker reports: €19.74)*	A sales team, not per order	Sponsored listings; *(assume 5%, €8.36)*
Contribution per unit *(illustrative)*	**about €79**	**about €36**	**about €62**

Line	Online D2C (Netherlands)	Wholesale (Bergzicht)	Marketplace (Amazon.de, seller-fulfilled)
What the line needs from the platform	Fast checkout, computed delivery date, one price per market, a returns flow that restocks and refunds once	Contract prices per account, electronic order capture, pallet picking with the labels the buyer demands, an invoice the buyer's system reads	A listing in the marketplace's format, stock per channel with a buffer, parity with the storefront price, the marketplace's return and performance rules

Read across, the table says why the four heads of business lines asked for different things. The D2C unit earns the most and spends it on marketing, parcels and returns, so its platform must convert, promise and take returns well. The wholesale unit earns less than half as much but ships 240 at a time, on one invoice, with no returns, so its platform must get prices and documents right and nothing else. The marketplace unit pays a fee for the traffic D2C buys with marketing, and lives under rules it did not write. **One jacket, three businesses, three sets of promises, and the architecture has to serve all three without pretending they are the same.**

Figure 4: *One Ridgewalker jacket, three businesses: about €79 per unit online, €36*

wholesale and €62 on a marketplace, each with its own promises for the platform to keep.

Seasonality and the Peak

Every consumer model has a peak, and the peak is the design load. The US Census Bureau's quarterly e-commerce release (August 2026) shows the shape in one number: e-commerce was 18.3% of retail sales in the fourth quarter of 2025, not seasonally adjusted, against 15.8% in the third. Adobe Analytics reported $44.2 billion of US online spending in Cyber Week 2025, with $16 million per minute between 8 and 10 pm on Cyber Monday (December 2025). Shopify reported $14.6 billion of GMV over the 2025 Black Friday weekend with a peak of $5.1 million per minute (December 2025), which is the moment a hosted platform earns its fee and a self-run one needs a load test.

Kestrel & Pine's peak is smaller and no gentler. Black Friday week 2026 brought **31,000** online orders, **10.6%** of the year's, and the peak hour on Friday 27 November carried **2,300** orders, **38 per minute**, with storefront traffic about **41 times** an average hour. The platform held. The stock sync chain and the warehouse queue did not, and the Black Friday Oversell followed. Two architectural facts come out of this. **Capacity is a property of the whole chain**, including every integration, not of the storefront, which Black Friday Every Year: Peak Load, Resilience and Recovery works. And **the peak sets the calendar**: Hobbycraft, a UK retailer, postponed a website relaunch from before Christmas 2012 to 2013 because, in its chief executive's words, "we felt it was too risky to launch the website in peak trading" (Retail Week, February 2013; evidence more than a decade old, and still the rule). Wholesale has its own seasons, set by retailers' buying calendars; Bergzicht's spring reorder in April is one.

The Four Scale Stages

The book compares every decision across four **scale stages**. This chapter owns the term: a scale stage is a **shape of business**, described by how many channels create orders, how many locations hold stock, how many countries and brands are served, and who runs the technology. The revenue bands are rough markers; the shape decides.

Stage	Typical size	Shape	Business models typically run	What becomes necessary	Warning signs of outgrowing it
1 — Starting	Under about €2 million	One channel, a hosted platform, founder-run operations, a 3PL or a back room	One model: D2C, or a single marketplace, or a small B2B shop	The platform's built-in features; an accounting export; a PSP	Orders keyed by hand into anything; the founder as the integration
2 — Growing	About €2–20 million	Two or three channels, first store or wholesale line, first integrations, a small team and an agency	A second model is added, usually wholesale, a marketplace or a store	An ERP or accounting system, a 3PL or warehouse system, a marketplace connector; one agreed order number	The first oversell between two systems; an agency script nobody owns

Stage	Typical size	Shape	Business models typically run	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel	About €20–200 million	Stores, wholesale, marketplaces and several countries; own warehouse; a dozen or more systems	Three or four models on one stock and one catalog	An order hub (an OMS, order management system), a product hub (a PIM, product information management system), an availability service, integration standards, one metric definition	Orders in four systems; two-hour stock lag; three definitions of “order”; support agents with five tabs

Stage	Typical size	Shape	Business models typically run	What becomes necessary	Warning signs of outgrowing it
4 — Group	Over about €200 million, or several brands or business units	Multiple brands, countries or units; platform teams; acquisitions	All models, per brand or country, with different mixes	Shared services where the benefit is measured; autonomy where it is not; vendor-scale concerns (tenancy, extensibility)	A brand forced onto a shared stack that does not fit, or kept apart at twice the cost

Kestrel & Pine passed through the first three: Starting in 2016–2018 with one Dutch web shop and a 3PL, Growing from 2019 with the first store, Belgium, Germany, the marketplace connector and the integration platform, and Omnichannel from 2023 with its own warehouse, the ERP, more stores and wholesale on the platform's business-to-business features. The Omnichannel stage is where its 2027 problems live, because that is the stage at which three or four models share one stock and one catalog, and the joins between a dozen systems become the business.

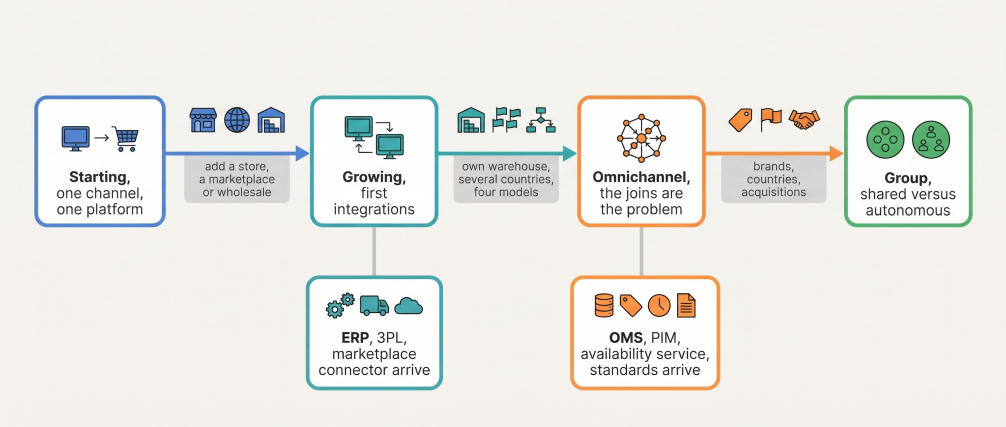

Figure 5: *Each stage transition adds a channel or a location and, with it, a join between systems; the hubs arrive at the Omnichannel stage because that is where the joins outnumber the boxes.*

Vendors confirm the stages in their price lists. Shopify's pricing page (n.d., accessed November 2026) runs from Basic at €28 a month (€21 paid yearly) to Plus "from €2,100" a month on one- or three-year terms, and its fee for using a third-party payment gateway falls from 2% on Basic to 0.2% on Plus. Shopware's pricing page (n.d., accessed November 2026) makes the Community Edition free until the shops linked to an account exceed €1 million of GMV in a year, after which a paid plan starts "from €600" a month, and states that pricing "is based on your gross merchandise value". Shopify's annual report says most merchants are on its two smallest plans while most GMV comes from Plus and enterprise. In plain words: a merchant crosses from a self-service plan to a negotiated contract at about the Growing-to-Omnichannel transition, and the fee cliffs and feature gates in a vendor's tiers are a map of which decisions the vendor expects at each size.

When the hubs arrive. Public sources do not establish a revenue threshold for buying an order, product or warehouse management system. Look at the work instead. Akeneo, a product information management (PIM) vendor, points to scattered content, slow manual channel launches and inconsistent product data (undated guidance, accessed November 2026). Practitioner guides for order management systems (OMS) emphasize multiple sales channels and stock locations.

My judgment is to consider a PIM when a third channel brings another content format: one approved product record can replace repeated manual corrections. An OMS becomes useful when orders need allocation across a second stock location; a warehouse management system (WMS) when receiving, picking and packing outgrow the ERP's stock module. These are reasons to investigate, not purchase thresholds. The coordination benefit must justify another system to operate. Kestrel & Pine's Waypoint program adds a PIM, an OMS and availability management at its Omnichannel stage. What Changes at Each Scale: From First Store to Group examines each transition.

The Choices: One Stack, Many Stacks or a Shared Core

A merchant running several business models has three realistic ways to arrange its systems. Each is right somewhere.

One stack for every line. The consumer platform also runs the wholesale portal, the stores on its own point-of-sale product and the marketplace listings through its channel features; everything shares one catalog, one stock figure and one order list. It is the simplest to run and the cheapest to integrate, and it fits while the lines are small and alike. It limits when one line's promises do not fit the platform's model: contract prices per account with approvals, pallets with the buyer's labels, fiscal receipt rules in a store, or marketplace order formats the connector does not speak.

A stack per line. Each line buys the systems its model needs, each with its own orders, prices and stock, and gets the best fit for its promises. The cost is the joins: four order records, four prices for one jacket and four stock figures, which is Kestrel & Pine in January 2027 and the origin of the Price Drift, the Three-Status Order and the Store Stock Mirage.

A shared core with line-specific edges. The facts every line shares, the product, the price list, the stock and the order record, are owned once, in systems that serve all lines: a product hub, a pricing source, an availability service and an order hub. The promises that differ per line are made at the edges: the consumer checkout, the B2B portal, the till, the marketplace listing. This is the Waypoint shape, and the one most Omnichannel merchants arrive at. It costs more systems and more integration discipline than one stack, and it only pays once the joins have become the problem.

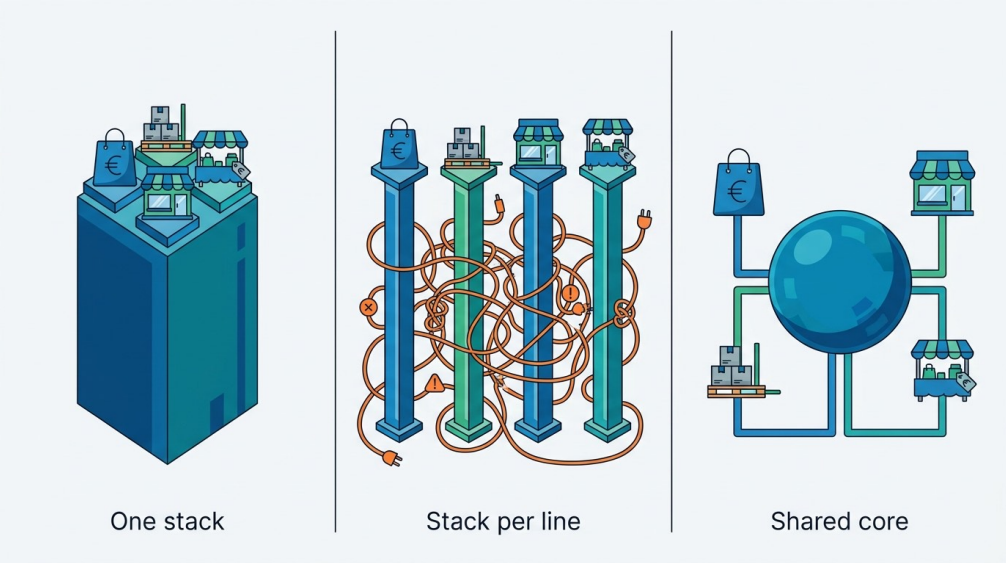

Figure 6: *One stack is simplest while the lines are alike, a stack per line multiplies the joins, and a shared core owns the common facts once while each line keeps its own edge.*

Criterion	One stack for every line	A stack per line	Shared core, line-specific edges
Number of order records	One	One per line	One, in the hub; the edges keep their own views
Fit of each line's promises	Only as good as the platform's model of that line	Best per line	Edges fit the line; the core fits all
Integration load	Lowest	Highest and unplanned	High but designed: every join has an owner and a contract
Typical stage	Starting, early Growing	Late Growing by accident	Omnichannel by decision; Group with per-brand edges
Warning sign	A line's team working around the platform in spreadsheets	Two prices for one jacket; support with five tabs	A core that becomes a bottleneck for every change

Criterion	One stack for every line	A stack per line	Shared core, line-specific edges
Kestrel & Pine	2016–2018	2021–2026	Waypoint, 2027–

My recommendation, with its conditions: **stay on one stack as long as every line's promises fit it, and move to a shared core the moment a second system starts writing a fact the platform also writes**, because that is the moment the joins start costing money. Do not move to a stack per line on purpose; it is only ever where a merchant ends up by buying one good system per need. And do not copy the shape of a marketplace operator, which runs tenancy, payouts and seller-of-record logic a brand never needs: the Giant's Boots.

In Practice: What the Public Record Says

Stores are the larger channel, and the architecture must treat them as one. Online sales were 13.5% of German retail in the latest HDE Online-Monitor (2026 edition, for 2025) and 17.1% of US retail in the second quarter of 2026 (Census Bureau, August 2026, seasonally adjusted). For a retailer with stores, the store is still where most sales happen; Target's annual report (March 2026) says its stores fulfilled 97.6% of its merchandise sales, including digital orders shipped from stores, picked up or delivered same-day. That is an order management decision at national scale, though Target does not name its system.

D2C is a channel mix with its own economics, and it can shrink. Nike's fiscal 2026 annual report (July 2026) reports NIKE Direct revenue of $17.7 billion against $18.8 billion the year before, "primarily driven by a decrease in traffic", and attributes its gross margin movement partly to "channel mix". VF's direct-to-consumer share is 44%. The lesson for an architect is that a brand's mix of D2C and wholesale moves year to year, and the platform must serve both without a replatform each time.

Marketplace operating is a different business. Amazon is not the seller of record for third-party sales and earned $172.2 billion from seller services in 2025 (annual report, February 2026); Marketplace Pulse estimated third-party sellers at about 61–62% of Amazon's units and 69% of an estimated $830 billion of 2025 GMV (February 2026; an analyst estimate). Zalando's partner business was 32.1% of its GMV in 2025 (annual report, March 2026). An operator's platform is built around sellers, listings, fees and payouts, which is why the book leaves it to Case: Amazon, From Monolith to Services to the Everything Store, Case: Zalando, Microservices, Platform Strategy and the Return to Discipline and LISTED.

Business customers order by message, not by web shop. Eurostat's enterprise statistics (data extracted February 2026) put EDI-type sales at 11.07% of EU enterprise

turnover against 8.39% for web sales; Forrester's 2022 forecast (evidence age: more than three years) expected US B2B e-commerce to reach about $3 trillion by 2027 while tracking EDI separately at about a fifth of transactions. A wholesale line meets EDI before it meets a B2B web shop, as Kestrel & Pine's did.

Growth has normalized. The Dutch Thuiswinkel Markt Monitor (March 2026) reported Dutch online consumer spending of €35.7 billion in 2025, down 1%, and the German bevh (January 2026) reported online goods sales of €83.1 billion, up 3.2%. With growth in single digits, the Growing and Omnichannel stages compete on margin, returns and operations rather than on traffic, which is why the chapters on orders, returns and data matter more than they did in 2021.

When It Goes Wrong

Each model fails in its own way when it shares systems with another, and Kestrel & Pine has lived most of them.

Two lines, one jacket, two prices. The Price Drift of October 2026: a price-list sync to the German storefront failed silently, and the jacket sold at €179 for nine days while the marketplaces and stores charged €199. 410 orders, about €8,200 of margin, and a price-parity warning from a marketplace whose rules forbid a seller's own site undercutting the listing. Detection came from the marketplace, not from Kestrel & Pine. The remedy is one price owner per market and a reconciliation that compares channels daily (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax).

Wholesale on a consumer platform. The platform's B2B features carry 62% of wholesale orders; the other 38% arrive by e-mail and phone and are keyed into the ERP by Lea's team, so a third of the wholesale line's orders are born in a system the consumer line never sees. The ASN Chargeback of Q3 2026, €14,200 deducted by Bergzicht because the pallet labels did not match the despatch advice (the electronic advance shipping notice, ASN, that tells the buyer what is on each pallet), is a wholesale promise (the labels the buyer's warehouse requires) that no consumer system knows how to make (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals, Documents Between Companies: EDI, Punchout and E-Invoicing).

The store as an afterthought. Store stock reached the ERP nightly and never reached the platform, so the Utrecht click-and-collect pilot promised 37 items in December 2026 that the store had already sold: the Store Stock Mirage. The store line carries no parcel and no return parcel per order, and its stock was invisible to the channel that could have sold it (Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment).

Returns across lines. 24% of online items come back, and in January 2027 212 of them were refunded twice, once by the returns portal and once by hand, because two tools, the returns portal and the PSP's console, could each issue a refund and neither could see the other's. A return rate that high is a business-model fact, and the platform's reverse flow

has to be designed for it, not bolted on (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics).

Detection in every case was the same: someone compared two systems' copies of the same fact. Recovery in every case was the same: decide who owns the fact and make the others consumers of it, which is Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record.

What to Take Away

Established.

- The models are defined operationally by the companies and agencies that report them: D2C as brand-operated channels (VF, Nike), wholesale as sales to resellers (Columbia), marketplace operating as not being the seller of record (Amazon), B2B2C as the partner owning the storefront while the brand keeps the offer and price (Zalando), omnichannel as stores fulfilling digital orders (Target, 97.6%).
- Business electronic ordering is mostly by structured message: EDI-type sales exceeded web sales as a share of EU enterprise turnover in 2024 (Eurostat).
- The order's economics are measurable and public benchmarks exist for each: conversion around 1–2%, cart abandonment about 70%, online return rates around a fifth and higher for apparel, interchange capped in the EU, PSP list prices published; the peak quarter carries a visibly larger share of sales (Census, Adobe, Shopify).

Depends on context.

- Whether one stack can serve several lines depends on whether each line's promises fit the platform's model; a small wholesale line on a consumer platform's business features works until contract pricing, approvals, pallets or EDI appear.
- Whether a merchant has reached the Omnichannel stage depends on shape, not revenue: three or four models on one stock, several locations and a dozen systems. Vendors' tier boundaries (Shopify Plus, Shopware's €1 million GMV threshold) are a reasonable proxy, not a definition.
- Whether payment fees are worth negotiating depends on volume and method mix; at a €92 basket and 293,000 orders, a half-point difference is material; at the Starting stage it is not.

The author's synthesis.

- I define a business model, for architecture purposes, as the answer to "who makes which promise": once that table is filled in for every line, the systems-of-record decisions mostly write themselves, and a vendor's model label stops mattering.

- I treat the unit-economics table as a map of where the architecture must be strong: a merchant with a 24% return rate must design the reverse flow first, and one with a 41-times peak hour must design the whole chain's capacity, not the storefront's.

Questions to Ask

1. For each business line, who captures the order, who sets the price, who is the seller of record, who holds the stock, who ships, who takes returns and who invoices?
2. Which facts do the lines share (product, price list, stock, order, customer), and which system owns each today?
3. What are the conversion rate, basket, margin, return rate, shipping cost and payment fee per line, and which of them has nobody measured?
4. What is the busiest hour of the year, how many times an average hour is it, and has every integration, not only the storefront, been tested at that load?
5. Which stage is the business in by shape, and which transition is next: a new channel, a new location, a new country, a new brand?
6. Where in the vendor's price list does the business sit, and which fee cliff or feature gate is nearest?
7. If a line's promises do not fit the shared stack, is the team working around it in spreadsheets, and who knows?
8. Which architecture is being copied, from whom, and does that company have the same business model?

The models and the numbers say what a commerce platform must do; they do not yet say what its parts are. The next chapter lays out the capability map: every part of a commerce platform, what each owns, and how they usually connect, so that the four business lines can be placed on it and the gaps seen. Continue with The Map of a Commerce Platform: Capabilities and Layers.

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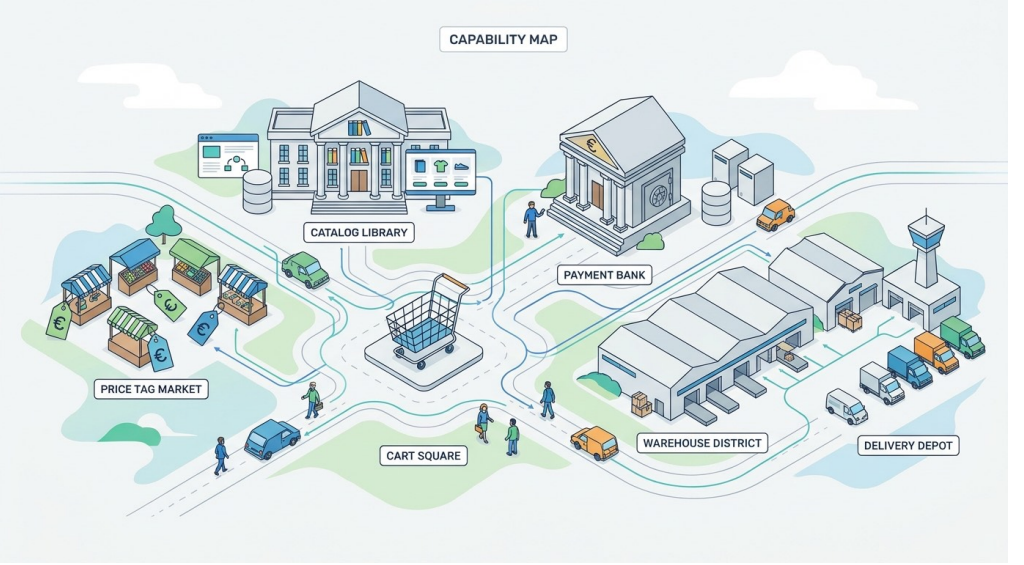
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

3. The Map of a Commerce Platform: Capabilities and Layers



IN THIS CHAPTER, YOU WILL: Learn the capabilities a commerce platform is made of, what each one owns, how they connect, and how to place any vendor or system on the map.

KEY POINTS:

- **A capability is a job the business needs done, not a product.** "Know what we can still sell" is a capability; the warehouse system, the finance system and an app are three systems that may each claim it. Draw the map in capabilities first, then place systems on it, and the gaps and overlaps appear by themselves.
- **Every capability owns one kind of business fact, and most of the others only copy it.** Product content, price, stock, order, payment, shipment, return, invoice and customer each need one home; the flows between capabilities are how the copies learn about changes. Where two systems both believe they own a fact, that is where the money leaks.

- **Vendors cover the map in three ways, and none covers all of it**. A suite covers many boxes with one vendor's data model; a platform plus apps covers the center and rents the edges; a best-of-breed assembly buys one product per box and makes the joins your problem. The right mix depends on the business shape and the stage, and a vendor's product page is read best with the map beside it.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Spaghetti Junction** — connectors multiply when nobody has drawn which capability feeds which, so every new need gets its own wire.
- **The Giant's Boots** — a map copied from a far larger company fills boxes that a smaller merchant does not need and cannot run.
- **The Lynchpin** — a capability that nobody has placed on the map is usually running through one vendor, one script or one person without anyone noticing.

When a merchant asks for store pickup, checkout, the store and the stock system all need to cooperate. Buying a pickup app will help only if it can get stock reserved and tell store staff what to prepare. To work out what must change, start with the business jobs involved. A **capability** is one such job, such as pricing a product or fulfilling an order; several systems may contribute to it.

The same approach helps with more frequent marketplace stock updates or one reliable order count. Which capability must change? Which system handles it now? Who depends on its answer? Without that shared view, teams can buy overlapping tools and leave the handoff between them unowned. Kestrel & Pine, the fictional outdoor brand used here, reached 34 integrations and 41 apps through individually reasonable decisions.

The map below groups the capabilities, names the facts they handle and shows their connections. Use it to identify gaps and overlaps before comparing vendors. Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record explains how to settle ownership when several systems hold a fact; Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026 covers vendor categories.

Where we are in the order's journey. Eva's jacket passes through search, pricing, checkout, delivery and possibly a return. Each step is a capability on this map. Each handoff is a connection that someone must build and own. Follow the Order: What Happens Between "Buy" and "Delivered" follows those steps in sequence.

Setting the Scene at Kestrel & Pine: A Whiteboard in February 2027

When Noor Visser joined Kestrel & Pine as Head of Engineering in January 2027, the company had thirteen systems and no shared picture of them. Lead integration engineer Ruben Alves tracked the 34 integrations. Storefront lead Milan Petrov knew the 41 apps. Finance Director Sanne de Wit knew what reached the enterprise resource planning system (ERP). Their knowledge needed to be brought together.

In February, Noor drew the jobs needed to sell a jacket: show it, price it, check stock, take the order and payment, pick, ship, accept a return, invoice and count the sale. Only then did she place the thirteen systems into those boxes. **Starting with the jobs made the gaps visible**.

Three things were visible within an hour that had not been visible in five years of buying systems one at a time:

- **Two boxes were empty**. No system held the enriched product record (a product information management system, PIM); product content lived in the platform, the item master of the ERP (the enterprise resource planning system: the books, purchasing and stock valuation), shared spreadsheets and the marketplace connector. No system held every order (an order management system, OMS); orders lived in the platform, the connector, the ERP and the point-of-sale system (POS, the store's till software).
- **One box had six writers**. Warehouse stock was written by the WMS feed, by hand in the ERP, by hand in the platform and by three apps, with the POS writing store stock separately. The Black Friday Oversell of November 2026 (1,340 units oversold, 1,120 orders canceled or delayed, about €96,000) and the App That Wrote Stock of March 2026 (phantom reservations on 600 variants) were both on that box.
- **Two more boxes were nobody's**. Customer identity (is the store customer the online customer?) and availability (what each channel may still sell) had no system and no owner. So, in a smaller way, did the delivery promise: a sentence on the web page, "ordered before 22:00, delivered tomorrow", computed from nothing and owned by the shipping box only on paper.

The whiteboard became the first working document of the Waypoint program approved in March. The rest of this chapter is that whiteboard, generalized.

What a Capability Is, and Why It Comes Before the Systems

A **capability** is a job the business needs done, named without reference to who does it. "Calculate the price a given customer pays for a given product today" is a capability.

So is "know how many units of a variant are physically on a shelf in Duiven". A **system** is a piece of software that does one or more of those jobs; a **vendor product** is a system someone sells, usually claiming several capabilities at once. A **layer** is a group of capabilities that share a purpose and a rhythm: the ones a shopper touches, the ones that describe what is for sale, the ones that move money and orders, the ones that move boxes, the ones that talk to partners, and the ones underneath everything.

The distinction matters because systems and vendor products overlap, and capabilities do not. The ERP, the WMS and the commerce platform all have a stock field; there is still only one job called "know what is physically on the shelf", and only one system should be doing it. When you draw the map in capabilities, an overlap shows as one box with two names in it, and a gap shows as an empty box. When you draw it in systems, overlaps and gaps are invisible, because every system looks complete from the inside.

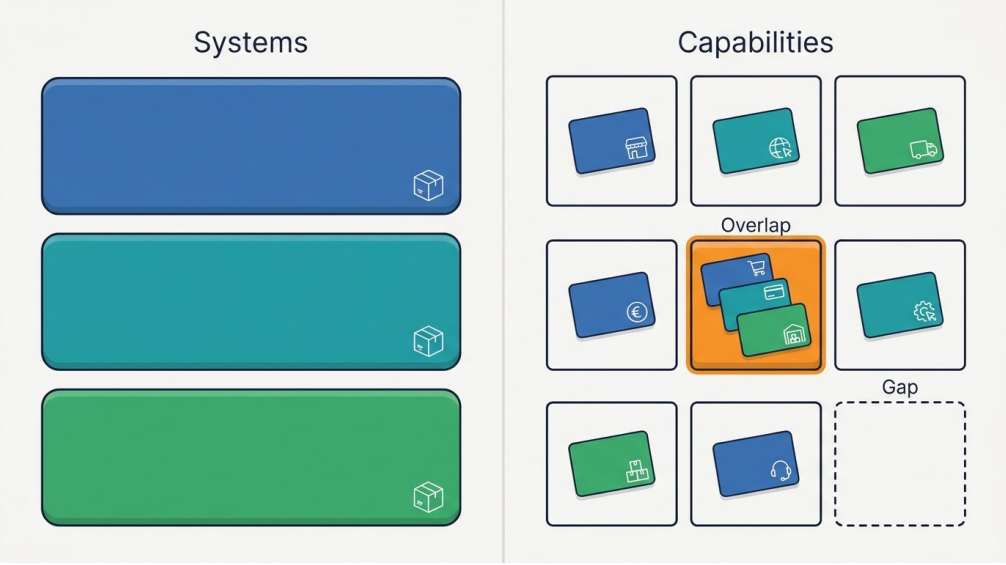


Figure 1: *Drawn as systems, every box looks complete; drawn as capabilities, an overlap shows as one box with several names and a gap as an empty box.*

Two more terms. A **business fact** is something the business needs to be true about the world: this variant has this GTIN (the Global Trade Item Number behind its barcode), this customer pays this price, this order has shipped. Most capabilities exist to create or keep one kind of fact. And a **flow** is the movement of a fact, or a request about it, from one capability to another: the order flows from checkout to order management to the warehouse; the stock level flows back. The map is boxes (capabilities) and arrows (flows), and later chapters are mostly about one box or one arrow each.

The Map: Six Layers

The map below is the author's synthesis, built from how the specialist vendors describe their own products (an OMS vendor's and a PIM vendor's definitions are quoted later in this chapter) and from the capability inventories of several commerce product families. It is a teaching map, not a standard: other authors cut the layers differently, and a vendor's architecture diagram will usually show its own product as the center.

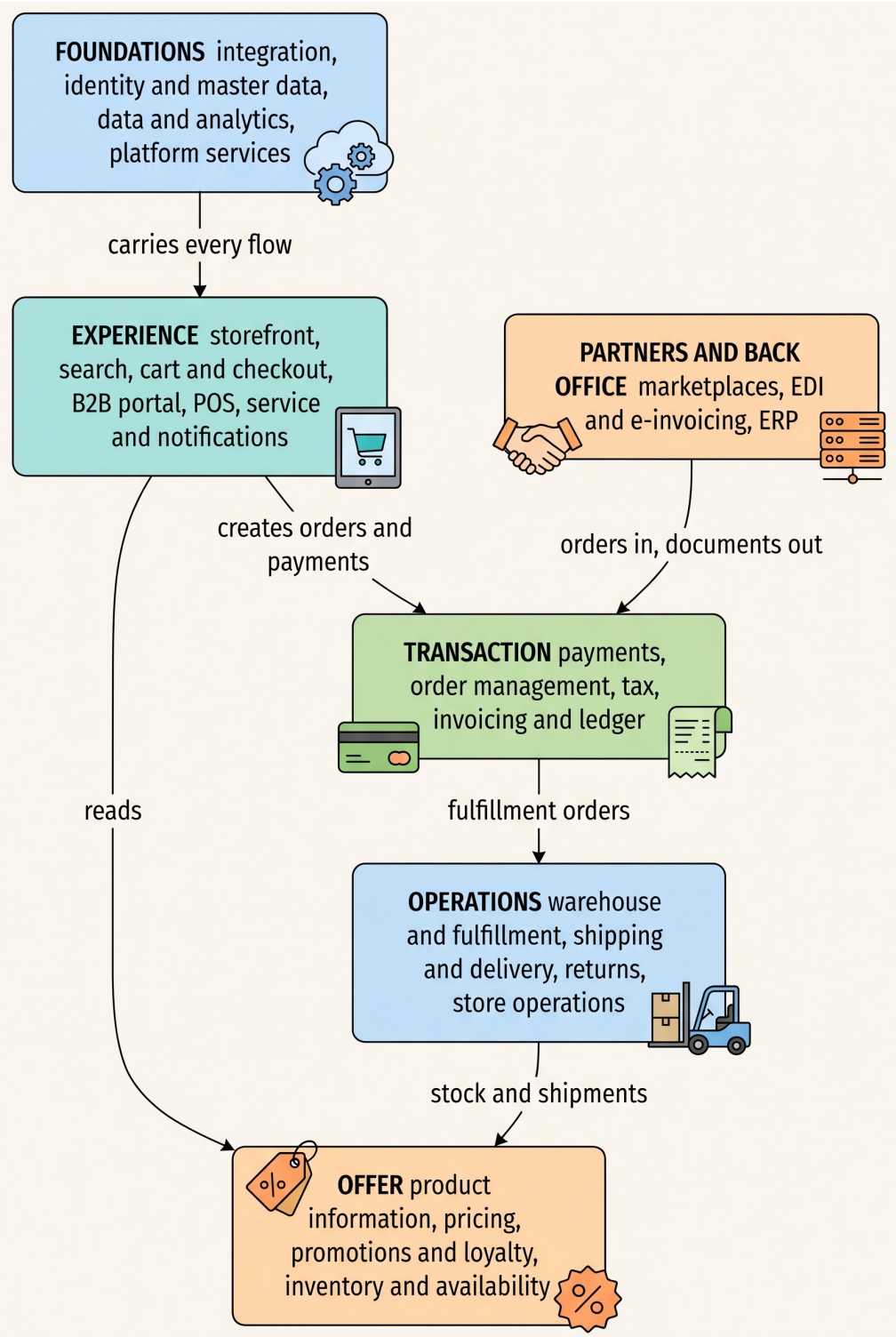

FOUNDATIONS integration, identity and master data, data and analytics, platform services
carries every flow
EXPERIENCE storefront, search, cart and checkout, B2B portal, POS, service and notifications
PARTNERS AND BACK OFFICE marketplaces, EDI and e-invoicing, ERP
creates orders and payments
orders in, documents out
TRANSACTION payments, order management, tax, invoicing and ledger
reads
fulfillment orders
OPERATIONS warehouse and fulfillment, shipping and delivery, returns, store operations
stock and shipments
OFFER product information, pricing, promotions and loyalty, inventory and availability

Figure 2: *Six layers of a commerce platform: the experience layer reads the offer layer and creates transactions, transactions become operations, operations feed stock back into the offer, partners inject orders and documents from outside, and the foundations carry everything.*

From the shopper inward: the **experience** layer is everything a shopper, store customer, wholesale buyer or service agent touches. The **offer** layer says what is for sale, at what price, under which promotion and whether it is available; it makes the promises the rest must keep. The **transaction** layer is the order and the money. The **operations** layer is the physical world: warehouse, carrier, store shelf, return parcel. The **partners and back office** layer holds systems that belong to other organizations or to finance: marketplaces, trading partners that speak EDI (electronic data interchange, the long-established standard for exchanging orders and invoices between companies' systems) and the ERP (enterprise resource planning system: the books, purchasing, stock valuation). The **foundations** are what no shopper sees and every layer depends on.

What each capability owns

This table is the chapter's main deliverable. For each capability it gives the job in plain words, the business fact it should own (the thing it is the natural home for), and the chapter that works it out. Terms used in the table are defined in those chapters; the one-line jobs are enough to place a system.

Layer	Capability	The job, in plain words	The fact it is the natural home for	Chapter
Experience	Storefront	Show products and content to shoppers on web and app, fast and accessible	Pages and themes (not the product data they show)	Delivering the Storefront: Themes, Headless, Rendering and Performance
Experience	Search and discovery	Let shoppers find products by query, filter and recommendation	The search index, a rebuildable copy of the catalog	Finding the Product: Search, Navigation and Merchandising

Layer	Capability	The job, in plain words	The fact it is the natural home for	Chapter
Experience	Cart and checkout	Hold a shopper's draft order and turn it into a paid order in a few steps	The cart (short-lived) and the checkout session	From Cart to Confirmation: Checkout Orchestration
Experience	B2B (business-to-business) buyer experience	Let a company's buyers order at their prices, with approvals, quotes and reorders	Company accounts, buyer roles, quotes	Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals
Experience	Point of sale	Sell in a store, offline if needed, with receipts and store stock	Store transactions and store stock	Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment
Experience	Customer service and notifications	Tell the customer what is happening and show an agent one view of it	Notification events sent; the service view (a copy)	Where Is My Order? Notifications, Service and the Customer Record
Offer	Product information	Hold products, variants, attributes, media and translations once, publish to every channel	Product content and the product identifiers	What Are We Selling? Products, Variants and the Catalog

Layer	Capability	The job, in plain words	The fact it is the natural home for	Chapter
Offer	Pricing	Compute the price a given customer pays for a given product in a given channel today	Price lists, contract prices, effective dates	What Does It Cost? Price Lists, Contract Prices, Currencies and Tax
Offer	Promotions and loyalty	Apply discounts, coupons, points and store credit without breaking price or margin	Promotion rules; store credit and gift card balances (liabilities)	Discounts, Coupons and Loyalty: Promotions Without Chaos
Offer	Inventory and availability	Know what is physically where, and compute what each channel may still sell	Available-to-promise per channel (derived from physical stock)	Is It in Stock? Inventory, Availability and the Promise You Can Keep
Transaction	Payments	Take the shopper's money, refund it, and reconcile it with payouts	Payment transactions (in the PSP, the payment service provider; the order keeps a reference)	Taking the Money: Payment Providers, Authorization, Capture and Refunds

Layer	Capability	The job, in plain words	The fact it is the natural home for	Chapter
Transaction	Order management	Hold every order from every channel, in one state model, and steer it to fulfillment and back	The order record and its state	The Order Record: Lifecycle, State and the Order Management System
Transaction	Tax, invoicing and ledger	Turn an order into tax, a legal invoice and entries in the books	Invoices, credit notes, ledger postings	Tax, Invoices and the Books: From Order to Ledger
Operations	Warehouse and fulfillment	Receive, store, pick, pack and confirm shipment of physical goods	Physical stock on hand and its movements	Pick, Pack, Ship: Warehouse Management and Fulfillment
Operations	Shipping and delivery	Choose carriers, print labels, track parcels, promise a delivery date	Shipments, tracking events, the delivery promise	Getting It There: Carriers, Labels, Tracking and the Delivery Promise
Operations	Returns and exchanges	Take goods back, inspect, grade, refund or exchange, restock	Return requests, inspection results, dispositions	The Way Back: Returns, Exchanges, Refunds and Reverse Logistics

Layer	Capability	The job, in plain words	The fact it is the natural home for	Chapter
Operations	Store operations	Run the shelf: store stock counts, transfers, click-and-collect, in-store returns	Store stock (shared with POS)	Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment
Partners	Marketplaces and channel integration	List, sell and fulfill on other companies' platforms, in their formats	Listings and marketplace orders (copies of product and order)	Selling on Someone Else's Platform: Marketplaces and Channel Integration
Partners	EDI and e-invoicing	Exchange orders, despatch advices and invoices with large partners and tax authorities as structured documents	Sent and received documents (immutable once sent)	Documents Between Companies: EDI, Punchout and E-Invoicing
Back office	ERP and back office	Keep the books, buy from suppliers, value the stock, hold the item and customer masters	Ledger, purchasing, stock value, item identity	The Back Office: ERP Integration and Who Is the Hub

Layer	Capability	The job, in plain words	The fact it is the natural home for	Chapter
Foundations	Integration	Move facts between systems reliably: styles, contracts, retries, reconciliation	The integration inventory and each flow's contract	APIs, Events, Files and Middleware: How Commerce Systems Talk, Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation, Promises Between Systems: API Contracts, Webhooks and Versioning
Foundations	Identity and master data	Decide that two records are the same customer or product, and keep masters in step	Customer identity; the cross-reference of product identifiers	One Customer, One Product: Identity and Master Data Across Systems
Foundations	Data and analytics	Count the same thing the same way everywhere	Metric definitions; the warehouse (a copy of everything)	Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics

Layer	Capability	The job, in plain words	The fact it is the natural home for	Chapter
Foundations	Platform services	Serve many merchants on shared infrastructure: tenancy, extensibility, peak capacity, security	The platform's own configuration, apps and limits	Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation, Apps, Plugins and Functions: Extensibility and Ecosystem Governance, Black Friday Every Year: Peak Load, Resilience and Recovery, Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers

Two columns deserve a note. The "fact it is the natural home for" column is the hinge to the next chapter: when two systems both hold a fact, the capability map says which one *should* be its home, and Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record says how to make that decision stick. And several entries say "a copy": the search index, the service view, the listing, the data warehouse. Those capabilities hold copies by design, which is fine as long as everyone knows they are copies and they are rebuilt from the home, never edited in place.

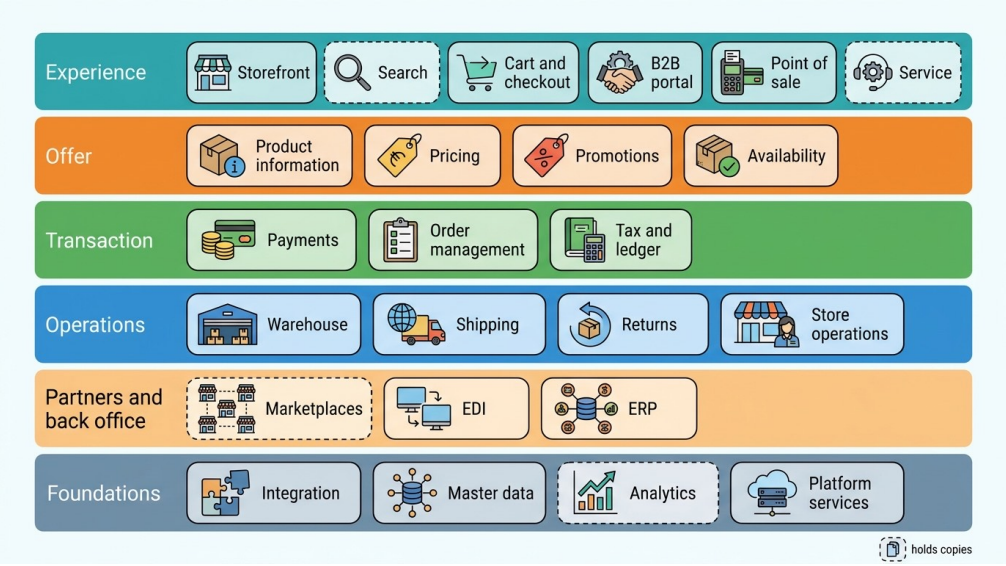


Figure 3: *The capability map as a checklist: twenty-four capabilities in six layers, each the natural home of one business fact; the dashed boxes (search, the service view, marketplace listings and analytics) hold copies by design and are rebuilt from the home, never edited in place.*

Facts that are derived, not recorded

Some capabilities own a calculation. **Availability** is one: the warehouse records physical stock, but a channel needs to know what it can still sell. That is **available-to-promise (ATP)**: stock on hand, minus commitments to unshipped orders, minus any buffer held back.

The calculation needs both warehouse stock and existing sales commitments. It may run inside an order management system (OMS), the commerce platform or a separate service. Give it one owner whichever product runs it. The delivery promise works similarly: it combines cut-off times, carrier lead times and location. Hand-editing either result bypasses the calculation and can make the storefront promise something the warehouse cannot deliver.

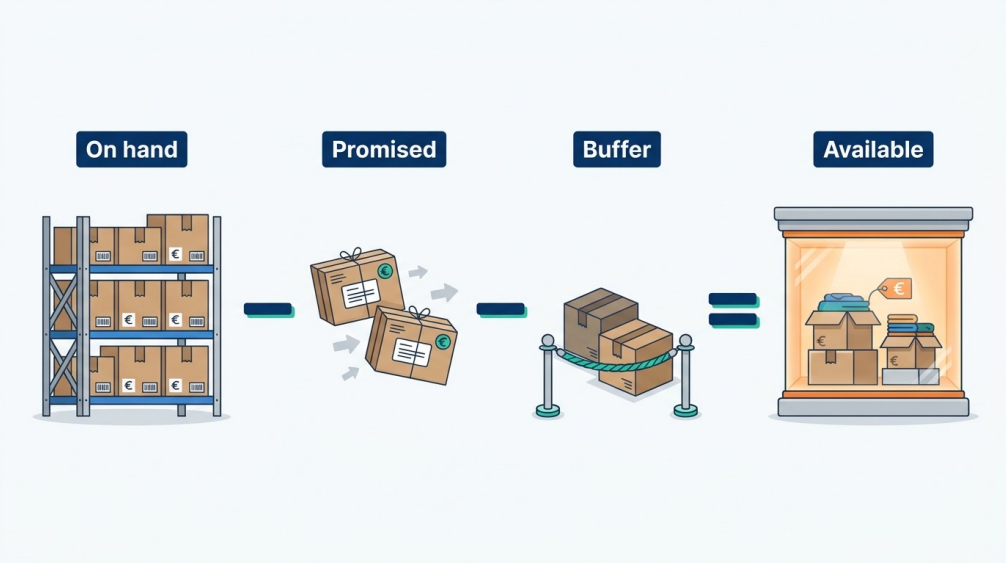


Figure 4: *Stock on hand is counted; what a channel may still sell is computed from on hand, what is already promised and the buffer, so its owner is the computation.*

How the Pieces Talk: The Typical Flows

Boxes are half the map. The arrows between them carry a small number of fact types, each with its own rhythm. The table below lists the flows almost every merchant above the first stage runs, the direction they usually take, and the requirements that decide how they should be built. "Freshness" is how old a copy may be before it causes harm; "consistency" is whether the copies must agree exactly or only eventually; the last two columns say what happens when the flow is slow or down. The requirements are the author's judgement from industry practice; the next chapters refine them.

Flow	From → to	Freshness needed	Consistency	If slow	If down
Product content	Product information → storefront, marketplaces, EDI partners	Minutes to hours	Eventual; completeness matters more than speed	A new product appears late	Channels keep yesterday's catalog

Flow	From → to	Freshness needed	Consistency	If slow	If down
Prices	Pricing → storefront, POS, marketplaces, B2B portal	Minutes; exact at a promotion's start and end	Exact across channels (a stale price is a broken promise, and on marketplaces a parity breach)	The Price Drift	Channels sell at the last known price
Stock and availability	Warehouse, stores, external locations → availability → every channel	Seconds to minutes at peak	Eventual, with a buffer and a reconciliation	Oversell window grows with the lag	Channels sell blind or stop selling
Orders	Checkout, marketplaces, EDI, POS → order management → warehouse, ERP	Seconds to minutes	Exact: never lost, never doubled	Late picking, missed cut-off	Orders queue; nothing ships
Payments	Checkout → PSP → order management, ledger	Seconds at authorization; days at settlement	Exact to the cent, reconciled against payouts	Checkout slows, conversion drops	Checkout stops

Flow	From → to	Freshness needed	Consistency	If slow	If down
Shipments and tracking	Warehouse and carriers → order management → notifications, service	Minutes	Eventual	"Where is my order" tickets rise	Customers unaware; service blind
Returns and refunds	Returns portal → warehouse, order management, PSP, ledger	Hours	Exact for money; eventual for stock	Refunds late, complaints	The Boomerang begins
Finance	Order management, PSP → ERP ledger	Daily batch is normal	Exact, reconciled per period	Month-end slips	Books lag, nothing else does

Two patterns stand out. **Only three flows need exactness in near real time: prices, orders and payments.** Everything else tolerates lag if there is a reconciliation (a job that compares two systems' copies and reports the differences) behind it. And **finance works in periods, not seconds**: the ledger can be fed nightly and nobody loses a sale, which is why The Back Office: ERP Integration and Who Is the Hub argues that the ERP's rhythm must never become the storefront's availability rhythm.

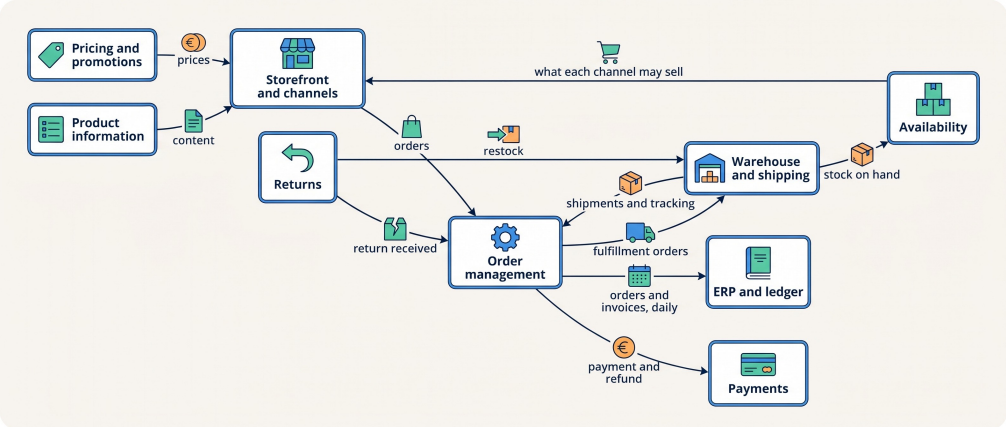


Figure 5: *The flows of a typical merchant: product, price and availability flow outward to the channels; orders flow inward to order management and down to the warehouse; stock, shipments and returns flow back; finance receives a daily batch.*

Every arrow is a place where one system's fact becomes another system's copy, and a copy can be late, lost or duplicated, which is why the joins, not the boxes, are where commerce platforms fail. Part VI — INTEGRATION: Making Separate Systems Behave Like One is about the arrows.

Three Ways to Cover the Map

No merchant builds all of this, and no vendor sells all of it, whatever the brochure says. In practice the map is covered in one of three ways, often mixed.

A suite. One vendor supplies most of the experience, offer and transaction layers, and sometimes order management and POS, in one data model. On 6 July 2026 Salesforce described its Agentforce Commerce release as one platform across B2C, B2B, point of sale and order management, with agents "natively connected to catalog, inventory, and orders". Adobe's October 2026 information center lists three ways to run Adobe Commerce (on-premises, on Adobe's cloud infrastructure, or as a fully managed cloud service) and a separate SaaS, Commerce Optimizer, that "can work alongside an existing Adobe Commerce or third-party commerce backend". The suite's promise is that the joins inside the covered boxes are the vendor's problem. Its limit is that the warehouse, the carriers, the PSP, the marketplaces, the EDI partners and the ERP are still outside, and the vendor's data model decides what "order" and "product" mean for everything inside.

A platform plus apps. A hosted SaaS platform (software the vendor runs for thousands of merchants at once, rented by the month) covers storefront, catalog, cart, checkout, basic inventory and basic order handling, and an app store rents the rest: reviews, returns, search, subscriptions, pre-orders, loyalty. Shopify's pricing page in November

2026 lists plans from €28 to a Plus tier "from €2,100" a month, with lower card rates on higher tiers; the platform is also a payments company, as is the POS vendor Lightspeed, whose fiscal 2026 results showed transaction-based revenue of $815.1 million against subscription revenue of $370.7 million. Hosted platforms tend to retreat from the operations layer: Shopify completed the sale of most of Shopify Logistics to Flexport on 6 June 2023 (an old event, but the direction has held). The center is excellent and cheap to start; the apps are many small vendors, each writing into your data (Kestrel & Pine's 41 apps, three of them writing stock), and the operations and partner layers are yours to assemble.

Best-of-breed, or composable. One product per box, chosen for that box, joined by APIs and an integration platform. The MACH Alliance (founded in June 2020 by commercetools, Contentstack, EPAM and Valtech; the acronym stands for microservices, API-first, cloud-native and headless; headless means the storefront is a separate application that talks to the commerce engine only through APIs) made this the named alternative to the suite. The specialist layers are real products with their own definitions: Manhattan describes an OMS as a platform "designed to track and manage the lifecycle of an order from start to finish", and Akeneo describes a PIM as a system "to centralize, manage, enrich, and distribute product information for commerce" (both n.d., accessed November 2026). Each box is best in its class and replaceable; the cost is the joins. The Alliance's own February 2026 survey of 600 enterprise technology leaders found 37% naming integration complexity as a primary concern, a figure against the interest of the people who published it.

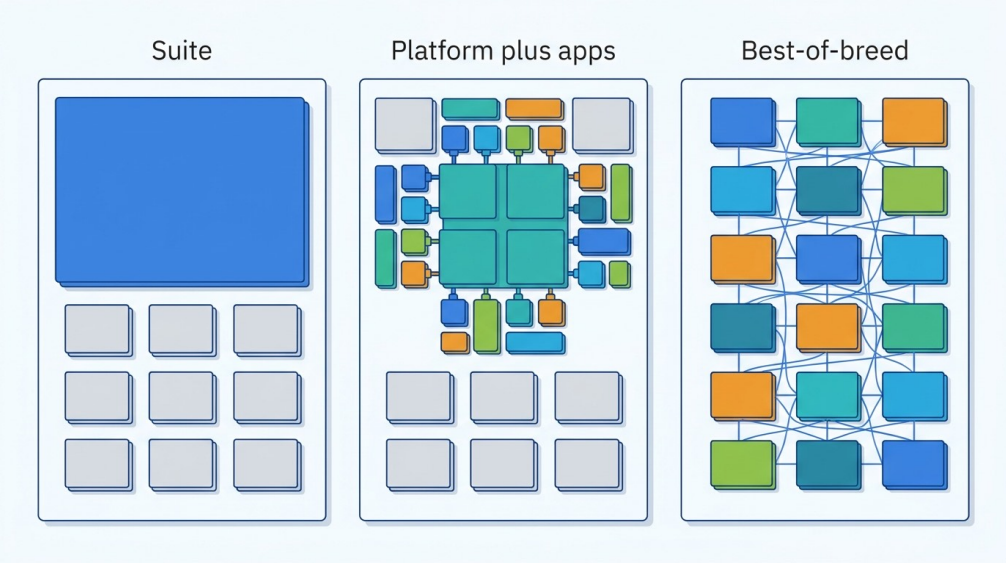


Figure 6: *Three ways to cover the same map: a suite covers many boxes with one data model, a platform covers the center and rents the rest as apps, and best-of-breed picks*

one product per box and pays for the joins.

Two variants are worth naming. **ERP-centered commerce** makes the ERP the center and the storefront a front end; Microsoft's documentation of 22 September 2026 says its Business Central Shopify Connector "is preinstalled for all new sign-ups and trials", which is the clearest public sign that ERP plus hosted storefront is a mainstream small-merchant shape. And a **specialist group** pre-integrates several best-of-breed products under one owner: Katapull, a Dutch group, announced on 31 March 2025 that five companies had joined forces and on 7 October 2026 added a WMS vendor as its eighth label, listing on its site an ERP-and-webshop product, a POS, a WMS, a marketplace integrator, an EDI provider, a returns product, a shipping product and an online-performance agency. On the map that is six operational and partner boxes from one group, with the storefront platform still bought elsewhere; the public pages give no figures beyond customer and user counts, and the book uses nothing else about it.

Trade-offs

	Suite	Platform plus apps	Best-of-breed	Specialist group
Boxes covered by one vendor	Many (experience, offer, transaction, often OMS and POS)	The center (storefront to basic orders)	One each	Operations and partners, pre-joined
Where the joins live	Inside the vendor, except operations and partners	Between platform and apps; all of operations and partners	Everywhere; the integration layer is the architecture	Inside the group; the join to the storefront is yours
Who defines "order" and "product"	The suite	The platform, with app fields bolted on	You, per box	The group's ERP or hub
Cost shape	Negotiated, often linked to sales	Subscription plus payment fees plus apps plus agency	Per product, per order, per request; plus integration	Per product

	Suite	Platform plus apps	Best-of-breed	Specialist group
Time to first sale	Months	Days	Months	Weeks
Fits	Group-stage and large omnichannel retailers with implementation partners	Starting and Growing merchants; Omnichannel merchants who surround the platform	Omnichannel and Group merchants with a platform team	Growing and early Omnichannel merchants in the group's home market
Limits	The vendor's model; cost at scale; upgrade projects when customized	App sprawl; the operations layer is unassembled; fees grow with sales	Integration cost and skills; vendor count	One owner across several layers; exit from the group

The table is the author's judgement. My rule for a merchant that is not a vendor: **cover the center with one product you do not customize, cover operations with products built for operations, and own the joins yourself**. Kestrel & Pine's Waypoint decision W4, stay on the SaaS platform and surround it with an OMS, a PIM and an availability service, is that rule applied; Build, Buy or Assemble: Choosing and Changing Platforms works the economics.

Across Scale Stages: When It Fits and When It Limits

The map does not change with scale. What changes is how many boxes have their own system, and which ones. The table uses the four stages defined in Who Sells What to Whom: Business Models and the Economics That Shape Architecture.

Stage	Boxes with their own system	Typical shape	Warning sign that the shape is ending
1 — Starting	Three or four: a hosted platform covers experience, offer and transaction; a PSP; an accounting package; the portal of a 3PL (a third-party warehouse that ships on the merchant's behalf)	The platform is the map	A second channel (first store, first marketplace) with its own order source
2 — Growing	Six to nine: an ERP or accounting system claims items and finance; a marketplace connector; a POS; an integration tool	Platform plus apps, with the ERP as a second center	Two systems writing stock; a spreadsheet as the product master
3 — Omnichannel	Twelve or more: own WMS, returns portal, shipping service, EDI provider, data warehouse; then OMS, PIM, availability	The joins become the architecture; the empty boxes (OMS, PIM, availability) must be filled	Support checks three systems for one order; a sync chain measured in hours
4 — Group	The same map per brand or country, with some boxes shared (PIM, OMS, data, integration) and some kept separate	Shared services versus brand autonomy	Brands forced onto one stack before the benefit is measured, or kept apart when they should share

Kestrel & Pine is at stage 3 with four boxes still empty. The judgement: fill a box with a dedicated system when the capability has more than one producer and more than one consumer and the system that holds it today cannot carry the workflow. A PIM is unnecessary at stage 1 and overdue at stage 3 with four channels; an OMS is overkill with one channel and one warehouse and overdue the day a second order source appears.

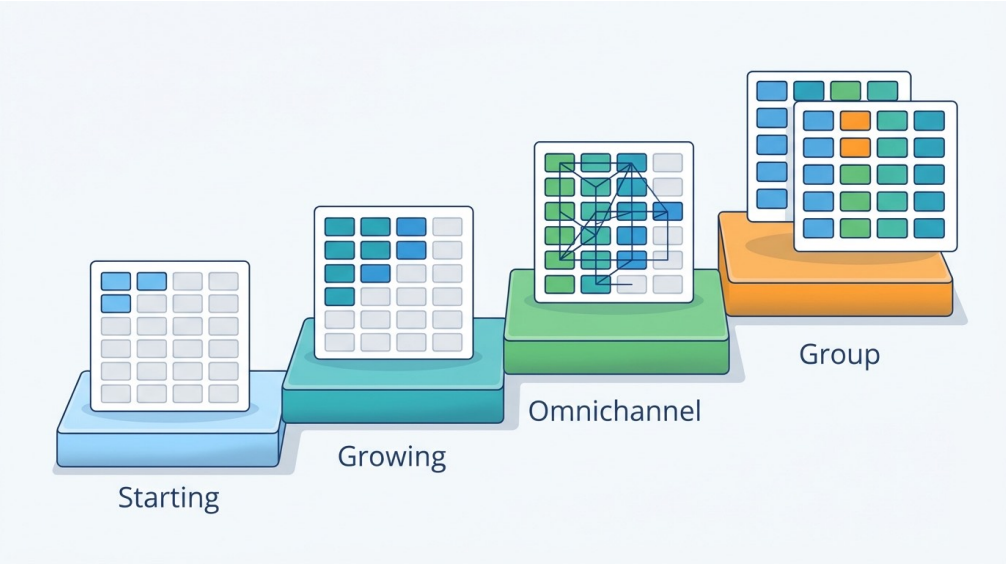


Figure 7: *The map does not change with scale; the number of boxes with their own system does, from three or four at the Starting stage to twelve or more at Omnichannel, and shared boxes across brands at Group.*

Reading a Vendor's Product Page Against the Map

A product page lists features. The map lets you turn a feature list into three questions.

1. **Which boxes does it claim?** Mark them. A channel-integration product that claims "order management" usually means it imports marketplace orders and pushes status back; that is a flow on the map, not the order-management box. A platform that claims "inventory" usually means it holds a stock number per location; that is a copy of the warehouse's fact, not the warehouse.
2. **Which fact does it own, and which does it copy?** For each claimed box, ask whether the product would be the home of that fact or a reader of it. Fluent Commerce, an OMS vendor, states the boundary on its own site: "your ERP stays the system of record" (n.d., accessed November 2026). A vendor that cannot answer the question for its own product is telling you something.

3. **What is the unit behind the numbers?** Vendor claims count different things. "1,300 marketplaces" (ChannelEngine's site, November 2026) counts connectors that exist, not connections that are maintained. "4,500+ customers" and "27,500+ users" (Katapull's about page) differ by a factor of six because users are seats. GMV (gross merchandise value, the value of orders passing through a platform before refunds and before the vendor's cut) is sales volume, not the vendor's revenue; Shopify's reported 2025 GMV of about $378 billion is roughly thirty times its revenue of about $11.6 billion. Ask for the unit, the date and the definition before comparing two vendors; Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026 has the full list of traps.

The same three questions apply inside your own company, to the systems you already have. Most landscapes contain a system that was bought for one box and has quietly grown into three.

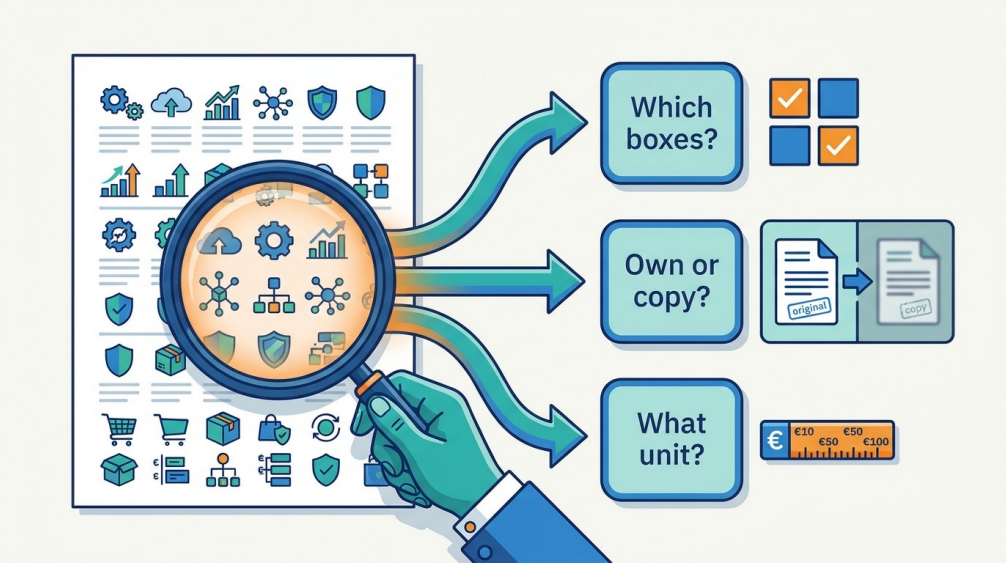


Figure 8: *Reading a vendor page against the map comes down to three questions: which boxes it claims, whether it owns each fact or copies it, and what unit sits behind its numbers.*

In Practice: Kestrel & Pine's Systems on the Map

This is Noor's February whiteboard as a table, reduced to the boxes that were not simply covered. The platform covers storefront, search, cart, checkout, promotions and the B2B portal; the WMS, the shipping service, the returns portal, the marketplace connector, the

EDI provider and the ERP each cover their own box; the platform vendor supplies the platform services. Systems are named by role, as the book does throughout.

Capability	System in January 2027	Status on the map
Product information	**Nobody**. Platform, ERP item master, spreadsheets, connector	**Empty box, four copies**
Order management	**Nobody**. Orders in the platform, the connector, the ERP, the POS	**Empty box, four order sources**
Inventory and availability	**Nobody computes availability**. Stock on hand in the WMS, ERP, platform, POS	**Empty box for availability; six writers of stock**
Identity and master data	**Nobody**	**Empty box**; Eva Janssen exists four times
Pricing	Platform price lists; ERP for wholesale; connector for marketplaces	Overlap: three writers; the Price Drift (€179 for nine days, about €8,200 lost)
Payments	The PSP online; a terminal provider in stores	Covered, but refunds also issued by hand in the PSP console (the Double Refund, €19,400)
ERP and back office	The ERP (since 2023)	Covered, and also used as order entry and stock master: five jobs in one box
B2B buyer experience	The platform's B2B features	62% of wholesale orders; 38% keyed into the ERP by hand
Point of sale, store operations	The POS (separate vendor, since 2019)	Store stock reaches the ERP nightly; no link to the platform
Shipping and delivery	The shipping service inside the WMS flow	The delivery promise is static text, computed from nothing

Capability	System in January 2027	Status on the map
EDI and e-invoicing	The EDI provider, for Bergzicht Outdoor and for Peppol (the European network for exchanging e-invoices)	One consultant knows the mappings (The Lynchpin)
Integration	The iPaaS (integration platform as a service, a hosted tool for building the flows between systems) plus agency scripts and vendor connectors; 34 integrations	Covered without standards
Data and analytics	The data warehouse and a BI (business intelligence) reporting tool	Three definitions of "order"

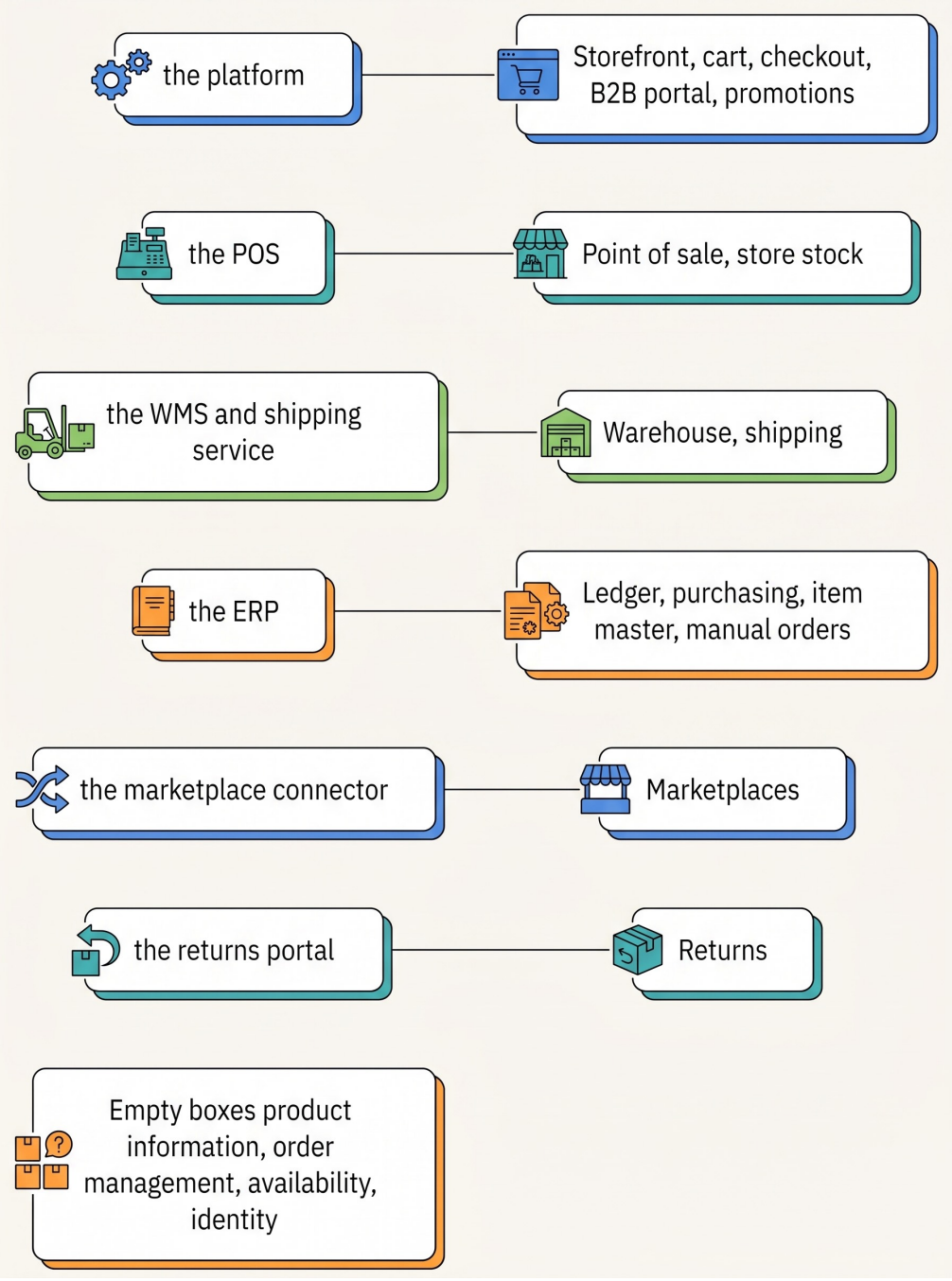


Figure 9: *Kestrel & Pine's thirteen systems fill most of the map, but four boxes have no system: product information, order management, availability and customer identity. Every 2026 incident sits on an empty box or on a box with several writers.*

The Waypoint decisions read as a plan to fill the empty boxes and thin the overlaps:

a PIM for product information (W3, Q3 2027), an OMS for order management with an availability service as one of its modules (W1 and W2, Q4 2027), the WMS as the single writer of physical stock (W1), and integration standards for the arrows (W5). Identity remains open; One Customer, One Product: Identity and Master Data Across Systems takes it up.

When It Goes Wrong

The map fails in four ways, and each shows up on the map before it shows up in the business.

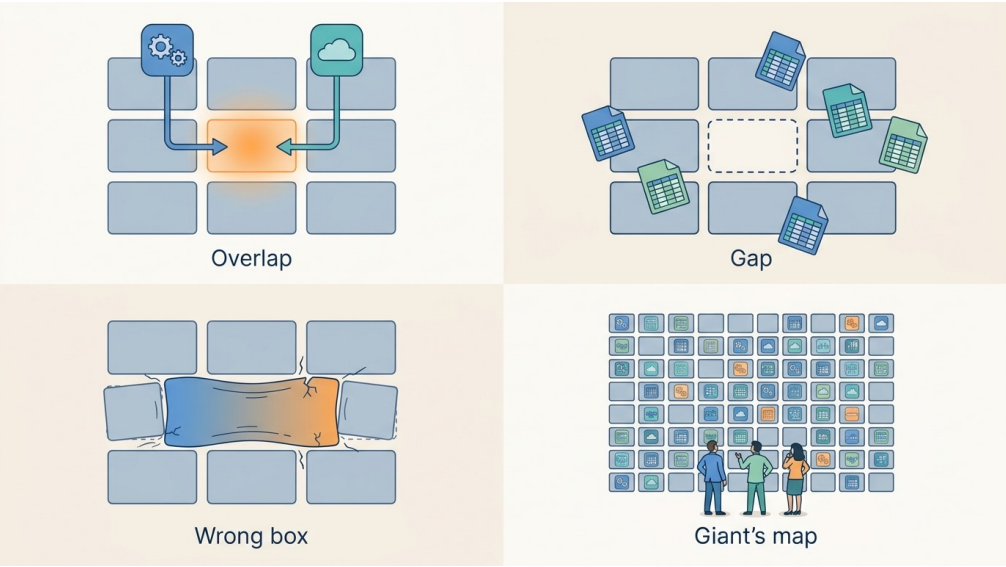


Figure 10: *The map fails in four ways, each visible on the map before it shows in the business: two writers for one fact, a fact with no home, a system holding boxes it was not bought for, and a map copied from a giant.*

An overlap: two systems own one fact. The App That Wrote Stock is the textbook case. A pre-order app adjusted inventory in the platform while the WMS feed also wrote it; for two weeks 600 variants carried phantom reservations. Detection is a *write audit*: for each box, list every system, app, script and person that can change its fact. If the list has more than one entry for a recorded fact, that is an overlap. Recovery is to name the single writer and re-publish the fact from it, which is exactly what the first weekly reconciliation in May 2027 did when it found 2.3% of active variants with a different quantity between the WMS and the platform (an illustrative figure).

A gap: nobody owns the fact. The gap fills itself with spreadsheets and with copies that start acting as records. Kestrel & Pine's product content lived in four places because no

box was named for it; its order status lived in four places because none was. Detection is the empty box on the map. Recovery is to name a home before buying a system for it, because a bought OMS fed by four unreconciled order sources is a fifth copy.

A box held by a vendor who sells a different box. A channel connector that is also used as the stock master, or an ERP used as the order hub, works at the stage it was bought in and becomes the bottleneck later, when the warehouse needs stock in minutes and the ERP posts hourly. Detection: compare the box the system was bought for with the boxes it now holds. The ERP that holds purchasing, the ledger, the item master, manual wholesale orders and the stock master is doing five jobs, two of them badly.

The map copied from a giant. A merchant at stage 2 that draws a stage-4 map, with a system per box and a platform team to run them, has bought the Giant's Boots. The boxes are the same; the number of systems that should fill them is not. Below the producer-and-consumer test above, the box stays inside the platform.

What to Take Away

Established.

- A commerce platform is a set of capabilities spread over several systems, not one system; the specialist layers (OMS, PIM, WMS, returns, search, channel integration, iPaaS, payments) are distinct product categories with their own vendors and definitions, as the vendors' own November 2026 pages show.
- No vendor covers the whole map. Suites stop at the warehouse, the carrier, the PSP, the marketplace and the ERP; hosted platforms have retreated from logistics (Shopify's 2023 sale of Shopify Logistics) and earn largely from payments (Lightspeed's fiscal 2026 revenue mix); composable assemblies make integration the architecture (the MACH Alliance's own 2026 survey: 37% cite integration complexity).
- Business-to-business ordering is a first-class capability, not an add-on: Eurostat's 2024 figures put EDI-type sales at 11.07% of EU enterprise turnover against 8.39% for web sales.

Depends on context.

- How many boxes get a dedicated system depends on the stage and the shape: one channel and one warehouse can live inside a hosted platform; four channels and an own warehouse need an OMS, a PIM and an availability service. The trigger is the producer-and-consumer test, not revenue.
- Suite versus platform-plus-apps versus best-of-breed depends on the team: a suite needs an implementation partner, best-of-breed needs a platform team, platform-plus-apps needs app governance. A fourteen-person team should expect to own joins, not boxes.

- Whether a specialist group (several pre-integrated labels under one owner) is a good deal depends on how much of the operations layer you need from one owner and what leaving the group would cost.

The author's synthesis.

- Draw the map in capabilities before systems, and mark three things on it: empty boxes, boxes with more than one writer, and boxes whose fact is derived. Those marks predict the next incident better than any vendor roadmap.
- For a merchant that is not a vendor, the durable shape is a center you do not customize, operations products built for operations, and joins you own. Covering a box is cheap; owning the joins between boxes is the work, and it is the work nobody can buy for you.

Questions to Ask

In a design review, a vendor demo or your first week:

1. Which capability on the map is this system or product for, and which other boxes has it quietly grown into?
2. For each fact it holds, is it the home or a copy? If a copy, where is the home and how does the copy learn about changes?
3. Which boxes on our map are empty, and what fills them today (a spreadsheet, a person, a habit)?
4. Which boxes have more than one writer? List every system, app, script and person that can change the fact.
5. Which of our facts are derived (availability, the delivery promise, a price after promotions), and which computation owns each one?
6. Which flows need exactness in near real time (prices, orders, payments), and which only need a reconciliation behind them?
7. What unit is behind each number on the vendor's page (GMV, merchants, users, implementations, connectors), and as of what date?
8. If this vendor disappeared, which boxes would go dark, and which joins would have to be rebuilt?
9. Which box would we fill next if we grew one stage, and what sign would tell us it is time?

The map says what the parts are and what each should own. It does not yet say what happens when two parts hold the same fact and disagree, which is Kestrel & Pine's actual condition in 2027 and the subject of Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record.

Sources and Evidence

Company, product and investor sources

- As AI Agents Transform Commerce, Salesforce Unleashes Its Biggest Agentforce Commerce Release Yet, Salesforce, 2026 — a suite covering B2C, B2B, POS and order management in one platform.
- Adobe Commerce: Platform and services, Adobe, 2026 — three deployment options and Commerce Optimizer working beside an existing backend.
- Shopify pricing, Shopify, n.d., accessed 2026-11 — plan tiers, card rates and transaction fees of a hosted platform.
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- Shopify Q4 2025 financial results, Shopify, 2026 — 2025 revenue; GMV as reported by Digital Commerce 360, 2026.
- Lightspeed fourth quarter and full year 2026 results, Lightspeed, 2026 — transaction revenue versus subscription revenue.
- Get started with the Shopify Connector, Microsoft, 2026 — the ERP connector preinstalled for new Business Central sign-ups.
- What is an OMS?, Manhattan Associates, n.d., accessed 2026-11 — an OMS vendor's definition of the order-management capability.
- Akeneo, Akeneo, n.d., accessed 2026-11 — a PIM vendor's definition of the product-information capability.
- Fluent Commerce order management system, Fluent Commerce, n.d., accessed 2026-11 — "your ERP stays the system of record".
- ChannelEngine, ChannelEngine, n.d., accessed 2026-11 — "over 1,300 marketplaces" as an example of a connector-count claim.
- Katapull and Over Katapull, Katapull, n.d., accessed 2026-11; Katapull breidt Commerce Hub uit met overname van Picqer, Katapull, 2026; Vijf softwarebedrijven bundelen krachten in Katapull, Dutch IT Channel, 2025 — the specialist-group shape, from public pages only.
- EPAM joins the MACH Alliance as a founding member, EPAM, 2020 — the founding of the MACH Alliance and the acronym's expansion.
- MACH Alliance Enterprise Technology Report, MACH Alliance, 2026 — 37% of respondents cite integration complexity.

Standards, regulation and official documentation

- E-commerce statistics, Eurostat, 2026 (2024 reference year) — EDI-type sales 11.07% versus web sales 8.39% of EU enterprise turnover.

Research and practitioner sources

- None beyond the vendor and official sources above; the layers, the flows table and the trade-offs are the author's synthesis.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

4. Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record

IN THIS CHAPTER, YOU WILL: Decide, for every business fact a commerce platform holds, which system is its record, which are copies, and what happens when they disagree.

KEY POINTS:

- **Copies are unavoidable; unowned copies are the failure.** A price, a stock level, an order status and a customer each live in several systems. The question is not how to have one copy but which one wins when they disagree, and only one may. That is a decision per fact, not per system and not per company.
- **"Order", "product" and "stock" mean different things in different systems, and that is not a bug.** A bounded context is the boundary inside which a word has one meaning; the warehouse's order and the ledger's order are different objects that share a number. Design the translation at the boundary instead of forcing one model on everyone.

- **Ownership changes with scale, and the finance system's role narrows.** At the start the platform owns everything but the books. By the omnichannel stage physical stock belongs to the warehouse, orders to an order hub, product content to a product information system, and the back-office system keeps finance. Kestrel & Pine's Waypoint program is that reassignment, written down.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Phantom Stock** — two channels sell the same last unit because each reads a different copy of the stock level and neither is the record.
- **Schrödinger's Order** — support finds three statuses in three systems because each system's "order" is a different object and nobody owns the view across them.
- **The Price Nobody Can Explain** — a price drifts between channels because three systems may write it and none is named as the source.

A customer asks where a parcel is. The storefront says *Delivered*, but the warehouse system says *In picking*. Support needs to know whether one record is stale or the two systems describe different parts of the order. Until that is clear, copying either status more quickly can spread the wrong answer and leave the customer no better informed.

A **system of record** is the system whose value is authoritative for a particular fact. Naming one gives support and other systems a place to resolve disagreements. It also requires decisions about who may change that fact, how copies receive updates and how errors are repaired. Two-way synchronization without those decisions can let each system overwrite the other's answer.

This chapter builds an entity–owner table: a list of business facts and the systems responsible for them. It explains how systems can use different meanings for "order", compares ownership choices and works through Kestrel & Pine's six stock writers. The Map of a Commerce Platform: Capabilities and Layers supplies the wider map; Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation explains how to detect and repair disagreements.

Where we are in the order's journey. Eva Janssen buys a Ridgewalker jacket from Kestrel & Pine, the fictional outdoor brand. The stock badge, €199 price and later shipment status all reach her through copies. The warehouse management system (WMS) confirms what shipped; the storefront displays that answer. This chapter decides which system each copy must answer to.

Setting the Scene at Kestrel & Pine: "Where Does Stock Live?"

In early March 2027, Head of Engineering Noor Visser asked who owned Kestrel & Pine's stock figure. CEO Mara Lindqvist was about to approve **Waypoint**, the company's commerce improvement program. Noor needed a clear answer before changing the integrations. The people running those systems gave her six.

Who	Where stock lives, in their words	What the number actually is
Daniel Okafor, Operations	"In the WMS. The pickers scan it."	Units on hand per bin at Duiven; updated at every scan
Sanne de Wit, Finance	"In the ERP. That is the valued stock."	A quantity per item in the ERP (the enterprise resource planning system: the books, purchasing, stock valuation), refreshed from the WMS every 15 minutes, plus manual adjustments, plus store stock loaded nightly
Milan Petrov, Ecommerce	"In the platform. That is what the storefront sells from."	A sellable quantity per variant, refreshed hourly from the ERP, also written by three apps and, occasionally, by hand
The store managers	"In the POS."	Units per store in the POS (the point-of-sale system, the store's till software), copied to the ERP overnight
Milan again, for marketplaces	"The connector pushes it to bol, Amazon.de and Zalando every hour."	A copy of the platform's copy

Who	Where stock lives, in their words	What the number actually is
Daniel again, for Germany	"Some of it is in the marketplace's warehouse."	Stock sent to the marketplace's fulfillment program in advance, in a location outside the company's control

Lead integration engineer Ruben Alves reconstructed the stock figures for the Ridgewalker women's M in moss at 20:00 on Black Friday, 27 November 2026. The warehouse had sold its last unit twenty minutes earlier. Six systems showed six quantities, from zero to a healthy double-digit storefront count *(illustrative reconstruction).*

The copying chain took up to two hours. Channels kept selling from old figures, producing the **Black Friday Oversell**: 1,340 units oversold, 1,120 orders canceled or delayed and about €96,000 in costs.

Noor's question was not "how do we sync faster". It was "**which of the six is right?**" Daniel answered for everyone: "The shelf." The WMS counts the shelf. Waypoint decision W1 followed: the WMS is the system of record for physical stock at Duiven, the POS for store stock, the marketplace program's stock is a tracked external location, and an availability service computes what each channel may sell from those three. Everything else becomes a copy.

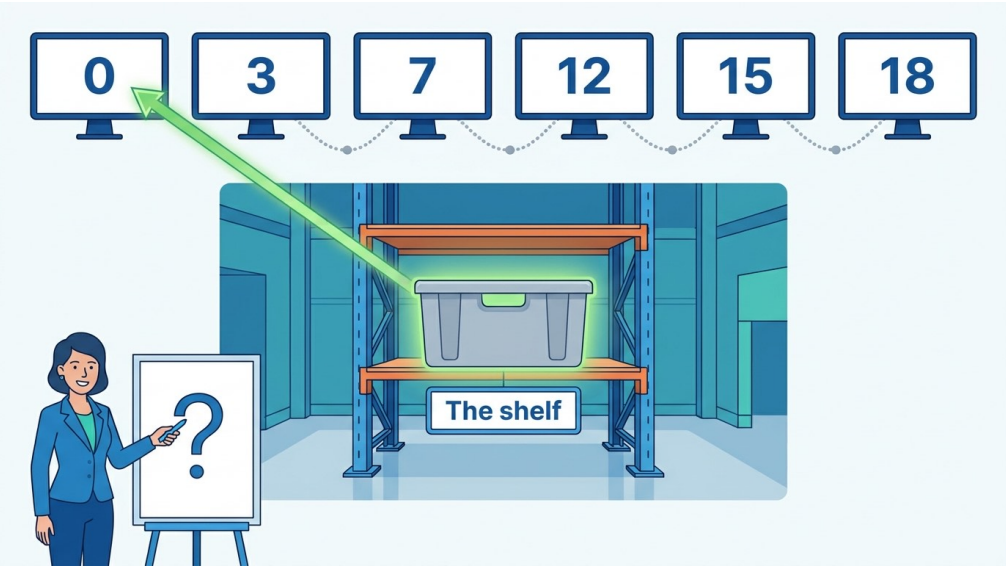


Figure 1: *Six systems, six honest numbers for one jacket at 20:00 on Black Friday; Noor's question was not how to sync them faster but which one is right, and the answer was the*

shelf.

Facts, Records and Copies

Three words carry this chapter.

A **business fact** is something the business needs to be true about the world: this variant has this barcode, this customer pays this price, there are 14 units of this size in Duiven, order #1072455 has shipped. The business acts on facts: it promises delivery, takes money, prints invoices.

A **system of record** for a fact is the one system whose value is treated as true when copies disagree, and the only system in which the fact is changed directly. Your bank's ledger is the record for your balance; the figure in your budgeting app is a copy. In commerce the shelf count in the WMS is the record for physical stock; the number on the product page is a copy. The finance book RECONCILED works the same idea for money in Systems of Record: Ending the Doppelgänger Record.

A **copy** is the same fact held in any other system. Copies are not a mistake. The storefront must hold a stock number to render a page in milliseconds; the data warehouse must hold every order to count them; the marketplace must hold the price to show it. The rule is only that a copy is known to be a copy: it is refreshed from the record, it is never edited to "fix" a disagreement, and when it differs from the record it is the copy that is wrong.

A **writer** is any system, app, script, job or person that can change a fact in a given system. The practical test of ownership is the writer list. If the record has one writer and the copies have none except the refresh from the record, ownership is real. If a copy has writers of its own, it is not a copy; it is a second record, and the two will diverge.

One phrase deserves a warning. "Single source of truth" is right only if read as *one record per fact*; it is wrong, and expensive, if read as *one system for every fact.* No system is the natural home of both the shelf count and the ledger posting. ("Master data", the slowly changing reference facts such as products, customers, suppliers and locations, follows the same rules; One Customer, One Product: Identity and Master Data Across Systems owns it.)

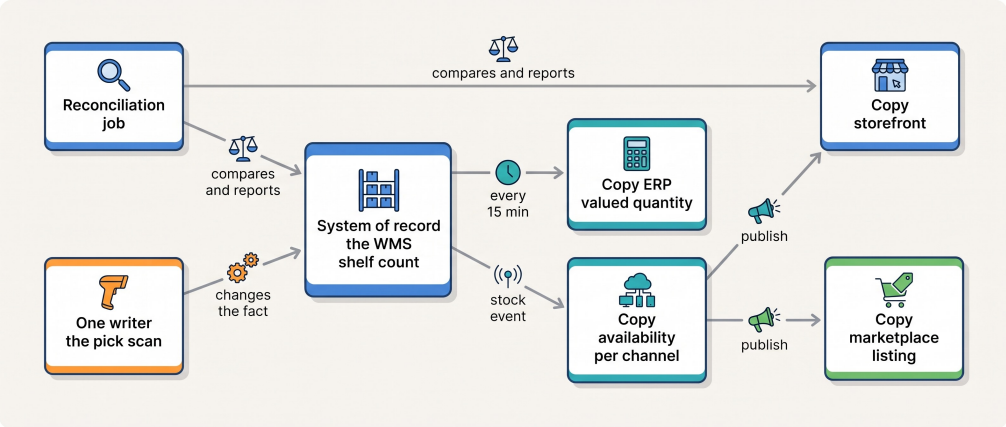


Figure 2: *One fact, one writer, one record and several copies: the shelf count changes only through the pick scan, every other system holds a copy refreshed from the record, and a reconciliation job compares record and copies so that drift is found instead of discovered by customers.*

Bounded Contexts: Why the Warehouse's Order Is Not the Ledger's Order

Ownership is harder than it looks because the same word names different things in different systems. **Bounded context** is the term Eric Evans gave this in *Domain-Driven Design* (2003): a boundary inside which a model, and the words in it, have one agreed meaning. Martin Fowler's 2014 note calls it "a central pattern in Domain-Driven Design", repeats Evans' claim that "total unification of the domain model for a large system will not be feasible or cost-effective", and adds, from experience, "Time and time again I see this confusion recur with polysemes like 'Customer' and 'Product' " (a polyseme is a word with several related meanings).

In plain words: the warehouse, the storefront and the ledger are three bounded contexts. Each has its own "order", and the three are not the same object.

Context	What "order" means there	What it contains	Its states
Storefront and checkout	What the customer bought, at what price, with what promise	Lines with consumer prices including VAT, promotions, a payment, a delivery promise, an address	Placed, paid, (customer-visible) shipped, delivered
Order management	The order's life across channels	One record per order from any channel, allocation to a location, state transitions, edits, cancellations	Placed, paid, allocated, picking, shipped, delivered, canceled, returned, refunded
Warehouse	Work to do	A fulfillment order: pick lines, bins, a packing station, one or more parcels; a split order is two of these	Released, in picking, packed, shipped
ERP	Something to invoice and post	A sales order with net amounts, VAT, a debtor, a receivable	Open, invoiced, paid

Kestrel & Pine's **Three-Status Order**, #1052877, generated 11 support tickets because its three statuses described different objects. The fleece shipped on 7 December 2026; the socks waited a week for replenishment. The platform showed *Delivered* after the fleece's carrier scan. The WMS showed *In picking* while the socks waited. The enterprise resource planning system (ERP) showed *Not invoiced* because it required the WMS's "order complete" message.

For six days, nobody could give the customer a combined view. **An order management system (OMS) must explain the whole order while retaining the separate facts**. Waypoint decision W2 assigns that job to a purchased OMS, with one state model and one order number family.

The same happens with "product" (content for the storefront; attributes and translations in a PIM, a product information management system; an item with cost and supplier

in the ERP; a SKU, a stock-keeping unit or sellable variant, with dimensions, weight and bin in the WMS) and with "customer" (an account with an e-mail in the platform; a loyalty card in the POS; a debtor with payment terms in the ERP; a contact in the returns portal). Eva Janssen exists four times, with no shared identifier, because four contexts each created the customer they needed. That is the Familiar Stranger, worked in One Customer, One Product: Identity and Master Data Across Systems.

Two design rules follow. First, **translate at the boundary** rather than forcing one model on every system: the connector between the OMS and the WMS turns an order into a fulfillment order, and the connector into the ERP turns a shipment into something to invoice; the translation is named work with an owner, not an accident of field mapping. Second, **share identifiers, not models**: the order number, the SKU and the GTIN (the Global Trade Item Number behind the barcode) let the contexts refer to the same thing while describing it differently. Promises Between Systems: API Contracts, Webhooks and Versioning treats the translations as contracts.

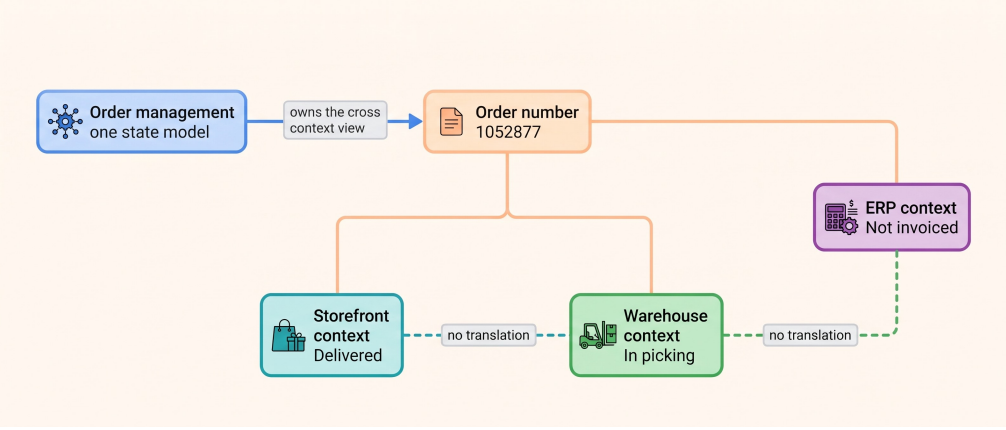


Figure 3: *The Three-Status Order: three bounded contexts each hold an honest status of their own object, share only an order number, and have no owner of the view across them; the order management system in Waypoint's design is that owner.*

The Entity–Owner Table

The deliverable of this chapter is a table with one row per business fact. For each fact it names the system of record, who may write it there, where copies are allowed, how the copies learn about changes, and how disagreement is detected. The version below is the author's recommendation for a merchant at the omnichannel stage with its own warehouse, stores, wholesale and marketplaces; the stage table later in the chapter says how it differs earlier and later. It assumes a PIM and an OMS exist; without them the

platform holds those rows.

Fact	System of record	Who may write	Copies live in	How copies learn	Disagreement found by
Product identity (SKU, GTIN, unit, supplier, cost)	ERP item master	Buying team	PIM, platform, WMS, connector, EDI partners (trading partners that exchange structured documents by electronic data interchange)	Scheduled publication on creation and change	Item count and GTIN check, weekly
Product content (names, attributes, media, translations)	PIM	Product and content teams; translators	Platform, marketplaces, EDI product catalogs	Publication on approval; nightly full sync as a safety net	Completeness and channel comparison, nightly
Consumer price per market	Pricing capability (the platform's price lists at this stage)	Pricing owner, with effective dates	Storefronts, POS, marketplace listings, connector	Event on change; exact at promotion start and end	Price comparison per channel, hourly

Fact	System of record	Who may write	Copies live in	How copies learn	Disagreement found by
Wholesale and contract price	ERP price lists	Wholesale team	B2B (business-to-business) portal, EDI price catalogs	Price events with effective dates	Per-account comparison, weekly
Physical stock per location	WMS (warehouse), POS (stores), tracked external location (marketplace program)	Scans, counts and receipts in that system only	Availability service, ERP, data warehouse	Stock events per SKU and location	Cycle counts in the record; WMS-to-platform reconciliation, weekly
Availability per channel (derived)	Availability service	The computation only; no manual edits	Storefronts, marketplaces, B2B portal	Publication on every stock or reservation event	Oversell and cancellation rate, daily
Order	OMS	Order creation from channels; state changes from fulfillment, payment and returns events	Platform order history, WMS, ERP, service tool, warehouse	Order events	Order count per channel per day, OMS versus each source, daily
Payment	PSP	The PSP, on instructions from checkout and the OMS	The order (a reference and a status)	Payment webhooks	Settlement-to-order reconciliation per payout

Fact	System of record	Who may write	Copies live in	How copies learn	Disagreement found by
Shipment and tracking	WMS and shipping service (shipment); carrier (events)	Pack confirmation; carrier scans	OMS, notifications, service tool	Shipment and tracking events	Shipped-not-delivered aging, daily
Return	Returns portal (request); WMS (receipt and grade); OMS (resolution)	The customer, the inspector, the OMS	ERP (credit note), PSP (refund)	Return events	Refunds versus credit notes versus receipts, daily
Invoice, credit note, ledger	ERP	Posting from OMS and PSP batches; finance	Reports	Daily batch	Period close
Customer (consumer)	Platform account	The customer; service	POS, newsletter tool, returns portal, warehouse	Sync on order; periodic merge	Duplicate rate, monthly
Customer (business)	ERP debtor	Wholesale and finance teams	B2B portal, EDI partner cards	Scheduled sync	Terms and credit-limit comparison, weekly

Fact	System of record	Who may write	Copies live in	How copies learn	Disagreement found by
Marketing consent	With the customer record, with evidence of when and where	The customer	Newsletter tool	Sync on change	Consent audit, quarterly

Several rows carry lessons that newcomers miss.

- **Stock has three records, not one**, because the warehouse, the stores and the marketplace's program each count a different shelf. Ownership is per fact *and per location.*
- **Availability has no human writer**. It is computed from the stock records and the open orders. A hand edit to a channel's available quantity is a second writer, and it is how the App That Wrote Stock produced phantom reservations on 600 variants.
- **Payment's record is outside the company**. The PSP (payment service provider, the company that takes the shopper's money and pays it out to the merchant) holds the transaction; the order holds a reference. A refund issued by hand in the PSP console, outside the returns flow, is a write to the record that the OMS does not know about, which is the Double Refund: 212 returns refunded twice, €19,400.
- **The business customer and the consumer have different records**. Microsoft's October 2026 documentation for its Business Central Shopify Connector says it plainly: decide "which system is the system of record for customer data", notes that most businesses manage customer records, payment terms and credit limits in the ERP, and warns that "Mixing both directions without a clear ownership rule can lead to conflicting records."

The "disagreement found by" column is the part most tables leave out. A record with copies and no reconciliation (a job that compares two systems' values and reports the differences) is a record on trust, and trust is what the Black Friday chain ran on. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation designs those jobs; here the rule is only that every copy row has one.

How fresh must a copy be?

Ownership decides who wins; it does not decide how quickly the copies must follow. That depends on what a stale copy costs, and it differs sharply by fact.

Fact	A copy may be stale for	Agreement required	Why
Availability shown to a channel	Seconds at peak; minutes normally	Eventual, within a buffer	Every minute of staleness is an oversell window; Kestrel & Pine's two hours produced 1,340 oversold units
Consumer price	Minutes; exact at a promotion's start and end	Exact across channels	A stale price is a broken promise, and on marketplaces a parity breach; the Price Drift ran nine days
Order existence	Seconds to minutes	Exact: never lost, never doubled	A missed order is a missed cut-off; a doubled one ships twice
Order status	Minutes	Eventual	Stale status creates "where is my order" contacts, not losses
Product content	Hours	Eventual	A late description is a missed sale, not a wrong one
Ledger postings	A day	Exact per period	Finance works in periods

The table is the author's judgement from industry practice and Kestrel & Pine's incidents. Its point is that the record for availability needs an event-driven copy (an **event** is a message sent the moment something changes, as opposed to a periodic copy) while the record for ledger postings can be fed by a nightly batch, and that choosing one rhythm

for all facts is how an ERP's hourly cycle became a storefront's availability latency.

Four Ways to Assign Ownership

Merchants assign ownership in one of four ways. Three are decisions; the fourth is what happens without one.

The ERP as hub. The ERP is the record for products, stock, orders and customers, and the web shop is a front end that reads from it and posts orders into it. Microsoft's connector documentation of October 2026 encodes this: "Business Central acts as the system of record for inventory", pushed to Shopify one way on a recommended cycle of every 15 to 30 minutes; customers are managed in Business Central and exported; orders flow the other way, from the shop into the ERP. This fits one channel, one warehouse and a wholesale-heavy business, because the ERP already holds the terms, the prices and the valued stock. It stops fitting when stores, marketplaces and a WMS each need stock in minutes and the ERP's batch rhythm cannot serve them.

The platform as hub. The commerce platform is the record for orders, customers and stock; the ERP subscribes to the result for finance. This is the natural shape of a direct-to-consumer (D2C) brand on a hosted platform with one or two channels: the platform's order is the order, its customer is the customer, its stock number is the stock. It stops fitting when orders are created elsewhere (a POS from another vendor, EDI from a retail chain, a marketplace connector) and when stock is counted elsewhere (an own warehouse on a WMS), because the platform's model cannot hold those orders well and its stock number becomes a copy with its own writers.

Ownership per fact, with an order hub. Assign physical stock to the WMS, orders and availability to the OMS, product content to the PIM, and identity, costs and finance to the ERP. The commerce platform keeps the storefront and checkout. The warehouse can then report what it counted while order management decides how much remains uncommitted. Separating those facts prevents a channel's sales estimate from overwriting a physical count.

Specialist vendors describe similar boundaries. Fluent Commerce says its OMS routes orders while "your ERP stays the system of record". OneStock describes the ERP as product master when no PIM exists, and its OMS as master for inventory availability (both undated vendor guidance, accessed November 2026). Read these as vendors explaining their own categories.

This arrangement fits a merchant with several order channels and stock locations. It adds a system and integrations to operate. I would not add that cost at an early stage without an actual coordination problem.

No decision. Systems are joined pairwise as needs arise, often with two-way syncs so that "both are always up to date". Every system becomes a writer of every fact it holds; disagreements are resolved by whoever notices, in whichever system they are looking

at. This is Kestrel & Pine in January 2027: 34 integrations, six writers of stock, four sources of orders, four copies of product data, three definitions of "order" in the data warehouse. It is the default because it is never chosen; each connector was a reasonable answer to one request.

	ERP as hub	**Platform as hub**	**Per fact, with an order hub**	**No decision**
Record for stock	ERP	Platform	WMS, POS, external location; availability derived	Everyone
Record for orders	ERP	Platform	OMS	Each channel's system
Record for product content	ERP	Platform	PIM	Spreadsheets
Record for finance	ERP	ERP	ERP	ERP
Fits	One channel, one warehouse, wholesale-heavy, small team	D2C-first on a hosted platform, one or two channels	Several channels, several stock locations, own warehouse	Nobody, but it is where most merchants are
Breaks when	Batch cadence meets real-time availability; several channels write one monolith	Orders and stock originate outside the platform	Too early: cost without orchestra-tion need	The first busy week
Copies tolerated	Few; the shop is a view	Few; the ERP is a ledger	Many, each named with a refresh and a rec-onciliation	Unknown

The table is the author's judgement, consistent with the vendor statements above. My

rule: **choose the hub by where orders and stock originate, not by which system is largest**. If every order starts in the platform and every unit sits in one warehouse the platform or the ERP can be the hub. The day a second order source or a second stock location appears, ownership per fact becomes necessary, and the longer the decision waits the more writers each fact collects.

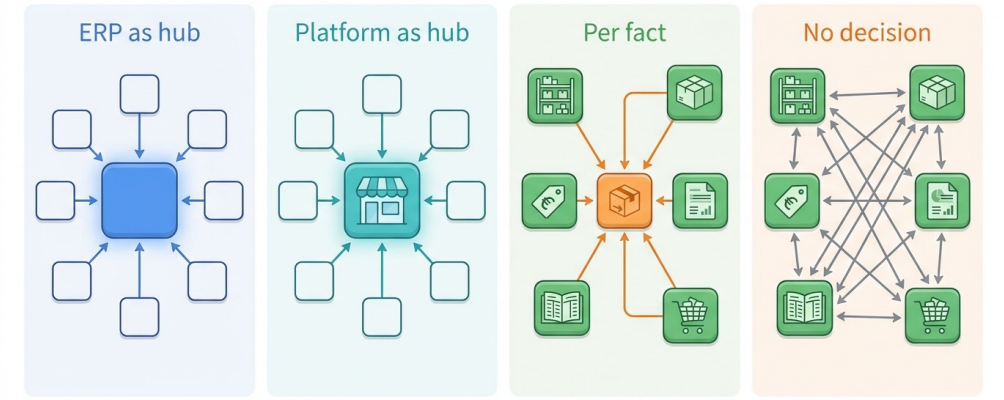


Figure 4: *Three of the four ways to assign ownership are decisions; the fourth is the tangle that forms when each connector answers one request, and the second order source or stock location is the moment to choose ownership per fact.*

Ownership in a Suite and Across Vendors

In a suite, the vendor has made most of these decisions for you. Salesforce's July 2026 release describes one platform across B2C, B2B, point of sale and order management with agents "natively connected to catalog, inventory, and orders": inside that boundary the suite's order is the order and its inventory is the inventory. A vendor-built connector makes the decisions configurable: the Business Central connector asks, per entity, which direction is the record (products: the merchant chooses; inventory: the ERP; orders: the shop). The entity–owner table becomes settings.

Two things remain yours even in a suite. First, the systems outside it: the PSP, the carriers, the marketplaces, a POS or WMS from another vendor, the EDI partners. Their facts are records the suite copies, and the suite's copies need the same refresh and reconciliation as anyone's. Second, the decision itself: the suite decides what it owns, not whether that is right for your business. Shopify's enterprise guidance of August 2026

states the discipline that holds across vendors in two sentences: “Document every field that moves between systems, and name the system of record for each” and “Decide in advance what happens when the two systems disagree.”

When Ownership Is Not Decided: The Five Truth Patterns

Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong names five patterns in a family it calls Truth. Each is a fact without an owner.

Pattern	The fact with no record	Kestrel & Pine’s incident	The owner Waypoint named
The Phantom Stock	Physical stock, and availability per channel	The Black Friday Oversell (1,340 units, about €96,000); the Store Stock Mirage (37 items promised that Utrecht had sold)	WMS and POS for stock; the availability service for what each channel may sell (W1)
Schrödinger’s Order	The order’s state across contexts	The Three-Status Order (#1052877, six days, 11 tickets)	The OMS, one state model, one order number family (W2)
The Familiar Stranger	Customer identity	Eva Janssen as four unlinked records	Not yet decided; One Customer, One Product: Identity and Master Data Across Systems
The Boomerang	The refund and its link to the return	The Double Refund (212 returns, €19,400 twice)	The OMS as the owner of the return’s resolution; refunds only through the flow (W2, W5)

Pattern	The fact with no record	Kestrel & Pine's incident	The owner Waypoint named
The Price Nobody Can Explain	The consumer price per market	The Price Drift (€179 for nine days, 410 orders, about €8,200)	A single price source per market, published by event (W5); What Does It Cost? Price Lists, Contract Prices, Currencies and Tax

The pattern that is not in the family, The Secret Agent, is the ownership failure seen from the writer's side: a script, app or person that changes a fact without being on the writer list. The App That Wrote Stock was a Secret Agent and a Phantom Stock at once.

Across Scale Stages: When It Fits and When It Limits

The entity–owner table is not written once. Each stage brings a new system that claims facts the previous owner held, and the moment of handover is when two writers exist. The stages are the book's four (What Changes at Each Scale: From First Store to Group).

Stage	Typical ownership	What becomes necessary	Warning signs of outgrowing it
1 — Starting	The platform is the record for product, price, stock, order and customer; an accounting package for the books; the portal of a 3PL (a third-party warehouse that ships on the merchant's behalf) holds a copy of stock	Nothing beyond one e-mail per customer and a monthly payout reconciliation	A 3PL or an accounting system starts writing stock or customers back
2 — Growing	An ERP arrives and claims items, business customers and valued stock; the platform keeps D2C customers and orders; a POS brings store stock and a second customer key	Decide the product master and the stock master now; one direction per entity; a daily order and stock reconciliation	Two systems write stock; a spreadsheet becomes the product master; the first oversell
3 — Omnichannel	The WMS owns physical stock, an OMS owns orders and availability, a PIM owns content, the ERP keeps identity, cost and finance, the platform keeps checkout	The full table with a refresh and a reconciliation per row; matching rules for customers; the ERP's rhythm decoupled from availability	Support checks three systems for one order; a stock sync chain measured in hours; refunds issued in two places

Stage	Typical ownership	What becomes necessary	Warning signs of outgrowing it
4 — Group	One table per brand or country, with shared records where sharing pays (PIM, customer master, OMS) and separate ones where it does not; survivorship rules per attribute	A master data function with a merge log; governance that decides what is shared before a stack is forced on a brand	One Size Fits None: brands forced onto one record before the benefit is measured, or kept apart when they should share

Kestrel & Pine moved from stage 2 to stage 3 in 2023 and kept stage 2's ownership for four years: the ERP as stock master for a company with its own WMS, the platform as order record for a company with four order sources. The judgement: **reassign ownership at the stage transition, not after the first incident**. The sign that it is time is always the same, a second system that originates orders or counts stock.

In Practice

Public examples. The vendor statements quoted above (Microsoft's connector documentation of 2026, Shopify's ERP guidance of August 2026, Fluent Commerce and OneStock) are the clearest public cases of records being named per entity. At a much larger scale, a 2020 account by a former Walmart engineering director on Confluent's blog describes Walmart's inventory backend partitioning its event stream so that "each item store always goes to a specific partition", which ensured "that only one consumer deals with that item store combination": one writer per entity, enforced by the plumbing. The account (May 2020) is six years old, is the author's own view with Walmart not endorsing it, and describes a group-stage system; the principle is the one in Figure 2.

Kestrel & Pine's stock, before and after W1. The table reconstructs the writer list for the Ridgewalker jacket's warehouse stock in January 2027 and under the Waypoint design from Q4 2027.

	January 2027	After W1 (Q4 2027)
Record for Duiven stock	None named; the ERP by habit	The WMS
Writers of Duiven stock	WMS feed; manual ERP adjustments; manual platform edits; three apps	Scans, counts and receipts in the WMS
Record for store stock	The POS, copied to the ERP nightly	The POS, publishing stock events
Record for stock in the marketplace's program	None; invisible until the marketplace reported it	A tracked external location fed by the marketplace's reports
Availability per channel	Each channel's copy of the ERP's copy of the WMS	Computed by the availability service from the three stock records and open orders, with a buffer per channel
How copies learn	WMS to ERP every 15 minutes, ERP to platform hourly, platform to connector hourly	Stock events, seconds; the sync chain retired
Disagreement found by	Customers, on 27 November 2026	Weekly WMS-to-platform reconciliation; cycle counts in the WMS

Replay the Black Friday evening against the right-hand column. At 19:40 the last Ridgewalker women's M moss is scanned at a packing station; the WMS emits a stock event; the availability service recomputes the channel quantities to zero within seconds and publishes them; the storefronts and the connector stop selling the variant before 19:41. The 1,340 oversold units become a handful, from orders already in checkout at 19:40, which is what the per-channel buffer exists to absorb. The first weekly reconciliation in May 2027 found 2.3% of active variants with a different quantity between the WMS and the platform (illustrative), most left behind by the three apps; the apps' write access was removed, and the number is the reconciliation's baseline.

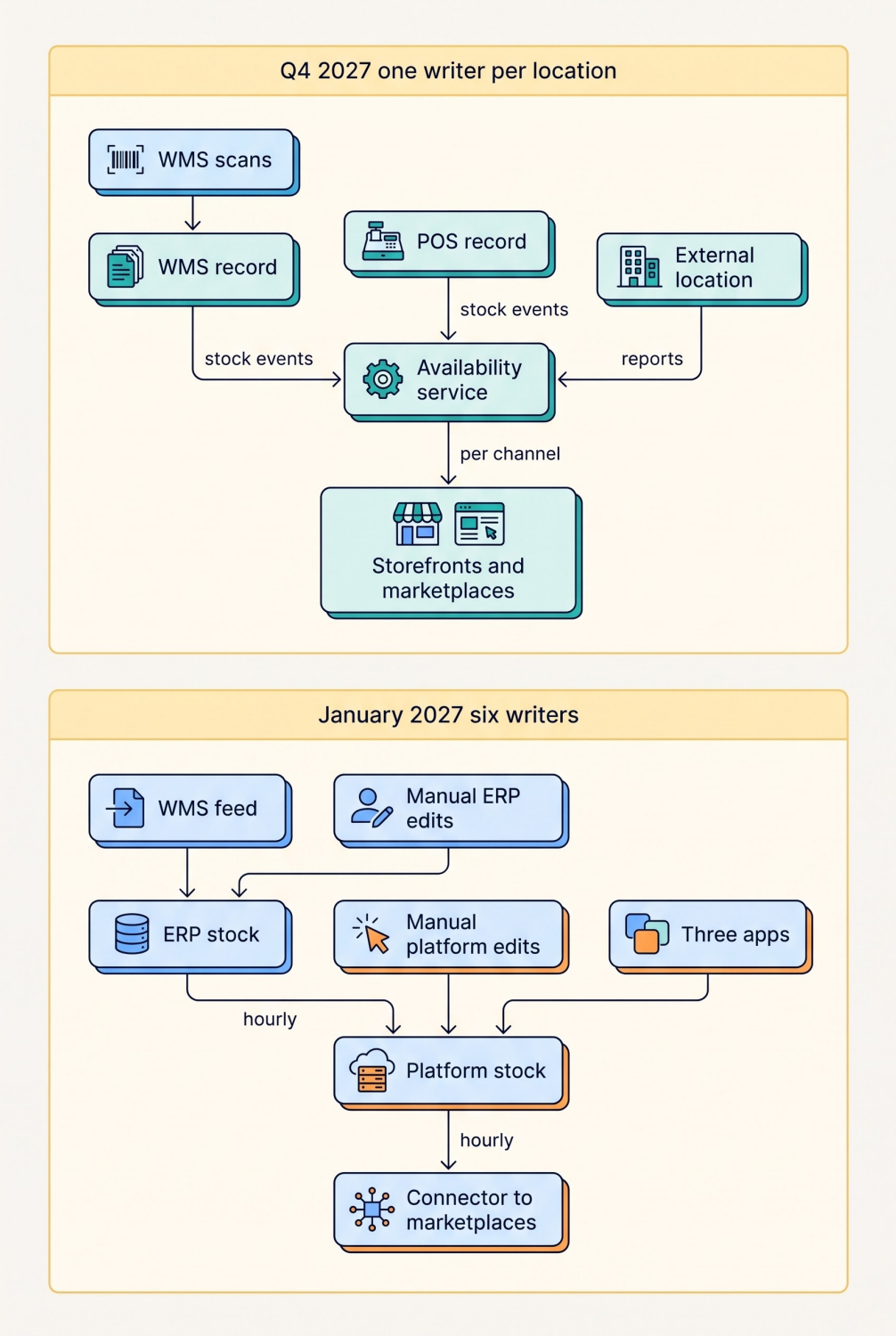

Q4 2027 one writer per location
WMS scans
WMS record
POS record
External location
stock events
stock events
Availability service
reports
per channel
Storefronts and marketplaces
January 2027 six writers
WMS feed
Manual ERP edits
ERP stock
Manual platform edits
Three apps
hourly
Platform stock
hourly
Connector to marketplaces

Figure 5: *Kestrel & Pine's warehouse stock before and after Waypoint decision W1: six writers feeding a two-hour chain of copies become one writer per location, an availability service that computes what each channel may sell, and copies that learn by event.*

When It Goes Wrong

Ownership decided is not ownership kept. Five things go wrong, each with a detection and a recovery.

A second writer appears. An app, a script, an integration "fix" or a well-meaning colleague starts changing a fact in a copy or beside the record. The App That Wrote Stock is the case. Detection is a periodic *write audit*: for each row of the entity–owner table, list every credential, app scope, job and role that can change the fact, and compare it with the writer column. Recovery is to remove the write, then re-publish the fact from the record to every copy.

A copy goes stale. The refresh breaks or slows; the copy keeps serving its last value. Kestrel & Pine's two-hour chain was stale by design, and the Deprecated Endpoint of February 2027 made the connector's copy stale for two days when an API version was retired. Detection is a *lag metric per copy* (time since last successful refresh, alarmed against the freshness table) and the reconciliation's difference count. Recovery is a full re-publication from the record once the refresh is repaired, not a hand correction of the copy.

A copy is edited by hand. Support changes an order's status in the platform to calm a customer; an operator types a stock quantity into the storefront; a refund is issued in the PSP console. Each is a write to a copy, or to a record outside the flow that owns it; the Double Refund was the last of these. Detection: change logs on copies, reviewed for human edits; refunds reconciled against credit notes daily. Recovery: route the action through the owner (the OMS issues the refund; the WMS corrects the count) and close the side door.

A copy is mistaken for the record. The data warehouse's order count is quoted as the company's; the connector's stock number is treated as the warehouse's. Detection: ask, for any disputed number, which row of the table it comes from. Recovery is the table itself, published where people can see it, and Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics's one definition of an order.

The record is wrong. The shelf count in the WMS differs from the shelf. No ownership rule fixes this; the record needs its own correction process (cycle counts, receipts, damage write-offs), performed in the record by its writers. Correcting the copy instead produces a right copy and a wrong record, and the next refresh undoes the correction.

What to Take Away

Established.

- Several systems holding one fact is normal, and the field's remedy is to name a system of record per fact and treat the rest as copies; vendors that document their integrations say so in those words (Microsoft 2026, Shopify 2026, Fluent Commerce and OneStock 2026).
- Different systems legitimately model "order", "product" and "customer" differently; Evans' bounded context (2003) and Fowler's 2014 restatement hold that unifying one model across a large system is not feasible or cost-effective, so the work is translation at the boundary and shared identifiers.
- One writer per entity is enforceable mechanically: a single refresh direction per entity in a connector, or a single consumer per item-and-location partition in an event stream (Walmart's 2020 account).

Depends on context.

- Which system is the hub depends on where orders and stock originate: the ERP or the platform can own everything at one channel and one warehouse; ownership per fact with an order hub becomes necessary at the second order source or the second stock location.
- Whether the business customer's record is the ERP or the platform depends on who manages terms and credit limits; the consumer customer's record is usually the platform until a store or a loyalty program adds a second key.
- How fresh a copy must be depends on the fact: seconds for availability at peak, exactness at promotion boundaries for price, a day for ledger postings; one rhythm for all facts is wrong in both directions.

The author's synthesis.

- Write the entity–owner table before buying any hub, and include the writer list and the reconciliation per row; a bought OMS fed by four unreconciled order sources is a fifth copy, and a written table with no reconciliation is a record on trust.
- Reassign ownership at the stage transition rather than after the incident: the day a second system originates orders or counts stock, the previous owner's claim has already expired, whatever the configuration still says.

Questions to Ask

1. For this fact, which system is the record, and is that written down anywhere a new colleague could find it?

2. Who, and what, can write this fact in that system and in every copy? Is there more than one entry on the list?
3. How does each copy learn about a change, how old may it be, and what alarms when it is older?
4. What job compares the record with each copy, how often, and who receives the differences?
5. When a customer sees one value and the record holds another, what does support do, and in which system?
6. Which of our facts are derived (availability, the delivery promise), and can anyone edit the derived number by hand?
7. What does "order" mean in each of our systems, and which system owns the view across them?
8. Which system claims a fact today because of the stage we were in when we bought it, not the stage we are in now?

With the parts mapped and their owners named, the next question is what the market sells to fill the boxes, who sells it, how it is priced, and how the landscape changed up to 2026, which is Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026.

Sources and Evidence

Company, product and investor sources

- Synchronize inventory with Shopify, Microsoft, 2026 — "Business Central acts as the system of record for inventory"; one-way push every 15–30 minutes.
- Synchronize items and products with Shopify, Microsoft, 2026 — choosing the record per entity ("To Shopify" or "From Shopify").
- Synchronize customers and companies with Shopify, Microsoft, 2026 — "decide which system is the system of record for customer data"; mixed directions cause conflicting records.
- Synchronize orders with Shopify, Microsoft, 2026 — orders originate in the shop and are imported into the ERP.
- ERP integration: what it is and how it works, Shopify, 2026 — "name the system of record for each" field; "Decide in advance what happens when the two systems disagree" (vendor guidance).
- Fluent Commerce order management system, Fluent Commerce, n.d., accessed 2026-11 — "your ERP stays the system of record" (vendor).
- OMS vs ERP: the architecture shift reshaping commerce, OneStock, n.d., accessed 2026-11 — the OMS as master for availability, the ERP for product data absent a PIM (vendor).

- As AI Agents Transform Commerce, Salesforce Unleashes Its Biggest Agentforce Commerce Release Yet, Salesforce, 2026 — a suite that owns catalog, inventory and orders inside one boundary.
- Real-time inventory management at Walmart using Kafka, Suman Pattnaik on Confluent's blog, 2020 — one consumer per item-store partition (older than three years; author's own views).

Standards, regulation and official documentation

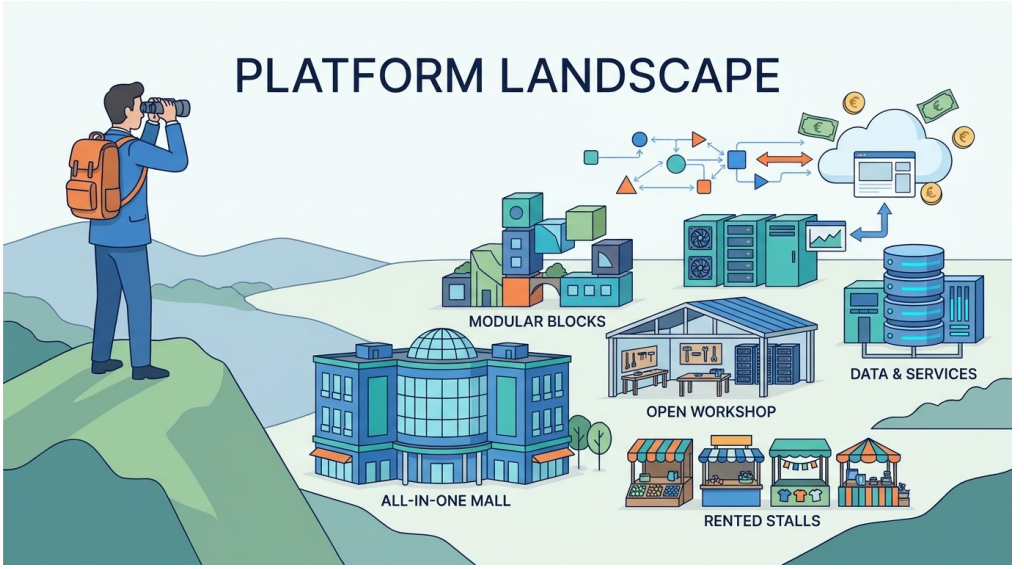
- None; this chapter's sources are vendor documentation and practitioner literature.

Research and practitioner sources

- Eric Evans, *Domain-Driven Design: Tackling Complexity in the Heart of Software*, Addison-Wesley, 2003 — the bounded context (older than three years; the concept is unchanged).
- BoundedContext, Martin Fowler, 2014 — bounded context as DDD's central strategic pattern; unification of a large model is not feasible or cost-effective; "Customer" and "Product" as polysemes (older than three years; the concept is unchanged).
- Systems of Record: Ending the Doppelgänger Record, RECONCILED, 2026 — the finance-side treatment of records and copies.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

5. Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026



IN THIS CHAPTER, YOU WILL: Learn the seven kinds of commerce platform, how each is priced, what changed between 2023 and 2026, and how to read a vendor's claims before comparing two of them.

KEY POINTS:

- **The category decides where the work lands.** A hosted platform keeps the core in the vendor's hands and pushes variation to apps; a suite keeps integration inside one vendor's data model; open source moves hosting and upgrades to you; composable moves the integration burden to you. Pick the category by who you have to carry that work.
- **Commerce platforms are priced on your sales, not on their software.** Shopify's FY2025 filing puts 76% of its revenue in fees that move with merchant sales; Lightspeed's FY2026 transaction revenue is more than twice its subscriptions; Shopware, Saleor and BigCommerce tie

their plans to gross merchandise value. The subscription is the entry ticket.
- **Between 2023 and 2026 the market converged on "keep your back-end, add a layer".** commercetools, Adobe and Salesforce now sell modules that sit beside an existing stack, Shopware absorbed integration and payments into the platform, and Commerce.com reported that replatforming demand is soft. Read every 2026 pitch, "composable" and "agentic" included, as a layer to add, not a reason to move.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Giant's Boots** — the landscape is full of architectures built for companies a hundred times larger; this chapter sorts them by the merchant they fit.
- **The Forked Platform** — the categories differ in who applies upgrades, and the ones that let you change the core are the ones that let you lose the upgrade path.
- **The Lynchpin** — a vendor's revenue mix, ownership and strategic reviews are public signals of how long a capability will be funded; this chapter shows where to read them.

Two commerce platforms can offer the same cart, catalog and checkout features while giving a merchant very different work to do. With one, the vendor runs the servers and storefront. With another, the merchant must operate a separate storefront and manage upgrades. Those responsibilities determine the team and budget needed after the purchase.

Pricing and customization add further differences. A percentage fee grows with revenue; changes to a platform's core can make upgrades expensive. A feature comparison can miss both. Evaluating the **operating model** means asking who runs each part, how costs change and what the merchant would need to move elsewhere.

The seven categories below describe those choices, their pricing models and documented changes from 2023 to 2026. Use the map to question vendor claims and identify which responsibilities your team would take on. Build, Buy or Assemble: Choosing and Changing Platforms provides the decision method; The Map of a Commerce Platform: Capabilities and Layers and Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record explain capabilities and ownership.

Where we are in the order's journey. Eva buys a jacket and socks for €235 from Kestrel & Pine, the fictional outdoor brand. Its platform runs the storefront, cart and checkout. The platform category determines which later jobs it can handle and which need separate systems.

Setting the Scene at Kestrel & Pine: Two Pitches in One Week

Kestrel & Pine has used a hosted **software-as-a-service (SaaS)** commerce platform since 2016. The vendor runs it; the merchant pays **€2,300 a month plus payment and transaction fees** on its enterprise tier. Three localized storefronts share one merchant account, or tenant, with 41 apps.

Around it sit separate finance and purchasing, warehouse, store till, marketplace, returns and integration systems. Total technology spend in 2026 was about **€2.6 million**, or 5.4% of revenue. The platform subscription is only one part of that cost.

In February 2027, Head of Engineering Noor Visser heard two proposals. An outside consultancy offered a **composable** system: separate commerce, storefront, search, content and integration products. Its illustrative estimate was **€3.2 million over eighteen months**.

The existing agency offered a **headless** storefront: a separate application calling the current platform through its API, hosted by the platform vendor, with a separate content system. That estimate was **€240,000 over nine months**. Both cited the Black Friday Oversell, when stale stock figures caused 1,340 oversold units and about €96,000 in costs.

Mara Lindqvist, the CEO, asked Noor three questions afterwards. *What kind of platform are we on now? What would each of these cost us in year three? And what would a move fix that nothing else fixes?* Noor did not have the vocabulary for the first question on the spot, which is why this chapter exists. By March her answer had become Waypoint decision **W4: stay and surround**. What follows is the map she drew.

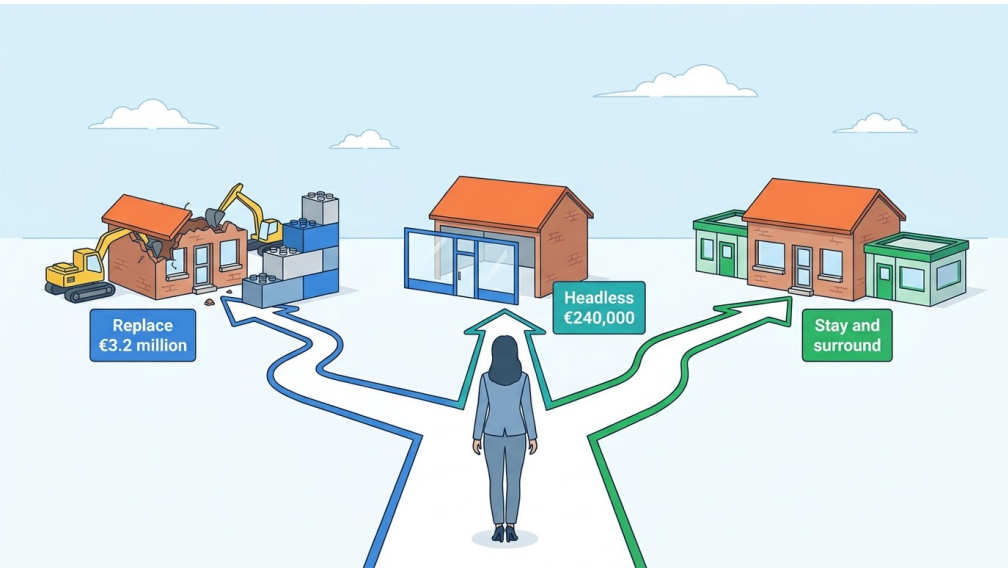


Figure 1: *Two pitches in one week: a €3.2 million replacement and a €240,000 headless rebuild, against the third answer Noor reached by March, Waypoint's W4, stay and surround.*

Seven Kinds of Platform

A **commerce platform** is the software that shows products, holds a cart, takes an order and the payment, and hands the order to whoever fulfills it. Some products stop there; others bundle warehouse, order management or finance too. The market sells this in seven shapes. The table is my synthesis; the categories overlap, and several vendors sell in more than one (Adobe sells on-premises, platform-as-a-service and SaaS).

Category	What you buy	Who runs it	Where the integration work lands
Hosted SaaS platform (Shopify, BigCommerce, Lightspeed, CCV Shop)	A storefront, cart, checkout and app store shared with many merchants	The vendor	On apps, webhooks and connectors you install

Category	What you buy	Who runs it	Where the integration work lands
Enterprise suite (SAP, Salesforce, Adobe)	One vendor's bundle of commerce, order management and content modules	The vendor's cloud, via a partner	Inside the vendor's data model, while you stay inside it
Open source and its forks (Magento and Mage-OS, WooCommerce, Shopware, PrestaShop)	Source code to host, extend and upgrade yourself; paid editions optional	You or your agency	On your agency, with upgrades as the recurring cost
Composable and headless vendor (commercetools, Elastic Path, Medusa, Saleor)	API-only commerce services; you assemble the rest	The vendor for the services; you for the rest	On you, by design
ERP-centered commerce (Odoo, JTL, PlentyONE, Business Central, NetSuite)	A storefront attached to the ERP that already owns stock, orders and finance	The vendor's cloud or your servers	Inside the ERP, while its pace is fast enough
Specialist layer (OMS, PIM, WMS, returns, search, channel, iPaaS, payments vendors)	One capability	The vendor	On the connectors between the layer and your platform
Specialist group (Katapull)	Several specialist products under one owner, pre-integrated	The group	Among the group's products; on you at its edge

Four terms carry the table. **SaaS** (software as a service) is software the vendor runs for many customers at once; you configure it and the vendor upgrades everyone on the same

day. A **suite** is one vendor's set of modules designed to work together. **Headless** means the storefront (what shoppers see) is separated from the commerce engine (catalog, cart, orders) and talks to it only through an **API** (application programming interface: the contract by which one program calls another). **Composable** means buying commerce as separate services, from one or several vendors, and assembling them. A **webhook**, which several tables mention, is a notification a platform sends to another system when something changes.

The column that matters most is the fourth. **Every category moves the integration work somewhere; none removes it.** Kestrel & Pine's 34 integrations exist because it chose a hosted platform and bought specialist layers around it. A suite would have fewer joins and one larger contract; a composable assembly more joins, all its own. The choice is who carries the work, at what price, with what freedom to change.

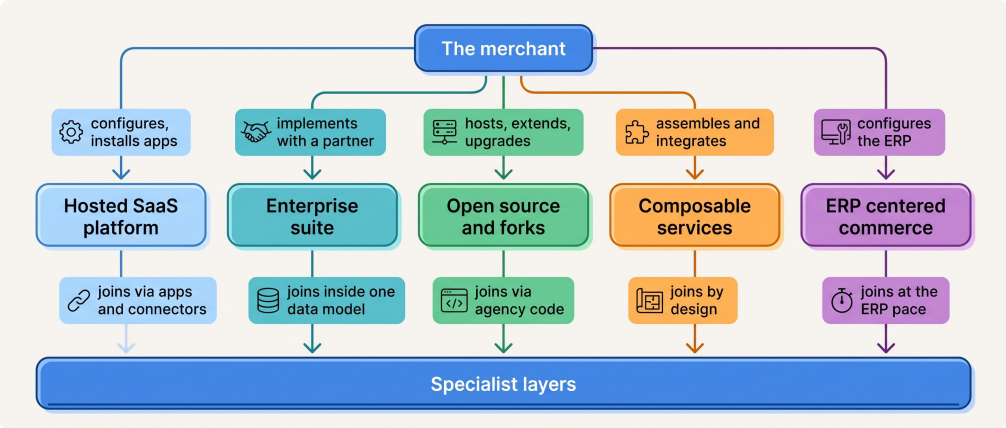


Figure 2: *Each platform category asks the merchant for a different kind of work and joins the specialist layers in a different place; the integration burden moves, it never disappears.*

Hosted SaaS platforms

The hosted platform is the default for a merchant that starts online, and the category with the most public evidence, because two of its vendors are listed. **Shopify**'s annual report for the year ended 31 December 2025 (filed 11 February 2026) reports gross merchandise volume (**GMV**, the total value of orders through the platform, which is not the vendor's revenue) of **$378.4 billion** and revenue of **$11.56 billion**. Its euro pricing page in November 2026 lists plans from Basic at €28 a month to Plus "starting at €2,100 a month", card rates from 1.9% + €0.25 down to 1.4% + €0.25, and a fee of 2% down to 0.2% on orders paid through another payment provider; larger Plus merchants may pay "a variable platform fee based on your revenue and business model". The same filing says most merchants are on the two cheapest plans while most GMV comes from Plus

and enterprise merchants: one product, two populations (Case: Shopify, the Modular Monolith That Serves Millions of Merchants takes this apart).

BigCommerce renamed its parent **Commerce.com** on 31 July 2025, keeping BigCommerce as the engine. Its November 2026 pricing page assigns plans by trailing-twelve-month GMV, with a 0.9% charge above $33,333 a month on the Scale plan: a subscription that upgrades you when you sell more is a GMV price in disguise. Its results for the quarter ended 30 June 2026 (filed 6 August 2026) showed revenue of $84.5 million, up 0.1%, and management saying B2C replatforming "remains softer than historically".

Lightspeed closed a strategic review on 6 February 2025 and chose to "focus on growing retail in North America and hospitality in Europe"; its FY2026 results (21 May 2026) showed transaction-based revenue of $815.1 million against subscription revenue of $370.7 million. Its Dutch eCom product still sells (€68 to €259 a month, November 2026).

> **INFERENCE, NOT DOCUMENTED:** Lightspeed has announced no change to its European e-commerce product. Reading the strategic review as a vendor-risk signal for a Dutch merchant on it, because European online retail is outside the vendor's stated growth engines, is my inference.

Below these sit the Dutch **CCV Shop** (€36 to €129 a month, "not a percentage of your sales") and the site builders.

Enterprise suites

A suite sells to large merchants through implementation partners, with negotiated prices, so its public evidence is thinner. **SAP Commerce Cloud** is described here only from its open-source storefront project Spartacus, which talks to the platform through its REST API (accessed November 2026), and from an October 2026 analyst note on SAP Connect 2026 that reports an "SAP Commerce Cloud, cloud ERP edition" for midsize organizations and warns buyers to "separate products that are available now from forward-looking demonstrations". SAP's product page refused automated fetching; I quote no feature list for it.

Salesforce announced on 6 July 2026 "its biggest Agentforce Commerce release yet": one platform across B2C, B2B, point of sale and order management, with three generally available agents "natively connected to catalog, inventory, and orders". Its pricing page shows "Contact for pricing" for every edition; the trade describes Salesforce as priced on a percentage of GMV, but the only written source here is a competitor's page, so treat that as industry belief.

Adobe Commerce is the clearest public example of one product in three deployment shapes. Adobe's information center (updated 21 October 2026) lists "Adobe Commerce on-premises" (you run everything), "Adobe Commerce on cloud infrastructure" (platform-as-a-service: Adobe runs the hosting, you own custom code) and "Adobe

Commerce as a Cloud Service" (fully managed SaaS with continuous updates). Beside them, **Adobe Commerce Optimizer** is a SaaS for storefront, catalog and merchandising that "can work alongside an existing Adobe Commerce or third-party commerce backend"; partner reports place both launches in June 2025. The middle option is where the **Forked Platform** pattern lives: the vendor runs the servers, the merchant owns the code, and deep customization can block every upgrade (Apps, Plugins and Functions: Extensibility and Ecosystem Governance).

Open source and its forks

Open source means the code is published under a license that lets you run and change it. It does not mean free: hosting, upgrades, security patches and the agency that applies them are the recurring price. Adobe bought Magento in 2018 and renamed Magento Commerce to Adobe Commerce in 2021; **Magento Open Source** continues, and a community **fork** (a copy maintained separately) called **Mage-OS** describes itself as an "independent, community-driven distribution of Magento Open Source", run by an association founded in 2022. Mage-OS 4.0.0 shipped on 30 October 2026; 3.6.0, two weeks earlier, was a security port of an Adobe patch, which is the dependency chain an open-source merchant lives with: Adobe publishes a fix, the distribution ports it, the agency applies it, the merchant tests it. **WooCommerce**, the open-source commerce plugin for WordPress, claims "more than 4 million stores" and prices hosting at $25 to $350 a month (November 2026).

Shopware, from Germany, has the most public evidence and the most instructive pricing. Its Community Edition is free; the paid plans (Rise "from €600 a month", Evolve "from €2,400 a month", November 2026) are "based on your gross merchandise value", and a fair-usage rule requires a paid plan once the shops on one account pass €1 million of GMV in a rolling year. So "open source" is free below the Growing stage and GMV-priced above it. The EHI Retail Institute's 2026 report found 12.5% of the 1,000 largest German B2C shops on Shopware, first place for the fifth year. In October 2025 PayPal raised its stake to more than 40%; on 10 June 2026 Shopware launched **Shopware Payments** on PayPal's infrastructure and **Nexus**, an orchestration layer it positions as replacing middleware. Case: commercetools, Shopware and the Composable Movement follows Shopware in detail.

Composable and headless vendors

commercetools, founded in Munich in 2006 and the best-known composable vendor, reported on 22 April 2026 that it had passed a "$100B GMV run-rate" with more than 600 million orders a year (a run-rate annualizes a recent period; it is not a full-year figure). On 26 February 2025 it had cut about 10% of staff after missing growth targets; on 21 January 2026 it launched **AgenticLift**, which lets companies "layer agentic commerce capabilities on top of your existing commerce solution"; on 15 July 2026 it began selling

its cart-and-order engine and its product catalog as standalone modules that integrate "with their existing technology stack". The flagship of composable now sells layers for stacks that are not composable. **Saleor**, an open-source entry, prices its cloud from $1,599 a month up to $200,000 of monthly GMV with 0.8% above (November 2026): API-only vendors price on GMV too.

MACH is the acronym a vendor alliance coined in June 2020 for Microservices, API-first, Cloud-native SaaS and Headless; the MACH Alliance's founders were commercetools, Contentstack, EPAM and Valtech. By November 2026 its principles page lists three principles (Open, Composable, Connected) and no longer defines microservices or headless. Here it is enough to know that "composable" and "MACH" name a category and a sales argument, not a verdict; the movement's rise, turn and own survey data are the subject of Case: commercetools, Shopware and the Composable Movement.

ERP-centered commerce

An **ERP** (enterprise resource planning system) keeps the books: purchasing, stock valuation, invoices, the general ledger. ERP-centered commerce makes the ERP the owner of products, stock and orders and treats the webshop as a front end. **Odoo** sells its eCommerce app free within a per-user ERP price (€19.90 or €35.90 per user per month, November 2026): the webshop costs nothing, the seats do. **JTL** (Germany) markets the ERP as the core with shop, WMS, marketplaces, EDI and POS around it; **PlentyONE** (formerly plentymarkets, renamed November 2024) bundles PIM, OMS, WMS and a free shop; **NetSuite SuiteCommerce** is described in Oracle's help pages as three products on one account. The strongest signal that this category is mainstream comes from Microsoft: its documentation dated 22 September 2026 says the Shopify Connector app "is preinstalled for all new sign-ups and trials" of Dynamics 365 Business Central. An ERP vendor shipping a connector to a hosted platform by default is "ERP as hub, storefront in front" declared as the expected shape.

The strength of this category at the Starting and Growing stages is one system and no integration problem. Its limit arrives at the Omnichannel stage, when stores, marketplaces and a warehouse each need stock in near real time and the ERP's batch cadence cannot serve them (The Back Office: ERP Integration and Who Is the Hub).

Specialist layers and specialist groups

Every category above is surrounded by **specialist layers**, each owning one capability and selling to every platform. The book's vocabulary comes from them, so the table defines each once.

Layer	What it owns, in plain words	Examples (vendor pages, accessed November 2026)
OMS (order management system)	The order after checkout: which warehouse or store fulfills it, what to promise, how to route a return	Manhattan Active Omni; Fluent Commerce ("150+ implementations"); Kibo
PIM (product information management)	The product record: attributes, completeness rules, enrichment, distribution to channels	Akeneo; inRiver ("1,600+ brands")
WMS (warehouse management system)	Physical stock and its movements: receiving, put-away, picking, packing	Manhattan; Picqer (NL, €379 a month with 3,000 orders, then €0.28 per order)
Returns	The reverse order: portal, rules, exchanges, labels, refunds	Loop ("5,000+ brands"); Narvar; Returnista (NL)
Search and discovery	Query understanding, ranking, merchandising rules	Algolia ($0.50 per 1,000 extra requests); Constructor
Channel integration	Listings, offers, stock and orders across marketplaces	ChannelEngine (NL, "over 1,300 marketplaces"); Rithum; Mirakl
iPaaS (integration platform as a service)	The flows between systems: connectors, mapping, retries, errors	Celigo; Boomi; Workato
PSP (payment service provider)	Taking the payment and settling the money	Adyen (interchange + 0.60% + $0.13); Stripe (1.5% + €0.25 on EEA cards in NL); Mollie (iDEAL €0.32)

OMS and PIM are the two layers Kestrel & Pine lacks and the two Waypoint buys (W2 and W3). An OMS decides which location fulfills an order and what to promise; a WMS decides where goods sit and how to pick them; The Order Record: Lifecycle, State and the Order Management System and Pick, Pack, Ship: Warehouse Management and Fulfillment keep the two apart.

A **specialist group** is a third shape beside suite and composable: several specialist products under one owner, kept as separate brands but pre-integrated. The Dutch group **Katapull** is the public example. Its site (accessed November 2026) names eight labels, one per role: Logic4 (ERP and webshop), Vendit (POS, point of sale: the store till software), Picqer (WMS), EffectConnect (marketplaces), DeliveryMatch (shipping), Returnista (returns), Transus (EDI, electronic data interchange: structured business documents between trading partners) and Highbiza (an online-performance agency). Five companies joined forces on 31 March 2025; Transus joined at a date no public page gives; Returnista followed on 16 February 2026 and Picqer, the eighth label, on 7 October 2026. The about page claims "4,500+ customers in 40+ countries"; no revenue, GMV or ownership figures are public, and I infer none. The group reduces the joins a merchant must build, at the cost of one owner across several layers.

How Platforms Are Priced, and What the Price Tells You

A platform's price is an architectural fact, because it tells you what the vendor's product really is. Seven pricing models recur across all seven categories.

Model	How it works	Dated examples	What it means for you
Flat subscription tier	Fixed monthly fee; limits rise with the tier	Shopify Basic €28 to Advanced €384; CCV Shop €36 to €129; Lightspeed eCom NL €68 to €259 (all Nov 2026)	Cheap to start; limits force upgrades
Subscription with GMV thresholds	Your tier is set by trailing sales; a percentage applies above a cap	BigCommerce 0.9% above $33,333 a month; Saleor 0.8% above $200,000 a month; Shopware paid above €1 million of GMV (Nov 2026)	Cost grows with revenue; the vendor takes a share once you are large

Model	How it works	Dated examples	What it means for you
GMV share (enterprise)	A negotiated percentage of sales, often with a floor	Shopify Plus "variable platform fee" above a $2,300 to $2,500 base (Nov 2026); Salesforce, widely described as 1% to 3% (unconfirmed)	Raises exit cost; wholesale (large orders, low margin) must negotiate separately
Payment fee share	The platform earns on card processing and charges a **transaction fee** per order when you use another PSP	Shopify 0.2% to 2% third-party fee; BigCommerce 0.6% to 2%; Lightspeed $815 million transaction revenue against $371 million subscriptions (FY2026)	Payments are the platform's core revenue; your PSP choice is constrained by the platform contract
Per order, **user**, **request**	A fee per unit above an included quota	Picqer €0.28 per extra order; Odoo €19.90 per user; Algolia $0.50 per 1,000 extra searches; Shopware Nexus €10 per 1,000 executions (Nov 2026)	Peak days and catalog size cost money

Model	How it works	Dated examples	What it means for you
Per transaction (payments)	One blended rate, or **interchange** (the card schemes' part of the fee) plus a margin	Adyen interchange + 0.60% + $0.13; Stripe NL 1.5% + €0.25; Mollie iDEAL €0.32 (Nov 2026)	Blended is simpler; interchange-plus is cheaper at volume but needs reconciliation (Taking the Money: Payment Providers, Authorization, Capture and Refunds)
Free core, paid services	Open-source software with hosting, extensions or editions sold	WooCommerce 0% revenue share; Mage-OS; Medusa ("no extra licenses or GMV-based tax")	Cost moves to the agency and to operations; you own upgrades

The pattern across the rows: **a vendor prices on the thing it sits in the path of**. GMV pricing assumes the vendor owns checkout; per-order pricing assumes it sees every order; per-request pricing assumes it is in the request path. This is why Shopify's FY2025 filing reports 76% of revenue as "merchant solutions" that are "directionally correlated with the level of GMV", with 65.6% of GMV through its own payment service, and why Lightspeed is, by revenue mix, a payments company with software attached.

A Kestrel & Pine illustration. Kestrel & Pine's records put the platform at €2,300 a month, €27,600 a year, "plus payment and transaction fees". Assume (an illustration, not one of Kestrel & Pine's records) that the platform's checkout carries the €27.0 million of D2C (direct-to-consumer) revenue plus the 62% of wholesale orders placed through the B2B (business-to-business) portal, about €7.4 million: roughly €34.4 million of GMV a year. At a 0.2% third-party transaction fee, the kind a top-tier hosted plan charges when the merchant brings its own PSP, that is about €69,000 a year, two and a half times the subscription; at a negotiated 1% platform fee on GMV, about €344,000, twelve times. Neither is what Kestrel & Pine pays; the point is that **the subscription line is the smallest line**, and a comparison that puts "€2,300 a month" against a competitor's list price compares entry tickets. Build, Buy or Assemble: Choosing and Changing Platforms builds the full cost model.

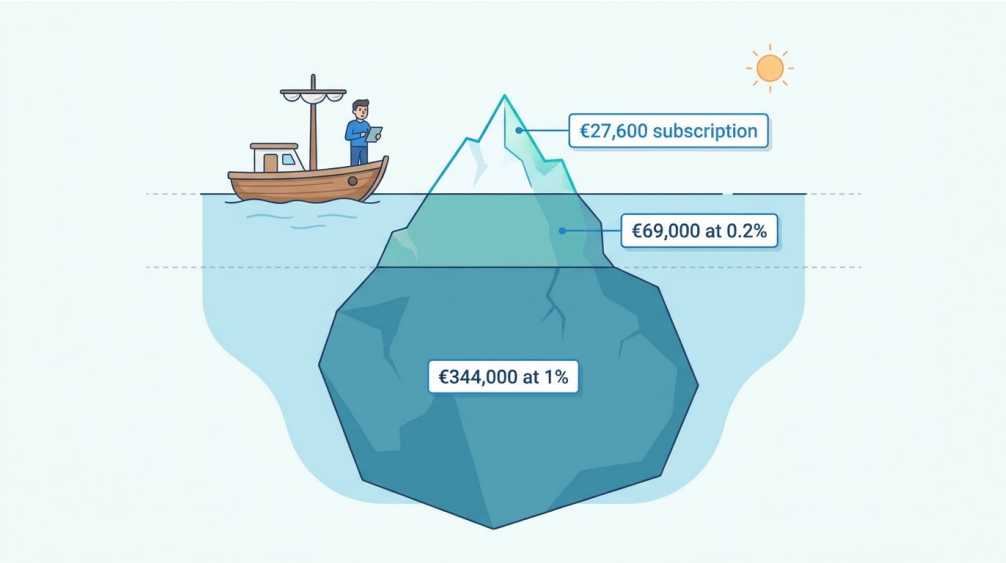


Figure 3: *The subscription line is the smallest line: on about €34.4 million of GMV a year, a 0.2% fee is two and a half times the €27,600 subscription and a 1% fee twelve times.*

What Changed, 2023 to 2026

The three years before this book's research cut-off (7 November 2026) moved in one direction: vendors retreated from physical logistics, pulled payments into the platform, and began selling layers beside a merchant's existing stack rather than replacing it.

Date	Event	What it tells a merchant
6 June 2023	Shopify completed the sale of most of Shopify Logistics to Flexport for a 13% equity stake	Platform vendors keep orchestration and push physical logistics to partners
26 February 2025	commercetools cut about 10% of staff after missing growth targets (TechCrunch)	Pure composable stopped selling itself

Date	Event	What it tells a merchant
June 2025	Adobe shipped Commerce as a Cloud Service and Commerce Optimizer, the latter "alongside an existing … third-party commerce backend"	A suite vendor sells SaaS layers beside a kept backend
21 January 2026	commercetools launched AgenticLift "on top of your existing commerce solution"	The composable founder sells a layer for non-composable stacks
10 June 2026	Shopware launched Nexus (orchestration replacing middleware) and Shopware Payments	An open-source platform absorbs integration and payments
6 July 2026	Salesforce shipped three agents "natively connected to catalog, inventory, and orders"	A suite vendor sells agents, not decomposition
15 July 2026	commercetools began selling its order engine and catalog as standalone modules "without replacing their existing platforms"	The composable flagship sells itself by the piece
6 August 2026	Commerce.com reported flat revenue and B2C replatforming "softer than historically"	Replatforming demand itself is weak

The newest surface is the **AI assistant as a channel**. Shopify's Agentic plan (accessed November 2026) has "no monthly fee" and syndicates products to ChatGPT, Gemini, Copilot and Perplexity so shoppers "check out in conversations", with "no Shopify online store required". For a merchant this is one more channel, priced as a channel; Assistants and Agents: AI in Commerce and What Is Real in 2026 separates what is available from what is announced.

How to Read a Vendor's Claims

Comparing two vendors starts with the **unit, the date and the definition** behind every number.

Claimed unit	Dated example	The trap
GMV	Shopify $378.4 billion in 2025; commercetools "$100B run-rate" in April 2026	A run-rate is not a full year; BigCommerce's plan GMV is "gross order value minus 10%", not its reported GMV
Merchants, customers, brands, users	Shopify's filing counts a merchant as a paying shop, so one brand with three stores counts three times; commercetools "500+ enterprises" in April 2026 and "more than 600" in July; Katapull "4,500+ customers" and "27,500+ users"	Dormant installs inflate open-source counts; "users" exceeds "customers" by a factor of five to ten
Marketplaces, integrations, apps	ChannelEngine "over 1,300 marketplaces"; Shopify "more than 21,000 apps" (31 December 2025)	A listing proves a connector exists, not that it is maintained or used
Surveys, analyst charts, performance claims	MACH Alliance, February 2026: 78% of mature adopters see "clear ROI from AI"; Gartner's 2025 quadrant rated 19 vendors, Forrester's Q3 2026 wave 10; Shopware Nexus "cut integration costs by 40%" (June 2026)	The vendor chose the sample; vendor counts change yearly and absence is not a verdict; unaudited claims are marketing

Market-share data need the same care. W3Techs, which crawls websites, reported on 7 November 2026 that WooCommerce runs 47.3% of sites with a detectable e-commerce technology, Shopify 32.5% and Shopware 0.2%; the EHI study of the 1,000 largest German shops put Shopware first. Both are true; they count different populations. One official statistic reframes the landscape: Eurostat's enterprise figures for 2024 (published February 2026) put web sales at 8.39% of EU enterprise turnover and sales by **EDI**-type messages (the business documents of Documents Between Companies: EDI, Punchout and E-Invoicing) at 11.07%. By value, more European enterprise e-commerce runs over EDI than over webshops, which is why Part V is not a side topic.

The Trade-offs Between Categories

The categories differ on five criteria a newcomer can check. The table is my judgment, from the evidence above and from patterns common among growing merchants.

Criterion	Hosted SaaS	Enterprise suite	Open source	Composable	ERP-centered	Specialist group
Time to first sale	Days	Months	Weeks to months	Months; a storefront must be built	Weeks	Weeks
Who upgrades	The vendor, everyone at once	The vendor or partner, per release	You or your agency	The vendor for services; you for what you built	The vendor	Each label's vendor
Integration burden	On apps and connectors	Low inside; high at the edge	On your agency	On you, by design	Low until the ERP's cadence is too slow	Low among labels; on you at the edge
Cost shape	Subscription plus a share of sales and payments	Negotiated, often GMV-linked, plus partner fees	Agency and hosting; GMV-priced editions above a threshold	Subscription with GMV caps plus the build	Per user plus ERP license	Per product
Exit cost	Moderate: export, rebuild theme and apps	High: one contract, one data model	Moderate to high: custom code tied to the core	High: every join is yours to rebuild	High: the ERP is the hub	High across several labels at once

The rule I use: choose the category by the team you have, not the team you want. A composable assembly needs people who build and run storefronts, integrations and services. A hosted platform with specialist layers needs people who own connectors and contracts. A suite needs a budget and a vendor manager. The category you cannot staff is the wrong one, whatever its feature list says.

Across Scale Stages: Which Categories Fit Which Stage

The four stages are the book's (Starting, Growing, Omnichannel, Group); Who Sells What to Whom: Business Models and the Economics That Shape Architecture defines them and What Changes at Each Scale: From First Store to Group works them.

Stage	Typical category	What becomes necessary	Warning signs of outgrowing it
1 — Starting (under about €2M)	Hosted SaaS or a site builder; an ERP-centered package for stock-heavy sellers; a 3PL (third-party logistics provider: a warehouse that ships for you)	The platform's own apps and payment service	The first store, wholesale account or marketplace; the first spreadsheet holding product data the platform cannot
2 — Growing (about €2M to €20M)	Hosted SaaS with the first specialist layers (accounting or ERP, a channel connector, a returns tool); open source with an agency	A named owner per integration; the platform's GMV tier and transaction fee reviewed against a PSP of your own	Pricing thresholds crossed in one year; the agency's customizations block an upgrade; stock in three systems

Stage	Typical category	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel (about €20M to €200M)	Hosted SaaS or open source surrounded by OMS, PIM, WMS and channel layers; a specialist group; a suite for the partner-led	Systems of record per fact; an OMS and a PIM; integration standards; composable layers per capability around a kept core	The joins fail at peak; a replatforming pitch after every incident; nobody knows how many apps write stock
4 — Group (over about €200M, or several brands)	A suite with a platform team, or a composable core with product teams	Tenancy and isolation; shared services against brand autonomy; versioned contracts for in-house APIs	One stack forced on every brand before the benefit is measured (One Size Fits None); a platform team that cannot keep the assembled services upgraded

The judgment: **at the Starting and Growing stages, "composable" is a vendor category, not an instruction**; at the Omnichannel stage it is something you do per capability, around a core you keep; only at the Group stage, with a platform team, is an API-only core realistic. The layers the 2026 vendors now sell beside existing stacks are the shape an Omnichannel merchant can absorb.

Kestrel & Pine's Place on the Map

Noor's answer to Mara's first question fits in one figure. The platform is a hosted SaaS product (€2,300 a month plus fees); the ERP (€3,900 a month) is an ERP-centered product used as the back office, not as the hub; the WMS (€2,100), the marketplace connector (€900), the returns portal (€650 plus €0.95 per return), the iPaaS (€1,200), the POS, the PSP, the EDI provider and the shipping service are specialist layers; and the two things Kestrel & Pine lacks, an OMS and a PIM, are specialist layers too.

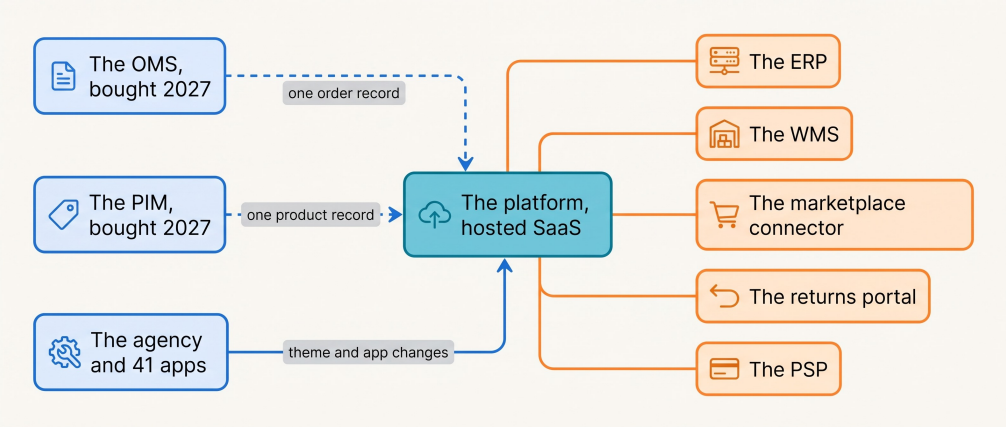


Figure 4: *Kestrel & Pine is a hosted platform surrounded by specialist layers; Waypoint adds the two layers it lacks beside the platform rather than replacing the center.*

Noor tested both proposals against that failure. The composable option asked a 14-person technology team, with three people on storefronts and apps, to operate several new systems and their integrations. Only its availability service addressed the oversell. The same capability could be bought with an order management system (OMS) and connected to the existing platform: Waypoint's stock-ownership decision, W1.

The proposal also assumed one cutover date. Together, the operating burden and migration risk illustrated **The Giant's Boots** and **The Burned Bridge**: adopting another organization's architecture without its capacity, then moving without a route back.

The headless proposal left the stock chain untouched. Its performance case weakened when Noor traced slow product pages to scripts from eleven apps. A quarter of agency time spent on performance fixed that problem (Delivering the Storefront: Themes, Headless, Rendering and Performance). She declined both proposals because neither justified its cost against the problems measured.

W4 says stay on the hosted platform, add the OMS, the PIM and the availability service beside it, and review in 2029. The honest cost is the one the trade-off table names: the fee grows with sales, the PSP choice is constrained by the platform contract, and every added layer is a join that W5 must govern. Noor's note to Mara ended: *we are not choosing a platform; we are choosing which work we are willing to own.*

When It Goes Wrong

A platform category fails slowly, in four ways the public record illustrates.

The vendor stops funding your region or product. Lightspeed's February 2025 strategic review named two growth engines, neither of them European online retail. Detection: read your vendors' results, strategic reviews and ownership changes once

a year, and watch the revenue mix; a vendor whose transaction revenue outgrows its subscriptions will steer product toward payments. Recovery: an exit plan per platform, written before it is needed (Build, Buy or Assemble: Choosing and Changing Platforms).

The core is forked and cannot upgrade. Platform-as-a-service deployments and every open-source platform let the merchant or its agency change the core; the **Forked Platform** is the result when each release becomes a project and each security patch arrives months late. Detection: ask how many vendor releases behind the platform is. Recovery: move extensions to sandboxed or API-based mechanisms (Apps, Plugins and Functions: Extensibility and Ecosystem Governance).

The price crosses three thresholds in one year. A merchant moving from Growing to Omnichannel typically crosses the platform's GMV tier, the PSP's volume pricing and its WMS and channel tools' per-order quotas at about the same time, and discovers that "€2,300 a month" was never the price. Detection: a yearly total-cost review listing every GMV-, order- and request-priced line. Recovery: negotiate on actual GMV, with the B2B share separated.

An acquisition changes the product. PayPal's stake in Shopware became Shopware Payments within eight months; Katapull added two labels in 2026. Neither is a failure, but each changes what the platform will push. Detection: ownership in the yearly vendor review. Recovery: contracts that name the capability you depend on and its support period, so **The Lynchpin** has a date attached.

What to Take Away

Established.

- The market sells commerce in seven shapes, and the shape decides who runs the servers, who applies upgrades and where the integration work lands (Adobe's three deployment options, October 2026; Shopware's editions, November 2026).
- Platforms are priced mainly on sales and payments, not on software: Shopify's FY2025 filing (76% of revenue correlated with GMV), Lightspeed's FY2026 revenue mix, and the GMV-based plans of BigCommerce, Saleor and Shopware (November 2026).
- Between 2023 and 2026 the leading vendors moved to selling layers beside existing stacks (Adobe Commerce Optimizer, June 2025; commercetools AgenticLift, January 2026, and standalone modules, July 2026; Salesforce agents, July 2026), while Commerce.com reported replatforming demand "softer than historically" (August 2026).

Depends on context.

- A hosted platform with specialist layers fits from Starting through Omnichannel **if** the merchant can staff integration ownership; it limits when the GMV fee and the PSP constraint outgrow the margin, a Group-stage question.
- ERP-centered commerce fits Starting and Growing merchants with heavy stock and few channels; it limits at the Omnichannel stage, when stores, marketplaces and a warehouse need stock faster than the ERP's batch cadence.
- A composable core fits a Group-stage merchant with product and platform teams; at any smaller stage, adopt composable per capability (search, OMS, PIM) around a kept core.
- A specialist group reduces joins and concentrates vendor risk in one owner; it fits where one exit plan covering several layers is acceptable.

The author's synthesis.

- Choose the category by the work you can own, because every category moves the integration burden rather than removing it, and the burden you cannot staff fails on your busiest day.
- Treat every 2026 pitch, "composable" and "agentic" included, as a layer to add beside the stack you have, because that is what the vendors themselves now sell and what a merchant below the Group stage can absorb.

Questions to Ask

1. Which of the seven categories is our platform, and on which deployment option: who runs the servers, and who applies upgrades?
2. What did we pay the platform last year in total, including transaction and payment fees, and what share of that is a percentage of sales?
3. When did we last read our platform vendor's annual results, ownership changes and strategic announcements, and what did its revenue mix tell us?
4. How many vendor releases behind is our platform, and what blocks the upgrade?
5. For every "500+ customers" or "1,300 marketplaces" in a vendor deck, what is the unit, the date and the definition?
6. Which capabilities do we lack as layers (OMS, PIM, availability), can each be added beside the platform, and who of the people we have would own the joins?

Part I ends here, with the business, the capability map, the ownership of facts and the market in view. Part II — CATALOG: Products, Prices and Promises starts the catalog: the product record, the price, the promotion, the search result and the stock figure that every channel shows, beginning with What Are We Selling? Products, Variants and the Catalog and the PIM Kestrel & Pine buys in Q3 2027.

Sources and Evidence

Company, product and investor sources

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part II — CATALOG: Products, Prices and Promises

KEY POINTS:

- **Catalog facts have different owners and different clocks.** Content tolerates a day's lag, prices carry legal weight, discounts are decided in the cart and stock changes by the second.
- **Every channel shows a copy, and a copy needs a source and a reconciliation.** The €179 German price and the marketplace selling an empty shelf were copies nobody compared with their source.
- **A promise is computed once, in one place, and then frozen.** The price is copied to the invoice, the discount allocated to every line, and the last unit committed to exactly one order.

A shopper reads a jacket's description, price and stock badge on the same page. The description may stay unchanged for a season; the last sellable unit can disappear while the page is open. Updating both on the same schedule can leave the page accurate about the product but wrong about whether it can be bought. This part explains how product facts, prices, discounts, availability and search reach the shopper with the timing each needs.

This part explains those responsibilities through **Kestrel & Pine, a fictional outdoor brand** whose product and pricing copies diverged across systems. Each chapter starts with the business cost of one such disagreement.

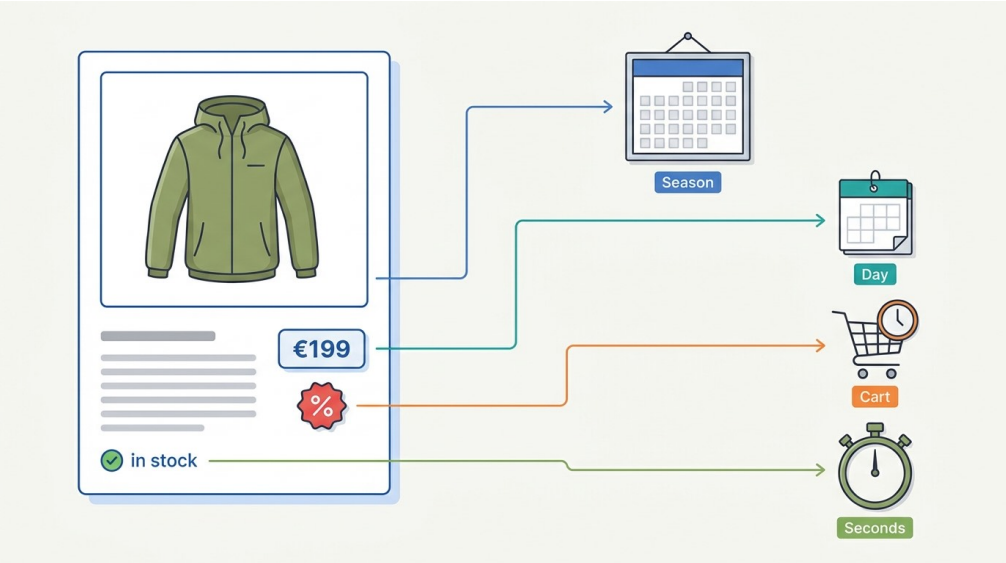


Figure 1: *Four facts that sit side by side on one product page run on four different clocks, which is why each needs its own owner and its own path to every channel.*

Where we are in the order's journey. Before Eva orders on 13 April 2027, the Ridgewalker women's M in moss needs a description, image and identifiers. Search must find it, pricing must select €199, and availability must confirm a unit can be sold. Her €235 basket also qualifies for free shipping. Follow the Order: What Happens Between "Buy" and "Delivered" provides the wider route.

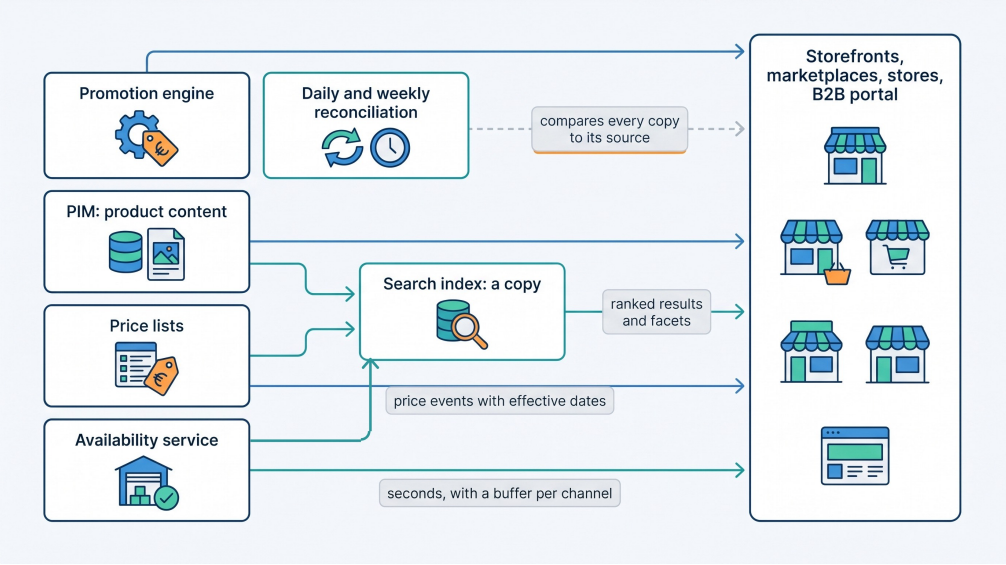


Figure 2: *The shape Part II recommends: four kinds of catalog fact with four owners and four clocks, a search index that is openly a copy of all of them, every channel a consumer, and a reconciliation that compares each copy to its source on a schedule.*

Chapters in this part

What Are We Selling? Products, Variants and the Catalog explains **which system owns product content and identifiers**. A season entered four times motivates a **product information management system (PIM)**. The chapter distinguishes the product, its sellable variants, internal stock codes and global barcode identifiers.

What Does It Cost? Price Lists, Contract Prices, Currencies and Tax establishes **who selects a price and how channels receive it**. A failed update left the German storefront at €179 for nine days instead of €199, losing about €8,200. Ownership, dated prices and comparisons address the failure.

Discounts, Coupons and Loyalty: Promotions Without Chaos explains **how offers combine and how discounts reach invoices and refunds**. A Black Friday setting let two discounts combine on 2,860 orders, costing about €52,000. The chapter separates promotional rules from gift-card and loyalty balances.

Finding the Product: Search, Navigation and Merchandising treats **the search index as a maintained copy** of product, price, stock and review data. Missing attributes and synonyms make a waterproof jacket hard to find; correcting them improves the fictional search results without replacing the engine.

Is It in Stock? Inventory, Availability and the Promise You Can Keep distinguishes **physical stock from what a channel may still sell**. The Black Friday oversell of

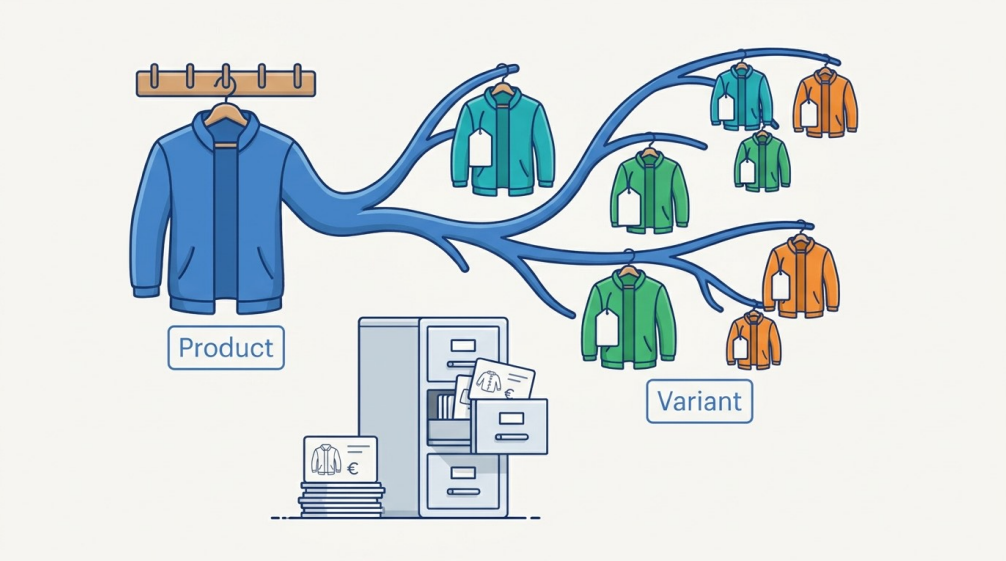
1,340 jackets shows the cost of delayed copies. The remedy assigns owners to stock and commitments, then calculates availability from both.

Begin with What Are We Selling? Products, Variants and the Catalog.

Sources and Evidence

Sources are cited in each chapter; this introduction makes no factual claim of its own. Kestrel & Pine is fictional, and its figures are the same in every chapter.

6. What Are We Selling? Products, Variants and the Catalog



IN THIS CHAPTER, YOU WILL: Learn how products, variants, identifiers and attributes are modeled, who should own each piece of product data, and when a PIM (product information management system) earns its place.

KEY POINTS:

- **A product is four things, and only two of them may ever change.** The product is what the shopper browses, the variant is what they buy, the stock keeping unit (SKU) is the merchant's code for it and the Global Trade Item Number (GTIN) identifies it across organizations. Titles and copy change freely; SKUs and GTINs never do, because orders, warehouses and marketplaces are keyed on them.
- **Product data has many producers and many consumers, so it needs one owner per attribute.** Purchasing, marketing, translators and suppliers write it; storefronts, marketplaces, search, the warehouse and wholesale read it. The design question is not which tool, but which system may change which field and how the others learn of it.

- **Every channel validates product data against its own rules and rejects quietly**. Google wants a stable id and a correct GTIN, bol wants a barcode identifier and a classification "chunk", a storefront wants a translation. Syndication needs a per-channel outbox, a rejection report and a loop back to the owner.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Spaghetti Junction** — each channel gets product data through its own hand-built import, feed or spreadsheet, each with its own owner or none.
- **The Pulled Rug** — a marketplace changes its content rules or retires an API version and listings go offline before anyone is told.
- **The Giant's Boots** — a product information system is bought because larger brands have one, before the merchant has more than one producer and one consumer of product data.

The Ridgewalker women's jacket comes in six sizes and four colors: twenty-four variants, each needing an internal code and barcode. Its descriptions, images and attributes must also reach several channels in three languages. If a corrected waterproof rating reaches only the web shop, marketplace buyers can still receive the wrong information about the same jacket.

Product data often begins wherever it is first needed: purchasing data in accounting, descriptions in the storefront, seasonal plans in spreadsheets and listings in a marketplace tool. The copies agree until someone renames a color, fixes a translation or adds a required field. **Give each field an owner and let the channels consume its value**. Otherwise every correction becomes another round of manual entry.

Where we are in the order's journey. Before Eva can find a jacket at Kestrel & Pine, the fictional outdoor brand, the Ridgewalker women's M in moss needs a title, image and Dutch description. It also needs a **SKU** (stock keeping unit, its internal variant code) and **GTIN** (Global Trade Item Number, the identifier encoded in its barcode). That record supports every later step in Follow the Order: What Happens Between "Buy" and "Delivered".

Setting the Scene at Kestrel & Pine: The Season That Was Typed Four Times

Kestrel & Pine sells **3,400 products** in **26,000 variants**, with about **17,000 active** and **1,200 new variants** each season. In January 2027, descriptions live in the platform, costs and purchasing data in the enterprise resource planning system (ERP), and translations and seasonal details in spreadsheets. The marketplace connector holds another copy.

There is no **product information management system (PIM)**: a shared place to prepare product content for publication to all channels.

In February 2027, Noor Visser, the new Head of Engineering, sat in on the Spring/Summer 2027 launch and wrote down what happened to one new variant of the Ridgewalker shell jacket on its way to being sold. The details are illustrative.

1. **Purchasing** created the item in the ERP: SKU, GTIN, supplier, landed cost (€58, what the item costs once delivered to the warehouse), purchasing unit. The color field said `MOSS GRN`.
2. **Teun Baas's product team** filled the "season sheet", a spreadsheet with the marketing name, the Dutch description, the fabric, the waterproof rating and the image file names. Color: `Moss`.
3. **The agency** imported the season sheet into the platform, a two-day job each season. The 20,000 mm waterproof rating stayed behind in the sheet; it never reached the platform as a field, so the storefront could not filter on it and search could not find "waterproof jacket" (Finding the Product: Search, Navigation and Merchandising).
4. **A translator** worked from a second spreadsheet. The German storefront launched with about **310 variants** still showing Dutch copy, for three weeks.
5. **Milan Petrov's marketplace team** re-mapped the new variants to bol's attribute model by hand in the connector. bol declined about **140 of the new variants** for missing mandatory attributes; the first anyone heard of it was the upload report, a day later. On the listings that went live, the color read `olive`, chosen from bol's list in 2024 and never revisited.

Four teams typed one color four times, in three forms. A marketplace rejected a tenth of the season. **Without an owner for product content, every channel created its own version.** Waypoint, Kestrel & Pine's improvement program, addresses this in decision W3: introduce a PIM in Q3 2027 for content and attributes. The ERP keeps costs and purchasing; the platform and marketplace connector consume published content.

Figure 1: *One color typed four times, three ways, by four teams: with no owner for product content, every consumer of it became a producer.*

What a Product Is: Four Things That Share a Name

The first thing a newcomer needs is that a "product" is a stack of four concepts, each identified differently and each stable to a different degree.

Term	What it is	Who assigns it	May it change?
Product	The sellable concept a shopper browses: "Ridgewalker shell jacket, women's"	The merchant	Title, copy and images change freely
Variant	One buyable combination of option values: size M, color moss	The merchant, from the options	Must stay stable once it has been sold
SKU	The merchant's own stock-keeping code for that variant, the thing you count on a shelf and put in a parcel	The merchant (usually the ERP)	Never

Term	What it is	Who assigns it	May it change?
GTIN	The Global Trade Item Number under the barcode, licensed from GS1, the barcode standards body, so that every company's systems mean the same item by it	A GS1 member organization	Never, and never reused for a different item

An **option** is a dimension of choice (size); an **option value** is one choice (M). A variant is one point in the grid the options define. The platforms and the standards all make this split. Shopify's developer documentation (n.d., accessed November 2026) describes the product as "the container for all product information (title, description, vendor)" and says "each variant represents a specific combination of option values, like 'Red / Small' ". The schema.org vocabulary, which search engines read from web pages, models the same thing in version 30.1 (September 2026): a `ProductGroup` is a family that differs "only along defined dimensions, such as size or color", each variant linked by `isVariantOf`.

The identifiers are where newcomers get hurt. A **SKU** is the merchant's code; nobody outside the company understands it, which is its strength. A **GTIN** is the global one. GS1, the standards body that runs the barcode system, says a GTIN is "unique, global and verifiable within the GS1 system" and that "only a GS1 Member Organisation can provide you with a GS1 GTIN" (GS1 support article, updated August 2024). The format on European retail barcodes is the 13-digit **EAN-13** (GTIN-13); a 14-digit GTIN-14 identifies a trade unit such as a carton of twelve, which is why wholesale and EDI need it and a web shop does not. Every distinct item and every packaging configuration gets its own GTIN. Kestrel & Pine has carried a GTIN on every own-brand SKU since 2021.

commercetools' product documentation (n.d., accessed November 2026) gives integrators the rule in one line: "treat the sku values as immutable identifiers". **The rule I use**: a SKU and a GTIN are promises to every other system, so they are assigned once, before the first sale, and a mistake is retired, never corrected in place. A reused SKU is a stock count that belongs to two items; a reused GTIN is a marketplace listing that shows the wrong product.

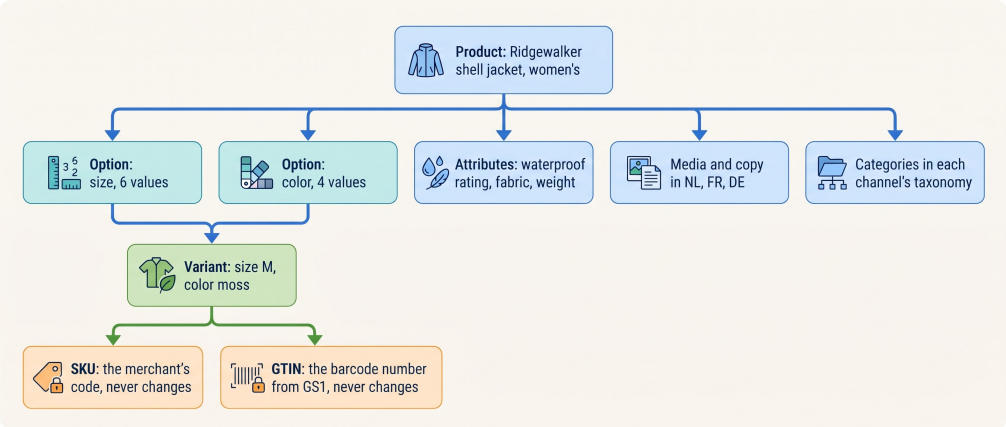


Figure 2: *One product is a grid of variants, each with two permanent identifiers, plus attributes, media and categories that change with the season; the model separates what may change from what may not.*

Attributes, attribute sets and taxonomies

An **attribute** is a typed field on a product or a variant: waterproof rating in millimeters, fabric, weight, care instructions, CE marking (the label that says a product meets EU safety rules). An **attribute set** is the list of fields a product of a given kind must carry: a jacket has a waterproof rating, a boot has a sole material, a stove has a fuel type. Attribute sets are the contract between the catalog and everything that reads it. A storefront filter, a search facet (Finding the Product: Search, Navigation and Merchandising) or a marketplace listing can only use an attribute that is filled consistently for every product of that kind. Kestrel & Pine's waterproof rating, living only in a spreadsheet, is invisible to all three.

A **taxonomy** is a fixed tree of categories a channel understands. Every channel has its own. Google's product taxonomy, the one its Shopping feeds use, is a file headed with version 2021-09-21 listing 5,595 categories; a feed may submit the numeric id or the full path such as "Apparel & Accessories > Clothing > Dresses" (Google Merchant Center help, n.d., accessed November 2026). bol, the largest Dutch and Belgian marketplace, describes its content model as a JSON file of more than 4,000 product classifications it calls "chunks", each defining the attributes a product must carry and which are mandatory (bol Retailer API documentation, n.d., accessed November 2026). The merchant therefore needs an internal taxonomy, the one it merchandises by, and a **mapping** per channel: a table that translates its categories and attribute values into each channel's. The mapping is data, not code, and it has an owner.

Bundles, digital goods, translations and lifecycle

Four product shapes need more than a size-and-color grid:

- A **bundle or kit** sells components together, such as a tent and footprint. Each component keeps its identity and stock. The sale must allocate the bundle price back to those components (Is It in Stock? Inventory, Availability and the Promise You Can Keep, Discounts, Coupons and Loyalty: Promotions Without Chaos).
- A **configurable product** groups purchasable variants under a parent. Adobe Commerce's October 2026 documentation distinguishes that parent from the simple products, each with its own SKU.
- A **digital good**, such as a download or license, skips warehouse and carrier processing and needs its own fulfillment and tax treatment.
- A **gift card product** creates a prepaid balance when sold. Finance records the obligation to honor that balance (Tax, Invoices and the Books: From Order to Ledger).

Translation is per product, per language, and often per channel. Shopify's documentation puts it from the merchant's side: "each published language gets its own URLs, and your store displays the translated version when translations exist" (Shopify Help Center, n.d., accessed November 2026). Kestrel & Pine sells in three countries and two languages in Belgium alone, so every description exists at least three times by design, and each translation has its own completeness status.

A product also has a **lifecycle**: draft, active, end of season, discontinued. A product is never deleted once it has been sold: orders, returns, invoices and analytics reference it for years, so "discontinued" is a state, not a deletion. commercetools enforces this mechanically: it separates staged from published data and "you can only delete unpublished products". And lifecycle is per channel: Google Merchant Center's `availability` takes `in_stock`, `out_of_stock`, `preorder` or `backorder`, and an item that is out of stock must still show its price on the landing page (product data specification, n.d., accessed November 2026). A variant can be end-of-season on the storefront and still listed on a marketplace clearing the last units.

Who Owns Which Field: Producers, Consumers and the System of Record

Now the architectural problem. Product data has **many producers**: purchasing creates the item; the product team describes it; translators localize it; suppliers provide specifications for third-party brands; the marketplace team adapts it to each channel. And it has **many consumers**: the storefronts, the search index, the marketplaces and feeds, the warehouse (which needs weight and dimensions to pick a box), the wholesale portal, the data warehouse, and, from 2027, AI agents that need accurate product data before they can shop (Assistants and Agents: AI in Commerce and What Is Real in 2026).

Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record introduces the **system of record**: for each fact, the one system whose value is the truth when copies

disagree. For product data, that decision must be made **per attribute**, not per product, because the attributes have different natural owners. The table below is the one the book uses; it is an industry-typical arrangement, not a law.

Data	System of record	Who produces it	Who reads it	Why here
Item identity: SKU, GTIN, supplier, purchasing unit, landed cost	**The ERP**	Purchasing	PIM, platform, WMS (warehouse management system), EDI (electronic data interchange with wholesale customers), finance	The item must exist and be valued before it can be bought, counted or sold
Logistic attributes: weight, dimensions, customs code	The ERP (or the PIM, if the ERP cannot hold them)	Purchasing, the warehouse at first receipt	WMS, carriers, customs	Measured once, needed for every parcel
Product content: names, descriptions, attributes, media, translations, internal categories	**The PIM** (the platform at the Starting and Growing stages)	Product team, translators, suppliers	Platform, marketplaces, feeds, search, wholesale, print	Many producers, many consumers; needs a workflow
Channel mappings: category and attribute translations per channel	The PIM (or the feed tool)	The marketplace team	Connector, feed tool	Changes when a channel changes its rules

Data	System of record	Who produces it	Who reads it	Why here
Storefront presentation: collections, theme blocks, landing pages	The platform	Ecommerce team	Storefront only	Belongs to one channel
Listings: the channel-specific product page	The channel, via the connector	Derived from the PIM and the mappings	Shoppers on that channel	A copy, by design
Price	See What Does It Cost? Price Lists, Contract Prices, Currencies and Tax			Changes at a different cadence
Stock	The WMS and the availability service, see Is It in Stock? Inventory, Availability and the Promise You Can Keep			Changes every minute

The last two rows explain why price and stock have their own chapters: they are product-related data with different freshness needs. A description can be a day old. A price that is a day old is a broken promise with legal weight. A stock figure that is an hour old is the Black Friday Oversell. **Product content, prices and stock travel on separate paths** even when they end up in the same marketplace feed.

Read against this table, Kestrel & Pine in January 2027 has no owner for the "product content" row. The platform holds content because the storefront needs it; the spreadsheets because the product team works there; the ERP a name because an item must

have one; the connector a re-mapped copy because bol needs its chunk. Four producers, no system of record. This is the **Spaghetti Junction**: every pair of systems has its own path, each with its own owner or none.

Requirements per flow

When product data moves between systems, the flows have different requirements (APIs, Events, Files and Middleware: How Commerce Systems Talk asks for this table whenever flows are discussed).

Flow	How fresh	How consistent	How fast	Availability need
New season into the ERP (identity, cost)	Weeks before launch	Exact: one SKU, one GTIN per variant	Batch is fine	Office hours
Enrichment into the PIM (content, media, translations)	Continuous during the season build	Workflow states: draft, enriched, translated, approved	Human speed	Office hours
Publication to the storefront and search	Minutes after approval, or on a scheduled date	Whole product at once: never a title without its variants	Minutes	Launch day matters
Syndication to marketplaces and feeds	Hours is usually acceptable for content	Per channel's rules; rejections reported back	Asynchronous, with a report	Launch and peak weeks

Two things stand out. Content tolerates lag but demands completeness: a product published with a missing size or an untranslated description is worse than one published an hour later. And syndication is **asynchronous by nature**: the channel decides later whether it accepts the product. bol's documentation says an upload report may take an hour; Google disapproves a product with an incorrect GTIN. The architecture must expect a "no" to arrive after the "sent".

How the Pieces Fit Together With a PIM

A **PIM** (product information management system) is the editorial system where product data is created, enriched, translated and approved before it is published to channels. **Enrichment** is adding the marketing, technical and media content a channel needs beyond what the ERP holds. **Syndication** is publishing product content to each channel in that channel's format and under its rules. A **feed** is a file or API stream of product records in a channel's format; **structured data** is product information embedded in the web page itself (schema.org) so that search engines can read it.

The vendor descriptions converge. Akeneo calls a PIM "a system or solution that helps businesses manage, centralize, and distribute product information", a "central source of truth" that collects "data from your ERP, MDM, PLM, and other systems" (master data management and product lifecycle management, the systems where products are designed), enriches it and manages "catalogs for every sales channel in one system of record" (Akeneo, n.d., accessed November 2026). Pimcore describes an "open core platform for Product Experience Management" (GitHub, version 2026.1, accessed November 2026). These are vendor pages; none gives a testable threshold for when a PIM pays off. The threshold comes later in this chapter.

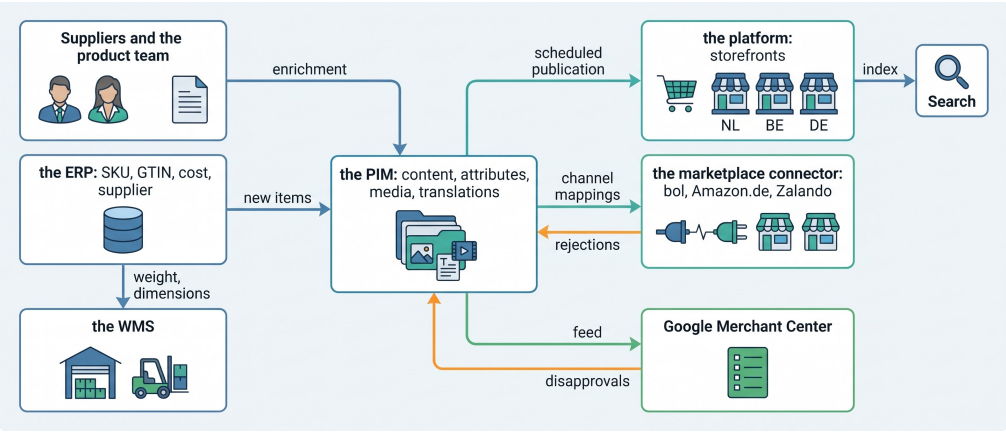


Figure 3: *With a PIM, the ERP still creates the item and keeps its cost, the PIM owns everything a shopper reads, every channel is a consumer, and each channel's rejections flow back to the owner instead of dying in an upload report.*

Three design rules make this picture work, and they are rules about ownership, not about the tool.

Rule 1: each attribute flows in one direction. The ERP creates the SKU; the PIM may not. The PIM owns the description; the platform may not edit it, only display it. Microsoft's documentation for the Business Central Shopify connector states this bluntly for a two-system case: "decide which system is the system of record for product data"

before configuring the sync, choosing the direction per field, with the Shopify SKU or barcode as the mapping key (Microsoft Learn, 16 October 2026). A field that can be edited in two places will be.

Rule 2: publish whole products on a schedule, not fields on change. Content changes constantly during a season build and nobody wants half-enriched products on the storefront. Waypoint's integration standard (W5) says products move "by scheduled publication" while stock, orders and prices move by **events** (a message sent the moment something changes). commercetools builds the same idea into its product model with staged and current data; Shopify's `productSet` mutation is designed for the case where "an external system is the source of truth" and replaces a product's options and variants in one operation (Shopify developer documentation, n.d., accessed November 2026). A PIM publication needs exactly that: a whole product, atomically.

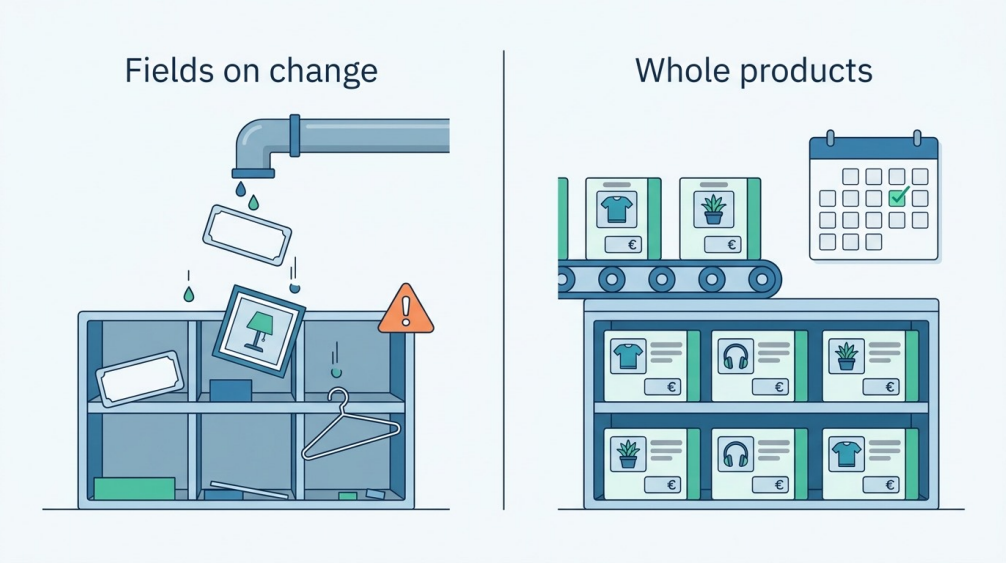


Figure 4: *Content changes all through a season build, so products are published whole on a schedule while stock, orders and prices move by events.*

Rule 3: every channel's answer comes back to the owner. A syndication path has an outbox per channel (what was sent, when, under which version of the channel's rules), a validation step that runs the channel's known rules before sending, and a rejection report that lands in the PIM as a task for a person (Selling on Someone Else's Platform: Marketplaces and Channel Integration owns the listing operations behind it). Without the loop, 140 declined variants are a line in a report nobody reads.

The Choices

There are four realistic ways to arrange product data.

1. **The platform is the catalog.** Product data is created and edited in the commerce platform; the ERP gets a copy or just a SKU and a price; marketplaces are fed from the platform through a connector. This is every merchant's starting point, and for one channel and one language it is right.

2. **The ERP is the product master.** The ERP creates the item and holds its name, attributes and sometimes its description; the platform receives items by connector (The Back Office: ERP Integration and Who Is the Hub). Common where the ERP came first (wholesalers, manufacturers). It works for identity and cost. It fails for content, because ERP item masters were designed for purchasing, not for marketing copy in three languages with images and a translation workflow; the content ends up in spreadsheets beside the ERP, which is Kestrel & Pine's situation.

3. **Spreadsheets with discipline.** The honest name for what most Growing-stage merchants do: a season sheet, an import, a translation sheet. Cheap and flexible. Its limit is not size but the number of hands: once more than two teams write to the sheet and more than two systems read from it, there is no record of who changed what, no workflow state and no validation.

4. **A PIM as the system of record for content.** The ERP keeps identity and cost; the PIM owns content, attributes, media, translations and channel mappings, with a workflow (enrichment status, approvals, completeness per channel) and syndication; the platform and the connector become consumers. This is Waypoint's W3. It is right under the conditions the next table makes explicit, and the Giant's Boots when they do not hold.

A fifth option, a full master data management platform spanning products, customers, suppliers and locations, belongs to the Group stage and to One Customer, One Product: Identity and Master Data Across Systems.

Trade-offs

	Platform as catalog	ERP as product master	Spreadsheets with discipline	PIM as system of record
Right when	One or two channels, one language, one team editing	ERP-first business, few channels, thin content (industrial parts, wholesale-only)	Two or three channels, a small product team, content richer than the platform holds	Several channels and languages, more than one producing team, channel rules to track
Cost	Included in the platform	Included in the ERP; connector by the ERP partner	Nearly free in licenses; expensive in people at every launch	A subscription *(assumption: a few hundred to a few thousand euros a month at mid-market)* plus implementation and migration
Workflow and approvals	Minimal (draft and active)	None for content	None; versions by file name	Built in: states, completeness, per-channel readiness
Translations	Per storefront, via apps or CSV	Rarely supported well	A second sheet	Native, per locale and channel
Channel validation	The connector's, after sending	None	None	Rules per channel before sending; rejection loop

	Platform as catalog	ERP as product master	Spreadsheets with discipline	PIM as system of record
Warning sign of outgrowing it	The first marketplace rejects products in bulk; the first second language	Content lives in spreadsheets beside the ERP	A launch takes longer than two weeks; nobody knows which sheet is current	Rare; the risk is buying it too early and running it as a spreadsheet with a login

My recommendation, with its conditions. A PIM becomes necessary when product data has **more than one producer** (buying, marketing, suppliers, translators) **and more than one consumer** (web, marketplaces, print, wholesale, stores), and the platform's product record can no longer hold the workflow between them. Below that line, the platform or a disciplined spreadsheet is cheaper and the PIM is the Giant's Boots: a workflow tool for a workflow that does not exist yet. Above it, the absence of a PIM shows up as channel-specific copies that drift, and the cost of the drift (declined listings, untranslated launches, a two-week season import) usually exceeds the subscription within a season. Kestrel & Pine crossed the line around 2021–2022, when it added marketplaces (2021) and a third country (2022) to a Belgian storefront that already ran in two languages.

Two assumptions sit under that recommendation. The ERP keeps item identity and cost, so the PIM is bought for content, not as a second item master. And the merchant can name one owner for product content with authority over the product team, the translators and the marketplace team at once; at Kestrel & Pine that is Teun Baas, who owns the catalog and the seasons. Without such an owner the PIM becomes a fifth copy.

Across Scale Stages: When It Fits and When It Limits

The four stages are the ones this book uses throughout (see Meet Kestrel & Pine and, in depth, What Changes at Each Scale: From First Store to Group).

Stage	What is typical	What becomes necessary	Warning signs you have outgrown it
1 — Starting (under about €2M; Kestrel & Pine 2016–2018)	The platform is the catalog; SKUs typed by hand; GTINs bought in a block; the platform's built-in Google feed; one language	SKU and GTIN discipline from day one: one SKU per variant, never reused, never corrected in place	Variants split into separate products to get around a limit; SKUs that encode the price or the season
2 — Growing (about €2M–€20M; 2019–2022)	An ERP with its own item master; the first marketplace through a connector; a season spreadsheet; the first second language	A written ownership rule per field between platform and ERP; a mapping table per channel; the item created in the ERP before it is sold anywhere	The first marketplace declines products in bulk; the same color has three spellings; the season import takes days; translations lag launches
3 — Omnichannel (about €20M–€200M; 2023 onward)	Stores, wholesale, three marketplaces, several countries; a dozen systems reading product data; four copies	A PIM as system of record for content, the platform demoted to a channel; attribute sets as a contract with search and filters; syndication with a rejection loop; product data as a prerequisite for EDI, marketplace rules and agents	A season launch is the slowest project of the year; a marketplace rule change takes listings offline before anyone knows; the data warehouse counts products three ways

Stage	What is typical	What becomes necessary	Warning signs you have outgrown it
4 — Group (over about €200M or several brands)	Several brands, each with a catalog; shared marketplaces and retailers; print and B2B (business-to-business) catalogs	A group taxonomy with brand extensions; shared syndication with brand-level channel accounts; a master data function that allocates identifiers across brands	Two brands list the same third-party product with different GTINs; one brand's PIM becomes everyone's bottleneck; a retailer receives two feeds from one group

The judgement: **identifier discipline does not change with scale; everything else does**. A SKU and a GTIN assigned correctly at the Starting stage save a migration at every later one. The ownership rule changes once, at the Growing-to-Omnichannel boundary, when content gets its own system; after that the question is how many brands share it.

In Practice

What the platforms and channels document

Variant limits shape catalogs. As of November 2026: Shopify's historic limit was 100 variants per product; new GraphQL product APIs supporting 2,048 shipped in API version 2024-04, and on 15 October 2025 the 2,048 limit opened to all merchants, with a warning that apps not on the supported APIs "may have a downgraded or broken experience" (Shopify changelog, 15 October 2025). commercetools' 2026 release notes describe two catalog models: the Classic model keeps variants embedded in the product with a 100-variant limit, while the Modular Catalog, generally available from 29 September 2026, makes variants first-class resources with up to 10,000 per product (commercetools release notes, 2026). Both are recent and will move. The lesson is older than either: an apparel catalog of sizes times colors routinely passes 100 combinations, and a merchant that split products to fit a limit poisoned its search, feeds and analytics for years after the limit was lifted.

Channels validate, and reject. Google Merchant Center's product data specification

requires `id` (at most 50 characters; "keep it stable"), `title`, `description`, `link`, `image_link`, `price` and `availability`; `gtin` is strongly recommended and "if you submit a product with an incorrect GTIN value, your product will be disapproved"; variants need `item_group_id`; and from 31 January 2027 a 500 × 500 pixel minimum image will be enforced (Google Merchant Center help, n.d., accessed November 2026). As of November 2026, bol requires an EAN for every product, declines content without one, makes its first two enrichment levels mandatory, takes products offline where a required CE marking is missing, and updates its data model "daily". Every one of these is a rule that changes on the channel's schedule, which is the Pulled Rug seen from the catalog side.

Ownership direction is a documented setting. The Business Central Shopify connector (Microsoft Learn, 16 October 2026) asks the merchant to choose a direction per product field ("If you manage item names, descriptions, and pricing in Business Central, choose the To Shopify direction. If your merchandising team maintains product content in Shopify … choose From Shopify") and maps by SKU or barcode. That page is this chapter's argument in two sentences.

Kestrel & Pine: what W3 means in practice

Waypoint W3 introduces a bought PIM in Q3 2027 as the system of record for product content and attributes. The sequence below is this chapter's reading of that decision, with illustrative dates consistent with the rest of the book.

Ownership first. In April 2027, building on the entity–owner table of March (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record), Noor, Teun and Milan wrote the attribute-level ownership table for the Ridgewalker as a one-page document before any vendor demo. The ERP keeps SKU, GTIN, supplier, landed cost (€58 for the Ridgewalker), purchasing unit, weight and dimensions. The PIM owns the marketing name, the descriptions in Dutch, French and German, the attribute set for jackets (waterproof rating, breathability, fabric, care, CE and other markings), media, the internal category and the mappings to Google's, bol's, Amazon.de's and Zalando's taxonomies. The platform owns collections and landing pages and nothing else about the product. The connector owns nothing; it derives listings. Color values are a controlled list in the PIM: `moss` has one spelling and a mapping to `olive` where a channel's list has no moss.

One direction per field. The ERP creates items and the PIM imports them nightly; the PIM cannot create a SKU. The PIM publishes whole products to the platform on a schedule, with a launch date per season, and the agency's season-sheet import is retired. The platform's product editor is read-only for owned fields.

Syndication with a loop. The PIM syndicates to the connector and to the Google feed; each channel's rejections return as tasks on the product, with the channel's rule quoted. A completeness score per channel (how many mandatory attributes are filled) is visible

before launch, so a repeat of bol's 140 declines becomes a list to clear before launch, not a surprise in the next day's upload report.

Migration consumer by consumer. The PIM is loaded in July 2027 from the four copies, and the load itself is a reconciliation: about **1,900 variants** with conflicting attributes are listed and resolved before anything is published. The DE storefront becomes the first consumer in August, the NL and BE storefronts and the connector follow in September, and the spreadsheets go read-only (Moving Without Burning the Bridge: Replatforming and Migration treats this as the small migration it is). The Spring/Summer 2028 season is the first built in the PIM from the start. The PIM is also the first prerequisite of **W8**, agent readiness: an AI shopping agent needs product data that is accurate, complete and the same in every channel, and the PIM is where that becomes checkable.

The numbers that matter are small. The season import drops from two agency days to a scheduled publication. The DE storefront launches with no untranslated variants because the completeness score blocks a product that is not translated. And one spelling of `moss` reaches four channels.

When It Goes Wrong

Product data fails quietly. The failures below are the ones to detect on purpose.

Silent rejection. A channel declines a product in an asynchronous report nobody reads, and the product is simply not for sale there. *Detection:* a listing status per variant per channel (sent, accepted, rejected with reason, live, offline) reconciled against what the channel reports; a daily count of variants that should be live but are not. *Recovery:* the rejection becomes a task on the owner's queue.

Identifier drift. A SKU reused for a new item, a GTIN with a wrong check digit, a variant deleted and recreated with a new id. The warehouse counts two items as one; a marketplace shows the wrong product; orders lose their link. *Detection:* a weekly cross-reference check (SKU to GTIN to platform id to connector id) across the ERP, the PIM, the platform and the connector. *Recovery:* retire the identifier, create a new one, map the old one for history; never edit in place.

The split product. Variants modeled as separate products to get around a limit, or because a color launched late. Search shows the jacket four times, `item_group_id` is wrong in feeds, analytics count four products. *Detection:* products that share a name and a brand but not a group id. *Recovery:* a merge that keeps the old URLs and ids mapped, which is a migration, not an edit.

Stale translation. A fixed Dutch description republished while the German one still carries the old claim. *Detection:* completeness per locale as a blocking state before publication. *Recovery:* publish the whole product again, in all locales.

The pulled rug. A marketplace changes its data model (bol's is updated daily), adds a mandatory attribute or retires an API version on a published date. Kestrel & Pine's

Deprecated Endpoint of February 2027 was stock, not content, but the mechanism is the same: the agency script that pushed stock to the connector called a retired API version, and bol's figures were stale for two days. *Detection:* the channel's changelog and deprecation notices read by a named owner, and a weekly listing status check that would show listings going offline. *Recovery:* the mapping is data, so a new mandatory field is a mapping change, not a code change; Promises Between Systems: API Contracts, Webhooks and Versioning owns the versioning side.

The common thread is the one Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation develops for every copy in the landscape: a copy is acceptable only with a reconciliation that compares it to its system of record on a schedule and lists the differences for a person.

What to Take Away

Established.

- A product is a product, its variants, their SKUs and their GTINs, and the identifiers are permanent. Shopify's product model, commercetools' "treat the sku values as immutable identifiers", schema.org's `ProductGroup` and GS1's GTIN rules all say so.
- Channels validate product data against their own rules and reject asynchronously: Google disapproves a wrong GTIN and requires a stable id, bol declines content without an EAN and takes products offline for missing markings. Syndication must expect and route the "no".
- Products with sales history are never deleted; "discontinued" is a lifecycle state.

Depends on context.

- The platform is the right catalog at the Starting stage and often at the Growing stage; it stops being right when there is more than one producing team and more than one consuming channel. That boundary, not revenue, is the test for a PIM.
- The ERP should own item identity and cost. Whether it also owns logistic attributes depends on whether it can hold them and whether the WMS already reads from it; at Kestrel & Pine it can and it does.
- Mappings to channel taxonomies live in the PIM when there is one and in the feed tool or connector when there is not; what must not change with the choice is that the mapping is data with an owner.
- Variant limits are a platform fact that changes (Shopify 2,048 since October 2025; commercetools 100 or 10,000 depending on model since September 2026); whether a limit matters depends on the catalog's shape, and apparel and footwear catalogs are the ones that hit it.

The author's synthesis.

- Decide the ownership table before choosing a PIM, because the PIM only works if each field has one owner and one direction; a PIM bought without the table becomes a fifth copy with a login.
- Treat syndication as an integration with an outbox, a validation step and a rejection loop, not as a feed; the "no" that arrives a day later is the message most worth routing.

Questions to Ask

1. For the product the shopper sees on your storefront, how many systems hold a copy of its description, and which one is right when they disagree?
2. Can anyone edit a product's name or description in two places? Which fields in the platform are read-only, and which should be?
3. When a SKU is typed wrong, is it corrected in place or retired and reissued? Has a SKU or a GTIN ever been reused?
4. Which attributes does a jacket (or your equivalent) carry as fields, and which live inside description text? Can the storefront filter on every attribute a shopper would filter on?
5. When a marketplace declines a product, who finds out, how long after, and where does the task land?
6. How long does a season launch take from the first item in the ERP to live listings on every channel, and which step is manual?
7. Which channel's rules changed in the last twelve months, and how did you learn of it?
8. If an AI shopping agent read your product data tomorrow, would it find the same jacket, in the same color, at the same completeness, in every channel?

With the product described once and owned per field, the next promise is the one the shopper reads under the title. A price is also product-related data, but it changes at a different cadence, it carries legal weight in Europe, and it is copied into more systems than any other fact. What Does It Cost? Price Lists, Contract Prices, Currencies and Tax takes it up with Kestrel & Pine's Price Drift of October 2026.

Sources and Evidence

Company, product and investor sources

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- Synchronize items and inventory, Microsoft Learn, 2026-10-16 — "decide which system is the system of record for product data"; mapping by SKU or barcode.
- Product Content API v10, bol, n.d., accessed 2026-11 — the EAN requirement, the 4,000+ "chunks", mandatory enrichment levels, asynchronous upload reports and the daily data-model update.

Standards, regulation and official documentation

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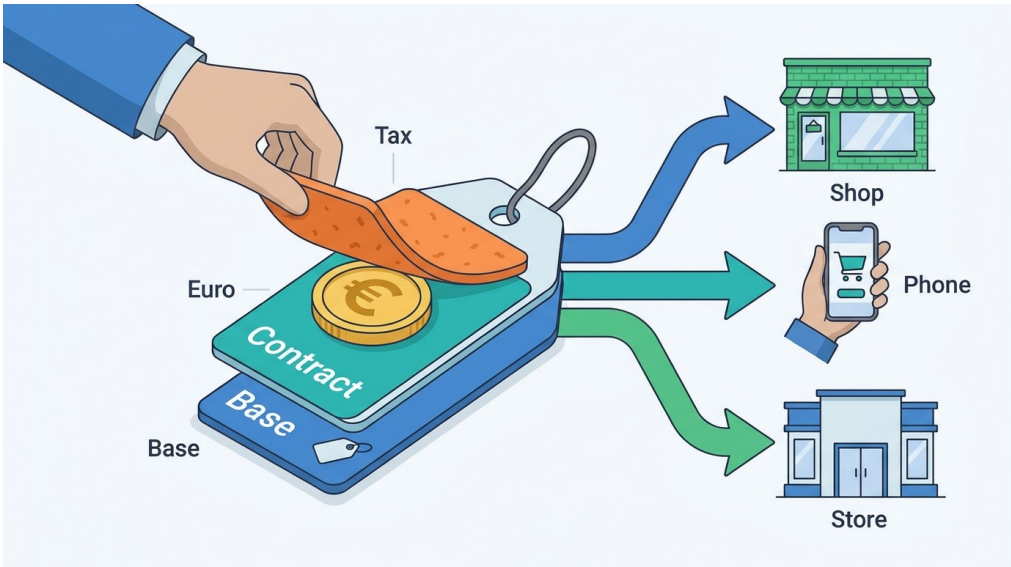
- Global Trade Item Number, Wikipedia, accessed 2026-11 — the GTIN-8, -12, -13 and -14 formats and the rule that each trade item and packaging configuration gets its own GTIN (secondary source; GS1's own site blocked fetching).
- Product data specification and google_product_category, Google Merchant Center help, n.d., accessed 2026-11 — required fields, the GTIN disapproval rule, `item_group_id`, availability values and the image minimum from 2027-01-31.
- Google product taxonomy, Google, version 2021-09-21 — the 5,595-category tree still in use.
- Merchant listing structured data, Google Search Central, n.d., accessed 2026-11 — variant URLs and `priceValidUntil`.

Research and practitioner sources

- None beyond the documentation above; no independent study of when a PIM pays off was found, and the threshold given in this chapter is the author's judgement.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

7. What Does It Cost? Price Lists, Contract Prices, Currencies and Tax



IN THIS CHAPTER, YOU WILL: Learn what a price is made of, where it should be selected, how it reaches every channel, and what European price rules require of its history.

KEY POINTS:

- **A price is a promise with a selection rule.** One stock keeping unit (SKU, a sellable variant) carries many prices (market, currency, customer group, channel, quantity, validity). What makes them one price is a deterministic, documented rule that picks one at a defined moment, and that moment must be the same for the storefront, the checkout and the invoice.
- **Separate the engine that selects a price from the copies that show it.** Channels that cannot call the engine (marketplaces, the point of sale, structured electronic orders from wholesale customers) receive copies, and every copy needs a source, an effective date and a reconciliation. Kestrel & Pine's Price Drift was a copy whose restore failed silently.

- **In the EU, price history is a legal record**. Consumer prices must be shown including tax, and any announced reduction must name the lowest price of the previous 30 days, so the pricing system must answer "what did we charge, where, when" for every SKU.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Price Nobody Can Explain** — the same product carries different prices in different channels and nobody can say which system set which price or why.
- **The Message in a Bottle** — a price change is sent to a channel once and trusted; the channel that missed it keeps selling at the old price.
- **The Secret Agent** — a repricing rule in a connector, an app or a spreadsheet import changes prices without a named owner, a floor or a person's approval.

At Kestrel & Pine, the fictional outdoor brand, the Ridgewalker jacket costs consumers **€199**, including local value-added tax (VAT), and wholesale buyers **€95**, excluding VAT, before negotiated terms. These are legitimate prices for different sales. The difficulty is ensuring that each channel selects the right one: a wrong price can either lose the sale or leave the merchant honoring an unintended discount.

The price rule therefore needs explicit inputs: buyer, country, channel, quantity and date. Marketplace commissions affect what the merchant keeps, while promotions affect what the buyer pays and which earlier price may be shown for comparison. Assign an owner to these rules so teams can explain a price and update it consistently. Typing separate prices into each system lets them drift as soon as one changes.

Where we are in the order's journey. Eva sees €199 on the product page and pays that jacket price at checkout on 13 April 2027. The platform records it on order #1072455. The invoice and any later refund must use that recorded price, even after the price list changes. What Are We Selling? Products, Variants and the Catalog describes the product, Discounts, Coupons and Loyalty: Promotions Without Chaos covers discounts, and Tax, Invoices and the Books: From Order to Ledger follows it into the books.

Setting the Scene at Kestrel & Pine: The Price Drift

In October 2026 the German storefront sold the Ridgewalker at **€179** for **nine days** while the marketplace listings and the stores charged **€199**. A price-list sync had failed silently. **410 orders** went out at the low price, about **€8,200** of margin was lost, and one marketplace sent a price-parity warning. Meet Kestrel & Pine records the incident; what follows is this chapter's reconstruction of the mechanism, with illustrative details.

The illustrative reconstruction starts with a German autumn campaign at €179, running from 25 September to 4 October. In 2026, the German consumer price list lived in the enterprise resource planning system (ERP), which also ran finance and purchasing. A connector copied it to the platform hourly.

Milan Petrov's storefront team entered the same campaign dates separately in the marketplace connector. The stores did not run the campaign; their point-of-sale (POS) tills received a weekly price file, and none of the stores was in Germany.

On 5 October, the update restoring €199 failed. The connector logged an error but reported success. Because it sent only changes since the last run, later runs never retried the missing price. The marketplace connector restored €199 from its own calendar, leaving only the German storefront at €179.

Nothing compared the prices. On 14 October, a marketplace warned Milan about the difference, and Finance Director Sanne de Wit's margin report exposed the loss. By then, 410 orders had each lost €20 of margin.

Figure 1: *The Price Drift: one restore failed silently, so for nine days one copy kept the campaign price while every other channel had moved back to €199.*

Noor Visser's April 2027 inventory of where the Ridgewalker's price lived made the pattern visible. The platform held three market price lists (NL, BE, DE) and four B2B price lists (wholesale NL, BE, DE and one for Bergzicht Outdoor). The ERP held the wholesale and contract prices again, for invoicing, and the consumer lists that fed the DE storefront and the POS file. The marketplace connector held a channel price per marketplace. The campaign spreadsheet held the dates. **One jacket, eleven prices, seven places, no rule saying which was the source of which.** That is the Price Nobody

Can Explain.

What a Price Is Made Of

A newcomer's first surprise is that a price is not a number on a product. It is a number selected from many by a rule. The dimensions below are the ones the major platforms document; the vocabulary in bold is the chapter's.

Dimension	Plain meaning	Kestrel & Pine example
SKU	The variant being priced	Ridgewalker women's, M, moss
Market (country, region)	Where the buyer is	NL, BE, DE
Currency	What the price is denominated in	Euro throughout
Customer group	Which kind of buyer	Consumer; wholesale; one named wholesale account
Channel	Through which sales path	Storefront, store, bol, Amazon.de, Zalando, B2B portal, EDI
Quantity tier	How many are bought	Wholesale price breaks per carton
Validity window	From when until when	Autumn campaign 25 September to 4 October 2026
Tax basis	Whether the number includes tax, and which tax	€199 includes 21% Dutch VAT; €95 excludes VAT

A **price list** (Shopify calls it a catalog, SAP Commerce a set of price rows, commercetools a set of standalone prices) is a named set of prices for a group of customers or a market. A **base price** is the list price for the general public. A **customer-specific price** applies to a customer group, such as wholesale. A **contract price** is agreed with one named account and outranks the list; Bergzicht Outdoor, Kestrel & Pine's largest wholesale customer, buys on a contract price list negotiated by Lea Moreau each season. A **tiered price** (volume pricing) changes with quantity. A **quote** is an offer at a stated price with a validity date that the buyer can accept into an order; it is the one place a price becomes a promise with an expiry written on it (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals owns quotes).

The selection rule

With these dimensions, several prices can match one purchase, so the platform needs a **price selection rule**: the order in which it tries the dimensions and what it does when more than one price still matches. commercetools documents its rule in full: a price is defined per SKU and per scope (currency, country, customer group, channel), selection walks an eight-step fallback from the most specific combination to the least, time-bound prices beat open-ended ones, and when several still match "the cheapest is selected" (commercetools pricing overview, n.d., accessed November 2026). Adobe Commerce scopes quantity tiers per website and customer group (Adobe Experience League, n.d., accessed November 2026), and SAP Commerce's long-lived Europe1 model matches price rows by product, customer group, quantity, currency and date range (SAP Javadoc for release 1811, 2018; eight years old, but the model persists).

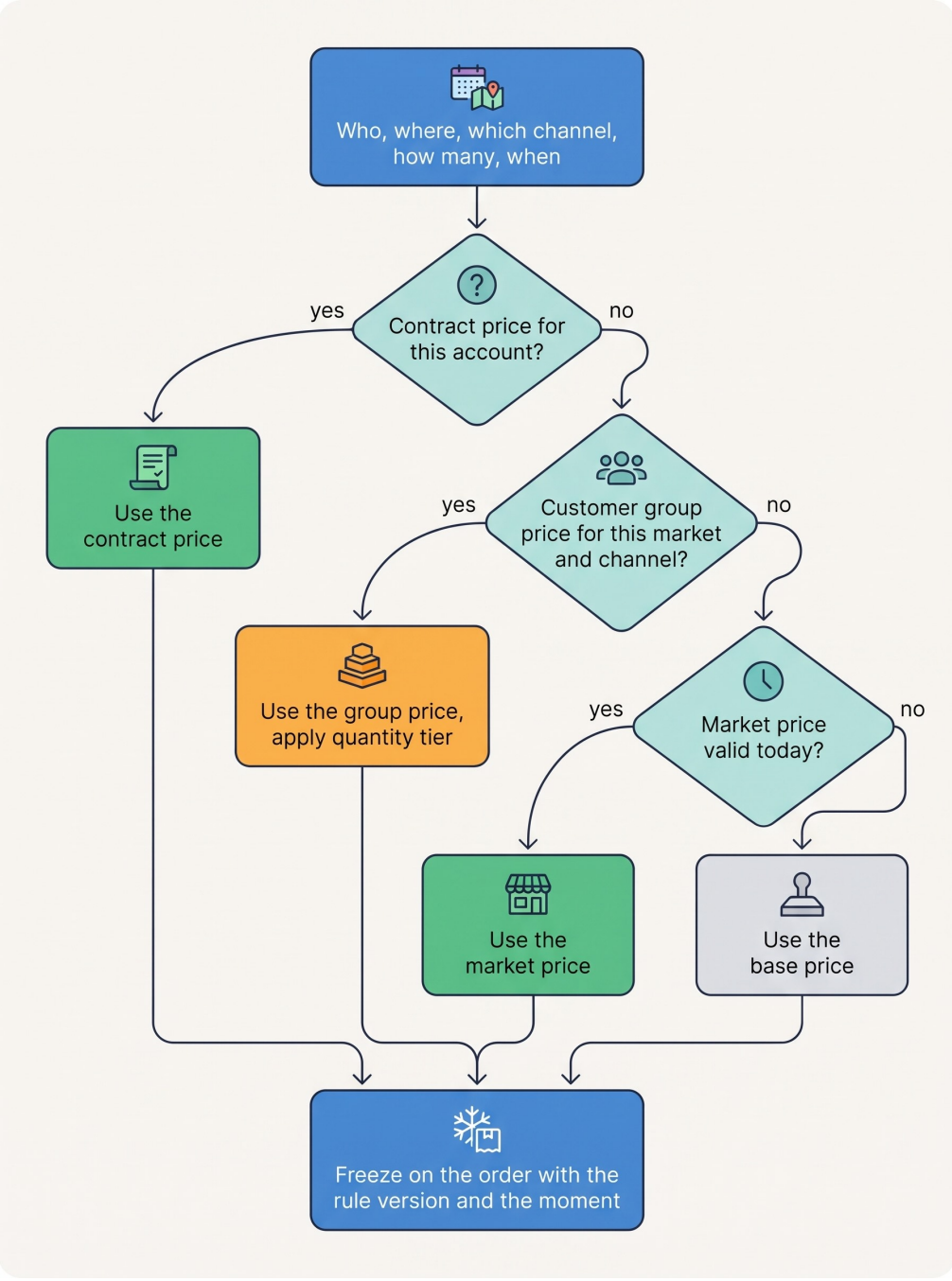


Figure 2: *A price is selected, not looked up: the rule tries the most specific price first, falls back to the base price, and freezes the result on the order together with when and by which rule it was chosen.*

The rule must be **deterministic** (the same inputs give the same price in every system

that selects) and **written down** where the people who type prices can read it, because "cheapest wins" and "most specific wins" give different answers when a campaign price and a contract price overlap.

The moment of selection

A price is also selected at a **moment**, and the moment is part of the promise. A storefront shows a price when the page loads; a cart holds it; checkout selects the price it will charge; the order freezes it; the invoice copies it. If any two of these select at different moments with different rules, the shopper sees one price and pays another.

SAP's support knowledge base records the generic bug: a campaign price that "does NOT expire after the validity end date has passed", because the validity is compared with the cart creation date, not the checkout date (SAP KBA 2267117, n.d., accessed November 2026). Adobe Commerce documents the opposite rule for purchase orders, keeping the original price if it changes after the order is created (Adobe Experience League, n.d., accessed November 2026): correct for a negotiated order, wrong for a cart.

The rule I use: display prices are selected at page load and may change; the price that counts is selected once, at order placement, from the same rule and the same data the storefront used, and is frozen on the order with the rule version and the timestamp. Quotes and purchase orders freeze earlier, for a stated validity. The invoice never selects; it copies.

Tax, Markets and Currencies: Why €199 Is Three Different Prices

In Europe a consumer price includes tax by law. Directive 98/6/EC of 16 February 1998 (published 18 March 1998) defines the selling price as "the final price for a unit of the product … including VAT and all other taxes" and requires that it "shall be indicated for all products" and be "unambiguous, easily identifiable and clearly legible". In the United States there is no federal equivalent, and sales tax depends on the delivery address, so it is added at checkout. Google's Merchant Center specification states the split operationally: for the US and Canada, prices "shouldn't include tax"; for all other countries, "include value added tax (VAT) or Goods and Services Tax (GST) in the price" (Google Merchant Center help, n.d., accessed November 2026).

Tax-inclusive means the shown price is what the consumer pays; **tax-exclusive** means tax is added later. The design consequence is that a stored price must say which it is. Kestrel & Pine sells the Ridgewalker at **€199 including VAT in all three countries**, but VAT differs: 21% in the Netherlands and Belgium, 19% in Germany (standard rates as of November 2026). So the net revenue per jacket is €164.46 in Nijmegen and €167.23 in Munich, and the margin differs by country at the same shelf price. The alternative, one

net price with the gross derived, would give the shopper €199.00 in Arnhem and €195.71 in Munich, and would break the one-price-everywhere story that marketplaces compare against. Either choice is defensible; a price stored without its tax basis is not, because then every consumer of the number guesses.

Which country's VAT applies is a tax question that Tax, Invoices and the Books: From Order to Ledger owns. The one fact a pricing design must carry is that since 1 July 2021 the EU's One Stop Shop (OSS) lets a merchant report the VAT of every EU country of delivery in one return at home, once its cross-border consumer sales pass €10,000 a year (European Commission OSS page, n.d., accessed November 2026). Kestrel & Pine passed that threshold years ago, so a €199 sale to Antwerp carries Belgian VAT inside the same €199. A wholesale sale to the same city with a valid VAT number carries none; the €95 is net and the invoice says why.

Currencies add a second axis. Kestrel & Pine prices in euros everywhere; a merchant selling in several currencies must choose between **converted prices** (base price times today's rate, rounded) and **fixed prices per market**; Shopify, for example, converts at the current exchange rate by default and rounds automatically, with fixed per-market prices as the alternative (Shopify Help Center, n.d., accessed November 2026). Converted prices move with the rate, which suits a storefront and not a marketplace that compares yesterday's price with today's. **Rounding is not cosmetic**: cents are what marketplace comparison and the 30-day reference rule see.

The Price Rules Europe Writes Into the System

Two European rules turn pricing from a merchandising choice into a record-keeping duty. Their dates matter, because "we did not know" is the Long Fuse.

The prior-price rule. Directive (EU) 2019/2161 of 27 November 2019, known as the Omnibus directive, inserted Article 6a into the 1998 price directive: "any announcement of a price reduction shall indicate the prior price", meaning "the lowest price applied by the trader during a period of time not shorter than 30 days prior to the application of the price reduction". It applies from **28 May 2022**; for widespread cross-border infringements the maximum fine must be at least 4% of annual turnover, or €2 million where turnover is unknown. Germany transposed it in § 11 of its 2022 price indication ordinance (PAngV). The Netherlands' consumer authority ACM supervised the rule from **1 January 2023**, with a worked example (6 December 2022): a price raised from €50 to €70 and cut by 35% after 25 days is really a 9% discount on the €50 reference. On 11 June 2024 ACM published fines against five web shops, among them Leen Bakker, JYSK and the Dutch Tommy Hilfiger store, because "consumers were therefore shown a fake discount" (ACM, 2024, translated).

The architectural reading is simple and demanding. **The pricing system must keep an effective-dated price history per SKU, per market and per channel, and must**

answer "lowest price in the last 30 days" at the moment a reduction is published. The reference price is itself a syndicated field, because marketplaces and feeds show the strike-through price too. A merchant whose prices live in seven places cannot answer the question reliably, which is the strongest legal argument for one source per kind of price in Europe.

Marketplace price comparison. Marketplaces compare a merchant's listing against its own web shop and against other sellers, continuously, and the legal boundary keeps moving. In November 2013 Amazon deleted its **price parity** clauses, which had forbidden sellers to sell cheaper elsewhere, after a German competition case (Bundeskartellamt, 2013; historical). On 5 February 2026 the Bundeskartellamt prohibited the opposite: Amazon's **price control mechanisms**, which removed or hid offers it judged too expensive; its case overview does not say whether the decision is final (accessed November 2026). Amazon's selling-partner API shows the mechanism from the seller's side: a `PRICING_HEALTH` notification says an offer is "ineligible for the Featured Offer (Buy Box) because its price is uncompetitive", the Buy Box being the default "add to cart" offer (Amazon SP-API documentation, n.d., accessed November 2026).

For the architecture, **channel price rules must be configuration, not code**, because the rules that bound them change on regulators' dates, and marketplace price-state events (Buy Box lost, offer suppressed, parity warning) are operational signals the pricing system should receive, not e-mails to the marketplace manager.

How the Pieces Fit Together

Four components make a pricing architecture, even when two of them live inside one product.

Component	What it is	What it owns	Example at Kestrel & Pine (2027 design)
Price data (price lists)	The numbers, with their dimensions and validity windows	The prices themselves, per kind	Market price lists in the platform; wholesale and contract lists in the ERP

Component	What it is	What it owns	Example at Kestrel & Pine (2027 design)
Pricing engine	The code that applies the selection rule to a request	The rule, its version, the moment	The platform's checkout and B2B portal; the ERP's order entry for manual and EDI orders
Price history	Every price, per SKU, market and channel, with effective dates	The 30-day reference, audit of who changed what	Kept with the price data; queried when a campaign is published
Channel price rules	Derived prices for channels that cannot call the engine	Marketplace price = base price plus uplift, within a floor and a ceiling	Configured in the connector, with a floor per product group set from the ERP's landed costs
Copies	What each channel actually shows	Nothing; they are promises made on the source's behalf	The POS price file, the marketplace listings, the EDI order confirmation

The distinction that organizes everything is **the engine versus the copies**. A channel that can call the engine at the moment of sale (the storefront, the B2B portal) needs no copy. A channel that cannot (the POS, which must work offline; a marketplace, which holds its own listing; an EDI partner, which sends its own price in the order) receives a copy, and **every copy is a promise that can drift**. The design question per channel is whether it can call or must receive, and if it must receive, from which source, with what effective date, and who reconciles it. At Kestrel & Pine the copies travel through the iPaaS (integration platform as a service, the hosted tool that runs its integrations).

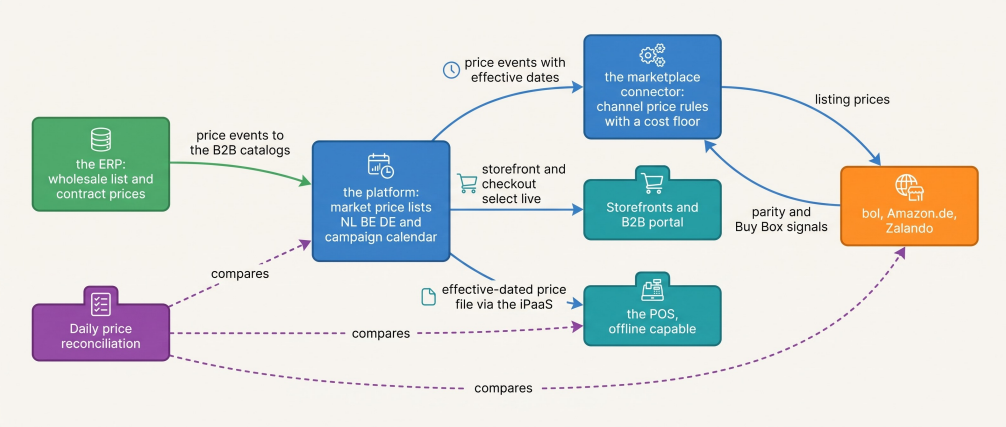


Figure 3: *Two sources, one per kind of price: the platform owns consumer prices and the campaign calendar, the ERP owns wholesale and contract prices; every other system receives effective-dated price events and is compared back to its source every day.*

Requirements per flow

Flow	How fresh	How consistent	How fast	When it fails
Storefront display	Seconds to minutes after a change	Must match what checkout will select	Page load	A campaign starts at midnight on the page and at 00:15 in checkout
Checkout selection	Live	Authoritative; frozen on the order	Under the checkout's clock, with tax and availability	A cart keeps a price a campaign ended
Marketplace listings	Minutes after a change *(assumption)*	Within channel rules and the floor	Asynchronous feeds, acknowledged	The Price Drift
Point of sale	Daily file, effective-dated	Must work offline with the last valid file	Overnight	A store sells yesterday's price for a day, by design

Flow	How fresh	How consistent	How fast	When it fails
EDI orders	The partner's price arrives in ORDERS (the EDI order message)	Must match the contract list or be flagged	Per document	A mismatch discovered on the INVOIC (the EDI invoice), as a deduction
Invoice	Copies the order	Immutable	Batch	An invoice recomputes and differs from the order

Two rows carry the design. Marketplace copies are asynchronous and must be **acknowledged**: the channel confirms the price it now holds, and that confirmation, not the sending, closes the loop. The POS copy is deliberately a day old, acceptable because it is effective-dated and reconciled, and because a till that cannot sell offline is worse than one that sells at yesterday's price.

Who may change a price

A **price change** should be an event with the SKU, the scope, the old and new values, the effective date, the author and the reason, published once by the source and consumed by every copy. Three kinds of author need rules.

Control every way a price can change. Merchandising sets consumer and campaign prices, wholesale sets contract prices, and finance approves prices below the cost floor. Give those rights in the source system and make copies read-only, so a channel cannot silently override an approved price and create a different offer.

Spreadsheet imports can overwrite prices just as an application can. Validate floors, dates and restoration rows, then review the result so a campaign file cannot quietly restore an obsolete price. Automated repricers need an identity, owner, log and limits. ChannelEngine describes 'margin guardrails' in its undated guidance, accessed November 2026. Bound the channel price using costs and fees while keeping the base price unchanged; otherwise a channel adjustment can become the price other channels inherit. The same discipline applies to agents (Assistants and Agents: AI in Commerce and What Is Real in 2026).

The Choices

Four arrangements cover what merchants actually do.

1. **The platform is the price source**. Prices are typed into the commerce platform's market and customer-group price lists; the ERP receives the price on each order; marketplaces and the POS receive copies from the platform. Right for a merchant whose prices are consumer prices and whose wholesale is small. Its limit is wholesale: contract prices, quantity breaks and the invoice's need for a price per account live naturally in the ERP, and a platform-only source forces them into a shape built for consumers.

2. **The ERP is the price source**. All price lists live in the ERP; the platform, the POS and the connector receive them. Common in ERP-first businesses, and the natural home of wholesale and contract prices. Its limit is the consumer side: campaigns with hourly start times, market-specific prices and promotion interplay need a cadence and a vocabulary ERPs do not have, and the copy to the platform becomes the slowest link at exactly the moments (a launch, Black Friday) when it matters most.

3. **A pricing service beside the platform**. A dedicated engine, bought or built, holds all price lists and the rule, exposes an API every channel calls, and publishes events to channels that cannot call. The cleanest answer and the most expensive: it adds a system, a call in the checkout path and an owner. Right when prices are genuinely complex (many markets and currencies, many customer groups, dynamic rules) or when many storefronts and apps must agree. The Giant's Boots for a merchant with three markets and one currency.

4. **Split by kind of price, with events**. Consumer prices and the campaign calendar in the platform; wholesale and contract prices in the ERP; each the system of record for its kind; everything else a consumer fed by effective-dated price events and compared daily. Two sources, but one rule and one history per kind. This is what Kestrel & Pine chose in 2027.

Per-channel pricing, where each channel owns its price, is what merchants drift into rather than choose; as a design it survives only as derived channel prices under arrangement 4.

Trade-offs

	Platform as source	ERP as source	Pricing service	Split by kind, with events
Right when	Consumer-led, wholesale small, one or two markets	Wholesale-led, few campaigns, ERP-first	Many markets, currencies, groups; many storefronts	Mixed consumer and wholesale; a hosted (SaaS) platform; Omnichannel stage
Consumer campaigns	Native: scheduled prices, markets	Slow; cadence is batch	Native	Native, in the platform
Contract and tiered prices	Possible (B2B catalogs), awkward at scale	Native	Native	Native, in the ERP
Price history for the 30-day rule	If the platform keeps it; often only current prices	Usually kept, per list	Designed in	Must be kept in both sources
Copies to reconcile	ERP, POS, connector	Platform, POS, connector	Only channels that cannot call	POS, connector, EDI
Cost	Included	Included	A system plus an owner *(assumption: the costliest option)*	Integration work and a reconciliation job
Warning sign of outgrowing it	Wholesale prices typed twice; invoices disagree with orders	Campaigns need a developer; the copy lags a launch	Rare; the risk is buying it early	The two sources start pricing the same thing (a consumer price in the ERP)

My recommendation, with its conditions, is arrangement 4 for a merchant like Kestrel & Pine, resting on three assumptions. The platform supports per-market and per-customer-group price lists with validity windows and keeps (or exports) a price history; without that, the 30-day rule has no data. The ERP invoices wholesale, so contract prices and invoices share a source. And every copy is fed by events with effective dates and compared daily, which is Waypoint's W5 applied to prices. Drop any of the three and arrangement 3 becomes the honest answer.

Across Scale Stages: When It Fits and When It Limits

The four stages are the ones this book uses throughout (Meet Kestrel & Pine; in depth, What Changes at Each Scale: From First Store to Group).

Stage	What is typical	What becomes necessary	Warning signs you have outgrown it
1 — Starting (under about €2M; Kestrel & Pine 2016–2018)	One price list in the platform; one country; tax-inclusive prices; campaigns typed by hand	A tax basis on every price; a written rule for what counts as a discount, because the 30-day rule applies from the first sale	A second country or a first wholesale customer priced in a spreadsheet
2 — Growing (about €2M–€20M; 2019–2022)	Market prices; the first wholesale list; a marketplace copy through a connector; the ERP arrives with its own lists	One source per kind of price; the first channel price rule (marketplace = base plus uplift); a reconciliation between platform and marketplace	The first marketplace shows how fast a copy drifts; the same price typed in the platform and the ERP; a campaign restore that someone has to remember

Stage	What is typical	What becomes necessary	Warning signs you have outgrown it
3 — Omnichannel (about €20M–€200M; 2023 onward)	Several markets, wholesale with contract prices, stores, three marketplaces; seven places that hold a price	Price events with effective dates; price history per SKU, market and channel; channel price rules as configuration with a cost floor; daily reconciliation; the moment of selection fixed at checkout	The Price Drift; a parity warning nobody routes; a Black Friday reference price computed by hand; the invoice differing from the order
4 — Group (over about €200M or several brands)	Several brands and currencies; per-country legal rules; group-wide marketplace accounts; a pricing team	A pricing service shared across brands with per-country rules as configuration; converted versus fixed prices per market as policy; price governance (who may change what, floors, approvals)	Two brands undercut each other on one marketplace; a currency move changes margins nobody sees; a regulator's rule implemented in three stacks

The judgement: **the tax basis and the 30-day history do not change with scale; the number of sources and copies does.** The source decision changes once, at the Growing stage, when the ERP arrives and the merchant must say which prices it owns; after that the work is in the copies.

In Practice

What the platforms document

Shopify's B2B feature prices by **catalog**, with percentage adjustments, fixed prices and volume pricing, and below the Plus plan allows "up to 3 active catalog assignments across active B2B markets" (Shopify Help Center, n.d., accessed November 2026; plan rules change). commercetools allows up to 100 embedded prices per variant or 50,000 standalone prices (commercetools documentation, n.d., accessed November 2026).

The pattern: **the number of prices grows multiplicatively** (SKUs times markets times groups times windows), so every platform moves prices out of the product into a separate store, and documents a selection rule because the growth makes one necessary.

Kestrel & Pine: the price of one jacket on 13 April 2027

In April 2027, Noor, Milan, wholesale lead Lea Moreau and Sanne agreed the owners. **The platform's market price lists own consumer prices and campaign dates** in the Netherlands, Belgium and Germany. The ERP retains wholesale list and contract prices; consumer prices no longer pass through it.

The ERP publishes wholesale price changes to the platform's business catalogs and, once live in Q4 2027, the order management system (OMS). Storefronts do not read the ERP directly. Price events carry effective dates through the integration platform, as required by Waypoint decision W5. Store tills and the marketplace connector consume those events. The connector's floors use landed costs from the ERP.

A daily comparison of all price copies starts in May 2027, and the old connector's price flow is retired.

On the day of Eva's order, the Ridgewalker women's, size M, moss, is priced like this:

Where	Price	Basis	Source	How it got there
NL storefront, checkout	€199	Incl. 21% VAT; net €164.46	Platform, NL market list	Selected live at 21:14, frozen on #1072455
DE storefront	€199	Incl. 19% VAT; net €167.23	Platform, DE market list	Selected live

Where	Price	Basis	Source	How it got there
bol, Amazon.de, Zalando	€199	Incl. VAT of the marketplace's country	Connector rule: the market price of the marketplace's country, uplift 0%, floor €95 *(illustrative, from the €58 landed cost plus fees)*	Price event, acknowledged by the connector
Six stores	€199	Incl. VAT of the store's country	Platform NL and BE market lists	Effective-dated daily file via the iPaaS
B2B portal, an independent shop	€95	Excl. VAT	ERP wholesale list NL	Price event to the platform's B2B catalog
Bergzicht Outdoor, by EDI	Contract price	Excl. VAT	ERP contract list	The ORDERS price is checked against the list on receipt
Eva's invoice	€199	Incl. 21% VAT	Copied from #1072455	Never recomputed

The last row makes Eva's exchange on 19 April work (Journey 2: The Jacket Comes Back): it reuses the frozen €199, whatever the list price is that day.

The Black Friday test. Suppose the NL autumn campaign of 2027 runs 20 October to 3 November at €179 (illustrative). A Black Friday reduction announced on 26 November must show the prior price as the lowest price of the previous 30 days, 27 October to 26 November, which includes a week at €179. The "was" price is **€179, not €199**, and a Black Friday price of €159 is an 11% reduction, not 20%. Before the 2027 design that answer required someone to remember the campaign; after it, the platform's price history answers it when the reduction is scheduled.

Figure 4: *The 30-day rule reads history, not memory: a campaign inside the window makes the was-price €179, not €199, so €159 is an 11% reduction, not 20%.*

What changed about the Price Drift. Under the 2027 design the German campaign is one row in the platform with a start and an end, so no hourly push has to carry the restore, and the daily reconciliation would have shown any channel still at €179 against the DE market list at €199 on the morning of 5 October. The margin at risk falls from nine days to one.

When It Goes Wrong

The silent restore. A campaign's end price never arrives, or arrives and is dropped. *Detection:* a daily reconciliation of every copy against its source; an expected-price check at campaign end; acknowledgement from each channel. *Recovery:* republish the current price as a new event; the copies are idempotent, so a repeat does no harm.

The repricer that undercuts the storefront. A connector rule chases a competitor's price below the merchant's own site, and the margin disappears. *Detection:* a floor from cost and fees; an alert when a channel price falls below the base price. *Recovery:* channel rules bounded by configuration, never writing to the base price.

Two moments of selection. The cart holds a campaign price that checkout no longer offers, or the reverse; the SAP cart-creation-date bug is the documented case. *Detection:* a test that opens a cart before a campaign boundary and checks out after it. *Recovery:* one rule, one moment, at order placement.

The reference price nobody kept. A reduction is announced against last week's list price when a cheaper campaign ran within 30 days. *Detection:* the pricing system

computes the reference from history when the reduction is scheduled and blocks a higher one. *Recovery:* correct the strike-through price in every channel, which is itself a syndicated field, and record why.

The EDI price that differs. Bergzicht's ORDERS carries the price it believes; the invoice carries the contract list's price; the difference becomes a deduction on payment. *Detection:* price validation on ORDERS receipt against the contract list; a mismatch is a task before shipment, not a discovery on the INVOIC (Documents Between Companies: EDI, Punchout and E-Invoicing).

Every one of these ends in the same component: a reconciliation that compares copies to their source on a schedule and gives a person a list. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation builds that component for every kind of copy in the landscape.

What to Take Away

Established.

- A price is selected by a rule from several candidates that differ by market, customer group, channel, quantity and validity; commercetools, Adobe Commerce and SAP Commerce each document such a rule, and every platform separates price data from the product as the count grows.
- In the EU, consumer prices must be shown including VAT and all other taxes (Directive 98/6/EC, 1998), and any announced reduction must name the lowest price of the previous 30 days (Directive 2019/2161, applied from 28 May 2022; enforced in the Netherlands from 1 January 2023 with fines published in June 2024).
- Marketplaces compare prices continuously and act on the result; Amazon dropped its price parity clauses in 2013 after a German competition case, its price-control mechanisms were prohibited in Germany in February 2026 (finality not stated as of November 2026), and the Buy Box is decided partly on price.

Depends on context.

- Whether the platform or the ERP is the source of a given kind of price depends on who prices it and who invoices it: consumer prices and campaigns belong with the storefront; wholesale and contract prices with the system that invoices wholesale. A merchant with no wholesale needs one source; a merchant with no consumer campaigns may need only the ERP.
- A dedicated pricing service is right for many markets, currencies and storefronts, and the Giant's Boots for a merchant with three euro markets.

- One gross price in every market (Kestrel & Pine's choice) simplifies marketplace comparison and costs margin where VAT is higher; one net price with derived gross does the reverse. The choice is a policy; the requirement is that the tax basis is stored with the price.

The author's synthesis.

- Treat every copy of a price as a promise made on the source's behalf: it gets a source, an effective date, an acknowledgement and a daily comparison, or it does not get made.
- Fix the moment of selection at order placement, freeze the price on the order with the rule version, and let the invoice copy; everything else about pricing can be reorganized later, but a price that differs between order and invoice cannot be explained to a customer or an auditor.

Questions to Ask

1. For one product, how many systems hold a price, and which one is the source for each kind of price (consumer, wholesale, contract, channel)?
2. Is the selection rule written down? When a campaign price and a contract price both apply, which wins, and does every system agree?
3. At which moment is the price that will be charged selected, and can a cart keep a price that checkout would no longer give?
4. Does every stored price say whether it includes tax, and which tax?
5. Can the system answer "the lowest price we charged for this SKU in this country in the last 30 days" at the moment a reduction is scheduled?
6. When a campaign ends, how does each channel learn, how do you know it learned, and what compares the channels the next morning?
7. Which automations change prices (repricers, apps, imports), who owns each, and what floor bounds them?
8. When a marketplace warns about a price, where does the warning land?

The price on the product page is only the start of what the shopper will pay. Coupons, bundles, free-shipping thresholds and loyalty points change it at checkout and must still respect the floor, the 30-day reference and the books. Discounts, Coupons and Loyalty: Promotions Without Chaos takes the frozen price from this chapter and shows what may be subtracted from it, in which order, and how the discount reaches the invoice.

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Research and practitioner sources

- None beyond the documentation above; the reconstruction of the Price Drift's mechanism and the 2027 ownership decision are the author's illustration.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

8. Discounts, Coupons and Loyalty: Promotions Without Chaos

IN THIS CHAPTER, YOU WILL: Learn how promotions, coupons, gift cards and loyalty points are modeled, where the final discount must be computed, and how a discount reaches the invoice, the refund and the books.

KEY POINTS:

- **A promotion is a rule evaluated in one place, at cart time.** The product page shows an estimate; the cart engine computes the final discount from the rules, the codes and the customer's eligibility, in a declared order, with a rule that says when stacking stops. Every platform that takes promotions seriously has converged on this shape.
- **A discount is not finished until it is allocated to lines.** Returns, partial refunds, invoices and tax all work per line, so an order-level discount must be spread across the lines at order time. Skip that step and every return refunds too much.
- **Gift cards, store credit and points are money you owe, not marketing.** They are ledgers with balances that finance must reconcile to

a liability account every month; the rules for when unredeemed value becomes revenue are accounting standards, not product decisions.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Price Nobody Can Explain** — a cart whose total nobody can reconstruct from the rules is a price with no owner.
- **The Boomerang** — refunds computed from the undiscounted line price give the discount back twice.
- **The First Snowfall** — Black Friday is the first time all the promotion rules meet all the customers at once.

A shopper uses a welcome code to buy a jacket and socks, then returns the jacket. How much should the merchant refund? The answer depends on how the discount was shared across the items. If the returns system uses the jacket's list price, it can refund more than the shopper paid for it. A promotion therefore needs a recorded calculation that checkout, returns and finance can all use.

Gift cards and loyalty points create a different obligation: track the balance customers can still spend. Treating that balance as an ordinary discount can lose the record of what the merchant owes.

Offers can also interact before payment. Two valid discounts may combine into a price below cost, and a cart may qualify for a different offer after the shopper adds an item or address. The promotion rules must specify which combinations apply so that the advertised saving, checkout total and later refund agree.

The chapter explains how a **promotion engine** evaluates offers, how **stacking** combines them, and how **discount allocation** shares a saving across items. It also separates discounts from stored balances the merchant owes customers. Related detail sits in What Does It Cost? Price Lists, Contract Prices, Currencies and Tax, From Cart to Confirmation: Checkout Orchestration, Tax, Invoices and the Books: From Order to Ledger, The Way Back: Returns, Exchanges, Refunds and Reverse Logistics and One Customer, One Product: Identity and Master Data Across Systems.

Where we are in the order's journey. Eva buys from Kestrel & Pine, the fictional outdoor brand, on 13 April 2027. Her €199 jacket and two €18 pairs of socks total €235. She gets no item discount, but qualifies for free shipping over €75. That is a promotion too: checkout must evaluate it and record the result.

Setting the Scene at Kestrel & Pine: The Black Friday Stacking Bug

Kestrel & Pine runs three countries' storefronts on a hosted software-as-a-service (SaaS) platform. For **Black Friday week, 23–29 November 2026**, Ecommerce and Marketplaces Manager Milan Petrov configured three promotions:

- an **automatic 20% discount on all outerwear** (jackets, fleeces, trail trousers), applied to every cart without a code, valid for the week;
- the standing **welcome code**, 10% off a first order, sent to every new newsletter subscriber;
- the standing rule of **free shipping on orders over €75**.

The welcome code was meant to give a first-time customer the better offer, without combining with the outerwear discount. But the platform's **discount combination setting** still allowed both. The agency had enabled it for an October fleece campaign and left it on. Nobody checked the setting against the new campaign plan. The figures below are illustrative.

What happened in the week of 23–29 November 2026	**Figure *(illustrative)***
Online direct-to-consumer (D2C) orders in the week (see Meet Kestrel & Pine)	31,000
Orders that received both the 20% outerwear discount and the 10% welcome code	2,860
Ridgewalker jacket price after both: €199 → €159.20 → €143.28	€15.92 extra discount per jacket
Margin given away beyond the plan (about €15 per affected order)	about €43,000
Orders whose discounted subtotal fell under €75 and were charged €4.95 shipping after the product page had promised it free	1,140
Shipping refunded by hand after complaints	about €5,600
Returned Black Friday jackets refunded at €159.20 instead of €143.28, because the order-level 10% had not been allocated to the lines the returns portal reads	190 returns, about €3,000 over-refunded
Total cost	about €52,000

Three things went wrong, and they are the three things this chapter is about. The **stacking** was not what the business intended, because the rule that decides it lived in a setting, not in the campaign. The **free-shipping threshold** was evaluated on the discounted subtotal in the cart while the product page promised it on the list price. And the **order-level discount was never allocated to the lines**, so the returns portal, which

reads line prices, refunded the discount back. Noor Visser, Head of Engineering since January 2027, put the incident next to the Price Drift of October 2026 and drew the same conclusion: **a price that nobody can reconstruct from the rules is a price with no owner**.

Figure 1: *The stacking bug: a combination setting left on from October let the welcome code stack on the outerwear discount, taking the jacket from €199 to €143.28.*

What a Promotion Is, and What It Touches

In plain words, a **promotion** is a rule that lowers the price a customer pays under stated conditions. A **coupon code** is a promotion the customer unlocks by typing a string; an **automatic discount** applies when the conditions are met with no code. **Eligibility** is the set of conditions: which products, customers, period, channel, minimum cart. **Stacking** is applying more than one promotion to the same cart or line. **Discount allocation** is spreading an order-level discount over the lines so that each line, and later each invoice line and return, carries its share.

The table lists the shapes every platform supports and what each needs from the rest of the system.

Promotion type	Example	What it needs to compute	What it changes
Percentage off a product or set	20% off outerwear	The product's current price and category	Line price
Fixed amount off a product	€10 off the Trailhead fleece	The product's price	Line price
Percentage or fixed amount off the order	10% off a first order with a code	The subtotal after product discounts; the customer's order history	Order total, to be allocated to lines
Buy X, get Y	Three pairs of socks for the price of two	Quantities per line; which unit is the free one	One line's price
Tiered	10% over €150, 15% over €250	The subtotal	Order total, allocated
Free or reduced shipping	Free shipping over €75	The subtotal and the shipping method	Shipping line
Bundle	Jacket plus fleece for €279	The presence of both items	Two lines, with a fixed total allocated between them

Every row touches three things the business cares about: the **price** the customer sees and pays, which in the EU must obey the prior-price rule described later; the **margin**, which disappears silently when promotions combine (the Ridgewalker at €143.28 still cleared its €58 landed cost, but gave up about a sixth of its planned Black Friday unit margin); and the **books**, because the invoice must show the net price per line with the VAT on it, and a refund must give back what was paid for that line, not what was listed.

Two things that look like promotions are not. **Gift cards**, **store credit** and **loyalty points** are value the merchant has been paid for or has promised and not yet delivered: money it owes. They are ledgers, not rules, and get their own section.

How the Pieces Fit Together

A promotion system has five parts, whether it is a platform feature, an app or a service. The diagram shows them around the decision that matters: the final discount is computed once, in the cart, and written to the order with its allocation.

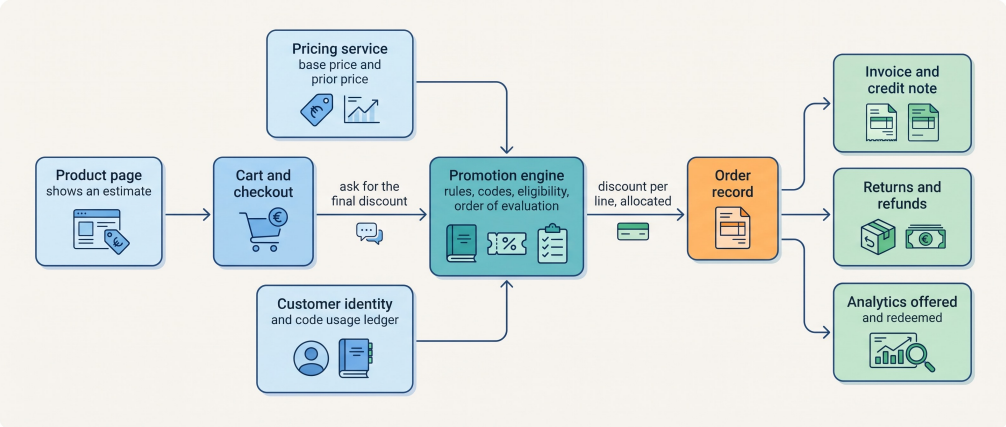

Figure 2: *The product page shows an estimate; the promotion engine computes the final discount once, at cart time, from the base price, the rules and the customer's eligibility, and writes it to the order allocated per line so that the invoice, the refund and the analytics all read the same number.*

The five parts

Promotion definitions. Each promotion is a record: its type, conditions, validity window, channel, budget or usage cap, and its position in the **order of evaluation**. The platforms that document this best make the order explicit. Shopify evaluates product discounts first, then order discounts on the revised subtotal, then shipping discounts; its help center (accessed November 2026) notes that two combined percentage *order* discounts are both calculated on the original subtotal, so 10% plus 20% on €100 gives €70, not €72. commercetools gives every cart discount a unique `sortOrder`, a stacking mode that can stop further discounts after a given one, and discount groups in which only the best saving applies. Adobe Commerce applies the highest-priority rule first and has a "Discard subsequent rules" switch. One idea, three vendors: **promotions are data with a declared order and a stop rule**.

Eligibility and identity. "First order", "members only" and "once per customer" all need to know who the customer is. On Kestrel & Pine's platform the welcome code's "first order" check looked at the e-mail address, so a customer with two addresses was a first-time customer twice. Promotions are one of the first places a merchant discovers it has no single customer identity: the Familiar Stranger pattern, owned by One Customer, One Product: Identity and Master Data Across Systems.

The code ledger. Every code has a usage record: how many uses are allowed in total and per customer, and how many have happened. It must be strongly consistent at redemption: two checkouts redeeming the last use of a single-use code in the same second must not both succeed. It is the one part of a promotion engine that behaves like a bank account, and the reason bulk-generated single-use codes with usage caps (Adobe

generates them in batches; Shopify allows 5 product or order codes and 1 shipping code per order) are the baseline defense against code sharing.

Evaluation in the cart. The engine takes the cart's lines, prices, shipping method and customer, applies the promotions in order, stops where the stop rule says, and returns a discount per line and per shipping line. Some conditions can only be evaluated late: Adobe's documentation notes that conditions on the shipping or payment method "are met only in checkout after the specific shipping/payment methods are selected". That is why the product page can only show an **estimate**, and why the estimate and the final must come from the same rules.

Allocation and events. The final step writes the discount to the order *per line*. commercetools makes the choice explicit: an absolute cart discount is distributed proportionately, evenly or individually across the lines. However it is distributed, each line then carries its net price, which the invoice and the refund read. The engine also emits events: promotion applied, code redeemed, promotion offered but not taken. The last is the one analytics needs most and platforms rarely record.

What each interaction needs

Promotions are evaluated at several moments with different requirements.

Interaction	Freshness	Consistency	Latency	Availability
Product page strike-through price	Start and end to the minute; prior price per the EU's 30-day rule	An estimate; may differ from the cart	Served from cache *(assumption)*	Must not block the page; show the list price if the engine is down
Cart evaluation	Current rules and base prices	Final; the same result on every re-evaluation	Under about 300 ms, on every cart change *(assumption)*	Checkout must not fail because promotions fail; degrade to no discount with a message, or hold the checkout, by business decision

Interaction	Freshness	Consistency	Latency	Availability
Code redemption	Current usage count	Strongly consistent; one redemption per allowed use	As the cart	As the cart
Gift card or store credit redemption	Current balance	Strongly consistent; idempotent per checkout attempt	As the cart	Decline the card if the balance service is down; never guess
Promotion publication	Scheduled start; tested against active promotions	All channels see the same rule set	Minutes	Stores and market-places receive a copy
Analytics	Daily	Offered and redeemed both recorded	Batch	—

The two rows that behave like money (code redemption and stored value) cannot be eventually consistent; everything else can be a cached copy with a known delay. It is the distinction Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record draws for every fact: decide which system holds the truth, and let the copies be copies.

How a discount reaches the invoice and the books

A discount leaves the order in three documents. The **invoice** shows each line at its net price after allocation, with VAT on that net amount; an order-level discount across lines with different VAT rates (a jacket at 21% and a book at 9%) must be allocated before the tax can be right, which makes allocation a tax problem as much as a promotions problem. The **credit note**, issued on a return, reverses the line at the same net price. The **ledger** records revenue net of the discount: a promotion is not an expense but revenue never earned. Tax, Invoices and the Books: From Order to Ledger walks the entries; RECONCILED's Payments-to-Ledger: Draining the Suspense Swamp goes deeper on reconciliation.

Here is the allocation on Eva's order, had she used a 10% code (she did not; the prices are her order's, the code is hypothetical):

Line	List price	Share of €235	Allocated discount	Net line price
Ridgewalker jacket, women's M, moss	€199.00	84.7%	€19.90	€179.10
Merino hiking socks × 2	€36.00	15.3%	€3.60	€32.40
Total	**€235.00**		**€23.50**	**€211.50**

When Eva returns the jacket, the refund is **€179.10**, not €199. The 190 over-refunded Black Friday returns were this calculation skipped: the returns portal read the line price (€159.20 after the product discount) because the order-level 10% had never been written to the lines, and gave back €15.92 per jacket the customer had never paid. That is the Boomerang in one table row.

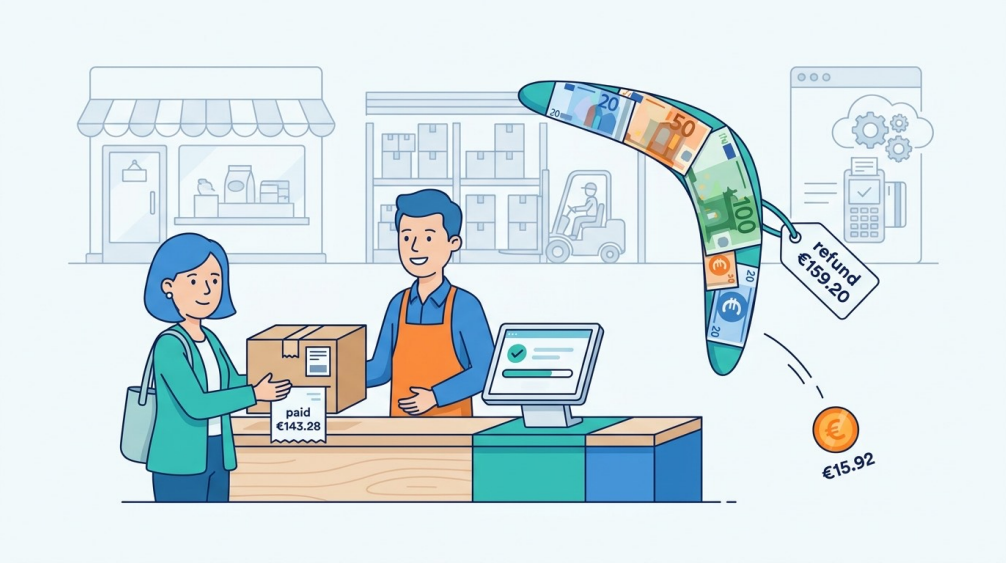

Figure 3: *The Boomerang: because the order-level 10% was never allocated to the lines, each Black Friday jacket return gave back €15.92 the customer had never paid.*

Gift Cards, Store Credit and Points: Ledgers, Not Promotions

A **gift card** is value the customer paid for in advance. **Store credit** is value the merchant granted, typically as a refund alternative or a goodwill gesture. **Loyalty points** are value the customer earned by buying. All three share a property that no promotion has: **they are a liability on the balance sheet**, money or value the merchant owes, until they are redeemed or expire.

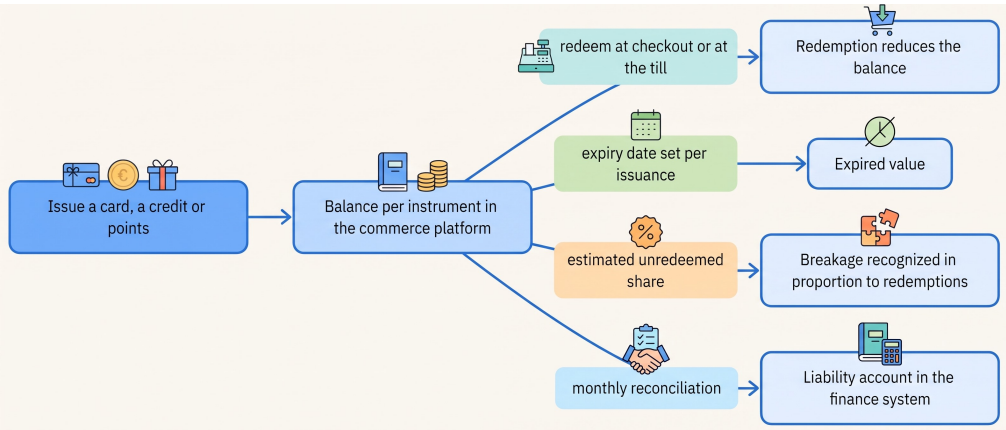

Figure 4: *Stored value is a ledger: every issuance, redemption, expiry and breakage estimate changes a balance per instrument in the commerce platform, and the sum of those balances must agree with the liability account in the finance system every month.*

The accounting is settled and drives the data model. Under IFRS 15, the international accounting standard for revenue (issued May 2014, effective from 1 January 2018, replacing IFRIC 13 on loyalty programs) and its US twin ASC 606, the sale of a gift card is a contract liability, not revenue; revenue is recognized on redemption. The part that will never be redeemed is **breakage**; the standard requires the merchant to *estimate* it and recognize it "in proportion to the pattern of exercised rights" rather than wait for cards to expire, and unclaimed value owed to a government is a liability, not revenue. Loyalty points that give the customer a **material right** (a benefit valuable enough to count as something they paid for) are a separate performance obligation, with revenue deferred until they are used.

Starbucks' FY2025 annual report (fiscal year ended 28 September 2025) shows what this means at scale: amounts loaded onto stored value cards are recorded as deferred revenue; the stored value card and loyalty liability stood at **$1,751.7 million**; breakage recognized in company-operated stores was **$200.4 million** in the year; Stars are deferred at their estimated selling price "as each Star is earned" and "generally expire after six to twelve

months, depending on the market". Starbucks is a coffee chain, but the obligation has the same shape for a merchant with €200,000 outstanding.

Four design rules follow:

- **Every instrument is a ledger**. Issuance, each redemption, each expiry and each breakage estimate is an immutable record per card, credit or points account; a balance is derived, never edited (RECONCILED's Never Update, Only Append: Designing Custom Ledgers gives the pattern).
- **Expiry is an attribute of the issuance, not a global setting**, because rules differ by country and, in the US, by state. Shopify's store credit documentation (accessed November 2026) models it that way: "Store credit expiration is set per issuance".
- **Redemption is idempotent**: a repeated request has the same effect as one. The idempotency key (a unique id that lets a retried request be recognized and applied once) that protects the payment (Taking the Money: Payment Providers, Authorization, Capture and Refunds) must also cover the stored-value debit.
- **The platform holds the balance, finance holds the liability, and they are reconciled monthly**; a difference is a bug or a fraud.

Paid membership programs (Prime-style: a fee in exchange for delivery and other benefits) are a different thing again, often confused with loyalty. They are a subscription with its own revenue line and its own promises (free delivery, early access) that the promotion engine reads as eligibility conditions. Amazon's annual report for 2025 (filed February 2026) reports subscription services net sales of $49.6 billion and describes Prime benefits as "effective worldwide marketing tools" it intends "to continue offering indefinitely"; Target's FY2025 report describes its paid Target Circle 360 membership with "unlimited same-day delivery on eligible orders over $35". For a merchant of Kestrel & Pine's size the architectural point is modest: a membership is a customer attribute with a start and end date, and a liability for benefits paid for but not yet delivered.

Kestrel & Pine's stored value in 2027

Kestrel & Pine sells gift cards online and through store point-of-sale (POS) tills. The systems are separate, so **neither can redeem the other's cards** *(illustrative)*. At 31 December 2026, outstanding balances were about **€186,000 online** and **€41,000 in stores**. Finance Director Sanne de Wit reconciles two exports to the liability account each month.

Stores have a loyalty list; the online shop has no loyalty program. Noor and Milan deferred one until they could recognize the same customer across channels. Otherwise points could accumulate in an account the redemption system cannot identify. A shared balance service remains a goal after Waypoint, to be considered in the 2028 store-system evaluation under decision W4.

The Choices

A merchant can run promotions in four ways. The fourth is listed because it is common, not because it is good.

The platform's built-in promotions. Hosted platforms ship discounts, codes, automatic promotions, gift cards and, increasingly, store credit, with documented combination rules and limits. Shopify allows at most 25 active automatic discounts (apps included); commercetools allows 100 active code-less cart discounts per project plus 100 per store, and 10 codes per cart. The limits are design signals: promotion interaction is the most common source of margin bugs. It fits any merchant whose promotions fit the platform's vocabulary.

The platform plus a promotion or loyalty app. When the business wants what the platform does not model (a tiered loyalty scheme, referral rewards, cross-channel vouchers), an app or a specialist SaaS adds it, usually by creating discounts in the platform on the customer's behalf. The app becomes a second writer of the price, with its own rules, its own view of the customer and its own failure modes. Kestrel & Pine's pre-order app wrote stock; a loyalty app is the same shape of risk, which is why Apps, Plugins and Functions: Extensibility and Ecosystem Governance treats apps as integrations.

A dedicated promotion engine shared across channels. A separate service (bought or, rarely, built) holds the definitions and evaluates them for the web storefront, the POS and the B2B portal through one API. It is the only option in which a store and the web shop run the same promotion from the same rule, and in which a wholesale contract price and a consumer promotion are kept apart by design. It costs an integration into every checkout and a team that owns the rules.

Rules hard-coded in the theme, a script or a marketplace listing. The agency makes a banner compute "20% off" in the browser; a script adjusts marketplace prices for a sale; store staff key their own discount into the POS. Each copy drifts, nobody can say what the price was on a given day, and the Price Nobody Can Explain follows. It is not a design but the absence of one.

Criterion	Platform built-in	Platform plus app	Dedicated engine	Hard-coded rules
Where the final discount is computed	Platform cart	Platform cart, rules written by the app	The engine, called by every channel	Several places
Order of evaluation and stop rule	Documented by the vendor; set per shop	Partly inside the app	Yours to define	Undefined

Criterion	Platform built-in	Platform plus app	Dedicated engine	Hard-coded rules
Same promotion in web, store and marketplace	Web only; stores and market-places get copies	Web and sometimes POS of the same vendor	Yes, by design	No
Discount allocation to lines	Vendor-defined	As the platform does it	Yours to define, must be exported	Usually missing
Stored value across channels	Within the platform's channels	App-dependent	A balance service	No
Fits	Starting, Growing; most Om-nichannel merchants	Growing merchants with one specific need	Omnichannel with stores and B2B on different systems; Group	Nobody

My rule: **use the platform's promotions until a promotion must be the same rule in two systems the platform does not control**. For Kestrel & Pine that moment has arrived for the stores (a separate POS) and will arrive for the B2B portal if wholesale promotions start; it never arrives for marketplaces, whose promotions stay in the marketplace's own tools. Until then the cheapest improvement is governance, not software: one owner for the combination settings, one calendar, and a rollout test against the promotions already active, which Shopify offers as a feature and any merchant can do in a staging copy.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
1 — Starting	A few codes and a free-shipping threshold in the platform; platform gift cards reconciled by finance from an export	A naming convention for codes and a note of who created each promotion	A code from an old campaign still works; nobody knows which promotions are active
2 — Growing	Automatic discounts plus codes; a welcome code; the first Black Friday with several promotions at once; store credit for refunds	An explicit combination decision per campaign; single-use bulk codes; a first breakage estimate with the accountant	The first stacking surprise; disputed refunds of discounted items; the "first order" check fooled by a second e-mail address

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
3 — Omnichannel	Web, stores, marketplaces and B2B each run their own promotions; stored value split by channel; two loyalty lists	One promotion calendar across channels; allocation exported to the OMS (order management system), the returns tool and the ERP (enterprise resource planning system, which invoices); a central stored-value balance, or a decision not to cross channels yet; identity resolved before a loyalty program	Store staff keying discounts the web shop does not know; a marketplace promotion undercutting the web shop; gift cards that work in one channel only
4 — Group	Several brands or countries with their own calendars; a shared loyalty platform; a liability ledger per legal entity	A promotion engine or shared service with per-brand rules; a group stored-value ledger reconciled per entity; holdouts as standard	Brands competing for one customer with conflicting offers; a group voucher no single system can honor; breakage estimated differently per country

The judgement: the promotion *engine* changes little with scale; evaluating a thousand rules is the same problem as evaluating ten. What changes is **how many systems must agree on the result** and **how much money the stored-value ledger holds**. Kestrel & Pine at stage 3 has the first problem acutely (three channels with their own discounts, two gift-card pools) and the second mildly (about €227,000 outstanding), so its order is

ownership and allocation first, a cross-channel balance second, a loyalty program last.

In Practice

Dated, sourced examples

The prior-price rule turns promotions into a data requirement. Directive (EU) 2019/2161, the Omnibus Directive, applied by member states from 28 May 2022, requires any announced price reduction to show the **prior price**: the lowest price the trader applied in the 30 days before the reduction. The Dutch regulator ACM announced on 6 December 2022 that it would supervise the rule from 1 January 2023, with a worked example: a price raised from €50 to €70 and cut by 35% after 25 days is really a 9% discount on the €50 reference. On 11 June 2024 ACM published fines against Leen Bakker, G-Star Raw eStore, JYSK, PVH Brands Netherlands (Tommy Hilfiger) and Koopjedeal for discounts shown against a misleading reference price; Germany's § 11 PAngV transposes the same rule. For architecture this means an effective-dated price history per SKU per country that can answer "lowest price in the last 30 days" when a promotion is published, and a strike-through price that is itself a syndicated field. Had Kestrel & Pine's Price Drift run into November, the German strike-through price for the Ridgewalker would legally have been €179, not €199.

Stored value is a liability with accounting rules. Starbucks' FY2025 10-K, above. PwC's revenue guide on unexercised rights (accessed November 2026) states the ASC 606 position that estimated breakage is recognized "in proportion to the pattern of exercised rights", that breakage not expected to occur is recognized only when the likelihood of redemption becomes remote, and that amounts due under unclaimed-property law are never breakage revenue. The IFRS Foundation's page confirms IFRS 15 replaced IFRIC 13 on customer loyalty programs; the matching IFRS 15 paragraphs B44–B47 are cited here from a secondary source, because the standard's own text was not checked for this chapter.

Promotion abuse is an identity problem. Ravelin's 2020 article on promotion abuse (older than three years, but the taxonomy holds) names multi-accounting for sign-up offers, self-referral and guessable codes as the main forms; its figure that 53% of merchants saw more abuse in a year is a vendor survey without a published sample, so treat it as directional.

Incrementality needs a holdout. Gubela, Lessmann and Jaroszewicz (EJOR, 2019) argue that coupon campaigns should be judged on revenue uplift, not conversions, because customers differ in spending and most exposed customers do not buy; Gubela and Lessmann (2021) report that of several coupon types tested, only a €15 coupon produced positive incremental revenue. Redemption rate is not incrementality, because many redeemers were already at checkout; the honest measure is the difference between an exposed group and a **holdout group** deliberately not given the offer. For the platform

this means recording which promotions were *offered*, flagging holdout customers in the engine, and one customer identity to assign them with; Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics takes it from there.

Kestrel & Pine's Black Friday 2027 plan

For Black Friday 2027 Milan and Noor wrote the promotion plan as a table, each row with an owner, an order of evaluation and a stop rule, and tested it against every active promotion in a staging copy of the platform in early November (illustrative):

Promotion	Type	Order	Combines with	Stop	Owner
20% off outerwear	Automatic product discount	1	Free shipping only	Yes	Milan
Welcome code 10%	Order code	2	Free shipping only; never with row 1 (the better one applies)	—	Milan
Free shipping over €75	Shipping	3	Everything; threshold on the *discounted* subtotal, stated on the product page	—	Daniel Okafor
Store promotion, same 20%	POS discount keyed by store managers from this table	—	—	—	Daniel

Promotion	Type	Order	Combines with	Stop	Owner
Marketplace deals	Each marketplace's own tool, priced at the web price after discount plus fee uplift	—	—	—	Milan

Two changes matter more. The order export to the WMS (warehouse management system), the returns portal and the ERP now carries the **allocated net price per line**, so a return refunds what was paid; Ruben Alves added it to the integration contract under decision W5. And every change to the platform's promotion settings, including the combination setting, needs a second person's approval. Neither cost software; both cost the week in which the bug was understood.

When It Goes Wrong

Unintended stacking. Detection: a daily report of orders with more than one discount, with the discount depth (discount as a share of list price) per order, and a threshold alert in the peak week. Recovery: honor the orders already placed, fix the setting, and compute the margin given away from the order lines, which is only possible if the allocation was written.

Estimate and final disagree. The page says free shipping, the cart charges €4.95; the page says "20% off", the cart shows 10% because the code replaced the automatic discount. Detection: log what the page rendered and what the cart computed per session, and watch abandonment at the shipping step (From Cart to Confirmation: Checkout Orchestration). Recovery: state thresholds in the terms the cart uses, and let the page ask the engine for an estimate when the customer is logged in.

The discount comes back twice. A return refunds the list price of a discounted item (the Boomerang), or a gift-card purchase is refunded to the card *and* to the payment method. Detection: a weekly reconciliation of refunds against the allocated net price of the returned lines. Recovery: a credit-note correction per case, and the allocation added to every order export.

Codes leak or are guessed. A generic code on a coupon site, or a sequential format a script can enumerate. Detection: redemptions per code per hour against the expected rate, and per identity signal (device, payment instrument, address) rather than per e-mail (LISTED's Scams, Fakes and Account Takeovers: Fraud Prevention for Listings).

Recovery: disable the code, issue single-use replacements, and accept that placed orders are placed.

Stored value debited twice, or not at all. A checkout retried after a timeout debits a gift card twice; a payment webhook (the message the payment provider sends when a payment completes) arrives late and the order is created without the debit. Detection: ledger entries per instrument that reference the checkout attempt, so a duplicate shows as two entries with one key. Recovery: one idempotency key per checkout attempt covering the payment and the stored-value debit, and a daily job matching debits to orders (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation).

A promotion ends but a copy does not. A cached page, a marketplace listing updated hourly, a price-comparison feed or a POS still showing last week's 20%. Detection: an end-of-promotion check that reads the price on every channel; a marketplace parity warning is the expensive version. Recovery: a scheduled end that every copy receives as an event, with its delivery promise in the integration contract.

What to Take Away

Established.

- Promotions are data with an explicit order of evaluation and a stop rule, computed once in the cart; the storefront shows an estimate. Shopify, commercetools and Adobe Commerce document this shape independently (accessed November 2026).
- An order-level discount must be allocated to lines at order time, because invoices, VAT, returns and partial refunds work per line.
- Gift cards, store credit and loyalty points are contract liabilities under IFRS 15 and ASC 606; breakage is estimated and recognized in proportion to redemptions (PwC's guide; Starbucks' FY2025 10-K as the worked example).
- In the EU, an announced price reduction must show the lowest price of the preceding 30 days (Directive 2019/2161, applied from 28 May 2022; Dutch fines published June 2024).

Depends on context.

- Use the platform's built-in promotions until one promotion must be the same rule in two systems the platform does not control (a separate POS, a separate B2B portal); then a shared engine earns its integration cost. Marketplace promotions stay the marketplace's own in every case.
- A central stored-value balance is worth building once customers expect to use a card in any channel and the outstanding balance makes two reconciliations a month a risk; below that, two pools and an honest sign at the till are acceptable.

- Launch a loyalty program only after the customer identity is unified across channels; a single-channel business already meets that condition.
- Measure incrementality with holdouts when the promotion budget is large enough that a 10% error in its effect costs more than withholding the offer from a few percent of customers.

The author's synthesis.

- The cheapest promotion architecture is governance: one owner for the combination settings, one calendar with an order of evaluation per campaign, and a rollout test against everything already active, because almost every promotion incident I have seen was a setting, not a system.
- Treat the allocated net line price as the order's primary price and the list price as metadata; every downstream system that reads the list price will eventually refund it.

Questions to Ask

1. Which system computes the final discount, and does the product page show that rule's estimate or its own?
2. What is the order of evaluation, where does stacking stop, and who may change that setting?
3. Does the order export carry the allocated net price per line to the warehouse, the returns tool and the ERP?
4. How does "first order" identify the customer, and what happens with a second e-mail address?
5. Can the pricing system answer "lowest price in the last 30 days" for any SKU and country when a promotion is published?
6. Where do gift-card and store-credit balances live, and who reconciles them to the liability account, how often?
7. Is a gift-card debit idempotent with the checkout attempt, or can a retry debit twice?
8. Which promotions were *offered* last quarter, not only redeemed, and could we run a holdout?
9. When a promotion ends, how does each copy (stores, marketplaces, feeds, cached pages) learn of it, and how do we check?
10. What did last Black Friday's promotions cost beyond the plan, and can we compute that from the order lines alone?

The price is now decided. The next question is how the customer found the product in the first place, and what has to be true of the catalog for a search box to answer "waterproof jacket": Finding the Product: Search, Navigation and Merchandising.

Sources and Evidence

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- ACM wil einde aan nepkortingen — Autoriteit Consument & Markt, 2022; supervision from 1 January 2023 and the worked example.
- ACM publiceert boetebesluiten webwinkels voor nepkortingen — Autoriteit Consument & Markt, 2024; the June 2024 fines.
- § 11 PAngV — Bundesministerium der Justiz, n.d., accessed 2026-11; the German transposition.
- IFRS 15 Revenue from Contracts with Customers — IFRS Foundation, n.d., accessed 2026-11; issue and effective dates, replacement of IFRIC 13.
- Unexercised rights (breakage), Revenue guide 7.4 — PwC Viewpoint, n.d., accessed 2026-11; the ASC 606 breakage rules.

- Revenue from customers' unexercised rights — IFRScommunity.com, 2024; secondary summary of IFRS 15 B44–B47 and material rights.

Research and practitioner sources

- Response Transformation and Profit Decomposition for Revenue Uplift Modeling — Gubela, Lessmann and Jaroszewicz, 2019; revenue uplift as the objective for coupon targeting.
- Interpretable Multiple Treatment Revenue Uplift Modeling — Gubela and Lessmann, 2021; several coupon types, one with positive incremental revenue.
- Promotion abuse in ecommerce — Ravelin, 2020; abuse taxonomy and a vendor survey figure.
- What is incrementality — Uniqodo, 2024; holdout method, vendor content.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

9. Finding the Product: Search, Navigation and Merchandising

IN THIS CHAPTER, YOU WILL: Learn how search, navigation, merchandising, recommendations and reviews help a shopper find a product, and what the catalog, prices and stock must provide for them to work.

KEY POINTS:

- **The search index is a copy, and every copy goes stale**. It is a separate, query-shaped copy of the catalog, the prices, the availability and the reviews, each of which changes at a different pace. Deciding what goes into the index and what is fetched fresh at query time is the central design decision.
- **Search is only as good as the data under it**. A zero-result query is usually a missing attribute, a missing synonym or an unindexed product, not a weak engine. Relevance tuning, merchandising rules and AI retrieval all sit on top of product data; none of them repairs it.
- **Reviews are discovery data with provenance rules**. A rating changes what ranks and what sells, so it belongs in the index; EU and US rules

require a merchant to know and be able to show whether a review came from a real purchaser.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Phantom Stock** — search results that say "in stock" from a copy of stock that is two hours old.
- **The Message in a Bottle** — index updates sent once and trusted, so a product changed in the catalog stays unchanged in the results.
- **The Lynchpin** — relevance configured inside one hosted vendor's console, by one agency, with no record of why.

A shopper filters for a waterproof jacket, but a suitable product disappears from the results because its waterproof rating exists only in a spreadsheet. A better ranking algorithm cannot fill that gap reliably. Search, category filters and recommendations all depend on which product facts reach the storefront and how those facts are structured.

Freshness matters too. Yesterday's price can mislead a shopper; an old stock figure can show a product the warehouse has sold out. The **search index**, the prepared set of records the engine searches, combines copies from several systems. Each copy needs an update and repair process, or the storefront can keep advertising a product it cannot sell on the displayed terms.

Reviews belong here for the same reason. A rating is a signal that ranks products and persuades shoppers, so it lives in the index next to the title and the price. But a review is also a claim about a purchase, and since 2022 EU law has required merchants to say whether reviews come from real purchasers and to check that they do. Reviews are discovery data with a legal provenance record.

The chapter covers the search index, **relevance** (how well results match the request), **merchandising rules** (how the merchant changes their order) and **facets** (filters built from product attributes). Product content, prices, promotions and stock come from their own owners: What Are We Selling? Products, Variants and the Catalog, What Does It Cost? Price Lists, Contract Prices, Currencies and Tax, Discounts, Coupons and Loyalty: Promotions Without Chaos and Is It in Stock? Inventory, Availability and the Promise You Can Keep. For deeper engine and ranking detail, see Search Architecture for Listings: Indices, Freshness and Scale and Ranking for Two Sides: Relevance, Freshness and Paid Visibility; agents are covered in Assistants and Agents: AI in Commerce and What Is Real in 2026.

Where we are in the order's journey. On 13 April 2027, Eva searches Kestrel & Pine's Dutch storefront for a jacket. Kestrel & Pine is the fictional outdoor brand used here. Before she can add the Ridgewalker to her cart, search must bring its description, price and stock status together.

Setting the Scene at Kestrel & Pine: The Waterproof Jacket Nobody Could Find

Kestrel & Pine uses its hosted platform's built-in search for 3,400 products and 26,000 variants across three storefronts. In March 2027, Data Lead Hanna Berg reviewed search results with Head of Engineering Noor Visser and catalog owner Teun Baas. The figures that follow are illustrative.

What the Q1 2027 search report showed *(illustrative)*	**Figure**
Sessions that used the search box	31%
Searches that returned no results (the **zero-result rate**)	9.4%
Rank of "waterproof jacket", "waterdichte jas" and "wasserdichte Jacke" among all queries	Top ten, in all three languages
Results for those queries	None
Why	The Ridgewalker is titled "shell jacket"; its 20,000 mm waterproof rating lives only in the seasonal spreadsheet; the platform's search indexes title, description, type and tags
Searches for "regenjas" (Dutch for rain jacket)	Zero results; no synonym
Ridgewalker shown "in stock" in search results after the warehouse sold out on 27 November 2026	Up to two hours, from the platform's stock copy

The report made three things visible. The best-selling product was invisible to the words shoppers used for it, because the attribute that answered their question was not in any system the search could read: Kestrel & Pine's missing PIM (product information management system, the system of record for product content), seen from the shopper's side. The engine had no synonyms, so the Dutch word for the product category returned nothing. And on the busiest evening of the year the results page had shown a copy of stock that was two hours old, which is the Black Friday Oversell as the shopper first met it.

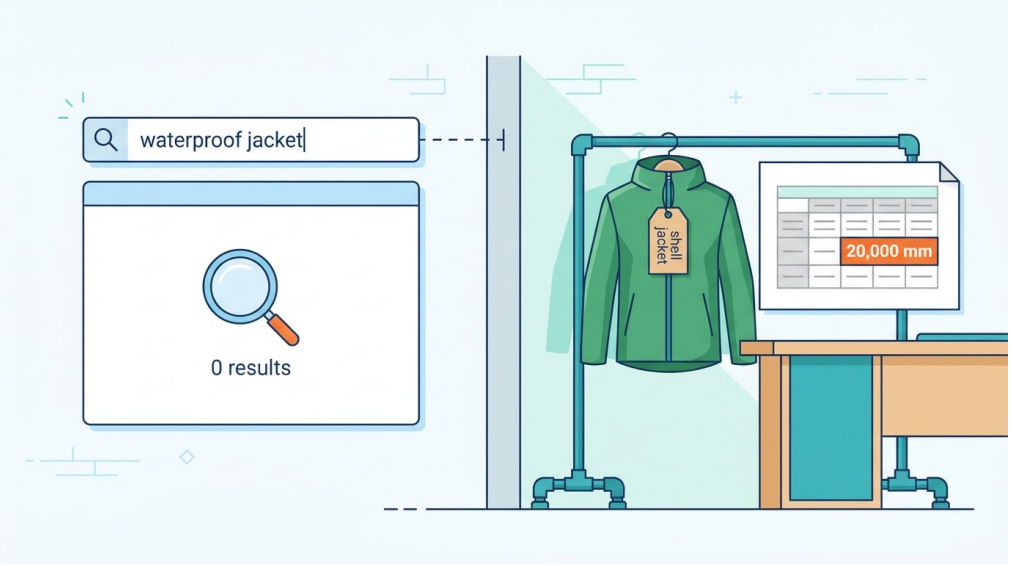

Figure 1: *Search ranks only what it can read: the best seller was titled shell jacket and its 20,000 mm rating lived in a spreadsheet, so waterproof jacket found nothing.*

The first fix was cheap: in March 2027 the storefront team added synonym sets per language (waterproof, rain, shell; regenjas, jas; Regenjacke, Jacke) in the platform's search settings. The second fix waited for the Waypoint program: when the PIM goes live in Q3 2027, "waterproofness" becomes an attribute every jacket carries and the search can filter on. By Q4 2027 the zero-result rate was 4.1%. The third fix is the subject of the next chapter: availability in search must come from the availability service, not from an hourly copy.

Three Ways to Find a Product

A shopper has a need; the storefront has 17,000 active variants. Discovery is whatever bridges the two.

Way of finding	What the shopper does	What the system needs	Who tunes it
Search	Types a query ("waterproof jacket M")	Indexed text and attributes; synonyms; typo tolerance; a ranking	The search team and merchandisers

Way of finding	What the shopper does	What the system needs	Who tunes it
Navigation	Clicks a category, then narrows with filters (size, color, price)	A category tree; complete attributes; **facets** (filters with counts) computed from the index	The catalog team
Recommendations	Accepts a suggestion ("often bought together")	Behavior data (views, purchases) and product relations	A model, with merchandiser overrides

Some vocabulary, in plain words. The **search index** is a separate copy of the catalog built for fast matching; it is not the catalog, and it is always slightly behind it. **Indexing** is the job that copies changes into it. **Relevance** is the engine's ordering of results for a query; **ranking** is the same thing seen as a list. A **facet** is a filter with counts ("Color: moss (14)") computed from indexed attributes. A **merchandising rule** is a human override of relevance for a query or category: pin a product to the top (boost), push one down (bury), hide one, redirect a query. A **synonym set** tells the engine that "jacket" and "coat", or "waterproof" and "rain", should match each other. **Typo tolerance** lets "ridgewalkr" find the Ridgewalker. **Personalization** reorders results for one shopper from what that shopper did before.

Baymard Institute's usability research, which benchmarks large retail sites against its own guidelines (article of September 2024, updated April 2026), reports that about half of test participants prefer search to navigation, that 56% of sites fail to adequately support users' search needs, and that failure rates rise with the query type: 12% of sites fail exact product-name searches, 39% fail feature searches ("waterproof"), 43% fail use-case searches ("jacket for hiking in rain") and 66% fail non-product searches ("returns"). These are expert reviews of mostly US and UK sites, not conversion measurements, so use them to name failure modes, not to size a business case. The pattern they show is the one at Kestrel & Pine: sites handle the product name and fail the words shoppers actually use.

How the Pieces Fit Together: The Index as a Copy

The diagram shows the search service as a consumer of four systems of record, each changing at its own pace, and the storefront asking it one question.

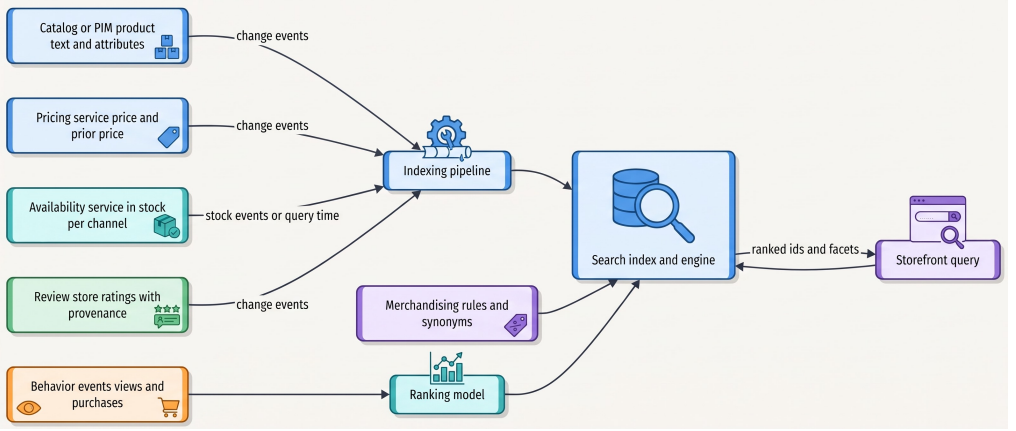

Figure 2: *The search index is a copy of the catalog, the prices, the availability and the reviews, fed by change events and overlaid with merchandising rules and a ranking model; the storefront gets back ranked identifiers and facets and fetches the fresh details separately.*

What goes in the index, and what is fetched fresh

The decision that shapes everything else is **which facts are in the index and which are joined at query time**. A fact in the index can be filtered and sorted on ("under €150", "in stock only", "best rated"), but it is only as fresh as the last indexing run. A fact fetched at query time from its system of record is fresh, but the engine cannot filter or sort on it. commercetools' Product Search (documentation accessed November 2026) shows the usual compromise: it indexes product fields, variant attributes declared searchable, prices and availability, returns product identifiers, and lets the client fetch full data separately; price filters work per currency, country, customer group and channel.

Fact	Changes how often	Freshness the shopper needs	Where it should live
Product title, description, attributes	Seasonally, with corrections	Hours	In the index
Category and assortment membership	Weekly	Hours	In the index

Fact	Changes how often	Freshness the shopper needs	Where it should live
List price and strike-through price	Daily; promotions by the minute	Minutes during a promotion	In the index for filtering; re-read from the pricing service on the product page
Availability per channel	Continuously; by the second at peak	Minutes, or at query time	A coarse flag in the index ("available"); the exact answer from the availability service at query time
Rating and review count	Daily	Hours	In the index
Behavior signals (views, purchases)	Continuously	Daily aggregates	In the ranking model, not the index

The availability row is where Kestrel & Pine was hurt. Its platform indexed the platform's own stock copy, which the stock sync chain updated hourly from the ERP (the enterprise resource planning system), which the WMS (the warehouse management system at Duiven) updated every fifteen minutes. On Black Friday evening the results page said "in stock" for up to two hours after Duiven ran out. The remedy is not a faster index; it is availability as a **stock event** (a message that stock changed, sent when it changes) from the availability service decided under Waypoint W1, with the index holding only a coarse flag and the product page asking for the exact number. Is It in Stock? Inventory, Availability and the Promise You Can Keep owns that design.

Figure 3: *The Phantom Stock seen from search: the index read a copy of a copy, so on Black Friday the results said in stock for up to two hours after the shelf at Duiven was empty.*

How the index is fed

Scheduled synchronization copies the catalog hourly or nightly. Its freshness depends on that interval and successful delivery. Algolia's documented Shopify business-to-business integration, for example, describes a daily resync, some changes taking up to two days, and product reindexing that is not automatic.

Event-driven indexing reacts to catalog, price and availability changes and can update affected records within seconds. It needs safe handling of repeated messages and a comparison job to find missed updates. Those are requirements of Kestrel & Pine's Waypoint integration standard, W5, explained in Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation.

Keep a **full rebuild** available with either approach. A new language, attribute or embedding model can require recreating the index as a whole.

Interaction	Freshness	Consistency	Latency	Availability
Search query	Index as of the last update	Eventual; the product page is the truth	Under about 200 ms for results and facets *(assumption)*	If search is down, navigation must still work
Attribute or text change	Minutes	Eventual	Seconds per event	Queue and retry
Price change during a promotion	Minutes	Eventual, with the pricing service re-read on the page	Seconds per event	Queue and retry
Stock change at peak	Minutes for the coarse flag	The availability service is the truth	Seconds per event; hundreds per minute at peak	Degrade to "check the product page"
Full rebuild	Hours	Atomic swap of the old and new index	Hours	Old index serves until the new one is complete

Relevance, merchandising and facets

A lexical engine matches words, with stemming (jackets → jacket), synonyms and typo tolerance; then it ranks the matches by text match, by popularity and by business rules. Elasticsearch's documentation (accessed November 2026) recommends search-time synonyms because the sets can be updated without reindexing; the distinction matters at Kestrel & Pine, where synonyms are per language and change every season.

Merchandising rules sit on top. Algolia's Rules (documentation accessed November 2026) can pin, hide, boost, bury, filter, rewrite the query or return custom data, carry up to 25 conditions, and have a validity period "so that promotions end automatically", with the vendor's own caveat that rules "are no substitute for a well-configured index". Shopify's Search & Discovery app allows at most 10 search terms per product boost. I treat merchandising rules as promotions: **data with an owner, a start, an end and a count**, reviewed on the same calendar as the campaigns they serve, because a rule that never expires is a price nobody can explain in a different costume.

Facets depend on attribute completeness, and the limits are concrete. Shopify's filters (documentation accessed November 2026) compute values and counts using only the first 1,000 results, show no filters on collections above 5,000 products, allow 25 filters per

store with up to 100 values each, and do not show a price filter in a currency other than the shop's default. A "waterproof" facet with half the jackets unfilled hides the other half from a shopper who ticks it, which is why the facet's quality is decided in the PIM, not in the search console. LISTED's Ranking for Two Sides: Relevance, Freshness and Paid Visibility covers ranking signals for marketplaces with two sides; a single-merchant storefront needs only a fraction of it.

Personalization and AI retrieval, in plain words

Collaborative filtering recommends items from patterns of what other shoppers viewed or bought together, without understanding the items. Amazon's item-to-item version was described in a 2003 paper and, by Amazon's own 2019 account of its history, had already been in use for six years; the same account notes that a deep-learning model for Prime Video first underperformed both the old method and a plain bestseller list, and only beat them after being reframed. The lesson is evidence, not nostalgia: **strong baselines are hard to beat, and reframing the problem beat scaling the model**.

Vector (semantic) search matches the meaning of a query to product text through numeric embeddings, so "something for rain" can match "shell jacket". **Hybrid search** runs a lexical and a vector query and merges the two ranked lists, commonly by reciprocal rank fusion, which rewards products that rank well in either. Elasticsearch (documentation accessed November 2026) runs both in one request and recommends approximate nearest-neighbor retrieval for production; OpenSearch introduced hybrid search in version 2.11; Algolia launched NeuralSearch on 2 May 2023; Shopify's semantic search reached its Shopify and Advanced plans on 24 June 2024. Amazon's 2019 research paper on semantic product search gives the reason: "pure lexical matching via an inverted index falls short" on hypernyms, synonyms, antonyms, morphological variants and spelling errors.

Two cautions. First, embeddings are a second derived artifact: Elasticsearch's documentation states that query vectors "must have the same dimension and be created with the same model as the document vectors", so a model change is a full reindex, and product text changes make embeddings stale just as they make keywords stale. Second, every conversion lift a search vendor publishes is a customer anecdote without a published control; Algolia's launch release reported a 65% drop in zero results and up to 17% conversion uplift for one customer in a four-week test, and Constructor's home page claims a 13% lift for one retailer with no baseline or method. Present them as "the vendor reports", and measure your own zero-result rate first.

Conversational search (a shopper describing a need in sentences) is the current frontier. Zalando announced a beta fashion assistant built on ChatGPT on 19 April 2023 for selected customers in four countries, and its Q3 2026 results release (3 November 2026) says, as of 2026, more than 22 million unique users visited the Zalando Assistant in the year. What such an assistant needs from the merchant (structured attributes, fresh prices

and availability, order status by API) is exactly what Waypoint W8 calls agent readiness; Assistants and Agents: AI in Commerce and What Is Real in 2026 takes it from there.

Search for wholesale: contract prices and assortments

For a wholesale account, search must answer "what can *this* buyer see, at *their* price". An **assortment** is the subset of the catalog an account may buy; a **contract price** is its negotiated price (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals owns both). Algolia's B2B guidance (accessed November 2026) gives the three designs: with fewer than about 100 pricing levels, store one price per tier or customer group in the index and sort by it; with more, keep per-customer prices out of the index and merge them at query time, accepting that they cannot be filtered or sorted on; or do both. Its Shopify B2B integration stores per-company-location prices in a hidden attribute that only the logged-in location can retrieve and recommends a ceiling of about 50 company locations before records hit size limits.

The rule that matters is about security, not speed: **assortment visibility must be enforced in the search service, with the filter bound to the account's token, not in the storefront**, because search results are an information leak. Kestrel & Pine's 1,100 accounts, on a handful of price tiers *(assumption)*, fit the segment-price design; a merchant with per-account prices for thousands of accounts does not.

Reviews and Ratings as Discovery and Trust Data

A review does two jobs. As **discovery data**, the average rating and the count rank products and persuade shoppers; schema.org's `Product` type (version 30.1, September 2026) carries `aggregateRating`, so the rating is also syndicated to search engines and shopping feeds. As **trust data**, a review is a claim that someone bought and used the product, and that claim is regulated.

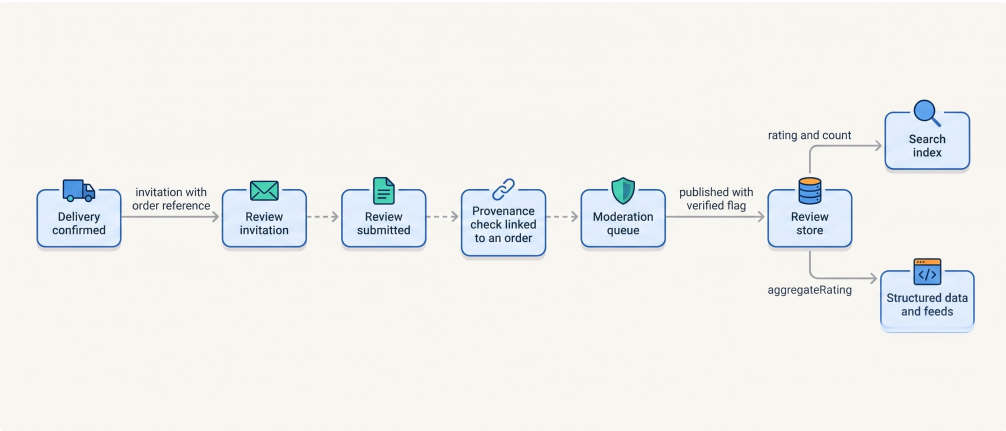

Figure 4: *A review enters through an invitation tied to a delivered order, carries its provenance through moderation, and reaches the index and the feeds as a rating with a verified-purchase flag the merchant can prove.*

Directive (EU) 2019/2161, applied from 28 May 2022, makes it material information whether reviews come from "consumers who have actually used or purchased the product", and blacklists both stating that reviews are from real purchasers "without taking reasonable and proportionate steps to check" and "submitting or commissioning another … to submit false consumer reviews"; its recital 19 defines ranking broadly, as the relative prominence given to offers "including resulting from the use of algorithmic sequencing, rating or review mechanisms, visual highlights, or other saliency tools", and the directive makes the main parameters determining ranking material information too. In the United States, the Federal Trade Commission's rule of 14 August 2024 bans fake or AI-generated reviews, compensation "conditioned on the writing of consumer reviews" with a particular sentiment, undisclosed insider reviews and review suppression, with civil penalties. Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation places these in the wider calendar.

Four design consequences:

- **Provenance per review**. Every review records how it was collected and, where it is marked verified, the order it is linked to. Kestrel & Pine's reviews app invites customers after delivery with the order reference, so each of its 6,800 reviews in 2026 is tied to an order (illustrative); a review form open to anyone can still exist, but its reviews cannot carry the verified flag.
- **Moderation is a queue with rules, not a judgement call**. Reject on policy (off-topic, personal data, abuse), never on sentiment; the FTC rule names suppression of negative reviews as a violation.
- **Incentives are disclosed and never conditional on sentiment**. A discount code for "a review" is lawful in most places if disclosed; a code for "a five-star review" is not.

- **Ranking parameters are documented.** If "best rated" or "most popular" sorts the results, the merchant must be able to say what the sort does; that is a one-page document, kept with the merchandising rules.

Marketplaces keep their own reviews under their own rules; a merchant selling on bol, Amazon.de and Zalando has three review pools it does not own, which Selling on Someone Else's Platform: Marketplaces and Channel Integration covers. For a merchant of Kestrel & Pine's size, reviews are a case for a specialist app or service with the provenance and moderation built in, and the architectural work is making sure the rating reaches the index and the feed, and that the order link survives a platform or app change.

The Choices

The platform's built-in search. Hosted platforms ship search, filters, synonyms, boosts and, since 2024 on some plans, semantic understanding. It is free, it indexes the platform's own copy of price and stock, and its limits are published (Shopify's 25 filters and first-1,000-results counts above). It fits a merchant whose catalog fits the platform's product model and whose promises can tolerate the platform's indexing delay.

A hosted search service fed by events. Algolia, Constructor, Bloomreach and others run the index and the engine as a service; the merchant feeds them from the catalog, the pricing service and the availability service, and merchandisers tune relevance in a console. This buys relevance tooling, analytics and hybrid retrieval, and costs an integration per source, a subscription, and relevance configuration that lives in a vendor's console unless it is exported and versioned, which is the Lynchpin risk.

A self-run engine. Elasticsearch or OpenSearch, managed or self-hosted, with the index design, analyzers, synonyms, ranking and the pipeline built by the merchant's team. It is the right choice when relevance is a competitive asset, the catalog does not fit a hosted service's record model, or the data cannot leave the merchant's control. It costs a team that owns search as a product.

A search module of a composable or marketplace stack. Larger merchants and marketplaces run discovery as its own platform, with per-brand indexes and ranking experiments; LISTED's Search Architecture for Listings: Indices, Freshness and Scale describes that shape. It is a Group-stage answer.

Criterion	Platform built-in	Hosted search service	Self-run engine
Who owns relevance	The vendor's defaults plus a few boosts	Merchandisers, in the vendor's console	Your team, in code and configuration
Who owns the data pipeline	The platform	You, per source	You, per source
Freshness of price and stock	The platform's copy and schedule	As fresh as your events	As fresh as your events
B2B prices and assortments	Limited	Segment prices in the index; per-account at query time	Any design
Hybrid or semantic retrieval	Plan-gated, English first	Built in since 2023–2024	Elastic kNN and hybrid; OpenSearch 2.11+
Exit cost	None	Relevance configuration and analytics history	The whole system
Fits	Starting, Growing	Growing, Omnichannel	Omnichannel with a search team; Group

My rule: **measure the zero-result rate and the top failing queries before buying anything**. Half of Kestrel & Pine's problem was fixed with synonym sets in the platform it already had; the other half waits for the PIM, which no search vendor can replace. A hosted service earns its subscription when the merchant has merchandisers who will use the console weekly and a catalog with the attributes to tune against. A self-run engine earns its team when search is a product, not a feature.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
1 — Starting	The platform's built-in search; one language; categories and a few filters; reviews from an app	A monthly look at zero-result queries; synonyms for the five words shoppers use that the catalog does not	Shoppers search for a word in the product's name and find nothing; a facet with half the products unfilled
2 — Growing	A second language or market; the first marketplace; a hosted search app with daily sync; review invitations after delivery	Synonym sets per language; attributes filled before a facet is switched on; a zero-result and click-position report per language	Search results show prices or stock from yesterday; merchandising rules that nobody remembers creating; reviews that cannot be linked to an order
3 — Omnichannel	Several markets, a PIM, a B2B portal, an app and store associates all querying products; promotions by the minute	Event-driven indexing from the PIM, pricing and availability services; a coarse availability flag with query-time truth; assortment and contract-price filtering enforced in the search service; one search service for storefront, app and store tools	A hosted service's record limit hit by B2B prices; "in stock" in results and sold out on the page; a vendor console nobody can reproduce in a new environment

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
4 — Group	A discovery platform shared by brands; per-brand indexes and relevance; ranking experiments with holdouts; a review platform with group-wide moderation	A search team that owns the pipeline and the experiments; a documented ranking policy per brand and market; model and index versioning for embeddings	Brands fighting over shared relevance defaults; an embedding model change that silently degrades one brand's results

The judgement: the engine matters less with each stage than the pipeline feeding it. A Starting merchant is limited by vocabulary and fixes it with synonyms; a Growing merchant is limited by staleness and fixes it with events; an Omnichannel merchant is limited by how many consumers (web, app, store associates, wholesale) must see the same products at their own prices and stock, and fixes it with one search service and query-time truth; a Group is limited by ownership. Kestrel & Pine sits in the third row with second-row tooling, which is the normal condition of a merchant that grew faster than its systems.

In Practice

Deprecation is a search risk too. commercetools' older Product Projection Search is deprecated: projects created after 31 August 2026 cannot activate it, and it deactivates after 30 days without calls (documentation accessed November 2026). A search integration built on a retired API is Kestrel & Pine's Deprecated Endpoint of February 2027 in another system; Promises Between Systems: API Contracts, Webhooks and Versioning covers the discipline.

Limits are documented, so read them. Shopify's filters use the first 1,000 results for counts and allow 25 filters per store; Algolia's Shopify B2B integration suggests about 50 company locations before records hit size limits and stores only the highest volume tier; commercetools indexes up to 10,000 variants and standalone prices combined per product in its Modular Catalog. Each limit is a design constraint a merchant discovers at the worst moment unless it is read first.

Metrics have definitions. Algolia's analytics (documentation accessed November 2026) define the no-results rate as the percentage of searches with no results, click-through

rate as the percentage of tracked searches with at least one click, conversion rate as the percentage with at least one conversion event, and average click position, where "smaller values are better"; keystrokes within 30 seconds aggregate into one search. Whatever engine a merchant uses, these four, per language and per channel, are the dashboard. Baymard's data show mobile and app sites performing worse than desktop, so a single blended number hides the worst channel.

Kestrel & Pine's 2027 search plan used the existing vendor. Noor added synonyms per language and a monthly review of the top 50 searches returning no results. The catalog team would model waterproof rating, fabric, insulation and fit in the new product information management system (PIM). Filters would appear only after 95% of jackets had values.

The plan also replaced indexed stock counts with a broad availability flag, leaving exact checks to the product page, and retained order links in exports of the 6,800 reviews. In this illustration, the zero-result rate fell from 9.4% to 4.1% by Q4, mainly through synonyms. The work took a few configuration days and an attribute model, using a PIM already approved for other needs.

When It Goes Wrong

The index says in stock; the warehouse says no. Detection: a daily sample of results pages against the availability service, and the rate of "add to cart" failures on items shown available. Recovery: a coarse flag in the index and the exact answer at query time; at peak, a rule that any product sold out in the last hour is re-indexed immediately. This is the Phantom Stock in its first outfit.

A change never reached the index. A product renamed in the catalog still shows its old name; a price change made at 09:00 appears at noon; a discontinued variant still ranks. Detection: a reconciliation job that compares a sample of index records with their sources every hour, the same job Waypoint W5 requires of every integration. Recovery: re-index the differing records, and fix the lost event, which is the Message in a Bottle.

A full rebuild is needed and nobody planned for it. A new attribute, a new language or a new embedding model requires rebuilding every record. Detection is easy; the failure is doing it in place and serving a half-built index. Recovery: build the new index beside the old one and swap atomically; keep the old one until the new one is verified.

Synonyms in the wrong language. A synonym set written for Dutch applied to the German index, so "jas" matches nothing and "Jacke" matches everything. Detection: zero-result and click-position metrics per language. Recovery: one synonym set per language index, owned by someone who speaks it.

A merchandising rule outlives its campaign. Last season's pinned jacket still tops "jacket" in April. Detection: a list of active rules without an end date. Recovery: validity windows on every rule, and the same monthly review as promotions.

Review provenance is lost. The reviews app is replaced, the export has the text and the stars but not the order reference, and the "verified purchase" badge can no longer be substantiated. Detection: a count of verified reviews whose order cannot be found. Recovery: keep the order link in your own store, not only in the app's; drop the badge where it cannot be proven.

The assortment leaks. A wholesale account's search returns products from another tier, or a retail visitor's search returns B2B prices, because the filter was applied in the storefront, not in the search service. Detection: a test account per tier that queries the service directly. Recovery: token-scoped filters in the service; treat it as a security incident, which it is (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers).

The hosted search service is down. Detection is immediate; the question is what the storefront does. Recovery: navigation and category pages must work without search, served from the platform, and the search box must fail with a message rather than an empty result, which shoppers read as "you do not sell it". Black Friday Every Year: Peak Load, Resilience and Recovery covers the general case.

What to Take Away

Established.

- The search index is a derived copy of the catalog, prices, availability and reviews; it is always behind its sources, and the design decision is which facts are in it (filterable, sortable) and which are fetched fresh at query time (commercetools', Algolia's and Shopify's documentation, accessed November 2026, all show this shape).
- Zero-result rate, click-through rate, conversion rate and click position, per language and channel, are the measures of search; a zero-result query is most often a data gap, not an engine fault (Baymard 2024–2026; Algolia's metric definitions).
- Hybrid retrieval (lexical plus vector, merged by reciprocal rank fusion) is now standard in Elasticsearch, OpenSearch (2.11), Algolia (2023) and Shopify (2024); embeddings are a second derived artifact that must be rebuilt when the model or the text changes.
- Since 28 May 2022, EU merchants must be able to show whether reviews come from real purchasers and must take reasonable steps to check; the main parameters determining ranking are material information (Directive 2019/2161). The FTC's 2024 rule bans fake, incentivized-by-sentiment and suppressed reviews in the US.

Depends on context.

- The platform's built-in search is enough while the catalog fits the platform's product model, the storefront is in one or two languages, and the indexing delay is shorter than the promises on the page; the first promotion-by-the-minute or the first B2B tier breaks that condition.
- A hosted search service pays for itself where merchandisers will tune relevance weekly and the catalog has the attributes to tune against; without a PIM or equivalent, buy the attributes first.
- Put availability in the index as a coarse flag only when the exact answer is available at query time from an availability service; without one, a stale "in stock" in the index is worse than no flag.
- Enforce B2B assortment and contract-price visibility in the search service once more than one tier exists; a single-tier wholesale business can do it in the storefront for a while.

The author's synthesis.

- Treat search as the first consumer of the PIM and the availability service, and fund those before any search vendor; every relevance feature I have seen bought ahead of its data was tuned against gaps.
- Keep merchandising rules, synonyms and ranking parameters as versioned data with owners and end dates, outside any vendor console, because they are the part of search a merchant cannot buy back.

Questions to Ask

1. What is our zero-result rate per language, and what are the top twenty queries that return nothing?
2. Which facts are in the search index, and which are fetched fresh at query time? How old can the price and the stock in a results page be?
3. How does a catalog change reach the index: on a schedule, or as an event with retry and reconciliation?
4. Can we rebuild the whole index beside the live one and swap, and when did we last do it?
5. Who owns the synonym sets per language, and where are the merchandising rules recorded, with their end dates?
6. Which attributes do our facets depend on, and what share of products carries a value for each?
7. For wholesale accounts, where is the assortment filter enforced, and has anyone tested it with a token from another tier?
8. For every published review, can we show how it was collected and, if it is marked verified, which order it came from?

9. If we sort by "best rated" or "most popular", can we write down in one page what that sort does?
10. If the search service is down for an hour on Black Friday, what does the storefront show?

Search has shown Eva a Ridgewalker in size M, moss, "in stock". The next chapter asks what that last phrase means, who computed it, and how to make it a promise the warehouse can keep: Is It in Stock? Inventory, Availability and the Promise You Can Keep.

Sources and Evidence

Company, product and investor sources

- Product Search and Product Projection Search (deprecated) — commercetools docs, n.d., accessed 2026-11; what is indexed, identifiers returned, price filters per scope, the deprecation dates.
- Limits — commercetools docs, n.d., accessed 2026-11; indexed variants and prices per product.
- Shopify B2B integration — Algolia docs, n.d., accessed 2026-11; daily resync, per-location prices, the 50-location ceiling.
- Personalized pricing for B2B — Algolia docs, n.d., accessed 2026-11; the three B2B price designs.
- Rules overview — Algolia docs, n.d., accessed 2026-11; merchandising rule consequences, conditions and validity periods.
- Search analytics metrics — Algolia docs, n.d., accessed 2026-11; metric definitions.
- Algolia launches NeuralSearch — Algolia, 2023; launch date and a vendor-reported customer result.
- Filters and Modifying search — Shopify Help Center, n.d., accessed 2026-11; filter limits, boosts, semantic understanding.
- Semantic search is now available on more plans — Shopify changelog, 2024; the 24 June 2024 rollout.
- Constructor — Constructor, n.d., accessed 2026-11; vendor claims, cited as such.
- Zalando launches fashion assistant powered by ChatGPT — Zalando, 2023; the April 2023 beta.
- Zalando Q3 2026 results — Zalando, 2026; 22 million unique Assistant users.
- The history of Amazon's recommendation algorithm — Amazon Science, 2019; item-to-item filtering and the Prime Video baseline story (older than three years; historical).

Standards, regulation and official documentation

- Directive (EU) 2019/2161 (Omnibus) — Publications Office of the EU, 2019; review provenance, fake-review blacklist, ranking as material information, recital 19.
- FTC announces final rule banning fake reviews and testimonials — Federal Trade Commission, 2024; the rule's prohibitions.
- schema.org Product — schema.org, 2026 (v30.1); `aggregateRating` and `offers`.
- kNN search, Hybrid search and Search with synonyms — Elastic docs, n.d., accessed 2026-11; vector model constraints, hybrid retrieval, search-time synonyms.
- Hybrid search — OpenSearch docs, n.d., accessed 2026-11; hybrid search since 2.11, normalization and reciprocal rank fusion.

Research and practitioner sources

- Semantic Product Search — Nigam et al., Amazon, KDD 2019; why lexical matching falls short and the reported gains.
- Ecommerce search query types — Baymard Institute, 2024, updated 2026; failure rates by query type and the search-vs-navigation preference.
- Ecommerce search report and benchmark — Baymard Institute, n.d., accessed 2026-11; autocomplete and misspelling findings.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

10. Is It in Stock? Inventory, Availability and the Promise You Can Keep

IN THIS CHAPTER, YOU WILL: Learn what "in stock" really means, who should compute it, and why Kestrel & Pine oversold 1,340 jackets on one Friday evening.

KEY POINTS:

- **"In stock" is a computed promise, not a count.** Physical stock (on hand) is a fact owned by whoever counts it. What a channel may still sell (available-to-promise) is derived: on hand minus units promised to orders minus units held back on purpose. Every oversell is a disagreement between the fact and the derivation.
- **Oversells come from writers and lag, not from bad luck.** Several systems writing the same stock number, periodic copying instead of events, and reservation windows let two channels sell the same last unit. The fix is one writer per stock state and stock events that travel in seconds.

- **One availability service, many channels, each with its own buffer.** With more than one location and more than one channel, availability belongs to a service that receives stock events, applies per-channel rules and publishes an available figure per channel. Kestrel & Pine buys it as a module of its order management system (Waypoint W1).

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Phantom Stock** — two channels sold the same last Ridgewalker because each learned of the other's sales up to two hours late.
- **The First Snowfall** — the stock sync chain had never met a Friday with 38 orders a minute; at that rate a one-hour copy interval is 2,300 orders of blindness.
- **The Secret Agent** — a pre-order app wrote inventory in the platform while the warehouse feed wrote it too, and for two weeks nobody owned the number.

A web shopper and a store customer try to buy the last jacket within the same minute. Both channels show it as available, but there is only one unit on the shelf. If each sale relies on its own stock count, both can succeed and the merchant must disappoint one customer. The channels need a shared way to decide which sale commits that unit.

A shelf count alone cannot answer the question. Some units already belong to unshipped orders; others are held back as buffers, damaged goods or wholesale allocations. Cart reservations and incoming deliveries add further conditions. **Availability must combine those facts before a channel commits to a sale.** Copying a stock number along a chain of scheduled updates leaves channels selling from different answers.

This chapter explains when stock becomes committed, how locations and channels share it, and when a merchant needs a separate availability service. Kestrel & Pine, the fictional outdoor brand, supplies the worked example: its worst evening of 2026.

Where we are in the order's journey. Eva is about to add a Ridgewalker jacket to her cart. Availability supplies the badge she sees, the cart's stock check and the commitment at checkout that prevents another shopper buying her unit. From Cart to Confirmation: Checkout Orchestration covers order creation; Pick, Pack, Ship: Warehouse Management and Fulfillment covers picking it.

Setting the Scene at Kestrel & Pine: Friday 27 November 2026, 19:40

In January 2027 Noor Visser, Kestrel & Pine's new Head of Engineering, asks Ruben Alves, the lead integration engineer, and Daniel Okafor, Head of Operations, to

reconstruct the Black Friday Oversell minute by minute. The mechanism was never written down. What they draw on the whiteboard is a chain.

Duiven's **warehouse management system (WMS)** knows which bin holds each unit and directs picking. It reduces physical stock when a unit is picked. In 2026, it sent counts every 15 minutes to the **enterprise resource planning system (ERP)**, which handles finance, purchasing and stock valuation. The ERP copied them hourly to the commerce platform; the platform copied them hourly to the marketplace connector for bol, Amazon.de and Zalando.

A storefront sale reduced the platform's count immediately. Other channels learned of it only after picking and the subsequent updates, up to two hours later. Marketplace sales reached the storefront through the same slow chain.

On Friday 27 November the Ridgewalker jacket, the hero product, was the deal of the week. The reconstruction is illustrative; the totals are those recorded in Meet Kestrel & Pine.

Time (27 Nov 2026)	What happened	What the channels believed about the Ridgewalker
19:40	The WMS picks the last units; physical on hand reaches zero	Storefronts and marketplaces: in stock
19:45	WMS sends stock to the ERP (15-minute run)	ERP: zero. Platform and marketplaces: in stock
20:00–21:00	Peak hour: 2,300 orders, 38 per minute	Storefronts and marketplaces keep selling it
about 20:30	ERP sends stock to the platform (hourly run) *(illustrative time)*	Storefronts: sold out at last
21:05	Milan Petrov, alerted by the warehouse through Daniel Okafor, sets the Ridgewalker to zero in the marketplace connector by hand; the hourly platform-to-connector run would have arrived later still	Marketplaces: sold out once the marketplaces process the push
Saturday	Support discovers what was sold against zero stock	1,340 units oversold, 1,120 orders affected

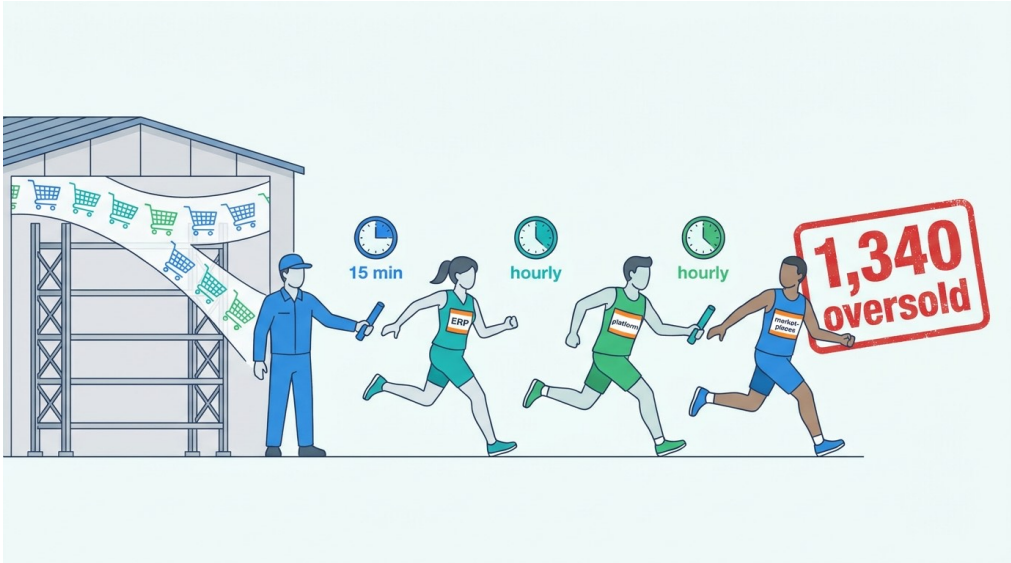

Figure 1: *The Black Friday Oversell as a relay: each hand-off waited for its schedule, so the channels kept selling for up to two hours after Duiven ran out, and 1,340 units were oversold.*

Two things made it worse. The order queue from the platform to the WMS backed up for four hours that evening (Pick, Pack, Ship: Warehouse Management and Fulfillment), so the WMS's count fell later than the sales it reflected. And three of the platform's 41 installed apps could write inventory. One, a pre-order app, had already caused the App That Wrote Stock in March 2026: two weeks of phantom reservations on about 600 variants, because the app and the WMS feed both wrote the same number and each undid the other.

Mara Lindqvist, the CEO, asked the question every CEO asks after an oversell: "Why did the website not know?" Noor's answer, which became Waypoint decision W1, was that the website could not know, because nobody had decided who owned the number and nothing carried changes fast enough. This chapter is that answer in full.

What "In Stock" Means: Five Numbers, Not One

Every system that tracks stock separates the units that are physically present from the units it may still promise. The names differ between vendors; the arithmetic does not.

- **On hand** is the number of units physically at a location right now. It is a fact, owned by whichever system counts it: a WMS, an ERP's warehouse module, a point-of-sale system (**POS**, the till software) for a store, or the platform itself for a merchant with a back room.

- **Committed** (also **allocated**) is the part of on hand already claimed by placed orders that have not shipped. Eva's jacket becomes committed when her order is confirmed and stays committed until it ships.
- **Reserved** is a temporary claim before an order is final: a unit in a cart during a limited-stock sale, or in a draft order a sales rep is building. Reservations expire; commitments do not.
- **Held back** covers units present but deliberately not for sale: a **safety stock** buffer against counting errors and sync lag, damaged units, units awaiting quality control, units set aside for a wholesale customer.
- **Incoming** is stock on the way in: a supplier's shipment or a transfer from another location. It is not on hand until received.

From these, the number a channel may actually sell is derived. In planning vocabulary that derived number is **available-to-promise (ATP)**: the uncommitted portion of stock, and sometimes of planned receipts, that can be promised to new orders. Commerce platforms implement a simple single-period version of it, usually called "available" or "salable". As industry practice, the formula is:

> available-to-promise = on hand − committed − reserved − held back (+ incoming, only if you promise against it)

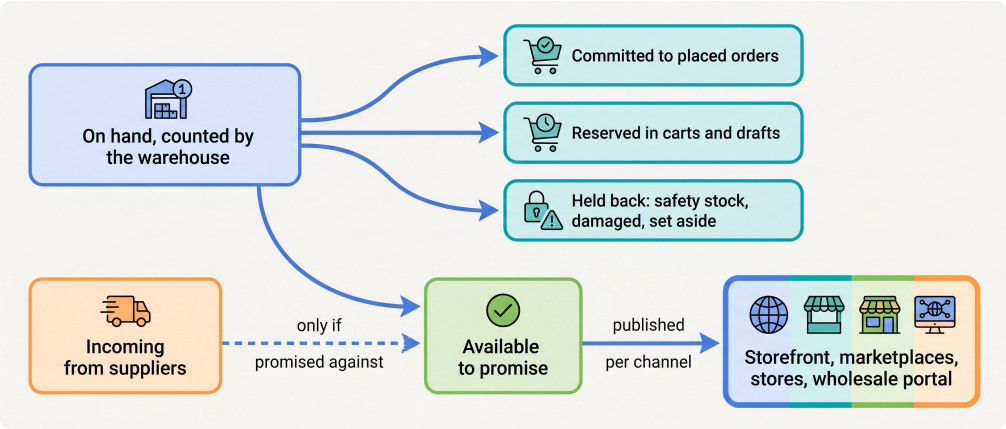

Figure 2: *On hand is the one counted fact; committed, reserved and held-back units are claims against it; available-to-promise is what remains, and it is a derivation that each channel should receive, not a number each channel should keep.*

The vendors' own vocabularies show that this is not one platform's idea. The table aligns four of them; the alignment is the author's, none of the vendors publishes such a mapping.

Concept	Shopify (developer docs, n.d.)	commercetools (API docs, n.d.)	Adobe Commerce (docs, updated 2026-08)	SAP Commerce (2022 API docs)
Physical units at a location	`on_hand`	`quantityOnStock`	stock quantity	actual stock level per warehouse
Promised to placed orders	`committed`	reserved, implicit	reservations against salable quantity	reserved via `reserve()`
Sellable now	`available`	`availableQuantity`	salable quantity per sales channel	available amount
Held back deliberately	`safety_stock`, `damaged`, `quality_control`	none built in	none built in	none built in
Location	location	supply channel	source, grouped into stocks	warehouse

Shopify's developer documentation (n.d., accessed November 2026) states the arithmetic plainly: on hand "equals the sum of inventory quantities that are in the available, committed, reserved, damaged, safety_stock, and quality_control states", and incoming is not part of on hand. Its POS gained "On hand" and "Unavailable" states on 9 February 2023, with the stated aim of reducing accidental overselling. And on 5 August 2026 Shopify moved the holds for draft orders and transfers from `reserved` to `committed`, leaving `available` and `on_hand` unchanged: an integration that subtracted `reserved` in its own calculation changed behavior that day without any code change. Even the meaning of a stock state can move under the systems that read it (Promises Between Systems: API Contracts, Webhooks and Versioning).

The warehouse owns on hand. The order-taking system owns committed. Nobody should own "available", because it is a computation. Most oversells begin when a system stores "available" as a fact and lets another system overwrite it.

When a Unit Becomes Promised: Reservations and Allocations

The platforms disagree about *when* a unit leaves the available number: when it goes into a cart, when the order is created, or when the warehouse allocates it to a pick. This is a business choice with a cost on each side, and the vendors expose it as configuration.

commercetools lets each cart choose an inventory mode (API documentation, n.d., accessed November 2026): `TrackOnly` lets an order be created "even if the Inventory Entry quantity is zero or negative"; `ReserveOnOrder` reserves line items "only for the duration of the ordering transaction" and returns an out-of-stock error if a line is gone when the order is created; `ReserveOnCart` reserves units at add-to-cart, releases them on removal and expires them after a configurable 1 to 44,640 minutes. The same documentation states that cart- and order-driven quantities are eventually consistent with the internal availability used for reservations, with a delay of up to 10 seconds. Adobe Commerce reserves "when submitting an order", creates a compensating reservation on cancellation and deducts from the specific source when the order ships (documentation, updated 20 August 2026). Shopify commits units at order creation and keeps them committed while an order is on hold (help center, n.d.).

Moment the unit is promised	Protects	Costs	Fits
Add to cart (reserve-on-cart, with expiry)	The shopper's promise: what is in the cart will be there at checkout	Stock tied up in abandoned carts; expiry and release logic	Scarce-stock drops, limited editions
Order creation (reserve-on-order; the default in all four platforms above)	Stock utilization: only real orders claim units	"Sold out" at the last click; a window of seconds in which two checkouts can both succeed	Most merchants, most of the time
Allocation at fulfillment (the warehouse claims units when it releases the pick)	Warehouse flexibility: late sourcing, substitutions	Orders accepted that cannot be fulfilled; the oversell surfaces at the pick	Wholesale and backorder flows where confirmation comes later

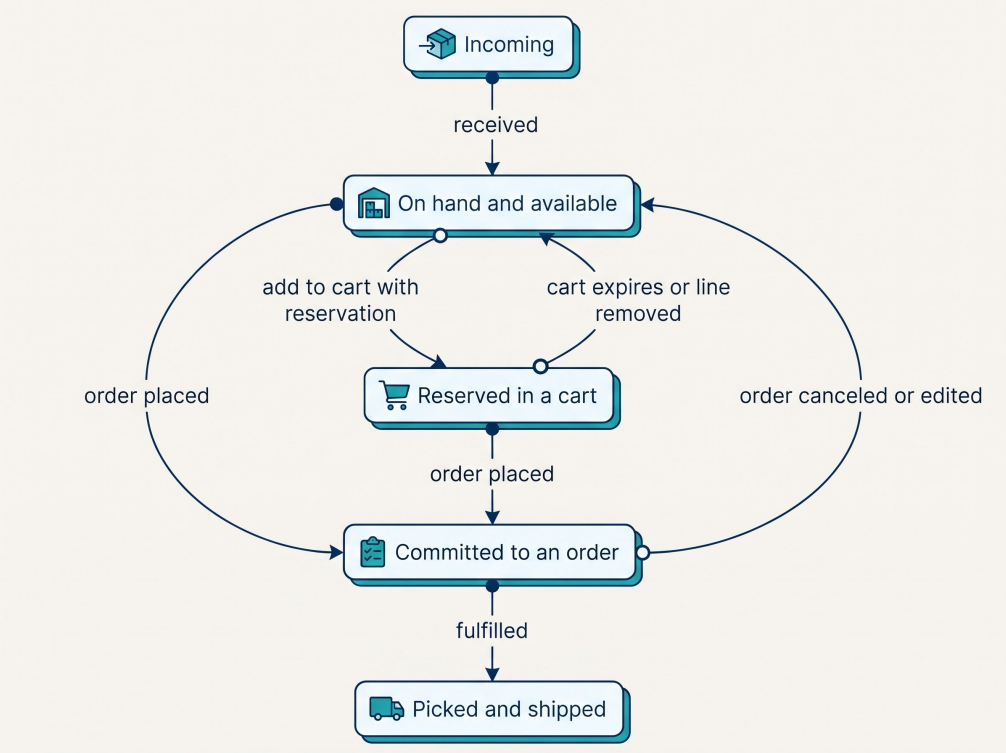

Figure 3: *A unit's life from the merchant's point of view: it is available until a cart reserves it or an order commits it, returns to available if the cart expires or the order is canceled, and leaves the count when it ships.*

commercetools' 10-second window is small, published and honest. It is also enough to oversell a hot item during a drop: at Kestrel & Pine's peak rate of 38 orders a minute, 10 seconds is six orders. **Every platform has such a window**; **most do not publish it**. Ask.

Many Locations, Many Channels

Everything above assumes one shelf and one shop. Kestrel & Pine has neither. Its stock sits at Duiven, in six stores of about 1,800 variants each, and in a marketplace's fulfillment program warehouse in Germany, which holds about 40% of the units it sells on that marketplace there. Two new concepts enter.

A **location** is any place from which stock can be promised and shipped: an own warehouse, a store, a third-party logistics provider (**3PL**, a company that stores and ships on the merchant's behalf), or a marketplace's fulfillment center. Each location has its own on-hand count and its own counting system. The marketplace's warehouse is the hardest case: the merchant sees the count the marketplace reports but cannot count it, so it must be modeled as an external location with its own buffer.

A **channel** is a place where a promise is made: a storefront, a marketplace listing, a store till, the B2B portal (business-to-business: the wholesale portal), a click-and-collect option (order online, pick up in a store). Availability is published per channel, and one physical pool may be promised to several channels at once. Three rules follow, all industry practice rather than any vendor's doctrine.

Availability per channel, with a buffer per channel. A marketplace oversell is not just a cancellation; it is a seller-performance penalty and, repeated, a suspended listing. So marketplaces usually get the largest safety buffer, so that the last units are sold where an oversell costs least. The buffer costs margin on the units it withholds, which is why it is a business setting, not an engineering constant.

Figure 4: *One pool, many promises: every channel reaches into the same stock, so availability is published per channel, with the largest buffer where an oversell costs most.*

Sourcing is a separate decision. Once a line can be fulfilled from more than one location, someone must decide which. Adobe Commerce ships two source-selection algorithms, source priority (a configured list) and distance priority (nearest source to the shipping address), per its documentation updated 20 August 2026. Shopify assigns groups of lines to a location through a *fulfillment order* (developer docs, n.d.). The moment sourcing needs capacity, cut-off times, carrier cost or split-shipment limits, the decision moves to an order management system, The Order Record: Lifecycle, State and

the Order Management System's subject.

Wholesale needs allocations, not just availability. A wholesale buyer who ordered 240 jackets for spring should not compete with Friday-evening shoppers for them. An **allocation** reserves part of the pool for a customer, a season or a channel in advance, so the storefront sees on hand minus committed minus allocations. None of the four platforms above holds per-customer allocations; they are order management territory (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals covers the commercial side).

Two further cases promise stock that is not on hand. A **backorder** accepts an order for a unit that is out but expected; a **pre-order**, for a unit not yet released. Both are legitimate only if the incoming quantity is tracked and the date is honest; Directive 2011/83/EU, Article 18, requires delivery within 30 days unless agreed otherwise, so an open-ended backorder is a legal risk too. **Inventory valuation**, what the stock is worth in the books, stays in the ERP; it needs cost prices and accounting rules the commerce side should never hold (The Back Office: ERP Integration and Who Is the Hub).

How the Pieces Fit: The Availability Service

With several locations and channels, the clean arrangement separates three responsibilities and connects them with events. A **stock event** is a message sent the moment a stock quantity changes, saying which stock-keeping unit (**SKU**, one sellable variant such as the Ridgewalker women's M in moss), at which location, changed by how much, why, and when. A **queue** holds events until each consumer has processed them.

Component	Responsibility	Writes	Never writes
Stock-counting systems (the WMS at Duiven, the POS in each store, the marketplace fulfillment program's reports)	Count physical units; emit stock events on receipt, pick, adjustment, count	On hand per SKU per location	Committed, available

Component	Responsibility	Writes	Never writes
Order-taking systems (the platform, the marketplace connector, later the order management system, OMS)	Commit units at order creation; release on cancellation; emit order events	Committed per SKU	On hand
Availability service	Hold the latest on hand and committed per SKU per location; apply reservations, allocations and per-channel buffers; compute and publish available per channel	Available per channel (a derived, republished value)	On hand, committed
Reconciliation job	Compare each counting system's on hand and each order system's committed with the service's copies; report differences	A variance report	The numbers it compares

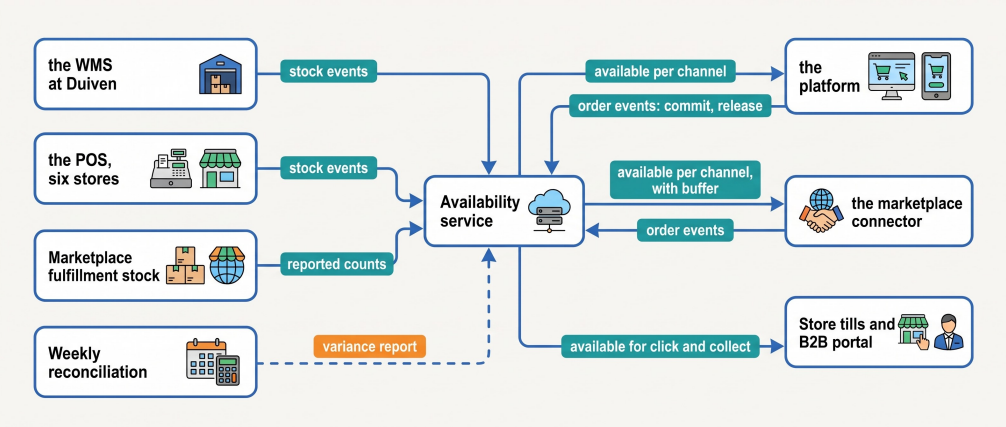

Figure 5: *The Waypoint target: counting systems send stock events, order-taking systems send commitments, one availability service derives available per channel and publishes it back, and a reconciliation job checks the copies against the counts.*

This is Waypoint decision W1 drawn as a diagram. The WMS is the system of record for physical stock at Duiven (the **system of record** is the one system whose value is the truth when copies disagree; Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record owns the idea). The POS is the system of record for store stock. The marketplace's program stock is a tracked external location. The availability service is bought as a module of the OMS Kestrel & Pine introduces under W2, not built, and the hourly chain is retired by Q4 2027.

Requirements per interaction

Different moments need different freshness, and over-specifying is as expensive as under-specifying. The targets below are the author's illustrative figures, not Kestrel & Pine facts.

Interaction	How fresh	How consistent	How fast	If the service is down
Product listing page, "In stock" badge	Seconds to a minute behind the service; stale is tolerable	Per-channel available, cached	Served from the storefront's cache; under 50 ms *(assumption)*	Show the cached value; never block the page

Interaction	How fresh	How consistent	How fast	If the service is down
Add to cart	Seconds	Check against the service's current available	Under 200 ms *(assumption)*	Allow the add; re-check at checkout
Checkout, order creation	Now	Commit atomically; the last unit goes to exactly one order	Under 500 ms *(assumption)*	Degrade to the platform's own committed count with a small buffer; alert
Marketplace stock push	Minutes; the marketplace sets the cadence	Available minus the marketplace buffer	Batched per rate limit	Push the last known value; widen the buffer
Click-and-collect at a store	Minutes behind the store till	Store on hand minus store commitments	Under 1 s *(assumption)*	Fall back to "call the store" for that store only

The checkout row carries the design. It is the only place where the last unit must go to exactly one order, so it is the one place that must call the service synchronously (ask and wait) and where the commit must be atomic (the order is created and the unit committed, or neither). Everything else can read a slightly stale copy. **Consistency where money changes hands**; **eventual consistency everywhere else**. Eventual consistency, in plain words, is the promise that if updates stop, all copies will agree soon, while accepting that for a short time they may differ; Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns the idea.

Events versus periodic sync

The difference between a stock event and an hourly sync is not only speed. A sync copies a snapshot, which must be applied carefully into a system that has its own claims (committed units). An event describes a change: it can be applied as a delta, kept in order per SKU and location, and deduplicated by its id.

Public examples show both ends. Shopify polls a fulfillment service's stock endpoint for on hand per SKU hourly (developer docs, n.d., accessed November 2026). Picqer, a Dutch

warehouse system for small and mid-size merchants, documents a `products.free_-stock_changed` webhook "to support real-time stock updates in a store" (n.d., accessed November 2026); a **webhook** is a push notification the source sends to a receiver's URL when something changes. Microsoft's inventory sync between Business Central and Shopify (documented 14 October 2026) is the careful middle: it compares "Shopify Stock" with "Last Calculated Stock" per variant and location and sends an update only when they differ.

At group scale the same design appears with industrial tooling. A 2020 account of Walmart's inventory backend, by its then director of engineering on Confluent's blog (4 May 2020; the author's views, not Walmart's endorsement), describes more than ten event sources normalized into one canonical inventory event, where "Each item-store combination always maps to one specific partition, so only one consumer processes it". The account is six years old; the idea, one ordered stream of changes per SKU and location, is the point.

The Choices

A growing merchant has four realistic ways to compute availability. Each is right for a stage, and each becomes a limit at the next.

1. **The platform's own stock**. The platform's inventory fields are the system of record for on hand, committed and available; apps, a same-vendor POS and a 3PL connector write to it. This is how Kestrel & Pine ran in 2016–2018 with its Tiel 3PL, and how most merchants start. It works while there is one location and the platform is the only system that takes orders.

2. **The ERP or WMS as the stock master, with periodic sync**. An ERP or WMS counts on hand; a connector copies it to the platform and from there to channels on a schedule. This is Kestrel & Pine's January 2027 landscape and the most common arrangement at the Growing stage, because every ERP and WMS vendor ships such a connector. Its limit is structural: the copy is a snapshot, the schedule is a lag, and each hop adds its own clock. The Oversell is what it does at peak.

3. **An event-driven availability service**. Counting systems emit stock events; order systems emit commitments; one service derives available per channel and publishes it. This is W1. It can be bought (a module of an OMS, a feature of an enterprise platform, a standalone inventory service) or built (a small service over a queue). The cost is integration work on every counting and ordering system, and the discipline of one writer per state.

4. **A fulfillment provider's shared pool**. The merchant hands physical stock to a logistics provider that promises it to several channels from one pool. Zalando's ZEOS announced in November 2024 that it would fulfill most of NEXT's continental-European direct-to-consumer orders through "a unified stock pool and an advanced

order management system"; by March 2026 it reported 24 markets and 18 sales channels (letter to shareholders, 12 March 2026). This moves the availability problem to the provider, at the price of the merchant's control over the count and the buffer.

Trade-offs

Criterion	1. Platform stock	2. Stock master plus sync	3. Availability service	4. Provider pool
Lag between a sale and the other channels knowing	Seconds (one system)	Minutes to hours, multiplied by hops	Seconds	Seconds within the provider; the merchant's own channels depend on the provider's feed
Number of writers of "available"	Platform plus apps (the risk)	Every connector in the chain (the risk)	One	One, not the merchant
Per-channel buffers and wholesale allocations	Basic or none	Usually none	Yes	The provider's rules
Integration cost	Low	One connector per hop	One event feed per counting and ordering system	One contract; high exit cost, the stock is physically there
When it breaks	Two apps write the same number	Peak day; a failed sync run	A lost or duplicated event; a stuck queue	The provider's outage or policy change

I would keep platform stock management until another system starts taking orders

or counting stock. Then assign owners for physical and committed stock before building the connector. Without that agreement, the connector can faithfully copy two incompatible answers and leave both channels selling the same unit.

Consider a separate availability service when any two conditions apply: multiple locations, multiple order channels, location-specific promises such as store pickup, or wholesale allocations. Kestrel & Pine met that test in 2023 but built the slow copying chain instead. Waypoint decision W1 replaces it with clear stock ownership and an availability service. A provider's stock pool can suit a brand selling mainly through that provider; it is less useful when the merchant's own channels dominate.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs the approach is outgrown
1 — Starting (Kestrel & Pine 2016–2018)	The platform's own stock fields; a 3PL or back room counts through the platform's app; reserve-on-order	A weekly count against the platform; at most one app with permission to write inventory	A 3PL or an accounting system starts "correcting" the platform's number by hand
2 — Growing (2019–2022)	An ERP or 3PL system counts on hand; a vendor connector syncs it to the platform on a schedule; the first marketplace gets a copy of the copy	A written owner for on hand and for committed; a daily reconciliation; a safety buffer on the marketplace	Out-of-stock cancellations cluster in busy weeks; two systems can both write inventory; the first "why did the website not know"

Stage	What is typical	What becomes necessary	Warning signs the approach is outgrown
3 — Omnichannel (2023–, now)	Own warehouse on a WMS, stores on a POS, a marketplace program holding stock, several channels; the chain of copies	One availability service (bought or built) fed by stock events; per-channel buffers; wholesale allocations; reconciliation weekly, daily at peak	An oversell at peak; click-and-collect checked by phone; apps writing stock; sync lag in hours; cycle counts finding double-digit variances
4 — Group	Several brands or countries, several warehouses, possibly several WMS vendors after acquisitions; a provider's pool for some markets	A shared availability service with per-brand segmentation; one ordered event stream per SKU and location; capacity-aware sourcing in the OMS	Brands competing for one pool with no allocation rules; the availability service becomes a bottleneck shared by every storefront

The approach changes with scale more than most topics in this book, because stock is the fact the most systems copy. What does not change: on hand is counted by one system per location, available is derived, and the last unit at checkout is committed atomically. A Starting merchant gets all three from the platform for free; an Omnichannel merchant must arrange them deliberately.

In Practice

The dated vendor evidence is above: Shopify's states added in 2023 and re-labeled in 2026, commercetools' 10-second window, Adobe's reservations and sourcing rules (2026), Business Central's compare-then-send sync (2026), Walmart's partition per item and store (2020). One more finding belongs here. **The counted fact is itself often wrong**. DeHoratius and Raman examined nearly 370,000 inventory records from 37 stores of one US retailer and found 65% inaccurate (*Management Science*, April 2008). The study is old, pre-dates RFID and concerns one retailer; use it as the field's classic finding, not a current rate. Later reviews cite 20% to 67% across studies. That is why

cycle counting (counting a small part of the stock on a rolling schedule, so errors are found continuously) and safety buffers exist.

Kestrel & Pine, worked

The reconstruction gave Noor three causes, and W1 answers each.

Cause 1: lag by design. Three hops, each on its own clock, up to two hours end to end. At 38 orders a minute, one hour of blindness is 2,300 orders, any of which could contain a Ridgewalker. W1 replaces the chain with stock events from the WMS to the availability service, carried by the iPaaS (the integration platform as a service that runs Kestrel & Pine's integrations) instead of passing through the ERP; the service republishes available to the platform and the connector within seconds. Of the 1,340 oversold units, about 790 were storefront orders and 550 marketplace orders (illustrative split); the marketplace share carried the penalties.

Cause 2: several writers. Six writers could change Duiven stock: the WMS feed (reaching the platform through the ERP's hourly sync), hand adjustments in the ERP, hand edits in the platform admin and three apps. Four of them were automated flows into the platform: the hourly sync and the three apps. W1 names one writer per state: the WMS writes on hand for Duiven, the POS for each store, the platform and the connector write committed, the service writes available. The apps lose their inventory permission; the pre-order app, if it survives the app review of Apps, Plugins and Functions: Extensibility and Ecosystem Governance, gets a pre-order allocation in the service instead.

Cause 3: no buffer, no reconciliation. Every channel sold down to the last unit, and nobody compared the WMS with the platform until the first weekly reconciliation in May 2027 found 2.3% of active variants with a different quantity, most left behind by the apps. After W1 the service holds a per-channel safety buffer, illustratively 3 units per SKU for marketplaces and 1 for the storefronts, and reconciliation runs weekly, daily in November.

Had W1 existed on 27 November 2026: the pick at 19:40 emits an event; the service recomputes the Ridgewalker's availability within seconds; the storefront badge flips to sold out and the connector pushes zero to the marketplaces on its next run, minutes later rather than two hours. The marketplace buffer would have pulled the listing a few sales before the shelf emptied. The oversell shrinks from 1,340 units to the handful that slip through the marketplace's own push interval. That is not zero. **No availability design reaches zero oversells; the goal is to make the window seconds wide and the buffer deliberate**.

When It Goes Wrong

Stock integrations fail in a small number of recurring ways. Each has a detection and a recovery.

Failure	What it looks like	Detection	Recovery
Duplicated stock event	The same pick applied twice; on hand too low; a sellable unit shown as sold out	Event id seen before; reconciliation variance in the "too low" direction	Idempotent handling: store processed event ids per SKU and location and skip repeats (the rule of Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation)
Out-of-order events	A receipt applied before the pick that preceded it; a brief negative or inflated count	Event timestamps or sequence numbers decreasing	Order events per SKU and location (one partition per key); reject events older than the last applied; re-fetch the current count from the counting system

Failure	What it looks like	Detection	Recovery
Snapshot overwrite	A periodic sync sets the platform's *available* to the warehouse's on hand, erasing the platform's own commitments; every sync run "un-commits" the hour's sales	Oversells that cluster right after sync runs	Set on hand, never available, from the outside, and let the platform derive available (Shopify's `inventorySetQuantities` can set either; set `on_hand`)
Two writers	The App That Wrote Stock: phantom reservations on 600 variants for two weeks	An app inventory with write-inventory permissions; variance that follows app activity	One writer per state; remove the permission; reconcile
Counting system outage	The WMS is down or its queue is stuck (the four-hour backlog); available goes stale	Queue depth; lag between order creation and WMS acknowledgment	The service keeps committing from order events and widens buffers; replay when the WMS returns
Identifier mismatch	`RW-M-MOSS` in the WMS and `rw-m-moss` in the platform; stock lands on the wrong or no variant	Events rejected or applied to unknown SKUs	One identifier per SKU across systems, mapped once (One Customer, One Product: Identity and Master Data Across Systems); Shopify's documentation warns that SKU strings are case-sensitive

Failure	What it looks like	Detection	Recovery
The count is wrong	The shelf has 3, the WMS says 5; the third order does not ship	Cycle counts; short picks	Cycle counting; safety stock sized to the measured error rate

One entry recurs in the detection column: the reconciliation job. **A stock integration without a reconciliation job is not finished**, because every delivery mechanism a merchant will meet (a webhook, a queue, a connector's retry) delivers at least once and sometimes not at all. Shopify's webhook documentation says delivery "isn't always guaranteed" and tells apps to "use reconciliation jobs to periodically fetch data" (n.d., accessed November 2026). Waypoint W5 makes the reconciliation job a condition of every integration.

What to Take Away

Established.

- On hand is a counted fact owned by one system per location; committed and reserved are claims; available is a derivation. All four vendor models in this chapter separate them, under different names.
- When a unit becomes promised is a configurable business choice (cart, order, fulfillment); the platforms default to reserve-on-order with a small eventual-consistency window, which commercetools publishes as 10 seconds.
- Counted stock is itself often wrong (65% of records in the 2008 DeHoratius and Raman study; 20–67% across later studies), so cycle counting and safety buffers are standard practice, not pessimism; and delivery of stock changes is at least once at best, so vendors instruct receivers to deduplicate and reconcile.

Depends on context.

- Keep stock in the platform while it is the only order-taking and counting system; this fails the day a 3PL, an ERP or a second channel arrives.
- A periodic sync from an ERP or WMS is acceptable at the Growing stage if the interval is minutes, the sync sets on hand rather than available, and a daily reconciliation exists; it fails at peak rates where an hour of lag is thousands of orders.
- Reserve-on-cart with short expiry is right for scarce-stock sales and wrong for ordinary catalogs, where it ties stock to abandoned carts.

- Buy the availability service as a module of an OMS when the OMS is coming anyway (Kestrel & Pine's case); build a small one when there is a strong integration team and no OMS on the horizon; use a provider's pool when the business sells mainly through that provider.

The author's synthesis.

- "Who owns the stock number" must be answered in writing before the first connector is built, because every oversell I have seen reconstructed came down to two writers or a copy on a schedule, never to a platform that could not count.
- Size the per-channel buffer from the measured count error and sync lag, and revisit it before every peak; an unmeasured buffer costs margin on quiet days and still oversells on busy ones.

Questions to Ask

1. Which system is the system of record for on hand at each location, and which system is the only writer of committed?
2. Which systems, apps and scripts can write inventory, and who owns each one?
3. At what moment does a unit leave "available": add to cart, order creation or fulfillment? What is the platform's documented consistency window?
4. How does a sale on one channel reach the others, by event or by schedule, and how long does it take at peak rate?
5. Does the outside feed set on hand or available? If available, what happens to the platform's own commitments on each run?
6. What is the safety buffer per channel, who set it, and from what measurement?
7. When did the last reconciliation between the counting system and the platform run, what variance did it find, and who fixed it?
8. What happens to checkout if the availability service is down for ten minutes?

Eva's jacket is in stock, and the unit she is about to buy will be committed to her order and to nobody else's. The next part follows that click: the cart as a draft, the checkout as a timed orchestration of every promise so far, and the payment that turns a promise into an order. Continue with Part III — ORDER: Cart, Checkout, Payment and the Order Record and From Cart to Confirmation: Checkout Orchestration.

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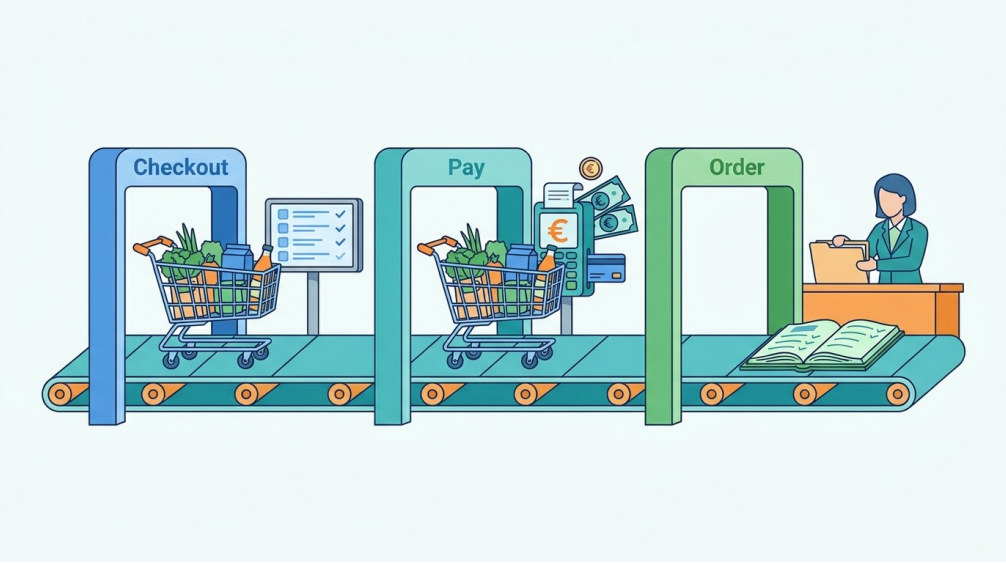
- Open-access review citing inventory inaccuracy rates of 20–67%, *Production* journal, 2018 — range across empirical studies.
- The alignment of vendor vocabularies and the grouping of oversell causes in this chapter are the author's synthesis, not any vendor's.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part III — ORDER: Cart, Checkout, Payment and the Order Record



KEY POINTS:

- **Checkout is the moment the shop stops promising and starts owing**. It asks every owner for a final answer and creates the order exactly once, while owning none of the facts it writes.
- **After checkout, the truth arrives by message, late, out of order and sometimes twice**. So the order keeps separate states for money, parcels and returns, and every message carries a key.
- **The order ends as documents, each issued by one system**. One writer per money movement, one invoice sequence per legal entity, and a monthly match of orders, payouts and the bank.

A shopper pays, but the order still appears unpaid because confirmation has not reached the shop. The warehouse cannot tell whether to dispatch it, and support cannot give a clear answer. Checkout, payment and order records describe different steps that happen at different times. This part explains how to connect them so the merchant can fulfill the purchase and account for the money.

A **payment service provider (PSP)** handles the money and reports results through a **webhook**, a server-to-server notification. The warehouse reports dispatch; the carrier reports delivery. These messages may arrive late or repeat. A shared order record must preserve the separate facts so a delivery update cannot hide an unresolved payment or a partial shipment. Kestrel & Pine, the fictional outdoor brand, lacked that view in 2026.

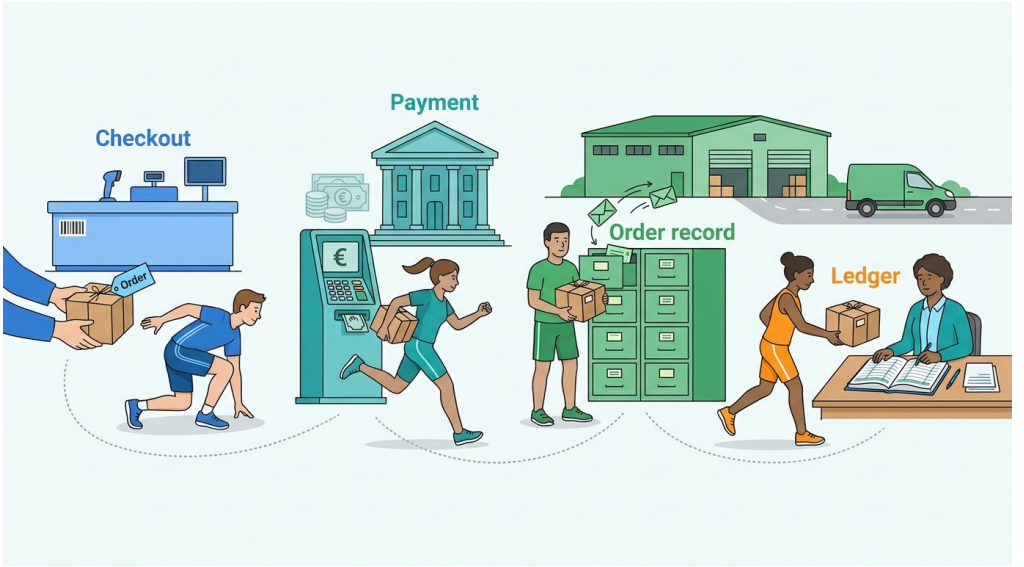


Figure 1: *One order passes through four owners in Part III: checkout creates it, the payment provider settles it, the order record holds it, and the ledger ends it.*

Where we are in the order's journey. Eva's €235 jacket-and-socks cart becomes order **#1072455 at 21:14 on 13 April 2027**, initially with payment pending. The provider confirms it seconds later. An invoice follows on 14 April and finance reconciles the payout at the May close.

Waypoint, the company's improvement program, later adds an **order management system (OMS)** to coordinate all channels. Physical fulfillment follows in Part IV.

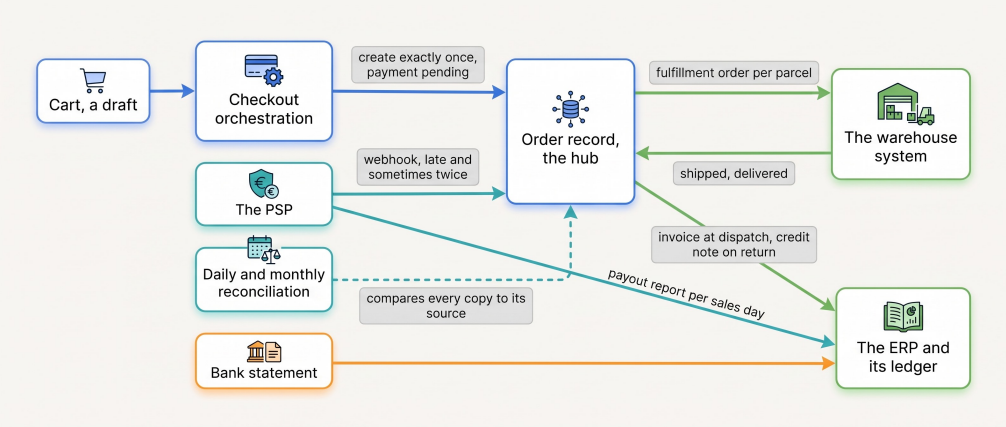


Figure 2: *The shape Part III recommends: checkout creates the order once; the order is one hub fed by messages from the payment provider, the warehouse and the carrier; the ledger is fed by documents, not events; and a reconciliation compares orders, payouts and the bank on a schedule.*

Chapters in this part

From Cart to Confirmation: Checkout Orchestration explains **how checkout coordinates its dependencies and creates one order despite retries**. Eva's payment redirect shows why order creation cannot depend on the thank-you page. The chapter also examines platform limits on checkout customization.

Taking the Money: Payment Providers, Authorization, Capture and Refunds follows **authorization**, **collection**, **settlement and payout**, showing how payment methods differ. January's 212 duplicate refunds, costing €19,400, explain why a refund needs one authorized writer and an identifier shared across retries.

The Order Record: Lifecycle, State and the Order Management System gives **the order a shared home with separate payment, shipment and return states**. A fleece delivered before its socks left one customer with three contradictory statuses and eleven tickets. The chapter shows how to represent partial completion honestly.

Tax, Invoices and the Books: From Order to Ledger connects **orders, invoices, payouts and the bank**. Eva's €235 splits into €194.21 net and €40.79 value-added tax. A Belgian wholesale example introduces Peppol, the network for structured invoices, and shows why finance needs confirmed documents and reconciliation.

Begin with From Cart to Confirmation: Checkout Orchestration.

Sources and Evidence

Sources are cited in each chapter; this introduction makes no factual claim of its own. Kestrel & Pine is fictional, and its figures are the same in every chapter.

11. From Cart to Confirmation: Checkout Orchestration

IN THIS CHAPTER, YOU WILL: Follow one cart through checkout, learn what checkout must call and in which order, see why platforms protect it, and choose between four ways to run it.

KEY POINTS:

- **A cart is a draft; checkout is a promise machine.** The cart reserves nothing and promises nothing. Checkout is the clocked sequence that asks pricing, promotions, tax, availability, shipping and payment for a final answer and turns the draft into an order the rest of the platform must keep.
- **Checkout should be thin, and it must create the order exactly once.** It composes calls and holds a short-lived state; it owns no prices, stock or payment state. The shopper may leave for the bank's app and never return, the payment provider's answer may arrive late and twice, so order creation is keyed on the cart and the payment, never on a page load.

- **Platforms withdraw the freedom to edit checkout, on purpose.** Shopify spent 2023 to 2026 replacing editable checkout templates and scripts with sandboxed extension points and dated deadlines. The lesson for a merchant is to customize checkout only through sanctioned mechanisms and to plan for the day the platform changes them.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Phantom Stock** — checkout promises availability from whatever stock figure it is given; on Black Friday 2026 that figure was up to two hours old.
- **The Message in a Bottle** — the payment provider's answer is a message that can arrive late, twice or never; a checkout that trusts one page load creates orders twice or not at all.
- **The Forked Platform** — a checkout customized by patching the platform's template or injecting scripts cannot follow the platform's upgrades, and the platform will one day switch it off.

While a shopper checks out, a price can change, another customer can buy the last item and the payment response can arrive late. Checkout must combine answers from several systems into one order the merchant can fulfill. Otherwise a successful-looking purchase can leave the customer with unavailable goods, an unexpected total or a second order after a retry.

Pricing supplies the price, inventory supplies availability, shipping supplies the delivery promise and the payment provider reports the money's state. Checkout coordinates these answers. It must check availability before committing stock and recognize repeated requests so a retry does not create another order or charge. The sequence and recovery behavior determine whether the promise survives a delay or failure. The supporting chapters are What Does It Cost? Price Lists, Contract Prices, Currencies and Tax, Discounts, Coupons and Loyalty: Promotions Without Chaos, Is It in Stock? Inventory, Availability and the Promise You Can Keep, Getting It There: Carriers, Labels, Tracking and the Delivery Promise and Taking the Money: Payment Providers, Authorization, Capture and Refunds.

Eva Janssen's purchase at Kestrel & Pine, the fictional outdoor brand, supplies the worked example. We then compare four ways to run checkout and use Shopify's documented 2023–2026 changes to examine customization limits.

Where we are in the order's journey. Eva has a jacket and socks in her cart. At 21:14 on Tuesday 13 April 2027, checkout creates order #1072455 with payment pending. The payment result arrives separately. Taking the Money: Payment Providers, Authorization, Capture and Refunds follows the money; The Order Record: Lifecycle, State and the Order Management System follows the order record.

Setting the Scene at Kestrel & Pine: Three Requests for the Checkout

In her second month, Noor Visser gets three requests about the same page.

Milan Petrov, who owns the storefronts, wants a delivery-date line in checkout ("delivered Thursday") instead of the static "ordered before 22:00, delivered tomorrow", and a gift-wrap option. The agency proposes an app for the first and a script injected into the checkout page for the second. Milan also wants fewer abandoned carts; about seven in ten carts never become orders *(illustrative, in line with Baymard's industry average)*.

Daniel Okafor, who owns the warehouse, wants to know why the checkout kept selling the Ridgewalker jacket for two hours after Duiven ran out on Black Friday evening 2026.

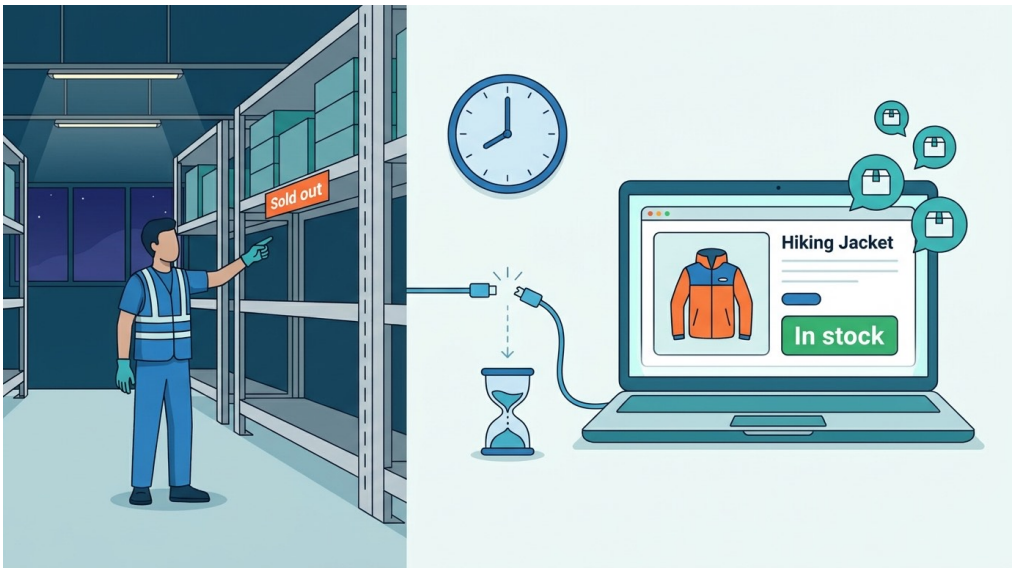


Figure 1: *The Phantom Stock: checkout promised availability from a stock figure up to two hours old, so it kept selling a jacket Duiven no longer had.*

Sanne de Wit, who owns finance, wants the checkout to stop creating orders the payment provider later reports as unpaid, each of which she explains at month-end, and asks why the Card-Testing Night of August 2026 cost €2,900 in fees for payments that were all declined.

Noor's first move is to ask a question none of the three had asked: **what, exactly, does the checkout call, and in what order?** The answer is the subject of this chapter; the agency's script proposal is the subject of its second half.

The Cart Is a Draft, the Checkout Is a Commitment

Three words carry the chapter, and they are easy to blur.

A **cart** is a draft of an order that the shopper can still change: product references and quantities, perhaps a coupon code, perhaps an address. It reserves nothing: a jacket in Eva's cart is still for sale to everyone else. It promises nothing: the price next to the jacket is an estimate that will be recomputed. Carts live for days or weeks; a signed-in shopper's cart follows her across devices, and a guest cart lives in the browser until it is merged into an account at sign-in. The cart also hides the first business decision about checkout: whether to **reserve stock when an item is added**. Most platforms do not; stock is committed only when the order is created, because a reservation held by every browsing shopper would make the "in stock" figure useless. Flash sales and ticketing are the exception, and they pay for it with visible countdowns.

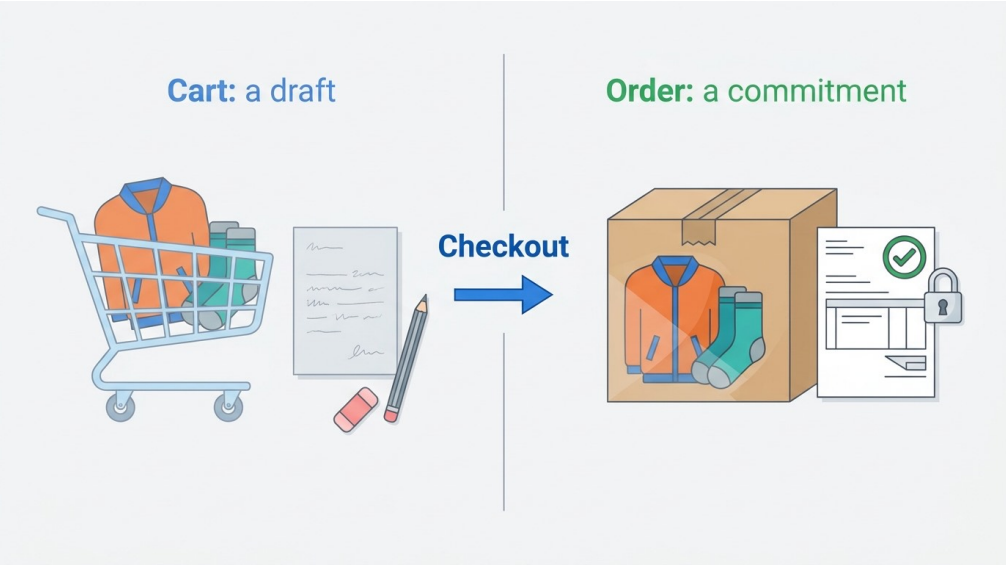


Figure 2: *The cart reserves and promises nothing; checkout turns it into an order the shop now owes.*

A **checkout** is the orchestrated sequence that turns the cart into an order. It is short (minutes), it has a clock, and it is the first moment the platform asks every other capability for a *final* answer rather than an estimate.

An **order** is the record that exists when checkout succeeds: what was bought, at what price, with what tax, delivered where and how, paid how. From that moment it belongs to order management (The Order Record: Lifecycle, State and the Order Management

System) and the checkout has no further say.

	Cart	Checkout	Order
Lifetime	Days to weeks	Minutes	Years (tax retention)
Can change	Freely, by the shopper	Only through the steps	Only by order edits with their own rules
Reserves stock	Usually not	Usually not until the final step	Yes, on creation
Price and tax	Estimates	Final computation	Frozen, with the basis recorded
Owner	Storefront or platform	Checkout orchestration	Order management

The distinction matters because every system downstream treats the order as a promise. If the checkout wrote a stale price or a stale stock figure into it, the warehouse, finance and the customer all inherit the error.

What Checkout Must Call

A checkout walks the shopper through a handful of steps, and each step is a call to a system that owns a fact. The table below is a reference-architecture view; vendor flows differ in order and in how many steps are visible, but the calls are the same. (The Baymard Institute's 2024 benchmark counted an average of 5.1 steps from cart to order review and 11.3 form fields, and argued that eight fields suffice.)

Step	What it must call	The promise it makes
Identity (guest or account)	Customer service; address book for returning customers	"We know whom to bill and where to deliver"
Address	Address validation; tax determination (the ship-to address decides the rate of VAT, value-added tax, Europe's sales tax)	"This is where it goes and what tax applies"

Step	What it must call	The promise it makes
Shipping options	Shipping-rate service or carrier rules; availability and the delivery promise	"Delivered by X for Y"
Price review	Pricing, promotions, tax engine	"You pay exactly this"
Payment	The PSP (payment service provider, the company that takes the shopper's money through many methods and pays the merchant one net amount), including authentication; fraud screening	"The money is reserved"
Confirmation	Order creation; stock commitment; notifications	"The order exists and will be kept"

Two things stand out. First, **checkout owns none of the facts it writes**. The price list lives in pricing, the rate in the tax engine, the stock figure in inventory, the delivery promise in shipping, the payment state in the PSP; checkout consumes all of them. Second, **the calls have different freshness needs**. A product description may be hours old without harm; the price should be minutes old at most; the availability figure must be current at the moment of commitment or the order is a Phantom Stock waiting to happen.

Call	How fresh	How consistent	How fast (illustrative targets)	If it is down
Price and promotions	Minutes; recomputed at the price review step	Final at order creation	Under 200 ms per cart *(assumption)*	Fail the step, never guess
Tax	Computed at price review from the ship-to address	Recorded on the order as the basis	Under 200 ms *(assumption)*	Fall back to the platform's rate table with a flag

Call	How fresh	How consistent	How fast (illustrative targets)	If it is down
Availability	Current at order creation	Committed by the order	Under 100 ms per line *(assumption)*	Degrade to "may be delayed", never to "in stock"
Shipping rates and promise	Current per address	Frozen on the order	Under 500 ms *(assumption)*	Offer the default service without a date
Payment	The PSP's answer only	Idempotent per order	Seconds, plus the authentication detour	Queue the order as pending; do not fulfil

This is why checkout is described as an **orchestration**: a coordinator that calls several services in sequence, holds a short-lived state between the calls, and completes or abandons the whole. In plain words, a conductor, not a musician.

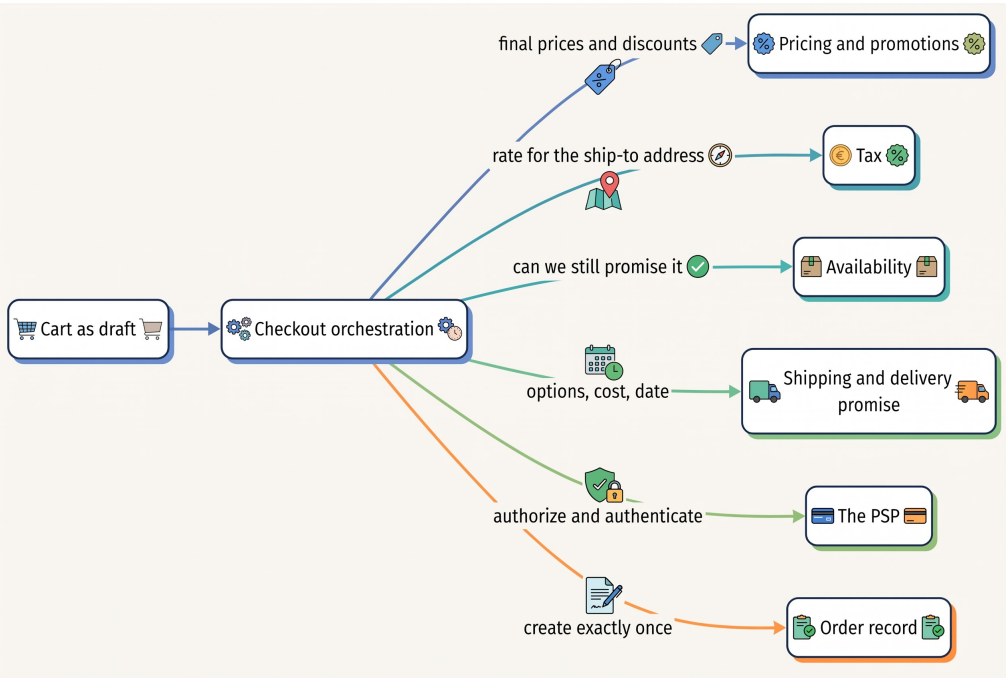


Figure 3: *Checkout owns none of the facts it writes into the order; it asks six systems for a final answer and creates the order exactly once.*

The Clock and the Detour

Checkout runs against two clocks it does not control.

The first is the **shopper's patience**. Baymard's 2026 list of abandonment reasons, from a survey of US adults who had abandoned a cart and were not merely browsing, puts "extra costs too high" at 40%, "delivery too slow" at 20%, "didn't trust the site with my card" at 19%, "forced to create an account" at 18% and "too long or complicated" at 17% (multi-select). Three of those five are architecture: showing total cost early means calling tax and shipping *on the cart*; showing a credible delivery date means calling the delivery promise; not forcing an account means the order needs a customer *reference*, not a customer *account*.

The second clock is the **payment provider's**. A payment session expires: Mollie's documentation (n.d., accessed November 2026) gives iDEAL (the Dutch method in which the shopper pays by bank transfer from her banking app) sessions 15 minutes, cards 30 minutes, PayPal 6 hours and Klarna 48 hours, and advises merchants not to predict expiry but to wait for the webhook. A card authorization, once granted, is valid for days (Stripe's and Adyen's documentation list 5 to 7 days for most card schemes). Checkout must finish inside the session; capturing the money can wait for the warehouse (Taking the Money: Payment Providers, Authorization, Capture and Refunds).

Inside the payment step sits a **detour**. For most European online payments the shopper leaves the merchant's page: to the bank's app for iDEAL, to a 3-D Secure challenge for a card (3-D Secure is the card networks' protocol for the strong customer authentication EU rules require; the shopper confirms in her banking app), to PayPal's page. She may lose her connection or never come back. Stripe's fulfillment guide states the consequence: "You can't rely on triggering fulfillment only from your checkout landing page, because it's not guaranteed customers visit that page." The platform learns the outcome from a **webhook**, an HTTP call the PSP makes to the merchant's server when something happened, and webhooks arrive late, out of order and sometimes twice (Stripe's documentation says so in as many words). Two rules follow.

Create the order exactly once, keyed on the cart and the payment. The commercetools checkout documentation (n.d., accessed November 2026) describes a flow in which the order is created with the payment transaction *pending* before the PSP's final answer; the webhook then moves it to success or failure, and the merchant should "start fulfilling an Order only if the Payment contains a Transaction" in a successful state. Stripe's guide tells the merchant to make the fulfillment function "correctly handle being called multiple times with the same Checkout Session ID". Both are the same discipline: an **idempotency key**, a unique token sent with a request so that a retry does nothing twice,

derived from the business event ("create order from cart X with payment Y"), not from a page load.

Treat the payment outcome as external truth. The order carries its own small payment vocabulary (pending, authorized, paid, failed) with the PSP's raw status beside it. Nothing in checkout decides that a payment succeeded; the PSP's signed webhook does, and a daily job asks the PSP about every order still pending after the method's expiry window, because webhooks stop retrying (Mollie after ten attempts over 26 hours, Stripe after three days). Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns the general pattern; checkout is its busiest application.

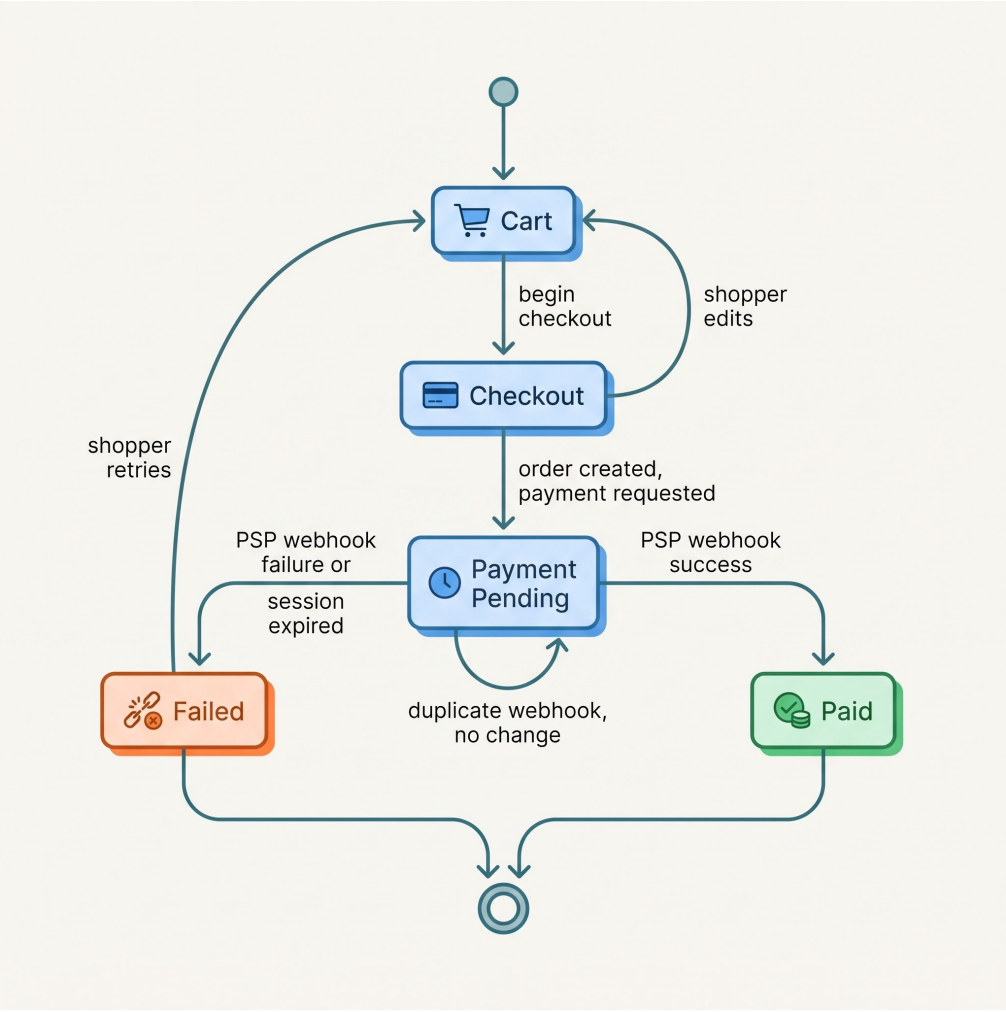


Figure 4: *The order exists before the payment outcome is known; the webhook, not the thank-you page, moves it to paid or failed, and a repeated webhook changes nothing.*

Guest, Account and the Express Wallets

Forced account creation is the fourth-largest stated reason for abandonment in Baymard's 2026 list. The architectural consequence is small and important: identity is optional at checkout and attachable afterwards. An order needs an e-mail address and a delivery address, not a login. The guest order can be claimed by an account later, one of the matching problems One Customer, One Product: Identity and Master Data Across Systems works out (Eva exists at Kestrel & Pine as an online account, a store loyalty customer, a newsletter subscriber and a returns-portal contact with no shared identifier).

Express wallets short-cut the steps. Apple Pay's developer overview (n.d., accessed November 2026) describes a purchase completed "from a product page with a single tap, without the need to add to a cart or fill out a form", with the merchant never receiving the card number. Google Pay's web documentation describes a payment sheet that returns a payment token to the merchant's server, which forwards it to the PSP. Shopify's Shop Pay stores e-mail, payment method and addresses; in Shopify's Black Friday–Cyber Monday reporting of 2 December 2025, Shop Pay carried 32% of orders across its merchants that weekend (a vendor-defined figure). The wallet supplies identity, address and a payment token in one callback. It does **not** remove the orchestration: pricing, tax, availability and shipping are still called, now from inside the wallet's callback and under a stricter clock.

Abandoned Carts and the Business Metric

Checkout has one business metric: **conversion**, the share of carts (or sessions) that become orders. Baymard's running list (updated 28 October 2026) gives an average documented abandonment rate of 70.22% across fifty studies from 2006 to 2025, with individual studies between 55% and 84%; the studies define "cart" and "abandonment" differently, so the figure is a direction, not a benchmark. Baymard's claim that fixing solvable usability issues could lift conversion by about 35% is its own estimate from usability testing, not a controlled experiment.

An **abandoned cart** is a cart started and not converted within some window. Recovering it (the reminder e-mail an hour later) has two preconditions: the cart is persisted server-side, not only in the browser, and it is attached to an identity the merchant may contact, which under EU privacy rules means a consent question (Whose Data Is It? Privacy, Consent and Retention). The architecture can do three things for conversion that no button color can: show the full price, including tax and shipping, before checkout begins (call tax and shipping rates on the cart page); show a delivery date the warehouse can keep (a computed promise, Getting It There: Carriers, Labels, Tracking and the Delivery Promise); and never fail the payment step because an internal call was slow.

The Choices: Four Ways to Run a Checkout

A merchant has four realistic options, and they map closely to the scale stages.

1. **The platform's hosted checkout, as is.** The platform renders, runs and protects the checkout; the merchant configures payment methods, shipping rules and branding. The merchant's PCI DSS scope (the card industry's security standard for anyone who touches card data) is the smallest possible, because the card fields are served from the PSP's domain and the merchant's servers never see a card number. Nothing to build and nothing to break.

2. **The hosted checkout with sanctioned extensions.** The platform exposes named places where apps may add a block, validate a cart, customize delivery or payment options or run a sandboxed function, each with a fixed input, a fixed output and a time budget. Shopify's checkout extensibility is the best-documented example (next section); commercetools' API Extensions are a synchronous variant, a call to the merchant's endpoint before a cart or order is saved, with a default two-second timeout after which the whole call fails. The merchant gets the delivery-date line and the gift-wrap option without owning the page.

3. **A custom or headless checkout on the platform's APIs.** The merchant or its agency builds the checkout pages on the platform's cart, checkout and payment APIs: full control of the experience, full responsibility for the orchestration rules above, for every script on the payment page and for keeping up with API versions (Promises Between Systems: API Contracts, Webhooks and Versioning). Card fields stay the PSP's hosted fields if the merchant wants the short PCI questionnaire.

4. **An own checkout service over composable capabilities.** The merchant runs a checkout service that calls separate pricing, promotion, tax, availability, shipping and payment services, often from different vendors, and creates the order in an order management system: the composable style of Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says applied to the most sensitive flow. It is right when one checkout logic must serve several storefronts, an app, a point of sale and a B2B portal, and it costs a team.

	1. Hosted as is	2. Hosted plus extensions	3. Custom on platform APIs	4. Own checkout service
Who runs the orchestration	The platform	The platform	The merchant, on the platform's APIs	The merchant

	1. Hosted as is	2. Hosted plus extensions	3. Custom on platform APIs	4. Own checkout service
Customization	Configuration and branding	Named extension points, time-boxed	Anything, at the merchant's risk	Anything
PCI DSS scope	Smallest (hosted page)	Smallest, if extensions cannot touch the payment fields	Small only with hosted fields; every page script is a liability	Depends on design; a service that touches card numbers is the long question-naire
Upgrade risk	None	Low: extensions are versioned	High: API versions, changed flows	Owned entirely by the merchant
Team needed	None	An app developer or agency	Front-end and integration engineers	A platform team
Fits	Starting; most Growing	Growing; Omnichan-nel on a SaaS platform	Omnichannel with a strong storefront team	Group; vendors

My rule: **take the most protected option that still gives you the two or three customizations the business needs, and get those through sanctioned extension points**. The reasons to leave option 2 are real (a checkout shared by web, app and store; a B2B flow the platform cannot express) and rarer than agencies suggest. Every freedom you take in checkout is a page you now keep working at 41 times normal load, under the card industry's rules, through every platform release.

How Protected Checkout Is: Shopify's Timeline, 2023–2026

The clearest public record of a platform taking checkout *away* from its merchants is Shopify's, and it is worth reading as a sequence of dates.

On 13 February 2023 Shopify's changelog announced that `checkout.liquid`, the template file that let Shopify Plus merchants edit the checkout's markup, would stop working for the information, shipping and payment pages on 13 August 2024; checkouts not upgraded "will automatically revert to a standard, uncustomized Shopify Checkout". The replacement, checkout extensibility, was described as "app-based, upgrade-safe, higher-converting, and integrated with Shop Pay".

The thank-you and order-status pages followed: Shopify's developer documentation (n.d., accessed November 2026) records that `checkout.liquid` and "additional scripts" were sunset for those pages on 28 August 2025 for Plus stores, and its help center names 26 August 2026 as the deadline for non-Plus stores. Shopify Scripts, the Ruby customizations that ran inside checkout, were retired on 30 June 2026 according to the help center.

As of November 2026, "advanced checkout customization features are available only on the Shopify Plus plan"; the mechanisms are UI extensions in defined slots, Shopify Functions (merchant logic compiled to WebAssembly and run by the platform inside the purchase flow, limited to 11 million instructions per run for carts of up to 200 lines, per the API reference for version 2026-10), web pixels for tracking and a branding API.

> **INFERENCE, NOT DOCUMENTED:** Shopify has not published its reasons beyond the changelog wording. The arc, free-form template and script access withdrawn over three and a half years with dated deadlines and replaced by sandboxed extension points with hard limits, is consistent with a platform that wants one checkout it can change under every merchant at once, keep inside the card industry's rules and keep fast at peak.

The timeline teaches three things that apply to any platform.

First, **a time budget is how a platform protects checkout from the merchant's own code**. Shopify's instruction count and commercetools' two-second extension timeout are the same idea: your logic may run inside checkout, but only for so long, and if it fails the platform decides what happens (commercetools fails the whole call; Shopify's Functions cannot call the network at all).

Second, **the payment page is where the card industry looks**. PCI DSS v4.0.1 added requirements for an inventory of scripts on the payment page and for tamper detection (6.4.3 and 11.6.1, effective 31 March 2025 per the PCI Security Standards Council's blog of 30 January 2025, which also removed them from the shortest self-assessment questionnaire in exchange for a declaration that the site is not susceptible to script attacks). The agency's gift-wrap script is exactly the kind of third-party script that rule exists for; a skimmer that compromises any script on that page reads every card typed into it (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers).

Third, **a platform will withdraw an extension point, and it will tell you the date**. The Forked Platform pattern is usually described for customized open-source cores, but a

checkout customized through a mechanism the vendor has announced it will retire is the same pattern on a SaaS platform (software rented by the month and run by the vendor). The remedy is an inventory of every checkout customization, its mechanism, and the vendor's support clock for that mechanism (Apps, Plugins and Functions: Extensibility and Ecosystem Governance).

What Changes When the Buyer Is a Company

A business customer changes four steps and removes one. Identity is a **company account** with several contacts and roles, not a person; the buyer may need an approver before the order is final. Prices come from a **contract price list**, and the price review shows amounts *excluding* VAT. The address step collects the company's VAT identification number, and a valid number from another EU country switches the tax treatment to the **reverse charge** (the seller charges 0% and the buyer accounts for the VAT itself; Tax, Invoices and the Books: From Order to Ledger).

The payment step often disappears: the order is placed on **payment terms** (net 30 days, at Kestrel & Pine) with a purchase-order number, and "payment" becomes a receivable in the ERP, not a PSP event. Shopify's Admin API (version 2026-10) types these terms as net, fixed date, on receipt or on fulfillment; Microsoft's documentation for its Business Central–Shopify connector (14 October 2026) tells merchants to decide which system owns payment terms and credit limits, because "mixing both directions without a clear ownership rule can lead to conflicting records".

A **credit check** against the view of the ERP (enterprise resource planning system: the back-office system that keeps the books and each business customer's credit limit) of the customer's balance is the one call a consumer checkout never makes.

Kestrel & Pine's B2B portal runs on the platform's company accounts, price lists and net terms; 62% of wholesale orders arrive through it. Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals works the whole picture.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting (under about €2M)	The hosted checkout as is; one PSP; guest checkout on; hosted payment page	Configuration only: home-market payment methods, shipping rules, legal texts	Requests to "edit the checkout page" for features apps provide; a script in the checkout
2 — Growing (€2M–€20M)	Hosted checkout plus apps and functions: delivery options, a validation rule, express wallets; a second country brings a second VAT rate and payment mix	Customizations only through sanctioned extension points, inventoried with the vendor's support dates; tax and shipping on the cart; abandoned-cart recovery with consent	Checkout slows at peak because of an extension; a plan-tier gate blocks a needed change; two apps both write to the order at creation
3 — Omnichannel (€20M–€200M)	Checkout as a channel-independent capability: web, app and B2B portal share pricing, tax and availability; order created before the PSP's final answer and settled by webhook	One availability service behind every channel's checkout; one owner for order creation; a daily payment reconciliation job; a peak-day degradation ladder that never includes the payment step	The Black Friday Oversell; orders that exist but stay unpaid for days; the Card-Testing Night

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
4 — Group	One checkout service with per-brand configuration, or several platform checkouts behind shared pricing, tax and payment services	A governed, versioned contract between checkout and the order system, owned by a platform team; the regulatory calendar for payment and card rules	Each brand's checkout re-implements VAT and promotion rules; a payment-flow change needs five deployments

Kestrel & Pine is a stage-3 merchant running a stage-2 checkout. Waypoint does not replace it (W4, stay and surround); it changes what the checkout *calls*: from Q4 2027 the availability answer at order creation comes from the availability service (W1), and from 13 October 2027 every storefront order is handed to the OMS, the order management system that becomes the order hub (W2), while the platform keeps the checkout. The chapter's advice holds at every stage; what changes is how many systems stand behind each call.

Eva's Checkout, Step by Step

The recurring order, with every system it touches. Times inside the window are illustrative; the order time, amounts and order number are canonical. Kestrel & Pine runs its platform's hosted checkout with two sanctioned extensions added in 2027, a delivery-date block and a gift-wrap option.

Time	Step	What checkout calls	Answer written into the order
21:09	Add to cart	Nothing beyond the storefront; the cart holds the jacket (women's, size M, moss) and two pairs of socks	Cart, no reservation

Time	Step	What checkout calls	Answer written into the order
21:10	Cart page	Pricing: €199 and 2 × €18; promotions: none; tax: NL 21% included in the consumer price; shipping: free over €75	Total €235 shown before checkout starts
21:11	Identity	Eva signs in to her online account; the address book returns her Nijmegen address	Customer reference, delivery address
21:12	Shipping options	The delivery-date block turns the static rule "ordered before 22:00, delivered tomorrow" into a date: "delivered Wednesday 14 April"; PostNL home delivery, €0	Delivery promise and carrier service
21:13	Price review	Pricing, promotions and tax recomputed for the final cart and address: net €194.21, VAT €40.79, total €235	The tax basis, frozen on the order
21:13	Payment	iDEAL selected; the PSP creates a payment session; Eva is redirected to her bank's app	Payment reference, status pending

Time	Step	What checkout calls	Answer written into the order
21:14	Confirmation	Order #1072455 created with the payment pending and the three units committed in the platform; the PSP's webhook arrives seconds later with the iDEAL result and the order becomes *paid*; the confirmation e-mail is sent	The order the rest of the book follows

Three details carry the design. **The order exists before the payment confirmation arrives.** If Eva never returns from her bank's app, it remains pending until the provider's server notification, or a daily comparison job, confirms the outcome. Repeating that notification does not create another order or payment.

Stock remains a weakness in this April 2027 example: checkout reads the platform's hourly copy from the enterprise resource planning system (ERP), itself fed every 15 minutes by the warehouse management system (WMS). On Black Friday, the same chain oversold 1,340 jackets.

The delivery date is also a static rule. Eva sees Wednesday 14 April, but receives the parcel on Thursday 15 April. Getting It There: Carriers, Labels, Tracking and the Delivery Promise and Journey 1: Eva's Jacket, From Search to Doorstep show how warehouse and carrier timings can produce a credible date instead.

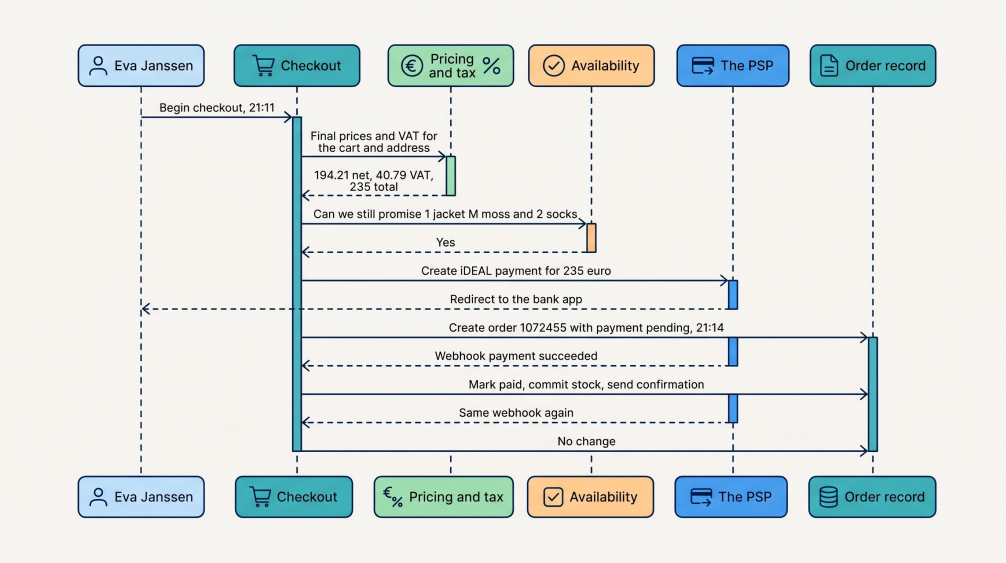


Figure 5: *Eva's checkout creates the order once, before the bank's answer, and lets the PSP's webhook move it to paid; a second delivery of the same webhook changes nothing.*

Daniel's question is now answerable. On 27 November 2026 at 19:40 the WMS knew the Ridgewalker was gone; the availability call was answered by the platform's copy, refreshed up to two hours earlier. Checkout trusted the answer it was given. The fix is not in checkout; it is in who answers the availability call, which is W1.

When It Goes Wrong

Checkout failures cluster into five kinds.

The duplicate order. A double-click, a browser retry after a timeout or a repeated webhook, and two orders exist for one cart, or one order is paid twice. Detect it with a report of orders sharing a cart id or payment reference; recover by canceling the duplicate before fulfillment and releasing the authorization rather than refunding a capture; prevent it with the idempotency key on order creation and a webhook handler that records every event id.

The lost outcome. The shopper paid and never returned, the webhook failed and the PSP stopped retrying, and the order sits pending while the money sits authorized. Detect it with a daily list of orders pending longer than the method's expiry window; recover by querying the PSP and settling the order either way. Mollie's documentation prescribes exactly this fetch-the-status pattern, and makes its webhook carry only an id so that a forged call cannot mark anything paid.

The stale answer. Availability, price or tax came from a copy that was behind. Detect it through oversold lines, price-parity warnings from marketplaces and invoices whose

VAT differs from the checkout figure. For stock, cancel or delay with an apology and a voucher (most of Kestrel & Pine's €96,000 Black Friday cost); for price and tax, honor the promise and correct the source. Prevent it with a current availability call at commitment, prices recomputed at the price review step and tax from the ship-to address.

The slow or failed dependency. An extension exceeds its time budget, a tax engine is unreachable, a shipping-rate call hangs. Watch checkout latency and error rate per step on peak day as closely as orders per minute (Black Friday Every Year: Peak Load, Resilience and Recovery), and decide the degradation ladder in advance: switch off the recommendation block, fall back to the platform's rate table with a flag for finance, offer the default carrier service without a date. The payment step and order creation are never on the ladder.

The attack. The Card-Testing Night: 48,000 bot attempts with stolen card numbers against Kestrel & Pine's guest checkout in one night in August 2026, €2,900 of PSP fees for declined authorizations and a 40-minute account pause. Stripe's documentation on card testing (n.d., accessed November 2026) names the symptoms (a spike in failed payments, nonsensical customer names) and the controls: rate limits per address, velocity rules, a CAPTCHA trigger on guest checkout and the PSP's risk tools fed with every signal the integration can send; Visa's May 2025 monitoring fact sheet tracks these enumeration attacks separately from disputes. The architectural control is a kill switch that disables guest card payments without a deploy (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers).

What to Take Away

Established.

- The cart is a draft that reserves nothing; checkout is the orchestration that asks every promise-making system for a final answer; the order is the record that results. The three have different lifetimes, owners and rules.
- The payment outcome is external truth that arrives by webhook, late, unordered and possibly twice; an order is created exactly once, keyed on the cart and the payment, and settled by the webhook or a reconciliation job, never by the thank-you page (Stripe's and commercetools' documentation, n.d., accessed November 2026).
- Platforms protect checkout by withdrawing free-form customization in favor of time-boxed extension points; Shopify's dated timeline from February 2023 to August 2026 is the documented case.
- Surprise costs, slow delivery and forced accounts lead Baymard's 2026 list of abandonment reasons; two of the three are answered by calling tax, shipping and the delivery promise on the cart.

Depends on context.

- The hosted checkout as is, or with sanctioned extensions, is the right choice for Starting, Growing and most Omnichannel merchants on a SaaS platform; a custom checkout is justified when several channels must share one checkout logic the platform cannot express, and only with a team that can keep it current and secure.
- Guest checkout should be on by default for consumers; for business customers the account *is* the checkout (contract prices, VAT treatment, terms, approvals).
- Creating the order before the PSP's final answer is the robust pattern where redirects and asynchronous methods dominate (iDEAL, Bancontact, 3-D Secure challenges); where every method answers synchronously it adds a pending state you must still reconcile.

The author's synthesis.

- Treat checkout as a conductor, never a musician: the moment it owns a price list, a stock figure or a payment state of its own, it has become a second system of record for a fact that already has one, and the Truth patterns follow.
- Keep an inventory of every checkout customization with its mechanism and the vendor's support date for that mechanism; the Forked Platform on a SaaS checkout is a deadline you were told about and did not write down.

Questions to Ask

1. What does our checkout call, in which order, and which of those answers come from a copy rather than from the system that owns the fact?
2. At the moment the order is created, how old can the availability figure be? What answered it on our busiest hour last year?
3. If the shopper pays and never returns to our site, what creates or settles the order, and when?
4. What happens if the PSP's webhook for one payment arrives twice? If it never arrives?
5. Which customizations run inside our checkout, through which mechanism, and when does the vendor's support for that mechanism end?
6. Which third-party scripts load on our payment page, and who approved each one?
7. Are tax and shipping shown on the cart page, or only after the shopper has typed an address?
8. Can we disable guest card payments, or a single extension, without a deploy, in the middle of a peak hour?
9. For business customers, which system answers the credit check and the VAT treatment?

Checkout hands the order to two systems at once. The order record goes to order management, where its lifecycle begins; the payment request goes to the PSP, where the money's own lifecycle, authorization, capture, settlement, payout and sometimes refund and dispute, begins. The next chapter follows the money: Taking the Money: Payment Providers, Authorization, Capture and Refunds.

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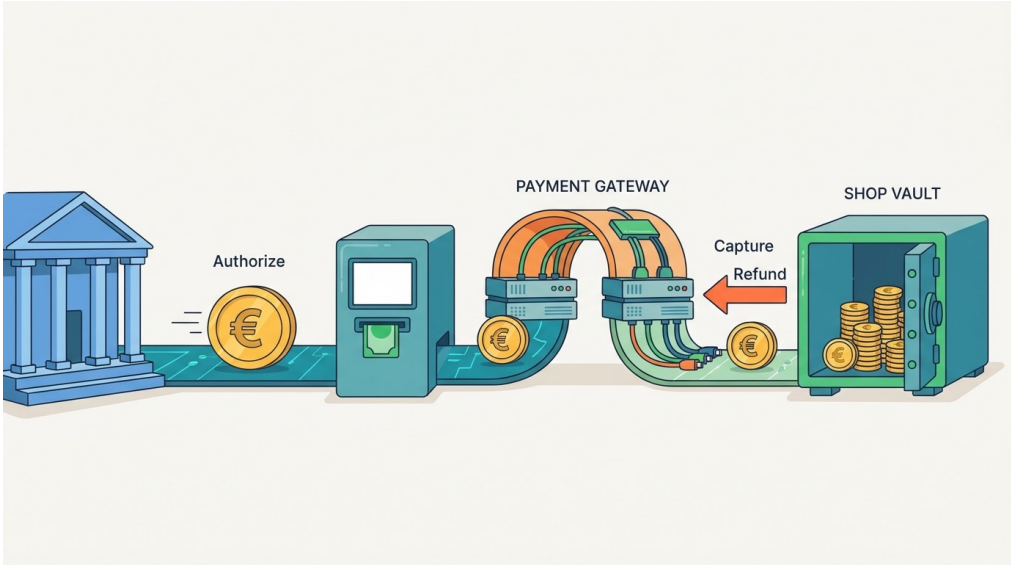
- Checkout flow: average form fields, Baymard Institute, 2024 — 11.3 fields and 5.1 steps on average; eight fields suffice.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

12. Taking the Money: Payment Providers, Authorization, Capture and Refunds



IN THIS CHAPTER, YOU WILL: Follow Eva's €235 from her bank to Kestrel & Pine's account and learn why the order must treat the payment provider's answer as the truth and every refund as a one-owner action.

KEY POINTS:

- **Payment is a state machine whose truth lives outside your systems.** The payment service provider, not the checkout, decides whether money moved, and its answer arrives by webhook: late, unordered and sometimes twice. The order keeps a small vocabulary (pending, authorized, paid, refunded, disputed) with the provider's references beside it, never a flag that says "paid: true".
- **Money moves in four stages, and each payment method bends them differently.** Authorization reserves, capture takes, settlement moves money between banks, payout lands it in the merchant's account.

A card authorization expires in days; iDEAL has no capture and no chargeback; buy-now-pay-later holds for weeks. A design that assumes one method's behavior breaks on the others.

- **Every action that moves money has one writer and one key**. The Double Refund cost Kestrel & Pine €19,400 twice because two systems could refund the same return and neither sent a key the provider could recognize. Idempotency keys, a daily reconciliation against the provider's records and a card number that never touches your servers are the three disciplines this chapter is built on.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Boomerang** — a refund issued from two systems for one return is reconciled by hand after the money has gone, and the second one cannot be pulled back.
- **The Message in a Bottle** — a payment result sent once and trusted, or a refund request retried without a key, produces unpaid shipments and paid-twice customers.
- **The Lynchpin** — one payment provider carries every online sale; when its account is paused for 40 minutes, the checkout is dark and nobody has rehearsed the switch.

A shopper approves a payment in a bank app and returns to the web shop. That return alone does not tell the merchant whether the money was reserved, collected or still pending. Payment providers and banks report those outcomes separately. The shop needs their confirmation to decide whether to release the order and avoid asking a customer to pay twice.

A missing confirmation can leave a paid order marked unpaid; a repeated refund request can send money twice. For payment methods that reserve funds before collecting them, the reservation can also expire. Checking unresolved payments against the provider's records makes these failures visible before they become customer complaints or unexplained gaps in the accounts.

The chapter explains payment service providers (PSPs), reserving and collecting money, customer authentication, payment notifications and refunds. From Cart to Confirmation: Checkout Orchestration covers the checkout sequence; Tax, Invoices and the Books: From Order to Ledger follows payments into the books. For deeper provider operations, see Payments for Finance Architects: Eleven Reasons the Numbers Disagree, Adyen: One Platform from Checkout to Settlement and Stripe: Every Movement Is a Balance Transaction.

Where we are in the order's journey. Eva buys a jacket and socks at Kestrel & Pine, the fictional outdoor brand. At 21:13 on 13 April 2027, checkout sends her to her bank's app to pay €235 through **iDEAL**, a Dutch bank-transfer payment method. We follow confirmation, settlement and payout, then a refund alternative to her later exchange.

Setting the Scene at Kestrel & Pine: Three Weeks in January

Kestrel & Pine's online **payment service provider (PSP)** offers iDEAL, cards, PayPal, Klarna, Bancontact and Apple Pay. It collects payments through their banks and networks and pays the merchant a net daily amount. The six stores use a separate terminal provider. Wholesale customers pay invoices by bank transfer on 30-day terms. Finance must account for all three routes.

In January 2027, Finance Director Sanne de Wit found duplicate credits in the enterprise resource planning system (ERP), which keeps the books. **212 returns had been refunded twice** over three weeks: automatically through the returns portal and manually by support in the PSP console. **€19,400** had gone out a second time.

Most was recovered by asking customers to return it; the bank-transfer refunds could not be recalled. Head of Engineering Noor Visser reconstructed the ordinary sequence behind the incident. The dates and details are fictional.

Day	What happened	Which system acted
0	A returned parcel arrives at Duiven; the customer sees PostNL's "delivered" scan	The carrier
0–3	The customer writes to support: "my return arrived, where is my refund?"	The customer
1	The agent opens the PSP console, sees the payment and no refund, and refunds €199 to be helpful	Support, in the PSP console
2–5	The warehouse grades the item resellable; the returns portal, which refunds on inspection, calls the platform's API; the platform instructs the PSP, which accepts a request that names only the order and an amount	The returns portal, the platform, the PSP
31	Finance finds two credits per order when matching the PSP's payout report to the ERP	The ERP

Two systems could refund, neither sent a key that named the business event, and the only place the two paths meet is the ledger. Noor's whiteboard question in February is

the chapter's: **who may move money, with what key, and how does the order find out what the PSP actually did?**

A second finding sharpened it. On the Card-Testing Night of August 2026, bots fired **48,000** stolen card numbers at the guest checkout, costing **€2,900** in fees for declined authorizations and a **40-minute** pause of the PSP account, and of every online sale. Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers owns the defense; this chapter owns the fact that one provider carried every sale.

What a Payment Service Provider Does

When Eva pays, at least five parties take part.

Party	In plain words	In Eva's iDEAL payment
The shopper	Pays with a card, a bank account or a wallet	Eva, in her bank's app
The issuer	The shopper's bank, which holds her money or her credit	Eva's Dutch bank
The scheme or network	The rules and rails a method runs on: Visa, Mastercard, iDEAL, SEPA, PayPal's own network	The iDEAL scheme
The acquirer	The licensed institution that accepts payments on the merchant's behalf and receives the money from the issuer	Part of the PSP
The PSP	The merchant's single counterparty: one API, many methods, fraud screening, authentication, a balance, payouts and reports	Kestrel & Pine's PSP

A PSP hides the other four behind one contract; most PSPs serving European merchants are also acquirers or partner with one (a reference-architecture description, not a fact about any one company). Adyen's, Stripe's and Mollie's documentation (all n.d.,

accessed November 2026) describe the same four-stage flow: **authorization**, **capture**, **settlement**, **payout**, with **refund** and **chargeback** flowing backward.

Even a platform that appears to take payments itself works this way: Shopify's processor list (n.d., accessed November 2026) names Stripe, PayPal and Adyen as the processors behind Shopify Payments in the Netherlands, Belgium and Germany, and its European payment terms state that no Shopify entity is "licensed as a bank, payment institution, or money services business". "The platform's own payments" is a commercial wrapper over PSPs, and the four stages apply underneath.

The four stages

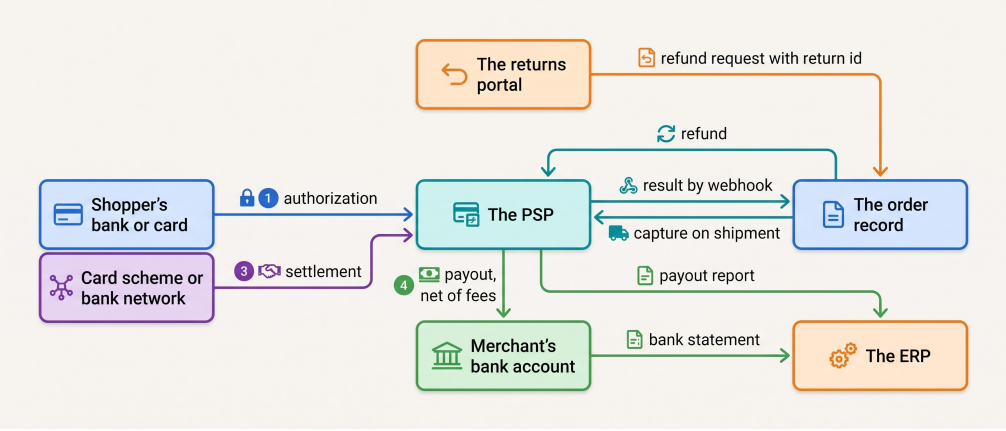


Figure 1: *Money moves in four stages the merchant does not control; the order learns the result by webhook, triggers capture and refunds through one path, and finance matches the payout to the orders it contains.*

Stage	Who acts	In plain words	What the order record keeps
Authorization	The issuer approves; the PSP reserves	"The money is there and held for you"	PSP reference, amount, method, authentication result, expiry

Stage	Who acts	In plain words	What the order record keeps
Capture	The merchant, or the PSP automatically	"Now take it"; Adyen: manual capture "lets you settle funds on fulfillment, when an order is shipped"	Captured amount, capture reference, link to the shipment
Settlement	The scheme or bank moves money to the PSP	The merchant's systems do nothing	Nothing
Payout	The PSP pays the merchant's bank	One batch per sales day, net of fees, refunds and chargebacks	Batch id, net amount, fees; matched by finance, not by the order system

The windows matter. Stripe lists card authorizations as valid for seven days for Visa, Mastercard, American Express and Discover when the shopper initiates an online payment; if the authorization expires before capture "the funds are released and the payment status changes to canceled". Klarna must be captured within 28 days at Stripe and Mollie. **"Capture on shipment" is a promise with a deadline**: if the warehouse ships on day eight, the card money is gone and the parcel is a gift.

Payout timing is separate: Adyen's default is "sales day payout", all funds from one sales day in one batch with a two-business-day delay, and Stripe's initial settlement timing in most EU countries is seven calendar days, moving to three business days. The batch report, listing the payments, refunds, chargebacks and fees it nets, is what finance matches against orders (Tax, Invoices and the Books: From Order to Ledger; in depth, Payments-to-Ledger: Draining the Suspense Swamp).

Methods are not interchangeable

Kestrel & Pine sells in three countries, and each pays differently; across the euro area cards still lead at 48% of online payments by number (the European Central Bank's SPACE 2024 study), so the card path is the one to get right first. The figures are the latest dated public ones; the last column is the author's judgement.

Country	Dominant online method	Evidence	What it means for the design
Netherlands	**iDEAL**, a transfer from the shopper's own bank account	Dutch Payments Association, February 2025: 1.47 billion iDEAL payments worth €141 billion in 2024; the ECB's SPACE 2024 study puts "e-payment solutions" at 76% of Dutch online payments by number	Instant and irrevocable, no separate capture, no chargeback, a 15-minute session; a refund is a new transfer back
Belgium	**Bancontact**, a card scheme paid mostly through its app	Bancontact Payconiq Company, February 2025: 382 million online payments in 2024, 90% from a smartphone	Card-like: authorized then captured; refunds and disputes follow card rules
Germany	**PayPal**, **purchase on account** (*Kauf auf Rechnung*) and direct debit	EHI's "Online-Payment 2024" study as reported by Ecommerce News, April 2024 (data for 2023): PayPal 27.7%, invoice 26.7%, bank transfer 16.7%	"Payment" may be a 30-day receivable or a buy-now-pay-later loan; the lender captures late and bears the credit risk

As of November 2026 the iDEAL scheme's own site brands itself "iDEAL | Wero", reflecting its move under the European Payments Initiative's wallet; what that changes for integrations is not documented there, so this chapter uses Mollie's documented semantics (instant charge, automatic capture, partial refunds for 365 days, no chargeback, 15-minute session) and flags the name as in motion.

Costs differ too. Adyen's indicative price list (n.d., accessed November 2026) charges "$0.13 + €0.22" for iDEAL, "$0.13 + Interchange++ + 0.60%" for Visa and Mastercard and "$0.13 + 2.99%" for Klarna in Germany; **interchange**, the fee the merchant's side

pays the cardholder's bank, is capped in the EU at 0.2% for consumer debit and 0.3% for consumer credit cards (Regulation (EU) 2015/751, in force since 9 December 2015). A buy-now-pay-later sale can cost ten times an iDEAL sale, a pricing decision before it is a technical one.

The Order Depends on a Message It Does Not Control

The rule the rest of the chapter builds on: **nothing in the merchant's own systems knows whether a payment succeeded**. The checkout sent a request. The shopper left for her bank's app or a 3-D Secure challenge (the card networks' extra authentication step, explained below) and may never come back. The PSP learns the outcome from the issuer and tells the merchant by **webhook**: an HTTP call to an address the merchant registered, saying "something happened to payment X". The order's job is to wait for that message and to be safe when it arrives twice or not at all.

The documentation is blunt about that message. Stripe (n.d., accessed November 2026): "Stripe doesn't guarantee the delivery of events in the order that they're generated"; "Webhook endpoints might occasionally receive the same event more than once"; and every event is signed, because "Without verification, an attacker could send fake webhook events to your endpoint to trigger actions like fulfilling orders." Mollie's webhook carries only an id, so "fake calls to your webhook will never result in orders being processed without being actually paid"; the merchant must fetch the status; and Mollie retries ten times over 26 hours before giving up.

Three consequences follow, and I would write them into the integration contract of any order system:

1. **Persist every event id and treat a repeat as a no-op**. The second "payment succeeded" changes nothing; so does the second "refund succeeded".
2. **When the decision matters, fetch the object rather than trust the payload**. The webhook is the doorbell; the PSP's record is the truth.
3. **Run a daily job for everything still pending**. Webhooks stop retrying, after 26 hours at Mollie and three days at Stripe. A job that lists every order whose payment has not reached a final state within the method's expiry window, and asks the PSP, is the floor under the design. From Cart to Confirmation: Checkout Orchestration placed the same job at the end of checkout; Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation generalizes it.

Figure 2: *The webhook is only the doorbell: the order trusts the PSP's record, ignores a repeated ring, and runs a daily job for every payment still pending.*

The order should keep its **own small payment vocabulary** and map the PSP's statuses (Mollie has seven, Stripe's PaymentIntent six) into it inside the adapter, so that a merchant that changes PSP changes an adapter, not every downstream system.

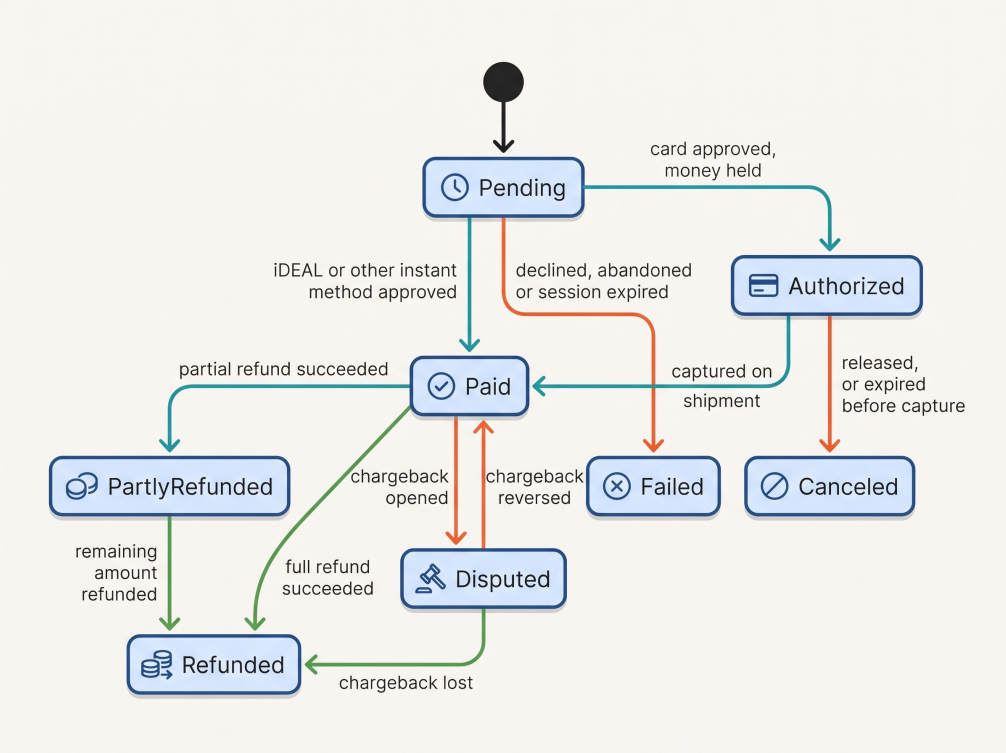


Figure 3: *The order's payment vocabulary: a few states every channel and system understands, moved only by the PSP's confirmed events, with the PSP's raw status and references stored beside them.*

Each flow between the order and the PSP has its own requirements; the latencies are assumptions for a merchant of Kestrel & Pine's shape.

Flow	How fresh	How consistent	How fast	If the other side is down
Payment outcome (webhook in)	As soon as the PSP knows	Applied once; duplicates ignored by event id	Seconds, but tolerate hours *(assumption)*	The daily pending-payments job settles it
Capture (out, on shipment)	Within the authorization window	One capture per fulfillment, keyed by it	Minutes *(assumption)*	Retry with the same key; alert before the window closes

Flow	How fresh	How consistent	How fast	If the other side is down
Refund (out, on inspection)	Within the legal 14 days (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics)	Exactly one per return, keyed by return id	Hours *(assumption)*	Never retry blindly; check the PSP's refund list first
Payout report (in)	Daily batch	Every line matched to an order, refund, fee or dispute	A day	Finance waits; nothing else does

Strong Customer Authentication and 3-D Secure

PSD2, the second Payment Services Directive (Directive (EU) 2015/2366, applied from 13 January 2018), defines **strong customer authentication** (SCA) in Article 4(30) as authentication using two or more of knowledge, possession and inherence, and Article 97 requires it when a payer initiates an electronic payment. The detailed rules are in Commission Delegated Regulation (EU) 2018/389, applied from **14 September 2019**, which sets the exemptions: contactless at the terminal up to €50 (Article 11), trusted beneficiaries (Article 13), recurring transactions after the first (Article 14), low-value remote payments up to €30 (Article 16) and transaction risk analysis up to €500 for PSPs with low fraud rates (Article 18).

For cards the protocol that performs SCA is **3-D Secure 2** (3DS): low-risk transactions pass "frictionless" and higher-risk ones require "a challenge, such as a one-time passcode" (EMVCo, n.d., accessed November 2026). Two rules follow from Stripe's SCA page. Exemptions belong to the PSP and the issuer, not the merchant: "Exemptions aren't guaranteed." And the **liability shift**: "If a cardholder disputes a 3DS payment as fraudulent, the liability typically shifts from you to the card issuer"; where an exemption was applied, "liability shift doesn't apply". An exemption buys conversion at the price of fraud liability, so the order should store the authentication result and the liability flag. A **merchant-initiated transaction** (MIT), one the merchant triggers later without the shopper present under a prior agreement, is what a split shipment captured late, a pre-order or "charge on dispatch" past the authorization window relies on; it needs an SCA-authenticated first interaction flagged for future use.

What comes next is not yet law. **PSD3** and the Payment Services Regulation reached

provisional agreement on 27 November 2025 (PSPs liable for customers' losses where fraud prevention was inadequate, mandatory payee-name checks, all charges disclosed before payment, per the European Parliament's press release of 21 November 2025); the Council published compromise texts on 23 April 2026, and both files were still open in its state of play of 29 October 2026. As of 7 November 2026 neither is in the Official Journal. The Instant Payments Regulation (Regulation (EU) 2024/886) is in force: euro-area PSPs had to receive instant credit transfers by 9 January 2025 and send them by 9 October 2025, with payee verification free of charge, which makes instant transfer a plausible checkout and refund rail.

Refunds, Cancels and Chargebacks: Money Flowing Backward

A refund looks like an undo. It is not. **A refund is a new payment in the opposite direction**, with its own state, failure modes and fees.

- **You can only refund what was captured.** Adyen: "You can only refund a payment after it has been captured"; an uncaptured authorization is *canceled* instead. Stripe's cost advice points the same way: businesses that refund soon after the sale should capture manually and cancel before capture "rather than processing a refund", because "Stripe's processing fees from the original transaction aren't returned."
- **A refund is asynchronous and can fail later.** Adyen: a webhook with `success: true` means the request went to the scheme, and "Although rare, a refund can fail after you received" it; for iDEAL, Bancontact and bank transfers a REFUNDED_-REVERSED event means the money came back because the account was invalid; funds can take "up to 40 business days" to reach the shopper. Stripe: the customer "sees the refund as a credit approximately 5-10 business days later".
- **Refunds go only to the original method, and partial refunds are bounded** (Stripe; Adyen). Store credit is a different instrument with its own ledger (Discounts, Coupons and Loyalty: Promotions Without Chaos).

The sequence shows how the Double Refund happened, and where one key would have stopped it.

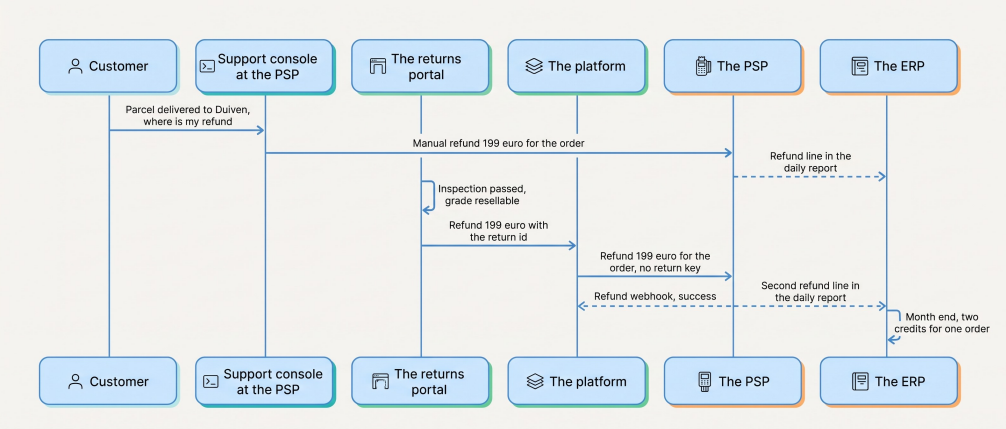


Figure 4: *The Double Refund: two writers, two paths, no shared key; the PSP cannot refuse the second request because nothing in it says it is for the same return, and the two paths meet only in the ledger.*

The fix Waypoint adopted in February 2027, which Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation describes from the integration side, has three parts. **One writer per money action**: refunds are issued by the returns portal through the platform, and support's PSP console access becomes read-only. **One key**: every refund request carries the return id as its **idempotency key**, a unique token sent with a request so that a retry does nothing twice; Stripe's implementation is the public reference (it "saves the status code and body of the first request made with a given idempotency key" and returns it on retries, and may prune keys after 24 hours, so the merchant's own key store must outlive the PSP's). **One daily check**: refunds issued by the PSP against refunds recorded by the platform, by order number and amount, with exceptions to Sanne de Wit's team.

Figure 5: *The fix for the Double Refund: one writer per money action and one key per business event, so the PSP refuses a repeat instead of paying twice.*

Chargebacks are the other backward flow, and not the merchant's decision: a dispute "occurs when a cardholder questions your payment with their card issuer"; the issuer "immediately reverses the payment", and the PSP "debits your balance for the payment amount and dispute fee" (Stripe). Adyen's flow runs from a notification of chargeback that opens the defense period through the chargeback, the supplied defense and a possible reversal, and Adyen auto-defends a dispute "that was already refunded". Cardholders "can dispute a charge up to 120 days after a payment was made", so a month's dispute ratio keeps changing for four months. The ratio has teeth: Visa's Acquirer Monitoring Program (fact sheet, May 2025) counts fraud reports plus disputes over settled transactions, with the excessive-merchant line lowered to 150 basis points on 1 April 2026. A slow refund path pushes customers to dispute instead, a second reason to get refunds right; iDEAL has no chargeback at all.

Fraud and Card Data: Keep the Attack Out and the Card Number Out

Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers owns the defense in depth and the sibling book LISTED covers scoring in Scams, Fakes and Account Takeovers: Fraud Prevention for Listings; this chapter keeps to what shapes the payment design.

Fraud screening is a ratio problem, so it belongs before authorization. Stripe's card-testing page (n.d., accessed November 2026) describes what hit Kestrel & Pine: attackers

"use scripts to test a large amount of card information at once"; the symptoms are "A spike in failed or blocked payments" and small payments "with nonsensical customer names and emails"; the costs are authorization fees, disputes and monitoring programs. Its controls: the PSP's rate limiters and models, CAPTCHA triggers, requiring login ("The easier it's for fraudulent actors to reach your payment form (for example, using guest checkout), the easier it's for them to execute card testing attacks"), per-IP limits and velocity rules. The addition I insist on is a **kill switch**: a setting, not a deploy, that disables guest card payments or one method while leaving iDEAL and logged-in customers alone.

Card data is a scope decision. **PCI DSS** is the card industry's security standard for anyone who stores, processes or transmits card data; version 4.0.1 was published on 11 June 2024, and its future-dated requirements took effect on 31 March 2025. The merchant's obligations depend on which systems can see a card number, and the design goal is that none of the merchant's can.

The mechanism is **hosted fields**: the PSP draws the card-number box inside the merchant's page from its own domain, or redirects to its own page, so the number never touches the merchant's code or servers. Stripe's PCI guide maps the integration to the self-assessment questionnaire (SAQ) the merchant signs: hosted checkout or hosted fields to **SAQ A**, a merchant-hosted form posting via the PSP's script to **SAQ A-EP**, direct API with raw card data to **SAQ D**, where a business "might be required to meet more than 300 security controls". Adyen adds the boundary: "Adyen is solely responsible for the security of cardholder data only as soon as Adyen receives the data."

Since 30 January 2025 the PCI Security Standards Council has replaced the two payment-page-script requirements in SAQ A with an eligibility statement that the site "is not susceptible to attacks from scripts", so every analytics, chat and testing script on the payment page remains the merchant's risk. The rule: **never store, log or transmit a card number**; **use the PSP's token**; **keep a list of every script on the payment page**.

The Choices: Four Ways to Take Payments

1. **The platform's built-in payments.** A hosted platform offers its own payment service, a wrapper over one or more PSPs, with a hosted checkout, automatic capture, refunds from the admin screen and payouts to the merchant's bank; Shopify Payments is the public example. Nothing to integrate, SAQ A, one payout feed. The limits: the method mix and fees are the platform's, and any second system that must move money goes through the platform's API or not at all.

2. **One external PSP through the platform's connector.** The merchant contracts with a PSP and uses its app or the platform's native integration; the PSP's hosted page or drop-in component renders the methods. Kestrel & Pine's setup since 2016: the PSP's local methods, pricing and risk tools, SAQ A if the page stays hosted, and ownership of the relationship, the webhook endpoint and the question of who else may call the PSP.

3. **A custom integration with hosted fields.** A headless or heavily customized checkout (Delivering the Storefront: Themes, Headless, Rendering and Performance) calls the PSP's API directly and embeds its hosted fields: full control of capture on shipment, partial captures and MITs; SAQ A if the eligibility criteria hold; and a checkout team that handles every property of the webhook itself.

4. **A payment service of the merchant's own, over several PSPs.** An internal service presents one payment API to every channel, routes each payment to one of several PSPs and reconciles across them. Payments for Finance Architects: Eleven Reasons the Numbers Disagree covers the trade-offs; it is worth its complexity only when volume, countries or resilience demand it.

Criterion	1. Platform's own payments	2. One PSP via connector	3. Custom with hosted fields	4. Own payment service, several PSPs
Time to first sale	Hours	Days	Weeks to months	Months
Method mix and pricing	The platform's	The PSP's, negotiable at volume	The PSP's	Best per method and country
PCI scope	SAQ A	SAQ A	SAQ A or A-EP	SAQ A-EP to SAQ D
Capture on shipment, partial capture, MIT	What the platform exposes	Via the connector's settings	Full	Full
Who else may move money	The platform's API only	Whoever holds the PSP's API key	The payment adapter	The payment service, by policy
Resilience to a provider outage	None	None	None without a second PSP	Routing to another PSP
Fits	Starting; one country	Growing; Omnichannel with discipline	Omnichannel with a checkout team	Group, or very high volume

My rule for a merchant of Kestrel & Pine's size: **stay on option 2 and spend the effort on discipline, not orchestration.** One PSP, local methods per country, capture on

shipment for cards, one refund path, verified webhooks handled by an adapter, a daily reconciliation, and a rehearsed fallback for one method rather than an orchestration layer a 14-person team would have to run.

Stores, Wholesale and Marketplaces: Four Feeds, Not One

Online is one of four ways money reaches Kestrel & Pine.

Stores take cards through the terminal provider, where SCA is chip-and-PIN or the contactless limit, the PCI questionnaire is a different one (SAQ C or a validated point-to-point encryption solution), and the payout is a separate daily feed into the ERP. A web order refunded at the till is the store's version of the Double Refund (Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment).

Wholesale customers do not pay at checkout. Bergzicht's spring reorder was invoiced on 9 April 2027 and is due on 9 May: "payment" is a **receivable**, an amount owed and tracked in the ERP, settled by a bank transfer that finance matches to the invoice (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals, Tax, Invoices and the Books: From Order to Ledger).

Marketplaces collect the shopper's money and pay the seller out, net of fees, on their own schedule: a third payout feed (Selling on Someone Else's Platform: Marketplaces and Channel Integration).

Online is the PSP's daily batch. Four feeds, four reconciliations, and the reason Waypoint's W5 says finance moves by daily batch: the books need completeness, not speed. Where four feeds meet, a payments sub-ledger earns its place (Never Update, Only Append: Designing Custom Ledgers).

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Signs the stage is ending
1 — Starting (under about €2M)	The platform's own payments or one PSP on a hosted page; automatic capture; refunds from the admin screen; one country's methods	Two-step authentication on the PSP and admin logins; someone who reads the decline rate	Refunds done by hand in the PSP console; a second country whose shoppers want a method the setup lacks
2 — Growing (about €2M–€20M)	One PSP with local methods per country; the first custom integrations (a returns portal, an ERP connector) that can call the PSP; card testing as the first incident	Capture on shipment for cards; one writer per money action; a webhook endpoint that is owned and verified; an edge rate limit and a kill switch for guest card payments	Two systems can refund; an agency script handles webhooks and nobody knows how; a dispute lost for want of a stored 3DS result
3 — Omnichannel (about €20M–€200M)	Four feeds: online PSP, terminal provider, marketplaces and wholesale receivables; payment states on an order hub	The order hub's payment vocabulary and adapter; refund keys; a rehearsed PSP fallback; a payment-page script inventory; a payments sub-ledger	Payout matching takes days each month; a PSP account pause stops every channel; fee lines nobody can explain per country

Stage	What is typical	What becomes necessary	Signs the stage is ending
4 — Group (over about €200M, or several brands)	Several PSPs or an orchestration layer; network tokens so saved cards are not locked to one PSP; a central regulatory calendar	A payment service with routing and one reconciliation pipeline; a deliberate decision whether any system may touch card numbers	Each brand negotiating its own PSP contract; a tokenized card base that cannot move

The judgement: the payment *mechanism* does not change with scale. What changes is **how many systems can move money and how many feeds finance must match**; stage 2 is where the Double Refund happens, because it is the stage at which the first integration beyond the platform gets a PSP key.

In Practice

Eva's €235, start to finish. The platform requests an iDEAL payment at 21:13 on 13 April 2027. At 21:14 it creates order #1072455 with payment pending. Seconds later, the PSP's server notification, or **webhook**, confirms success and the order becomes paid (From Cart to Confirmation: Checkout Orchestration). iDEAL has no separate capture step or card chargeback.

Assume a €0.30 fee and payout on Thursday 15 April, two business days later. These are illustrative assumptions; the cited Adyen indicative price is $0.13 + €0.22. The batch includes that day's sales, less fees and earlier refunds. Sanne's team matches it to orders using the PSP reference.

With a card, the example would instead authorize on Tuesday and capture on Wednesday when the label prints, within the cited seven-day Visa window. The fulfillment order id would identify that capture so a retry could not collect it twice.

The refund variant. Had Eva chosen a refund instead of size L on 19 April, the returns portal would have waited for the inspection on 22 April, then asked the platform to refund €199 with the return id as the key; the platform would have instructed the PSP; the PSP would have made a new iDEAL transfer back to Eva's account and reported it by webhook; the ERP would have booked a credit note against the invoice (Tax, Invoices and the Books: From Order to Ledger). The refund appears as a deduction in a later payout batch, which is why the batch for 22 April does not equal the sales of 22 April,

and why matching is by line, not by total. Support, since February 2027, could have seen all of this in the returns portal and issued nothing in the PSP console.

When It Goes Wrong

The duplicate refund. Two systems, no key. Detection: a daily list of refunds at the PSP against refunds in the platform by order and amount. Recovery: ask for the money back, because bank transfers cannot be recalled, and block the writer. Prevention: one writer, the return id as idempotency key, read-only console access for everyone else.

The lost outcome. The shopper paid and never returned; the webhook failed; the PSP stopped retrying. Detection: the daily job for payments pending beyond their method's expiry (15 minutes for iDEAL, 30 for cards at Mollie). Recovery: query the PSP, settle the order either way.

The expired authorization. The warehouse shipped on day eight and the capture failed. Detection: an alert when an authorized order approaches its window uncaptured. Prevention: capture at label print, and a fulfillment hold when the window is close.

The provider outage or account pause. The Card-Testing Night: 40 minutes with no online sales. Detection: a checkout health check that watches the PSP's error rate, not only its status page. Recovery: a rehearsed fallback for one method, a kill switch for the attacked method, an honest storefront message. Prevention: edge rate limits and velocity rules before authorization, so the PSP never has to pause you.

What to Take Away

Established.

- Money moves in four stages, authorization, capture, settlement and payout, with refunds and chargebacks flowing backward; capture must happen inside an authorization window of days, and a refund is a new payment that can fail later (Adyen, Stripe and Mollie documentation, n.d., accessed November 2026).
- The payment outcome arrives by webhook, at least once, in no guaranteed order, and must be verified; the merchant persists event ids, fetches the object when the decision matters, and runs a daily job for everything still pending (Stripe and Mollie documentation).

Depends on context.

- Capture on shipment is right for cards where cancellations before dispatch are common, because it saves refund fees and ties money to goods; it does not exist for iDEAL and other instant methods, and it is a liability if the warehouse cannot ship inside the window.

- One PSP with disciplined integration is right up to the Omnichannel stage; orchestration across PSPs is right at Group scale or where one provider's outage would stop several brands, and it changes the PCI scope if the merchant's service ever sees a card number. **The author's synthesis**.
- I treat payments as a problem of writers and keys before a problem of providers: name one system that may refund, one that may capture, one that may create a payment, give each action a business key the PSP will remember, and reconcile daily; everything else about the PSP can be changed later.
- The order's payment vocabulary should be the merchant's, not the PSP's: five or six states, moved only by confirmed events, with the provider's raw status beside them, so that changing provider changes one adapter.

Questions to Ask

1. Which systems hold a credential that can refund, capture or create a payment at the PSP, and who approved each one?
2. What is the idempotency key for a refund, a capture and a payment creation, and what does the PSP do when it sees the same key twice?
3. Where does the order store the PSP's reference, the authentication result and the liability-shift flag?
4. How long is the authorization window for each method, and what happens when a shipment approaches it uncaptured?
5. If the PSP's account is paused for an hour, what still sells, and who flips which switch?
6. How many payout feeds does finance match each month, by line or by total, and how long does it take?
7. Which regulatory changes (PSD3, 3-D Secure versions, PCI DSS) are on a calendar with an owner?

The money is in, and the webhook has told the order so. The order now has a life of its own, across the platform, the warehouse, the carrier and, when Waypoint delivers it, an order management system that holds every channel's orders in one record. The Order Record: Lifecycle, State and the Order Management System follows the order from "paid" onward.

Sources and Evidence

Company, product and investor sources

- Capture, Adjust authorisation, Refund, Sales day payout, Dispute process and flow, Webhooks, Batch-level reconciliation, PCI DSS compliance guide and SAQ A

eligibility, Adyen documentation, n.d., accessed 2026-11 — authorization and capture semantics, expiry windows, asynchronous refunds, payout batches, dispute stages, the PCI responsibility boundary.
- Pricing, Adyen, n.d., accessed 2026-11 — indicative per-method fees (volatile).
- Place a hold on a payment method, Refunds, Disputes, Measuring disputes, Monitoring programs, Strong Customer Authentication, CITs and MITs, Idempotent requests, Webhooks, Payouts, Payout reconciliation, Security guide, Card testing and PCI compliance guide, Stripe documentation, n.d., accessed 2026-11 — authorization windows, refund behavior and fees, dispute mechanics and thresholds, SCA exemptions and liability shift, MITs, idempotency keys, webhook delivery, settlement timing, reconciliation reports, SAQ mapping, card-testing controls.
- iDEAL, Place a hold for a payment, Refunds, Webhooks, Handling payment status and Settlement report, Mollie documentation, n.d., accessed 2026-11 — iDEAL semantics, hold periods, duplicate-refund blocking, id-only webhooks with ten retries over 26 hours, status model and expiry per method, settlement contents.
- Shopify Payments Terms of Service (Ireland) and Processor list, Shopify, n.d., accessed 2026-11; Form 10-K for fiscal 2025, Shopify Inc., 2026-02-11; Shopify Payments supported countries, Shopify Help Center, n.d., accessed 2026-11 — a platform's payments as a wrapper over licensed PSPs; Payments penetration; country availability.
- Annual turnover iDEAL and PIN payments, Dutch Payments Association, 2025-02-18 — 2024 iDEAL volumes; iDEAL | Wero, Currence iDEAL, n.d., accessed 2026-11 — the scheme's current branding.
- 2.5 billion payments with Bancontact and Payconiq in 2024, Bancontact Payconiq Company, 2025-02-12 — Belgian online volumes.
- PayPal remains most popular online payment method in Germany, Ecommerce News reporting EHI's "Online-Payment 2024", 2024 — German method shares for 2023 (secondary source).

Standards, regulation and official documentation

- Directive (EU) 2015/2366 (PSD2), Official Journal, 2015-12-23 — the SCA definition (Article 4(30)) and requirement (Article 97); applied from 13 January 2018.
- Commission Delegated Regulation (EU) 2018/389, Official Journal, 2018-03-13 — the RTS on SCA, applied from 14 September 2019, and its exemptions.
- Regulation (EU) 2015/751 on interchange fees, Official Journal, 2015-05-19 — the 0.2% and 0.3% caps (older than three years; still in force).
- Regulation (EU) 2024/886 on instant credit transfers, Official Journal, 2024-03-19 — receive and send deadlines, free payee verification.
- Payment services deal: more protection from online fraud and hidden fees, European Parliament, 2025-11-21, and State of play of legislative files, Council of the EU, 2026-10-29 — PSD3 and PSR content and status.

- EMV 3-D Secure, EMVCo, n.d., accessed 2026-11 — frictionless and challenge flows.
- Just published: PCI DSS v4.0.1, PCI Security Standards Council, 2024-06-11; Important updates announced for merchants validating to SAQ A, PCI SSC, 2025-01-30; PCI DSS, PCI SSC, n.d., accessed 2026-11 — versions, dates, scope, the SAQ A change.
- Visa Acquirer Monitoring Program fact sheet, Visa, 2025-05 — VAMP ratio and thresholds (volatile).
- Study on the payment attitudes of consumers in the euro area (SPACE 2024), European Central Bank, 2024-12 — online payment shares by method and country.

Research and practitioner sources

- Payments for Finance Architects: Eleven Reasons the Numbers Disagree, Adyen: One Platform from Checkout to Settlement, Stripe: Every Movement Is a Balance Transaction, Payments-to-Ledger: Draining the Suspense Swamp, Never Update, Only Append: Designing Custom Ledgers and Reconciled by Design: Traceability and Audit Across Boundaries in RECONCILED — the deeper treatment of parties, pricing, orchestration, tokens, the payments sub-ledger and reconciliation this chapter points to.
- Scams, Fakes and Account Takeovers: Fraud Prevention for Listings in LISTED — scoring and velocity controls beyond the basics given here.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

13. The Order Record: Lifecycle, State and the Order Management System

IN THIS CHAPTER, YOU WILL: Learn what an order record holds, which states it moves through, how it is routed and split, and when an order management system becomes necessary.

KEY POINTS:

- **An order is one record with several state machines, not one status.** Money, goods and returns each move on their own, and the goods state belongs to each shipment, not to the order. A single "status" string is where Kestrel & Pine's Three-Status Order began.
- **Routing and splitting are decisions, and they need a home.** Which location ships which line, whether to split, and what a split may cost are rules over stock, capacity, cut-off times and carrier cost. Platforms route by priority or distance; the rest is order management.

- **An order management system is the order hub once two of four things are true**: several channels create orders, several locations fulfill them, promises depend on location, or service cannot see one order across systems. Kestrel & Pine has all four and buys an order hub under its improvement program.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Schrödinger's Order** — three systems each held an honest status of a different object and nobody held the order's.
- **The Spaghetti Junction** — four order sources (platform, marketplace connector, finance system, store tills) each with its own connector to the warehouse and the books.
- **The Message in a Bottle** — one shipment confirmation, sent once and trusted, turned a two-parcel order into a delivered one.

A customer orders a jacket and socks, but the jacket ships first. The warehouse knows which items left; the payment provider knows what was paid; support needs to explain when the socks will arrive. If those facts remain separate, a single 'shipped' status can mislead both the customer and the team handling the order.

The merchant needs a record that brings those facts together: what was bought, at what price, where it goes and when it is due. A hosted platform may be enough for a small shop. Adding marketplaces, wholesale, stores and warehouses creates more orders and copies elsewhere. An **order management system (OMS)** becomes useful when it can coordinate those commitments, route work and track partial completion across systems.

Kestrel & Pine, the fictional outdoor brand this book follows, illustrates the problem with its December 2026 Three-Status Order: different systems gave different answers about the same purchase.

Where we are in the order's journey. Eva buys a jacket and socks from Kestrel & Pine, the fictional outdoor brand. Order #1072455 is created at 21:14 on 13 April 2027 and paid through iDEAL, the Dutch bank-transfer method. Order management tracks it through delivery and the replacement jacket's dispatch on 22 April. From Cart to Confirmation: Checkout Orchestration, Pick, Pack, Ship: Warehouse Management and Fulfillment and The Way Back: Returns, Exchanges, Refunds and Reverse Logistics cover those operations.

Setting the Scene at Kestrel & Pine: Order #1052877

In the first week of December 2026, a customer ordered a Trailhead fleece (€119) and a pair of merino hiking socks (€18) on the Dutch storefront: order **#1052877**, €137.

The fleece was picked and shipped from Duiven on Monday 7 December. The socks were short in their pick location and waited for replenishment from bulk stock, so the warehouse management system (**the WMS**, the software that runs picking, packing and shipping) kept the order open as a second pick.

The agency's shipment-confirmation integration could not represent a partial shipment. When the fleece left, it marked the whole platform order fulfilled and attached that parcel's tracking number. PostNL delivered it on 8 December, triggering the *Delivered* status and email.

Meanwhile, the **enterprise resource planning system (ERP)** waited for the warehouse's "order complete" message before invoicing. With the socks still unpicked, it showed *Not invoiced.* The **warehouse management system (WMS)** showed *In picking.* Each system exposed a different part of the situation.

System	What it held	Its status	True of what
The platform	The customer's order, one tracking number	Delivered	The first parcel
The WMS	A warehouse order with two lines, one still open	In picking	The second pick
The ERP	A copy awaiting completion	Not invoiced	The invoice run
The customer	A box with a fleece and no socks	"Where are my socks?"	The order

Figure 1: *Order #1052877: the platform, the WMS and the ERP each held an honest status of a different object, and nobody held the order.*

Eleven tickets followed before the socks shipped on 14 December (Where Is My Order? Notifications, Service and the Customer Record). Reviewing the case in January, Head of Engineering Noor Visser found no owner for the combined view. The platform held the customer-facing order, the WMS its execution and the ERP its finances. Other orders began in store tills, marketplace connectors and electronic trading-partner feeds.

Noor and CEO Mara Lindqvist made the remedy decision W2 of **Waypoint**, the company's improvement program: **choose one home for every order and specify what it must explain**.

What an Order Record Holds

An order is more than a list of items. Stripped to what every downstream system needs, the record has six parts.

Part	What it holds	Who reads it most
Header	Order number, channel, customer reference, placed-at time, currency, market, legal seller	Everyone

Part	What it holds	Who reads it most
Lines	Each SKU (a stock-keeping unit, one sellable variant), quantity, frozen unit price, discounts allocated, tax	Warehouse, finance
Money	Payment method, authorization and capture references from the payment service provider (**PSP**, the company that takes the payment), amounts paid, refunded, owed	Finance, service
Promises	Delivery address and method, the promised date, pickup store if any	Warehouse, carrier, service
Goods	Which lines are assigned to which location, in which shipment, with which tracking number	Warehouse, carrier, service, customer
References	The marketplace's order id, the EDI document number, the invoice number, return ids	Integration, finance

Three distinctions trip up newcomers, and Follow the Order: What Happens Between "Buy" and "Delivered" draws them in full. **A cart is not an order**: a cart is a draft that commits nobody. **An order is not a shipment**: one order can leave in two parcels from two locations. **An order is not an invoice**: the invoice is a legal document issued when goods ship or when finance says so (Tax, Invoices and the Books: From Order to Ledger).

Order numbers and identity

The order number is the one thing every system, the customer and the carrier label share. Two rules make it useful. First, **one number family across channels**: whether an order came from the storefront, Amazon.de or an EDI message, it gets a number from the same sequence, and the channel's own id (the marketplace order id, the EDI document number) is kept as a reference, not used as the key. Second, **the order number is the idempotency key for creating the order anywhere else**. Idempotent handling means that receiving the same message twice has the same effect as once; keyed on the order number, a duplicated "order placed" event finds the existing order and does nothing. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns the pattern; this chapter needs only the rule, because the ERP, the WMS and the OMS will each receive that event at least once.

Several State Machines on One Record

A **state machine** is a fixed set of states an object can be in and the allowed moves between them. The mistake behind the Three-Status Order was to assume an order has one. It has at least three, and the platforms say so in their own documentation.

Shopify keeps three status families on one order (help center, n.d., accessed November 2026). The *order status* is Open, Archived or Canceled. The *payment status* has eight values, from Authorized and Paid to Partially refunded, Refunded, Voided (which "releases the hold on the customer's payment method", distinct from canceling) and Expired. The *fulfillment status* has six: Unfulfilled, In progress, On hold, Scheduled, Partially fulfilled, Fulfilled.

Below the order, each *fulfillment order* (the instruction to one location to ship a group of lines) has seven statuses of its own plus a request status for the handshake with the warehouse (developer docs, n.d.).

commercetools keeps an `OrderState` (Open, Confirmed, Complete, Cancelled), a `ShipmentState` with eight values including *Partial* ("items in the Order are shipped in more than one shipment") and *Backorder*, and a `PaymentState`; a fourth field can point to a custom state machine, with transitions left unvalidated when "an external system runs the workflow" (API docs, n.d., accessed November 2026). Its return items carry two further states each.

Machine	Typical states	Lives at	Who moves it
Order lifecycle	Open, confirmed, complete, canceled	The order	The order system
Money	Authorized, paid, partially refunded, refunded, voided	The order	The PSP's events
Goods	Requested, accepted, in progress, shipped, delivered, canceled, incomplete	Each fulfillment order (shipment), not the order	The warehouse and the carrier

Machine	Typical states	Lives at	Who moves it
Return	Requested, received, graded, refunded or exchanged	Each return	The returns portal and the warehouse

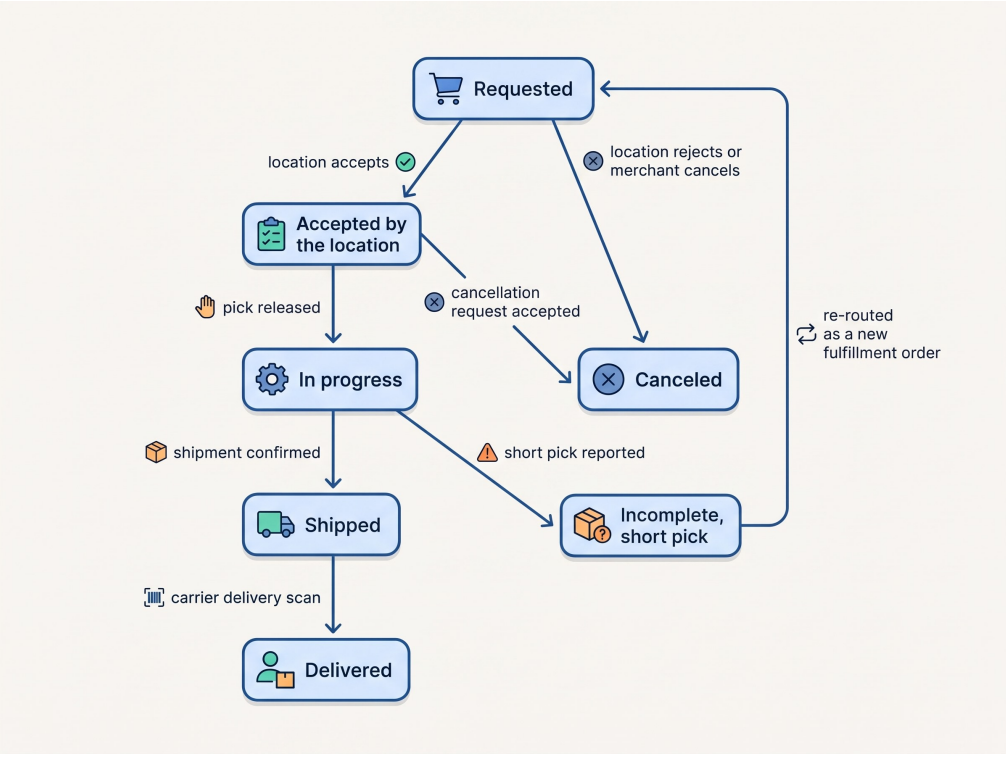


Figure 2: *The goods state machine lives on each fulfillment order, not on the order: a two-parcel order has two of these, and "delivered" is true only when both reach the last state.*

The order-level words a customer sees ("Delivered", "Shipped") are summaries computed from the machines below, and two rules keep the summary honest. **"Fulfilled" is not "delivered"** unless the carrier's events are fed back; both Shopify's and commercetools' models leave that gap for the merchant to close (Getting It There: Carriers, Labels, Tracking and the Delivery Promise). And **the summary of a split order is the state of its least-advanced shipment**, which is the rule the agency's flow lacked: with it, #1052877 would have read "Partially shipped: one parcel delivered, one in preparation".

What "Delivered" on the order says	What it does not say
At least one carrier has scanned at least one parcel as delivered	That every line of the order has been delivered
The platform received a delivery event	That the invoice was issued or the ERP knows
The customer-facing summary has been computed	That a return, if any, has started

Changing a Placed Order

An order is edited, held or canceled far more often than newcomers expect: wrong size, wrong address, a line out of stock, a fraud hold, a customer who changed their mind. Each change is a transaction against the money, goods and stock commitments, not a form update.

Edits. Shopify's order editing (developer docs, n.d., accessed November 2026) begins a *calculated order* that tracks changes, lets a line's quantity be set to zero to remove it, and commits the edit; "You can edit only unfulfilled line items", and "If an edit changes the total order value, then a balance might need to be collected from or refunded to the customer". The constraint is the rule everywhere: **the edit window closes when the warehouse accepts the work**. In the fulfillment-order handshake that Shopify documents, once a warehouse has accepted a request "the merchant can no longer cancel it directly and must submit a cancellation request". Every edit must also reach the systems that already copied the order: the WMS's pick, the ERP's invoice copy, the availability service's commitment. An edit that reaches the platform and not the warehouse ships the wrong size.

Holds. A hold stops fulfillment without canceling: a fraud review, an address check, a customer's request. The stock stays committed; Shopify documents that an order on hold keeps its inventory reserved while fulfillment waits.

Cancellations. Before acceptance, a cancellation releases stock and voids or refunds the payment. After shipment, a cancellation is not possible; what the customer calls "cancel" becomes a return, and the undo of a shipment is a business process (return label, receipt, inspection, refund) rather than a database rollback. This is a **compensation**: a later step that reverses an earlier one when the whole cannot be undone, the vocabulary of the saga pattern that Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation explains. The order system is, in practice, the coordinator of that sequence.

Routing, Sourcing and Splits

Once an order can be fulfilled from more than one location, three decisions appear that a single-warehouse merchant never makes. **Sourcing** is choosing which location fulfills which line. **Routing** is the rule set that makes that choice. A **split shipment** is one order shipped in more than one parcel, usually from more than one location or at more than one time.

Platforms route by simple rules. Adobe Commerce ships source priority (work through a configured list) and distance priority (the nearest source to the shipping address), per its documentation updated 20 August 2026. Shopify assigns lines to a location through fulfillment orders, and the assigned location may change only while the fulfillment order is still open, scheduled or on hold. These rules know about stock and geography. They do not know about capacity, cut-off times, carrier cost, marketplace fees or a maximum number of parcels.

Order management vendors describe what the fuller decision looks like, and the descriptions are worth reading as a specification rather than as marketing. Kibo's routing API (documentation, n.d., accessed November 2026) returns a *suggestion* built from *candidates*, "a single, possible way to fulfill an item or group of items", logs why locations were considered or rejected, and "does not create shipments or deduct inventory"; when only part of a shipment is in stock, "the remaining unavailable quantity is automatically placed into a new child shipment for re-routing". Fluent Commerce's split logic "analyses inventory and costs across split scenarios, aiming to fulfil the maximum number of items while using the minimum number of shipments", and merchants may "enforce a maximum allowable cost or number of shipments for any given order" (documentation, n.d., accessed November 2026). Manhattan says an OMS assigns each order "to the fulfillment center that can meet the promise most profitably" (n.d., accessed November 2026).

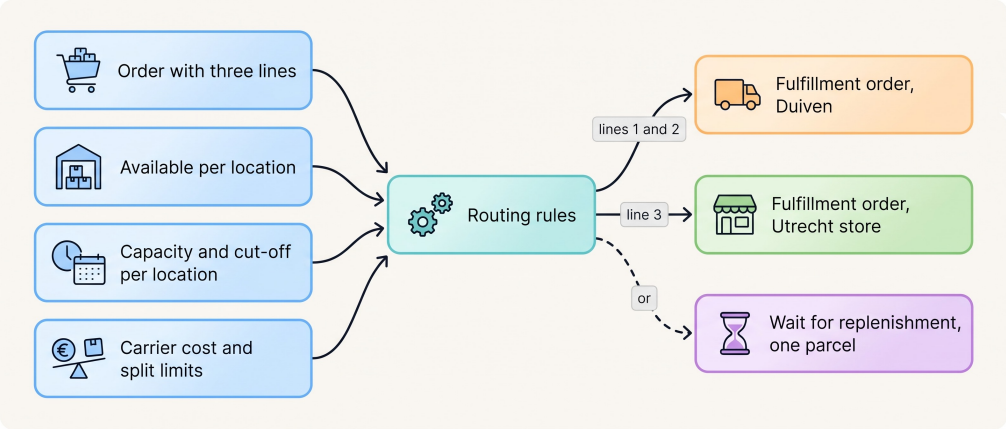


Figure 3: *Routing turns one order into one or more fulfillment orders using stock per location, capacity, cut-off times and cost; the same inputs decide whether to split now or wait and ship once.*

The business trade-off inside a split is concrete. Shipping the fleece now and the socks later keeps the promise on one line and breaks it quietly on the other, and costs a second parcel. Waiting ships once but late. **A split is a cost decision that needs the carrier rates, the promise and the customer's channel as inputs**, and whichever system makes it must also tell the customer "two parcels", the message nobody at Kestrel & Pine could send in December 2026.

A **backorder**, a line accepted for a unit that is out of stock but expected, is a split in time rather than space, and needs the same honesty: a date from Is It in Stock? Inventory, Availability and the Promise You Can Keep's incoming quantities, and a message.

Many Channels, One Record

Kestrel & Pine's orders begin in four systems. Storefront and B2B (business-to-business) portal orders are created by the platform. Marketplace orders are pulled by the marketplace connector. Bergzicht's EDI orders land in the ERP, and 38% of wholesale orders are keyed into the ERP by hand. Store sales are POS receipts. Each source has its own connector to the warehouse, its own path to the books and its own status vocabulary. That is the Spaghetti Junction, the normal result of adding one channel at a time.

The alternative is one order hub: every channel's order is created in one system, with one state model and one number family, and that system drives the warehouse and informs the books. Which system that should be is the real question, with three honest candidates.

The order lives in	What it is good at	What it lacks	Common at
The platform	The customer-facing record: account history, status page, checkout context, returns through its API	Orders from channels the platform did not sell through; routing beyond priority and distance; service view across systems	Starting and Growing

The order lives in	What it is good at	What it lacks	Common at
The ERP	The financial record: invoice, credit note, receivable, cost; wholesale order entry	Customer-facing state; parcel-level goods state; speed (the ERP runs batches); the ERP partner becomes the integration owner	Growing merchants with strong wholesale
An OMS	The operational record: every channel, one state model, routing and splits, holds and edits, the service view	The checkout (stays on the platform) and the books (stay in the ERP); it is a third system to integrate	Omnichannel and Group

Fluent Commerce states the boundary plainly on its own site: "your ERP stays the system of record" for finance, while the OMS decides where orders go and writes back (n.d., accessed November 2026). The three records are not rivals; they are three truths about the same order, and the design decision is which system owns which truth and how the others are told (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record owns the general idea). Waypoint W2 is that decision written down: the OMS becomes the order hub, every channel's order is created there, the platform keeps the checkout and the storefront order history, and the ERP keeps the financial record.

The OMS as a Product

An **order management system** takes orders from every channel, decides which location fulfills each line, tracks each order through fulfillment, delivery and returns, and gives service agents one view. **Orchestration** is its job of sequencing those steps across systems. The vendors describe the same shape: Manhattan lists inventory visibility across sources, promise dates at checkout, sourcing, "end-to-end fulfillment execution and tracking" and post-purchase changes (n.d., accessed November 2026); Kibo lists "distributed order orchestration across every store, warehouse, and vendor" with rules such as fewest shipments, lowest cost or closest distance (n.d.); IBM Sterling lists ship from store, in-store pickup and a call-center add-on "for complex order modifications" (n.d.).

What the OMS owns, and what it leaves alone, is the point.

Component	Responsibility	System of record for	Not for
Order intake	Create one order per channel event, keyed on the order number; attach channel references	The operational order	Checkout, pricing, payment
Availability module	Available per channel and location from stock events (Is It in Stock? Inventory, Availability and the Promise You Can Keep)	Available, committed	On hand
Routing	Choose locations, decide splits, create fulfillment orders	The sourcing decision	Picking
Orchestration	Drive the money, goods and return machines from events; request, accept, cancel with the warehouse	The order's state	The warehouse's internal states
Service view and edits	One screen per order across channels; edits validated against state	Edits and holds	Refund execution (the PSP), invoice (the ERP)

Component	Responsibility	System of record for	Not for
Events out	Status events to the platform (customer history), the ERP (invoice triggers), notifications, the data warehouse	Nothing; it publishes	Anything it did not change

Requirements per interaction

The figures are the author's illustrative targets for an Omnichannel merchant of Kestrel & Pine's size, not Kestrel & Pine's records.

Interaction	How fresh	How consistent	How fast	If the OMS is down
Order creation from a channel event	Seconds after confirmation	Exactly one order per order number	Under 2 s end to end *(assumption)*	Events queue; the platform's checkout keeps working; nothing is lost
Fulfillment order to the WMS	Before the next wave release; minutes at peak	Exactly one pick list per fulfillment order	Minutes	The WMS continues on what it has
Status event back to the storefront and service	Seconds to a minute	Per shipment, never collapsed to one word	Under 1 min *(assumption)*	The customer sees the last known state, dated
Edit or cancellation	Now	Validated against the goods state; propagated to WMS and ERP	Under 2 s *(assumption)*	Refuse the edit rather than apply it in one system only

Interaction	How fresh	How consistent	How fast	If the OMS is down
Invoice trigger to the ERP	Per shipment, at dispatch (Tax, Invoices and the Books: From Order to Ledger sets the rule)	Idempotent per shipment id	Daily batch is acceptable	Resume from the last acknowledged shipment

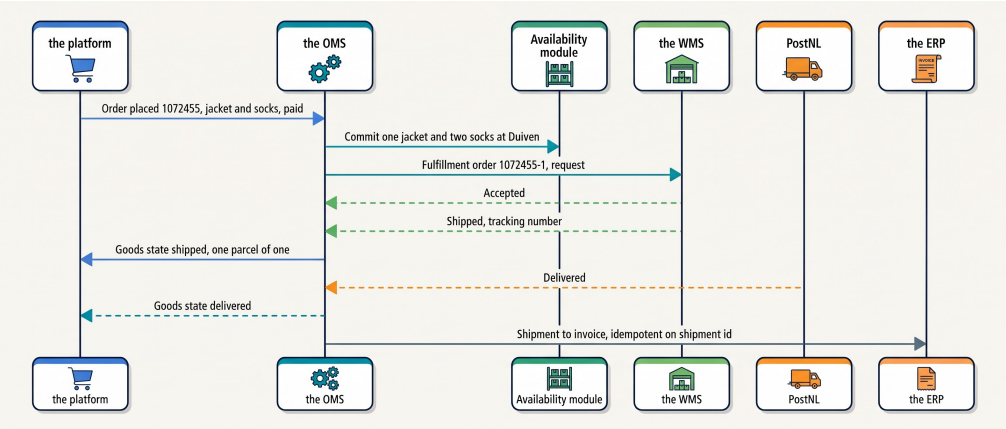


Figure 4: *Eva's order under Waypoint: the platform confirms and pays, the OMS commits stock, requests fulfillment, receives the warehouse's and the carrier's events, computes the goods state per parcel and publishes it to the storefront and the ERP.*

The rule I use for when an OMS is needed: **at least two of these four**. Several fulfillment locations. Several channels creating orders. Promises that depend on location (click-and-collect, ship-from-store). Service agents who cannot see one order across systems. One of them can be handled by the platform plus a connector; two make the joins the problem. Kestrel & Pine has had all four since 2023.

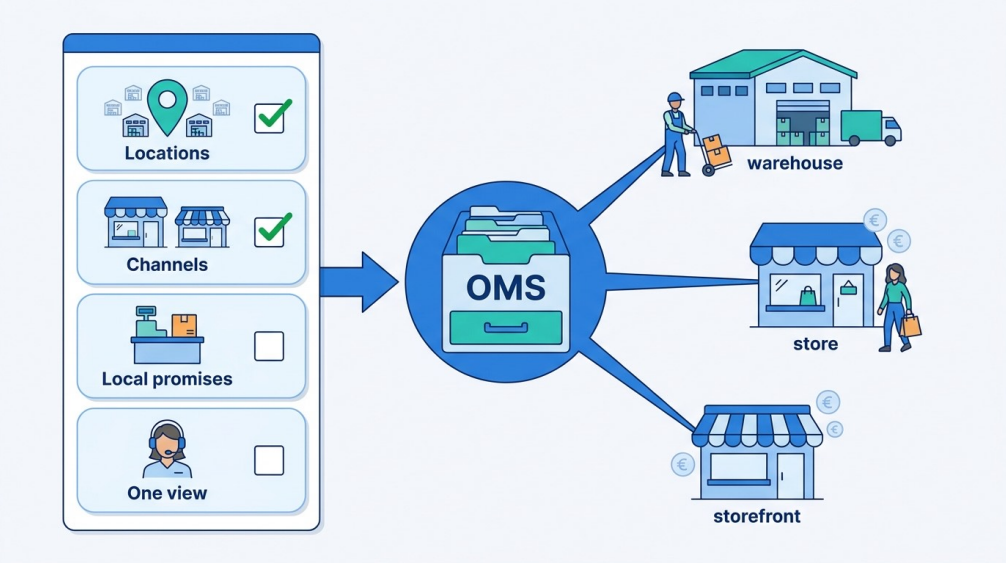


Figure 5: *An OMS becomes the order hub once two of four conditions hold: several locations, several channels, location-based promises, or service that cannot see one order.*

The Choices

1. **The platform's order is the record**. The platform creates and holds every order; marketplace apps and a POS from the same vendor create orders in it; a 3PL (a third-party logistics provider, a warehouse that ships for the merchant) or WMS connector reads them and writes fulfillments back. Shopify's own features, fulfillment orders with a request-and-accept handshake, holds, edits, returns and multi-location assignment, cover this well; what its documentation does not describe is rule-based sourcing on cost, capacity or cut-off, and that absence is the limit (an inference from the documentation, not a vendor statement).

2. **The ERP as the order hub**. Every channel's order is created in the ERP, which drives the warehouse and the books. Attractive where wholesale dominates and the ERP already holds most orders, and where the ERP partner is strong. The costs are speed (ERPs run in batches), a customer-facing state the ERP cannot show, and a hub owned by the finance system's vendor. The Back Office: ERP Integration and Who Is the Hub works through the hub debate.

3. **A bought OMS as the hub**. The platform keeps checkout; an OMS holds every order, routes, orchestrates and serves; the ERP keeps finance. This is W2. The price is a third core system, its subscription (illustratively €4,500 a month for Kestrel & Pine's OMS including its availability module, with a one-time implementation of about €320,000; see Build, Buy or Assemble: Choosing and Changing Platforms), and the integration work to re-point every order source and the warehouse.

4. **A built order service**. A small team writes its own order record, state machines and routing over a queue. Right for a vendor whose product is commerce, or a group with unusual rules and a platform team; wrong for a 14-person technology team with a warehouse to run. The order record is where custom code is most expensive to get wrong, because every other system depends on it.

Trade-offs

Criterion	1. Platform	2. ERP hub	3. Bought OMS	4. Built service
Orders from every channel	Only channels sold through the platform or its apps	Yes, often by hand or batch	Yes, by event	Yes
State model	The platform's; summaries collapse on the way out	The ERP's, financial first	One model, money, goods and return separate	Whatever is designed
Routing and splits	Priority and distance	Basic	Rules over stock, capacity, cut-off, cost	Custom
Service view across channels	The platform's orders only	The ERP's screen	One screen	Must be built
Time to value	Days	Months, with the ERP partner	Months (Kestrel & Pine: August to December 2027)	Quarters
Running cost	Included	ERP licenses and partner	Subscription plus integration	Team
Exit cost	Low	High, entangled with finance	Medium, the order events are reusable	High

Criterion	1. Platform	2. ERP hub	3. Bought OMS	4. Built service
Breaks when	A second channel or location arrives	Peak day; a customer asks the ERP a parcel question	A missed event; a routing rule nobody owns	The team changes

My judgement: stay with the platform's order until the second order source or the second location, then choose between the ERP and an OMS by asking who should answer a customer's parcel question. If the answer is "not the finance system", it is an OMS. Build only when order management *is* the product.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs the approach is outgrown
1 — Starting (Kestrel & Pine 2016–2018)	The platform's order is the only order; a 3PL app reads it; the accounting tool imports a daily summary	Order number continuity; the platform's own states used as they are	Orders keyed by hand into anything
2 — Growing (2019–2022)	The platform's order plus copies in the ERP and the 3PL; a marketplace app creates marketplace orders in the platform; support checks two systems	Idempotent order creation in the ERP keyed on the order number; a daily reconciliation of orders between platform and ERP; one status mapping written down	The ERP's order count differs from the platform's; the first partial shipment that reads "fulfilled"; store orders invisible online

Stage	What is typical	What becomes necessary	Warning signs the approach is outgrown
3 — Omnichannel (2023–, now)	Four order sources, four status vocabularies, connectors from each to the warehouse and the books	An OMS as hub with one number family and separate money, goods and return machines; routing rules with owners; status events per shipment to the storefront, service and ERP; edits validated against state	The Three-Status Order; eleven tickets; routing done by warehouse staff by phone; click-and-collect checked by e-mail
4 — Group	Several brands or countries, several warehouses, perhaps several OMS instances after acquisitions	One state model and one number family per unit, with shared definitions; routing across brands' stock with allocation rules; the OMS as the saga coordinator for every channel	Brands' orders routed to each other's stock without rules; the service tool reads two OMSs

The shape of the record does not change with scale: an order has lines, money, promises, goods per shipment and references at every stage. What changes is how many systems create orders and how many fulfill them, and the moment either count passes one is the moment the order needs a hub.

In Practice

The dated vendor evidence is above: Shopify's three status families and fulfillment-order handshake, commercetools' separate states, Adobe's sourcing algorithms (2026), Kibo's candidates and child shipments, Fluent's split rules and ERP boundary, Manhattan's and IBM's OMS descriptions, all accessed November 2026. One figure frames the volume: Shopify processed 13 billion webhooks in the period its CTO described for 2021

(engineering blog, 20 January 2022; a source more than three years old, so the figure is historical), the scale at which "your app might receive the same webhook more than once" becomes the design assumption behind idempotent order creation.

Kestrel & Pine, worked: W2 and the order that had three statuses

Waypoint W2 is being implemented through 2027, seam by seam (Moving Without Burning the Bridge: Replatforming and Migration describes the method). The OMS shadow-copied all orders in August, ran in parallel from 1 to 14 September (differences on about 1.4% of orders in the first week, mostly status mapping and timing, illustrative), took marketplace orders on 15 September, storefront orders on 13 October, paused for a change freeze around Black Friday, and takes EDI and manual wholesale orders on 8 December; the old order paths retire in January 2028. From then on every order, whatever its channel, is created in the OMS by an event keyed on its order number, gets a number from one family continuing the platform's sequence, and keeps the channel's own id as a reference.

Replay #1052877 through that design. The platform's order event creates the OMS record. An illustrative routing rule allows a consumer order under €150 to wait one working day for replenishment before splitting. When the socks remain unavailable on Monday, the OMS creates two fulfillment orders and tells both the status page and notification service: **ships in two parcels**.

The fleece's dispatch and delivery update its own fulfillment order, 1052877-1. The order summary becomes "Partially delivered: 1 of 2 parcels". Each dispatch also requests an invoice, using the shipment id to prevent duplicates. Finance issues one invoice for the fleece on 7 December and another for the socks on 14 December; both retain the original order number (Tax, Invoices and the Books: From Order to Ledger).

Support can now see both parcels. The customer still waits for the socks; that stock problem remains (Is It in Stock? Inventory, Availability and the Promise You Can Keep). What improves is the explanation and the need to contact support repeatedly.

When It Goes Wrong

Failure	What it looks like	Detection	Recovery
Duplicate order	The "order placed" event arrives twice; the ERP or WMS creates two orders; the customer receives two parcels	Two records with one order number; a daily count of orders per source against the platform's count	Idempotent creation keyed on the order number; cancel the duplicate before it ships; refund if not
Lost confirmation	The parcel left; the confirmation never arrived; the order reads unfulfilled, the ERP never invoices	A daily reconciliation of shipped-in-WMS against shipped-in-OMS	Replay from the WMS; never re-key by hand (Pick, Pack, Ship: Warehouse Management and Fulfillment)
Collapsed status	A partial shipment reads "fulfilled"; a delivery scan reads "delivered" for the whole order	Orders with more than one fulfillment order and a single-word status	Goods state per fulfillment order; summary from the least-advanced shipment
Out-of-order events	"Delivered" arrives before "shipped"; a cancellation arrives after the pick started	A state transition the machine does not allow	Tolerant handlers: store the early event, apply it when its predecessor arrives; carrier events keyed on the tracking number
Edit after acceptance	The platform shows the new size; the box holds the old one	Edits committed while the fulfillment order is accepted or in progress	Lock edits at acceptance; turn later changes into a cancellation request plus a new order

Failure	What it looks like	Detection	Recovery
Cancel after shipment	Support "cancels" a shipped order; stock is released, money refunded, the parcel still arrives	Cancellations on orders with a shipped fulfillment order	Cancel becomes return; refund on receipt (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics)
Routing nobody owns	Lines split across three locations for a €40 order; a store asked to ship what the warehouse has	Parcels per order and split cost per channel, tracked weekly	Named owner for routing rules; a maximum shipments per order; a split cost cap
Stuck invoice trigger	The ERP waits for "order complete" that a short pick prevents; nothing is invoiced for a week	Shipped-not-invoiced older than one day	Invoice per shipment, idempotent on the shipment id; an exception queue with an owner

Most rows end the same way: an identity on every message, a state machine that refuses impossible moves, and a daily comparison between the systems that hold copies. **An order hub does not remove the joins; it gives each join one key, one owner and one check.**

What to Take Away

Established.

- An order carries several state machines: Shopify keeps order, payment and fulfillment statuses on one order plus statuses on each fulfillment order; commercetools keeps order, shipment and payment states plus return states; in both, "fulfilled" means "delivered" only when carrier events are fed back (documentation accessed November 2026).
- The fulfillment order, with a request, accept or reject, in-progress, shipped and cancellation-request lifecycle, is the documented contract between order

management and the warehouse; edits are possible only on unfulfilled lines, and cancellations after acceptance are requests (Shopify documentation, n.d.).
- Order management vendors converge on one boundary: the OMS holds the operational order and the sourcing decision, the ERP the financial record, the warehouse the execution (Fluent, Kibo, Manhattan, IBM, accessed November 2026).

Depends on context.

- The platform's order is enough while one channel creates orders and one location fulfills them; the second of either makes an order hub necessary.
- The ERP can be the hub where wholesale dominates, the ERP partner is strong and no customer ever asks the finance system where a parcel is; an OMS is the hub where consumers, stores and marketplaces do.
- Invoice per shipment, not per completed order, wherever orders can split (Kestrel & Pine's rule); a merchant that never splits may invoice per order, but either trigger must be idempotent on the shipment or order id, or a short pick blocks the books.
- Split now or wait is a cost rule that depends on carrier rates, marketplace fees and the promise; it needs an owner and a cap, not a default.

The author's synthesis.

- The order's home should be chosen by asking who answers the customer's parcel question, because the system that can say "two parcels, one delivered" is the only one that can also route, edit and reconcile honestly; everything else follows from that record.
- Buy the OMS and keep the checkout on the platform and the books in the ERP; the order record is the worst place for a small team's custom code, because every other system fails when it does.

Questions to Ask

1. Which systems create orders today, and which single system is the order's home for operations, for the customer and for finance?
2. Does one order number family cover every channel, and is the order number the idempotency key for creating the order in every other system?
3. How many state machines does the order record keep, and is the goods state held per shipment or collapsed into one word?
4. What computes the customer-facing summary of a split order, and what message goes out when an order ships in two parcels?
5. When does the edit window close, and how does an edit reach the warehouse, the ERP and the stock commitment?

6. What are the routing rules, who owns them, and what is the maximum number of parcels or the maximum split cost per order?
7. What triggers the invoice, and what happens to it when one line is short-picked?
8. Which daily reconciliation compares orders and shipments across the platform, the hub, the warehouse and the ERP, and who reads it?
9. If the order hub is down for an hour, can checkout continue, and how are the missed events replayed?

Eva's order now has one home, one number, three honest state machines and a warehouse that has accepted the work. The next chapter follows the money and the paperwork that this record generates: the tax it carries, the invoice it becomes and the entries it posts in the books. Continue with Tax, Invoices and the Books: From Order to Ledger.

Sources and Evidence

Company, product and investor sources

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- What is an OMS? and Manhattan Active Order Management, Manhattan Associates, n.d., accessed 2026-11 — OMS definition and capabilities; Forrester Wave Q1 2025 placement as reported by the vendor.
- IBM Sterling Order Management, IBM, n.d., accessed 2026-11 — omnichannel fulfillment and the call-center add-on.

Standards, regulation and official documentation

- None cited directly; the invoice rules this chapter defers to are in Tax, Invoices and the Books: From Order to Ledger.

Research and practitioner sources

- Saga, Chris Richardson, microservices.io, n.d., accessed 2026-11 — compensation and orchestration vocabulary used for "cancel after shipment".

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

14. Tax, Invoices and the Books: From Order to Ledger

IN THIS CHAPTER, YOU WILL: Follow one order from the tax shown at checkout to the invoice, the credit note and the ledger entries, and learn what finance needs from the shop.

KEY POINTS:

- **Tax is computed from facts the shop owns and rules it should not own.** The ship-to address, the product's tax category and the customer's status (consumer or VAT-registered business) decide the rate; the rates, thresholds and mandates belong to a tax engine or one rate table, maintained once for every channel.
- **The invoice is a legal document, not a receipt.** It carries a gap-free sequential number, cannot be changed once issued, and is corrected only by a credit note that references it; one system per legal entity issues it, and in Belgium since January 2026 a business invoice travels as structured data over Peppol.

- **The books agree with the shop only when someone proves it**. Orders, the payment provider's payout reports and the bank statement are three views of the same money; a monthly three-way match, keyed on stable identifiers, turns a commerce platform's promises into accounts an auditor will sign.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Boomerang** — refunds, credit notes and restocking that are matched by hand after the return, and sometimes twice, because no document references the one it corrects.
- **Schrödinger's Order** — the order that is *Delivered* in the platform and *Not invoiced* in the finance system for six days, because the invoice waits for an "order complete" message that a short pick prevents, and nobody watches for it.
- **The Long Fuse** — an e-invoicing mandate with a published date that becomes a project only when the first business customer refuses a PDF.

A customer pays for an order, but the amount reaching the merchant's bank may be smaller because the payment provider deducted a fee. A later refund changes the position again. Finance needs to connect the tax calculation, invoice, fee, refund and bank receipt to explain the difference. Without those links, accountants must reconstruct it in spreadsheets and may struggle to distinguish normal deductions from missing money.

Kestrel & Pine, the fictional outdoor brand used here, must do this for 293,000 online orders, 43,000 marketplace orders, 118,000 store transactions and 9,400 wholesale orders a year, across three countries. Manual explanation does not scale with that workload.

We follow the value-added tax (VAT) within Eva's €235 order into the enterprise resource planning system (ERP), which holds the books. A Belgian wholesale example shows invoicing over **Peppol**, a network for exchanging structured invoices. Further detail sits in Taking the Money: Payment Providers, Authorization, Capture and Refunds, Documents Between Companies: EDI, Punchout and E-Invoicing, The Back Office: ERP Integration and Who Is the Hub, Payments-to-Ledger: Draining the Suspense Swamp and Never Update, Only Append: Designing Custom Ledgers.

Where we are in the order's journey. Eva's jacket order produces financial records from checkout onward: an invoice at dispatch, a provider payout and ledger entries, followed by any return adjustments. Finance reconciles those records and reports the tax in the Dutch VAT return.

Setting the Scene at Kestrel & Pine: Sanne's April Close

Finance Director Sanne de Wit closes April 2027 in the second week of May. Four payment routes must reconcile with the ERP:

- The online **payment service provider (PSP)** pays daily batches after fees, refunds and chargebacks, the reversals made by cardholders' banks. Assume about thirty batches of 800 orders each.
- A terminal provider settles card payments from six stores separately.
- Three marketplaces pay on different schedules after commission.
- **1,100 wholesale accounts** pay invoices on 30-day terms. Some pay short, twice or with references the ERP cannot match.

Each route needs its own matching rules before Sanne can trust the close.

Two winter incidents remain unresolved. January's **Double Refund** sent **€19,400** twice across **212 returns**, once through the portal and again through support. The ERP credit notes were correct; nobody compared them with the actual refunds.

December's **Three-Status Order**, #1052877, showed *Delivered* while the ERP waited six days to invoice. The warehouse could not report "order complete" until missing socks were replenished. No report flagged the dispatched fleece as uninvoiced.

A third thing is new. Since 1 January 2026, businesses established in Belgium must exchange invoices with each other as structured electronic documents over the Peppol network (a European network for structured electronic invoices). Whether the obligation reaches a supplier with no Belgian establishment is not settled by the sources this book relies on (Documents Between Companies: EDI, Punchout and E-Invoicing reads the scope), but Kestrel & Pine's Belgian business customers' bookkeeping expects a Peppol invoice, so its ERP has sent them one since then through the access point of its EDI provider, the service that relays electronic business documents between companies.

In April Ardennes Trail Co., a wholesale customer in Namur, rejected an invoice because the buyer reference it had asked for was missing. Sanne asks Noor two questions: **which system is allowed to issue an invoice, and how do we prove, every month, that the orders, the payouts and the bank agree without reading them one by one?**

What the Commerce System Owes Finance

In business terms the shop owes finance three outcomes. **The right tax**: every sale carries the VAT (value-added tax, Europe's consumption tax, charged at every sale and reclaimed by business buyers) of the right country at the right rate; tax authorities

enforce it. **A valid invoice**: numbered in sequence, never edited and kept for years; auditors and business customers, whose VAT deduction depends on it, enforce it. **Books that match the bank**: the ledger, the PSP's reports and the bank statement agree; the auditors and the board enforce it at every close.

A handful of documents carry those outcomes, and each has one owner. This is the chapter's version of the entity–owner table from Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record.

Document	Created when	Owner (system of record)	References
Order	Checkout completes	The platform (from Q4 2027 at Kestrel & Pine, the OMS, the order management system)	Cart, payment reference
Tax calculation	At checkout (estimate) and at invoicing (final)	The tax engine or rate table; recorded on the order and the invoice	Ship-to address, product tax category, customer status
Invoice	At dispatch of each shipment (goods) or at order (digital goods)	The ERP, one numbering sequence per legal entity	Order number, shipment id
Credit note	When a return is accepted or an invoice is wrong	The ERP, same sequence family	The invoice it corrects
Payment event	When the PSP authorizes, captures, refunds	The PSP; mirrored on the order	Order, invoice or credit note
Payout report	When the PSP pays the merchant's bank	The PSP	Every payment, refund, fee and chargeback in the batch
Ledger entries	When documents are posted	The ERP's general ledger	All of the above, through their identifiers

Two rules follow from the table and carry the rest of the chapter. **The shop does not issue invoices**: it issues order confirmations, which look like invoices to a consumer and are not; the ERP issues the legal document. **Nothing posts to the ledger from a webhook** (the HTTP call a PSP makes to the merchant's server when a payment event happens): payment events update the order; the ledger is fed from documents, in batches, with identifiers that let every entry be traced back.

VAT in Plain Words

Five facts about VAT carry the chapter.

Rates differ by country and by product category. EU law sets a minimum standard rate of 15% and lets each Member State apply reduced rates to listed categories (European Commission VAT rates page, n.d., accessed November 2026). Kestrel & Pine's three markets:

Country	Standard rate	Reduced rates	Source
Netherlands	21%	9%; 0% for exports and intra-EU supplies	business.gov.nl, n.d., accessed 2026-11
Germany	19%	7%	§ 12 UStG, consolidated text, accessed 2026-11
Belgium	21%	12%, 6%, 0%	PwC Worldwide Tax Summaries, September 2026 (the official page could not be fetched)

Kestrel & Pine's clothing and equipment carry the standard rate everywhere.

Consumer prices include VAT; business prices exclude it. Stripe Tax's documentation (n.d., accessed November 2026) puts it plainly: "Tax-inclusive prices are common for B2C sales in many markets outside the US", "Tax-exclusive prices are common in the US and Canada and for B2B sales in other countries", and once a price's tax behavior is set it cannot be changed. Eva's €199 jacket is tax-inclusive; the wholesale €95 is tax-exclusive. The pricing model is a property of the market and the customer type, not of the product (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax).

Cross-border consumer sales are taxed where the customer is. Once a merchant's sales to consumers in other EU countries exceed €10,000 a year, it charges the *destination* country's VAT, and since 1 July 2021 the EU's One Stop Shop (OSS) lets it report all of that

VAT in one quarterly return at home (European Commission OSS page, n.d., accessed November 2026). Kestrel & Pine's €8.5 million of Belgian and German online sales passed that threshold years ago, so a Belgian shopper pays 21% Belgian VAT, a German one 19% German VAT, and both are reported through the Dutch OSS return. The shop must know the destination rate per product category *at checkout*.

Business sales to another EU country carry no VAT. With a valid customer VAT number, checked in the EU's VIES service, the seller charges 0% and the buyer accounts for the VAT in its own return, the **reverse charge** (Your Europe cross-border VAT page, n.d., accessed November 2026). The invoice states both VAT numbers and the reverse-charge mention. Ardennes Trail Co.'s order, below, is this case.

The United States is the contrast. There is no VAT; each state levies its own sales tax, and since *South Dakota v. Wayfair* (21 June 2018) a state may require a remote seller to collect it once sales into the state pass a threshold (South Dakota's was $100,000 or 200 transactions), regardless of physical presence. Rates vary by state, county and city, which is why **tax engines** exist: Avalara's product page claims rules across "12,000+ U.S. sales and use tax jurisdictions" (n.d., accessed November 2026). A Dutch merchant that opens a US storefront inherits all of this on day one.

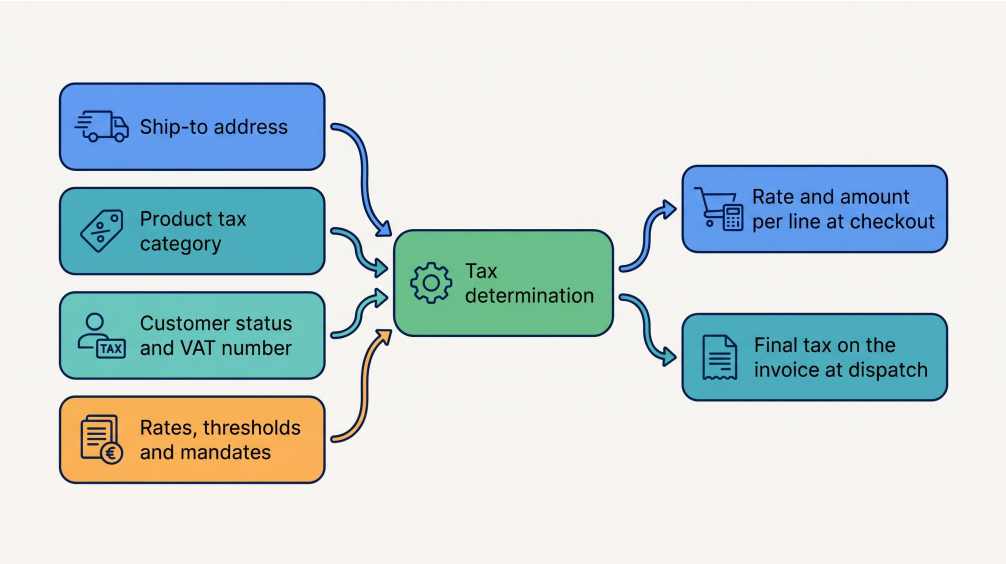

Figure 1: *Three facts the shop owns and one set of rules it should not own decide the tax, once as a promise at checkout and once as the legal amount on the invoice.*

Tax at Checkout and Tax at Invoice

The tax a shopper sees at checkout is a **promise**, computed from the ship-to address, the product category and the customer's status at that moment. The tax that matters legally is the one on the **invoice**, computed when the supply happens, which for goods is dispatch. Between the two, a line may be removed, an order split or a rate changed on 1 January. The order system has two honest options: re-run the determination at invoicing and treat the checkout figure as an estimate, or freeze the checkout result as the tax basis on the order and reconcile any difference explicitly. It must never let the two drift silently. Tax engines are commonly used the same way: a calculation at checkout that records nothing, and a committed document at invoicing.

Flow	How fresh	How consistent	How fast	If it is down
Tax at checkout	Per address and cart, at the price review step	Recorded on the order as the basis	Under 200 ms *(assumption)*	Fall back to the platform's rate table, flagged for finance
Tax at invoicing	At dispatch, from the shipped lines	Final; the invoice is immutable	Batch	Hold the invoice, not the shipment

When a cart mixes rates, the shipping charge and any basket-wide discount must be split across lines; Stripe Tax documents two methods, proportional allocation and the highest rate. Discount allocation is as much a tax problem as a promotions problem (Discounts, Coupons and Loyalty: Promotions Without Chaos): it decides what the credit note for one returned line is worth.

The Invoice as a Legal Document

An **invoice** states what was sold, to whom, when, for how much and with how much VAT, and its content is prescribed. The Dutch tax authority's invoice requirements page (n.d., accessed November 2026) lists both parties' names and addresses, the supplier's VAT identification number (and the customer's for reverse-charge and intra-EU supplies), the invoice number and date, the date of supply, the quantity and nature of the goods, and per rate the net amounts, the rate and the VAT amount. Germany's § 14 UStG (consolidated text, accessed November 2026) requires the same and spells

out the number: "eine fortlaufende Nummer mit einer oder mehreren Zahlenreihen", a consecutive number from one or more number series, unique to the invoice.

Three properties follow, and each is an architectural rule.

The number is sequential and gap-free within its series. So it is assigned by the system of record at the moment the invoice is posted, never pre-allocated by checkout, never reused, and never taken from an order-number sequence that has cancellations in it.

The invoice is immutable. Odoo's accounting documentation (version 18.0, accessed November 2026) describes the standard ERP behavior: an invoice is created in draft, receives its number from a sequence on confirmation, and "once confirmed, an invoice can no longer be updated". Any system that lets a user edit an issued invoice is not issuing invoices.

Corrections are documents. The same documentation states that "issuing a credit/debit note is the only legal method for canceling, refunding, or modifying a validated invoice". A **credit note** is the invoice's mirror image: it references the invoice, reverses some or all of its lines, and carries its own sequential number. A return never edits the invoice; it produces a credit note that references it, and a refund that references both, and the three reconcile to zero. The Double Refund happened because the second refund referenced nothing.

Figure 2: *An issued invoice is immutable: a return never edits it but produces a credit note that references it and carries its own number.*

One invoice per shipment, not per order. When an order ships in two parcels, Kestrel & Pine issues two invoices, each for the lines in its parcel, dated at its dispatch

and carrying the one order number; wholesale follows the same rule, one invoice per delivery. The supply happens per shipment, so an invoice per shipment carries the right date of supply, while an invoice that waits for the whole order waits for the slowest line, which is how #1052877 sat *Not invoiced* for six days. The trigger is the warehouse's shipment confirmation, idempotent on the shipment id. Invoicing per order works only for a merchant that never splits.

Invoices are kept for years: seven for basic records in the Netherlands (Belastingdienst retention page, n.d., accessed November 2026), eight for invoices in Germany (§ 14b UStG) and ten for books (§ 147 AO). The invoice must outlive the commerce platform, one reason the ERP, not the shop, is its system of record (Moving Without Burning the Bridge: Replatforming and Migration).

E-Invoicing at a Glance

An **e-invoice** is an invoice issued, transmitted and received as structured data (XML) that the receiver's software can process; a PDF by e-mail is not one. The European standard EN 16931 defines what it contains; **Peppol** is the open network through which a business reaches any other participant, each side connecting through its own accredited **access point** (OpenPeppol, n.d., accessed November 2026). Credit notes travel the same way.

The dates are the requirement, and they differ per country. Belgium made structured e-invoices mandatory for domestic business-to-business transactions from 1 January 2026, over Peppol (einvoice.belgium.be, 18 September 2024). Germany has required every business to be able to *receive* e-invoices since 1 January 2025 and phases in the obligation to *issue* them from 2027, or 2028 for issuers with prior-year turnover up to €800,000 (Federal Ministry of Finance FAQ, accessed November 2026). The Netherlands has no domestic mandate; the cabinet announced on 11 September 2026 that it intends one from 1 July 2030. The EU's ViDA package, adopted in March 2025, requires digital reporting based on e-invoicing for cross-border business transactions from 1 July 2030 (European Commission, accessed November 2026). The Long Fuse is lit for the German one. Documents Between Companies: EDI, Punchout and E-Invoicing owns the formats and the mandates; Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation owns the calendar.

What matters here is ownership. An e-invoice has a transmission status (sent, delivered, rejected by the receiver's validator) that a PDF never had, so the system that issues the invoice must also hold that status and the archive. At Kestrel & Pine that is the ERP; Ardennes Trail Co.'s rejection was a missing buyer reference (Peppol's BT-10 field) that the B2B portal never asked for, a checkout change rather than a finance one.

From Order to Ledger: What Gets Posted

A **ledger** (the general ledger) is the book of accounts in which every transaction is recorded twice, as a debit and a credit, so that the books always balance. The ERP posts from documents, not from events. For a consumer order paid through a PSP the standard entries are these (account names vary by company):

Event	Debit	Credit
Invoice at dispatch	Accounts receivable, PSP (gross)	Revenue (net), VAT payable
PSP payout received	Bank (net); payment fees (expense)	Accounts receivable, PSP (gross)
Credit note after a return, refund deducted from a later payout	Revenue (reversal), VAT payable (reversal)	Accounts receivable, PSP
Chargeback	Dispute expense (or a receivable while defended), dispute fee	Accounts receivable, PSP
Gift card sold	Bank or PSP receivable	Gift-card liability (no revenue yet)

Three accounting rules sit under the table.

Revenue is recognized when control of the goods passes to the customer, not when the order is placed or paid. IFRS 15, effective from 1 January 2018, recognizes revenue when a performance obligation is satisfied, which for goods is the transfer of control, in practice dispatch or delivery depending on the terms (IFRS Foundation, n.d., accessed November 2026).

Kestrel & Pine recognizes consumer revenue at delivery and wholesale revenue at delivery to the customer's shop or distribution center; "shipped" is the operational milestone, not the accounting one.

Because the invoice is issued at dispatch, finance makes a cut-off adjustment at month end that moves parcels invoiced but not yet delivered into the next month: an order shipped on 30 April and delivered on 3 May is May revenue on an April invoice. This is why the shop's "sales today" dashboard and the ledger never agree to the euro (Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics).

A right of return is a liability. With a 19% order return rate, Kestrel & Pine expects one order in five to come back in part; under IFRS 15 the expected refunds are a liability estimated at period end (the author's reading of the standard).

Gift cards are liabilities until redeemed. Selling a €50 gift card is €50 owed, not revenue, until the card pays for goods; the part never redeemed (breakage) follows rules that depend on the jurisdiction (RevenueHub's summary of the US rules, n.d.). The gift-card balance is a finance fact, and only one system may change it (Discounts, Coupons and Loyalty: Promotions Without Chaos).

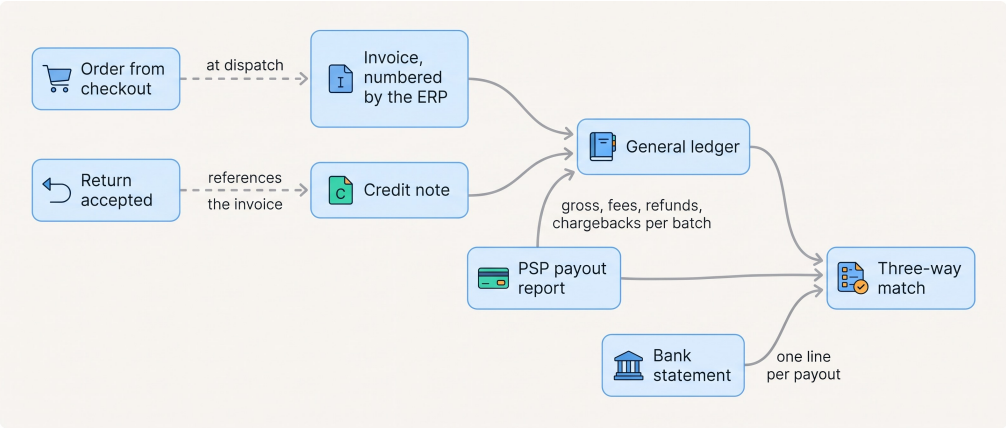

Figure 3: *Documents, not events, feed the ledger: the invoice at dispatch, the credit note on return, the payout report per batch, and the bank line that the monthly match ties them to.*

Reconciliation: Orders, Payouts and the Bank

Reconciliation is proving that two records of the same money agree, line by line, and explaining every difference. Commerce has a three-way version: the ERP's invoices and credit notes, the PSP's payout reports, the bank statement.

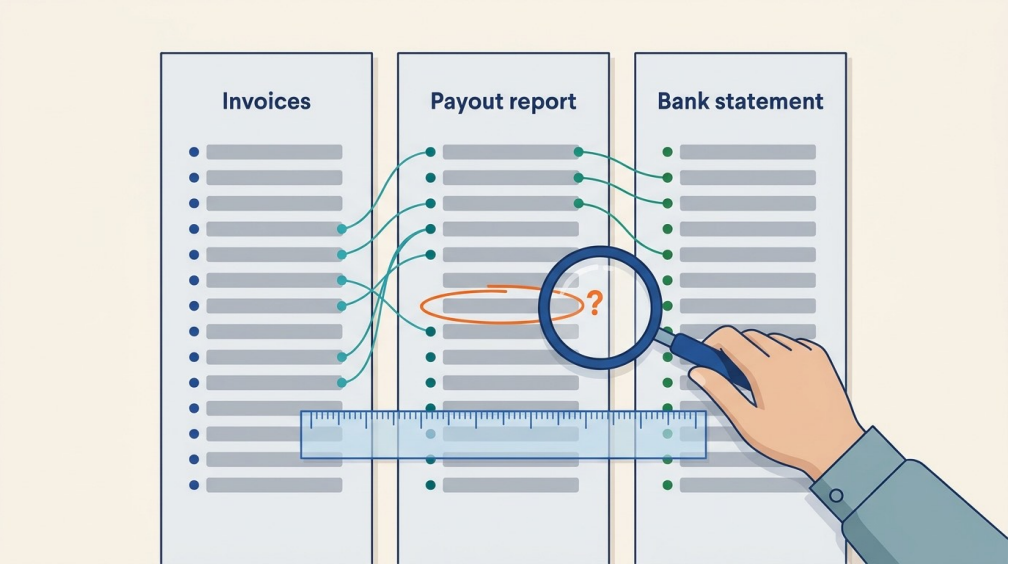

Figure 4: *Reconciliation lines up the ERP's documents, the PSP's payout report and the bank statement on shared keys, and every unmatched line needs an explanation.*

The PSP's side is the best documented. Stripe's payout reconciliation report (n.d., accessed November 2026) lists "every payment, refund, dispute, fee, and other balance transaction included in the payout", can carry the merchant's own order id, and gives a `trace_id`, "a unique identifier generated by bank for the Stripe payouts", that matches the bank line. Adyen's batch-level reconciliation keys everything on a batch number, and Mollie's settlement report does the same per settlement. In all three, the join keys are the merchant's order reference on the way in and the PSP's batch or trace reference on the way out. **A merchant that does not pass its order number to the PSP cannot reconcile at the order level**; it can only reconcile totals.

What does not match	Why	How to catch it
Captured in April, paid out in May	Settlement delay (Adyen's default two business days; Stripe's EU default several days)	Reconcile by sales day, not by calendar month alone
Payout smaller than the day's sales	Fees netted inside the batch; earlier refunds and chargebacks deducted	Post fees from the report, not from a rate card
A refund in the bank with no credit note	Someone refunded in the PSP console (the Double Refund)	A daily job listing PSP refunds without a matching credit note

What does not match	Why	How to catch it
A credit note with no refund	The return was accepted but the refund call failed	The same job, in the other direction
A failed refund returning weeks later	Adyen notes refunds can fail after acceptance; Stripe says the money returns in up to 30 days	A refund state on the order that is not terminal until the PSP says so
Four payout feeds	Online PSP, terminal provider, marketplaces, B2B receivables	One reconciliation job per feed, one owner (W5)

The Double Refund is the table's third row. The fix had two parts: one writer for refunds (the returns portal, through the platform; the ownership rule from The Way Back: Returns, Exchanges, Refunds and Reverse Logistics), and a daily comparison of PSP refunds against credit notes, a reconciliation job in the sense of Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation. RECONCILED's Reconciled by Design: Traceability and Audit Across Boundaries treats the discipline in full.

The Choices

Three decisions shape the handoff, each with three realistic answers.

Where is tax computed? In the platform's rate tables, in a **tax engine** (Avalara, Vertex, TaxJar and others) called from checkout and invoicing, necessary for the United States, or in the ERP at invoicing only, which is wrong for a consumer shop that must show the final price.

Where is the invoice issued? By the platform, by the ERP, or by an invoicing service in front of several ERPs, the Group answer where some countries run **state clearance** (the tax authority's platform must accept an invoice before it is valid, as in Italy and Poland).

How does the ledger learn? From one posting per paid order, from documents in a daily idempotent batch (the W5 rule at Kestrel & Pine), or from summary journals with the detail in a **sub-ledger**, a detailed book of every order whose totals feed the general ledger (Never Update, Only Append: Designing Custom Ledgers).

Decision	Simplest answer and its risk	Usual answer and its risk	Scaled answer
Tax computed	Platform rate tables; a rate maintained in four places	Tax engine or one rate service behind every channel; cost per call	Engine plus nexus tracking per entity
Invoice issued	The platform; a second numbering authority once the ERP exists	The ERP, one sequence per legal entity; connector failures hide invoices	An invoicing service in front of several ERPs
Ledger fed	Posting per order on payment; revenue before dispatch, duplicates on retry	Documents in a daily idempotent batch; a failed batch must be noticed	Summary journals plus a sub-ledger; detail outside the general ledger

My recommendation for a merchant at Kestrel & Pine's stage is the usual answer in each row: destination rates maintained once, invoices and credit notes from the ERP only, documents posted in a daily idempotent batch with a job that lists the orders that should have been invoiced and were not. The Three-Status Order was six days of *Not invoiced* because that last job did not exist.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting (under about €2M)	The platform's tax tables and tax-inclusive home-market prices; the platform's invoice or an accounting package fed by order export; one PSP payout feed matched by the bookkeeper	Gap-free, immutable invoice numbers; the order number passed to the PSP	The €10,000 OSS threshold crossed without noticing; invoices edited after issue
2 — Growing (€2M–€20M)	OSS registration; destination rates per market; an ERP or accounting system as the invoice's system of record, fed by a connector; VIES checks and reverse-charge invoices for business customers	The connector idempotent per shipment and per return; invoices at dispatch, not at payment; a daily job for orders not invoiced; refunds from one writer	Month-end takes a week; refunds reconciled by hand; revenue posted before dispatch; the first Peppol request from a Belgian customer

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel (€20M–€200M)	Four payout feeds (online PSP, terminals, marketplaces, B2B); invoicing out of the shop entirely, one sequence per legal entity across every channel; a tax engine or one rate service behind every channel; Peppol from the ERP	A reconciliation job per feed with a named owner; order-level matching or a sub-ledger so the general ledger stays readable; a regulatory calendar for three countries' mandates	The Double Refund; three definitions of "revenue"; marketplace VAT booked as if the merchant collected it
4 — Group	Several legal entities and countries, some with state clearance systems; US nexus tracking; an invoicing service in front of several ERPs; a finance data pipeline consuming PSP reporting APIs	The order-to-ledger contract versioned and tested like any integration; one chart-of-accounts mapping per entity; intercompany invoices for stock moved between entities	Each brand's shop computes VAT its own way; month-end depends on one spreadsheet; an acquisition brings a second numbering authority

The shape of the handoff does not change with scale: documents, identifiers, one issuer, one match. What changes is the number of feeds and legal entities, and whether anyone can still do the match by eye. Kestrel & Pine's answer under Waypoint is W5: finance moves by daily batch, with a reconciliation job per flow.

Eva's Order in the Books, and Ardennes Trail Co.'s Peppol Invoice

Eva's jacket. Order #1072455, placed Tuesday 13 April 2027 at 21:14 on the Dutch storefront: a Ridgewalker jacket at €199 and two pairs of merino socks at €18, €235 in

all, free shipping, paid by iDEAL (the Dutch bank-transfer method most online shoppers there use). Dutch consumer prices include 21% VAT, so the tax basis frozen on the order at checkout is:

Line	Gross	Net	VAT (21%)
Ridgewalker jacket, women's M, moss	€199.00	€164.46	€34.54
Merino hiking socks × 2	€36.00	€29.75	€6.25
Shipping	€0.00	€0.00	€0.00
Total	**€235.00**	**€194.21**	**€40.79**

The payment: iDEAL has no separate authorization and capture; the money moves when Eva confirms in her bank app at 21:14, and the PSP's webhook marks the order paid. Assume a PSP fee of about €0.30 for an iDEAL payment *(assumption, in the range of published PSP price lists).*

The invoice: on Wednesday 14 April the WMS (the warehouse management system) confirms the shipment; that evening the ERP connector's daily batch creates invoice 2027-041822 *(illustrative number)* from the order, with the order number on it, and the ERP posts: debit PSP receivable €235.00; credit revenue €194.21; credit VAT payable €40.79. April revenue, because PostNL delivered the parcel in April, on 15 April.

The payout: the PSP pays out each sales day in one batch two business days later *(assumption, matching one PSP's documented default).* Thursday 15 April's payout for the 13 April sales day contains Eva's €235 among some 800 orders, net of fees. Finance posts it from the PSP's report: debit bank and payment fees, credit PSP receivable. The bank line carries the PSP's batch reference, which is how the May close ties the three together.

The exchange: on 22 April the size M jacket is received and graded resellable and the size L ships. In the books this is a credit note referencing invoice 2027-041822 for the jacket line (net €164.46, VAT €34.54) and a new invoice for the size L at the same amounts; no money moves. In the refund variant the credit note is the same, the returns portal triggers a €199 iDEAL refund through the platform, the PSP deducts it from a later batch, and the daily refund-versus-credit-note job sees one of each.

Ardennes Trail Co.'s invoice. On Wednesday 21 April 2027 Ardennes Trail Co. (fictional), an outdoor shop in Namur, Belgium, orders 40 Ridgewalker jackets at the wholesale price of €95 through the B2B (business-to-business) portal, €3,800 excluding VAT *(illustrative order, at Kestrel & Pine's wholesale price)*; it ships from Duiven by bpost. The portal validates its Belgian VAT number in VIES; the order is an intra-EU business supply, so the VAT line is 0% with the reverse-charge mention, and payment is on 30-day terms, so no PSP is involved.

The ERP issues the invoice at dispatch on 23 April from the same Dutch sequence, with

both VAT numbers, the buyer reference (BT-10) and the order number, and sends it as a Peppol BIS Billing 3.0 invoice through the EDI provider's access point, recording the delivery receipt.

The entry is debit trade receivable €3,800, credit revenue €3,800 (April revenue once bpost delivers), no VAT; the receivable is due 23 May and is cleared when the bank transfer arrives quoting the invoice number. The Dutch VAT return reports the supply in the intra-EU sales listing.

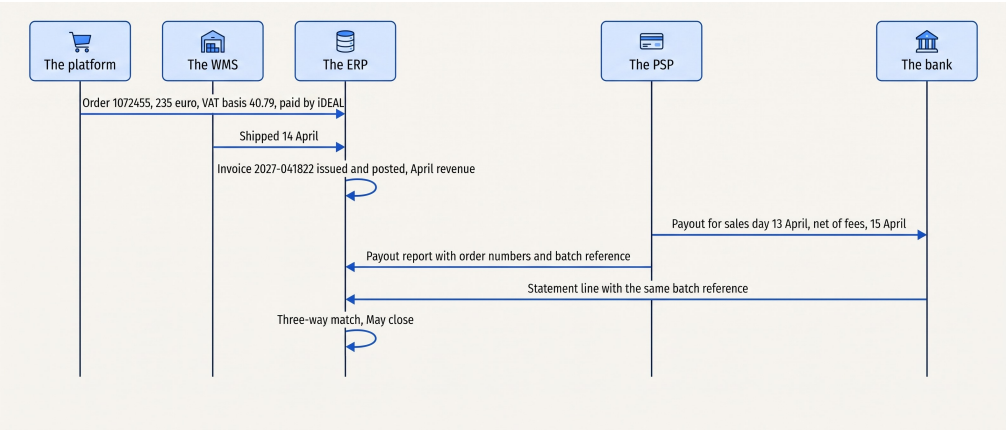

Figure 5: *Eva's order reaches the ledger as an invoice at dispatch and a payout batch two days later; the order number and the batch reference are the keys that let the May close prove the three records agree.*

When It Goes Wrong

The invoice that never came. The invoice waits for a trigger that does not arrive: an "order complete" message that a short pick prevents (the Three-Status Order), or a connector batch that fails on one order and continues with the rest. The order is *Delivered* in the platform and *Not invoiced* in the ERP for days. Detect it with a daily list of shipped orders older than one day with no invoice, the single most valuable report in this chapter; recover by invoicing the listed shipments, which requires the trigger to be idempotent per shipment.

The invoice that came twice. The connector retried after a timeout and the ERP created two invoices for one order. Detect it as two invoices referencing one order number; recover with a credit note against the duplicate, never a deletion, because the sequence must stay gap-free; prevent it with an idempotency key per shipment on the invoice-creation call.

The tax that differs. The invoice VAT differs from the checkout VAT because the address, the lines or the rate changed in between. Compare the order's frozen tax basis

with the invoice at posting and list the differences. The invoice is right by definition; if the customer was promised a lower gross price, the difference is a commercial discount, not a tax correction.

The refund that doubled, and the payout that does not match. Both are the reconciliation table above: one writer for refunds, a daily job comparing PSP refunds with credit notes in both directions, and every payout residual investigated in the same month, because a residual carried forward becomes next quarter's write-off.

The mandate discovered late. A Belgian or German customer rejects a PDF, and the ERP partner quotes four months to add Peppol. A regulatory calendar with the mandates per country, their dates and an owner (Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation) is the detection; an access point in front of the existing ERP is usually faster than the partner's quote. The Long Fuse is a governance failure.

What to Take Away

Established.

- VAT follows the destination for cross-border consumer sales above the €10,000 OSS threshold, is reverse-charged for business sales to another EU country with a valid VAT number, and differs by country and category (European Commission and national tax authority pages, accessed November 2026).
- An invoice has prescribed content and a sequential gap-free number, is immutable once issued and is corrected only by a credit note that references it (Belastingdienst; § 14 UStG; Odoo documentation describing standard ERP behavior).
- Revenue for goods is recognized when control transfers, at dispatch or delivery depending on the terms, not at order or payment (IFRS 15; Kestrel & Pine uses delivery); gift cards are liabilities until redeemed.
- Belgium requires structured business-to-business e-invoices over Peppol since 1 January 2026; Germany phases in issuing from 2027; the Netherlands plans 2030; ViDA requires cross-border digital reporting from 1 July 2030.

Depends on context.

- The platform's own tax tables suffice for one or two countries with standard-rate products; a tax engine becomes necessary with many categories, many countries or any US sales, at a cost per call.
- The platform may issue invoices at the Starting stage if numbering is gap-free and immutable; from the Growing stage, one issuer per legal entity (usually the ERP) for every channel, because stores, marketplaces and EDI customers otherwise bring their own numbering.

- Summary journals with a sub-ledger are the right ledger shape at high volume or across several entities; at Kestrel & Pine's volume, order-level postings in a daily batch keep the general ledger readable.
- Recomputing tax at invoicing is right when addresses and baskets change between checkout and dispatch (split shipments, B2B); freezing the checkout basis is right for a consumer shop that ships whole orders within a day.

The author's synthesis.

- The order-to-ledger handoff is a small set of documents with stable identifiers, nothing more: shipment → invoice at dispatch, return → credit note, payout report → bank. Every failure in this chapter is a document issued by the wrong system, twice, or without a reference.
- The report that lists shipped orders with no invoice, and the job that compares PSP refunds with credit notes, are worth more than any finance dashboard; build them in the first month, before the OMS and the PIM.

Questions to Ask

1. Which system issues our invoices, and is there exactly one numbering sequence per legal entity across web, stores, marketplaces and EDI?
2. Is the invoice created at dispatch or at payment? What does our revenue figure mean on the last day of the month?
3. Which facts decide the tax at checkout, and in how many places are the rates maintained?
4. Are we registered for the One Stop Shop, and does the checkout charge the destination country's rate per product category?
5. Do we pass our order number to the PSP, and does its payout report come back with it and with the bank's batch reference?
6. How many payout feeds do we reconcile, who owns each, and how long did the last close take?
7. Which e-invoicing mandates apply to our customers by country, from which dates, and who owns the calendar?

Eva's jacket has now been promised, paid, picked, delivered, exchanged and accounted for. The promise was made in Part III; keeping it is the work of the warehouse, the carriers, the stores and the returns flow, where the next part begins: Part IV — FULFILLMENT: From Warehouse to Doorstep and Back, starting with Pick, Pack, Ship: Warehouse Management and Fulfillment.

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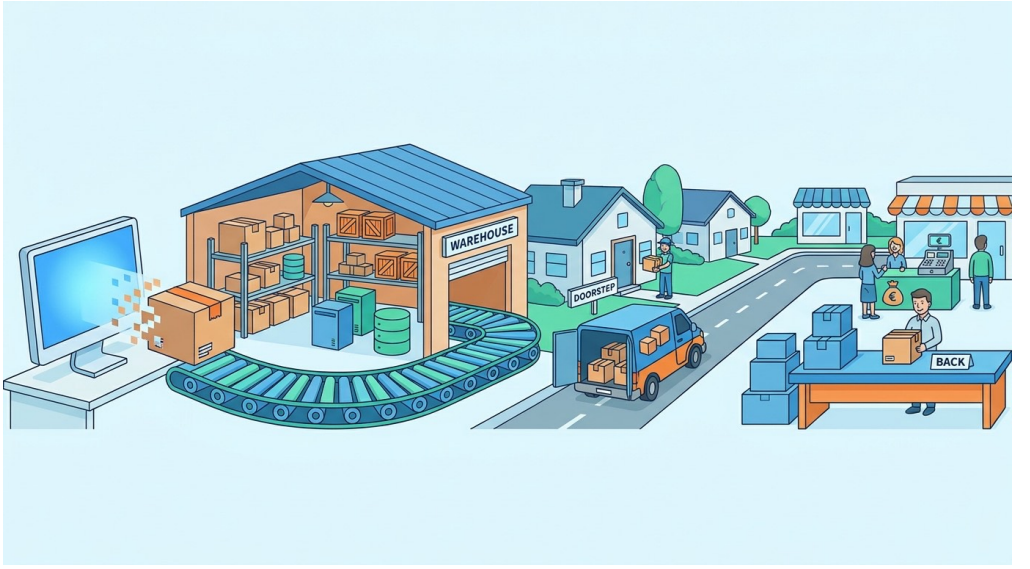
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part IV — FULFILLMENT: From Warehouse to Doorstep and Back



KEY POINTS:

- **The physical world has its own systems of record.** The warehouse counts shelves, the carrier's scan proves delivery and the returns bench knows what came back; nothing upstream may overwrite them.
- **A promise about the physical world is a computation, not a sentence.** "Delivered tomorrow" needs the pick wave, the backlog and the carrier's transit time; every incident in this part is a promise made without its inputs.

- **One writer per fact, one source per view.** One refund writer keyed by the return, stock that rises only on the warehouse's verdict, and every customer-facing screen reading the same order record.

Checkout can confirm an order in seconds, but someone still has to pick and pack the goods before the carrier's collection. A late instruction or an empty shelf can push delivery back a day. Returns and store services create similar dependencies on people, space and time. This part explains how the systems planning that physical work supply dates and statuses the customer can rely on.

Eva's jacket order illustrates the sequence. She buys on 13 April 2027; Duiven picks and packs on 14 April; PostNL delivers on 15 April, a day later than promised. She requests a larger size on 19 April. The warehouse receives and inspects the return and ships the replacement on 22 April. Store pickup and counter exchange provide alternative routes.

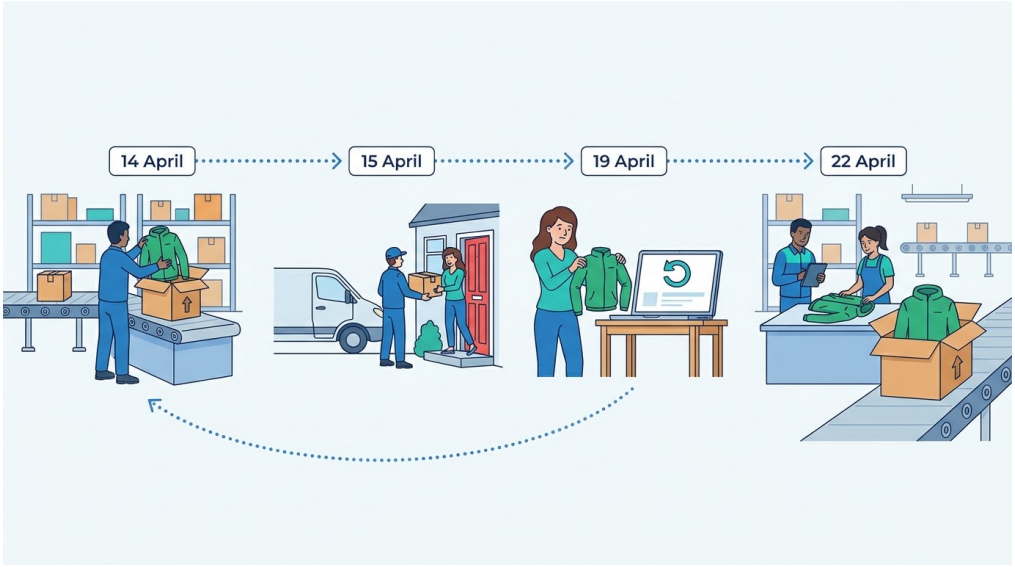


Figure 1: *Eva's jacket in Part IV: packed at Duiven on 14 April, delivered in Nijmegen on 15 April, size L requested on 19 April, and the exchange shipped on 22 April.*

Kestrel & Pine is a fictional outdoor brand shipping 336,000 parcels and 9,400 wholesale shipments annually. It operates one warehouse, uses four carriers, runs six stores on separate till software and receives 72,000 return parcels.

Its **Waypoint** improvement program assigns stock to the systems that count it, establishes one shared order record and calculates delivery promises from actual operations. Customer messages must reflect those records. The chapters below show why each decision matters.

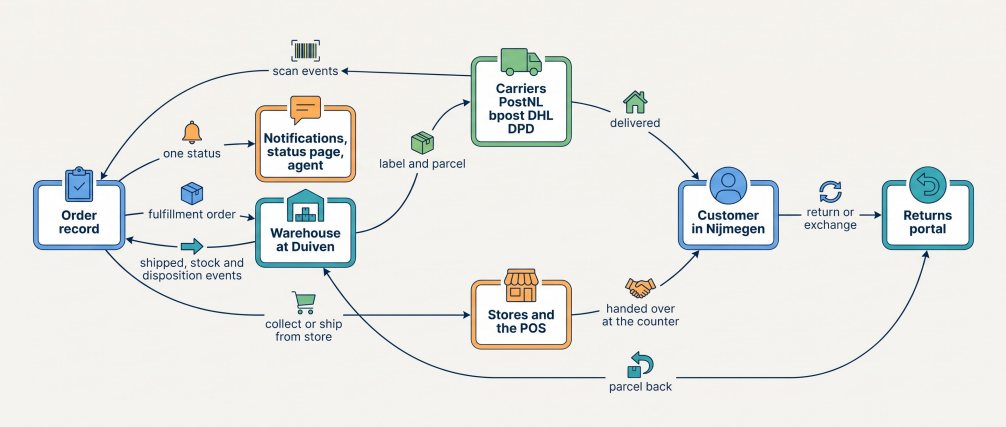


Figure 2: *The systems of Part IV: the order record instructs the warehouse and the stores, the warehouse and the carriers move the parcel there and back, and every physical fact returns as an event to the one record that the notifications, the status page and the agent read.*

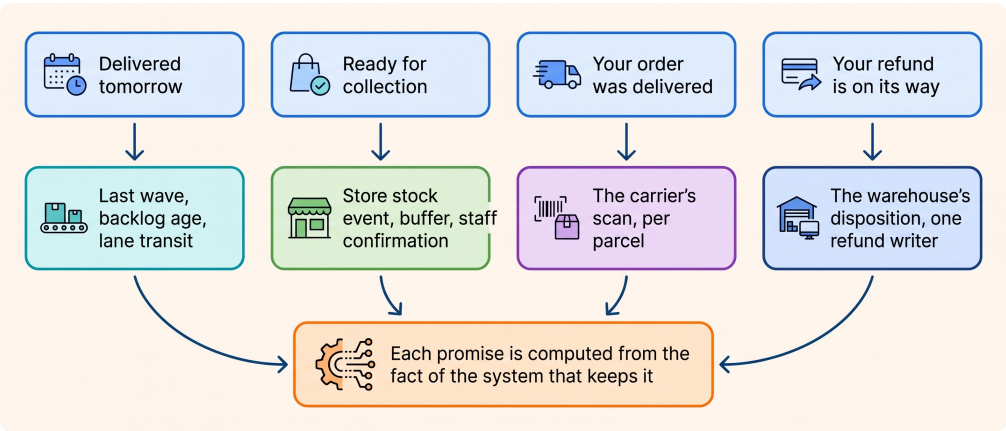


Figure 3: *The four promises of Part IV and the inputs each needs; the Queue That Waited, the Store Stock Mirage, the Three-Status Order and the Double Refund are the four made without them.*

Chapters in this part

Pick, Pack, Ship: Warehouse Management and Fulfillment defines **the work instruction and response between order management and the warehouse**. A four-hour Black Friday order backlog shows why a warehouse can have spare picking capacity while customers wait for orders that never reached it.

Getting It There: Carriers, Labels, Tracking and the Delivery Promise explains **carrier integration and credible delivery dates**. Eva is promised Wednesday and receives Thursday because a static storefront rule ignores warehouse timing. Collection schedules, backlog, location and carrier performance must inform the promise.

Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment treats **the store till as another source of orders, stock and customer records**. Utrecht's pickup pilot promises 37 items already sold. The chapter examines the data needed before offering cross-channel services or replacing the till.

The Way Back: Returns, Exchanges, Refunds and Reverse Logistics separates **the return request, inspection, stock decision and money movement**. Eva's size exchange and January's 212 double refunds show why each action needs its own state and authorized owner.

Where Is My Order? Notifications, Service and the Customer Record aligns **messages, the status page and the support view**. Eleven tickets about one split order could have been reduced by the missing explanation: the order ships in two parcels. The chapter builds that explanation from shared facts.

Begin with Pick, Pack, Ship: Warehouse Management and Fulfillment.

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine is fictional; recommendations and labeled assumptions are the author's judgement.

15. Pick, Pack, Ship: Warehouse Management and Fulfillment

IN THIS CHAPTER, YOU WILL: Follow an order through receiving, picking, packing and shipping, and learn what the warehouse system must be told, what it must own, and what it must report back.

KEY POINTS:

- **The fulfillment order is the contract between the order world and the warehouse**. It says "ship these lines of this order from this location by this service", carries nothing else, and has its own lifecycle: requested, accepted or rejected, in progress, shipped or closed. Cancellations after acceptance are requests, not commands.
- **The warehouse system owns the count of what is physically there, and nothing upstream should overwrite it**. Receiving, put-away, picking, cycle counts and returns change on-hand stock inside the warehouse system; the platform and the availability service derive what may still be sold from warehouse events.

- **Capacity, cut-off times and the wave schedule are architecture inputs, not warehouse trivia.** A promise made at checkout is kept or broken by whether the order reaches the warehouse before the last wave of the day leaves, which is why the queue between the platform and the warehouse system deserves the same attention as checkout.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The First Snowfall** — the order feed into the warehouse is the integration most likely to meet real load for the first time on the busiest evening of the year.
- **Schrödinger's Order** — when the warehouse's shipment confirmation does not reach the order record, the platform, the warehouse and finance systems each tell a different story about the same parcel.
- **The Phantom Stock** — a warehouse that is not the single writer of on-hand stock, or that writes it hours late, lets every channel sell units that are no longer on the shelf.

A picker walks to the shelf for a paid order and finds no jacket. The storefront has already confirmed the sale, but the warehouse now needs to investigate, find a replacement unit or report a failed pick. Those steps take people and time, and may push the parcel past its dispatch cutoff. A stock error has become a delivery problem.

A missed order edit can similarly put the wrong size in the box; an unrealistic delivery date can commit work the building cannot finish in time. This chapter explains the **warehouse management system (WMS)**, which directs picking and packing, and the **fulfillment order**, its instruction to ship specific items. Accurate instructions and timely results let the shop make promises the warehouse can keep.

Where we are in the order's journey. Eva buys a jacket and socks from Kestrel & Pine, the fictional outdoor brand, at 21:14 on 13 April 2027 and pays through iDEAL, the Dutch bank-transfer method. On 14 April, Duiven's warehouse picks, packs and confirms the shipment, then prints a PostNL label. Getting It There: Carriers, Labels, Tracking and the Delivery Promise follows the parcel onward.

Setting the Scene: Duiven's Day

Kestrel & Pine opened its **8,500-square-meter Duiven warehouse** near Arnhem in 2023. Operations head Daniel Okafor runs it with 88 staff and a standalone cloud WMS. Before that, a **third-party logistics provider (3PL)** in Tiel stored and shipped online orders using its own warehouse software. Owning the building brought direct control and responsibility for its daily capacity.

Head of Engineering Noor Visser visits Duiven in April 2027 to understand incidents involving its integrations. The warehouse's daily rhythm below is illustrative; company figures come from the fictional scenario introduced in Meet Kestrel & Pine.

Duiven, a normal working day in April 2027	**Figure**
Parcels shipped (D2C and marketplace orders)	about **1,300**
Wholesale shipments (cartons and pallets)	about **35**
Units picked	about **7,500**
Pick waves released	**06:30**, **10:00**, **13:00**, **16:00**, **19:00** and a final wave at **21:00**
Carrier collections	PostNL, bpost, DHL and DPD between **18:00 and 22:30**
Inbound deliveries from suppliers	4 to 8 trucks, received against purchase orders
Returns received and inspected	about **280** parcels

Daniel walks Noor through receiving, putting goods away, picking, packing and daily stock counts. Purchase orders arrive from the **enterprise resource planning system (ERP)**, which handles finance, purchasing and item records. The WMS groups work so one picking route can serve several orders.

Then Daniel shows the Black Friday chart. The platform stayed up and the warehouse had enough staff, but **orders took four hours to reach the WMS**. An order placed at 20:00 arrived after midnight and missed the 21:00 picking wave. The storefront kept promising next-day delivery for orders before 22:00.

Daniel calls the incident **the Queue That Waited**. The promise ignored the warehouse's actual work queue. Noor needs to establish what information the warehouse must receive and what it must report back.

What a Warehouse Management System Does

A WMS runs a warehouse the way a checkout runs a purchase: as a sequence of steps, each of which changes the state of something. The vocabulary is small and worth learning once.

Step	What it is, in plain words	What it changes
Receiving	Checking an inbound delivery against the purchase order (or the supplier's advance shipping notice) and booking the units in	On-hand stock goes up
Put-away	Moving received goods to a storage **location** (a shelf, bin or pallet position with its own barcode)	Units get an address
Slotting	Deciding where each SKU (stock-keeping unit: one sellable variant, the thing you count on a shelf) should live: fast movers near the packing stations, bulky items on the floor	Walking distance per pick
Wave or batch release	Grouping orders for one picking run, typically by carrier cut-off, channel or zone, so one walk serves many orders	Which orders ship today
Picking	Walking or driving the planned route, scanning each location and unit	Units leave their locations
Packing and cartonization	Choosing the box (cartonization is the software's choice of box size from the items' dimensions), packing, weighing	A shipment exists
Shipping confirmation	Printing the carrier label, closing the pick list and reporting the shipment with its tracking number	The order is fulfilled; on-hand goes down

Step	What it is, in plain words	What it changes
Cycle counting	Counting a small part of the stock on a rolling schedule instead of one annual full count, correcting the record continuously	Record accuracy

Vendors describe these steps in their own words. Odoo's inventory module (n.d., accessed November 2026) lists wave picking "per aisle, or area, and regroup at packing zone", cluster picking (several orders in one trip) and batch picking (several orders in one picking, consolidated at packing), plus cycle counts and barcode scanning. Picqer, a Dutch WMS for the small and mid-sized segment (n.d., accessed November 2026), processes pick lists "in batches", routes pickers "through the warehouse in sequence" and sends labels and tracking codes back to the online store "in one step"; its customers "range from about 25 to 5,000+ orders per day".

Manhattan's warehouse product (n.d., accessed November 2026) describes "Order Streaming" that handles "wholesale, retail, and direct orders at the same time". Infios, formerly Körber Supply Chain Software (n.d., accessed November 2026), frames the market by Gartner's warehouse complexity levels "from 2 (rudimentary locator) to 5 (highly automated warehouse)"; on that scale Duiven is a level-3 building, with locations, waves and scanning but no automation.

Three facts about warehouse work shape every integration decision in this chapter:

- **It runs in batches**. Orders are collected into waves, so "did the order reach the warehouse?" has a deadline: the release time of the last wave that can still make the carrier's collection.
- **It has capacity**. A building with 88 people picks a certain number of units per hour. An order that arrives after the day's capacity is spoken for ships tomorrow, whatever the storefront promised.
- **Its facts are physical**. The WMS's on-hand number is checked against shelves every day by pickers and counters. No other system has that property, which is why Is It in Stock? Inventory, Availability and the Promise You Can Keep and Waypoint decision W1 make the WMS the **system of record** for physical stock at Duiven (the one system whose value counts as the truth when copies disagree).

How the Warehouse Connects to Everything Else

Seen from outside, a warehouse has four interfaces: instructions in, confirmations out, stock events out, and inbound goods in. The instructions come from whichever system

holds the orders: today Kestrel & Pine's platform, and from Q4 2027 the **OMS** (order management system: the one place that holds every channel's orders and steers each through fulfillment; The Order Record: Lifecycle, State and the Order Management System). The figure shows the interfaces around the physical steps.

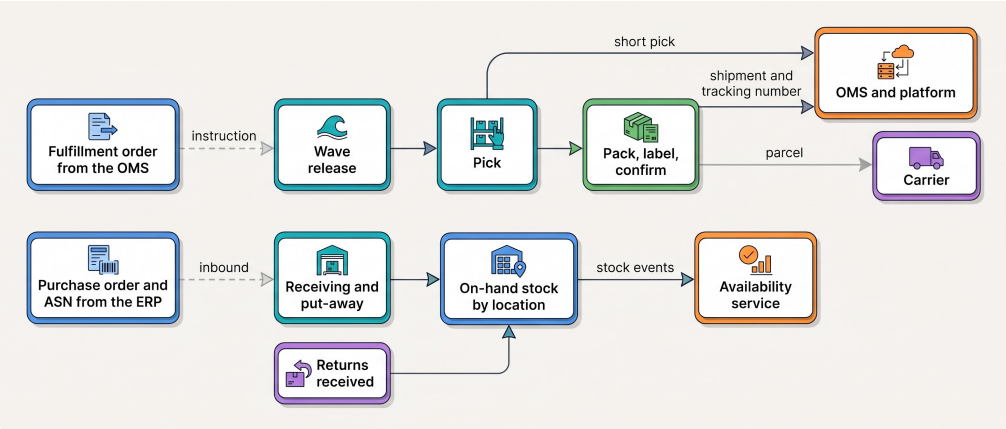


Figure 1: *The warehouse has four interfaces: instructions in as fulfillment orders, confirmations and short picks out to the order record, stock events out to the availability service, and inbound goods in against purchase orders and returns.*

Each interface has different requirements for freshness, consistency, speed and availability. The table states them for a merchant of Kestrel & Pine's shape; the latencies are assumptions, not vendor facts.

Flow	How fresh	How consistent	How fast	How available
Fulfillment order into the WMS	Before the next wave release; at peak, within minutes *(assumption)*	Exactly once; a repeat must not create a second pick list	Must keep pace with the order rate at peak	If the WMS is down, orders queue; nothing is lost

Flow	How fresh	How consistent	How fast	How available
Shipment confirmation out	The moment the label prints	One per shipment, carrying the tracking number	Within a minute, so the "shipped" message and the invoice follow *(assumption)*	Queued if the OMS is down; replayed in order
Stock events out	Each change, not an hourly total	The WMS's count is the truth; consumers derive, never write back	Seconds *(assumption)*	A reconciliation job catches a lost event
Cancellation request in	Before picking starts	A request with an answer, not a command	Minutes	If unanswered, the order is still in progress
Inbound receipt	Same day as the truck	One receipt per purchase-order line	Hours	Can wait

Two rows carry the chapter's argument: the fulfillment order and the stock events. The next two sections take them in turn.

The fulfillment order: the contract between the order world and the warehouse

A **fulfillment order** is the instruction "ship these lines of this order from this location, by this service level, with these notes". It is not the customer's order. The customer's order has prices, discounts, a payment, a customer history and a return policy; the warehouse needs none of that. The fulfillment order has exactly what the warehouse needs and has **its own lifecycle**, separate from the order's: requested, accepted or rejected, in progress, shipped (possibly in several parcels) or closed incomplete.

Shopify's developer documentation (n.d., accessed November 2026) is the clearest public description of this seam, because Shopify had to formalize it for thousands of third-party warehouses and 3PLs. A fulfillment order is "an item or a group of items in an order that are to be fulfilled from the same location"; each warehouse or 3PL "should have its

own fulfillment service" mapped to a location; Shopify sends the service a `FULFILLMENT_REQUEST` or a `CANCELLATION_REQUEST`; the service accepts or rejects; and "fulfillments can't be created for a fulfillment order before the fulfillment service accepts the fulfillment request". The request status moves through `SUBMITTED`, `ACCEPTED` or `REJECTED`, and later `CANCELLATION_REQUESTED`, `CANCELLATION_ACCEPTED` or `CANCELLATION_REJECTED`.

Once the warehouse has accepted the work, 'the merchant can no longer cancel it directly and must submit a cancellation request'. Picking or packing may already have started. Changing the storefront's status cannot stop that physical work; the warehouse must confirm whether it can still cancel before the shop promises the customer that nothing will ship.

Kibo's order management documentation (n.d., accessed November 2026) describes the same shape from the OMS side: shipments move through Accept Shipment, Validate Stock (where substitutions happen), Ready and Fulfilled, and "can be transferred or reassigned between different fulfillment locations".

The sequence shows the handshake as Waypoint designs it for late 2027, when the OMS sends fulfillment orders, using Eva's order as the example. It is carried by the iPaaS (integration platform as a service: the hosted tool that runs the flows between systems) with an idempotency key (a reference that lets the receiver recognize a repeated message), and with a cancellation request arriving too late.

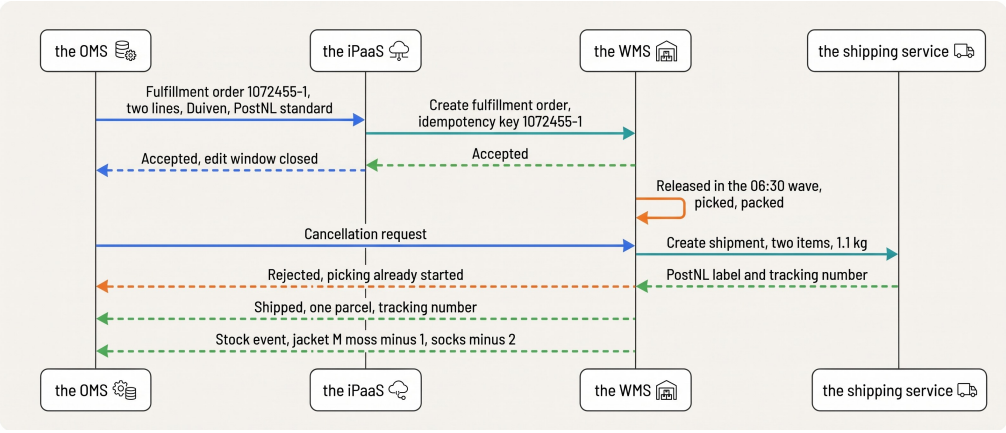


Figure 2: *The fulfillment-order handshake: the warehouse accepts before anyone may count on it, a cancellation after acceptance is a request the warehouse may refuse, and the shipment confirmation and the stock event travel back as separate messages the moment the label prints.*

Why insist on this shape rather than simply "send the order to the WMS"? Four reasons, each tied to a cost.

- **It carries the right data and nothing else.** A WMS that receives prices and payment details holds data it must protect and that changes for reasons it does not care about.
- **It supports partial and split shipments.** One order can have two fulfillment orders (Duiven and the marketplace's fulfillment center), and one fulfillment order can ship in two parcels. The Order Record: Lifecycle, State and the Order Management System owns splits; the warehouse needs each instruction to be complete on its own.
- **The accept step is the lock.** Until the warehouse accepts, the order can be edited and canceled freely. After acceptance, every change is a request with an answer, and that answer is what a service agent needs to tell the customer the truth.
- **It makes retries safe.** Networks and queues guarantee that the same instruction will sometimes arrive twice. If it carries the fulfillment order's identity and the WMS treats a repeat as the same instruction (**idempotent** handling: receiving a message twice has the same effect as once), the retry is harmless. If the WMS creates a pick list per message, the retry ships the jacket twice.

The fulfillment order is also a stable seam when the systems on either side change. Kestrel & Pine's platform sends orders to the WMS today; under Waypoint decision W2 the OMS will send fulfillment orders by Q4 2027, and the WMS need not know which system is on the other side.

Stock: what the warehouse owns and what it reports

The second interface is stock. Is It in Stock? Inventory, Availability and the Promise You Can Keep owns the concepts; this chapter needs three of them. **On hand** is what is physically in the building. **Committed** (or allocated) is the part of on hand that placed orders have already claimed. **Available** (available-to-promise, ATP) is what a channel may still sell: on hand minus committed minus whatever the merchant deliberately holds back. The warehouse's job is the first number only.

The rule I give every merchant: **one writer per stock state.** The WMS writes on hand, because it is the only system that counts shelves. The order system writes committed, because it is the only system that knows which orders exist. The availability service derives available. Nobody else writes any of the three. Kestrel & Pine's "App That Wrote Stock" of March 2026, in which a pre-order app and the WMS feed both adjusted the platform's inventory, and the first reconciliation of May 2027, which found 2.3% of active variants with different quantities in the WMS and the platform, are what happens when the rule is broken.

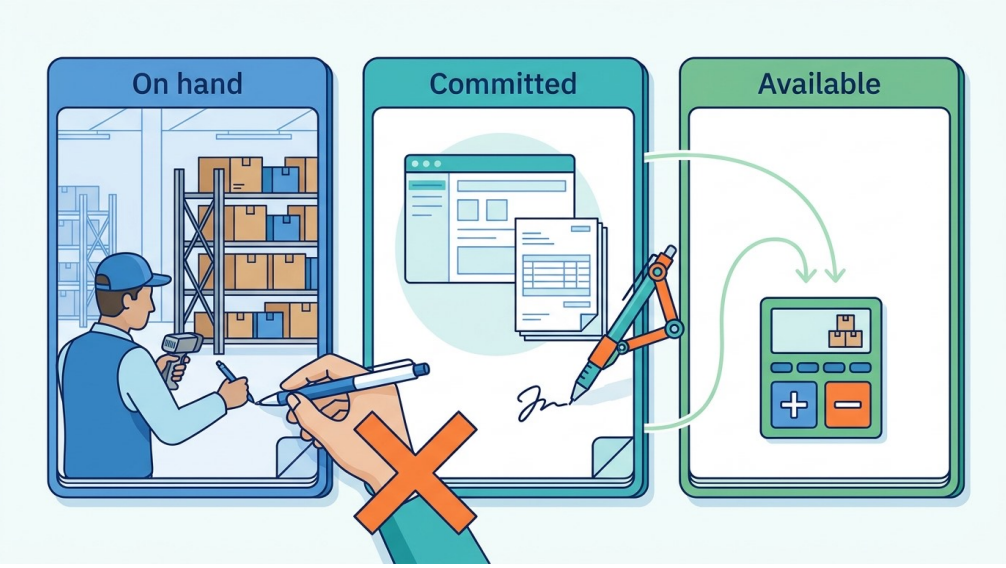


Figure 3: *One writer per stock state: the WMS writes on hand, the order system writes committed, the availability service derives available, and no other hand may write any of the three.*

On hand changes inside the warehouse constantly: a receipt, a pick, a cycle-count correction, a damaged unit moved to quarantine, a return restocked. How the rest of the landscape learns is the architectural question, and two patterns exist.

- **Periodic full sync**. The WMS exports every SKU's on hand on a schedule and consumers overwrite their copy. Shopify's documentation for fulfillment services (n.d., accessed November 2026) describes exactly this: Shopify polls the service's stock endpoint for on hand per SKU hourly. Kestrel & Pine's chain (WMS to ERP every 15 minutes, ERP to platform hourly, platform to the marketplace connector hourly) is three of these in series, which is how a sale took up to two hours to reduce the stock shown elsewhere on Black Friday evening.
- **Stock events**. The WMS emits a message for each change ("SKU X at location Y: minus 1, reason pick, timestamp"). Picqer's API (n.d., accessed November 2026) emits a `products.free_stock_changed` webhook "to support real-time stock updates in a store"; a **webhook** is a call the vendor's system makes to your URL when something happens. Waypoint decision W1 retires the sync chain in favor of this pattern by Q4 2027.

Events are better once order volume makes lag visible, and they are harder: a lost event leaves a stale copy until a **reconciliation** (a job that compares the two systems and lists differences) catches it. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns that discipline; the warehouse's part is to emit every change with a reason and to publish a daily full snapshot as the baseline.

Capacity, cut-off times and the wave schedule

The third thing the warehouse owns is its **day**. Every promise the platform makes about timing rests on three warehouse facts that rarely appear in any API:

- **The cut-off** is the latest moment an order can reach the warehouse and still ship today: the release time of the last wave that makes the last carrier collection. At Duiven the last wave is released at 21:00 and PostNL collects at 22:30 (Kestrel & Pine illustrations). The storefront's "ordered before 22:00" line dates from the Tiel 3PL, whose collection was later; nobody updated it in 2023. The promise and the building disagree by an hour on every weekday.
- **Capacity** is units or parcels per hour at the day's staffing. A wave with more work than the time until collection allows leaves orders behind, whatever their order time.
- **The wave policy** decides which orders go first: by carrier collection, by service level, by channel (marketplace orders carry late-shipment penalties).

These three belong in the data the delivery promise is computed from. Until they are, the promise is a sentence; Getting It There: Carriers, Labels, Tracking and the Delivery Promise is about making it a computation. On Black Friday evening 2026 the warehouse's capacity was never the limit; the order feed was.

Wholesale fulfillment: pallets, SSCC labels and the despatch advice

A parcel for Eva and a pallet for Bergzicht Outdoor leave the same building through different processes. Wholesale shipments are picked to cartons and pallets, and a large retail customer expects two things a consumer never sees: a label on each logistics unit and an electronic message announcing what is on it.

The **SSCC** (serial shipping container code) is, in the words of the GS1 General Specifications, Release 26.0 (January 2026), "the GS1 identifier for an individual logistics unit" and "the only GS1 key that SHALL be used as the identifier of a logistic unit": the 18-digit number on the pallet label. The **despatch advice** (DESADV in EDIFACT, the UN message standard most European EDI uses; also called an ASN, advance shipping notice) tells the receiver what is coming, pallet by pallet, so receipt can be scanned instead of counted. Documents Between Companies: EDI, Punchout and E-Invoicing owns the messages; this chapter owns where they are generated.

The retailers' rules are public and strict. bol's supplier page (n.d., accessed November 2026) states that "as of December 31, 2021, all logistic shipping units delivered to bol must be provided with GS1 label(s)" and that "for suppliers with an EDI connection, the SSCC must be included in the DESADV message". Jumbo's 2025 guidelines for its

mechanized distribution center say that "a delivery without an accompanying DESADV message cannot be processed and is therefore refused" and that "the SSCC label forms the link with the DESADV despatch message"; its purchasing conditions of 9 April 2026 add €70 per message that does not comply with GS1 standards.

The architectural consequence is one sentence: **the DESADV must be generated from what the warehouse actually packed, never from the order**. The pallet's SSCC and its contents are known only at pack time, in the WMS. Kestrel & Pine's ASN Chargeback of Q3 2026 (€14,200 deducted by Bergzicht for despatch advices whose SSCC labels did not match what arrived) happened because the DESADV was built from the ERP's view of the order while the WMS packed something slightly different. Waypoint decision W6 moves SSCC generation to the WMS at pack time and sends the DESADV from the WMS confirmation. Bergzicht's spring reorder of April 2027 (240 units across 18 SKUs on three pallets, DESADV at pack confirmation, delivered to Zwolle on 9 April) is the first cycle under the new rule; Journey 3: A Chain Reorders by EDI, From Order to Invoice follows it end to end.

The Choices: Who Runs the Warehouse Software

A merchant has four realistic ways to get warehouse execution, and most use two at once.

The ERP's warehouse module. Mid-market ERPs include receiving, locations, picking and shipping; Odoo's module is the public example. It is one system, one integration and one vendor fewer. It fits one building, modest volume and simple routes; it strains when waves must be planned around carrier cut-offs, when labor needs managing or when the warehouse serves more than one company.

A dedicated WMS. Two segments exist. Small and mid-sized WMS products (Picqer: "about 25 to 5,000+ orders per day") are cloud services integrated by order push and stock webhook, with carrier labels included. Enterprise WMS products (Manhattan, Blue Yonder, Infios) add wave optimization, labor management, slotting, automation interfaces and multi-site operation; Blue Yonder notes (n.d., accessed November 2026) its placement as a Leader in Gartner's 2026 Magic Quadrant for the eighteenth consecutive time, and Infios cites "1,600+ WMS customers". The step between segments is a project measured in quarters.

A 3PL. The provider runs the building on its own WMS, often in a multi-client mode. The merchant integrates with it exactly as with its own WMS: fulfillment orders in, confirmations and stock events out. The difference is who owns the people, the lease and the process. A 3PL is right when volume, geography or season does not justify a building, and wrong when the merchant's process (gift wrapping, kitting, strict brand packaging) is the product.

A marketplace's fulfillment program. Amazon's FBA ("we'll pick, pack, and ship

orders, as well as handle customer service and returns", n.d., accessed November 2026) and Zalando's ZFS, launched in autumn 2017 and used by "more than 100 partner brands" by June 2019, are 3PLs run by the marketplace. The seller sends stock in advance; the marketplace stores, ships and handles returns; the seller pays per unit, per order and per day of storage. Zalando's ZFS invoice structure (updated 16 February 2026) charges outbound "per order, regardless of package count", storage "per item per day" and a peak surcharge of €0.15 per outbound order from 15 October to 31 January; the amounts are in the partner contract.

In October 2023 Zalando productized the network as **ZEOS** for brands "whether or not they sell on Zalando"; in November 2024 it announced it would fulfill most of NEXT's continental-European orders "through a single shared stock pool and an order management system", and its letter to shareholders of 12 March 2026 reports ZEOS in "24 markets" serving "18 sales channels". That shared pool is the availability problem of Is It in Stock? Inventory, Availability and the Promise You Can Keep run by the provider instead of the merchant.

For the merchant, marketplace-held stock is a **location it can see but not count**: a separate location in the availability model with its own buffer, fed by a purchase-order-like inbound flow. Kestrel & Pine sends about 40% of its German marketplace units to such a program.

Choice	Fits when	Limits when	What the integration looks like
ERP warehouse module	One building, simple routes, modest volume	Waves around cut-offs, labor planning, several clients or sites	Orders already in the ERP; stock is the ERP's own table
Dedicated WMS, small or mid-sized	Own building, up to a few thousand orders a day	Automation, labor management, multiple sites	Order push by API; stock and shipment webhooks back
Dedicated WMS, enterprise	Several sites, automation, wholesale and parcels in one building	Needs its own specialists	Fulfillment-order contract, event streams, EDI from pack confirmation

Choice	Fits when	Limits when	What the integration looks like
3PL	No building of your own; seasonal or regional volume; a new country	Brand-specific packing, returns grading, cost at high volume	Same contract as a WMS; the 3PL's portal is not the record
Marketplace fulfillment program	Marketplace share is large and its delivery promise sells	Stock you cannot count; fees per order and per day; returns graded by the marketplace	An external location with inbound shipments and stock reports

My recommendation follows the requirements, not the product. One location and one channel: use what the platform or the ERP gives you. One building, several channels and a carrier schedule to plan around: a dedicated WMS, behind the contract above. Several buildings, or a wholesale flow with SSCC and DESADV obligations: an enterprise WMS or a 3PL that can meet them, behind the same contract. At no stage let the warehouse system become a second order system that holds prices, customer data and status for support; that is how Kestrel & Pine arrived at three definitions of an order.

Across Scale Stages: When It Fits and When It Limits

Stage	Typical setup	What becomes necessary	Warning signs of outgrowing it
1 — Starting	A back room or a 3PL; the platform prints labels; stock is the platform's own fields	A 3PL with an order feed the day the founders cannot pack everything; tracking numbers back on the order	Orders re-typed into the 3PL's portal; stock counted in a spreadsheet; the first parcel shipped twice after a retry

Stage	Typical setup	What becomes necessary	Warning signs of outgrowing it
2 — Growing	A 3PL's WMS or a small WMS; orders pushed by a connector; stock back by webhook or nightly file	The accept-or-reject handshake; stock events instead of a nightly file; a cut-off the storefront and the warehouse agree on	The 3PL's portal is the only place to see shipment status; next-day promises missed on busy days; the first retail chargeback
3 — Omnichannel	Own building or a 3PL per region; the WMS as system of record for on hand; an OMS sending fulfillment orders; waves by carrier cut-off and channel; wholesale and parcels in one building; marketplace fulfillment stock as an external location	SSCC and DESADV from pack confirmation; a queue between the order hub and the WMS sized for the peak hour; WMS-to-platform reconciliation; capacity and cut-off exposed to the delivery promise	The order feed backing up at peak; WMS and platform status disagreeing for days; one connector per channel into the WMS; a DESADV built from the order
4 — Group	A WMS per site, often from different vendors after acquisitions; routing among sites; automation; shared returns	One fulfillment-order contract for every site, whatever its vendor; a network-level view of capacity	Each brand's warehouse integrated its own way; nobody can say how many parcels the network can ship tomorrow

The judgement: the warehouse software can be bought, rented or outsourced at any stage, and the choice can change. What should not change from Growing onward is the

contract: fulfillment orders in with an accept step, confirmations and stock events out, the WMS as the only writer of on hand. Kestrel & Pine bought a good WMS in 2023 and integrated it as if it were a 3PL portal: orders pushed one by one, status pulled by the ERP, stock copied hourly. The building was Omnichannel; the integration was Starting.

In Practice

Public evidence

Three details from the public record sharpen the argument above. Shopify's fulfillment-service documentation (n.d., accessed November 2026) warns that "SKU strings are case-sensitive", a detail that has produced many a phantom unit at the seam between two catalogs. Picqer's API (n.d., accessed November 2026) publishes a rate limit of about 500 requests per minute, the kind of number an order feed must be designed against before the peak hour arrives. And DeHoratius and Raman's 2008 study in *Management Science* found, "examining nearly 370,000 inventory records from 37 stores of one retailer", that 65% were inaccurate; the study is old, from one US retailer's stores and before RFID, so read it as the field's reminder that a record drifts from the shelf unless someone counts.

Jumbo's conditions also carry delivery-performance penalties (€250 per order below 90% delivered on time and in full on an order line). Chargebacks thus come in two families: delivery performance, a logistics problem, and data and label compliance, an architecture problem.

Kestrel & Pine: Eva's jacket at Duiven, and the Queue That Waited

Eva's order #1072455 is placed on Tuesday 13 April 2027 at 21:14. In April 2027 the platform still sends it to the WMS through the iPaaS; the OMS of Waypoint decision W2 arrives in Q4. The flow creates the order in the WMS at 21:16 (illustrative). The last wave was released at 21:00. The order waits for the 06:30 wave on Wednesday 14 April, is picked with about 180 other orders on one route (illustrative), packed into one box of 1.1 kg with a PostNL label and confirmed at 08:40. The confirmation carries the tracking number; the stock event says jacket M moss minus one and socks minus two. PostNL collects it that evening and delivers on Thursday 15 April. The storefront had promised Wednesday. The warehouse did nothing wrong; the promise was not computed from the warehouse. Getting It There: Carriers, Labels, Tracking and the Delivery Promise starts from that gap.

Black Friday evening, Friday 27 November 2026, is the same gap at scale. In the peak hour, 20:00 to 21:00, the storefront took 2,300 orders, 38 a minute. The iPaaS flow that creates orders in the WMS made several calls per order and was throttled by the WMS

vendor's rate limit (illustrative mechanism); it fell behind, and by 21:00 the backlog was four hours long. Orders placed at 20:00 reached the WMS after midnight; the 21:00 wave contained the orders of the early afternoon. Roughly 8,000 orders placed between 17:00 and midnight (assumption) missed next-day delivery while the storefront promised it. The warehouse's staffing that night was never tested.

What would have prevented it is not a faster warehouse. It is a queue between the platform and the WMS that is **allowed to grow** without failing checkout, a flow that creates orders in **batches** within the WMS's rate limit, a monitor on the age of the oldest unprocessed order, and a cut-off that **moves** when that age grows, so the promise on the page tells the truth. All four are on Waypoint's list under decisions W2 and W5; Black Friday Every Year: Peak Load, Resilience and Recovery and Journey 4: Black Friday, Hour by Hour work the whole evening.

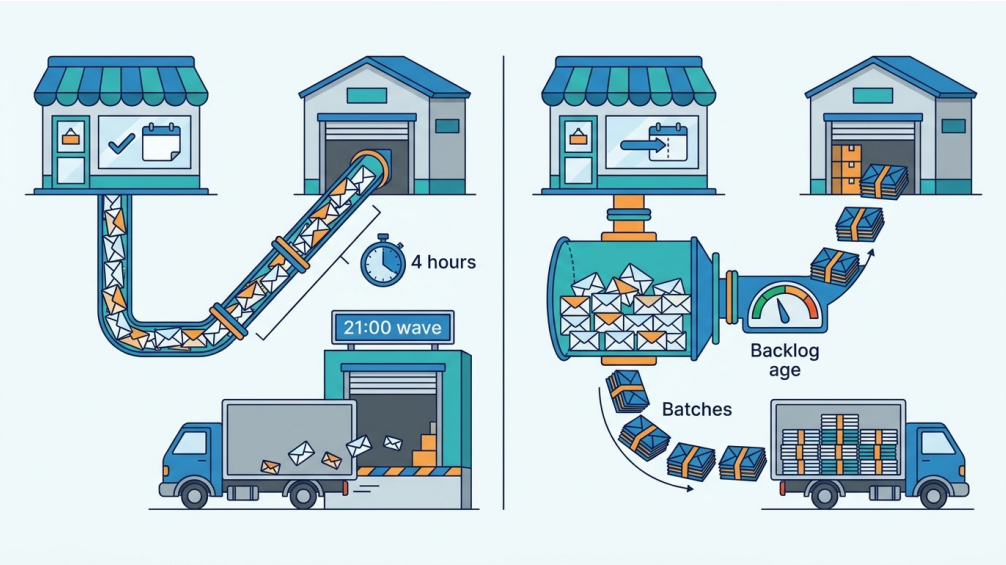


Figure 4: *The Queue That Waited and its cure: on Black Friday 2026 the order feed fell four hours behind while the page still promised next day; a queue allowed to grow, batch creation, a backlog-age monitor and a moving cut-off keep the promise honest.*

When It Goes Wrong

The warehouse interface fails in a small number of recognizable ways. Each has a detection and a recovery.

Duplicate fulfillment orders. A retry after a timeout delivers the same instruction twice and the WMS creates two pick lists; the customer receives two jackets. Detection: a fulfillment-order identity on every message and a daily check for two WMS orders

with the same identity. Recovery: idempotent creation in the WMS (same key, same order) and an alert rather than a second pick list.

The cancellation that arrived after the pick. Support cancels the order in the platform; the parcel ships anyway because the WMS had already accepted it. Detection: the cancellation must be a request with an answer, and the platform must show the answer. Recovery: the order stays in progress, support tells the customer, and the item comes back through The Way Back: Returns, Exchanges, Refunds and Reverse Logistics.

The lost shipment confirmation. The parcel left; the confirmation was lost; the platform says "unfulfilled" and the ERP never invoices. This is Kestrel & Pine's Three-Status Order (#1052877: *Delivered* in the platform, *In picking* in the WMS, *Not invoiced* in the ERP for six days) seen from the warehouse. Detection: a daily reconciliation of shipped-in-WMS against fulfilled-in-platform. Recovery: replay the confirmation from the WMS; never re-key it by hand.

Stock events out of order or missing. A pick event arrives before its receipt event, or not at all, and the available figure is wrong until the next snapshot. Detection: sequence numbers on events, and a weekly reconciliation such as Kestrel & Pine's May 2027 run (2.3% of active variants differing). Recovery: apply the daily snapshot as the baseline and replay events since.

The short pick. The record said one; the shelf has none. Detection: the WMS records a short pick against the fulfillment order. Recovery: the WMS tries another location; if none, it reports a partial shipment or closes the fulfillment order as incomplete, and the order system decides (ship the rest, wait, or cancel the line and refund). A short-pick rate per SKU tells you where to cycle count.

SKU mismatch. The platform's variant identifier and the WMS's differ by a suffix, a case or a leading zero; the fulfillment order is rejected or maps to the wrong product. Detection: rejects per SKU and a daily comparison of the two catalogs. Recovery: one identifier family owned by the product system (What Are We Selling? Products, Variants and the Catalog), published to both.

The WMS outage. The cloud WMS is unavailable for an hour on a peak evening. Detection: the age of the oldest queued order. Recovery: queue, do not fail checkout; when the WMS returns, release in batches within its rate limit; move the cut-off while the backlog is older than the time to the last wave.

What to Take Away

Established.

- A warehouse runs in batches with finite capacity and a daily cut-off set by carrier collections; wave, cluster and batch picking are the standard methods (Odoo, Picqer, Manhattan, n.d., accessed November 2026).

- The fulfillment order, with request, accept or reject, in-progress, shipped and cancellation-request states, is the documented contract between order management and fulfillment in Shopify's and Kibo's models: nothing ships before acceptance, and cancellations after acceptance are requests.
- Stock records drift from the shelf without counting (the classic 2008 study found 65% of records inaccurate in one retailer's stores); cycle counting is the standard remedy.
- Large retail customers require SSCC-labeled logistics units announced by a matching DESADV (GS1 General Specifications 26.0; bol since 31 December 2021; Jumbo 2025–2026), with refusal or fees when they do not match.

Depends on context.

- An ERP's warehouse module is enough for one building with simple routes and modest volume; a dedicated WMS becomes necessary when waves must be planned around cut-offs, labor must be managed, or the building serves several companies or channels.
- A 3PL or a marketplace fulfillment program is right when volume, geography or season does not justify a building, and wrong when packing or returns grading is part of the brand. Marketplace-held stock must be modeled as a location you cannot count.
- Periodic stock sync is acceptable while the lag is shorter than the time in which a popular unit sells out; stock events become necessary the moment that is no longer true, which for a brand with a hero product is the first peak.
- Generating the DESADV from the WMS pack confirmation is mandatory for a merchant with EDI retail customers and irrelevant for one without.

The author's synthesis.

- Design the warehouse integration as a contract that survives every system on either side: fulfillment orders in with an accept step, confirmations and stock events out, the WMS the only writer of on hand. Buy, rent or outsource the WMS freely; never let the contract depend on the vendor.
- Cut-off, capacity and wave schedule are data the delivery promise must read, not operations folklore. A merchant whose storefront states a cut-off its warehouse does not keep has already broken the promise; it just has not measured how often.

Questions to Ask

1. What exactly is sent to the warehouse per order, and does it carry anything the warehouse does not need?

2. Is there an accept-or-reject step before the warehouse counts on an order, and what happens to a cancellation after acceptance?
3. If the same instruction is delivered twice, does the warehouse create one pick list or two?
4. Which system writes on-hand stock, and can anything else (an app, a script, the ERP) also write it?
5. How does the rest of the landscape learn of a stock change, and how long is the lag at peak?
6. When was the last reconciliation between the warehouse's stock and the platform's, and what share of SKUs differed?
7. What is the release time of the last wave, and does the storefront's cut-off agree with it?
8. How many orders per minute can the order feed into the warehouse sustain, and what happens to checkout when it cannot keep up?
9. For wholesale customers with EDI: which system generates the SSCC label and the DESADV, and from what?
10. If the warehouse system is unavailable for an hour on the busiest evening, what do customers see?

The parcel is packed, labeled and confirmed. What happens next is outside the building and outside the merchant's control: a carrier's network, a scan at a depot, a van on Eva's street, and the date she was shown at checkout. Getting It There: Carriers, Labels, Tracking and the Delivery Promise is about choosing the carrier, producing the label, reading the tracking events, and making a delivery promise the warehouse can keep.

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Research and practitioner sources

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

16. Getting It There: Carriers, Labels, Tracking and the Delivery Promise

IN THIS CHAPTER, YOU WILL: Learn how a parcel gets a carrier, a label and a tracking history, and what data a delivery promise needs before you may show it at checkout.

KEY POINTS:

- **Carriers are many, and each speaks its own language**. Every carrier has its own API generation, label format, manifest rule and status codes, and each migrates on its own calendar; a normalizing layer, bought or built, is what keeps carrier change out of the order record.
- **A tracking event is a fact about a parcel; the order's status is derived from it**. Map every carrier's scans onto one small vocabulary, let the order status page and notifications read from that vocabulary, and measure carrier performance per lane from the same events.

- **A delivery promise is a computation, not a sentence.** It needs the stock location, the warehouse's cut-off and capacity, the carrier's transit time for the lane and a margin for the tail; a promise written into the page is kept by luck and broken without anyone noticing.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Pulled Rug** — carriers retire API versions, rename products and change manifest rules on published dates; a label integration that nobody watches breaks on the morning the old version is switched off.
- **The Message in a Bottle** — a tracking number that does not reach the order record, or a scan that arrives twice, leaves the customer without a "shipped" message or with two.
- **The First Snowfall** — a static promise is right on an ordinary Tuesday and wrong on the one evening a year when the queue to the warehouse is four hours long.

A shopper sees 'delivered tomorrow' after the warehouse's last dispatch has left. Even a carrier that delivers every parcel on time cannot keep that promise. The date shown at checkout must account for the work before collection as well as the journey afterward: warehouse capacity, collection cutoffs, carrier services and sometimes customs. A fixed sentence in a storefront cannot adjust when those conditions change.

Kestrel & Pine, the fictional outdoor brand, uses PostNL in the Netherlands, bpost in Belgium, DHL in Germany and DPD as backup. Each has different labels, interfaces, collection rules and tracking terms. Putting those differences behind a shared shipping interface lets checkout request a delivery option and support read a consistent status without each maintaining its own carrier rules. The choice is who builds and operates that interface.

Where we are in the order's journey. Eva's jacket shipment is confirmed at Duiven at 08:40 on 14 April 2027. PostNL collects that evening and delivers on 15 April. This chapter follows the label and tracking messages, then asks whether checkout could have promised that date accurately. Pick, Pack, Ship: Warehouse Management and Fulfillment covers packing.

Setting the Scene: A Sentence in the Theme

In April 2027, Ecommerce Manager Milan Petrov asks Head of Engineering Noor Visser to add delivery dates to the German and Belgian storefronts. The Dutch one already says "ordered before 22:00, delivered tomorrow".

Operations head Daniel Okafor points out the problem. Carriers collect at different times and transit times vary by destination. Even the Dutch promise ignores Duiven's

last picking wave at 21:00. Those timings are illustrative, as described in Pick, Pack, Ship: Warehouse Management and Fulfillment.

Nothing computes the promise. It is text written when an outside logistics provider in Tiel handled fulfillment. The agency can change the words in an hour; nobody can say how often they are true.

Eva ordered at 21:14 on 13 April, was promised Wednesday and received her parcel Thursday. During Black Friday 2026, the same promise stayed visible while orders waited four hours to reach the warehouse.

Waypoint, the company's improvement program, therefore requires availability and delivery dates through APIs in decision W8. A service agent, marketplace or future shopping agent needs the same computed answer as the storefront. Milan's request exposes that shared need.

The Vocabulary of Getting It There

Shipping has a small vocabulary a newcomer needs once.

Term	In plain words
Carrier	The company that moves the parcel: PostNL, bpost, DHL, DPD. Each sells several **services** (standard, next-day, evening, pickup point, signature).
Carrier API	The carrier's interface for creating a shipment, getting a label, closing the day and reading tracking.
Label	The sheet on the parcel with the carrier's barcode and routing data; the barcode is the parcel's identity in the carrier's network.
Manifest	Closing the day: handing the carrier the list of parcels it is collecting. DHL's Parcel DE Shipping API states that "shipments are transferred to DHL backend systems only when manifested", with an automatic close-out at 17:45 by default (API reference updated 3 November 2026).

Term	In plain words
Tracking event	One scan with a time and place: accepted, at depot, out for delivery, delivered, attempted, exception.
Pickup point, locker	A delivery address that is not the customer's home: a shop counter, a parcel locker (DHL calls its lockers Packstations and addresses them by a `lockerID`).
Delivery option	What the customer chooses at checkout: home, pickup point, locker, a time window, a date.
Cut-off	The latest order time that still ships today, set by the warehouse's last wave and the carrier's collection (Pick, Pack, Ship: Warehouse Management and Fulfillment).
Lead time, transit time	Days from order to shipment (handling) and from collection to delivery (transit), per lane: an origin, a destination area and a service.
Delivery promise, estimated delivery date (EDD)	The date or window shown before purchase; the EDD is the same thing computed rather than written.
Rate shopping	Choosing the cheapest or fastest carrier service for each parcel, given its size, weight, destination and the promise made.
Multi-carrier platform	A service that sits between the merchant and many carriers: one API for labels and tracking, one event vocabulary, often one contract.

How the Pieces Fit Together

Shipping touches three moments of the order: the promise at checkout, the label at packing, and the tracking after collection. The figure shows the systems around them.

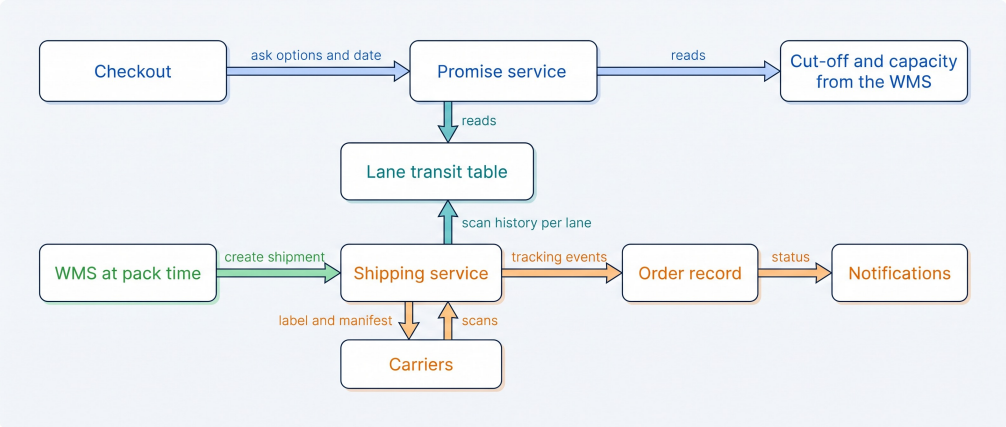

Figure 1: *Three moments, one shipping layer: the promise reads the warehouse's cut-off and the lane table before checkout, the WMS creates the shipment through the shipping service at pack time, and the carriers' scans come back through the same service as normalized events that drive status, notifications and, over time, the lane table itself.*

The requirements differ by moment. The latencies below are assumptions for a merchant of Kestrel & Pine's shape, not vendor figures.

Flow	How fresh	How consistent	How fast	How available
Delivery options and date at checkout	Reflects today's cut-off and the current backlog *(assumption: minutes)*	Same answer for the same address and basket within a session	Under a few hundred milliseconds; checkout waits for it *(assumption)*	Must degrade to a safe default ("2–3 working days") rather than block checkout
Pickup point search	Carrier's list, cached hours	Points that exist and are open	Sub-second	Degrade to home delivery
Label creation at pack	Immediate	Exactly one label per parcel; a retry must not create a second shipment	Seconds per parcel; the packing station waits	If a carrier's API is down, fall back to the backup carrier or queue the parcel

Flow	How fresh	How consistent	How fast	How available
Manifest	Before the carrier's close-out	Every labeled parcel on exactly one manifest	Minutes	Missing the close-out means the carrier does not know the parcels exist
Tracking events in	Within minutes of the scan	Events idempotent by carrier, barcode and event id; order preserved	Minutes	Late is fine; lost must be caught by polling or reconciliation
Tracking number to the order	At label print	One per parcel, linked to the fulfillment order	Within a minute, so the "shipped" message is right	Queued and replayed

Labels and manifests: the carrier's side of the contract

Creating a shipment is a transaction with the carrier: it returns a barcode and a label, and from that moment the carrier expects the parcel. Three details decide whether the integration is reliable.

- **Idempotency** (a retried call must have the same effect as the first). A timeout after the carrier created the shipment, followed by a retry, yields two barcodes for one parcel; one of them will be manifested and never collected, and the customer may receive two tracking numbers. The shipment call must carry the merchant's shipment reference and the integration must look up before it creates.
- **Manifesting**. A label is not a booking. DHL's rule above is explicit; PostNL's current "Warehouse V4" API family includes a separate Confirming step (developer portal, n.d., accessed November 2026). A packing flow that prints labels but misses the close-out produces parcels that leave the building with no record in the carrier's network.
- **Version drift**. PostNL's developer portal says that "in 2025, we started rolling out the renewed APIs in phases", with Shipment V4, Return V4, Checkout V4 and Warehouse V4 next to the "legacy" Send & Track and Checkout APIs. DHL's Parcel DE Shipping API is at version 2.2.0 (8 October 2026), replaced the Warenpost product with "DHL Kleinpaket (from 1.1.2025)", and scheduled a TLS certificate

change for 2–4 November 2026. bpost's Shipping Manager API "moved from v3 to v5" on 17 April 2023. Four carriers, four calendars; a merchant that integrates directly owns four migrations.

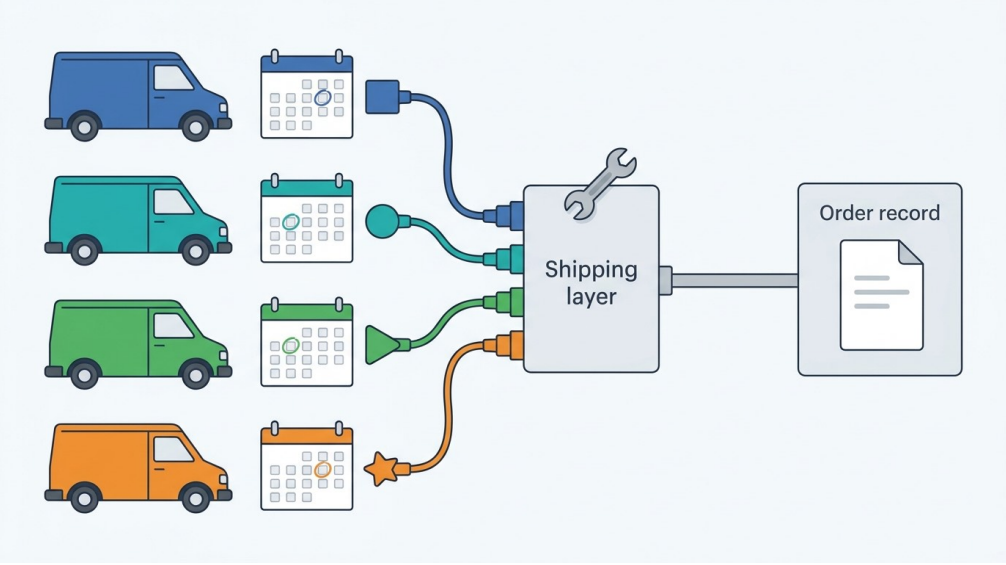

Figure 2: *Four carriers retire API versions on four calendars; a normalizing shipping layer absorbs each migration so the order record never notices.*

Tracking events: one vocabulary for many carriers

Every carrier publishes its own status codes. DHL's Shipment Tracking Unified API (launched 1 October 2018, changelog through September 2026) says that status codes "vary by service" and that clients "should treat unrecognized values as undefined", which is honest and inconvenient. DPD Belgium's Predict notifications fire "when the parcel arrives at the depot, and when it leaves the depot for the receiver" (n.d., accessed November 2026). The merchant needs one vocabulary.

EasyPost's tracker object (n.d., accessed November 2026) is a reasonable public model: ten statuses, `unknown`, `pre_transit`, `in_transit`, `out_for_delivery`, `delivered`, `available_for_pickup`, `return_to_sender`, `failure`, `cancelled` and `error`, with a scan history of status, message, time and location, and webhooks (calls the vendor's system makes to your URL when something happens) on each update. Shopify's order status page (n.d., accessed November 2026) shows that the customer needs even fewer: "Confirmed, On its way, Out for delivery, Delivered, and Attempted delivery", and only for supported carriers, because "if you fulfill an order without a

tracking number, then no shipping updates are received".

Normalized event	What it means	Who consumes it
`pre_transit`	Label created; the carrier has not scanned the parcel	Nobody yet; a parcel stuck here for a day is a missed manifest
`in_transit`	Scanned into the carrier's network, moving between depots	Status page: "On its way"; notification: shipped
`out_for_delivery`	On the van	Status page and notification
`delivered`	Handed over or left as instructed	Status page; the 14-day return clock of The Way Back: Returns, Exchanges, Refunds and Reverse Logistics starts
`available_for_pickup`	At the pickup point or locker	Notification with the location and the holding period
`failure`, attempted	Not delivered: absent, refused, address problem	Service queue; second attempt or re-route
`return_to_sender`	Coming back without a return request	Warehouse receiving; refund or re-ship decision
`exception`	Lost, damaged, customs hold	Service and claims

Two rules follow. **The order's shipping status is derived from these events**, never set by hand, and the status page, the notifications and the service agent read the same derived status (Where Is My Order? Notifications, Service and the Customer Record). And **every event is stored with its lane**, because the scan history is the only data that will ever tell you how long PostNL really takes from Duiven to Nijmegen on a Wednesday in April.

The delivery promise: from sentence to computation

A static promise is a business rule with no data behind it. A computed one is an estimate with inputs, and the inputs are the chapter's argument.

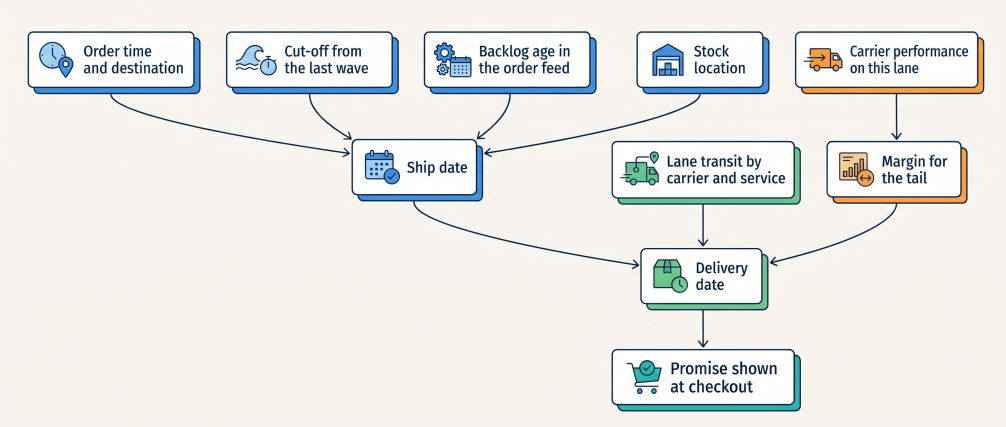

Figure 3: *A computed promise: the ship date comes from the order time, the warehouse's cut-off, the backlog in the order feed and where the stock is; the delivery date adds the carrier's transit for the lane and a margin set by how that lane has actually performed.*

In words, for Eva on Tuesday 13 April at 21:14: the stock is at Duiven; the last wave was 21:00, so the ship date is Wednesday; PostNL's standard service from Duiven to Nijmegen takes one day on that lane; the margin is small because the lane has been reliable; the promise is **Thursday 15 April**. That is the day the parcel arrived. The static sentence said Wednesday. A computed promise would have been honest, and it would have known at 20:00 on Black Friday that the ship date had already slipped by at least a day.

The public evidence shows that even the best-resourced operators treat the promise as a model with a measured effect. Amazon's SPEEDY paper (ACM SIGSPATIAL, 2025) learns from "multiple heterogeneous views of order-to-delivery data" to recommend "dynamic, faster same-day delivery time slots", predicting narrower slots for "60%+ eligible orders" in its test locale, and reports a 21-day online A/B test with +17 basis points in units for the treatment group: a real, small, measured gain from a sharper promise.

A 2025 arXiv paper by Ye, Hijazi and Van Hentenryck forecasts order fulfillment time as a distribution rather than a point, with "spatiotemporal features tied to fulfillment location and carrier performance", because "traditional rule-based approaches often fail to capture the inherent uncertainties in delivery operations"; on an industrial dataset it reports "up to 14% higher accuracy and up to 75% better identification of late deliveries" than the rule-based system.

The point for a newcomer is not the models. It is that **the promise has a tail**, and a point estimate hides it; the margin in the figure is where the tail lives.

Vendor case studies claim that showing an estimated date lifts conversion by around 10–20% (Maude via Wonderment, Natch via Hive, 2024–2026); they are vendor-reported A/B tests without peer review, so read them as "the effect is positive and worth measuring",

not as a number to plan on. Consumer law sets the outer bound: Directive 2011/83/EU (2011), Article 18, requires delivery "without undue delay, but not later than 30 days from the conclusion of the contract" unless otherwise agreed, and Article 20 passes the risk of loss or damage to the consumer only when they have "acquired the physical possession of the goods". Until the scan says delivered, a lost parcel is the merchant's loss.

Delivery options, cross-border and the cost lever

Options at checkout. Pickup points and lockers are a second delivery network with its own lookup: PostNL's Checkout V4 bundles Locations and TimeFrame APIs so a shop can "add delivery options to your checkout"; multi-carrier platforms aggregate them (Sendcloud lists "390K+" service points and nShift "1.2 million pick-up and drop-off points", both n.d., accessed November 2026, both marketing counts). The option chosen must be stored on the order, because it changes the label, the promise, the notifications and, under Article 13(2) of the same directive, the refund: a merchant need not reimburse the extra cost of a delivery "other than the least expensive type of standard delivery".

Cross-border. Kestrel & Pine ships only within the EU, so no customs applies. The moment a merchant ships outside the customs union, three things appear: an import declaration (since 1 July 2021, "required for all goods entering the EU, regardless of their value", with a reduced data set for consignments up to €150, per the European Commission's customs pages, n.d., accessed November 2026), the **IOSS** (Import One-Stop Shop: the scheme that lets a seller charge destination-country VAT at checkout on imported consignments up to €150 and remit it monthly, so the parcel is not stopped for VAT at the border; a 2026 Council document proposes deleting the €150 threshold, so check its status), and customs data per SKU (stock-keeping unit: one sellable variant): a commodity code and a country of origin, which are catalog attributes (What Are We Selling? Products, Variants and the Catalog). DHL's Parcel DE API added paperless customs documents to production on 3 November 2026. Cross-border makes the label call depend on catalog data quality; Tax, Invoices and the Books: From Order to Ledger owns the VAT.

Cost. Shipping is a business lever as much as a logistics fact. Kestrel & Pine's D2C average order is €92 and shipping is free above €75, so most parcels carry a cost the margin absorbs; assume €4.50 to €6.00 per domestic parcel for a merchant of this size (assumption; carrier contracts are confidential). Rate shopping across carriers and services, right-sized boxes from the WMS's cartonization, and steering customers to pickup points (cheaper for most carriers) are the three levers the architecture can offer; the platform's checkout must be able to price options differently and the shipping service must be able to choose per parcel.

The Choices

Who talks to the carriers

Direct integration per carrier. The merchant's WMS or shipping module calls each carrier's API. It gives the best access to carrier-specific features and negotiated rates; it costs one integration per carrier per country, each with its own migration calendar, label format and status codes. It fits a merchant with volume commitments to two or three carriers and an engineer who owns the integrations; it limits a merchant with a small team the first time two carriers migrate in the same quarter.

A multi-carrier platform. A service such as Sendcloud ("30,000+ e-commerce businesses across Europe", "170+" carriers on its marketing page of 6 November 2026; its developer site says "160+", which is the usual inconsistency of marketing counts), nShift ("over 1,000 carrier services and 450+ platforms", n.d.) or ShipStation sits between the merchant and the carriers with one API, one label flow and one tracking vocabulary, and often its own carrier contracts. It absorbs version drift and gives small merchants rates they could not negotiate. It limits a merchant whose carrier contract includes services the platform does not expose, and it is one more vendor between the packing station and the truck.

Through the WMS or the 3PL. Most WMS products and every 3PL bundle carrier integration; Picqer sends "carrier labels and tracking codes" back to the store in one step. The merchant integrates nothing and sees only tracking numbers. It fits Starting and Growing merchants completely; it limits a merchant who wants to offer delivery options at checkout that the WMS cannot price, or who wants the scan history for its own promise.

Through the commerce platform. Hosted platforms sell labels and tracking in their admin for the markets they cover. Simplest of all for a single-country merchant; limited the moment a warehouse system owns the packing station.

Choice	Fits when	Limits when	Who absorbs carrier change
Direct per carrier	Volume contracts, carrier-specific services, an owner for each integration	Small team, many countries, two migrations in one quarter	The merchant

Choice	Fits when	Limits when	Who absorbs carrier change
Multi-carrier platform	Several carriers and countries; no negotiated rates yet; delivery options at checkout wanted	Contract services the platform does not expose; one more vendor in the critical path	The platform vendor
WMS or 3PL bundle	One warehouse owns packing; the merchant wants no shipping code	Checkout options and the promise need data the WMS does not expose	The WMS or 3PL vendor
Commerce platform's labels	One country, one warehouse, no WMS	A WMS or 3PL enters	The platform vendor

The combination Kestrel & Pine runs (a multi-carrier shipping service inside the WMS flow, with carrier contracts of its own for PostNL and DHL) is common at the Omnichannel stage: direct contracts for price, a shipping service for labels and tracking normalization. My rule is that **the normalizing layer is not optional from the second carrier onward**; what is optional is whether you buy it or build it.

How the promise is made

A static rule. "Ships in 1–2 days." "Ordered before 22:00, delivered tomorrow." Costs nothing, needs no data, and is right as often as the warehouse and the carrier happen to agree with it. Fits a Starting merchant with one carrier and one cut-off; limits anyone with two storefronts or one peak.

A transit table. A table of lead days per carrier, service and destination area, plus the warehouse's cut-off, maintained by operations. Honest on an ordinary day, blind to backlog and to the carrier's bad week. Fits a Growing merchant.

A promise service. A component that reads the warehouse's cut-off and backlog, the stock location, the lane table and a margin from recent carrier performance, and answers "options and date for this basket and this address" to the checkout, to marketplaces and, later, to agents. This is what Waypoint W8 means by "delivery promise through APIs". Fits the Omnichannel stage; its cost is that it must be fed, and that a promise service nobody measures is a static rule with more moving parts.

A predictive model. Distributional forecasts of fulfillment time from scan history, as in the 2025 papers above. Right for a Group-stage operator with the data volume to train

and the analysts to watch it; a Giant's Boots for everyone else.

Approach	Inputs	Honest when	Blind to
Static rule	None	Warehouse and carrier happen to agree	Everything
Transit table	Cut-off, lead days per lane	Ordinary days	Backlog, carrier bad weeks, stock location
Promise service	Cut-off, backlog, stock location, lane table, performance margin	Most days including peak, if fed and measured	Events it has no data for yet
Predictive model	Scan history at scale	At volume, with monitoring	Its own drift, without a back-test

Across Scale Stages: When It Fits and When It Limits

Stage	Typical setup	What becomes necessary	Warning signs of outgrowing it
1 — Starting	The platform's labels or a multi-carrier plugin; one carrier; a static promise; a tracking link in the e-mail	A tracking number on every order the moment the label prints; one person who knows the carrier's close-out time	Addresses typed into the carrier's portal; "where is my order" e-mails answered by looking up the carrier site by hand

Stage	Typical setup	What becomes necessary	Warning signs of outgrowing it
2 — Growing	A multi-carrier platform; a second carrier or country; pickup points at checkout; a transit table; the platform's notifications driven by tracking	The delivery option stored on the order; one event vocabulary; the promise text agreed with the warehouse's cut-off; a watch on carrier API changelogs	A different promise on each storefront; labels failing the morning a carrier version is retired; a lost parcel nobody notices until the customer does
3 — Omnichannel	A shipping service in the WMS flow with direct contracts for volume; a normalizing layer for tracking; a promise service reading cut-off, backlog and lane performance; the promise exposed by API to marketplaces; an exceptions queue	Scan history stored per lane; the promise measured (promised vs delivered, per lane, per week); a backup carrier that can be switched on; cross-border data in the catalog if shipping leaves the EU	A static promise during peak; marketplace promise and storefront promise disagreeing; tracking numbers missing on 3PL or marketplace-fulfilled orders; carrier migrations found by an outage
4 — Group	The promise as a shared service with per-brand policies; group-level carrier contracts; carrier invoice audit against scan data; lane performance across brands	One promise API for every brand and channel; an owner for carrier relationships separate from the warehouse	Each brand with its own carrier stack and promise logic; nobody can say the group's on-time rate

The judgement: shipping is the one area where a Growing merchant should buy the normalizing layer early and without regret, because carrier change is external, frequent and uninteresting to the merchant's customers. The promise is the opposite: it should stay simple (a transit table agreed with the warehouse) until the merchant has the cut-off, backlog and lane data to feed a real service, and then it should be built or bought as a component with an API, because by then three other systems need the same answer.

In Practice

Public evidence

The carriers' own developer pages document the drift that this chapter asks the architecture to absorb: PostNL's phased V4 rollout from 2025 with legacy APIs still listed; DHL's Parcel DE Shipping API v2 (current 2.2.0, 8 October 2026), its product rename to DHL Kleinpaket from 1 January 2025 "with automatic correction of old requests", its 17:45 default manifest close-out and its November 2026 certificate change; its tracking API's initial access of 250 calls per day, one every 5 seconds, which rules out polling as the primary tracking source at any real volume; bpost's Shipping Manager API move from v3 to v5 on 17 April 2023 and its bpack integration manual at version 3.3.35 (26 May 2026). Each of these is a Pulled Rug for a merchant that integrated directly and did not subscribe to the changelog.

On what carriers deliver, the public measures are thin. PostNL's Annual Report 2024 (published 2025) reports "delivery quality parcels in NL" of 97% in both 2023 and 2024 and a parcel volume of 371 million items in 2024; that is PostNL's own key performance indicator, and no independent parcel on-time measure for the Netherlands, Belgium or Germany was found. The figures of the Dutch regulator ACM (Authority for Consumers and Markets) concern letters (86% within 24 hours in 2024 against a 95% legal requirement, per DutchNews, 13 June 2025; a fine of nearly €7 million for 2023, reported 12 May 2026). The lesson for an architect is practical: **the only on-time rate you will ever trust is the one you compute from your own scan history**, which is why the events must be stored per lane from the first day.

On the promise, Amazon's 2025 SPEEDY paper and the 2025 conformal-forecasting paper above are the clearest public statements that a promise is a model with a measured conversion effect and a tail; Shopify's five customer-facing states and its "no tracking number, no updates" rule are the clearest statement of what the customer actually needs to see.

Kestrel & Pine: Eva's PostNL parcel, and the promise that should have said Thursday

At 08:40 on 14 April 2027, the WMS confirms Eva's shipment. The shipping service creates a PostNL parcel: 1.1 kg, standard delivery to Nijmegen. The fulfillment order reference acts as an **idempotency key**, identifying the request so retries cannot book another shipment.

The label prints and the tracking number returns through the integration platform within a minute. Eva receives a shipped email. At 18:30, the shipping service includes her parcel in the day's manifest of 1,100 PostNL parcels; collection is at 22:30.

Tracking scans follow: depot acceptance at 23:40, out for delivery at 07:50 Thursday and delivered at 11:20. These timings and batch size are illustrative. The service translates each scan into a common status, records the route and forwards it to the platform for customer updates. Kestrel & Pine shipped 336,000 parcels in 2026 through this process.

What did not work was Tuesday evening. At 21:14 the storefront showed "ordered before 22:00, delivered tomorrow". A promise service would have read the last wave (21:00, already gone), the backlog (none), the stock location (Duiven), the lane (one day, reliable) and said Thursday. Eva would have seen Thursday and received the parcel on Thursday. The difference between the two is not the parcel; it is one "where is my order" (WISMO) ticket, one disappointed customer and, multiplied by every weekday order between 21:00 and 22:00, a measurable share of the year's complaints that nobody has measured. Under Waypoint W8, the promise becomes an API by 2028; the first version Noor specifies in May 2027 is the transit table plus the WMS's cut-off and backlog, measured weekly as promised-versus-delivered per lane (illustrative plan). Journey 1: Eva's Jacket, From Search to Doorstep follows Eva's order end to end; Journey 4: Black Friday, Hour by Hour shows the same promise on Black Friday evening.

Figure 4: *Eva's promise written versus computed: the theme's sentence said Wednesday, while a promise computed from the 21:00 wave, the stock at Duiven and the one-day lane would have said Thursday 15 April, the day the parcel arrived.*

When It Goes Wrong

The retired API version. A carrier switches off an API version on a published date; the shipping flow fails at the first label of the morning. Detection: subscribe to every carrier's changelog, and test the label call in the carrier's sandbox against the new version before the date. Recovery: switch to the backup carrier for the day (Kestrel & Pine keeps DPD for exactly this), queue the parcels, fix the integration. The multi-carrier platform's job is to make this its problem instead of yours.

Two labels for one parcel. A timeout and a retry create two shipments. Detection: two barcodes against one fulfillment order; a `pre_transit` event that never progresses. Recovery: cancel the second shipment with the carrier before the manifest; make the shipment call idempotent by reference.

The missed manifest. Parcels left on the truck without a close-out; the carrier scans them as unknown or not at all. Detection: parcels in `pre_transit` for more than a few hours after collection. Recovery: manifest late if the carrier allows; otherwise the parcels travel on the barcode alone and the tracking is late or absent. Prevention: the close-out is a scheduled job with an alert, not a button.

The lost or duplicated scan. A webhook delivery fails or arrives twice. Detection: events keyed by carrier, barcode and event id; a daily poll of parcels with no event in 24 hours. Recovery: idempotent event handling; a status derived from the full history, not from the last message received.

The tracking number that never reached the order. The label printed; the confirmation lost its tracking number on the way; the customer sees "unfulfilled" and the status page shows nothing, which is Shopify's "no tracking number, no updates" rule in action. Detection: shipments in the shipping service with no tracking number on the order. Recovery: replay from the shipping service.

The promise at peak. A static promise stays on the page while the order feed backs up (the Queue That Waited of Pick, Pack, Ship: Warehouse Management and Fulfillment). Detection: the age of the oldest unprocessed order, and the promised-versus-delivered measure per day. Recovery: a promise that reads the backlog, or at minimum a peak switch that changes the text; a goodwill voucher policy decided before the day.

Undeliverable, lost, damaged. The parcel comes back (`return_to_sender`), disappears (`exception` with no further scans) or arrives broken. Detection: the normalized events and the absence of a `delivered` within the lane's expected time plus margin. Recovery: a service workflow with the carrier's claims process, and a decision rule (reship from stock, refund) owned by service, with the stock side handled by The Way Back: Returns, Exchanges, Refunds and Reverse Logistics. Under Article 20 of the Consumer Rights Directive the loss is the merchant's until delivery.

The marketplace parcel. Orders fulfilled by the marketplace's program (about 40% of Kestrel & Pine's German marketplace units) produce tracking the merchant never sees unless the connector imports it. Detection: marketplace orders with no tracking in the order record. Recovery: import tracking from the marketplace as events; Selling on Someone Else's Platform: Marketplaces and Channel Integration owns the connector.

What to Take Away

Established.

- Carriers version and retire their APIs, rename services and set manifest rules on their own calendars (PostNL V4 from 2025; DHL Parcel DE v2 and Kleinpaket from 2025; bpost v5 from 2023), and their status codes differ and change (DHL: treat unknown values as undefined).
- A tracking number must reach the order record for any status or notification to exist (Shopify, n.d.); a small normalized event vocabulary such as EasyPost's ten statuses covers every carrier and the customer needs five of them.
- Consumer law bounds the promise: delivery within 30 days absent agreement (Directive 2011/83/EU, Art. 18), risk with the merchant until physical possession (Art. 20), and refund of only the cheapest standard delivery (Art. 13(2)).
- The promise has a tail; published research (Amazon 2025; Ye, Hijazi and Van Hentenryck 2025) treats it as a distribution with a measured conversion effect, not a rule.

Depends on context.

- Buy a multi-carrier platform from the second carrier or country onward unless you have volume contracts, carrier-specific services and an engineer who owns each integration; at Omnichannel scale, direct contracts for price and a normalizing layer for labels and tracking commonly coexist.
- Keep the promise a transit table agreed with the warehouse until you have cut-off, backlog and lane data to feed a service; build or buy the service when marketplaces, service agents or shopping agents need the same answer by API.
- Cross-border customs data (commodity codes, origin, IOSS) matters only if you ship outside the customs union, and then it is a catalog-quality problem before it is a shipping problem.
- Pickup points and lockers are worth offering where the carrier networks are dense (the Benelux and Germany) and are a cost lever as well as a convenience.

The author's synthesis.

- A promise you cannot compute is a promise you cannot measure, and a promise you cannot measure is broken more often than you think. The first step is not a model; it is making the storefront's sentence agree with the warehouse's last wave, and counting how often the parcel arrives on the day shown.
- Store every scan with its lane from the first day. It is the cheapest data a merchant will ever collect and the only source of an on-time rate it can trust.

Questions to Ask

1. What exactly is the delivery promise computed from today, and who would know if it stopped being true?
2. Does the storefront's cut-off agree with the warehouse's last wave and the carriers' collection times, on every storefront?
3. Which system talks to the carriers, and who is notified when a carrier announces an API version retirement?
4. If the label call times out and is retried, can one parcel get two barcodes?
5. When and how is the manifest closed, and what alerts if it is missed?
6. Which event vocabulary does the order status use, and is the status derived from events or set by hand?
7. How long after the label prints does the tracking number reach the order, and what happens to the "shipped" message if it does not?
8. Is the scan history stored per lane, and can you say last month's on-time rate for Duiven to Antwerp by bpost?
9. What does the customer see at checkout when the promise service is down?

10. Which carrier is the backup, and when was it last used?

The parcel is on Eva's doormat in Nijmegen and the status page says delivered. For most orders this is the end. For Kestrel & Pine, with six stores, a click-and-collect pilot and store stock that reaches the ERP (the enterprise resource planning system that keeps the books and the item master) only nightly, the next question is what changes when the delivery address is a shop and the stock is on a store shelf. Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment takes the order into the store.

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

17. Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment

IN THIS CHAPTER, YOU WILL: Learn what a store adds to the architecture, how click-and-collect, ship-from-store, return-in-store and endless aisle work, and when to change the point of sale.

KEY POINTS:

- **A store is a second system of record, not a sales channel.** It creates orders, holds stock, knows customers and takes money, often while offline and under fiscal rules of its own. The architecture has to model it as a stock location, an order source and an identity source, each with a named owner.
- **Store stock may be promised only with a buffer, a confirmation or a count behind it.** The classic field study found about two-thirds of store inventory records wrong, and a nightly copy makes a wrong number older. The Store Stock Mirage is what promising at face value from a nightly copy produces.

- **"Unified commerce" is a goal state, not a product.** One record per order, per stock unit and per customer across channels can be reached with one platform or with an order management system and an identity layer over several. Which path fits depends on the scale stage, which is why Kestrel & Pine fixed the joins first and deferred its point-of-sale decision to 2028.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Phantom Stock** — store stock copied nightly into the back office is promised online as if it were counted this minute.
- **Schrödinger's Order** — a click-and-collect order exists in the web platform, in an e-mail to the store and as a till receipt, and none of the three knows about the others.
- **The Familiar Stranger** — the store's loyalty list, the web shop's accounts and the newsletter each count the same shopper as a different person.

A shopper brings a web order to a nearby store for an exchange. The store's **point of sale (POS)**, its till software and hardware, must find the purchase, work out the amount paid and record the returned item. If the online platform holds those facts separately, staff may have to reconstruct the transaction by hand. A service that sounds simple to the customer requires the systems to cooperate.

Separate store and online records can work while each channel sells only its own stock; a nightly report may be enough for finance. Store pickup, shipping from a store or exchanging a web order at the till creates a new dependency. **One channel now acts on another channel's stock or order.** Those facts must arrive before staff or software make the next commitment, or the merchant can sell the same unit twice or refund the wrong amount.

The chapter explains those cross-channel services, the limits of store stock records and the choice between a platform's own POS and a separate product. Is It in Stock? Inventory, Availability and the Promise You Can Keep covers sellable quantities, The Order Record: Lifecycle, State and the Order Management System coordinates orders, and One Customer, One Product: Identity and Master Data Across Systems links customer records.

Where we are in the order's journey. Eva's jacket from Kestrel & Pine, the fictional outdoor brand, travels from the Duiven warehouse to her Nijmegen home. The company's Nijmegen store could offer alternatives: pickup, dispatch from its shelf or a counter exchange from M to L. This chapter explains what those alternatives would require.

Setting the Scene at Kestrel & Pine: The Utrecht Pilot and a Deferred Decision

Kestrel & Pine's six stores hold about **1,800 variants each**. Their **118,000 transactions** produced **€6.0 million** in 2026, about **€51 per receipt**. They use a cloud POS chosen in 2019 and a separate card-terminal provider.

Stock reaches the **enterprise resource planning system (ERP)**, which runs finance and purchasing, once a night. **No integration connects the POS to the online platform**. The stores are in Arnhem, Utrecht, Amsterdam, Nijmegen, Eindhoven and Antwerp.

The Utrecht **click-and-collect** pilot lets a customer order online and collect at the counter. In 2026 it relied on people: the platform checked last night's ERP stock copy and emailed the store. Staff checked the shelf and contacted the customer.

In December, **37 promised items had already sold** before the nightly update. This **Store Stock Mirage** is a stale-stock failure: the platform could accept an order without knowing what the till had sold that day.

In February 2027, Noor Visser and operations head Daniel Okafor considered replacing the POS with the platform's own till. Their **Waypoint** improvement program instead repairs the information flow first:

- **W1**: make the POS authoritative for store stock and send changes as they happen. An availability service calculates what each channel may sell.
- **W2**: bring channel orders into a purchased **order management system (OMS)**, adding stores later.
- **W4**: keep the hosted platform and evaluate its POS in 2028.

This sequence fixes what the store tells the other systems before committing to new tills.

What a Store Adds to the Architecture

In plain words, a store adds four things the online architecture must now take into account, and two constraints it has never had to live with.

What the store adds	What it means for the architecture	Chapter that owns the concept
A second order source	Every receipt is an order created outside the platform, with its own number, its own line items and its own tax treatment	The Order Record: Lifecycle, State and the Order Management System
A stock location per store	Six stores are six more places stock can be promised from, each counted by hand and sold from by people	Is It in Stock? Inventory, Availability and the Promise You Can Keep
A second customer list	The POS knows a customer by loyalty card, phone number or e-mail captured at the till, if at all; most store sales are anonymous	One Customer, One Product: Identity and Master Data Across Systems
A second payment path	Terminal payments settle through the terminal provider, with their own references and payout files	Taking the Money: Payment Providers, Authorization, Capture and Refunds
Offline operation (constraint)	The store must keep selling when the connection drops; for those hours it is its own system of record, reconciled later	this chapter
Fiscal rules (constraint)	Some countries prescribe how a sale is recorded and what a receipt must contain, down to a signed transaction counter	this chapter, Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation

The two constraints make a POS a different kind of product from a web platform. A web checkout can refuse to work when a dependency is down; a till cannot. A web order can be a row in a database; in Germany a store sale must be signed by a certified security device and carry a sequential number that makes gaps detectable. That is why "just use the platform's POS" is sometimes right and sometimes not available at all.

Figure 1: *A store is a second commerce system, not a channel: like the web shop it creates orders, holds stock, knows customers and takes money, and it must also keep selling offline and follow fiscal rules.*

The offline rule

Vendors document what their POS does without a connection. Shopify's help pages (n.d., accessed November 2026) say its POS needs the internet to sync with the admin, e-mail receipts and capture card payments; cash sales work offline, and card payments taken offline are "captured after your device reconnects to the internet", with a merchant-set daily limit per device, a declined payment that surfaces only after reconnection, and advice to reconnect "ideally within 24 hours". Square's help page (n.d., accessed November 2026) goes further: offline payments "expire after 72 hours if you don't reconnect", and "You're responsible for any expired, declined, or disputed payments accepted while taking offline payments."

For the hours a store is offline, it is an independent system of record for sales and stock. Its transactions need locally generated identifiers that never collide; its stock changes arrive late and in a batch; the merchant, not the payment provider, carries the risk of a declined card. Any online promise made against that store's stock during those hours is a guess.

The fiscal rule

Fiscalization is a legal requirement that every sale be recorded in a way the tax authority can verify. Its form differs by country, and the differences decide which POS products

can be used where.

Country	Mandate for retail stores	Mechanism	Dates and source
Germany	Yes, every electronic cash-register-like system	A certified **technical security device** (TSE: security module, storage medium, uniform interface) signs each transaction; a sequential transaction number so that "gaps in transaction records are recognizable"; a receipt obligation; registration with the tax office within one month	§ 146a Fiscal Code; KassenSichV of 26 September 2017 (last amended 14 January 2026); registration via ELSTER (the tax administration's online portal) since 1 January 2025
Belgium	Not for shops; the "white cash register" (GKS) applies only to restaurant and catering services above €25,000 turnover	For hospitality: a certified register, a certified fiscal data module ("black box") and a VAT signing card; GKS 2.0 mandatory for new hospitality businesses from 1 July 2026	FPS Finance GKS site, 2026; records kept seven years
Italy	Yes	Electronic storage and telematic transmission of daily receipts to the tax agency	From 1 July 2019 (turnover above €400,000) and 1 January 2020 (all others), Agenzia delle Entrate release of 29 June 2019

Country	Mandate for retail stores	Mechanism	Dates and source
Netherlands	No certified register	Register data are part of the administration, kept seven years; a voluntary private quality mark	Belastingdienst brochure, n.d., accessed November 2026

The table is my compilation of the sources listed at the end. For Kestrel & Pine the point is specific: its five Dutch stores faced no certified-register mandate when the POS was chosen, and Antwerp's receipts follow Belgian VAT and seven-year record-keeping rules, not the hospitality-only black box. A first **German store** would need a POS with a certified TSE, registered with the tax office, and that requirement alone can rule a product out. A store opening in a new country is a POS decision as much as a logistics one.

How the Pieces Fit Together

The January 2027 landscape at Kestrel & Pine is the shape many Growing-stage merchants have when they open stores: a POS chosen for the stores, an ERP as the meeting point, and the web platform on the other side of a nightly copy.

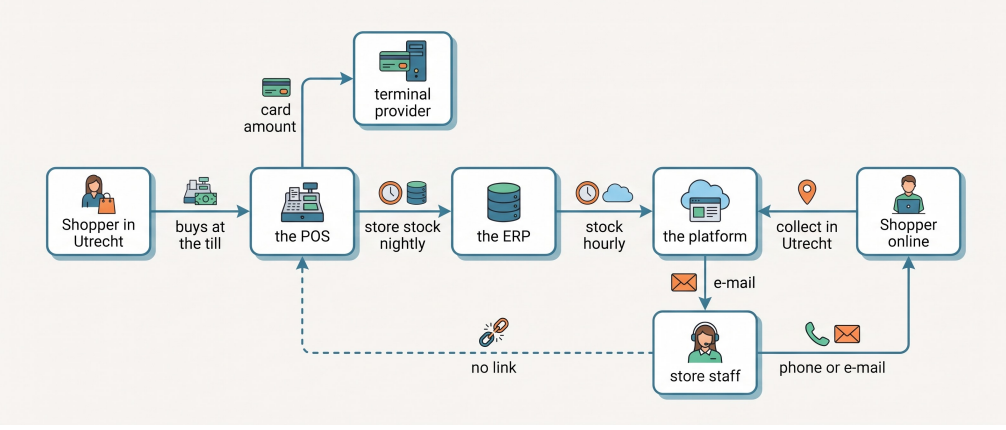

Figure 2: *Kestrel & Pine's stores in January 2027: a sale at the till reaches the platform the next day by way of the ERP, while a click-and-collect order reaches the store as an e-mail and never reaches the POS at all.*

The target shape is different in two places: store stock flows as events into one

availability calculation, and store orders of every kind pass through one order hub. The components and their responsibilities:

Component	Responsibility	Owns (system of record for)	At Kestrel & Pine
The POS	Ring up sales, take terminal payments, issue receipts, handle returns at the counter, count stock	Store transactions; store stock (W1)	The cloud POS from 2019
The terminal provider	Card acceptance in the store	The card transaction and its reference	Separate from the online PSP (the payment service provider for online payments)
The availability service	Compute what each channel may promise per location from stock events, with buffers	The available-to-promise number (what a channel may still sell: physical stock minus what is promised, minus buffers)	A module of the OMS, from Q4 2027
The OMS	Hold every channel's orders in one state model; route each order line to a location (**sourcing**); hand work to the warehouse or a store	The operational order record	Bought, Q4 2027; store receipts imported "later"

Component	Responsibility	Owns (system of record for)	At Kestrel & Pine
Store staff tooling	Show staff what to pick, confirm, mark ready and hand over; look up stock elsewhere; start a return	Nothing; a client of the OMS and the POS	E-mail and phone today
The customer record	Recognize the same person across till, web and newsletter	Customer identity	Four unlinked lists; One Customer, One Product: Identity and Master Data Across Systems

The four omnichannel services

The vocabulary first, because vendors use several names for each.

Service	Order placed	Stock used	Handed over	What it needs from the architecture
Click-and-collect (BOPIS, buy online pick up in store)	Online	Store stock, or warehouse stock sent to the store	At the counter	Store stock visible to availability; an order line routed to a store; staff tooling to confirm, hold and hand over; a notification when ready

Service	Order placed	Stock used	Handed over	What it needs from the architecture
Ship-from-store	Online	Store stock	Carrier collects at the store	Everything above plus labels, packaging and a carrier pickup at the store; routing rules that weigh store cost and capacity
Return-in-store	Online	None	Return accepted at the till	The till can find an online order, accept the item, trigger one refund through one path, and restock to the store or send it back
Endless aisle	In store	Warehouse or another store's stock	Carrier to the customer	The till can see availability elsewhere, create an order in the order hub, and take payment for goods that are not in the building

Each row is a promise made against stock somewhere else, and **all four are the same three problems in a different order**: knowing what stock is where (availability), deciding which location serves the order (sourcing), and giving the store the tools to

do its part (staff tooling). The requirements differ by service:

Flow	How fresh must store stock be	How consistent	How fast	What happens when the link is down
Store sale reduces online availability	Minutes, as an event; nightly is the Mirage	Eventually consistent is fine; the buffer absorbs the lag	Not latency-critical	The store keeps selling; availability holds the last known number minus a larger buffer
Click-and-collect promise	Minutes, plus a per-store buffer	The order must be visible to the store within minutes and confirmed by a person before "ready" is sent	The confirmation, not the promise, is the commitment	Offer collection "within two hours" rather than "now", or route to the warehouse
Ship-from-store routing	Minutes	The chosen store must accept the fulfillment order before the carrier label exists	Routing in seconds at order time	Fall back to the warehouse
Return-in-store	Order lookup in seconds	Exactly one refund, from one system	Seconds at the counter	Accept the item, record the return, refund when the link returns, never twice

Flow	How fresh must store stock be	How consistent	How fast	What happens when the link is down
Endless aisle	Seconds, from the shared availability number	The order is created in the hub, not re-keyed later	Seconds at the counter	Do not promise; take the customer's details and call back

The target flow for click-and-collect looks like this, with the OMS and the availability service in place:

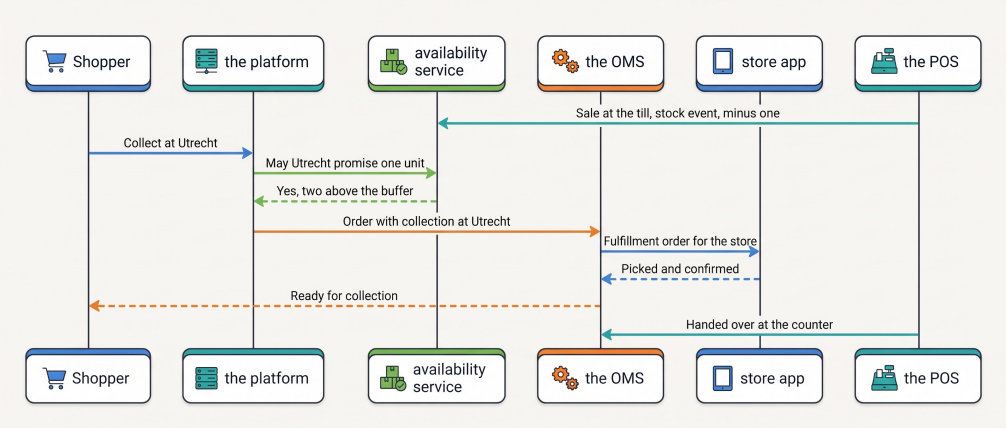

Figure 3: *Click-and-collect done with the joins in place: the till reports every sale as a stock event, the availability service promises only above a buffer, the order hub routes the work to the store, and a person's confirmation, not the stock number, is what turns the promise into "ready".*

Notice where the commitment sits. The storefront's "available for collection" is a **promise with a buffer**; the store's "picked and confirmed" is the **commitment**; the customer hears "ready" only after the second. That ordering is the whole design, and the Utrecht pilot lacked it.

What store stock is actually worth as a promise

The buffer and the confirmation are not caution for its own sake. Store inventory records are wrong far more often than warehouse records, and the research is old but unambiguous. DeHoratius and Raman, in *Management Science* in 2008, examined

"nearly 370,000 inventory records from 37 stores of one retailer" and found **65% to be inaccurate**; auditing reduced inaccuracy, store complexity increased it.

The data are from one US retailer in the 2000s, before RFID (radio-frequency identification: a tag on each item that a reader counts without line of sight), so take the mechanism rather than the number.

Auburn University's RFID Lab reported in 2018 (Project Zipper, eight brands, five retailers) that order audits using barcode data found inaccuracies "in 69% of cases" against "less than 0.01%" with RFID. Both sources are more than three years old; nothing newer with an open methodology turned up in this book's research.

The ways to make a store-stock promise honest, in rising order of cost:

1. **A per-store safety threshold**: never promise the last *N* units of a variant at a store. Cheap, loses some sales on slow movers, and is the first thing to do.
2. **A confirmation step**: the store confirms the pick before the customer is told to come. Costs a few minutes of staff time per order; removes almost all broken promises.
3. **Exception-driven cycle counting**: when a store cannot find an item the system said it had, that variant is counted that day, and the count corrects the record. Turns every miss into a data fix.
4. **RFID**: tags on every item and readers at the door and the counter. Expensive; the evidence above is why fashion retailers at Group stage do it anyway.

Kestrel & Pine's interim fix for Utrecht, until the availability service arrives, is the first two: a threshold of two units per variant per store and a staff confirmation before the "ready" message. It costs nothing to build and would have prevented most of the 37.

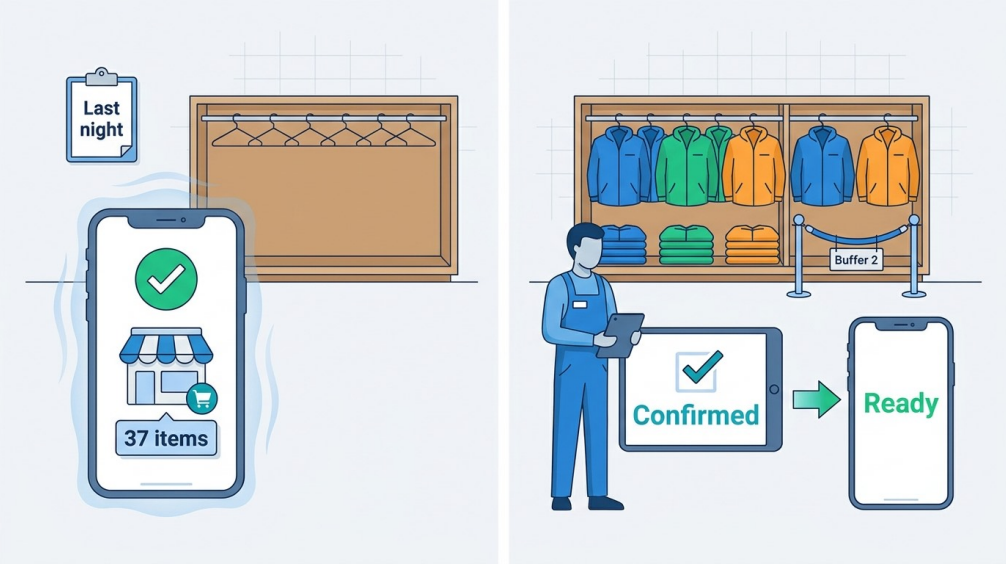

Figure 4: *The Store Stock Mirage and its interim fix: in December 2026 a nightly copy promised 37 items already sold; a buffer of two units and a staff confirmation before "ready" would have prevented most of them.*

The Choices

A merchant with stores has four realistic ways to connect them to the rest of the landscape. Each is a real product category, and each is right somewhere.

1. **The platform's own POS.** The web platform's vendor sells a till on the same back end: one catalog, one stock record per location, one order list, one customer list. Shopify's POS pages (n.d., accessed November 2026) describe "buy online, pickup in store, or buy in store, ship to customer", staff roles and "Returns and exchanges at any location"; POS Pro is €79 per location per month on its euro page as of November 2026.

The Dutch vendor Vendit makes the same promise from the POS side: a till "from the same database as your webshop", so that "if you sell the last item at the till, the article is immediately at zero in your webshop" (n.d., accessed November 2026, translated). The joins disappear.

The limits: the platform's POS must be available and fiscally certified in every country you have stores in, it inherits the platform's offline rules, and it couples the stores to the platform contract, so changing the platform later means changing the tills too.

2. **A specialist POS, integrated by events through the order hub.** The stores keep a POS built for stores (fiscal certification, offline operation, loyalty, counts), and three integrations connect it: stock events to the availability service, store receipts imported into the OMS as orders, and fulfillment orders from the OMS to a store app. This is

where Waypoint takes Kestrel & Pine: the best POS for the stores and the best platform for the web, at the cost of three contracts and the reconciliation jobs behind them.

3. **A specialist POS, integrated by nightly batch through the ERP**. The January 2027 state. It works until someone promises store stock outside the store; then it produces the Mirage.

4. **The ERP's POS**. Some mid-market ERPs include a till module, so store stock, store sales and the books are one record and the web platform is the odd one out. Attractive for a wholesaler with a few outlet stores; less so for a brand whose stores are a growing channel, because the ERP's till rarely has the customer and omnichannel features of a specialist POS (The Back Office: ERP Integration and Who Is the Hub).

	Platform's POS	**Specialist POS by events through the OMS**	**Specialist POS by nightly batch**	**ERP's POS**
Store stock visible online	Immediately, one record	Within minutes, as events	Next day	Same record as the ERP; the web gets a copy
Store orders in the order record	Native	Imported, with their fiscal identifiers	Only as ERP postings	Native to the ERP, not to the OMS
Fiscal coverage	Where the platform vendor supports it	Chosen per country	Chosen per country	Where the ERP supports it
Offline operation	The platform's rules and limits	The POS's rules	The POS's rules	The ERP's rules
Integrations to own	Few	Three contracts plus reconciliation	One batch job	Web-to-ERP only
Exit cost	The tills change with the platform	Each part replaceable	Low	The tills change with the ERP

	Platform's POS	Specialist POS by events through the OMS	Specialist POS by nightly batch	ERP's POS
Fits best	Starting and Growing; one country; web-first brands	Omnichannel; several countries; stores that matter	Stores with no omnichannel promises	Wholesalers with outlet stores

My rule: **choose by the number of countries and the weight of the stores, not by the vendor's demo**. A web-first brand with two stores in one country should take the platform's POS and enjoy the one record. A merchant whose stores earn a serious share of revenue, in more than one fiscal regime, should keep a store-grade POS and spend the money on the events and the order hub. Kestrel & Pine sits between: six stores in two countries, **12.5% of revenue**, one fiscal regime that is lenient and one that is lenient for shops. That is why W4 defers rather than decides: the 2028 evaluation will have an availability service and an OMS in place, and the question will have shrunk to "which till", not "how do stores join the architecture".

"Unified commerce", and who is speaking

You will hear **unified commerce** in every vendor meeting. Shopify's 2026 blog post defines it as connecting "sales channels, inventory, and customer data through a single back-end platform", against omnichannel, which "doesn't connect back-end systems". Adyen's point-of-sale documentation (n.d., accessed November 2026) says the same from the payments side: in-person payments "processed on the same platform as online payments", with shopper recognition across channels. No analyst definition could be fetched for this book; every definition above comes from a vendor with a platform to sell.

I use the term for the **goal state**: one record per order, per stock unit and per customer, visible to every channel in near real time. It can be reached with one platform (choice 1) or with an OMS and an identity layer over several systems (choice 2). When a vendor says "unified commerce", ask which they mean, and whether their definition would survive your store in Germany.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
1 — Starting	No store, or a pop-up on the platform's POS; stock is the back room; store sales are a line in the monthly accounts	Nothing beyond the platform's own POS; one location record	Store count and platform count disagree every month-end; staff keep a notebook of "web orders to pick up"
2 — Growing	One to three stores, often on a specialist POS chosen by the store manager; POS to accounting or ERP nightly; no omnichannel promises	A decision on which system owns store stock; a per-store threshold before any click-and-collect pilot; one customer key (loyalty id or e-mail) captured at the till	Click-and-collect run by e-mail and phone; the first Mirage; store managers who cannot say whether online refunds come off their sales
3 — Omnichannel	Several stores in more than one country, a visible share of revenue; stock events from the POS; an OMS that routes to stores; staff tooling; return-in-store	The availability service with store buffers; store receipts imported into the order hub with fiscal identifiers; one refund path; reconciliation of POS, OMS and ERP; a fiscal checklist per country	Store-stock promises without a confirmation step; ship-from-store launched before routing knows store capacity; a POS API deprecation nobody owns

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
4 — Group	Dozens to hundreds of stores across fiscal regimes; several POS products after acquisitions; RFID in fashion; stores as fulfillment nodes with their own cut-offs and carrier pickups	One fulfillment-order contract every POS and WMS honors; a registry of which store does what; accuracy targets; per-brand decisions on shared versus separate POS	A single POS forced on brands with different fiscal needs (One Size Fits None); store fulfillment that undercuts store sales because the economics were never modeled

The judgement: the store joins the architecture at the Growing stage, with the first omnichannel promise, and the cheapest moment to decide who owns store stock is before that promise, not after the first broken one. The POS itself rarely needs to change until the merchant opens stores in a country whose fiscal rules its POS cannot meet, or until the stores' weight justifies a single record. Kestrel & Pine has reached the first point with its joins and not the second with its till.

In Practice

Ship-from-store on someone else's marketplace: Zalando Connected Retail, 2021 to 2027. Zalando's Connected Retail program let physical shops list their store stock on Zalando and ship the parcels from the store; trade press reported more than 3,400 active stores in nine markets in early 2021. In April 2026 TextilWirtschaft reported a Zalando spokesperson confirming that the program ends by the end of June 2027, to bundle all partners in one central model. Two lessons transfer: ship-from-store works at scale when the store's stock feed and the return path are built properly, and programs retire. A merchant that exposed store stock through its own store-stock feed rather than a program-specific tool faces a connector change, not a redesign (Selling on Someone Else's Platform: Marketplaces and Channel Integration).

Store orders in one order model. Microsoft's documentation for its Business Central connector to Shopify (published 14 October 2026) notes that orders created through Shopify POS carry a staff identifier that the connector maps to a salesperson. Small detail, large point: on a platform with its own POS, a store sale and a web order travel to the ERP by the same road. A merchant with a separate POS builds that road itself,

and the import must carry the store's own identifiers (receipt number, fiscal transaction number, staff id) rather than re-create the sale as a web order.

Routing to stores. Adobe Commerce's inventory documentation (updated August 2026) ships two source-selection rules, a fixed priority and a distance rule that picks "the closest source". Shopify binds work to a location through a **fulfillment order** whose assigned location may change only while it is open, scheduled or on hold (n.d., accessed November 2026). Kibo's documentation (n.d., accessed November 2026) describes routing that returns a suggestion, logs why locations were rejected, "does not create shipments or deduct inventory", and lets a pickup order trigger a transfer shipment to the store. That is the ladder: simple rules in the platform, then an explainable routing decision separate from the deduction of stock, which is the OMS's job once stores, cost and capacity enter the rule (The Order Record: Lifecycle, State and the Order Management System).

Store payments and PCI scope. With **point-to-point encryption** (P2PE; the PCI Security Standards Council lists validated solutions by exact version), the card number is encrypted inside the terminal and decrypted only at the payment provider, so a **semi-integrated** terminal that receives an amount and returns a result keeps the POS out of the scope of PCI DSS (the Payment Card Industry Data Security Standard) (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers). Keep the terminal provider's transaction reference on the receipt; it later matches the payout file (Taking the Money: Payment Providers, Authorization, Capture and Refunds).

Adoption, a decade ago. The only open, dated adoption figures found are old and UK-weighted: a 2016 UPS survey of over 700 European retailers put stores used as online fulfillment centers at 27%; a 2018 Savills survey of UK stores found 47% offering click-and-collect, about one in ten in ship-from-store, and nearly one in three store managers in one category who "did not know whether online refunds were deducted from store sales". Treat the percentages as history; keep the last finding. "Whose sale is it" must be answered before the first store ships a web order.

Kestrel & Pine, worked: the Utrecht pilot, before and after

The numbers below are Kestrel & Pine illustrations; where they are not among Kestrel & Pine's recorded figures they are labeled as assumptions.

	December 2026 (the Mirage)	**Interim fix, from April 2027**	**Target, from Q4 2027**
Store stock seen by the platform	ERP copy, up to a day old	Same copy, minus a threshold of 2 units per variant *(assumption)*	POS stock events into the availability service, minutes old, per-store buffer
Promise shown online	"Available for collection today"	"Collection today, we confirm within two hours"	"Ready in two hours" after confirmation, computed per store
How the store learns of the order	E-mail from the platform	E-mail, with a reply template that confirms or declines	Fulfillment order in the store app from the OMS
Commitment to the customer	At order placement	At the store's confirmation	At the store's confirmation
Broken promises	**37 items** in the month	Illustrative target under 5 *(assumption)*	Measured per store as "promised but not found"
Store receipt for the handover	A POS sale at €0 keyed by hand *(local detail)*	Same	Handover recorded against the OMS order; the POS imports it with its receipt number

Two details deserve a sentence. The €0 till sale is how many pilots record a collection, and it is why the order exists three times: in the platform (paid), in an e-mail (picked) and in the POS (a zero sale); that is *Schrödinger's Order* with a store in it. And the interim promise is deliberately weaker than the original: a two-hour window kept every time beats "today" broken once in thirty.

When It Goes Wrong

Stores fail in ways online systems do not, because people and buildings are involved. The ones that matter most, with how to notice them and how to recover:

- **The till and the web sell the same last unit**. A store sale and an online reservation

race for the final unit before the store's event reaches availability. *Detection*: "promised but not found" per store, counted daily. *Recovery*: the buffer makes it rare and the confirmation step catches it before the customer travels; when it still happens, the OMS reroutes to the warehouse or cancels with a notification, and the variant is counted.

- **Offline session replay creates duplicates**. A store reconnects after three hours and uploads its sales; the upload is retried after a timeout, and the same receipts arrive twice. *Detection*: daily sales count against the receipt-number range. *Recovery*: every import is **idempotent** on the POS transaction number (receiving the same receipt twice has the same effect as once), which is also what Germany's fiscal transaction counter is for (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation).
- **The nightly job fails silently**. The POS-to-ERP copy stops on a Friday; by Monday the ERP's store stock is three days old and the platform has promised against it all weekend. *Detection*: a monitor that alarms when the copy is older than its schedule, not only when it errors (APIs, Events, Files and Middleware: How Commerce Systems Talk). *Recovery*: rerun and reconcile the weekend's orders by hand; the cure is events with a freshness check.
- **Return-in-store refunds twice**. A web order is returned at the Nijmegen counter; the till refunds to the card through the terminal provider, and the returns portal, seeing the item "received", refunds through the platform's API as well: the store's version of *The Double Refund* (212 returns, €19,400, January 2027). *Recovery*: one refund path per order, decided once; a return recorded at the till closes the portal request before anything is paid (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics).
- **The store charges a different price**. During the Price Drift of October 2026 the German storefront sold the Ridgewalker at €179 while the stores charged €199; the stores were right by accident, because the POS has its own hand-loaded price list. *Recovery*: one pricing source publishing to the POS as to the storefronts (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax).
- **The loyalty list and the web account never meet**. Eva Janssen is a POS loyalty customer and an online account with no shared identifier (the *Familiar Stranger* pattern). Return-in-store still works, because the till finds the order, not the person; store purchases simply never reach her online history. *Recovery*: a declared matching rule with a confidence level (One Customer, One Product: Identity and Master Data Across Systems).
- **A fiscal rule arrives with a store**. A German lease is signed before anyone checks whether the POS has a certified TSE (*The Long Fuse*). *Recovery*: store openings go on the regulatory calendar (Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation); a POS per country is acceptable, a store without a compliant till is not.

What to Take Away

Established.

- A store's POS is a system of record for store transactions and, for the hours it is offline, for store stock; vendors document offline capture, limits and merchant-borne risk (Shopify and Square help pages, accessed November 2026).
- Store inventory records are far less accurate than people assume: 65% inaccurate in the 2008 DeHoratius and Raman study; 69% of audited orders with barcode-data errors against under 0.01% with RFID in the 2018 Auburn study. The numbers are old; the mechanism is not.
- Fiscalization differs by country and constrains POS choice: Germany requires a certified security device and sequential transaction numbers (§ 146a AO, KassenSichV 2017); Italy requires telematic transmission (2019–2020); Belgium's certified register applies to hospitality only; the Netherlands requires seven-year records and no certified device.
- Platforms and order management systems separate routing from the deduction of stock and bind work to a location through a fulfillment order that can move only before work starts (Shopify, Kibo and Adobe Commerce documentation, 2026).

Depends on context.

- The platform's own POS is the right choice for web-first brands with a few stores in one lenient fiscal regime; a specialist POS integrated by events is the right choice once stores carry weight or span fiscal regimes. The line moves with the platform vendor's country coverage.
- Click-and-collect against store stock needs at least a threshold and a confirmation step; ship-from-store needs routing rules that know store capacity and cost; both can wait until an availability service exists, and both can run earlier with weaker promises.
- Whether store sales are imported into the OMS from day one or "later" (Kestrel & Pine's W2) depends on whether anyone needs a store receipt in the order hub: returns-in-store and customer history do; finance does not, because the ERP already has the nightly postings.

The author's synthesis.

- Fix what the store tells the other systems, and how fast, before changing the till: a store-stock event feed, a buffer and a confirmation step remove most broken promises, and they are the same work whichever POS you end up with.
- Promise only what a person has confirmed. The storefront's availability number is a forecast; the staff member's "picked" is the commitment; the customer should hear from the second, not the first.

Questions to Ask

1. Which system is the system of record for store stock, and how old is the copy that the storefront promises from?
2. What does the POS do when the connection drops, for how long, and who carries the risk of a declined card taken offline?
3. Which countries will we have stores in within three years, and does the POS meet each country's fiscal rules today?
4. When a click-and-collect order is placed, what exactly is the customer promised, and at what moment does a person confirm it?
5. How does a store receipt reach the order record: imported with its own identifiers, re-keyed, or not at all?
6. When a web order is returned at the till, how many systems can issue the refund?
7. Does the POS receive prices from the same source as the storefronts and the marketplaces?
8. If we replaced the platform in five years, would the tills have to change with it?

The store is where a web order can come back to the counter, and the return is where stock, money and the books have to agree again. The Way Back: Returns, Exchanges, Refunds and Reverse Logistics follows the jacket on its way back.

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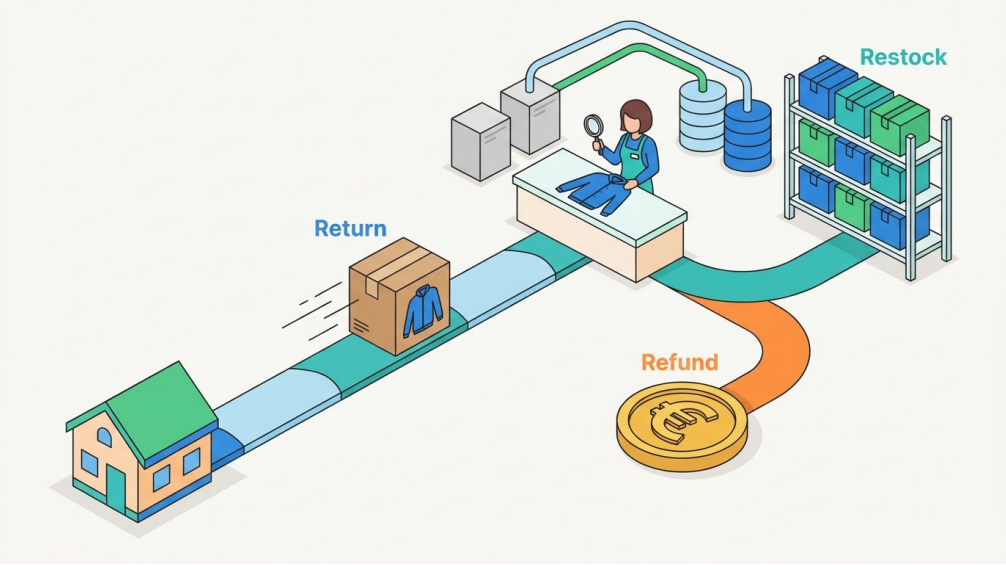
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

18. The Way Back: Returns, Exchanges, Refunds and Reverse Logistics



IN THIS CHAPTER, YOU WILL: Follow Eva's jacket back from Nijmegen to Duiven and learn how a return keeps the stock, the money and the books in agreement when one item in four comes back.

KEY POINTS:

- **A return is two records, not one.** The *return* (what comes back, why, and whether the customer gets a refund or an exchange) belongs to the order side; the *reverse fulfillment* (receiving, inspection, the decision what to do with the item) belongs to the warehouse. The warehouse's decision, not the customer's request, is the event that releases money and stock.
- **Money, stock and books move on different events, each exactly once.** The refund is issued by one system with the return id as its key; stock rises only on a "restocked" decision; the credit note follows the

refund. EU law sets the clocks: 14 days to withdraw, 14 days to refund, and the right to wait for the goods or proof of sending.
- **At a 24% item return rate the return path is a main road, not a side street.** Kestrel & Pine takes back 72,000 parcels a year; 28% of returns are exchanges, which are new orders held against a return; marketplace and store returns arrive as events the merchant did not start. Design for all of them, not for the happy path plus exceptions.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Boomerang** — refunds, restocking and credit notes are reconciled by hand after the return arrives, and sometimes twice, because no single event and no single writer were ever agreed.
- **Schrödinger's Order** — a customer asks "where is my refund?" and support checks the returns portal, the warehouse and the payment console, which disagree about whether the parcel has arrived.
- **The Phantom Stock** — a returned item is booked back into sellable stock before anyone has looked at it, or never booked back at all.

A customer sends back a jacket that does not fit and asks for the next size. The returns portal knows the request, the warehouse must inspect the original and find a replacement, and finance must record any adjustment. Marking the return complete in one system does not complete the others' work. The customer can still be waiting while the returned jacket or the money is counted incorrectly.

Kestrel & Pine, the fictional outdoor brand used here, receives almost one in four online items back. At that volume, disconnected decisions become a routine source of errors: two systems can refund the same return, or stock can be offered for sale before inspection. A coordinated return process records each step and authorizes the next, including whether the outcome is a replacement or a refund.

The chapter separates the return authorization (**RMA**, return merchandise authorization), the item's **disposition** or inspection outcome, the replacement order and the refund. Taking the Money: Payment Providers, Authorization, Capture and Refunds explains money movement, Pick, Pack, Ship: Warehouse Management and Fulfillment the physical work and Tax, Invoices and the Books: From Order to Ledger the credit note. Store and marketplace variants appear in Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment and Selling on Someone Else's Platform: Marketplaces and Channel Integration; Journey 2: The Jacket Comes Back traces a complete example.

Where we are in the order's journey. Eva receives her Ridgewalker jacket on 15 April 2027 and requests an M-to-L exchange on 19 April. Duiven receives and inspects it on 22 April, then dispatches L. A separate refund alternative returns €199 through iDEAL, the Dutch bank-transfer method. Keeping those alternatives distinct matters: an exchange is not automatically a cash refund.

Setting the Scene at Kestrel & Pine: One Item in Four

Kestrel & Pine received **72,000 return parcels** after shipping 336,000 parcels in 2026. Its online **item return rate was 24%**; **19% of orders** had at least one return. Partial returns and **bracketing**, ordering two sizes to keep one, explain why those measures differ.

28% of returns were exchanges and **61%** of returned items were resold as new. In the illustrative daily workload, two people process about **280 parcels**; after Black Friday, five staff the bench for three weeks.

Since 2025, a hosted returns portal has checked eligibility, collected reasons, offered exchange or refund, and issued carrier labels. After warehouse inspection, it can request a refund through the platform's API. It costs €650 monthly plus €0.95 per return.

Kestrel & Pine's commercial policy offers free returns within 14 days of delivery in all three countries. The statutory withdrawal rules are explained below.

The portal was bought to fix a returns process run by e-mail. It did, and it created the Double Refund. In January 2027 **212 returns** were refunded twice, **€19,400** in total, most of it recovered. Taking the Money: Payment Providers, Authorization, Capture and Refunds told the story from the money side; here is the returns side. Customers could see on PostNL's tracking that their parcel had reached Duiven. Some wrote to support the same day asking for their refund. The agent could see the original payment in the console of the PSP (payment service provider: the company that takes the shopper's money and settles it to the merchant) and no refund, because the portal refunds on inspection, which took two to five days. The agent refunded by hand. Two to five days later the portal refunded again. Nothing in either request named the return, so the PSP accepted both.

When Noor Visser, Head of Engineering since January, reviewed the incident in February, she drew three columns on the whiteboard, money, stock and books, and asked for each: **on which event does it move, and which system moves it?** Nobody could answer for all three. That is the chapter's question.

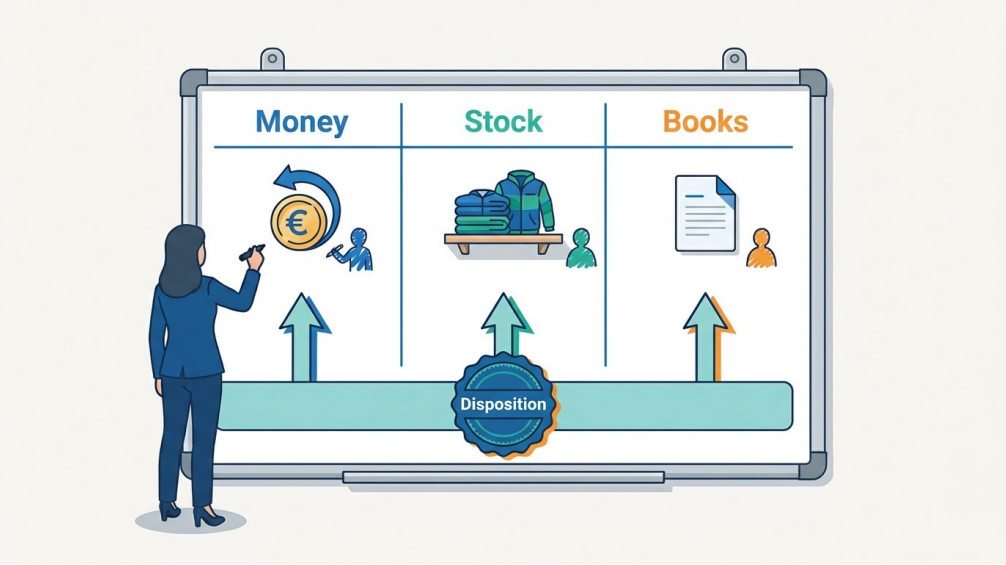


Figure 1: *Noor's question as a whiteboard: for money, stock and books, which event moves each and which one system moves it; the answer is the warehouse's disposition, releasing the refund, the restock and the credit note once each.*

What a Return Is, Step by Step

A return passes through nine steps, and each step has a record, an owner and a way to go wrong. The vocabulary is small and worth learning once.

Step	What happens, in plain words	Record	Who acts	What can go wrong
Request	The customer says which items are coming back and why	A **return request**, or **RMA** (return merchandise authorization: the merchant's record that this return of these items was approved)	The customer, in the portal or with an agent	Eligibility computed from a stale copy of the order
Eligibility	Is the item returnable, is it inside the window, who pays shipping	Rules evaluated against the order and the product	The portal or the order system	A sealed hygiene product accepted; a 15-day-old order refused that the law still covers
Label	A prepaid label or a drop-off code is issued	A return shipment with a tracking number	The portal via the carrier API	A label addressed to the old third-party warehouse in Tiel
Transit	The carrier scans the parcel on its way back	Carrier tracking events	The carrier	"Delivered to Duiven" read as "received"
Receiving	The warehouse opens the parcel against the RMA	A **reverse fulfillment** record: the work to process this return	The warehouse	A parcel with no RMA; an empty box
Inspection	Someone looks at the item and grades it	An inspection result per item	The warehouse	No check at all; a grade nobody can explain

Step	What happens, in plain words	Record	Who acts	What can go wrong
Disposition	The decision what happens to the item: back to stock, needs processing, discard, missing	A **disposition** per item	The warehouse	Restocked but not resellable; restocked to the wrong location
Settlement	Refund, exchange or store credit, and the credit note	A refund with the return id; an exchange order; a credit note	One system for money; the ERP (enterprise resource planning system, the books)	Two writers; a refund before receipt; the wrong amount after a promotion
Stock	The resellable unit becomes available again	A stock event from the WMS (warehouse manage-ment system)	The WMS to the availability service	An increment on request instead of on disposition

Three terms carry the chapter. The **RMA** is the merchant's side of the promise "you may send this back". The **disposition** is the warehouse's verdict, and it is the most important event in the whole flow, because it is the first moment the merchant knows what it actually got back. An **exchange** is a return paired with a new order for a different size, color or item, with any price difference settled as a balance. **Reverse logistics** is the trade's name for the whole backward operation: labels, carriers, receiving, grading, resale, recycling.

Return rates as an economic fact

The rate decides how much of this deserves engineering. The US National Retail Federation (NRF) with Happy Returns put US retail returns at **$890 billion** for 2024 (December 2024) and projected **$849.9 billion** for 2025, with an estimated online return

rate of **19.3%** of online sales and **9%** of all returns judged fraudulent (October 2025); its 2024 survey found that **76%** of consumers "consider free returns a key factor in deciding where to shop". Apparel is higher: Coresight's survey of 100 US apparel decision-makers (April 2023, a small survey and now three years old) estimated a **24.4%** return rate on online apparel orders, with size and fit the reason in **53%** of cases. Kestrel & Pine's 24% is apparel-typical. Definitions matter: NRF's rates are shares of sales value across all US retail; Coresight's is a share of online apparel orders; Kestrel & Pine quotes items and orders separately. Quote the basis with every number.

What a return costs is mostly invisible in the order system: the label, the handling at the bench, the grading, the repackaging, the markdown on the 39% not resold as new, and the capital tied up in a jacket that spends ten days in transit and inspection. Policy is the lever that moves the rate: better size guidance and photos, charging for returns on some items, and steering toward exchanges. All three need data the return record must capture: the reason, the size ordered against the size kept, the disposition.

How the Pieces Fit Together

Seen from outside, a return is a loop through six systems. The figure shows the target design at Kestrel & Pine after Waypoint, with the OMS (order management system, the one record of every channel's orders that Waypoint introduces in Q4 2027) as the order record; in April 2027 the platform still plays that role.

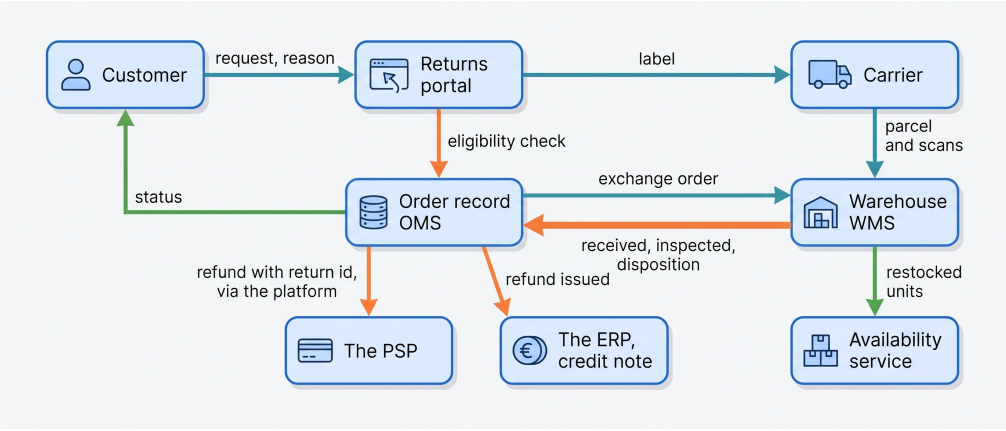


Figure 2: *A return loops through six systems; the warehouse's disposition is the one event that releases the refund, the restock and the credit note, and the order record is the only system that may ask for a refund.*

Two records: the return and the reverse fulfillment

The clearest public model of this split is Shopify's developer documentation (n.d., accessed November 2026). A **return** "represents the intent of a buyer to ship one or more items from an order back to a merchant"; it moves through `REQUESTED`, `OPEN`, `DECLINED`, `CLOSED` ("when all items have been processed and a restock decision has been made") and `CANCELED`.

Separately, a **reverse fulfillment order** "represents the work that's required to process a return", and a **reverse delivery** is "a set of items to be packaged together and sent back to the merchant". The dispositions on the reverse fulfillment order are `RESTOCKED`, `PROCESSING_REQUIRED` ("An item that requires further processing before being restocked or discarded"), `NOT_RESTOCKED` and `MISSING` ("An item that was expected but absent").

commercetools' order API (n.d., accessed November 2026) keeps the same two concerns as two states per return item: a shipment state that starts at `Advised` or `Returned` and may move to `BackInStock` or `Unusable`, and a payment state that starts at `NonRefundable` or `Initial` and moves to `Refunded` or `NotRefunded`.

The split is the pattern to teach. The first record is about intent and money and belongs to the order side; the second is about work and stock and belongs to the warehouse. They meet at the disposition.

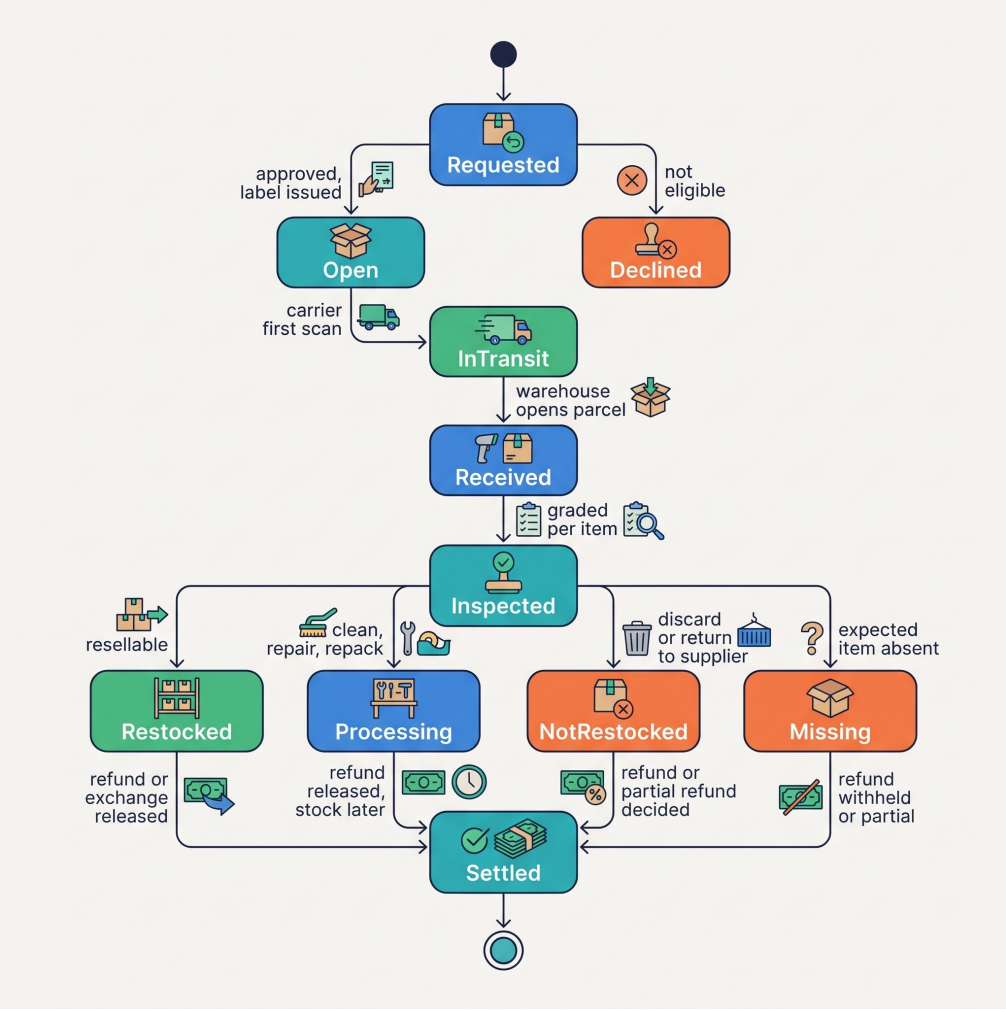


Figure 3: *The return's lifecycle: the request and the label are the customer's side, receiving and inspection are the warehouse's, and settlement waits for a disposition.*

Requirements per flow

Flow	How fresh	How consistent	How fast	If the other side is down
Eligibility check (portal reads the order)	The order as it is now, including earlier returns on the same line	Must see the delivery date and the delivery option chosen	Seconds, the customer is waiting	Refuse politely and retry; never guess
Received and disposition (WMS to order record)	The moment the bench scans the RMA	One event per return item, idempotent by return id and line	Minutes *(assumption)*	Queue; the refund waits with it
Refund (order record to PSP)	On disposition, inside the legal 14 days	Exactly one per return, keyed by return id	Hours *(assumption)*	Queue; check the PSP's refund list before any retry
Restock (WMS to availability)	On a restocked disposition only	Never on request or carrier scan	Minutes *(assumption)*	A weekly reconciliation catches a lost event
Credit note (order record to ERP)	After the refund is confirmed	One credit note per refund, at the allocated line price	Daily batch is enough	Finance waits
Status to the customer	On each state change	The same state the agent sees	Minutes	The status page shows the last known state

The Law: Fourteen Days, Three Times Over

For a consumer buying at a distance in the EU, the right to return is not the merchant's policy; it is law, and the architecture has to compute its clocks. The **Consumer Rights Directive**, Directive 2011/83/EU of 25 October 2011, published in the Official Journal on 22 November 2011, applies to contracts concluded from **13 June 2014**. Its text, as consolidated by the Publications Office, says:

- **Article 9(1):** the consumer "shall have a period of 14 days to withdraw from a distance or off-premises contract, without giving any reason, and without incurring any costs other than those provided for in Article 13(2) and Article 14". For goods the period runs from the day the consumer "acquires physical possession of the goods", and for several goods delivered separately, from possession "of the last good" (Article 9(2)(b)).
- **Article 10:** if the trader did not inform the consumer of the right, the period extends to twelve months.
- **Article 13(1):** the trader "shall reimburse all payments received from the consumer, including, if applicable, the costs of delivery without undue delay and in any event not later than 14 days from the day on which he is informed of the consumer's decision to withdraw", "using the same means of payment as the consumer used". **Article 13(2):** the trader need not refund extra costs "if the consumer has expressly opted for a type of delivery other than the least expensive type of standard delivery". **Article 13(3):** the trader "may withhold the reimbursement until he has received the goods back, or until the consumer has supplied evidence of having sent back the goods, whichever is the earliest".
- **Article 14(1):** the consumer must send the goods back "not later than 14 days from the day on which he has communicated his decision to withdraw" and "shall only bear the direct cost of returning the goods unless the trader has agreed to bear them or the trader failed to inform the consumer that the consumer has to bear them". **Article 14(2):** the consumer is liable only for diminished value "resulting from the handling of the goods other than what is necessary to establish the nature, characteristics and functioning of the goods".
- **Article 6(1)(h) and (i):** before the purchase, the trader must state the conditions and model form for withdrawal and that the consumer bears the return cost where that applies.
- **Article 16:** no withdrawal right for, among others, goods "made to the consumer's specifications or clearly personalised", goods "liable to deteriorate or expire rapidly", and "sealed goods which are not suitable for return due to health protection or hygiene reasons and were unsealed after delivery".

Read as requirements, the directive asks for four dates per order line: the delivery date (possession), the withdrawal notice date, the goods-sent evidence date and the refund date; the three 14-day clocks run from the first three. It asks the order to keep the delivery option chosen at checkout and its standard-rate equivalent, because the refund amount depends on them. It asks for product attributes (personalized, perishable, sealed hygiene) that eligibility rules can read, which is product-information work (What Are We Selling? Products, Variants and the Catalog). And it makes Article 13(3) the legal basis for Kestrel & Pine's "refund on inspection" policy: the trader may wait for the goods, and then has the balance of its 14 days from the notice to pay. The wider regulatory calendar, including the price-reduction rules that decide what a refunded promotion is worth,

is in Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation.

The Choices

Four decisions shape a returns architecture; each has a few realistic answers.

1. **Who runs the customer's side**. *Manual*: an e-mail address and a refund button in the platform's admin; the Starting-stage answer. *The platform's own returns*: the platform models the return and the reverse fulfillment, as Shopify does, and may issue labels.

A returns portal: a specialist SaaS product. Loop (n.d., accessed November 2026) was "Originally built for Shopify", steers shoppers "toward exchanges or store credit instead of refunds" and claims "5,000+ brands"; Narvar sells returns beside tracking and delivery-date promises and claims "1,500+ global brands"; Returnista, from Amsterdam, offers labels, exchanges "for the right size", store credit, rules-based automation and an API with webhooks for "1,000+ brands in Europe"; Happy Returns, "a UPS company", runs "10,000 Return Bar locations" for box-free drop-off and consolidates returns before they reach the merchant. Multi-carrier shipping platforms bundle the same portal with their labels.

The OMS: order management products such as Manhattan and Fluent list returns orchestration and cross-channel returns as modules (The Order Record: Lifecycle, State and the Order Management System). The vendor claims are undated and will change.

2. **Which event releases the refund**. This is the decision that caused the Double Refund, and it is a policy choice with a legal floor.

Trigger	Customer waits	Merchant's risk	Legal position (Directive 2011/83/EU)	Fits
On request	Not at all	Pays for items that never come back or come back worn	Allowed; the merchant gives up Article 13(3)	Low-value goods; trusted repeat customers

Trigger	Customer waits	Merchant's risk	Legal position (Directive 2011/83/EU)	Fits
On carrier scan (first scan or "delivered to warehouse")	A day or two	Empty boxes, wrong items, worn goods; the 9% fraud share	Allowed; "evidence of having sent back"	Fast-fashion scale with fraud scoring
On receipt	Transit plus a day	Wrong or damaged items refunded before anyone looks	Allowed; goods received	Standardized goods with little grading
On inspection	Transit plus grading, two to five days at Duiven	Lowest; the refund matches the disposition	Allowed; the merchant then has the balance of 14 days from the notice to pay	Apparel and equipment, Kestrel & Pine's choice

3. **Who issues the refund**. The portal through the platform's API (Kestrel & Pine today), the OMS through the same platform refund API once it holds the return records (after Waypoint, Q4 2027), or the ERP as part of the credit note. Any of the three works; **two of them at once is the Boomerang**. Taking the Money: Payment Providers, Authorization, Capture and Refunds gave the money-side rules: one writer, the return id as idempotency key, and a daily comparison of the PSP's refunds with the platform's.

4. **How exchanges work**. *Exchange as a held order*: a new order for the replacement is created at request time, reserving stock, and ships when the return's disposition is resellable; Kestrel & Pine's model, and the right one where the method cannot hold money, as iDEAL cannot. *Instant exchange*: the replacement ships immediately and the merchant holds the customer's card for the original price until the return arrives, which needs a card, an authorization that survives the transit window and a merchant-initiated transaction if it does not (Taking the Money: Payment Providers, Authorization, Capture and Refunds). *Exchange with a balance*: a different item at a different price, settled as a partial refund or an extra charge. Shopify's model lets the merchant "Release the exchange and collect a balance". Loop's and Returnista's pitch, steering to exchanges

and store credit, reflects the economics: an exchange keeps the revenue and costs one more parcel.

Choice	Manual	Platform's own returns	Returns portal	OMS returns module
Who computes eligibility	A person	The platform, from its order	The portal, from a copy of the order	The OMS, from the order record
Labels and carriers	By hand	Platform's carriers	Many carriers, drop-off points	Via the shipping service
Exchanges	By hand	Basic	Strong, with balances and store credit	Strong, cross-channel
Who may refund	A person in the admin	The platform	The portal through the platform's API	The OMS
Marketplace and store returns	By hand	Mostly outside	Some connectors	Designed for it
Fits	Starting	Starting to Growing on one platform	Growing to Omnichannel	Omnichannel to Group

My rule: **choose the refund trigger and the refund writer before choosing the product**, write both into the integration contract, and let the warehouse's disposition be the trigger unless the goods are cheap enough that fraud costs less than waiting. At Kestrel & Pine's rate and prices, inspection is the only defensible trigger, and the €199 jacket that comes back worn is the reason.

Returns From Marketplaces and Stores

A return the merchant did not start still has to reconcile. On Amazon, for seller-fulfilled orders, Amazon's seller guidance (April 2025) says most items "can be returned within 30 days of the order's 'estimated delivery date' "; US return requests covered by Amazon's policy are authorized automatically and prepaid labels are issued through Amazon's

shipping; sellers should "respond to requests within 24 hours" when their authorization is needed and issue refunds "within two days of receiving returns, or Amazon may issue an automatic refund"; a claim under the A-to-z Guarantee can be charged to the seller.

For orders fulfilled by the marketplace's own program, the marketplace receives and grades the return and charges the seller a returns-processing fee; Zalando's logistics arm ZEOS described "around 20 returns centres" in its November 2024 announcement.

The merchant's return record must therefore accept RMAs it did not create and refunds it did not issue, keyed by the marketplace's return id. Otherwise support can issue a second refund or finance can miss the first one. The 24-hour and two-day clocks also become deadlines the integration must expose, so staff can act before the marketplace acts for them (Selling on Someone Else's Platform: Marketplaces and Channel Integration).

Store returns are the mirror case. A web order brought to the Nijmegen counter is returned at the till: the till must find the online order, accept the item, record the return and restock to the store or send it to Duiven, and the portal's request, if one exists, must be closed before anything is paid. Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment sets that rule; until the POS (the stores' point-of-sale system) and the platform are linked, Kestrel & Pine's stores refund web orders only after the portal request is closed by a support agent, by hand *(illustrative)*.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Signs the stage is ending
1 — Starting (under about €2M)	Returns by e-mail; a refund button in the platform; the third-party warehouse or the back room restocks by hand; refund on receipt	The withdrawal information in the terms and on the product page (Article 6); one person who owns refunds	Refunds take longer than the legal 14 days in busy weeks; "where is my refund" is the top ticket

Stage	What is typical	What becomes necessary	Signs the stage is ending
2 — Growing (about €2M–€20M)	A returns portal with labels and reasons; refund on inspection; exchanges as held orders; the ERP books credit notes from an export	The refund writer and key decided once; "received" and "disposition" as events from the warehouse; a return-aging report against the 14-day clock	Two systems can refund (the Double Refund); returns restocked without grading; the first marketplace return with no record
3 — Omnichannel (about €20M–€200M)	Returns from storefronts, marketplaces and stores into one record; dispositions feeding availability; returns analytics by reason, size and product; a returns bench that scales for peak	An order hub that accepts external RMAs and external refunds; one refund path per channel; a daily refund reconciliation; returns in the Black Friday plan	Support checks three systems for one return; marketplace refunds counted twice in the books; the bench backed up for weeks in December
4 — Group (over about €200M, or several brands)	Shared reverse logistics (returns centers, consolidation, resale channels) with dispositions per brand; return policy as a per-brand setting on shared infrastructure; return fraud scoring	A returns service with per-brand rules; cross-brand customer return history (One Customer, One Product: Identity and Master Data Across Systems); negotiated carrier return networks	Each brand running its own portal and carrier contracts; dispositions defined differently per warehouse

The judgement: the return's *shape* (request, transit, inspection, disposition, settlement)

does not change with scale, and neither does the law. What changes is the number of channels a return can come from and the number of systems that could act on it. Stage 2 is where the refund trigger and writer must be fixed, because it is the stage at which a portal, a warehouse and an ERP first all hold a copy of the return.

In Practice

Public examples. Shopify's return and reverse-fulfillment model, with its four dispositions, is the most complete public description of the two-record pattern (n.d., accessed November 2026). commercetools' per-item shipment and payment states show the same separation in a headless platform. Amazon's seller rules (April 2025) show a marketplace owning the policy, the authorization and sometimes the refund. NRF's 2024 and 2025 reports put numbers on why it matters: free returns as a deciding factor for 76% of consumers, and 9% of returns fraudulent.

Kestrel & Pine: Eva's exchange, 19 to 22 April 2027. The table says what each system knew and did; the days are canonical, the hours are illustrations.

When	Event	System	Record
Mon 19 Apr, 20:40	Eva opens the portal from her order e-mail, selects the jacket, reason "too small", exchange for size L	The returns portal	Return R-20270419-1, status requested, 4 days after delivery, inside the 14-day window
Mon 19 Apr, 20:41	The portal checks eligibility against order #1072455 (delivered 15 April, standard delivery, apparel, no earlier return on the line) and issues a prepaid PostNL label	The portal, the platform	Return open; a return shipment with a tracking number

When	Event	System	Record
Mon 19 Apr, 20:41	The portal asks the platform to create the exchange order for size L, held until the return is graded	The platform	Order #1072455-X1 for one jacket, women's L, moss, €0 balance; one unit of L committed
Tue 20 Apr	Eva drops the parcel at a PostNL point; first scan	PostNL, the shipping service	Return in transit
Thu 22 Apr, 10:15	The bench scans the RMA, opens the parcel, inspects the jacket	The WMS	Reverse fulfillment received; inspected; disposition restocked
Thu 22 Apr, 10:16	The WMS reports the disposition; the platform closes the return and releases the held order	The WMS, the integration platform, the platform	Return closed; exchange order released; stock of M moss plus one in the WMS
Thu 22 Apr, 13:00 wave	Size L is picked, packed and labeled	The WMS	Exchange shipped; Eva's shipping confirmation
Night of 22 Apr	The ERP receives the return, the credit note for the size M invoice line and the new invoice for size L, netting to zero	The ERP	Books agree (Tax, Invoices and the Books: From Order to Ledger)

The refund variant. Had Eva chosen a refund, step three would not exist. On the disposition at 10:16 on 22 April the portal would have asked the platform to refund €199 with return id R-20270419-1 as the idempotency key; the platform would have instructed the PSP; the PSP would have sent a new iDEAL transfer to Eva's account and reported it by webhook; the ERP would have booked a €199 credit note. Eva's notice was 19 April,

so the legal deadline was 3 May; the refund went out on day three. Had support, on 21 April, seen "delivered to Duiven" on the tracking and refunded in the PSP console, that would have been the Double Refund; since February 2027 the console is read-only for support and an agent opens the return in the portal instead, where the pending refund and its trigger are visible.

The amount. A refund is the price the customer paid for that line, after the discounts that applied to it. The promotions chapter records what happens otherwise: in November 2026, 190 returned jackets were refunded at €159.20 instead of €143.28 because the order-level discount had not been allocated to lines in the export. Since Waypoint's W5 the order export to the returns portal, the WMS and the ERP carries the allocated net price per line (Discounts, Coupons and Loyalty: Promotions Without Chaos).

When It Goes Wrong

The double refund. Two writers, no key. Detection: the daily comparison of the PSP's refunds with the platform's by order and amount; two for one return is a hit. Recovery: ask for the money back; block the second writer. Prevention: one writer, the return id as key, read-only console access, and a portal the agent can see.

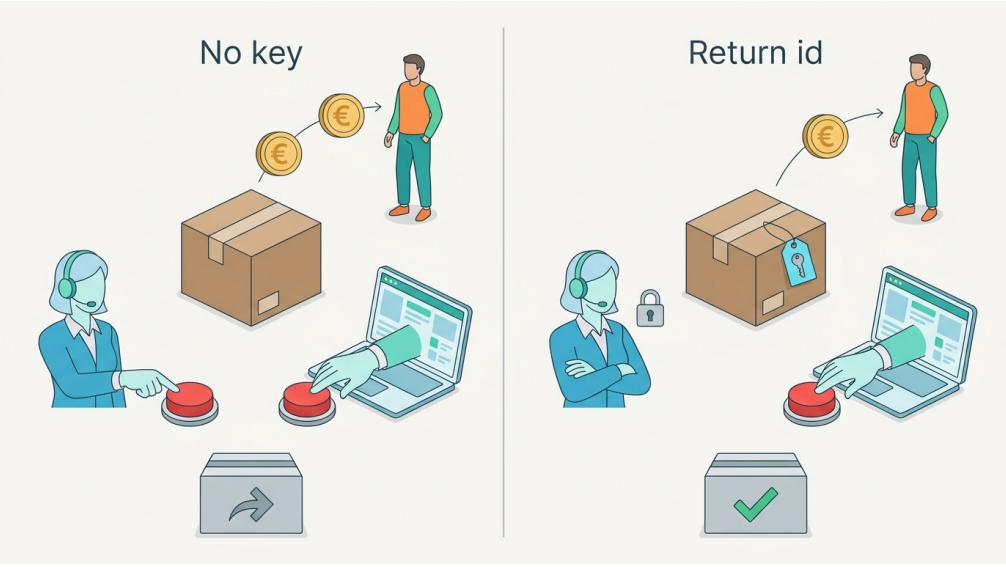


Figure 4: *The Double Refund in one picture: two writers and no key let the payment provider accept two refunds for one return; one writer using the return id as its key lets it accept exactly one.*

The refund without a return. Refund on request or on scan, and the parcel never arrives or holds a different item. Detection: refunds older than N days with no "received" event;

dispositions of `MISSING` or `NOT_RESTOCKED` against refunded returns. Recovery: a customer conversation and, at scale, a fraud score per customer. Prevention: refund on inspection where the goods are worth it.

The phantom restock. Stock is incremented on the request, or on the carrier scan, or on receipt before grading, and a worn jacket is sold again. Detection: the weekly WMS-to-availability reconciliation; complaints about "new" items. Prevention: stock moves only on a `RESTOCKED` disposition from the WMS, which is the single writer of on-hand stock (Is It in Stock? Inventory, Availability and the Promise You Can Keep, Waypoint W1).

The return with no record. A parcel arrives at Duiven with no RMA: a marketplace-authorized return the connector did not import, or a customer who used an old label. Detection: a "no RMA" queue at the bench with a daily count. Recovery: match by tracking number or order number, create the RMA after the fact, settle under the channel's rules. Prevention: every marketplace return event acknowledged into the order record with the marketplace's return id (Selling on Someone Else's Platform: Marketplaces and Channel Integration).

The exchange that ships before the return exists. An instant exchange, or a held order released by a "delivered to warehouse" scan rather than a disposition; the return never comes. Detection: exchange orders shipped with the paired return not closed after N days. Prevention: release on disposition; for instant exchanges, hold the card and know the window.

The missed deadline. The refund is issued on day sixteen because the bench was three weeks behind in December. Detection: a return-aging report against the Article 13 clock, by notice date. Recovery: refund now, apologize, and staff the bench for peak. Prevention: returns in the Black Friday plan (Journey 4: Black Friday, Hour by Hour).

The three-system status. The customer asks "where is my refund?"; the portal says open, the WMS says received, the PSP says nothing. Detection and recovery: one status, built from the same events, shown to the customer and the agent alike (Where Is My Order? Notifications, Service and the Customer Record).

What to Take Away

Established.

- A return is two records: the return (intent and money) and the reverse fulfillment (work and stock), with the disposition as the event that joins them (Shopify's and commercetools' public data models, n.d., accessed November 2026).
- EU consumers may withdraw within 14 days of receiving the goods, the trader must refund within 14 days of the notice by the same means, and may withhold the refund until the goods or proof of sending arrive (Directive 2011/83/EU, Articles 9, 13 and 14, applicable since 13 June 2014).

- Return rates in apparel run around a quarter of items, free returns drive where consumers shop, and a measurable share of returns is fraudulent (NRF 2024 and 2025; Coresight 2023), so the return path is a first-class flow.
- A marketplace owns the return policy, the authorization and sometimes the refund for orders sold through it, with clocks the seller's integration must meet (Amazon seller guidance, April 2025).

Depends on context.

- Refund on inspection is right for apparel and equipment at Kestrel & Pine's prices; refund on carrier scan is right for low-value goods at high volume with fraud scoring; refund on request is right only where trust is worth more than the goods.
- A returns portal is right from the Growing stage when the platform's own returns lack labels, exchanges or carriers; an OMS returns module is right once returns come from three channels and the OMS is the order record.
- Exchanges as held orders are right where the payment method cannot hold money or the goods are expensive; instant exchanges are right where cards dominate and a slow exchange loses the sale.

The author's synthesis.

- I decide three things before buying any returns product: which event releases the money, which single system may issue the refund, and that stock rises only on a restocked disposition; the product is then a matter of labels, carriers and screens.
- I treat the 14-day clocks as data requirements, not as legal footnotes: four dates per line, the delivery option kept, and an aging report that someone reads every week.

Questions to Ask

1. On which event does the refund go out: request, carrier scan, receipt or inspection, and who decided it?
2. Which single system may ask the PSP for a refund, and what key does the request carry?
3. What does the warehouse send back when a return arrives, and does the restock wait for a disposition?
4. Which four dates does the order keep per line, and can the system compute the 14-day clocks from them?
5. Does the eligibility rule know the delivery option chosen and the product's exemption attributes?
6. How does a marketplace-authorized return, or a store return of a web order, enter the return record?

7. How is an exchange modeled, and what releases the replacement?
8. What is the refund amount on a line that carried an order-level discount?
9. How many returns are older than ten days without a refund, and who sees that list?

The item is back on the shelf, the money is back in Eva's account or a size L is on its way, and the books agree. What Eva saw of all this was a few messages: your return is registered, your parcel is on its way, we have received it, your exchange has shipped. Those messages, the status page behind them and the agent who answers when they are not enough are the subject of Where Is My Order? Notifications, Service and the Customer Record.

Sources and Evidence

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Standards, regulation and official documentation

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Research and practitioner sources

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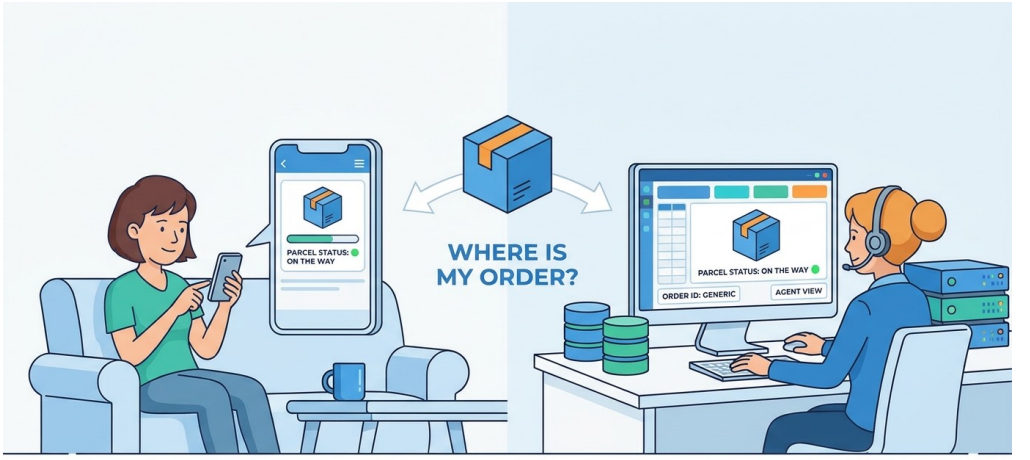
- The True Cost of Apparel Returns, Coresight Research, 2023-04-04 — the online apparel return rate and the size-and-fit reason (small survey; evidence age).
- Payments for Finance Architects: Eleven Reasons the Numbers Disagree in RECONCILED — refunds as new payments and the reconciliation that follows; Scams, Fakes and Account Takeovers: Fraud Prevention for Listings in LISTED — scoring returns abuse at scale.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

19. Where Is My Order? Notifications, Service and the Customer Record



IN THIS CHAPTER, YOU WILL: Learn how order events become notifications, a status page and an agent's view, why the three must share one status, and how to read service load as a signal.

KEY POINTS:

- **Notifications, the status page and the agent's screen are three views of one status, and they must be built from the same events.** The platform emits "confirmed", "canceled" and "refunded"; the warehouse emits "shipped" with a tracking number; the carrier emits "out for delivery" and "delivered". When each view reads a different system, the customer and the agent see different orders.
- **An order has several status machines, not one.** Money, goods and returns each move on their own; a single "status" string collapses them

and loses what the customer is asking about. Kestrel & Pine's Three-Status Order was a two-parcel order whose first parcel turned the whole order "Delivered".

- **"Where is my order" is a measurement, not a cost of doing business**. The one vendor dataset that states its basis puts it at about a fifth of tickets; measured per shipped order, it rises when tracking events go missing, promises slip or shipments split without explanation. Self-service that is gated on the fulfillment state removes the contacts that need no judgement.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Schrödinger's Order** — the customer, the status page and the agent each read a different system and get a different answer.
- **The Message in a Bottle** — a shipping notification sent once, from a label printed rather than a parcel handed over, is trusted by the customer and wrong.
- **The Familiar Stranger** — the agent searches by e-mail and finds the online account, not the store purchases, the newsletter consent or the open return.

A customer receives a shipping email, checks a page that says 'Delivered' and then hears from support that the order is still being picked. The customer has three answers and still does not know when to expect the parcel. Conflicting messages create extra contacts and weaken confidence in promises that may otherwise have been kept.

The disagreement starts with facts held in different places. The platform knows the order and payment; the warehouse knows the shipment; the carrier knows delivery; the returns process knows what came back. Notifications, the status page and support need a shared interpretation of those facts, including partial shipments, so they can give the same useful answer.

This chapter connects three views: **transactional notifications**, messages about an actual order; the customer's **order status page**; and the agent's **service view**. They draw on The Order Record: Lifecycle, State and the Order Management System, Getting It There: Carriers, Labels, Tracking and the Delivery Promise and The Way Back: Returns, Exchanges, Refunds and Reverse Logistics. One Customer, One Product: Identity and Master Data Across Systems explains how to recognize the same customer across channels.

Where we are in the order's journey. Eva buys a jacket from Kestrel & Pine, the fictional outdoor brand. We follow the messages from order #1072455's confirmation on 13 April 2027 to her replacement jacket's dispatch on 22 April, the page she checks between messages and the record support needs if she contacts them.

Setting the Scene at Kestrel & Pine: Eleven Tickets, Six Days, One Order

Kestrel & Pine's **Three-Status Order**, #1052877, produced **11 support tickets over six days** in December 2026. The platform said *Delivered*, the warehouse management system (WMS) said *In picking*, and the enterprise resource planning system (ERP) said *Not invoiced*. The Order Record: Lifecycle, State and the Order Management System explains the shared-record problem. Here we examine its cost to the customer and support team.

The €137 order contained a €119 fleece and €18 socks. The fleece shipped on 7 December; the socks awaited replenishment. The integration treated the first shipment as fulfillment of the whole order. When PostNL delivered the fleece on 8 December, the page and email said *Delivered*.

The WMS kept the order *In picking* while the socks waited. The ERP could not invoice until the WMS sent "order complete". **Nobody told the customer to expect two parcels**. These details are the fictional reconstruction used throughout the book.

Day	Channel	What the customer asked	What the agent could see	What the agent did
Tue 8 Dec, evening	E-mail	"Delivered, but the socks are not in the box."	Platform: Delivered, one tracking number	Asked for a photo of the packing slip
Wed 9 Dec	E-mail	Photo sent; "where are my socks?"	Same	Checked PostNL tracking: delivered; replied that it was delivered
Wed 9 Dec	Chat	"I know it was delivered, the socks were not in it."	Same	Opened a "missing item" case; said someone would check with the warehouse

Day	Channel	What the customer asked	What the agent could see	What the agent did
Thu 10 Dec	E-mail	"Any news?"	Same; the warehouse question sat in an inbox	Replied "still checking"
Thu 10 Dec	Phone	Asked for a refund of the socks	Platform: paid €137; refund possible	Offered the refund; the customer said she wanted the socks
Fri 11 Dec	E-mail	"Status still says Delivered. Is this a scam?"	A WMS reply by e-mail: order "In picking", socks on replenishment	Explained that the socks would ship; could not say when
Fri 11 Dec	Chat	"Why does the page say Delivered if it is being picked?"	Both answers, in two tabs	Apologized; could not change the status page
Sat 12 Dec	E-mail	Asked whether she had been charged for both	Platform: paid €137; ERP not visible to support	Asked finance by e-mail on Monday
Sun 13 Dec	E-mail	Asked again	Nothing new	No reply until Monday
Mon 14 Dec	Phone	"Just tell me one thing that is true."	Socks shipped that morning; new tracking number	Gave the second tracking number by phone
Mon 14 Dec	E-mail	Asked for the tracking number in writing	Same	Sent it, with a €10 voucher *(local detail)*

The useful answer was simple: **the order ships in two parcels, and the second leaves on the 14th**. No system held that answer. Agents checked five tabs: platform, warehouse, carrier tracking, payment provider and returns portal. Finance alone could

see the ERP.

Waypoint, Kestrel & Pine's improvement program, addresses this with a purchased **order management system (OMS)** by Q4 2027 under decision W2, and order status through an API under W8. The goal is one explainable record for customers and agents.

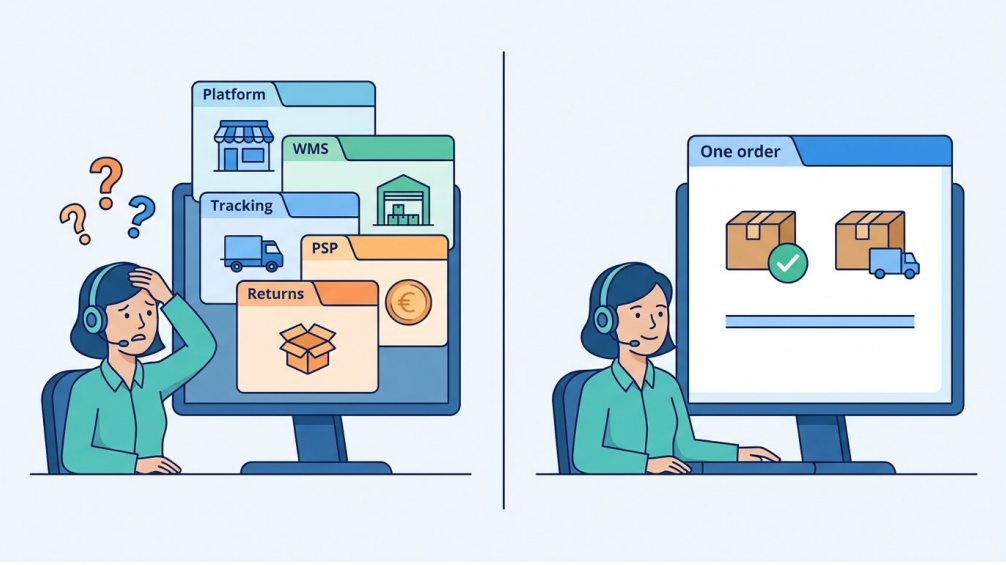


Figure 1: *The five tabs: each system told the agent a different truth about one order, so the message that would have ended the eleven tickets, two parcels, existed on no screen; a service view over one order record shows it.*

What the Customer Needs to Know, and Which System Knows It

A **transactional notification** is a message triggered by an event on the customer's own order, as opposed to marketing: confirmed, shipped, delivered, refunded. The list of notifications a merchant sends is, read the other way, a list of events some system must emit. Shopify's help page on customer notifications (n.d., accessed November 2026) is a usable inventory: order confirmation (which cannot be deactivated), shipping confirmation at fulfillment, shipping update "when tracking information is updated", out for delivery and delivered, each triggered "when a carrier or fulfillment app reports" the event, order refund and order canceled.

Customer-facing message	Event behind it	Which system emits it	Typical lag the customer tolerates
Order confirmed	Order placed and payment authorized	The platform, after the PSP answers	Seconds
Payment problem	Authorization failed or pending (bank transfer, buy-now-pay-later)	The platform, from the PSP's webhook (a message the PSP pushes when a payment changes state)	Minutes
Shipped, with tracking	Parcel handed to the carrier, label scanned	The WMS or the 3PL (third-party logistics provider), via the shipping service	Minutes to an hour
Ships in two parcels	A fulfillment order split, or one line delayed	The OMS; nothing emits it in a landscape without one	Before the first parcel arrives
Out for delivery; delivered; attempted	Carrier scan events	The carrier, normalized by the shipping service	Minutes
Ready for collection	Store confirmation of a click-and-collect pick	The OMS from the store app (Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment)	Minutes

Customer-facing message	Event behind it	Which system emits it	Typical lag the customer tolerates
Return received; refund issued	Return inspected; refund executed	The WMS and the platform or PSP, via the returns portal (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics)	Hours to a day

The middle row is the one a newcomer should stare at. "Ships in two parcels" is the message that would have ended the eleven tickets, and it has no emitter at Kestrel & Pine in 2026, because no system holds the order's lines and their fulfillment state together. It is also the kind of message that an order management system produces as a side effect of doing its job (The Order Record: Lifecycle, State and the Order Management System).

The order status page

The **order status page** is the self-service page where the customer sees where the order is. It is the cheapest answer to "where is my order" the merchant will ever give, and it is a **read model**: a view assembled from the order record and the tracking stream for the purpose of being read, not the place where either is written.

Shopify's documentation of its page (n.d., accessed November 2026) shows the design choices in miniature: five customer-facing states, "Confirmed, On its way, Out for delivery, Delivered, and Attempted delivery", updated only for supported carriers; "If you fulfill an order without a tracking number, then no shipping updates are received"; and an access rule under which a visitor who is not logged in sees items, total and status but not the address, the payment method or the tracking number, with link access lasting three weeks in the same browser.

Three rules follow. The page must be fed by the **same** status the notifications and the agent use. It must receive tracking numbers the moment the parcel is handed over, or it goes silent. And it is a privacy boundary: a link in an e-mail is a credential, and the page should show a stranger holding it as little as possible (Whose Data Is It? Privacy, Consent and Retention).

"Where is my order" as a measurement

WISMO ("where is my order") is the service contact that asks for delivery status. Gorgias, a helpdesk vendor, reported from its own customer base in 2023 that WISMO

requests "account for 18% of incoming requests on average" (an article updated in November 2023, now approaching three years old, drawn mostly from small Shopify merchants). Other vendor figures range from 10% to 50% and rise at peak; a 2026 critical reading on the Branvas blog points out that these are shares of *tickets*, not of *shipped orders*, and are not comparable across vendors. No independent study was found for this book.

The safe statement: WISMO is consistently reported as the largest single contact category in online retail service, roughly a fifth of tickets in the one dataset that states its basis. The useful statement is architectural: **WISMO per shipped order is a direct measure of the notification and tracking layer.** It rises when tracking numbers arrive late, the promise slips, shipments split without a message, or the status page reads a system that is behind. Measured that way, it is a dashboard for the integrations, not a staffing problem.

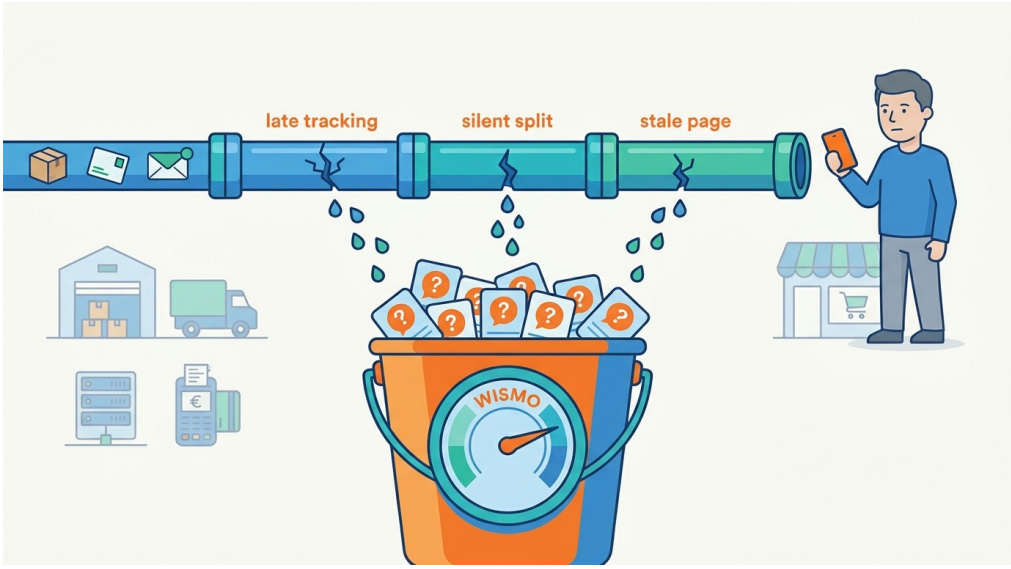


Figure 2: *WISMO per shipped order reads the health of the notification and tracking layer: every late tracking number, silent split or stale status page leaks out as a where-is-my-order ticket.*

How the Pieces Fit Together

The target shape has one order record (the OMS, from Q4 2027 at Kestrel & Pine) emitting events; a notification service that turns events into messages per channel and country; a status page and a service tool that read the same record; and self-service actions that write back through the order record's own rules.

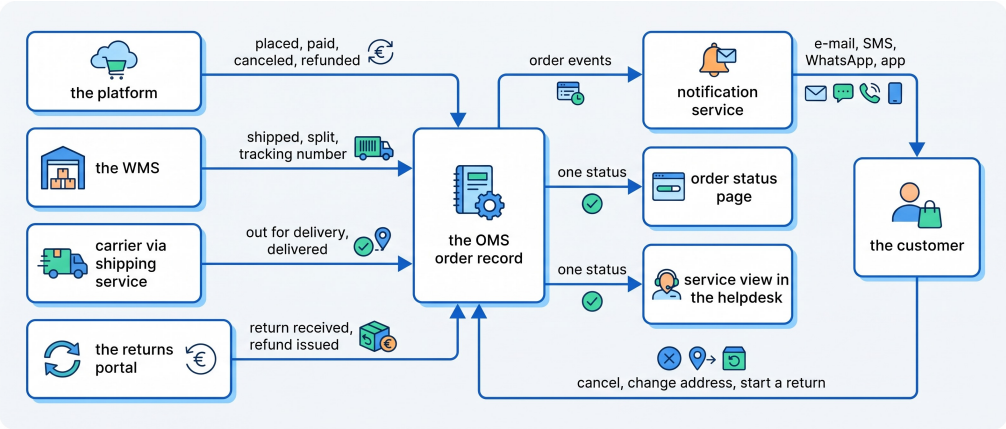


Figure 3: *One order record, fed by the platform, the warehouse, the carrier and the returns portal, drives the notifications, the status page and the agent's view alike; self-service actions go back through the same record and its rules, so the customer and the agent can never read two different orders.*

Component	Responsibility	What it must never do	At Kestrel & Pine, 2027
Order record (OMS)	Hold every channel's order with separate state machines for money, goods and returns; emit an event on every change	Collapse the state machines into one string	Bought; order hub by Q4 2027 (W2)
Notification service	Subscribe to order events; choose channel and language per customer and country; render templates; send through providers; record what was sent	Send from a label printed rather than a parcel handed over; send twice for one event	The platform's built-in e-mails today; a subscriber to the OMS later

Component	Responsibility	What it must never do	At Kestrel & Pine, 2027
Order status page	A read model of the order record plus the tracking stream, with a privacy boundary	Read a different system than the agent's tool	The platform's page today, fed by the iPaaS; W8 adds status by API
Service view	Show the agent one order across systems; offer actions (refund, cancel, change address, reship) gated on the order's state	Issue a refund through a second path	Five tabs and a helpdesk that reads the platform; a helpdesk over the OMS later
Self-service	Let the customer cancel before picking, change the address before the label, and start a return, each gated on the fulfillment state	Allow an edit the warehouse has already acted on	A returns portal; no cancel or address change
Customer record	Tell the agent who this is: online account, store customer, wholesale contact, consent	Guess a match	Four unlinked lists; One Customer, One Product: Identity and Master Data Across Systems

Requirements for the flows

Flow	How fresh	How consistent	How fast	How available
Order confirmation	Seconds after payment	Must match the order the customer sees in their account	Under a minute	Queued if the mail provider is down; never lost
Shipped notification	Minutes after the carrier scan	Tracking number must already resolve at the carrier	Within the hour	Degrades to "shipped, tracking to follow"
Status page	Minutes behind the record	Identical to the agent's view	Under two seconds to load	Serve the last known status with its timestamp when the record is unreachable
Service view	Seconds	Identical to the status page; actions gated on state	Under two seconds per lookup	Read-only fallback when the OMS is down; refunds queued, not re-keyed
Self-service cancel	Immediate check against the fulfillment state	Exactly once; reversed stock commitment	Seconds	Refuse rather than guess when the state is unknown

The consistency column is the point. Freshness can be minutes; two views of one order can never differ, because the difference is what the customer writes in about.

The state machines behind "status"

The Three-Status Order was not a bug in one system; it was three systems each telling the truth about a different thing. Platforms model this openly. Shopify's help page on order status (n.d., accessed November 2026) keeps three status families on one order: an

order status (open, archived, canceled), a payment status (eight values from authorized to refunded) and a fulfillment status (six values including partially fulfilled), with seven more statuses on the fulfillment order below it. commercetools' orders API (n.d., accessed November 2026) keeps an order state, a shipment state (with *Partial* for "items in the Order are shipped in more than one shipment" and *Backorder*) and a payment state, plus return item states. In both, "fulfilled" does not mean "delivered" unless carrier events are fed back.

The lesson for the service layer: the agent's screen and the status page should show the **goods** state per line (the socks are on backorder), the **money** state for the order (paid €137, nothing refunded) and the **return** state when there is one. A single word is a subject line for the customer's inbox, not what the agent or the page should read.

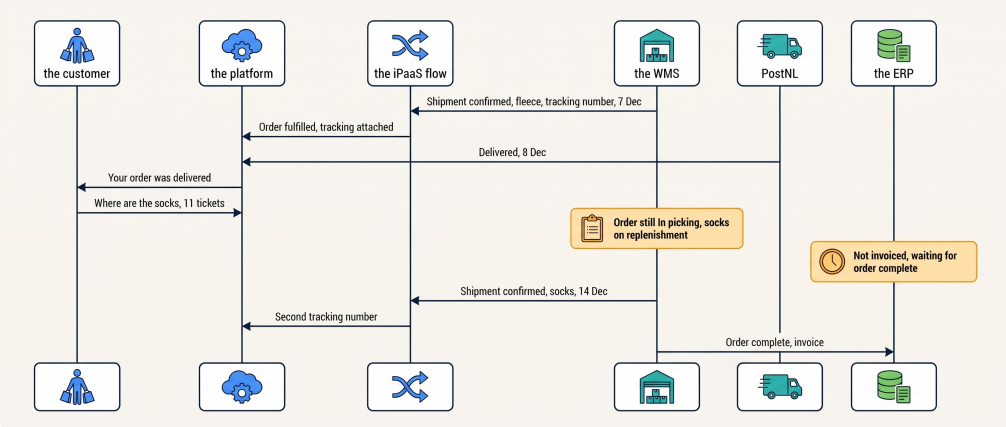


Figure 4: *The Three-Status Order: a shipment confirmation for the first parcel turned the whole order "fulfilled", the carrier's delivery scan turned it "Delivered", and the socks stayed "In picking" and the invoice "Not invoiced" until the second parcel shipped six days later; the one message the customer needed, "two parcels", had no system to come from.*

The Choices

A merchant has four realistic ways to build the service layer, and they line up with the scale stages.

1. **The platform's built-in notifications and a helpdesk that reads the platform.** The platform sends its standard e-mails; a helpdesk such as Zendesk or Gorgias shows the platform's order next to the ticket. Zendesk's documentation of its Shopify integration (n.d., accessed November 2026) describes the shape: orders matched by the requester's e-mail or phone; date, value, payment status, fulfillment status and tracking numbers in the sidebar; agents can refund in any fulfillment state and cancel only unfulfilled orders.

This is the right choice while the platform is the only system that fulfills. It stops being enough the day a warehouse system or an OMS holds a state the platform does not.

2. **A tracking-and-notifications vendor over the carrier stream.** Multi-carrier and post-purchase vendors (Sendcloud, nShift, Narvar and others) sell branded tracking pages and shipping notifications driven by carrier events; Narvar claims its pages "convert 10% higher and reduce support costs" (vendor claim, n.d., accessed November 2026). They solve the carrier half well and the order half not at all: they know where the parcel is, not whether there is a second one or whether the refund went out.

3. **A notification service and a service view over the order management system.** Once an OMS is the order record, the notification service subscribes to its events, the status page reads it, and the helpdesk's order sidebar is backed by it. OMS vendors sell this view: Fluent Commerce says agents "see the same order data the platform used to make its decisions"; Kibo lets representatives edit, credit and refund "without leaving the order screen" (both n.d., accessed November 2026). This is where Waypoint takes Kestrel & Pine.

4. **A built notification and service layer.** A merchant with engineers and unusual needs (many countries, brands and channels) builds the event subscriber, the template engine and the service screens itself. Right at Group stage; the Giant's Boots for anyone smaller (Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says).

	Platform notifications and helpdesk	**Tracking vendor**	**Notification service and service view over the OMS**	**Built layer**
Knows the carrier events	Only for supported carriers with tracking numbers	Yes, normalized	Yes, via the shipping service	Yes
Knows splits, backorders and returns	No	No	Yes	Yes
Agent actions gated on state	Refund any state; cancel unfulfilled	None	Yes, by the OMS's rules	As built

	Platform notifications and helpdesk	Tracking vendor	Notification service and service view over the OMS	Built layer
Channels and languages	The platform's templates	The vendor's	Configurable per country	Anything
Integrations to own	None	One (carrier stream)	OMS events, templates, helpdesk connector	Everything
Risk of two refund paths	Helpdesk and returns portal both refund through the platform	None	Low if the OMS owns the refund command	As built
Fits best	Starting, Growing	Growing, as a complement	Omnichannel	Group

My rule: **the service view must read the system that owns the goods state**. While the platform owns it, choice 1 is right and cheap. The moment an OMS or a WMS owns it, every view that still reads the platform is a Schrödinger's Order waiting to happen, and the helpdesk must be re-pointed at the OMS even if nothing else changes. The notification channels are a second-order question; the source of truth is the first.

Self-service, gated on state

The contacts that need no judgement (cancel before picking, change the address before the label, start a return) should not reach a person. Each is an edit to a placed order, and each must check the fulfillment state first. Shopify's order editing documentation (n.d., accessed November 2026) states the constraint plainly: "You can edit only unfulfilled line items", and its fulfillment order has a request status (unsubmitted, submitted, accepted) after which "the merchant can no longer cancel it directly and must submit a cancellation request". In plain words: once the warehouse has accepted the work, a cancellation is a request to the warehouse, not a change to a record. A self-service cancel button that does not know this will cancel orders that are already in a box.

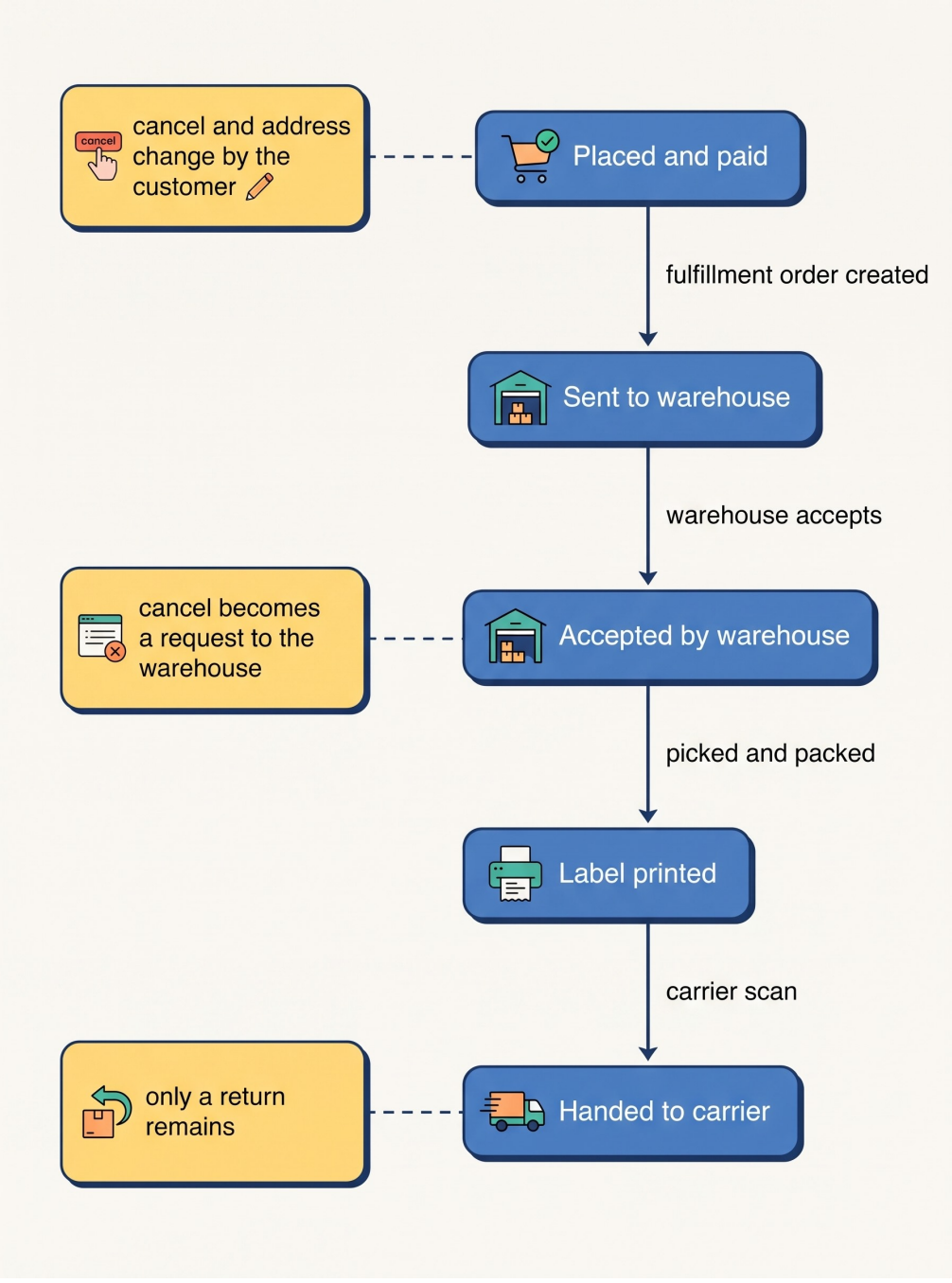


Figure 5: *Which self-service actions are safe depends on the goods state: before the warehouse accepts, the customer may cancel or change the address; afterwards a cancel is a request; once the carrier has the parcel, only a return remains.*

Service for marketplace orders

On a marketplace, the customer conversation belongs to the marketplace, and the merchant answers under its rules and clocks. bol's partner standards (n.d., accessed November 2026) include a per-seller "dynamic standard" for customer questions as a share of orders, a response-time standard and a strike process in which weeks below standard accumulate toward closure; Amazon's order performance policy (European policy document, May 2026) sets the Order Defect Rate, the share of orders with negative feedback, a guarantee claim or a chargeback, at "under 1%" over 60 days.

Import marketplace messages with their deadlines into the helpdesk, and let the service view find an order by the marketplace's id as well as its own. That connects the customer's question to the right purchase and makes the response deadline visible. Leaving messages in a separate portal makes both easy to miss.

Personal data is the other difference: Amazon's SP-API (Selling Partner API) returns a buyer's name and address only through a restricted data token, so the service view holds only what fulfillment needed (Selling on Someone Else's Platform: Marketplaces and Channel Integration, Whose Data Is It? Privacy, Consent and Retention).

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
1 — Starting	The platform's standard e-mails; a shared mailbox; the founder answers WISMO by looking at the carrier's site	Tracking numbers on every shipment; the platform's status page turned on; one reply template per common question	The same question asked ten times a week; orders shipped without a tracking number

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
2 — Growing	A helpdesk reading the platform's order; a 3PL or WMS that confirms shipments by integration; the platform's notifications plus a carrier's own messages	The shipment confirmation from the 3PL must carry the tracking number within the hour; a WISMO-per-shipped-order measure; a single refund path between the helpdesk and the returns tool	Agents keep a second tab open for the 3PL; "shipped" e-mails that precede the handover; refunds issued from two consoles
3 — Omnichannel	Orders from web, marketplaces, stores and wholesale; an OMS as order record; a notification service with channels per country; a service view over the OMS; self-service gated on state; marketplace messages in the helpdesk with their deadlines	One status source for the page, the messages and the agent; split-shipment and backorder messages; a customer record that joins the online account to the store customer; service metrics per channel	Five tabs; the Three-Status Order; WISMO rising at peak while order volume is flat; marketplace strikes for late replies

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
4 — Group	Several brands and countries; a shared notification platform with per-brand templates; AI-assisted agents and chat; a customer data layer feeding service	Per-brand service levels on a shared order record; consent and data residency per country in the notification layer; audit of what was sent to whom	A shared template that breaks one brand's tone or one country's law; an assistant answering from a stale copy of the order

The judgement: the layer changes shape exactly once, when the goods state leaves the platform. Everything before that point can stay on the platform's built-in tools; everything after it must read the order hub. Kestrel & Pine is at that point now, and the eleven tickets are what the delay cost.

In Practice

Events, not polling, is also the rule for the messages customers receive. DPD's Predict notifications in Belgium (n.d., accessed November 2026) are sent by the carrier itself when the parcel "arrives at the depot, and when it leaves the depot for the receiver". EasyPost's tracker object (n.d., accessed November 2026) reduces every carrier's codes to ten statuses, from pre-transit to delivered, with webhooks on update. A notification service needs five of the ten for the customer and all ten for the agent, pushed, not polled (APIs, Events, Files and Middleware: How Commerce Systems Talk).

Channels change their economics on published dates. Meta's WhatsApp Business Platform pricing page (accessed November 2026) classifies templates as marketing, utility ("template messages for transactional or informational objectives") and authentication, states that per-message pricing arrived on 1 July 2025, when utility messages sent inside the 24-hour customer service window became free, and that from 1 October 2026 those utility messages are charged as regular messages again, with a free tier of 1,000 a month that applies to service messages only. A notification service that chooses the channel per message, with cost as an input, survives such changes; one that hard-codes "WhatsApp for shipped" does not.

Order status by API is a prerequisite, not a feature. Waypoint's W8 lists order status by API among what an AI shopping agent would need before acting for a customer (Assistants and Agents: AI in Commerce and What Is Real in 2026). The same endpoint

feeds the status page, the helpdesk and the carrier vendor's tracking page: one read model, every consumer.

The service tool's refund button is another refund path. Zendesk's Shopify integration lets an agent refund from the ticket in any fulfillment state; Kestrel & Pine's returns portal refunds through the platform's API; its PSP console allows a refund by hand. In January 2027 the last two produced the Double Refund (**212 returns**, **€19,400** refunded twice, most recovered); the helpdesk's button is a third. The rule: the order record owns one refund command and every tool invokes it, with the order and return ids as the **idempotency key** (a key that makes a repeated request have the same effect as one) (Taking the Money: Payment Providers, Authorization, Capture and Refunds, Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation).

Kestrel & Pine, worked: Eva's jacket from the customer's side

The dates are Kestrel & Pine's records; the message times and channels are illustrative details. The "2026 landscape" column is what Eva would have received had she ordered in December 2026; the "target" column is the Q4 2027 design.

When	What happened	2026 landscape: what Eva receives	Target: what Eva receives
Tue 13 Apr 2027, 21:14	Order #1072455 placed, €235 by iDEAL	Confirmation e-mail from the platform within a minute	Same, with "delivery expected Thursday" computed, not static (Getting It There: Carriers, Labels, Tracking and the Delivery Promise)
Wed 14 Apr, 08:40; collected 22:30	Picked, packed, PostNL label; collected that evening	"Shipped" e-mail when the iPaaS flow marks the order fulfilled, sometimes before the parcel leaves	"Shipped" when the carrier scan arrives, with a tracking link that already works

When	What happened	2026 landscape: what Eva receives	Target: what Eva receives
Thu 15 Apr, morning	Out for delivery	A PostNL message, if Eva opted in with PostNL	A message from Kestrel & Pine, same event, her chosen channel
Thu 15 Apr, 11:20	Delivered	Status page: Delivered	Status page: Delivered, per parcel
Mon 19 Apr	Exchange for size L requested in the returns portal	Portal e-mail with the label; the platform's page knows nothing of it	"Return registered, send within 14 days"; the order's return state visible on the page
Thu 22 Apr	Return received, graded resellable; size L ships	Two e-mails from two systems, in two designs	"Received; your size L ships today", one message, one design
If Eva writes in	"Has my return arrived?"	The agent opens the returns portal tab and the WMS tab	The agent's view shows the return received at 10:15 and the exchange's tracking number

Nothing in the target column needs a new channel. It needs one record emitting the events and one read model serving the page, the messages and the agent. The messages that matter most are the ones that pre-empt the ticket: "two parcels", "received", "ships today".

Measuring service as an architecture signal

A small set of measures turns the service desk into the platform's most honest monitor. Targets are illustrative.

Measure	Definition	Why it is an architecture signal	Kestrel & Pine, 2026 *(assumption)*	Illustrative target
Contacts per 100 orders	All tickets, calls and chats divided by shipped orders	The total cost of every gap	9	Under 5
WISMO per 100 shipped orders	Delivery-status contacts only	The notification and tracking layer	3.5, rising past 6 in Black Friday week	Under 1.5, flat at peak
Contacts per split shipment	Tickets on orders shipped in more than one parcel	Whether the split message exists	Not measured	Under 0.2
Status disagreements found	Orders whose page and agent view differ, sampled weekly	Whether the views read one source	Not measured	Zero
Tickets per order (max)	The worst order of the month	The long tail that becomes a social-media story	11	Under 4
First reply with a true answer	Share of WISMO tickets answered with the correct goods state on first reply	Whether the agent can see the order	Unknown	Over 95%

Hanna Berg's data team can produce most of these from the helpdesk and the order record once there is one; the first and the last are the ones to put on Noor's monthly page (Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics).

When It Goes Wrong

- **The notification is sent twice**. Events are delivered at least once; a retried "delivered" event sends a second e-mail, and the customer wonders whether there were two parcels. *Detection*: sent-message log keyed by order and event. *Recovery*: the notification service is idempotent on the event id; it records what it sent before it sends.
- **The notification precedes the event**. "Shipped" goes out when the label prints, and the parcel waits a day for the carrier (Kestrel & Pine's Black Friday queue backed up for four hours); "delivered" goes out when the carrier's scan says so, and the parcel is with a neighbor. *Detection*: WISMO tickets that quote the notification. *Recovery*: trigger "shipped" on the handover scan, and say "delivered to a neighbor" when the carrier says so; the carrier's vocabulary has that state, and the message should too.
- **No tracking number, no updates**. A fulfillment confirmed without a tracking number leaves the status page frozen at "confirmed". *Detection*: shipments without tracking, counted daily. *Recovery*: the WMS integration contract requires the number; the notification service sends "shipped, tracking to follow" and then the number.
- **The split that nobody announced**. The Three-Status Order. *Detection*: contacts per split shipment. *Recovery*: an order record that holds goods state per line, a "ships in two parcels" message, and an agent view that shows both parcels.
- **The carrier is down**. No scans for a day; every parcel looks stuck. *Detection*: a monitor on the carrier stream's freshness. *Recovery*: a banner on the status page ("carrier updates delayed") and a hold on "attempted delivery" messages until the stream resumes (Black Friday Every Year: Peak Load, Resilience and Recovery).
- **The agent and the page disagree**. The helpdesk reads the platform; the page reads the OMS. *Detection*: the weekly status-disagreement sample. *Recovery*: re-point the helpdesk; never run two sources.
- **The template is wrong for the country**. A Belgian customer in French receives a Dutch "shipped" with a Dutch legal footer. *Recovery*: templates selected by storefront, language and country, with the legal text owned by whoever owns the regulatory calendar (Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation).
- **The agent cannot find the person**. Eva writes from the e-mail on her store loyalty account, which differs from her online account's; the agent finds no order (the *Familiar Stranger* pattern). *Recovery*: search by order number first, by any known identifier second, with a matching rule of declared confidence (One Customer, One Product: Identity and Master Data Across Systems).

What to Take Away

Established.

- Platforms model an order with several status machines (Shopify's order, payment and fulfillment statuses plus the fulfillment order's own; commercetools' order, shipment and payment states), and "fulfilled" means "delivered" only when carrier events are fed back (vendor documentation, accessed November 2026).
- Transactional notifications are triggered by events from three different emitters: the platform (placed, canceled, refunded), the warehouse or 3PL (shipped with tracking) and the carrier (out for delivery, delivered, attempted); a status page without a tracking number receives no updates (Shopify documentation, accessed November 2026).
- Order edits and cancellations are safe only before the warehouse accepts the work; afterwards they become requests (Shopify order editing and fulfillment order documentation, accessed November 2026).
- "Where is my order" is consistently reported as the largest single contact category in online retail service; the only figure with a stated basis is 18% of tickets (Gorgias, 2023), and vendor figures are shares of tickets, not of orders.

Depends on context.

- While the platform owns the goods state, its built-in notifications and a helpdesk that reads the platform are the right design; once a WMS or an OMS owns it, every view must read the order hub. The trigger is the ownership of the goods state, not the order volume.
- Whether to add a tracking-and-notifications vendor depends on how many carriers and countries the merchant ships to; it complements the order record and never replaces it.
- The channel mix (e-mail, SMS, app, WhatsApp) depends on country and on pricing that changes on published dates; choose per message with cost as an input.

The author's synthesis.

- Build the status page, the notifications and the agent's screen as three read models over one order record, and treat any difference between them as a defect, because every difference is a ticket with the merchant's name on it.
- Measure WISMO per shipped order and put it next to the integration monitors, not next to the staffing plan: it is the customer's reading of the platform's joins, and the one that arrives first.

Questions to Ask

1. Which system does the order status page read, and which system does the service agent's tool read? Are they the same?
2. Which event triggers the "shipped" message: the label, or the carrier's scan?
3. If an order ships in two parcels, what does the customer receive, and when?
4. How many systems can issue a refund on one order, and which of them is the one that should?
5. What can a customer do without contacting anyone (cancel, change address, start a return), and what state check guards each action?
6. How many WISMO contacts per hundred shipped orders did we have last month, and how did the number move in peak week?
7. How many tabs does an agent open to answer "where is my order"?
8. How do marketplace messages reach the agent, with which deadline, and under which order id?
9. What happens when the customer writes from an e-mail the platform has never seen?

Part IV ends here, with the order delivered, returned if it must be, and explained to the customer. The next part leaves the merchant's own channels: Part V — CHANNELS: Selling Everywhere begins with selling on someone else's platform, in Selling on Someone Else's Platform: Marketplaces and Channel Integration, where the customer conversation, the deadlines and the order id all belong to the marketplace.

Sources and Evidence

Company, product and investor sources

- Customer notifications, Shopify Help Center, n.d., accessed 2026-11 — the notification list and the events and emitters behind each message.
- Understanding order status pages, Shopify Help Center, n.d., accessed 2026-11 — the five customer-facing states, the tracking-number rule and the access rules without login.
- Order status, Shopify Help Center, n.d., accessed 2026-11 — the three status families on one order.
- Edit orders and FulfillmentOrder object, Shopify developer documentation, n.d., accessed 2026-11 — edits only on unfulfilled items; cancellations as requests after acceptance.
- Orders (OrderState, ShipmentState, PaymentState), commercetools, n.d., accessed 2026-11 — separate order, shipment and payment states, including Partial and Backorder.

- Using the Shopify integration in Support and Chat, Zendesk, n.d., accessed 2026-11 — the helpdesk's order sidebar, refund in any state, cancel only unfulfilled.
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- Tokens API use case guide, Amazon SP-API, n.d., accessed 2026-11 — personal data available only through a restricted data token.

Standards, regulation and official documentation

- Directive 2011/83/EU on consumer rights, Publications Office of the EU, 2011 — the 14-day withdrawal and refund clocks that the return messages refer to.

Research and practitioner sources

- WISMO benchmarks, Branvas, 2026 — a critical reading of vendor WISMO figures as shares of tickets rather than of orders.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part V — CHANNELS: Selling Everywhere

KEY POINTS:

- **Every outside channel is a way into the same order record**. Marketplace, portal, email and electronic trading-partner orders should all create one kind of order, flagged by channel.
- **The intermediaries translate; they must not own**. Connectors and electronic document providers carry facts between owners, and each fact keeps one honest source.
- **Outside parties keep score and set the calendar**. Marketplaces grade, retail chains charge back and mandates arrive with legal dates, so availability must never wait on a scheduled finance-system update.

A merchant can sell the same jacket through its own shop, a marketplace or a retail chain, yet fulfill each sale under different conditions. A marketplace can penalize late dispatch; a chain can require specific pallet labels and invoices. Reusing the consumer checkout process without those requirements can turn extra sales into deductions or delayed payment. This part follows the systems needed to honor each channel's agreement.

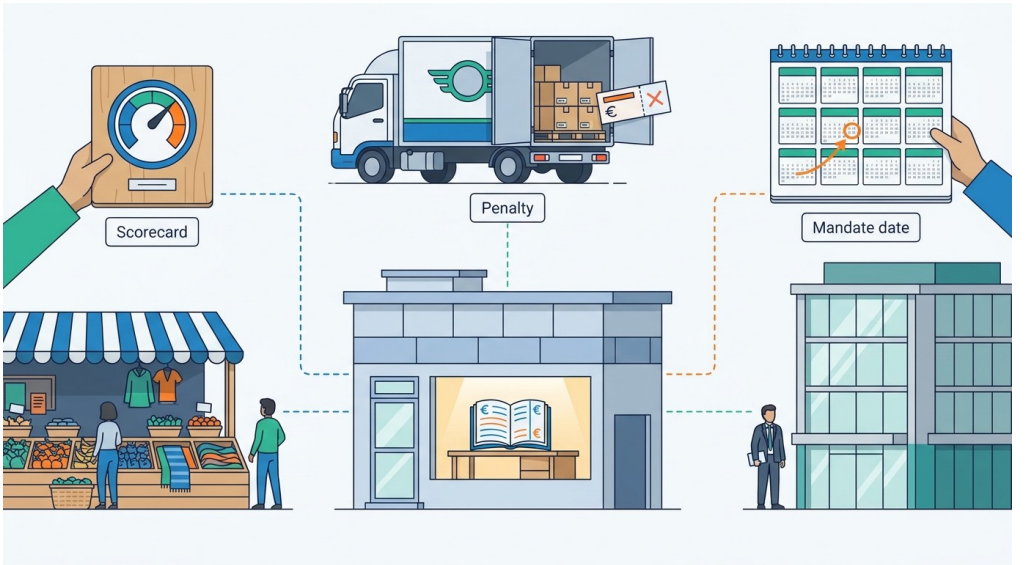

Figure 1: *Each outside channel comes with an outside party that sets the rules: a marketplace keeps a scorecard, a retail chain charges penalties, and a mandate arrives with a legal date.*

The examples are different routes for the same jacket: two Amazon.de orders, one fulfilled by the merchant and one by the marketplace; a shop's pre-season purchase; and Bergzicht Outdoor's **240-unit reorder on three pallets**. Consumer Eva's order uses the same warehouse and financial systems.

Kestrel & Pine, **the fictional outdoor brand**, earns €3.0 million through marketplaces and €12.0 million from 1,100 wholesale accounts in 2026. Its **business-to-business (B2B)** portal receives 62% of wholesale orders; staff enter 38% by hand in the **enterprise resource planning system (ERP)**, which keeps finance and purchasing records. Bergzicht uses **electronic data interchange (EDI)** to send structured orders.

The shared decision is where each order becomes authoritative. Connectors should translate channel requirements while preserving ownership of prices, stock and orders.

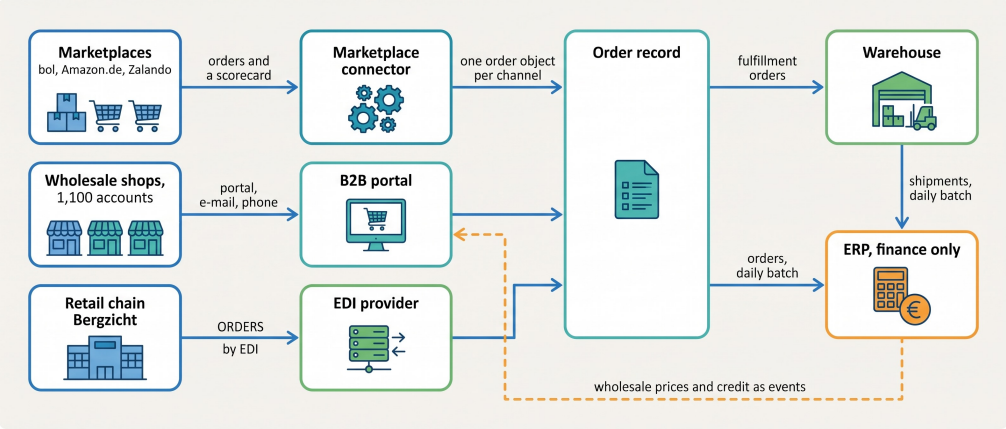

Figure 2: *The outside channels of Part V: three kinds of outside party, three intermediaries that translate, one order record that every channel's order is created in, and an ERP that receives daily batches and feeds no channel directly: its wholesale prices and credit limits leave it only as events.*

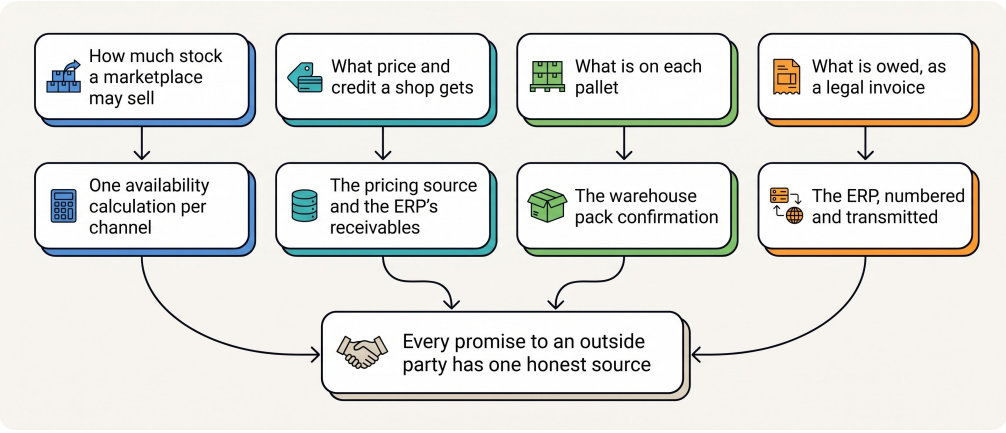

Figure 3: *The four promises Part V makes to outside parties and the one system allowed to make each; every chargeback, strike and keyed-in order in the part is a promise that came from the wrong system.*

Chapters in this part

Selling on Someone Else's Platform: Marketplaces and Channel Integration explains **listings, imported orders, stock allocation and seller performance**. Stale stock causes cancellations and penalties; a retired interface leaves one channel outdated for two days. The connector needs both a technical contract and a responsible owner.

Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals adds **company accounts**, **negotiated prices**, **payment terms and credit limits**. De Kaap Buitensport's purchase shows why accepting an order requires more than checking stock. Finance's terms must reach the portal reliably.

Documents Between Companies: EDI, Punchout and E-Invoicing follows **structured orders**, **despatch advices and invoices**. Bergzicht's €14,200 deduction for mismatched pallet labels shows why the shipment notice must describe what was packed. The chapter also separates trading-partner requirements from dated e-invoicing mandates.

The Back Office: ERP Integration and Who Is the Hub assigns **the ERP a clear financial and purchasing role**. A connector that took three weeks to understand illustrates the cost of undocumented ownership. Finance can consume complete batches while stock availability needs a faster route.

Begin with Selling on Someone Else's Platform: Marketplaces and Channel Integration.

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine is fictional; recommendations and labeled assumptions are the author's judgement.

20. Selling on Someone Else's Platform: Marketplaces and Channel Integration

IN THIS CHAPTER, YOU WILL: Learn how a merchant lists, sells and ships on a marketplace, what a channel connector does and must not do, and how to allocate stock and import orders so that marketplace promises are kept.

KEY POINTS:

- **A marketplace is a channel with a scorecard.** It measures your cancellations, late shipments and tracking against public thresholds (Amazon: cancellation rate under 2.5%, late shipment rate under 4%; bol: at most 2% cancellations a week), so every hour of stock or shipment latency inside your integration is a commercial risk, not a technical detail.
- **The connector translates; it must not own.** Listings should come from the product system of record, the stock number from one availability calculation, prices from the pricing source, and every marketplace

order should land in the merchant's one order record. A connector that keeps its own copies becomes a fourth system of record, which is what Kestrel & Pine has today.
- **The number you send is a promise**. Decide how much stock each channel may sell (a buffer at first, later a computed available-to-promise figure: what can still be sold after existing orders and reservations), model marketplace-held stock as a location you do not control, and own the deprecation calendar of every marketplace API, because versions retire on published dates whether or not anyone is watching.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Phantom Stock** — the stock figure sent to a marketplace is a copy of a copy, hours old, and the last unit sells twice.
- **The Pulled Rug** — marketplaces and connectors retire API versions on published dates; an unowned script finds out two days later.
- **Schrödinger's Order** — a marketplace order that lives only in the connector cannot be found by support, finance or the warehouse report.

Listing a jacket on Amazon, bol or Zalando can put it in front of shoppers who never visit the merchant's own site. The merchant pays a commission, a share of the sale, and must follow the marketplace's rules for prices, fulfillment and customer service. That additional reach brings obligations its own storefront does not control.

The marketplace also grades performance: cancellations, late dispatch and valid tracking can affect visibility or the right to sell. Those measures depend on the data your integrations send. An old stock figure can become a canceled order; a late shipment message can damage the seller's score even when the parcel left on time.

This chapter follows product **listings**, marketplace orders and status updates through a **channel connector**, the service translating between merchant and marketplace systems. Its permissions matter because a connector that changes prices or stock independently can make a different promise from the web shop. Give it authoritative values to distribute and define which incoming marketplace changes it may accept.

Where we are in the order's journey. At Kestrel & Pine, the fictional outdoor brand, Eva buys from the merchant's own storefront. Two illustrative Amazon.de orders on the same day, 13 April 2027, show different routes. One ships from Kestrel & Pine's Duiven warehouse; the other from stock already held by the marketplace. Both must reach the merchant's records. Follow the Order: What Happens Between "Buy" and "Delivered" provides the wider journey.

Setting the Scene at Kestrel & Pine: Two Days of Stale Stock

Kestrel & Pine sells through bol, Amazon.de and Zalando using a rented marketplace connector costing €900 a month. In 2026, those channels brought **43,000 orders and €3.0 million**, averaging **€70 per order** against €92 on its own storefronts. Marketplace fulfillment programs ship about **40% of its German marketplace units**.

Ecommerce and Marketplaces Manager Milan Petrov owns the connector. In March 2027, he walks Head of Engineering Noor Visser through two incidents.

The **Black Friday Oversell** began when Duiven ran out of Ridgewalker jackets at 19:40 on 27 November 2026. Stock traveled from the warehouse management system (WMS) to the enterprise resource planning system (ERP) every 15 minutes, then hourly to the platform and hourly to the connector. Marketplaces kept selling for up to two hours.

In this illustration, about **160** of the 1,120 canceled or delayed orders were marketplace orders, and Kestrel & Pine canceled **140**. Those cancellations counted against seller standards. Bol issued a formal warning, or **strike**, for the week. Marketplace penalties form part of the incident's €96,000 cost.

The second is the **Deprecated Endpoint**. In February 2027 the connector retired an older version of its API on a date it had published months earlier. The agency script that pushed the platform's stock figures into the connector still used that version. Nobody had read the notice. For **two days** the connector kept sending bol the last figures it had, and bol sold from them.

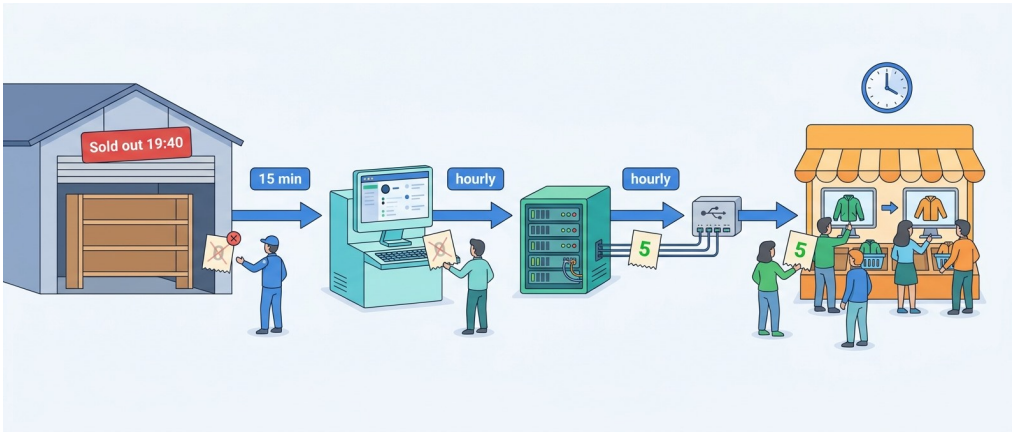

Figure 1: *The Black Friday Oversell: the stock figure reached the marketplaces as a copy of a copy, so the Ridgewalker jacket sold out at Duiven at 19:40 and the marketplaces kept selling it for up to two hours.*

Noor's questions are the chapter's questions. Which stock number should each marketplace be told, and who computes it? Where does a marketplace order live, so that support, the warehouse and finance see the same record? And who owns the API calendar of a vendor Kestrel & Pine pays €900 a month and depends on for €3.0 million?

What a Marketplace Is, and the Two Ways to Sell on One

Selling on a marketplace means publishing an **offer** against a product the marketplace knows: your price, your stock and your delivery promise for an item identified by its **GTIN** (Global Trade Item Number, the number behind the barcode; What Are We Selling? Products, Variants and the Catalog owns it). Several sellers can offer the same product, and the marketplace shows one of them in the **buy box**, the single "add to basket" button on the product page; performance and price decide who wins it.

Every large marketplace offers the same two logistics models under its own names.

Model	Who holds the stock	Who ships	Who handles returns	Amazon name	bol name	Zalando name
Seller-fulfilled	The merchant	The merchant, with its own carriers or the marketplace's labels	The merchant, under marketplace rules	FBM (Fulfilled by Merchant), "Merchant Fulfilled Network"	Own shipping, or "Shipping via bol"	"Own Logistics"

Model	Who holds the stock	Who ships	Who handles returns	Amazon name	bol name	Zalando name
Marketplace-fulfilled	The marketplace's warehouse, stocked in advance	The marketplace	The marketplace	FBA (Fulfillment by Amazon)	"Logistics via bol" (LvB)	ZFS (Zalando Fulfillment Solutions); ZEOS (Zalando E-commerce Operating System) for multi-channel

The names are from the vendors' own pages (bol partner platform and Zalando partner university, n.d., accessed November 2026; Amazon's order performance policy, 2026). bol describes Logistics via bol as "We handle storage, shipping, and returns", counts stock daily and charges for it, with a 40% surcharge on stock costs in November and December, as of November 2026.

The architectural difference is larger than the operational one. In the seller-fulfilled model, a marketplace order must enter your order record and your warehouse flow with the marketplace's deadline attached, and your ship confirmation must travel back in time to count. In the marketplace-fulfilled model, your stock sits in **a location you do not control**: you learn its quantity from the marketplace's reports, its sales from the marketplace's notifications, and its returns after the marketplace has processed them. Is It in Stock? Inventory, Availability and the Promise You Can Keep treats such stock as an external location with its own feed, and Kestrel & Pine's Waypoint decision W1 says the same.

The marketplace is also a competitor. The German competition authority's 2013 press release on Amazon's price-parity clauses states it plainly: Amazon, "as the largest online retailer, competes directly with" the sellers on its marketplace (Bundeskartellamt, 26 November 2013). Everything you send a marketplace is also market intelligence, and the design question is which attributes and which stock granularity leave the building.

What it costs

Fees differ by marketplace, category, price band and fulfillment model, and change on published dates. As of November 2026, Amazon's US schedule charges $39.99 a month for a professional account plus a referral fee of mostly 8–15% of the price (clothing 5% up to $15, 10% to $20, 17% above); bol charges no subscription but a fixed amount per item plus a percentage of the price including VAT, by category; Zalando publishes no figures. The lesson is not the numbers but their shape: **fees are dated reference data**, not constants. Margin per channel belongs in the analytics layer (Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics) with the fee schedule versioned by effective date, and the channel price floor (below) is computed from it.

What the Marketplace Measures You On

Marketplaces publish the metrics they grade sellers on. Amazon's European order performance policy (PDF dated May 2026) and its seller blog (October 2024) give these thresholds:

Metric	What it counts	Threshold	Window	Applies to
Order Defect Rate	Orders with negative feedback, an A-to-z Guarantee claim or a chargeback	Under 1%	60 days	All orders
Cancellation rate (pre-fulfillment)	Orders the seller canceled; customer cancellations excluded	Under 2.5%	7 days	Seller-fulfilled
Late shipment rate	Shipping confirmations completed after the expected ship date	Under 4%	10 or 30 days	Seller-fulfilled

Metric	What it counts	Threshold	Window	Applies to
Valid tracking rate	Shipments with a valid tracking number	Above 95%	30 days	Seller-fulfilled
On-time delivery rate	Tracked units delivered by the promised date	90% or higher expected on Amazon.de; consistently under 80% after a warning may lead to restrictions	Rolling 14 days	Seller-fulfilled

bol's partner pages (n.d., accessed November 2026) set an "Items on time" minimum of 90%, a cancellation standard of at most 2% a week (from three or more items canceled), and a **strike process**: a strike is a warning for a week below standard on items on time, cancellations or response time; strikes count over 22 weeks; at five the shop is temporarily closed pending an improvement plan; at seven the account is closed for at least a year. bol also cancels an unconfirmed order automatically on the third day after the promised delivery date, and counts that against the seller.

Read the table as an integration specification. **An oversell becomes a seller cancellation**, and a merchant with a few hundred marketplace orders a week needs only a handful to cross 2.5%. **A warehouse confirmation that reaches the marketplace a day late is a late shipment**, however promptly the parcel left. Each row maps to a flow in the next section and to a freshness requirement on it.

How the Pieces Fit Together

A **channel integration platform** (connector, integrator, "marketplace middleware") sits between the merchant's systems and many marketplaces and translates four kinds of data both ways. Vendors describe the same four functions: ChannelEngine calls itself "One platform for listings, orders, inventory, and pricing across 1,300+ channels" (n.d., accessed November 2026); Shopify's own Marketplace Connect app does the same for Amazon, Walmart, Target Plus and eBay from inside the platform (Shopify Help Center, accessed November 2026).

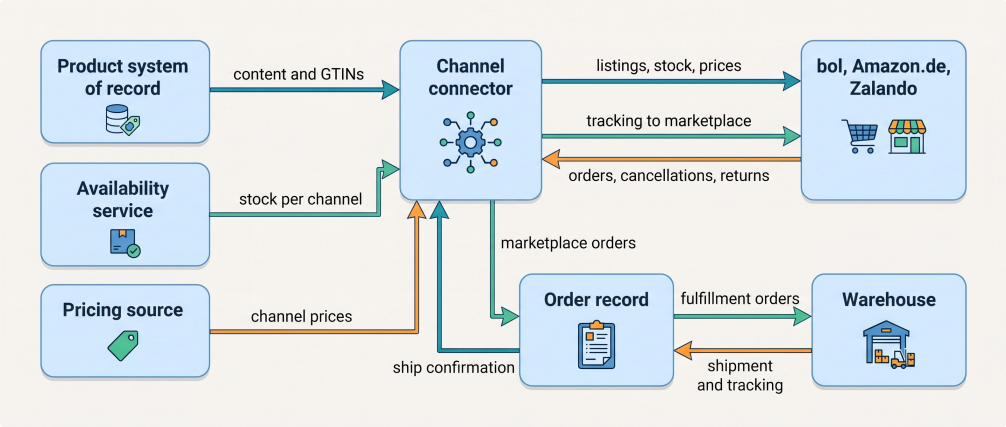

Figure 2: *The connector translates five flows between the merchant's systems of record and the marketplaces; it owns none of the data it carries.*

The five flows and what each needs:

Flow	Direction	How fresh	How consistent	What goes wrong when it is late
Listings (content, images, attributes, GTINs)	Merchant to marketplace	Hours to a day; seasonal	Must match the product system of record; incomplete content is rejected	A listing goes offline or shows last season's text
Stock (quantity per offer)	Merchant to marketplace	Minutes; as events happen	One availability figure per channel, computed once	Oversell, seller cancellation, a strike
Prices	Merchant to marketplace	Minutes to hours	Derived from the pricing source, bounded by a floor	Price drift between channels, a parity warning

Flow	Direction	How fresh	How consistent	What goes wrong when it is late
Orders (import, acknowledge, cancel)	Marketplace to merchant	Minutes; before the ship deadline	One order record, keyed on the marketplace order id, idempotent	Auto-cancellation, a duplicate shipment
Status back-flow (ship confirmation, tracking, returns)	Merchant to marketplace	Within the day of shipment	Carrier and tracking from the shipping service, once	Late shipment rate, invalid tracking rate

The two fast rows, stock and orders, are why Waypoint's W5 moves them by **events** (a message sent the moment something changes) rather than by periodic copies; APIs, Events, Files and Middleware: How Commerce Systems Talk picks a style per flow from requirements like these.

Order import, acknowledgement and back-flow

The marketplace order lifecycle has the same shape everywhere, and the connector's job is to keep two order records in step. The steps below are my synthesis of Amazon's, bol's and ChannelEngine's documented interfaces; marketplace state names differ.

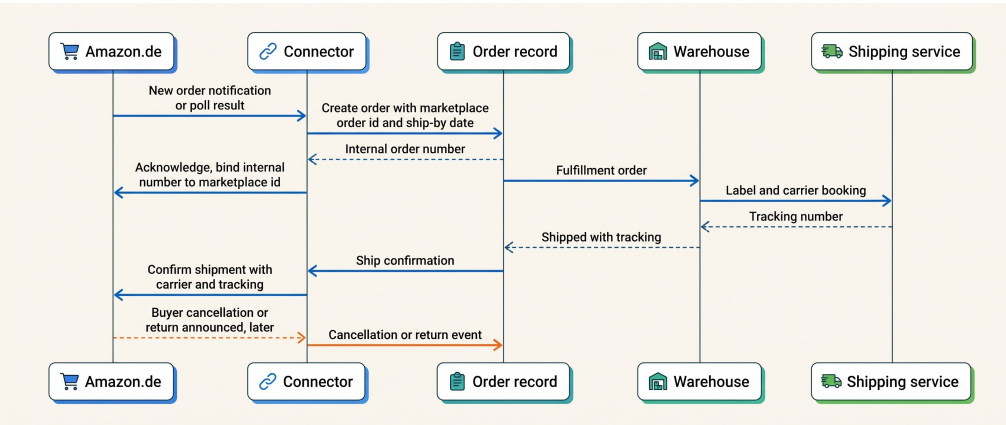

Figure 3: *A seller-fulfilled marketplace order: the order is created once in the merchant's record, acknowledged back with the binding between the two order numbers, and its shipment confirmed with tracking before the marketplace's deadline.*

Three points on this diagram carry the chapter.

Acknowledgement is a binding. ChannelEngine's documentation calls it "the critical step that links the ChannelEngine order to your internal system"; bol auto-cancels unacknowledged orders on day three after the promised delivery date. The acknowledgement ties the marketplace order id to your order number, and that pair is the key for everything after: cancellations, returns, disputes, payouts.

Delivery can be duplicated. Amazon's notifications push to a queue you own; you can also poll; a connector retries. All three can deliver the same order twice, so the receiving side must be **idempotent**: creating the same marketplace order a second time has no effect, because the order record enforces uniqueness on the marketplace order id (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation).

Ship confirmation is a deadline, not a courtesy. The late shipment rate is computed from the confirmation's timestamp, so the path WMS → order record → connector → marketplace must complete on the day of shipment. When Kestrel & Pine's WMS connector queue backed up for four hours on Black Friday 2026, every delayed confirmation was also a late shipment on a scorecard.

What the marketplace APIs actually impose

The marketplaces' own documentation shows that an integration cannot be designed as a tight loop of calls.

Amazon's Selling Partner API (SP-API), the only programmatic route since the older MWS (Marketplace Web Service) was retired on 1 April 2024, uses a token-bucket **rate limit** per operation: each call draws a token, tokens refill at a published rate, and a throttled request returns HTTP 429 and must be retried later. The Orders API's `getOrders` has a rate of 0.0167 requests a second, one call a minute, with a burst of 20 (SP-API documentation, accessed November 2026). Amazon's guidance is explicit: "Don't hardcode timers" and "Respond to events rather than polling on a loop, for example, update prices on price notifications".

Listings, prices and stock go up as **feeds**, asynchronous bulk uploads whose result you fetch later. Acceptance of the upload does not confirm that the listing changed. The connector must check the outcome so a rejected update does not leave the marketplace selling at an old price or from an old stock count.

Customer names and addresses are **restricted data** that need a separate token. Keep only the personal data fulfillment needs in the order record: every extra copy adds another place to secure and another record to find when handling a customer request (Whose Data Is It? Privacy, Consent and Retention).

bol's Retailer API (as of November 2026) warns that "most calls are asynchronous; confirm the result afterwards", publishes its rate limits, and advises sending "only changes that need synchronizing" and "events as they happen instead of large batches". Its lifecycle policy: when a new major version is released, the previous one is deprecated and "scheduled for removal in 12 months". On 3 November 2026 bol's partner news announced that Offers API v10 stops on 1 February 2027.

That last sentence is the Deprecated Endpoint in public form. **A deprecation schedule is a contract clause with a date**, and someone at the merchant must own the calendar even when the connector vendor does most of the work; Promises Between Systems: API Contracts, Webhooks and Versioning owns the general discipline.

Channel Allocation: The Number You Send Is a Promise

The stock figure a connector sends to a marketplace is not a report. It is **the quantity that marketplace is now entitled to sell**, and the marketplace will sell it without asking again. Amazon even publishes a notification (`LISTINGS_ITEM_MFN_QUANTITY_CHANGE`) so a seller can see what Amazon currently believes its quantity to be.

Channel allocation is the decision of how much of your available stock each channel may promise. Connectors support several policies, and the choice follows the merchant's stage:

Policy	How it works	Fits	Breaks when
Shared pool	Every channel is told the full available quantity	One channel, or very deep stock	Two channels sell the last units in the same window; the Oversell
Fixed quantity per channel	Each marketplace gets a set allocation, replenished by hand	A few hero products, a small team	Stock sits unsold in one channel while another runs out
Buffer	Each channel is told available minus a safety margin; zero below it	Hourly or slower sync; the interim fix	Long tails: the last units never sell anywhere

Policy	How it works	Fits	Breaks when
Percentage split	Each channel gets a share of available stock	Channels with predictable demand ratios	Demand shifts (a promotion on one marketplace) and the split is stale
Computed available-to-promise per channel	One availability service computes what each channel may still sell, from stock events, reservations and channel rules, and publishes changes as they happen	Omnichannel and Group; several warehouses and external locations	Only if the events it depends on are late or missing

Figure 4: *The number you send is a promise: several channels draw on one stock, so one availability calculation decides how much each channel may still sell, instead of each channel reaching into the jar.*

The last row is what Is It in Stock? Inventory, Availability and the Promise You Can Keep builds and what Waypoint's W1 buys as a module of the OMS (order management

system, the hub that holds every channel's orders). The principle behind it is simple: **compute available-to-promise once, from the system of record for physical stock, and publish it to every channel from that one computation**. A connector that holds its own stock copy and applies its own rules is a fourth system of record, which is what Kestrel & Pine has in January 2027: the connector keeps a copy of product data for listings and a copy of stock that is up to two hours old.

Marketplace-fulfilled stock adds a second rule. Units in a marketplace's warehouse are **an external location** with its own feed (Amazon's `FBA_INVENTORY_AVAILABILITY_CHANGES` notification exists for exactly this), never a copy of your warehouse figure; the availability service tracks the location, and the marketplace promises from it.

Kestrel & Pine after the Oversell

Milan Petrov's first fix, set in December 2026, was a buffer. The connector now publishes the platform's figure **minus ten units per variant** to each marketplace, and zero below ten *(illustrative)*. With a two-hour sync lag, ten units is enough for most variants; for the Ridgewalker on a Black Friday evening it would have held for minutes. The buffer has a cost: across roughly 17,000 active variants, the last few units of each are never offered on marketplaces at all.

The target, under W1, is this: the WMS is the system of record for stock at Duiven; the marketplace's fulfillment stock in Germany is a tracked external location; the availability service computes available-to-promise per channel from stock events and publishes changes to the connector as they happen; the connector forwards them to each marketplace within minutes, as bol's rate-limit guidance asks. The hourly chain is retired by Q4 2027, and the buffer shrinks to what the event latency justifies.

A channel rule can then say, for example, "hold the last 20 units of each variant for the storefront, where the margin is highest" *(assumption)*: the connector sends the marketplaces zero while the storefront keeps selling. None of that is possible when the connector only knows "the platform said 410 an hour ago".

Prices across channels

The same discipline applies to prices. **Price parity** clauses, which required a seller's marketplace price to be no higher than its price elsewhere, were dropped by Amazon in the EU in 2013 after the German competition authority's proceedings (Bundeskartellamt, 26 November 2013; an old source, but the origin of today's rules). Marketplaces still compare prices to decide visibility: Amazon's `PRICING_HEALTH` notification tells a seller that an offer is "ineligible for the Featured Offer (Buy Box) because its price is uncompetitive", and bol's pages say good performance "improves your chance of appearing in the buy box".

Repricing is an automated rule that moves a listing's price within bounds in reaction to competitors; ChannelEngine sells it as "dynamic pricing, margin guardrails, and cost-based insights to adjust prices per channel" (n.d., accessed November 2026). The architectural rule: repricing acts on a **channel price** derived from the merchant's pricing source, never on the base price list, and it is bounded by a **floor** computed from cost and fees.

For the Ridgewalker at €199 including 21% VAT, with an assumed 15% commission (the most common referral rate on Amazon's US pricing page; European rates differ) and €6 of shipping, about €128 remains against a landed cost of €58 *(assumptions)*, so the floor sits well above cost.

A connector rule allowed below the floor can undercut the brand's own storefront, which is one way the Price Drift of October 2026 (€179 in Germany for nine days, and a parity warning from a marketplace) happens. What Does It Cost? Price Lists, Contract Prices, Currencies and Tax owns the channel price as a derived price with a source, an effective date and a floor.

The Choices

A merchant has four realistic ways to connect to marketplaces, and the order-record decision cuts across all four.

1. **The marketplace's own seller tools**. Upload listings by spreadsheet, read orders in the seller portal, key them in, confirm shipments by hand. Right at a few orders a day on one marketplace; wrong when a second marketplace arrives or the late-shipment clock starts biting.

2. **The platform's native channel app**. Hosted platforms ship or list channel apps (Shopify's Marketplace Connect and its social and Google channels, for example). Orders arrive in the platform's own order list and stock is the platform's stock. Cheap, quick and adequate at the Growing stage for the marketplaces the app covers; the reviews are the warning (Marketplace Connect's listing shows a 4.1 rating with 15% one-star reviews citing sync and pricing errors, accessed November 2026). The limit is structural: the app treats the platform as the system of record for stock and orders, true at stage 2 and false at stage 3.

3. **A channel integration platform**. A specialist SaaS (software as a service: rented, hosted software) product (ChannelEngine, EffectConnect, Rithum, Linnworks and others) with ready-made marketplace connections, listing mapping, allocation rules, repricing and order import into your platform, OMS or ERP. This is Kestrel & Pine's choice and the common one for Omnichannel merchants in Europe: tracking dozens of marketplaces' content rules and APIs is the vendor's business. The risk is the fourth-copy problem: the connector *can* hold your stock, prices and products, and will if you let it.

4. **Direct integration**. Build against the marketplaces' APIs yourself. Right for Group-stage merchants with the volume to justify engineers on marketplace APIs, or where no connector expresses a rule the business needs. Below that, you inherit every deprecation, feed quirk and rate limit, per marketplace.

Across all four runs the question **where the marketplace order lives**: either the connector (or app) creates the order in your order record and mirrors status back, or the order lives in the connector and your systems read it there. The first gives one order record with one state model; the second is cheaper at the Growing stage and produces the Three-Status Order the moment stores and EDI arrive. Kestrel & Pine's W2 chooses the first.

	Marketplace tools by hand	**Platform channel app**	**Channel integration platform**	**Direct integration**
Marketplaces covered	Any, one at a time	The few the app supports	Dozens to hundreds	Those you build
Who tracks API changes	Nobody	The platform vendor	The connector vendor; you own the connector's own versions	You
Stock allocation	By hand	Platform stock, sometimes per-channel availability	Rules per channel; risk of a private stock copy	From your availability service
Order record	Keyed in	The platform's order list	Configurable: create in the OMS or hold in the connector	Your OMS
Cost	Staff time	Low subscription or per-order fee	Subscription (Kestrel & Pine: €900 a month)	Engineers
Fits	Starting	Growing, D2C-first (direct-to-consumer)	Omnichannel	Group, or a rule no vendor expresses

My recommendation for a merchant at the Omnichannel stage is the third option with two conditions written into the contract and the design: the connector creates orders in the order record and does not keep them; and the stock number it publishes comes from the merchant's availability calculation, not from a rule inside the connector. Both conditions assume the merchant has, or is getting, an order record and an availability service. Without them, the connector will own those facts by default, because something has to.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	One marketplace, or none; seller portal and spreadsheets; orders keyed into the shop; one fulfillment model	A decision whether the marketplace is a test or a channel; GTINs on every variant before listing	More than an hour a day copying orders; the first late-shipment warning
2 — Growing	Two or three marketplaces through the platform's channel app or a connector; the platform's stock is the figure sent everywhere; orders mirrored into the platform	A buffer per channel; one owner for the seller account and its scorecard; the deprecation calendar on someone's list	A second order source (a store or wholesale) makes the platform's order list incomplete; a strike or a restricted category

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel	A channel integration platform over several marketplaces and countries; both fulfillment models; marketplace-held stock	Available-to-promise per channel from an availability service; marketplace orders created in the OMS; channel prices with floors; ship confirmation by event from the WMS; reconciliation of connector stock against the system of record	Oversells at peak; the connector holds the only copy of marketplace orders; nobody can say what a marketplace believes the stock is
4 — Group	Many marketplaces per brand and country; direct integrations where volume justifies; a marketplace as logistics provider for the brand's own shop; perhaps operating a marketplace	A channel syndication service over the PIM (product information management system) with reduced, channel-specific views; fee schedules and performance data in the data platform; seller-identity obligations if operating a marketplace	Each brand runs its own connector with its own rules; margin per channel unknown; a marketplace program change forces a re-architecture

The judgement: the mechanism scales with the number of *order sources*, not with order volume. One marketplace feeding the platform's order list works at stage 2 at any volume the platform handles. The moment a store, a wholesale portal or a second marketplace also creates orders, the platform's order list is no longer the order record, and the marketplace integration must be re-pointed at whatever is. Kestrel & Pine crossed that

line in 2023 and noticed in December 2026.

In Practice

Marketplaces retire interfaces on their own calendar. Amazon's older MWS interface was no longer available from 1 April 2024, after a migration schedule that had already moved once (SP-API changelog, 2023–2024); bol's Offers API v10 stops on 1 February 2027. Programs retire too: Zalando announced in April 2026 that Connected Retail, under which "stationary retailers sold branch stock directly through the platform and shipped orders from their stores", ends by the end of June 2027 (TextilWirtschaft, 24 April 2026). A merchant whose store-stock exposure was built on Connected Retail's tool rebuilds it; one whose store stock is a location in its own availability service changes a connector (Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment).

The marketplace is becoming infrastructure. Zalando's ZEOS, announced on 17 October 2023 as "one stock pool, one carrier integration and one user interface", was by December 2025 available in 24 markets and serving 18 sales channels, including brands' own web shops (Zalando corporate releases, 2023 and 2025). For a merchant that means a third kind of stock location, marketplace-held but multi-channel, and a dependency on the marketplace's APIs for its *own* storefront's orders.

Social channels switch between feed and order source. Meta moved Facebook and Instagram Shops from in-app checkout to website checkout in 2025 (trade press, June 2025), so for most merchants it is again a **shopping feed** destination: a structured product file with title, price, availability, GTIN and image.

TikTok Shop went the other way, opening in-app checkout in Germany, France and Italy on 31 March 2025 (widely reported) and in the Netherlands, Belgium, Austria and Poland from 15 June 2026 (TikTok newsroom, 28 May 2026).

A social channel may send shoppers to the merchant's checkout or take orders itself. Supply its catalog from the product system of record and route any orders through the marketplace import, so those purchases join the same stock and fulfillment controls. **Retail media**, sponsored marketplace listings, adds a per-channel cost and a reporting stream; it does not create a separate order flow.

Operating a marketplace is a different book. Mirakl sells the operator side (seller onboarding, per-seller offers, order splitting, payouts) to over 450 companies (Mirakl and Wikipedia, accessed November 2026). The Digital Services Act adds the legal core: since 17 February 2024, Article 30 requires platforms that let consumers buy from traders to obtain and verify the trader's identity, payment account and register details before they can sell, and to suspend traders who do not supply them (Regulation (EU) 2022/2065); that is why every marketplace onboarding asks a merchant for identity documents and bank details. The operator's architecture is the subject of LISTED; start with Rules of the

Marketplace: DSA, DAC7 and Vertical Regulation and Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics.

Kestrel & Pine: the marketplace variant, 13 April 2027

Two Amazon.de customers order a Ridgewalker jacket on the day of Eva's order. Here is each order's path in April 2027, before W1 and W2 land, and what changes after.

Step	Seller-fulfilled order (shipped from Duiven)	Marketplace-fulfilled order (shipped by Amazon)
Where the order appears	The connector pulls it from Amazon and creates a connector order; the iPaaS (integration platform as a service, which runs Kestrel & Pine's data flows) passes it to the WMS; it reaches the ERP only in the daily finance batch	The connector sees it in Amazon's reports; no warehouse work; it reaches the ERP only in the finance batch
Stock effect	Duiven stock falls when the WMS picks; the ERP learns within 15 minutes, the platform within an hour, the connector within two	The German external-location quantity falls by one; Kestrel & Pine learns it from the marketplace's inventory feed
Shipment	Picked and packed 14 April, DHL label from the shipping service; tracking flows WMS → iPaaS → connector → Amazon the same day	Amazon ships and sends tracking to the customer; Kestrel & Pine is told after the fact
Scorecard	Cancellation, late shipment, valid tracking and on-time delivery all apply	Only the order defect rate applies

Step	Seller-fulfilled order (shipped from Duiven)	Marketplace-fulfilled order (shipped by Amazon)
Return	Announced by Amazon, acknowledged by the connector, received at Duiven	Handled by Amazon from the external location
After W1 and W2	Created in the OMS by the connector with the Amazon order id and ship-by date; available-to-promise reduced by event; ship confirmation by event from the WMS	Created in the OMS as a marketplace-fulfilled order for finance and analytics; the external location adjusted by event; no fulfillment order

The second column is the one most merchants get wrong: because the marketplace does all the work, the order never enters the order record, and revenue, returns and the external location's stock are reconstructed from the marketplace's statements at month-end. Hanna Berg's three definitions of "order" (W7) are partly this.

When It Goes Wrong

Failure	How it shows	Detection	Recovery
Duplicate order delivery (notification and poll both deliver it; the connector retries)	Two internal orders, two parcels, one payment	Uniqueness on the marketplace order id at creation; a daily count of marketplace orders by id against internal orders	Cancel the duplicate before pick; if shipped, treat as a return; make the import idempotent

Failure	How it shows	Detection	Recovery
Stale stock (sync failure, a retired API version, a feed rejected)	The marketplace keeps selling a sold-out variant; the cancellation rate rises	An **age monitor** per channel: alert when the last accepted stock update is older than the agreed interval; read every feed result (Amazon keeps them 28 days); compare what the marketplace believes with available-to-promise	Push a full stock snapshot; fix source data in the product system of record, not in the connector; cancel oversold orders early
Late ship confirmation (warehouse queue, iPaaS outage)	The late shipment rate climbs though parcels left on time	Compare carrier scan time with confirmation time daily	Send confirmations by event from the WMS; backfill from the shipping service's records
External-location drift (marketplace-fulfilled stock)	Your figure and the marketplace's differ; replenishment is planned on the wrong number	Weekly reconciliation of the external location against the marketplace's inventory report	Adopt the marketplace's figure as the location's truth and investigate the difference
Return announced by the marketplace, unknown to the returns portal	The unit arrives at Duiven with no record; the marketplace refunds while a second refund is prepared internally	Every marketplace return acknowledged into the order record with the marketplace return id	One refund path per channel; The Way Back: Returns, Exchanges, Refunds and Reverse Logistics

Failure	How it shows	Detection	Recovery
A retired API version	Everything stops, quietly, on a published date	The deprecation calendar in the integration inventory with an owner per entry; the age monitor above	Upgrade; meanwhile a manual stock snapshot through the seller portal

The **age monitor** is the cheapest control in the chapter: for each channel, the time since the marketplace last accepted a stock update. It would have caught the Deprecated Endpoint within the hour instead of after two days. **Early cancellation** runs against instinct: once an oversell is certain, canceling at once costs a cancellation-rate point, while waiting for the stock to "turn up" costs a late shipment and often a defect as well.

What to Take Away

Established.

- Marketplaces grade sellers on published performance metrics with consequences; Amazon's thresholds (cancellation rate under 2.5% over 7 days, late shipment rate under 4%, valid tracking above 95%, order defect rate under 1%) and bol's standards and strike process (at most 2% cancellations a week; five strikes close the shop) are documented as of 2026.
- Marketplace APIs are rate-limited and asynchronous, and versions retire on published dates: Amazon's `getOrders` allows one call a minute and Amazon recommends notifications over polling; bol removes deprecated major versions after 12 months. An integration must be event-driven, idempotent and owned.
- The two fulfillment models are architecturally different: seller-fulfilled orders enter the merchant's order and warehouse flow with a deadline; marketplace-fulfilled stock is a location the merchant does not control and learns about from the marketplace's feeds.

Depends on context.

- A platform's native channel app is enough while the platform is the only source of orders and holds the only stock figure (Starting and Growing stages). Once a store, wholesale or a second system creates orders, marketplace orders must be created in whatever the order record is.
- A stock buffer per channel is the right interim allocation when sync is hourly; computed available-to-promise per channel is right when stock moves by events and several channels and locations compete. The buffer's cost (units never offered) grows with the catalog's long tail.

- Direct marketplace integrations are right only with the volume to keep engineers on the marketplaces' APIs, or a rule no connector expresses; below that, rent the connector and own the contract.

The author's synthesis.

- The connector's job is translation, and the one design rule I insist on is that it owns nothing: not the product content, not the stock figure, not the order. Every incident in this chapter is a connector that was allowed to own a copy because nothing else did.
- Treat the marketplace scorecard as an integration SLA (service-level agreement) you did not negotiate: derive a freshness requirement per flow from the metrics, and monitor the age of the last accepted update per channel before anything else.

Questions to Ask

1. Which system computes the stock figure each marketplace is told, and does the connector apply any rule of its own on top of it?
2. Where is a marketplace order created, and can support, the warehouse and finance find it by the marketplace order id?
3. Who owns the seller account's scorecard, and who sees the cancellation and late shipment rates weekly?
4. How long does a ship confirmation take to reach the marketplace, and what happens when the iPaaS is down?
5. For marketplace-fulfilled stock, which system holds the external location's quantity, and how often is it reconciled against the marketplace's report?
6. Who holds the deprecation calendar for the connector's own API and for each marketplace's API, and when is the next retirement date?
7. Is the price sent to each marketplace derived from the pricing source with a floor, and can a repricing rule go below it?
8. If a marketplace closed our account tomorrow, what revenue and which stock would be stranded?

Marketplaces are one kind of order that arrives from outside the shop. The next chapter turns to the other kind Kestrel & Pine lives on: the 1,100 wholesale accounts whose buyers are companies, with contract prices, payment terms and credit limits that no consumer checkout knows how to express. Continue with Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals.

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

21. Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals

IN THIS CHAPTER, YOU WILL: Learn what is different when the buyer is a company, the records every business-to-business (B2B) platform needs, who should own accounts, prices and credit, and the choices for building a wholesale channel.

KEY POINTS:

- **The customer stops being flat.** A business buyer is a company with locations, several people and roles, an assortment, a price agreement, payment terms and a credit limit. Shopify, commercetools, SAP, Adobe and BigCommerce all model it that way, under different names, and the model must have one owner.
- **A B2B price is a promise with a validity date.** List, tiered, contract and quoted prices coexist and must say which wins; a quote freezes a price, and the invoice must match what the portal showed, or the wholesale team keys orders in by hand to make it so.

- **Credit and terms are finance facts.** Payment terms and credit limits usually belong to the ERP (enterprise resource planning system, the back office for finance, purchasing and invoicing), which is why a portal that cannot read them at checkout cannot replace e-mail and phone; the order record, not the ERP's order entry, is where a wholesale order should live.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Price Nobody Can Explain** — the portal shows one wholesale price, the ERP invoices another, and the negotiated pre-season discount lives in an e-mail.
- **Schrödinger's Order** — portal orders live in the platform, keyed orders in the ERP, EDI (electronic data interchange) orders in the EDI provider's mailbox; one customer, three order lists.
- **The Spaghetti Junction** — each large buyer asks for its own way in (a portal login, a punchout session, an EDI mailbox, a spreadsheet by e-mail), each with its own connector.

A consumer buys a jacket and pays before dispatch. An outdoor shop may order sixty jackets at a negotiated price, request delivery to two locations and pay thirty days after invoicing. The buyer, approver and bookkeeper may be different people. **Business-to-business (B2B) commerce** must represent that agreement, or checkout can accept an order the seller cannot fulfill on the promised terms.

The agreement includes an account price, quantity breaks, **payment terms** specifying when the invoice is due, and a **credit limit** on what the customer may owe. Orders may require approval or delivery in agreed size combinations before a season starts. Checkout must check those conditions before making a commitment.

Those checks depend on knowing **who owns the company account**, **contacts**, **prices**, **terms**, **credit and orders**. If checkout reads an old credit balance or a different price list, having the right features will not protect the agreement. Evaluate how the systems obtain and enforce the same answers.

Where we are in the order's journey. Kestrel & Pine, the fictional outdoor brand, sells to consumers such as Eva and to wholesale customers. This chapter follows independent shops ordering through its portal or by email. Retail chain Bergzicht Outdoor sends structured orders through **electronic data interchange (EDI)**; Documents Between Companies: EDI, Punchout and E-Invoicing and Journey 3: A Chain Reorders by EDI, From Order to Invoice follow that route.

Setting the Scene at Kestrel & Pine: The 38%

Wholesale lead Lea Moreau manages **1,100 active accounts**: 640 in the Netherlands, 210 in Belgium and 250 in Germany. Most are independent outdoor shops; Bergzicht is the one retail chain using EDI. Wholesale produced **€12.0 million** in 2026 through **9,400 orders**, averaging **€1,277**, on 30-day terms.

Since 2023, shops have used the platform's **B2B portal**, with company accounts, price lists and payment terms in the same platform account as the consumer storefronts. It receives **62% of wholesale orders**.

The remaining **38% arrive by email or phone** and are keyed into the **enterprise resource planning system (ERP)**. That is where Finance Director Sanne de Wit's team manages terms, credit limits and invoices. The portal needs those facts to handle more of the work.

When Noor Visser asks Lea in February 2027 why the 38% has not fallen since 2024, the answer is not that shops dislike the portal. It is that the portal cannot express what those shops have agreed *(illustrative)*:

- The portal shows one **wholesale price list per country**. The accounts that order by e-mail are mostly the larger ones, whose **pre-season terms** (an extra discount on orders placed before the season, agreed per account) live in the ERP's customer price lists and in Lea's e-mail, not in the platform.
- The portal cannot see **credit**. A shop near its limit can place a portal order that finance then holds, by hand, a day later; the team keys larger orders into the ERP because the ERP checks the limit at entry.
- A portal order and a keyed order are **two different records**: one in the platform, one only in the ERP. To see one shop's orders, Lea opens both.

That is the chapter's problem in one company. The portal is good at what the platform owns (the catalog, the cart, the checkout) and weak at what the ERP owns (terms, credit, negotiated prices, invoices), and neither holds the order record for the whole channel. Waypoint's answer, W2, is that every wholesale order, portal, keyed or EDI, is created in the OMS (order management system, the one record of every channel's orders) from Q4 2027, and the ERP keeps finance.

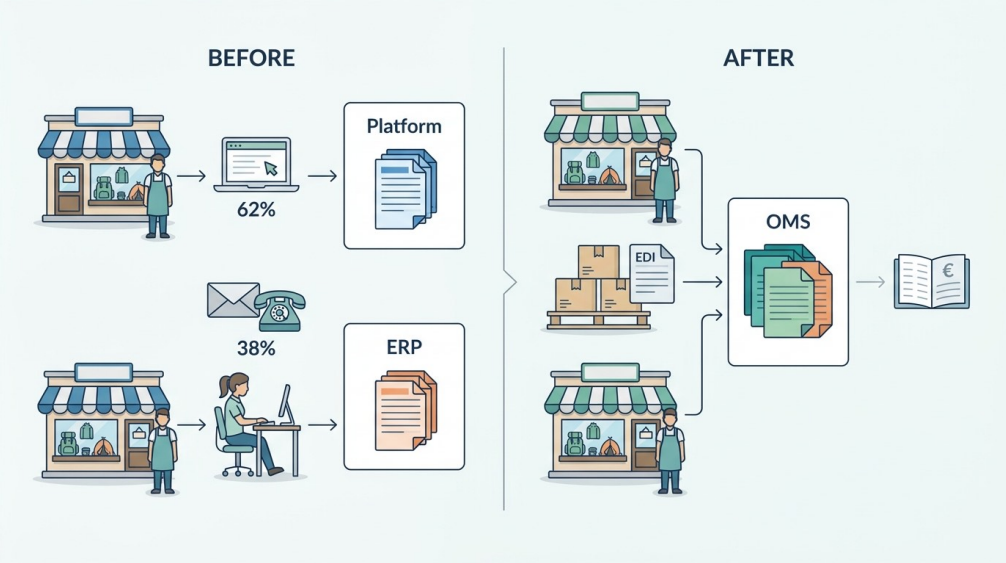

Figure 1: *Today 62% of Kestrel & Pine's wholesale orders arrive through the portal and 38% are keyed into the ERP, so one shop's orders live in two records; under W2 every wholesale order is created once in the OMS.*

What Is Different When the Buyer Is a Company

The contrasts below are industry practice; the vocabulary is defined as it appears.

	Consumer (B2C)	Business (B2B)
Who buys	One person, one account	A **company account**: an organization with several locations, several users and roles (who may order, who may approve, who may see invoices)
Price	One list price, promotions on top	**Contract pricing** (a price agreed with one customer or group), **tiered pricing** (price breaks by quantity), customer-specific assortments; prices quoted **excluding VAT**

	Consumer (B2C)	Business (B2B)
How the price is agreed	Fixed	Often negotiated; a **quote** (the seller's offer at a stated price with a validity date) that the buyer accepts into an order
Payment	At checkout, before shipment	On **payment terms** (net 30: due 30 days after the invoice) within a **credit limit**; sometimes card or direct debit
Order size and units	One or two items	Dozens to hundreds of units in cartons, inner packs and size runs; sometimes pallets
When orders are placed	Any time	In apparel, a **pre-season order** months ahead for delivery in a window, then **in-season reorders** of what sells
Who may commit	The buyer	The buyer, subject to the buyer's own **approval workflow** (a manager must approve above a limit)
Document that matters	The order confirmation	The **invoice**, matched by the buyer against its purchase order and the delivery

Each row introduces a fact that has to live somewhere and be readable by the checkout at the moment of the promise. Three of them, **the company account**, **the price and the credit limit**, are the ones the ERP and the platform fight over.

The vocabulary the platforms share

Whatever the vendor, the documented object models converge on the same shapes. The table is my synthesis of the vendors' documentation as of November 2026; a blank cell means "not verified", not "absent".

Concept	Shopify	commercetools	SAP Commerce (Spartacus docs)	Adobe Commerce B2B	BigCommerce B2B Edition	OroCommerce
The buying organization	Company, Company-Location, Company-Contact	Business Unit (Company and Divisions, up to five levels) with Associates	B2BUnit tree with B2BUser	Company account, company hierarchies	Company account with users	Customer, customer group, customer user
Who may do what	Contact permissions (ordering only, location admin)	Associate Roles	Roles including approver and administrator; purchase limits	Company roles; administrators bypass rules	Roles controlling Buyer Portal access	Customer user roles
Customer-specific prices and assortment	Catalogs (price lists plus product sets) assigned to company locations	Stores and prices per Business Unit	Cost centers, budgets	Shared catalogs with custom pricing	Price lists per company	Price lists with scheduled adjustments

Concept	Shopify	commercetools	SAP Commerce (Spartacus docs)	Adobe Commerce B2B	BigCommerce B2B Edition	OroCommerce
Negotiation	Draft orders for review and approval	Quote Request, Staged Quote, Quote	Quotes	Quotes	Sales quotes	RFQ (request for quote) to Quote, with approval workflow
Spending control	— (apps)	Approval Rules and Approval Flows	Order approval, budgets, purchase limits	Purchase orders with approval rules	—	Quote approval workflows
Pay later	Payment terms per company location (net, fixed date, on receipt, on fulfillment)	— (payment integrations)	—	Payment on Account	Net terms and available credit	—

Shopify's developer documentation defines a Company as "Information about the business entity that makes a B2B purchase", a CompanyLocation as "A single location or branch of the company" to which catalogs, tax exemptions and payment terms are assigned, and a CompanyContact as "A person that acts on behalf of the company"

(shopify.dev, accessed November 2026). commercetools says "Business Units are representations of Companies or Divisions", nesting up to five levels, with Associates whose roles "grant Associates different permissions" (commercetools documentation, accessed November 2026). The structure is the same: **organization**, **places**, **people**, **permissions**, with prices and terms attached to the organization or the place, not to the person.

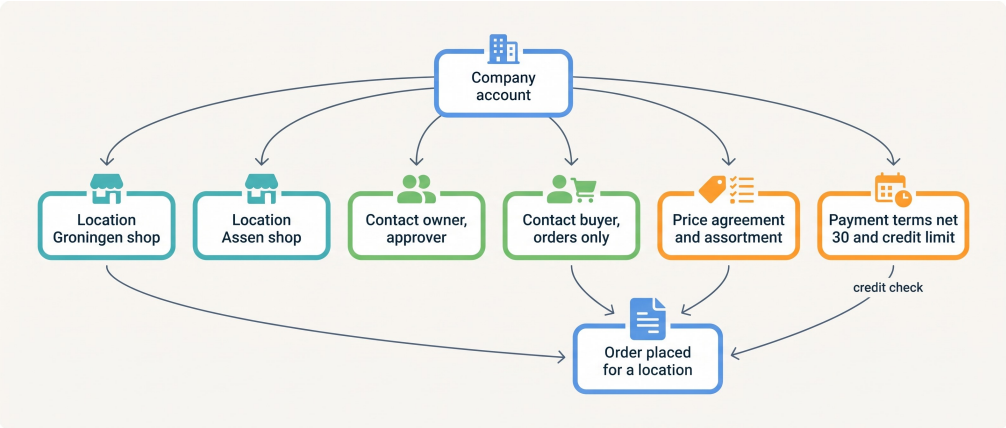

Figure 2: *The B2B object model every platform shares: an organization with locations and people, with prices and terms attached to the organization, and an order that draws on all of them at checkout.*

Who Owns What

The company account is the first place the customer record stops being flat, and the first place two systems claim it. The commerce platform needs the company, its contacts and its catalog to run the portal. The ERP needs the customer, its terms and its credit to invoice and to manage receivables. Both can hold all of it; neither should.

Microsoft's documentation for its Business Central to Shopify connector states the problem and the usual answer without hedging: merchants must "decide which system is the system of record for customer data. Most businesses manage customer records, payment terms, and credit limits in Business Central and export them to Shopify … Mixing both directions without a clear ownership rule can lead to conflicting records" (Microsoft Learn, 14 October 2026). The connector imports company locations with payment terms and tax IDs and can create a B2B catalog per exported company so that portal prices follow the ERP customer's price group. Shopify's own 2026 guidance on ERP integration says the same in general: "Document every field that moves between systems, and name the system of record for each" (Shopify enterprise blog, updated 28 August 2026).

My ownership table for a merchant with an ERP, a commerce platform with B2B features and, at the Omnichannel stage, an OMS:

Fact	System of record	Readers	Why there
Company identity, legal name, VAT number, locations	The ERP (or a master-data system at Group stage)	Platform, OMS, invoicing, EDI	The ERP must know the customer before it can invoice; the VAT number drives tax treatment
Contacts, logins, roles, permissions	The platform (or an identity service)	Portal, service tools	Logins and sessions are the platform's job
List and tiered wholesale prices	The pricing source: ERP price lists or a pricing engine (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax)	Platform, OMS, EDI	One source, published to channels; the platform's catalogs are a copy with an effective date
Contract and pre-season agreements	The same pricing source, as customer-specific price lists	Platform, OMS	If they live only in e-mail and the ERP's order entry, the portal cannot show them, and the 38% stays
Quotes	The platform or OMS, as an explicit object with a validity date	Portal, sales reps, OMS	A quote is a frozen price; the order is created from it
Payment terms, credit limit, open balance	The ERP (receivables)	Platform or OMS at checkout, by a credit check	Finance facts; the ERP knows what is owed

Fact	System of record	Readers	Why there
The customer's own approval rules	The platform or OMS	Portal	The buyer's internal control; invisible to the ERP
Orders (portal, keyed, EDI, punchout)	The OMS at stage 3 and beyond; the platform at stages 1 and 2; never only the ERP's order entry	ERP (for posting), WMS (warehouse management system), service	One order record with one state model (The Order Record: Lifecycle, State and the Order Management System)
Invoices, credit notes	The ERP	Peppol (the European e-invoice network) or EDI gateway, portal (for display)	Numbering, archive and legal weight live here (Tax, Invoices and the Books: From Order to Ledger)

The pattern is that **the ERP owns the money facts and the platform owns the session facts**, with prices sourced once and published. For wholesale and contract prices that source is the ERP: it publishes them as price events with effective dates to the OMS and the platform's B2B catalogs, and feeds no channel directly, so no storefront reads the ERP live. It is the split SAP describes for its own suite: master data replicated from the ERP to the commerce system, storefront orders sent to the ERP (SAP Learning, accessed November 2026). The Back Office: ERP Integration and Who Is the Hub works out when the ERP should also be the live source at checkout and when it should not.

What the flows need

Interaction	Freshness	Consistency	Latency at checkout	When it fails
Price lookup for a company	Effective-dated; changes published hours ahead	The portal price must equal the invoice price for the same date and quantity	Precomputed per catalog; millisec-onds	Recompute at invoicing and hope: The Price Nobody Can Explain
Credit check at order placement	Open balance as of today, including unposted shipments	One answer from the receivables system	Under a second if syn-chronous *(assump-tion)*; otherwise place the order and hold it	Nightly copy of the balance: an order accepted at 09:00 against yesterday's balance
Order creation	Immediate	One record, one number, idempotent on the buyer's purchase order number	Seconds	Keyed duplicates of e-mailed orders
Invoice on shipment	Same day as shipment	Generated from what shipped at the price the order carried	Batch	Invoice from the order rather than the shipment: short shipments invoiced in full

The credit-check row decides whether the portal can replace e-mail. If the ERP's open balance is copied to the platform nightly, the portal is a day behind and finance keeps holding orders by hand. If the platform or OMS asks the ERP at order time, or receives balance changes as events, the portal can say "this order needs approval from Kestrel & Pine" at once. Stock for wholesale follows the same logic: it is promised from available-to-promise (stock not yet promised to another order) with a wholesale allocation, so that wholesale and consumer channels do not sell the same units (Is It in Stock? Inventory,

Availability and the Promise You Can Keep).

How a Price Becomes a Promise

B2B pricing is layered, and every layer must say whether it beats or adds to the one below.

1. **List wholesale price**: the price in the seasonal wholesale price list, excluding VAT, per country or currency. Kestrel & Pine's Ridgewalker is €95, the Trailhead fleece €57, merino socks €8.50.
2. **Tiered or volume price**: breaks by quantity. Shopify's catalogs support "discounted price breaks when a customer buys a set quantity" and quantity rules per order (Shopify Help Center, accessed November 2026).
3. **Contract or customer-group price**: a price agreed with one account or group, usually a percentage off list or a fixed price per product, with an effective period. Shopify expresses it as catalogs assigned to company locations; Adobe as shared catalogs "with custom pricing for different companies" (Adobe B2B guide, 1 October 2026); OroCommerce as price lists with scheduled adjustments (OroCommerce documentation, version 7.0).
4. **Quote**: a one-off price for a specific request, with a validity date.

The rule that matters is which layer wins. Commonly the most specific wins (quote over contract over tier over list), and discounts do not stack unless the agreement says so. Write it down and implement it once in the pricing source, so the portal, the OMS and the invoice compute from the same rule; a portal that applies the rule its own way produces the invoice dispute.

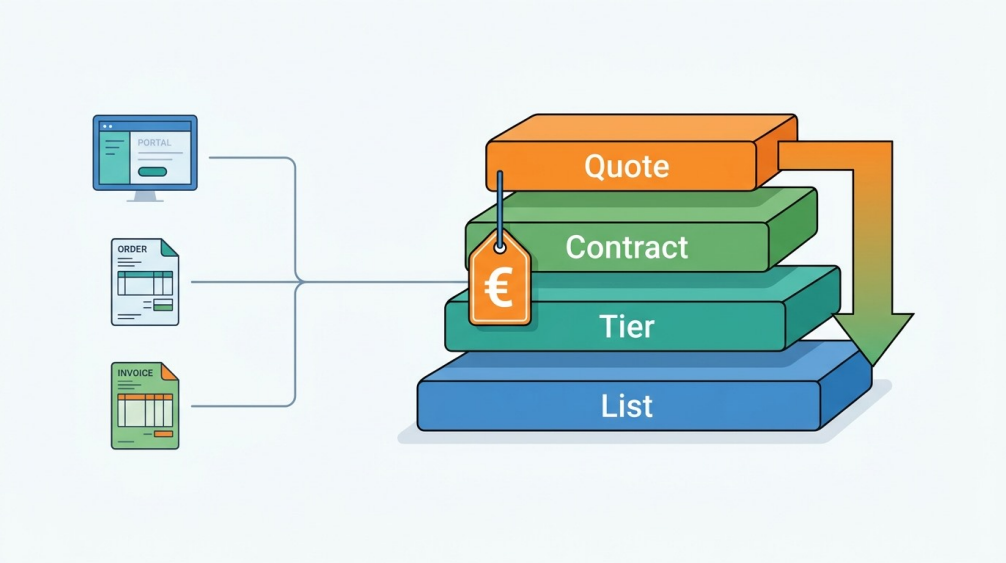

Figure 3: *B2B prices are layered: the most specific layer wins, and the portal, the order and the invoice must all compute the price from that one rule.*

A quote freezes the price. commercetools models negotiation as three objects, Quote Request, Staged Quote (with a `validTo` date) and Quote, and says "Quotes are legally binding custom offers presented to a Buyer" (commercetools documentation, accessed November 2026). Adobe's purchase-order flow states the consequence for the order built from one: "If the price of an item changes after the PO is created, the original price is used" (Adobe Commerce documentation, accessed November 2026).

On Shopify, no native "request a quote" page was found in November 2026; the documented mechanism is the **draft order** a merchant creates "for contacts to review and approve", into which third-party quote apps convert accepted quotes.

Under every name the decision is the same: the moment a seller states a price to a named buyer with a validity date, that price is stored as an object the order is created from, not recomputed later from whatever the price list then says.

VAT follows the buyer. B2B prices are shown and invoiced excluding VAT, and the treatment depends on who the buyer is and where: a Dutch shop is invoiced with Dutch VAT; a German or Belgian business customer with a valid VAT identification number is commonly invoiced without Dutch VAT under the EU rules for cross-border business supplies, and accounts for the tax itself. Tax, Invoices and the Books: From Order to Ledger owns the rules; this chapter only insists that the VAT number is master data with an owner and a validation date, not a free-text field on the portal signup form.

Credit, Terms and the Buyer's Own Controls

Payment terms are typed, not free text. Shopify's Admin API (version 2026-10) models them as `NET` ("Payment is due a set number of days after the invoice is issued"), `FIXED` (a calendar date), `RECEIPT`, `FULFILLMENT` and `UNKNOWN`, with templates carrying the number of days. The due date is then computed from the invoice date, and the invoice date is the ERP's. This is why terms belong to the ERP and are *mirrored* to the platform: the platform needs them to show "net 30" and skip card payment at checkout; the ERP needs them to age receivables and chase payment.

The credit limit is the seller's control: the most a customer may owe. A credit check at order time compares the open balance (unpaid invoices, plus shipped orders not yet invoiced, plus accepted orders not yet shipped, depending on policy) with the limit, and accepts, holds for approval or rejects. BigCommerce's B2B Edition documents "available credit, and net terms" in payment settings (accessed November 2026); Microsoft's connector can carry credit limits to Shopify as metafields (custom fields), which is a mirror, not a check.

The buyer's approval workflow is the opposite control: the buyer's own rule that an order above an amount needs a manager. commercetools' Approval Rules are tiered ("Each tier contains one or more roles that are required to approve an Order before it can proceed to the next tier"), inherited down the Business Unit hierarchy, and "a single instance of rejection will lead to the rejection of the entire Approval Flow". SAP Commerce's organization model adds cost centers, budgets and purchase limits (Spartacus documentation, which describes an unmaintained storefront version); Adobe lets company administrators "bypass all approval rules". For a brand selling to independent shops, approval workflows are rarely needed; for one selling to a chain's 38 branches, or to corporate buyers, they are the difference between an order and a request.

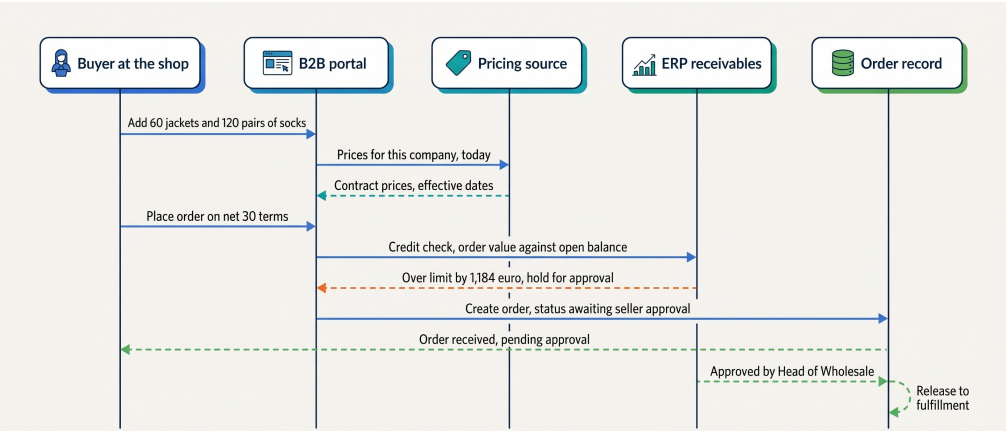

Figure 4: *A portal order at the Omnichannel stage: prices come from the pricing source,*

the credit decision from the ERP, and the order is created once in the order record, where the approval and the release are states, not e-mails.

The Choices

A merchant that sells to businesses has four realistic ways to give them a channel, and the choice depends less on features than on who the buyers are and which system is fast enough to answer at checkout.

1. **The B2C platform's B2B features.** Shopify's B2B runs "in one store or use a separate B2B-only store", on the Basic, Grow, Advanced and Plus plans (Shopify Help Center, accessed November 2026; the expansion beyond Plus is recent and undated, and plans below Plus allow at most three active catalog assignments). Adobe Commerce ships company accounts, shared catalogs, quick order, purchase orders, quotes and requisition lists as a B2B module (Adobe guide, 1 October 2026); BigCommerce's B2B Edition serves its APIs from a separate domain, the trace of a partner-built origin.

This is Kestrel & Pine's choice: one tenant, one catalog, one checkout, with company accounts and price lists added. It fits a brand whose wholesale customers are many small shops buying the same products as consumers, and it limits when terms, credit and negotiated prices must come from the ERP in real time and the platform's model cannot express them.

2. **A dedicated B2B platform.** OroCommerce (open source, for "distributors, wholesalers, and manufacturers") models requests for quote, quotes with approval workflows, price lists with scheduled adjustments and order entry by matrix form (OroCommerce documentation, version 7.0); SAP Commerce carries budgets and cost centers from its procurement heritage. These fit a wholesaler or manufacturer whose whole business is B2B, with quotes as the normal path and buyers with their own controls; they are a second platform to run beside a consumer storefront, with a second catalog copy to govern.

3. **An ERP-driven web store.** Sana Commerce's pitch is "built on your SAP or Dynamics ERP", so that "your buyers order online with the prices you agreed and the stock you really have" (sana-commerce.com, accessed November 2026). The store is a window on the ERP: no price copy, no terms copy, no separate order list. It fits a B2B-first company with one ERP and modest traffic; it limits when consumer traffic arrives, when the ERP's response time becomes the storefront's, and when the business adds channels the ERP cannot model.

4. **The buyer's system as the channel.** Large buyers do not log into a supplier's portal. They send EDI orders (Documents Between Companies: EDI, Punchout and E-Invoicing), or they **punch out**: their procurement system opens the supplier's catalog in a session, the buyer fills a cart, and the cart returns to the procurement system as a requisition that later comes back as a purchase order.

The cXML (commerce XML, a procurement message standard) User's Guide (version 1.2.037, dated 2018 and still the "current" link) describes it: "procurement applications display a button instead of product or pricing details … When users are done selecting items, they click a button that returns the order information to the procurement application." SAP's OCI (Open Catalog Interface) is the equivalent for SAP procurement.

A punchout cart returns the buyer's selections for approval; it does not yet authorize the supplier to ship. Wait for the subsequent cXML OrderRequest or EDI ORDERS and bring it into the same order record as a portal order. Building punchout is justified when a buyer requires that approval workflow; otherwise its buyer-specific integration adds work without replacing a missing capability.

	B2C platform's B2B features	**Dedicated B2B platform**	**ERP-driven web store**	**EDI and punchout**
Fits	Brands selling to many small resellers alongside consumers	B2B-first wholesalers and manufacturers with quotes and buyer controls	B2B-first companies with one ERP and modest traffic	A few large buyers who dictate the channel
Where prices live	Platform catalogs, mirrored from the pricing source	The platform's price lists; ERP mirrored	The ERP, read live	The buyer's contract; the supplier confirms
Credit check	Mirror of the limit; a live check needs integration	Built in or integrated	Live, in the ERP	In the ERP on order receipt
Order record	The platform (stages 1–2); the OMS (stage 3)	The B2B platform	The ERP	The OMS or ERP

	B2C platform's B2B features	Dedicated B2B platform	ERP-driven web store	EDI and punchout
Limits	Negotiated terms, approvals and credit beyond the platform's model; catalog caps on lower plans	A second platform and catalog copy; weak consumer features	ERP latency becomes storefront latency; no multichannel model	Per-buyer mapping cost; no self-service
Kestrel & Pine	Today, for 1,100 shops	Not needed	Not chosen: three consumer storefronts share the tenant	Bergzicht by EDI

My recommendation for a brand like Kestrel & Pine is to stay with the platform's B2B features and fix the three gaps that keep the 38% alive: publish contract and pre-season price lists from the ERP into the platform's catalogs as price events with effective dates; add a credit check at order placement against the ERP's open balance (synchronous if the ERP answers in under a second, otherwise accept-and-hold with an event-driven release); and create every wholesale order, however it arrives, in the OMS. That assumes the ERP exposes prices and balances by API and the platform's catalogs can carry per-account lists; where either fails, the ERP-driven store is the better answer for the wholesale channel alone.

Sales reps and B2B2C. The rep's tooling is the seller-side portal: create an order or a draft on behalf of a shop (Shopify's draft orders; BigCommerce's "Masquerade" lets staff act as a buyer), see terms and open balance, convert a quote. A rep's order is the same order object as the shop's own, with a different creator; a rep tool that writes straight into the ERP is another unowned writer. **B2B2C** names a flow, not a product: either a brand feeds product content and stock to a retailer who sells to consumers (a data and document problem, What Are We Selling? Products, Variants and the Catalog), or a business customer's own end users buy under the business's terms, as in a dealer portal, which is the model above with the company holding the prices and terms and its end users as contacts.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	A few resellers served by e-mail and a spreadsheet price list; orders keyed into the shop or the accounting package; payment by invoice at the founder's discretion	Wholesale prices excluding VAT in one list; a written rule for which price wins; GTINs on every variant for the first chain	More than a few wholesale orders a week keyed by hand; a reseller's invoice disputed over a price
2 — Growing	The platform's B2B features or a simple portal: company accounts, one price list per country, net terms; the ERP or accounting system holds customers and invoices; a partner-built connector between them	One system of record for the company account (usually the ERP), mirrored to the portal; typed payment terms; a credit limit per account in the ERP, checked by hand	Negotiated accounts bypass the portal; the portal's and the ERP's prices drift; two order lists for one customer

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel	Hundreds to thousands of accounts in several countries; contract and pre-season price lists; the first EDI buyer; wholesale competing with consumer and marketplace channels for the same stock	Prices published from one source with effective dates; a credit check at order placement; quotes as objects; wholesale orders in the OMS; wholesale allocation in the availability service; punchout for the buyers who require it	The share of keyed orders stops falling; credit holds by hand; a chain's chargeback for a document the ERP generated; the wholesale team cannot see a customer's orders in one place
4 — Group	Business units per brand or country; buyers with their own hierarchies and approval rules; key accounts on EDI and punchout; distributors and dealers as B2B2C	Business units with inherited rules; per-brand price lists under one pricing service; an ERP-fed credit check shared across brands; a master-data service for company identity and VAT numbers	Each brand runs its own portal and terms; the same retailer is three customers in three ERPs; approvals re-implemented per platform

The judgement: the B2B channel outgrows the platform's features not when the account count grows but when **the facts the checkout needs stop living in the platform**: negotiated prices, credit, approvals. Kestrel & Pine's 38% is that signal in numeric form, and it has held steady since 2024 because the facts behind it never moved.

In Practice

The platforms converge on one model, and none resolves ownership for you. Shopify, commercetools, Adobe and BigCommerce (whose older B2B authentication

token was deprecated on 30 September 2025) added B2B objects to consumer platforms; OroCommerce and SAP Commerce came from the B2B-first side with RFQs, budgets and cost centers. Every one of them can hold terms and credit, and the advice to pick one owner appears in the connector documentation (Microsoft's, above), not in the platforms'.

The market is large by any definition, and the definitions differ. Forrester's June 2022 release put US B2B e-commerce sales at $1.7 trillion in 2021, forecast to exceed $3 trillion by 2027, when it would be 24% of US B2B sales (a four-year-old forecast counting manufacturer and distributor sales to businesses). Digital Commerce 360 counted digital B2B sales "on ecommerce sites, password-protected sites, and apps", excluding EDI, at about $2.1 trillion in 2023 (7 February 2024), against total US manufacturing and wholesale sales of $15.12 trillion in 2025 (26 January 2026). No European figure meeting this book's sourcing standard was found. The point for an architect is the share that is *not* on a portal yet.

Kestrel & Pine: De Kaap Buitensport's pre-season order

Let me work one order through the portal as it is in spring 2027 and as it should be after Waypoint. **De Kaap Buitensport** is a fictional independent outdoor retailer with two shops, in Groningen and Assen, one of Kestrel & Pine's larger Dutch accounts, on net 30 terms with a **credit limit of €15,000** and pre-season terms of **5% off list** on orders placed before 1 March *(illustrative).* On **25 February 2027** its buyer wants 60 Ridgewalker jackets (a size run across women's and men's) and 120 pairs of merino socks for delivery in the first week of April.

	What happens in February 2027	**What happens after Waypoint**
Price	The portal shows list: 60 × €95 = €5,700 and 120 × €8.50 = €1,020, total €6,720 excl. VAT. The 5% pre-season discount exists only in the ERP's customer price list and Lea's e-mail, so the buyer e-mails the order instead	The pre-season list is published from the ERP as a price event into the account's portal catalog, effective to 1 March; the portal shows €6,384

	What happens in February 2027	What happens after Waypoint
Credit	The wholesale team keys the order into the ERP, which flags that open invoices of €9,800 plus this order exceed the €15,000 limit by €1,520 (at list) and holds it for Lea	The portal asks the ERP at placement; the order is created in the OMS as *awaiting seller approval*, and the buyer sees that status at once
Approval	Lea approves by e-mail on 27 February because a €4,100 invoice is due on 2 March; the team releases it in the ERP	Lea approves in the OMS; the release is an event; the ERP posts the order
Order record	The order exists in the ERP only; the portal history for De Kaap shows nothing for February	One order, one number family, visible to the shop, Lea, finance and the warehouse
Stock	The ERP reserves 60 jackets from a figure 15 minutes behind the WMS and an hour ahead of the platform	The availability service reserves against wholesale allocation; consumer channels see the reduced figure within minutes
Invoice	On shipment in April, from the ERP, at the price keyed in; if it differs from the e-mail, a credit note follows	From the ERP, from what shipped, at the price the order carried

The after-Waypoint column assumes the ERP can answer a credit check in under a second at the portal's volume and that the platform's catalogs can carry per-account, effective-dated lists; both should be tested before the 38% is promised to fall.

When It Goes Wrong

Failure	How it shows	Detection	Recovery
Portal price differs from invoice (the ERP recomputes at invoicing; a contract list expired)	Credit notes, disputes, shops asking for the "real" price	Compare order-line price with invoice-line price daily; alert on any difference	Invoice at the order's price; publish price lists with effective dates and recalculate only on explicit change
Stale credit balance (nightly copy)	Orders accepted that finance later holds, or rejected against an outdated balance	Age of the balance the portal used, logged per order	Credit check against the ERP at placement, or accept-and-hold with an event-driven release
Duplicate orders (the shop e-mails and then uses the portal; the rep keys an order the buyer already placed)	Double shipments, double invoices	Uniqueness on the buyer's purchase order number per company; a same-day duplicate check on lines	One order record for every entry path; a rep tool that creates the same order object
Wrong unit of measure (12 pieces ordered as 12 cartons)	A twelve-fold shipment and invoice	Order-line quantity against pack sizes and the account's history	Units as master data on the product; the portal confirms pack sizes
Expired or invalid VAT number	The wrong tax treatment on an invoice; a correction months later	Validate at account creation and periodically; store the validation date	The VAT number as master data with an owner
Quote accepted after it expired, or at a changed price	A dispute over which price was agreed	Quote state and `validTo` enforced when the order is created	Create orders only from valid quotes; expire the rest

The first two rows are the Kestrel & Pine story. The others arrive with scale, and all are cheaper to prevent in the object model than to fix at month-end.

What to Take Away

Established.

- Every major platform's B2B model has the same shape: an organization with locations, people and roles; customer-specific prices and assortments; quotes; approval rules; and payment terms, documented by Shopify, commercetools, SAP, Adobe, BigCommerce and OroCommerce as of 2026.
- A quote freezes a price. commercetools' "legally binding custom offers" with a validity date and Adobe's "the original price is used" are the same design decision, and the order must be created from the quote.
- Payment terms and credit limits are receivables facts; connector vendors' own documentation (Microsoft, 2026) says to manage them in the ERP and name one system of record for customer data.

Depends on context.

- The B2C platform's B2B features are the right channel for a brand selling to many small resellers, provided prices are published from one source and a credit check reaches the ERP at order time; once negotiated terms, approvals and credit live outside the platform's model, an ERP-driven store or a dedicated B2B platform fits better.
- A synchronous credit check is right when the ERP answers in under a second at the portal's volume; otherwise accept the order and hold it, with the hold released by an event, rather than copy the balance nightly.
- Punchout and EDI are worth building only for the buyers who require them; for everyone else they are cost without benefit, but the orders they produce must still enter the same order record.

The author's synthesis.

- The share of wholesale orders keyed in by hand is the best single measure of a B2B architecture, because it counts the facts the portal cannot see; Kestrel & Pine's 38% is a list of missing integrations (contract prices, credit, one order record), not a list of missing features.
- Decide ownership of the company account, the price and the credit limit before choosing or configuring any B2B platform; every platform can hold all three, and the one that does by default is the one nobody decided on.

Questions to Ask

1. Which system is the system of record for the company account, and which fields are mirrored to the portal, in which direction?
2. Where do contract and pre-season prices live, and can the portal show them to the right account on the right dates?
3. Which price wins when a quote, a contract price, a volume break and a list price all apply, and is that rule implemented once?
4. Is the credit limit checked at order placement against today's balance, or against a copy, and how old is the copy?
5. What share of wholesale orders arrive outside the portal, and for each reason, which fact is the portal missing?
6. Are quotes objects with a validity date, and is an order created from the quote or recomputed from the price list?
7. Is the invoice generated from the shipment at the order's price, and how often do portal price and invoice price differ?
8. Do punchout and EDI orders enter the same order record as portal orders, with the buyer's purchase order number as a uniqueness key?

Portal orders are the wholesale channel's self-service path. The largest buyers do not use it: they send documents. The next chapter opens the EDI envelope, from the ORDERS message to the despatch advice whose pallet labels cost Kestrel & Pine €14,200, and the e-invoicing mandates that now reach every business customer in Belgium. Continue with Documents Between Companies: EDI, Punchout and E-Invoicing.

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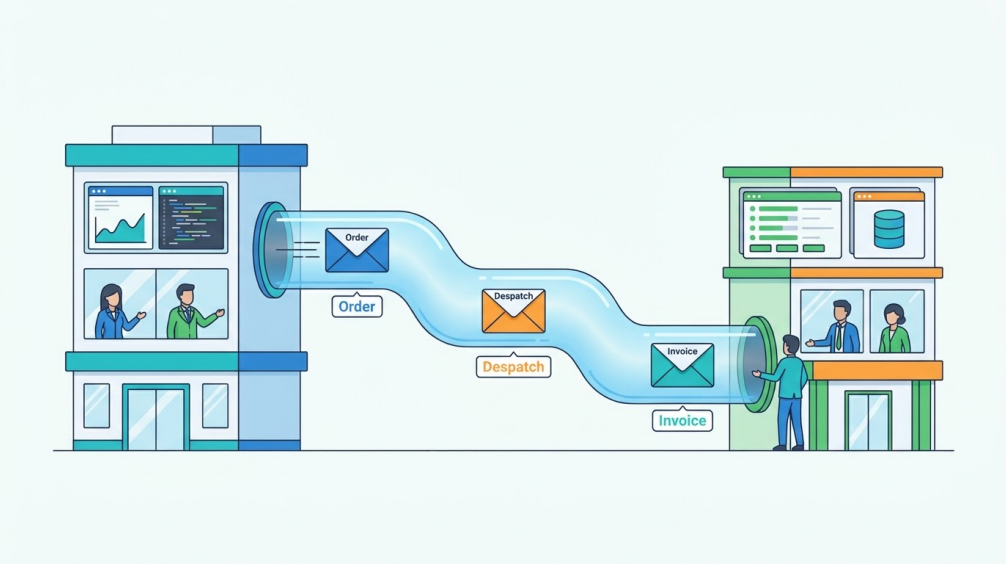
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

22. Documents Between Companies: EDI, Punchout and E-Invoicing



IN THIS CHAPTER, YOU WILL: Learn how companies trade by exchanging standard documents, why each document must come from a different system, and which e-invoicing deadlines already apply.

KEY POINTS:

- **Electronic data interchange (EDI) is a channel into the order record, not a separate order system.** An EDI order should create the same order object as a web order, flagged by channel, so allocation, invoicing and the books see one shape.
- **Each document has its own system of truth.** The order comes from the buyer, the despatch advice must come from what the warehouse packed and labeled, and the invoice must come from what was shipped and priced. Generating any of them from the wrong system is how chargebacks happen.

- **E-invoicing is a legal calendar, not a feature.** Belgium has required structured e-invoices over Peppol for domestic business-to-business invoices since 1 January 2026, Germany's issuing obligation starts in 2027, and the Netherlands plans 2030. The invoice becomes a transmitted document with a delivery status the finance system must hold.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Pulled Rug** — a trading partner changes its message guideline, a label rule or a legal format, and the merchant learns from a rejected delivery.
- **The Lynchpin** — the EDI mappings live in one consultant's head or one provider's portal, and nobody inside the company can read them.
- **The Long Fuse** — an e-invoicing mandate with an official date is discovered when the first customer's system refuses a PDF.

A retail chain sends a purchase order from its own purchasing system. It expects a shipment notice describing each pallet and an invoice its accounting software can read. Sending the goods is only part of fulfilling that order: a mismatched pallet label or unreadable invoice can hold up receiving or payment and trigger a supplier penalty. The buyer's document requirements therefore shape the merchant's systems.

Electronic data interchange (EDI) exchanges those structured business documents between companies without retyping them. Selling to a chain may require it even when the merchant already has a capable web shop.

A second kind of document has joined it. Since 1 January 2026, a business in Belgium that invoices another Belgian business must send a **structured electronic invoice** over a public network called Peppol; a PDF by e-mail no longer counts. Germany, France, Italy and Poland have their own rules and dates, and the Netherlands has announced 2030. **The invoice is becoming a transmitted legal document with a delivery receipt**, and the system that issues invoices has to know whether each one arrived.

Where we are in the order's journey. Kestrel & Pine, the fictional outdoor brand, supplies retail chain Bergzicht Outdoor. Bergzicht's 6 April 2027 reorder arrives electronically, while consumer Eva's jacket order starts online. Both use Duiven's stock and **warehouse management system (WMS)** for picking and packing, then reach the accounts. Journey 3: A Chain Reorders by EDI, From Order to Invoice follows Bergzicht's order end to end.

Setting the Scene at Kestrel & Pine: A Deduction Instead of a Payment

Bergzicht Outdoor is a fictional Dutch chain with 38 shops and a distribution center in Zwolle, worth €2.1 million annually to Kestrel & Pine. Since 2025 it has ordered through an **EDI provider**, a service that relays trading-partner documents, also called a value-added network (VAN). The three main messages are the order (**ORDERS**), despatch advice (**DESADV**) and invoice (**INVOIC**).

In October 2026, Wholesale Lead Lea Moreau receives a payment **€14,200 short**. Bergzicht has charged for nine third-quarter deliveries whose pallet labels did not match the despatch advice. Its terms allow a compliance fee plus manual counting costs.

This is the **ASN Chargeback**: ASN means advance shipping notice, another name for the despatch advice. It is a supplier-compliance deduction, distinct from a card-payment dispute.

The mechanism took Lea and Ruben Alves, the integration engineer, two weeks to reconstruct, because the only person who knew the EDI mappings was an external consultant engaged in 2025. **The ERP** (enterprise resource planning system: the back-office system that keeps the books, purchasing and invoicing) generated the despatch advice the moment a wholesale order was released to the warehouse, before anyone had packed anything, with pallet numbers from an ERP number range and contents as the order lines suggested. At Duiven, the WMS printed its own pallet labels at pack time. When pickers merged two half-pallets or split a heavy one, the labels on the truck and the message in Bergzicht's system described different pallets, and Bergzicht's scanners rejected them.

Figure 1: *The ASN Chargeback: the ERP announced pallets before anyone had packed them, the WMS labeled the pallets it actually built, and the mismatches on nine deliveries cost €14,200.*

Noor Visser's question in the Waypoint proposal is the one this chapter answers: **which system knows the truth each document claims to carry?** The ERP knew the order. Only the WMS knew the pallets. Waypoint decision **W6** fixed it: pallet labels generated by the WMS at pack time, the despatch advice sent from the WMS pack confirmation, and one EDI partner card per trading partner so that no mapping lives in one person's head again.

What EDI Is and Why It Has Not Gone Away

In plain words, EDI is a set of agreed message formats plus a way of delivering them. A **message** is a document such as an order; a **standard** says which fields it has and in which order; a **transport** is how the file travels; a **mapping** is the translation between the standard's fields and the fields in your own system.

Two standards dominate. **UN/EDIFACT** is the United Nations standard, maintained by UN/CEFACT, whose syntax became ISO 9735 in 1987; European retail uses the **GS1 EANCOM** subset, which GS1's manual (2002 edition, last updated 2021) describes as the UN/EDIFACT messages and fields relevant to retail. **ANSI X12** is the North American standard. A European brand selling to a Dutch chain and a US chain needs both; the message pairs are functional equivalents, not field-for-field translations.

GS1's 2021 implementation survey, the most recent public one, reported ORDERS, INVOIC and DESADV as the three most implemented EANCOM messages and AS2

as the most used transport. The survey is five years old; the standards it describes have barely changed since, which is the point.

Why does a format from the 1980s still carry wholesale orders in 2027? Four reasons, stated as industry practice. **The buyer chooses**: Jumbo, the Dutch supermarket chain, writes in its purchasing conditions of 9 April 2026 that suppliers "undertake to conduct all communication concerning the flow of goods (being order, packing slip (DESADV) and invoice) via EDI", forfeit €70 per message otherwise, and may have deliveries without DESADV and SSCC (the pallet's serial number, defined below) refused.

The documents carry meaning: an order is a purchase commitment, a despatch advice a receiving plan, an invoice a payment claim, and their stability is a feature. **Network effects**: once a chain has mapped hundreds of suppliers, changing format costs more than keeping it. **Regulators accommodate it**: Germany lets an EDI procedure that does not yet meet the e-invoice requirements run to the end of 2027 rather than banning it.

The Documents: Order, Despatch Advice, Invoice

Document	EDIFACT	X12	Sender	What it says	Internal system of truth
Order	ORDERS	850	Buyer	Items, quantities, delivery date and address	The buyer's purchasing system; it creates an order record on the seller's side
Order response	ORDRSP	855	Seller	What will be delivered, line by line, including shortages	The seller's order record after allocation
Despatch advice (**ASN**)	DESADV	856	Seller	What is physically coming, pallet by pallet	The WMS pack confirmation

Document	EDIFACT	X12	Sender	What it says	Internal system of truth
Invoice	INVOIC	810	Seller	What is owed for what was shipped, at agreed prices, with VAT (value-added tax)	The ERP, from shipment and price, never from the order
Syntax acknowl-edgment	CONTRL	997	Receiver	"I could parse your file", not "I accept it"	The EDI translator

The **ASN** (advance shipping notice) deserves its own sentence. It is a message sent at despatch telling the receiver what is coming, pallet by pallet, so receipt can be scanned instead of counted. Jumbo's guidelines for its mechanized distribution center (October 2025) state the consequence: every delivery "must be preceded by a DESADV message", completed to pallet level, "a delivery without an accompanying DESADV message cannot be processed and is therefore refused", and "the SSCC label forms the link with the DESADV despatch message". The ASN is the receiving plan; the pallet label is the key into it.

A CONTRL or 997 says only that the file was syntactically readable, not that the order was accepted, the delivery received or the invoice will be paid; those answers come in ORDRSP, a receiving advice and, eventually, payment. **Treating a syntax acknowledgment as acceptance is a common and expensive mistake**.

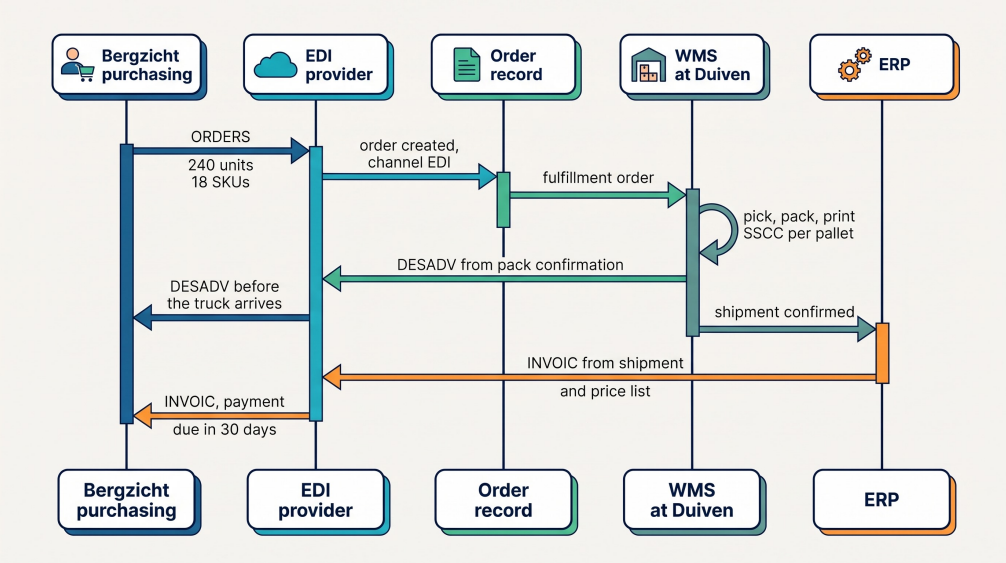


Figure 2: *Bergzicht's spring reorder as a document cycle: the order creates a normal order record, the despatch advice comes from the WMS once the pallets are labeled, and the invoice comes from the ERP once the shipment is confirmed. Three documents, three systems of truth.*

Identifiers: What, Where and This Pallet

EDI works because both sides use the same numbers for things, places and shipments. GS1's General Specifications (Release 26.0, ratified January 2026) define all three:

- A **GTIN** (Global Trade Item Number) identifies *what*: a sellable product, the number in the barcode on a jacket's hang tag; What Are We Selling? Products, Variants and the Catalog owns the term.
- A **GLN** (Global Location Number) identifies *who and where*: a legal entity or a physical place, such as Bergzicht's Zwolle distribution center. Messages address parties by GLN; Jumbo requires that the supplier GLN in the DESADV equal the GLN the order was sent to.
- An **SSCC** (Serial Shipping Container Code) identifies *this pallet*: an 18-digit serial number on a logistics unit's label. GS1 states that the SSCC "is the only GS1 key that SHALL be used as the identifier of a logistic unit" and "is used with EDI communications to enable identification and traceability". bol has required SSCC labels on every logistics unit since 31 December 2021 and requires EDI suppliers to include the SSCC in the DESADV.

The ASN ties the three together: each SSCC lists the GTINs and quantities it holds, addressed from one GLN to another. Hence the rule Kestrel & Pine paid for: **the SSCC**

must be generated by the system that physically builds the pallet. An ERP that invents pallet numbers before packing produces an ASN that is wrong whenever the warehouse deviates from the plan.

Transport: Point-to-Point or Network

Transport	What it is	You connect to	Typical use
AS2	A protocol (RFC 4130, 2005) for signed, encrypted documents over HTTP with a signed receipt proving delivery	Each partner: one certificate and endpoint per partner	Large retailers; the most used channel in GS1's survey
SFTP	Secure file transfer to a server, polled on a schedule	Each partner	Smaller partners, batch flows
VAN (value-added network)	A provider-operated mailbox network that relays to any partner on it and often translates formats	The provider, once	The common choice for a mid-size brand; Kestrel & Pine's EDI provider is one

Transport	What it is	You connect to	Typical use
Peppol	An open network run by OpenPeppol, a Brussels non-profit, in which parties connect through any accredited **access point** and reach any participant; it mandates document standards and has a public directory	Your access point, once	E-invoices where governments mandate them; Belgian business invoices since 2026

AS2 and SFTP are point-to-point: every partner is a relationship to manage. A VAN and Peppol are networks: connect once, reach many. Peppol alone has a public directory and mandates what the documents look like. OpenPeppol calls it the **four-corner model**: the sender hands the document to its access point, which delivers it to the receiver's access point, which delivers it to the receiver, with no bilateral agreement.

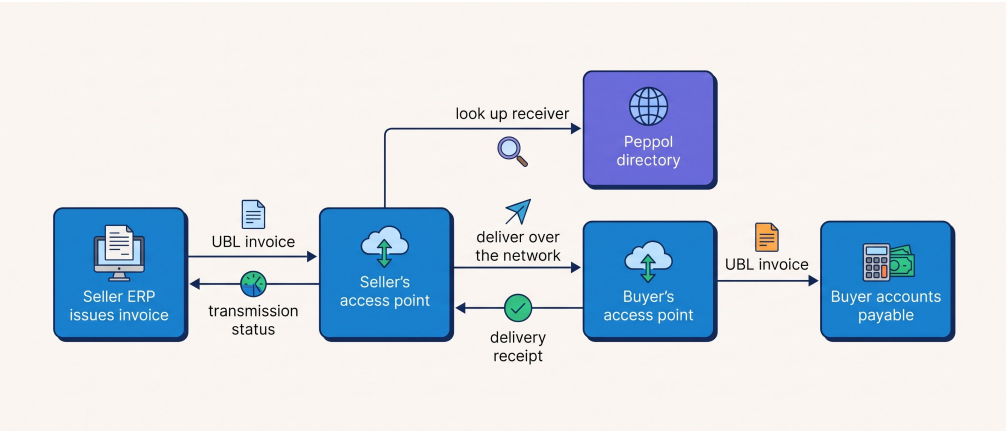


Figure 3: *Peppol's four-corner model: each party connects once to an access point of its choice; the sender's access point finds the receiver in a public directory and delivers; the transmission status flows back to the system that issued the invoice, which must store it.*

Mapping, Onboarding and the Chargeback Economy

The standards are public; the cost is per partner. Every chain publishes its own **message implementation guideline**: which optional fields it requires, which codes it accepts, how deep the DESADV must go. **Mapping** is the translation between a partner's guideline and your own order, shipment and invoice records, done and maintained per partner. That is the business model of the EDI providers: SPS Commerce's annual report for 2025, filed in February 2026, describes a network that lets companies "connect once and immediately transact with thousands of trading partners without negotiating standards, building integrations, or maintaining" them, and reported $751.5 million of revenue, 96% of it recurring. Jumbo tells small suppliers that its WebEDI portal is "the simplest, and cheapest in the short term", while an integrated connection "requires translating EDI messages to and from your internal system, with effort from IT, a solution provider, and system users".

Chargebacks are how the buyer enforces the guideline. They come in two families. **Delivery performance**: Walmart's on-time-in-full program charged a 3% fee on orders not delivered in full or on time when it launched in 2017 (the thresholds have risen since; no official Walmart page states them), and Jumbo's 2026 conditions add €250 per order when less than 90% of an order line arrives as agreed. **Data and label compliance**: a missing ASN, an unscannable label, an ASN quantity that does not match receipt; Jumbo's €70 per message is this family. The first family is a logistics problem. **The second is an architecture problem**: the ASN is only as right as the system that generated it.

How EDI Documents Reach the Order Record and the Ledger

Here is the rule this chapter exists to state. **An EDI order is a channel into the order record, not a separate order system.** An ORDERS message should create the same order object a web order creates, flagged by channel, so that stock allocation, invoicing and the books see one shape. The common failure is the opposite: EDI orders sit in the ERP or the provider's portal, web orders in the platform, marketplace orders in the connector, and nobody has one list. That was Kestrel & Pine's January 2027 state, and The Order Record: Lifecycle, State and the Order Management System explains why Waypoint's OMS (order management system: the one place every channel's order is created, with one state model) takes EDI orders from the ERP in Q4 2027.

Each document then has a different internal source:

Document	Produced from	Never from	Why
Order record	The ORDERS message after mapping and validation: item codes resolved, prices checked against the contract price list, delivery date feasible	A retyped e-mail	The buyer's commitment enters the system of record once, unchanged
Order response	The order record after allocation (stock set aside for this order; Is It in Stock? Inventory, Availability and the Promise You Can Keep)	The ORDERS echoed back	A response that promises everything promises nothing
Despatch advice	The WMS pack confirmation: which SSCC holds which GTINs in which quantity	The order or an ERP number range	The receiver scans what is physically on the truck
Invoice	The ERP, from the confirmed shipment and the contract price, with VAT, in the legal numbering sequence	The order (quantities may have changed) or the WMS (prices and VAT live elsewhere)	The invoice is a payment claim for what was delivered; Tax, Invoices and the Books: From Order to Ledger owns its legal shape

The flows have different requirements, which is the argument against forcing them through one system:

Flow	How fresh	How consistent	How fast	How available
ORDERS in → order record	Minutes; chains commonly expect a response within hours *(assumption)*	Exactly once: a duplicate ORDERS must not create two orders	Batch acceptable	A provider outage must queue, not lose
DESADV out	Must leave before the truck arrives	Must match the labels exactly	Minutes after pack confirmation	A late ASN is a refused delivery
Invoice out	Same day as shipment, or per month-end rules	One invoice per shipment; immutable once numbered	Daily batch acceptable	Rejection must be a tracked state
Transmission status back	Within hours	Per document	Not critical	Needed as dispute evidence

Punchout: When the Buyer's System Owns the Order

Some corporate buyers send no ORDERS at all; they shop in your web shop from inside their own procurement system. This is **punchout**. The cXML User's Guide, the protocol behind SAP Ariba's punchout, describes it plainly: procurement applications "display a button instead of product or pricing details", the buyer's browser then shows "pages from the supplier's local website", and when the user is done, a button "returns the order information to the procurement application". SAP's OCI (Open Catalog Interface) is the equivalent for SAP procurement; a 2026 integrator's description notes that OCI posts cart lines back to a hook URL and lacks native purchase-order and invoice automation.

Punchout **inverts the usual flow**. The buyer's procurement system owns the requisition and the purchase order; the supplier's shop is only a catalog session. The real order arrives later as a cXML order request or an EDI ORDERS, so **the shop must never treat a punchout cart as an order**. Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals covers punchout as a feature; here it matters as another channel into the same order record.

E-Invoicing: The Invoice Becomes a Transmitted Legal Document

The European Commission defines an **electronic invoice** as one "issued, transmitted and received in a structured data format which allows for its automatic and electronic processing"; a PDF attached to an e-mail is not one. Three layers to keep apart:

1. **The semantic model.** **EN 16931** is the European standard listing what a core invoice contains: seller and buyer with VAT identifiers, a document type (380 for an invoice, 381 for a credit note), lines each with a VAT category, a VAT breakdown per category and rate, payment information.
2. **The syntax.** **UBL** (Universal Business Language, an XML format) is the one most e-invoices use; national formats such as Germany's XRechnung are profiles of the same model.
3. **The network and profile.** **Peppol BIS Billing 3.0** is a usage specification of EN 16931 for the Peppol network. Its current release, 3.0.21, was published on 20 May 2026 and became mandatory on 17 August 2026: even the invoice format has a version calendar.

An EDI INVOIC and a Peppol invoice carry the same claim and differ in almost everything else:

	EDI INVOIC (EANCOM or X12 810)	**Peppol / EN 16931 invoice**
Who defines the content	A bilateral guideline per retailer, on top of the standard	The European standard, with a published usage specification and validator
Purpose	Supply-chain settlement: match to order and receipt	A tax-compliant legal invoice any receiver's accounts-payable system can read
Addressing	GLNs agreed per partner	A Peppol participant identifier in the public directory
Transport	AS2, VAN or SFTP, per partner	Any accredited access point

	EDI INVOIC (EANCOM or X12 810)	Peppol / EN 16931 invoice
Validation	Syntax acknowledgment and partner-side rejection	Validation rules at the access point; state clearance in Italy and Poland
Legal weight in 2026	Tolerated in Germany through 2027; in Belgium only if it meets the structured-invoice requirement	The format the mandates require

Three things change for the architecture. First, **the system that numbers and issues invoices, usually the ERP, must also hold the transmission status, rejections and archive**. Second, in **clearance** countries (Italy's SdI, Poland's KSeF) the state platform sits in the invoice path and the invoice is valid only once accepted, so rejection becomes a first-class state. Third, **commerce platforms do not issue tax invoices in these regimes**; they produce orders and confirmations, and the storefront's "invoice" PDF is a courtesy copy. Tax, Invoices and the Books: From Order to Ledger explains the invoice as a legal document; The Back Office: ERP Integration and Who Is the Hub explains why its home is the ERP.

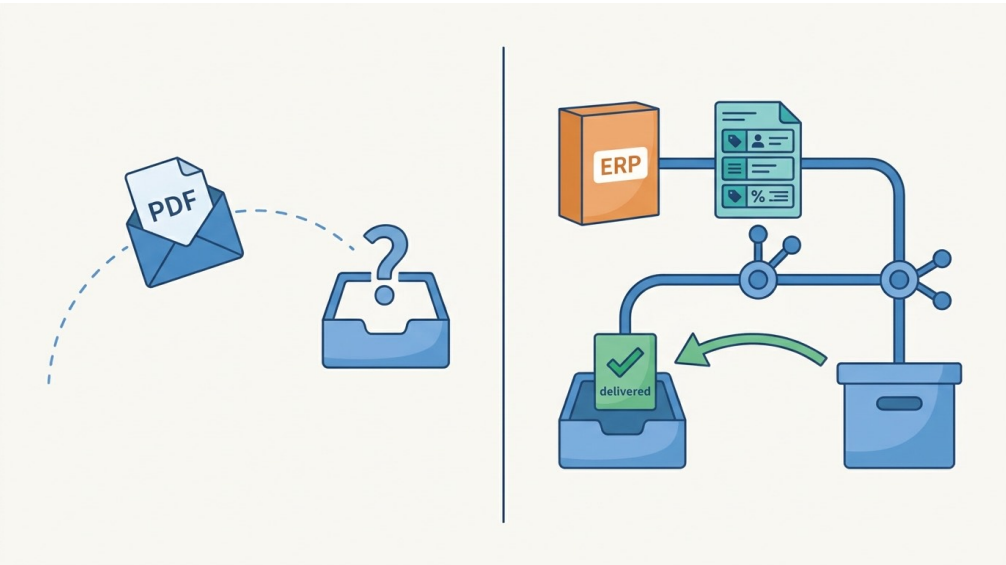


Figure 4: *An e-invoice is not a PDF in an e-mail but a structured, transmitted document with a delivery state, and the system that issues invoices must hold that state.*

The Mandate Calendar

The dates below are the official ones as of November 2026, from national tax administrations and the Commission. They are the most volatile facts in this chapter: check the source before relying on any of them, and read Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation for how to keep a regulatory calendar.

Jurisdiction	What is mandatory	Since or from	Format and channel	Touches Kestrel & Pine?
Belgium	Structured e-invoices for domestic business-to-business invoices between VAT-registered businesses	**1 January 2026**; tolerance on penalties in the first three months of 2026 only for businesses that can show timely steps; "under no circumstances" a general postponement	Peppol network, Peppol BIS at minimum, credit notes included; fines of €1,500, €3,000 and €5,000 for lacking the technical means	Indirectly: the bookkeeping of its 210 Belgian wholesale accounts expects Peppol invoices; whether the obligation reaches a Dutch supplier is not settled (see below)

Jurisdiction	What is mandatory	Since or from	Format and channel	Touches Kestrel & Pine?
Germany	Receive e-invoices (all domestic businesses); issue after transition periods	Receive since **1 January 2025** ("an e-mail inbox is already sufficient"); issue from **1 January 2027**, or **1 January 2028** if prior-year turnover was at most €800,000; EDI procedures tolerated to end 2027	EN 16931 formats such as XRechnung and ZUGFeRD from 2.0.1; no network mandated	Indirectly: covers transactions between businesses established in Germany
France	Receive (all); issue plus transaction reporting (large and mid-size)	**1 September 2026**; small and medium-sized enterprises (SMEs) and micro-enterprises issue from **1 September 2027**	Accredited platforms	No sales in France
Italy	Business-to-business and business-to-consumer e-invoicing via the state platform	Since **1 January 2019**; size exemptions removed **1 January 2024**	FatturaPA XML via the Sistema di Inter-scambio; EU derogation to 31 December 2027	No

Jurisdiction	What is mandatory	Since or from	Format and channel	Touches Kestrel & Pine?
Poland	Structured invoices via the national system KSeF	**1 February 2026** (2024 sales above PLN 200 million), **1 April 2026** (others), **1 January 2027** (smallest)	KSeF 2.0 central platform	No
Netherlands	None for business-to-business today; e-invoices to central government required	Cabinet plan of 11 September 2026: invoicing from **1 July 2030**, reporting from **1 July 2031**; bill before summer 2027	EN 16931; network not yet decided	Yes, from 2030: 640 Dutch accounts
EU (ViDA, "VAT in the Digital Age")	Digital reporting based on e-invoicing for cross-border business-to-business supplies	Adopted 11 March 2025; in force 14 April 2025; reporting from **1 July 2030**; national systems aligned by 1 January 2035	EN 16931	Yes, from 2030: every Belgian and German wholesale invoice

Two readings for a newcomer. The mandates are **domestic** in scope: Belgium's FAQ states that VAT-registered persons not established in Belgium and without a permanent establishment there are not subject to the obligation (the FAQ adds that the law is still being amended on this point), and Germany's covers transactions between domestic companies. A Dutch brand invoicing a Belgian shop is not automatically obliged, but the Belgian customer's bookkeeping expects a Peppol invoice, and many customers ask for

one; whether a foreign supplier with a Belgian store counts as established is a question for a tax adviser. The second reading: **the cross-border case arrives in 2030 with ViDA**, so a brand selling into several EU countries will soon send every business invoice as structured data. Building the capability once, for the first country that requires it, is cheaper than building it per deadline.

The Choices

A provider's web portal. You log in to the provider's site, read the chain's order, key the shipment and the invoice. No integration. Jumbo recommends it for small suppliers. It is also retyping, which EDI was meant to remove, and it does not scale past a few orders a day.

A provider integrated with your systems. The provider's translator maps the chain's guideline to your order record, WMS and ERP, over the provider's network, often through an iPaaS (integration platform as a service: a hosted tool for building and running the flows between systems). The provider owns the mappings and the partner connections; you own the contract for what each document means inside your company. This is the common choice at Kestrel & Pine's size.

In-house EDI. Your own translator software and AS2 endpoints, your own mappings under version control. Full control, full cost: certificates, onboarding, guideline changes and a person who understands EDIFACT. It makes sense when EDI is the business, as for a distributor with hundreds of partners.

An e-document gateway. One internal component, bought or assembled, that produces and receives every structured document (EDI, Peppol, punchout, clearance formats) from the same internal order, shipment and invoice records, with per-partner and per-country rules as configuration. The integrated provider grows into this as partners and countries multiply.

	Provider portal	**Provider, integrated**	**In-house**	**E-document gateway**
Time to first partner	Days	Weeks to months	Months	Months
Running cost	Lowest subscription; staff time per order	Subscription plus mapping per partner	Staff and software	Highest fixed cost

	Provider portal	Provider, integrated	In-house	E-document gateway
Who owns the mappings	Provider	Provider; you hold a copy and a partner card	You	You, under version control
Chargeback exposure	High: retyping; ASN rarely from the WMS	Low if the ASN comes from the WMS	Low if staffed	Lowest
Right when	A handful of orders a week, one partner	One to a dozen partners; a WMS exists	EDI is core to the business	Several countries, many partners, e-invoicing in more than one regime

My recommendation for a brand at the Omnichannel stage is the integrated provider, with two conditions that are not optional: the ASN is generated from the WMS pack confirmation, and the company holds a **partner card** per trading partner in its own repository, listing the partner's GLNs, transport, message versions, guideline version, mapping version, contacts, compliance rules and open disputes. The provider may hold the mapping; the company must hold the knowledge. That is W6's third clause.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	No EDI; business customers order by e-mail; invoices as PDF from an accounting package	A provider portal if the first chain demands EDI; a Peppol-capable accounting package if any customer sits in a mandate country	The first purchasing conditions with an EDI clause arrive
2 — Growing	First chain forces EDI; a provider connector into the ERP; invoices numbered in the ERP and, where required, sent through a Peppol access point	A decision on which system produces the ASN (if a 3PL, a third-party logistics provider, packs, its system must)	Orders retyped from the portal into the ERP; the first chargeback; a second chain with a different guideline
3 — Omnichannel	Integrated EDI through the provider into the OMS via the iPaaS; SSCC labels from the WMS; Peppol invoices from the ERP to customers in mandate countries	A partner card per partner; a compliance scorecard (ASN match rate, on-time-in-full); disputes backed by pack and label evidence; a regulatory calendar	Mappings one person understands; EDI orders in the ERP while web orders live in the platform; a deduction nobody can explain

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
4 — Group	An e-document gateway shared across brands and countries; mappings under version control; clearance integrations where required	One invoicing service behind which country entities handle different clearance systems; mandate monitoring as a standing task	Each brand with its own provider contract; the same chain onboarded three times

Kestrel & Pine sits at stage 3 with a stage 2 setup: one chain, one provider, mappings built by one consultant, the ASN from the wrong system. The deduction was the warning sign.

In Practice

Belgium as the first mandate the reader will meet. The Belgian government's e-invoicing site announced on 18 September 2024 that from 1 January 2026 "all Belgian enterprises liable to VAT will have to use structured electronic invoices" for transactions with each other, that "sending a PDF invoice by e-mail or via a platform will no longer be enough", and that invoices travel "via a decentralised network called Peppol". On 18 December 2025 it confirmed a three-month tolerance on penalties for businesses that could show timely steps and refused any general postponement.

Bergzicht's spring reorder under the corrected flow. On 6 April 2027, the EDI provider delivers an ORDERS message for **240 units across 18 SKUs**, totaling **€19,860 excluding VAT**. A SKU identifies one sellable variant. The ERP resolves Bergzicht's item codes through shared product identifiers (GTINs), checks contract prices and creates the order. The planned OMS does not go live until Q4.

The warehouse reserves the units and an ORDRSP order response confirms the lines. At packing, the WMS prints one **Serial Shipping Container Code (SSCC)** label for each of three pallets and records their contents. The DESADV comes from that same packing confirmation and reaches Zwolle before delivery on 9 April.

The ERP issues INVOIC for the shipped quantities at contract prices, due 9 May. This is Waypoint decision W6 in practice: **the labels and shipment notice describe the same packed pallets**. In Q3 2026 they came from different systems.

The Belgian variant. Ardennes Trail Co., a fictional outdoor shop in Namur and one of the 210 Belgian accounts, orders 40 Ridgewalker jackets at €95, **€3,800** excluding VAT,

through the platform's business-customer (B2B) portal on Wednesday 21 April 2027.

The jackets leave Duiven by bpost, and at dispatch on 23 April the ERP issues the invoice in the same numbering sequence: an intra-EU supply at 0% VAT with the reverse-charge mention, payment due 23 May, rendered as a UBL (Universal Business Language, the XML invoice format Peppol carries) document under Peppol BIS Billing 3.0 and sent through the EDI provider, which is also Kestrel & Pine's Peppol access point; the ERP stores the transmission status.

Kestrel & Pine has sent Peppol invoices to all Belgian business customers since January 2026 by choice, not because the mandate is known to bind a Dutch supplier (see above): its larger Belgian customers' bookkeeping expects Peppol, and one rule was cheaper than exceptions.

When It Goes Wrong

A duplicate ORDERS. The provider re-sends a file after a timeout that was actually a success. Without idempotent handling (receiving a message twice has the same effect as once; Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns the term), two orders for 240 units are allocated. Detection: the buyer's order number plus GLN is the natural key; a second ORDERS with the same key is a duplicate or a change, never a new order. Recovery: cancel before allocation; if shipped, a return and a credit note.

A changed guideline. The chain moves to a new DESADV level or SSCC rule and notifies the address on file, which is the consultant's. The first the merchant hears is a refused truck: the Pulled Rug. Detection: the partner card names an internal owner who receives partner notices; the scorecard shows the match rate dropping. Recovery: version the mapping, test against the chain's test environment, keep the old version until the switch date.

The ASN from the wrong system. The Kestrel & Pine case. Detection: compare every DESADV against the WMS pack confirmation before sending and alert on any difference; a match rate below 100% is a defect, not a statistic. Recovery: re-send a corrected DESADV before arrival where the chain allows it; otherwise dispute the deduction with the pack record and the label print log as evidence, which is why both must be kept.

A rejected e-invoice. The receiver's access point rejects a UBL file because a VAT category code is missing on one line, or because Peppol's validation rules changed with a release. The invoice is numbered in the ERP but was never received. Detection: the transmission status is a field on the invoice, and an "issued but not delivered" report runs daily. Recovery: correct and re-send, or a credit note and a new invoice; which one depends on the country and belongs in the ERP's configuration, not in someone's memory.

A provider outage, or the consultant leaves. The VAN is down for a morning: nothing is lost if every document is persisted before sending and carries a reference the provider can deduplicate, and a heartbeat per partner turns silence from a chain that orders daily into an alert. The consultant leaves: the mappings are in the provider's portal and the test cases in an inbox, which is why the partner card and a mapping export belong in the company's own repository.

What to Take Away

Established.

- EDI is a buyer-driven channel with its own standards (UN/EDIFACT and GS1 EANCOM in Europe, ANSI X12 in North America), identifiers (GTIN, GLN and SSCC, defined in GS1's General Specifications) and transports (AS2 per RFC 4130, VANs, SFTP, Peppol); the order, despatch advice and invoice are its most implemented documents, per GS1's survey.
- The ASN is a receiving plan keyed by the SSCC label, and large receivers refuse deliveries without it, as Jumbo's 2025 guidelines and 2026 purchasing conditions and bol's supplier rules document.
- Structured e-invoicing is law with dates: Belgium since 1 January 2026 over Peppol, Germany's issuing obligation from 2027, Italy since 2019 and Poland since 2026 through state clearance, the Netherlands planned for 2030, and ViDA's cross-border reporting from 1 July 2030.

Depends on context.

- A provider's portal fits a handful of orders a week with one partner; an integrated provider fits from the first chain with a WMS onward; in-house EDI fits only where EDI is the business; a shared e-document gateway is a Group-stage answer.
- Whether a foreign supplier is obliged under a domestic mandate depends on establishment rules; the decision to send structured invoices anyway depends on what customers can receive and on the cost of exceptions.
- EDI orders belong in the OMS where one exists; at the Growing stage, before an OMS, the ERP is an acceptable home, provided the ASN still comes from the system that packs.

The author's synthesis.

- Each document has exactly one honest source, and the architecture should make it impossible to generate it anywhere else: the order from the buyer's message, the ASN from the pack confirmation, the invoice from the ERP's confirmed shipment. Every compliance chargeback in the public rule sets above is a document that came from the wrong place.

- The e-invoice turns the invoice into a message with a delivery state, so the ERP needs the discipline the order channel needs: persistence before sending, idempotent re-sends, a status per document and a daily report of what was issued but not received.

Questions to Ask

1. Which of our customers require EDI, under which purchasing conditions, and what do those conditions charge for a missing or wrong ASN?
2. Which internal system generates the DESADV, and is it the system that physically packs and labels the pallets?
3. Does an EDI order create the same order record as a web order, or does it live in the ERP or the provider's portal?
4. For each trading partner, where is the partner card: GLNs, transport, guideline version, mapping version, internal owner, open disputes?
5. What is our ASN match rate per partner, who sees it weekly, and who receives the chain's notices of guideline changes?
6. In which countries do we invoice business customers, which mandate applies to each, and from what date?
7. Which system numbers and issues invoices, and does it store the Peppol transmission status, rejections and archive?
8. Who inside the company can read our EDI mappings without calling a consultant?

The documents in this chapter end in one place: the ledger. The next chapter asks what the system that keeps the ledger, the ERP, should own in a commerce business, and whether it should be the hub everything else connects through. Continue with The Back Office: ERP Integration and Who Is the Hub.

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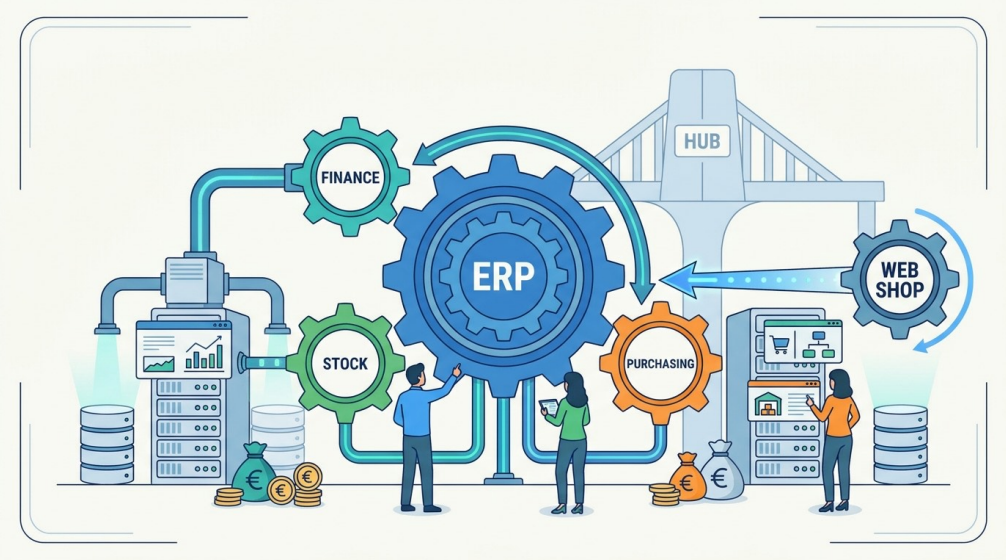
- The scale-stage table, the "one honest source per document" rule and the partner-card recommendation are the author's synthesis of the sources above and of common practice; the Kestrel & Pine latencies and Bergzicht's compliance clause are illustrative.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

23. The Back Office: ERP Integration and Who Is the Hub



IN THIS CHAPTER, YOU WILL: Learn what an enterprise resource planning (ERP) system owns in a commerce business, what flows between it and the commerce systems, and how to decide which system is the hub.

KEY POINTS:

- **The ERP keeps the books, not the shop.** Its four jobs are the ledger, purchasing, inventory valuation and master data. The WMS (warehouse management system) knows what is on the shelf; only the ERP knows what it is worth.
- **"Who is the hub" is a scale-stage decision.** ERP-as-hub fits one channel and one warehouse, commerce-as-hub fits a direct-to-consumer (D2C) brand, OMS-as-hub (an order management system at the center) fits omnichannel, and an integration platform with no hub at all is Kestrel & Pine's thirty-four integrations and three definitions of "order".
- **Finance can be batch; availability cannot.** Push orders, shipments and payments as events from the systems that own them and let the

ERP post them daily, but never let the ERP's cycle time become the storefront's stock latency. The partner-built connector that carries these flows is a contract to own, not a black box to rent.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Spaghetti Junction** — every system was connected to the ERP by whoever implemented it, and the ERP became a hub by accident, with no owner for any flow.
- **Schrödinger's Order** — the ERP's copy of the order lags the platform's and the warehouse's, so finance, support and the customer see three statuses.
- **The Lynchpin** — the platform-to-ERP connector was built by the ERP partner, is documented nowhere, and only the partner can change it.

When finance closes the books, it must connect sales, payments, fees and refunds to explain what the business earned and is owed. An **enterprise resource planning system (ERP)** keeps that ledger, alongside purchasing, stock valuation and master records for items, customers and suppliers. Its broad coverage makes it a natural place to look for data, but different consumers need that data at different speeds.

Because the ERP arrives early and holds so much data, other systems tend to connect through it. Storefronts send orders, warehouses send stock, and trading partners send wholesale documents through **electronic data interchange (EDI)** (Documents Between Companies: EDI, Punchout and E-Invoicing).

The ERP can become the hub by accident. Its scheduled updates then determine how quickly a shopper learns that a jacket has sold out. **The storefront's need for fresh stock cannot wait for the accounting cycle**.

Decide which facts the ERP owns, which it consumes and how often each must move. This chapter compares it with a commerce platform or order management system as the hub. Where selling needs faster answers than accounting can provide, separating those responsibilities lets checkout use current operational facts while the ERP remains responsible for the books.

Where we are in the order's journey. Eva buys a jacket and socks for €235 from Kestrel & Pine, the fictional outdoor brand. Her iDEAL bank-transfer payment, provider fee, shipment and any later refund become ERP postings. Finance Director Sanne de Wit needs those records to agree when she closes April 2027.

Setting the Scene at Kestrel & Pine: The Connector Nobody Could Read

Kestrel & Pine has used a cloud ERP since 2023 for finance, purchasing, valuation, item records and invoices. Wholesale staff also enter the **38% of orders received by email**

or phone there. An external implementation firm, **the ERP partner**, built its connector to the commerce platform.

In January 2027, Head of Engineering Noor Visser asks the partner which fields move, in which direction, how often and which system wins a disagreement. She receives a configuration export and a bill for explaining it. She and integration engineer Ruben Alves spend three weeks reconstructing the behavior.

The connector copies platform orders hourly. Invoicing waits for the WMS's "order complete" message. Stock arrives from the warehouse every fifteen minutes, then goes to the platform hourly; German consumer prices use the same connector. Repeated messages can produce repeated effects: the flow is not **idempotent**. Failed runs wait until tomorrow.

Three of 2026's incidents ran through this connector. The **Black Friday Oversell** happened because the ERP sat between the WMS (warehouse management system: the software that runs receiving, picking, packing and shipping at Duiven) and the platform in the stock chain, adding up to an hour of delay on top of the WMS's fifteen minutes and the connector's hour. The **Three-Status Order** (#1052877, December 2026) showed "Not invoiced" in the ERP for six days because the ERP invoices only on a WMS "order complete" message, and the second line of a two-line order shipped a week after the first. The **Price Drift** of October 2026 was a price-list sync through the same connector that failed silently for nine days.

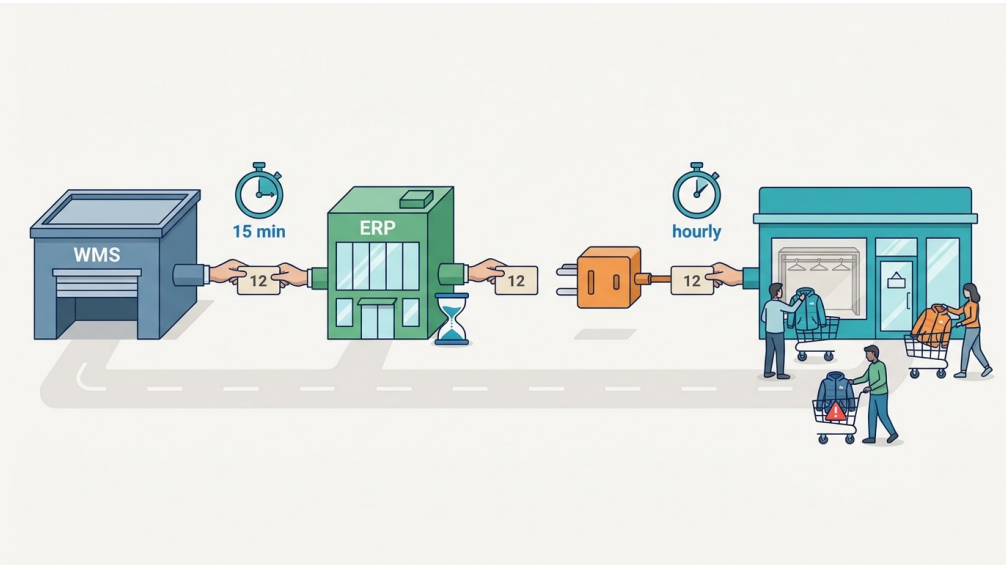


Figure 1: *With the ERP in the stock chain, a stock figure from Duiven waited on the WMS's fifteen minutes, the ERP and the connector's hour before a storefront saw it, which is how the Black Friday Oversell happened.*

Sanne's month-end problem was quieter and larger. The ERP held platform, EDI and manual wholesale orders, but marketplace orders arrived as a weekly summary from the marketplace connector and store sales as a nightly total from the POS (point of sale: the stores' till software), each in a different shape; January's close took eleven working days. Her question to Noor was the chapter's question from the finance side: **what should the ERP be the truth for, and what should it just receive?**

Waypoint, the company's improvement program, narrows the ERP's role. It remains authoritative for finance, purchasing, valuation, costs, wholesale contract prices, business terms and credit limits. An **order management system (OMS)** takes channel orders; a **product information management system (PIM)** takes descriptions and attributes. The WMS owns physical stock, and an availability service calculates what channels may sell.

The ERP consumes orders, shipments, payments and refunds in daily batches. It publishes wholesale prices, terms and credit changes for the OMS and business catalogs to consume; **storefronts do not read it live**. These are Waypoint decisions W1, W2, W3 and W5. The partner connector is retired as the OMS takes over order flows in late 2027.

What an ERP Does in a Commerce Business

In plain words, the ERP has four jobs, and a commerce architect should be able to name them:

Job	What it means	What it needs from commerce	What commerce needs from it
The ledger	Every order, payment, payment fee, refund, VAT (value-added tax) amount and shipping charge becomes a posting; month-end, VAT returns and the annual accounts are produced from it	Orders, payments, refunds and shipments, complete and once	Nothing in real time

Job	What it means	What it needs from commerce	What commerce needs from it
Purchasing	Supplier orders, goods receipts, supplier invoices, landed cost (purchase price plus freight and duty)	Sales history and forecasts	Expected stock (what is on order) for pre-orders and planning
Inventory valuation	What the stock is worth: a cost per unit under a costing method, and the balance-sheet value	Stock movements and quantities, in batch	Cost per SKU (a sellable variant, such as one jacket in one size and color), for margin reports
Master data	Items (identity, cost, supplier, purchasing units), customers (terms, credit limits), suppliers	New customers from the storefront; new items from merchandising	The item must exist here before it can be bought or valued; business-customer (B2B) terms and credit limits

The Dutch mid-market ERPs describe themselves in exactly these terms. Exact's trade edition page (accessed November 2026) promises to "manage stock, orders and accounting in one system", with purchasing, multi-warehouse stock, sales orders, returns and "(e-)invoicing" included; AFAS describes one integrated system with one database in which "data only needs to be entered once, and processes run automatically" (both translated from Dutch). The pitch is true at the size the pitch is written for, which is the size at which the ERP *is* the commerce back end.

Inventory valuation is the job newcomers underrate. IAS 2, the international accounting standard for inventories, measures them "at the lower of cost and net realisable value" under a cost formula such as first-in, first-out or weighted average. Microsoft's Business Central documentation (March 2026) lists five costing methods and warns that "you can't change an item's costing method if item ledger entries exist for the item". None of this has anything to do with whether Eva can buy a jacket tonight. **The commerce platform knows stock quantities; only the ERP knows what the stock is worth.** Keeping the two apart, quantity in the WMS and value in the ERP, is the first boundary this chapter draws. Is It in Stock? Inventory, Availability and the Promise

You Can Keep owns the quantity side.

What Flows Between Commerce and the ERP

Six kinds of data cross the boundary. The direction and the cadence are the design, not details:

Flow	Direction	System of record	Cadence that fits	Why
Orders	Commerce → ERP	The OMS (or the platform before an OMS exists)	Event on creation, consumed by the ERP in a daily batch	The ERP needs every order once; it does not need it within the minute
Shipments	WMS → ERP	The WMS	Event, consumed daily	Triggers the invoice; revenue is recognized at delivery, not shipment (Tax, Invoices and the Books: From Order to Ledger)
Payments, fees, refunds, payouts	PSP and platform → ERP	The PSP for money movements; the platform for refund decisions	Daily batch, reconciled against PSP payouts	Finance works in periods

Flow	Direction	System of record	Cadence that fits	Why
Invoices and credit notes	ERP → customer, EDI, Peppol (the European e-invoice network)	The ERP	On shipment; daily	Legal numbering and archive live here; see Documents Between Companies: EDI, Punchout and E-Invoicing
Products	ERP ↔ PIM ↔ commerce	ERP for identity and cost; PIM for content	Scheduled publication; nightly full sync as safety net	Content tolerates minutes of lag; the item must exist in the ERP first
Stock quantity	WMS → ERP	The WMS	Batch, for valuation; **never the ERP → storefront** at the Omnichannel stage	Availability needs events; valuation needs totals
Customers and terms	ERP → commerce (B2B); commerce → ERP (D2C)	The ERP for business accounts, terms and credit limits; the platform for consumer accounts	On change	Payment terms and credit limits are finance facts

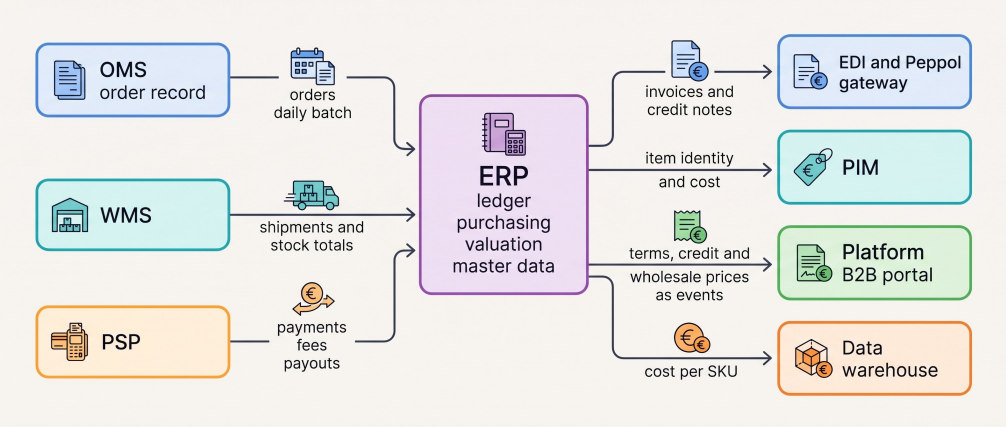


Figure 2: *The finance-only ERP: orders, shipments and money flow in from the systems that own them, in daily batches; invoices, item identity and cost, business terms and cost per SKU flow out, as events or batches. No channel reads stock, availability or prices from the ERP live.*

The flows have different requirements, which is why they should not share one connector and one schedule:

Flow	How fresh	How consistent	How fast	How available
Orders to ERP	Same day	Exactly once per order; one ERP document per order number	Batch window of minutes	A missed night must be replayable
Shipments to ERP	Same day	Matches the WMS pack record	Batch	Replayable

Flow	How fresh	How consistent	How fast	How available
Payments and payouts to ERP	Daily, aligned to the PSP's payout cycle	Every payout reconciled to its orders; Payments-to-Ledger: Draining the Suspense Swamp covers the method	Batch	Replayable
Business terms to the portal	Minutes after a credit limit changes *(assumption)*	The ERP's value wins	Event or short poll	The portal must degrade to "order pending review", not block
Availability to the storefront	Seconds to a minute	Per SKU and location	Event	**Not from the ERP**

The last row is the whole argument in one line.

Who Is the Hub? Three Answers and a Non-Answer

An **integration hub** is the one system every other system connects through and that owns the shared records. There are three honest answers to which system that should be, and one accident.

ERP as hub. Every system integrates with the ERP; the ERP holds stock, prices and customers, routes orders to the warehouse and feeds the storefront. The purest product form is a web store that reads the ERP directly: Sana Commerce positions itself (page accessed November 2026) as "the B2B e-commerce platform for manufacturers, built on your SAP or Dynamics ERP", where buyers "order online with the prices you agreed and the stock you really have". ERP-centered suites such as Odoo and JTL do the same from the ERP side. This fits one channel, one warehouse, a B2B-heavy business and a small team, because it avoids the integration problem by not having separate systems.

Commerce platform as hub. The platform is the system of record for orders, customers and stock; the ERP is a finance subscriber that receives what happened. Microsoft's

Business Central Shopify connector documentation (October 2026) describes exactly this division for a D2C merchant: orders flow from Shopify into Business Central as sales orders, with the "Shopify Order Id" kept on every document. It fits a D2C brand with one or two channels. It breaks when stores, wholesale and marketplaces each have their own order source, because a storefront platform's order model holds EDI and POS orders badly.

OMS as hub. An order management system (the one place every channel's order is created, with one state model, which The Order Record: Lifecycle, State and the Order Management System explains) owns the order record, allocation and availability across channels; the ERP owns finance and valuation; the platform owns the storefront. The OMS vendor OneStock argues the case in a 2026 blog post, which favors its own category and should be read that way: "ERPs update inventory in batch cycles, hourly or nightly in many implementations", the ERP "was designed for process consistency, not fulfillment optimization", and "ERP isn't broken. It's just not built for everything." It fits the Omnichannel and Group stages, and it is wrong at the Starting and Growing stages, where it is a second integration surface nobody needs.

No hub: an iPaaS. An **iPaaS** (integration platform as a service: a hosted tool in which you configure the flows between systems instead of coding them) carries point-to-point flows, and no system owns the cross-channel order record. This is not a design; it is what happens when each system is connected as it arrives. Kestrel & Pine's thirty-four integrations and three definitions of "order" are the result; APIs, Events, Files and Middleware: How Commerce Systems Talk explains the iPaaS as a tool, not an architecture.

Hub	What it means	Fits	Breaks when
ERP as hub	Every system integrates with the ERP; it holds stock, prices, customers and routes orders	One channel, one warehouse, B2B-heavy, small team	Peak traffic and real-time availability meet batch cycles; several channels write into one monolith; change is slow
Commerce platform as hub	The platform owns orders, customers, stock; the ERP subscribes for finance	D2C-first brands on a hosted platform, one or two channels	Stores, wholesale and marketplaces each bring their own order source

Hub	What it means	Fits	Breaks when
OMS as hub	The OMS owns orders, allocation and availability; ERP owns finance; platform owns the storefront	Omnichannel and Group	Cost and a second integration surface; wrong at stages 1–2
iPaaS, no hub	Point-to-point flows through an integration platform; nobody owns the cross-channel record	Growing stage with a few flows	Thirty-four integrations and three definitions of "order"

Waypoint's finance-only rule is the OMS-as-hub position written as an ownership rule; it does not move the ledger anywhere. Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record explains why ownership has to be decided per fact, not per system; this chapter applies that rule to the back office.

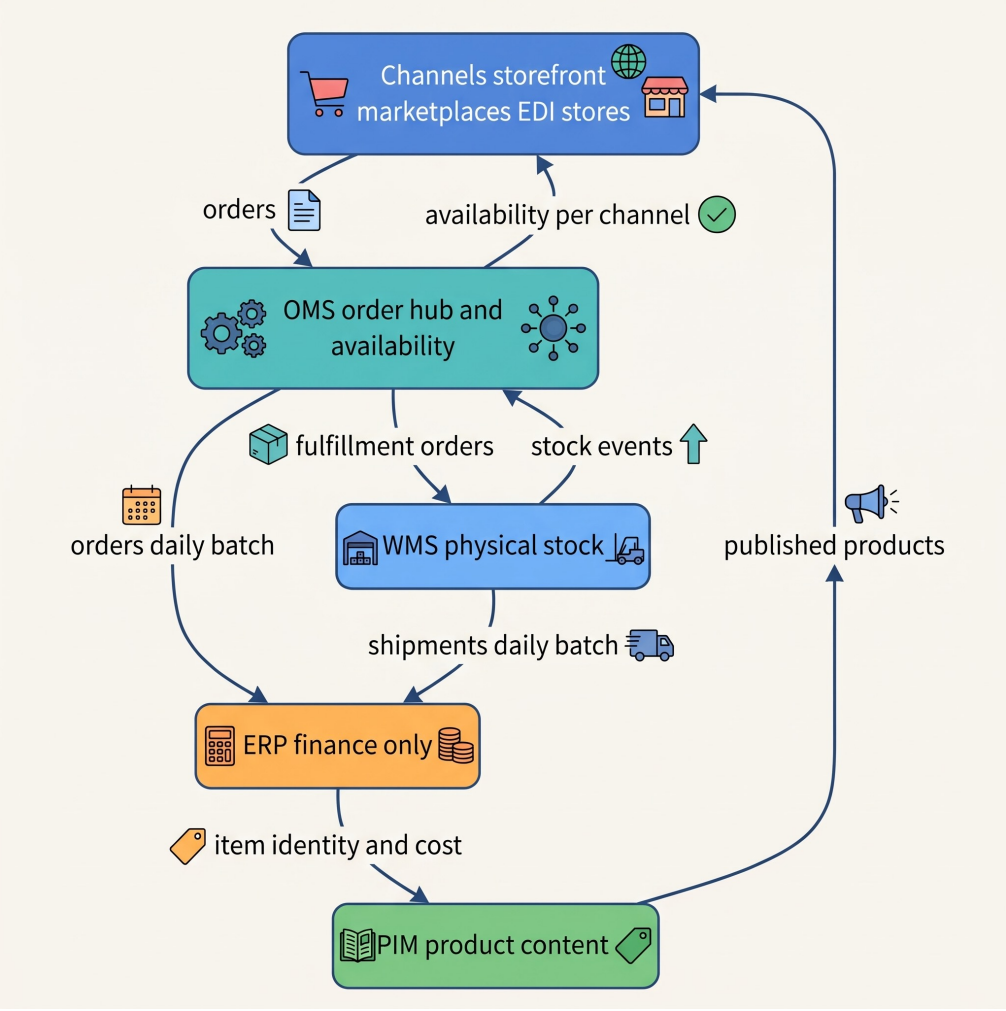


Figure 3: *Kestrel & Pine's target shape after Waypoint: the OMS is the hub for orders and availability, the WMS owns physical stock, the PIM owns content, and the ERP receives daily batches and feeds no channel directly. The ERP is no less important; it is no longer in the path of a sale.*

Batch or Events: The ERP's Rhythm

ERP integration is batch by default, and the vendors say so. Shopify's enterprise guidance on ERP integration (updated August 2026) says financials "sync in real time or through a daily batch". Business Central's connector runs its customer, company, order and inventory synchronizations as scheduled job-queue entries and recommends "every 15-30 minutes" for inventory. OneStock's "hourly or nightly" is the same observation from a competitor's side.

Finance can often work in batches; availability needs fresher answers. A complete, reconciled daily feed is more useful to finance than a fast feed that silently loses orders. Let each fact's owner publish changes, then let consumers process them at the cadence their work requires.

Kestrel & Pine's 2026 stock chain broke that rule. Routing warehouse changes through scheduled ERP, platform and marketplace updates left channels selling stale stock. The Black Friday Oversell exposed the delay. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation explains message safety and completeness checks.

Figure 4: *Finance can be batch, availability cannot: the ERP consumes what happened in its own daily rhythm and is never the system a storefront waits on.*

The business reason to keep the ERP out of the real-time path is not performance but **change**: a finance system is changed carefully, in releases, with auditors watching, while a storefront's availability logic changes weekly, and coupling them means the slow one sets the pace.

Connectors as Products

Between a hand-built integration and a general iPaaS sits a third option: a **connector** built and maintained by the vendor of one of the two systems. Shopify's Global ERP Program, announced on 14 October 2021, let selected ERP vendors build direct integrations into its app store; the announcement is five years old and the program's shape has changed since. The best-documented result is Microsoft's **Business Central Shopify Connector**, whose documentation (dated September and October 2026) is

unusually explicit about ownership and worth reading as a dated example of a vendor encoding the entity–owner table into configuration:

- It is "preinstalled for all new sign-ups and trials" of Business Central.
- For products, the merchant must "decide which system is the system of record for product data" before configuring the direction.
- For inventory, "Business Central acts as the system of record for inventory"; stock is pushed one way, as projected available balance or free inventory, every 15 to 30 minutes.
- For customers and B2B companies, "most businesses manage customer records, payment terms, and credit limits in Business Central and export them to Shopify", and "mixing both directions without a clear ownership rule can lead to conflicting records".
- For orders, a real-time notification from Shopify triggers a batch import job; each sales document keeps the Shopify order id; refunds create credit memos and a refund "still pending in Shopify" is retried later.
- The connector "uses expiring offline access tokens" that it refreshes automatically, with a refresh token valid for up to 90 days without a successful refresh.

The last bullet is the small print of what "a connector as a product" actually maintains: tokens, retries, version changes, mapping options. Shopify's 2026 guidance adds that certified ERP apps "undergo a 100-checkpoint review" and that "custom integrations introduce more risk and maintenance overhead".

SAP's learning material for integrating S/4HANA Cloud with SAP Commerce Cloud describes the suite-era division from the ERP side: master data replicated from the ERP to the commerce platform, storefront orders sent automatically to the ERP.

> **INFERENCE, NOT DOCUMENTED:** The Business Central inventory rule ("Business Central acts as the system of record for inventory") is the ERP-as-hub position for stock, and it is correct for the merchant the connector is built for: one warehouse, one storefront, no WMS. The same rule applied at Kestrel & Pine's size, with a WMS, six stores and three marketplaces, is the stock chain that oversold. Which rule is right is a property of the merchant, not of the connector.

Master Data: Who May Create and Change What

Master data is the slow-changing reference data (items, customers, prices, locations) that many systems copy; **master data ownership** says which system may create and change each record. The table below is my recommendation for a merchant at the

Omnichannel stage; at the Starting and Growing stages, the platform takes the PIM's and OMS's rows.

Data	Owner	Readers	Note
Item identity (SKU, GTIN barcode number), cost, supplier, purchasing units	ERP	PIM, platform, WMS, EDI	The item exists in the ERP before it can be bought or valued
Product content (names, descriptions, attributes, media, translations)	PIM	Platform, marketplaces	Kestrel & Pine's "three places" problem was the absence of this owner; What Are We Selling? Products, Variants and the Catalog
Wholesale list and contract prices	ERP price lists, published as price events with effective dates	OMS, platform B2B catalogs, EDI	What Does It Cost? Price Lists, Contract Prices, Currencies and Tax
Consumer prices	The platform's market price lists	POS, marketplace connector	The Price Drift was a consumer price copied through the ERP connector
Business accounts, payment terms, credit limits	ERP	Platform B2B portal, EDI, invoicing	Business Central's advice; Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals

Data	Owner	Readers	Note
Consumer accounts	Platform	ERP (for invoicing), service tools	One Customer, One Product: Identity and Master Data Across Systems for the matching problem
Stock quantity and availability	WMS for physical stock; OMS or availability service for available-to-promise (what a channel may still sell)	Platform, marketplaces, EDI	Never the ERP at the Omnichannel stage
Stock value	ERP	Finance, data warehouse	IAS 2 and the costing method
Orders	OMS; the platform before an OMS; never the EDI portal	ERP for posting, WMS	One order record
Invoices, credit notes, the ledger	ERP	EDI and Peppol gateway, finance	Numbering and archive live here

The product and pricing vendor Akeneo describes a PIM (product information management system) as "a central source of truth" for product content that "collects data from your ERP, MDM, PLM, and other systems" (MDM and PLM are master data management and product lifecycle management systems); OneStock's post says "ERP is the master for product data (if no dedicated PIM system exists)". Both are vendor sources, and both say the same thing from different sides: the ERP owns the item's identity and cost; something else owns what the shopper reads.

Trade-offs

Criterion	ERP as hub	Commerce as hub	OMS as hub	iPaaS, no hub
Integration surfaces to maintain	One, the ERP's	One, the platform's	Two: OMS and ERP	One per pair of systems
Availability latency	The ERP's batch cycle	The platform's	Seconds, by events	Whatever the slowest hop is
Order record across channels	The ERP's, if every channel feeds it	The platform's; weak for EDI and POS	The OMS's, by design	None
Cost	Lowest licenses; highest customization when channels multiply	Platform plus connector	Highest licenses; lowest custom code	Low licenses; high and hidden labor
Right when	One channel, one warehouse, B2B-heavy, under about €20M *(illustrative)*	D2C-first, one or two channels	Stores, wholesale, marketplaces and several countries	Never as a destination; acceptable as a transition

My recommendation follows the requirements table: if any channel needs availability in seconds and more than one channel creates orders, the ERP cannot be the hub, whatever the ERP vendor's connector assumes. If one channel creates orders and one warehouse holds stock, the ERP-centered suite is the cheapest honest architecture and an OMS would be the Giant's Boots.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	An accounting package or a small ERP connected by a vendor connector; the platform owns orders and products; the ERP owns the ledger and purchasing	Nothing more; a preinstalled connector (Business Central's is the dated example) is enough if the entity–owner choices in its configuration are made consciously	The founder posts marketplace sales by hand; the first "which number is right" argument
2 — Growing	The ERP owns items, business customers and terms; the platform owns consumer customers; a connector or iPaaS carries orders in and stock out; the ERP partner builds the first integration and becomes a dependency	A written field-level contract for the connector; a daily reconciliation of orders between platform and ERP	Stock to the storefront routed through the ERP; a second order source (wholesale, marketplace) living only in the ERP

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel	The ERP retreats to finance, purchasing, valuation and the cost side of the item master; an OMS owns orders and availability; a PIM owns content; the partner's connector is replaced by owned contracts	The finance-only rule; events from the owners, daily batches into the ERP; an e-document gateway for invoices; the ERP out of every real-time path	A peak day where the ERP's batch is the bottleneck; month-end over ten days; an order status that finance, support and the warehouse disagree on
4 — Group	A finance ERP per legal entity, or one ERP with company codes; shared OMS and PIM across brands; a shared integration platform with ownership rules	A single posting interface the OMS and WMS write to, with per-entity routing; master data governance as a function	Each brand with its own ERP connector to the same OMS; intercompany flows done by hand

Kestrel & Pine is at stage 3 with a stage 2 back office: the ERP owns too much, the connector is the partner's, and the ERP sits in the stock path. The month-end that takes eleven days is the finance-side warning sign; the Oversell is the commerce-side one.

In Practice

Microsoft's connector as a written entity–owner table. The Business Central Shopify documentation (September and October 2026) is the clearest public example of a vendor making the merchant answer the ownership question in configuration: products either direction, inventory in Business Central, business terms in Business Central, orders from Shopify. Read it as a template for the contract your own connector should have, not as the right answer for your merchant.

Shopify's two sentences. On ownership, Shopify's enterprise guidance (updated 28 August 2026) gives two sentences I would put on the wall of any integration team: "Document every field that moves between systems, and name the system of record for each" and "Decide in advance what happens when the two systems disagree." The second is the one most integration contracts lack.

Kestrel & Pine: Eva's order in the books. Eva Janssen's order #1072455 of 13 April 2027 is €235 including Dutch VAT at 21%: a Ridgewalker jacket at €199 and two pairs of merino socks at €18. Under Waypoint's target flow, the OMS sends the order to the ERP in the night batch of 13–14 April; the WMS sends the shipment confirmation of 14 April in the next batch; the PSP's payout file, received a few days later, carries the €235 iDEAL payment and the PSP's fee (assume an illustrative fee of about €0.30 for an iDEAL transaction).

The ERP posts revenue of €194.21, recognized at delivery on 15 April, and VAT payable of €40.79, relieves stock at the jacket's landed cost of €58 plus the socks' cost, records the PSP fee as an expense, and reconciles the payout to the order through the PSP's reference, which Payments-to-Ledger: Draining the Suspense Swamp in the finance book explains in depth.

When Eva exchanges the jacket for size L on 19 April, the exchange is a return and a new order in the OMS; the ERP receives a credit note for €199 and a new sale for €199 in the batch of 22 April, and the two net to zero in revenue while stock moves twice. Nothing in this paragraph needed the ERP within the minute, and nothing in it is visible to Eva.

Kestrel & Pine: the wholesale side. Bergzicht's spring reorder of 6 April 2027 (240 units, €19,860 excluding VAT) enters the ERP as an EDI order today and the OMS from Q4 2027. Either way, the ERP issues the INVOIC (the EDI invoice message) on 9 April from the WMS's shipment confirmation, posts a receivable due 9 May, and runs dunning (payment reminders) if payment is late; Bergzicht's credit limit and 30-day terms live in the ERP and are read at order acceptance. The ERP's job did not shrink here.

Kestrel & Pine: retiring the connector. The partner-built connector is replaced in two steps. In Q3 2027 Ruben writes the field-level contract it should have had, as the specification for the OMS-to-ERP batch: one row per field, with direction, cadence, system of record and the rule when the two disagree. Then the connector is switched off flow by flow as each order path moves through the OMS (marketplace orders in September, storefront orders in October, EDI and manual wholesale in December 2027), with the old paths retired in January 2028 and stock already out of the ERP's path under W1. The partner remains the ERP's implementation partner; it stops being the owner of an integration nobody else could read.

What Never Belongs in the ERP, and the Partner as a Dependency

Some things should never run in the ERP, because they are in the path of a sale and change at the storefront's pace: carts, search, promotion logic, real-time availability for checkout, marketplace listings, customer-facing order status. Some things always belong there: the posting of every order, payment, fee, refund and VAT amount; the cost of stock; the legal invoice and credit note; receivables and dunning; supplier orders and landed cost. The grey area is master data, and the ownership table above settles it.

The **ERP implementation partner** is an architectural dependency most merchants do not list. The connector is usually theirs, undocumented, and priced into the ERP relationship. That is the Lynchpin: a capability customers depend on, running through one firm, with no exit plan. The remedy is not to fire the partner but to treat the connector as a contract the merchant owns: a field-level specification, a named internal owner, idempotent handling, a reconciliation job and a monitored failure alert, which is Waypoint's W5 applied to one integration. Who Builds and Who Owns: Teams, Vendors, Agencies and Governance covers the partner relationship as a management question.

When It Goes Wrong

Duplicate sales orders. The platform's webhook fires twice, or the nightly batch is re-run after a partial failure, and the ERP holds two sales orders for #1072455. Finance sees double revenue until month-end. Detection: one ERP document per order number is a rule the batch enforces, and a daily count of orders in the OMS against sales orders in the ERP. Recovery: cancel the duplicate before posting; after posting, a reversing entry with a reference to the original.

Posted before paid. The order reaches the ERP as a sale before the payment is captured, and the customer's bank declines. Detection: the order event carries payment status, and the batch posts only captured orders. Recovery: a credit note for the uncaptured sale, and a rule that the PSP's capture event, not the order event, is what makes a sale postable.

Refunded twice. Kestrel & Pine's Double Refund of January 2027: 212 returns refunded once by the returns portal through the platform's API and once by support in the PSP console, €19,400 twice. The ERP received and posted both, which is how finance found it. Detection: a daily reconciliation of refunds against PSP refund records by return number; the ERP is where the two paths meet. Recovery: The Way Back: Returns, Exchanges, Refunds and Reverse Logistics owns the fix; the ERP's role is to be the reconciliation point.

The item that does not exist. Merchandising creates a variant in the platform; the ERP has no item for it; the order batch fails on that line and, because nothing is idempotent,

the whole night's batch. Detection: the batch reports per line, not per file. Recovery: create the item in the ERP, replay the line; and fix the cause, which is an item created outside its owner.

The batch window meets the peak. The nightly order batch, sized for about 800 orders, meets Black Friday's 11,500 (31,000 in the week) and runs into the morning while the stock sync waits behind it. Detection: a batch duration alert. Recovery: split the batch by flow so a slow one does not block a fast one, and take the ERP out of the stock path for good; Black Friday Every Year: Peak Load, Resilience and Recovery covers the peak plan.

The partner changes the connector. The ERP's new release changes a field; the partner updates the mapping to keep the connector running and tells nobody, and a flow starts failing quietly. The Price Drift shows the cost: one failed push of the German price list was reported as a success, and the storefront sold at €179 for nine days. Detection: a reconciliation that compares prices across systems daily. Recovery: the contract, under version control, with a change notice as a condition of the partner's work.

What to Take Away

Established.

- An ERP's jobs in a commerce business are the ledger, purchasing, inventory valuation and master data; inventory valuation is an accounting function governed by IAS 2 and the costing method, distinct from stock availability.
- ERP integration is batch by default: the vendors' own documentation (Business Central's job queues, Shopify's "real time or through a daily batch", OneStock's "hourly or nightly") says so, and batch is the right rhythm for finance.
- Ownership must be named per field, with a rule for disagreement, as Microsoft's connector documentation and Shopify's 2026 guidance both state.

Depends on context.

- ERP-as-hub is right for one channel, one warehouse and a B2B-heavy business; it becomes the bottleneck when stores, wholesale and marketplaces each need stock in seconds, because the ERP's batch cycle becomes the availability latency.
- An OMS as hub is right at the Omnichannel and Group stages and wrong before them, where it is cost and a second integration surface nobody needs.
- A vendor connector is right when the merchant matches the merchant the connector was built for (its documented ownership rules are the test); otherwise its configuration encodes someone else's architecture.

The author's synthesis.

- The finance-only rule: the ERP is the system of record for finance and nothing else, consumes everything by daily batch from the systems that own the facts, and feeds no channel directly: what it owns for channels (wholesale prices, terms, credit) leaves it as events. It gives the ERP a smaller job and a more important one, because a ledger that is complete and reconciled is worth more than one that is fast.
- The connector your ERP partner built is the most important undocumented integration you own; write its contract before you replace it, because the contract is what you are really buying.

Questions to Ask

1. For each of orders, shipments, payments, products, stock and customers, which system is the system of record, and is that written down with a rule for disagreement?
2. Does any channel read stock or availability from the ERP? If so, what is the end-to-end delay from a sale to the storefront's number?
3. Who built the platform-to-ERP connector, who can change it, and where is the field-level specification?
4. Is the ERP in the path of any real-time decision (checkout, availability, order status to the customer)?
5. How does the ERP learn about marketplace orders and store sales, and in what shape?
6. What reconciles orders in the commerce systems against sales orders in the ERP, how often, and who reads the result?
7. Is a sale posted on the order event or on the payment capture, and what happens to a declined payment?
8. If the ERP partner ended the relationship tomorrow, which integrations would you be unable to change?

The ERP is one of a dozen systems that must behave like one. The next part is about how they talk: the styles of integration, the events and reconciliations that let separate systems disagree and recover, the contracts between them, and the identity and data questions that follow. Continue with Part VI — INTEGRATION: Making Separate Systems Behave Like One.

Sources and Evidence

Company, product and investor sources

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- Synchronize and fulfill sales orders, Microsoft Learn, 2026: order import, the Shopify order id, refunds and credit memos.
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- Shopify launches Global ERP Program, Shopify, 2021: the program and its launch partners (older than three years).
- ERP Integration: Methods, Benefits, and Examples, Shopify Enterprise blog, 2026: "real time or through a daily batch", the review checkpoints and the two ownership sentences.
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- ERP software, AFAS, n.d. (accessed 2026-11): one system, one database (translated from Dutch).
- What is a PIM?, Akeneo, 2026 (accessed 2026-11): the PIM as source of truth for content, collecting from the ERP (vendor source).

Standards, regulation and official documentation

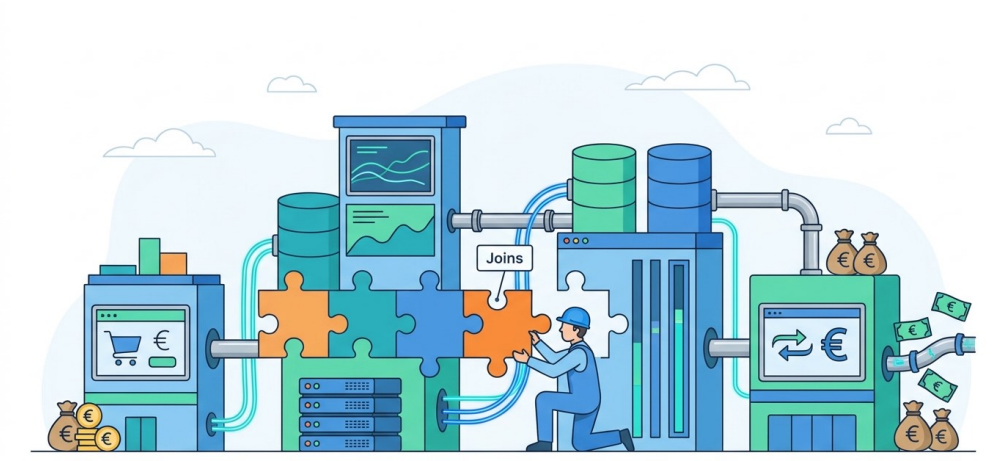
- IAS 2 Inventories, IFRS Foundation, n.d. (accessed 2026-11): measurement at the lower of cost and net realisable value; FIFO and weighted average.

Research and practitioner sources

- The hub comparison, the finance-only rule, the master data ownership table and the scale-stage table are the author's synthesis of the sources above and of common practice; Kestrel & Pine's connector behavior, month-end duration and the iDEAL fee are illustrative.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part VI — INTEGRATION: Making Separate Systems Behave Like One



KEY POINTS:

- **Integration is decided per flow, and every flow needs an owner.** Events for stock and orders, scheduled publication for products, batch for finance; in January 2027, 13 of Kestrel & Pine's 34 integrations had nobody to call.
- **Separate systems are built to disagree, so design them to recover.** Idempotency keys, outboxes, compensations and reconciliation jobs, plus a version and sunset date on every contract.

- **Moving the data is half the work; meaning the same thing by it is the other half.** One owner per entity and per metric is what makes joined systems behave like one.

An order reaches the warehouse twice, a stock update arrives after the last unit has sold, or an invoice names a customer the finance system cannot identify. The systems at either end can work correctly while the exchange fails. This part explains how to make those handoffs timely, understandable and recoverable, so an interrupted message does not become a missing shipment or a second charge.

Figure 1: *Every handoff between systems is a place where an order, a stock level or a price can arrive late, twice, in the wrong shape or not at all.*

Where we are in the order's journey. Eva's purchase at Kestrel & Pine crosses from checkout to warehouse and finance. Payment success arrives separately. Picking changes what channels may sell; her later exchange adds return and replacement records. Each step depends on messages that must be understood and checked at both ends.

Kestrel & Pine is a fictional outdoor brand with thirteen systems and 34 integrations. Its **Waypoint** improvement program introduces integration standards, shared metric definitions and customer identity rules. These decisions make the flows explainable and recoverable; the integration tool alone cannot supply that ownership.

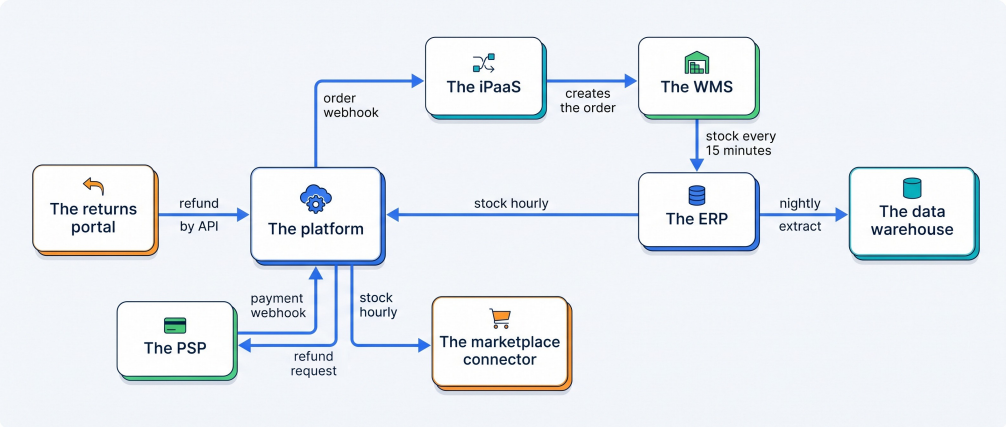


Figure 2: *The joins Eva's order crosses in April 2027: eight systems, four styles, and a stock figure that takes up to two hours to travel from the warehouse to the marketplaces.*

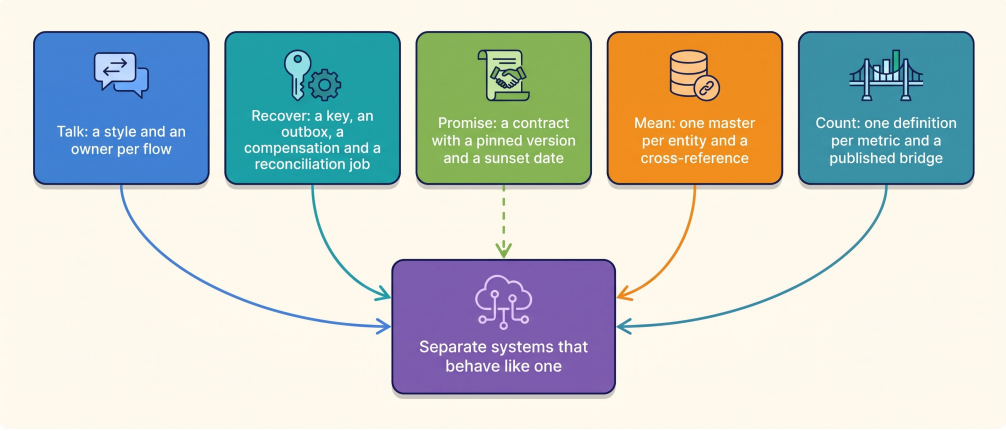


Figure 3: *The five questions of Part VI and the answer each chapter gives; every incident in the part is a flow that had no answer to one of them.*

Chapters in this part

APIs, Events, Files and Middleware: How Commerce Systems Talk chooses **files, shared databases, direct calls or messages according to the flow's needs**. The worked inventory finds 13 unowned integrations. Each flow needs a documented purpose, owner, failure behavior and check.

Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation explains **safe retries and recovery from disagreement**. It introduces idempotency, making repeats harmless; an outbox, saving an event with its underlying change; and rec-

onciliation, comparing records to find gaps. Duplicate refunds and stock differences demonstrate their roles.

Promises Between Systems: API Contracts, Webhooks and Versioning manages **interface changes and retirement dates**. A forgotten 2021 script stops sending stock when a vendor retires its API version in February 2027. A recorded version, deadline, owner and upgrade check would have made that change actionable.

One Customer, One Product: Identity and Master Data Across Systems connects **records referring to the same customer or product**. Eva appears separately in storefront, loyalty, newsletter and returns systems. The chapter defines authoritative identities, cross-references, matching and reversible merges.

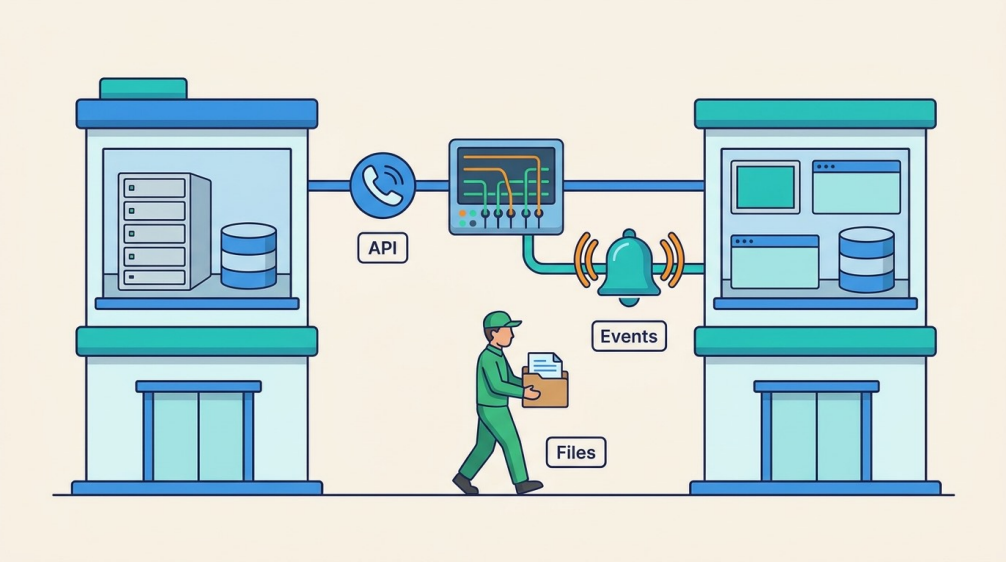
Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics explains **why valid reports can disagree**. Platform, finance and warehouse report 21,050, 22,710 and 24,140 February orders because their definitions and coverage differ. A common definition makes genuine missing data distinguishable from expected differences.

Begin with APIs, Events, Files and Middleware: How Commerce Systems Talk.

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine is fictional; recommendations and labeled assumptions are the author's judgement.

24. APIs, Events, Files and Middleware: How Commerce Systems Talk



IN THIS CHAPTER, YOU WILL: Learn the four ways commerce systems exchange data, the mechanisms that trigger each exchange, how to choose a style per flow, and why an inventory of integrations is the first management tool.

KEY POINTS:

- **There are four styles and five triggers, and the right one is chosen per flow, not per company.** A file, a shared database, a direct call or a message moves the data; a call, a webhook, a poll, a schedule or a database log decides when. Stock and orders want events plus a safety net, products want scheduled publication, finance wants a daily batch.
- **Point-to-point connectors multiply with every system; a hub only helps when it comes with ownership rules.** A hosted integration platform centralizes mapping, retries and monitoring, but a hub whose

flows have no owner, contract or reconciliation recreates the same tangle in one place.
- **The integration inventory turns integration debt into a list**. One row per flow, with source, target, style, owner, contract version, failure handling and reconciliation. Kestrel & Pine's inventory found 34 flows, 13 of them without an owner in January 2027, and none with a check on stock or orders.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Spaghetti Junction** — every pair of systems has grown its own connector, with its own owner or none, because each was built for the need of the moment.
- **The Message in a Bottle** — a scheduled sync or a webhook is sent once and trusted, so a silent failure stays silent until a customer finds it.
- **The Lynchpin** — a flow that only an agency script, one consultant or one vendor connector understands is a single point of failure the business does not know it has.

A storefront accepts an order, but the warehouse can only pick it after its own system receives the instruction. If that message is lost, the customer can be charged while the parcel never enters the picking queue. If it arrives twice, the warehouse may prepare two parcels. The reliability of the handoff directly affects what the customer receives.

An **integration** exchanges data or triggers work between systems. Its design determines how fresh the stock badge is, what happens during an outage and who investigates a missing order. Kestrel & Pine, the fictional outdoor brand used here, oversold for up to two hours on Black Friday because three scheduled updates added their delays together.

Different flows need different timing and recovery. A stock update during a sale can be urgent, while finance may accept records in a later batch. Choosing an exchange method for those needs avoids paying for speed where it adds little or tolerating delay where it loses sales. This chapter compares files, shared databases, direct calls and messages, plus the tools that connect them. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation explains recovery from duplicates and gaps; Promises Between Systems: API Contracts, Webhooks and Versioning covers changes to the agreements.

Where we are in the order's journey. Eva's jacket order crosses from the platform to the warehouse after checkout on 13 April 2027. The payment provider separately reports success. After picking, stock changes travel back toward the storefronts and marketplaces. This chapter follows those messages and the responsibilities at each crossing.

Setting the Scene at Kestrel & Pine: Thirty-Four Lines Nobody Had Drawn

Kestrel & Pine had thirteen systems and **34 integrations** in January 2027. New Head of Engineering Noor Visser asked lead integration engineer Ruben Alves for an inventory. Only fourteen flows were listed: those running in the **integration platform as a service (iPaaS)**, a hosted tool for configuring data flows. Scripts and other connectors were missing.

Noor spent February building one row per flow: source, destination, data, method, frequency, owner, recovery and checks. Three flows emerged only because data arrived that none of the listed integrations could explain. The finished inventory had 34 rows:

What the inventory found (February 2027, illustrative)	Count
Flows that move data by **files or batch exports** (EDI documents, that is electronic data interchange, the standard exchange of business documents between companies' systems; Peppol e-invoices on the European e-invoicing network; the nightly stock export from the POS, the store till system; data-warehouse extracts; spreadsheet imports)	9
Flows that **read another system's database directly** (the BI tool, for business intelligence reporting, on the ERP's reporting copy)	1
Flows that are **direct API calls** (one system calling an operation another exposes through its application programming interface: checkout to the PSP, the returns portal to the platform's refund API, the WMS to the shipping service, the connector to three marketplaces, the stock pulls)	11

What the inventory found (February 2027, illustrative)	Count
Flows that are **events, webhooks or queues** (the platform's order webhooks into the iPaaS, PSP payment webhooks, WMS shipment events, the connector's order imports)	13
Running on the iPaaS / as agency scripts / as vendor-built connectors or plugins	14 / 9 / 11
Flows with **no named owner** (as of January 2027; 5 still unowned in April 2027, after the ownership map)	13
Flows with a **reconciliation job** (something that compares both ends): PSP payouts, EDI invoices with Bergzicht, store cash	3
Flows that **write stock into the platform**	4 (the hourly ERP sync and three apps); with hand edits in the ERP and the platform admin, six writers of Duiven stock in all

Two numbers made the business case for Waypoint's integration standards (decision W5, below). In January 2027, thirteen flows had nobody who would be called when they stopped, and nothing compared stock or orders between any two systems. The Black Friday Oversell, the Price Drift and the Deprecated Endpoint had all happened in flows with no owner and no check. Noor's summary to Mara Lindqvist, the CEO: "We have 34 integrations and 34 different answers to the question 'how do we know it worked?' "

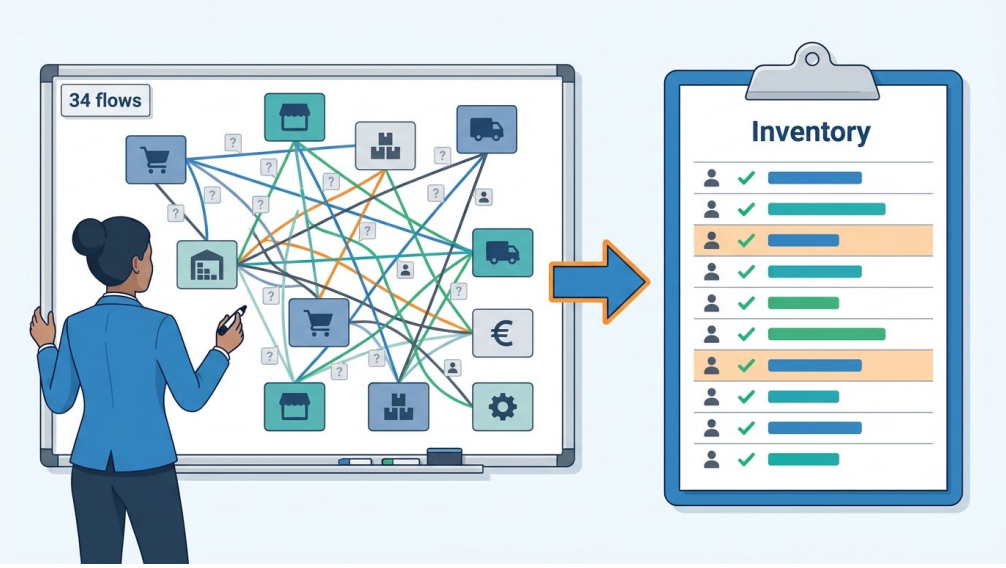


Figure 1: *Before the inventory, 34 integrations ran between thirteen systems, and 13 of them had nobody who would be called when they stopped; the inventory turned the tangle into a list.*

What an Integration Is, and Why the Joins Cost Money

Two systems need to exchange data when a fact one of them owns is needed by the other. The WMS knows how many Ridgewalker jackets in size M are on the shelf; the storefront needs that number to show "in stock". The platform knows Eva ordered one; the warehouse needs the order to pick it. The fact has a **system of record** (the one system whose value counts as the truth when copies disagree; Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record owns the idea), and every other system holds a **copy**.

An integration moves the fact from the owner to the copies, or sends a request the other way ("refund €199 on order #1072455"). Four properties of that movement matter to the business:

- **Freshness**. How old is the copy? A stock copy an hour old sells units that are gone; a product description an hour old is fine.
- **Reliability**. When the move fails, is the fact sent again, lost, or sent twice?
- **Coupling**. How much does each side need to know about the other? A flow that depends on the exact field names of the WMS's interface breaks when the WMS vendor changes them.

- **Ownership**. Who understands this flow, is told when it fails, and decides what to change?

The style is the main lever on the first three; the inventory is the only lever on the fourth.

The Four Styles: File, Shared Database, Direct Call, Message

The vocabulary most practitioners still use comes from Gregor Hohpe and Bobby Woolf's *Enterprise Integration Patterns* (2003; the book is old, the styles have not changed, the technology under them has). It names four **integration styles**:

1. **File transfer**. One system writes a file; the other reads it later. The file is the contract. Commerce examples: the nightly product feed to a marketplace, the EDI order Bergzicht's system sends through the EDI provider, the POS export of store stock to the ERP.
2. **Shared database**. Two systems read and write the same tables. Common inside a suite where the web shop and the ERP are one product; risky across vendors, because any schema change breaks the other side and there is no record of who changed what.
3. **Direct call** (Hohpe and Woolf say *remote procedure invocation*). System A calls an operation System B exposes over the network, usually through an **API** (application programming interface: the set of operations a system lets other software perform), and waits for the answer. Commerce examples: the checkout asks the PSP to authorize €235; the returns portal asks the platform to refund €199.
4. **Messaging**. System A puts a message on a shared channel (a **queue** or **broker**: software that holds messages until the receivers take them) and continues without waiting; B, C and D read it when they are ready. Commerce example: "order #1072455 placed" is published once and consumed by the WMS, the ERP connector and the e-mail service.

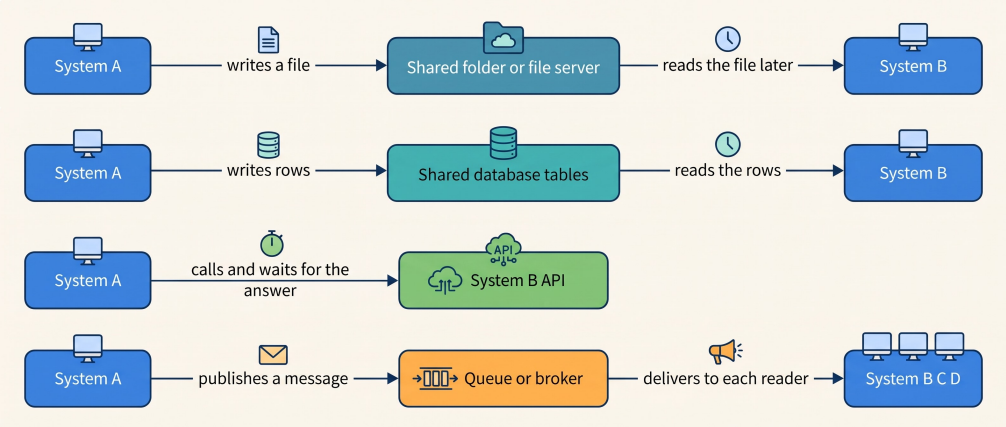


Figure 2: *The four integration styles. Only the direct call makes A wait for B; the other three let the two systems work at different times, which is what makes them resilient and what makes their copies stale.*

Two words recur. A **synchronous** exchange makes the caller wait for the answer (the direct call): the checkout cannot continue until the PSP says yes or no. An **asynchronous** exchange lets the sender continue (files, messages): the storefront does not wait for the warehouse to confirm it received the order. Synchronous is simple to reason about and fragile under load: if B is slow, A is slow; if B is down, A fails. Asynchronous is resilient and introduces a delay during which the two sides disagree. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation is about living with that delay.

Hohpe and Woolf's warning is the one to remember: "Tightly coupled applications make numerous assumptions about how the other applications work." A direct call to the WMS assumes the WMS is up, fast and unchanged; a message assumes only that the broker will hold it.

Five Ways to Trigger an Exchange

The style says how the data moves. A second choice says **when**: who notices that something changed and starts the exchange. Five mechanisms do that work, and most merchants run all of them.

Mechanism	Who starts it	Typical delay	What the receiver must handle	Typical commerce use
Direct API call	The caller, when it needs the answer	Milliseconds	Timeouts, partial success, retry storms under load	Price, tax or stock check at checkout; payment authorization; a refund request
Webhook	The source, when something happens: one HTTP request to a URL you registered	Seconds	Missed, duplicated and out-of-order deliveries; must answer fast	Order placed, payment captured, shipment created
Polling	The receiver, on a timer: "what changed since my last visit?"	Minutes	Rate limits, large pages, clock differences	Catching up after an outage; the safety net under webhooks
Scheduled batch or file	Either side, on a clock	Hours to a day	Late files, partial files, stale by design	Product feeds, EDI documents, nightly stock and finance exports
Queue or event stream	The source publishes; each consumer pulls at its own pace	Seconds	Duplicates (delivery is "at least once"); ordering only per key	Stock and order events with several consumers

Mechanism	Who starts it	Typical delay	What the receiver must handle	Typical commerce use
Change data capture (CDC)	The database's own change log, read by a connector	Seconds	A row change is not a business event; someone must translate it	Turning an old ERP or order management system (OMS) into an event source without rewriting it

(The author's synthesis of the vendor documentation cited under "In Practice" and in the next chapter.)

Two of these need plain words. A **webhook** is a notification, not a guarantee: the source makes one HTTP request to your address when something happens, and if your server was down, slow or busy, the request is retried for a while and then dropped. Shopify's developer documentation (n.d., accessed November 2026) says its HTTPS webhooks are retried eight times over four hours, that a subscription created through the API is deleted after eight consecutive failures, and that delivery "isn't always guaranteed." Stripe's (n.d., accessed November 2026) retries for up to three days. Both tell integrators the same thing: run a periodic **reconciliation job** that asks the source for everything changed since the last run. The webhook is a speed-up on top of polling, never a replacement for it.

Change data capture reads a database's transaction log, the record every database keeps of each change, and turns it into a stream of "row inserted, updated, deleted" events. Debezium, the most used open-source tool for this (documentation for version 3.7, accessed November 2026), describes itself as a set of source connectors that use each database's own change-capture features. CDC is the honest way to get events out of a system never built to publish them. Its limit is meaning: "row updated in the item ledger table" is not "stock changed for SKU X at location Y" (a SKU being one sellable variant, one size in one color), and someone has to write that translation and keep it current.

"Event-driven" itself means several things. Martin Fowler's 2017 article separates **event notification** (the source says "order #1072455 changed" and receivers call back for details; low coupling, hidden flow) from **event-carried state transfer** (the event carries the data; receivers keep their own copy and never call back; faster, but every receiver now holds a copy that can go stale). A **stock event** in this book's sense (Is It in Stock? Inventory, Availability and the Promise You Can Keep) is the second kind: "SKU 2113-M-moss at Duiven: 14 on hand, 3 reserved, as of 21:14:07." Fowler's other two, event

sourcing and CQRS (command query responsibility segregation), are internal design choices of a platform or an OMS, not integration styles.

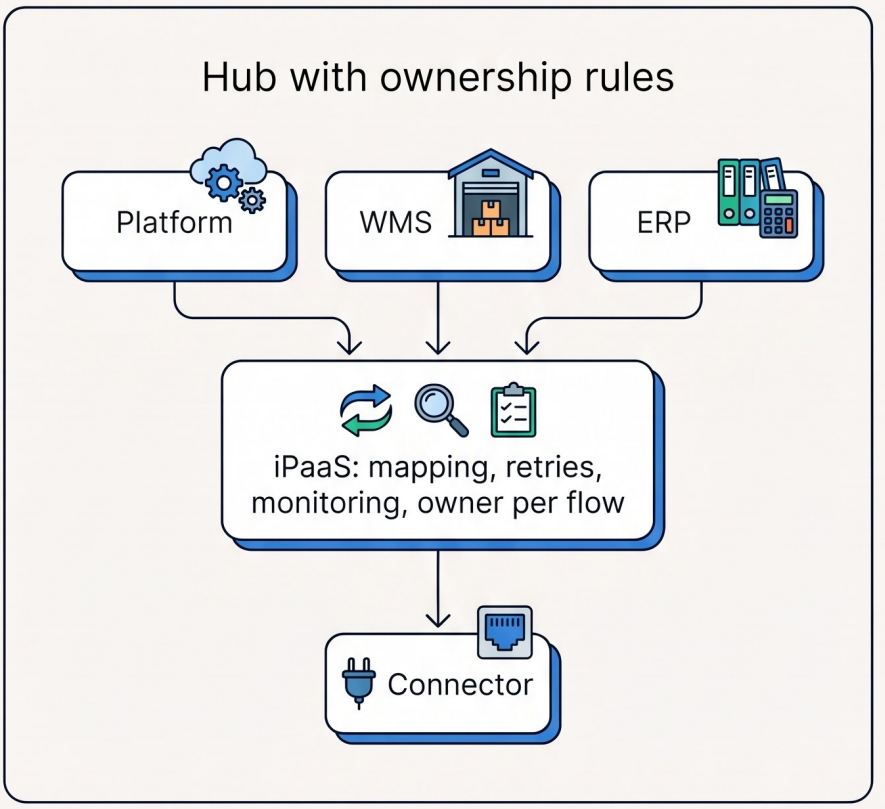
Point to Point, Hub, or the Vendor's Connector

With thirteen systems there is a second question: how the whole landscape is wired. There are three answers, and most Omnichannel merchants run all three at once.

Point to point. Each pair of systems that needs to talk gets its own connector, built by whoever needed it: the agency, the ERP partner, an engineer who has since left. The first three are cheap and fast. By the tenth, every change to one system means finding and testing every connector that touches it, and the mapping between two data models (which WMS field means which platform field) lives in code one person understands. This is The Spaghetti Junction, and it is what buying one good system per need, one need at a time, produces.

A hub. An integration platform sits in the middle; every system talks to the hub, which holds the mapping, the schedule, the retries and the monitoring in one place. The hosted form is the iPaaS. Gartner's definition, as relayed by an encyclopedia entry (2015 retrieval; the Gartner page itself could not be fetched), is "a suite of cloud services enabling customers to develop, execute and govern integration flows between disparate applications," with prebuilt connectors so that "no software development" is needed for common pairs. The catch is that a hub does not create ownership. Kestrel & Pine's fourteen iPaaS flows were the best-documented of the 34 and still had three with no owner.

The vendor's connector. Between the two sits a connector built and maintained by the vendor of one of the systems. Microsoft's documentation for Dynamics 365 Business Central's Shopify Connector (published October 2026) shows what such a connector decides for you. It tells the merchant to "decide which system is the system of record for product data" before configuring, states that for inventory "Business Central acts as the system of record" and pushes stock one way at a recommended interval of 15 to 30 minutes, and for orders runs a batch import that a real-time notification from Shopify merely *triggers*. That last detail is the webhook-plus-polling pattern built into a product. A vendor connector encodes an ownership model; if it matches yours, it is the cheapest reliable option, and if not, no configuration will fix it.


Hub with ownership rules
Platform
WMS
ERP
iPaaS: mapping, retries, monitoring, owner per flow
Connector

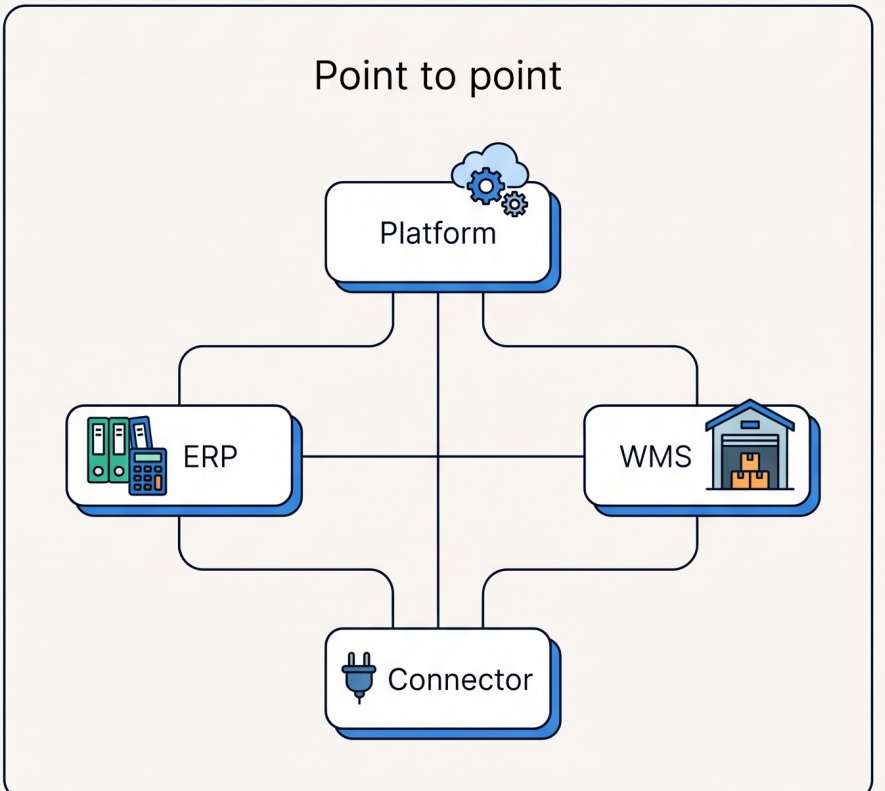

Point to point
Platform
ERP
WMS
Connector

Figure 3: *Four systems need six point-to-point links; a hub needs four. The saving grows with every system added, but only if each flow through the hub still has an owner, a contract and a check.*

The arithmetic: n systems that all need to talk need up to $n(n-1)/2$ point-to-point links, 78 for thirteen systems; through a hub they need 13. The real saving is one place to find a flow's mapping, schedule, failure handling and log.

Hub costs are hard to compare. Fetched in November 2026: MuleSoft packages "start at $2,000 per month, billed annually" on usage-based credits, with a $50,000 minimum for its pre-commit model; Boomi's pay-as-you-go starts at $99 a month plus usage; Workato and Celigo publish no prices; Make bills in credits with a free plan of 1,000 a month; n8n charges per workflow execution, from €20 a month for 2,500 to €667 for 40,000, billed annually.

The lesson is that the metering unit (credits, executions, connections, concurrency) differs per vendor, so the only honest comparison is per flow at your volume: Kestrel & Pine's roughly 950 orders a day (D2C, marketplace and wholesale, the 2026 average), each triggering five or six flow executions, costs very differently under per-execution and per-connection pricing.

The low end is priced for a Growing-stage merchant's first three integrations; a $50,000 pre-commitment signals a Group-stage buyer.

> **INFERENCE, NOT DOCUMENTED:** Every iPaaS pricing page in November 2026 sells "AI agents" beside integration. What a merchant buys underneath is the same four functions as in 2020: connectors, mapping, scheduling and monitoring. I would evaluate on those and treat the AI framing as a feature that may not survive the next pricing page.

Choosing Per Flow

The mistake I see most often is choosing a strategy for the company ("we are event-driven now"). Flows differ in what they need, and the style follows from the needs. "Consistency" below means how closely the copies must agree with the system of record; "freshness" is how old a copy may be.

Flow	System of record (typical)	Freshness	Consistency	Latency of the exchange	Availability
Stock	WMS, POS, 3PL (a third-party warehouse); an availability service for what may be sold	Seconds to minutes; the oversell window is the sync interval	High per SKU and location; a wrong copy sells a unit that is gone	Seconds	Must survive a consumer being down: queue it
Orders	Platform or OMS	Seconds for the warehouse, minutes for finance	Never lost, never doubled; one order number family	Seconds	Must survive the WMS being down: queue it
Prices	The platform, the ERP or a pricing service	Minutes; promotions with start times want the minute	A stale price is a broken promise with legal weight	Minutes	Fail loudly, never silently
Products and content	PIM, the product information management system (or the ERP item master before one exists)	Hours; a season is published, not streamed	Completeness over speed	Hours	Can wait a day

Flow	System of record (typical)	Freshness	Consistency	Latency of the exchange	Availability
Customers	Platform, CRM (customer relationship management system) or an identity service	Hours	Matching is the hard part, not transport (One Customer, One Product: Identity and Master Data Across Systems)	Hours	Can wait
Finance (invoices, payments, payouts)	ERP and the PSP	Daily; finance works in periods	Exact to the cent; reconciled, not trusted	Hours	Complete, not fast

(Typical values; the author's recommendation, not a standard.)

From those needs, the rule I use:

Flow	At the Growing stage	At Omnichannel and beyond	Why
Stock	The inventory owner pushes to the channels every 15–30 minutes (the interval Business Central's connector recommends, October 2026)	Stock events per SKU and location into an availability service, plus a daily reconciliation and a safety buffer	The oversell window equals the sync interval; at 38 orders a minute in the peak hour, fifteen minutes is 570 orders

Flow	At the Growing stage	At Omnichannel and beyond	Why
Orders	A webhook starts the import; a polling sweep catches what it missed	Order events to the OMS, WMS and ERP connector; every consumer idempotent (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation)	Orders must never be lost or doubled; a webhook alone cannot promise that
Prices	Scheduled sync, hourly or faster, with an alert when it fails	Events for promotion start and end; scheduled sync as the net	The Price Drift was a scheduled sync that failed without an alert
Products	Scheduled publication from the system of record	Event on change for the storefront, plus a nightly full publication	A nightly full copy heals any missed change
Customers	Copied with each order; periodic merge	An identity service or customer data platform	The problem is deciding whether two records are one person
Finance	Nightly batch to the ERP, reconciled against the PSP's payout report	Same; reconciliation becomes a data-platform job across brands	A daily batch that is checked beats an event stream nobody balances

This is what Waypoint's decision **W5** says for Kestrel & Pine: stock, orders and prices move by events; products by scheduled publication; finance by daily batch; and every flow has an owner, a contract, idempotent handling and a reconciliation job. The next chapter explains the last two terms; the reason the rule is per flow is this table.

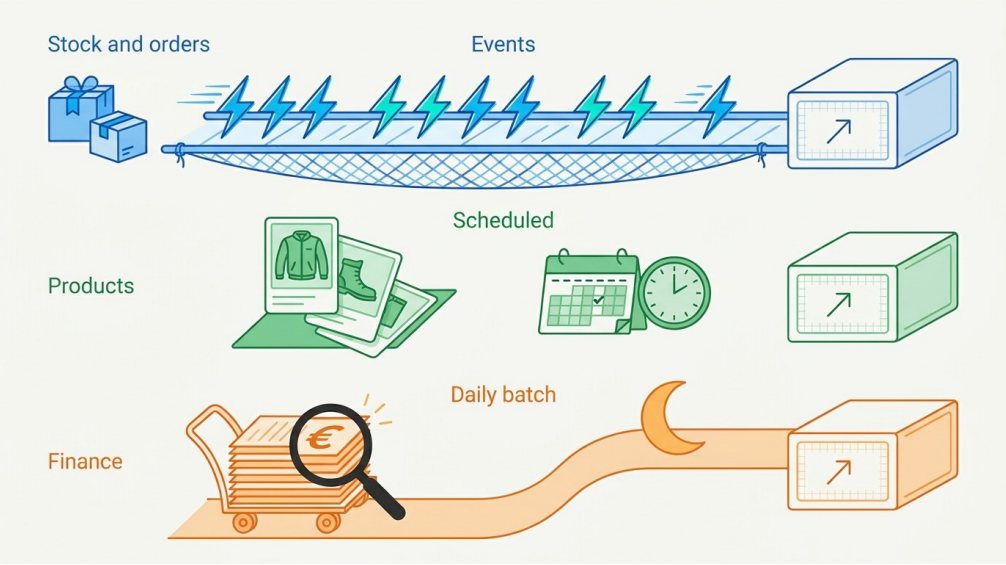


Figure 4: *The style follows the flow: stock and orders move as events with a reconciliation net beneath them, products by scheduled publication, finance by a daily batch that is checked.*

The Trade-Offs

	Direct API call	Webhook plus polling	Queue or event stream	Scheduled file or batch	Shared database
Freshness	Immediate	Seconds, gaps healed by the poll	Seconds	Hours to a day	Immediate
When the other side is down	The caller fails or waits; under load, a retry storm	Deliveries are retried for hours, then dropped	Messages wait in the queue	The file is late	Both sides fail

	Direct API call	Webhook plus polling	Queue or event stream	Scheduled file or batch	Shared database
Coupling	High: on the callee's availability and interface	Medium: on the payload format and the callback address	Low: on the message format	Low: on the file format	Highest: on the schema, with no record of who changed what
Duplicates and order	None if the call is idempotent	Expected; the receiver deduplicates	Expected; delivery at least once, order per key	A rerun file is the risk	None
Who owns the mapping	The caller	The receiver	Each consumer, or the hub	Whoever writes the file spec	Nobody, which is the problem
Fits	Decisions that need an answer now: payment, tax, a stock check at checkout	Receiving changes from a SaaS vendor: the platform, the PSP, the returns portal	Facts with several consumers that must not be lost: stock, orders	Facts that change by season or by day: products, finance, EDI	A suite from one vendor; nothing across vendors
Limits	Every dependency's outage becomes yours	A deleted subscription stops the flow silently	Needs the discipline of the next chapter	Stale by design; a late file is a quiet failure	Breaks on the first upgrade

Three judgements follow. **Use a direct call only where the answer is needed before the next step can happen**: whether the card is authorized, what the tax is, whether this unit may be promised; everything else should be asynchronous, so one system's outage does not become every system's outage. **Treat a webhook as a notification and polling as the truth**; the vendors say so in their own documentation. **A scheduled batch is not old-fashioned**; it is right for finance and product publication, provided a missed run

raises an alert.

Across Scale Stages: When It Fits and When It Limits

The four stages are the book's (see What Changes at Each Scale: From First Store to Group).

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	The platform's apps and one or two vendor connectors (shop to accounting, shop to the 3PL); the merchant owns no integration code	An inventory anyway, even with four rows, and a weekly look at each app's failure log	A second channel whose orders are keyed in by hand; a stock number maintained in two places
2 — Growing	The first point-to-point scripts and the first iPaaS; an agency builds most flows; stock and orders move by scheduled sync every 15–60 minutes	Choose the style per flow; a reconciliation job for stock and for orders; a named owner per flow, even when the agency built it	An oversell in the first busy week; a flow only the agency can explain; a sync interval that is "fine" until the first promotion

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel	A dozen or more systems, 20–40 flows in three or four styles, several builders; the joins are the incident source	The integration layer becomes a product with an owner, standards (idempotency, contract versions, monitoring, reconciliation) and a queue or event stream for stock and orders; an OMS as the order hub	Nobody can draw the landscape from memory; the same fact is written by more than one flow; a vendor's API retirement is discovered by a customer
4 — Group	Several brands or countries, each with its own landscape, or a shared platform with a central team	A governance decision: shared hub with service levels, or per-brand autonomy with shared standards; a schema registry for events; integration as a platform team's product	Every brand queues behind one central hub; a brand builds its own connectors because the hub is too slow; hub flows with no owner, The Spaghetti Junction inside one tool

The judgement: the styles do not change with scale; the **discipline around them** does. A Starting-stage merchant and a Group should both choose events for stock and batch for finance. What a Group needs on top is ownership at team level, contracts enforced by tooling, and a hub run as a product, not bought as a box. Kestrel & Pine is at the point where that discipline becomes necessary, which is what W5 is.

In Practice

Dated public evidence. Shopify's webhook documentation (n.d., accessed November 2026) is the clearest public statement of the webhook-plus-reconciliation rule: it does not guarantee ordering within a topic, warns that an app "can miss or mishandle events," and tells apps to "use reconciliation jobs to periodically fetch data from Shopify" with

`updated_at` filters. Stripe's webhook documentation (n.d., accessed November 2026) says the same about ordering and duplicates and asks the receiver to answer with a success code "before any complex logic" and to process events from a queue, because spikes "might overwhelm your endpoint hosts." Microsoft's Business Central connector (October 2026) shows a vendor connector encoding ownership into configuration; Debezium (3.7) shows CDC as a product category.

Eva's order before the change. At 21:14:03 on 13 April 2027, the platform creates #1072455 and sends a **webhook**, a server-to-server notification, to the iPaaS. The integration translates it and calls the warehouse management system (WMS). Assume three retries at one-minute intervals before it stops if the warehouse does not answer.

The payment service provider (PSP) sends payment success separately. An hourly connector copies the order to the enterprise resource planning system (ERP) for finance. After picking, stock takes fifteen minutes to reach the ERP, up to an hour more to reach the platform and another hour to reach marketplaces. Nobody has recorded who responds if a message is missed.

Under Waypoint, the company's improvement program, each flow gets an owner and a reason for its timing. In the 2028 target, the platform's order notification reaches an **order management system (OMS)** through the iPaaS. The OMS becomes the shared order home and supplies warehouse, finance and notification flows.

Warehouse stock events feed the availability service, which recalculates sellable quantities for each channel. The ERP leaves the stock path. A nightly comparison finds missing orders; stock comparisons run weekly and daily in November. The technology choices are familiar. The responsibilities and checks are now explicit.

Kestrel & Pine flow (inventory excerpt, illustrative)	Style and trigger	Owner (Feb 2027 / under W5)	Reconciliation (Feb 2027 / under W5)
Platform → iPaaS → WMS: orders	Webhook, then direct call	Ruben / Ruben	None / nightly order count and list

Kestrel & Pine flow (inventory excerpt, illustrative)	Style and trigger	Owner (Feb 2027 / under W5)	Reconciliation (Feb 2027 / under W5)
WMS → ERP → platform → connector: stock	Scheduled pulls, 15 min then hourly	Ruben, the ERP partner, Milan / retired; WMS stock events through the iPaaS to the availability service, which publishes to the platform and the connector	None / weekly per SKU and location, daily in November
PSP → platform: payment status	Webhook	Nobody named / Milan	Monthly payout matching by finance / daily
Returns portal → platform: refunds	Direct API call	Nobody named / Sanne's team	None / daily refund matching against the PSP
Bergzicht → EDI provider → ERP: order, despatch advice and invoice messages (ORDERS, DESADV, INVOIC)	Files in EDIFACT, the UN EDI message standard	The EDI consultant / Lea, with the DESADV sent from the WMS (W6)	Invoice matching by Bergzicht / outbound document log checked weekly
POS → ERP: store stock	Nightly file	The store team / Daniel; an event feed once the POS link exists	None / daily
ERP → data warehouse: orders and invoices	Nightly extract	Hanna / Hanna	Three order counts / one definition (W7)

The Integration Inventory and Integration Debt

No vendor-neutral source defines "integration debt," so this book defines it operationally: **the set of integrations whose owner, contract version, failure alert and reconciliation job are unknown**. Each unowned flow is cheap today and charges interest as the next silent failure, the next vendor retirement, the next person who leaves with the only understanding of a script.

The interest is real, and the vendors document it. Shopify deletes an API-created webhook subscription after eight consecutive failed deliveries, so an unmonitored integration silently stops receiving events. Shopify declared its REST Admin API legacy as of 1 October 2024 and removes deprecated fields on a published schedule; Stripe and Shopify both retire API versions on dates they choose (Promises Between Systems: API Contracts, Webhooks and Versioning). An unmaintained integration breaks on the vendor's schedule, not yours.

The inventory makes the debt a list. One row per flow:

Column	What goes in it	Why it matters
Source → target, and what moves	"WMS → ERP: stock on hand per SKU"	The flow's identity
Style and trigger	Scheduled pull, every 15 min	Decides freshness and failure mode
System of record	The WMS	Which side wins when they disagree
Owner	A named person, not a team or a vendor	Who is called at 20:30 on Black Friday
Contract version	The API version or file format, and the vendor's retirement date	The Pulled Rug has a date; put it in the row
Volume and expected rate	About 950 orders a day; 38 a minute at peak	Sizes the queue and the rate-limit budget
Failure handling	Retries, dead-letter queue, alert	Whether a failure is loud or silent
Reconciliation	What is compared, how often, who gets the exceptions	Whether anyone knows it worked
Last incident	Date and one line	The row's history

Noor's rule, which I would adopt anywhere: **no flow goes live without a row, and no row without an owner**. By April 2027 her ownership map (Who Builds and Who

Owns: Teams, Vendors, Agencies and Governance) had named owners for 8 of the 13; 5 were still unowned. The inventory is reviewed quarterly, and a flow whose owner has left is an incident, not a note. Apps inside the platform are inventoried separately (Apps, Plugins and Functions: Extensibility and Ecosystem Governance); the integration inventory counts flows between two systems, and an app that moves data to another system appears in both lists on purpose.

When It Goes Wrong

The silent stop. A scheduled sync fails and nothing says so. The Price Drift (October 2026) was a price-list sync to the German storefront that stopped; the storefront kept €179 for nine days while the stores and the marketplace charged €199: 410 orders, about €8,200 of margin and a price-parity warning. The fix is not a different style; it is an alert when a scheduled run does not happen, or produces no changes when changes were expected.

The dropped notification. A webhook is retried for hours, then dropped, or the subscription is deleted after repeated failures. The receiver believes nothing happened. Detection is the polling sweep; recovery is the reconciliation job that lists orders present in the source and absent in the target.

The backed-up queue. On Friday 27 November 2026 the WMS connector queue backed up for four hours. No order was lost, but orders reached the warehouse late. A queue's depth and its oldest message's age are health metrics; unwatched, a resilience feature becomes a delay nobody can explain (Black Friday Every Year: Peak Load, Resilience and Recovery).

The retired contract. In February 2027 the marketplace connector retired an API version on a published date; the agency script that read it had not been updated, and stock to bol was stale for two days. A contract-version column with the retirement date puts it on someone's calendar (Promises Between Systems: API Contracts, Webhooks and Versioning).

The second writer and the vendor outage. The App That Wrote Stock (March 2026) adjusted platform inventory while the hourly ERP sync, carrying the WMS's figure, also wrote it: two weeks of phantom reservations on 600 variants. Two flows writing one fact is a failure of the entity-owner table (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record), visible in the inventory's "system of record" column. And when the PSP paused Kestrel & Pine's account for 40 minutes during the Card-Testing Night, every direct call from the checkout failed; that is what a synchronous dependency means, and the only responses are a short call list, quick timeouts and a decision made in advance about what the checkout does when no answer comes (Taking the Money: Payment Providers, Authorization, Capture and Refunds).

What to Take Away

Established.

- The four integration styles (file transfer, shared database, remote call, messaging) and the criteria for choosing between them have been stable since Hohpe and Woolf named them in 2003; the technology changes, the trade-offs do not.
- Webhooks are a notification, not a guarantee. Shopify and Stripe both state in their current documentation that deliveries can be missed, duplicated and reordered, and both tell integrators to run periodic reconciliation.
- A vendor's connector encodes an ownership model (Business Central's Shopify connector, documented October 2026, has the ERP own inventory and the shop own orders); it fits when that model matches yours.

Depends on context.

- At the Starting and Growing stages a scheduled stock sync every 15–30 minutes is normal and acceptable; it stops being acceptable when the sync interval times the peak order rate exceeds the stock buffer you can afford to oversell. At 38 orders a minute, fifteen minutes is 570 orders.
- An iPaaS pays for itself when the number of flows and builders makes point-to-point unmanageable, usually at the Omnichannel stage; below that, two vendor connectors and a short inventory are cheaper. Compare vendors per flow at your volume, because each meters a different unit.
- Whether the hub is the iPaaS, the OMS or the ERP depends on which system owns the order (The Back Office: ERP Integration and Who Is the Hub argues this out); the style per flow does not change with that choice.

The author's synthesis.

- Choose the style per flow from its needs, never for the company: stock and orders by events with a reconciliation net, products by scheduled publication, finance by daily batch, direct calls only where the next step needs the answer.
- The integration inventory with a named owner per row is worth more than any integration tool, because every incident in this chapter happened in a flow that had no row.

Questions to Ask

1. How many integrations do we have, and can anyone produce the list with an owner on every row?

2. For each flow that moves stock: how old can the copy be, who writes it, and what checks that the two ends agree?
3. Which flows are synchronous calls, and what does the caller do when the other side does not answer within two seconds?
4. For each webhook we receive, what polls the source to catch what it missed; for each scheduled sync, what alerts when it does not run?
5. Which vendor connector are we relying on, and which ownership model does it encode for products, stock and orders?
6. What does our iPaaS meter, and what would our peak-day volume cost under it?
7. Which flows does only the agency, one consultant or one engineer understand, and which API versions retire when?

Choosing the style is half the work. The other half is living with what every asynchronous style introduces: a window in which two systems disagree, messages that arrive twice or out of order, and failures that leave one side updated and the other not. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation shows how to make that window safe, with idempotency, the outbox, retries, compensation and reconciliation as a first-class component, and Kestrel & Pine's first stock reconciliation as the worked example.

Sources and Evidence

Company, product and investor sources

- Webhooks, Shopify developer documentation, n.d., accessed 2026-11 — ordering not guaranteed, deliveries can be missed, reconciliation jobs with `updated_at` filters recommended.
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- MuleSoft Anypoint pricing, Boomi pricing, Workato pricing, Celigo pricing, Make pricing, n8n pricing, all n.d., accessed 2026-11 — iPaaS metering units and the public price points quoted.

Standards, regulation and official documentation

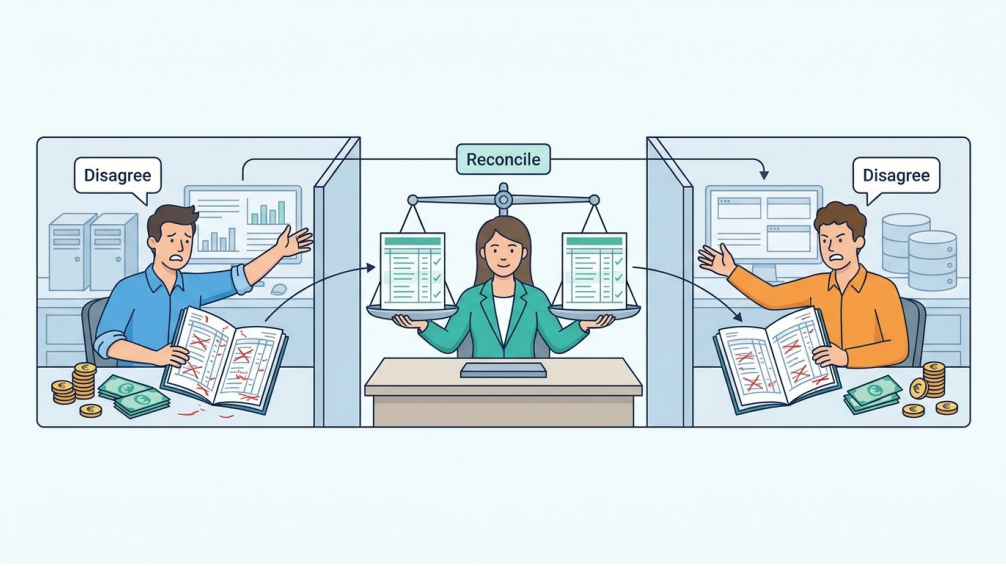
- Cloud-based integration (iPaaS), Wikipedia, relaying Gartner's definition retrieved 2015, accessed 2026-11 — the iPaaS definition; the Gartner page itself was not reachable.

Research and practitioner sources

- Integration Styles, Gregor Hohpe and Bobby Woolf, *Enterprise Integration Patterns*, 2003 (site updated 2023) — the four styles and the criteria for choosing; the source is more than twenty years old and the styles are unchanged.
- What do you mean by "Event-Driven"?, Martin Fowler, 2017 — event notification, event-carried state transfer, event sourcing and CQRS.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

25. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation



IN THIS CHAPTER, YOU WILL: Learn why copies of a fact always disagree for a while, and the five mechanisms that make the disagreement safe: idempotency, the outbox, retries, compensation and reconciliation.

KEY POINTS:

- **Every delivery you will meet is "at least once"; "exactly once" is something the receiver builds.** Webhooks, queues and integration-platform retries all deliver duplicates and reorder messages. An idempotency key, a business identifier such as the order number, makes a repeat harmless, and the receiver must be able to say what its key is.
- **The outbox makes "the order was saved" and "the event was sent" one act; compensation undoes a step that cannot be rolled back.** A cancellation after shipment is not an undo but a return and a refund: a business process with its own owner.

- **Reconciliation is a component, not a cleanup**. A job that compares two systems' records on a schedule, with a key, a tolerance and a named recipient for the exceptions, is the only way to know that the other mechanisms worked. Kestrel & Pine's first one found 2.3% of active variants with the wrong stock figure.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Message in a Bottle** — a message sent once and trusted produces exactly the duplicates and gaps that idempotency and reconciliation exist to catch.
- **The Boomerang** — the Double Refund is what happens when a money action has two writers and no shared key.
- **The Phantom Stock** — a stock copy that drifts from the warehouse's truth, with nothing comparing the two, sells units that are gone.

A warehouse accepts an order, but its reply is lost. The sending system cannot tell whether the work started, so it tries again. Unless the warehouse recognizes the repeated instruction, one purchase can become two shipments. Separate systems need a way to recover from uncertainty without repeating the business action. APIs, Events, Files and Middleware: How Commerce Systems Talk compares exchange methods; this chapter explains that recovery.

A database transaction can keep changes together inside one system. It cannot cover a hosted storefront, another vendor's warehouse software, a finance system and an external payment provider as one shared operation. Each completes its own work and can fail independently. The merchant must discover which steps succeeded, then finish the rest or undo commitments that can no longer be kept.

Five mechanisms do that work: recognize repeated messages; save an outgoing event with the fact it describes; retry without overwhelming the receiver; reverse completed actions when necessary; and compare records to find gaps. We explain **idempotency**, **outbox**, **backoff**, **compensation** and **reconciliation** through those jobs.

Where we are in the order's journey. Eva buys a jacket from Kestrel & Pine, the fictional outdoor brand, on 13 April 2027. Order, payment and warehouse updates arrive at different times. Her later exchange adds return and stock messages; choosing a refund instead would add a payment reversal. Each message may repeat or fail to arrive. What must the next system do?

Setting the Scene at Kestrel & Pine: The First Reconciliation

Kestrel & Pine had never compared warehouse and storefront stock. The **warehouse management system (WMS)** sent counts to the **enterprise resource planning system**

(ERP) every fifteen minutes, and the ERP copied them to the platform hourly. People assumed the chain kept the figures aligned.

In **May 2027**, as part of Waypoint's integration standards (decision W5: every integration gets an owner, a contract, idempotent handling and a reconciliation job), Ruben Alves ran the first weekly **reconciliation**: a job that reads the WMS's on-hand quantity and the platform's available quantity for every active variant and lists the ones that differ. Of about 17,000 active variants, **2.3%**, about **390**, had a different figure in the two systems *(illustrative).* The breakdown, over the following week:

Cause of the difference (May 2027, illustrative)	Variants (about)	What it was
Three platform apps that wrote inventory	240	The pre-order app from the App That Wrote Stock incident of March 2026 and two others each adjusted platform stock on their own, on top of the ERP sync
In-flight units	90	Returns the WMS had received and restocked that the hourly chain had not yet carried to the platform when the job ran
Manual edits	40	Corrections typed into the platform's admin by the e-commerce team after customer complaints, never passed back to the WMS
Unexplained	20	No writer found; overwritten from the WMS and watched

Head of Engineering Noor Visser drew two conclusions. First, nobody had known about the differences. Second, many came from systems writing stock they did not own. Faster messaging would not fix competing writers.

Waypoint decision W1 gives the WMS authority over physical stock and introduces an availability service in Q4 2027. The three apps lose stock-writing permission; the pre-order app receives an explicit allocation instead. The comparison job remains as a check, weekly and daily through November.

I would run it daily all year. Leaving about 390 unexplained differences for a week

leaves room for more overselling. Noor put more effort into the event feed. That is the tradeoff: faster propagation reduces drift; frequent comparison reveals what still escaped.

The second scene is older. In **January 2027**, over three weeks, **212 returns** were refunded twice: once by the returns portal through the platform's API (application programming interface: the operations a system lets other software perform), which instructed the PSP, and once by support agents in the PSP's own console, who refunded by hand when customers whose parcels had reached Duiven asked where their money was. **€19,400** went out twice, most of it recovered. Two writers, no shared key, no job comparing refunds issued with refunds due. The rest of this chapter is the toolkit that stops both scenes from recurring.

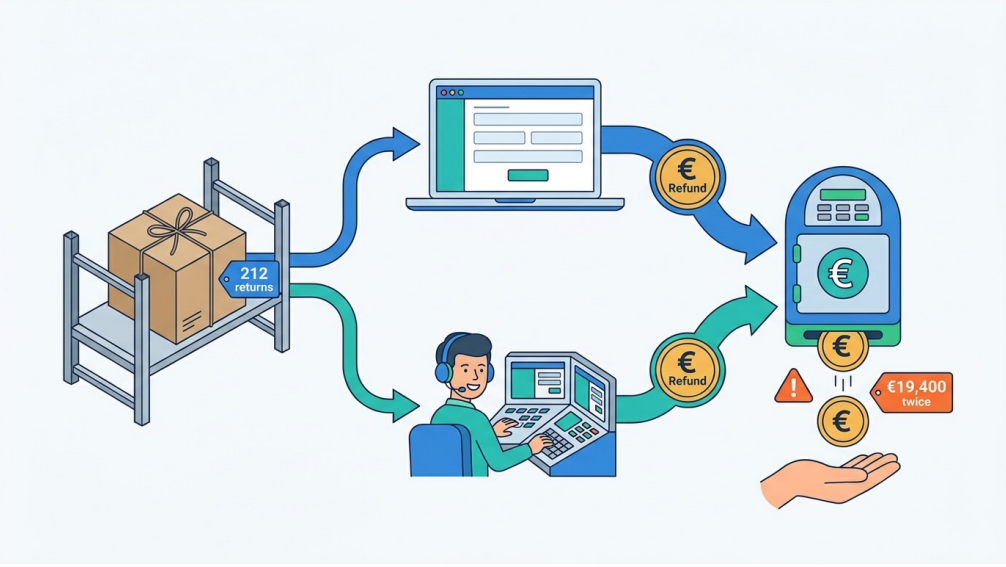


Figure 1: *Two writers and no shared key: 212 returns were refunded twice, once by the returns portal and once by hand in the payment provider's console, and €19,400 went out twice.*

Eventual Consistency in Plain Words

A **copy** of a fact is consistent with its **system of record** (the system whose value counts when they disagree; see Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record) when both show the same value. Across separate systems joined by messages, they cannot always show the same value, because the message takes time. **Eventual consistency** is the promise that is possible: if updates stop, all copies will eventually show the last value. Werner Vogels, then Amazon's CTO, defined it that way

in a December 2008 article (old, but the definition has not changed), together with the smaller promises a designer can add on top, such as "a reader always sees its own writes."

Gregor Hohpe's 2004 essay "Starbucks Does Not Use Two-Phase Commit" made the point most practitioners still quote: the cashier takes the order and the money while baristas make drinks asynchronously; drinks finish out of order, so the name on the cup matches drink to customer; and when something goes wrong the shop does not freeze the queue; it writes off the drink, remakes it or refunds the money. Starbucks, he wrote, "is primarily interested in maximizing throughput of orders." So is a commerce platform at 38 orders a minute. The three recoveries he named, **write-off**, **retry and compensating action**, are still the whole menu.

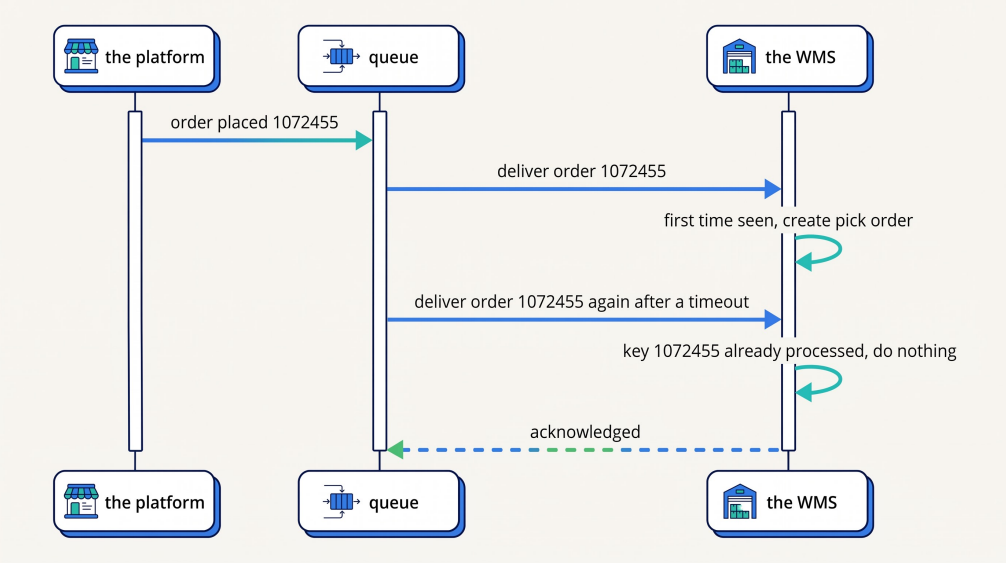


Figure 2: *The same order event delivered twice. The queue is doing its job (it would rather repeat than lose); the WMS is safe only because it remembers the order number it has already processed.*

The window of disagreement is normal; a window that never closes, or that nobody can see, is not.

At Least Once: Why Duplicates Are the Default

Every delivery mechanism a merchant meets prefers repeating a message to losing it. Apache Kafka's design documentation (version 4.4, accessed November 2026) states that Kafka "guarantees at-least-once delivery by default"; exactly-once is available inside Kafka's own processing and with transactions, but destinations outside Kafka "need cooperation from those systems." Shopify's webhook documentation (n.d., accessed

November 2026) says it "minimizes duplicate deliveries, but your app might receive the same webhook more than once." Stripe's (n.d., accessed November 2026) says endpoints "might occasionally receive the same event more than once." Mollie, a Dutch PSP, calls a webhook up to ten times over 26 hours until it gets a success response (documentation accessed November 2026).

At-least-once delivery means exactly what it says: the message will arrive one or more times. The alternative, at-most-once, means it may not arrive at all, which for an order or a refund is worse. "Exactly once" is not a property you receive from a queue or a vendor; it is a property the receiver builds by recognizing repeats. The CloudEvents specification (a standard of the Cloud Native Computing Foundation, CNCF, for the envelope around an event; graduated January 2024) builds this in: every event carries a `source` and an `id`, a re-sent duplicate keeps the same `id`, and consumers may treat events with the same `source` and `id` as duplicates.

Ordering is the second thing not to expect. Shopify "doesn't guarantee ordering within a topic," so a product update can arrive before the product's creation; Stripe "doesn't guarantee the delivery of events in the order that they're generated" and warns against using the event's timestamp to decide order, because distinct events can share one. Kafka orders messages only within a partition, one slice of a stream. A design that assumes "shipped" arrives after "allocated" will one day process them the other way round.

Idempotency: Making a Repeat Harmless

An operation is **idempotent** when doing it twice has the same effect as doing it once. Reading a stock level is; "create an order", "refund €199" and "subtract one from stock" are not, unless you make them so. The way to make them so is an **idempotency key**: a unique token for the business action, sent with the request and remembered by the receiver, so that a second request with the same key is recognized and answered without doing the work again.

The PSPs document it best, because money is where duplicates hurt most. Stripe's API (documentation accessed November 2026) accepts an `Idempotency-Key` header on every request that creates or changes something; it "saves the status code and body of the first request made with a given idempotency key" and returns the saved response to retries, errors if the retry's parameters differ from the original's, suggests random unique identifiers (UUIDs) as keys, and may prune keys after 24 hours. Shopify's GraphQL Admin API (accessed November 2026) takes an `idempotencyKey` on mutations that are not naturally idempotent, so that "repeated requests with the same parameters are executed only once, no matter how many times the request is retried." On the receiving side, Shopify gives each webhook delivery an `X-Shopify-Webhook-Id` and Stripe each event an `id`, and both tell you to store the ids you have processed and skip repeats.

The rule I use for choosing the key is that **it must name the business action, not the**

attempt. A timestamp or a random number per attempt defeats the purpose, because the retry gets a new one. The table gives the keys for the actions every merchant performs:

Action	Idempotency key	Why that key
Create the order in the WMS, the OMS (order management system, the order hub) or the ERP	The order number (#1072455)	One order, one record, however many times the event arrives
Refund a return	The return id, or (order, line, quantity, reason)	The same return refunded twice is the Double Refund
Capture a payment	The order number plus "capture"	A capture retried after a timeout must not capture twice
Apply a stock event	(SKU, location, event id); a SKU is one sellable variant, one size in one color	The same movement applied twice doubles the change
Send a shipping confirmation e-mail	(order, shipment id)	The customer should get one
Create a customer	E-mail or external id, with care (One Customer, One Product: Identity and Master Data Across Systems)	Matching is harder than deduplication

The Double Refund fails this test twice over. The returns portal and the support console were two writers with no shared key; the PSP would have declined the second refund had the second request carried the return id the first one used. Waypoint's fix was two rules: **one writer per money action** (refunds go through the returns portal and the platform; support no longer refunds in the PSP console) and the return id as the key on every refund request. A receiver that cannot say what its key is cannot be made safe; that is the first question to ask of any integration that creates, moves or pays.

Out-of-order events need a second defense, because a stale update is not a duplicate. Three responses, in rising cost: make the handler tolerant (a "shipped" event that arrives before "allocated" is stored and re-evaluated when the missing state arrives); carry a version or `updated_at` per entity and drop updates older than what is stored; or route all events for one entity through one consumer, as Walmart's inventory design described on Confluent's blog in May 2020 does, one partition per item and store so that "only one consumer processes it." Stripe's own advice is the simplest of all: when an event arrives

early, fetch the current object from the API rather than trusting the payload.

The Outbox: Never Losing the Event Between the Database and the Queue

There is a quiet way to lose an event. A service saves the order to its database, then publishes "order placed" to the queue. If the process dies between the two steps, or the queue is briefly unreachable, the order exists and nobody is told. Doing it the other way round (publish first, then save) risks telling everyone about an order that was never saved. This is the **dual-write problem**, and no amount of retry fixes it, because the sender does not know which half failed.

The **transactional outbox**, as cataloged by Chris Richardson (microservices.io, accessed November 2026), solves it by making the two writes one. The service writes the business change *and* a row in an **outbox table** ("event to send: order placed, #1072455") in the same database transaction, so either both exist or neither. A separate relay reads the outbox and publishes to the queue, either by polling the table or by tailing the database's transaction log (the change data capture of APIs, Events, Files and Middleware: How Commerce Systems Talk). The relay "might publish a message more than once," Richardson notes, which is why "a message consumer must be idempotent": the outbox and idempotency are two halves of one design.

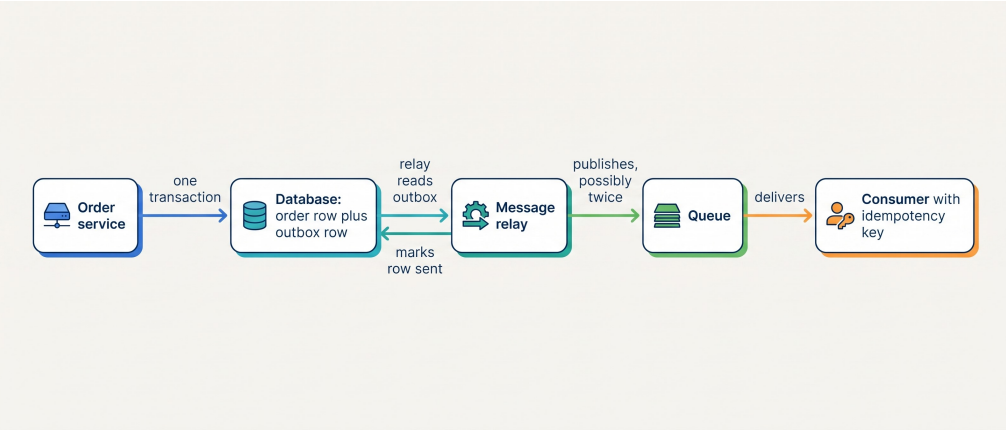


Figure 3: *The outbox: the order and the promise to announce it are saved together, so a crash between saving and publishing loses nothing; the relay may repeat, and the consumer's key makes the repeat harmless.*

A merchant on a SaaS platform cannot build or inspect the platform's outbox. What plays the same role from the outside is the pair the previous chapter described: the platform's webhook plus a polling sweep of everything changed since the last run. The sweep is the merchant's substitute for a relay it cannot inspect.

Retries, Backoff and the Dead-Letter Queue

When a call or a delivery fails, the obvious response is to try again. Done naively, retries make an outage worse: a thousand integrations all retrying at the same second hit the recovering system together. **Exponential backoff** waits longer after each failure (one second, two, four, eight). Marc Brooker of AWS showed by simulation in March 2015 that backoff alone is not enough, because clients still cluster: "The solution isn't to remove backoff. It's to add jitter," meaning randomness in the wait, which in his simulation with 100 contending clients cut total calls by more than half. Stripe's rate-limit guidance (accessed November 2026) says the same for its `429` responses: back off exponentially and "add randomness to the backoff schedule to avoid a thundering herd effect." Shopify's recommended backoff after a throttle is one second.

Retries must end somewhere. A **dead-letter queue** (DLQ) is where a message goes after it has failed more than a set number of times, so that it is neither lost nor retried forever. The documentation of Amazon SQS (Simple Queue Service; accessed November 2026) describes the `maxReceiveCount` after which a message moves, advises setting it "high enough to allow for sufficient retries," and recommends giving the dead-letter queue a longer retention than the source, because a moved message keeps its original timestamp. Messages can be moved back ("redrive") once the cause is fixed.

A dead-letter queue with nobody watching it is a silent order-loss mechanism. The operational rule: **every DLQ has an alarm, an owner and a written replay procedure**, and the number of messages sitting in dead-letter queues is a health metric beside orders per minute (Black Friday Every Year: Peak Load, Resilience and Recovery). On Black Friday 2026 no order was lost, but an order that had exhausted the integration platform's (the iPaaS's) three retries would have gone nowhere a person would look: the inventory of APIs, Events, Files and Middleware: How Commerce Systems Talk found that no flow had a DLQ at all.

Sagas and Compensation: Undoing What Cannot Be Rolled Back

Placing an order touches several systems: reserve the stock, authorize the payment, create the order, confirm to the shopper. Inside one database you would wrap those in a **transaction**, a unit of work that either completes entirely or leaves no trace. Across the availability service, the PSP and the OMS there is no such thing. If the payment is declined after the stock was reserved, the reservation must be released by a second, deliberate action.

The name for this is a **saga**. Hector Garcia-Molina and Kenneth Salem introduced it in a 1987 paper (old, but it is still the definition in use): a long-lived transaction written as a sequence of smaller transactions, where the system guarantees "that either all the

transactions in a saga are successfully completed or compensating transactions are run to amend a partial execution." A **compensating action** is the business undo: release the reservation, void the authorization, cancel the pick. Richardson's catalog adds the two ways to coordinate one: **choreography**, where each step publishes an event that triggers the next (no central brain, hard to see the whole flow), and **orchestration**, where one component tells each participant what to do and tracks where the saga is.

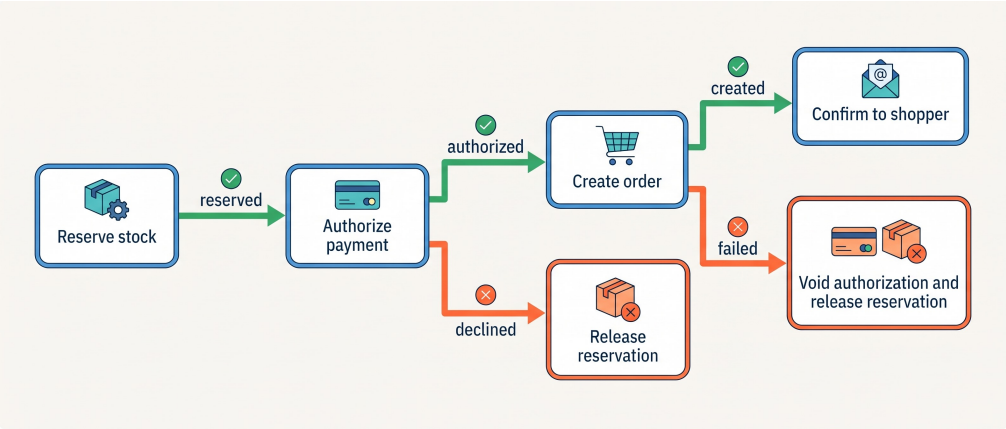


Figure 4: *The place-order saga: each step that fails triggers the compensations for the steps before it; nothing is rolled back, everything is undone on purpose.*

The commerce sagas worth naming are three. **Place order**: reserve, authorize, create, confirm, with the compensations in the figure. **Cancel after shipment**: there is no undo for a parcel already with PostNL; the compensation is a return and a refund on receipt, which is a business process with a parcel, a warehouse grade and a credit note, not a database operation (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics). **Split shipment**: each location's allocation is its own step with its own release. The drawback Richardson names, "lack of isolation," is real: between steps, another order can see stock that is reserved-but-not-yet-paid, which is exactly why an availability service counts reservations, not just physical units (Is It in Stock? Inventory, Availability and the Promise You Can Keep).

In practice the saga orchestrator for a merchant's order is the **OMS**, the one place that holds every order and steers it through payment, fulfillment, delivery and return (The Order Record: Lifecycle, State and the Order Management System). That is a large part of what Waypoint's decision W2 buys: a component whose job is to know which step of which order has happened, and what to do when the next one fails.

Reconciliation as a First-Class Component

Everything above reduces the chance of disagreement. None of it proves there is none. **Reconciliation** is the job that compares two systems' records on a schedule and reports or repairs the differences. In finance it is centuries old; in commerce integration it is usually an afterthought.

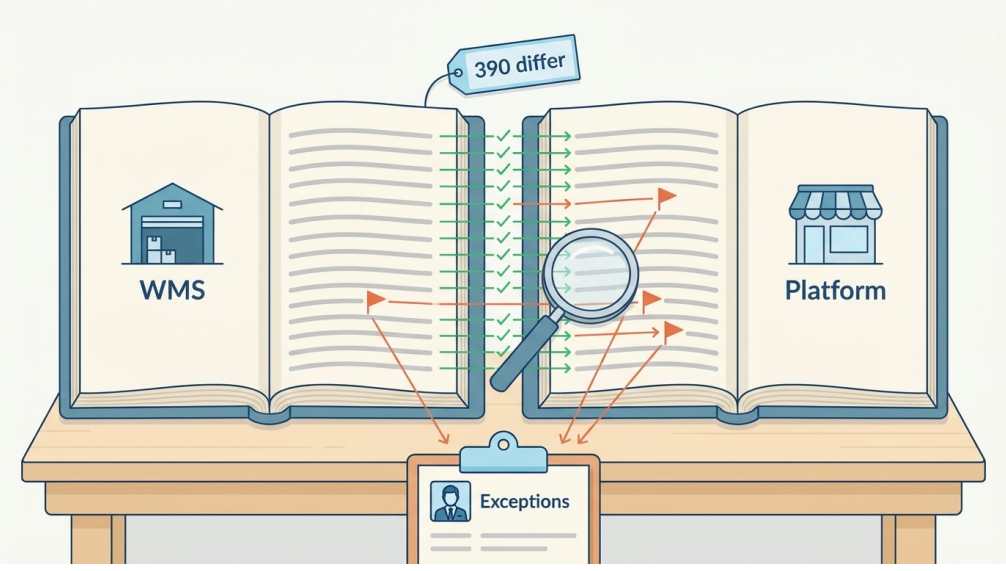


Figure 5: *Reconciliation is two ledgers matched line by line: in the first stock run, about 390 of some 17,000 variants differed between the WMS and the platform, and each difference went on an exception list to a named owner.*

The vendors give it precise shape. Stripe's payout reconciliation report (accessed November 2026) helps "match the payouts you receive in your bank account with the batches of payments and other transactions that they relate to," itemizing "every payment, refund, dispute, fee, and other balance transaction included in the payout" with an id per line and an `order_id` for matching, and it warns that "a payout might arrive before its reconciliation data is ready." Shopify tells apps to run periodic sweeps with `updated_at` filters so they stay consistent with Shopify's data. Microsoft's Business Central Shopify connector (documentation of October 2026) reconciles stock continuously: it compares "Shopify Stock" with "Last Calculated Stock" per variant and location and sends an update only when they differ, keeping "Last Synced On" and "Last Calculated On" timestamps for troubleshooting. That compare-and-send loop is a reconciliation job built into a product.

Every reconciliation job can be written on one card, and I would require the card before

the job goes live:

Card field	Stock (Kestrel & Pine, from May 2027)	Refunds (from February 2027)	Orders (from Q4 2027)	PSP payouts
What two sets are compared	WMS on-hand minus reservations vs platform available, per active variant	Refunds issued by the PSP vs refunds recorded by the platform, per day	Platform orders of the day vs OMS orders of the day	PSP payout lines vs ERP receivables
Matching key	SKU and location	Order number and amount	Order number	Balance transaction id or order id
Tolerance	Zero per variant	Zero	Zero	Zero to the cent
Cadence	Weekly, Sunday 03:00; daily in November; the availability service closes the gap by events in between	Daily	Nightly	Per payout, and monthly close
Who gets the exceptions	Ruben's team and Milan	Sanne's team	Ruben's team	Finance
The fix	Overwrite the platform from the WMS (W1), find and remove the writer	Reverse the duplicate, block the writer	Replay the missing event	Book the difference, open a dispute

Two design points matter. **The system of record decides the direction of the fix.**

Because W1 names the WMS for physical stock, the platform's figure is overwritten and the platform-side writer is hunted, never the reverse. And **a zero tolerance is a choice, not a default**: for stock it is zero because one wrong variant at peak is one Phantom Stock incident; for a copy of product descriptions it could be a count threshold. No public source gives a typical disagreement rate between a WMS and a shop; the 2.3% is Kestrel & Pine's illustration and has no benchmark behind it.

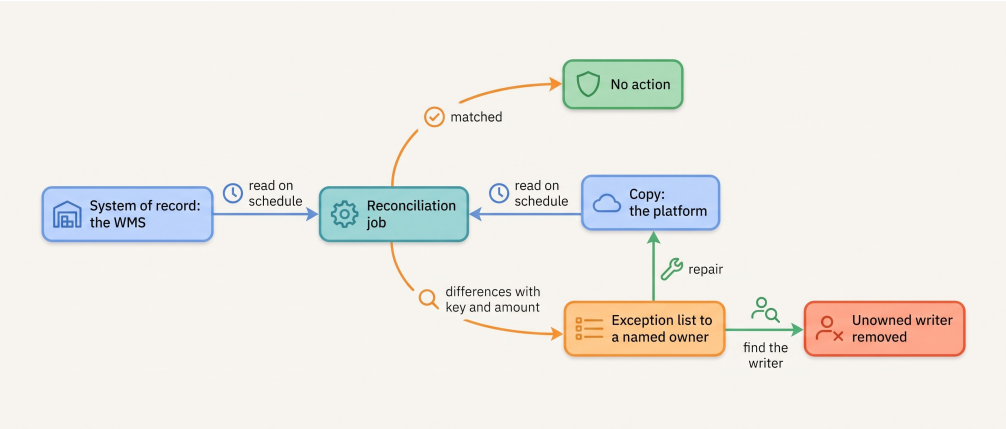


Figure 6: *A reconciliation job is a loop with a recipient: the copy is corrected from the system of record, and the exception list leads to the writer that should not have been writing.*

Reconciled by Design: Traceability and Audit Across Boundaries in the sibling book RECONCILED covers the finance version for ledgers; Taking the Money: Payment Providers, Authorization, Capture and Refunds covers the PSP's reports.

What Each Flow Needs

The mechanisms are chosen per flow, like the styles. The table states, for the flows this chapter touches, how long disagreement is tolerable and which mechanism closes the gap.

Flow	Tolerable window of disagreement	Freshness needed	Consistency on arrival	Availability	Mechanisms that earn their place
Stock events (WMS to availability, availability to channels)	Seconds; minutes at most	Seconds	Per SKU and location, latest version wins	Queue; channels keep selling from the last value	Idempotency on (SKU, location, event id); version per entity; reconciliation with zero tolerance, daily in peak months
Order placed (platform to OMS, WMS, ERP)	Minutes for the warehouse; an hour for finance	Seconds	Never lost, never doubled	Queue; the warehouse works from what it has	Order number as key; outbox or webhook plus sweep; nightly order reconciliation; DLQ with owner
Refunds (returns portal, platform, PSP)	Hours	Minutes	Exactly once, to the cent	Can queue; a late refund is a complaint, a double one is a loss	One writer; return id as key; daily refund reconciliation

Flow	Tolerable window of disagreement	Freshness needed	Consistency on arrival	Availability	Mechanisms that earn their place
Shipment confirmed (WMS to platform, OMS, notifications)	Minutes	Minutes	One confirmation per shipment	Can queue	(order, shipment) as key; tolerant handler for early arrival
Prices and promotions	Minutes; a promotion start to the minute	Minutes	Latest version wins	Fail loudly	Version per price list; alert on a missed run (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax)
Payment status (PSP to platform)	Minutes; method expiry windows for the long tail	Seconds to minutes	Terminal state must be reached	Webhook retried by the PSP for a day or three	Event id as key; fetch the status from the PSP when it matters; daily sweep of orders not in a terminal state

(The author's recommendation; the Kestrel & Pine cadences are illustrative.)

The Choices and Their Trade-Offs

There are four ways to keep two systems in agreement, and a merchant will use more than one.

	Distributed transaction (two-phase commit)	Events with idempotent consumers and reconciliation	Scheduled compare-and-send	Manual correction
How it works	Both systems commit together or not at all	Each change is published; receivers deduplicate; a job checks the ends	On a timer, read both sides, send what differs	A person fixes what a customer reported
Window of disagreement	None	Seconds, closed by the job	The timer interval	Days
Coupling and availability	Both must be up; one locks the other	Each works alone; the queue holds the rest	Each works alone	None
Cost to build	High; needs both vendors' cooperation, which SaaS vendors do not give	Medium: keys, outbox or sweep, DLQ, card per job	Low	None, then everything
Fits	One database, one vendor	Stock, orders, refunds, shipments across vendors	Products, prices, finance, and as the safety net under events	Never as the design; always as the last resort

	Distributed transaction (two-phase commit)	Events with idempotent consumers and reconciliation	Scheduled compare-and-send	Manual correction
Limits	Hohpe's warning: it "can also hurt the free flow of messages"; unavailable across SaaS	Needs the discipline of this chapter	Stale by design; the Price Drift was a compare-and-send that stopped	The Double Refund was manual correction without a key

And between the two ways to run a saga:

	Choreography	Orchestration
Who decides the next step	Each participant, by reacting to events	One orchestrator, usually the OMS
Visibility	The whole flow is in nobody's code	One place shows where every order is
Coupling	Lowest	The orchestrator knows every participant
Fits	Two or three steps, few participants	The order lifecycle at Omnichannel scale, where support must answer "where is my order"

My judgement: for a merchant, orchestration by the OMS wins once there are more than three systems in the order's path, because the question support asks ("where is this order and what happens next?") has to be answerable from one screen. Choreography is right for the edges: notifications, analytics, anything that listens and never has to be undone.

Across Scale Stages: When It Fits and When It Limits

The four stages are the book's (see What Changes at Each Scale: From First Store to Group).

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	The platform and its apps handle retries and duplicates; the merchant never sees a queue	Check the apps' failure logs weekly; reconcile PSP payouts to the books monthly	A refund issued twice "by accident"; an order the 3PL (the third-party warehouse) never received, found by the customer
2 — Growing	The first custom integration (ERP connector, returns portal, marketplace sync) built by an agency or a partner, usually without keys or a DLQ	Idempotent writes on every create, refund and stock change; retry with backoff; a daily reconciliation for orders and for stock	An oversell in the first busy week; a connector that "sometimes" doubles orders; nobody can say what a flow's key is
3 — Omnichannel	A dozen systems, 20–40 flows; a double-refund or a stock-drift incident has already happened	Written standards: a key per entity, an event envelope with id, source and type, a DLQ with an owner per flow, a reconciliation card per flow, four signals per flow; an OMS orchestrating the order saga	Reconciliation findings that nobody owns; a DLQ nobody has opened; sagas that live in three people's heads

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
4 — Group	A shared event bus and schema registry across brands; a platform team owns the standards	Reconciliation as a data-platform job across brands (Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics); replay tooling; service levels for the bus	One brand's bad event schema stalling another's consumers; exceptions routed to a central team that does not know the business

The judgement: idempotency and reconciliation do not change with scale; a Starting-stage merchant needs them from its PSP and its 3PL, a Group needs them from its own bus. What changes is who builds them. Below the Omnichannel stage they are bought inside products (the PSP's idempotency keys, the connector's compare-and-send); from the Omnichannel stage the merchant has enough flows of its own that the discipline must be written down and owned, which is what W5 is.

In Practice

Dated public evidence. Stripe's idempotent-requests documentation (accessed November 2026) is the clearest public model of a key-based design: one key per request, the first response saved and replayed, a mismatch in parameters rejected. Mollie's webhook design (accessed November 2026) shows a different defense against lies and duplicates: the webhook carries only an `id`, never the status, so "fake calls to your webhook will never result in orders being processed without being actually paid," and the receiver must fetch the current status.

Microsoft's Business Central connector (October 2026) ships a compare-and-send reconciliation for stock. Stripe's payout reconciliation report (accessed November 2026) is the finance-side template for the reconciliation card.

Walmart's 2020 inventory design, described on Confluent's blog (six years old; it may describe a replaced system), shows one consumer per item and location at a scale no merchant in this book needs; the idea is the same at any size. Google's Site Reliability Engineering book (2017) names the four golden signals, latency, traffic, errors and saturation, that the monitoring section below applies to flows.

The refund path after the fix. Suppose Eva chooses a refund rather than a replacement jacket. In the illustrative design introduced in February 2027, inspection on 22 April triggers the portal's refund request. The return id identifies the request all the way to the payment provider, so a retry cannot send another €199.

The provider's success notification is stored once, and finance receives the credit note in its nightly batch. Support can inspect the refund in the platform but can no longer issue a second one in the provider's console.

Each morning, a job compares provider and platform refunds by order and amount and sends unmatched entries to finance. January's 212 duplicate refunds would have appeared the next morning.

Monitoring the flow. The SRE book's four signals become, per integration: end-to-end lag (order placed to order in the WMS), events per minute against the expected rate, failed deliveries and DLQ depth, and rate-limit headroom. The reconciliation job's variance is the black-box check, "what's broken"; the connector's own log is the white-box check, "why." The alert most teams forget is traffic at zero while the shop is busy: a deleted webhook subscription produces no errors at all.

When It Goes Wrong

Duplicates. The Double Refund: two writers, no key. Detection is the daily refund reconciliation; recovery is a reversal and a block on the second writer. The same shape appears as a double pick when a WMS keys on its own sequence number instead of the order number.

Gaps. A webhook retried for hours and then dropped; a subscription deleted after repeated failures; an event stuck in a DLQ nobody opens. Detection is the polling sweep and the DLQ alarm; recovery is replay. Stripe allows manual resend of events for 15 days from its dashboard, 30 by its command-line tool (documentation accessed November 2026); replay is safe only because of idempotency.

Stale copies. The 390 variants of May 2027, and the Price Drift of October 2026. Detection is the reconciliation job and the alert on a missed run; recovery is overwriting from the system of record and finding the writer. A stale copy with no job is found by a customer.

Retry storms. A recovering ERP hit by every connector at once. Backoff with jitter on every caller, and a circuit breaker (stop calling after repeated failures, try again later) on the ones that matter at peak (Black Friday Every Year: Peak Load, Resilience and Recovery).

Partial failures. The order exists in the WMS and not in the ERP, or the reservation exists and the payment was declined. The saga's compensations handle the second; the nightly order reconciliation finds the first. A partial failure nobody designed for becomes

the Three-Status Order (The Order Record: Lifecycle, State and the Order Management System).

Vendor outages. When the PSP paused Kestrel & Pine's account for 40 minutes in August 2026, webhooks stopped too. Webhooks resume with retries, but only within the vendor's window (26 hours at Mollie, three days at Stripe); a daily sweep of orders whose payment has not reached a terminal state, querying the PSP directly, is the net under the net.

What to Take Away

Established.

- Delivery across systems is at least once and unordered: Kafka's design documentation, Shopify's and Stripe's webhook documentation and Mollie's retry schedule all say so in their own words. Exactly once is built by the receiver.
- An idempotency key must identify the business action; Stripe's and Shopify's API documentation describe the mechanism, and CloudEvents standardizes `source` plus `id` as the duplicate key for events.
- The transactional outbox (Richardson) and sagas with compensating transactions (Garcia-Molina and Salem, 1987) are the standard answers to losing events and to undoing partial work; backoff with jitter (Brooker, 2015) is the standard retry.

Depends on context.

- Below the Omnichannel stage the merchant buys these mechanisms inside products (the PSP's keys, a connector's compare-and-send) and needs only to check payouts monthly and failure logs weekly; from the Omnichannel stage, with its own flows, it must write the standards down and own them.
- Orchestration by an OMS is right when more than three systems sit in the order's path and support must answer "where is it" from one screen; choreography is right for listeners that never have to be undone.
- Reconciliation cadence follows the cost of a wrong copy: daily for refunds, nightly for orders, per payout for money, weekly for content; for stock, weekly is tolerable only while an event feed closes the gap in between, and daily in the peak months.

The author's synthesis.

- Treat the reconciliation card as part of the integration, not as its audit: no flow that creates, moves or pays goes live without one, because the other mechanisms reduce disagreement and only reconciliation proves it.
- Most drift has a writer that should not have been writing; when a reconciliation finds differences, the first question is "who else writes this?", not "which queue dropped it."

Questions to Ask

1. For each integration that creates, refunds, captures or adjusts stock: what is its idempotency key, and where are processed keys stored?
2. What happens when the same order event arrives twice, and when "shipped" arrives before "allocated"?
3. Which flows have a dead-letter queue, who is alerted when it is not empty, and how is a message replayed?
4. For each money action: how many systems can perform it, and is it one?
5. Which reconciliation jobs run, on what key, with what tolerance, and who receives the exceptions?
6. When a reconciliation finds a difference, which side is corrected, and is that written down as the system of record?
7. If the platform's webhooks stopped for a day, what would notice?
8. Which component orchestrates the order saga, and what compensations exist for a declined payment after a reservation and for a cancellation after shipment?

The mechanisms in this chapter protect a flow against the messages it receives. They do not protect it against the other side changing the shape of those messages: a field renamed, an API version retired, a webhook format revised on a date the vendor chose. Promises Between Systems: API Contracts, Webhooks and Versioning is about those promises between systems, with Kestrel & Pine's Deprecated Endpoint as the worked example.

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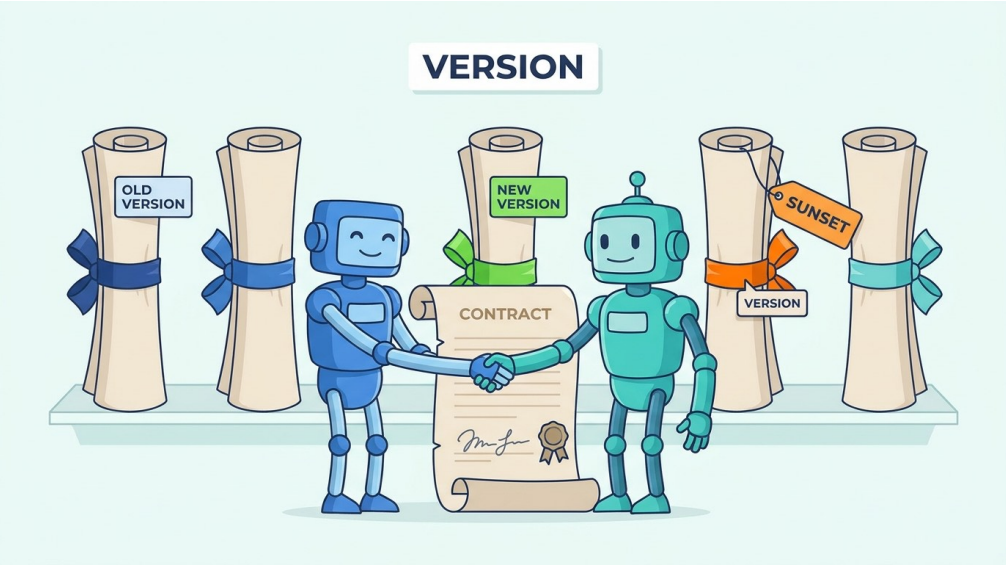
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

26. Promises Between Systems: API Contracts, Webhooks and Versioning



IN THIS CHAPTER, YOU WILL: Learn what an API contract promises, how dated versions and deprecation dates let a system change without breaking its consumers, and how to receive webhooks safely.

KEY POINTS:

- **A contract is a written promise with a lifetime.** It says what a system accepts and returns, how fast and how often you may call it, and until when. Shopify's quarterly dated versions and Stripe's dated versions are two public models; both make the consumer responsible for pinning a version and scheduling the upgrade.
- **Change by adding; retire by date, with consent and monitoring.** Google's and Zalando's published rules agree that adding optional fields is safe and removing or renaming anything is not. A deprecation is a date pair, announced and removed, plus a check of who still calls the

old version; a "deprecated" label without a date is a warning, not a deprecation.
- **A webhook is a signed hint, not a ledger.** Shopify and Stripe both document that deliveries can be missed, duplicated and reordered. The receiver verifies the signature, answers fast, deduplicates on the delivery id and runs a periodic sweep for what it missed.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Pulled Rug** — a vendor's API version or a partner's document format changes on a date nobody at the merchant had written down.
- **The Message in a Bottle** — a webhook is received once and trusted, so a missed, duplicated or reordered delivery becomes a missing or doubled order.
- **The Lynchpin** — the only person who knew which version a script called, and where the vendor's notices went, was one developer at the agency.

An order connector can keep sending the same requests while the platform it calls changes. If the platform stops accepting a field or changes what a response means, orders may stop reaching the warehouse even though nobody changed the connector. Preventing that disruption requires an agreement about both the interface and how it will evolve.

An **application programming interface (API)** defines how one program asks another for data or action. Its contract covers accepted fields, meanings, responses and failure behavior. Providers need to add features and fix problems; consumers need time to adapt. A managed transition lets both happen without interrupting sales or stock updates.

Kestrel & Pine, the fictional outdoor brand, consumes vendor contracts in most of its 34 integrations: commerce platform, payment provider, carriers, returns portal and marketplace connector. It also provides interfaces to its agency and trading partners. It needs to track vendor deadlines and give its own consumers notice.

Where we are in the order's journey. Eva's jacket reaches the warehouse through a platform notification. Her return uses a portal API. Marketplace stock uses another API. Every exchange depends on a contract, and versioned contracts can expire. APIs, Events, Files and Middleware: How Commerce Systems Talk covers communication methods; Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation covers message recovery.

Setting the Scene: The Deprecated Endpoint

On 10 February 2027, Ecommerce Manager Milan Petrov receives a bol escalation: Kestrel & Pine has canceled several jacket and fleece orders for lack of stock. Lead

integration engineer Ruben Alves finds that the stock update script stopped working on Monday.

The agency wrote it in 2021, before the connector's native stock feature existed. Its API version was fixed in code. The connector retired that version on **8 February 2027**, after publishing the date in November and emailing the technical contact, an agency developer who had left in 2025.

Each subsequent call returned an error. The script logged it and tried again the next hour. For two days, bol showed Monday's stock. The dates and script history belong to the fictional scenario.

The provider followed its published schedule; the consumer had no owner tracking it. **A deprecation notice is useful only if someone acts on it.** Head of Engineering Noor Visser adds pinned versions and retirement dates to the contracts required by Waypoint decision W5, alongside a named integration owner.

What a Contract Promises

An **API contract** is the written description of what a system promises to accept and return: the operations you may call, the fields each one takes and gives back, the errors it can answer with, the limits on how often you may call it, and the period for which all of that stays true. In plain words, it is what lets two teams, or a merchant and a vendor, build against each other without reading each other's code. It has four parts, and each breaks in its own way.

Part of the contract	What it covers	How it breaks	Example in this chapter
Shape	The operations, fields, types and required values	A field is removed, renamed or made required	A connector drops a field the script sends
Meaning	What a value means and what the call does	The same field starts meaning something else	"available" changes from physical stock to available-to-promise
Timing	Rate limits, timeouts, retries, delivery guarantees	A limit is lowered; a retry window changes; ordering was never promised	A stock sync is throttled at peak because another app used the budget

Part of the contract	What it covers	How it breaks	Example in this chapter
Lifetime	Version, support window, deprecation and sunset dates	The version is retired on a date the consumer did not track	The Deprecated Endpoint

Shape is the part people write down; the other three are where incidents come from. A contract that documents only the fields is a lease that names the apartment but not the rent or the end date.

Figure 1: *A contract has four parts: shape is the part people write down, while meaning, timing and lifetime are where incidents come from.*

Contracts are written in machine-readable formats so that tools can check them: **OpenAPI** for HTTP APIs (version 3.2.1 as of September 2026), **AsyncAPI** for message-driven APIs where one system publishes events and others subscribe (version 3.0.0, December 2023), **JSON Schema** (2020-12) for the shape of a single document, and **CloudEvents**, a vendor-neutral envelope that gives every event an id, a source, a type and a time so that a consumer can deduplicate and route it without reading its body. A merchant writes none of these from scratch; it reads its vendors' documents and, when it becomes a provider, publishes its own. The reason to write the contract down is not ceremony: **a written contract can be diffed**, and a diff between two versions is the only reliable way to know whether your integration is affected.

A merchant is on both sides of the table, and the obligations differ. As a **consumer**, the merchant cannot prevent change; its obligations are to know which version each

integration uses, when that version expires, and to test against the next version before the old one disappears. As a **provider**, the merchant owes its consumers what it wants from its vendors: a published document, additive change, and a dated, monitored deprecation. At Omnichannel scale the provider role grows: once Kestrel & Pine's OMS (order management system, the single record of every order, bought under Waypoint decision W2) exposes order status by API, the agency, the returns portal and later a shopper-facing agent all consume a contract that Kestrel & Pine owns.

Requirements differ by flow

Not every contract needs the same guarantees. The figures below are illustrative.

Flow	Direction	How fresh	How consistent	How fast	What the contract must say
Order placed, platform to WMS	Webhook, then queue	Seconds to minutes	Exactly one order in the WMS per order number	2xx (an HTTP success code) within the sender's timeout	Signature, delivery id, retry schedule, payload version
Stock level, platform to marketplace connector	API call, hourly today; events under W1	Minutes	Last write wins per SKU and location	Must finish within the rate budget	Version, sunset date, batch size, cost per call
Availability check at checkout	Synchronous API	Live	Read of the system of record	Under a few hundred milliseconds *(assumption)*	Timeout, degraded answer, version

Flow	Direction	How fresh	How consistent	How fast	What the contract must say
Order status, OMS to agency and portal	API provided by the merchant	Minutes	Read from the order hub	Under a second *(assumption)*	Published document, additive change, deprecation policy

Two rows carry the chapter. The order webhook needs the receiving discipline of the webhook section; the stock push needs the version discipline of the next two sections. The Deprecated Endpoint was the second row with no contract card.

Changing Without Breaking

The cheapest way to version an API is not to. If every change a provider makes is one a consumer written against the old shape can survive, there is nothing to retire and nothing to pull. Two published rule sets define "survivable", and they agree.

Google's API Improvement Proposal 180 on backwards compatibility (created July 2019, updated October 2025) states that existing client code "must not be broken" by a minor or patch release. It allows adding fields, enumeration values and methods, provided new fields are optional and default to the old behavior, and classes as breaking: removing or renaming anything, changing a field's type "even to a wire-compatible type", changing default values or visible behavior.

Zalando's RESTful API Guidelines, published on GitHub and owned by its API Guild, with changelog entries to 16 March 2026, make the same rules mandatory: rule #106, **MUST** not break backward compatibility ("Change APIs, but keep all consumers running"); #107, **SHOULD** prefer compatible extensions; #108, **MUST** prepare clients to accept compatible extensions; #113, **SHOULD** avoid versioning. Rule #108 is the consumer's half of the bargain: a **tolerant reader** is a client that ignores fields it does not know, so the provider can add fields without asking anyone. A client that rejects a payload because an unexpected field appeared has made itself impossible to extend.

The combination is the workhorse of API change in commerce: **the provider only adds**; **the consumer ignores what it does not understand**. Most quarterly change in a platform's API is of this kind, and a tolerant consumer never notices it.

Additive change has limits. You cannot add your way out of a field whose meaning was wrong or a whole generation of the API built on a different model; Shopify's REST

Admin API, "a legacy API as of October 1, 2024", with all new public apps on GraphQL since 1 April 2025, is change of that size. For it, you need versions.

Versioning: Dated, Semantic or in the Media Type

Versioning is naming generations of a contract so that a consumer can say which one it relies on. Three strategies are in common use.

Dated versions. The version is a date, the consumer states it on every call, and the provider supports each date for a published window. Shopify's documentation (undated, accessed November 2026) is the clearest public example: Shopify "releases a new API version every three months at 5pm UTC on the first day of the quarter", named by date (the latest stable version at writing is `2026-10`, accessible until 16 October 2027); "Each stable version is supported for a minimum of 12 months", with at least nine months of overlap between consecutive versions; the version is written in the request URL. Deprecations are announced in the developer changelog with migration guidance, and "Deprecated fields or types are removed in a subsequent release".

Stripe uses dates with a different support promise. Its 2017 engineering post by Brandur Leach (nine years old; the mechanism is still the documented one) explained that each account is pinned to the version current at its first request, that a caller can override it per request with the `Stripe-Version` header, and that breaking changes are implemented as "version change modules" applied in reverse until the caller's version is reached; Stripe had then "maintained compatibility with every version of our API since the company's inception in 2011". Since October 2024 Stripe releases monthly versions with no breaking changes and two named major releases a year; the current version at writing is `2026-10-28.endive`.

Semantic versions. A number in three parts, `major.minor.patch`, where only a major change may break consumers: the convention for software libraries, which Stripe applies to the software development kits it ships with each major release.

Versioning in the media type. The version travels in the `Accept` header, as part of the content type the client asks for. Zalando's guidelines require it (#114) and forbid versions in the URL (#115), because a resource's address should not change when its representation does.

Strategy	Where the version lives	Who moves	Public example	Fits when
Additive only, no versions	Nowhere; one contract that only grows	Nobody	Zalando's preferred path (#113)	Changes are small and the provider controls meaning
Dated versions, forced upgrade	URL; supported for a window, then removed	Every consumer, at least yearly	Shopify, quarterly, 12-month minimum	Many consumers; a provider that must keep its surface small
Dated versions, consumers may stay	Account default or header; old versions kept alive by transformation code	Only consumers who want new features	Stripe, every version since 2011	Payments, where silent change is unacceptable
Media type versioning	`Accept` header	Consumers, per representation	Zalando (#114)	Internal and partner APIs where URL stability matters
Semantic versions	Package or URL prefix such as `v2`	Consumers at each major	SDKs; many internal APIs	Libraries; coarse-grained APIs with few consumers

Two details of Shopify's design decide what an expired version does to you. First, the **fall-forward rule**: "If your app targets an inaccessible version, Shopify falls forward and responds using the oldest accessible stable version", and a response header says which one answered. The call does not fail; it is answered by a newer contract that may have removed or changed what you asked for. That is kinder than an error and more dangerous, because it is silent.

Second, **webhook payloads follow the version of the subscription** that created them, so a subscription made in 2025 keeps sending 2025-shaped payloads while the app's other calls have moved on.

The marketplace connector in the Deprecated Endpoint answered with an error, the loud failure; a fall-forward would have kept the script running against a changed contract, the quiet version of the same incident.

My reading is that Shopify and Stripe offer the same contract with a different default: Shopify makes consumers move every year; Stripe lets them stay but ships breaking changes only behind opt-in. For a merchant, both mean two obligations: **pin the version explicitly** (in the URL for Shopify; in the header and on each webhook endpoint for Stripe) and **put the upgrade on a calendar**. A merchant that pins nothing is on the vendor's default, which changes when the vendor says so.

Deprecation: A Date Pair, Monitoring and Consent

Deprecation is the announced, dated withdrawal of part of a contract; the **sunset** is the date it stops working. The two dates are the whole mechanism, and the web has standard headers for them. RFC 9745, a Proposed Standard of the Internet Engineering Task Force (IETF, the body that publishes the web's protocol standards as numbered RFCs) from 2025, defines the `Deprecation` HTTP response header, with a date that may lie in the future or the past; the older `Sunset` header (RFC 8594) says when the resource is expected to stop responding, and its date "MUST NOT be earlier than the one given in the Deprecation header field". A deprecation with no Sunset "doesn't imply the resource will go offline". In plain words: **"deprecated" alone is a warning; "deprecated on this date, gone on that date" is a deprecation**.

Zalando's guidelines show what a provider owes beyond the headers: #185, **MUST** obtain approval of clients before API shut down ("all clients have given their consent on a sunset date"); #187, **MUST** reflect deprecation in the specification with the sunset date and migration guidance; #188, **MUST** monitor usage of a deprecated API scheduled for sunset; #189, **SHOULD** add `Deprecation` and `Sunset` headers; and, for the consumer, #191, **MUST** not start using deprecated APIs. The organization behind the rules matters as much as the text: Zalando's September 2015 engineering post (eleven years old; the guild may have changed since) described an API Guild of about twenty people spending a tenth of their week on it, "a sounding board and review resource, not an approval board", with the business reason in one sentence: "We cannot afford to break our APIs".

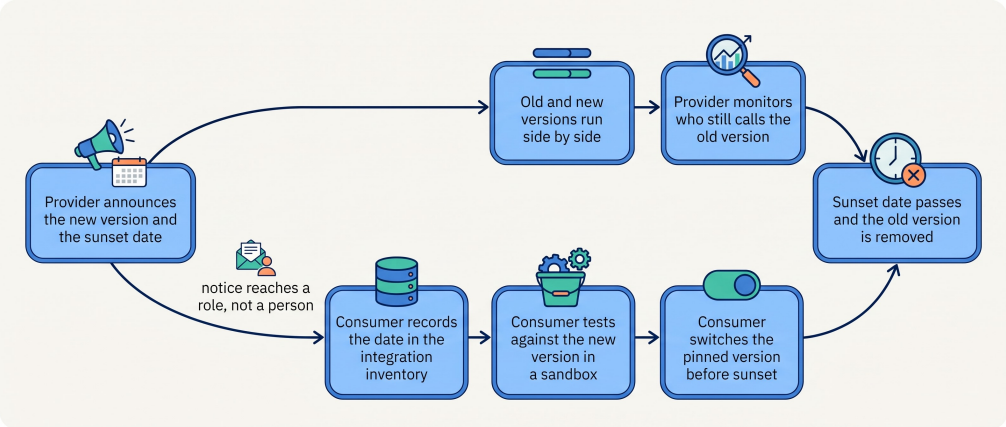


Figure 2: *A deprecation is two timelines that must meet before the sunset date: the provider announces, overlaps, monitors and removes; the consumer records, tests and switches. The Deprecated Endpoint is the bottom line with nobody on it.*

The consumer's half is the half a merchant controls, and it is small. The **integration inventory** that APIs, Events, Files and Middleware: How Commerce Systems Talk introduced (one row per flow: source, target, style, owner, volume, reconciliation) gains two columns: the pinned version and its sunset date. The vendor's changelog is read monthly from a role mailbox, not a person's. Each sunset date becomes a calendar entry three months ahead, assigned to the flow's owner. That is the whole fix for The Pulled Rug on the consuming side; it costs an afternoon to set up and an hour a month to keep. Vendors signal the same obligation from their side: Shopify's app requirements state that "apps using APIs that will be deprecated within 90 days can't be submitted" to its App Store.

Webhooks Done Well

A **webhook** is an HTTP request the source system sends to a URL you registered when something happens: an order is placed, a payment is captured, a shipment is created. It is the most common way a hosted platform tells the rest of the landscape that the world changed, and the mechanism most often trusted beyond what it promises. The two best-documented providers say the same things about their own webhooks.

Shopify's documentation (undated, accessed November 2026) states that "Webhook delivery isn't always guaranteed, and your app can miss or mishandle events for other reasons", that Shopify "doesn't guarantee ordering within a topic, or across different topics for the same resource", and that apps should "use reconciliation jobs to periodically fetch data from Shopify so that your app stays consistent with Shopify's data". "If Shopify receives no response or an error, it retries 8 times over the next 4 hours"; and "After 8 consecutive failures, the subscription is automatically deleted if it was configured using the Admin API". Every delivery carries a signature computed from

the raw body and the app's secret and a webhook id unique per delivery, because "your app might receive the same webhook more than once".

Stripe's documentation (undated, accessed November 2026) says Stripe "attempts to deliver events to your destination for up to three days with an exponential back off in live mode", "doesn't guarantee the delivery of events in the order that they're generated", that endpoints "might occasionally receive the same event more than once", and that handlers must "quickly return a successful status code (2xx) before any complex logic". Its signature is an HMAC (a keyed hash that proves who sent the message and that it was not altered) over a timestamp and the payload, with a five-minute tolerance against replay.

The Standard Webhooks specification (version 1.0.0; its implementations include Svix, Kong, Twilio, OpenAI and Anthropic) generalizes the pattern into three headers: `webhook-id`, unchanged across retries; `webhook-timestamp`; and `webhook-signature` over `id.timestamp.body`. It tells consumers to "Use the `webhook-id` header as an idempotency key", tells senders to retry over "multiple days, with an exponential backoff", and does not address ordering.

Property	Shopify	Stripe	Standard Webhooks
Signature	HMAC-SHA256 of the raw body	HMAC-SHA256 of `timestamp.body`	HMAC-SHA256 of `id.timestamp.body`
Replay protection	Triggered-at header (advisory)	5-minute tolerance by default	Timestamp tolerance, value unspecified
Duplicate key	`X-Shopify-Webhook-Id`	`event.id`	`webhook-id`
Retries	8 over 4 hours; subscription deleted after 8 consecutive failures when API-created	Up to 3 days, exponential backoff	Multiple days, exponential backoff with jitter
Ordering	Not guaranteed	Not guaranteed	Not addressed
Response expected	2xx within 5 seconds in total	2xx quickly, before business logic	Not specified

Read the table as a contract: **delivery is at least once at best**; **order is not promised**; **the signature is the only proof of origin**; **the receiver owns deduplication and a periodic sweep**. The receiving design that follows is the same for every provider.

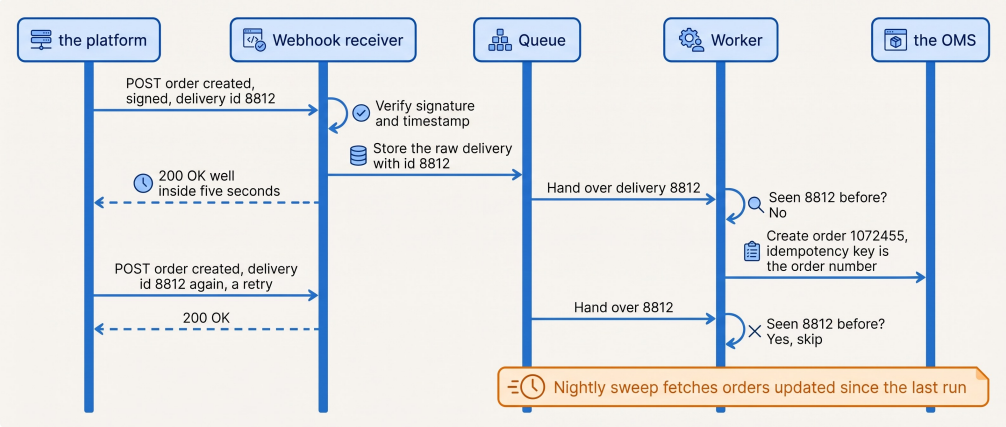


Figure 3: *A safe webhook receiver verifies, stores, answers fast and processes later; a duplicate delivery is recognized by its id, a duplicate order by its number, and a nightly sweep catches whatever was never delivered.*

Four rules make webhook delivery manageable:

- **Authenticate the delivery** before trusting its contents (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers).
- **Store it before acknowledging it.** Return success after durable receipt, then process it from a queue. A slow back-office system should not hold the provider's request open.
- **Deduplicate the delivery and the business action.** A delivery id catches retries; an order or return key catches different notifications about the same action. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation explains the distinction.
- **Compare periodically with the provider's records.** Fetch changes since the last successful run to detect missed deliveries.

Notifications reduce delay. The comparison job checks completeness.

Rate Limits and Partner APIs

A **rate limit** is the maximum number of calls, or the maximum "cost" of calls, a consumer may make against an API in a period; above it, the provider answers with an error instead of data. It is a timing clause of the contract, and it decides whether an integration can finish before its next run starts.

Shopify's GraphQL Admin API meters calls in cost points: a bucket per app and per store refills at 100 points per second on the Standard plan, 200 on Advanced, 1,000 on Plus and 2,000 on its enterprise offering; an object costs 1 point, a mutation (a write)

10, and "A single query may not exceed a cost of 1,000 points, regardless of plan limits". Stripe's live-mode limits are 100 requests per second per account and 25 per endpoint, answered with `429 Too Many Requests`.

Two consequences follow, and the first is arithmetic. Kestrel & Pine's platform is not named, so apply Shopify's published numbers to its volumes purely as an illustration: a stock push that updates each of the roughly 17,000 active variants with one mutation at 10 points needs 170,000 points, about 28 minutes at 100 points per second and under three minutes at 1,000. An hourly job that takes 28 minutes and shares its bucket with 40 other apps will sometimes not finish, which is why batch operations are a necessity rather than an optimization, and why a flow's contract card records its expected call rate. The second consequence is that **limits are per store and shared by every app**: at peak, a chatty analytics app can starve the WMS connector (Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation, Black Friday Every Year: Peak Load, Resilience and Recovery).

Partner APIs, the ones a marketplace, a carrier or a wholesale customer exposes to the merchant, carry the same four parts and usually a thinner document. The EDI relationship with Bergzicht Outdoor is a contract too, and the ASN Chargeback of 2026 was a shape-and-meaning failure of exactly this kind: the despatch advice said one thing and the pallets said another (Documents Between Companies: EDI, Punchout and E-Invoicing).

Who Owns a Contract

Inside a company, the provider team owns the contract: it publishes the document, keeps the compatibility rules, announces deprecations and monitors usage. Zalando's account in InfoQ (December 2016; ten years old) put it as "We see API definitions as a source truth, part of a contract between a service provider and consumers", written in OpenAPI before any code ("API First"). The often-told origin story, an Amazon mandate from around 2002 that teams communicate only through service interfaces with "no direct reads of another team's data store", exists in public only as Steve Yegge's 2011 recollection, never confirmed by Amazon in any page this book cites; its rule is the useful part: **the contract is the only door**.

For a merchant of Kestrel & Pine's size, the rule applies to vendors before teams: integrations go through a vendor's documented API, never through a direct read of the ERP's database or a scrape of an admin page, because an undocumented path has no contract and therefore no deprecation notice.

With vendors, ownership has three parts. The **commercial contract** names a minimum deprecation notice period, in months, and a technical contact that is a role. The **integration inventory** records the pinned version and the sunset date. And a **consumer test** runs against the vendor's sandbox on each new version before it becomes the

default, checking that the fields the integration relies on are still there and still mean the same.

That is as far as **consumer-driven contracts** can go with a vendor. The idea, coined by Ian Robinson in June 2006 (twenty years old and unchanged), says a provider's contract should be shaped by the union of what its consumers actually use; the Pact tool (documentation updated August 2026) generates the contract from the consumer's tests and verifies the provider against it. Between services inside one company it works well; with a SaaS vendor, the vendor's schema is the contract.

The Choices and Their Trade-offs

As a consumer, the merchant has three postures per integration; the agency script took the second one, with a version string nobody treated as a pin.

Posture as consumer	What it means	Maintenance cost	Failure mode	Right when
Pin and schedule	State the version, record its sunset date, test the next version in a sandbox, switch before sunset	Low, continuous	A missed calendar entry	The flow carries orders, stock or money
Float on the default	State no version; take whatever the vendor serves	None, then an incident	Silent behavior change or sudden error	A read-only report; never orders, stock or money
Delegate to a connector	The vendor's own integration or an iPaaS connector tracks versions for you	Connector fee; its own calendar	Connector lags the vendor; mapping changes unnoticed	The merchant lacks engineers to own the flow

As a provider, the choices are the versioning strategies above, with one judgement: a merchant providing its first APIs should take Zalando's compatibility rules wholesale (additive change, tolerant readers, avoid versions) and defer the full deprecation machinery until it has consumers it cannot call on the phone.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	The platform's apps and a vendor connector are the integrations; the merchant owns no code	A list of which apps and connectors exist and which version each pins	An app stops working after a platform release and nobody knows why
2 — Growing	First custom scripts and an iPaaS; webhooks from the platform to an ERP or 3PL; an agency writes most of it	Pinned versions with sunset dates in the inventory; webhook handlers that verify, deduplicate and sweep; a role mailbox for vendor notices	A script with a version string from the year it was written; a webhook subscription that vanished after an outage

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel (Kestrel & Pine)	A dozen systems and dozens of flows; the merchant becomes a provider too (order status to agencies, portals, marketplaces)	A published document for every merchant-owned API, additive change, dated deprecation with monitoring, signed webhooks, a sandbox test per vendor version, notice periods in vendor contracts	The Deprecated Endpoint; one app's calls starving another at peak; a partner asking which version of "your" API to use
4 — Group	Several brands or units share platforms and partners; external partners depend on the group's APIs	API guidelines in the Zalando mold, a schema registry for events, contract tests in the delivery pipeline, a guild that reviews APIs	Two brands publishing incompatible versions of the same concept; a partner broken by a change one unit made

The topic never disappears; who owns it moves. At Starting, the vendor owns everything and the merchant's duty is to read. At Growing, the merchant's first scripts create its first consumer obligations, and this is where most Pulled Rugs are planted. At Omnichannel the merchant becomes a provider; at Group the duties become a function with a name. My judgement is that the Growing-stage habit pays most: a merchant that pins and schedules from its first script never has to learn the rest in an incident.

In Practice: The Contract Card for the bol Stock Flow

After February 2027, Ruben rewrote the inventory row for the flow that failed. Under Waypoint decision W5, every integration gets a contract; here is the card for the stock

push to bol, in the form Kestrel & Pine uses from March 2027. All details are fictional.

Field	Value
Flow	Stock levels, platform to marketplace connector, bol listings
Owner	Milan Petrov (business), Ruben Alves (technical)
Style	Scheduled API push, hourly, until the availability service's events replace it in Q4 2027 under W1
Contract	The connector's stock API, pinned to the connector's current 2026 release; its changelog read monthly from the integrations role mailbox
Sunset of the pinned version	Published by the connector; recorded in the inventory; calendar entry three months ahead, assigned to the owner
System of record	The WMS for Duiven stock (W1); the push carries available-to-promise per channel once the availability service exists
Idempotency	Last write wins per SKU and location; each push carries its snapshot time so an older push cannot overwrite a newer one
Failure handling	An error raises an alert within fifteen minutes; two consecutive failed runs page the owner; the job never exits silently
Reconciliation	A daily comparison of the connector's stock per SKU against the platform's, with the variance published
Expected rate	About 17,000 variants per run, in batches within the connector's limit

The card is ordinary, and that is the point. Nothing on it required a new system; the flow now has a person, a version, a date, an alarm and a check. When the connector publishes its next retirement date, it lands on Milan's calendar and Ruben's test plan, not in the mailbox of someone who has left. By the end of 2027 the inventory shows the pinned version and sunset date of all 34 integrations on one page *(illustrative)*, which is what Noor shows the board when asked what the integration standards bought: not a platform, a calendar.

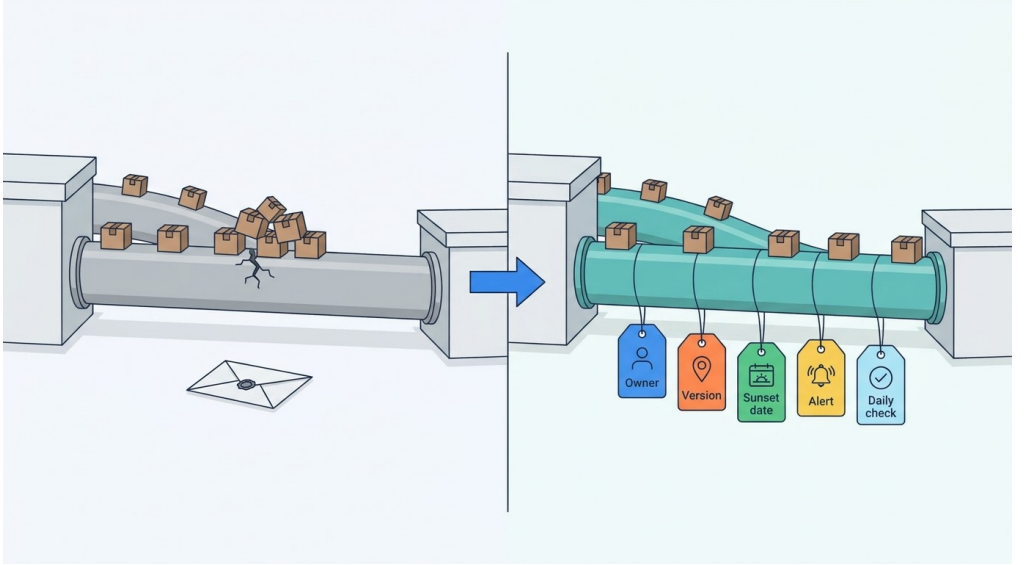


Figure 4: *The contract card adds no system: the stock flow now has a person, a version, a date, an alarm and a check.*

When It Goes Wrong

Failure	How you notice	What to do
The version expires (the Deprecated Endpoint)	Errors in a log nobody reads; a partner's complaint; a zero-traffic alert on the flow	Alert on errors and on silence; pin and calendar every version; test the next version early
Silent fall-forward	Nothing fails; a field is missing or means something else; a reconciliation variance grows	Watch the version header the provider returns; validate responses against the schema you expect
Webhook subscription deleted after an outage on your side	Orders stop arriving while the shop is busy; the sweep finds a backlog	Alert on traffic dropping to zero; re-create subscriptions automatically; rely on the sweep

Failure	How you notice	What to do
Duplicate or reordered deliveries	Two orders in the WMS with one order number; a cancellation processed before its order	Deduplicate on delivery id and business key; order by the provider's timestamp; apply updates as state, not as steps
Meaning changes without a version	Numbers shift on a date; no error anywhere	Read the changelog; keep a reconciliation that would notice; raise it with the vendor as a contract matter

Recovery is the same in every row: the sweep and the reconciliation job find what the contract failure hid, and the inventory says who fixes it.

What to Take Away

Established.

- Additive change is safe and removal is not: adding optional fields keeps existing clients working; removing, renaming, retyping or changing meaning breaks them (Google AIP-180, 2019, updated 2025; Zalando rules #106 to #108).
- A deprecation is a date pair plus monitoring: the `Deprecation` and `Sunset` headers (RFC 9745, 2025; RFC 8594) and Zalando's rules #185 to #189 make the announced date, the removal date, usage monitoring and consumer consent the standard parts.
- Webhooks are at least once, unordered and signed, and the receiver owns deduplication and a periodic sweep: Shopify's and Stripe's documentation say so in their own words, and Standard Webhooks 1.0.0 standardizes the headers.

Depends on context.

- Whether to force consumers to upgrade (Shopify) or carry old versions indefinitely (Stripe) depends on how many consumers you have and what a silent change costs: for payments, carrying old versions is worth the transformation code; for a catalog API with many apps, a forced annual upgrade keeps the surface small.
- Consumer-driven contract tests pay between services one organization owns; with a SaaS vendor, a sandbox validation per version is the practical substitute.

- Delegating version tracking to an iPaaS or a vendor connector is right for a merchant without integration engineers; it moves the obligation rather than removing it.

The author's synthesis.

- The cheapest fix for The Pulled Rug is a calendar: two columns in the integration inventory, a role mailbox for vendor notices and a rule that no flow goes live against a version with less than six months of support left. I have not seen a merchant need more until it becomes a provider.
- Treat your first self-provided API as if a stranger would call it, because within a year one will: publish the document, change only by adding, and deprecate with dates, before anyone asks.

Questions to Ask

1. For each integration that moves orders, stock or money: which contract version does it pin, when does that version expire, and who is on the calendar for the upgrade?
2. Where do the vendor's deprecation notices go: a role mailbox someone reads monthly, or a person who may leave?
3. Does our webhook receiver verify the signature, answer within the provider's timeout, deduplicate on the delivery id, and run a periodic sweep for what it missed?
4. If the provider answers a retired version by falling forward instead of failing, would we notice?
5. Which APIs do we provide to agencies, portals, partners or stores, and is each described in a published document with a stated compatibility and deprecation policy?
6. When a flow goes silent, who is alerted, within how long, and by what signal other than a partner's complaint?

Contracts decide how systems speak to each other; the next question is whether they are speaking about the same thing. When the platform, the POS and the returns portal each hold a record called "Eva Janssen" with no shared identifier, no contract can tell you they are one customer. That is the subject of One Customer, One Product: Identity and Master Data Across Systems.

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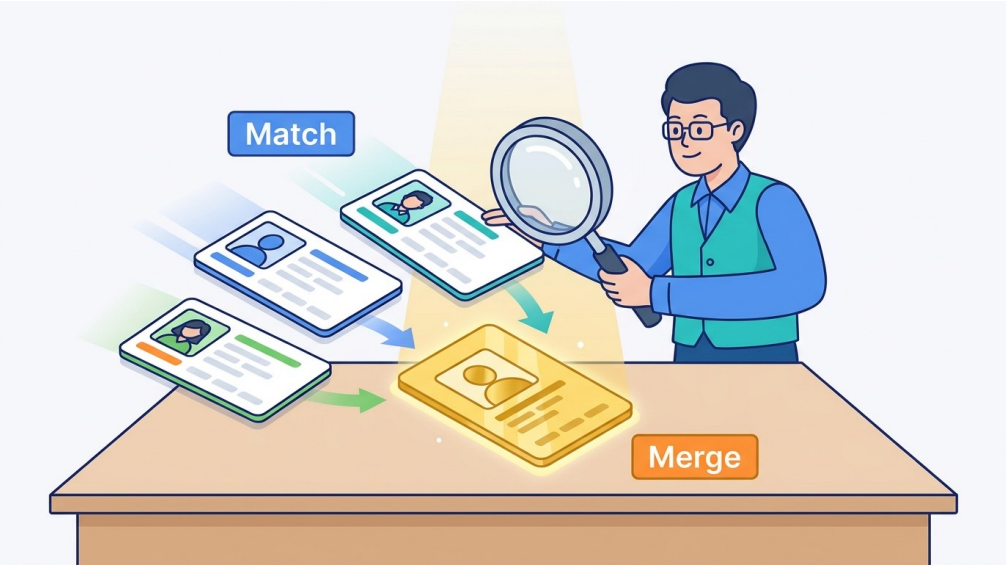
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

27. One Customer, One Product: Identity and Master Data Across Systems



IN THIS CHAPTER, YOU WILL: Learn why one person or product becomes four records, how matching and merging decide that they are one, and who should own the result.

KEY POINTS:

- **Every system keys the same person or product its own way, so duplicates are the normal case.** An online account is keyed by e-mail, a store customer by phone or loyalty card, a wholesale contact by company plus person, a product by an internal stock code in the warehouse and a global product identifier on the marketplace. Nothing links them unless you build the link.
- **Identity is a decision before it is software.** For each master entity, name the system of record, keep a cross-reference of every other system's identifier for it, write the matching rules and the survivorship

rules down, and make every merge reversible. A merge that joins two different people is a privacy incident, not a data-quality blemish.
- **The marketing copy is not the master**. A customer data platform builds a fuzzy, useful profile for audiences; the record that drives orders, refunds, invoices and deletion requests must be exact and owned. Keep the two apart and let the exact one feed the fuzzy one.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Familiar Stranger** — each system counts customers its own way, so nobody can say whether the store customer and the online customer are the same person.
- **The Spaghetti Junction** — every connector maps product and customer identifiers in its own way, so the cross-reference between systems lives inside the connectors, with no owner.

A customer buys online, joins a store loyalty scheme and later contacts the returns portal. Each system may create its own record for that person. At Kestrel & Pine, the fictional outdoor brand, platform accounts, store loyalty entries, newsletter subscribers and returns contacts all overlap. Without reliable matching, support can miss part of a customer's history and reports can count one buyer several times.

Products have the same problem. One Ridgewalker women's M in moss appears as a platform variant, finance item, warehouse item and several marketplace offers. Each has a local identifier. If only the connectors know how those identifiers match, replacing a connector can lose the relationship.

This chapter explains **master data**, the shared records transactions refer to, and **matching**, the decision that two records describe the same entity. A **golden record** is the maintained, authoritative view assembled from those records. Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record covers ownership, Whose Data Is It? Privacy, Consent and Retention covers personal-data rules and What Are We Selling? Products, Variants and the Catalog explains the product model.

Where we are in the order's journey. Eva buys a jacket online on 13 April 2027. She has previously bought socks through the Nijmegen store's loyalty account. Her exchange request creates a returns contact on 19 April. The identity problem is whether those systems can establish that their records describe the same customer.

Setting the Scene at Kestrel & Pine: Four Evas

In March 2027, Ecommerce Manager Milan Petrov proposes an autumn online loyalty program. Head of Engineering Noor Visser asks how it would recognize a store customer online. Nobody can reliably count eligible customers or connect the accounts.

Ruben Alves, the lead integration engineer, pulled the counts. The figures are illustrative.

List	System	Records (illustrative)	Key the system uses
Customer records	The platform (storefront, cart, checkout)	about 410,000, including guest checkouts	Customer id; e-mail address
Loyalty list	The POS (point of sale: the till software in the six stores)	about 38,000	Phone number; loyalty number
Subscribers	The newsletter tool (one of the 41 platform apps)	about 165,000	E-mail address
Contacts	The returns portal	about 61,000	E-mail address plus order number

The four lists add up to about 674,000 records. Nobody believes Kestrel & Pine has 674,000 customers; nobody can say how many it has.

Eva Janssen shows why. She is one person in Nijmegen. In Kestrel & Pine's systems she is four:

Where	What the record says	What it does not know
The platform	An account from 2024 under her personal e-mail; three orders, the latest #1072455 of 13 April 2027; an address in Nijmegen	That she shops in the Nijmegen store
The POS loyalty list	"E. Janssen", joined at the Nijmegen store in 2023 with her mobile number; four purchases, the latest socks in February 2027	Her e-mail; her online orders
The newsletter tool	A subscriber since 2022 under an older e-mail address	That she is a customer at all

Where	What the record says	What it does not know
The returns portal	A contact created on 19 April 2027 from order #1072455, with her personal e-mail	Nothing else about her

No field is shared by all four records. **The Familiar Stranger** is the book's name for this problem: a merchant repeatedly encounters a person its systems cannot reliably connect. Milan and Noor defer online loyalty until that relationship can be maintained (Discounts, Coupons and Loyalty: Promotions Without Chaos).

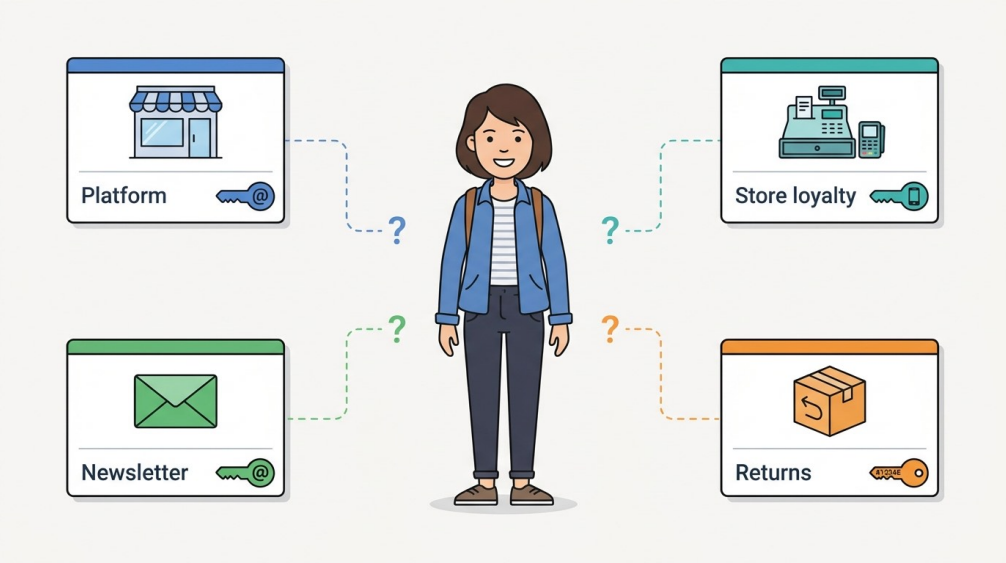


Figure 1: *One shopper, four records, four keys: no field is shared by all four, so the four lists add up to about 674,000 records and nobody can say how many customers there are.*

Why One Customer and One Product Matter

The vocabulary first, gently. An **identifier** is a value a system uses to tell one record from another: a customer id, an e-mail address, an order number, a barcode number. **Customer identity** is the set of identifiers that different systems use for the same person, plus the rules that link them. **Master data** is the slowly changing reference data that transactions point to: customers, products, suppliers, locations. A **system of record** is the one system whose version of a fact is the truth (the idea comes from Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record); for master data it is usually called the **master**.

Why does a merchant lose money or trust when identity is undecided?

- **Marketing**. Eva gets the new-subscriber offer at her older e-mail while her account is three orders old; a store customer who opted out is e-mailed because the opt-out lives only in the POS. Consent belongs to a person, and a person the systems cannot see cannot be respected (Whose Data Is It? Privacy, Consent and Retention).
- **Service**. The agent answering Eva's exchange question cannot see her store purchases. Where Is My Order? Notifications, Service and the Customer Record describes the five tabs an agent works from; identity is why there is no sixth that joins them.
- **Loyalty and promotions**. "First order", "members only" and "once per customer" need to know who the customer is (Discounts, Coupons and Loyalty: Promotions Without Chaos); a program built on one list rewards that list and ignores the rest.
- **Returns and refunds**. A refund goes back to the order's payment method, which is safe; a store credit or an exchange raised against the wrong record goes to the wrong person.
- **Analytics**. "Customers", "new customers" and "repeat rate" are undefined while the person is (Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics).
- **Privacy**. A person who asks to be forgotten must be found in every system within one month; without an index of where Eva is, the request cannot be fulfilled (Whose Data Is It? Privacy, Consent and Retention).

Products have their own version of each loss: a marketplace listing without the GTIN of its warehouse item becomes a second product, and an ERP item whose variant code maps to no platform variant is a sale the ERP cannot cost.

The identifiers a commerce estate uses

For customers:

Identifier	Who assigns it	Stable?	Unique to one person?	Who else understands it
Customer id	The platform, the POS, the ERP, each its own	Yes, inside that system	Yes, inside that system	Nobody else, unless carried in a cross-reference

Identifier	Who assigns it	Stable?	Unique to one person?	Who else understands it
E-mail address	The person	Changes with jobs and providers	Often shared by a household; one person has several	Every system, which is why it is the usual match key
Phone number	The person	Changes rarely	Usually one person, sometimes a family	Stores and carriers
Loyalty number	The POS or a loyalty program	Yes	Yes, but one person may hold two cards	Only the program
Postal address	The person	Changes with moves	Shared by a household	Carriers; useful for matching, never alone
Company id plus contact	The ERP or the B2B portal	Yes	A company has many contacts; a contact moves between companies	EDI (electronic data interchange) partners by GLN (the Global Location Number that GS1, the barcode standards body, assigns to companies and sites; see Documents Between Companies: EDI, Punchout and E-Invoicing)

For products, the lesson of What Are We Selling? Products, Variants and the Catalog is that a SKU (the merchant's own code for a sellable variant) and a GTIN (the Global Trade Item Number under the barcode, licensed from GS1 so that every company means the same item by it) are assigned once and never change. Here is what each identifier is good for, and what it is not:

Identifier	Who assigns	Scope	Good for	Not
GTIN (EAN-13, the 13-digit barcode number, in Europe)	The brand owner, through GS1	Global, one per sellable item	Marketplace listings, retail scanning, matching across companies	An internal key: resellers of the same item share it, and packaging variants differ
SKU	The merchant	Internal, one per variant	Warehouse, stock, merchandising, the join key between the merchant's own systems	Stable across systems unless governed: connectors overload it
Internal id	Each system	That system only	Joins inside that system	Meaningful anywhere else
Marketplace id	The marketplace (an ASIN, Amazon's own product number; an offer id on bol)	That marketplace	That channel's listings and orders	Yours; the marketplace can change what it refers to

The table is the author's synthesis of the GS1 US definition of the GTIN (n.d., accessed November 2026) and of Microsoft's documentation for its Business Central connector for Shopify (October 2026), which matches items on SKU or barcode and, when it finds an existing variant, links rather than creates. That one sentence of documentation is a product-identity policy: **match on the governed identifiers, link, never duplicate**.

How the Pieces Fit Together

Identity is not one system but a set of responsibilities spread across the systems a merchant already has.

Responsibility	What it does	Where it usually lives	Kestrel & Pine, January 2027
The customer master	The one record per consumer the business treats as true: name, contact details, addresses, consent	The platform for consumer merchants; the ERP or a CRM (customer relationship management system) for business customers	Undecided; four lists
The product master	Holds the one record per product and variant, with its SKU and GTIN	The PIM (product information management system) where there is one; the ERP's item master where there is not	Three copies plus the connector's; the PIM arrives in Q3 2027 (W3)
The cross-reference and merge log	For each master record, every other system's identifier for it, and every link or merge made, reversibly	A table owned by the master's owner; often in the integration platform or the data warehouse	Inside each connector's mapping
Matching	Decides whether two records are the same entity when they share no identifier	A batch job, a platform feature, a customer data platform (CDP) or a master data management (MDM) tool	Nothing

Responsibility	What it does	Where it usually lives	Kestrel & Pine, January 2027
Survivorship	When two records merge, which value wins per field	Rules written down; enforced by the merge tool	Nothing
Consent	The person's marketing choices, with when, where and what they agreed to	On the customer master, synchronized outward	In the newsletter tool and the POS, separately
The customer data platform (CDP)	A unified profile per person built from every touchpoint, for audiences and personalization	A separate product, fed by the master and the event streams	None

The design rule I use: **the master owns the record, the cross-reference owns the links, and every other system keeps its own record but carries the master's id once matched**. The POS still needs its loyalty list and the returns portal its contacts; what changes is that each record points at one master, and the master knows where its copies are.

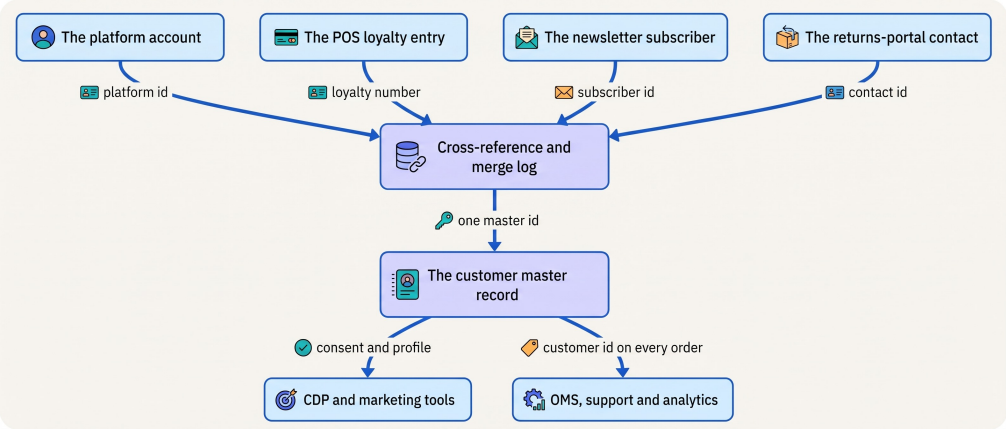


Figure 2: *Four systems keep their own records of Eva; a cross-reference links each to one master record, which feeds marketing copies and the operational systems.*

How a new record finds its master

The sequence below shows what should happen when Eva buys socks in the Nijmegen store in February 2027, in the design Kestrel & Pine adopts later in the chapter.

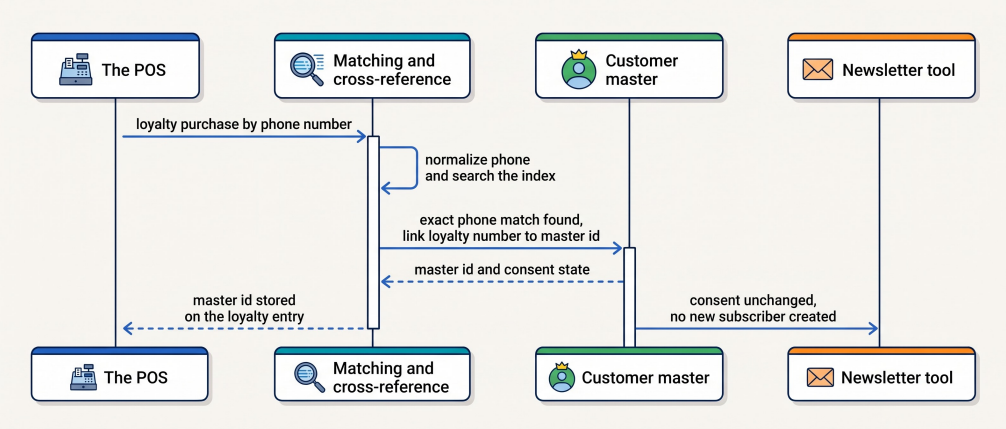


Figure 3: *A store purchase is matched on a normalized phone number; the loyalty entry is linked to the master, and consent is read from the master rather than recreated.*

Three things carry the design: the POS keeps its own entry and gains a master id; the match is on an exact, normalized key, so no human is needed; and consent is read from the master, which is why the newsletter tool does not quietly gain a subscriber from a store sign-up.

Requirements per flow

Identity flows differ in how fast and how sure they must be.

Flow	How fresh	How consistent	How fast	How available
Online checkout, logged in	Immediate	The account id is the key; no matching needed	Part of checkout	Must not block checkout

Flow	How fresh	How consistent	How fast	How available
Guest checkout	Immediate	Creates a record keyed by e-mail; matched to an account later, never during checkout	Part of checkout	Must not block checkout
Store loyalty sign-up or purchase	Hours	Exact match on phone or e-mail; otherwise a new master record	Batch is fine	Can wait; the store never waits on it
Newsletter sign-up	Hours	Exact match on e-mail; consent recorded on the master with time, place and wording	Batch is fine	Can wait
Returns-portal contact	Minutes	Keyed by order number, so the match is exact	Part of the return start	Can degrade to "unlinked"
New product variant	Weeks before launch	One SKU and one GTIN per variant, created once in the master, published outward	Batch	Office hours

The pattern: **the flows that face a customer are exact or deferred, never fuzzy**. Nothing in the checkout path waits for a matching job; probabilistic matching, explained next, runs in the background and sends its uncertain cases to a person.

Matching and Merging

Matching is deciding that two records with no shared identifier refer to the same real-world thing. The field is called record linkage, and it is older than commerce computing: Dunn described it in 1946, Newcombe and colleagues showed in 1959 that agreeing on a rare value (an unusual surname) is stronger evidence than agreeing on a common one, and Fellegi and Sunter gave the model most tools still use in 1969.

In plain words: for each candidate pair, add up evidence field by field (the same e-mail is strong, the same first name weak), and compare the total with two thresholds. Above the upper one, a match; below the lower one, not; between them, a person decides. The 1969 paper is paywalled, so the citation is secondary.

There are two families of rule. **Deterministic matching** uses exact rules: the same normalized e-mail or phone means the same person. It is cheap, explainable and safe, and it misses every person who used two e-mail addresses. **Probabilistic matching** scores partial agreement (name, address, postal code) and tolerates typos. It finds more matches and makes more mistakes, which is why it needs thresholds and a review queue.

Before either, **normalization**: lower-case the e-mail and strip spaces; write phone numbers in the international format; write postal codes in the national format. Most "duplicates" in a commerce estate are the same key spelled two ways. And for any estate larger than a few thousand records, **blocking**: compare only pairs that already agree on something (the same postal code, the same surname), because comparing every record with every other grows with the square of the count; two lists of a million records would make a million million pairs.

The rules I use for a merchant's customer matching:

1. Normalize first; match on exact e-mail or exact phone automatically.
2. Score name plus address plus postal code; auto-link above a high threshold, queue the middle band for a person, and never link on name plus city alone.
3. Never merge during a customer-facing flow; link in the background.
4. Keep every merge reversible, in a log that records which records became one, when, by which rule and who approved it.
5. Treat a wrong merge as a privacy incident and rehearse the unmerge: it shows one person the other's orders and addresses, a personal-data breach (Whose Data Is It? Privacy, Consent and Retention).

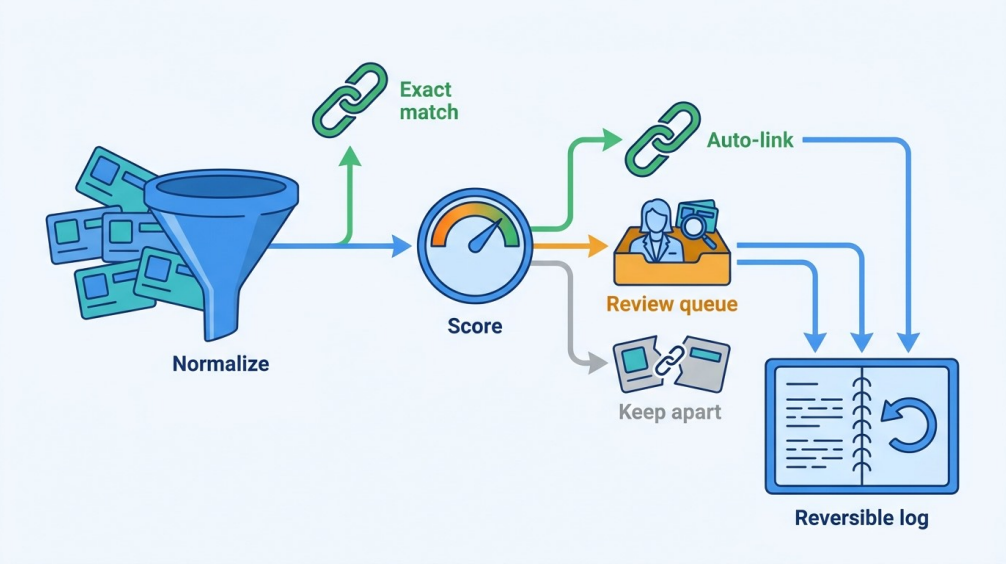


Figure 4: *Customer matching in the order the chapter recommends: normalize, link exact e-mail or phone matches automatically, score the rest, send the middle band to a person, and keep every merge reversible.*

Survivorship is the rule that decides which value wins when two merging records disagree: two names, two addresses, two consent states. The usual rule types are source priority (the platform's address beats the newsletter tool's), recency (the latest address wins) and completeness (the record with a phone number wins), as the MDM vendor Profisee describes them (August 2025). The rule is per field, not per record: the newest address, the strictest consent, the earliest created-date.

Shopify documents its own survivorship, which makes it the clearest dated example. Its `customerMerge` mutation (API version 2026-10) says the two ids "don't guarantee which customer is kept": the caller may name the customer whose e-mail survives; otherwise the one with an e-mail wins, then an enabled account, then the marketing-consent state; first and last name can be overridden per field; the merge runs as a job whose resulting id must be treated "as authoritative", and a preview shows the outcome first. Most platforms have a comparable rule and do not publish it. **That is why the entity–owner table must name the customer master: if the platform merges by its rules and the ERP by others, the two masters drift apart on their own.**

For products, matching is simpler and stricter. The GTIN is the match key across companies, the SKU inside one. A product record with no GTIN is matched by hand or not at all. GS1's rule, relayed by secondary sources, is that every distinct trade item and packaging configuration gets its own GTIN and a GTIN is never reused for a different item; What Are We Selling? Products, Variants and the Catalog carries the detail. The product cross-reference (GTIN, SKU, each system's internal id, each marketplace's id) is the product master's companion table. It belongs to the PIM where there is one, and

to the ERP's item master where there is not.

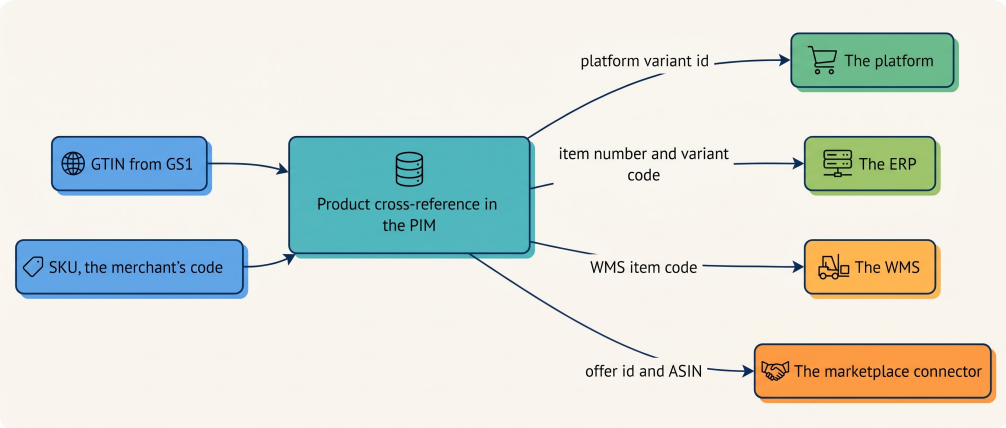


Figure 5: *One product, two governed identifiers and four internal ones; the cross-reference in the PIM is where every other system's key for the Ridgewalker women's M moss is recorded.*

Consent travels with the identity

The General Data Protection Regulation (GDPR), the EU's personal-data law, defines consent as a freely given, specific, informed and unambiguous indication of the person's wishes (Article 4(11)) and requires the controller to be able to demonstrate that consent was given (Article 7(1)). "Demonstrate" is a data requirement: the record must hold when, where and to what the person agreed, which is why Shopify's consent mutation records "when and where the customer gave or withdrew consent". A merge must carry that evidence forward, with its source, rather than recompute consent from whichever record survived. The legal side belongs to Whose Data Is It? Privacy, Consent and Retention.

The Choices

A merchant has four realistic ways to own customer identity, and the product side has two.

The platform's customer record is the master, with periodic deduplication. Every other system's record is linked to a platform customer id; matching is the platform's own merge feature plus a scheduled job for the other lists. This fits a consumer merchant whose platform already holds most customers, which is most merchants below the Group stage.

The ERP or a CRM is the master. The finance or sales system holds the customer; the platform's accounts are copies. This fits wholesale-heavy businesses, where the customer is a company with terms and a credit limit that the ERP already owns (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals). For consumers it usually fails: the ERP's customer record was designed for invoicing, not for consent, addresses and preferences.

A cross-reference and matching service beside the existing systems. Nothing moves; a small service, or tables in the integration platform or the data warehouse, holds the master id, the links and the merge log, and runs matching. It is the lightest way to make several masters behave as one, and the design Kestrel & Pine adopts. Its weakness: it is "just tables" until a wrong merge happens.

A customer data platform as the master. The CDP Institute (2026 wording) defines a CDP as software that "creates and maintains a persistent, unified customer record that is accessible to other systems" and that "assumes primary responsibility for defining and maintaining customer identity". Twilio Segment's identity resolution (documentation of June 2026) merges each person's history across web, mobile, server and third-party sources into one profile through an identity graph of cookie ids, device ids, e-mails and external ids, with merge protection against non-unique anonymous ids.

These are strong marketing tools. My judgement, labeled as mine: do not make the CDP the master that service agents, the OMS and the ERP read. Its profile is built for audiences and may be fuzzier than a record that drives invoices, refunds and deletion requests. Let the exact master feed the CDP, and let the CDP's matches come back as suggestions, not merges.

For products: **the ERP item master or the PIM.** Until product content needs enrichment, translation and syndication, the ERP's item master with a SKU and a GTIN per variant is enough, and the cross-reference to channel ids lives beside it. From that point the PIM is the product master and the ERP keeps costs and purchasing, which is Waypoint's W3 (What Are We Selling? Products, Variants and the Catalog).

A fifth option exists for the Group stage: a **master data management (MDM)** platform. Informatica describes MDM as creating "a single master record for each person, place, or thing in a business", the record "also known as a 'golden record' or 'best version of the truth' ", with software that matches and merges across systems and keeps an audit trail (n.d., accessed November 2026). A **golden record** is that one assembled record per customer or product, built by merging sources under survivorship rules. Below the Group stage I keep MDM small: a decision per entity (owner, survivorship rule per field, merge process), not a product. Buying the platform is a move driven by several brands, acquisitions or countries, each with a master to reconcile.

Trade-offs

Option	Fits when	Strengths	Costs and limits	Warning sign
Platform as customer master	Consumer merchant; most customers come through the storefront	Already exists; merge and consent features; one id on every online order	Store and wholesale records are second-class; survivorship rules are the vendor's	The POS list is never matched because "the platform is the master"
ERP or CRM as master	Wholesale-heavy; company customers with terms	Terms, credit and invoices already there	Consumer fields fit badly; marketing tools cannot read it	Consumer records keyed by invoice address
Cross-reference and matching service	Several systems holding customers; no replatforming	Nothing moves; reversible; cheap to start	Needs a named owner and a review queue; easy to leave untended	A wrong merge nobody can undo
CDP as master	Marketing needs audiences across many touchpoints	Identity graph across anonymous and known; activation to advertising and e-mail	Probabilistic; not built for refunds, invoices or deletion; cost scales with events	Service agents working from the CDP's view
MDM platform	Group stage: several brands, countries or acquisitions	Survivorship per field across masters; audit trail; stewardship workflow	License plus a data-governance function; a year to implement	Bought before the second brand exists

The product side has the same shape in miniature: the ERP item master until enrichment and syndication need a PIM, and never two masters for the same identifier.

Across Scale Stages: When It Fits and When It Limits

The four stages are the book's (see What Changes at Each Scale: From First Store to Group).

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	One platform; its customer record is the only one; SKUs typed by hand, GTINs bought in a block	One e-mail per customer; a periodic duplicate merge in the platform; one SKU and one GTIN per variant, never reused	Guest checkouts outnumber accounts and nobody merges them; a SKU corrected in place after the first sale
2 — Growing	The first store, wholesale account and marketplace each bring a new key: loyalty number, company plus contact, marketplace id	Decide the customer master (platform, ERP or CRM) and the product master (ERP or PIM); a product cross-reference before the first marketplace; consent on the master	A loyalty list nobody can join to online accounts; a marketplace listing created as a second product for want of a GTIN

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel	A dozen systems holding customers; four product copies; a mapping inside each connector; marketing asking for a CDP	Matching with thresholds, a review queue and a merge log; consent with evidence on the master, synchronized outward; a PIM as product master; a CDP, if needed, fed from the master	A wrong merge found by a customer; "how many customers" answered with four numbers; a deletion request that cannot find every copy (Whose Data Is It? Privacy, Consent and Retention)
4 — Group	Several brands or countries, each with its own masters; acquisitions bring more	An MDM function or platform across brands; survivorship per field; a merge log privacy can audit; one consent model with national differences as configuration	One brand's customers e-mailed under another's consent; two GTINs for one item in two catalogs

The judgement: the identifiers do not change with scale, and the cheapest decisions are the earliest: one SKU and one GTIN per variant from the first day, one e-mail per account, a named customer master before the first store. What changes is how many systems hold a copy, and so how much linking there is to own. Kestrel & Pine is a stage-3 merchant that skipped the stage-2 decision, which is why it has four lists and no answer.

In Practice

The public evidence for this chapter is documentation, not case studies: vendors describe their identity features, and no merchant publishes its duplicate rate.

- **Shopify (2026)** documents what a platform-as-master design gives a merchant: `customerMerge` with published survivorship rules and a preview, and a consent

mutation that records when and where consent was given. Both operate only on the platform's own records. Orders from Shopify's own POS land in the same customer and order model (Microsoft's connector documentation of October 2026 maps their staff id to a salesperson): the suite answer to the Familiar Stranger is one vendor for till and storefront. Kestrel & Pine's POS is from another vendor; the platform's own POS is evaluated in 2028 (W4).

- **Microsoft's Business Central connector (October 2026)** matches items on SKU or barcode and links rather than creates; its SKU field may hold the ERP item number, the item number plus a variant code, or the vendor's item number, depending on a setting. A merchant with two such connectors and two settings has two cross-references.
- **The CDP category (2013–2026)**: named by David Raab in 2013, defined by the CDP Institute (founded 2016), implemented with identity graphs by products such as Twilio Segment (June 2026).
- **The MDM vocabulary**: Informatica's and Profisee's pages (2025–2026) define the golden record and survivorship; DAMA's Data Management Body of Knowledge (revised 2024) is the formal reference.

Kestrel & Pine's identity decision (2027, illustrative)

In **June 2027**, at the weekly architecture hour, Noor Visser, Milan Petrov and Ruben Alves adopted a customer-identity rule, recorded as an addition to Waypoint rather than a new W-decision. The figures are illustrative.

- **The platform's customer record is the master for consumers**. The POS loyalty list, the newsletter tool and the returns portal keep their own records and carry the platform customer id once matched.
- **The ERP is the master for business customers and contacts**, as it already was for terms and credit (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals).
- **The cross-reference and merge log live in the data warehouse**, owned by Hanna Berg's data pair, because the warehouse already receives every list nightly and W7's one definition of a customer needs the same table.
- **Matching runs weekly**. Exact matches on normalized e-mail or phone link automatically; name plus address plus postal code goes to a review queue worked by the application support specialists; the platform's own merge is used only after review, with its preview.
- **Consent is read from the master**. The newsletter tool's subscription state and the POS's opt-in flag are copied to the platform's consent record with source and date; the strictest state wins where they disagree.

The first run, in **July 2027**, linked about **21,000** of the 38,000 loyalty entries to a platform customer by exact phone or e-mail, sent about **4,100** to the review queue and left the

rest unlinked. Of the 165,000 newsletter subscribers, about **96,000** matched a platform customer.

Eva Janssen's four records became one master with three links: her loyalty entry by phone, her returns contact by order number, and her older newsletter address by a reviewer who saw the same name, street and postal code.

The product side waited for the PIM: its July 2027 load from the four copies of product data found 1,900 variants with conflicting attributes (Moving Without Burning the Bridge: Replatforming and Migration), and the cross-reference of GTIN, SKU, platform, ERP, WMS and marketplace ids was built in the PIM at the same time (W3).

What the decision bought: one answer to "how many customers" (the master count, with unlinked store entries reported separately), a loyalty program that could start in 2028 on one list, a service screen that could show Eva's store purchases next to her online orders once the OMS arrived (W2), and an index of where every person's data lives, which Whose Data Is It? Privacy, Consent and Retention needs.

When It Goes Wrong

Failure	How it shows	Detection	Recovery
The wrong merge	A customer logs in and sees another person's orders or address	A complaint; a review of recent merges near the threshold	Unmerge from the log; notify the affected people; assess it as a personal-data breach (Whose Data Is It? Privacy, Consent and Retention); raise the threshold
Duplicates from guest checkout	One person, six platform records, six "first order" discounts (Discounts, Coupons and Loyalty: Promotions Without Chaos)	Duplicate rate on normalized e-mail, reported monthly	Scheduled exact-match merge; a "sign in to use your discount" rule

Failure	How it shows	Detection	Recovery
Consent lost in a merge	A person who opted out is e-mailed because the surviving record said "subscribed"	A consent reconciliation between the master and each marketing tool	Strictest state wins; re-sync outward; keep the evidence of the original opt-out
The overloaded SKU	A connector writes the ERP's item-plus-variant code into the platform's SKU field; the WMS expects the plain SKU; pick lists fail	A cross-reference reconciliation: one SKU and one GTIN per variant in every system	Fix the connector setting; rebuild the cross-reference from the master
The missing GTIN	A marketplace creates a second listing for the same item; stock and reviews split between them	A listing-to-GTIN reconciliation per marketplace	Merge the listings where the marketplace allows; never list without a GTIN
The connector that owned the mapping	A connector is replaced (Promises Between Systems: API Contracts, Webhooks and Versioning) and the cross-reference goes with it	Discovered at migration	Keep the cross-reference in the master's companion table, never only in a connector; export it before any connector change

The wrong merge is what makes matching dangerous; the safeguard is not a better algorithm but a reversible log and a person in the middle band. The overloaded SKU is the product-side Spaghetti Junction: the estate's identifier policy expressed as a drop-down in one connector's settings.

What to Take Away

Established.

- Record linkage is a mature field: exact rules for shared keys, probabilistic scoring with two thresholds and a review band (Fellegi and Sunter, 1969), normalization and blocking to make it feasible.
- Platforms and connectors implement survivorship and cross-reference policies their own way and sometimes publish them: Shopify's `customerMerge` rules (API 2026-10) and Business Central's item matching on SKU or barcode (October 2026).
- The GTIN is the only product identifier every external party understands; the SKU is the only one the merchant controls; both are assigned once and never reused (GS1; What Are We Selling? Products, Variants and the Catalog).
- Consent must be demonstrable (GDPR Article 7(1)), so it is a record with time, place and wording that travels with the identity through every merge.

Depends on context.

- The platform is the right customer master for a consumer merchant below the Group stage and the ERP for business customers; a merchant with both has two masters and one cross-reference, which is fine as long as both are named.
- A CDP earns its place when marketing needs audiences across anonymous and known touchpoints; it does not earn the role of master for service, orders and deletion, whatever the vendor's definition says.
- Probabilistic matching is worth its review queue from the Omnichannel stage; below it, exact matching on normalized e-mail and phone catches most duplicates at no risk.
- An MDM platform is a Group-stage purchase, triggered by a second brand or an acquisition, not by the first duplicate list.

The author's synthesis.

- Identity is the box the capability map leaves blank the longest, because every system works without it; make the decision before the first store, and if you did not, make it the way Kestrel & Pine did: nothing moves, the links are owned, every merge can be undone.
- Keep the exact record and the fuzzy profile apart and let the exact one feed the fuzzy one; the day they are the same table, a marketing match becomes an invoice error.

Questions to Ask

1. For each master entity (consumer, business customer, product, supplier, location): which system is its master, and is that written down?
2. Where is the cross-reference between the master's id and every other system's id kept, and is it anywhere other than inside a connector?
3. What are the matching rules, what is matched automatically, and who works the review queue?
4. Is every merge logged and reversible, and when was an unmerge last rehearsed?
5. Which value wins, per field, when two records merge, and does consent carry its evidence?
6. What is the duplicate rate on normalized e-mail, and who reports it?
7. Does every sellable variant have one SKU and one GTIN in every system, and which job checks?
8. Could you list, today, every system that holds a given person's data?

The last question is where this chapter hands over. Once a merchant can say who its customers are and where their records live, it can count them. Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics takes the master and the cross-reference built here and asks why three reports still show three different order counts, and how to build analytics the business trusts. The question of what the merchant may do with the records, and how it forgets them, waits for Whose Data Is It? Privacy, Consent and Retention.

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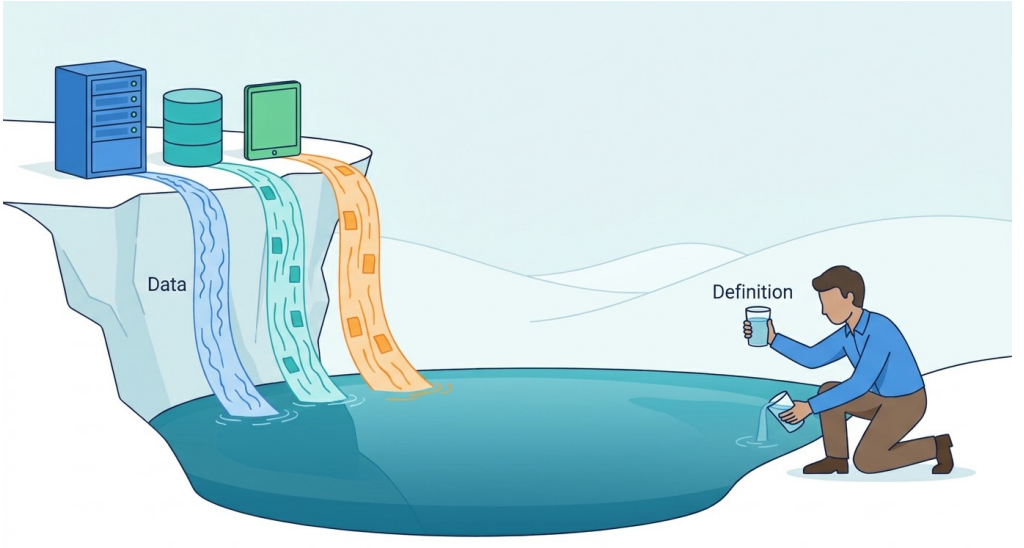
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

28. Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics



IN THIS CHAPTER, YOU WILL: Learn why honest systems count orders differently, how a data platform collects every system's data, and how one owned definition per metric makes a number trustworthy.

KEY POINTS:

- **Three order counts have three honest causes: definition, collection and time.** Each system counts what it exists to handle (orders placed, orders invoiced, parcels shipped), each loses or gains records in the copying, and each uses its own clock. None is wrong; none is the company's number until someone defines it.
- **Keep operational and analytical data apart, and copy everything into one place.** Operational stores answer "where is this order" for one order; analytical stores answer "how many orders by channel" over all of them. A warehouse or lakehouse fed by extract-and-load jobs and event streams is where the counts can be compared.

- **Define each metric once, with an owner, and publish the bridge**. A metrics layer holds "order", "net sales" and "return" as code so every dashboard computes them the same way. Waypoint decision W7 gives Kestrel & Pine's definitions to the Data Lead and puts the reconciliation between the platform's count and the warehouse's count next to the reported number.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Schrödinger's Order** — when the operational systems disagree about an order, every report built on them disagrees too, and the dashboard becomes a fourth opinion.
- **The Message in a Bottle** — an order edit or refund that one connector missed is a permanent difference between the warehouse and the platform until a reconciliation finds it.
- **The Familiar Stranger** — "customers" cannot be counted while four systems each count their own, so the customer metrics inherit the identity problem of the previous chapter.

Finance and operations can bring different order totals to the same meeting and both have counted correctly. Finance may be counting invoices while operations counts shipments; merchandising may be looking at products sold. Calling each figure 'orders' hides those differences. Leadership can then mistake a change in definition for a change in performance and make the wrong staffing or purchasing decision.

The source systems serve different jobs. The storefront records orders placed, the enterprise resource planning system (ERP) records invoices, and the warehouse records parcels. Their totals should sometimes differ. The analytical system must explain which differences follow from definitions and which reveal missing or incorrect data.

At Kestrel & Pine, the fictional outdoor brand, decision W7 of the Waypoint improvement program gives the Data Lead ownership of common order, revenue and return definitions. I would settle those definitions before adding dashboards: they let people distinguish a real business change from a reporting difference and investigate missing records against an agreed expectation.

Where we are in the order's journey. Eva's €235 purchase on 13 April 2027 becomes order #1072455, invoice lines and a parcel dispatched on 14 April. Her exchange adds a credit note and a second parcel on 22 April. The reporting question is what each event counts toward, rather than which system's total is inherently right.

Setting the Scene: Three Answers to One Question

At the March 2027 leadership meeting, CEO Mara Lindqvist asks how many orders arrived in February. Ecommerce Manager Milan Petrov reports **21,050**, Finance Director

Sanne de Wit **22,710**, and Operations Head Daniel Okafor **24,140**. Daniel's report comes from a **business intelligence (BI)** tool reading the **warehouse management system (WMS)**. These fictional figures take forty minutes to reconcile.

Hanna Berg, the Data Lead, had the explanation by Wednesday, and it is worth reading slowly because every merchant has a version of it.

What each count includes	**Platform: 21,050**	**ERP: 22,710**	**Warehouse report: 24,140**
What is counted	Orders created in the platform in February	Sales orders invoiced in February	Parcels and shipments leaving Duiven in February
Date used	Order creation, Amsterdam time	Invoice date	Ship date
D2C (direct-to-consumer: the brand's own storefronts) orders	18,900 created, including 290 later canceled and 40 test or draft orders	18,420 invoiced (canceled and test orders excluded; 150 fewer from month-end timing)	19,100 parcels, including 640 second parcels of orders split across two boxes
Marketplace orders	Not here; they live in the marketplace connector	3,310, from the connector's weekly summary	2,520 shipped from Duiven; the 790 fulfilled by the marketplace's program never pass through the warehouse
Wholesale orders	610 placed through the B2B portal	980: the 610 portal orders plus 370 keyed in by the wholesale team or received by EDI	980 shipments
Exchange orders from the returns portal	1,540, created through the platform's API	None; an exchange is a credit note and a replacement line	1,540 exchange parcels

What each count includes	Platform: 21,050	ERP: 22,710	Warehouse report: 24,140
Store transactions	Not here	A nightly sales total, not orders	Not here

Three differences explain the totals. **Definition**: do canceled orders, exchanges, parcels and store receipts count? **Coverage**: marketplaces reach the ERP as weekly summaries and do not reach the platform. **Timing**: an order placed late on 28 February becomes a March invoice and shipment. A useful report makes those choices visible.

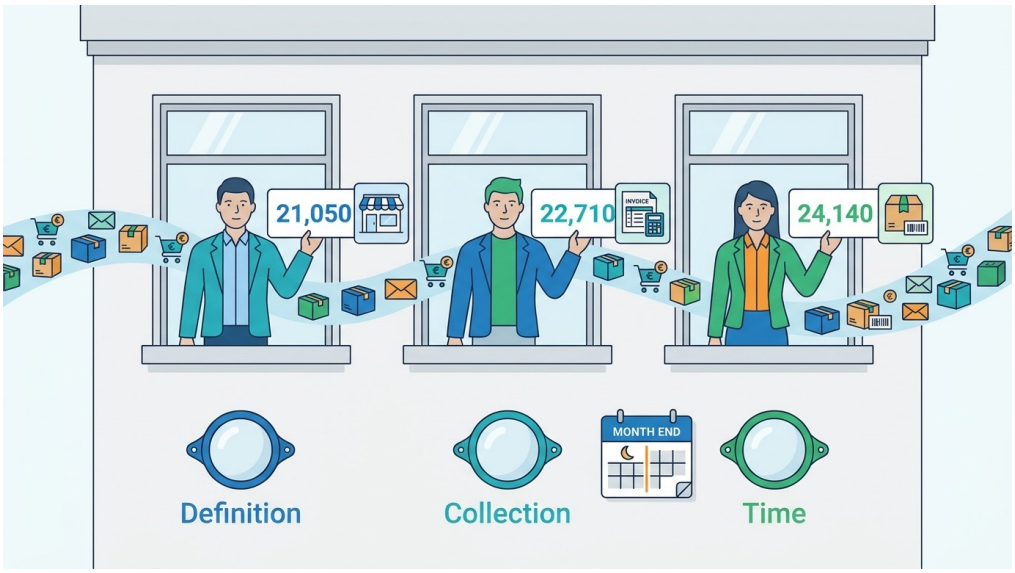


Figure 1: *The platform's 21,050, the ERP's 22,710 and the warehouse report's 24,140 are honest counts that differ by definition, collection and time; none is the company's number until someone owns the definition.*

Mara's reaction was the right one. She did not ask which number was correct. She asked who owned the definition, and the answer, in March 2027, was nobody.

Operational Data and Analytical Data

The first idea a newcomer needs is that the same facts are used in two different ways, and that the two ways want different stores.

Operational data is the live state of the business: this order's status, this SKU's stock at Duiven, this customer's address. Operational systems (the platform, the OMS, the WMS, the ERP, the POS) read and write it one record at a time, thousands of times a

minute, and must answer fast. **Analytical data** is the history of the business laid out for questions across all records: orders by channel by week, return rate by product by size, stock cover by category. Analytical questions read millions of rows at once and can tolerate hours of lag.

The two must be kept apart for a practical reason. A report that scans every order of the year, run against the platform's own database on a Friday evening in November, competes with the checkout for the same machine. The rule is simple: **never run an analytical question against an operational store, and never trust an analytical store for an operational answer.** "The dashboard says it shipped" is not a shipment (Black Friday Every Year: Peak Load, Resilience and Recovery has the Black Friday version).

The analytical store has had two names. A **data warehouse** is a database built for large analytical queries, into which copies of every system's data are loaded and modeled into tables analysts can use. Snowflake describes its architecture as separating storage from compute so that one team's queries do not slow another's; Google's BigQuery calls itself "a fully managed serverless data warehouse" with the same separation (both from current documentation, accessed November 2026).

A **lakehouse** keeps the raw copies in open file formats in cloud storage and runs warehouse-style queries over them; the term comes from a 2021 paper by Databricks' founders (Armbrust, Ghodsi, Xin and Zaharia, CIDR 2021) that argued the two-tier design of a data lake feeding a separate warehouse suffered from staleness, duplication and cost.

For a merchant of Kestrel & Pine's size the difference is a vendor choice, not an architecture choice; what matters is that one analytical store exists and that every system lands in it.

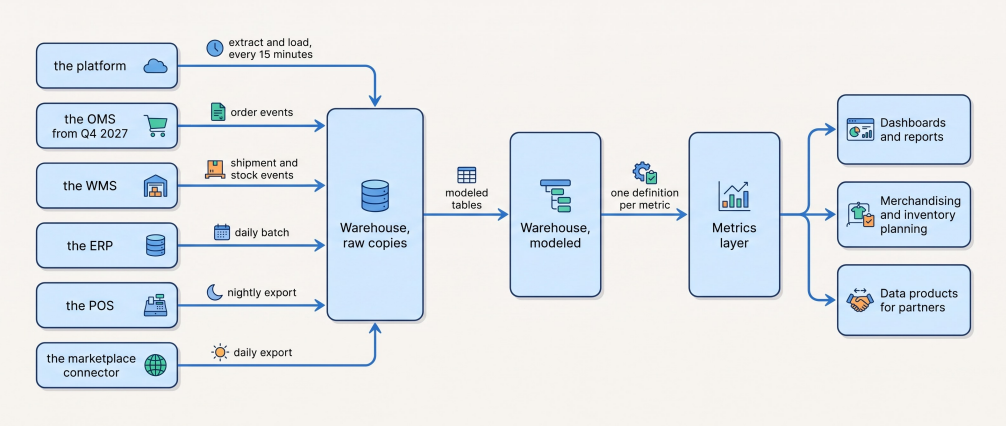


Figure 2: *Every operational system is copied into one analytical store, modeled once, and served through a metrics layer, so that every dashboard, plan and partner feed computes "order" the same way.*

Getting the Data Out

Three mechanisms carry data from the operational systems into the analytical store, and a merchant usually uses all three.

Extract-and-load (ELT: extract, load, then transform inside the warehouse) copies a system's tables on a schedule through a connector. Hosted tools sell this by volume: Fivetran prices by "monthly active rows", the rows inserted, updated or deleted in a month, with a free tier of 500,000 and 15-minute syncs on its Standard plan (pricing page, accessed November 2026; prices change).

The connectors do what the previous chapters taught: Airbyte's Shopify source (version 4.3.5, changelog entry of 3 November 2026) reads orders, refunds, products, variants, customers, transactions, fulfillments and inventory levels, syncs incrementally using each record's update time as a cursor, offers a "lookback window" to catch late updates, captures deleted products in a separate stream, and backs off on rate-limit errors.

Read that list as a reconciliation sweep (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation) applied to analytics: an order edited after the connector's last run is picked up by the lookback window or not at all, and a missed edit becomes a permanent difference between the warehouse and the platform.

Event streams carry each change as it happens. A system publishes an event (order placed, stock changed), a durable log keeps it, and the warehouse reads the log at its own pace alongside the OMS and the WMS. The best-known public account is from a Walmart engineer on Confluent's blog (May 2020; six years old), describing more than ten event sources normalized into one inventory event feed, with each item and store pair always handled by one consumer so that no two updates to the same stock figure race. APIs, Events, Files and Middleware: How Commerce Systems Talk owns the style; for analytics the point is that an event stream gives the warehouse the same facts the operational systems act on, at the same moment, with no second extraction.

Platform exports exist to spare the API. Shopify's bulk operations run a query asynchronously and deliver a file; Stripe recommends its Data Pipeline "for a complete export of API data" rather than read-heavy integrations (both from documentation accessed November 2026). A merchant that pulls a year of orders through the same API its checkout uses, one page at a time, is spending the rate budget Promises Between Systems: API Contracts, Webhooks and Versioning described on a report.

The requirements differ by what the data is for; the targets below are illustrative.

Data flow	How fresh	How complete	Who may disagree with it	Typical mechanism
Orders and shipments into the warehouse	Minutes to an hour	Every order, including edits and cancella-tions after the fact	The platform's own count, within a published tolerance	Events from the OMS; ELT with a lookback window
Stock snapshots for planning	Hourly, or a daily snapshot	Every SKU and location, including the marketplace program's stock	The WMS, which is the system of record	Stock events; a nightly snapshot
Finance figures	Daily	To the cent; reconciled with the ERP and the payouts of the PSP (payment service provider)	Nothing; the ERP wins	Daily batch from the ERP; PSP settlement reports
Operational dashboards (orders per minute, checkout errors, queue depth)	Seconds	Approximate is fine	Nobody; they are rates for action, not records	Direct from the operational systems' monitoring
Board and manage-ment reports	Monthly, revised	Defined and reconciled	The ERP for revenue	The metrics layer

The last two rows are the ones that get confused. An orders-per-minute dashboard and a monthly order count are different objects: one is a rate used to act now, the other a record that will be revised and reconciled. Showing both on one screen with one label is how a dashboard becomes the fourth opinion.

One Definition: Metrics and the Metrics Layer

A **metric definition** is the written rule for computing a business number: what is counted, which records are included or excluded, which date decides the period, which currency and tax treatment apply, and what happens when the underlying record changes later. The public definitions of large companies show how many decisions hide in a word like "order".

Source	What it defines	What it says, in short
Shopify's ShopifyQL sales schema, API version 2026-10	`orders`, `gross_sales`, `net_sales`, `total_sales`	An order is counted once "even when the order contains multiple items"; gross sales exclude "taxes, shipping, duties, and fees"; net sales are gross sales minus discounts minus "sales reversals" (refunds, returns, cancellations, edits); "a single order can produce multiple rows if refunds or adjustments follow"
Shopify's FY2025 annual report (February 2026)	GMV (gross merchandise volume)	"the total dollar value of orders facilitated through our platform … net of refunds"
Zalando's 2025 annual report (March 2026)	GMV, average basket	GMV after cancellations and returns, including VAT (value-added tax); the average basket of €62.8 is GMV per order on that basis

Source	What it defines	What it says, in short
Google Analytics 4 e-commerce reference (accessed November 2026)	the `purchase` event	Requires a `transaction_id` to "avoid getting duplicate events for a purchase"; `value` is the sum of price times quantity, excluding shipping and tax; a `refund` is a separate event

Four sources, four units: an order net of reversals as later rows, an order value net of refunds, an order value after cancellations and returns including tax, and a browser event with its own value rule. Each is internally consistent; none is interchangeable with another. This is why analytics vendors who reconcile a platform's sales against web analytics publish lists of the usual gaps (orders created by apps or subscriptions that never fire a browser event, point-of-sale and draft orders, ad blockers and refused consent, refunds netted by one side and not the other, currency and tax inclusion, attribution windows); those lists are widely reported in vendor help pages rather than in a neutral source, and the direction of each gap depends on the merchant.

The architectural answer to definitional drift is the **metrics layer**: one place where each metric is defined as code, from which every dashboard, report and tool computes it. Benn Stancil's 2021 essay named the gap: "there's no central repository for defining a metric", so metric logic lives in individual dashboards, is re-derived by each analyst, and drifts.

The products that followed implement the idea in different ways: dbt's Semantic Layer lets a team define metrics on top of its models so that "If a metric definition changes in dbt, it's refreshed everywhere it's invoked" (documentation updated September 2026); LookML, Looker's modeling language, describes "dimensions, aggregates, calculations, and data relationships" once; Cube's semantic layer adds access control, caching and APIs so that applications and AI tools query governed metrics (all accessed November 2026; product scope changes often).

The tool matters less than two properties: **the definition lives outside any one dashboard, and it has an owner**.

What a definition should contain, in plain words, is a table like the one Hanna wrote for W7 (the In Practice section below). The hard rows are always the same: what to do with cancellations, returns, exchanges, test orders and split shipments; which date; which currency and tax basis; and how a late change (an order edited in March) is reported against February.

What the Data Is For

The same warehouse serves four audiences with different tolerances, and a design should name them.

Reporting is the monthly and weekly record: orders, net sales, margin, return rate, by channel, country and category. It must be defined, reconciled and revised with a stated policy.

Experimentation is the comparison of two versions of something (a checkout step, a delivery promise, a price) on randomly split traffic. It needs event-level data with a stable customer or session key, and it needs the same metric definitions as reporting, or the winning variant will be judged by a number nobody else recognizes. Finding the Product: Search, Navigation and Merchandising and Delivering the Storefront: Themes, Headless, Rendering and Performance use it; this chapter only insists on the shared definitions.

Data science builds forecasts and models: demand per SKU and week, return probability per product, customer value. It reads the modeled tables and writes back predictions; it is the main reason a merchant keeps raw history rather than only aggregates.

Merchandising and inventory planning is the most commercially important use at Kestrel & Pine's stage. Two numbers drive a season. **Sell-through** is the share of the units bought for a season that have been sold at a point in time (units sold divided by units received); it tells Teun Baas whether the moss Ridgewalker is a hit or a markdown waiting to happen. **Weeks of cover** is the stock on hand divided by the average weekly sales rate; it tells Daniel Okafor and the buyers when to reorder and when to stop. Both depend on one definition of a sale and one view of stock across Duiven, the six stores and the marketplace program's location, which is why Is It in Stock? Inventory, Availability and the Promise You Can Keep and W1 are prerequisites for trustworthy planning, not just for selling. What Are We Selling? Products, Variants and the Catalog supplies the attributes (category, season, size) the numbers are grouped by.

Data products for partners appear at the Omnichannel stage: sell-through of a third-party brand's products to that brand, listing performance to a marketplace team, a stock feed to a wholesale customer such as Bergzicht. A **data product** is a dataset with an owner, a definition, a contract and a consumer, and the warehouses now make delivery cheap: Snowflake's Secure Data Sharing lets a consumer query a provider's data in place with "no actual data … copied or transferred between accounts", and Databricks documents an open sharing protocol that works "regardless of the computing platforms" the other organization uses (both accessed November 2026; Databricks' product naming for this was changing at the time of writing). Cheap delivery moves the cost to definitions and consent: a sell-through feed to a brand is also a disclosure of the merchant's sales, and a customer-level feed is personal data (Whose Data Is It? Privacy, Consent and Retention).

Ownership follows from the audiences. Zhamak Dehghani's 2019 data mesh article argued that domains should "host and serve their own datasets" and treat them as products, with a shared self-serve platform and global standards so that cross-domain data fits together. The commerce reading is modest: the order domain (the OMS team after W2) owns the order dataset and its definitions; the inventory domain owns stock snapshots; finance owns revenue recognition; the data team owns the platform, the keys, the time zones and the currency rules that make the domains' data joinable. A central data team that owns every definition becomes a bottleneck; domain teams with no shared standards produce four incompatible "orders" again, inside the warehouse this time.

Real-Time Dashboards and When They Mislead

A real-time dashboard shows a number that will be revised. Google Analytics' export documentation says its intraday data is best-effort and may have gaps and that some attribution lags "by up to 24 hours"; Stripe's reconciliation documentation says a payout "might arrive before its reconciliation data is ready"; Shopify's sales schema records a refund as a later row against an earlier order. None of these is a defect. They are what it means to count while the facts are still changing.

The rule I use separates two kinds of number by what they are for. **Operational signals** (orders per minute, checkout error rate, queue depth, the integration lag from order placed to order in the WMS) are rates used to act in the next ten minutes; show them live, approximately, from the operational systems' monitoring (Black Friday Every Year: Peak Load, Resilience and Recovery). **Business metrics** (revenue, conversion, return rate, margin) are records; show them on a lag, with a definition, a revision policy and the reconciliation beside them. A screen that mixes the two with one label invites the question "why does the live dashboard say 2,300 orders and the report say 2,210", to which the answer is always "because 90 of them were canceled, refunded, edited or duplicated after the dashboard counted them".

Figure 3: *Operational signals are live rates for acting in the next ten minutes; business metrics are records shown on a lag with a definition and a revision policy, which is why a live 2,300 orders becomes a reported 2,210.*

The Choices and Their Trade-offs

Four realistic designs exist for a merchant's analytics, and the right one is mostly a function of stage.

The platform's reports plus web analytics. The commerce platform's own sales reports and a web analytics tool such as GA4. No warehouse, no engineering. Two definitions of "order" from day one, and no way to count marketplaces, wholesale or stores alongside.

A BI tool on operational copies. A business intelligence tool pointed at exports or read replicas of the ERP, the WMS and the platform. Cheap and quick; much of Kestrel & Pine's reporting in early 2027 still worked this way, beside a warehouse with no owned definitions. Every report embeds its own definition, and a report that is wrong is wrong invisibly.

A warehouse with extract-and-load, a modeling tool and a metrics layer. Every system copied into one store on a schedule, modeled into shared tables, with metrics defined once and reconciled against the source systems. The standard Omnichannel design; it needs one or two data engineers and a named owner per domain.

An event stream feeding a lakehouse and the operational systems. Order and stock events carried once, to the OMS, the WMS, the ERP connector and the warehouse alike,

with the lakehouse keeping full history for data science and partner products. The Group-stage design, and Walmart's 2020 account is its large-scale version.

Design	Cost and skills	Definitions	Freshness	Right when
Platform reports plus web analytics	None beyond subscriptions	Two, both the vendors'	Live, unreconciled	One channel; Starting stage
BI on operational copies	A BI tool; an analyst	One per report; drift is the norm	Hours; risks loading operational systems	A stopgap; never for a multi-channel merchant
Warehouse, ELT, metrics layer	ELT fees by volume; one or two data engineers; domain owners	One per metric, owned, reconciled	15 minutes to a day by flow	Two or more channels; Omnichannel stage; Kestrel & Pine from 2027
Event stream and lakehouse	Streaming platform; a data platform team	As above, plus schemas on events	Seconds for events; history for models	Several brands or units; data science and partner products at scale

My recommendation for a merchant at Kestrel & Pine's stage is the third design, with one addition the vendors do not sell: a **reconciliation published with the number**. Each monthly report shows the platform's count, the OMS's count and the warehouse's count and the bridge between them. A number with its bridge is believed; a number alone is argued with.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	The platform's reports and web analytics; the founder knows the numbers by heart	Knowing that the two tools define "order" differently and which one the accountant uses	Marketing's conversion rate and the platform's order count stop agreeing and nobody can say why
2 — Growing	A BI tool on exports; spreadsheets for the season; a first ERP report	A warehouse fed by an ELT connector, three or four agreed metric definitions, and a monthly check of warehouse orders against platform orders	Three order counts in the leadership meeting; a buying decision made on a report nobody can reproduce
3 — Omnichannel (Kestrel & Pine)	Many systems, several channels, a warehouse that exists but has no owned definitions	An event stream or OMS feed into the warehouse, a metrics layer, domain ownership of definitions, data products for key partners, operational dashboards separated from business metrics	Reports revised after the board saw them; partners asking for feeds built by hand; the warehouse used as a system of record for stock

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
4 — Group	Several brands or units, each with its own warehouse or definitions	A shared data platform with federated ownership, warehouse-native sharing with partners, a governance function that also serves privacy	Two brands reporting "net sales" on different bases to one board; group-level forecasts that cannot be reconciled to any unit's books

The subject changes shape with scale more than most. At Starting there is nothing to build and one thing to know. At Growing, the warehouse is worth building before the second channel, not after, because the second channel is what creates the second count. At Omnichannel the problem is ownership, not technology: Kestrel & Pine had a warehouse for three years before it had a definition. At Group the problem becomes federation across units that each solved it their own way.

In Practice: Waypoint W7 at Kestrel & Pine

Under W7, Hanna Berg published Kestrel & Pine's metric definitions in June 2027 and the warehouse's monthly report adopted them from the July close. The definitions and the bridge below are fictional and illustrative.

Metric	Definition	Decisions it settles
Order	A confirmed customer purchase with one order number, in any sales channel: D2C storefronts, the B2B portal, manual and EDI wholesale, and marketplaces including orders fulfilled by the marketplace's program. Counted at the moment of confirmation in Amsterdam time. Test and draft orders are excluded. Canceled orders are counted as orders and reported with a cancellation status; "orders net of cancellations" is a second, named metric. An exchange order created by the returns portal is a return outcome, not an order. An order split into two parcels is one order.	Exchanges, cancellations, test orders, split shipments, the marketplace program, the period
Store transaction	A POS receipt. Reported beside orders, never added to them, until the POS joins the OMS (evaluated in 2028 under W4).	Whether a €51 receipt is an order

Metric	Definition	Decisions it settles
Net sales	Merchandise value excluding VAT, after discounts, minus the value of returns and cancellations in the month they are refunded; shipping charges and payment fees excluded. Recognized when control passes to the customer under IFRS 15 (the international accounting standard for revenue): at delivery for D2C and marketplace orders, and at delivery to the customer's distribution center for wholesale; "shipped" is the operational milestone, not the accounting one. This is the ERP's basis, so the warehouse's net sales reconcile to the ERP to the cent each month.	Tax, shipping, returns timing, recognition
Return	A returned item received and accepted at Duiven. Item return rate is returned items over shipped items (24% in 2026); order return rate is orders with at least one return over orders (19%). Exchanges are returns with an exchange outcome (28% of returns).	What a return is, and which rate is meant

Metric	Definition	Decisions it settles
Customer	Deferred to the identity work of One Customer, One Product: Identity and Master Data Across Systems; until then, customer counts are reported per channel and labeled as such.	Whether Eva is one customer

With the definitions, the February numbers became one number and a bridge. Hanna's report for February 2027, recomputed in July, reads: **orders 23,150** (18,860 D2C after excluding the 40 test orders and keeping the 290 canceled, 3,310 marketplace, 980 wholesale), **22,860 net of cancellations**, and 9,200 store transactions reported beside them.

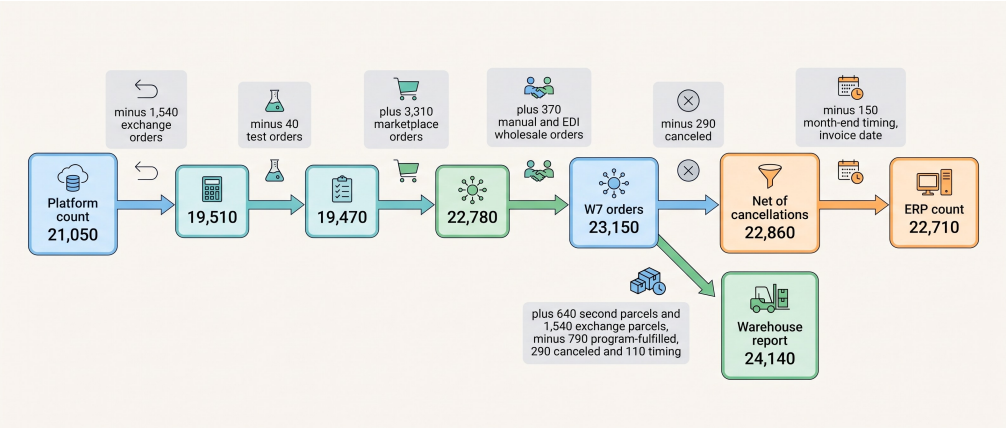


Figure 4: *The bridge from the platform's February count to the W7 definition, and from there to the ERP's and the warehouse's counts: each count is right about what it counts, and the bridge is what makes the company's number believable.*

The bridge, not the number, is what changed the meeting. From the July close, the monthly report shows the W7 order count, the platform's own count, the ERP's invoiced count and the warehouse's shipment count on one page, with the bridge between them; a difference that the bridge does not explain is an exception for Hanna's team, which is how a missed order edit or a connector gap is found. The first such exception, in August 2027, was 23 marketplace orders the connector's weekly summary had dropped during a vendor outage *(illustrative)*.

Two things made W7 cheap. The definitions took Hanna three weeks of conversations, not three months of engineering; the warehouse and the BI tool already existed. And

the OMS, from Q4 2027, makes the order count's collection problem smaller, because every channel's order is created in one place with one number family (W2); W7 is still needed, because one record does not settle whether an exchange is an order.

When It Goes Wrong

Failure	How you notice	What to do
A late edit or cancellation is missed by the extract (the Message in a Bottle, in the warehouse)	The bridge shows an unexplained difference; a refund in the PSP has no match	A lookback window on incremental loads; a monthly full compare of order ids and amounts against the source
Two clocks (store time versus UTC; order date versus invoice or settlement date)	Month-end orders move between months; the ERP and the warehouse disagree by a day's volume	One period rule per metric, written in the definition; store the source timestamp and the time zone
Refunds netted on one side only	Net sales in the warehouse exceed the ERP's by the month's refunds	Define the refund month in the metric; reconcile net sales to the ERP every close
A report built straight on an operational database	The platform or ERP slows during reporting; worst on peak days	Route every analytical query to the warehouse; revoke reporting access to operational stores
Browser analytics as the order count	Marketing's count is lower by the ad-blocked and consent-refused share, and misses app-created orders	Treat web analytics as a behavior source, never as the order record; reconcile it against the platform monthly
A metric redefined in one dashboard	Two dashboards with the same title show different figures	Definitions only in the metrics layer; dashboards consume, never compute
The warehouse used as a system of record	Someone adjusts stock or an order "because the dashboard said"	The warehouse is a copy; corrections go to the owning system and flow back

Failure	How you notice	What to do
A partner feed without a contract	A brand or wholesale customer disputes its sell-through figure	A data product with an owner, a definition, a schedule and a version, like any API (Promises Between Systems: API Contracts, Webhooks and Versioning)

What to Take Away

Established.

- Operational and analytical workloads belong in different stores: the warehouse and lakehouse vendors' own architecture descriptions (Snowflake, BigQuery; the 2021 lakehouse paper) rest on separating the two, and a report run against a checkout database is a known peak-day failure.
- Public metric definitions differ and are explicit about it: Shopify's sales schema (2026-10), Shopify's and Zalando's annual-report GMV definitions and GA4's purchase event each count an order on a different basis, which is why a merchant's own definition must be written down.
- Incremental extraction from a SaaS platform is a reconciliation sweep: connectors use update timestamps and lookback windows, and a missed late edit is a permanent discrepancy until a compare finds it (Airbyte's Shopify source documentation, 2026).

Depends on context.

- Whether to feed the warehouse by scheduled extract-and-load or by an event stream depends on whether the operational systems already emit events: once an OMS and an availability service publish order and stock events (W1, W2), the warehouse should read the same stream; before that, ELT with a lookback window is enough.
- Whether definitions are owned centrally or by domains depends on team size: under about five data people, one owner (the Data Lead) with domain reviewers; beyond that, domain ownership with central standards for keys, time and currency.
- A metrics layer product is worth its cost when more than one tool consumes the metrics; a merchant with one BI tool can keep the definitions as versioned models in the modeling layer and add the product later.

The author's synthesis.

- The number that gets believed is the number published with its bridge: show the platform's, the ERP's and the warehouse's counts beside the defined one every month, and treat an unexplained difference as an incident, not a footnote.
- Define before you build. Kestrel & Pine's warehouse was three years old when W7 was decided; the definitions took three weeks and did more for trust than the warehouse had.

Questions to Ask

1. If I ask three people how many orders we took last month, do I get one number, and if not, who owns the definition?
2. Where is each metric defined: in a metrics layer with an owner, or inside the dashboards that show it?
3. What happens to February's numbers when an order is canceled, refunded or edited in March, and does the report say so?
4. Which systems are not in the warehouse at all (marketplaces, stores, the returns portal), and whose count is missing because of it?
5. Does any report run against an operational database, and what happens to it at 20:00 on Black Friday?
6. How does the warehouse learn about orders edited after the extract ran, and when was the last full compare against the platform?
7. Which numbers on the live dashboard are rates for action, and which are records that will be revised, and are they labeled?
8. Which partners receive data from us, under what definition, on what schedule, and who signed off on what it discloses?

Part VI has been about making separate systems behave like one: how they talk, how they recover, what they promise, who they mean, and what they count. The next part turns from the joins to the platforms themselves, and asks how commerce platforms are built, why their vendors made the choices they made, and what a merchant should copy and what it should not. Continue with Part VII — PLATFORM: How Commerce Platforms Are Built.

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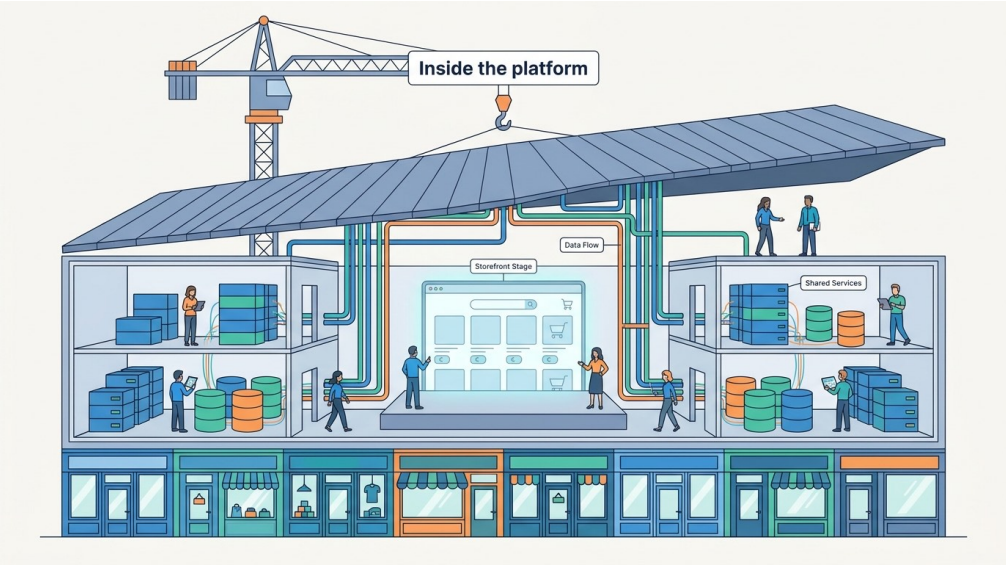
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part VII — PLATFORM: How Commerce Platforms Are Built



KEY POINTS:

- **The platform's internals are the vendor's; the code around it is yours.** The giants' architectures describe thousands of engineers, and most merchants need one deployment unit.
- **Renting a shared system means governing your share of it.** Know which API limits each app uses, give it an owner and restrict which facts it can change.
- **The storefront renders facts it does not own, and the peak shows it.** Cache the page, fetch price and stock live, and measure the whole chain in orders per minute.

A hosted commerce platform can keep accepting orders while the merchant's warehouse connector falls behind. Renting the core application transfers some operating work to the vendor, but extensions, storefront customizations and integrations can still interrupt sales or fulfillment. This part separates those responsibilities so the merchant can invest in the component actually limiting the order flow.

This part follows the machinery behind a jacket purchase: application structure, shared infrastructure, extensions, page delivery and peak capacity. It distinguishes vendor decisions from the operating choices a merchant's team can make.

Kestrel & Pine, **the fictional outdoor brand**, faces two rebuild proposals in early 2027: about €3.2 million for separate commerce services and €240,000 for a separate storefront. Yet the platform handled 2,300 orders in the Black Friday peak hour. Order export managed about seven per minute, and stock updates lagged up to two hours.

Head of Engineering Noor Visser directs attention to those measured failures. The five chapters explain how to evaluate the alternatives without taking on more systems than the team can operate.

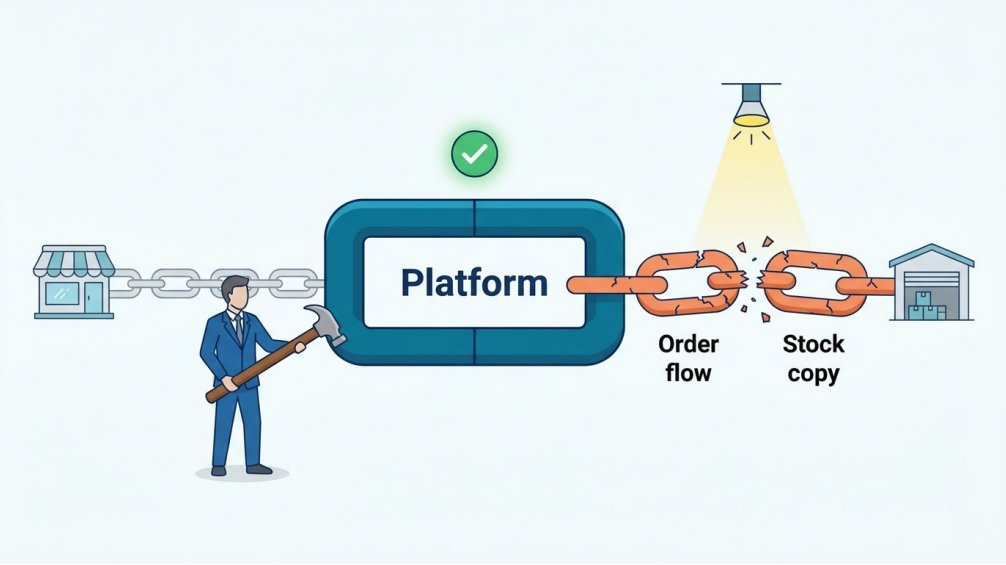


Figure 1: *On 27 November 2026 the rented platform held at 2,300 orders in the peak hour; the merchant's own joins, an order flow of about seven orders a minute and stock up to two hours old, were the weak links the rebuild proposals left for phase two.*

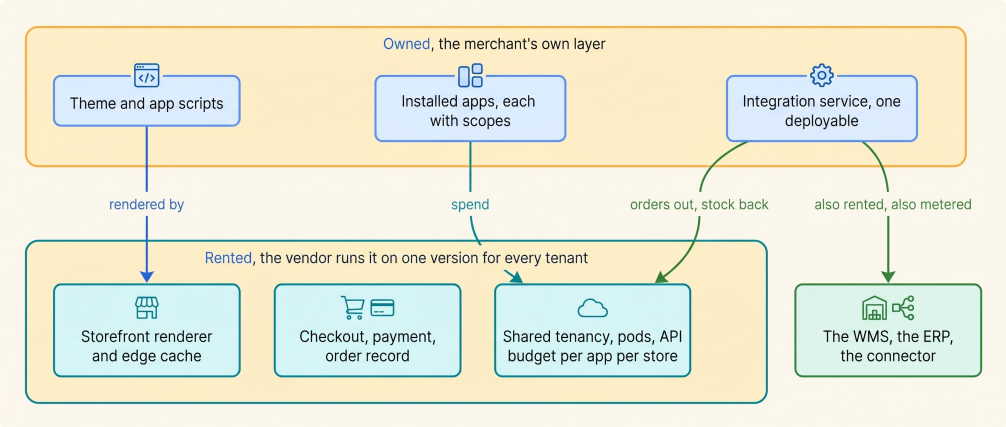


Figure 2: *A merchant on a hosted platform rents the renderer, the checkout and a share of a multi-tenant system, and owns the theme, the apps and the integration code around them; the owned layer and the budget it spends are where the peak day is won or lost.*

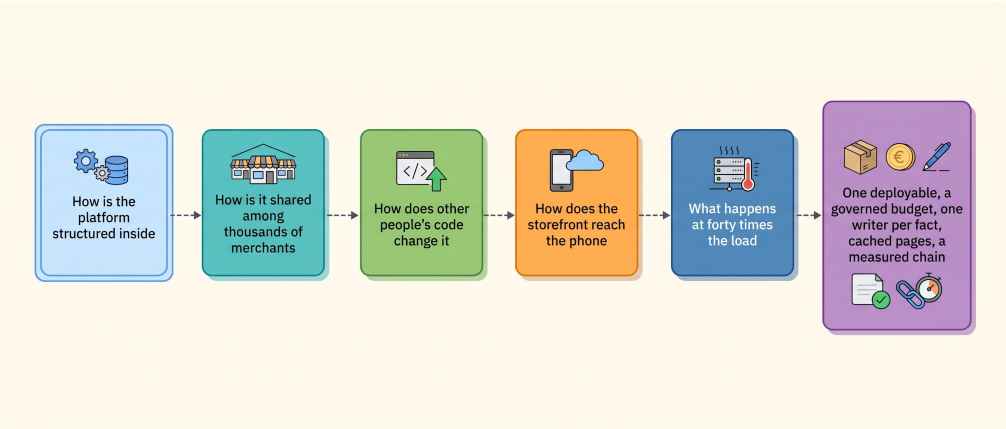


Figure 3: *The five questions of Part VII in reading order, from the inside of the platform outward to the day it is tested; each answer narrows what a merchant should treat as its own.*

Chapters in this part

Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says compares **how software is divided into separately deployed programs**. Dated company accounts show why release independence can justify a split, and why each additional service also adds operating work.

Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation explains **how vendors share infrastructure while separating customers**. API call budgets can

constrain order export even when the storefront is healthy. Kestrel & Pine's peak logs show why both ends of an integration need capacity checks.

Apps, Plugins and Functions: Extensibility and Ecosystem Governance establishes **an owner and permission boundary for each extension**. An inventory of 41 apps finds six unused and three writing stock. The chapter follows the permissions and maintenance decisions that reduce the conflicts.

Delivering the Storefront: Themes, Headless, Rendering and Performance compares **platform-rendered pages with a separate, headless application**. Measurements trace slow interaction to eleven app scripts. Removing that cause improves the existing theme and weakens the case for a rebuild.

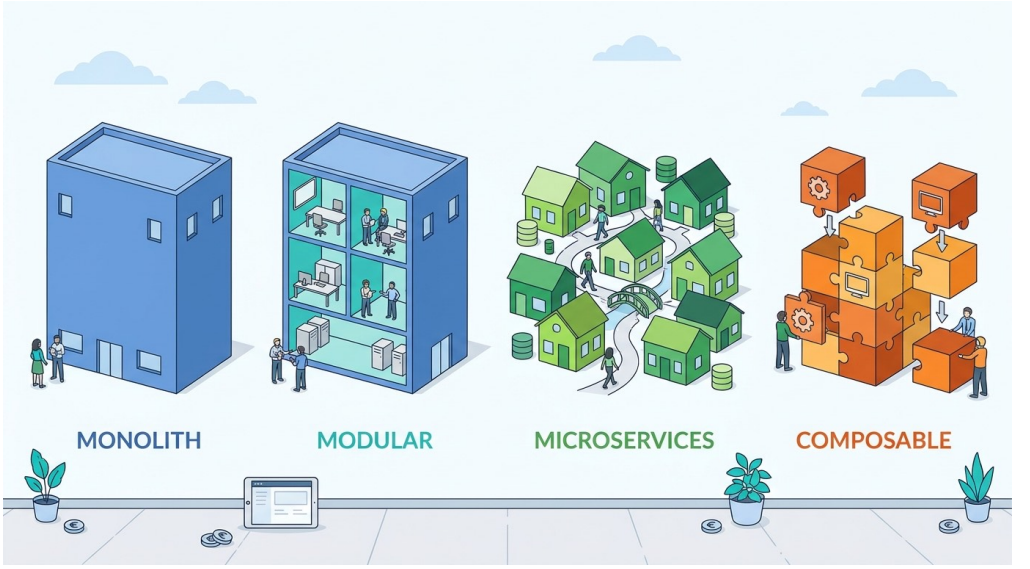
Black Friday Every Year: Peak Load, Resilience and Recovery prepares **the entire order path for concentrated demand**. The 2027 plan forecasts 60 orders per minute and tests every link at 80. It includes monitored queues, optional work to pause, named responders and reconciliation after the peak.

Begin with Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says.

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine is fictional; recommendations and labeled assumptions are the author's judgement.

29. Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says



IN THIS CHAPTER, YOU WILL: Learn the four ways a commerce system can be structured inside, what Amazon, Zalando and Shopify documented, and how to size modularity to your team.

KEY POINTS:

- **A style is a choice about deployment units and boundaries, not about quality.** Monolith, modular monolith, microservices and composable differ in how many separately deployed pieces exist, who owns which data, and how the boundaries between them are enforced. The same commerce logic can live in any of them; what changes is where the cost lands.
- **The famous examples document reasons and costs, not a recipe.** Amazon split at a scaling wall in 2001; Zalando split for team autonomy in 2015 and spent a decade re-standardizing; Shopify refused to split

and spent seven years learning that boundaries drawn on an org chart do not match how code runs. Every account is dated and describes a company with thousands of engineers.

- **For a merchant, the style is mostly the vendor's; what is yours is the integration layer**. Size modularity to the number of teams that must release without blocking each other, decide who owns each fact before splitting anything, and treat a microservices or all-custom composable rebuild by nine engineers as The Giant's Boots.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Giant's Boots** — a microservices or all-custom composable rebuild copied from a company with thousands of engineers, without its reasons.
- **The Spaghetti Junction** — composable moves the distribution costs the big companies paid into the merchant's own integration layer, where every join needs an owner.
- **The Burned Bridge** — a rebuild that replaces the platform on one date, when every documented migration took years and was done piece by piece.

A commerce team wants to release checkout changes without waiting for a catalog release. Separating the programs may help, but each new connection must carry data and handle failures. The benefit depends on whether independent releases are worth that extra work for this team. A large company's successful redesign cannot answer that question on the merchant's behalf.

Read a company's account for its date, scale, problem and limits. Then identify what you control. On a hosted platform, the vendor decides most internal architecture; your team owns surrounding connectors, custom apps and data pipelines. A separation can be useful when those programs need different release schedules or operating capacity. Without that need, it adds coordination without removing a practical obstacle.

An **architecture style** describes how a system's running programs divide work and data. A **deployment unit** is a program built, tested and released as one piece. Splitting it creates another program to operate and another boundary across which data and failures must be handled.

Where we are in the order's journey. Eva's jacket order at Kestrel & Pine, the fictional outdoor brand, crosses the vendor's platform code and the merchant's integrations. Every network crossing can add delay or fail independently. This chapter examines which separations earn that cost.

“Are We Behind?” at Kestrel & Pine

Kestrel & Pine reviewed two February 2027 proposals: a consultancy’s **composable rebuild**, assembling separate commerce services for about **€3.2 million over eighteen months**, and the agency’s separate storefront on the existing platform. Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026 and Delivering the Storefront: Themes, Headless, Rendering and Performance describe them.

The review also considered an enterprise suite and **stay and surround**: keep the platform and add missing capabilities. It chose the latter as decision W4 of **Waypoint**, the improvement program launched in March with **€1.1 million for 2027–2028** (Build, Buy or Assemble: Choosing and Changing Platforms).

In April 2027, a board member forwarded CEO Mara Lindqvist an article about a fashion retailer with roughly three thousand technology staff that had abandoned its monolith. Mara asked Head of Engineering Noor Visser whether Kestrel & Pine had chosen correctly. The question was reasonable; the team-size difference mattered.

Noor had fourteen people in technology, nine of them engineers. On Black Friday 2026 the platform itself had held at 2,300 orders in the peak hour; what failed was the chain of connectors between the platform, the ERP (enterprise resource planning system: finance, purchasing, invoicing), the WMS (warehouse management system) and the marketplace connector, which let the Ridgewalker jacket sell for two hours after the warehouse ran out. The rebuild option would have rebuilt the part that worked while leaving the joins that failed to “phase two”. Noor’s memo to the board is the rest of this chapter.

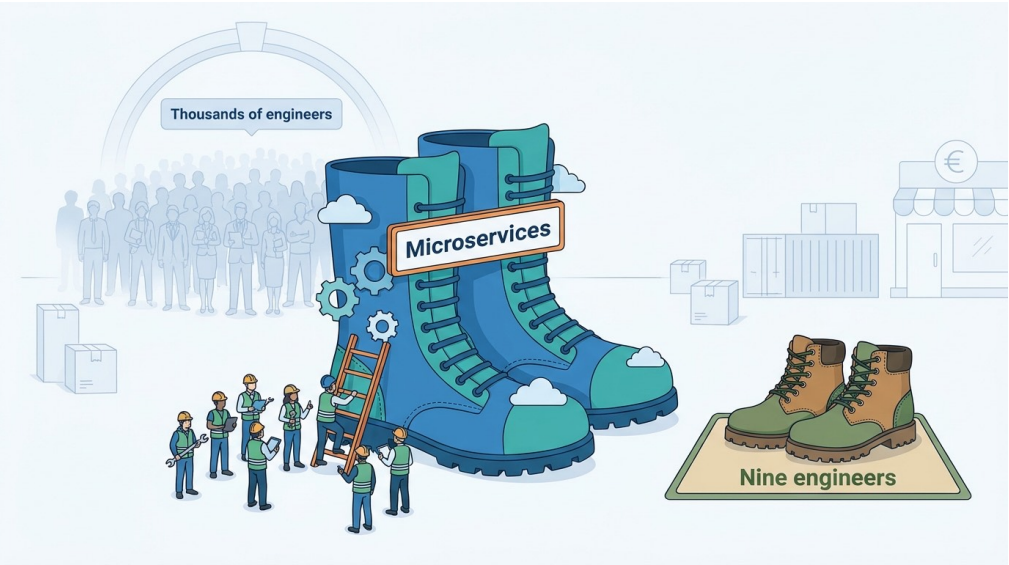


Figure 1: *The famous accounts describe companies with thousands of engineers; a microservices rebuild copied from them by nine engineers is The Giant's Boots, paying the costs without the reasons.*

Four Styles in Plain Words

Three ideas separate the styles: the **deployment unit**, defined above; **data ownership**, which program may read and write a given table or record directly; and the **boundary**, the rule that one part of the system may only reach another through a defined interface, and how that rule is enforced (by a tool, by a review, or not at all).

Style	What is deployed	Who owns the data	How boundaries are enforced	Documented example
Monolith	One program holding every domain: catalog, cart, orders, payments, admin	One shared database; any code can read any table	Usually not enforced; conventions and reviews	Zalando's Java shop "Jimmy" 2010–2015; Shopify's Rails core before 2017
Modular monolith	Still one program, divided into named modules with a public interface each	One database, but each module is meant to own its tables	A static check in the build that blocks new cross-module references	Shopify 2019–2024 (Packwerk); Medusa's module layout (n.d., accessed 2026-11)
Microservices	Many small programs, each built, released and scaled on its own	Each service owns its own database; nobody else reads it	The network: a service can only be reached through its API	Amazon from about 2001 (described 2006); Netflix 2008–2016; Zalando from 2015

Style	What is deployed	Who owns the data	How boundaries are enforced	Documented example
Composable (MACH)	Separately *bought* API-first SaaS services plus a storefront and an integration layer the merchant owns	Each vendor owns its data; the merchant owns the copies between them	Vendor APIs and contracts	commercetools and the MACH Alliance vendors (2020–2026)

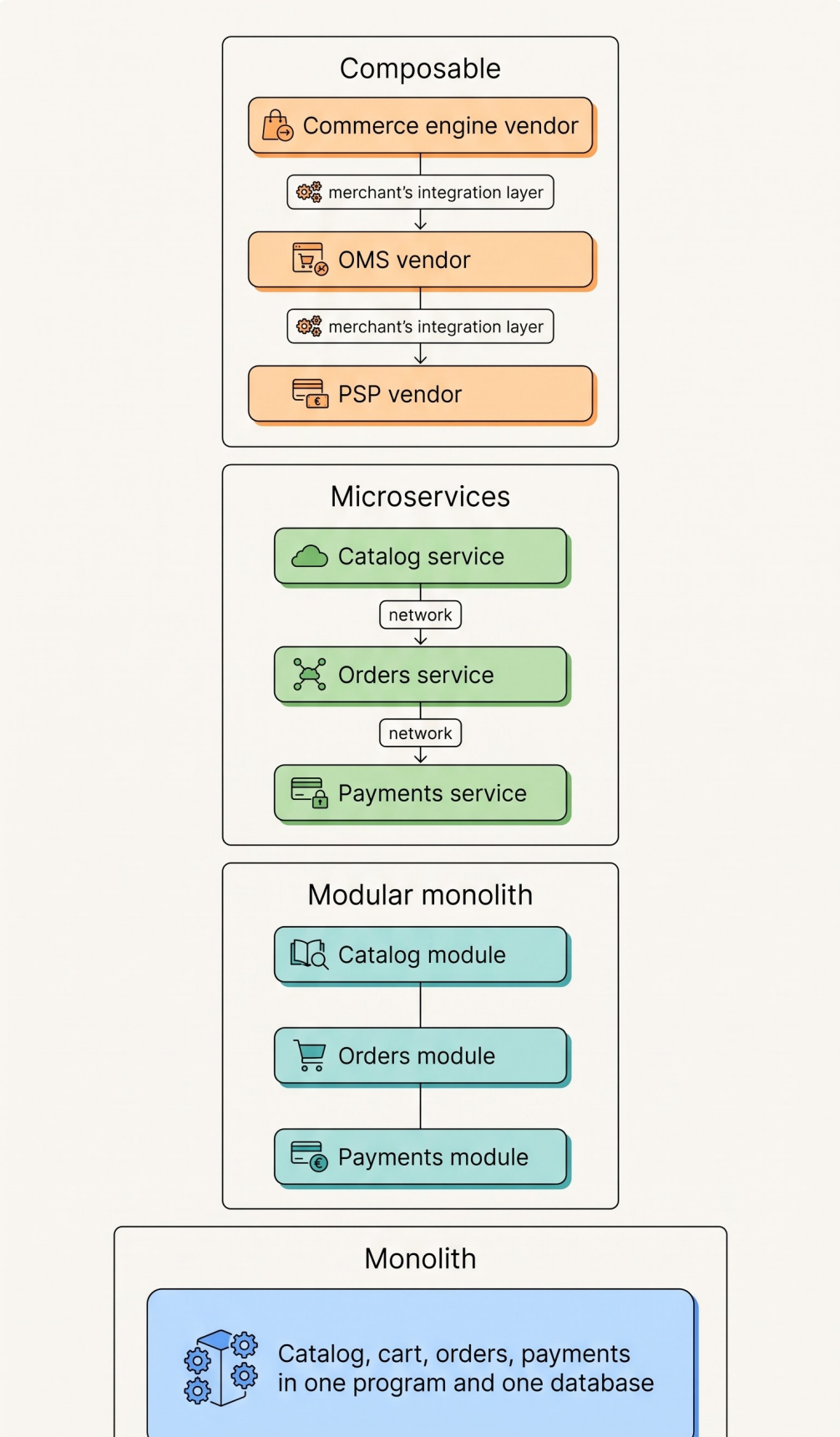

Composable
Commerce engine vendor
merchant's integration layer
OMS vendor
merchant's integration layer
PSP vendor
Microservices
Catalog service
network
Orders service
network
Payments service
Modular monolith
Catalog module
Orders module
Payments module
Monolith
Catalog, cart, orders, payments
in one program and one database

Figure 2: *The four styles differ in how many separately deployed pieces exist and who owns the joins between them; in a modular monolith the joins are function calls inside one program, in microservices they are network calls between a company's own services, and in composable they are integrations between vendors that the merchant must own.*

Two clarifications a newcomer needs. First, "**monolith" is not an insult**. It is one program with one database, which is also the shape in which every transaction is cheapest and every fact is consistent. Second, **composable is a different kind of thing from the other three**. They describe how one organization structures code it writes. Composable describes how a merchant *buys*: several API-first products (an engine, a search service, a PIM, an OMS, a payment provider) assembled behind a storefront the merchant builds or buys. The merchant does not write the services; it writes the joins, and the distribution costs the big companies document for microservices transfer to that integration layer almost unchanged.

Two terms from earlier chapters recur. A **system of record** (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record) is the one system allowed to be right about a fact: the WMS for physical stock, the OMS (order management system) for the order. **Eventual consistency** (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation) means a copy of a fact in another system may be out of date for a while, and will catch up. Every style with more than one deployment unit lives with it, and with **idempotent** handling: processing the same message twice must have the same effect as processing it once.

What the Famous Examples Actually Show

The accounts below are listed with their dates because the architecture they describe has certainly changed since. Treat each as evidence that a pattern existed and worked at that date in that company, not as a description of today's systems.

Amazon: a scaling wall, then hundreds of services (2001–2007)

The primary account is a 2006 conversation between Jim Gray and Werner Vogels, Amazon's chief technology officer, in *ACM Queue*. As digested by High Scalability in 2007, Amazon's front end was a "two-tier monolith" and "in 2001 it became clear that the front-end application couldn't scale anymore"; Amazon moved to "a fully-distributed, decentralized, services platform" that grew into hundreds of services, with a single page request touching between 100 and 150 of them. The same source carries the two cultural rules the industry still quotes: "you build it, you run it" (the team that writes a service also operates it, confirmed in AWS's own 2006 relay of the interview) and the "two-pizza team", a team small enough to be fed by two pizzas. The 2007 Dynamo paper gives the scale: the shopping cart service served "well over 3 million checkouts in a single day".

What this shows: a company that split because one program could no longer serve its traffic, and organized the split around small teams owning services end to end. What it does not show: a timeline a smaller company could copy, or anything about Amazon's retail architecture today. The evidence is two decades old.

Zalando: a split for autonomy, then a decade of discipline (2015–2025)

Zalando's record is unusually complete and unusually honest. Henning Jacobs's 2020 retrospective of a decade in Zalando's technology gives the sequence: a PHP shop on the Magento stack, rewritten in Java in 2010 because the stack hit scaling limits; in-house order and payment services in 2012; and in March 2015, under the banner "Radical Agility", a move to Amazon Web Services (AWS), Amazon's cloud, with containerized microservices owned by autonomous teams, each with its own AWS account. Zalando's 2015 post on the shop rebuild gives the motive: the monolith, "Jimmy", meant that "all of the Shop teams shared the same code base" and teams "couldn't move quickly because of Jimmy's slow startup time and our overly complicated deployment processes". The trigger was not traffic. It was too many teams in one program.

By 2020 Zalando had, per Jacobs, more than 260 engineering teams, more than 1,300 developers and more than 4,000 applications. The same post lists the costs: team autonomy over technology produced "similar CRUD REST services written in six languages" (simple create-read-update-delete web services), which "hurts productivity at scale"; a standard stack would have been "a major win"; and in 2020 Zalando "still dealt with monoliths from 2010", with some functionality implemented twice. The 2015 engineering principles repository is now marked deprecated, noting that they "no longer reflect the current reality (2020) in all aspects".

The discipline that followed is documented step by step. In July 2020 Zalando narrowed its technology radar, dropping Clojure, Haskell and Rust for new applications because "a broad freedom of choice leads to increased difficulties in supporting software long-term". In 2023 one developer portal replaced "more than 100 disconnected interfaces". In March 2025 it reported that its event-driven pipeline for price and stock updates "became a bottleneck": more than 90% of each payload was unchanged, during Cyber Week stock and price events "could be delayed by up to 30 minutes", and teams that consumed the stream adapted product data for their own needs, producing what the post calls "competing sources of truth"; it was replaced by a synchronous read API. Zalando's 2025 annual report (published March 2026) puts about 3,000 employees in its technology departments.

What this shows: microservices gave Zalando team autonomy at hundreds of teams, and the bill came as technology sprawl, tooling sprawl and, in one critical path, a boundary that had to be redrawn. What it does not show: that microservices "failed". Each later fix narrowed a choice; none reversed the architecture.

Shopify: the monolith that refused to split, and what it learned (2019–2024)

Shopify is the counter-example, and its own account runs from 2019 to 2024. In February 2019 it described its Rails codebase as one of the largest in existence, worked on for over a decade by more than a thousand developers, and said that around 2016 "the downsides of the monolith were outweighing the benefits". It rejected microservices explicitly: they would add test and deployment pipelines and infrastructure per service, network calls that add latency and reduce reliability, and coordinated deploys for large refactors; the team "wanted more modularity without more deployment units". Instead it regrouped code into business components (orders, shipping, inventory, billing), each meant to have a public interface and "exclusive ownership of its associated data", and built a tool, Wedge, to score each component's isolation. The payoff it claimed: the isolation work let it replace a legacy tax engine.

By September 2020 the core had "over 2.8 million lines of Ruby code", 37 components and an Architecture Guild of about 400 members, and Shopify said it split out services "only for good reasons", naming storefront rendering and credit-card vaulting. The same post is candid: "every component depended on over half of all the other components", with "lots of circular dependencies". Packwerk, open-sourced that month, is a static check that records existing boundary violations in a list so that new ones are blocked without fixing everything at once. Shopify's FY2025 annual report (filed February 2026) confirms the broad shape today: a multi-tenant platform, Ruby on Rails, "single branch" software on which all merchants run the latest version, and about 7,600 employees.

The February 2024 Packwerk retrospective is the latest first-party account and the most useful. Privacy checks "were removed from Packwerk with the release of version 3.0" because they pushed teams into API design instead of dependency management. Several common ways of loading code are "untracked and invisible to the tool", so "a package can show zero violations and still fail at runtime". Isolating one base package with zero violations and zero dependencies "took many months of hard work".

The rate at which the tool found problems "vastly outpaced" teams' capacity to fix them; "we have discussed removing Packwerk from our monolith", and it stayed because it "has provided value in holding the line against new dependencies at the base layer".

Most important: "domains and the boundaries between them do not reflect the way Shopify's code actually functions in practice", so components became organizational groupings rather than enforced units. Two quoted lessons: "Packwerk is a sharp knife, and it must be wielded with care" and "running code, more than any metric, will always be the best indicator of real progress".

The one documented case of Shopify carving a hot path out of the monolith is the Storefront Renderer, rewritten from 2019 because storefront code "has much stricter performance requirements" than, say, payment processing; by August 2020 it reported

server response times four to six times faster. It was carved for latency, behind a verifier that compared old and new responses on live traffic, not for team autonomy.

What this shows: a company with a thousand developers kept one deployment unit, got real value from ownership and from one cheap gate at the base layer, and found that enforced domain boundaries did not match runtime coupling. The honest reading is not "Shopify modularized its monolith" but "Shopify spent seven years learning which modularity was worth paying for".

The consolidations: Segment (2018) and Prime Video (2023)

Two accounts are often quoted as "microservices failed" and support something narrower. Segment wrote in 2018 that it had grown to over 140 services, each with its own queue, that three full-time engineers spent most of their time keeping them running, and that shared-library versions diverged; it consolidated them into one service and wrote "the solution for us was a monolith", while listing the trade-offs (weaker fault isolation; a shared upgrade can break several destinations).

In 2023 Prime Video's video-quality team replaced a serverless, orchestrated pipeline (small functions run on demand by the cloud provider, chained by a workflow service) that "supported only around 5% of the expected load" with a single process, cutting operational cost by 90%; the original post has been taken down, and InfoQ's contemporaneous summary is the surviving source.

Both are consolidations of *one subsystem* that had been cut too fine. The transferable claim is "right-size the unit of deployment to the data flow across it", not "reverse the architecture".

Composable and MACH: a style whose definition moved (2020–2026)

The MACH Alliance was announced in June 2020 by commercetools, Contentstack, EPAM and Valtech, with the acronym spelled out as "Microservices, API-first, Cloud-Native, and Headless". It passed 100 members in December 2023. In August 2025 it listed five guiding principles without restating the acronym, and as of November 2026 its principles page states three (Open, Composable, Connected) and no longer defines microservices or headless at all.

Its own February 2026 survey of 600 enterprise technology decision-makers reports that 37% cite integration complexity as a primary concern, a finding against interest from a vendor alliance whose respondents are its adopters.

In the same period commercetools, the alliance's flagship vendor, cut about 10% of staff in February 2025 after missing growth targets (widely reported), and in January 2026 launched a product it describes as layering agentic capabilities "on top of your existing commerce solution". The composable champions now sell composable incrementally.

What a composable vendor sells as a unit is visible in its documentation. A commercetools Project is "the top-level container" for a merchant's data, isolated from other Projects, with published limits (25 API Extensions per Project as of November 2026). Its API Extensions, synchronous call-outs to the merchant's own code before a cart or order is saved, carry a plain warning: "if it fails or takes a second longer to return a result, the whole API call fails or takes a second longer." That sentence is the distribution cost of microservices, delivered to the merchant's doorstep. And none of the accounts above describes a company with fourteen people in technology.

What Each Style Costs

Martin Fowler's 2015 "Microservice Trade-Offs" names the benefits as strong module boundaries, independent deployment and technology diversity, and the costs as **distribution** (remote calls are slower and can fail), **eventual consistency** ("because of their laudable insistence on decentralized data management") and **operational complexity** ("there's just no way to handle dozens of services without the automation and collaboration that continuous delivery fosters"). His "MonolithFirst" of the same year adds the point that decides most cases: "even experienced architects working in familiar domains have great difficulty getting boundaries right at the beginning", so "the premium of microservices is a drag you should do without" early on.

In commerce those costs land on specific promises. The table takes three steps of Eva's order and shows what changes when the pieces sit inside one deployment unit versus across several.

Promise in Eva's order	Inside one deployment unit	Across deployment units (microservices or composable)	Who pays
The price at checkout matches the product page	One function call; one transaction; always the same number	A network call to a pricing service, or a copy that can be stale (the Price Drift of October 2026 was a stale copy between systems)	The merchant, in margin and in a reconciliation job

Promise in Eva's order	Inside one deployment unit	Across deployment units (microservices or composable)	Who pays
The last unit is sold once	One row, one lock, one decrement	A reservation message; two channels can both read "one left" before either decrement lands (the Black Friday Oversell)	The customer, in cancellations; the merchant, in goodwill
The order reaches the warehouse	Not applicable; the WMS is always a separate system	A message that can be delayed, duplicated or lost; needs idempotent handling, retries and reconciliation	The operations team, in late orders

The third row is the honest one: some boundaries exist whatever style you choose, because the WMS, the PSP (payment service provider) and the carriers are other companies' systems. The question a style answers is how many *additional* boundaries of that kind you create inside your own code.

The people cost is the one the pitch decks omit. Netflix's own account of its move to cloud microservices runs from August 2008 to January 2016: seven years. Uber reported in 2020 that with about 2,200 critical microservices a root-cause investigation could run through about 50 services owned by 12 teams, a simple feature needed 10 touchpoints, services had a half-life of about 1.5 years, and ownership had blurred into "networked monoliths" that must be deployed together; its remedy grouped services into about 70 domains behind gateways, the modular-monolith discipline rebuilt on top of microservices. Segment's three engineers keeping 140 services alive is the same cost at a company a hundredth of Uber's size.

	Monolith	Modular monolith	Microservices	Composable
Teams it serves well	One, up to roughly ten engineers	A few teams on one release train; Shopify ran it with a thousand developers	Many teams releasing on their own schedules	A merchant with an integration team that can own every join
Consistency of price, stock, order	Transactional	Transactional	Eventual; needs idempotent handling, reliable publishing and reconciliation	Eventual, between vendors; reconciliation is the merchant's
Release independence	None	None by default; ownership per module	Full	Per vendor, on the vendor's schedule
Operating cost	One pipeline, one runtime	Plus one boundary check	A pipeline, monitoring and on-call per service; a platform team	An integration platform, contracts and reconciliation per vendor pair
Typical failure	Nothing changes without touching everything	Boundaries drift; the tool reports more than teams can fix	The distributed monolith; sprawl of languages and tools	The Spaghetti Junction; a vendor's limit becomes the merchant's outage
Right when	One team, one product, early	Several teams, one product, a decade of change ahead	Hundreds of engineers; the organization is the bottleneck	Capabilities differ in pace and the merchant can staff the joins

Modularity Sized to the Team

Melvin Conway wrote in 1968 that organizations "are constrained to produce designs which are copies of the communication structures of these organizations". Every account above is Conway's observation at work: Amazon's two-pizza teams produced services; Zalando's 260 teams produced 4,000 applications; Shopify's one release train produced one deployment unit with internal ownership. Team Topologies, the 2019 book by Matthew Skelton and Manuel Pais, rests on the premise that "teams can only handle so much complexity before breaking down".

The rule I use follows from that. **Create one deployment unit per team that must release without waiting for another team.** Not per domain, flow or capability: per team with its own release need. A four-person integration team gets one deployable with modules inside it; a storefront team gets the theme and its apps. When those two teams block each other's releases, that is the evidence for a second unit, and not before.

Figure 3: *One deployment unit per team that must release without waiting for another: a small team gets one deployable with modules inside, and a second unit only when two teams block each other's releases.*

Three disciplines make a single unit work for years, all in the public record.

Decide who owns each fact before you split anything. Fowler's warning about boundaries and Shopify's finding that domains and their boundaries "do not reflect the way Shopify's code actually functions in practice" are the same warning. In commerce the boundaries that matter are the systems of record: who owns stock, the order, the

product. Kestrel & Pine's Waypoint decisions W1 to W3 are exactly that, made before a line of custom code was moved.

Enforce boundaries cheaply and gate new violations rather than chase zero. Packwerk's design, a list of existing violations and a block on new ones, is the transferable idea, and Shopify's retrospective says where it earned its keep: holding one base layer clean. A merchant's version is a dependency check in the build of its integration code that says, for example, the stock module may not import from the orders module.

Split a hot path only on measured evidence, behind a verifier. Shopify's Storefront Renderer is the model: a measured latency need, a rewrite, a comparison of old and new responses on live traffic, a fallback. If you cannot name the measurement, you do not have the evidence for the split.

The vendor's view and the merchant's view differ, and the pitch deck blurs them.

Question	A platform vendor	A merchant on a hosted platform
Who must be modular?	Thousands of engineers shipping one product to millions of tenants	A dozen engineers shipping connectors, apps and pipelines around a bought core
What is the unit of change?	A component or service inside the platform	A connector, an app, a theme, a data pipeline
Where does distribution cost land?	In the vendor's engineering budget	In the merchant's integration layer, as integration debt (APIs, Events, Files and Middleware: How Commerce Systems Talk)
What does "microservices" buy?	Release independence for many teams	Nothing directly; the merchant does not deploy the platform
What does "composable" buy?	A product category to sell	Replacing one capability (search, OMS, PIM) without replacing the core, at the price of owning the joins

A vendor's internal style is also not something a merchant can verify. A WMS vendor's page saying it is "built on microservices" (Manhattan's does, as of November 2026) says nothing about uptime, release cadence or the API contract, which are the things to ask

about (Promises Between Systems: API Contracts, Webhooks and Versioning).

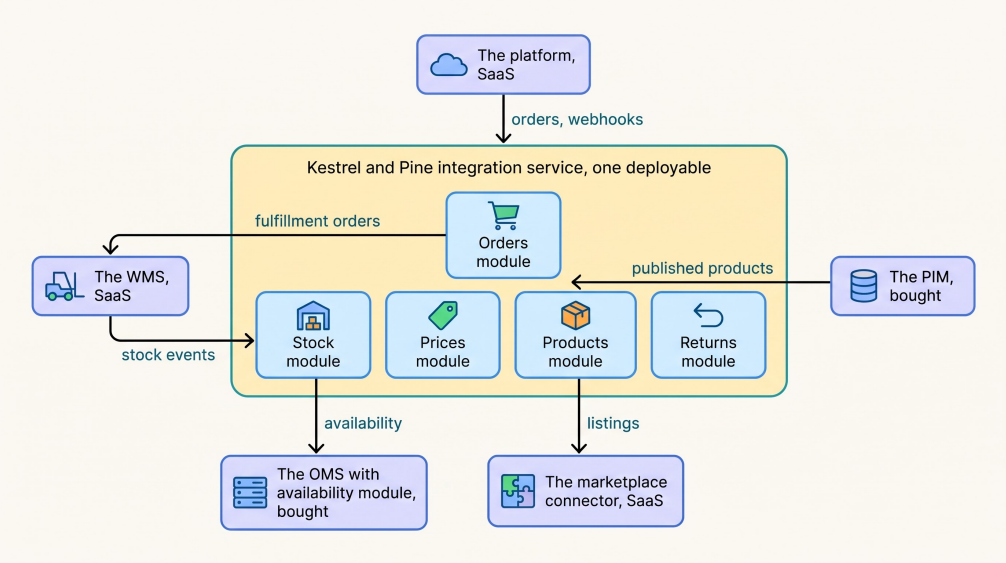


Figure 4: *Kestrel & Pine's own code in 2027 is one deployment unit with a module per flow, surrounded by bought systems; the boundaries that matter are the systems of record on the outside, not service boundaries on the inside.*

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting (under about €2M)	A hosted platform and a handful of apps; no custom code beyond the theme; the style is the vendor's	An inventory of apps with their owners; nothing more	A custom integration with no owner; an agency proposes "headless" before the first store or wholesale account

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
2 — Growing (about €2M–€20M)	A first ERP or accounting connector, a 3PL feed, a marketplace tool; custom code as agency scripts	One repository and one deployable for the merchant's own integration code; boundaries drawn around systems of record	Scripts in three languages on three hosts that nobody can list; a connector fails and its author has left
3 — Omnichannel (about €20M–€200M)	A dozen systems; an OMS and PIM beside the platform; an integration team of two to five; a modular monolith for own code	Modules per flow with a dependency check; OMS and PIM bought rather than built; a second deployable only when two teams block each other	A rebuild proposal sized for a company a hundred times larger; a "service" per flow with one engineer each; releases that need three teams to coordinate
4 — Group (over about €200M, or several brands)	Platform teams; shared storefront, integration and data services across brands; some services the group writes itself	Standard stacks, API guidelines and a paved road, a supported default toolset (Zalando's 2020 and 2023 lessons); a deployable per stream-aligned team (a team owning one flow of business change end to end); a boundary gate in every shared codebase	Six languages for similar services; a service half-life under two years; a feature that needs ten teams; functionality built twice

The judgement: through stage 3 the right number of deployment units for a merchant's own code is one, perhaps two, and the energy goes into boundaries around systems of record. Microservices become a real question only at stage 4, when the organization rather than the traffic is the bottleneck, and even then the documented path is incremental and takes years. Composable read as "replace one capability at a time around a kept core" fits stage 3; read as "an API-only core and a custom storefront" it is a stage 4 choice that needs a platform team.

In Practice: Noor's Memo

Noor's memo to the board (May 2027) ran to two pages. The numbers below are Kestrel & Pine illustrations.

She started with the headcount. Nine engineers, shaped from April 2027 as a "Buy" team of three storefront and apps engineers, a "Fulfil and Return" team of two back-end engineers, an integration pair under Ruben Alves and a data pair (Who Builds and Who Owns: Teams, Vendors, Agencies and Governance). Three of those groups write code, and none has yet waited on another to release. By the rule above, Kestrel & Pine's own code needs one storefront codebase and one integration service, shared by the integration pair and the Fulfil and Return team until the day they block each other. The rebuild option had described about a dozen services for nine engineers.

She then matched the rebuild against the incidents. The Black Friday Oversell, the Three-Status Order, the Price Drift and the Store Stock Mirage were all failures of ownership and of the joins, not of the platform's internal structure; an API-only engine would have recreated every one of those joins in new code. Waypoint decides ownership (W1 to W3), buys the missing hubs and makes the copies trustworthy (W5); none of it depends on the platform's internals.

	The composable rebuild option *(illustrative)*	**Stay and surround as approved**
Cost over 2027–2029	About €3.2M, 18 months	About €1.33M including subscriptions; the €1.1M Waypoint budget
What is rebuilt	Storefront, commerce engine, order service, every integration to them	Nothing; the platform stays (W4)

	The composable rebuild option *(illustrative)*	**Stay and surround as approved**
What is bought	An API-only engine; the rest custom	OMS with availability module, PIM (product information management system); standards on the existing iPaaS (integration platform as a service)
Deployment units the team must run	About a dozen services plus a storefront	One integration service; vendors run the rest
Incidents addressed in 2027	None until "phase two"	Oversell (W1), Three-Status Order (W2), Price Drift and Double Refund (W5 contracts and reconciliation)
Reversibility	One-date cutover of storefront and checkout: The Burned Bridge	Each hub added beside the platform; decision revisited in 2029

The one piece of the rebuild option Noor kept was the vocabulary. She wrote the integration rule into the engineering handbook: one repository, one deployable, a module per flow (stock, orders, prices, products, returns), a dependency check in the build that blocks new cross-module imports, and a measured latency problem as the only accepted reason to split anything. The agency scripts that move in-house during 2027, and every new connector outside the iPaaS, land in that one service. The answer to "are we behind?" was no: Kestrel & Pine stood where every documented company stood before it split, and those companies split for reasons Kestrel & Pine did not have.

When It Goes Wrong

The distributed monolith. Services that must be deployed, tested and rolled back together, with the network added for nothing; Uber named it "networked monoliths" in 2020. Detection: a release that needs three teams in one meeting; a change to one service that always needs a change to another. Recovery: merge the pair, or redraw the boundary around the data that actually flows, as Uber's domains did.

The half-finished migration. Zalando in 2020 "still dealt with monoliths from 2010" and had some functionality implemented twice. Detection: two systems that both claim

to own an order or a price; a legacy program nobody dares to switch off. Recovery: the strangler approach (Moving Without Burning the Bridge: Replatforming and Migration), replacing the old system one capability at a time with a retirement date for each; never the one-date cutover of The Burned Bridge.

The boundary in the wrong place. Zalando's price-and-stock pipeline (described in 2025) reprocessed payloads that were more than 90% unchanged and fell up to 30 minutes behind during Cyber Week, and consumers' adapted copies became competing sources of truth; Prime Video's pipeline paid for every step between stages that belonged in one process. Detection: consumers behind at peak; copies that disagree; a bill that scales with hops, not with orders. Recovery: a synchronous read API beside the events for "what is the price now", and a merged boundary where the data flow is tight.

The over-decomposed integration layer. Segment's 140 services for 140 destinations is the shape a merchant's 34 integrations take when each is its own deployable with its own queue. Detection: engineers who spend their weeks keeping connectors alive; library versions that differ per connector. Recovery: one integration service with a module per flow and recorded traffic replayed as tests, as Segment did.

Composable without a platform team. The MACH Alliance's own 2026 survey puts integration complexity first among its adopters' concerns. Detection: every vendor pair joined by a different script with a different owner; a vendor's call-out timeout failing your checkout. Recovery: an integration inventory (APIs, Events, Files and Middleware: How Commerce Systems Talk) and a contract per join (W5).

What to Take Away

Established.

- The costs of splitting a system into separately deployed pieces are distribution, eventual consistency and operational complexity, and boundaries are hard to get right at the start (Fowler, 2015); every documented commerce case confirms both.
- The documented splits were driven by a traffic wall (Amazon, 2001) or by too many teams in one codebase (Zalando, 2015), and each took years; the documented refusal to split (Shopify, 2019–2024) served a thousand developers on one deployment unit.
- System structure follows team structure (Conway, 1968); teams can only carry so much complexity (Skelton and Pais, 2019).
- Composable is a buying style, not an internal architecture; its own alliance dropped "microservices" from its principles (2026) and its own survey ranks integration complexity as the top concern (2026).

Depends on context.

- One deployment unit for a merchant's own code is right through the Omnichannel stage; a second when two teams measurably block each other's releases; a service per team at Group stage, with a platform team and a standard stack to carry it.
- Composable per capability (replace search, add an OMS, add a PIM) fits a merchant that can staff the joins; an API-only core with a custom storefront fits a group with platform teams and a differentiated front end, and not before.
- Splitting a hot path out of a single unit is right when a measured latency or load requirement demands it, behind a verifier and fallback; it is wrong as a reorganization tool.

The author's synthesis.

- The number of deployment units in a merchant's own code should equal the number of teams that must release independently, and for most merchants below group scale that is one; the modularity that matters is the ownership of facts across bought systems, not service boundaries inside custom code.
- Every famous architecture account should be read with three questions: what size was the company, what was the trigger, and what did they pay; a proposal that cannot answer those three for your company is The Giant's Boots.

Questions to Ask

1. How many teams do we have that must release without waiting for another team? That is our ceiling for deployment units.
2. Which facts (stock, order, price, product, customer) have a named system of record, and which do not? Boundaries come from that list, not from a diagram.
3. For any proposed split: what is the measured problem, in latency, load or blocked releases, that the split solves?
4. For any proposed rebuild: which of our incidents of the last year would it have prevented, and by when?
5. What is the reference company's size in engineers, what triggered its split, and what did it report paying?
6. Where will the distribution costs (retries, duplicates, stale copies, reconciliation) land, and who owns them?
7. If we adopt composable for one capability, who owns the join to the core, and what are the vendor's call-out timeout and failure rule?
8. What is the rollback if the new piece is wrong: a feature switch, a parallel run, or The Burned Bridge?

On a hosted platform the internal structure is the vendor's problem; what a merchant shares with the vendor, every day and especially on its peak day, is the platform itself.

The next chapter asks how one platform serves thousands of merchants at once, and why rate limits, noisy neighbors and pods are a merchant's concern: Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation.

Sources and Evidence

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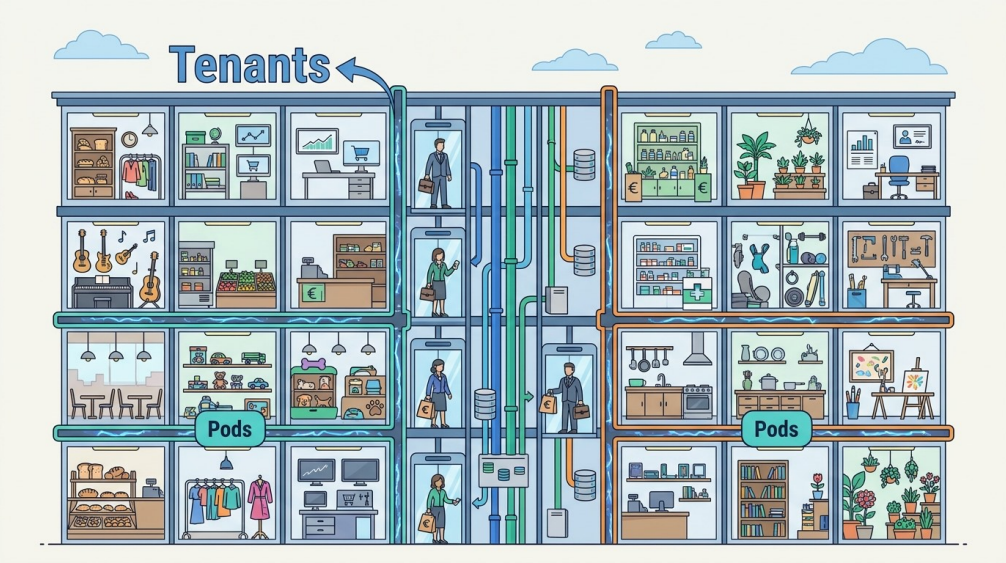
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

30. Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation



IN THIS CHAPTER, YOU WILL: Understand how a SaaS (software-as-a-service) platform serves thousands of merchants on shared systems, and what that sharing means for your rate limits, neighbors and peak day.

KEY POINTS:

- **Tenancy is the vendor's design; its consequences are yours**. A hosted platform pools merchants on shared servers and databases, keeps them apart with a tenant identifier, a routing layer and, at the largest scale, isolated groups of databases; the unit of isolation is invisible to you and the vendor may move you between units.
- **The rate limit is the tenancy mechanism you meet**. A platform protects shared capacity by giving each app a budget of calls per store, tiered by plan. Every connector and app you install spends from that budget, so your noisiest neighbor on a peak day is often your own stock-sync app, and the vendor may lower the budget "to protect platform stability".

- **Every SaaS you rent is a tenancy arrangement, and the peak day is shared.** The warehouse, returns and product information systems and the marketplace connector pool you with strangers whose Black Friday is the same Friday; one tenant or several for three countries is a real decision with costs both ways.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The First Snowfall** — the API budget, the checkout throttle and the vendor's shared capacity are met for the first time on the busiest day of the year.
- **The Spaghetti Junction** — every connector and app draws on the same per-store budget, and nobody has added up what they spend.
- **One Size Fits None** — three countries or several brands are forced into one tenant, or split across several, before anyone has measured what the sharing costs.

During a sale, a merchant's order export may be delayed because the platform refuses further requests until its usage allowance resets. The shop can still accept purchases while warehouse instructions wait. Hosted platforms impose limits to protect shared capacity, so renting a platform does not remove the merchant's need to plan how its integrations use that capacity.

Sharing one running system among several customers is **multi-tenancy**; each customer is a tenant. It spreads infrastructure costs, but requires **isolation** to stop one customer's load, data or failures affecting others. Shopify's FY2025 annual report, filed in February 2026, describes millions of merchants on one codebase. Vendors may split databases into **shards**, group tenants into isolated **pods**, and impose **rate limits**, budgets for API use. Those choices determine which work can proceed when capacity is tight.

The chapter uses dated vendor accounts to explain the mechanisms, then focuses on the merchant's decisions: call budgets, priorities at peak and whether several countries should share one tenant. Ask the same questions of warehouse and returns vendors; they share infrastructure too.

Where we are in the order's journey. Eva orders a jacket at Kestrel & Pine, the fictional outdoor brand. Once the platform records it, integrations must fetch and send the order within both platform and warehouse API limits. An ordinary April evening leaves spare capacity. Black Friday can exhaust it.

Two Rate Limits on One Friday Evening

Kestrel & Pine's platform handled **38 orders per minute** at the 2026 peak. Its export flow created warehouse orders one at a time, using several calls per order. The **warehouse**

management system (WMS) vendor throttled those calls, and the backlog reached four hours by 21:00. Black Friday Every Year: Peak Load, Resilience and Recovery follows the wider incident.

In June 2027, Lead Integration Engineer Ruben Alves examined logs from both ends before planning the next peak. The quantities below are illustrative.

On the **WMS side**, the vendor's API allows each customer about 30 calls a minute. The order flow made four calls per order, one order at a time, so it could carry about seven orders a minute on any day, and 38 were arriving. The WMS is a multi-tenant SaaS like the platform; the limit is how its vendor keeps one customer's peak from slowing the warehouses of its other customers, several of which were also having their Black Friday.

On the **platform side**, the illustrative account shows nine integration consumers: order export, finance, marketplaces, returns, three stock-writing apps, reviews and email. Rate-limit scope matters: a budget may apply to an app, a store or another vendor-defined unit, so consumers do not necessarily share one quota.

The reconstruction attributes about 55% of recorded usage to the stock apps and full-catalog marketplace sync, and about 20% to orders. Order export was throttled about 1,900 times, adding one-second waits. **Nobody had set priorities or checked the relevant limits before the peak.**

Neither finding was a bug; both vendors did what their documentation said. **The rate limit is the shared platform's way of being fair**; **the merchant's job is to decide how its own share is spent**. The rest of the chapter explains the machinery behind the error code.

Why Platforms Share: Tenancy in Plain Words

A **tenant** is one customer of a shared software platform, with its own data and configuration: at a commerce platform, one shop. **Multi-tenancy** is the design in which one running system serves many tenants. Vendors choose it for two reasons a merchant benefits from every day. The first is cost: one fleet of servers, one database estate and one operations team serve everyone, which is how a hosted platform can be rented by the month. The second is a single version: Shopify's FY2025 annual report calls its platform "single branch" software, meaning "all of our merchants use the latest version"; there is no upgrade project because there is nothing to upgrade.

The Amazon Web Services (AWS) SaaS Lens, a published reference for SaaS vendors (n.d., accessed November 2026), names three ways to arrange tenants. In the **silo** model "tenants are provided dedicated resources", such as a separate database or a separate stack per tenant, while identity, onboarding and operations stay shared. In the **pool** model "tenants share resources": "the more classic notion of multi-tenancy", one set of servers and tables holding everyone's rows with a tenant identifier on each. The **bridge** model mixes them: "some microservices in your architecture might be implemented

with silo and others might use pool", chosen by regulatory profile, noisy-neighbor risk, agility and cost.

The term **noisy neighbor** is also the SaaS Lens's: "a user of a system could place load on the system's resources that could have an adverse effect on other users". In a pool, one tenant's flash sale is every tenant's load unless the vendor meters it. In a silo, the noisy tenant only hurts itself, at a cost per tenant that few commerce subscriptions can carry.

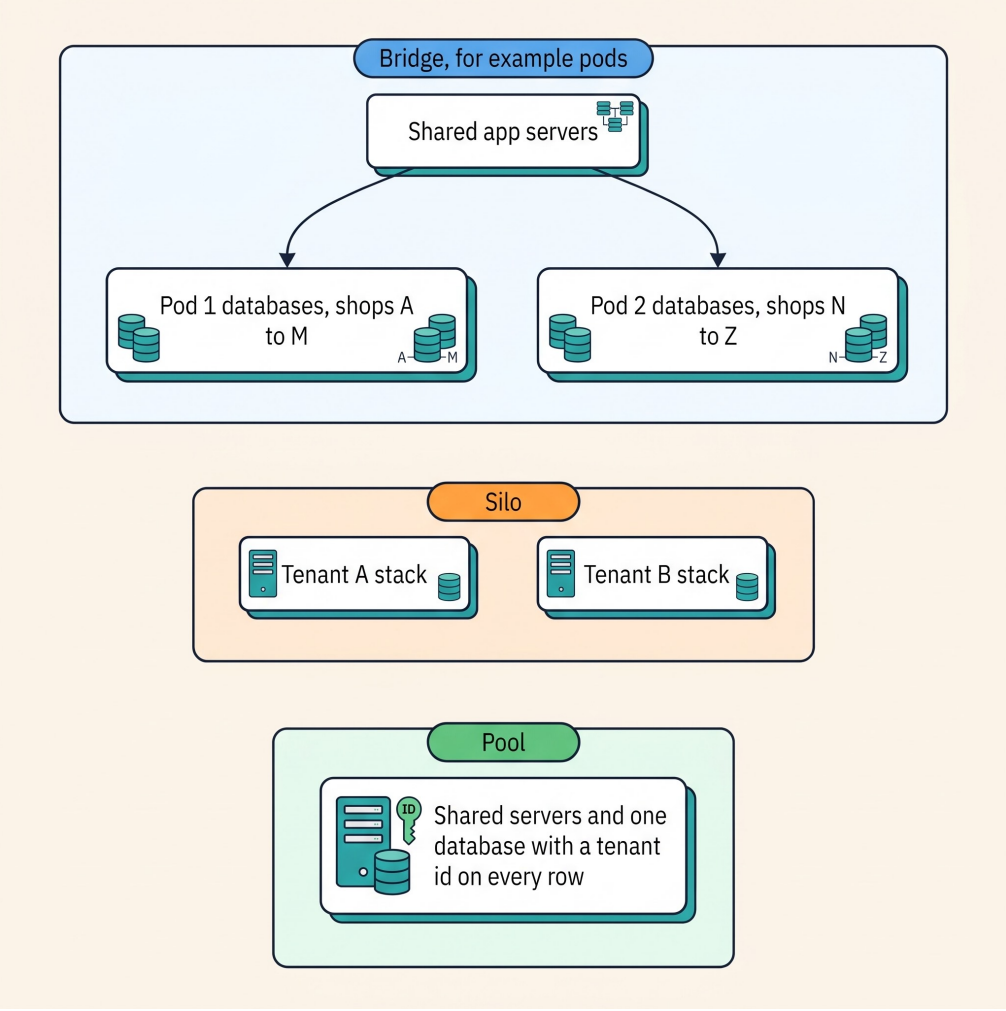


Figure 1: *Three ways to arrange tenants: a pool shares everything and relies on identifiers and limits, a silo dedicates a stack per tenant at a cost per tenant, and a bridge pools the compute while grouping tenants into isolated sets of databases.*

A merchant can see the spectrum in what vendors sell. Adobe's information center for

Adobe Commerce (updated October 2026) lists three deployment options: on-premises, where the merchant or its host runs everything (a silo the merchant operates); "on cloud infrastructure", where Adobe runs the platform and the merchant maintains its own code (a silo Adobe operates); and "as a Cloud Service", a fully managed SaaS (a pool). Shopware draws the same line from the extension side: plugins that run inside the core process "have direct access to the database" and therefore "are not supported by Shopware cloud", while apps run outside and talk through APIs. In a pool, nobody's code may touch the shared database, which is why hosted platforms extend by API and not by plugin (Apps, Plugins and Functions: Extensibility and Ecosystem Governance).

How a Shared Platform Keeps Tenants Apart

The public evidence comes from three vendors who have described their mechanisms, each at a date.

Shopify: a tenant identifier, a router and pods (2018–2021). By 2015, per Shopify's 2018 engineering post, its database "could no longer be scaled by buying a bigger server", so it **sharded**: split the one database into many, each holding a set of shops. Sharding alone reduced resilience, because one failed shard could take an action down for everyone.

In 2016 Shopify reorganized into **pods**: "a pod consists of a set of shops that live on a fully isolated set of datastores". Application servers, job workers and load balancers stay shared, but "all shared resources can only ever communicate to a single pod at a time"; every web request or background job is assigned to one pod, and a routing component ("Sorting Hat") matches each request to its pod and adds a header the application uses to pick the right datastore.

Each pod has an active and a recovery data center, and a "Pod Mover" could move a pod between them in about a minute without dropping requests. That post is eight years old and describes a pre-cloud topology.

The 2021 post on shard balancing fills in the tenant mechanics. Each shop's data lives in one pod with its own MySQL database shard plus Redis and Memcached caches; the shop is the sharding unit, with a `shop_id` on every shop-owned table; a routing table mapping domains to pods "lives in a separate database that isn't sharded".

Shops land on shards arbitrarily and grow at different rates, so the least- and most-loaded shards differed by "almost four times" before rebalancing.

Moving a shop uses Ghostferry (open-source): rows are copied and changes tailed while the shop stays live, a short application-level write lock covers the cutover, the routing table is updated and stale data pruned; "the only opportunity for downtime is during the cutover". The post's worked example is two flash-sale merchants on the same pod whose bursts coincided, one of which was moved; placement uses measured utilization, not shop counts.

> **INFERENCE, NOT DOCUMENTED:** In SaaS Lens terms a Shopify pod is a bridge: siloed datastores under pooled compute. Shopify has not described the pod topology after its move to Google Cloud; the 2019 backup post and the 2021 balancing post both describe shards on Google Cloud, so pods and the cloud coexisted at least to 2021. Nothing public after 2021 gives pod or shard counts.

Salesforce: one database, one identifier, governor limits (n.d., accessed November 2026). Salesforce's architecture documentation describes the purest pool in commercial use: a "multitenant and metadata-driven" platform in which tenants' definitions and data sit in shared tables, each organization identified by an OrgID; "when the kernel accesses the database, it uses this unique identifier to ensure that each org's activities are private". Because everything is shared, the platform protects itself with **governor limits**, published caps on what one tenant's code may consume. In a pool, limits are the isolation mechanism.

commercetools: projects and regions (n.d., accessed November 2026). commercetools documents a Project as "the top-level container" for a merchant's data and configuration, isolated from other Projects, with published per-Project limits (for example 25 API Extensions and 50 Subscriptions). It hosts in five Regions (two in North America, two in Europe, one in Australia) that "are completely isolated from each other, and no data is transferred between them". **Data residency**, the question of which country's servers hold your data, is therefore decided at sign-up by choosing a region, and changing it later is a migration (Whose Data Is It? Privacy, Consent and Retention).

Mechanism	What it keeps apart	What the merchant sees	Evidence
Tenant identifier on every row	One tenant's data from another's in a shared table	Nothing, unless it fails; the design rule to copy in any shared system you build	Shopify 2021; Salesforce n.d.
Router in front of the application	Requests, so that each one touches only its tenant's datastore	Nothing	Shopify 2018

Mechanism	What it keeps apart	What the merchant sees	Evidence
Pod or shard	One group's database load and failures from another's	A move: a short write lock at cutover; a neighbor's flash sale, if you share a pod	Shopify 2018, 2021
Governor and resource limits	Shared compute from any one tenant's code	Published caps on scripts, queries and extensions	Salesforce n.d.; commercetools n.d.
Region	One jurisdiction's data from another's	The residency choice at sign-up; a migration to change it	commercetools n.d.
Rate limit	Shared API capacity from any one app on any one store	The budget, the HTTP 429 ("too many requests") or throttled response and the retry	All platforms (next section)

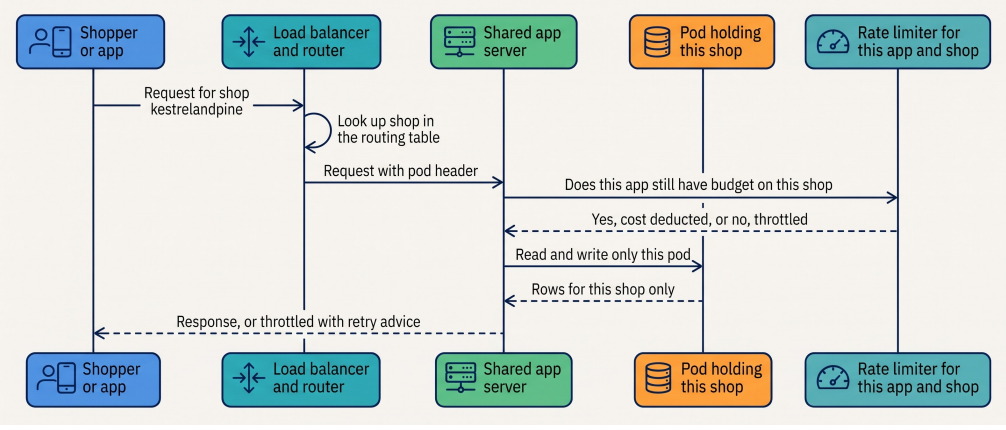


Figure 2: *A request on a pooled platform is routed to the one group of databases that holds the shop, and an API request is first checked against the budget of the app that made it for that shop; the merchant never sees the pod, and always sees the budget.*

Rate Limits: The Tenancy Tool You Meet

A **rate limit** is the maximum number of API calls, or units of query cost, that a platform lets one caller make in a time window. On a hosted commerce platform the caller is usually an **app** (any program that talks to your store through its API, whether installed from an app store or built for you) and the window is per second or per minute. Three vendors' published models, all volatile and all as of November 2026, show three philosophies.

Platform	Model	Published figures (November 2026, subject to change)
Shopify GraphQL Admin API	A "leaky bucket" of query cost per app and store, refilling at a plan-dependent rate; a write costs ten times a simple read; no single query may exceed 1,000 points	100 points per second on the standard plan, 200 on Advanced, 1,000 on Plus, 2,000 for enterprise; bulk operations exempt from the per-query cost
Amazon Selling Partner API	A token bucket per operation with a rate and a burst, looked up from the caller's credentials; "standard" plans are static, "dynamic" plans are "automatically adjusted to each selling partner based on the current and historical business needs"	Example on the page: one request per second with a burst of two
BigCommerce	A request quota per time window, by plan, reported in response headers	60,000 requests an hour on the Scale plan; 20,000 on Growth and Core
commercetools	No published request rate; instead caps on data and query shape per Project	500 elements per query, 500 update actions per request, a GraphQL complexity ceiling, 25 API Extensions

Shopify charges by the shape of the query; Amazon charges by operation and adapts to the seller's history; BigCommerce counts requests; commercetools caps capacity. For a merchant the consequence is the same across all four: **the budget belongs to a tenant, and everything you install spends from it**. Three rules from Shopify's documentation apply, in spirit, to every hosted platform:

- The budget is per app per store, "both the bucket size and the restore rate vary by API and by plan", and the recommended wait after a throttle is one second. A merchant that buys a higher plan buys a larger bucket; a merchant that installs another app adds another spender.
- "Shopify may temporarily reduce API rate limits to protect platform stability." The budget is not a contract; on the platform's own worst day it can shrink.
- Buyer traffic is treated differently from app traffic: "requests from real buyers aren't subject to a fixed request-per-minute limit", bots are limited, and checkout creation is "throttled per minute", returning a throttled response that the caller should queue and retry. The platform protects the shopper and the checkout first, and the merchant's integrations second.

Webhooks, the notifications a platform sends when something changes, are the other half of the budget story, because they are how a merchant avoids spending it on polling. They are also a tenancy mechanism in reverse: the platform spends its capacity to call you. Shopify's documentation promises eight retries over four hours if your endpoint fails, and warns that after eight consecutive failures a subscription created through the Admin API "is automatically deleted"; an integration that was down for an evening can wake up unsubscribed (Promises Between Systems: API Contracts, Webhooks and Versioning).

Flow in Eva's order	Who owns the capacity	What is metered	What the merchant controls	On a peak day
Browsing and search on the storefront	The platform; served largely from caches and the content delivery network	Not rate-limited for real buyers; bots are	Theme weight, app scripts	The platform's peak, not yours; it is smoothed across tenants

Flow in Eva's order	Who owns the capacity	What is metered	What the merchant controls	On a peak day
Checkout and payment	The platform and the PSP (payment service provider)	Checkout creation throttled per minute; the PSP has its own limits	Nothing but plan tier and warning the PSP of a flash sale	Shoppers queue briefly; payment is separated from order creation under load (Shopify, 2018)
Order to the warehouse through a connector	Your share of the platform's API budget and of the WMS vendor's	Calls per app per store on one end; calls per customer on the other	Which apps share the budget, batching, webhooks instead of polling	The first link to fall over, and the one you can govern
Stock sync to channels	The same two budgets plus the market-place's	Writes cost more than reads	Events instead of full-catalog polls; bulk operations	The loudest spender of the budget unless redesigned

The table shows where tenancy bites. The platform's own peak is shared across every tenant and smoothed, which is why the storefront held; the merchant's integrations draw on budgets that are per tenant and fixed, which is why the order flow did not (Black Friday Every Year: Peak Load, Resilience and Recovery).

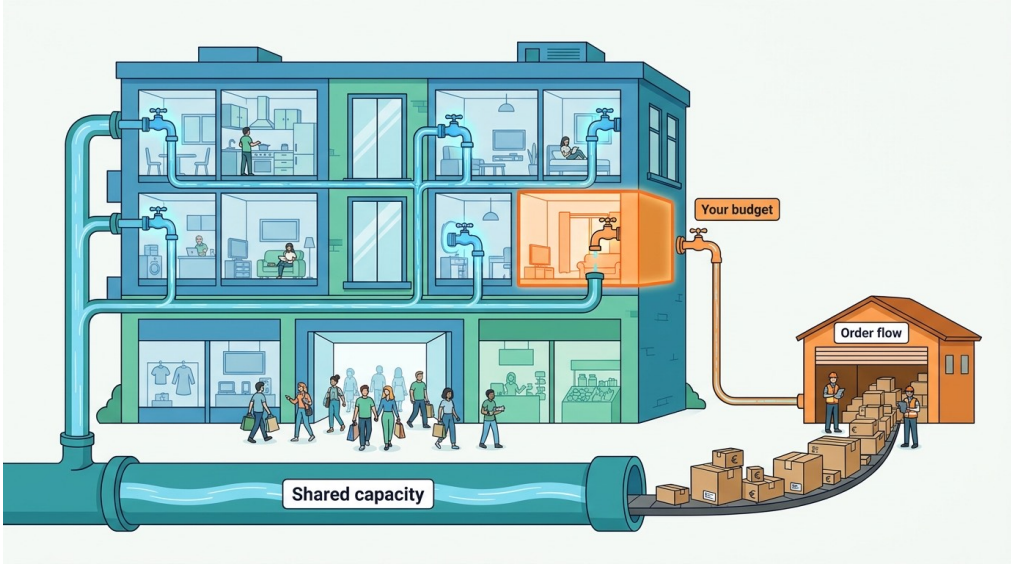


Figure 3: *The platform's own peak is shared across every tenant and smoothed, which is why the storefront held; the merchant's integrations draw on a fixed budget per tenant, which is why the order flow did not.*

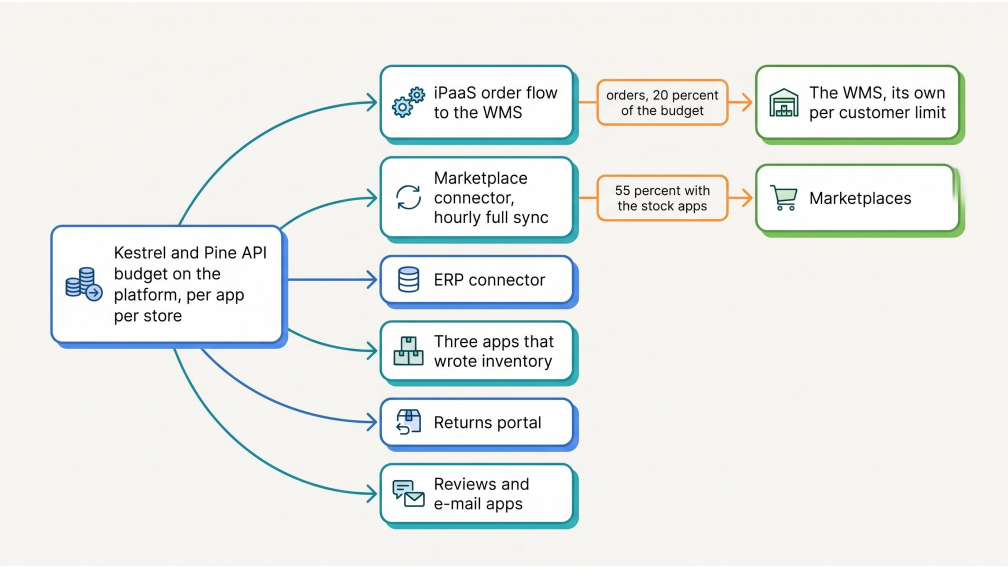


Figure 4: *On Black Friday 2026 nine consumers spent one store's API budget with no order of priority; the order flow, the only one with a customer waiting on it, got a fifth of the budget on one end and met the WMS vendor's own limit on the other (illustrative shares).*

One Tenant or Many

A merchant with several countries or brands faces a tenancy decision of its own: one tenant for everything, or one per country, brand or business unit. Kestrel & Pine runs three storefronts (Netherlands, Belgium in Dutch and French, Germany) in **one tenant**. The choice has costs both ways.

Hosted platforms have built features to keep multi-country merchants in one tenant. Shopify's Markets (n.d., accessed November 2026) lets one shop tailor "currency, language, pricing, and theme content" by customer location, with per-market settings for domains, product availability and taxes; its Managed Markets variant makes a partner the merchant of record (the legal seller on the invoice) but, as of November 2026, is offered to merchants in the United States, Canada and the United Kingdom, not the EU.

Option	What is shared	What multiplies	Fits when	Breaks when
One tenant, several markets (Kestrel & Pine)	Catalog, customers, apps, integrations, the API budget, the blast radius	Nothing	One brand, one assortment, one operations team; the platform supports markets	One country's flash sale starves another's order export; one country needs a legal entity, catalog or checkout the platform cannot vary per market
One tenant per country	Nothing but the vendor contract	Catalogs, apps, integrations, reconciliation, reporting	Separate legal entities, assortments or fulfillment per country; regulatory separation	Product data drifts between tenants (The Price Nobody Can Explain across countries); the integration count triples

Option	What is shared	What multiplies	Fits when	Breaks when
One tenant per brand (Group stage)	Possibly a shared OMS, PIM and integration layer beneath	Storefronts and apps per brand	Brands with different assortments, audiences and teams	A group forces shared apps and release dates on brands that do not share customers (One Size Fits None); or keeps brands apart that share a warehouse and a customer

My rule: **stay in one tenant as long as the platform can vary what differs by market and the shared budget can carry every market's peak with the integrations you have**. The first condition is a product question (markets, catalogs per market, per-market taxes); the second is the budget analysis above, repeated per market. Split tenants when a legal entity, a separate assortment or a separate operations team makes the sharing a fiction, and when you do, put the PIM and the OMS beneath all tenants so product and order truth is not multiplied (What Are We Selling? Products, Variants and the Catalog, The Order Record: Lifecycle, State and the Order Management System). Group-stage merchants with several brands become a tenant of tenants and should ask the vendor what they ask of themselves: which unit can be isolated, moved and restored alone (What Changes at Each Scale: From First Store to Group).

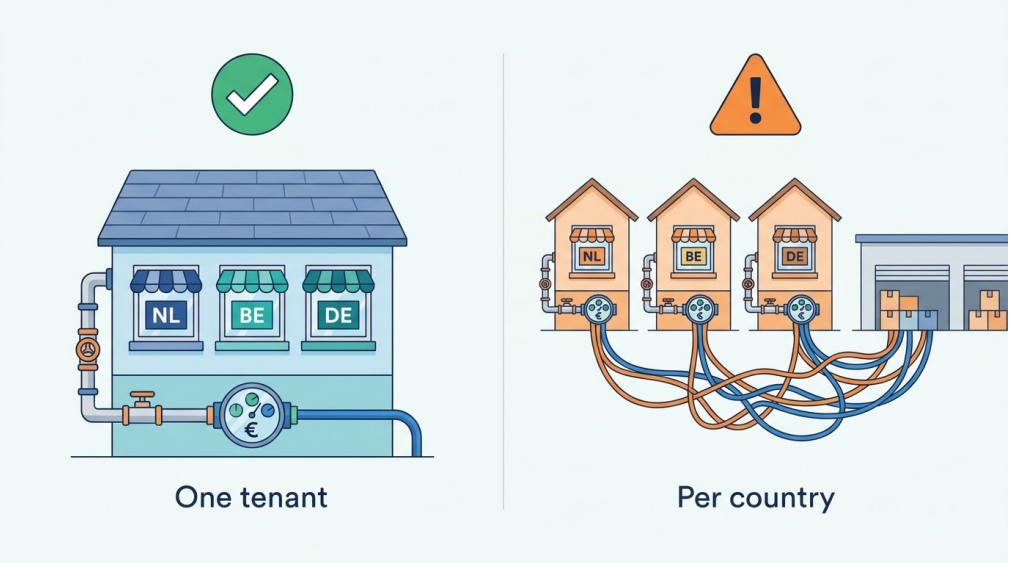


Figure 5: *Kestrel & Pine keeps its three storefronts in one tenant, sharing one catalog, one set of integrations and one API budget; one tenant per country would multiply catalogs, apps and integrations.*

Tenancy in the SaaS You Also Rent

The platform is the most visible tenancy arrangement but not the only one. Kestrel & Pine's WMS, returns portal, marketplace connector, iPaaS, shipping service and, from 2027, OMS (order management system) and PIM (product information management system) are each a multi-tenant SaaS in which Kestrel & Pine is one tenant among the vendor's other customers. Three consequences follow.

The peak day is shared. A WMS vendor's other customers are retailers whose Black Friday is the same Friday. The vendor's capacity planning is your risk, and its per-customer rate limit is its tool for being fair when every customer peaks at once. The WMS limit that throttled Kestrel & Pine's order flow was doing for that vendor's other tenants what the platform's budget does for its own.

The pricing unit tells you what the vendor meters. Operational SaaS commonly prices by volume: Picqer, a Dutch WMS, publishes plans from €379 a month including 3,000 orders and €0.28 per extra order (n.d., accessed November 2026). A per-order price means the vendor's capacity model is per order, and a peak day costs money as well as calls (Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026).

Residency is per component. Each SaaS was provisioned in a region at sign-up. A European merchant should record which region holds the data at the platform, the PIM, the OMS, the returns portal and the search service; the answer differs per vendor and moving is a migration (Whose Data Is It? Privacy, Consent and Retention).

The questions for every vendor are the same four: what is your unit of isolation, and can you move or restore my tenant alone; what are my limits, per what, and do they change under load; which region holds my data; and what happened on your busiest day last year.

The Merchant's Choices and Their Trade-Offs

A merchant cannot change how its vendor pools tenants, but it has four realistic responses to sharing.

Accept the pool and govern the budget. Keep the tenant and the plan; inventory every spender of the API budget; decide an order of priority; replace polling with webhooks and events; use bulk operations for catalog work; remove what nobody uses. It is cheap, and it is where Kestrel & Pine starts.

Buy a larger share. Move up a plan tier for a bigger bucket, or negotiate an enterprise agreement with higher or, as some vendors advertise for their largest customers, no fixed limits. It is a subscription line, not an engineering project, and it does nothing for the WMS vendor's limit or the marketplace's.

Rent or run a silo. A platform-as-a-service or on-premises edition (Adobe Commerce's two non-SaaS options are the public example) gives dedicated capacity and your own database at the price of running upgrades, capacity and security yourself (Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says).

Split tenants. One tenant per country or brand, as in the table above.

	Govern the budget	Buy a larger share	Rent or run a silo	Split tenants
Cost	Engineering days; an inventory and some redesign	A plan tier or enterprise contract	An operations capability and a larger team	Integrations, catalogs and reconcilia-tion per tenant
Protects against	Your own apps starving your order flow	Running out of budget at your own peak	Other tenants' peaks; vendor-side limit changes	One market's peak starving another's

	Govern the budget	Buy a larger share	Rent or run a silo	Split tenants
Does not protect against	Other tenants' load; the vendor lowering limits	The WMS or marketplace limits; the vendor lowering limits	Your own capacity mistakes; it is now your problem	Shared vendors beneath the tenants
Right when	Always first; every stage	The budget is measured and still short	Group stage with platform teams, or a regulatory reason	Legal entities, assortments or teams genuinely differ

The recommendation is the order of the table: govern first, buy second, and treat silos and splits as decisions with their own chapters' costs. It assumes the merchant can measure its budget consumption; a merchant that cannot see the logs should get them first.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting (under about €2M)	A pooled tenant on a standard plan; a handful of apps; the vendor's region chosen by default	A list of installed apps with their permissions; knowing which region holds the data	A 429 or throttled response in a connector log that nobody reads; an app that polls the whole catalog every hour

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
2 — Growing (about €2M–€20M)	A first ERP connector, a 3PL (third-party logistics provider) feed, a marketplace tool, all spending the same budget; the first peak that throttles	An inventory of budget spenders; webhooks and events for stock and orders; one owner per connector; a question to each vendor about its limits	The order export falls behind on a promotion; two apps write stock; the agency cannot say how many calls a flow makes
3 — Omnichannel (about €20M–€200M)	One tenant for several countries; a dozen multi-tenant SaaS products; the budget is the peak-day bottleneck	A budget allocation with the order flow first; batching within the WMS limit; bulk operations for catalog; a decision on one tenant or several per market; residency recorded per component	A market's peak starving another's; a vendor's limit change breaking a flow; a country that needs what the platform cannot vary per market
4 — Group (over about €200M, or several brands)	Several tenants or brands; enterprise agreements; some dedicated capacity; shared OMS and PIM beneath	Vendor-side isolation in the contract (pods, dedicated instances, restore of one tenant); the group's own shared services built with a tenant identifier on every row; peak capacity reserved per brand	One brand's sale starving another's on a shared connector; a group service that cannot restore one brand alone; a vendor that cannot say what its isolation unit is

The judgement: tenancy does not change what a merchant must do so much as how much of it. At every stage the first move is the same, know who spends your budget and put the order flow first; what grows is the number of budgets, the number of markets sharing them and, at Group stage, the merchant's own duty to isolate its brands the way its vendors isolate their tenants.

In Practice: Kestrel & Pine's API Budget for 2027

The peak plan in Black Friday Every Year: Peak Load, Resilience and Recovery forecasts 60 orders a minute for Black Friday 2027 and tests every link at 80. Its tenancy decisions, from Ruben's June 2027 analysis (all figures illustrative):

Spender of the platform's API budget	Black Friday 2026	Black Friday 2027 plan
Order flow to the warehouse	One order at a time through the iPaaS, about 20% of the budget, throttled about 1,900 times in two hours	Orders created in the OMS (W2), which sends fulfillment orders to the WMS in batches within the WMS limit; webhooks from the platform instead of polling; first call on the budget
Marketplace connector stock sync	Hourly full-catalog write of about 17,000 active variants, the largest single spender	Stock by events from the availability service (W1), changed variants only; if the service misses its 15 October go/no-go, the fallback is the old chain with the ten-unit marketplace buffer
Three apps writing inventory	Each polled and wrote stock on its own schedule	None may write stock after W1; the pre-order app uses an allocation in the availability service; six unused apps removed in the audit

Spender of the platform's API budget	Black Friday 2026	Black Friday 2027 plan
ERP connector	Hourly	Daily batch for finance (W5); nothing in the peak hours
Returns portal, reviews, e-mail	Small, unmanaged	Small, listed, each with an owner and a scope
WMS vendor's own limit	About 30 calls a minute per customer; four calls per order, one at a time, about seven orders a minute	Fulfillment orders in batches within the limit; tested at 120 orders a minute with marked orders the WMS skips

Two decisions carry over. The three storefronts stay in **one tenant**: the platform varies price, language and tax per market, the catalog is one assortment, and the budget analysis showed that governing the spenders, not splitting the markets, was what the order flow needed. And the plan records the **region** of every SaaS component Kestrel & Pine rents, the platform's EU region among them, so that the privacy work in Whose Data Is It? Privacy, Consent and Retention starts from a list rather than a search.

When It Goes Wrong

A neighbor's flash sale on your pod. The documented case is Shopify's 2021 example of two flash-sale merchants on one pod with synchronized bursts; the vendor's remedy was to move one. Detection: slow responses on a day that is not your peak, with your own logs quiet. Recovery: there is none on your side; report it, and ask the vendor what its isolation unit is and whether it rebalances.

A noisy neighbor inside your tenant. Your own stock app consumes the budget your order flow needs. Detection: throttled responses in the order flow's logs while another app's call count climbs; the gauge that was missing in 2026, queue depth with rate-limit headroom. Recovery: pause the app (a switch on the degradation ladder), then redesign it to events and bulk operations.

The vendor lowers the limit. "Shopify may temporarily reduce API rate limits to protect platform stability" is a published possibility, and other vendors have the same right in their terms. Detection: throttles at call rates that worked yesterday. Recovery: a flow that reads the budget headers and slows itself, rather than one that assumes a fixed rate.

The webhook that unsubscribed itself. Your endpoint failed eight times in a row

during an outage and the platform deleted the subscription. Detection: silence; a reconciliation job that finds orders in the platform and none in your system. Recovery: re-subscribe, replay from the platform's API, and keep the reconciliation job (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation).

The shop move. A vendor moves your shop between shards; the documented mechanism takes a short write lock at cutover and fails the move if the lock cannot be acquired in time. Detection: a brief burst of write failures with no change on your side. Recovery: idempotent retries; it should pass in seconds, and the vendor's status page should say so.

The shared peak at a specialist vendor. Your WMS, returns portal or connector slows because every tenant peaks at once. Detection: throttles and latency at the vendor's end, correlated with the calendar rather than with your load. Recovery: batch within the limit, queue behind the order, and put the vendor's busiest-day record into next year's contract talk.

What to Take Away

Established.

- Multi-tenant SaaS arranges tenants as pool, silo or bridge, and the noisy-neighbor problem is inherent to pooling (AWS SaaS Lens, n.d.); the documented commerce mechanisms are a tenant identifier on every row, a router, isolated database groups, resource limits and regions (Shopify 2018 and 2021, Salesforce n.d., commercetools n.d.).
- Rate limits on commerce platforms are granted per app and per store, tiered by plan, and can be reduced by the vendor under load (Shopify documentation, 2026); buyer traffic and checkout are protected ahead of integrations.
- Webhook delivery is not guaranteed, is unordered, and a subscription can be deleted after repeated failures; consumers must be idempotent and reconcile (Shopify documentation, 2026).

Depends on context.

- One tenant for several countries is right while the platform can vary what differs by market and one budget can carry every market's peak; it stops being right when a legal entity, assortment or operations team genuinely differs.
- Buying a larger budget is right when consumption is measured and still short; it is wrong as a substitute for removing the spenders, because it does nothing for the WMS's or the marketplace's limits.
- A silo (platform-as-a-service or self-hosted) is right at Group stage with a platform team or under a regulatory requirement; below that it trades a vendor's shared operations for your own.

The author's synthesis.

- Tenancy is the vendor's architecture, but the budget is the merchant's: treat the API budget per store as a capacity you plan, with the order flow first and a named owner for every spender, because on a pooled platform that budget is the only isolation you control.
- Every SaaS you rent should be asked the same four questions (unit of isolation, limits and their changes, region, busiest day), and the answers belong in the vendor card next to the price, because a vendor's tenancy is a promise about your peak day.

Questions to Ask

1. Which apps and connectors spend our platform's API budget, how much each, and in what order of priority on a peak day?
2. What is our plan's budget per app per store, and may the vendor lower it under load? Do our flows read the budget headers and slow down, or assume a fixed rate?
3. For each multi-tenant SaaS we rent (WMS, OMS, PIM, returns, connector, iPaaS): what is its limit per customer, and how did it behave on its busiest day last year?
4. What is each vendor's unit of isolation? Can it move, restore or rebalance our tenant alone, and how would we know it happened?
5. Which region holds our data at each vendor, and what would it take to change it?
6. Are our markets or brands one tenant or several, and which condition (legal entity, assortment, operations team, budget) would change that answer?
7. Which flows poll where they could subscribe, and which write the whole catalog where they could write only the changes?
8. If a webhook subscription were silently deleted tonight, which reconciliation job would notice by morning, and is there a tenant identifier on every row of every shared service we have built ourselves?

The budget a merchant spends on a shared platform is spent mostly by apps, and apps are the next chapter's subject: how platforms let merchants and partners extend them without breaking them, what an app may touch, and how a merchant with forty-one apps, six unused and three writing stock, should govern them: Apps, Plugins and Functions: Extensibility and Ecosystem Governance.

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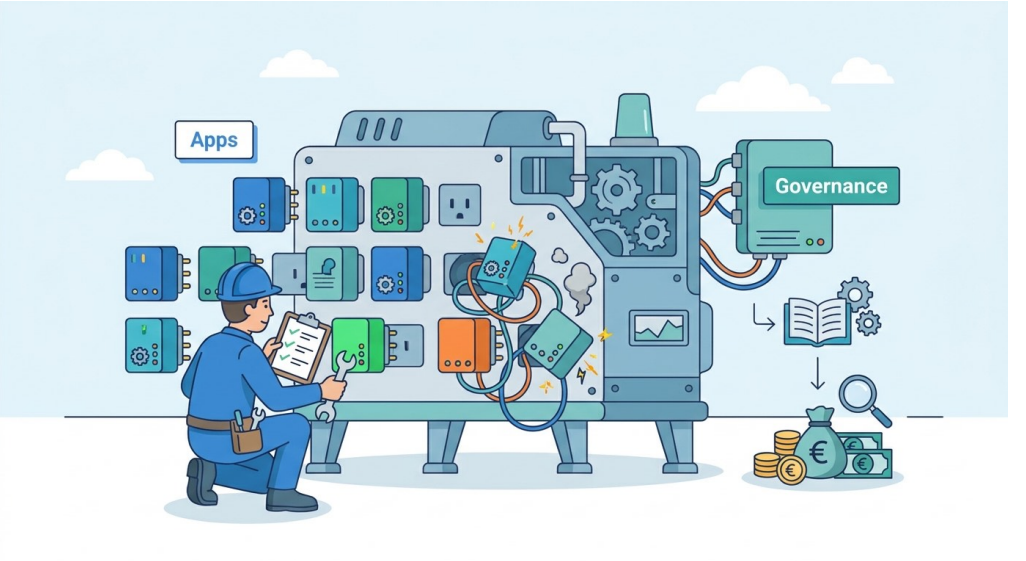
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

31. Apps, Plugins and Functions: Extensibility and Ecosystem Governance



IN THIS CHAPTER, YOU WILL: Learn how platforms let other people's code change them, what each mechanism may touch, why installed apps sprawl, and how to govern the apps a merchant runs.

KEY POINTS:

- **Extension mechanisms differ by where the code runs and what it may touch.** A plugin inside the platform's process can reach anything, including the database; an app outside it may do only what its granted permissions allow; a function at a named extension point gets a fixed input, a fixed output and a time budget. The deeper the code sits, the more every vendor release costs.
- **The app store is the vendor's business model and the merchant's governance problem.** Scopes, app review and dated API versions protect the platform and the shopper; none of them tells the merchant

which of its installed apps writes stock, prices or orders. That inventory is the merchant's own work, and the App That Wrote Stock is what happens when nobody does it.
- **Govern by owner, scope and writer, and keep custom behavior out of the core.** One named owner per app, its scopes on a card, one writer per business fact, a quarterly uninstall review, and custom logic in the platform's bounded extension points rather than in a modified core. A platform you cannot upgrade is a platform you have already left.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Secret Agent** — an app that writes stock, prices or orders without a named owner, its own identity or anyone's approval is this pattern in its most common form.
- **The Forked Platform** — in-process customization of a platform's core is the mechanism that makes a platform un-upgradable, and the vendors describe the cost in their own documentation.
- **The Spaghetti Junction** — forty-one apps, each with its own path into the platform's data, behave like forty-one hand-built connectors, many of them without an owner.

A merchant needs a size guide, a pre-order flow or a loyalty scheme that its platform does not provide. An app or plugin can add that behavior quickly. To do its job, the extension may also need access to customer records or permission to change prices and stock. Choosing the feature therefore includes deciding which business facts another piece of software may control.

The risk becomes clear when extensions change the same fact. A stock badge only reads inventory; a pre-order app may change it while a warehouse feed overwrites the same number. Both can work as designed and still give shoppers an unreliable availability promise. Assigning permissions and an owner at installation gives the merchant a way to prevent that conflict and someone to investigate when it occurs.

Kestrel & Pine, the fictional outdoor brand, supplies the example: **41 apps, 6 unused and 3 writing stock**. Head of Engineering Noor Visser introduces an ownership rule in April 2027.

Where we are in the order's journey. Eva's jacket page includes a size guide, stock badge and review widget; a notification app later sends tracking. Each extension reads data, and some can change it. This chapter establishes what they may touch and who answers for the result.

Setting the Scene at Kestrel & Pine: Forty-One Apps and One Spreadsheet

In April 2027, Noor and storefront manager Milan Petrov inventory all 41 apps. Milan knows about twenty well. The agency installed 22 over several years; others came from campaigns, pilots and former staff.

The illustrative bill is about **€2,900 monthly**, spread across platform and agency invoices and three company cards. Recording each app's purpose, owner and permissions lets Noor identify unused subscriptions and conflicting stock writers. A count of installed apps alone would reveal neither which costs to stop nor which behavior needs changing.

Category	Apps	Unused	Write platform data	Example
Storefront features (size guide, reviews, badges, pop-ups, bundles)	14	3	0	A "low stock" badge that reads inventory 40 times a minute
Marketing and analytics (pixels, e-mail, referral, A/B testing)	11	2	1 (customer tags)	An e-mail app writing customer tags for segmentation
Operations (pre-orders, returns, shipping rates, invoices, stock sync)	9	0	5 (stock, orders, fulfillment)	The pre-order app and an inventory-sync app the agency installed
Finance and compliance (tax, cookie consent, fiscal receipts)	4	0	1 (order metadata)	The Belgian cookie-consent banner
Agency utilities and pilots	3	1	0	A search pilot nobody switched off
Total	**41**	**6**	**7**	**3 of the 7 write inventory**

Three apps can write inventory, separately from the main warehouse feed. That

feed copies **warehouse management system (WMS)** counts through the **integration platform as a service (iPaaS)**, the hosted flow tool, into the platform hourly (APIs, Events, Files and Middleware: How Commerce Systems Talk).

One app copies stock adjustments between locations. A pre-order app reserves units by altering the same number the warehouse feed later overwrites. This caused **the App That Wrote Stock** incident in March 2026: about 600 variants showed phantom reservations for two weeks. A third app still has write permission for a sales channel no longer used. All three need an explicit decision about their access.

Noor's question to Milan was not "which app did it". It was: **who in this company is allowed to let a program write stock, and how would we know it had?** Nobody was, and nobody would. That is the question this chapter answers.

Five Ways to Extend a Platform

A commerce platform is a promise-keeping machine that many businesses share. Shopify's annual report for the year ended 31 December 2025 (filed 11 February 2026) describes its platform as " 'single branch' software, which means that all of our merchants use the latest version", serving "millions" of merchants with "more than 21,000 apps" in its App Store at year end. One codebase cannot carry a million businesses' particular needs, so the long tail of behavior lives in apps; at a different scale, that is why SAP Commerce describes itself as "a flexible, modular architecture based on extensions".

Three things decide how dangerous an extension is: **where it runs** (inside the platform's process or outside it), **what it may touch** (everything the platform can reach, or only the data its permissions cover) and **how tightly it is coupled to the platform's version** (whether a release can break it silently, or whether it talks to a versioned interface with a stated support window).

In plain words, an **extension point** is a named place in the platform's flow where the platform agrees to run, or call, someone else's code. Platforms offer five kinds.

Mechanism	Runs where	Can touch	Coupling to the platform's version	Dated examples
In-process plugin, **module**, **extension**, **cartridge**	Inside the platform's program, in its language	Anything the code can reach, including the database	Tight: re-tested on every release	Adobe Commerce plugins; SAP Commerce extensions; WordPress hooks under WooCommerce; Shopware plugins; Salesforce B2C Commerce cartridges
Theme code	Inside the storefront renderer (templates and scripts)	Presentation, and through scripts the whole page	Medium: theme updates can conflict with edits	Liquid themes (Shopify); Stencil themes (BigCommerce); SFRA, the Storefront Reference Architecture (Salesforce B2C Commerce)
App over APIs and webhooks	Outside, hosted by the app's vendor	Only what its granted scopes permit	Loose: versioned APIs with a support window	Shopify, BigCommerce, Shopware and Saleor apps; Adobe App Builder

Mechanism	Runs where	Can touch	Coupling to the platform's version	Dated examples
Serverless function at an extension point	Inside the platform's request path, in a sandbox	A fixed input and a fixed output	Loose, and time-boxed	Shopify Functions
Synchronous call-out	The platform calls the merchant's endpoint before saving	Validate or amend one resource	Loose; latency becomes the merchant's problem	commercetools API Extensions

In-process: the plugin can do anything, which is the problem

An **in-process plugin** runs inside the platform program itself, in the same language, and can therefore do anything the platform can. Adobe Commerce's developer documentation describes its plugins (also called interceptors) as code that changes a public method's behavior by running *before*, *after* or *around* it without modifying the class; an "around" plugin that never calls the original method skips it entirely, several plugins on one method are ordered by a `sortOrder` setting or else by module load order, and around plugins "should be avoided unless they're required, because they add to stack traces and can hurt performance" (n.d., accessed November 2026). Read that as a newcomer: a plugin can replace the platform's own logic for computing a price or saving an order, and two plugins can disagree about who goes first.

WooCommerce's extension guide adds the warning that matters most: "backwards compatibility between WooCommerce releases is not guaranteed" (n.d., accessed November 2026). Salesforce B2C Commerce layers **cartridges** (packages of templates and code) over a base storefront and tells developers "you never edit `app_storefront_base` directly", which "keeps the backward compatibility guarantee intact" (Trailhead, n.d., accessed November 2026): extend by layering over the core, never by editing it. Shopware draws the line most explicitly, because it sells both kinds: plugins "are executed within the Shopware Core process", "have direct access to the database", and therefore "are not supported by Shopware cloud"; apps "are not executed within the process of the Shopware Core" and interact through webhooks and the admin API (n.d., accessed November 2026). A vendor that runs the platform for you cannot let your code run inside it.

Theme code: the storefront's own extension point

Theme code is the set of templates and scripts the platform renders into pages; Delivering the Storefront: Themes, Headless, Rendering and Performance owns it. It is a second place where apps land: before Shopify's Online Store 2.0 (announced 29 June 2021), apps typically injected code into theme files and uninstalling could leave "ghost code" behind, which theme app extensions were introduced to end. Agency-edited theme files are the storefront's version of the Forked Platform: the more templates edited by hand, the more a theme update becomes a merge.

Apps over APIs: outside the process, limited by scopes

An **app** in the hosted-platform sense is a separate program, run elsewhere, that talks to the platform through its APIs and receives **webhooks** (push notifications the platform sends to the app's address when something changes; Promises Between Systems: API Contracts, Webhooks and Versioning). It may only do what its granted **scopes** allow: a scope is one named permission the app asks for at install, such as "read orders" or "write inventory", shown to the merchant on the consent screen. Shopify's documentation gives `read_inventory` and `write_inventory`, `read_orders` and `write_orders` as examples and notes that order scopes "cover orders created within the last 60 days" unless a broader scope is approved (n.d., accessed November 2026). Saleor describes its apps the same way, as external services over webhooks and the API.

Apps are loosely coupled because the API is versioned. Shopify "releases a new API version every three months", supports each stable version "for a minimum of 12 months" and, if an app targets a retired version, "falls forward and responds using the oldest accessible stable version" (n.d., accessed November 2026). The price of looseness is that every app carries an upgrade obligation of its own.

Functions and call-outs: a fixed input, a fixed output, a time budget

The newest mechanism runs the merchant's small program at a named moment in the platform's own flow, with a fixed input, a fixed output and a limit on how long it may take. Shopify Functions plug into targets such as discounts, delivery and payment customization, cart and checkout validation and order routing. The function is compiled to WebAssembly (a portable binary format that runs in a sandbox), receives JSON the platform produces, returns a list of operations, cannot be called by URL and has no network access; the documented limits for API version 2026-10 are 11 million instructions per run for carts of up to 200 lines, 10 MiB of memory and a 256 KiB module. The earlier mechanism, Scripts (Ruby code running inside checkout for Plus merchants), was retired on 30 June 2026 (help center, accessed November 2026).

commercetools' **API Extensions** are the synchronous call-out variant: after a create or update on a cart, order, payment or customer is processed but before it is saved, the platform calls the merchant's endpoint, which may reject the change or return update actions. The documentation is candid: "if it fails or takes a second longer to return a result, the whole API call fails or takes a second longer." The default budget is 2 seconds to connect and 2 seconds to respond, configurable to 10 with a support request; the recommended target is about 50 milliseconds; a project may have 25 extensions; a failure fails the API call and is not retried (n.d., accessed November 2026).

The common design is the lesson. **An extension point is a named moment with a fixed input, a fixed output, a time budget and a failure rule.** The time budget is the vendor protecting checkout from the merchant's code, which is the architectural fact behind the claim in From Cart to Confirmation: Checkout Orchestration that checkout is the most protected part of a platform.

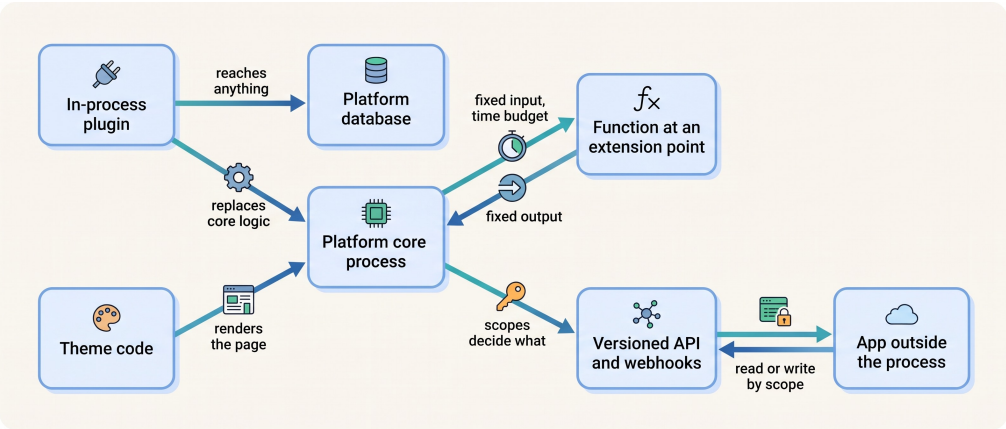


Figure 1: *Where an extension runs decides what it can touch: a plugin inside the process reaches the database and can replace core logic; a function gets a fixed input and a time budget; an app outside the process reads and writes only what its scopes allow.*

What each interaction needs from the platform

The requirements differ by interaction; time budgets are the vendors' documented ones where given and assumptions otherwise.

Interaction	Mechanism	Time budget	On failure	Who owns the outcome
Checkout validation or discount	Function or call-out	Milliseconds (Shopify: an instruction count; commercetools: 2 s default)	The platform fails the call or skips the function by its documented rule	The merchant; the shopper waits
Stock write from the warehouse	App over API	Seconds, asynchronous *(assumption)*	Retried; stale stock until it succeeds	The integration owner
Order created notification to an app	Webhook	Shopify: 1 s to connect, 5 s total; 8 retries over 4 hours	After 8 consecutive failures the subscription is deleted	The app's owner, and the merchant
Core method replaced (price, tax, save order)	In-process plugin	Whatever the plugin takes	The platform does what the plugin does, right or wrong	The merchant, entirely

Every row but the last has a platform rule for failure. The in-process row has none, because the platform no longer knows what it is doing.

The App Store: Business Model and Governance

An **app store** is where a platform vendor lists third-party apps, reviews them, bills for them and takes a share. It is a business and a governance mechanism, and the two are not the same.

As a business. Shopify's revenue-share page states that developers keep all of the first $1,000,000 of lifetime gross app revenue (counting revenue earned on or after 1 January 2025) and pay 15% above it (n.d., accessed November 2026). The app store aligns the vendor's and the developers' interests around installs, which is why it is full, and why nothing in it is designed to reduce the number of apps a merchant runs.

As governance. The review requirements are the platform's and the shopper's protection, and read like a list of the harms the vendor has seen:

- **Least scope**. Shopify asks that an app request only the scopes it needs. BigCommerce states that "Apps configured with all scopes aren't approved", that a checkout-content scope triggers a security assessment, and that apps handling card data need a PCI review (against the card industry's Payment Card Industry security standard) (both n.d., accessed November 2026).
- **Performance and currency**. A Shopify public app "shouldn't lower a store's Lighthouse performance score by more than 10 points" (Lighthouse is Google's page-speed audit tool), and apps using APIs deprecated within 90 days cannot be submitted.
- **Clean uninstall**. BigCommerce requires that uninstalling "return the control panel and storefront to their pre-install state" and remove storefront scripts.
- **Privacy**. Every Shopify App Store app must implement three compliance webhooks (customer data request, customer redaction, shop redaction) and complete the request "within 30 days". The deletion request that touches nine of Kestrel & Pine's systems in Whose Data Is It? Privacy, Consent and Retention reaches its apps this way.
- **Billing through the platform**, so the merchant sees charges on one invoice.

Now read what the list does not contain. **Nothing in app review tells the merchant that two of its installed apps both write inventory**. Each app passed review on its own; each asked for `write_inventory` for a reason; the merchant approved each consent screen months apart. The platform cannot know that, in your business, the warehouse feed is the system of record for stock and the pre-order app is not, because that is a decision about your business (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record). The App That Wrote Stock is a governance gap on the merchant's side, and no vendor can close it.

What an App May Touch: The Writer Problem

The most important question about any extension is not what it does but **what it writes**. A reader can be slow, wrong or expensive. A writer can change a fact that other systems trust. In March 2026 Kestrel & Pine's platform had two writers of one fact: the WMS feed **set** stock levels every hour to what the warehouse counted, and the pre-order app **adjusted** stock by the quantity pre-ordered. Each operation was correct alone. Together they fought, and the number the storefront showed drifted away from both.

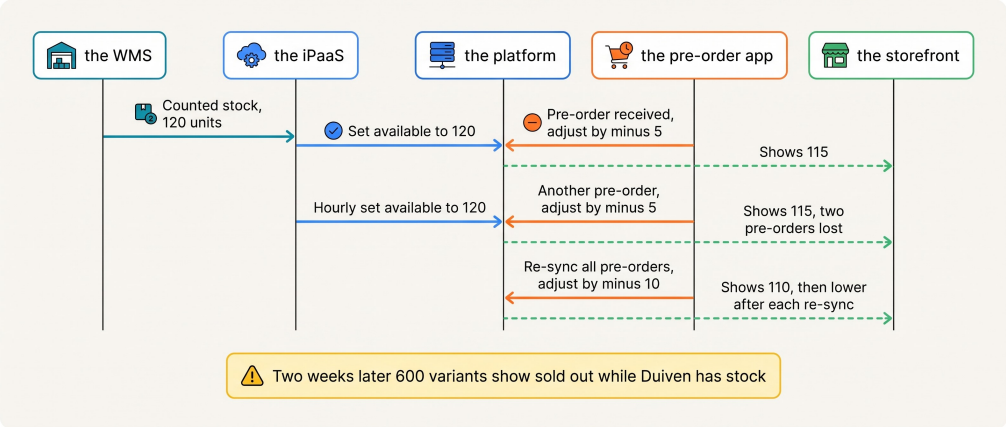


Figure 2: *Two writers of one fact: the warehouse feed sets stock to the counted quantity, the pre-order app adjusts it by reservations, and the number the storefront shows drifts away from both.*

Shopify's inventory documentation (n.d., accessed November 2026) shows the distinction cleanly: its API offers `inventoryAdjustQuantities` (a change per state) and `inventorySetQuantities` (an absolute quantity), and models stock per location in states such as *available*, *committed* (set aside for placed orders) and *on hand*.

Kestrel & Pine's platform is not named and the mechanism is general: **set and adjust do not mix on the same number**, and a reservation belongs in its own state, not subtracted from the physical count. Is It in Stock? Inventory, Availability and the Promise You Can Keep owns the states and the availability service that Waypoint's decision W1 introduces; this chapter owns the rule that got Kestrel & Pine there: **one writer per fact**.

For each fact the platform holds there is one system of record and at most one path by which it is written into the platform; every app that holds the write scope on that fact is either that path or a mistake.

Fact in the platform	System of record (Kestrel & Pine, after Waypoint)	The one writer into the platform	Apps that may only read
Physical stock and availability	The WMS for physical stock; the availability service for what each channel may promise (W1)	The WMS feed through the iPaaS, until the availability service replaces it (Q4 2027)	Badges, feeds, the pre-order app

Fact in the platform	System of record (Kestrel & Pine, after Waypoint)	The one writer into the platform	Apps that may only read
Prices	The platform's market price lists for consumer prices; the ERP's (enterprise resource planning system's) for wholesale (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax)	Price events with effective dates from the price lists	Repricing and marketplace apps
Product content	The PIM, product information management system (W3)	The PIM's publication job	Search-engine optimization (SEO) and feed apps
Orders	The OMS, order management system (W2); the platform keeps checkout and order history	The checkout, and the OMS for status	Notification, invoice and analytics apps

Writing is not forbidden (the returns portal creates exchange orders through the platform's API, with an owner and a reason); every writer is named.

The Forked Platform: When the Core Has Been Changed

The Forked Platform is a platform whose core or theme has been customized so deeply that it can no longer be upgraded, so every vendor release is a project and, eventually, no release is taken. The vendors who offer in-process mechanisms describe it best.

Adobe Commerce's upgrade guidance says: "The amount of custom work and the number of installed third-party modules strongly affects the complexity of an upgrade", "The quality of the custom work and modules can determine whether an upgrade goes smoothly", skipping releases makes the next upgrade harder, and if upgrades cost more than expected, audit the project (n.d., accessed November 2026).

Adobe ships an Upgrade Compatibility Tool that checks a customized instance's modules and core code against a target version and reports "critical issues, errors, and warnings" (updated 8 October 2026).

Its reasoning for its newer out-of-process model is the same lesson from the vendor's side: in-process code "must stay compatible with each Commerce version and its runtime", while App Builder apps can be "developed, deployed, and operated on their own schedule"; the trade-off is that in-process extensibility offers "deep, synchronous control of the Commerce runtime" that out-of-process does not (n.d., accessed November 2026).

The clock that turns a postponed upgrade into a forced migration is the support policy. Adobe's lifecycle page (accessed November 2026) gives each release line roughly three years of standard support from general availability. Version 2.4.4 reached general availability on 12 April 2022 and left standard support on 12 April 2025; 2.4.7 (general availability 9 April 2024) is supported to 31 May 2027. From 1 June 2027 Adobe says it will no longer maintain cloud environments running unsupported versions, including "suspending traffic to the affected infrastructure", so the storefront "will go offline". A merchant still on 2.4.4 has a dated deadline and a bill that grows with every module.

Hosted platforms cannot be forked, but they withdraw extension points, which is the same risk from the other direction. Shopify's checkout template (`checkout.liquid`) stopped working for in-checkout pages on 13 August 2024, by a notice of 13 February 2023; non-upgraded checkouts were "automatically upgraded to an un-customized, out-of-the-box Shopify Checkout"; Scripts followed on 30 June 2026 (From Cart to Confirmation: Checkout Orchestration tells the story). **An extension point is a promise with a date**, and the merchant's inventory must record which date each app depends on.

So open source and suites put the upgrade risk in the merchant's hands (the merchant may change the core, and must manage module count, module quality and an upgrade calendar under the vendor's support clock), while hosted SaaS puts the deprecation risk there instead (the merchant cannot change the core, and must manage app scopes, API versions per app and the vendor's deprecation calendar). Neither is safe, and both are managed by the same inventory.

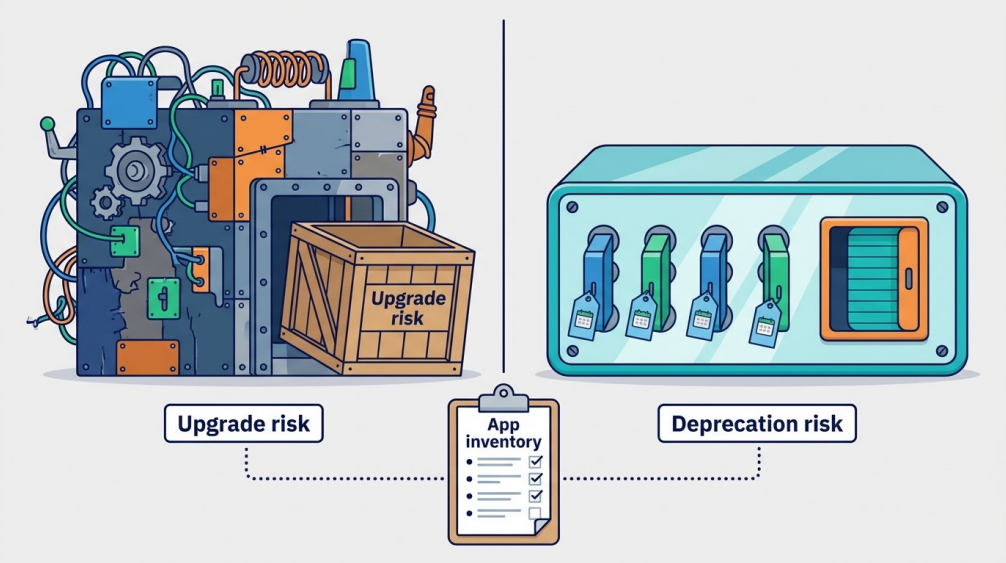


Figure 3: *Open-source and suite platforms put the upgrade risk in the merchant's hands and hosted platforms put the deprecation risk there instead; an extension point is a promise with a date, and one app inventory manages both.*

The Choices: Where Custom Behavior Should Live

When a merchant needs the platform to behave differently, there is a ladder of places to put the change. I climb it from the top and stop at the first rung that works: **configuration** (most "customizations" are settings the platform already has; Pete Hodgson's 2017 toggle taxonomy helps decide whether a switch is a release, ops or permissioning toggle, and so whether it belongs in configuration at all); **a platform-native extension point** (a function, a call-out or a workflow tool such as Shopify Flow, which "monitors your store for events" and runs actions); **a public app** for commodity capabilities; **a custom app over the API** for integrations and special behavior, with an owner and an API-version pin; **an in-process plugin** only where the platform allows it and the behavior needs synchronous control of the core; and **editing the core**, never.

Choice	Upgrade coupling	What it may touch	Latency risk to checkout	Who maintains it	When it is right
Configuration	None	What the vendor exposes	None	The merchant's admin	Always try first

Choice	Upgrade coupling	What it may touch	Latency risk to checkout	Who maintains it	When it is right
Native extension point (function, call-out, workflow)	Loose, versioned	Fixed input and output	Bounded by the vendor	The merchant or agency	Checkout, discount and routing rules
Public app	Loose	Its scopes	Theme scripts can slow pages	The app vendor	Commodity capabilities
Custom app	Loose	Its scopes	None, if asynchronous	The merchant (named owner)	Integrations; special behavior
In-process plugin	Tight	Anything	Whatever it takes	The merchant, on every release	Rarely; synchronous core control
Core edit	Total	Anything	Whatever it takes	Nobody can	Never

The recommendation assumes a merchant with a small team that does not sell software; a Group-stage merchant with a platform team on a suite can afford a disciplined, budgeted in-process layer, and Kestrel & Pine, with three storefront-and-apps engineers and an agency, cannot.

Across Scale Stages: When It Fits and When It Limits

The mechanisms do not change with scale; the governance does, and the stage decides which failure arrives first.

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
1 — Starting (Kestrel & Pine 2016–2018)	A hosted platform and ten to twenty public apps installed by the founder or the agency; no custom code beyond the theme	A list of installed apps with their scopes, kept from day one; one person who approves installs	Two apps for the same job; an app billed to a card nobody checks; theme edits by hand
2 — Growing (2019–2022)	The first custom app or connector (ERP, 3PL or third-party logistics provider, marketplace); thirty apps; agency-built theme features; on open-source platforms, the first in-process modules	A named owner per app and connector; the API version each targets, written down; one writer each for stock and prices; out-of-process mechanisms preferred even where in-process ones are allowed	Stock or price disagreements traced to an app; an upgrade postponed because "the modules might break"; the agency as the only one who knows what an app does
3 — Omnichannel (2023–now)	Forty or more apps across storefront, marketing, operations and compliance; apps that write stock, orders and customers; several countries in one tenant	A governance card per app; a quarterly uninstall review; scopes revoked when a system of record moves (W1, W2, W3); on in-process platforms, an upgrade calendar set by the vendor's support clock and a module budget	A phantom-stock or price incident caused by two writers; app subscriptions nobody can account for; a deprecation discovered after its date; a platform release more than a year behind

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
4 — Group	Several brands, each with its own app list; platform teams; possibly in-process customization on a suite	A shared extension catalog across brands; one upgrade calendar for every vendor; app security review inside vendor management (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers); the same contract discipline for the group's own APIs	Each brand solving the same need with a different app; a module written for one brand blocking the group's upgrade; **One Size Fits None** when brands are forced to share apps that fit none of them

The judgement: extensibility is cheap to adopt and expensive to govern, and the bill arrives at the Omnichannel stage as two writers of one fact. A merchant that keeps the list from the Starting stage pays almost nothing for it; one that starts at forty apps pays a week of archaeology, as Kestrel & Pine did.

In Practice

Dated public examples

- **A hosted platform withdrawing in-process code (2021–2026).** Shopify announced WebAssembly-based Functions in beta on 29 June 2021, deprecated its checkout template by a notice of 13 February 2023 effective 13 August 2024, and retired Scripts on 30 June 2026. Over five years the platform replaced free-form code inside checkout with sandboxed, time-budgeted extension points; read that as the direction every hosted platform takes as it grows.
- **A suite vendor moving extensibility out of process (2025–2026).** Adobe's Commerce as a Cloud Service, whose release notes begin in November 2025,

extends through App Builder, API Mesh, events and webhooks rather than in-process modules, with updates delivered continuously and 30 days to evaluate them in a sandbox (Adobe, 20 October 2026). The vendor that documented the upgrade problem most candidly built its SaaS edition without the mechanism that causes it.

Kestrel & Pine: the app governance rule

Noor and Milan turned the April 2027 spreadsheet into a rule, approved by Mara Lindqvist alongside Waypoint and recorded on one page per app (the **app governance card**, reproduced in the Toolkit: Cards and Templates for Commerce Architecture Decisions). The rule has five lines.

1. **Every app has a named owner** in the business (Milan for storefront and marketing apps, Daniel Okafor's team for fulfillment apps, Sanne de Wit's team for tax and invoicing apps, Ruben Alves for connectors); the agency installs nothing without an owner's approval.
2. **Every app's scopes are recorded** on its card, with the API version it targets and the date that version leaves support.
3. **One writer per fact.** The WMS feed through the iPaaS is the only writer of stock into the platform until the availability service takes over in Q4 2027 (W1); consumer prices change only through the platform's market price lists and wholesale prices only through the ERP's (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax); the OMS will be the only writer of order status (W2). Every other write scope on those facts is revoked.
4. **A quarterly uninstall review.** Any app without an owner, without use in the quarter, or duplicating another is removed.
5. **Custom behavior goes to the lowest rung that works:** configuration, then the platform's functions and workflows, then a public app, then a custom app with an owner.

The results: the six unused apps were uninstalled in April 2027; inventory write access was withdrawn from all three stock-writing apps by June 2027, after the first weekly reconciliation in May had measured the damage (2.3% of active variants with a different quantity in the WMS and the platform, most from those apps); the pre-order app kept read access and records pre-orders without adjusting stock until the availability service handles pre-order quantities. Thirty-four apps remained in July 2027, at about €2,400 a month (illustrative). The saving is not the point. The point is that Noor can now answer, on one page, the question nobody could answer in March 2026: who is allowed to let a program write stock.

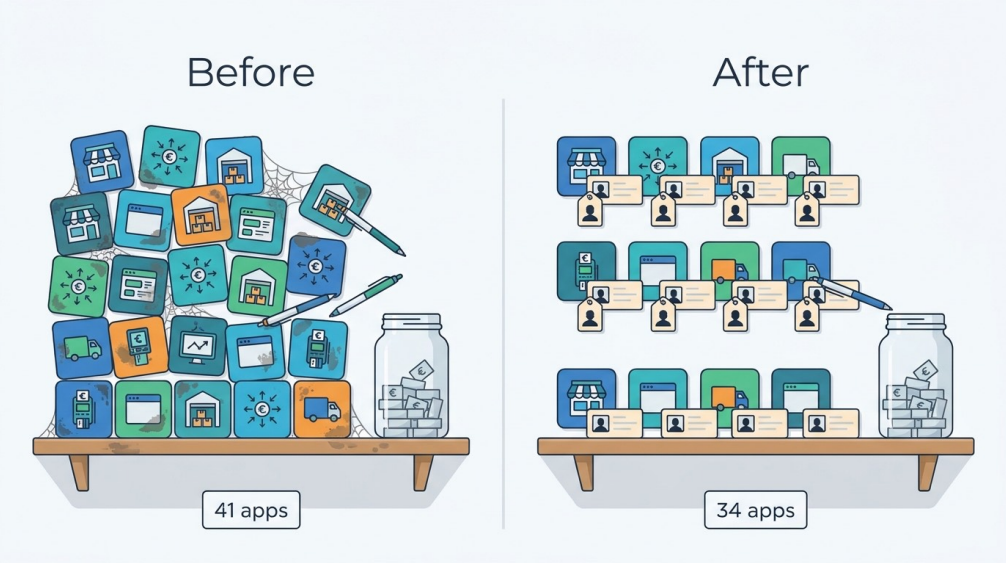


Figure 4: *The April 2027 spreadsheet found 41 apps with nobody's name on them; an owner per app, scopes on a card, one writer per fact and a quarterly uninstall review left 34 by July.*

App governance card	Kestrel & Pine example: the pre-order app
App, vendor, owner	Pre-order app, public, installed 2025; owner Milan Petrov
Purpose	Take pre-orders for next season's Ridgewalker colors
Scopes granted	read_products, read_inventory, write_orders (pre-order tags)
Scopes revoked and why	write_inventory, June 2027: the WMS feed is the only writer of stock (W1)
API version and support end	2027-01; supported at least to January 2028 *(illustrative; mirrors the 12-month rule of hosted platforms)*
Review date	October 2027
Exit plan	Pre-orders handled by the availability service and the OMS from Q4 2027; app removed after the spring 2028 season

When It Goes Wrong

Extensions fail the way integrations fail, with one difference: the merchant often does not know the extension exists.

Failure	What it looks like	Detection	Recovery
Two writers of one fact (the App That Wrote Stock)	Stock, price or status drifts between the platform and its system of record	The reconciliation job (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation) lists variants whose quantity differs; the platform's audit log, where it exists, names the last writer	Revoke the extra write scope; re-set the fact from the system of record; keep the reconciliation weekly
An app lost its webhook	An app stops receiving orders after an outage on its own side; Shopify, for example, deletes a subscription "After 8 consecutive failures" over 4 hours	Orders missing in the app; the platform's webhook delivery log	Re-subscribe; backfill from the API; webhooks need a reconciliation job, not trust

Failure	What it looks like	Detection	Recovery
The withdrawn extension point (the Pulled Rug)	Checkout customizations stop on the vendor's announced date; an app on a retired API version "falls forward" to a newer one with different fields	The deprecation calendar on the cards; the vendor's changelog	Move the logic to the replacement mechanism before the date; after it, the platform's default behavior runs, usually safe and always surprising
The un-upgradable core (the Forked Platform)	Each release is a project; releases are skipped; the support end approaches	The upgrade compatibility report; the module count; the version gap	Audit modules; move what can move out of process; plan the upgrade as a migration (Moving Without Burning the Bridge: Replatforming and Migration)
An app as the attacker's path, or an app nobody maintains	An app with broad scopes, or its vendor, is compromised (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers); or a capability customers use runs through an app whose vendor has gone quiet (the Lynchpin)	Scope review; vendor assessments; the review date and API-version lag on the card	Revoke and rotate credentials; the exit plan on the card

Every row is found faster, and fixed faster, by a merchant that can list its apps, their owners, their scopes and the facts they write.

What to Take Away

Established.

- Where an extension runs decides what it can touch and what it costs at upgrade time: in-process plugins can reach anything (Adobe, Shopware and WordPress documentation); apps over APIs only what their scopes allow (Shopify, BigCommerce, Saleor); functions and call-outs get a fixed input, a fixed output and a time budget (Shopify Functions, commercetools API Extensions).
- Upgrade cost scales with the number and quality of in-process customizations, and the vendor's support clock turns a postponed upgrade into a forced migration (Adobe Commerce's upgrade guidance and lifecycle policy).
- Hosted platforms withdraw extension points on published dates; Shopify's checkout template and Scripts timeline (2023–2026) is the reference case.
- App review protects the platform and the shopper (least scope, performance, clean uninstall, privacy webhooks); it does not tell a merchant which installed apps write the same fact.

Depends on context.

- Whether in-process extension is acceptable depends on who runs the platform and the size of the team: a merchant on a hosted platform has no choice; one on a suite with a platform team can run a disciplined, budgeted in-process layer; one with a small team and an agency should treat every in-process module as debt.
- How formal app governance must be depends on stage: a list with owners and scopes at Starting; a card per app, a quarterly review and a one-writer rule at Omnichannel; a shared extension catalog and an upgrade calendar at Group.
- Whether to build a custom app or buy a public one depends on whether the need is special: commodity capabilities belong to the store; integrations and business rules belong to a custom app or a function with a named owner.

The author's synthesis.

- I treat every installed app as an integration and give it what Waypoint's W5 gives every integration: an owner, a contract (its scopes and API version), idempotent handling where it writes, and a reconciliation job for the facts it touches. Forty apps are forty integrations.
- I would rather lose a feature than keep a second writer. When a business need seems to require a second app writing stock or prices, the need is real and the mechanism is wrong: the answer is a reservation state, a price list or an availability service, not another writer.

Questions to Ask

1. How many apps, plugins or extensions are installed, and can someone produce the list with an owner per app in an afternoon?
2. For each fact the platform holds (stock, price, product content, order status, customer), which single path writes it, and which apps hold a write scope on it anyway?
3. Which mechanism does each custom behavior use: configuration, a function or call-out, a public app, a custom app, an in-process module, or an edit to the core?
4. On an open-source or suite platform: how many in-process modules are installed, which version is running, and when does standard support for it end?
5. On a hosted platform: which API version does each app and connector target, and which deprecations fall in the next twelve months?
6. What time budget does the platform give our checkout customizations, and what happens to the shopper when they time out?
7. Who approves a new app, and if its vendor stopped maintaining it tomorrow, what is the exit plan on its card?

The apps that extend a platform mostly land on one surface: the storefront, where theme code, app blocks and scripts meet the shopper's browser. The next chapter follows that surface from the platform to the phone, compares themes with headless storefronts, and works through the headless proposal Noor declined in February 2027. Continue with Delivering the Storefront: Themes, Headless, Rendering and Performance.

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

32. Delivering the Storefront: Themes, Headless, Rendering and Performance

IN THIS CHAPTER, YOU WILL: Follow a product page from the platform to the shopper's phone, compare themes with headless storefronts, and learn when owning the storefront is worth it.

KEY POINTS:

- **A storefront is a rendering of facts other systems own.** It shows the catalog, prices, promotions, availability, search results and the cart, and hands the shopper to checkout; it owns none of them. Three rendering models exist (a theme rendered by the platform, a headless storefront on the vendor's runtime, a headless storefront anywhere else), and under all three the checkout stays on the platform.
- **Cache the shell, fetch the promise.** Product and category pages can be cached per market and language, which is what survives a peak day; the price for a logged-in account, live stock, the cart and the checkout

cannot, and are fetched after the cached page renders. The platform's full-page cache and a headless framework's regeneration are the same idea at two layers.
- **Go headless for a need the theme cannot meet, never for speed alone**. The performance evidence is directional, dated and self-selected; a themed storefront with disciplined app scripts can pass Google's thresholds. Headless is justified by a shared front end across brands, a content-led site or a business-customer portal with negotiated prices, and it costs a second deployable that a small team must run.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Giant's Boots** — headless storefront architectures are copied from brands with front-end teams of fifty, by merchants with three.
- **The Forked Platform** — a theme edited by hand for years is the storefront's version of a core that cannot take the vendor's updates.
- **The First Snowfall** — the storefront meets forty times its normal traffic once a year, and whether its pages are cacheable decides whether it stands.

A slow product page can lose a sale before the shopper reaches checkout. Replacing the code that builds the page may help, but the delay might come from a price service, a stock lookup or an app script. A new page that waits for the same slow answers can inherit the same problem. Establish what is delaying the shopper before choosing a replacement.

The **storefront** produces the pages shoppers use to find products and buy. A platform can render them from a theme; a **headless storefront** is a separate application calling the commerce engine through APIs. The latter gives the merchant more control over how pages combine catalog, price, search and cart data. It also makes the team responsible for running that application and keeping its displayed answers fresh.

Kestrel & Pine, the fictional outdoor brand, runs three localized storefronts from one agency-maintained theme. It has 41 installed apps and peak traffic 41 times an average hour. The example follows Head of Engineering Noor Visser's decision to decline a February 2027 headless proposal.

Where we are in the order's journey. Eva sees the Ridgewalker jacket on her phone before completing checkout at 21:14 on 13 April 2027. Price, stock and the next-day delivery promise shape her decision. From Cart to Confirmation: Checkout Orchestration, Taking the Money: Payment Providers, Authorization, Capture and Refunds and The Order Record: Lifecycle, State and the Order Management System follow the obligations created by that page.

Setting the Scene at Kestrel & Pine: The Headless Proposal

The agency proposes a separate storefront application on the platform vendor's **edge runtime**, servers near shoppers that execute frontend code, plus a separate content system. It promises better speed, more flexible brand content and future readiness.

The fictional estimate is **€240,000 and nine months**, followed by another application to maintain. Noor asks which need the existing theme cannot meet.

She starts with real-user measurements. In the illustrative February 2027 mobile results, the main content appears in **2.1 seconds**, response to interaction takes **310 milliseconds**, and layout shift scores **0.08**. These are 75th-percentile measurements: three quarters of visits perform at least as well.

Content loading and layout stability already meet Google's cited good ranges. Responsiveness does not. The agency traces it to scripts from eleven apps. That finding points to specific work before an architectural replacement.

She declined the rebuild. The reasons form the rest of this chapter.

What a Storefront Does, and What It Does Not Own

A storefront produces pages: home, category (also called collection) pages, product pages, search results, content pages, the cart, the account pages and, at the end, the checkout. To produce them it needs facts from almost every other part of the platform. **It owns none of those facts.** It owns the templates, the layout, the brand, and the scripts that run in the browser.

What the page shows	Who owns the fact	Chapter
Product name, description, images, attributes	The catalog, or the PIM (product information management system) once there is one	What Are We Selling? Products, Variants and the Catalog
Price, and the tax-inclusive display for EU shoppers	The pricing engine or price list	What Does It Cost? Price Lists, Contract Prices, Currencies and Tax

What the page shows	Who owns the fact	Chapter
Discounts and badges	The promotion engine	Discounts, Coupons and Loyalty: Promotions Without Chaos
Search results, facets, sort order	The search index	Finding the Product: Search, Navigation and Merchandising
"In stock", "only 3 left", pre-order	The availability service, or the platform's stock until there is one	Is It in Stock? Inventory, Availability and the Promise You Can Keep
The delivery promise	The shipping service, or a static rule	Getting It There: Carriers, Labels, Tracking and the Delivery Promise
The cart and the checkout	The platform's checkout	From Cart to Confirmation: Checkout Orchestration
Editorial content (stories, guides, campaign pages)	A content management system (CMS), the tool editors use to write and publish pages without engineers	this chapter
Who the shopper is, and which prices they see	Identity and the customer record	One Customer, One Product: Identity and Master Data Across Systems

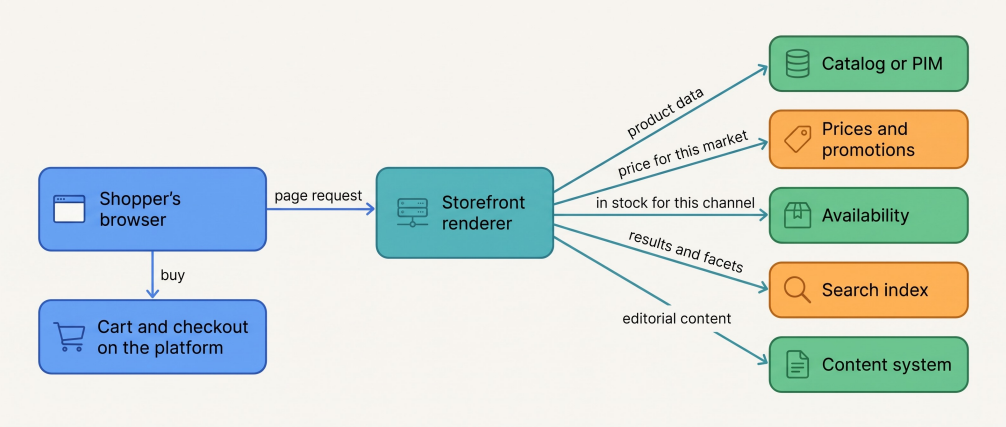

Figure 1: *The storefront renders facts it does not own: product data, prices, availability, search results and content come from other systems, and the buy button hands the shopper to a checkout the platform runs.*

That table is the reason the storefront question is an architecture question. A headless storefront does not move any of those facts; it moves the **rendering** and the **glue** between them into the merchant's hands. Whether that is a gain depends on whether the merchant wanted either.

Three Rendering Models

Two words first. **Server-side rendering** means a server builds the complete page as HTML and sends it; the browser shows it at once. **Client-side rendering** means the server sends a mostly empty page plus JavaScript, and the browser fetches data and builds the page itself. Early headless storefronts were client-rendered single-page applications, which is where their reputation for slow first loads comes from; today's headless frameworks render on the server by default, so the distinction between the models is no longer about rendering technique but about **who runs the renderer and where the templates live**.

Model 1: the theme, rendered by the platform

A **theme** is a set of templates (and, on hosted platforms, the settings and scripts around them) that the platform's own renderer turns into pages. Shopify's Liquid themes, BigCommerce's Stencil ("BigCommerce's Handlebars-based theme engine") and the templates of Salesforce's Storefront Reference Architecture (SFRA) are the dated examples. The platform owns rendering, caching, delivery, security and scale; the merchant owns the templates.

Shopify described its renderer in August 2020 (evidence now six years old): the Storefront Renderer is "a server-side application that loads a Shopify merchant's storefront

Liquid theme" and returns HTML, rewritten from scratch from January 2019 because storefront code "has much stricter performance requirements" than, say, payment processing. It reads from dedicated database replicas and caches at three layers: a query cache in a key-value store, an in-memory cache of the most frequent queries, and a full-page cache. Reported results: server response times "4x to 6x faster than the legacy implementation". In January 2022 the CTO added that the separate renderer could "deploy in regions worldwide without deploying the full Rails monolith".

The theme model's ceiling is the template language and the platform's page types. Its floor is high: a merchant inherits caching, edge delivery and a renderer tuned by a vendor whose business depends on it. Shopify's Online Store 2.0 (announced 29 June 2021) moved theme content into JSON templates with sections on every page, added theme app extensions (so apps add blocks instead of editing templates) and set a performance bar for themes.

Model 2: headless on the vendor's runtime

The platform vendor supplies a storefront **API** and a hosting runtime, and the merchant builds a separate front-end application on it. Shopify's Hydrogen is "Shopify's headless toolkit", not a framework: it adds Shopify-specific clients for "the Storefront API and Customer Account API only" to a framework the merchant chooses (React Router 7 is the starter), deployed to Oxygen, Shopify's hosting (documentation version 2026-10, accessed November 2026); Shopify's February 2023 Editions post made Oxygen available on its main plans "at no extra cost". Salesforce's PWA Kit is "a storefront technology for headless commerce using Salesforce Commerce APIs and React", deployed to a Managed Runtime. Adobe's Commerce as a Cloud Service uses a headless storefront on Edge Delivery Services as its only storefront (Adobe, 20 October 2026).

What stays on the platform in this model is instructive. Hydrogen's documentation says it covers the Storefront and Customer Account APIs and "doesn't include checkout UI, because checkout stays on Shopify"; the headless storefront builds the cart and hands off to the platform's checkout. The vendor keeps the part it is paid for and the part it must keep secure.

Model 3: headless anywhere

The merchant builds the storefront on a general-purpose framework and hosts it where it likes, against the platform's storefront API, possibly against several platforms. Vercel's Next.js Commerce template, "A high-performance, server-rendered Next.js App Router ecommerce application" with Shopify as reference and forks for BigCommerce, Medusa, Saleor and Shopware, was archived and moved to a successor repository (accessed November 2026), a reminder that this layer churns. Alokai (formerly Vue Storefront) sells a **unified data layer** that normalizes "product data from each platform into this common format" through a middleware, for BigCommerce, commercetools, Magento,

SAP and Salesforce (documentation, accessed November 2026): the merchant's front end talks to one shape of product, whichever engine is behind it, which is the Group-stage reason to go headless.

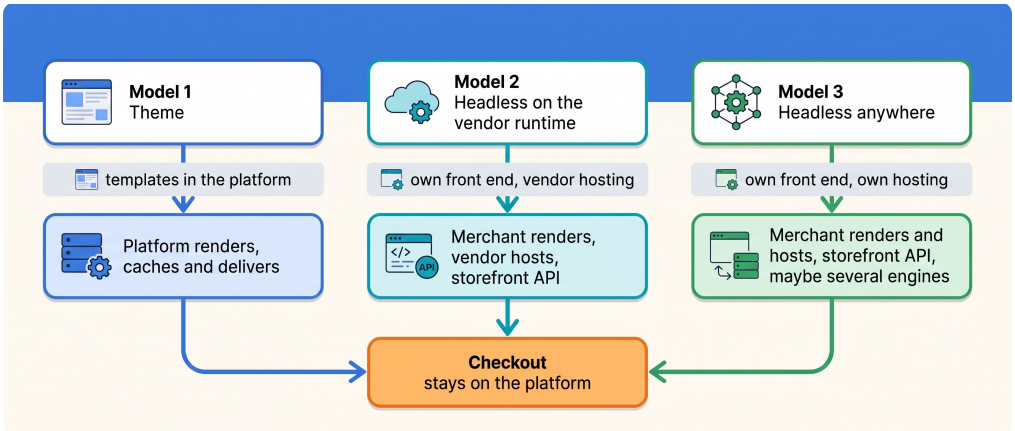

Figure 2: *Three rendering models differ in who renders, who hosts and how many engines the front end can face; under all three the checkout stays on the platform.*

Cache the Shell, Fetch the Promise

A storefront's performance on an ordinary day and its survival on a peak day both come from one split. Some of a page can be computed once and served many times; some must be computed for this shopper, now.

Cacheable: the product page's content, images, attributes and layout, per market and language; category pages; editorial pages; search facets to a degree. These change when the catalog or the campaign changes, not when a shopper arrives.

Not cacheable: the price for a logged-in wholesale account; live stock and the "only 3 left" badge; the cart; the checkout; anything personalized. These are **promises**, in this book's vocabulary, and a cached promise is a lie waiting for its moment, which is exactly what the Price Drift of October 2026 and the Black Friday Oversell were from the page's side.

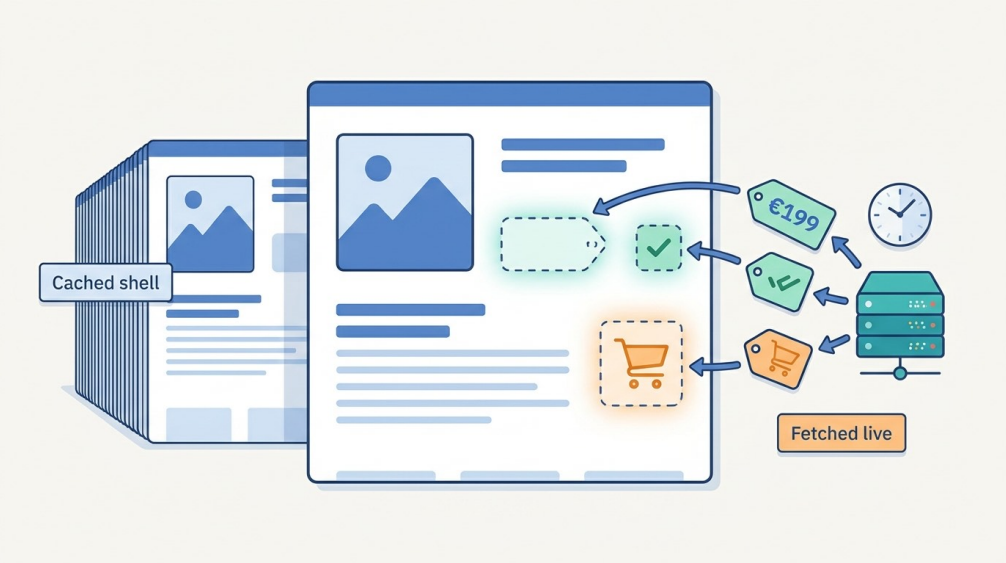

Figure 3: *Cache the shell, fetch the promise: the page content is computed once and served many times, while the price, the stock badge and the cart are fetched for this shopper, now.*

The pattern every model implements is **cache the shell, fetch the promise**: serve the cached page immediately, then fetch the live facts into it. Shopify's full-page cache is this at the platform layer.

At the headless layer the same idea appears as **incremental static regeneration**: Vercel's documentation (updated 28 August 2026) describes serving the cached page while regenerating it in the background on a timer or on demand, names e-commerce catalogs "that need current pricing and availability without rebuilding the entire site" as the first use case, collapses concurrent requests for an uncached page "into one function invocation per region, protecting your backend during traffic spikes", and serves stale content with a retry if regeneration fails.

Request collapsing is the peak-day feature: when 41 times the normal traffic hits one hero product at once, the backend sees one request, not thousands.

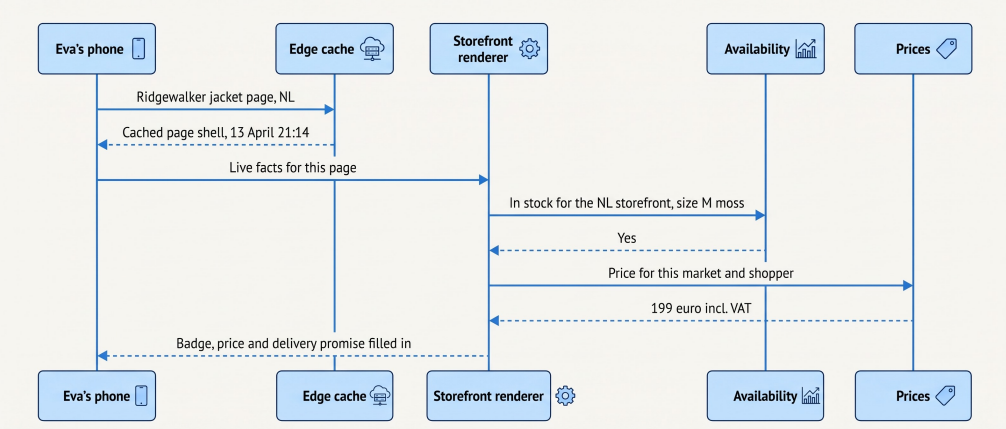

Figure 4: *Eva's product page arrives from the edge cache first; the storefront then fetches the facts that must be true now, availability and price, and fills them in.*

Requirements per page type

Page	How fresh	How consistent	How fast	On a peak day
Product and category content	Minutes after a catalog publish	Same content in every channel (the PIM's job after W3)	Cached at the edge; server response in tens of milliseconds *(assumption)*	Must be fully cacheable; request collapsing on misses
Price shown	Within the price change's effective time; What Does It Cost? Price Lists, Contract Prices, Currencies and Tax	One price per market and channel	Fetched live or cached for seconds, by market	Cache per market, never per shopper, for anonymous visitors

Page	How fresh	How consistent	How fast	On a peak day
Stock badge and delivery promise	Seconds to a minute after a stock event *(assumption)*	Same answer the checkout will give	Fetched after the shell	Degrade to "in stock / out of stock" or hide the count (Black Friday Every Year: Peak Load, Resilience and Recovery)
Search results	Minutes after a catalog publish	Index matches the catalog	Under 300 ms *(assumption)*	Fewer facets, simpler sorts
Logged-in B2B (business-to-business, wholesale) pages	Live	Contract price per account	Never cached across accounts	Rate-limited by account
Cart and checkout	Live	One cart per session; the checkout's rules win	Platform-owned	Never on the switch-off list

Performance: What the Evidence Says, With Dates

Core Web Vitals are Google's three field metrics for page experience, measured from real visitors and assessed at the 75th percentile of page loads, mobile and desktop separately (web.dev, updated 31 October 2024): **largest contentful paint** (LCP, when the main content has loaded; "good" is 2.5 seconds or less), **interaction to next paint** (INP, how quickly the page responds to a tap or click; good is 200 milliseconds or less) and **cumulative layout shift** (CLS, how much the page jumps while loading; good is 0.1 or less).

INP replaced the older first-input-delay metric on 12 March 2024. Google's Search documentation says it "highly recommend[s]" good Core Web Vitals "for success with Search" and that they "align with what our core ranking systems seek to reward", without stating a weight (accessed November 2026).

The evidence that speed moves money is consistent in direction and weak in precision:

- **Deloitte Ireland for Google (24 March 2020; six years old)**: across mobile sites in retail, travel, luxury and lead generation, a 0.1-second improvement in site speed went with +8.4% retail conversions and +9.2% retail average order value. Caveats the study carries: commissioned by Google, the number of sites not disclosed, and a correlation, not an experiment.
- **web.dev's case collection (undated, accessed November 2026)**: Vodafone Italy improved LCP 31% and saw sales up 8%; Lazada tripled LCP speed and saw mobile conversion up 16.9%; Cdiscount improved all three metrics and reported +6% revenue in a Black Friday sale. Every case is a self-selected success story, and none separates speed from the redesign that came with it.
- **A vendor's admission (Shopify, accessed November 2026)**: the theme-performance page says performance "directly influences conversion rates, repeat business, and search engine rankings" without citing a figure; the Theme Store requires a minimum average Lighthouse performance score of 60; public apps "shouldn't reduce Lighthouse performance scores by more than 10 points"; and the admin's performance report warns that theme code and apps "can slow down your site".

The honest reading: **faster is better, nobody can tell you by how much, and the usual culprit is not the rendering model but the scripts.** A rule that caps each app at ten points is a vendor admitting that apps are where storefront speed goes. Kestrel & Pine's INP of 310 milliseconds was eleven apps' scripts; the rendering model was innocent.

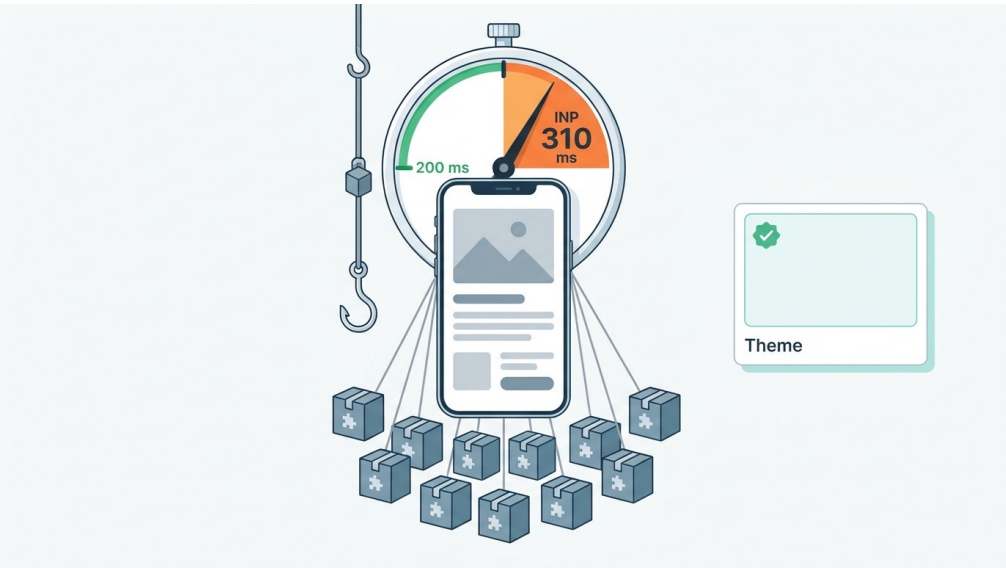

Figure 5: *Kestrel & Pine's interaction to next paint of 310 milliseconds was eleven apps' scripts, not the rendering model; good is 200 milliseconds or less.*

Accessibility is a requirement, not a performance option

WCAG 2.2, the Web Content Accessibility Guidelines of the World Wide Web Consortium (W3C), is a Recommendation dated 12 December 2024, with three conformance levels (A, AA, AAA) and four principles (perceivable, operable, understandable, robust).

The **European Accessibility Act** (Directive (EU) 2019/882) applies to services "provided to consumers after 28 June 2025" and lists "e-commerce services" explicitly, defined as a service provided at a distance "with a view to concluding a consumer contract"; microenterprises (under 10 people and up to €2 million turnover) are exempt.

Kestrel & Pine, with 310 employees and three consumer storefronts, has been in scope since June 2025; its B2B portal, serving businesses, is not a consumer service. National enforcement differs; Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation owns the calendar.

An accessible theme can still contain an app block that a shopper cannot navigate or use. Apply accessibility criteria to both the theme and the inserted components, and test their combined purchase flow. The shopper needs to complete the whole task, whichever rendering model or supplier produced each piece.

Mobile apps and progressive web apps

A **progressive web app** (PWA) is a website "built and enhanced with modern APIs" that can be installed like an app, work offline and send push notifications (web.dev, accessed November 2026). For a merchant it is mostly the headless question in disguise: the same server-rendered storefront can be made installable. A **native app** adds a second storefront that must show the same prices, stock and promises as the first, and is justified by loyalty, push and in-store features, not by speed.

The Choices and Their Trade-Offs

The realistic options for a merchant on a hosted platform are four.

1. **The theme, disciplined.** Keep the platform's renderer; use sections and app blocks instead of edited templates; police app scripts with a performance budget; let the platform's cache and edge do the peak-day work.
2. **The theme plus a content system.** Keep the theme for commerce pages and feed editorial content from a CMS through the platform's content features or a small integration; the brand team gets its content model without a new renderer.

3. **Headless on the vendor's runtime.** Own the front end; let the vendor host it, serve the storefront API and run checkout. The team now owns rendering, caching rules and the glue to search, prices and availability.
4. **Headless anywhere, possibly over several engines.** Own everything in front of the APIs; add a unified data layer if several engines or brands sit behind one front end.

Option	Who runs rendering and caching	Flexibility	Peak-day posture	Team it needs	When it is right
Theme, disciplined	The vendor	The template language and page types	The vendor's cache and edge; apps are the risk	One or two front-end engineers plus an agency	Starting through Omnichannel, unless a need below applies
Theme plus CMS	The vendor, plus a content feed	Content model of the CMS; commerce pages as the theme allows	As above	The same, plus editors	A content-led brand on a themed platform

Option	Who runs rendering and caching	Flexibility	Peak-day posture	Team it needs	When it is right
Headless on the vendor runtime	The merchant renders, the vendor hosts	Full control of the front end; checkout stays the vendor's	The merchant's caching rules; request collapsing if used	Three or more front-end engineers who can own a production application	A design system shared across brands or countries; a B2B portal with account pricing that the theme cannot express
Headless anywhere	The merchant	Full, including several engines behind one front end	Entirely the merchant's	A front-end platform team	Group stage: several brands or engines behind one storefront

The recommendation is tied to assumptions. Options 1 and 2 assume the platform's page types and checkout fit the business, which for a direct-to-consumer (D2C) brand with a B2B portal on the same platform they usually do. Options 3 and 4 assume a team that can run a production application on its own: on-call, caching bugs, framework upgrades every year, and the storefront API's versioning calendar (Promises Between Systems: API Contracts, Webhooks and Versioning). The agency can build it; the question is who runs it in year two.

Three needs justify headless, in my experience: a **shared front end across brands or countries** that the theme model cannot express because each tenant has its own theme; a **content-led site** where the editorial model, not the catalog, drives the page structure; and a **B2B portal** with account pricing, assortments and approval flows that the consumer theme cannot represent (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals). Speed is not on the list, because a disciplined theme reaches the thresholds and an undisciplined headless storefront misses them.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
1 — Starting (Kestrel & Pine 2016–2018)	A purchased theme with light customization; a handful of apps; the platform's defaults for caching and delivery	Field measurements at the 75th percentile, read monthly; a rule that every app's script is justified	A theme edited by hand "just this once"; LCP drifting past 2.5 seconds after each app install
2 — Growing (2019–2022)	An agency-customized theme; two or three storefronts for countries; thirty apps; the first content pages fought over by brand and engineering	A performance budget per page type; sections and app blocks instead of edited templates; a content model decision (theme sections or a CMS)	Theme updates skipped because the merge is painful; brand asking for page types the theme lacks; a headless proposal arriving before any need is measured

Stage	What is typical	What becomes necessary	Warning signs of outgrowing the approach
3 — Omnichannel (2023–now)	Several localized storefronts in one tenant; a B2B portal; forty apps; a peak hour many times normal; accessibility obligations in force since June 2025	Cacheability per page type as a design rule; the stock badge and the delivery promise fed by the availability service (W1) and the shipping service rather than static text; an app-script audit each quarter; accessibility testing on theme and app blocks; a headless decision made against the three needs, not the pitch	Peak-day slowness traced to uncacheable pages; the B2B portal needing account logic the theme cannot express; a second brand or a native app that must share the design system
4 — Group	Several brands or engines; a front-end platform team; design systems; possibly a unified data layer over several commerce engines	One front-end platform (framework, design system, normalization layer) across brands; headless as the shared surface, with each brand's engine behind it; a performance and accessibility standard enforced across brands	Each brand running its own storefront stack; a group-wide campaign needing five implementations; the One Size Fits None pattern when a brand with a different model is forced onto the shared front end

The judgement: at Starting and Growing stages the theme is the right answer and the

work is discipline; at the Omnichannel stage headless becomes a real question only when one of the three needs appears; at the Group stage headless becomes the usual answer, because the thing being shared is the front end itself.

In Practice

Dated public examples

- **A platform carving out its renderer for latency (Shopify, 2020–2022).** The Storefront Renderer rewrite is the one documented case of Shopify separating a hot path from its monolith, and it was done for storefront latency, not for team autonomy. The rollout pattern transfers to any merchant replacing a storefront: a verifier compared "status code, headers, and body" of old and new responses on sampled production traffic, with fallbacks to the legacy path. Replace a storefront behind a comparison, not on a date.
- **Checkout stays home under every model (Shopify Hydrogen, docs 2026-10; Adobe Cloud Service, October 2026).** Hydrogen "doesn't include checkout UI"; Adobe's SaaS edition ships a headless storefront but keeps the commerce engine and checkout. A merchant's headless storefront is a new front door, not a new shop.
- **Regeneration as the peak-day mechanism (Vercel, updated 28 August 2026).** Request collapsing to "one function invocation per region" and serving stale content on failed regeneration are the documented behaviors that make a headless storefront survivable; a merchant choosing model 3 should ask its hosting for the equivalent and test it before November.
- **Accessibility as a dated obligation (EU, 28 June 2025; W3C, 12 December 2024).** The European Accessibility Act's application date and WCAG 2.2 are the storefront's two external deadlines, and they apply to the theme and to every app's block alike.

Kestrel & Pine: why Noor said no, and what she said yes to

Noor's answer to the agency in late February 2027 had four parts, recorded as a short decision note that later folded into Waypoint's W4 (stay and surround; no replatforming, with the platform's own point-of-sale (POS) system evaluated in 2028 and the whole decision revisited in 2029).

1. **The speed argument did not survive measurement.** LCP 2.1 seconds and CLS 0.08 were already "good"; INP of 310 milliseconds was the problem, and it was app scripts. The agency's retainer (€14,000 a month) was redirected for one quarter to a theme performance pass and an app-script audit, run alongside the app governance

review of April 2027 (Apps, Plugins and Functions: Extensibility and Ecosystem Governance). By June 2027 INP on product pages measured 180 milliseconds (illustrative), inside the threshold, after six unused apps and four redundant scripts were removed.

2. **The content argument was real, and headless was the wrong remedy.** Teun Baas's brand team wanted seasonal guides and stories that the theme's page types handled badly. The remedy was option 2: theme sections for commerce pages, and a content model that waits for the PIM (W3, Q3 2027) so product stories draw on one product record rather than a fifth copy.
3. **"Ready for the future" named no need.** None of the three needs applied: one brand, three countries sharing one design, a B2B portal that the platform's B2B features serve for 62% of wholesale orders, and no native app. A headless rebuild would have been the Giant's Boots: an architecture borrowed from brands with front-end teams of fifty, run by three engineers who also own the apps.
4. **The peak-day argument pointed the other way.** The 2026 peak hour (2,300 orders, 41 times an average hour) was survived by the platform's cache; what failed was the stock sync chain behind the badge. The fix for the badge is the availability service (W1), not a new renderer. Until W1 lands in Q4 2027, the 2027 peak plan hides the stock count on product pages and shows "in stock / low stock" from a per-channel buffer (Black Friday Every Year: Peak Load, Resilience and Recovery, Journey 4: Black Friday, Hour by Hour).

What she said yes to cost about €38,000 of agency time over the quarter (illustrative) instead of €240,000 and nine months, and left the storefront on the one rendering model her team can run. The decision is revisited at the 2029 review together with the POS question, because the one need that could change the answer, a second brand or a store-mode app sharing the design system, is a Group-stage question Kestrel & Pine has not yet reached.

When It Goes Wrong

Failure	What it looks like	Detection	Recovery
A cached promise	A price change or a sell-out shows on the page minutes or hours after the fact; a shopper buys at the old price or orders the last unit twice	Compare what the page showed with what checkout computed; the Price Drift and the Oversell are both this	Cache content, never promises; purge on publish; fetch price and stock after the shell
The uncacheable peak	Pages that depend on a per-shopper fact are rendered fresh for every visitor; the renderer or the APIs behind it saturate at 41 times normal	Cache hit rate per page type in the load test (Black Friday Every Year: Peak Load, Resilience and Recovery)	Make product and category pages cacheable per market; collapse requests on misses; degrade the badge
Script sprawl	INP and LCP drift with every app install; checkout steps slow or break when a script errors	Field metrics at the 75th percentile, monthly; the platform's app performance rule	A performance budget; the quarterly app-script audit; app blocks instead of injected scripts
The forked theme	Theme updates skipped for years; security and feature updates unavailable; every change a merge	The gap between the installed and the current theme version; the count of hand-edited templates	Rebuild the customizations as sections and app blocks on a current theme; treat the theme like code, with versions

Failure	What it looks like	Detection	Recovery
The headless storefront that drifted	A front end pinned to an old storefront API version breaks on the vendor's retirement date; a framework upgrade stalls for a year	The API version on the integration card (Promises Between Systems: API Contracts, Webhooks and Versioning); the framework's support calendar	Treat the storefront as the integration it is: an owner, a version pin, an upgrade calendar
Bots at the front door	Crawlers and AI agents read the catalog at a rate that costs money or starves real shoppers; Commerce.com's (formerly BigCommerce) second-quarter 2026 filing reported lower gross margin, and an earnings-call summary attributed part of it to hosting costs from AI crawlers and agents (6 August 2026)	Traffic by source; the storefront API's bot limits (Shopify limits bots "most strictly when unsigned")	Bot rules and signed agent access; a catalog feed for agents instead of page scraping (Assistants and Agents: AI in Commerce and What Is Real in 2026)
An inaccessible storefront	A complaint or an enforcement notice under the Accessibility Act; checkout unusable with a screen reader	Automated and manual WCAG 2.2 testing of theme and app blocks	Fix in the theme and in every app block; add accessibility to the app review on the governance card

What to Take Away

Established.

- Core Web Vitals are defined and thresholded by Google (LCP 2.5 s, INP 200 ms, CLS 0.1, at the 75th percentile; INP since 12 March 2024) and are recommended for Search success; the business evidence (Deloitte 2020; web.dev cases) is consistent in direction and imprecise in size.
- Under every rendering model the vendor keeps checkout: Hydrogen ships no checkout UI; Adobe's Cloud Service keeps the engine behind its headless storefront.
- Cacheable content and live promises must be split; the platform's full-page cache (Shopify, 2020) and incremental regeneration with request collapsing (Vercel, 2026) are the two documented mechanisms.
- The European Accessibility Act applies to consumer e-commerce services since 28 June 2025, with WCAG 2.2 (December 2024) as the technical reference; app scripts are the usual cause of storefront slowness (Shopify's ten-point rule).

Depends on context.

- Whether to go headless depends on one of three needs being present (a shared front end across brands or countries, a content-led site, a B2B portal the theme cannot express) and on a team that can run a production application; absent both, the theme is right at every stage up to Group.
- Where editorial content lives depends on how much of the site the brand team drives: theme sections suffice for a catalog-led brand; a CMS feeding the theme suits a content-led one; a headless front end is the answer only when content drives the page structure of the whole site.
- How aggressively to cache depends on the business model: an anonymous D2C storefront can cache nearly everything per market; a B2B portal with account prices can cache almost nothing across accounts.

The author's synthesis.

- I treat the storefront as an integration consumer with a brand on top: it owns no fact, so the architectural work is deciding which facts are fetched live, which are cached, and who runs the renderer. Decide that before the framework.
- I decline headless proposals that lead with speed. The speed is in the scripts and the cache, which the theme already has; the real reasons for headless are organizational (brands, content, B2B), and a merchant that does not have one should keep the renderer the vendor runs and spend the money on the availability service that makes the badge true.

Questions to Ask

1. What do the field Core Web Vitals say at the 75th percentile on mobile for the product and category pages, and when were they last read?
2. Which rendering model is in use, who runs the renderer, and who is on call when it fails on a Friday evening in November?
3. Which facts on the product page are cached, for how long, and which are fetched live; is the stock badge one of the live ones?
4. How many hand-edited theme templates exist, and when was the theme last updated from the vendor's version?
5. How many apps add scripts to the storefront, and what is the performance budget each must meet?
6. If a headless storefront is proposed, which of the three needs (shared front end, content-led site, B2B portal) does it serve, and who will run it in year two?
7. Which storefront API version does the front end target, and when does that version leave support?
8. Has the storefront, including every app block and the checkout, been tested against WCAG 2.2, and who owns the result?
9. What does the page show for stock and delivery when the availability service or the shipping service is down?

The storefront is where the peak arrives first: forty-one times the usual traffic hits the pages this chapter described, and whether they are cacheable decides whether the shop stands. The next chapter takes that day apart, from load testing and backpressure to the order in which features are switched off, and returns to Kestrel & Pine's Black Friday hour by hour. Continue with Black Friday Every Year: Peak Load, Resilience and Recovery.

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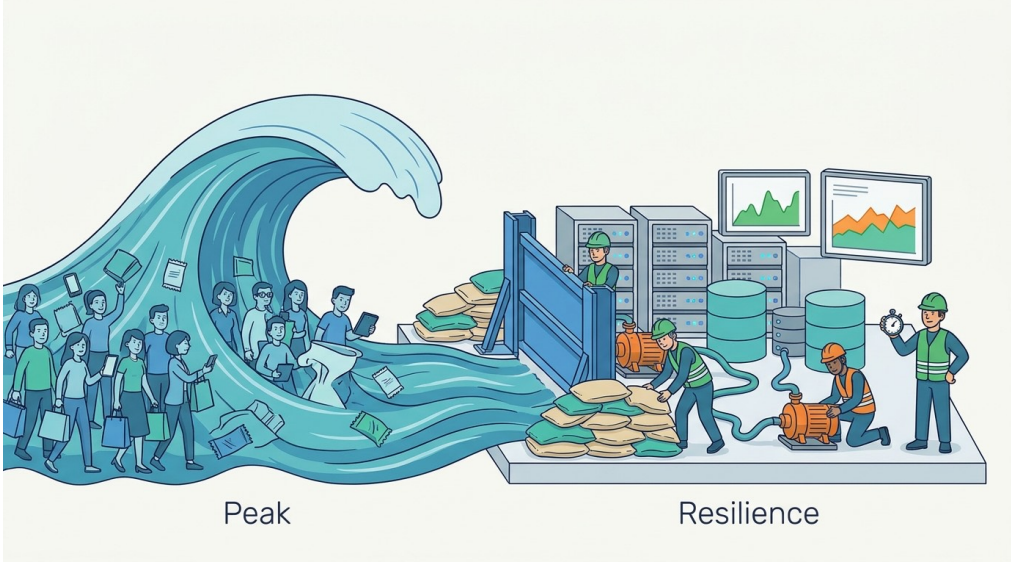
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

33. Black Friday Every Year: Peak Load, Resilience and Recovery



IN THIS CHAPTER, YOU WILL: Learn how a peak day breaks a commerce chain, how to size, test, protect and observe that chain in orders per minute, and how to recover afterwards.

KEY POINTS:

- **The platform's peak is not your peak.** A hosted platform smooths thousands of merchants' spikes into a gentle curve; your own connectors, warehouse and finance systems see your spike alone. The storefront usually holds; the chain behind it is where the day is lost.
- **Measure the whole chain in one unit, orders per minute, and protect it with queues and switches.** Every link gets a capacity figure, a load test and a place in a written switch-off order. Checkout, payment and the order record are never on that list; everything behind them may wait.
- **Recovery is reconciliation.** A peak leaves orders accepted but not exported, stock sold twice and messages dropped in retries. The plan

names, before the day, who compares which two systems the morning after and how the differences get fixed.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The First Snowfall** — the peak is the first time the integrations, not just the storefront, meet real load, and nobody has measured what they can carry.
- **The Phantom Stock** — every sync interval that is harmless on a quiet day becomes, at forty times the pace, a window in which every channel sells the same last unit.
- **The Message in a Bottle** — retries, throttles and timeouts at peak produce the duplicates and gaps that quiet days never show.

Kestrel & Pine, the fictional outdoor brand, takes about **800 online orders on an ordinary day**, roughly 33 an hour. On Black Friday, it reaches **2,300 in one hour**. An integration that clears its queue between ordinary purchases can accumulate a backlog when that many orders arrive together. The store may look healthy while promised deliveries become impossible to meet.

Order export, stock updates and picking each have their own capacity, even when the hosted storefront handles the traffic. A slow step can produce an unavailable jacket sold, a late confirmation or a missed dispatch. Testing the whole order path at the busiest hour's rate exposes those bottlenecks while there is still time to add capacity, reduce demand or change the promise.

Where we are in the order's journey. Eva's April 2027 jacket order reached the warehouse within minutes. Here we ask what happens when thousands of orders take that path in an hour. The steps stay the same; queues and delays become decisive.

Setting the Scene at Kestrel & Pine: Reading the Post-Mortem

In March 2027, Head of Engineering Noor Visser reads the post-mortem for **27 November 2026**. CEO Mara Lindqvist wants to know whether the next Black Friday will repeat it. The book calls this failure pattern **The First Snowfall**: discovering capacity limits on the day they matter most.

During **23–29 November 2026**, the storefronts took **31,000 orders**, 10.6% of the year's total. Friday's **20:00–21:00** hour brought **2,300 orders**, about **38 per minute**, with traffic **41 times** an average hour. The platform handled checkout. The chain behind it accumulated these failures:

- The **stock sync chain** ran as it always ran. The WMS (warehouse management system, the software that runs the Duiven warehouse) reported stock to the ERP (enterprise resource planning system: the books, purchasing and the item master) every 15 minutes; the ERP copied it to the platform hourly; the platform copied it to the marketplace connector hourly. A sale at Duiven took **up to two hours** to reduce the stock shown anywhere else. The Ridgewalker jacket, the hero of the week's 20% outerwear promotion, sold out in the warehouse at **19:40**; the storefronts and marketplaces kept selling it for up to two hours. **1,340 units** were oversold and **1,120 orders** canceled or delayed, at a cost of about **€96,000** including goodwill vouchers and marketplace penalties.
- The **order flow to the warehouse**, a flow on the iPaaS (integration platform as a service, the hosted tool that runs the data flows between systems), created orders in the WMS one at a time, several API calls each, and was throttled by the WMS vendor's **rate limit** (the maximum number of calls a vendor lets one customer make in a time window). By 21:00 the backlog was **four hours** long; orders placed at 20:00 reached the warehouse after midnight, and the "ordered before 22:00, delivered tomorrow" promise on the Dutch storefront, a sentence on a page rather than a computed figure, was broken for roughly 8,000 orders (a Kestrel & Pine figure from Pick, Pack, Ship: Warehouse Management and Fulfillment).

The systems ran at their configured rates, but **nobody had compared those rates in orders per minute**. The team could not identify the slowest link or reduce work before it overwhelmed the next one. Journey 4: Black Friday, Hour by Hour gives the hour-by-hour account and 2027 plan; this chapter explains the design decisions.

What a Peak Is, and Why the Merchant's Is Worse Than the Platform's

A **peak** is a period, planned or not, in which traffic and orders are many times their normal level. Peaks have different shapes, and the shape decides what breaks.

Peak profile	Shape	What it stresses most
Seasonal peak (Black Friday, Singles' Day, pre-Christmas)	Days of elevated load with hour-long spikes; a known date	Everything in sequence: storefront, checkout, order export, warehouse, carriers, returns three weeks later

Peak profile	Shape	What it stresses most
Launch or drop	Minutes of extreme load on one product at a known moment	Stock accuracy and fairness: one SKU (a sellable variant), thousands of buyers, a few hundred units
Flash sale	A short promotion concentrating demand on a few products	Checkout throughput, promotion rules, stock on the discounted items
Unplanned moment (a TV mention, a viral post)	No warning	Whatever has no headroom; usually the first integration

The figures that circulate about peaks come from the largest operators, and they teach one lesson above all: **each reports in its own unit**. Alibaba's cloud arm reported **583,000 orders per second** 26 seconds after midnight on 11 November 2020, without defining "order" (Alibaba Cloud, 2020). Zalando reported **7,200 orders per minute** at its 2019 Cyber Week peak (Zalando Engineering, 2020; six years old, so historical). Shopify reported **$5.1 million of sales per minute** at 12:01 p.m. EST on Black Friday 2025 against a weekend total of **$14.6 billion**, with **489 million requests per minute at the edge** and **117 million on its app servers**, all unaudited (Shopify, December 2025). Adobe put US Cyber Monday 2024 at **$13.3 billion**, with **$15.8 million per minute** in the peak hours (Adobe, December 2024).

Two things in Shopify's figures matter more than their size. First, the ratio of edge to app-server requests, about **four to one**, means roughly three quarters of peak traffic was absorbed by the content delivery network (**CDN**, the layer of servers near the shopper that serves cached pages and images) before it reached application code. Second, the peak minute ($5.1 million) against the weekend average (about $2.7 million a minute over roughly 5,400 minutes) is a **peak-to-average ratio of about two**. Kestrel & Pine's ratio in orders was about **seventy** (2,300 in the peak hour against 33 in an average one). That is the central fact of this chapter:

> **The platform's peak is not the merchant's peak.** A hosted platform serving a million merchants smooths their spikes into a curve. Each merchant's own connectors, ERP, WMS and warehouse staff see that merchant's spike, alone and in full.

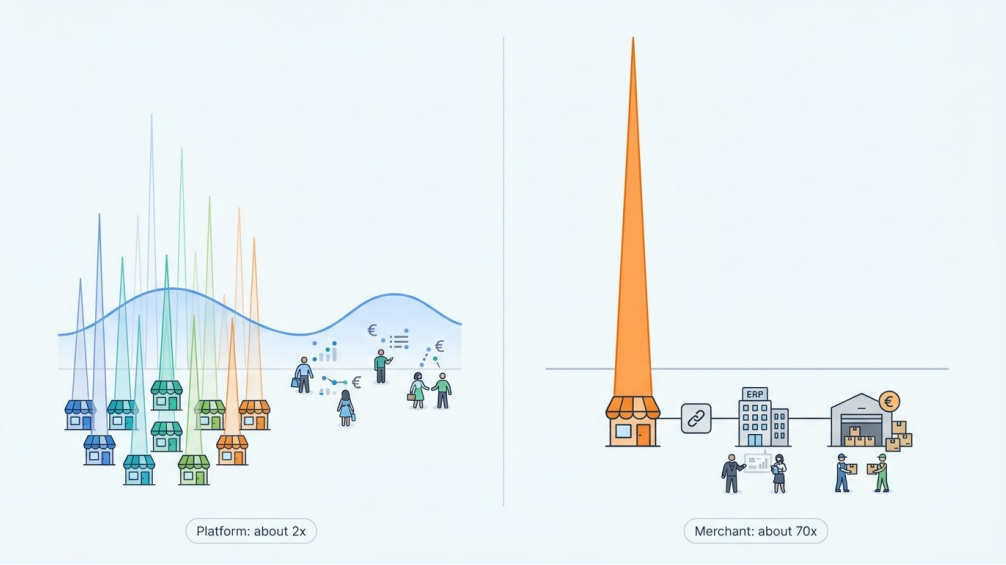


Figure 1: *The platform's peak is not the merchant's peak: a hosted platform smooths a million merchants' spikes to a peak about twice its average, while Kestrel & Pine's own chain saw about seventy times an average hour.*

Multi-tenancy (Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation) gives the platform its smoothing; the merchant's chain gets none, and it is what this chapter sizes.

The Chain Under Load: Where a Peak Day Breaks

A commerce chain is a sequence of handoffs, and a peak day runs down it from the front. The figure shows Kestrel & Pine's chain in November 2026 with the capacity each link turned out to have; the per-link figures are illustrative reconstructions consistent with Kestrel & Pine's recorded totals.

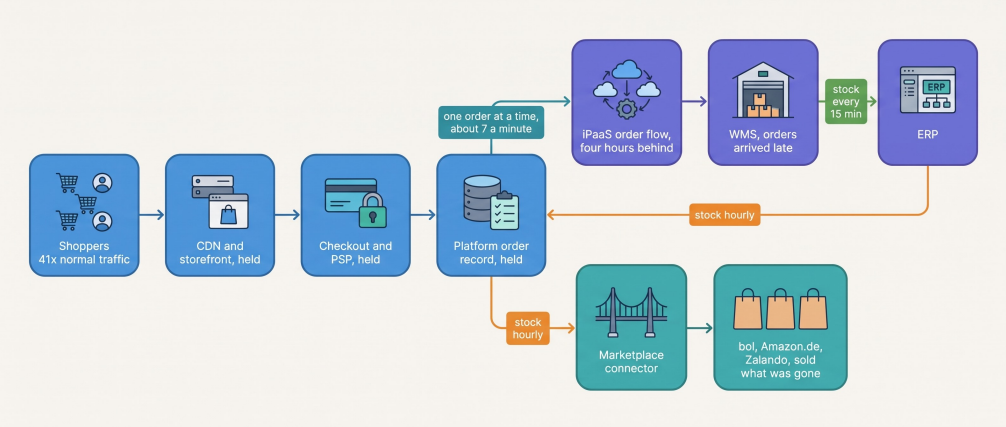


Figure 2: *Kestrel & Pine's chain on 27 November 2026: the vendor-hosted links at the front held at 38 orders a minute, the hand-built order flow behind them carried about seven, and the stock copied back around the loop was up to two hours old by the time the marketplaces saw it.*

Reading the chain from the front:

The storefront. Catalog pages, images and scripts are mostly cacheable, and a hosted platform's CDN serves them without touching the application. This is why "the platform held" is the normal outcome at Omnichannel scale and why the storefront is the wrong thing to worry about first. Delivering the Storefront: Themes, Headless, Rendering and Performance covers what can be cached: a product page, yes; the stock badge and a wholesale account's price, only with care.

Checkout and payment. Checkout is the most protected part of any platform, because it is where the money is. The **PSP** (payment service provider, the company that takes the shopper's payment) has its own rate limits: Stripe's documentation asks merchants planning flash sales that may exceed them to tell support in advance (Stripe, n.d., accessed November 2026). Shopify described in a 2018 changelog that under high sales volume "processing payments at checkout happens separately from creating orders", so the shopper finishes checkout at once and the order record and confirmation e-mail catch up afterwards (Shopify changelog, November 2018; eight years old, and today's "checkout throttle" is not publicly described). The idea transfers whatever platform you use: **take the money and the commitment fast; let the record-keeping queue.** From Cart to Confirmation: Checkout Orchestration and Taking the Money: Payment Providers, Authorization, Capture and Refunds own the details.

The order export. Here the vendor-hosted part ends and the merchant's part begins. The order has to reach the WMS, usually through a connector or an iPaaS flow the merchant or its agency built. This link is almost always the first to fall over, for three reasons that compound: it processes orders one at a time with several API calls each; it runs under rate limits at both ends, the platform's (shared with every installed app, which are busiest when the store is) and the WMS vendor's; and it was tested, if at all,

at the rate of an ordinary afternoon. A flow carrying seven orders a minute against 38 arriving builds a queue of 31 a minute, invisible unless someone built a gauge for it.

The WMS and the people. The warehouse system is usually a hosted product with capacity to spare; the constraint is pickers and carrier collections. Pick, Pack, Ship: Warehouse Management and Fulfillment owns this, and Getting It There: Carriers, Labels, Tracking and the Delivery Promise owns the delivery promise, which in 2026 no system could withdraw when the warehouse fell behind.

The stock loop. Stock has to come back around the loop to every channel before the next sale. On a quiet day a two-hour loop sells nothing twice, because nothing sells out in two hours. On a peak day the loop is the oversell. Is It in Stock? Inventory, Availability and the Promise You Can Keep owns **available-to-promise** (ATP: physical stock minus what is already promised) and the fix; here the point is that *every* sync interval is a peak-day risk in proportion to the sales rate.

The ERP, the data warehouse and the notifications come last and can wait longest. An invoice posted at 03:00 instead of 21:00 costs nothing; a confirmation e-mail an hour late costs a support ticket. They are where queues should grow.

The requirements table makes the ordering explicit; the latencies are the author's illustrative targets for an Omnichannel merchant.

Flow	Freshness needed at peak	Consistency	Latency target *(assumption)*	If it cannot keep up
Product page and search	Minutes (cached)	Eventual	Under 1 s	Serve cached; drop personalization
Stock shown on the product page	Minutes, with a buffer	Eventual, but bounded	Under 2 min from the WMS event	Show "low stock" from the buffer; never "in stock" from a stale copy
Checkout, payment authorization, order record	Immediate	Strong (one order, one payment)	Under 3 s	**Never degrade**; queue everything after it

Flow	Freshness needed at peak	Consistency	Latency target *(assumption)*	If it cannot keep up
Order to the WMS	Minutes	At-least-once, idempotent	Under 10 min from order to warehouse	Let the queue grow; widen the delivery promise
Stock from the WMS to every channel	Minutes	Eventual with a per-channel buffer	Under 2 min	Pause marketplace listings of low-stock SKUs
Notifications, ERP import, invoicing	Minutes to hours	At-least-once; daily batch for finance	Under 15 min for e-mails; overnight for the ERP	Delay; never duplicate

Preparing: Capacity, Forecasts and Load Tests

Capacity planning is deciding, before the day, how many orders a minute each link must carry and proving it can. The input is a forecast; the proof is a **load test**, deliberately sending realistic artificial traffic to find where the chain breaks before customers do.

The two best-documented programs are a platform's and a retailer's. Shopify's 2025 readiness post describes preparation that "started in March with capacity planning", five full-scale tests from April to October on production infrastructure at around **150% of the previous year's load**, scripted browse, cart and checkout flows with flash-sale bursts layered on top, and **Game Days** that injected faults into "critical journeys" (checkout, payment processing, order creation, fulfillment, search) and recorded the findings in a Resiliency Matrix of vulnerabilities, runbooks and recovery objectives.

Early tests saw "checkout queues backed up", and adding authenticated checkout "exposed rate-limit paths that anonymous browsing never touches" (Shopify Engineering, November 2025).

Its 2023 post explains why tests "bypass the CDN for a portion of the traffic": so that the origin, not the cache, is measured (Shopify Engineering, January 2023).

Zalando load-tested its production website end to end in 2019 because team-level tests could not show whether the whole system held together: recorded shopper journeys were replayed to about **130,000 requests per second**, above the real Cyber Week peak, and broke on replay because sessions expired, items sold out and rate limiting blocked

requests, which is itself a finding (Zalando Engineering, April 2019; seven years old). Its 2020 retrospective adds the practice a merchant can copy most directly: **order-flow simulators that place test orders which fulfillment skips**, exercising inventory and payments without shipping anything (Zalando Engineering, 2020).

For a merchant on a hosted platform, the translation is short:

1. **Forecast in orders per minute**, from last year's peak hour and this year's promotion plan. One and a half times last year's peak is a defensible figure for a growing merchant; the forecast's owner is whoever owns the promotion calendar, because the promotion makes the peak.
2. **Test the journeys, not the servers**. Place marked orders at the forecast rate and watch them reach the WMS, the stock come back, the e-mails go out and the ERP import run. Mark the orders so the warehouse skips them and finance excludes them.
3. **Test the authenticated and integrated paths**. Logged-in wholesale accounts, the marketplace connector's order pull and every app that writes stock take API budget that anonymous browsing does not.
4. **Rehearse failure on the money paths**. Switch off the WMS connector for thirty minutes in a test and see whether anyone notices and whether replay works; have the PSP decline every tenth test payment; zero the stock feed. Each exercise finds a gap or proves a runbook.
5. **Fix the slowest link and repeat** until every link carries the forecast with headroom.

A **freeze** (no new apps, theme edits, connector or configuration changes after a set date) is the cheapest reliability measure available, because the most common cause of a peak-day incident is a change made the week before.

Running the Peak: Queues, Backpressure, Caches and Switches

Preparation removes most surprises; it does not remove all of them. Four mechanisms let a chain survive the surprises that are left.

Queues and backpressure

A **queue** is a buffer in which work waits until the next system can take it. **Backpressure** is the discipline of letting a slow downstream system slow the producers, or letting work pile up in a queue, instead of failing the customer-facing step. The AWS Builders' Library states the principle: a system given more work than its resources support becomes slow

and should "start rejecting excess requests so that it can focus on the requests it decides to let in"; what matters is **goodput**, "the subset of the throughput that is handled without errors" and fast enough to be useful (AWS, December 2019).

In a commerce chain the principle has a specific shape. **Accept and persist the order fast**; **queue everything downstream**: the confirmation e-mail, the ERP export, the WMS release, the marketplace stock push. Zalando's 2015 practice of pausing non-essential asynchronous processing to protect order acceptance is this rule in production (Zalando Engineering, 2020). Kestrel & Pine's order flow *had* a queue in 2026, on the iPaaS; what it lacked was a gauge, an alarm, an owner and a plan for when the gauge rose. A queue nobody watches is a delay nobody announces.

Queues also create the duplicates and gaps of **The Message in a Bottle**. A flow that times out and retries delivers the same order twice unless the receiver is **idempotent** (handling a repeated message has the same effect as handling it once), with the order number as the key. A message that fails every retry lands in a **dead-letter queue** (a holding queue for messages that could not be processed, so a person can look at them), which needs an alarm and an owner or it is a silent order-loss mechanism. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns these mechanics.

Caching

A **cache** is a stored copy served instead of recomputing or refetching. On a peak day, what not to cache is the question: a cached product page is a gift; a cached stock figure is a Phantom Stock waiting to happen. My rule: a cached stock badge may be minutes old only if it reads "low stock" under a threshold, never "in stock" from a copy older than the sales rate allows.

Waiting rooms

A **waiting room** is a virtual queue in front of the storefront that admits shoppers at a set rate when traffic exceeds a threshold. Cloudflare launched its Waiting Room in January 2021 after vaccine registration sites failed under demand; it is configured with a maximum number of active users, a rate of new users per minute and a session duration (Cloudflare, January 2021; configuration reference accessed November 2026). Queue-it, a specialist vendor, redirects visitors only when traffic exceeds a set capacity and lets operators adjust the outflow during the event (Queue-it, n.d., accessed November 2026).

A waiting room converts an unknown spike into a known outflow rate. That is useful only if the merchant knows its end-to-end capacity in orders per minute, and it protects the storefront, not the chain: **it does nothing for an order export that falls over at 2,000 orders an hour**. For a hosted-platform merchant it is a launch-day tool more than a Black Friday one, and its admission rate should be set from the slowest tested link, usually behind the platform.

Graceful degradation and the switch-off order

Graceful degradation is switching off non-essential features in a planned order so the essential ones keep working. The mechanism is a **feature switch** (also called a feature flag or toggle: a setting that turns a behavior on or off at runtime without a deployment). Pete Hodgson's taxonomy calls the relevant kind **ops toggles** and notes that long-lived kill switches let operators "gracefully degrade non-vital system functionality" under high load (Hodgson, martinfowler.com, October 2017).

The documented case of degradation under fire is Amazon's Prime Day 2018, as reported by CNBC from internal documents: within fifteen minutes of the noon launch the site switched to a simplified fallback front page, at 12:15 international traffic was cut off to relieve a failing internal system, and at 12:37 the default page was restored for a quarter of traffic (CNBC, July 2018; a press account, eight years old). The fallback page and the traffic cut were switches that existed before the day. The team had somewhere to go.

A merchant's switch-off order is shorter. The figure gives the ladder I recommend for an Omnichannel merchant on a hosted platform; Journey 4: Black Friday, Hour by Hour gives Kestrel & Pine's 2027 version with owners and triggers.

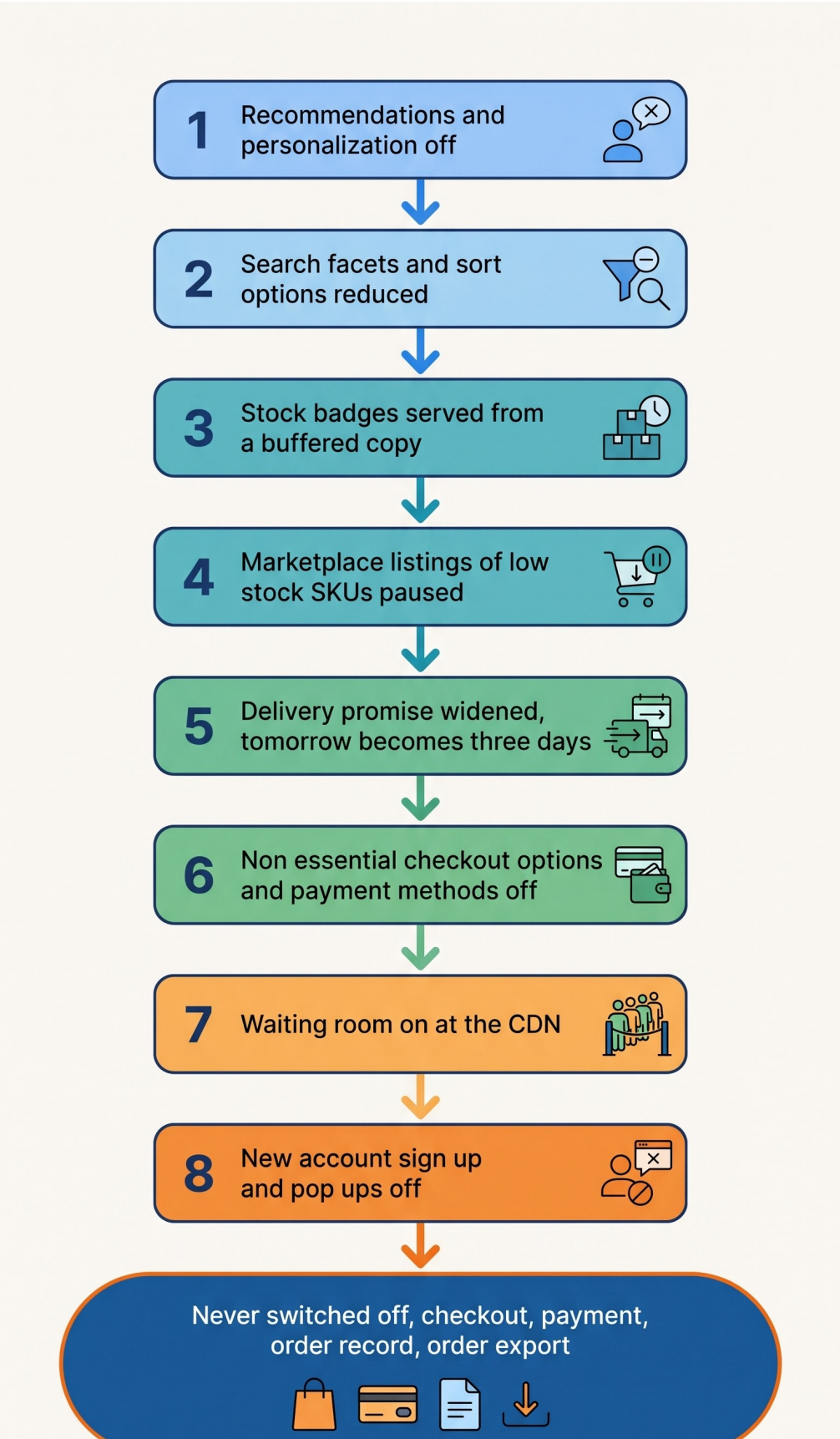

1
Recommendations and personalization off
2
Search facets and sort options reduced
3
Stock badges served from a buffered copy
4
Marketplace listings of low stock SKUs paused
5
Delivery promise widened, tomorrow becomes three days
6
Non essential checkout options and payment methods off
7
Waiting room on at the CDN
8
New account sign up and pop ups off
Never switched off, checkout, payment, order record, order export

Figure 3: *A degradation ladder for a hosted-platform merchant: each rung removes load or risk while keeping the promise that matters, and the bottom rung names what is never switched off.*

Three rules make a ladder work: every switch has been flipped in a test before November, because a switch that has never been used is a hope; every switch has one named owner who may flip it without a meeting; and the order is decided in a calm month by the people who own the revenue, not in the peak hour by whoever is on call.

The Choices and Their Trade-offs

Peak preparation has four levels, each adding responsibility:

- **Trust the platform**: reasonable for a small merchant whose hosted platform and logistics provider handle the full path, until custom integrations appear.
- **Protect the front door**: add caching, traffic controls or a higher service tier for concentrated launches.
- **Protect the chain**: measure every link in orders per minute, test integrations, monitor queues and define which work to pause. Kestrel & Pine adopts this for 2027.
- **Operate the platform yourself**: take responsibility for the broader capacity and fault-testing program described by companies such as Shopify and Zalando.

Choose the level that protects the order path you actually operate. If the warehouse connector is the bottleneck, a faster storefront will only feed its queue more quickly. Additional controls need a team that can maintain them; otherwise the apparent protection can disappear as systems change.

	Trust the platform	**Protect the front door**	**Protect the chain**	**Build a peak-grade stack**
What it protects	Nothing beyond the vendor's own uptime	The storefront and checkout	Every link from order to warehouse and back	Everything, including the platform itself

	Trust the platform	Protect the front door	Protect the chain	Build a peak-grade stack
Cost	None	Low: CDN, waiting room, a tier upgrade	Months of integration work, test tooling, one owner's time	A platform team and a year of calendar
Fails when	The first connector exists	The order export is the bottleneck	Nobody owns it after the first year	Copied by a merchant a hundred times smaller (The Giant's Boots)
Right for	Starting	Growing with launch-driven peaks	Omnichannel	Group, or a platform vendor

My recommendation for a merchant with its own warehouse and more than one channel is the third posture, with one addition: **the peak plan has a named owner all year**, not a project team that disbands in December. It does not hold for a merchant whose peak is under five times its normal hour (ordinary reliability work is enough) or whose integrations are all vendor-built and vendor-tested, which is rare.

Observing the Day: Orders per Minute as the Health Signal

Server metrics tell you that the servers are fine; in 2026 every server in Kestrel & Pine's chain was fine while the business lost €96,000. **The health signal for a commerce platform is orders per minute**, plotted against the same weekday and hour of recent weeks, with a sustained drop as the alarm. The operators of the largest peaks report their health in business units (orders per second at Alibaba, orders per minute at Zalando, checkouts and sales per minute at Shopify), and a merchant should do the same.

Figure 4: *The health signal is orders per minute against the same weekday and hour of recent weeks: in 2026 every server in the chain was fine while the business lost €96,000.*

I would organize the peak dashboard around **traffic**, **latency**, **errors and saturation**, Google's four monitoring signals, applied to each business flow (Google, 2017). For order export, show incoming orders per minute, the oldest unprocessed order, failed deliveries and remaining API capacity. For stock, show event rate and lag. For payment, show authorization time and decline rate against normal.

Queue age and spare capacity expose trouble before customers do. Payment declines reveal failures a healthy server cannot explain. Also alert when events stop arriving while the shop remains busy; a silent flow can look deceptively healthy.

Targets belong in business units too. Google's SRE book defines a **service level objective** (SLO) as a target value for a measured aspect of service (Google, 2017); a merchant's peak SLOs are checkout success rate, order-confirmation latency, time from order to WMS release and stock lag to marketplaces. The vendors' service agreements cover their own uptime; **the merchant's integration SLOs are nobody else's**.

Both documented programs staff a room through the peak: Zalando's Situation Room with incident commanders (2020), Shopify's round-the-clock incident coverage (2025). A merchant's version fits six people in one room, but needs the same elements: an **incident commander** who decides, one person per link, the vendors' emergency contacts, a **runbook** (a written list of symptoms, checks and actions, including which switch to flip and who may flip it) and a communication plan. The runbook is written in September and rehearsed in November; a runbook written during the incident is a diary.

Across Scale Stages: When It Fits and When It Limits

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting (one channel, hosted platform, a 3PL)	The platform's peak is not yours; the 3PL's cut-off and the PSP's limits are the real risks	Know the 3PL's peak capacity and cut-off; confirm the PSP's limits; a promotion calendar	The first hand-built connector; a launch that sells more than the 3PL can ship in a day
2 — Growing (two or three channels, an ERP link, a marketplace)	The first integrations are the first bottlenecks; peaks found by surprise; the agency on the phone	Test the order export at ten times normal; a queue between the platform and the ERP; a written switch-off list; marked test orders	A sync interval that oversells; a queue nobody can see; a peak that the agency, not the merchant, runs
3 — Omnichannel (stores, wholesale, marketplaces, own warehouse)	Kestrel & Pine in 2026: the storefront holds and the chain behind it fails in sequence; marketplaces and store stock are in the blast radius	A peak plan with a named owner; capacity per link in orders per minute; integration load tests; gauged queues with dead-letter alarms; the degradation ladder with owners; a situation room; a reconciliation owner	Vendors' service level agreements (SLAs) quoted as the merchant's SLOs; the same post-mortem two years running; an OMS (order management system) or availability service bought but not load-tested before November

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
4 — Group (several brands or countries, platform teams)	Shared peak tooling across brands; some brands on different platforms; a group calendar	Shared load tests, dashboards and runbooks; vendor SLAs read against group SLOs; failover decisions; a platform team that owns the year-round plan	One brand's flash sale starving another's order export on a shared connector; peak readiness that depends on one engineer

The judgement: the practices do not change across stages, the ownership does. At Starting and Growing a peak plan is a checklist the founder or the agency runs; at Omnichannel it is a program with a name, a budget line and an owner who does nothing else in October and November; at Group it is a platform team's product. Kestrel & Pine ran its 2026 peak with a Growing-stage plan (the agency watching the storefront) for an Omnichannel-stage chain.

In Practice: Kestrel & Pine's 2027 Peak Plan

Noor's answer to Mara's question was a plan built from the dated practices above, approved with Waypoint in March 2027 and worked out in full in Journey 4: Black Friday, Hour by Hour. Its spine, with the figures the plan commits to (all fictional):

- **One unit and one forecast**. Every link is measured in orders per minute. The forecast for Black Friday 2027 is **60 orders a minute** in the peak hour, about one and a half times 2026, and every link is tested at **80**.
- **The order path rebuilt**. By the peak, marketplace orders (since 15 September 2027) and direct-to-consumer (D2C) orders (since 13 October) are created in the new OMS (Waypoint W2), which sends fulfillment orders to the WMS in batches within its rate limit, with the order number as the idempotency key, a gauge on the age of the oldest unsent order, an alarm at ten minutes and a dead-letter queue with an owner. Tested at 120 orders a minute with marked orders that the WMS skips.
- **The stock loop replaced**: the availability service, a module of the OMS (W1), computes available-to-promise per channel from WMS stock events within a minute, with a buffer per channel. Go/no-go for the peak: **15 October 2027**; if

it is not ready, the old chain runs with the ten-unit marketplace buffer in force since December 2026.
- **A degradation ladder** of eight switches, each with an owner and a trigger, rehearsed in October.
- **A freeze** on apps, theme, connector and OMS changes from **15 November to 5 December 2027**, after an app audit removed the six unused apps and reduced the systems writing stock to one.
- **A situation room** on Friday 26 November from 16:00 to midnight with Noor as incident commander, and a **reconciliation report** at 08:00 on Saturday comparing orders in the platform, the OMS and the WMS, stock in the WMS and the availability service, and payments captured against orders.
- **Deliberately after the peak**: wholesale orders by EDI (electronic data interchange, the structured order messages a retail chain sends) and by hand move to the OMS on 8 December. Nothing that has not run through a load test goes live in November.

The plan costs about four months of one integration engineer's time, a load-testing tool and Noor's October and November. Against €96,000 in one evening, Mara approved it in twenty minutes.

When It Goes Wrong: Recovery and Reconciliation

A peak plan reduces incidents; it does not eliminate them. What distinguishes a prepared merchant is that it knows, before the day, what each failure leaves behind and how to clean it up.

Failure at peak	What it leaves behind	Detection	Recovery
Order export falls behind	Orders in the platform not yet in the WMS; a broken delivery promise	Queue depth; order-to-WMS lag alarm	Let the queue drain; widen the promise; notify customers once
Connector retries after timeouts	The same order twice in the WMS; two parcels	Duplicate check on the order number; the morning reconciliation	An idempotent receiver prevents it; otherwise cancel the duplicate before picking

Failure at peak	What it leaves behind	Detection	Recovery
Messages exhaust their retries	Orders in the dead-letter queue, invisible to the warehouse	Dead-letter alarm	Replay after the cause is fixed; confirm each order arrived once
Stock copy stale	Units sold in several channels	Orders for a SKU above available stock	Pause listings; allocate what exists to the earliest orders; cancel or delay the rest with a voucher
PSP degraded or one method failing	Declines rising; carts abandoned	Decline rate vs. normal	Switch the method off; confirm every captured payment has an order

Recovery after a peak is **reconciliation**: comparing two systems' records and fixing the differences. The platform's own documentation makes the case: Shopify tells apps that "webhook delivery isn't always guaranteed" and to run reconciliation jobs that periodically fetch changed data (Shopify developer docs, accessed November 2026). The morning-after report compares orders across systems by order number, stock per SKU between the WMS and the availability service, and payments captured against orders; the owner of each difference is named in the plan, not found on the day. Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns the reconciliation card.

The last step is the **post-mortem**, written within two weeks, blameless, with the timeline in orders per minute and every action assigned. Kestrel & Pine's became the Waypoint proposal; a post-mortem that becomes nothing is the surest predictor of the same incident next year.

What to Take Away

Established.

- A hosted platform's multi-tenancy smooths its peak to roughly twice its average (Shopify's 2025 figures), while a single merchant's chain sees that merchant's full spike; the storefront is the link most likely to hold and the integrations the links most likely to fail (Shopify 2025, Zalando 2020).

- Preparation at the documented operators follows one shape: forecast, scale tests at forecast load that bypass caches and include checkout and authenticated paths, fault injection on the money paths, a freeze, a staffed room and a runbook (Shopify 2023 and 2025, Zalando 2019 and 2020).
- Under overload, persisting the customer-facing commitment fast and queuing everything behind it preserves goodput (AWS 2019); planned degradation through ops toggles keeps the essential path alive (Hodgson 2017); and because delivery between systems is at-least-once, a peak's retries produce duplicates and gaps unless receivers are idempotent and a reconciliation runs afterwards (Shopify and Stripe documentation, 2026).

Depends on context.

- A waiting room pays off for launches and flash sales where the storefront is the bottleneck and the admission rate can be set from a known end-to-end capacity; for a hosted-platform merchant whose bottleneck is the order export it adds little.
- One and a half times last year's peak is a reasonable forecast for a growing merchant with a stable promotion plan; a merchant adding a marketplace, a country or a television campaign should forecast from the plan, not the history.
- The full readiness calendar belongs to platform owners and Group-stage merchants; an Omnichannel merchant that copies it wears the Giant's Boots, and one that skips its integration half meets the First Snowfall.

The author's synthesis.

- Measure the whole chain in orders per minute, because it is the only unit that every link, every vendor and every executive can read, and the only one in which the slowest link is visible before the day.
- The peak plan is a year-round product with one owner, not a November project; the switch-off ladder is written by the people who own the revenue, in a calm month, and the engineers' job is to make sure every rung exists and has been flipped once before it matters.

Questions to Ask

1. What was our peak hour last year, in orders per minute, and what is the forecast for this year? Who owns that forecast?
2. For each link from checkout to warehouse and back, what is its tested capacity in orders per minute, and when was it last tested at that rate?
3. Which apps and connectors share the platform's API budget, and what are the WMS's and the PSP's rate limits?

4. Where are the queues in our chain, who can see their depth, and what happens to a message that fails every retry?
5. What is our switch-off order, who owns each switch, and when was each last flipped?
6. Can we withdraw the delivery promise on the page when the warehouse falls behind, and who decides?
7. Does the dashboard show orders per minute against last week, order-to-warehouse lag and payment decline rate, and who is the incident commander on the day?
8. The morning after, which two systems do we compare, by which key, and who fixes the differences? When is this year's freeze?

The peak is the day the chain is honest. The next part turns to the people who exploit that honesty on purpose, the card testers, account thieves and bots that arrive with the shoppers, and to the rules that govern what a commerce platform may do with the data it holds. Continue with Part VIII — TRUST: Security, Privacy and Regulation.

Sources and Evidence

Company, product and investor sources

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- Black Friday Cyber Monday 2024 data, Shopify News, 2024 — BFCM 2024 load figures for comparison.
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- Exponential backoff and jitter, AWS Architecture Blog, 2015 — why retries need randomness.

Research and practitioner sources

- Feature Toggles (aka Feature Flags), Pete Hodgson, martinfowler.com, 2017 — the toggle taxonomy and ops toggles as kill switches for graceful degradation.
- Starbucks Does Not Use Two-Phase Commit, Gregor Hohpe, 2004 — accept the order fast and let the work queue; write-off, retry and compensation as the error strategies.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part VIII — TRUST: Security, Privacy and Regulation

KEY POINTS:

- **Attackers use the shop as built.** Card testing, credential stuffing and skimming are the checkout and login at machine speed; the merchant secures every identity, key, app and script it adds.
- **The order and the person are different things in law.** The order stays for years under contract and tax law; the profile goes when the person asks, so the transaction must outlive the person.
- **Most rules are data requirements with a date.** Each belongs in a system of record, with a test and a calendar row that names an owner.

The checkout that accepts a real purchase can also receive automated payment attacks. The address used for delivery can remain in a forgotten export, and a pricing rule

can require data the shop has never collected. Security, privacy and regulatory work each follow information across several systems. This part explains how to identify the responsible owners and controls before the gap harms a customer or interrupts trading.

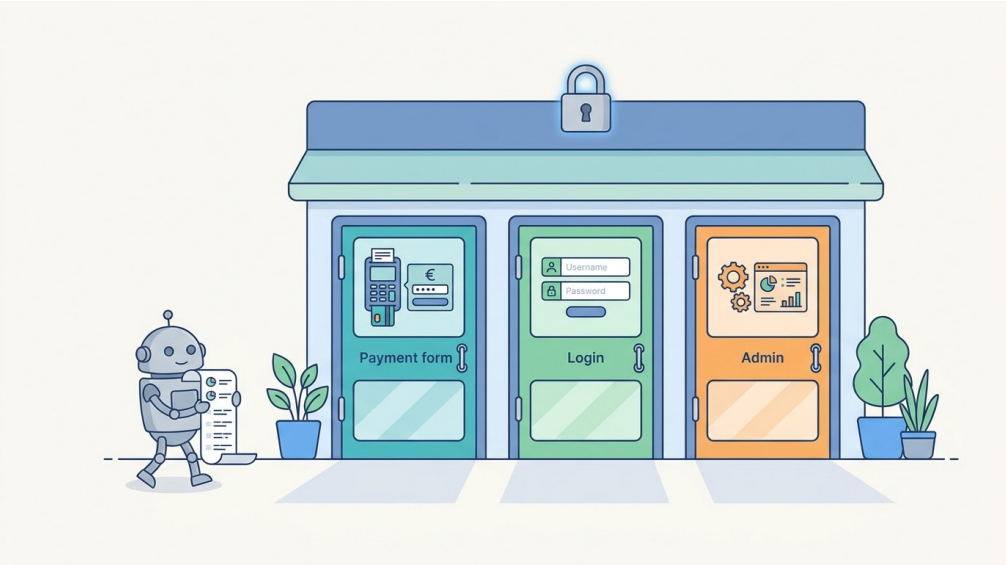


Figure 1: *A shop is three public forms, a payment form, a login and an admin, and none of them has to be broken to be abused.*

Where we are in the order's journey. Eva buys a jacket from Kestrel & Pine, the fictional outdoor brand. Her purchase passes through checkout, payment, warehouse, carrier and finance. Each needs appropriate access and data handling. Product information, pricing, invoicing and the later return process must also meet the rules applicable to that transaction.

Three examples show the work: **48,000 abusive checkout attempts** in August 2026; a customer's May 2027 deletion request; and regulatory changes reaching engineering through unrelated channels. The response begins with a map of systems, integrations and owners. Add credentials, data purposes and retention, then applicable requirements and deadlines. The map makes responsibilities actionable; each topic still needs its own controls.

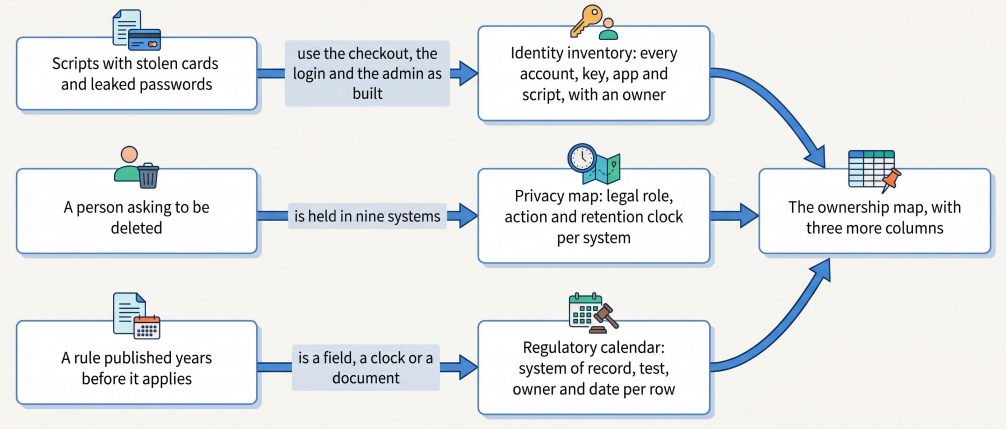


Figure 2: *Three things reach a shop from outside, a script, a person with rights and a rule with a date, and each is answered by a list with owners; the three lists are columns of the same ownership map, not a separate discipline.*

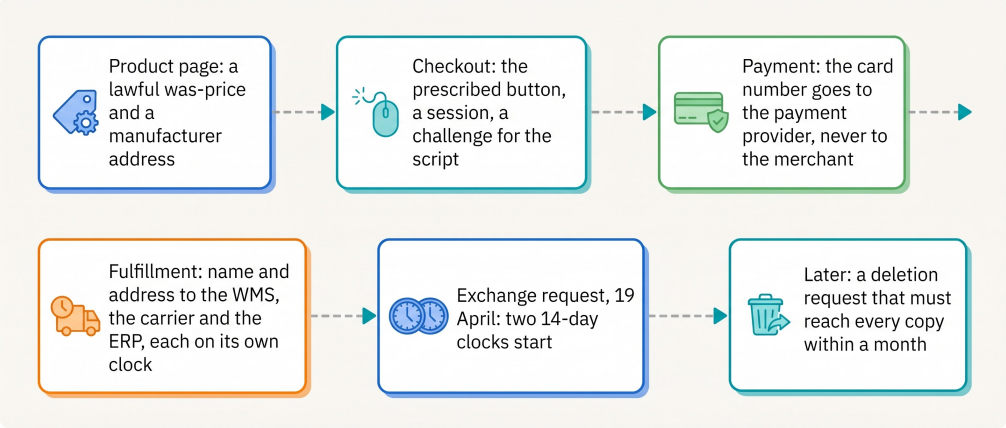


Figure 3: *Where Part VIII meets Eva's jacket: security at the checkout and the payment, regulation at the price, the button, the product page and the clocks, privacy in every system her name reaches and in the request that one day asks for it back.*

Chapters in this part

Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers examines **card testing, account takeover, checkout scripts and integration credentials**. The attack costs €2,900 in declined-payment fees and interrupts sales. The chapter assigns controls to the merchant and provider and establishes incident recovery.

Whose Data Is It? Privacy, Consent and Retention maps **customer records, their purposes and the actions required by a request**. A customer in Ghent appears in

nine systems. Identity links and a coordinated workflow make later requests easier to complete and verify.

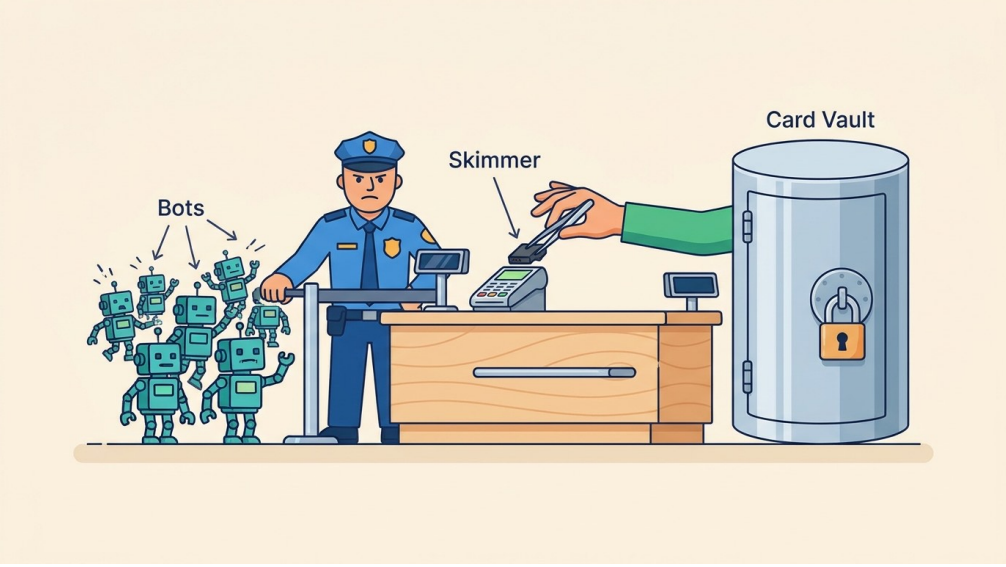
Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation turns **an applicable rule into owned implementation work**. Identify scope, required data or behavior, authoritative system, verification and deadline. The worked calendar covers product information, accessibility, electronic invoicing and unsold goods, with dates and evidence in the chapter.

Begin with Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers.

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine is fictional; recommendations and labeled assumptions are the author's judgement.

34. Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers



IN THIS CHAPTER, YOU WILL: Learn what attackers take from a shop, where the card-data standard draws its line, which controls stop card testing, takeover and skimming, and what stays your job on a hosted platform.

KEY POINTS:

- **Attackers do not break in; they use the shop as built.** Card testing, credential stuffing, coupon abuse and inventory hoarding are the payment form, the login, the promotion engine and the cart used at machine speed. The control is to make every endpoint that can trigger a payment or a login authenticated, rate-limited and challengeable, and to watch the decline rate as closely as the order rate.
- **Keep the card number out of your systems, and own everything around the box that takes it.** With a hosted checkout or hosted fields the card number never touches your code and the lightest PCI

DSS self-assessment applies. Since 31 March 2025 the standard's script requirements (6.4.3 and 11.6.1) make the scripts on the payment page your inventory to keep, because skimmers run in the shopper's browser on a page you serve.

- **The platform secures the platform; you secure the identities**. Every vendor's shared-responsibility statement draws the same line: the vendor patches and certifies the service; the merchant owns admin accounts, API keys, apps, scripts and integrations. An integration without its own identity and a least-privilege key is the Secret Agent waiting for a leak.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Secret Agent** — a script, app or integration that can change orders, prices, stock or money without its own identity cannot be traced when its key leaks, and a leaked key is the attacker's favorite entry.
- **The Lynchpin** — one shared admin password, one payment account and one person who knows where the keys are is a security architecture that fails in a single step.
- **The Spaghetti Junction** — thirty-four integrations (hosted integration flows, agency scripts and vendor connectors), each holding a credential nobody inventoried, are thirty-four doors with no record of who holds the key.

A checkout must accept payment attempts from strangers. An attacker can use that same behavior to test stolen card details, while automated logins can try stolen passwords and bot carts can reserve stock. The shop may process these requests exactly as designed. Security must distinguish a legitimate purchase from misuse of the same functions, as well as prevent unauthorized access.

The damage appears as fees for declined payments, a paused payment account, stolen customer access or stock held in bot carts. Malicious checkout scripts can also steal card details. The cited British Airways case, a 2018 breach followed by a £20 million penalty in 2020, shows the consequences of that browser-side exposure.

For a merchant renting its platform, security begins with the responsibilities it still holds: access, integrations, custom scripts and abuse controls. This chapter covers payment-page exposure, card testing and account takeover. Whose Data Is It? Privacy, Consent and Retention covers personal data, Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation the wider rules and Scams, Fakes and Account Takeovers: Fraud Prevention for Listings fraud models.

Where we are in the order's journey. Eva buys a jacket at Kestrel & Pine, the fictional outdoor brand, in April 2027 using iDEAL, a Dutch bank-transfer method. The same checkout received 48,000 abusive attempts one night in August 2026. This chapter examines how to distinguish legitimate purchases from automated misuse.

Setting the Scene at Kestrel & Pine: The Card-Testing Night

Kestrel & Pine's **Card-Testing Night** runs from **18 to 19 August 2026**. Its figures belong to the fictional scenario; the mechanism below is an illustrative reconstruction.

Between about 23:40 and 04:10, a script makes **48,000 guest-checkout attempts**, using stolen card numbers and the cheapest catalog item. Its purpose is to discover which cards still work. This is **card testing**, or carding.

The Open Worldwide Application Security Project (**OWASP**) lists this automated abuse as OAT-001 in its Automated Threats project (undated guidance, accessed November 2026). A successful-looking payment form has become a testing service for attackers.

Three things happened that the shop was not built to notice.

- The **PSP** (payment service provider: the company that takes the shopper's payment and settles the money to the merchant) charged a fee for every authorization request, declined or not: **€2,900** on the September statement for sales that never happened.
- At **02:05** the PSP's own monitoring saw an authorization rate and a decline rate it had never seen from this account and **paused the account for 40 minutes**. For 40 minutes nobody could pay by card on any storefront, and the e-mail that said so went to a shared mailbox nobody read until 07:30.
- About **260** of the 48,000 cards were live (illustrative). The platform created real orders for them, which Milan Petrov, who owns the storefronts, canceled and refunded by hand; **21** came back as disputes, and the share of genuine card payments declined by issuers rose from about 4% to about 7% for two weeks, because issuers had seen a burst of fraud from this merchant (illustrative; Stripe's card-testing documentation describes the mechanism, n.d., accessed November 2026).

Most attempts used cards issued outside Europe (illustrative), where **strong customer authentication** (SCA, the European rule that a shopper must prove their identity at payment; Taking the Money: Payment Providers, Authorization, Capture and Refunds) does not bind the issuer, so the usual challenge never fired. The agency switched on the platform's bot protection and the PSP's velocity rules that week; no runbook was written. In March 2027 Noor Visser, Head of Engineering since January, reads the incident report for the Waypoint program and asks a different question: **not "how do we stop card testing", but "which of our doors are open to a script, and who holds the key to each".**

What Attackers Want From a Shop

The table lists the threats in roughly the attacker's order of preference, with the OWASP Automated Threat identifier where one exists (OWASP, n.d., accessed November 2026).

Threat	What it is, in plain words	Who loses	Where the control lives
Card testing (OAT-001; OAT-010 when the expiry or security code is guessed)	Many small payments or card-save attempts to find which stolen cards work	The merchant: fees, disputes, a paused account, a worse decline rate for months	Checkout, PSP rules, rate limits, challenges
Credential stuffing (OAT-008) and **account takeover**	Password pairs leaked from one site tried on yours at machine speed; any route that ends with an attacker logged in as a customer or employee	The customer: saved addresses and cards; the merchant: refunds and trust	Login, multi-factor authentication (MFA), bot management
Web skimming (Magecart)	Malicious script on the checkout page that copies what the shopper types into the card fields	Every shopper who paid; then the merchant, through penalties and reputation	The scripts on the payment page: inventory, policy, tamper detection
Coupon abuse (OAT-002)	Guessing or sharing codes; stacking; one-time codes used many times	Margin	The promotion engine (Discounts, Coupons and Loyalty: Promotions Without Chaos)

Threat	What it is, in plain words	Who loses	Where the control lives
Refund and returns abuse	Refunds to a different card, empty-box returns, serial "not received" claims	Margin and stock	The returns and order systems (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics)
Inventory hoarding and scalping (OAT-021, OAT-005)	Bots fill carts to hold stock during a launch, or buy it all to resell	Real customers, then the brand	Cart reservation rules (Is It in Stock? Inventory, Availability and the Promise You Can Keep), bot management
Admin compromise	An attacker with a staff login or a leaked API key: payout account changed, scripts added, customers exported	Everything	Identity: MFA, least privilege, key management
Supply-chain risk	An app, plugin, theme or hosted script that is compromised and runs with your privileges	Everything the app could touch	App governance (Apps, Plugins and Functions: Extensibility and Ecosystem Governance), script policy, patch cadence

The last two rows bypass every other control: a skimmer does not need to beat the PSP, and an attacker with the admin password does not need to guess. And half of the rows are not intrusions but **policy abuse**, whose controls live in the order, promotion and returns systems; a bot-management product covers the top of the table and nothing else. Vendor reports put automated traffic at about half of all web requests (Imperva, 2025, measuring its own customers); whatever the true share, **on most shops a large part of the requests are not people**.

Shared Responsibility: What the Platform Does and What You Must Do

The line between what the vendor does and what the merchant does is published. Adobe's shared-responsibility model for Adobe Commerce on cloud infrastructure (n.d., accessed November 2026) is the most explicit: Adobe maintains the core code, the operating system, global firewall rules and PCI certification for the infrastructure; the merchant secures "custom code and integrations with third-party applications", applies patches, manages user access and handles its own incidents.

Shopify states that it is "certified Level 1 PCI DSS compliant" and that the certification "extends by default to all stores powered by Shopify", while adding that "there are also steps you will need to take on your own" (Shopify, n.d., accessed November 2026).

Stripe says the same about payments: "PCI compliance is a shared responsibility and applies to both Stripe and your business" (Stripe, n.d., accessed November 2026).

Read together: the vendor covers the service's code, hosting, patching and certification and the built-in tools with their defaults; the merchant covers the code, scripts and apps it adds, who holds an admin account and whether MFA is enforced, which API keys exist, and incident response for the shop. **The hosted platform removes the patching burden. It never removes the identity, script and endpoint burden**, which is the rest of this chapter.

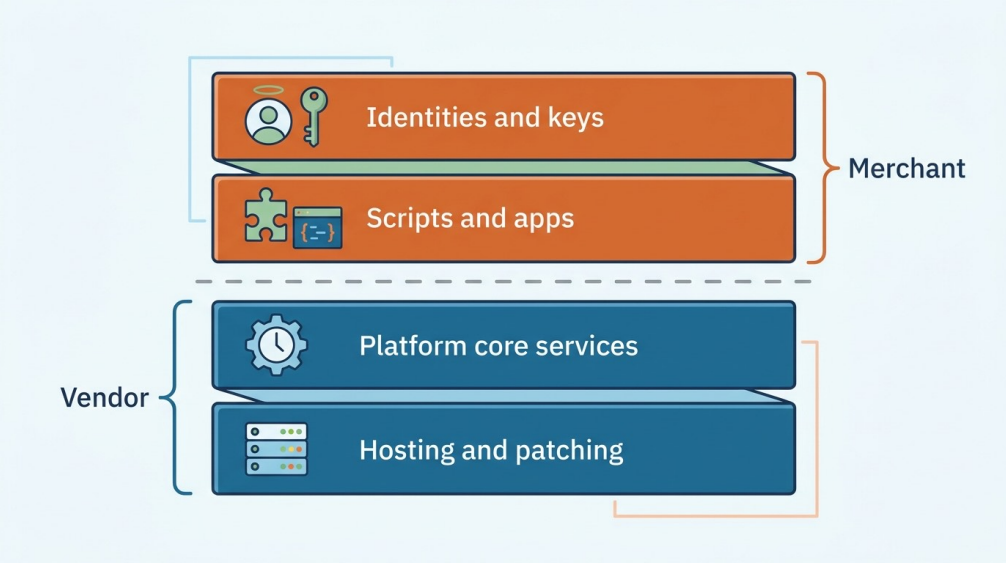


Figure 1: *The hosted platform removes the patching burden; the identities, keys, scripts and apps the merchant adds stay the merchant's to secure.*

Card Data: Where PCI DSS Draws Its Line

PCI DSS (Payment Card Industry Data Security Standard) is the card brands' security standard for anyone who stores, processes or transmits card data, a contractual obligation through the merchant's acquiring bank (the acquirer: the bank that accepts card payments on the merchant's behalf) rather than a law. Its **scope** is the set of systems that touch card data, and the design goal of a modern checkout is to make that set as small as possible: ideally, none of the merchant's own.

The dates matter because the standard changed recently: PCI DSS v4.0 was published in March 2022; v3.2.1 was retired on 31 March 2024; a limited revision, **v4.0.1**, was published on 11 June 2024, and v4.0 was retired on 31 December 2024. The requirements that v4.0 introduced as "best practice" became mandatory on **31 March 2025**, and v4.0.1 did not move that date (PCI Security Standards Council blog, 11 June 2024).

How much of the standard applies to you

The answer depends on how the card number, the **PAN** (primary account number), reaches the PSP. Smaller merchants attest compliance through a **self-assessment questionnaire** (SAQ), and which SAQ applies follows the integration pattern. The table is the author's synthesis of Stripe's and Adyen's guides (n.d., accessed November 2026); the acquirer, not a vendor's documentation, sets the SAQ.

Integration pattern	Who sees the card number	Typical SAQ	What the merchant still owns
PSP-hosted checkout page (the shopper is redirected to the PSP's page)	The PSP only	**SAQ A**, the lightest	Site integrity (the script statement below), admin access, order data
Hosted fields (card input boxes served by the PSP inside an iframe on the merchant's page)	The PSP only	SAQ A, if the eligibility criteria are met	Everything on the page around the iframe: scripts, headers, policy

Integration pattern	Who sees the card number	Typical SAQ	What the merchant still owns
Merchant-hosted form posting through the PSP's JavaScript	The merchant's page collects, the PSP tokenizes	SAQ A-EP, considerably heavier	Page security, scripts, tamper detection
Direct API with raw card data	The merchant's servers	SAQ D or a full assessment; "the most challenging SAQ there is" (Stripe)	The full cardholder-data environment: more than 300 controls

Hosted platforms with their own checkout put every merchant in the first two rows. **This is the largest security decision a merchant makes, and most make it by default.** A custom checkout that collects card numbers itself, as some headless projects do for design reasons (Delivering the Storefront: Themes, Headless, Rendering and Performance), moves the merchant down the table and multiplies the controls it must prove.

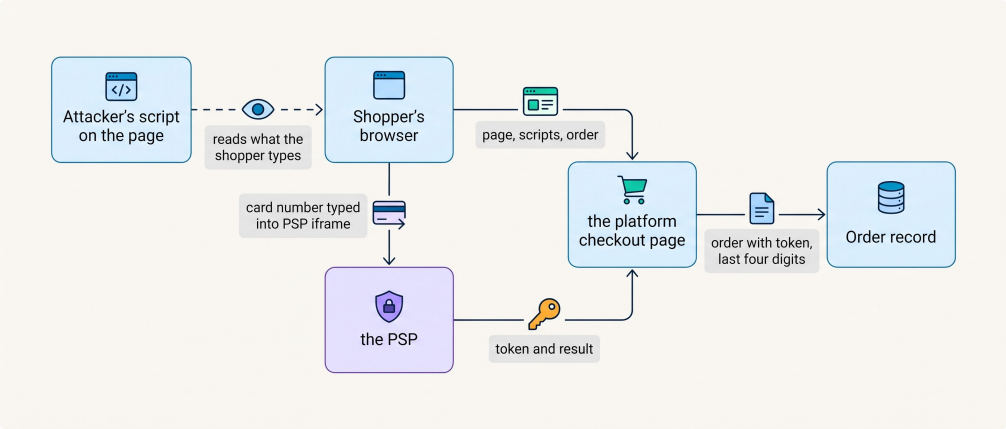


Figure 2: *With hosted fields the card number travels from the shopper's browser straight to the PSP and the merchant's systems hold only a token; the remaining exposure is any script on the page that can watch the shopper type, which is why the page's scripts are the merchant's responsibility.*

The card brand, last four digits and expiry returned by the PSP are out of scope and may be stored (Stripe security guide, n.d.); everything else is replaced by a **token**, a reference the PSP issues that can charge or refund that card again without the merchant

ever holding the number.

The two script requirements and the SAQ A statement

Two requirements exist because of skimmers. **Requirement 6.4.3**: every script on a payment page is authorized, integrity-checked and inventoried with a written business justification. **Requirement 11.6.1**: a mechanism detects and alerts on unauthorized changes to the payment page's content and security-relevant HTTP headers. Both were best practice from March 2022 and are required since **31 March 2025** (PCI SSC, 26 March 2025).

The Council then **removed 6.4.3 and 11.6.1 from SAQ A** and added an eligibility criterion instead: an SAQ A merchant must confirm that its site "is not susceptible to attacks from scripts that could affect the merchant's e-commerce systems", which it can do by applying the techniques itself, through a third party, or by obtaining its PCI-compliant payment processor's confirmation that the hosted solution includes them when implemented as instructed.

As the Council wrote, "these requirements were only removed from SAQ A, and they are still in the standard" (PCI SSC, 26 March 2025). In plain words: **a merchant on a hosted checkout is no longer asked to prove a script inventory, but it is asked to state that its page cannot be skimmed, and it had better know why that statement is true**.

Adyen's guidance describes the work: deploy a **Content Security Policy** (CSP, an HTTP header that tells the browser which script sources a page may load), first in report-only mode, then enforcing; keep the inventory with business reasons; use **Subresource Integrity** (SRI, a hash on a script tag so the browser refuses a changed file) on the PSP's scripts; and treat CSP violation reports as "valid alerts and 'indicators of compromise' " (Adyen docs, n.d., accessed November 2026).

Two other requirements recur. **Requirement 12.8** makes the merchant responsible for its service providers and for holding each one's attestation of compliance ("Adyen is solely responsible for the security of cardholder data only as soon as Adyen receives the data", Adyen docs, n.d.). **Requirement 8.4.2**, also mandatory since 31 March 2025, requires multi-factor authentication for all access into the cardholder-data environment: for a hosted merchant, the PSP console and the platform admin.

Card Testing: The Attack and the Controls

Stripe's documentation is the clearest public description of the attack (Stripe, "Protect yourself from card testing", n.d., accessed November 2026). Attackers prefer **saving a card** to paying with it, because card-setup authorizations "don't typically show up on cardholder statements". The symptoms are a spike in failed payments, a spike in HTTP

402 responses (the status code a payment API returns for a declined charge) and low-value payments "often with nonsensical customer names and emails". The consequences outlast the night: in Stripe's words, "a high decline rate might damage the reputation of your business with card issuers and card networks".

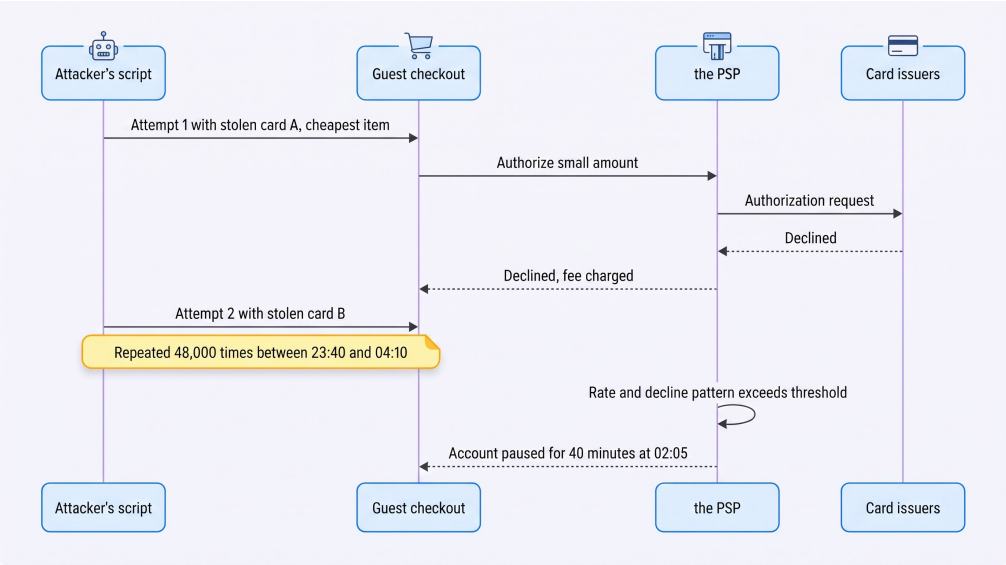


Figure 3: *Card testing on the Card-Testing Night: every attempt is a normal checkout to the platform and a normal authorization to the PSP; only the count, the pace and the decline rate reveal it, and the first system to react was the PSP, by pausing the account.*

Stripe states that "simple firewall rules or filters based on a single heuristic such as IP addresses are usually not sufficient", because attackers rotate addresses. The **leaked secret key** is the worst version: "card testers can create payments and set up cards using your secret key", with no checkout involved at all.

The control ladder

The controls form a ladder; the right rung depends on how much of the checkout the merchant built.

1. **Use the hosted checkout or hosted fields and accept the PSP's defaults**; do not bypass them with a custom flow.
2. **Switch on what you already pay for**: the platform's bot protection on checkout, the PSP's velocity rules (attempts per card, address and hour), a CAPTCHA validated on the server.
3. **Require a session before a payment**. Stripe recommends login or session validation before payment, limits on cards per account and rate limits such as new

customers per address per day. A guest still has a session that loaded a product page and a cart; a request that reaches the payment step with no history is the script.

4. **Alert on the decline rate, not only the order rate.** Black Friday Every Year: Peak Load, Resilience and Recovery makes orders per minute the health signal; the security twin is **declines against normal**. Kestrel & Pine's night was visible at 23:55 as a 99% decline rate against a usual 4%.
5. **Have a kill switch:** a setting, not a deploy, that disables guest card payments while iDEAL, Bancontact and PayPal keep running (From Cart to Confirmation: Checkout Orchestration).
6. **Never retry the cards the attacker saved.** Stripe warns that retrying saved cards "repeats the original attack"; a dunning job (an automatic retry of failed payments) that retries every failed card is a card-testing machine with your name on it.

Requirements per control point

Controls sit on the request path and have budgets like any other call (figures are assumptions).

Control point	Latency budget	False-positive tolerance	Consistency	When the control itself fails
Challenge on guest checkout	Under 300 ms at the payment step *(assumption)*	Low: each false challenge costs a conversion	Stateless	Fail open for methods with their own authentication (iDEAL, cards verified by 3-D Secure, the issuer's own confirmation step), fail closed for plain card
Login rate limit and MFA	Under 100 ms *(assumption)*	Low for customers; zero for staff	Per-account and per-address counters	Fail closed for staff, degrade to a challenge for customers

Control point	Latency budget	False-positive tolerance	Consistency	When the control itself fails
Webhook signature check	Microseconds	Zero: an unsigned webhook is never processed	Idempotent handling by event id (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation)	Queue and alert, never process

A challenge that fires on 2% of genuine shoppers costs more than most card-testing nights, which is why challenges are triggered by risk signals (no session history, velocity, reputation) rather than shown to everyone.

Account Takeover, Bots and Scraping

Credential stuffing is "the automated injection of stolen username and password pairs" to gain access to accounts, and OWASP names multi-factor authentication as a primary countermeasure (OWASP, n.d., accessed November 2026). In commerce: a shopper reuses a password that leaked elsewhere; a script tries it; the attacker changes the delivery address and orders with the saved card; the merchant sees refunds weeks later.

Three controls carry the weight: **rate limits and bot management on the login**, an edge service that classifies requests as human, good bot or bad bot from device, behavior and reputation signals; **a challenge for new devices and risky logins**, with passkeys (a phishing-resistant login using the device's own unlock) as the platforms' direction; and **notification of change**, an e-mail when an address or payment method changes, which catches the takeover after the fact and costs nothing. Customer MFA is rarely enforced because it costs conversion; staff MFA is never optional.

Inventory hoarding turns the cart into an attack surface: if a cart holds stock for 30 minutes, a bot can hold a launch's units for 30 minutes at no cost. The control is the reservation model, not the edge: a short hold, extended only by progress through checkout, and a limit per account during launches (Is It in Stock? Inventory, Availability and the Promise You Can Keep). **Scraping** is tolerated selectively: comparison and search engines scrape, so the typical policy allows product pages, challenges cart and checkout, and rate-limits the storefront API.

Skimmers: The Attack in the Shopper's Browser

The British Airways breach of 2018 remains the clearest public record of web skimming, though it is eight years old and predates PCI DSS v4. The attacker entered BA's network on 22 June 2018 with a stolen third-party login that had no multi-factor authentication, found a saved privileged credential, and between 14 and 25 August 2018 added 22 lines of JavaScript to the payment pages that sent card details to an attacker-controlled site; roughly 429,612 people's data was potentially accessible, and the UK Information Commissioner's Office issued a £20 million penalty on 16 October 2020 (Wikipedia, citing the ICO penalty notice, 2020).

Two lessons survive. **The skimmer did not break the PSP**; **it ran in the shopper's browser**, **on a page the merchant served**, **loaded by a script the merchant or a vendor put there**. And the entry was identity: a third-party login without MFA.

The supply-chain version is newer. On 25 June 2024 Sansec reported that `cdn.polyfill.io`, a script embedded by more than 100,000 sites, had been sold in February 2024 and was injecting code that redirected mobile users to a betting site, skipping admin users to evade detection (Sansec, 2024). Nobody who embedded that script had changed anything. For self-hosted platforms the equivalent is the patch: Adobe fixed CosmicSting (CVE-2024-34102) on 11 June 2024, and Sansec counted thousands of unpatched stores breached that October and November (Sansec, 2024). Stripe's guide puts the principle in one sentence: "Including JavaScript from other sites makes your security dependent on theirs."

The controls are the ones 6.4.3 and 11.6.1 encode: a **script inventory** with an owner and a reason per entry, a **CSP** that reports violations, **SRI** on scripts that should never change, and **tamper detection** against the known page. On a hosted platform the checkout page is the vendor's; the merchant's exposure is the scripts it adds through the theme, apps and tag managers, and the inventory is still the merchant's.

Admin Accounts, Keys and the Apps

The commerce admin is the most valuable login on the estate: it can change the payout bank account, add a script to the checkout, install an app, export every customer and issue refunds. **An attacker with it needs no other technique**, which is why identity is the center of a merchant's security program.

Shopify's documentation shows how platforms think about it (Help Center, n.d., accessed November 2026): staff "can set up two-step authentication for their own individual accounts", the store owner "can't activate two-step authentication for any other staff member" on standard plans, Plus organizations can require it for all users, and without it on the Payments account "your payouts might be placed on hold". The

platform makes MFA available; **enforcing it is the merchant's decision**, on many plans a policy to police rather than a switch to flip.

Keys are the other half. Stripe's key guidance (n.d., accessed November 2026) applies to every system: never put secret keys in source code, because "bad actors continuously scan public repositories" for them; use a secrets vault; create **restricted keys** with the least privilege the job needs (a refund-only key for the returns portal); rotate; audit; treat any exposure as compromise. A **webhook** (a message the PSP or platform sends to your endpoint when something happens) is an unauthenticated request until its signature is verified; an unsigned "payment succeeded" accepted by an order system is a free order.

Kestrel & Pine's January 2027 inventory found **34 integrations**, **13 without owners**, and **41 apps**, including three with stock-writing permission. Five integrations remained unowned in April (APIs, Events, Files and Middleware: How Commerce Systems Talk, Who Builds and Who Owns: Teams, Vendors, Agencies and Governance). Each credential needs someone who can explain its use, limit its permissions and replace it when staff or suppliers change.

My rule is **one integration identity, narrow permissions and a named owner who can rotate its key**. Separate identities let the team trace an action and revoke one integration without disabling all the others. Narrow permissions limit what a stolen or misused credential can change. Record the identity, permissions and owner in the inventory; resolve an email app's request for inventory-write access before installation (Apps, Plugins and Functions: Extensibility and Ecosystem Governance).

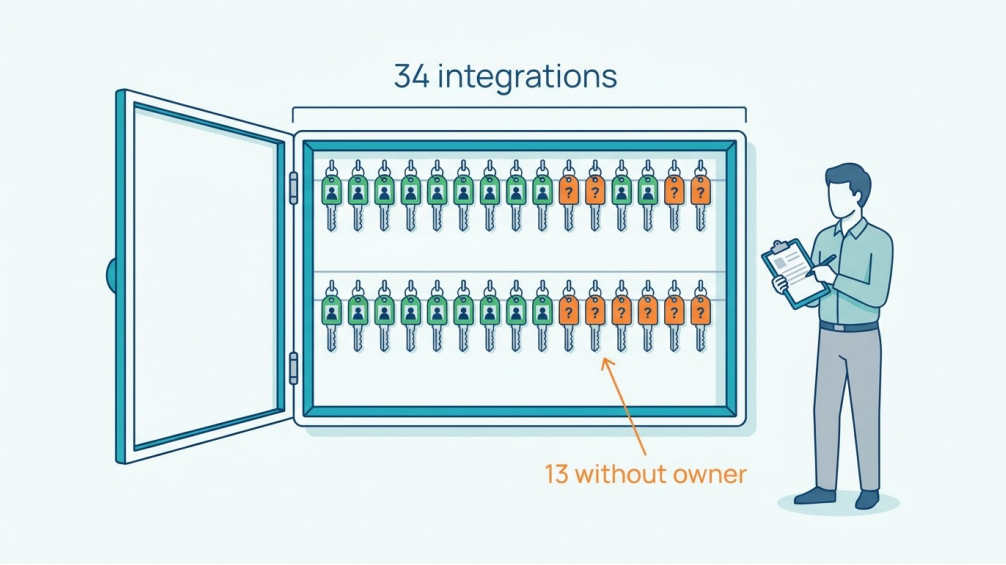


Figure 4: *Kestrel & Pine's inventory found 34 integrations, 13 without a named owner: every one holds a credential, and a key without an owner cannot be rotated, scoped or*

traced.

The Choices: Four Security Postures

The four are **platform defaults**, **defaults plus the tools you already pay for**, **edge policy and identity management** (a CDN, the content delivery network in front of the storefront, with bot management, per-endpoint policies, a CSP, a vault, keys with owners, attestations on file, app review in procurement) and **a security function** (central identity, maintained inventories, monitoring with on-call). The last is right for a group or a vendor and the Giant's Boots for a merchant with fourteen engineers.

Posture	Cost (illustrative)	Closes	Leaves open	Right when
Platform defaults	Nothing	Card-data scope (SAQ A); the service itself	Card testing on guest checkout, takeover, keys, scripts, apps	One channel, no integrations, few staff
Defaults plus paid tools	Days of configuration; attention	Card testing, most takeover, the admin	Integrations' keys, scripts, apps, terminals	Two or three channels, first integrations
Edge policy and identity	A CDN tier, a vault, a quarter of an engineer *(assumption)*	Keys, scripts, apps, per-endpoint abuse	Monitoring depth, cross-brand consistency	Stores, wholesale, marketplaces, many integrations
Security function	Headcount; tooling	Nearly everything in the table	Cost, and controls nobody can explain	Group or vendor scale

The recommendation is dull and I hold it firmly: **move one rung at the moment the landscape gains the thing the rung protects**. The first integration with a key is the moment for the vault and the inventory; the first custom checkout is the moment for the CSP and script inventory; the first store terminal is the moment for the attestation file. Earlier buys controls nobody operates; later buys an incident.

Across Scale Stages: When It Fits and When It Limits

Stage	Typical posture	What becomes necessary	Warning signs of outgrowing it
1 — Starting (under about €2M)	Hosted checkout, SAQ A, the PSP's defaults, the founder's admin with MFA	Removing unused apps; no shared admin account; MFA on the PSP console	A custom checkout "for the brand"; a developer's key in a repository; the first decline spike
2 — Growing (about €2M–€20M)	Defaults plus paid tools; an agency with admin access; first vendor connectors	Challenges on guest checkout and login; velocity rules; MFA enforced for staff and agency; a decline-rate alert; a kill switch; a key per integration	Keys shared between agency scripts; nobody can list the apps; a PSP pause noticed at 07:30 (Kestrel & Pine, 2026)
3 — Omnichannel (about €20M–€200M)	Edge policy and identity management; store terminals; dozens of integrations	Per-endpoint bot policy; CSP with reporting; a vault and rotation; attestations for every provider; app review in procurement; three runbooks	Integrations without owners (13 of 34 at Kestrel & Pine in January 2027); a leaked key that reaches every channel; vendor logs the contract does not grant
4 — Group	A security function; central identity; one PCI program or a token vault across brands	Vendor allow-lists; a shared CSP policy; supplier clauses with incident clocks; monitoring with on-call	Each brand with its own PSP keys and no central revocation

The shape of the control barely changes with scale: identities, keys, scripts and endpoints are the same problem at every size. What changes is **the number of them**, **and whether anyone has a list**. A Starting-stage merchant holds the list in its head; an Omnichannel merchant with 34 integrations and 41 apps cannot, and that is the moment the Lynchpin applies.

In Practice: Kestrel & Pine's 2027 Security Baseline

In April 2027 Noor Visser and Milan Petrov write a one-page security baseline: the second posture, plus the parts of the third the integration inventory demands. All figures are illustrative.

Item	What was true in August 2026	What is true from Q2 2027
Card-data scope	Hosted checkout, SAQ A; store terminals through the terminal provider; nobody held the providers' attestations	SAQ A confirmed with the acquirer; the script statement answered with the platform's confirmation; attestations for every payment provider on file, reviewed yearly
Guest checkout	No challenge, no velocity rule, no alert	Platform bot protection on checkout; PSP velocity rules per card, address and hour; a server-validated challenge when a payment request has no session history
Decline-rate alert and kill switch	Neither	Alert at three times the 30-day median to the on-call phone; a platform setting that disables guest card payments, tested quarterly

Item	What was true in August 2026	What is true from Q2 2027
Staff and vendor identity	31 platform admin accounts, 4 shared, MFA optional; the agency on one shared login	MFA enforced on the platform, the PSP console, the ERP (finance and invoicing), the WMS (warehouse management system) and the iPaaS (the integration platform); no shared accounts; agency engineers named; leavers removed within a day
Integration identities	34 integrations, 13 without owner in January 2027 (5 still in April), keys in scripts and the iPaaS, never rotated	One identity and one restricted key per integration, owner in the inventory, stored in a vault, rotated yearly or on exposure
Apps	41, of which 6 unused and 3 writing stock	6 removed; scopes reviewed; every new app through a procurement check of scopes and data access; a quarterly review
Checkout scripts	Unknown; the agency's theme loaded a tag manager	A script inventory with an owner and reason per script; CSP in report-only mode from May, enforcing from September
Runbooks	None	Three: a card-testing wave, a skimmer found on checkout, an admin or key compromised; each names who is called, the first three actions and the vendor contacts

The baseline reads the Card-Testing Night backwards: the alert fires at 23:55, the

challenge on session-less payment requests stops most of the 48,000, the kill switch ends the rest in a minute, and the runbook makes Milan's phone ring instead of a shared mailbox. The €2,900 was never the cost that mattered; the paused account and the two weeks of raised declines were.

When It Goes Wrong

A card-testing wave. Detection: the decline-rate alert, a 402 spike, orders with nonsense names, a PSP e-mail. Response, in order: kill switch for guest card payments; cancel and refund the orders that succeeded before they reach the warehouse; rotate every key if the attempts did not come through the checkout, because then they came through a leaked credential; afterwards watch the decline rate for weeks. Never retry the cards the attacker saved. If the PSP pauses the account, it is doing its job; have the contact and the evidence ready.

A skimmer found on checkout. Detection: a CSP violation report, a tamper alert, a shopper's bank, or a researcher's e-mail, which is how many are found. Response: remove the script through whatever loaded it (theme, app, tag manager) or switch to the PSP's redirect page so no script of yours is on the page where the card is typed; preserve the evidence; find how it got there, which is usually an identity rather than a vulnerability; then the legal clock of Whose Data Is It? Privacy, Consent and Retention starts, because card numbers are personal data.

An admin account or key compromised. Detection: a payout account changed, a script added, a bulk export in the audit log, a vendor's notice that a key was seen in public. Response: revoke, then investigate; rotate every key the account could see; check the payout account and the staff list; read the audit log from the first suspicious login. A merchant that cannot say which integration a leaked key belonged to must rotate everything.

Refund and coupon abuse. Not intrusions, so no edge control helps. Refund only to the original payment method; inspect before refunding above a threshold; bind one-time codes to a customer; give the returns portal a refund-only key. The Double Refund of January 2027 (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics) is the same lesson from the inside: one writer per money movement.

What to Take Away

Established.

- Keep the card number out of your systems: a hosted checkout or hosted fields puts the merchant in PCI DSS SAQ A with tokens and the last four digits as the only card data held (Stripe and Adyen guidance, n.d.; PCI DSS v4.0.1, 2024).

- The scripts on the payment page are the merchant's responsibility: requirements 6.4.3 and 11.6.1 are mandatory since 31 March 2025, and SAQ A merchants must state that their page is not susceptible to script attacks (PCI SSC, 2024 and 2025); British Airways and Polyfill are why.
- Card testing and credential stuffing are defeated by layered controls, not single filters (Stripe documentation; OWASP, n.d.).
- Shared responsibility is published and consistent: the platform secures the service; the merchant owns identities, keys, apps, scripts and integrations (Adobe, Shopify, Stripe, n.d.).

Depends on context.

- How far down the posture ladder to go depends on what the landscape contains; the trigger is the first key, the first custom checkout, the first terminal.
- Whether to challenge guest checkout depends on the payment mix: where most payments carry their own authentication (iDEAL, 3-D Secure cards), a challenge on card-only, session-less requests costs little; where plain card dominates, the false-positive cost is real and the PSP's velocity rules should carry more of the load.
- On a self-hosted platform the patch cadence is a security control in its own right (CosmicSting, 2024); on SaaS the saved effort should go into identities and scripts.

The author's synthesis.

- For a merchant that rents its platform, security is an **inventory problem before it is a technology problem**: a list of every identity, key, app, script and endpoint, with an owner, is worth more than any product, because every incident in this chapter started with something nobody had on a list.
- The decline rate belongs next to orders per minute on the dashboard: the two signals catch the two ways a checkout is used in bad faith.

Questions to Ask

1. Which SAQ does our acquirer say applies to us, and who holds the attestation of compliance for every provider that touches payments?
2. Can anyone list the scripts that load on our checkout page, with an owner and a reason for each, and would a CSP tell us if a new one appeared?
3. Which endpoints can trigger a payment authorization or a login, and which has no challenge, rate limit or session requirement?
4. Is multi-factor authentication enforced, not merely available, on every admin console, including the agency's and the partner's accounts?

5. Does every integration and app have its own identity and a scoped key, with an owner who can rotate it today?
6. If the PSP paused our account at two in the morning, who would know, and what would they do first?
7. What is our normal decline rate, would anyone be alerted if it tripled, and can we switch off guest card payments without a deploy?
8. Which apps can write inventory, prices, orders or refunds, and did anyone review those scopes before installation?

Security decides who may act on the shop. The next chapter asks the mirror question about the data the shop holds on people: who may use it, for what, for how long. Continue with Whose Data Is It? Privacy, Consent and Retention.

Sources and Evidence

Company, product and investor sources

- Protect yourself from card testing — Stripe Docs, n.d., accessed 2026-11; the attack's mechanism, symptoms, consequences for the decline rate, controls, the retry warning and the leaked-key vector.
- Integration security guide — Stripe Docs, n.d., accessed 2026-11; shared responsibility, what card data may be stored, third-party JavaScript, webhook signatures.
- Best practices for managing secret API keys — Stripe Docs, n.d., accessed 2026-11; restricted keys, vaults, rotation, exposure handling.
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- PCI DSS compliance guide — Adyen Docs, n.d., accessed 2026-11; where the PSP's responsibility starts, attestations and requirement 12.8.
- Script security for ecommerce — Adyen Docs, n.d., accessed 2026-11; CSP, script inventory, SRI and violation reports as the 11.6.1 alert source.
- Security and PCI compliance — Shopify, n.d., accessed 2026-11; Level 1 certification extending to all stores, and the merchant's own steps.
- Two-step authentication — Shopify Help Center, n.d., accessed 2026-11; who can enforce MFA, SMS retirement, payouts on hold.
- Shared responsibility — Adobe Experience League, n.d., accessed 2026-11; the merchant's and Adobe's duties on Adobe Commerce cloud infrastructure.
- Polyfill supply chain attack hits 100K+ sites — Sansec, 2024-06-25; the compromise, its evasion logic and the CSP advice.
- CosmicSting — Sansec, 2024, accessed 2026-11; the Adobe Commerce flaw, its fix dates and the breach campaigns.

- 2025 Bad Bot Report — Imperva, 2025; vendor figures on automated traffic, quoted as such.

Standards, regulation and official documentation

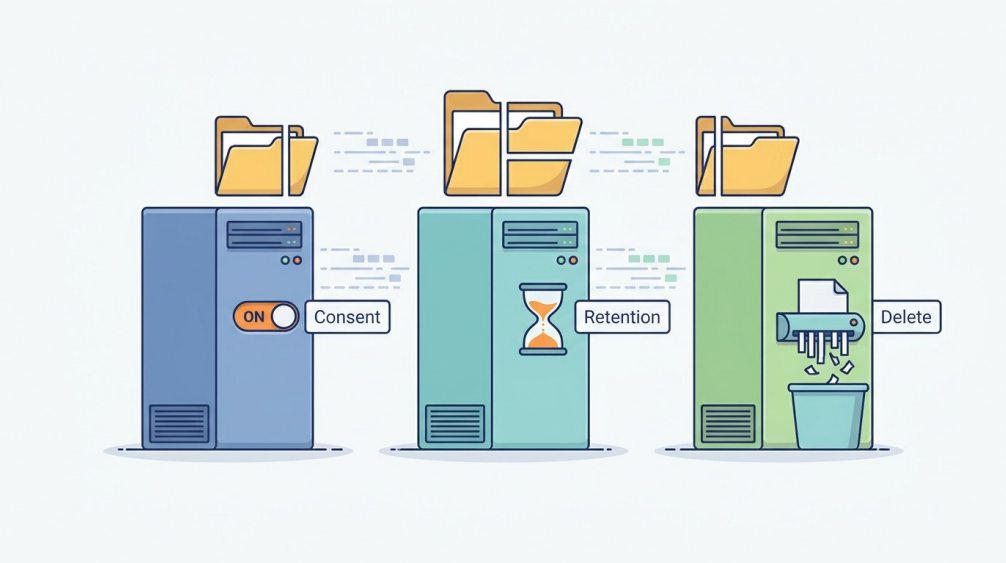
- Just Published: PCI DSS v4.0.1 — PCI Security Standards Council, 2024-06-11; publication of v4.0.1, retirement of v4.0 on 31 December 2024, v4.0's March 2022 publication, the unchanged 31 March 2025 effective date.
- Guidance for PCI DSS e-commerce requirements effective after 31 March 2025 — PCI Security Standards Council, 2025-03-26; requirements 6.4.3 and 11.6.1, their removal from SAQ A and the eligibility statement.
- OAT-001 Carding — OWASP Automated Threats to Web Applications, n.d., accessed 2026-11; the threat identifiers used in this chapter.
- Credential stuffing — OWASP, n.d., accessed 2026-11; definition and MFA as countermeasure.

Research and practitioner sources

- British Airways data breach — Wikipedia, citing the ICO penalty notice of 16 October 2020, accessed 2026-11; the entry point, the script, the numbers and the penalty. Eight years after the incident; the ICO's own pages could not be fetched.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

35. Whose Data Is It? Privacy, Consent and Retention



IN THIS CHAPTER, YOU WILL: Learn where personal data lives in a commerce estate, which rules govern each use of it, and how to design consent, retention and deletion across many systems.

KEY POINTS:

- **Every integration is also a data flow with a legal role.** The merchant decides why and how customer data is used, so it is the controller; the platform, the warehouse system and most apps act on its instructions as processors; carriers, marketplaces and payment providers usually decide for themselves. The system-of-record map needs that column.
- **The order and the marketing profile are two different things in law, even when they describe one person.** The order rests on the contract and on tax law and must be kept for years; the profile rests on consent and must go when the person says so. Design them to separate: pseudonymize the transaction, erase the person.

- **A deletion request is a fan-out with a deadline**. One month, every system that holds the person, a different right answer in each, and a record of what was done. Without an identity index and an orchestrator it is a seventeen-day hunt through nine systems; with them it is a job.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Familiar Stranger** — a person the systems cannot recognize as one cannot be deleted, told the truth about, or respected when they opt out.
- **The Secret Agent** — an app, script or export that copies customer data without a named owner or a contract is a processor nobody appointed.
- **The Long Fuse** — a data subject request has a legal clock of one month, and a cookie or transfer rule can change with a court decision; both are discovered late by merchants with no map.

A customer asks a merchant to remove their personal data. Their account is in the commerce platform, but addresses, purchases and contacts have also reached warehouse, carrier, payment, marketing and support systems. Removing the storefront profile does not settle what happens to those copies. The merchant needs to know where the data went, why it is held and which records can be removed or must be retained.

That is why the architecture needs a map of data copies, purposes, access and retention rules. A forgotten export can retain a record, and an unchanged integration can recreate one after removal. A request must reach the systems that hold the data and produce a result the merchant can verify. This chapter offers an architect's reading of the rules; legal details and national requirements still matter.

The chapter explains who decides the use of data (**controller**), who processes it for that party (**processor**), how consent is recorded, how long records are retained and how requests are handled. One Customer, One Product: Identity and Master Data Across Systems connects customer identities; Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation provides the wider calendar.

Where we are in the order's journey. Eva buys a jacket from Kestrel & Pine, the fictional outdoor brand, on 13 April 2027. Fulfillment, payment and invoicing distribute her details to different systems. Notifications need an email address; newsletter enrollment needs its own basis. Her 19 April exchange adds the returns portal. Each copy needs a purpose and retention rule.

Setting the Scene at Kestrel & Pine: The Request From Ghent

On **10 May 2027**, a Belgian customer, called Lotte Claes in this fictional example, asks Kestrel & Pine to delete her data and stop emailing her.

Lotte has two storefront orders, a marketplace order, a return, a newsletter subscription and an Antwerp store loyalty entry. Head of Engineering Noor Visser must identify the records behind those relationships and decide the action for each system. Deleting the storefront account alone will not complete the request.

Ruben Alves and the application support team took seventeen working days to find out. She was in nine of the thirteen systems:

System	What it held	What they did	What they could not do
The platform	Account, addresses, two orders, consent flag, the apps' copies	Deleted the account; the platform anonymized the orders and notified its apps	Confirm that every installed app complied
The marketplace connector	A copy of the bol order with name and address	Anonymized the copy	Touch bol's own record; bol is a separate controller
The ERP	Two consumer invoices and a credit note	Nothing: invoices are tax records	Erase; retention rules apply
The WMS	Three shipments with name and address	Deleted addresses from shipments older than a year	Erase recent shipments while a claim is possible
The shipping service	Labels and tracking for three parcels	Deleted after the claims window	Reach bpost's own delivery record
The PSP	Payment records for two orders	Removed shopper details from the console view	Erase the PSP's own records
The POS	A loyalty entry with phone number and five purchases	Deleted the entry; receipts kept as fiscal records	Unlink receipts already exported to the ERP
The returns portal	A contact and one return	Deleted the contact; anonymized the return	Nothing further

System	What it held	What they did	What they could not do
The data warehouse	Rows keyed by e-mail in four tables and a BI extract	Replaced identifiers with a hash	Reach a spreadsheet export made in 2026

The newsletter tool, one of the platform's apps, unsubscribed her through the platform's notification; the iPaaS (the integration platform) held her in three days of logs that expired on their own; nobody checked the backups. The reply went out on **4 June 2027**, inside the one-month limit with six days to spare. It had taken three people and a shared spreadsheet. **The request was answered. The next one would arrive before anyone had written down how.**

Figure 1: *One customer's deletion request found her in nine of the thirteen systems and took seventeen working days, three people and a shared spreadsheet to answer.*

What Personal Data a Commerce Platform Holds, and Under Which Rules

The vocabulary, because the words carry the design. **Personal data** is any information relating to an identified or identifiable person: a name, an e-mail, an order with an address, a cookie id tied to a browser. The **General Data Protection Regulation (GDPR)**, Regulation (EU) 2016/679, has applied across the European Union since 25 May

2018; it gives the person rights (access, rectification, erasure, objection, portability) and the organizations that handle the data duties.

A **controller** is the organization that decides why and how personal data is used; a **processor** handles data only on the controller's instructions (Article 4(7) and 4(8)). A **data processing agreement (DPA)** is the contract Article 28 requires between them. A **lawful basis** is the legal reason a controller may use the data at all; Article 6 lists six, of which four matter here: consent, performance of a contract, a legal obligation and the controller's legitimate interests. A **data subject request (DSR)** is a person exercising a right: send me my data, correct it, delete it. **Retention** is how long data is kept, and the law pulls two ways: no longer than the purpose needs, and at least as long as tax law demands.

The fine ceiling for the core infringements is €20 million or 4% of worldwide annual turnover (Article 83(5)). For Kestrel & Pine, with €48.0 million of 2026 revenue, 4% would be €1.9 million; commoner sanctions are orders to stop processing and the public finding.

What a merchant like Kestrel & Pine holds, by purpose:

Data	Where it lives	Purpose	Usual lawful basis	How long it is needed
Name, delivery and billing address	Platform, OMS (order management system), WMS, shipping service, ERP, carriers	Deliver and invoice	Contract; legal obligation for the invoice	Order: the fiscal period; shipment: the claims window
E-mail and phone	Platform, notification service, returns portal, POS, newsletter tool	Order messages; service; marketing	Contract for order messages; consent or the soft opt-in for marketing	Service: while the account exists; marketing: until withdrawn
Order history, returns, refunds	Platform, OMS, ERP, returns portal, PSP, data warehouse	Fulfill, account, support, warranty	Contract; legal obligation; legitimate interests for fraud	Fiscal period; then pseudonymized

Data	Where it lives	Purpose	Usual lawful basis	How long it is needed
Browsing and behavior	Analytics, advertising tags, CDP (customer data platform), personalization	Measurement; marketing	Consent (the ePrivacy Directive for the device; GDPR for the profile)	Until withdrawn or aggregated

The right-hand columns are the design. The same person appears in every row, but each row has its own basis and its own clock. **The order record and the marketing profile need separate lawful bases, separate retention clocks and separate deletion behavior, even though they refer to the same person.**

Controller, Processor and the Commerce Supply Chain

Who is the controller for Eva's data? The merchant: Kestrel & Pine decides that orders are taken, delivered and invoiced, that a newsletter exists, that a warehouse system is used. The guidelines of the European Data Protection Board (EDPB) on the two roles (version 2.0, July 2021) draw the line: decisions on the purpose of processing "are clearly always for the controller to make"; the "essential means" (which data, for how long, who receives it, whose data) are the controller's; the "non-essential means" (the choice of hardware or software, the detailed security measures) may be left to the processor.

Applied to the commerce supply chain, the usual outcome of that test, as the author reads it and not as a rule:

System	Role for the merchant's customer data	Why
The hosted platform, the WMS, the OMS, the iPaaS, the returns portal, the data warehouse	Processor	Tools the merchant chose, used for the merchant's purposes under its instructions and a DPA

System	Role for the merchant's customer data	Why
Most apps	Processor	Installed by the merchant to act on its data; each needs a DPA
The PSP	Processor for the merchant's order data; usually an independent controller for the payment itself	Payment providers have their own legal duties (anti-money-laundering, fraud) that they decide on
The carrier; the marketplace	Independent controllers	The carrier decides how it delivers and keeps its own records; the shopper is the marketplace's customer
Advertising platforms	Often joint or independent controllers	They use the data for their own purposes too

Platforms encode this in their app rules. Shopify requires every App Store app to subscribe to three compliance webhooks: `customers/data_request`, `customers/redact` (sent "10 days after the deletion request") and `shop/redact` (48 hours after a store owner uninstalls the app); apps must complete the action within 30 days and may decline redaction where the law requires them to retain data (developer documentation, n.d., accessed November 2026). Its protected-customer-data program gates name, address, phone and e-mail behind approval and requires minimum data, stated purposes, respect for consent, retention periods and encryption. That is a platform acting as the merchant's processor and policing the sub-processors on its behalf.

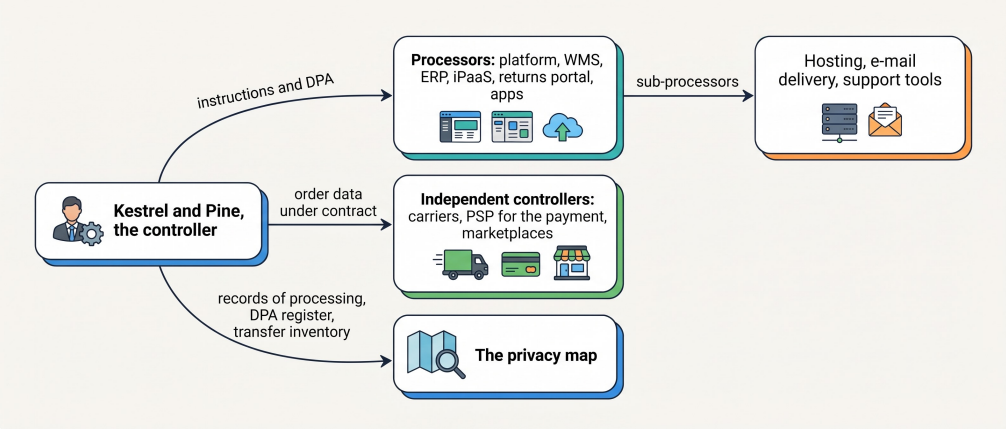


Figure 2: *The merchant is the controller; most of its systems are processors it must instruct and contract; carriers, payment providers and marketplaces decide for themselves.*

The consequence: **the system-of-record map from Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record needs a column for each system's legal role, and the integration inventory from APIs, Events, Files and Middleware: How Commerce Systems Talk is also the inventory of data flows.** The thirteen integrations with no named owner in the integration inventory as of January 2027 were thirteen data flows with nobody accountable for them; five were still unowned in April 2027, after the ownership map.

Lawful Bases: The Order Versus the Marketing

In commerce terms, the four bases divide the work like this:

Use	Basis	In practice
Taking, delivering, invoicing and supporting the order; warranty; returns	**Contract**	Needed to do what the customer asked; no consent needed; cannot be withdrawn while the contract runs
Keeping invoices, accounting records, VAT evidence	**Legal obligation**	Kept for the statutory period even if the customer objects

Use	Basis	In practice
Fraud screening; security; basic analytics; defending claims	**Legitimate interests**	Allowed where the merchant's interest is not overridden by the person's; documented; can be objected to
Newsletters; profiling for personalized marketing; cross-site tracking	**Consent**, or for e-mail to existing customers about similar products, the ePrivacy **soft opt-in**	Freely given, specific, informed, recorded, and as easy to withdraw as to give

The soft opt-in deserves a sentence because every merchant asks about it. The ePrivacy Directive (Article 13(2), 2002 text consolidated in 2009) lets a business that obtained a customer's e-mail "in the context of the sale of a product or a service" use it "for direct marketing of its own similar products or services", provided the customer can object, free of charge, at collection and in every message. National transpositions differ; the lawyer confirms the local reading. The architectural point holds either way: **the order's e-mail address and the marketing permission are two facts, and the second must be recorded with its basis**.

Consent, Cookies and Tracking

Consent lives in two laws. The GDPR defines it (Article 4(11)) as a freely given, specific, informed and unambiguous indication of the person's wishes by a statement or a clear affirmative action, and requires the controller to be able to demonstrate it (Article 7(1)). The ePrivacy Directive (Article 5(3), as amended in 2009) adds a rule about the device: storing information on, or reading information from, a user's device is allowed only with consent after clear information, except for what is strictly necessary to transmit a communication or to provide a service the user explicitly asked for.

In plain words: the cart cookie and the language choice need no consent; the analytics tag, the advertising pixel and the personalization script do, before they fire. The regulators Kestrel & Pine answers to agree on the substance:

- The **EDPB's consent guidelines** (May 2020) forbid cookie walls and say scrolling or swiping "will not under any circumstances" count as consent; its technical-scope guidelines (version 2.0, October 2024) bring tracking pixels, URL tracking and IP-only tracking within the device rule.
- The **Dutch regulator** (Autoriteit Persoonsgegevens, n.d., accessed November 2026) exempts functional cookies and "limited analytical cookies" with little privacy impact, requires consent for tracking cookies, and calls cookie walls prohibited.

- The **Belgian regulator** (Gegevensbeschermingsautoriteit, November 2023) exempts only strictly necessary functional and technical cookies (naming the language choice, cookie preferences and the shopping basket), rejects pre-ticked boxes and continued browsing, and requires cookies to be grouped by purpose, with own and third-party advertising as separate purposes.

A **consent management platform (CMP)** is the banner and the record behind it: it stores each visitor's choices per purpose and tells each tag whether it may run. **Server-side tagging**, in which the merchant's own server receives the browser's events and forwards them to vendors, changes who receives the data first, not whether consent is needed: the browser is still told to emit identifiers, so the device rule still applies. Present it as a governance improvement, never as a consent bypass; that reading follows the EDPB's technical-scope guidelines and is the author's.

Two moving parts. The Commission's **Digital Omnibus** proposal of November 2025 would move the device rule into the GDPR with four no-consent purposes (transmission, requested service, audience measurement, security), require a one-click refusal and oblige controllers to honor machine-readable browser signals; one law firm's reading (Osborne Clarke, December 2025) is that the end of cookie banners "is not yet in sight". It is a proposal, not law, as of November 2026. In the United States the model is opt-out: California's law lets consumers opt out of the "sale or sharing" of their data, including through the Global Privacy Control browser signal, and the 2025 chart of the IAPP (International Association of Privacy Professionals) lists nineteen state laws of that family.

The design consequence joins this chapter to the previous one. **Consent is state that must travel**: from the CMP to the tag layer, to the newsletter tool and the CDP, to the POS where staff ask for a loyalty sign-up, and to every app that sends a message. One Customer, One Product: Identity and Master Data Across Systems put it on the customer master with its time, place, wording and source; here is why: the merchant must demonstrate it per person and per purpose, and withdraw it everywhere at once.

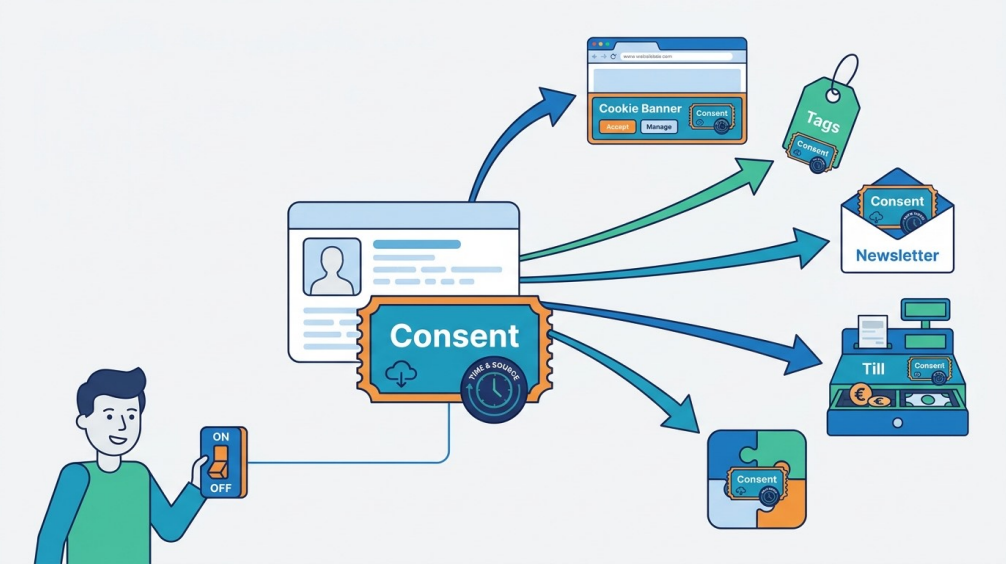


Figure 3: *Consent is state that must travel: recorded once with its evidence on the customer master, it must reach every tag, tool, till and app, and a withdrawal must reach them all at once.*

Requirements per flow

Flow	How fresh	How consistent	How fast	How available
Consent capture (banner, checkout box, store sign-up)	Immediate for the device; hours to reach every outbound tool	One record per person per purpose, with evidence	Must not slow the page or the till	Degrade to "no consent" when the CMP is down, never to "consent"
Consent withdrawal	Immediate for the channel used; same day everywhere	Strictest state wins across tools	Minutes	Works even if the master is unreach-able; reconcile later

Flow	How fresh	How consistent	How fast	How available
Data subject request	One month, extendable by two	Every system that holds the person answers; answers recorded	Days	Batch; the orchestrator can queue
Retention purge	Daily or weekly	Pseudonymize transac- tions; erase profiles; log what ran	Hours	Batch; a failed run must be visible

Data Subject Requests Across Many Systems

The GDPR requires the controller to act on a request "without undue delay and in any event within one month", extendable by two months for complex or numerous requests (Article 12(3)). What makes it hard is not the law but the fan-out: one person, many systems, a different right answer in each. Three parts make it a job.

An identity index: which systems hold this person, under which identifiers. This is the cross-reference One Customer, One Product: Identity and Master Data Across Systems built for another reason; a deletion request is its second customer. Without it, every request starts with a search of every system by e-mail, phone and name, which is where Lotte's seventeen days went.

A request orchestrator: a small workflow that verifies the requester (asking for "all my data" with someone else's e-mail is a way to steal it), fans the request out, records each answer and produces the reply. Shopify's compliance webhooks are the platform-side half of one; for the systems outside the platform, the merchant builds or buys the other half.

A per-system action table, decided in advance:

Answer	When	Example at Kestrel & Pine
Erase	The purpose has ended and no law requires the data	The platform account, the loyalty entry, the newsletter subscription, the returns contact, old shipment addresses

Answer	When	Example at Kestrel & Pine
Pseudonymize	The record must stay for accounting, statistics or evidence, but the person need not be identifiable from it	Orders in the platform and the OMS, warehouse rows, data-warehouse rows: name and address replaced by a token the merchant cannot reverse
Retain	A legal obligation applies or a claim is still possible	Invoices and credit notes in the ERP; recent shipments during the carrier's claims window; fraud evidence

Two answers lie outside the merchant's reach: **refer** (the carrier, the marketplace and the PSP are told, and the person must ask them too, because they are controllers in their own right) and **except** (backups, under a documented, bounded exception: erased data is not restored, and if a backup is restored the deletion is replayed).

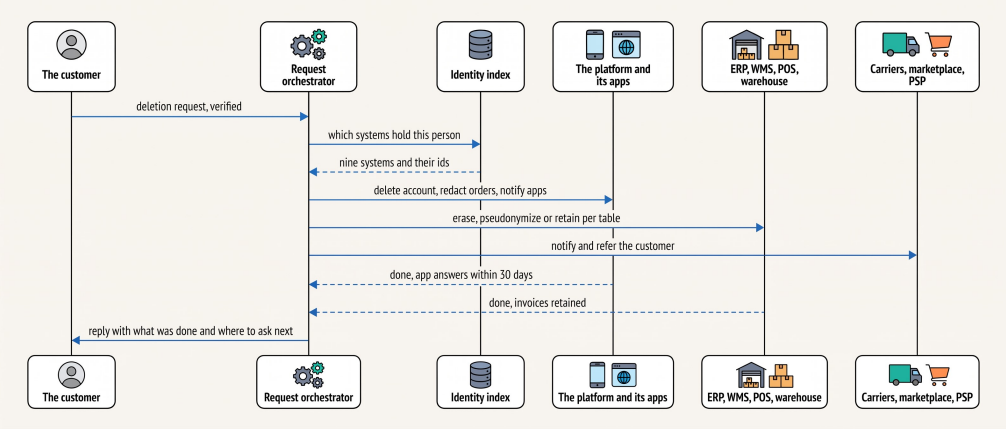


Figure 4: *A deletion request is verified once, resolved against the identity index, fanned out with a predetermined action per system, and answered with a record of what each system did.*

The reply should say, in plain words, what was erased, what was kept and why, and whom else to ask; the record behind it is what a regulator asks for.

Retention: The Person Versus the Transaction

Tax law makes the merchant keep what privacy law makes it release. One design move reconciles them: **split the person from the transaction.** The retention facts, with their

status:

- **Netherlands**. The government's business portal states that businesses "must keep your business records for at least 7 years", naming the general ledger, stock records, purchase and sales records and debtor and creditor accounts, with at least 10 years for data on immovable property, and requires digital files to remain accessible for inspection (business.gov.nl, n.d., accessed November 2026).
- **Germany**. §147 of the Abgabenordnung requires books, records, inventories and annual accounts to be kept for ten years, accounting vouchers for eight, and other records including commercial letters for six, from the end of the calendar year of the record (gesetze-im-internet.de, current text, accessed November 2026).
- **Belgium**. Ten years for accounting records and invoices is widely cited; the official text was not verified for this chapter, and Sanne de Wit treats ten years as the Belgian rule pending confirmation.

So Lotte's invoice from November 2026 stays until at least the end of 2036 under the Belgian reading, whatever she asks; "legal obligation" is a lawful basis. The conflict arises only when the invoice *is* the customer record, as it is in many ERPs and platforms: the order carries name, address and e-mail, and the customer master is the list of people with orders. Deleting the person then means deleting the order, which tax law forbids, so nothing is deleted.

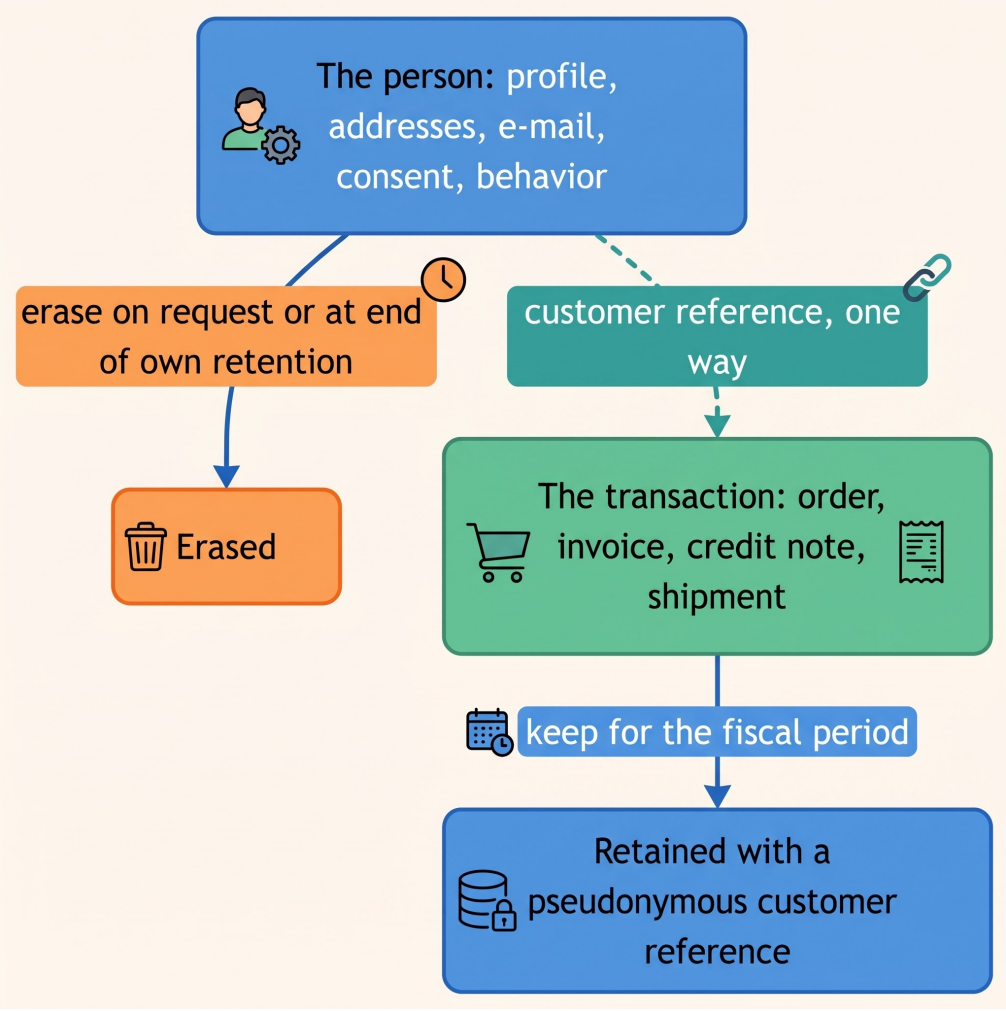


Figure 5: *Profiles are erasable; transactions are kept with a reference that no longer points to a person once the profile is gone.*

In practice: the order and the invoice carry a customer reference and the delivery address (part of the transaction, kept for the fiscal period); the profile (e-mail, phone, preferences, consent, browsing, loyalty) is a separate, erasable record; and the analytics tables carry a hash of the customer reference, not the e-mail. **Retention periods become a property of each record type, not of the person**, and a nightly job purges what has expired without anyone deciding per customer.

Backups need a sentence. A deletion cannot reach into last month's backup; the accepted practice is a bounded, documented exception: backups are kept for a stated period, never used to resurrect erased data, and if one is restored the deletion log is replayed against it.

Sharing, Transfers and the Store

Sharing with partners. Every order sends personal data to a carrier, the PSP, sometimes a marketplace, and to apps and connectors. The rules are minimization (the warehouse does not need the e-mail, the carrier does not need the order history), a recorded role per recipient, and a DPA or terms that cover the transfer. Kestrel & Pine's marketplace connector copied everything the marketplace sent; after Lotte's request it copies what the warehouse needs.

International transfers. Most commerce stacks send data to the United States somewhere: analytics, e-mail delivery, support tooling, hosting. The lawful routes are an adequacy decision, standard contractual clauses (SCCs) or binding corporate rules. The EU-US Data Privacy Framework (DPF), Commission Implementing Decision (EU) 2023/1795 of 10 July 2023, is the adequacy decision for certified US companies; the modern SCCs are Decision (EU) 2021/914.

The General Court dismissed Philippe Latombe's challenge to the DPF on 3 September 2025; an appeal (Case C-703/25 P) is pending as of November 2026; law firms advise keeping SCCs as a fallback.

The architectural asset is a **transfer inventory** keyed to the integration map: per vendor, where the data goes and under which mechanism, so that a court decision becomes a filter on a list rather than a discovery project.

The store. The POS holds a loyalty list and receipts; a store may also run cameras, Wi-Fi analytics and tablet sign-ups. Consent asked at a till by a member of staff is easy to ask badly and leaves no evidence. Kestrel & Pine's rule from mid-2027: the POS loyalty sign-up shows the same purposes and wording as the web, records the time and the store, and carries the master's consent once matched (One Customer, One Product: Identity and Master Data Across Systems).

The vendor's side. For a commerce software vendor, the obligations are a processor's: a DPA you can sign, sub-processors listed and announced, data kept in the regions you promise, the ability to export and erase one customer of one merchant, security you can evidence, and breach notification to the merchant without undue delay (Article 33(2)) so that it can meet its own 72-hour deadline.

The Choices

For the data subject request, four realistic designs:

Choose the coordination method for the number of systems and requests:

- **A runbook**: list the action and responsible person for each system. This can suit a small merchant with few requests.

- **Platform-led handling**: use the platform's erasure and app notifications, with a runbook for outside systems.
- **An integration flow**: verify the request, resolve identities, send the required actions, record results and reply. Kestrel & Pine adopts this in autumn 2027.
- **A privacy management product**: automate more intake, discovery and records across a larger group.

I would map the data before choosing the product. Automation still needs to know which systems and records it must reach.

Option	Fits when	Strengths	Costs and limits	Warning sign
Runbook by hand	Few systems, few requests	No build; forces the map to be written	Slow; no record unless disciplined	The runbook is a person's memory
Platform-centered	Most data in the platform and its apps	Erasure and app fan-out come with the platform	ERP, WMS, POS and warehouse are outside it	"The platform handles GDPR"
Orchestrator plus identity index	A dozen systems; weekly requests	Repeatable; a record per request; reuses the identity index	A flow to own; per-system steps; a human verification step	Requests queued behind other integration work
Privacy management product	Several brands, countries, entities	Intake, verification, discovery, consent and records in one place	License and configuration; connectors still map to your systems	Bought before the map exists

For consent the ladder is shorter: the platform's built-in banner and checkout box (Starting), a CMP with per-purpose consent synchronized to the customer record (Growing), and consent as an attribute of the master with evidence, synchronized to every outbound tool (Omnichannel onward).

Across Scale Stages: When It Fits and When It Limits

The four stages are the book's (see What Changes at Each Scale: From First Store to Group).

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
1 — Starting	The platform's standard DPA, cookie banner, checkout consent box and app vetting; requests handled in the platform admin	A privacy notice that matches the stack; a DPA for the 3PL and the e-mail tool; one person who answers requests	An app with no DPA; a newsletter list built from every order without a soft opt-in notice
2 — Growing	The first store, ERP and marketplace bring data outside the platform; a CMP with per-purpose consent	A DPA register and the rule that nothing goes in without one; a runbook for requests; consent on the customer record; a retention rule per record type	A request that takes two weeks; POS consent that never reaches the newsletter tool; tags firing before the banner answers

Stage	What is typical	What becomes necessary	Warning signs of outgrowing it
3 — Omnichannel	A dozen systems, 20–40 flows, requests weekly; a data warehouse keyed by e-mail	A data-flow map kept with the integration inventory; an identity index and a request orchestrator; the person split from the transaction; a transfer inventory; a bounded backup exception	A request that touches nine systems by hand; an erased customer resurrected by a backup; a court decision that forces a vendor-by-vendor investigation
4 — Group	Several brands, countries and legal entities, each a controller; a privacy function	One consent model with national differences as configuration; a privacy management product; a data protection officer where required; group-wide records of processing	One brand's consent used by another; two brands answering the same person differently

The judgement: the duties do not change with scale; a Starting-stage merchant is as much a controller as a Group. What changes is where the data is. Below the Omnichannel stage the platform holds most of it and its DPA, banner and app rules do most of the work; from the Omnichannel stage the person is in a dozen systems and the merchant must own the map, the index and the orchestrator itself. Kestrel & Pine learned the stage-3 lesson from one request from Ghent.

In Practice

Shopify's compliance webhooks and protected customer data (accessed November 2026) are the clearest public example of a platform acting as the merchant's processor

and policing its sub-processors: three mandatory webhooks, a 30-day completion window, and a gated program for the four identifying fields.

Sephora, California, August 2022. A case more than three years old, but still the reference example: the Attorney General's settlement (US$1.2 million) found that third-party trackers on Sephora's site and app amounted to a sale of personal information without notice, and that Global Privacy Control signals were not honored. The tag layer is a legal surface, and a browser signal must be read as well as a banner.

The DPF and its appeal (2023–2026). A merchant whose transfer inventory names each US vendor's mechanism can respond to the outcome in an afternoon.

Kestrel & Pine's deletion design (autumn 2027, illustrative)

After Lotte's request, two changes. In **June 2027** the identity rule linked every system's record to the customer master (One Customer, One Product: Identity and Master Data Across Systems), and the July matching run produced the index.

In **October 2027** Ruben Alves built the request orchestrator as an iPaaS flow with a human verification step: the request is logged, the requester verified through the platform's account or a signed e-mail reply, the index queried and the action table applied.

The platform deletes the account, anonymizes the orders and notifies its apps; the OMS (live for marketplace orders from 15 September) pseudonymizes its order copies; the WMS and the shipping service purge addresses older than their claims window of **twelve months** *(assumption)*; the ERP keeps invoices and credit notes and marks the customer reference as erased; the POS deletes the loyalty entry; the returns portal deletes the contact; the data warehouse replaces e-mail and name with a hash.

The carrier, the marketplace and the PSP are notified and the customer told to ask them directly. Backups are kept for **35 days** *(assumption)* and the deletion log is replayed after any restore.

The second request through the flow, in November 2027, took **two working days** *(illustrative)*. The map that made it possible, one row per system with its legal role, data, action and retention clock, is now a column of Noor's ownership map, reviewed quarterly.

When It Goes Wrong

Failure	How it shows	Detection	Recovery
The missed copy	An erased customer receives a newsletter because nobody reached the newsletter tool	A complaint; a reconciliation of erased master ids against each tool's list	Erase; add the tool to the index and the action table; ask who else was missed
The resurrected customer	A restore from backup brings back erased records	The deletion log replayed after a restore	Replay the log; document the backup exception
The impostor	Someone requests "all my data" with another person's e-mail	A verification step that fails	Never answer an unverified request with data; verify through the account or a signed reply
The over-deletion	An invoice or a fraud record is erased with the profile	The ERP's retention check; a tax audit	Restore from the retained copy; fix the action table: transactions retained, the person erased
Consent without evidence	Marketing cannot show when or how a person consented	A consent audit sampling records	Treat unevidenced consent as absent; re-ask where the soft opt-in does not apply
The invalidated transfer	A court or a decision removes a transfer mechanism	A transfer inventory naming the mechanism per vendor	Switch to SCCs or an EU region, vendor by vendor from the list

The missed copy is the one that recurs, and its cause is a system added after the map was made; that is why the privacy map lives with the integration inventory and is reviewed when a row is added.

What to Take Away

Established.

- The merchant is the controller of its customers' data; its platform, warehouse system, integration platform and most apps are processors that need a contract; carriers, payment providers for the payment itself, and marketplaces usually decide for themselves (GDPR Articles 4 and 28; EDPB Guidelines 07/2020).
- Orders rest on the contract and on tax law; marketing rests on consent or the soft opt-in; consent must be demonstrable, and cookie walls, scrolling and pre-ticked boxes do not produce it (GDPR Articles 6 and 7; EDPB Guidelines 05/2020; the Dutch and Belgian regulators).
- Requests must be answered within one month, extendable by two (Article 12(3)); business records must be kept at least seven years in the Netherlands and ten, eight or six years in Germany depending on the record (business.gov.nl; §147 AO).
- Platforms have operationalized the fan-out: Shopify's three compliance webhooks and 30-day window are a documented, dated example.

Depends on context.

- Below the Omnichannel stage the platform's DPA, banner, consent box and app rules do most of the work; from the Omnichannel stage the merchant must own a data-flow map, an identity index and an orchestrator, because the platform's reach ends at its apps.
- The soft opt-in depends on national transposition and on the "similar products" test; take the lawyer's reading per country.
- The retention period per record type is a legal and finance decision per country; the architecture needs only that it be a property of the record type, not of the person.

The author's synthesis.

- Privacy in a commerce estate is an integration problem wearing a legal coat: the map, inventory and identity index that fix the Phantom Stock and the Familiar Stranger make a deletion request a job rather than a hunt.
- Split the person from the transaction in the data model before the first request, because every system that stores the name on the order otherwise forces a choice between the tax law and the person.

Questions to Ask

1. Which systems hold personal data, and does the system-of-record map say, per system, whether it is a processor or a controller?
2. Is there a signed DPA for every processor, including every app, and who keeps the register?
3. Where is consent recorded, with what evidence, and how long does a withdrawal take to reach every outbound tool?
4. Do tags fire before the banner is answered, and what happens when the CMP is down?
5. How would you find every record of one person today, and how long did the last request take?
6. Can an order outlive its customer in your data model, or does deleting the person delete the invoice?
7. Which vendors receive data outside the EU, under which mechanism, and is that a list or a memory?
8. What happens to an erased person when a backup is restored?

A deletion request is one legal clock among many. Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation lays out the others, from the right of withdrawal and the Omnibus price rules to e-invoicing, accessibility, product safety and the platform rules for marketplaces, and shows how Kestrel & Pine turned them into a regulatory calendar for 2027 so that no deadline is discovered only months before it applies.

Sources and Evidence

Company, product and investor sources

- Privacy law compliance, Shopify developer documentation, n.d., accessed 2026-11 — the three compliance webhooks, the 10-day and 48-hour delays, the 30-day completion window.
- Protected customer data, Shopify developer documentation, n.d., accessed 2026-11 — minimum data, purpose limitation, consent, retention, encryption and level-2 controls for apps.
- Attorney General Bonta announces settlement with Sephora, California Department of Justice, 2022-08-24 — trackers as a sale, Global Privacy Control, the US$1.2 million settlement.
- Latombe's challenge to EU-US data transfers fails, Lewis Silkin, 2025-09-03 — the General Court judgment and the advice to keep SCCs as a fallback.

- Digital Omnibus reshapes EU cookie rules, leaves banner fatigue largely intact, Osborne Clarke, 2025-12-10 — a reading of the November 2025 proposal.

Standards, regulation and official documentation

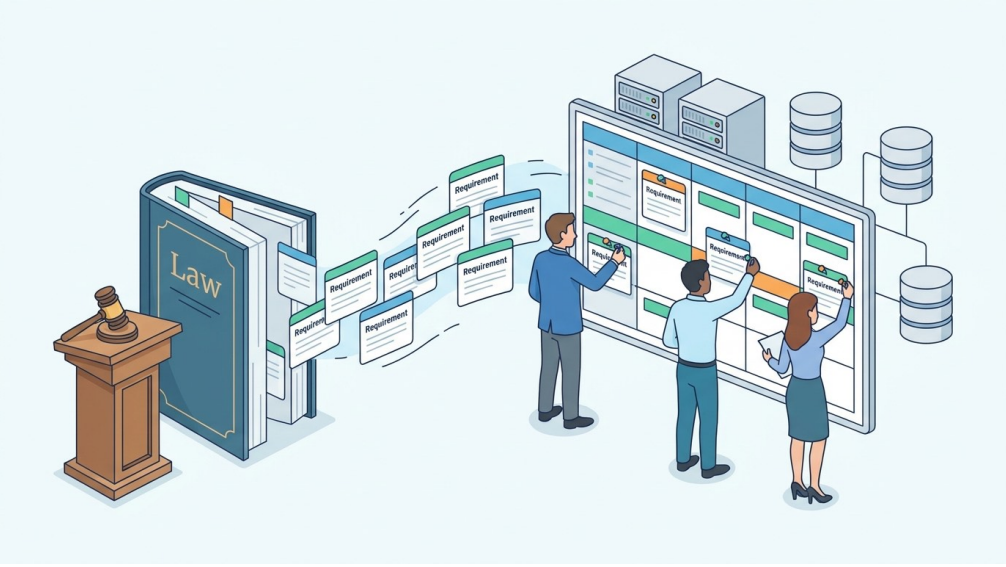
- Regulation (EU) 2016/679 (GDPR), Official Journal text, 2016 — Articles 4, 6, 7, 12(3), 28, 33 and 83(5); application from 25 May 2018.
- Directive 2009/136/EC, Official Journal text, 2009 — the amended Article 5(3) device rule of the ePrivacy Directive.
- Directive 2002/58/EC (ePrivacy), consolidated 2009, EUR-Lex — Article 13(2), the soft opt-in; text more than three years old, national transpositions differ.
- Guidelines 07/2020 on the concepts of controller and processor, European Data Protection Board, version 2.0, 2021-07-07 — purposes, essential and non-essential means.
- Guidelines 05/2020 on consent, European Data Protection Board, 2020-05-04 — cookie walls, scrolling.
- Guidelines 2/2023 on the technical scope of Art. 5(3) of the ePrivacy Directive, European Data Protection Board, version 2.0, 2024-10-07 — pixels, URL tracking, IP-only tracking.
- Cookies, Autoriteit Persoonsgegevens (Dutch), n.d., accessed 2026-11 — functional and limited analytical cookies, tracking cookies, cookie walls.
- Cookies en andere traceringsmiddelen, Gegevensbeschermingsautoriteit (Belgium, Dutch), 2023-11-17 — exempt categories, pre-ticked boxes, purpose grouping.
- Keeping business records, business.gov.nl (Netherlands Enterprise Agency), n.d., accessed 2026-11 — at least seven years; ten for immovable property; digital files accessible.
- §147 Abgabenordnung, gesetze-im-internet.de, current text, accessed 2026-11 — ten, eight and six years.
- Commission Implementing Decision (EU) 2023/1795 (EU-US Data Privacy Framework), Official Journal text, 2023 — the adequacy decision.
- Commission Implementing Decision (EU) 2021/914 (standard contractual clauses), Official Journal text, 2021.
- Case C-703/25 P, Latombe v Commission, Official Journal C/2025/6610, 2025-12-22 — the pending appeal.
- California Consumer Privacy Act, California Department of Justice, n.d., accessed 2026-11 — thresholds, rights, Global Privacy Control; US State Privacy Legislation Tracker, IAPP, updated 2026-09-08, and its 2025 chart — nineteen state laws.

Research and practitioner sources

- One Customer, One Product: Identity and Master Data Across Systems — the identity index this chapter's orchestrator depends on; Scams, Fakes and Account Takeovers: Fraud Prevention for Listings in the sibling book LISTED for the legitimate-interest side of risk screening.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

36. Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation



IN THIS CHAPTER, YOU WILL: Learn which EU and US rules shape a commerce platform, what each requires in data terms and by which date, and how to keep a regulatory calendar that stops deadlines arriving as surprises.

KEY POINTS:

- **Most rules are data requirements, not page requirements.** The prior-price rule needs 30 days of price history per SKU (stock-keeping unit: one sellable variant) and country; the withdrawal right needs refund clocks started by the right events; product-safety and marketplace rules need manufacturer and seller fields on every listing; e-invoicing needs a structured invoice over a network. Each belongs in a system of record, not in a template.

- **Dates slip, scope does not; transposition multiplies the calendar.** The AI Act's high-risk dates moved by more than a year, the Dutch e-invoicing mandate moved to 2030 before it was a bill, and one directive becomes 27 national laws with different dates. Plan to the scope, schedule to the date, and keep the calendar live.
- **The platform absorbs some of it, never all of it.** A hosted platform brings a compliant checkout button, cookie banner and card-data scope; price history, product-safety fields, packaging reports, deletion fan-out and every vendor contract clause stay with the merchant. The practical artifact is one calendar with an owner, the systems touched and a date per row.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Long Fuse** — every rule in this chapter was published years before it applied, and the merchant that discovers it months away pays for speed instead of planning.
- **The Pulled Rug** — a marketplace turns a regulation into a mandatory listing field on a date of its own, and the connector starts rejecting products before anyone reads the notice.
- **The Price Nobody Can Explain** — a discount claim is lawful only against a 30-day price history, and a shop with seven copies of a price cannot produce one.

A rule about displaying an earlier price can require price history that the merchant has never stored. Engineering then needs a data source, a collection process and a checkout or product-page change. Learning about the rule just before it applies leaves little time for that work, even if the deadline has been public for years. This book calls that avoidable delay **The Long Fuse**.

Commerce rules also affect returns, product details, invoices and unsold goods. For each applicable requirement, ask **which data it needs, who owns that data, how the behavior will be tested, and when it must work**. These questions turn a legal deadline into work the team can schedule and verify, reducing the risk of a rushed release or interrupted selling. Legal interpretation and national differences still require care. Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers and Whose Data Is It? Privacy, Consent and Retention cover related controls.

Where we are in the order's journey. Eva buys a jacket from Kestrel & Pine, the fictional outdoor brand. Rules affect the displayed price, product information, checkout and later return process. This chapter connects those obligations to the systems that must implement them.

Setting the Scene at Kestrel & Pine: Four Rules Through Four Doors

Kestrel & Pine is a Dutch company with 310 employees selling in three countries. In February 2027, Head of Engineering Noor Visser finds that four regulations reached engineering through four unrelated channels. The legal dates below are sourced; the company's discovery of them is fictional.

- **The listing rejections (December 2024).** The **General Product Safety Regulation** (GPSR, Regulation (EU) 2023/988) applied from 13 December 2024 and requires every online offer to show the manufacturer's name and address, a responsible person in the EU where the manufacturer is outside it, product identification and warnings. Amazon.de and Zalando turned those into mandatory listing fields, the connector began rejecting third-party equipment products, and Milan Petrov, who owns the marketplaces, filled the gaps from a spreadsheet over three weeks.
- **The accessibility audit (summer 2025).** The **European Accessibility Act** applied from 28 June 2025 to e-commerce services, exempting only microenterprises. The agency audited the three storefronts in May 2025 (illustrative: about 140 findings, mostly contrast, labels and keyboard navigation in the customized theme) and fixed them over the summer. The checkout page was the platform's problem; the theme was Kestrel & Pine's.
- **The refused PDF (December 2025).** Belgium made structured electronic invoices mandatory between Belgian VAT-registered businesses from 1 January 2026. Whether that binds a Dutch supplier without a Belgian establishment is not settled by the sources; what mattered was that the Belgian customer's bookkeeping now expected a Peppol invoice. Kestrel & Pine learned of it when a larger Belgian account announced it would no longer process PDFs (Tax, Invoices and the Books: From Order to Ledger tells it from finance's side); the EDI (electronic data interchange) provider, already an access point to Peppol, the European network for exchanging structured business documents, was connected to the ERP (the finance and invoicing system) in six weeks, and Peppol invoices have gone to all Belgian business customers since January 2026.
- **The newsletter (June 2026).** The **Ecodesign for Sustainable Products Regulation** (ESPR, Regulation (EU) 2024/1781) prohibits, from 19 July 2026, the destruction of unsold apparel and footwear by large enterprises, and its definition of "unsold" includes goods returned under the right of withdrawal. Teun Baas, who owns the seasons, had an August routine for end-of-line stock and returns not resold as new (about 39% of returned items); it now needs a documented disposition per unit and a yearly disclosure. Sanne de Wit, the Finance Director, found the rule in an auditor's newsletter in June 2026, weeks before it applied.

An error log, an agency, a customer and a newsletter delivered the warnings. The product-safety, accessibility and sustainable-product rules had been published years earlier. Noor adds a regulatory calendar to **Waypoint**, the company's improvement program, with a named owner for every applicable requirement.

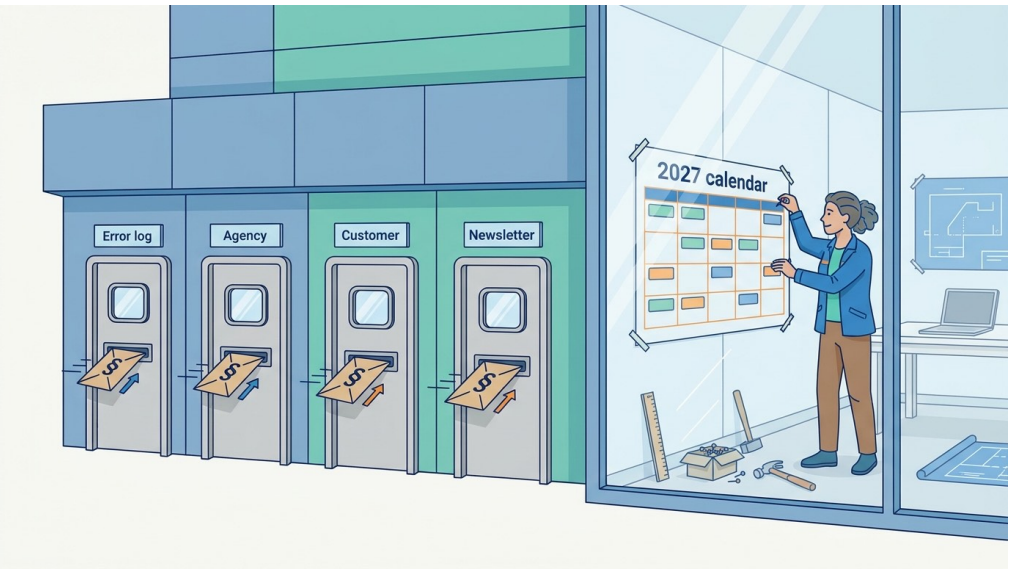


Figure 1: *Four rules, four doors: an error log, an agency, a customer and a newsletter brought the law in, and none of them was a door the company owned; the 2027 calendar is the door of its own.*

How to Read a Law as a Requirement

A legal text is written for courts, not engineers; the translation follows four questions.

1. **Who is in scope, and are we?** By role (trader, marketplace, producer, payment provider), by size (microenterprises are exempt from the Accessibility Act; small and medium enterprises are exempt or deferred under the ESPR) and by place (established in a member state, selling into it, or both).
2. **What does it require, in data terms?** Nearly every obligation is a field, a history, a document or a clock: the lowest price in the last 30 days; a manufacturer's address per product; a structured invoice; a refund within 14 days of a dated notice.
3. **Which system of record holds that, and which channels must show it?** The field belongs where the fact is owned (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record); a manufacturer field kept only in the marketplace connector is a copy with no owner.

4. **When does it apply, and how will we know we comply?** The application date, the national transposition date for a directive, and a test: a query, a report or a check in the release process.

The table uses **OMS** for order management system, **PIM** for product information management and **WMS** for warehouse management. **ViDA** means VAT in the Digital Age, **PPWR** the Packaging and Packaging Waste Regulation, and **WCAG** the Web Content Accessibility Guidelines. Each row links a rule to the records and controls it requires.

Rule	Data requirement	System of record	Channels that must show it	Test
Prior price (Omnibus, Art. 6a)	Lowest price per SKU per country in the 30 days before a reduction	Pricing engine or price history (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax)	Storefronts, marketplaces, stores	Every strike-through price equals the 30-day minimum
Right of withdrawal (Consumer Rights Directive)	Delivery, notice and goods-received dates; refund within 14 days of the notice	OMS and returns portal (The Way Back: Returns, Exchanges, Refunds and Reverse Logistics)	Returns portal, service tool	No refund older than 14 days from notice

Rule	Data requirement	System of record	Channels that must show it	Test
Product safety (GPSR, Art. 19)	Manufacturer and responsible-person name and address, identification, warnings in the shopper's language	PIM (What Are We Selling? Products, Variants and the Catalog)	Storefronts, marketplaces	No product published without the fields
E-invoicing (Belgium, Germany, ViDA)	EN 16931 invoice over Peppol or an accredited platform, with transmission status	ERP and access point (Documents Between Companies: EDI, Punchout and E-Invoicing)	Business customers	Every in-scope invoice has a delivered status
Packaging and producer responsibility (PPWR, national registers)	Packaging weight by material per unit sold, per country	PIM and WMS	Reporting only	A yearly report per country reconciles with units shipped
Unsold-goods destruction (ESPR, Art. 24–25)	Disposition per unsold or returned unit, reasons, yearly disclosure	WMS and returns disposition	Reporting only	No "destroy" disposition for apparel without a derogation

Rule	Data requirement	System of record	Channels that must show it	Test
Accessibility (European Accessibility Act)	Storefront, checkout, identification and payment meet WCAG 2.1 AA	Theme and embedded widgets (Delivering the Storefront: Themes, Headless, Rendering and Performance)	Storefronts	An audit per release; vendor statements on file
AI transparency (AI Act, Art. 50)	Disclosure that a shopper talks to an AI; machine-readable marking of generated content	Any assistant or content tool (Assistants and Agents: AI in Commerce and What Is Real in 2026)	Storefronts, service	No undisclosed assistant; generated images marked

The pattern to notice is the third column: **most rows land in a system Kestrel & Pine did not have or did not own in 2026** (the PIM, the OMS, the price history, a disposition record). Regulation is one more reason the Waypoint hubs exist.

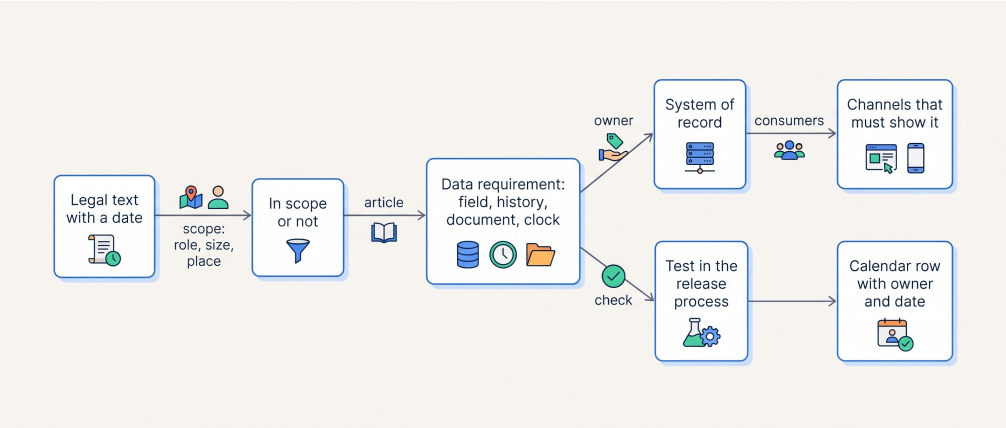


Figure 2: *Reading a law as a requirement: scope first, then the field or clock it demands,*

the system that owns it, the channels that show it, a test, and a dated row with a name on it.

The Rules, by What They Touch

Consumer law: withdrawal, the button, the prior price, reviews

The **Consumer Rights Directive** (2011/83/EU, applying to contracts from 13 June 2014) gives a consumer "a period of 14 days to withdraw from a distance or off-premises contract, without giving any reason", extended to 12 months if the trader fails to inform them; the trader must refund "not later than 14 days from the day on which he is informed of the consumer's decision to withdraw", and may withhold it until the goods are back; and an order button must be labeled "only with the words 'order with obligation to pay' or a corresponding unambiguous formulation" (Directive 2011/83/EU, 2011). Three clocks and one string. The clocks need the right events recorded: delivery (the carrier's scan), the withdrawal notice (the returns portal) and receipt of the goods (the WMS). Eva's exchange request of 19 April 2027 started the clock that day, not on 22 April when the jacket reached Duiven.

The **Omnibus Directive** (2019/2161, applied from 28 May 2022) added the **prior-price rule**: "any announcement of a price reduction shall indicate the prior price", meaning "the lowest price applied by the trader during a period of time not shorter than 30 days prior to the application of the price reduction" (Directive (EU) 2019/2161, 2019). It also made ranking parameters and review provenance material information, blacklisted claiming reviews are genuine "without taking reasonable and proportionate steps to check", and required disclosure of prices personalized by automated decision-making. What Does It Cost? Price Lists, Contract Prices, Currencies and Tax works the Black Friday case where an autumn campaign at €179 makes the lawful "was" price €179, not €199; Finding the Product: Search, Navigation and Merchandising covers reviews. **A 30-day price history per SKU per country is a legal record**, and a merchant whose prices live in seven copies, as Kestrel & Pine's did in 2026, cannot produce it.

Payments: strong customer authentication and what is coming

The second **Payment Services Directive**, **PSD2** (Directive (EU) 2015/2366, applying from 13 January 2018) requires strong customer authentication when a payer initiates an electronic payment; the technical standards (Delegated Regulation (EU) 2018/389, from 14 September 2019) allow exemptions for remote transactions up to €30 and for low-risk transactions below thresholds. For the checkout this means a challenge in the middle of paying by card and an exemption strategy for saved cards (Taking the Money: Payment Providers, Authorization, Capture and Refunds).

Since 9 October 2025 euro-area banks also apply **verification of payee** (Regulation (EU) 2024/886): the name on a transfer must match the account, so the company name on a B2B invoice must match the bank account.

PSD3 and the Payment Services Regulation are pending as of November 2026: provisionally agreed on 27 November 2025, confirmed by member states' representatives on 22 April 2026, and still open in the Council's state of play of 29 October 2026 (Council of the EU, 2026). Providers that fail to implement fraud prevention "will be liable for covering customers' losses" (European Parliament, 21 November 2025). The dates will follow adoption; the scope is known.

Invoicing: the mandates and ViDA

Documents Between Companies: EDI, Punchout and E-Invoicing owns the mandate calendar in full. For a Dutch brand selling to businesses in three countries, as of November 2026: **Belgium** requires structured e-invoices over Peppol between VAT-registered businesses since 1 January 2026, with fines of €1,500 to €5,000 for lacking the technical means (FPS Finance, 2024 and 2025); **Germany** requires receiving since 1 January 2025 and issuing from 1 January 2027 for German-established issuers above €800,000 prior-year turnover, from 2028 for all (Federal Ministry of Finance FAQ, n.d.); **the Netherlands** plans a domestic mandate from 1 July 2030 and reporting from 1 July 2031 (Rijksoverheid, 11 September 2026); and **ViDA** requires e-invoicing-based reporting for cross-border business-to-business supplies from 1 July 2030 (Council Directive (EU) 2025/516). The mandates are domestic in scope: a Dutch supplier is not automatically obliged in Belgium or Germany, but its customers there can demand the format, and many do.

Platforms and marketplaces: the DSA and product safety

The **Digital Services Act** (Regulation (EU) 2022/2065, applying from 17 February 2024) requires a marketplace, before a trader may sell, to obtain the trader's identity and contact details, an identification document, "the payment account details of the trader", the trade-register number and a self-certification to offer only compliant products, and to verify them (Art. 30); it bans deceptive interfaces (Art. 25) and requires the interface to let traders provide compliance and product-safety information (Art. 31). For a merchant selling on three marketplaces this is an onboarding data set to keep current three times; for one operating a marketplace or dropship program, a seller-onboarding workflow with verification (LISTED's Rules of the Marketplace: DSA, DAC7 and Vertical Regulation treats the operator's side).

The **GPSR** (Regulation (EU) 2023/988, applying from 13 December 2024) treats a product offered online as made available in the EU if the offer targets EU consumers, and requires every distance offer to show the manufacturer's name and address, the responsible person's where the manufacturer is outside the EU, identification including a picture,

and warnings in a language the consumer understands (Art. 19); marketplaces must let traders provide this per product (Art. 22(9)). In data terms: **four mandatory attributes per product**, **owned by the PIM and syndicated everywhere**, with publication blocked when they are missing.

Product and packaging: destruction bans, producer responsibility, the passport

The **ESPR** (Regulation (EU) 2024/1781) prohibits, "from 19 July 2026, the destruction of unsold consumer products as listed in Annex VII", which covers apparel, clothing accessories and footwear; the prohibition "shall not apply to micro and small enterprises" and applies to medium-sized enterprises from 19 July 2030 (Art. 25(1)); and economic operators that discard unsold products must disclose yearly the number and weight discarded by category, the reasons and the share sent to reuse, recycling, recovery or disposal (Art. 24).

The regulation also defines the **digital product passport**, product data reachable through a data carrier on the item, with a registry of identifiers due by 19 July 2026 (Art. 13); which product groups come first depends on delegated acts not yet adopted, and the expectation that textiles come early (around 2027) is an assumption.

For an apparel brand: **the disposition of every unsold or returned unit is a record with legal weight**, and the PIM becomes the passport's source.

The **Packaging and Packaging Waste Regulation** (Regulation (EU) 2025/40) applies from 12 August 2026 and makes producers responsible for packaging they first place on a member state's market (Art. 45), with harmonized labels from 12 August 2028 or later (Art. 12). National registers already collect the fees (Germany's LUCID: "if you are distributing packaged goods in Germany, you have to be registered", n.d., accessed November 2026; Dutch and Belgian schemes not verified here). The fee is computed from packaging weight by material per country of sale: the product needs its sales packaging, the shipment its carton, the order its destination.

Accessibility

The **European Accessibility Act** (Directive (EU) 2019/882) covers e-commerce services (websites and apps through which consumers conclude contracts) from **28 June 2025**, exempts microenterprises providing services (fewer than 10 persons and at most €2 million turnover), lets products lawfully in use before that date continue until 28 June 2030, and requires "identification methods, electronic signatures, security, and payment services which are perceivable, operable, understandable and robust" (Directive (EU) 2019/882, 2019). The technical content in practice is WCAG 2.1 AA. Because the obligation covers identification and payment, **every embedded widget on the path to the order is in scope**: the hosted checkout, the PSP's iframe, the consent banner.

Vendor conformance statements become a procurement item, and the theme is audited per release.

Cyber and AI: NIS2, the Cyber Resilience Act and the AI Act

NIS2 (Directive (EU) 2022/2555, applied nationally from 18 October 2024) covers listed entity types that are medium-sized or larger, among them "providers of online marketplaces"; significant incidents require an early warning within 24 hours, a notification within 72 hours and a final report within a month (Art. 23(4)). The Netherlands brought it into force as the **Cyberbeveiligingswet** on 15 August 2026 (Rijksoverheid, 7 July 2026). An ordinary retailer is usually out of scope; a marketplace operator or commerce SaaS vendor is in. The merchant meets NIS2 through its suppliers, as incident-notification clauses in the platform's, the WMS vendor's and the iPaaS vendor's contracts, so the 24-hour and 72-hour clocks become terms the merchant's incident process (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers) must meet. The **Cyber Resilience Act** (Regulation (EU) 2024/2847) adds vendors' duties from 11 December 2027.

The **AI Act** (Regulation (EU) 2024/1689) applied generally from 2 August 2026. Its live obligations for commerce are transparency: a system "intended to interact directly with natural persons" must say it is an AI unless that is obvious, and generated content must be "marked in a machine-readable format and detectable as artificially generated" (Art. 50). The **Digital Omnibus on AI** (Regulation (EU) 2026/1744, July 2026) moved the high-risk obligations to 2 December 2027 (Annex III) and 2 August 2028 (Annex I). A search or fraud model is not high-risk; worker monitoring in warehouse software is the case to watch. Kestrel & Pine's W8 decision, agent readiness without agents, keeps it outside most of this in 2027 (Assistants and Agents: AI in Commerce and What Is Real in 2026).

Data: the Data Act and transfers

The **Data Act** (Regulation (EU) 2023/2854) applies from 12 September 2025; its switching chapter covers cloud contracts concluded after that date and, from 12 September 2027, older ones (Art. 29). For a merchant it is an exit right: the OMS and PIM contracts Kestrel & Pine signs in 2027 must allow switching and data retrieval, the legal floor under Moving Without Burning the Bridge: Replatforming and Migration. Transfers belong to Whose Data Is It? Privacy, Consent and Retention.

The Regulatory Calendar for 2027

Kestrel & Pine's calendar as Noor and Sanne write it in March 2027, with application dates as of November 2026 from the official sources in this chapter. *Maintain* rows are in force and need a test; *prepare* rows have a date ahead; *watch* rows are pending. Owners and statuses are illustrative; dates are documented.

Date	Rule (official source)	What it requires of Kestrel & Pine	Owner	2027 status
13 Jun 2014	Consumer Rights Directive 2011/83/EU	14-day withdrawal and refund clocks; the "order with obligation to pay" button	Daniel Okafor	*Maintain*: refund-age report monthly
14 Sep 2019	PSD2 RTS on SCA (2018/389)	Challenge or exemption on every card payment	Milan Petrov	*Maintain*: exemption rate from the PSP
28 May 2022	Omnibus Directive 2019/2161	30-day prior price per SKU and country; review provenance; ranking disclosure	Milan Petrov	*Maintain*: strike-through check before every campaign
17 Feb 2024	Digital Services Act, Art. 30	Trader data and self-certification current on bol, Amazon.de, Zalando	Milan Petrov	*Maintain*: yearly review
13 Dec 2024	GPSR 2023/988, Art. 19	Manufacturer, responsible person, identification and warnings on every offer	Teun Baas	*Maintain*: publication blocked without the fields (PIM rule from Q3)

Date	Rule (official source)	What it requires of Kestrel & Pine	Owner	2027 status
1 Jan 2025 / 1 Jan 2027 / 1 Jan 2028	Germany e-invoicing (BMF FAQ)	Receive since 2025; German-established issuers above €800k issue from 2027, all from 2028; EDI tolerated to end 2027	Sanne de Wit	*Prepare*: offer XRechnung (Germany's structured invoice format) over Peppol to German accounts that ask; not obliged as a Dutch issuer
28 Jun 2025	European Accessibility Act 2019/882	Storefronts, identification and payment steps accessible; not a microenterprise	Milan Petrov	*Maintain*: audit per theme release; vendor statements on file
12 Sep 2025	Data Act 2023/2854	Switching and data-retrieval terms in cloud contracts	Noor Visser	*Prepare*: clause check before signature

Date	Rule (official source)	What it requires of Kestrel & Pine	Owner	2027 status
1 Jan 2026	Belgium e-invoicing (FPS Finance)	Structured invoices over Peppol to Belgian business customers, as they expect (whether the mandate binds a Dutch supplier is unsettled); fines of €1,500–5,000 for Belgian businesses lacking the means	Sanne de Wit	*Maintain*: delivered-status report; 210 accounts
19 Jul 2026	ESPR 2024/1781, Art. 24–25	No destruction of unsold or returned apparel (large enterprise); yearly disclosure	Daniel Okafor	*Maintain*: disposition codes; first disclosure for 2026
2 Aug 2026	AI Act 2024/1689, Art. 50	Disclose AI assistants; mark generated content	Noor Visser	*Watch*: applies the day an assistant or image generator is used

Date	Rule (official source)	What it requires of Kestrel & Pine	Owner	2027 status
12 Aug 2026	PPWR 2025/40; national registers (LUCID)	Producer responsibility per country; packaging data by material	Sanne de Wit	*Maintain*: yearly reports NL, BE, DE
15 Aug 2026	Dutch Cyberbeveiligingswet (NIS2)	Out of direct scope; vendor clauses with 24/72-hour clocks	Noor Visser	*Maintain*: clause inventory
2 Dec 2027	AI Act high-risk, Annex III	Worker monitoring in warehouse software would be in scope	Daniel Okafor	*Watch*: ask the WMS vendor
11 Dec 2027	Cyber Resilience Act 2024/2847	Vendors' product security duties (reporting since 11 Sep 2026)	Noor Visser	*Watch*
Pending	PSD3 and PSR; Digital Omnibus (data); Digital Fairness Act; DPF appeal C-703/25 P	Fraud liability and payee checks; cookie rules; dark patterns; transfers	Noor Visser	*Watch*: quarterly review of status

Date	Rule (official source)	What it requires of Kestrel & Pine	Owner	2027 status
1 Jul 2030 / 1 Jul 2031	ViDA 2025/516; Netherlands e-invoicing and reporting (planned)	Cross-border e-invoicing and reporting; domestic Dutch mandate for 640 accounts	Sanne de Wit	*Prepare*: the Peppol path already exists; watch the Dutch bill in 2027
Not yet set	ESPR delegated act for textiles; digital product passport	Resolvable identifier and passport data per product *(assumption on timing)*	Teun Baas	*Watch*: verify the Commission's working plan each quarter

Three readings. **Most rows are maintenance, not projects. The 2027 projects are few**: Data Act clauses in two contracts, the GPSR publication rule when the PIM arrives, the first ESPR disclosure. And **the owners are mostly not engineers**, because they own the facts; Noor owns the calendar and the vendor rows. The toolkit card is in Toolkit: Cards and Templates for Commerce Architecture Decisions.

Requirements for the compliance data flows

The rows become flows between systems, with requirements like any other (APIs, Events, Files and Middleware: How Commerce Systems Talk).

Flow	Freshness	Consistency	Latency	Availability
Price history for the prior price	Every price change, with effective dates	One record per SKU, market and channel; the strike-through computed from it	Batch; the check runs before a campaign	The campaign does not launch without it
Withdrawal and refund clocks	Each event as it happens (delivery scan, notice, receipt)	One order record with dated events (the OMS)	Minutes	A missed event is a missed deadline; reconcile daily

The Long Fuse: Why Deadlines Arrive Late

Every row had the same life: a proposal, an adopted text, often a national transposition, an application date, then enforcement. Publication to application is usually two to four years.

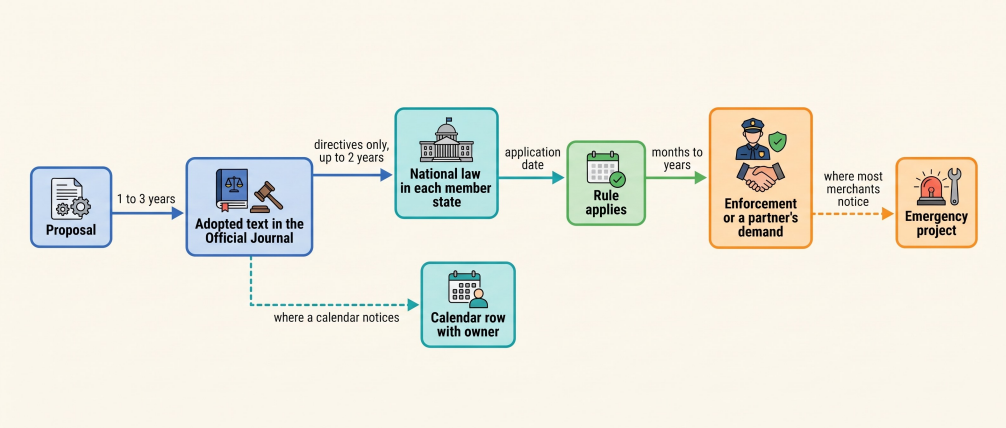


Figure 3: *The Long Fuse: a rule is public years before it applies, a calendar notices it at adoption, and a merchant without one notices it when a customer, a marketplace or a regulator enforces it.*

Five lessons the evidence supports.

Dates slip; scope does not. The AI Act's high-risk dates moved by more than a year (Regulation (EU) 2026/1744); the PCI Council softened its script requirements weeks

before they took effect (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers); the Dutch e-invoicing mandate moved to 2030 before it was a bill. Plan the capability to the scope and schedule it to the date.

Transposition multiplies the calendar. A directive becomes 27 laws with 27 dates: e-invoicing in Belgium (2026), Germany (2025–2028) and the Netherlands (planned 2030) under one EU framework; NIS2 in the Netherlands in August 2026. A merchant selling in three countries keeps three calendars.

Partners enforce before regulators do. Marketplaces turn the GPSR and the DSA into listing fields with their own deadlines; business customers turn a mandate into a refused PDF; retail chains write label standards into purchasing conditions with penalties (Documents Between Companies: EDI, Punchout and E-Invoicing). The Pulled Rug and the Long Fuse are often one event seen from two sides.

Courts and appeals are calendar events too. *South Dakota v. Wayfair* created sales-tax obligations for remote sellers in a day (21 June 2018); the Eighth Circuit vacated the FTC's click-to-cancel rule in July 2025; the pending appeal against the EU–US Data Privacy Framework (C-703/25 P) could change every US transfer. A "watch" row is a real row.

The platform absorbs some of it, never all of it. A hosted platform ships the button text, the cookie banner, the withdrawal flow, the card-data scope of its checkout and the accessibility of the pages it renders. It does not ship the merchant's price history, product-safety fields, packaging reports, disposition records or deletion fan-out. The rows that remain are the data rows.

The Choices: Four Ways to Run Compliance

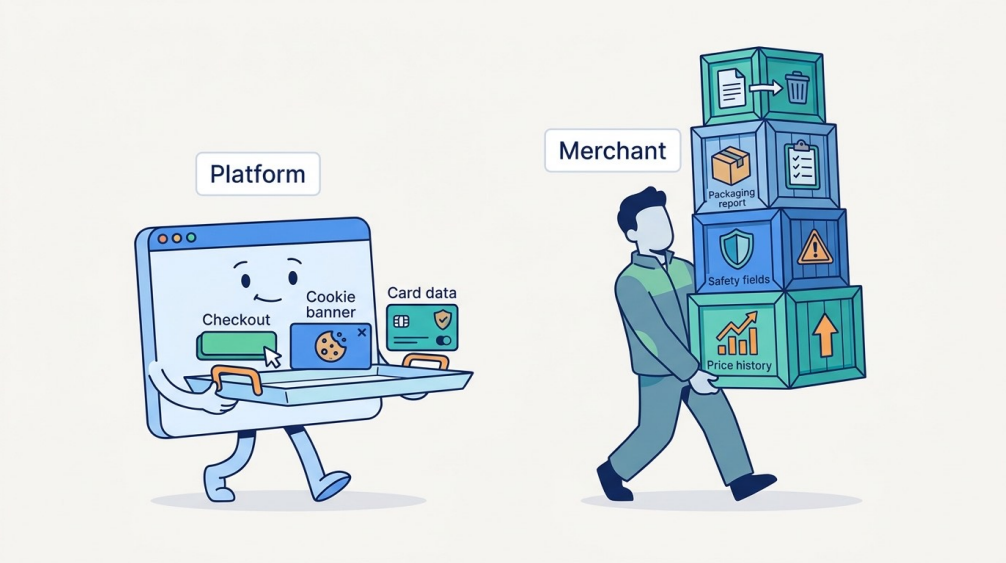


Figure 4: *The platform absorbs some of it, never all of it: the button, the banner and the checkout's card scope travel with the platform, while the data rows stay with the merchant.*

React when someone complains. No calendar; rules arrive through customers, marketplaces, auditors and fines. The default, and nearly adequate for a Starting-stage shop on a hosted platform in one country, because the platform carries most of what applies.

One owner, one calendar, reviewed quarterly. A named person keeps the table, reads the official source per row and brings new instruments to a quarterly review. Cheap; it removes the surprise but does not put the fields in the systems.

Calendar plus requirement cards tied to the system-of-record map. Each row becomes a card (scope, data requirement, system of record, channels, test, date, owner, as in Figure 2), attached to the entity–owner table of Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record, with the tests in the release process. Where an Omnichannel merchant should be, because that is when rules start landing in systems it owns.

A compliance function with counsel and tooling. Legal and compliance staff, tracking subscriptions, controls testing, local advisers. Right for a group or a vendor; at Kestrel & Pine's size the Giant's Boots.

Approach	Cost (illustrative)	Catches	Misses	Right when
React	Nothing, then emergencies	Whatever a partner or regulator enforces	Every deadline's lead time	One country, one channel, hosted platform
One owner, one calendar	Days a quarter	Published dates; pending files	Whether the data actually exists in the systems	Two or three channels, first mandates
Calendar plus requirement cards	A card per row; tests in releases	Dates and the data behind them	Deep national detail; litigation risk	Omnichannel; several countries; owned hubs
Compliance function	Headcount; subscriptions	Nearly everything	Cost; distance from engineering	Group, marketplace operator, vendor

My recommendation: **the third approach, built from the second, the quarter the merchant first owns a PIM, an OMS or a price history**, because from then on the rules are its fields. The calendar is a management artifact; the cards are what engineering can test.

Across Scale Stages: When It Fits and When It Limits

Stage	Typical approach	What becomes necessary	Warning signs of outgrowing it
1 — Starting (under about €2M)	React; the hosted platform's defaults; one country	Checking the defaults against home-country consumer law; reading a marketplace's listing requirements before listing	A second country; the first business customer who wants a structured invoice; a microenterprise exemption that stops applying
2 — Growing (about €2M–€20M)	One owner, one calendar; the agency fixes what the platform does not cover	Price history as a record; GPSR fields in the catalog; a refund-age report; the first e-invoicing route; an accessibility audit of the theme	Rules arriving through marketplaces' error logs (December 2024); a customer's refused PDF (December 2025)
3 — Omnichannel (about €20M–€200M)	Calendar plus requirement cards on the system-of-record map; owners in finance, operations and merchandising	Fields owned by the PIM and OMS; tests in releases; vendor clauses with incident clocks; packaging and disposition reporting; a Data Act check on new contracts	Three national calendars in people's heads; a rule found in a newsletter (June 2026); a report rebuilt by hand each year
4 — Group	A compliance function tied to the map; per-country entities	Country clearance systems behind one invoicing service; one consent model across brands; a regulatory watch with counsel	Each brand reading the law separately; a marketplace business bringing DSA and NIS2 duties of its own

Here **scope follows size**: a microenterprise is exempt from the Accessibility Act, a small enterprise from the ESPR ban, a Dutch company below the small-business threshold from the planned e-invoicing mandate, and a retailer that becomes a marketplace operator picks up the DSA's and NIS2's duties that day. Growth is a change of scope, and a row should say which size or role triggers it.

In Practice

The United States, as the contrast

A European merchant selling into the United States meets a different shape of law: few federal rules, many state ones, and lawsuits where the EU has regulations. **Sales tax**: since *South Dakota v. Wayfair* (21 June 2018) a state may tax remote sellers without physical presence (thresholds of $100,000 of sales or 200 transactions were upheld), and marketplace-facilitator laws make the marketplace collect, so US sales tax is a tax-engine problem (Tax, Invoices and the Books: From Order to Ledger).

Marketplace sellers: the INFORM Consumers Act (effective 27 June 2023) requires marketplaces to collect and verify bank, tax and contact data from high-volume sellers and to disclose sellers above $20,000 on listings, with penalties of $53,088 per violation (Federal Trade Commission, FTC, n.d., accessed November 2026), a mirror of DSA Art. 30.

Reviews: the FTC rule of 14 August 2024 bans fake or AI-generated reviews. **Accessibility**: no federal rule for private sites; exposure is litigation (a vendor counted 5,114 suits in 2025, 70% against e-commerce; UsableNet, 2025). **Privacy**: an opt-out model across at least 19 state laws (IAPP, 2025) against the EU's opt-in.

A platform serving both regions needs a tax engine, a consent record that expresses both models, and the same onboarding data twice.

Kestrel & Pine's 2027 worked example

Take one row through the method. **GPSR, Art. 19**. Scope: a trader offering products to EU consumers at a distance, so every product, own-brand and third-party. Data: manufacturer name and address (Kestrel & Pine B.V. for own brand; the supplier for third-party equipment), a responsible person where the manufacturer is outside the EU, a picture and an identifier (the GTIN, the global trade item number behind the barcode, that every own-brand SKU has carried since 2021), warnings in Dutch, French and German.

System of record: the PIM from Q3 2027 (W3); until then the platform and the connector, which is why the 2024 fix lived in a spreadsheet. Channels: three storefronts, three marketplaces.

Test: a PIM rule that blocks publication without the four fields, run on the 1,200 new variants each season. Owner: Teun Baas. Cost (illustrative): two days of PIM configuration and a clean-up of about 850 third-party variants.

Three weeks of spreadsheet work in 2024 becomes a validation rule in 2027, because the system that should own the field now exists.

When It Goes Wrong

A deadline discovered late. Detection: a customer, marketplace or auditor says so. Recovery: the capability first, the evidence second; Belgium's tolerance of December 2025 spared only businesses that could show timely steps ("under no circumstances" a general postponement), and a dated plan, a signed provider contract and a test log are that evidence. Then ask which other rows have no owner.

A clock that nobody started. A withdrawal notice received by e-mail and typed into the returns portal four days later; the clock ran from the e-mail. Detection: a refund-age report using the notice date. Recovery: every notice channel creates the event with its real date.

A partner's deadline before the law's. A marketplace requires a field months before a regulation applies, or a chain demands a label standard the law never mentions. Detection: partner notices read by the row's owner. Recovery: the partner's date becomes the row's date, because the partner is the enforcer that matters.

What to Take Away

Established.

- The core EU rules for a shop are documented facts with dates: 14-day withdrawal and refund (2014); the 30-day prior price (28 May 2022); strong customer authentication (14 September 2019); DSA trader traceability (17 February 2024); GPSR product information (13 December 2024); accessibility (28 June 2025); the ESPR destruction ban for large enterprises (19 July 2026); e-invoicing in Belgium (1 January 2026) and Germany (2025–2028); ViDA from 1 July 2030.
- Most of these are requirements on data held in a system of record: price history, dated order events, product attributes, structured invoices, disposition and packaging records.
- Directives are transposed with national differences; regulations apply directly; scope depends on role, size and place; pending files (PSD3/PSR, the Digital Omnibus, the Digital Fairness Act, the DPF appeal) are rows with no date yet.

Depends on context.

- Whether a mandate applies depends on establishment and size, not on where your customers are: a Dutch issuer is not bound by the German e-invoicing mandate, the sources do not settle whether the Belgian one reaches it, and customers in both countries will ask for the format; a microenterprise is exempt from the Accessibility Act; a small or medium enterprise is exempt or deferred under the ESPR ban.
- A US presence changes the model from regulation to litigation and from opt-in to opt-out, and adds a tax engine and seller disclosures; a merchant without US sales leaves those rows out.

The author's synthesis.

- **Regulation is a second source of the system-of-record map**, not a separate discipline: most rules in the method table land in a PIM, an OMS or a price history, so the Waypoint hubs are compliance infrastructure too.
- The right unit of work is the calendar row with an owner who owns the fact, and the right cadence is a quarterly read of the official source per row; the rest is ordinary engineering.

Questions to Ask

1. Do we have a regulatory calendar, who owns it, and when was each row's source last read?
2. For each row, which system of record holds the data, and which channels consume it?
3. Can we produce the lowest price of the last 30 days for any SKU in any country, and does the strike-through on the storefront compute from it?
4. Which event starts our 14-day refund clock, and is it the customer's notice or the day someone typed it in?
5. Are manufacturer, responsible-person and warning fields mandatory in our catalog, and is publication blocked without them?
6. Which of our size or role thresholds (microenterprise, small or medium enterprise, marketplace operator) are close, and which rows switch on when we cross them?
7. Which vendors pass NIS2 or Cyber Resilience Act clauses to us, and do this year's contracts carry Data Act switching terms?
8. Which marketplaces or customers enforce a rule earlier than the law does, and who reads their notices?

With security, privacy and regulation in place, Part VIII is complete, and the book turns from the parts to the whole: four journeys that follow one order, one return, one wholesale order and one peak day through every system at once. Continue with Part IX — JOURNEYS: One Order, End to End.

Sources and Evidence

Company, product and investor sources

- Sephora settlement — California Attorney General, 2022-08-24; the CCPA settlement and the Global Privacy Control requirement.
- 2025 Year-End Digital Accessibility Lawsuit Report — UsableNet, 2025-12; US lawsuit counts, quoted as a vendor figure.
- US State Privacy Legislation Tracker chart — IAPP, 2025; the count of comprehensive state privacy laws.

Standards, regulation and official documentation

- Directive 2011/83/EU (Consumer Rights) — Publications Office of the EU, 2011; withdrawal, refund, the button label.
- Directive (EU) 2019/2161 (Omnibus) — Publications Office of the EU, 2019; the prior price, reviews, ranking, personalized prices.
- Directive (EU) 2015/2366 (PSD2) and Delegated Regulation (EU) 2018/389 (RTS on SCA) — Publications Office of the EU, 2015 and 2018; strong customer authentication and exemptions.
- Regulation (EU) 2024/886 (Instant Payments) — Publications Office of the EU, 2024; verification of payee by 9 October 2025.
- Council document 15005/26, state of play for ECOFIN — Council of the EU, 2026-10-29; PSD3/PSR still open. Payment services deal — European Parliament, 2025-11-21; the agreement's content.
- Structured electronic invoices between companies compulsory in 2026 (2024-09-18), FAQ on fines (n.d., accessed 2026-11) and Tolerance during the first three months of 2026 (2025-12-18) — FPS Finance Belgium.
- FAQ E-Rechnung — Bundesministerium der Finanzen, n.d., accessed 2026-11; German dates and thresholds (translated).
- Kabinet kiest voor invoering e-facturatie en rapportage — Rijksoverheid, 2026-09-11; the Dutch 2030 and 2031 decision (translated).
- Council Directive (EU) 2025/516 (ViDA) — Publications Office of the EU, 2025; application from 1 July 2030.
- Regulation (EU) 2022/2065 (Digital Services Act) — Publications Office of the EU, 2022; Art. 25, 30 and 31.
- Regulation (EU) 2023/988 (General Product Safety) — Publications Office of the EU, 2023; Art. 4, 19 and 22(9).
- Regulation (EU) 2024/1781 (Ecodesign for Sustainable Products) — Publications Office of the EU, 2024; Art. 13, 24 and 25 and Annex VII (text verified 2026-11).

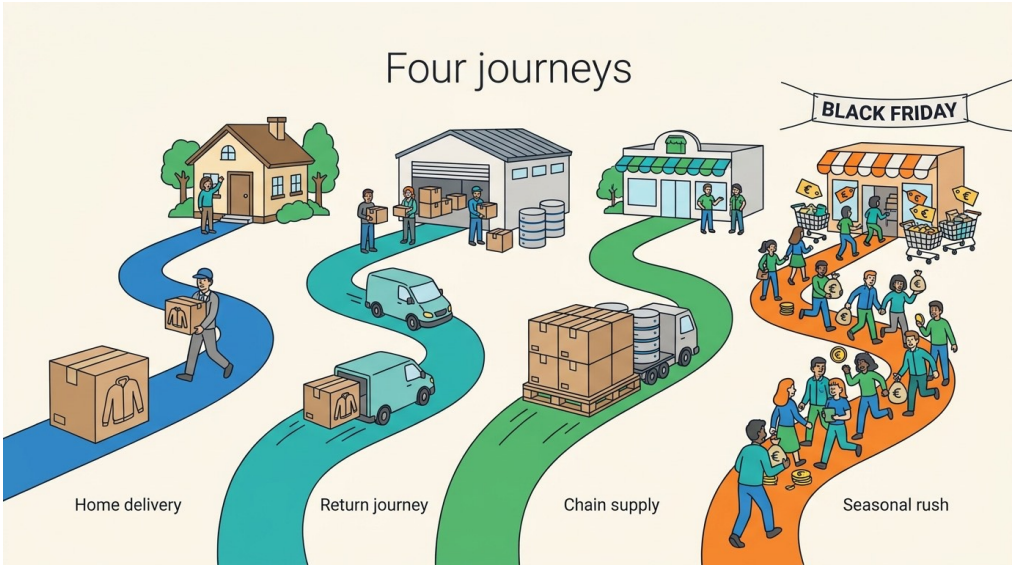
- Regulation (EU) 2025/40 (Packaging and Packaging Waste) — Publications Office of the EU, 2025; application date, Art. 12 and 45. LUCID Packaging Register — Zentrale Stelle Verpackungsregister, n.d., accessed 2026-11.
- Directive (EU) 2019/882 (European Accessibility Act) — Publications Office of the EU, 2019; scope, dates, the microenterprise exemption, Annex I.
- Directive (EU) 2022/2555 (NIS2) — Publications Office of the EU, 2022; scope and reporting clocks. Cyberbeveiligingswet in force 15 August 2026 — Rijksoverheid, 2026-07-07. Regulation (EU) 2024/2847 (Cyber Resilience Act) — Publications Office of the EU, 2024.
- Regulation (EU) 2024/1689 (AI Act) and Regulation (EU) 2026/1744 (Digital Omnibus on AI) — Publications Office of the EU, 2024 and 2026; Art. 50 and the revised dates.
- Regulation (EU) 2023/2854 (Data Act) — Publications Office of the EU, 2023; switching rights and dates.
- Appeal C-703/25 P, Latombe v Commission — Official Journal C/2025/6610, 2025; the pending DPF appeal.
- Guidance for PCI DSS e-commerce requirements effective after 31 March 2025 — PCI Security Standards Council, 2025-03-26; the SAQ A change.
- South Dakota v. Wayfair, Inc. — Supreme Court of the United States via Cornell LII, 2018-06-21; economic nexus.
- Informing Businesses about the INFORM Consumers Act — FTC, n.d., accessed 2026-11; seller thresholds and penalties. Final rule banning fake reviews — FTC, 2024-08-14.
- ADA Title II web rule — US Department of Justice, 2024, updated 2026; scope limited to governments.

Research and practitioner sources

- FTC Relaunches Negative Option Rulemaking — Kelley Drye via JD Supra, 2026-03-13; the vacated click-to-cancel rule.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part IX — JOURNEYS: One Order, End to End



KEY POINTS:

- **An order is a chain of promises, each with one owner.** Many systems holding a piece of one order is normal; a fact with two writers is not.
- **The same four stories run on two architectures.** The dates and amounts stay fixed; what changes between the January 2027 landscape and Waypoint is whether the promises hold.
- **Operations by event, finance by batch, and every join with a key and a net.** Each message carries an idempotency key, and a daily reconciliation catches what the webhooks miss.

A warehouse can report a parcel shipped while the storefront still shows it being prepared. Knowing how each component works does not explain whether their

combined answers keep the customer's promise. These four walkthroughs follow a purchase, an exchange, a wholesale shipment and a peak day across the handoffs, showing where information or responsibility can be lost. All use **Kestrel & Pine, a fictional outdoor brand**; its people, amounts and events are teaching examples.

Figure 1: *An order is a chain of promises: each system carries the order for one leg, hands it on in a marked zone, and owns only its own leg.*

Eva's jacket travels from search to the books, then back for a larger size. Retail chain Bergzicht orders from the same stock under a different document contract. Black Friday puts the shared order path under 2,300 orders an hour. Each chapter repeats the facts needed to follow its example.

How to read the journeys. The first three assign an authoritative system to each fact, then trace the steps, messages and failure recovery. Tables and illustrations show what each handoff needs. The peak chapter first reconstructs the 2026 incident, then presents the 2027 plan.

The first three journeys replay April orders through the late-2027 design. Waypoint, the company's improvement program, adds an availability service for sellable stock, an **order management system (OMS)** for shared orders and a **product information management system (PIM)** for catalog content. Warehouse packing records supply wholesale shipment notices.

The dates and amounts stay in April, but the design is deliberately later. Each walkthrough also compares the earlier landscape, where scheduled stock copies and separate order records caused failures.

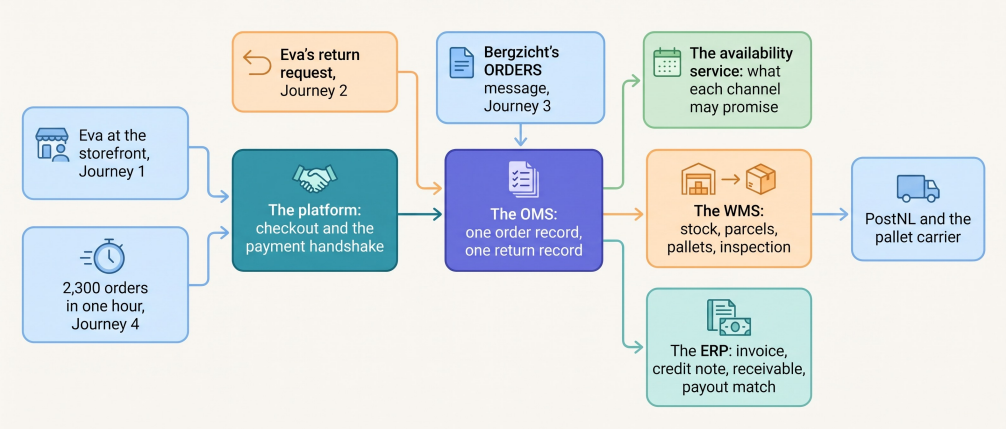


Figure 2: *Four journeys, one chain: a consumer, a return, a retail chain and a peak evening enter the same systems through different doors, and every arrow is a handoff with one sender and one owner.*

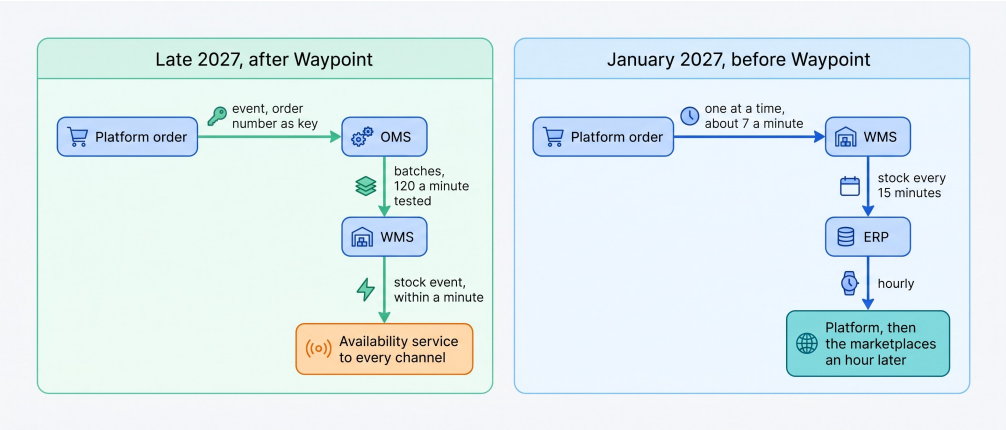


Figure 3: *What the "before Waypoint" comparison in each journey shows: in January 2027 orders crawled to the warehouse one at a time and stock came back through three scheduled copies up to two hours late; in late 2027 orders move as keyed events in batches and stock returns as events within a minute.*

Chapters in this part

Journey 1: Eva's Jacket, From Search to Doorstep follows **order #1072455 from search to accounting**. Eva buys a €199 jacket and two €18 pairs of socks at 21:14 on 13 April 2027, pays through the iDEAL bank-transfer method and receives the parcel on 15 April. Each step identifies who owns its facts and how the next system learns them.

Journey 2: The Jacket Comes Back follows **the M-to-L exchange requested on 19 April**. Inspection on 22 April determines what can be restocked and releases the replacement. A separate refund branch shows how one authorized request avoids January's duplicate-payment failure.

Journey 3: A Chain Reorders by EDI, From Order to Invoice follows **Bergzicht's 240-unit, €19,860 reorder** through structured order, pallet notice, invoice and payment. Three pallets arrive on 9 April; payment is due 9 May. An earlier €14,200 deduction explains why the notice must describe actual packing.

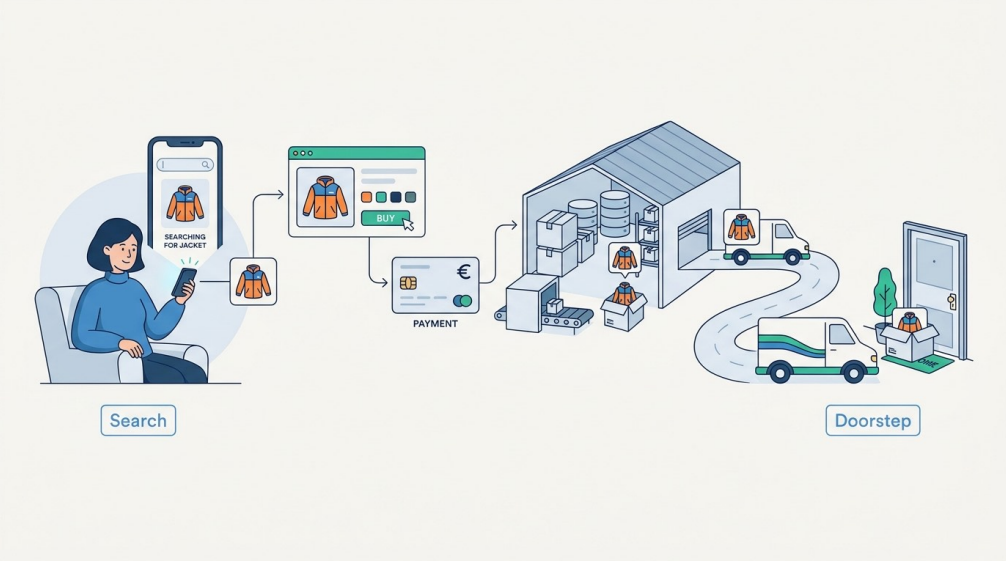
Journey 4: Black Friday, Hour by Hour reconstructs **Black Friday 2026 and the next year's preparation**. A four-hour order backlog and delayed stock updates produce 1,340 oversold units and about €96,000 in costs. The 2027 plan tests capacity, monitors queues and gives each protective action and recovery check an owner.

Begin with Journey 1: Eva's Jacket, From Search to Doorstep.

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine and Bergzicht Outdoor are fictional; recommendations and labeled assumptions are the author's judgement.

37. Journey 1: Eva’s Jacket, From Search to Doorstep



IN THIS CHAPTER, YOU WILL: Follow one consumer order through every system it touches, from search to the ledger, and see which system makes each promise and what breaks when it cannot keep it.

KEY POINTS:

- **One order is fifteen handoffs, and each handoff is a promise.** The storefront promises a product, a price, a stock level and a delivery day; the payment provider promises money; the warehouse promises a parcel; the carrier promises a doorstep; the finance system promises a correct invoice. Nearly everything that goes wrong is a promise one system made and another could not keep.
- **Each promise has exactly one owner in the target architecture.** Product information management owns what the jacket is, the availability service owns whether it can be sold, the platform owns the checkout and the payment handshake, order management owns the order from the moment it exists, the warehouse system owns physical

stock and the parcel, the finance system owns the invoice and the books. Copies are allowed; a second writer is not.
- **Every join delivers at least once and sometimes twice, late or not at all**. The order number is the key that makes a repeated message harmless, and a daily reconciliation is the net under every webhook. A design that assumes each message arrives exactly once is the one that produces the duplicates and the gaps customers find.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Schrödinger's Order** — an order whose status lives in three systems has no status; this journey shows where one record replaces three.
- **The Phantom Stock** — the storefront's "in stock" is a copy of the warehouse's count, and the journey shows how long that copy may be allowed to lag.
- **The Message in a Bottle** — the order crosses six joins, and the chapter names, at each, what happens when the message arrives twice or not at all.

Eva Janssen buys a jacket and socks from **Kestrel & Pine, a fictional outdoor brand**. To deliver that simple purchase, search, checkout, payment, warehouse, carrier and finance systems must pass records to one another. A wrong stock answer or a late warehouse instruction can break the delivery promise even when checkout succeeds. Following one order makes those dependencies visible.

Kestrel & Pine takes 293,000 online orders annually. At that volume, even a failure in one order per thousand affects customers almost daily. This walkthrough follows the ordinary path from search to the books, showing what each handoff must preserve and how a failure reaches the customer. The figures, dates and systems are invented for teaching; Meet Kestrel & Pine introduces the company.

Where we are in the order's journey. The full purchase, from search to the books. Journey 2: The Jacket Comes Back continues with the jacket coming back; Follow the Order: What Happens Between "Buy" and "Delivered" offers a shorter overview.

The Order, the Setting and the Rules of This Walk

The order. At **21:14 on 13 April 2027**, Eva Janssen in Nijmegen buys a **€199 Ridgewalker women's jacket, M, moss**, and **two €18 pairs of socks**. Order **#1072455** totals **€235 including value-added tax (VAT)**, with free shipping over €75. She pays through **iDEAL**, a Dutch bank-transfer method. Duiven picks and packs on **14 April**; PostNL delivers on **15 April**.

The setting is a replay through the late-2027 design. Waypoint, Kestrel & Pine's improvement program, adds an availability service to calculate sellable stock (**W1**), an **order management system (OMS)** to coordinate orders (**W2**), and a **product information management system (PIM)** for catalog content (**W3**).

None was live for Eva's actual April order. Storefront orders move to the OMS on 13 October. We keep her dates and amounts but replay the order through the later design; a section near the end explains the original route.

The systems. The table names each system by its job and identifies its authoritative facts. That makes it an **entity–owner table**. If copies disagree, the assigned **system of record** supplies the value to trust (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record).

Fact about the order	System of record (late 2027)	Who else holds a copy
What the Ridgewalker is: name, attributes, images, sizes, GTIN	**the PIM** (product information management system)	the platform, the search index, the marketplace connector
Consumer price €199	**the platform's** price list for the NL storefront	the search index, the OMS (as the price sold at)
Physical stock at Duiven	**the WMS** (warehouse management system)	the availability service, the ERP (valuation only)
What may still be sold to the NL storefront	**the availability service** (a module of the OMS)	the platform, the search index
The cart and the checkout	**the platform**	nobody
The payment	**the PSP** (payment service provider), with its reference stored on the order	the platform, the OMS, the ERP
The order: lines, state, customer reference, promises made	**the OMS** (order management system)	the platform (customer-facing history), the ERP, the data warehouse
The fulfillment order and the parcel	**the WMS**, with the shipping service for the label	the OMS
Tracking events	**PostNL**, normalized by the shipping service	the OMS, the platform's order status page

Fact about the order	System of record (late 2027)	Who else holds a copy
The invoice, VAT and the ledger entries	**the ERP**	the data warehouse

Eight systems hold a piece of one order, and that is the normal condition, not a problem to fix. The problem is a fact with two writers, and the table has none.

Fifteen Steps in One Table

Here is the whole journey at a glance. Each row names the step, the system that performs it, the data that moves, the promise made to Eva or to another system, the failure that belongs to this step, and the chapter that owns it. Times within a day are illustrative; the ledger fixes the dates.

#	Step	System	Data	Promise made	What could fail here	Chapter
1	Search "waterproof jacket"	the platform's search index	query, ranked products	"These products exist and match"	Stale index: a discontinued or renamed product shows	Finding the Product: Search, Navigation and Merchandising

#	Step	System	Data	Promise made	What could fail here	Chapter
2	Product page	the platform, reading PIM content	attributes, sizes, images, price €199	"This is the jacket, at this price"	Wrong attribute; price differs from another channel (The Price Drift)	What Are We Selling? Products, Variants and the Catalog, What Does It Cost? Price Lists, Contract Prices, Currencies and Tax
3	Availability check, size M moss	the availability service	available-to-promise for channel NL	"You can buy this one"	Stale copy, two writers, wrong buffer (The Phantom Stock)	Is It in Stock? Inventory, Availability and the Promise You Can Keep
4	Add to cart, 21:09	the platform	cart lines, quantities	none: a cart is a draft	Cart lost on login; abandonment	From Cart to Confirmation: Checkout Orchestration

#	Step	System	Data	Promise made	What could fail here	Chapter
5	Checkout: identity, address, delivery option	the platform, calling the delivery-promise service	address, shipping method, promised date	"Delivered Thursday 15 April"	Static promise contradicts the warehouse cut-off	From Cart to Confirmation: Checkout Orchestration, Getting It There: Carriers, Labels, Tracking and the Delivery Promise
6	Price review and VAT	the platform	€235 incl. 21% VAT, free shipping	"You pay exactly this"	Promotion stacking error; tax basis differs from invoice	Discounts, Coupons and Loyalty: Promotions Without Chaos, Tax, Invoices and the Books: From Order to Ledger
7	Payment by iDEAL, 21:13–21:14	the PSP, Eva's bank	payment reference, amount, status	"The money is paid"	Eva never returns from the bank app; webhook late or duplicated	Taking the Money: Payment Providers, Authorization, Capture and Refunds

#	Step	System	Data	Promise made	What could fail here	Chapter
8	Order created and confirmed, 21:14	the platform, then the OMS	order #1072455, lines, customer and payment references	"The order exists and will be kept"	Created twice; created in the platform but never in the OMS	The Order Record: Lifecycle, State and the Order Management System
9	Stock committed	the availability service	committed +1 jacket, +2 socks	"Nobody else can buy these units"	Commit event lost; oversold on a hot SKU	Is It in Stock? Inventory, Availability and the Promise You Can Keep
10	Fulfillment order sent and accepted	the OMS → the WMS via the iPaaS	lines, destination, service level, promised date	"Duiven will ship this"	Request duplicated or never acknowledged; WMS queue backs up	Pick, Pack, Ship: Warehouse Management and Fulfillment, APIs, Events, Files and Middleware: How Commerce Systems Talk

#	Step	System	Data	Promise made	What could fail here	Chapter
11	Pick and pack, Wednesday morning	the WMS	pick list, packed units, carton	"These exact units are in this box"	Short pick (shelf count wrong); wrong size scanned	Pick, Pack, Ship: Warehouse Management and Fulfillment
12	Label, manifest, handover to PostNL	the shipping service, PostNL	tracking number, label, manifest	"The parcel is with the carrier"	Label call fails; manifest closed before the parcel	Getting It There: Carriers, Labels, Tracking and the Delivery Promise
13	Shipped event, confirmation and tracking	the WMS → the OMS → the platform	shipment, tracking number, carrier events	"You can see where it is"	Tracking number not passed on: no updates, a WISMO ticket	Where Is My Order? Notifications, Service and the Customer Record
14	Delivered, Thursday 15 April	PostNL → the OMS	delivered event, time	"It is at your door"	Delivered to a neighbor; lost; damaged	Getting It There: Carriers, Labels, Tracking and the Delivery Promise
15	Invoice, revenue and payout	the ERP, the PSP	invoice, VAT €40.79, PSP payout line	"The books agree with the bank"	Order posted twice or not at all; payout unmatched	Tax, Invoices and the Books: From Order to Ledger

Every "promise made" column entry has a different owner from the row above it. That is why the joins between rows are where Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong locates most of its sixteen patterns.

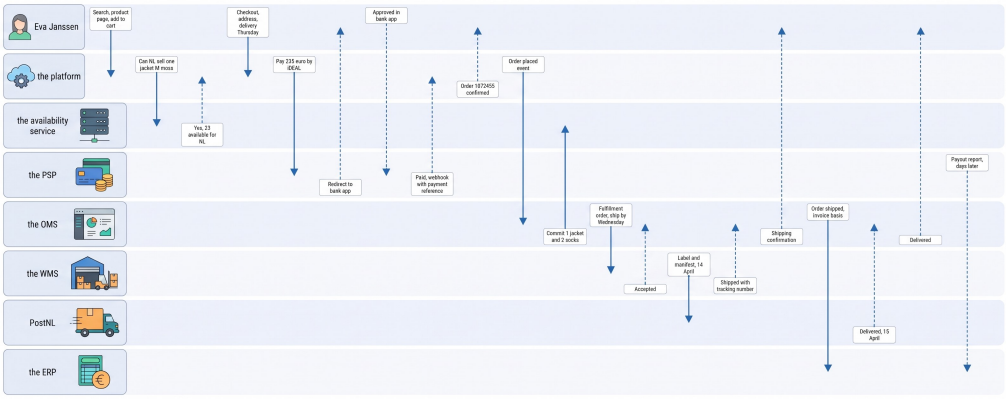


Figure 1: *Eva's jacket in the late-2027 architecture: the platform sells and takes the money, the OMS owns the order from the moment it exists, the WMS and PostNL move the parcel, and the ERP books the invoice and matches the payout; every arrow is a handoff with one sender and one owner.*

Before the Buy: Search, Product Page and "In Stock"

At 21:02 Eva types "waterproof jacket" into the Dutch storefront. The storefront does not search the catalog; it searches a **search index**, a copy of the catalog rebuilt from the PIM whenever product content is published. A copy is fast and a copy goes stale, so the first promise of the journey ("these products exist and match") belongs to the publication schedule: under W5, **products move by scheduled publication**, the PIM publishes and the index consumes. Finding the Product: Search, Navigation and Merchandising owns the index; What Are We Selling? Products, Variants and the Catalog owns the PIM.

The product page makes the second and third promises, and they have different owners. **What the jacket is** (the name, the fabric attributes, the sizes, the moss color, the GTIN, the Global Trade Item Number behind the barcode on every own-brand SKU since 2021) comes from the PIM. **What it costs** (€199, including 21% Dutch VAT, as every consumer price in the EU is shown) comes from the platform's price list for the Dutch storefront. Before the PIM the same jacket was described in four places, and the Price Drift of October 2026, nine days of the German storefront selling it at €179 while every other channel charged €199, is what a silent copy costs (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax).

"In stock" is the most expensive words on the page. When Eva selects size M in moss, the storefront asks the availability service for **available-to-promise** for the NL channel: in plain words, the units that may still be promised to a new order. The formula the industry uses, as practice rather than as any one vendor's definition, is on-hand stock minus units already committed to orders minus any buffer the merchant holds back, per location and per channel.

Every platform exposes some version of it: Shopify's inventory documentation (n.d., accessed 2026-11) separates `on_hand`, `committed` and `available` per location; commercetools documents that its `availableQuantity` is "eventually consistent" with the quantity used for reservations "with a delay of up to 10 seconds" (n.d., accessed 2026-11).

Ten seconds is small; on a hot SKU during a launch it is still enough to sell the last unit twice, which is why Is It in Stock? Inventory, Availability and the Promise You Can Keep treats the buffer and the reservation path as design decisions.

In the target architecture the answer comes from one place. The WMS is the system of record for physical stock at Duiven (W1) and emits a **stock event**, a message sent the moment a count changes, for every receipt, pick and adjustment. The availability service consumes those events, subtracts commitments and the per-channel buffer (one unit per SKU for the storefronts), and publishes an available number per channel; the storefront reads it and never computes its own. Assume the moss jacket in size M shows 23 available for NL at 21:05 *(assumption)*. "In stock" is then a copy seconds old rather than up to two hours old, the January 2027 figure behind the Black Friday Oversell.

Checkout: Six Promises Under a Clock

Eva adds the jacket and two pairs of socks at 21:09. A **cart** is a draft: it reserves nothing and promises nothing, and most carts never become orders; the Baymard Institute's running average of fifty published studies puts abandonment at 70.22% (updated 28 October 2026). Kestrel & Pine's platform, like most, reserves stock at order creation rather than at add-to-cart, so an abandoned cart costs nothing but a lost sale.

Checkout is the ninety seconds in which the platform turns the draft into promises it can keep. It is an orchestration: a sequence of calls to systems that own the facts, run under the clock of Eva's patience, and it owns none of those facts itself (From Cart to Confirmation: Checkout Orchestration). In order:

1. **Identity.** Eva signs in to her account. An order needs a customer *reference*, not an account; forcing one is the fourth-largest stated reason to abandon in Baymard's list. Her account is still one of four records of her, which is One Customer, One Product: Identity and Master Data Across Systems's problem, not this chapter's.

2. **Address**. Nijmegen, the Netherlands. The address decides the tax: a Dutch consumer pays Dutch VAT at 21%; a Belgian one would pay Belgian VAT under the EU's One Stop Shop rules once the merchant's cross-border consumer sales pass €10,000 a year (Tax, Invoices and the Books: From Order to Ledger).
3. **Delivery option and promise**. The Dutch storefront's standing promise is "ordered before 22:00, delivered tomorrow". In January 2027 that was a sentence in the page template with no data behind it. In the target architecture the checkout asks a **delivery-promise service**, which W8 lists as a prerequisite, and that service knows the warehouse's real cut-off. Duiven releases its last pick wave at 21:00 and PostNL collects at 22:30 (Kestrel & Pine illustrations from Pick, Pack, Ship: Warehouse Management and Fulfillment): an order placed at 21:14 misses the last wave, is picked Wednesday and delivered Thursday, so the checkout shows **Thursday 15 April**, and that is the day the parcel arrives. The dates are the same under both architectures; what changes is whether the promise matched them. Getting It There: Carriers, Labels, Tracking and the Delivery Promise owns the computed promise.
4. **Price review**. €199 plus 2 × €18 is €235; shipping is free above €75; no coupon applies. The platform freezes these amounts on the order. The price is now a promise, and the invoice in step 15 must match it.
5. **Payment**. See the next section.
6. **Confirmation**. The order exists.

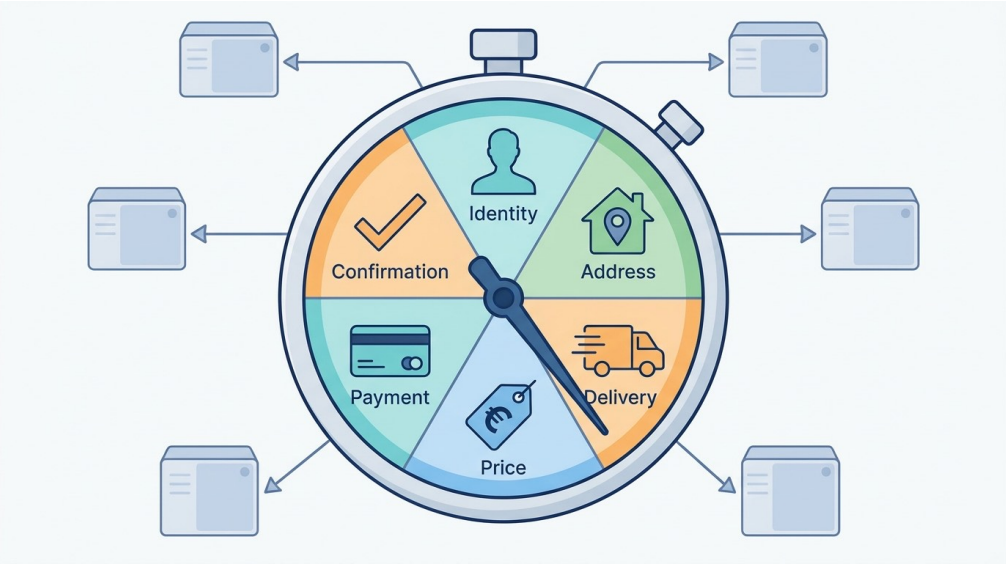


Figure 2: *Checkout is six promises under a clock: the platform orchestrates them in about ninety seconds, but every fact it promises is owned by another system.*

None of these calls should be patched into the checkout's markup. Shopify's dated history is the public lesson: in February 2023 it announced that the `checkout.liquid` file merchants had edited would stop working for the main checkout pages on 13 August 2024, replaced by sandboxed extensions. Kestrel & Pine's agency customized the theme, not the checkout, and the decision to stay on the platform (W4) depends on keeping it that way.

The payment: iDEAL, a detour through the bank

Eva chooses iDEAL. It matters which method she chose, because the money's state machine differs by method, and the order system must model the one it is given rather than a generic "paid: true" (Taking the Money: Payment Providers, Authorization, Capture and Refunds).

iDEAL is a bank-initiated transfer: the shopper is redirected to her own bank's app, approves the amount there, and the bank pushes the money. The Dutch Payments Association reported 1.47 billion iDEAL payments worth €141 billion in 2024 (18 February 2025). Mollie's method documentation (n.d., accessed 2026-11) states the properties that shape the architecture: an "instant charge", **auto-capture only**, no chargeback risk, a session that expires after 15 minutes. Three consequences follow.

- **There is no separate capture.** For a card, the PSP first *authorizes* (reserves the money) and the merchant later *captures* (takes it), often when the parcel ships. For iDEAL the money moves at approval; "capture on shipment" does not apply, and a refund is a new transfer back, not the release of a hold.
- **The shopper leaves and may not come back.** Between 21:13 and 21:14 Eva is in her bank's app. Stripe's fulfillment guide (n.d., accessed 2026-11) states the rule for every PSP: "You can't rely on triggering fulfillment only from your checkout landing page, because it's not guaranteed customers visit that page." The platform learns the outcome from the PSP's **webhook**, an HTTP call the PSP makes to the merchant's system when the status changes, not from Eva's browser.
- **The webhook is a notification, not a fact.** Mollie's webhook carries only an identifier, "the status is not transmitted", so a forged call cannot mark an order paid; the receiver fetches the status from the API. Mollie retries an unanswered webhook ten times over 26 hours; Stripe retries for up to three days and does not guarantee order. The platform records the payment reference and the fetched status on the order, never the webhook's claim.

At 21:14 the platform creates order **#1072455** with the payment *pending* and shows Eva the confirmation; seconds later the PSP's webhook arrives, the platform fetches the status, and the order is marked paid. The order confirmation e-mail, the one notification Shopify's documentation says cannot be switched off (n.d., accessed 2026-11), goes out from the platform in the same minute.

The Order Becomes a Record: Platform to OMS to WMS

Until 21:14 the order was the platform's. From 21:14 it belongs to the OMS, and this handoff is the one Waypoint (W2) was created to fix.

The platform emits an "order placed" event carrying the order number, the lines, the frozen prices, the customer reference, the delivery promise and the PSP's payment reference. The iPaaS, the integration platform through which Kestrel & Pine's flows run, delivers it to the OMS. The OMS creates the order with **one state model** for every channel: the money state (paid, by iDEAL, with its reference), the goods state (unfulfilled) and later the return state are three separate machines on one record, as Shopify's and commercetools' order models keep them (both n.d., accessed 2026-11), because "fulfilled" never means "delivered" and "paid" never means "invoiced". The platform keeps the order history Eva sees in her account; the OMS keeps the operational truth; the ERP will keep the financial one.

The order number is the idempotency key. The event may arrive twice: Shopify's documentation says plainly that "your app might receive the same webhook more than once" and that ordering is not guaranteed (n.d., accessed 2026-11), and every queue and iPaaS retry behaves the same way. **Idempotency** means that handling a message twice has the same effect as handling it once; the OMS keys order creation on #1072455, so the second copy finds the order and does nothing. Stripe's API applies the same principle in the other direction, saving the first response to a client-supplied idempotency key so a retry returns it instead of charging again (n.d., accessed 2026-11). W5 makes this a standard for every Kestrel & Pine integration.

Stock is committed. The OMS tells the availability service to move one jacket and two pairs of socks from available to committed for the Duiven location. On-hand does not change; the shelf still holds the units. Available drops from 23 to 22 for the moss jacket in M, and the next shopper's product page says so within seconds.

The fulfillment order is the contract with the warehouse. The OMS now creates a **fulfillment order**, an instruction to one location to ship these lines to this address by this service level, and sends it to the WMS through the iPaaS. It carries what the warehouse needs (lines, destination, carrier service, promised date) and nothing it does not (prices, payment, customer history). The handshake follows the best-documented public model: in Shopify's fulfillment-order design the merchant *requests* fulfillment, the warehouse *accepts* or *rejects*, and once accepted the merchant must *request* cancellation rather than cancel (shopify.dev, n.d., accessed 2026-11). The WMS accepts at 21:15; the order is now **in progress**, the state the Three-Status Order of December 2026 could never agree on.

Wednesday morning at Duiven. The order joins a **pick wave**, a group of orders released together so one walk through the warehouse serves many of them, at the first release of the day, 06:30 (illustrative). A picker scans the jacket's and the socks' GTINs

into a tote; the scan is the moment the WMS confirms that the units it promised exist. Packing chooses the carton and closes the lines. Two things then leave the WMS for different places: a **stock event** (on-hand minus three units) to the availability service, which also clears the commitment, and a **shipped event** to the OMS once the label exists.

The label comes from the **shipping service**, the multi-carrier layer inside the WMS flow. It asks PostNL's API for a label and a tracking number, prints it, and at the end of the day **manifests**: closes the day's list of parcels and hands it to the carrier. Manifesting is a carrier-specific rule integrations forget; DHL's German parcel API transfers shipments "only when manifested", with an automatic close-out at 17:45 (developer.dhl.com, updated 2026-11), and PostNL's own APIs are mid-migration to a renewed generation rolled out from 2025 (n.d., accessed 2026-11). The shipping service absorbs these differences so the WMS sees one label call and one tracking vocabulary (Promises Between Systems: API Contracts, Webhooks and Versioning).

At 18:30 the shipping service closes Wednesday's PostNL manifest, and at 22:30 the PostNL truck collects (times illustrative). Eva's jacket has left the building.

In Transit: Tracking, Notifications and "Where Is My Order"

From the collection scan onward the parcel's state is PostNL's to know and Kestrel & Pine's to relay. A **tracking event** is one carrier scan with a time and a place: accepted, in transit, out for delivery, delivered, attempted. Carriers each have their own codes; a multi-carrier layer maps them to one small vocabulary, such as the ten statuses EasyPost documents for its tracker object (n.d., accessed 2026-11). The shipping service feeds normalized events to the OMS, which updates the order's goods state and raises the notifications.

Eva receives three messages with three origins: the **shipping confirmation** with the tracking link when the WMS reports the label (Wednesday afternoon), **out for delivery** when PostNL's morning scan arrives (Thursday 07:50), and **delivered** at 11:20 on Thursday (times illustrative). The order status page in her account is a read model over the OMS's order and the tracking stream, the same record a service agent sees; built from different sources, the two would disagree on the day it matters.

The measure of this part of the journey is a service metric. "Where is my order" (WISMO) is consistently the largest single category of e-commerce support contacts; Gorgias's helpdesk data put it at 18% of incoming requests (updated 13 November 2023). The number is an architecture signal: a tracking number that never left the WMS, a static promise, a split shipment without an explanation. Shopify's help documentation states the failure in one sentence: "If you fulfill an order without a tracking number, then no shipping updates are received" (n.d., accessed 2026-11). Where Is My Order? Notifications, Service and the Customer Record owns the service view.

Delivery is also a legal event. Under the EU Consumer Rights Directive (2011/83/EU), the risk of loss passes to the consumer when she "has acquired the physical possession of the goods"; until 11:20 on Thursday the parcel is Kestrel & Pine's risk, which is why the OMS needs the delivered event and not only the shipped one.

The Money and the Books

The parcel's journey ends at the door. The order's does not. Three documents still have to agree: the invoice, the payment and the bank.

The invoice comes from the ERP, not the shop. The OMS sends a "shipped" event to the ERP on Wednesday, and the ERP issues the invoice from it. The reason is numbering. A Dutch invoice must carry a unique, sequential invoice number and date, the parties, the goods, the net amount per VAT rate and the VAT amount in euros (Belastingdienst, n.d., accessed 2026-11); "sequential" means gap-free within a series, so the number is assigned by one system at the moment of posting, never pre-allocated by a checkout.

Once issued, an invoice is immutable: Odoo's documentation, representative of any ERP, states that "Once confirmed, an invoice can no longer be updated" and that a credit note "is the only legal method for canceling, refunding, or modifying a validated invoice" (18.0 documentation, accessed 2026-11). Eva's exchange next week will produce a credit note, not an edit (Journey 2: The Jacket Comes Back).

For #1072455 the invoice reads, net of 21% VAT: jacket €164.46, two pairs of socks €29.75, total net **€194.21**, VAT **€40.79**, gross **€235.00**, arithmetic on the ledger's prices. The amount matches the checkout because nothing changed between the two; when something does (a short pick, a canceled line) the invoice follows what shipped, not what was ordered (Tax, Invoices and the Books: From Order to Ledger).

Revenue is recognized at delivery, not at payment. Under IFRS 15 (issued 2014, effective from 2018), revenue is recognized when control of the goods passes to the customer, which for a parcel is delivery. Kestrel & Pine was paid on Tuesday, invoiced on Wednesday and earned the revenue on Thursday, and a 24% item return rate means a refund liability sits beside the revenue until the withdrawal period runs out. Sanne de Wit's month-end depends on these being three different fields.

The payout arrives days later and is matched by finance, not by the order system. The PSP pays out in batches: Adyen's default "sales day payout" groups a day's funds into one batch with a delay of two business days (n.d., accessed 2026-11); Stripe's default settlement in most EU countries moves from seven calendar days to three business days once an account is established (n.d., accessed 2026-11).

One bank deposit therefore contains hundreds of orders, minus fees, refunds and chargebacks. The join is a **payout reconciliation report**: Stripe's lists "every payment, refund, dispute, fee, and other balance transaction included in the payout" with the order id and a `trace_id` the bank also sees (n.d., accessed 2026-11).

The ERP matches Eva's €235 inside Tuesday's batch to the invoice, posts the PSP's fee (for iDEAL a fixed amount per transaction on Adyen's indicative price list, n.d., accessed 2026-11), and the three-way match of invoices, PSP report and bank statement closes. The deeper mechanics belong to the sibling book RECONCILED (Payments-to-Ledger: Draining the Suspense Swamp, Reconciled by Design: Traceability and Audit Across Boundaries).

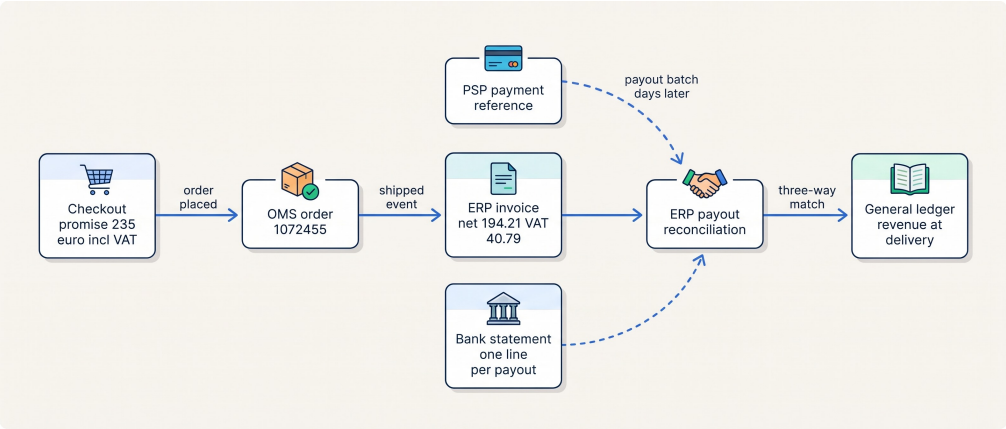


Figure 3: *One order, three financial records: the checkout's promise, the ERP's invoice issued on dispatch, and the PSP's payout matched against the bank statement; the ledger closes only when all three agree.*

What Each Handoff Requires

The style of every join follows from four questions: how fresh the data must be, how consistent, how fast, and how available. The latency figures are the author's illustrative targets for a merchant of Kestrel & Pine's size, not ledger facts.

Handoff	Freshness	Consistency	Latency	Availability	Style (W5)
WMS → availability service → storefront	Seconds	Eventually consistent, one writer per quantity	Under 10 s *(assumption)*	Storefront keeps the last value, with a buffer	Events

Handoff	Freshness	Consistency	Latency	Availability	Style (W5)
Checkout → PSP → platform	Redirect immediate; webhook within minutes	Status fetched, never assumed	Under 2 s per call *(assumption)*	Fails closed: no order without a fetched status	Call plus webhook
Platform → OMS	Seconds	One order per order number	Under 30 s *(assumption)*	OMS catches up by replay	Events, idempotent
OMS → WMS (fulfillment order)	Minutes	Request, accept, reject, cancel-request	Before the next pick wave	Queue absorbs an outage; the cut-off is the deadline	Events with acknowledgement
WMS → shipping service → carrier	At pack	One label per parcel; manifest once	Under 5 s per label *(assumption)*	Fallback carrier (DPD) if the label call fails	Synchronous call
OMS → ERP (invoice, posting)	Same day	One invoice per shipment, one posting per invoice	Daily batch	Reconciliation count per day	Daily batch
PSP → ERP (payout)	Days	Batch total equals bank line	Days	Monthly close	File or reporting API

Two rows cannot be batch, availability and the fulfillment order, because an hour's delay on the first is the Black Friday Oversell and on the second a missed cut-off. Two rows should be batch, invoice posting and payouts, because finance tolerates a day and gains a reconciliation count from it. "Finance can be batch; availability cannot" is the rule The Back Office: ERP Integration and Who Is the Hub states, and this journey demonstrates it.

The Same Order in January 2027

Run #1072455 through the landscape Noor Visser found in January, and four things change while the dates stay the same.

Step	Late 2027	January 2027
"In stock" on the product page	Seconds old, from the availability service on WMS events	Up to two hours old: WMS → ERP every 15 minutes, ERP → platform hourly (The Phantom Stock)
Delivery promise at checkout	Computed: Thursday 15 April	Static "delivered tomorrow": Wednesday 14 April, which the parcel missed by a day
The order record	One, in the OMS, with one state model	Three: the platform's order, the WMS's pick, the ERP's posting, each with its own status (Schrödinger's Order)
The invoice	From the ERP on the shipped event	From the ERP through the partner-built connector, only on the WMS's "order complete" message, so a split order showed "Not invoiced" for days

The parcel still shipped on Wednesday and arrived on Thursday; Eva would have noticed only that the promised day was wrong. The Three-Status Order of December 2026, which showed *Delivered*, *In picking* and *Not invoiced* for six days and cost eleven tickets, is this January path with one split shipment: the iPaaS flow marked the whole two-line order fulfilled at the first parcel, the WMS still showed the second line in picking, and the ERP waited for an "order complete" message that could not come until the second parcel shipped. Nothing in that landscape was broken; the joins had no owner, no idempotency and no reconciliation, and W5 is the sentence that changed that.

When It Goes Wrong Along This Path

The step table names a failure per step. Five of them deserve the mechanism, the detection and the recovery, because they recur at every merchant.

The duplicate order. The "order placed" event arrives at the OMS twice, or the iPaaS replays a day after an outage. Detection: by design, the OMS keys on the order number and the second copy is a no-op; a count of "orders received" against "orders created" per hour shows the duplicates without harm. Without the key, the WMS ships two parcels and the customer keeps one: the Message in a Bottle in its most expensive form.

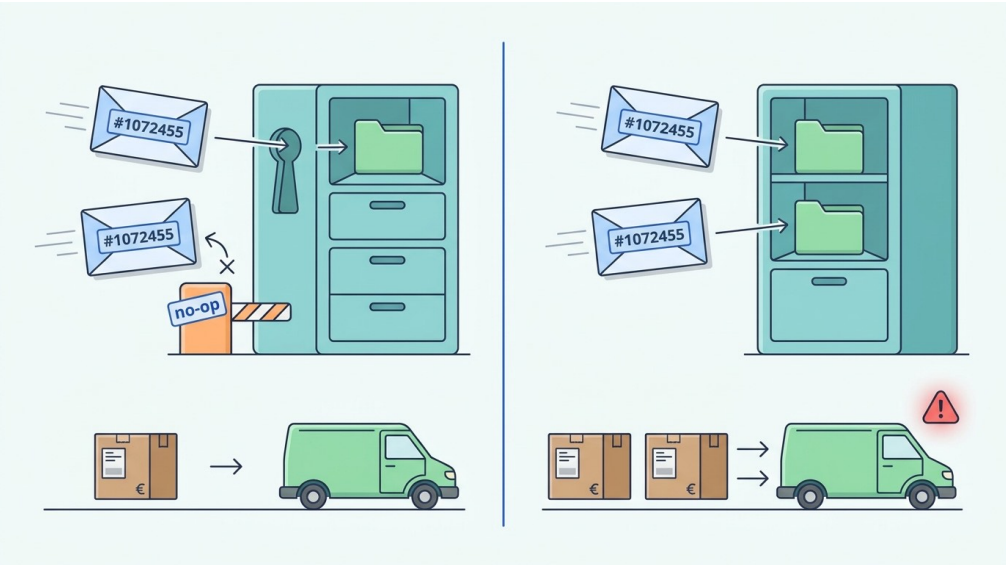


Figure 4: *The order number is the key that makes a repeated message harmless: the second copy of #1072455 is a no-op, and without the key the warehouse ships two parcels.*

The order that was paid and never confirmed. The PSP's webhook fails for 26 hours and then stops; the shopper paid and the order still says *pending*. Detection: a daily sweep that lists checkouts whose payment reached no terminal state within the method's expiry window and queries the PSP, the job Stripe's and Mollie's documentation both ask merchants to run. Recovery: mark the order paid from the fetched status, late but once, and release it to the OMS. Webhooks are a latency optimization over a reconciliation, never a replacement (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation).

The short pick. The picker finds no moss jacket in M where the WMS says there are 22. The classic study of inventory-record accuracy found 65% of about 370,000 records inaccurate at one US retailer's stores (DeHoratius and Raman, *Management Science*, 2008, dated but still the reference point); cycle counting keeps the number low, not zero.

Detection: at the scan. Recovery: the WMS reports the line short, the OMS splits or holds it, the availability service corrects on-hand, and the customer is told before the promise date. A WMS that silently ships two of three items produces a wrong invoice and a WISMO ticket. The queue that backed up for four hours on Black Friday 2026 is the same path under load (Journey 4: Black Friday, Hour by Hour).

The lost parcel. No scan for 48 hours after collection. Detection: a rule on the tracking stream, not a customer e-mail. Recovery: re-ship from stock (the risk was still the merchant's) and claim against the carrier; the OMS records a second shipment on the same order, and the invoice does not change. PostNL reports a parcel "delivery quality" of 97% in 2023 and 2024 (annual report, 2025); at Kestrel & Pine's volumes the remaining 3% is about ten thousand parcels a year that need this path.

The payout that does not match. Tuesday's batch is short by €235. Causes, in the order finance checks them: a refund deducted from a later batch, a fee netted differently, a failed refund returned days later, or an order refunded twice, the Double Refund of January 2027 (€19,400, twice). Detection: the three-way match. Recovery: a journal correction with the PSP's line as evidence and, for the double refund, a rule that only one system may call the PSP's refund API (Taking the Money: Payment Providers, Authorization, Capture and Refunds).

What to Take Away

Established.

- The order's journey crosses systems that each own one fact, and the handoffs between them are where commerce failures concentrate; Shopify, commercetools, Stripe and Mollie document the same model from four sides: separate state machines for money and goods, order creation driven by fetched payment status, request-and-accept handshakes with the warehouse.
- Every delivery mechanism between those systems is at-least-once in practice: Shopify, Stripe and Mollie all document duplicate and out-of-order deliveries and ask the receiver to deduplicate and reconcile. Exactly-once is a property the receiver builds with an idempotency key, not one it receives.
- The invoice is issued by one numbering authority at dispatch, is immutable once issued, and is corrected only by credit note; revenue is recognized at delivery under IFRS 15. The order system makes promises; the ERP keeps the books.

Depends on context.

- Whether the OMS should exist at all. At Kestrel & Pine's stage (four channels, one warehouse, stores, marketplaces, EDI) one order record is necessary; a Starting or Growing merchant with one channel and a 3PL gets the same journey from

the platform's own order and fulfillment-order model, and an OMS would be the Giant's Boots.
- Whether the delivery promise should be computed. A static rule is honest at one warehouse with one carrier and a cut-off the warehouse actually meets; compute the promise when the warehouse's collection time and the page's cut-off differ, as they did here, or when there is more than one lane.
- Whether capture should follow shipment. For cards, capturing at dispatch saves refund fees and ties money to goods; for iDEAL and other instant methods there is no capture step, and the design must not assume one.

The author's synthesis.

- I read any commerce architecture by walking one order through it with this chapter's step table, and I stop at every row where the promise and the system of record belong to different systems with no event, no key and no reconciliation between them. Those rows are next year's incidents.
- I would rather have eight systems with one owner per fact than three systems with two owners for stock. Count writers, not boxes.

Questions to Ask

1. For each fact on an order (product, price, stock, promise, payment, status, invoice), which single system is the system of record, and which systems hold copies?
2. What is the idempotency key for "create order", "create fulfillment order", "refund" and "post invoice", and what happens when the same message arrives twice?
3. When the PSP's webhook never arrives, which job notices, how long after, and who owns it?
4. Where does the delivery promise shown at checkout come from, and is the warehouse's real cut-off one of its inputs?
5. Does the storefront's "in stock" come from an availability service fed by stock events, or from a periodic copy, and how old can the copy be at peak?
6. When the warehouse picks short, who tells the customer, when, and does the invoice follow what shipped?
7. Does the order status page read the same record the service agent reads?
8. Which system issues the invoice number, and can anyone edit an issued invoice?
9. How is one PSP payout matched to the orders it contains, and what were last month's unexplained differences?

Eva's jacket arrived on Thursday, and on Monday she will ask for a size L. The way back crosses many of the same systems in the opposite direction, with a refund and a credit note where the payment and the invoice were, and it is where Kestrel & Pine paid €19,400 twice. Journey 2: The Jacket Comes Back follows it.

Sources and Evidence

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- Orders and Inventory — commercetools API documentation, n.d., accessed 2026-11; order, shipment and payment states; the 10-second availability consistency window.
- Fulfill orders with Checkout, Webhooks, Idempotent requests, Payouts and Payout reconciliation report — Stripe documentation, n.d., accessed 2026-11; fulfillment from webhooks, retries for three days, idempotency keys, settlement timing, payout matching.
- iDEAL, Webhooks and Handling payment status — Mollie documentation, n.d., accessed 2026-11; iDEAL's instant charge and auto-capture, the id-only webhook, ten retries over 26 hours.
- Capture, Sales day payout and Pricing — Adyen documentation and price list, n.d., accessed 2026-11; authorization versus capture, payout batching and delay, indicative iDEAL fee.
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- Parcel DE Shipping API v2 — DHL developer portal, updated 2026-11; manifesting as a carrier rule.
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- Trackers — EasyPost documentation, n.d., accessed 2026-11; a canonical tracking-status vocabulary.
- PostNL Annual Report 2024 — PostNL, 2025; the self-reported 97% parcel delivery quality.
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- Customer invoices and Credit notes — Odoo 18.0 documentation, accessed 2026-11; sequential numbering, immutability, credit notes as the only correction.

Standards, regulation and official documentation

- Directive 2011/83/EU on consumer rights — Official Journal, 2011-11-22; delivery within 30 days (Art. 18) and passing of risk (Art. 20).
- Invoice requirements — Belastingdienst, n.d., accessed 2026-11; mandatory invoice contents in the Netherlands.
- VAT rates and exemptions — business.gov.nl, n.d., accessed 2026-11; the 21% Dutch standard rate.
- One Stop Shop — European Commission, n.d., accessed 2026-11; the €10,000 cross-border threshold.
- IFRS 15 Revenue from Contracts with Customers — IFRS Foundation, issued 2014, effective 2018; revenue when control transfers.
- Study on the payment attitudes of consumers in the euro area (SPACE 2024) — European Central Bank, 2024-12; e-payment solutions at 76% of Dutch online payments.

Research and practitioner sources

- Cart abandonment rate statistics — Baymard Institute, updated 2026-10-28; the 70.22% average and the reasons list.
- Are inventory records inaccurate? Evidence from retail — DeHoratius and Raman, *Management Science* 54(4), 2008; the 65% record-inaccuracy finding (dated).
- Starbucks does not use two-phase commit — Gregor Hohpe, 2004; write-off, retry and compensation as the three error strategies behind every handoff in this chapter (dated, still the clearest analogy).

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

38. Journey 2: The Jacket Comes Back

IN THIS CHAPTER, YOU WILL: Follow one returned jacket from the request to the restock, the exchange or the refund, and the credit note, and see which system owns each step and where the money goes out twice.

KEY POINTS:

- **A return is three flows that must end in agreement, not one.** The goods come back through the carrier and the warehouse, the money through the payment provider, and the books record a credit note. Each flow has its own owner and clock, and the return is finished only when stock, money and books agree.
- **The refund is triggered by what the warehouse found, and it has one writer and one key.** The decision to refund belongs to the disposition event from inspection, not to the request or the carrier scan; exactly one system may call the payment provider, with the return id as the key that makes a repeated call harmless. The Double Refund of January 2027 is what two writers and no key cost.

- **An exchange is a return plus a new order, never an edit.** The exchange order carries the original frozen price, commits stock in the availability service and ships through the normal path; the books record a credit note and a new invoice that net to zero. Marketplace and store returns are the same flow with a different first step.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Boomerang** — refunds, restocking and credit notes reconciled by hand after the return arrives, and sometimes twice; this journey shows the one order of events that makes the three agree.
- **The Message in a Bottle** — the refund request, the disposition event and the refund confirmation each cross a join that can deliver twice or not at all, and the chapter names the key at each.
- **Schrödinger's Order** — a return whose state lives in the portal, the warehouse and the payment console has no state; the chapter shows where one return record replaces three.

A customer returns a jacket and asks for a larger size. Receiving the parcel does not tell the merchant whether the original can be sold again, whether the replacement is reserved or whether a financial adjustment is due. Those decisions belong to different systems. If each treats its own step as a completed return, the customer can receive a refund twice or wait for a replacement that was never secured. This walkthrough connects the steps; Journey 1: Eva's Jacket, From Search to Doorstep covers the original purchase, with the facts needed here repeated below.

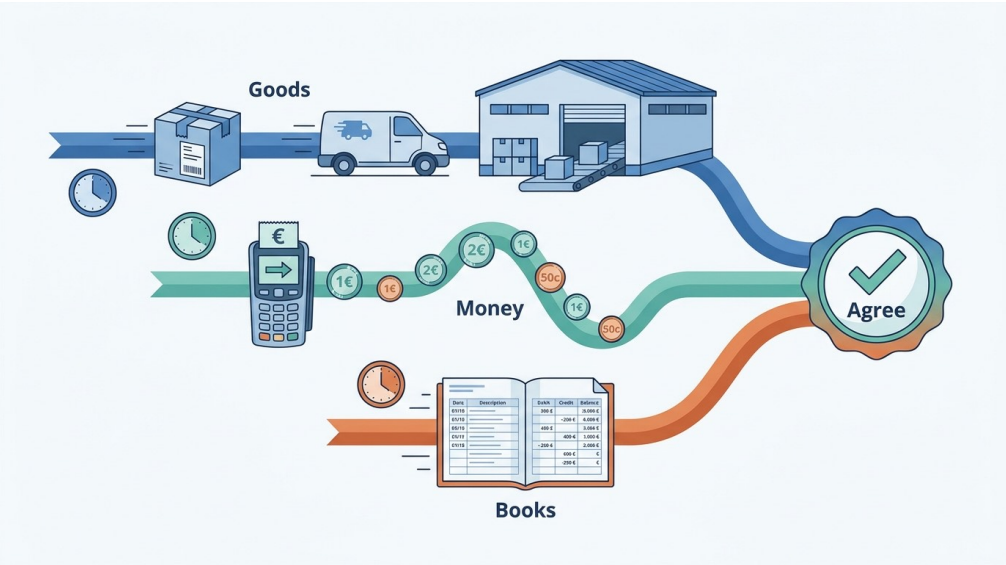


Figure 1: *A return is three flows, not one: the goods, the money and the books each have their own owner and clock, and the return is finished only when the three agree.*

Kestrel & Pine is the fictional outdoor brand used in this example. In 2026 it received **72,000 return parcels** after shipping 336,000. Online returns affected **24% of items** and **19% of orders**; **28%** were exchanges and **61%** of returned items could be resold as new. For comparison, the US National Retail Federation reported a **19.3% online return rate** for 2025, with **9% of returns fraudulent** (October 2025). These measures describe different populations and should retain their definitions.

Where we are in the order's journey. Eva received her jacket on **15 April 2027** and asks for a larger size on **19 April**. We follow authorization, transport back, inspection, replacement and accounting. The Way Back: Returns, Exchanges, Refunds and Reverse Logistics examines the choices behind each step.

The Return, the Setting and the Rules of This Walk

The return. Eva Janssen in Nijmegen exchanges her **€199 Ridgewalker women's M jacket in moss for L**, because it is too small. She keeps the two €18 pairs of socks from the order. Duiven receives and inspects the jacket on **22 April**, grades it resellable and sends the replacement that day.

The alternative is a **€199 refund** through iDEAL, the Dutch bank-transfer method she used, with a credit note. It is a separate branch of the example. Times within a day, identifiers and carrier scans are illustrative.

The setting is a replay through late 2027. Waypoint, the company's improvement program, adds availability management (**W1**), a shared **order management system (OMS)** with return states (**W2**), and **product information management (PIM)** (**W3**).

Eva's actual April return used the earlier landscape, where portal, platform and warehouse held separate pieces. We replay it through the target design, then compare the earlier route near the end.

The systems. The table assigns an owner to each fact: the return request, inspection outcome, stock change and refund. The owner's value is authoritative when copies disagree, making it the **system of record** (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record).

Fact about the return	System of record (late 2027)	Who else holds a copy
The original order, its lines, frozen prices and payment reference	**the OMS** (order management system)	the platform, the returns portal
Whether this line may be returned, by when, at what cost	**the OMS**, from the delivery date, the policy and the product's attributes in the PIM	the returns portal
The return record: lines, reason, requested outcome, state	**the OMS**	the returns portal, the WMS, the ERP
What arrived and what it is worth: inspection and **disposition**	**the WMS** (warehouse management system)	the OMS, the data warehouse
Physical stock at Duiven, including returned units	**the WMS**	the availability service
What may be sold again, per channel	**the availability service** (a module of the OMS)	the platform, the marketplace connector
The refund: amount, method, state	**the PSP** (payment service provider), with its reference on the return record	the OMS, the platform, the ERP
The credit note and the new invoice	**the ERP**	the data warehouse

A fact with two writers is the problem, and in January 2027 the refund had two.

Twelve Steps in One Table

The whole way back at a glance: the step, the system, the data that moves, the promise made, the failure that belongs to the step, and the chapter that owns it.

#	Step	System	Data	Promise made	What could fail here	Chapter
1	Return request, Monday 19 April	the returns portal, reading the OMS	order #1072455, jacket line, reason, outcome "exchange for L"	"You may return this, free, until 29 April"	Eligibility from a stale order copy	The Way Back: Returns, Exchanges, Refunds and Reverse Logistics, Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation

#	Step	System	Data	Promise made	What could fail here	Chapter
2	Held exchange order created	the OMS, asking the availability service	order #1072455-X1 for one L moss at €0 balance, held; one unit of L committed for NL	"Size L is yours when the M is back"	Size L promised with no stock; unit committed and never released	Is It in Stock? Inventory, Availability and the Promise You Can Keep, The Order Record: Lifecycle, State and the Order Management System
3	Return record created	the OMS	return R-20270419-1, requested → approved, linked to #1072455 and to the held order	"This return exists and will be tracked"	Created twice from a retried request; held only in the portal	The Order Record: Lifecycle, State and the Order Management System

#	Step	System	Data	Promise made	What could fail here	Chapter
4	Label issued and return announced	the returns portal via the shipping service; the OMS → the WMS	PostNL label with the return reference; reverse fulfillment order	"This label brings it to Duiven, and Duiven expects it"	Label without a reference; announcement lost	Getting It There: Carriers, Labels, Tracking and the Delivery Promise, Pick, Pack, Ship: Warehouse Management and Fulfillment
5	Drop-off and transit, 20–22 April	PostNL → the shipping service → the OMS	accepted, in transit, delivered-to-warehouse scans	"It is on its way" and, legally, "evidence of having sent back"	No scan: a return cannot be told from a lost parcel	Getting It There: Carriers, Labels, Tracking and the Delivery Promise
6	Receiving and inspection, Thursday 22 April	the WMS	return reference, GTIN, grade "resellable"	"This is the jacket, and it can be sold again"	Restocked without inspection; wrong item scanned	Pick, Pack, Ship: Warehouse Management and Fulfillment

#	Step	System	Data	Promise made	What could fail here	Chapter
7	Disposition and stock event	the WMS → the availability service	disposition restock; on-hand +1 for M moss	"One more M moss can be sold"	Stock raised for an unsellable unit; event delivered twice	Is It in Stock? Inventory, Availability and the Promise You Can Keep
8	Disposition event to the order	the WMS → the OMS	return R-20270419-1 received, line disposition	"The merchant may now settle"	Event lost: nothing released; duplicated: settled twice	Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation
9a	Held exchange order released and shipped	the OMS, then the WMS and PostNL	#1072455-X1 released, L moss at the frozen €199, fulfillment order	"Size L ships today"	Priced at today's price; released by a scan rather than a disposition	The Order Record: Lifecycle, State and the Order Management System, What Does It Cost? Price Lists, Contract Prices, Currencies and Tax

#	Step	System	Data	Promise made	What could fail here	Chapter
9b	Refund requested (the alternative)	the OMS → the platform's refund API → the PSP	one request keyed on R-20270419-1, €199, original method	"Your €199 is on its way"	Two writers; a retry without the key; a refund that fails days later	Taking the Money: Payment Providers, Authorization, Capture and Refunds
10	Notification	the OMS → e-mail	registered, received, exchange shipped, refund issued	"You know what is happening"	"Refund issued" before the PSP confirmed; silence at receipt	Where Is My Order? Notifications, Service and the Customer Record
11	Credit note and new invoice	the ERP	credit note –€199 against the invoice of #1072455; invoice +€199 for #1072455-X1	"The books show what happened"	Invoice edited instead of credited; refund without credit note	Tax, Invoices and the Books: From Order to Ledger

#	Step	System	Data	Promise made	What could fail here	Chapter
12	Reconciliation	the ERP, the PSP's reports, a daily job	PSP refunds matched to OMS refunds and to credit notes	"Money, stock and books agree"	The match runs monthly, and 212 double refunds wait three weeks	Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation

Rows 7, 9 and 11 are stock, money and books; one event, row 8, releases all three.

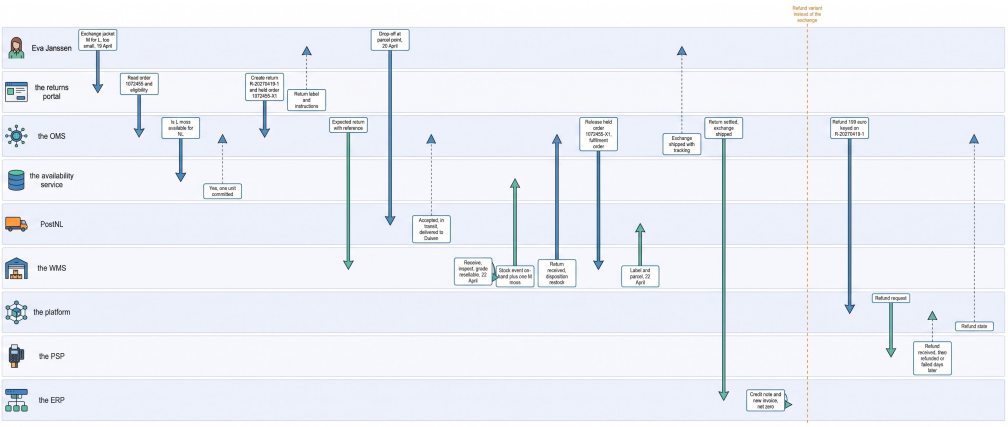


Figure 2: *The way back in the late-2027 architecture: the returns portal creates the return in the OMS, PostNL and the WMS bring the jacket to inspection, and one disposition event releases the stock, the exchange or the refund, and the credit note; the refund, when there is one, has a single writer and a single key.*

The Request: Eligibility, Reason and the Choice

A **returns portal** is the customer-facing page where a return is requested, a reason chosen, a label issued, and an exchange or a refund selected. Kestrel & Pine has rented one since 2025 (€650 a month plus €0.95 per return). In the target architecture the portal owns the conversation with Eva and nothing else: the facts it shows come from the OMS, and the return it creates lives there. The industry calls the merchant's record of

an approved return an **RMA** (return merchandise authorization); this book says "return record".

Eligibility is computed from three facts, each with an owner. The delivery date (15 April) is on the OMS's order, from PostNL's delivered scan in Journey 1. The policy is the merchant's: Kestrel & Pine offers free returns within 14 days of delivery in all three countries, with a prepaid label from the portal (illustrative). The product's attributes come from the PIM: EU law lets a merchant refuse withdrawal for goods made to the consumer's specification, perishable goods and sealed hygiene goods once unsealed, so "returnable" is a product attribute, not a portal setting (What Are We Selling? Products, Variants and the Catalog).

Under the policy sits the law. The EU Consumer Rights Directive (Directive 2011/83/EU, 2011) gives a consumer **14 days** from taking physical possession of the goods to withdraw from a distance contract without giving a reason (Article 9); Eva's window runs to 29 April. She must send the goods back within 14 days of telling the merchant (Article 14). The merchant must refund within 14 days of being told, but may withhold the refund until it has the goods back **or evidence that they were sent**, whichever comes first (Article 13). The consumer bears the cost of return only if the merchant said so before the sale (Article 14). Four dates must therefore be computable per order line: delivery, notice, evidence of sending, refund. Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation owns the law; this chapter needs the clocks. The reason, "too small", is data too: collected per SKU, it feeds size guidance.

The choice: exchange or refund. Eva chooses an exchange, as 28% of Kestrel & Pine's returning customers do. Portals are built to steer that way, because a refund gives back €199 and an exchange keeps it. An exchange needs stock. Kestrel & Pine models an exchange as a **held order**: at request time the OMS asks the availability service for one L moss for the NL channel, creates order **#1072455-X1** for it at a €0 balance, commits the unit and holds the order until the return is graded. The size is Eva's from the moment she asks; the parcel waits for the inspection.

When the held order is released is a policy with three settings. *Release on disposition* (Kestrel & Pine's: the L ships on 22 April, after the M is inspected) is cheapest and slowest. *Release on scan* ships the exchange when the carrier accepts the return parcel, a day or two earlier, at the risk of an empty box. *Instant exchange* ships at once and holds a payment authorization until the return arrives; Shopify's returns model calls the mechanics "release the exchange and collect a balance" (n.d., accessed 2026-11). The rule I use: release on disposition until inspection results are measured per SKU, then move the hero products to release on scan.

At 20:40 on 19 April the portal creates return **R-20270419-1** in the OMS, state *requested*, linked to #1072455, its jacket line and the held order, with the reason, the outcome and the four dates; the policy approves it automatically. The return state machine is the third on the OMS's order record, beside money and goods (The Order Record: Lifecycle, State and the Order Management System).

The Way Back: Label, Drop-off and the Scan That Ends a Right

The portal asks the shipping service, the multi-carrier layer in Kestrel & Pine's WMS flow, for a **return label**: a PostNL label addressed to Duiven, with the return reference printed as a barcode. The reference is the one thing the label must carry; a parcel that arrives without it is a jacket from nobody.

In the same minute the OMS tells the WMS to **expect** the return: a **reverse fulfillment order**, the mirror of Journey 1's fulfillment order, with the lines, the return reference and the tracking number. Shopify's model names the split this chapter teaches: a *return* "represents the intent of a buyer to ship one or more items from an order back", a *reverse fulfillment order* "represents the work that's required to process a return" (n.d., accessed 2026-11). The first is the OMS's, the second the warehouse's. The message may arrive twice; the return reference makes the second copy a no-op (W5).

Eva drops the parcel at a PostNL point in Nijmegen on **Tuesday 20 April** at 17:50 *(illustrative)*. The acceptance scan reaches the OMS within the hour, and the return moves to *in transit*. **That scan is a legal event**. It is "evidence of having sent back the goods" under Article 13(3): from here the merchant may no longer withhold the refund for lack of proof. The refund deadline, 14 days from Eva's notice on 19 April, does not move, so a week in a depot eats the merchant's margin. The OMS computes the deadline from the notice and the end of the withholding right from the scan.

PostNL delivers the parcel to Duiven on **Thursday 22 April**. A parcel on the dock is not yet a return.

At the Dock: Receiving, Inspection and Disposition

Returns receiving is a separate flow in the warehouse, with its own dock, its own bench and one rule: **nothing is restocked that has not been looked at**. A receiver scans the return reference, which finds the reverse fulfillment order, then the jacket's GTIN (Global Trade Item Number, the number behind the barcode on every own-brand SKU since 2021), which confirms the right item came back. Then an inspection: the jacket that was sent, unworn, tags on? Eva tried it on, which Article 14(2) allows without liability.

The result is a **disposition**, the warehouse's decision about the returned unit. Shopify's reverse-fulfillment model names four, and they are the industry's four under any name: *restocked*, *processing required* (clean, repair, re-label before a decision), *not restocked* (discard, donate, return to the supplier) and *missing* (n.d., accessed 2026-11). Kestrel & Pine resold 61% of returned items as new in 2026, 27% after processing at a reduced price, 9% were recycled or donated and 3% were missing or the wrong item (illustrative

split), and **none of the 39% that were not graded new may increase sellable stock at full price**. A WMS that increments on-hand at the dock scan, before the grade, is the mechanism behind "restocked but not resellable": the availability service promises a jacket that is in a repair bin.

At 10:15 on 22 April the inspector at the returns bench, which handles about 280 parcels a day with two people (illustrative), grades Eva's jacket resellable, and the WMS emits two events. A **stock event** goes to the availability service (on-hand +1 for M moss at Duiven), and the next shopper sees one more M within seconds. The **disposition event** goes to the OMS: return R-20270419-1, jacket line, received, restock. That second event is the hinge of the whole journey.

Figure 3: *Nothing is restocked that has not been looked at: the inspector's disposition sorts every returned unit, and only the 61% graded new goes back to sellable stock at full price.*

The Exchange Order

On the disposition event the OMS's return moves to *received* and releases the outcome Eva asked for. **An exchange is a return plus a new order**. The held order **#1072455-X1**, created on Monday, has one line, Ridgewalker women's L moss, at **€199**: the price frozen on the original order, not today's list price, the rule Kestrel & Pine adopted in April 2027 for invoices and exchanges (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax). It references the return, carries Eva's address and the original payment reference, and shows a balance of **€0**. Had Eva exchanged the jacket for a €119

Trailhead fleece, the balance would be –€80 and the refund path below would carry it; a dearer item would have been paid for through the platform's checkout before approval.

From the release the exchange is Journey 1 from step 10: the OMS creates a fulfillment order for the already committed unit; the jacket joins the 13:00 pick wave; the shipping service prints a PostNL label; the parcel leaves with the evening collection on 22 April. Eva gets the shipping confirmation that afternoon and the jacket on **Friday 23 April** *(illustrative).* The original order's goods state stays *delivered*, its return state reads *received, exchanged*, and the exchange order has its own goods state: three state machines, one customer-facing line, "Your exchange has shipped."

Why not edit the line from M to L? Because the M was picked, invoiced and paid for; an edit would leave an invoice for a jacket Eva no longer has.

The Refund Alternative: One Writer, One Key

Suppose instead that Eva asks for her money. The return record carries outcome *refund*; everything up to the disposition event is identical; what the event releases is a refund of €199.

The refund is triggered by the disposition, not by the request and not by the scan. Refund on request is the most generous policy and the most exposed to fraud; refund on scan meets the 14-day clock early; refund on inspection, Kestrel & Pine's policy, keeps money and goods in one causal chain: no inspection, no refund. The deadline from the notice (19 April) runs to 3 May; the inspection on 22 April leaves eleven days of margin, enough for a week in transit and a dock that inspects within a day. A merchant whose dock is three days behind in January must know that number, because the law does not.

One writer. On the disposition event the OMS raises exactly one refund request, through the platform's refund API, because the platform holds the PSP integration and the payment reference from the checkout, and the PSP is the only system that can move money. Support cannot refund, the returns portal cannot, the ERP cannot. In February 2027, after the Double Refund, Kestrel & Pine made the returns portal the single writer through the same platform API and made support's PSP console access read-only; when the OMS took over return records in Q4 2027 the writer moved with the record. What never changed is the count.

One key. The request carries the return id, R-20270419-1, as its **idempotency key**: a token sent with a request so that a retry does nothing twice. Stripe's API saves the first response to a key and returns it to any retry (n.d., accessed 2026-11); Mollie's documentation says the PSP "blocks duplicate refunds made in a short time frame" and asks that after a connection error the merchant "do not retry automatically" but check the payment's refunds through the API first (n.d., accessed 2026-11). The PSP's protection is a backstop; the merchant's key is the design.

Refunds are asynchronous, and they fail late. Adyen's documentation states that "the refund process is asynchronous", that "although rare, a refund can fail after you received" the success webhook, sometimes days later, and that for iDEAL a *refunded-reversed* event means the funds came back because the shopper's account was invalid (n.d., accessed 2026-11). Stripe's says the customer sees a refund "approximately 5-10 business days later" and that the original processing fees are not returned (n.d., accessed 2026-11). For iDEAL there was no capture step in Journey 1 and there is no hold to release now: the refund is a new transfer to Eva's bank account.

So the return's money state is a small machine of its own, *requested* → *accepted by the PSP* → *refunded* or *failed*, driven by the PSP's webhooks and confirmed by fetching the refund's status, as the payment was in Journey 1. "Refund issued" is Eva's message when the PSP accepts the request, with "it can take a few business days to show"; "refund failed" is a case for an agent, not a silent retry. Taking the Money: Payment Providers, Authorization, Capture and Refunds owns the mechanics.

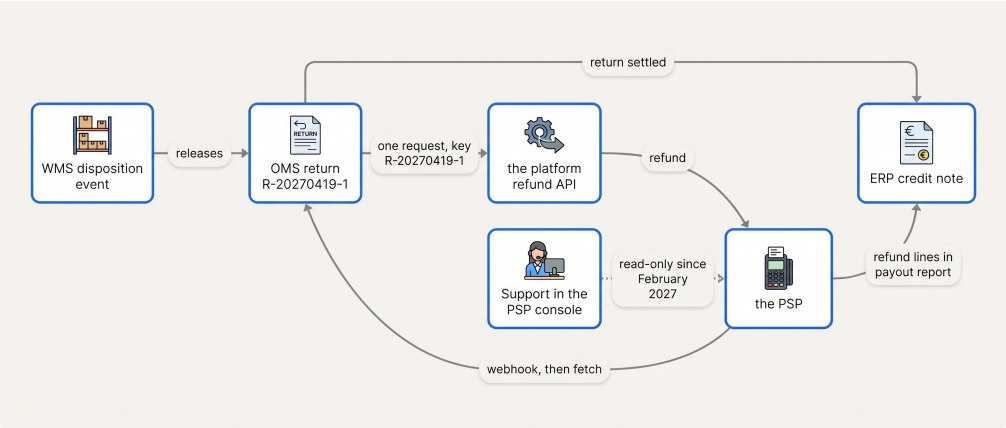


Figure 4: *One writer and one key: the disposition event releases a single refund request keyed on the return id, the PSP's answer comes back by webhook and is confirmed by a fetch, and the ERP matches the credit note to the PSP's payout lines; the dotted path is the second writer the Double Refund closed.*

The Books: Credit Note, New Invoice, Reconciliation

An invoice is never edited. Journey 1 left an invoice for #1072455 in the ERP: net €194.21, VAT €40.79, gross €235.00. As Journey 1 showed, a credit note is the only way to correct an issued invoice. On the "return settled" message the ERP issues a **credit note** against the original invoice for the jacket line: gross **–€199.00**, net –€164.46, VAT –€34.54, referencing the invoice number and the return id. For a Belgian business customer the credit note would travel through Peppol like the invoice (Documents

Between Companies: EDI, Punchout and E-Invoicing); for Eva it is a PDF and a ledger entry (Tax, Invoices and the Books: From Order to Ledger).

An exchange is two documents that net to zero. The exchange order produces a new invoice on dispatch, gross €199.00 with the same VAT split. Credit note minus €199, invoice plus €199, no refund: Eva's bank statement shows nothing, and the ledger shows the full history.

The refund variant pairs the credit note with a refund at the PSP; the credit note reverses the revenue recognized at delivery in Journey 1.

Reconciliation is daily, not monthly. Since February 2027 a daily job matches the PSP's refunds to the merchant's on order number and amount and sends exceptions to the finance team; the monthly payout reconciliation of Journey 1: Eva's Jacket, From Search to Doorstep remains the net under everything. In the target architecture the match has three legs: every refund at the PSP has a return in the OMS with state *refunded*, every *refunded* return has a credit note, every credit note traces to an invoice. A return that fails a leg is an exception by name, the same day. The sibling book RECONCILED works the ledger side (Payments-to-Ledger: Draining the Suspense Swamp, Reconciled by Design: Traceability and Audit Across Boundaries).

Two Variants: The Marketplace Return and the Store Return

The same jacket can come back through two other doors. The first step changes; the rest does not.

The marketplace return. On 13 April an Amazon.de customer bought a Ridgewalker that Kestrel & Pine shipped itself from Duiven, and another bought one the marketplace's fulfillment program shipped from stock sent there in advance. On a marketplace, **the return policy is the marketplace's, not the seller's.**

Amazon's 2025 guidance for seller-fulfilled orders, written for the US, says most items can be returned within 30 days, that sellers should respond within 24 hours where their authorization is needed, and that they should refund "within two days of receiving returns, or Amazon may issue an automatic refund" (15 April 2025; European terms sit behind Seller Central and are not verified here).

For the marketplace-fulfilled jacket the marketplace receives, grades and refunds, and charges the seller: Zalando lists "return shipping" and "return processing" among its fulfillment fees, Amazon a "returns processing" fee for FBA (partner and seller pages, 2026, subject to change).

The consequence for the OMS: **it must accept returns it did not create and refunds it did not issue.** The marketplace connector creates the return in the OMS from the marketplace's authorized-return event, keyed on the marketplace's return id, and marks

the money state *refunded externally* when the marketplace reports its refund; the 24-hour and two-day clocks are service levels on that integration. The seller-fulfilled jacket then follows steps 4 to 8 unchanged. The marketplace-fulfilled jacket never comes to Duiven; its disposition arrives as a report line that adjusts the external location's stock, and the money as a deduction in the marketplace payout (Selling on Someone Else's Platform: Marketplaces and Channel Integration).

The store return. Eva lives in Nijmegen, which has had a Kestrel & Pine store since 2023. In January 2027 a store could only take the jacket and phone head office, because the POS had no link to the platform.

In the target architecture the store's tooling looks the order up in the OMS, the POS records the return against #1072455, and three things differ. The **inspection** happens at the counter, with the same four dispositions.

The **stock** lands in the Nijmegen store location, whose system of record is the POS (W1), so the availability service learns of one more M moss in Nijmegen, not Duiven.

And the **refund** still goes to Eva's original iDEAL payment through the online PSP, raised by the OMS exactly as above, never as cash from the till, because the directive's "same means of payment" rule and the reconciliation both depend on it. From 2028 store receipts are imported into the OMS with their numbers (Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment); matching Eva's store and online records is One Customer, One Product: Identity and Master Data Across Systems's problem.

What Each Handoff Requires

How fresh, how consistent, how fast and how available each join must be; the figures are the author's illustrative targets, not ledger facts.

Handoff	Freshness	Consistency	Latency	Availability	Style (W5)
Returns portal → OMS (create return)	Immediate	One return per request; return id as key	Under 2 s *(assumption)*	Portal fails closed: no label without a return record	Synchronous call, idempotent

Handoff	Freshness	Consistency	Latency	Availability	Style (W5)
OMS → WMS (expected return)	Minutes	Request and acknowledgement; return reference as key	Before the parcel arrives	Queue absorbs a WMS outage	Events with acknowledgement
WMS → availability service (stock event)	Seconds	Single writer per quantity; restock only on disposition	Under 10 s *(assumption)*	Weekly reconciliation as the net	Events
WMS → OMS (disposition)	Seconds	One settlement per return line	Under 30 s *(assumption)*	Daily job lists returns received but not settled	Events, idempotent
OMS → platform → PSP (refund)	Immediate	One writer, one key; state fetched, never assumed	Under 5 s *(assumption)*	Queued if the PSP is down; never retried blindly	Synchronous call plus webhook
OMS → ERP (credit note, invoice)	Same day	One credit note per settled return line	Daily batch is acceptable	Reconciliation count per day	Daily batch

The stock event and the disposition event cannot be batch, because a returned unit and Eva's exchange both wait on them; the credit note can be. **Operations by event, finance by batch**, as in Journey 1.

The Same Return in January 2027

Run R-20270419-1 through the January 2027 landscape, and the dates stay the same while four things change.

Step	Late 2027	January 2027
The return record	One, in the OMS, with its own state	Three: the portal's return, the platform's order note, the WMS's receipt, each with its own status
The exchange	A held order in the OMS, committing a unit in the availability service, released by the disposition	A held order in the platform, committing a unit in the platform's stock copy, which could be two hours stale
The refund	One writer (the OMS), one key (the return id), released by the disposition	Two writers, the portal through the platform's API and support in the PSP console, and no key naming the return
Stock from returns	WMS stock event to the availability service in seconds	WMS restock → ERP every 15 minutes → platform hourly; about 90 of the in-flight units in the May 2027 reconciliation were returns on their way through this chain

The jacket still came back on 22 April and the L still shipped the same day. What the January landscape could not do was tell one agent, in one screen, that the refund had already been sent.

When It Goes Wrong Along This Path

The step table names a failure per step. Four deserve the mechanism, the detection and the recovery.

The Double Refund. In January 2027, **212 returns were refunded twice**, sending **€19,400** out a second time. In the illustrative mechanism, customers saw the carrier's warehouse-delivery scan and contacted support. The portal waited two to five days for

inspection. An agent saw no refund yet and issued one manually; the portal later issued another.

Neither request identified the return, so the provider accepted both. Finance Director Sanne de Wit discovered duplicate ERP credits at month-end. Most money was recovered by asking customers to transfer it back.

From February, one authorized path uses one return key for refunds, and a daily comparison checks platform against provider records. Receipt, inspection and refund remain separate events; support can now see which has actually happened.

The refund that failed days later. The PSP accepted the request, Eva received "refund issued", and three days later a *refund failed* or *refunded-reversed* webhook arrived because the bank account could not take the transfer. Detection: the webhook, if the receiver handles it; a daily job that lists refunds not in a terminal state after the method's typical window, if it does not. Recovery: a service case and a new refund to a confirmed account, never a blind retry. The return's money state must be able to go backwards.

The restock that was not resellable. On-hand went up at the dock scan; the grade "processing required" came an hour later and did not reverse it; the availability service promised a jacket in a repair bin, and a shopper's order was short-picked. Detection: restocked units reconciled against dispositions, and the short-pick rate on returned SKUs. Recovery: the stock event is emitted on the disposition, never on the scan, and the count is corrected from the bin.

The wrong refund amount. During Black Friday 2026 an automatic 20% outerwear discount stacked with a 10% welcome code, and 190 returned jackets were refunded at €159.20 instead of the €143.28 actually paid, about €3,000, because the order-level discount had never been allocated to lines in the order export. Detection: a refund that exceeds the line's paid amount, which the PSP's ceiling on partial refunds catches only at order level. Recovery: the export to the returns portal, the WMS and the ERP now carries the allocated net price per line, a W5 contract field since 2027, and the refund is computed from it (Discounts, Coupons and Loyalty: Promotions Without Chaos).

What to Take Away

Established.

- A return is a stock flow, a money flow and a book-keeping flow with three owners, and the public models agree on the split: Shopify separates the return (intent and money) from the reverse fulfillment order (work and stock) and names four dispositions; commercetools keeps goods-side and money-side return states apart; every ERP corrects an invoice only by credit note.
- Refunds are asynchronous, can fail days after acceptance and reach the customer days to weeks later; Adyen, Stripe and Mollie document this from three sides and

ask the merchant to use idempotency keys, fetch status rather than assume it, and never retry blindly.
- EU consumer law fixes the clocks: 14 days to withdraw from possession, 14 days to refund from notice, with the right to withhold until the goods or proof of sending arrive.

Depends on context.

- Whether the refund waits for inspection. Refund on inspection is right when the dock inspects within a day or two and the return rate makes fraud and unsellable returns matter; refund on scan is right for slow transit lanes and a tight 14-day margin; refund on request is a marketing choice that needs a fraud model behind it.
- Whether the OMS owns the return record. With returns arriving from a portal, three marketplaces and six stores, one return record is necessary; a Starting or Growing merchant with one channel gets the same journey from the platform's own return model, and the portal may be the record.
- Whether stores take online returns at all. The flow needs the POS to look up the order and the refund to go to the original payment method; a merchant whose POS can do neither should route the refund through the online path by hand, and know that it is doing so.

The author's synthesis.

- I count the systems that can call the PSP's refund API, and if the answer is not one, I stop the review there; nothing else about the return path matters until it is.
- The disposition event is the most important message in the return path and the least often designed; I would make it the only trigger for stock, settlement and the credit note, and measure the time from dock scan to disposition every day, because that number is both the customer's wait and the merchant's legal margin.

Questions to Ask

1. Which systems can issue a refund, and what is the idempotency key on the request?
2. What triggers a refund: the request, the carrier's acceptance scan or the warehouse's disposition, and who chose that?
3. Does a returned unit increase sellable stock at the dock scan or at the disposition?
4. How are the four legal dates (delivery, notice, evidence of sending, refund) computed per order line, and what is the median time from drop-off scan to refund?
5. Is an exchange a new order at the original frozen price, with its own invoice, or an edit?

6. Can the OMS accept a return created by a marketplace, a store or the warehouse, and a refund issued by someone else?
7. What matched last month's PSP refunds to credit notes, and how many exceptions did it find?

Eva has her size L, the M is back on the shelf, and the books show a credit note and an invoice that net to zero. The next journey leaves the consumer behind: on 6 April, a week before Eva's order, a retail chain's computer sent Kestrel & Pine an order for 240 units, and nobody typed anything. Journey 3: A Chain Reorders by EDI, From Order to Invoice follows it from the EDI message to the invoice.

Sources and Evidence

Company, product and investor sources

- Returns apps and ReverseFulfillmentOrderDispositionType — Shopify developer documentation, n.d., accessed 2026-11; return versus reverse fulfillment order, return statuses, the four dispositions, exchanges with a balance.
- Orders — commercetools API documentation, n.d., accessed 2026-11; goods-side and money-side return item states.
- Refund — Adyen documentation, n.d., accessed 2026-11; asynchronous refunds, late failures, refunded-reversed for iDEAL, up to 40 business days.
- Refunds and Idempotent requests — Stripe documentation, n.d., accessed 2026-11; fees not returned, 5–10 business days, double-refund risk on bank debits, idempotency keys.
- Refunds — Mollie documentation, n.d., accessed 2026-11; duplicate blocking and "do not retry automatically".
- Credit notes — Odoo 18.0 documentation, accessed 2026-11; invoice immutability and the credit note as the only correction.
- How to manage Amazon customer returns for seller-fulfilled orders — Amazon, 2025-04-15; the 30-day window, 24-hour response, two-day refund and automatic refunds (US guidance).
- Fulfillment by Amazon — Amazon, n.d., accessed 2026-11; returns processing as an FBA fee category.

Standards, regulation and official documentation

- Directive 2011/83/EU on consumer rights — Official Journal, 2011-11-22; Articles 9, 13, 14 and 16: the 14-day clocks, withholding until goods or proof, return costs, diminished value, exemptions.
- IFRS 15 Revenue from Contracts with Customers — IFRS Foundation, issued 2014, effective 2018; revenue reversal and the refund liability.

Research and practitioner sources

- 2025 Retail Returns Landscape — NRF with Happy Returns, 2025-10-15; 19.3% online return rate, 9% of returns fraudulent.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

39. Journey 3: A Chain Reorders by EDI, From Order to Invoice

IN THIS CHAPTER, YOU WILL: Follow Bergzicht's 240-unit spring reorder from the structured electronic order message to the paid invoice, and see why the despatch advice, not the order, is where wholesale money is lost.

KEY POINTS:

- **A wholesale order is a conversation of documents with a contract behind it.** An order message comes in, a despatch advice goes out with every pallet, an invoice follows, and a bank transfer arrives thirty days later. Each document is a promise written into the buyer's purchasing conditions, usually with a penalty attached.
- **Each document has its own system of truth.** The order record belongs to order management, the despatch advice to the warehouse system (what was actually packed on which pallet), the invoice to the finance system. Kestrel & Pine lost €14,200 in one quarter because its despatch advices were generated by the wrong system.

- **The money arrives as a bank transfer that must be matched, not as a payment-provider event**. Payment terms turn a sale into a receivable; credit limits, cash application and reminders are finance-system work. E-invoicing mandates such as Belgium's change the invoice's format and transport, not its owner.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Pulled Rug** — the buyer's message guideline is a contract that changes on the buyer's calendar, and a supplier that does not track it finds out from a deduction.
- **The Lynchpin** — Kestrel & Pine's electronic document mappings were understood by one external consultant; the chapter shows what that costs and what replaces it.
- **The Message in a Bottle** — an order message or a despatch advice sent once and trusted is the mechanism behind duplicate orders and refused deliveries.

A retail chain orders pallets of clothing and expects its receiving system to recognize what arrives. The supplier must send the shipment notice in the agreed format and label the pallets so they match it. A correct physical delivery can still be held up or penalized when those records disagree. **Electronic data interchange (EDI)** carries the structured orders, shipment notices and invoices between the companies' systems; its accuracy affects receiving and payment.

Figure 1: *A wholesale order is a conversation of documents with a contract behind it: the order comes in, a despatch advice goes out with the pallets, an invoice follows and the money arrives weeks later.*

Kestrel & Pine, the fictional outdoor brand, earns a quarter of its revenue from **1,100 wholesale accounts**. Its largest customer orders through EDI. Volumes are smaller than the web shop's, but average order values are about fourteen times larger. A mislabeled pallet or unreadable notice can therefore turn a technical error into a substantial deduction.

The walkthrough follows one wholesale order from receipt to payment. The core jobs remain order acceptance, stock commitment, dispatch and accounting; the documents, approvals and timing differ from a consumer purchase. Journey 1: Eva's Jacket, From Search to Doorstep provides the comparison.

Where we are in the order's journey. Bergzicht's order commits **240 units at Duiven on 6 April 2027**. Those units reduce what the online shop may offer other buyers, including Eva, who orders a jacket on 13 April. Wholesale and consumer channels compete for the same stock.

The Order, the Setting and the Rules of This Walk

The buyer. Bergzicht Outdoor is a fictional Dutch chain with **38 shops** and a distribution center in **Zwolle**. It buys **€2.1 million annually** from Kestrel & Pine and has used EDI since 2025. Buyer Pieter Smit works with Kestrel & Pine's Wholesale Lead Lea Moreau.

The order. On **6 April 2027**, an **ORDERS** message requests **240 units across 18 stock keeping units (SKUs)**, each identifying a sellable variant. Ridgewalker jackets and Trailhead fleeces total **€19,860 excluding value-added tax (VAT)** at contract prices.

Three pallets receive **Serial Shipping Container Code (SSCC)** labels. Packing confirmation generates the **DESADV** despatch advice describing their contents. They reach Zwolle on **9 April**, the invoice (**INVOIC**) follows that day, and payment is due **9 May**. Intraday times and credit figures are illustrative.

The setting. As in Journey 1: Eva's Jacket, From Search to Doorstep, the order is run through Kestrel & Pine's **target architecture of late 2027**: the availability service live (W1), the OMS as order hub (W2; EDI and manual wholesale orders route through it from 8 December 2027), the PIM live (W3) and, specific to this chapter, **W6 in force**: SSCC labels generated by the WMS at pack time, the DESADV sent from the WMS's confirmation, and one EDI partner card per trading partner. In April 2027 the order ran through the January landscape; a section near the end shows the difference. The ASN Chargeback of Q3 2026 is walked as it happened, on that older landscape.

The documents and the identifiers

EDI has a vocabulary of its own, and newcomers meet it as a wall of capital letters. The table gives the six messages this journey uses, in the UN/EDIFACT standard most European retail uses (as a GS1 subset called EANCOM) and in the ANSI X12 standard of North American retail, with the system each is generated from. GS1's implementation survey (2021) found ORDERS, INVOIC and DESADV the three most implemented EANCOM messages (Documents Between Companies: EDI, Punchout and E-Invoicing owns this in depth).

Message	X12 twin	In plain words	Sent by	Generated from
ORDERS	850	The purchase order: what, how many, at which agreed price, delivered where and when	Bergzicht	Bergzicht's purchasing system
CONTRL	997	A syntax receipt: "your message parsed", nothing about its meaning	Either side	The EDI provider's translator
ORDRSP	855	The order response: accepted, accepted with changes, or rejected	Kestrel & Pine	The OMS

Message	X12 twin	In plain words	Sent by	Generated from
DESADV	856	The despatch advice, also called the ASN (advance shipping notice): what is on its way, pallet by pallet, so the receiver can scan instead of count	Kestrel & Pine	**The WMS's pack confirmation**
INVOIC	810	The invoice: a legal payment claim for what was delivered, at the agreed prices, with VAT	Kestrel & Pine	**The ERP**
RECADV	861	The receiving advice: what the buyer counted in; optional, and Bergzicht does not send one	Bergzicht	Bergzicht's warehouse system

Three GS1 identifiers tie the documents together, all defined in the GS1 General Specifications (Release 26.0, January 2026): the **GTIN** (Global Trade Item Number) says *what* a unit is, the number behind the barcode on every own-brand SKU; the **GLN** (Global Location Number) says *who and where*, the key for Bergzicht's Zwolle DC and Kestrel & Pine's Duiven warehouse in every address segment; the **SSCC** (Serial Shipping Container Code) says *this pallet*, an 18-digit serial number the specification calls "the only GS1 key that SHALL be used as the identifier of a logistic unit", one that "is used with EDI communications to enable identification and traceability".

A despatch advice is the document in which each SSCC lists the GTINs and quantities it carries. **A warehouse system that cannot print an SSCC per pallet and report that pallet's contents cannot produce a compliant despatch advice**, and that sentence is the whole of the ASN Chargeback.

Fifteen Steps in One Table

#	Step	System	Data	Promise made	What could fail here	Chapter
1	ORDERS received, Tuesday 6 April 09:12	the EDI provider (a value-added network: a provider-run mailbox between trading partners)	PO number, GLNs, 18 lines of GTIN, quantity, price, delivery date	"Bergzicht has ordered this"	Delivered twice; sent to a retired mailbox	Documents Between Companies: EDI, Punchout and E-Invoicing
2	CONTRL returned	the EDI provider	syntax acknowledgement	"We received a well-formed message"	Negative acknowledgement nobody reads	Documents Between Companies: EDI, Punchout and E-Invoicing

#	Step	System	Data	Promise made	What could fail here	Chapter
3	Mapping to the order record	the iPaaS, per the Bergzicht partner card	PO fields → order fields; GTIN → SKU; GLN → address	"The message means this order"	Guideline changed (The Pulled Rug); one person can fix it (The Lynchpin)	Documents Between Companies: EDI, Punchout and E-Invoicing, Promises Between Systems: API Contracts, Webhooks and Versioning
4	Validation: items, contract prices, credit	the OMS, reading the ERP's price list and credit limit	18 SKUs resolved; prices equal the contract; receivables within limit	"We accept this at these prices"	Price differs from the contract; limit exceeded	Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals, What Does It Cost? Price Lists, Contract Prices, Currencies and Tax

#	Step	System	Data	Promise made	What could fail here	Chapter
5	Order created, channel EDI	the OMS	order in the shared number family, PO reference, NET 30	"The order exists, once"	Created twice from a resent ORDERS; created in the ERP or the EDI portal instead	The Order Record: Lifecycle, State and the Order Management System
6	ORDRSP sent	the OMS → the EDI provider	accepted lines, confirmed dates	"You will get this, then"	Response promises what stock cannot keep	Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals
7	Allocation of 240 units	the availability service	committed quantities per SKU at Duiven	"These units are Bergzicht's"	Commit invisible to the storefront (The Phantom Stock)	Is It in Stock? Inventory, Availability and the Promise You Can Keep
8	Fulfillment order, pallet service level	the OMS → the WMS via the iPaaS	lines, ship-to GLN, packing rules, delivery date	"Duiven ships this by Thursday"	Request lost; parcel rules applied to a pallet order	Pick, Pack, Ship: Warehouse Management and Fulfillment

#	Step	System	Data	Promise made	What could fail here	Chapter
9	Pick to pallet, Wednesday and Thursday	the WMS	cartons picked and built onto three pallets	"These units are on these pallets"	Short pick; mixed pallets not recorded	Pick, Pack, Ship: Warehouse Management and Fulfillment
10	SSCC labels printed at pack	the WMS	one SSCC per pallet, contents recorded	"This barcode is this pallet"	Pallet numbers taken from the order system, not the pack (the ASN Chargeback)	Documents Between Companies: EDI, Punchout and E-Invoicing
11	DESADV at pack confirmation, Thursday 8 April	the WMS → the iPaaS → the EDI provider	pallet SSCCs, cartons, GTINs and quantities, PO reference	"This is exactly what arrives"	Sent after the truck; rejected on syntax; edited after sending	Documents Between Companies: EDI, Punchout and E-Invoicing
12	Transport and delivery, Friday 9 April	the shipping service, a pallet carrier	booking, delivery slot at Zwolle	"It arrives Friday"	Slot missed; delivery refused without a DESADV	Getting It There: Carriers, Labels, Tracking and the Delivery Promise

#	Step	System	Data	Promise made	What could fail here	Chapter
13	Receipt at Zwolle	Bergzicht's warehouse system	SSCC scans matched to the DESADV	"What we scanned is what you said"	Scan does not match: manual count, deduction	Documents Between Companies: EDI, Punchout and E-Invoicing
14	INVOIC, Friday 9 April	the ERP → the EDI provider	invoice number, lines as shipped, €19,860 net, €4,170.60 VAT, due 9 May	"You owe this, by then"	Invoiced from the order, not the shipment	Tax, Invoices and the Books: From Order to Ledger
15	Payment and cash application, by 9 May	Bergzicht's bank → the ERP	a SEPA transfer with remittance information	"The receivable is settled"	Short payment; remittance without invoice numbers	Tax, Invoices and the Books: From Order to Ledger, The Back Office: ERP Integration and Who Is the Hub

Rows 10 and 11 are the ones the chapter is about: the two steps where a document is written about physical goods by a system that may or may not have touched them.

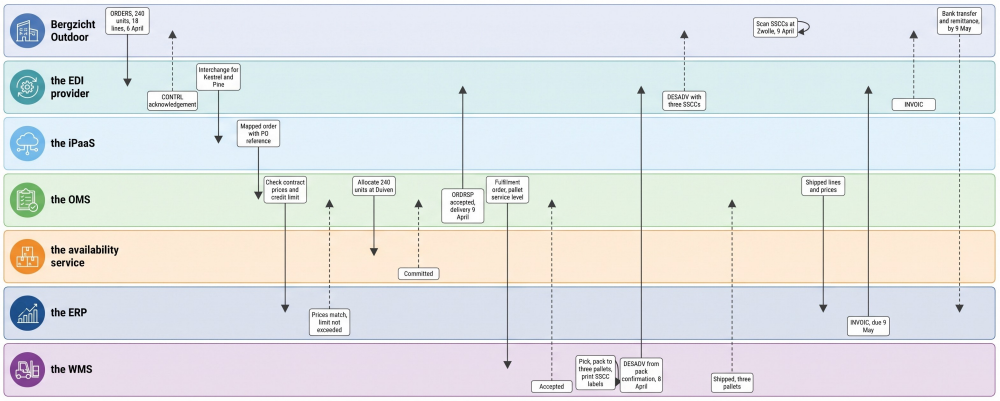


Figure 2: *Bergzicht's reorder in the late-2027 architecture: the order message becomes an OMS order, the availability service commits the stock, the WMS packs and writes the despatch advice from what it packed, the ERP invoices what shipped, and the money comes back thirty days later as a bank transfer to be matched.*

ORDERS In: From Message to Order Record

At 09:12 on Tuesday an interchange lands in Kestrel & Pine's mailbox at the EDI provider. A **VAN** (value-added network) is a provider-run mailbox: each trading partner connects once and the provider relays to everyone. The alternative is a direct point-to-point link, most often **AS2**, an internet protocol for signed and encrypted business documents defined in 2005 (RFC 4130); a direct link means a certificate and an endpoint per partner, which is why mid-sized suppliers buy the network. The provider's translator checks the syntax and returns a **CONTRL** acknowledgement within minutes. It says "parsed", not "agreed"; a supplier that treats CONTRL as acceptance has made its first mistake.

Mapping is the translation between Bergzicht's message implementation guideline, the document that says which of EDIFACT's many segments Bergzicht uses and how, and Kestrel & Pine's order record. Every retailer has its own guideline, so the cost of EDI is per partner, not per standard, which is the business model of every EDI provider: SPS Commerce's annual report for 2025 (filed 19 February 2026) sells a network that lets suppliers "connect once and immediately transact with thousands of trading partners". In the target architecture the mapping runs in the iPaaS and is described on the **EDI partner card** that W6 requires: GLNs, guideline version, messages in each direction, test status, contacts on both sides, and the owner at Kestrel & Pine, Ruben Alves. In 2026 all of that lived in one external consultant's head.

Validation turns a message into an acceptable order. Three checks run in the OMS against data the ERP owns:

- **Items.** Each of the 18 GTINs resolves to a SKU in the item master; a retired or

mistyped GTIN rejects the line in the order response rather than creating a line for nothing.

- **Prices**. An ORDERS message carries the price the buyer expects. Bergzicht's contract prices differ from the list wholesale prices (€95 for the Ridgewalker, €57 for the Trailhead) and live in the ERP's price list for this customer, because the ERP owns B2B terms: Microsoft's documentation for its Business Central-to-Shopify connector (October 2026) states that "Most businesses manage customer records, payment terms, and credit limits in Business Central and export them" and warns that mixing directions "can lead to conflicting records". A price mismatch is answered in the ORDRSP, never silently accepted, because an unexpected invoice price is a short payment four weeks later (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax).
- **Credit**. The ERP holds Bergzicht's credit limit and open receivables. Assume a limit of €450,000 and €188,000 open on 6 April *(assumption)*; the order passes. A credit hold is a visible state of the order, not a silent queue.

The order is created in the OMS, with channel EDI, Bergzicht's PO number as a reference, NET 30 terms, and a number from the same family as every web and marketplace order (W2). It is not created in the ERP, and not in the EDI provider's web portal, where EDI orders commonly end up and "no single order record" quietly begins. The ERP receives the order for posting; the OMS owns it. Shopify's Admin API (version 2026-10) shows how a platform types such terms, defining `NET` as "due a set number of days after the invoice is issued".

Idempotency protects this step: an idempotent operation has the same effect whether performed once or twice. The VAN may redeliver an interchange, and Bergzicht's system may resend an ORDERS after a timeout; the OMS keys creation on Bergzicht's GLN plus its PO number, so the second arrival finds the order and does nothing. Stripe's API documents the same discipline for payments (n.d., accessed 2026-11). Without the key, Kestrel & Pine ships 480 units and invoices twice: the Message in a Bottle at pallet scale.

The **ORDRSP** goes back on Tuesday afternoon: all 18 lines accepted, delivery 9 April. It is the first promise Kestrel & Pine makes in this journey, and the availability service is what lets the OMS make it honestly.

Allocation: Whose Stock Is It?

A wholesale order takes stock differently from a web order. Eva's jacket is one unit committed at order creation and picked the next morning. Bergzicht's 240 units are **allocated** three days before they ship, while the storefronts, the marketplaces and the B2B portal keep selling from the same Duiven shelves.

In the target architecture there is one answer: the **availability service** computes available-to-promise per channel (what a channel may still promise: on-hand stock,

minus units committed to orders, minus a buffer) from the WMS's stock events (W1). The OMS commits Bergzicht's 240 units on Tuesday morning and every channel's "in stock" drops within seconds. Assume the order includes 12 women's moss Ridgewalkers in size M, on-hand at Duiven was 36 and the storefront buffer is one unit *(assumption)*: the NL storefront's available number for that SKU falls to 23 on 6 April, the number Eva sees on 13 April in Journey 1: Eva's Jacket, From Search to Doorstep. In January 2027 the wholesale order sat in the ERP and the platform learned of it an hour later at best; one such hour during a season launch is an oversell with a chain's pallet behind it.

Two B2B rules follow. **Large accounts get allocations, not promises**: some merchants reserve a share of a season's stock for key accounts, so a web surge cannot consume a chain's reorder (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals). And **shortfalls are negotiated, not canceled**: a missing line is answered in the ORDRSP with a changed quantity or a later date, because a chain plans its shelves on the response.

Figure 3: *One shelf, two buyers: Bergzicht's 12 moss jackets and Eva's storefront draw on the same 36 at Duiven, so the storefront may promise 23 once the allocation and the buffer are taken.*

Pick to Pallet, Label and Despatch Advice: The Document That Costs Money

On Tuesday afternoon the OMS sends the WMS a **fulfillment order**, the instruction to ship these lines to this address by this date, with the wholesale service level: cartons

of fixed quantities, built onto pallets, delivered to a DC with a booked slot. Warehouse systems run parcel and pallet work side by side (Manhattan describes its warehouse product as handling "wholesale, retail, and direct orders at the same time", n.d., accessed 2026-11); at Duiven the wholesale wave runs in the afternoon after the morning's parcel waves *(a chapter-local detail)*.

Packing is the moment of truth. Over Wednesday and Thursday the pickers build three pallets. As each pallet is closed, the WMS prints its **SSCC label** and records which cartons, and therefore which GTINs in which quantities, are on it. This is W6's first rule: the SSCC is generated by the WMS at pack time, from the pallet as built, not from the order, which says only what should ship. If a carton of fleeces ended up on pallet two, the label and the record say so.

The DESADV is written from that record. At pack confirmation on Thursday 8 April *(assumption; the ledger says "at pack confirmation")*, the WMS produces the despatch advice: the shipment, then each pallet by SSCC, then the cartons and the GTIN quantities inside, with ship-from and ship-to GLNs and Bergzicht's PO number. The iPaaS hands it to the EDI provider, which sends it to Bergzicht before the pallets leave Duiven. This is W6's second rule: the DESADV is sent from the WMS's confirmation, not from the ERP.

Why the buyer cares is best read in a buyer's own words. Jumbo, a Dutch supermarket chain, publishes its receiving rules: "Every delivery to our mechanized DC must be preceded by a DESADV message", "A delivery without an accompanying DESADV message cannot be processed and is therefore refused!", and "The SSCC label forms the link with the DESADV despatch message" (guidelines version October 2025, translated from Dutch).

Its purchasing conditions of 9 April 2026 add the price: a supplier that fails to use EDI with GS1-compliant content "forfeits €70 … per message", Jumbo may refuse a delivery without DESADV and SSCC, and €250 applies per order line delivered less than 90% on time and in full.

bol has required SSCC labels on every logistic unit since 31 December 2021 (n.d., accessed 2026-11); Walmart's on-time-in-full program carried a 3% non-compliance fee at its 2017 introduction (trade press, dated). Bergzicht's conditions are fictional; their shape is this.

The pallets travel on Thursday night with a pallet carrier booked through the shipping service and reach Zwolle on **Friday 9 April**. Receiving scans each SSCC; the scan finds the pallet in the DESADV and books its contents in without counting. Bergzicht does not send a RECADV, so Kestrel & Pine's only evidence of what was received is what was not deducted a month later; the partner card records that a receiving advice has not been agreed.

Invoice, Terms and the Money Thirty Days Later

The invoice is the ERP's, and it is written from what shipped. On Friday the OMS passes the shipped lines and their contract prices to the ERP, and the ERP issues the **INVOIC**: a gap-free sequential number, the dates of issue and supply, both parties with VAT numbers, 18 lines at contract prices, **€19,860** net, **€4,170.60** VAT at the Dutch 21% rate, **€24,030.60** gross, due **9 May**. The Dutch tax authority's invoice requirements list exactly these elements (Belastingdienst, n.d., accessed 2026-11), and an issued invoice is immutable: a credit note "is the only legal method for canceling, refunding, or modifying a validated invoice", as Odoo's documentation puts it for any ERP (18.0, accessed 2026-11). Had pallet two shipped a carton short, the invoice would show the shipped quantity, because an invoice that exceeds a receipt is a short payment by design (Tax, Invoices and the Books: From Order to Ledger).

On Bergzicht's side the invoice meets the four-way match: purchase order, despatch advice, receipt scan, invoice. A line that agrees on all four is paid; one that does not is paid short, and the difference arrives as a deduction with a code, not as a conversation.

Revenue and the receivable. The ERP posts accounts receivable of €24,030.60 and revenue of €19,860; under IFRS 15 (effective from 2018) revenue is recognized when control passes, which for a delivered pallet is 9 April. For thirty days the sale is a receivable, an amount owed, and the risk Kestrel & Pine carries is Bergzicht's credit, not a card network's.

The money arrives as a bank transfer. On or before 9 May Bergzicht pays by SEPA credit transfer, with remittance information listing the invoice numbers it settles, often several at once. Since October 2025 euro-area banks must be able to send instant credit transfers (Regulation (EU) 2024/886), so the money can land in seconds; the work is **cash application**, matching the transfer to open invoices in the ERP, and a remittance that names no invoice numbers turns a ten-second transfer into an afternoon's e-mail. A late payment enters a reminder cycle, commonly called dunning; a typical pattern is a reminder at the due date, a second after a week or two, then a hold on new orders and a conversation between Lea Moreau and Pieter Smit. The order hold is a state the OMS must be able to show.

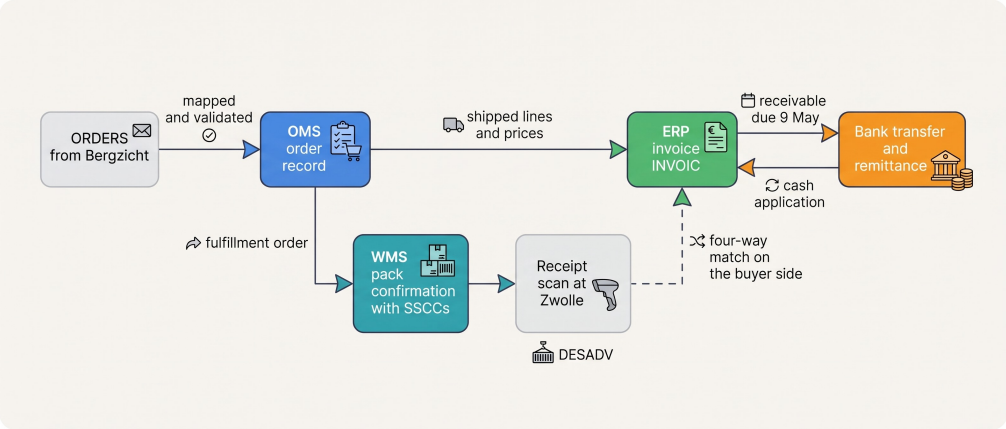


Figure 4: *Each wholesale document has its own system of truth: the order belongs to the OMS, the despatch advice to the WMS's pack confirmation, the invoice to the ERP, and the money to the bank; a document written by any other system is the one the buyer deducts for.*

What Each Handoff Requires

The four questions the book asks of every join (how fresh, how consistent, how fast, how available) give wholesale a different profile from the consumer journey: slower almost everywhere, stricter on consistency, with one hard deadline, the truck. Latencies are the author's illustrative targets.

Handoff	Freshness	Consistency	Latency	Availability	Style (W5)
EDI provider → iPaaS → OMS (ORDERS)	Minutes	One order per GLN plus PO number	Under 15 min *(assumption)*	Mailbox queues; replay after an outage	Events, idempotent
OMS → ERP (prices, credit)	At validation	Price list and credit limit owned by the ERP	Seconds	Hold the order on ERP outage, never guess	Synchronous call

Handoff	Freshness	Consistency	Latency	Availability	Style (W5)
OMS → availability service (allocation)	Seconds	One writer per committed quantity	Under 10 s *(assumption)*	Storefront keeps last value, with buffer	Events
OMS → WMS (fulfillment order)	Minutes	Request, accept, reject, cancel-request	Before the wholesale wave	Queue absorbs an outage; the slot is the deadline	Events with acknowledgement
WMS → EDI provider → Bergzicht (DESADV)	At pack confirmation	Matches the pallets exactly; corrections reference the original	Before the truck leaves	Hold the truck rather than ship without it	Event, acknowledged by CONTRL
OMS → ERP → Bergzicht (INVOIC)	Day of delivery	One invoice per shipment, from shipped lines	Daily batch	Reconciliation count per day	Daily batch
Bank → ERP (payment)	Days	Matched to invoices by remittance	Days	Monthly close	File or bank feed

The DESADV row is the one with no tolerance: freshness, consistency and the deadline all bind at once, and the only safe failure mode is to hold the truck.

The Failure Path: The ASN Chargeback

This is what happened in the third quarter of 2026, on the January landscape, and why it cost **€14,200**.

Why the shipment notice failed. The old ERP flow generated DESADV as soon as it released the order to the warehouse. It assigned pallet numbers and expected contents before packing. The WMS then packed according to the actual goods and printed its own labels.

Bergzicht could receive a known pallet number with different contents, an unknown number, or a notice arriving after the truck. Each required manual counting and triggered a contractual fee. Nine Q3 deliveries produced the October 2026 deduction. Documents Between Companies: EDI, Punchout and E-Invoicing explains the corrected flow: generate the labels and notice from the same packing confirmation.

Why nobody saw it for a quarter. The deductions arrived as unexplained short payments on invoices that were themselves correct. Sanne de Wit's team saw receivables that did not close; Lea Moreau heard of "label problems" from Pieter Smit; Daniel Okafor's warehouse had shipped every unit. The three views were never joined, and only the consultant could read the mapping. **The Lynchpin** was the consultant who built the mappings in 2025; **The Pulled Rug** was the contract itself: supplier conditions that demand pallet-level DESADV content (the level Jumbo's public rules also demand), which nobody at Kestrel & Pine owned or checked against what the systems sent.

Detection, in the target architecture. Three signals replace the silence: a per-partner **compliance scorecard** on the partner card (DESADVs sent before departure, CONTRL rejections, deductions per delivery), an alarm on any negative acknowledgement, and a rule in cash application that a short payment opens a case naming the invoice, the DESADV and the deduction code.

Recovery. A deduction can be disputed only with evidence that the DESADV described the pallets as shipped, which is what W6 produces: the SSCC from the WMS at pack, the DESADV from the pack confirmation, the partner card with the guideline version. The €14,200 was not disputed because the evidence did not exist.

What can still go wrong after W6. A DESADV sent after the truck leaves because the manifest closed first (fix: the DESADV precedes the manifest, and the truck waits). A negative CONTRL that stays in a mailbox (fix: the alarm). A quantity changed after the DESADV was sent (fix: a corrected DESADV that references the original, never a silent edit). An SSCC serial reused because two systems could print labels (fix: one serial source, the WMS). And the next guideline revision, which the partner card's change contact now receives.

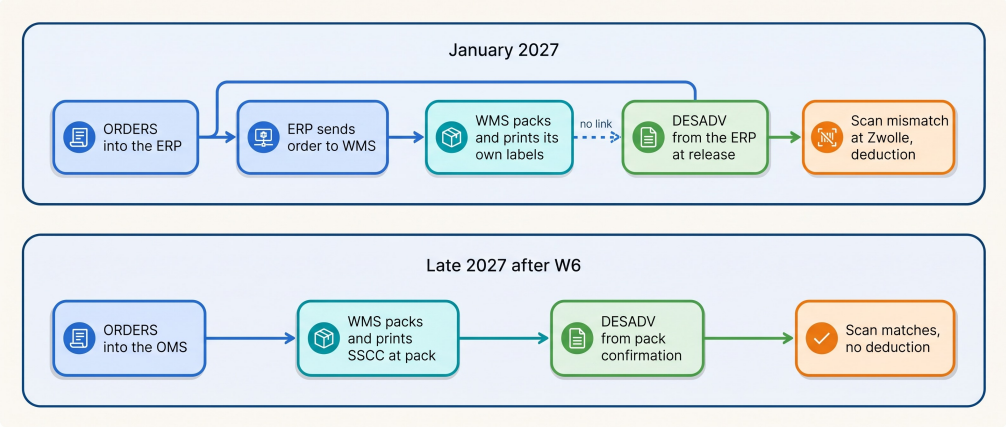


Figure 5: *The ASN Chargeback was an architecture failure, not a warehouse one: in January 2027 the despatch advice described the order, while the pallets described the pack; after W6 one system writes both.*

The Variant: A Belgian Shop's Peppol Invoice

Not every wholesale customer is a chain with an EDI mailbox. Two weeks later, on Wednesday 21 April, Ardennes Trail Co. (fictional), an outdoor shop in Namur, orders 40 Ridgewalker jackets at €95 through the B2B portal, €3,800 excluding VAT *(illustrative; the same order Tax, Invoices and the Books: From Order to Ledger books).* The order enters the OMS with channel "B2B portal" and ships from Duiven by bpost on 23 April, invoiced that day and due on 23 May; up to the invoice it is Eva's journey with a company account and payment terms.

Belgium mandates structured e-invoices. From **1 January 2026**, Belgian businesses liable to VAT must issue invoices to each other as structured electronic invoices over the **Peppol** network; "Sending a PDF invoice by e-mail or via a platform will no longer be enough" (einvoice.belgium.be, 18 September 2024). A business that lacks the technical means to issue or receive them faces fines of €1,500, €3,000 and €5,000 for successive offenses; the tax administration announced tolerance for the first three months of 2026 but "under no circumstances" a general postponement (einvoice.belgium.be, December 2025). Whether the obligation reaches a Dutch supplier without a Belgian establishment is not settled by the sources this book has read. Kestrel & Pine sends Peppol e-invoices to all its Belgian business customers since January 2026 either way, because their bookkeeping now expects invoices in their accounts-payable system, not in an inbox.

Peppol in plain words. Peppol is an open network and a document standard. Each company connects through an **access point**, a service provider, and reaches any other participant through theirs: the four-corner model (supplier, supplier's access point, customer's access point, customer), as OpenPeppol describes it (n.d., accessed 2026-11). The document is a **Peppol BIS Billing 3.0** invoice: XML in the UBL syntax implementing

the European semantic standard **EN 16931**, which lists what any invoice must contain. Release 3.0.21 was published on 20 May 2026 and became mandatory on 17 August 2026; it uses type code 380 for an invoice and 381 for a credit note and requires a VAT category code on every line.

What the ERP produces for Ardennes Trail is the same invoice as Bergzicht's in substance and different in two ways. First, the VAT: the goods leave the Netherlands for a business customer in Belgium, an intra-EU supply at 0% with the reverse-charge mention, valid only if the customer's VAT number was validated in VIES, the EU's registry; the buyer accounts for Belgian VAT itself (europa.eu, n.d., accessed 2026-11). Second, the transport: the ERP hands the document to Kestrel & Pine's access point, which delivers it to the customer's access point and so to the customer's software. **Transmission becomes a state of the invoice**: sent, delivered, or rejected by the network's validation rules. A rejected e-invoice is not an invoice; the ERP must hold that state and retry or correct, a new failure mode for a finance system that used to attach a PDF.

	EDI INVOIC to Bergzicht	**Peppol invoice to Ardennes Trail**
Who defines the content	Bergzicht's guideline, on top of the EANCOM standard	EN 16931, with a published validator
Purpose	Settlement matched to order and receipt	A tax-compliant invoice any receiver's software can read
Addressing	GLNs agreed per partner	A Peppol participant identifier in a public directory
Transport	The EDI provider's network	Any accredited access point, four corners
Validation	CONTRL and the partner's rejection	Schematron rules at the access point
Legal position, November 2026	Accepted in the Netherlands, where e-invoicing is planned for 1 July 2030 (cabinet announcement, 11 September 2026)	The format Belgium requires between Belgian businesses

Germany, Kestrel & Pine's third market, sits between the two: every German business must be able to receive e-invoices since 1 January 2025, and the Finance Ministry's FAQ tolerates existing EDI procedures through 2027 (accessed 2026-11). EU-wide, the ViDA package adopted on 11 March 2025 brings digital reporting for cross-border B2B from 1

July 2030. Documents Between Companies: EDI, Punchout and E-Invoicing and Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation keep the calendar; the point here is that the invoice's owner did not change, only its format and its delivery receipt.

The Same Order in January 2027

Step	Late 2027	January 2027
Where the order lives	The OMS, channel EDI, one number family	The ERP's wholesale order entry, beside the platform's web orders and the connector's marketplace orders
Who can read the mapping	Ruben Alves, from the partner card	The external consultant (The Lynchpin)
Stock	Committed in the availability service within seconds	Visible to the platform an hour later at best, through the sync chain
SSCC and DESADV	From the WMS at pack	DESADV from the ERP at release, with ERP pallet numbers; labels printed separately by the WMS (the ASN Chargeback)
Invoice	From the ERP, from shipped lines	From the ERP, from order lines
Deductions	A case per short payment, with DESADV evidence	Unexplained short payments; €14,200 in Q3 2026

When It Goes Wrong Along This Path

Beyond the ASN path, five failures recur in wholesale, each with a detection and a recovery.

The duplicate ORDERS. Detected by the GLN plus PO key; recovered by doing nothing. **The order change after allocation.** Bergzicht sends a change (an ORDCHG message, or a phone call to Lea); the OMS re-validates and re-allocates, and the ORDRSP restates the whole order, not only the changed line. **The short pick.** A carton missing at pack;

the DESADV and the invoice show the shipped quantity, and Lea tells Pieter before the truck arrives. **The remittance that names nothing**. A transfer for €23,214.80 against three invoices and two deductions; recovered by a cash-application rule that matches on amount patterns and opens a case rather than parking the money. **The guideline revision**. Detected by the partner card's change contact and a test cycle with the EDI provider before the effective date; the alternative is the Pulled Rug, discovered by deduction.

Gregor Hohpe's 2004 Starbucks essay names the three recoveries all of these reduce to: write-off, retry, or compensating action. In wholesale the compensating action is a credit note, and the write-off is a deduction nobody disputed.

What to Take Away

Established.

- EDI between retailers and suppliers is a buyer-defined document exchange with penalties in the contract; the public conditions of Jumbo (2026) and bol (2021) and Walmart's program (2017) all tie deliveries to a despatch advice and SSCC labels, and refuse or charge for non-compliance.
- The GS1 identifiers divide the work: GTIN for the item, GLN for the location, SSCC for the logistic unit; the GS1 General Specifications (Release 26.0, 2026) make the SSCC the only identifier of a pallet and the one the DESADV refers to.
- An invoice is issued once, by one numbering authority, from what was delivered, and corrected only by credit note; revenue is recognized at delivery (IFRS 15). Structured e-invoicing is mandatory in Belgium since 2026, receivable in Germany since 2025, planned in the Netherlands for 2030, and EU-wide for cross-border B2B from July 2030 under ViDA.

Depends on context.

- Whether EDI orders should enter an OMS or the ERP. At Kestrel & Pine's stage, with four channels and one warehouse, one order record in the OMS is right; a Growing merchant with one EDI partner and no OMS does better putting the order in the ERP and the DESADV in whichever system packs than adding a hub for one partner.
- Whether to run a direct AS2 link or a provider's network. A provider is right while partners are few and mappings change rarely; a direct link pays off only for a supplier with many partners and an EDI team.
- Whether to send Peppol invoices to customers who are not obliged to receive them. Yes where their bookkeeping already expects it, as in Belgium; elsewhere, have the ERP's access point ready before the local mandate makes it the only route.

The author's synthesis.

- I treat the despatch advice as the most dangerous document a merchant sends, because it is a claim about physical goods that only one system can verify. Whatever system packs the pallet writes the DESADV; no exceptions, no mapping that "knows" how the warehouse packs.
- I would put every trading partner on a one-page partner card before touching a message, and I would consider an EDI relationship that only a contractor can explain to be an outage that has not happened yet.

Questions to Ask

1. Which system creates the order record for an EDI order, and does it carry the same number family and state model as a web order?
2. What is the duplicate key for an incoming ORDERS message, and what happens when the same interchange arrives twice?
3. Which system prints the SSCC label, and which system generates the DESADV: the one that packed the pallet or the one that holds the order?
4. Is the DESADV sent before the truck leaves, and what happens to the truck if it is not?
5. For each trading partner, where are the guideline version, the GLNs, the test status and the change contact written down, and who at the company can change the mapping?
6. How is a short payment or deduction detected, which documents are attached to the dispute, and how many were disputed last quarter?
7. Does the invoice come from the shipped lines or from the order lines, and who owns the contract price it uses?
8. Which countries' e-invoicing obligations apply to the company's customers today and in 2027 and 2030, and does the ERP hold the transmission status of each e-invoice?

Bergzicht's pallets and Eva's parcel left the same warehouse in the same April week, on a landscape built for an ordinary day. The next journey is the day that is not ordinary: Friday 27 November 2026, when the Ridgewalker sold out at 19:40 and the storefronts kept selling it for two hours. Journey 4: Black Friday, Hour by Hour walks it hour by hour.

Sources and Evidence

Company, product and investor sources

- Handout richtlijnen gemechaniseerd DC — Jumbo, version 2025-10 (translated from Dutch); DESADV required before every delivery, the SSCC label as the link, refusal without it.
- Algemene inkoopvoorwaarden — Jumbo, 2026-04-09 (translated from Dutch); the €70 per-message forfeit, refusal at receipt, the €250 per-order-line penalty.
- EDI — Jumbo supplier pages, accessed 2026-11; a retailer's VAN-only channel and the GLN rule for DESADV.
- SSCC for EDI suppliers — bol, n.d., accessed 2026-11; SSCC labels mandatory since 31 December 2021 and required inside the DESADV.
- Walmart U.S. gives suppliers new order fulfillment guidelines — Talk Business & Politics, 2017-02-23 (dated); the 3% on-time-in-full fee at introduction.
- How to dispute Walmart OTIF chargebacks — SPS Commerce, updated 2024-05-06; ASN quantity mismatch as a disputable reason.
- SPS Commerce Form 10-K for 2025 — SEC filing, 2026-02-19; the "connect once" network model and customer count.
- Synchronize customers and companies — Microsoft Learn, 2026-10-14; customer records, payment terms and credit limits owned in the ERP.
- PaymentTermsType — Shopify Admin API 2026-10, accessed 2026-11; how a platform types NET and other B2B terms.
- Warehouse management — Manhattan Associates, n.d., accessed 2026-11; wholesale, retail and direct orders handled in one warehouse system.
- Idempotent requests — Stripe documentation, n.d., accessed 2026-11; the idempotency-key discipline applied here to order messages.
- Customer invoices and Credit notes — Odoo 18.0 documentation, accessed 2026-11; immutable invoices and credit notes as the only correction.

Standards, regulation and official documentation

- GS1 General Specifications, Release 26.0 — GS1, 2026-01; definitions of GTIN, GLN and SSCC and the SSCC's role in EDI.
- GS1 EANCOM 2002, Edition 2016 Upd. 2021, Part I — GS1, 2021 (the standard dates from 2002); EANCOM as a subset of UN/EDIFACT.
- EDI implementation survey — GS1, 2021 (dated); ORDERS, INVOIC and DESADV as the most implemented messages, AS2 the most used channel.
- X12 transaction sets — X12, n.d., accessed 2026-11; the 850, 855, 856, 810 and 997 equivalents.
- RFC 4130: MIME-based secure peer-to-peer business data interchange using HTTP, Applicability Statement 2 (AS2) — IETF, 2005 (dated, still in production use); the AS2 protocol.
- About Peppol — OpenPeppol, n.d., accessed 2026-11; the four-corner model and the role of access points.

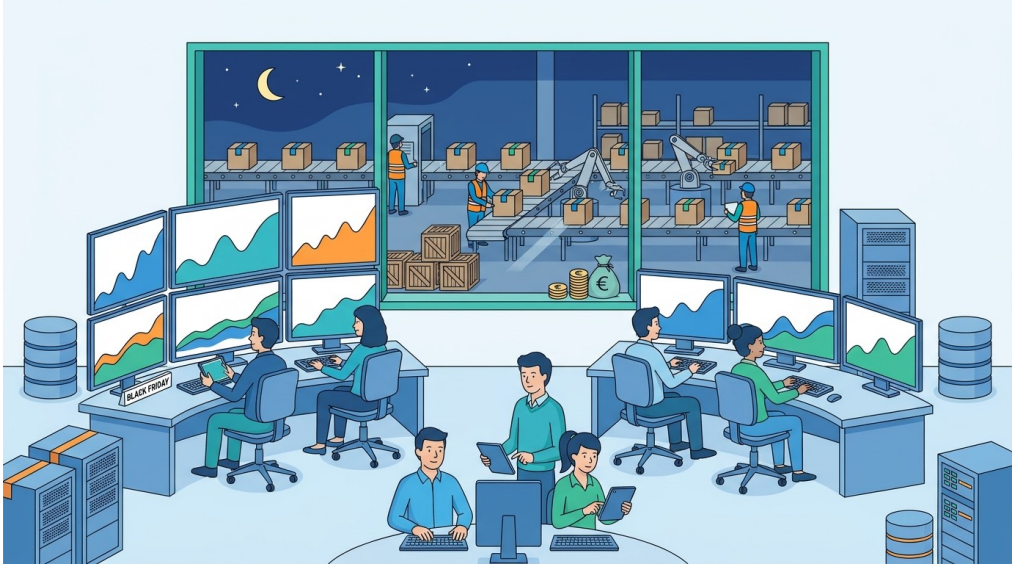
- Peppol BIS Billing 3.0 and release notes — OpenPeppol, version 3.0.21, 2026-05-20 (mandatory from 2026-08-17); the invoice specification, type codes 380 and 381, VAT category per line.
- Structured electronic invoices between companies are compulsory in 2026 — FPS Finance (Belgium), 2024-09-18; the mandate and the Peppol network.
- General questions B2B and Specific questions about e-invoicing — FPS Finance (Belgium), n.d., accessed 2026-11; non-established suppliers out of scope, fines of €1,500, €3,000 and €5,000.
- Period of tolerance during the first three months of 2026 — FPS Finance (Belgium), 2025-12-18; tolerance without postponement.
- FAQ E-Rechnung — German Federal Ministry of Finance, n.d., accessed 2026-11 (translated from German); receiving obligation since 2025 and EDI tolerated through 2027.
- Kabinet kiest voor invoering e-facturatie en rapportage voor bedrijven — Dutch government, 2026-09-11 (translated from Dutch); the planned 2030 and 2031 dates.
- VAT in the Digital Age (ViDA) — European Commission, n.d., accessed 2026-11; adoption on 11 March 2025 and the 1 July 2030 reporting date.
- Cross-border VAT — European Commission, n.d., accessed 2026-11; 0% intra-EU B2B supplies with a valid VAT number and VIES.
- Invoice requirements and VAT rates and exemptions — Belastingdienst and business.gov.nl, n.d., accessed 2026-11; mandatory invoice contents and the 21% rate.
- Regulation (EU) 2024/886 on instant credit transfers — Official Journal, 2024-03-19; sending instant euro transfers mandatory from 9 October 2025.
- IFRS 15 Revenue from Contracts with Customers — IFRS Foundation, issued 2014, effective 2018; revenue when control transfers.

Research and practitioner sources

- OMS vs ERP: the architecture shift reshaping commerce — OneStock, 2026 (vendor source); the argument for the OMS owning the order and the ERP owning finance.
- Starbucks does not use two-phase commit — Gregor Hohpe, 2004 (dated); write-off, retry and compensation as the three recoveries.

Kestrel & Pine, Bergzicht Outdoor and Ardennes Trail Co. are fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

40. Journey 4: Black Friday, Hour by Hour



IN THIS CHAPTER, YOU WILL: Follow Kestrel & Pine's Black Friday week of 2026 hour by hour, see what each system and person did, and read the 2027 plan built from it.

KEY POINTS:

- **The evening was lost in the afternoon.** The order flow to the warehouse fell behind at about 13:00, when nobody was watching, because it created orders one at a time under the warehouse vendor's rate limit. By the peak hour it was four hours behind, and no gauge showed it.
- **The sell-out at 19:40 took two hours to reach the shelves that were still selling.** The warehouse knew at 19:40, the finance system at 19:45, the storefronts at about 20:30 and the marketplaces at about 21:35, because stock traveled by three scheduled copies. 1,340 units were sold that did not exist.

- **The 2027 plan changes the chain, not the platform.** A forecast of 60 orders a minute, an order path through the order management system tested at 120, stock events to an availability service within a minute, eight switches with owners and triggers, a freeze, a situation room and a reconciliation report at 08:00 on Saturday.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The First Snowfall** — the order flow, the stock chain and the warehouse's evening shift met forty times their normal load for the first time on the day itself.
- **The Phantom Stock** — three channels sold the same last jacket for up to two hours, because each learned of the warehouse's zero by a scheduled copy.
- **The Message in a Bottle** — a queue with no gauge, no alarm and no owner turned a resilience feature into a four-hour delay nobody could explain.

In Black Friday week 2026, **Kestrel & Pine, the fictional outdoor brand**, takes **31,000 orders**, including **2,300 in one hour**. Its platform keeps accepting purchases, but integrations cannot send orders to the warehouse or return stock updates fast enough. Customers can therefore buy jackets that have already sold out while the storefront still appears to be working. Measuring the page's availability would miss the failure that matters to those customers.

We follow the evening twice: first the **November 2026 incident**, then the **November 2027 plan**. The scenario fixes the totals: a 19:40 sell-out, 1,340 units oversold, 1,120 orders canceled or delayed, €96,000 in costs and a four-hour backlog. Detailed hourly figures are labeled reconstructions.

Black Friday Every Year: Peak Load, Resilience and Recovery explains the methods used in the plan: capacity in orders per minute, controlled queues, optional work that can be paused, an incident team and checks after recovery.

Where we are in the order's journey. Product page, checkout, order export, picking and stock updates all run under peak pressure. The handoffs that take minutes for an ordinary April order now take hours.

The Week Before the Evening

Ecommerce Manager Milan Petrov plans a **20% outerwear discount for 23–29 November**, reducing the Ridgewalker jacket from €199 to **€159.20**. The agency builds campaign pages. Operations Head Daniel Okafor books temporary pickers and an extra PostNL collection. Stock for German marketplace fulfillment has already been sent in October.

Discounts, Coupons and Loyalty: Promotions Without Chaos examines the campaign's discount settings.

The preparation misses the integrations. There is no throughput forecast, load test, list of optional work to pause, integration on-call owner or dashboard comparing accepted orders with warehouse intake. The team plans to watch only the platform's status page. The process still fits a smaller, simpler business (What Changes at Each Scale: From First Store to Group).

The week built as the table shows. The daily figures are illustrative; the weekly total is the ledger's.

Day	D2C orders *(illustrative)*	What happened
Monday 23 Nov	2,100	Promotion live at 00:00; the storefront's busiest Monday of the year; nothing strained
Tuesday 24 Nov	2,300	Marketplace listings carry the discount; the connector's hourly stock push is unchanged
Wednesday 25 Nov	2,700	Duiven ships Monday's and Tuesday's orders on schedule; the first "where is my order" tickets arrive
Thursday 26 Nov	3,900	Evening traffic double Wednesday's; the order flow to the WMS keeps up, with no slack left
Friday 27 Nov	**11,500**	The day this chapter tells
Saturday 28 Nov	3,900	The queue drains overnight; support finds the oversell; cancellations begin
Sunday 29 Nov	4,600	Reconciliation by hand; the marketplace cancellations are filed; Cyber Monday is tomorrow
Week	**31,000**	10.6% of the year's D2C orders

Thursday evening is the warning nobody read: the order flow kept up with Thursday's evening rate only because that rate was below its ceiling. Nobody knew what the ceiling was.

Friday 27 November 2026, Hour by Hour

Friday brings **11,500 orders** *(illustrative).* Three clocks shape the outcome:

- **Order export**: the hosted integration tool (**iPaaS**, integration platform as a service) creates orders one at a time in the **warehouse management system (WMS)**. Its reconstructed capacity is about **seven orders per minute**; nobody measured it before the day.
- **Stock updates**: WMS to the **enterprise resource planning system (ERP)** every 15 minutes, then hourly to the storefront platform and hourly onward to marketplaces.
- **Picking and collection**: the last normal picking wave is 21:00 for PostNL at 22:30. Checkout promises next-day delivery until 22:00. Daniel adds a 23:00 collection for Black Friday (Pick, Pack, Ship: Warehouse Management and Fulfillment).

Time	Orders *(illustrative)*	What the systems did	What the team saw and did
00:00–12:00	900	Everything on schedule; the order flow idles between orders	Nothing; a normal busy morning
12:00–15:00	1,400 (about 8 a minute)	At about 13:00 the arrival rate passes the flow's ceiling of about 7 a minute; the WMS vendor's rate limit throttles the calls; the queue on the iPaaS begins to grow and nothing shows it	Nothing. The platform admin shows orders arriving; the WMS shows orders arriving; nobody compares the two

Time	Orders *(illustrative)*	What the systems did	What the team saw and did
15:00–17:00	1,200 (10 a minute)	The queue reaches about 500 orders by 17:00; the 15:00 and 16:00 waves at Duiven are thinner than the storefront's day suggests	The warehouse lead notices quiet pick lists on a day the radio says is Black Friday, and assumes the evening will bring the rush
17:00–18:00	900 (15 a minute)	Queue about 1,000; Ridgewalker stock at Duiven falling fast as the afternoon's orders reach the WMS	Milan watches the campaign dashboard: revenue ahead of plan
18:00–19:00	1,000 (17 a minute)	Queue about 1,600; the ERP's 18:45 stock snapshot still shows Ridgewalkers in most sizes	Daniel's evening shift starts; temporary pickers are briefed
19:00–20:00	1,400 (23 a minute)	**19:40: the WMS picks the last Ridgewalker units; physical stock zero.** 19:45: the WMS tells the ERP. Queue about 2,500 by 20:00	19:50: the warehouse lead sees the Ridgewalker pick location empty with orders for it still in the wave, and calls Daniel

Time	Orders *(illustrative)*	What the systems did	What the team saw and did
20:00–21:00	**2,300 (38 a minute)**	The peak hour. The storefronts and marketplaces still show the Ridgewalker in stock. About 20:30: the ERP's hourly push sets the platform's Ridgewalker quantities to zero; the storefronts stop selling it. Queue about 4,400 by 21:00; the oldest unprocessed order is four hours old	20:20: Daniel calls Milan: "we are out of Ridgewalker and the orders keep coming." 20:35: Milan calls Ruben Alves. 20:50: Ruben opens the iPaaS and sees the queue for the first time
21:00–22:00	1,500 (25 a minute)	About 21:15: the platform's hourly push sends zero to the marketplace connector; about 21:35 the marketplaces show sold out. The queue keeps growing	21:05: Milan sets the Ridgewalker to zero by hand in the connector, minutes before the push would have. 21:30: Ruben raises an emergency ticket with the WMS vendor; nobody answers on a Friday night until 22:40

Time	Orders *(illustrative)*	What the systems did	What the team saw and did
22:00–24:00	900	The storefront's 22:00 promise expires; a late wave releases for the extra 23:00 collection Daniel booked, with thousands of orders placed before 22:00 still on the iPaaS	Daniel decides the night shift will keep picking whatever arrives; Mara Lindqvist is told at 22:15
Saturday 00:00–03:00	—	About 00:00: with the vendor's limit raised, Ruben switches the flow to the WMS's batch import; throughput about 40 a minute; the queue is empty by about 02:30	Orders placed at 20:00 reach the WMS after midnight. Saturday's shift picks through the day; PostNL's Saturday collection carries them; most are delivered Monday

Three things in the table deserve a closer look, because they are the mechanisms the 2027 plan is built against.

The queue nobody could see

The queue prevents checkout failures and lost orders, but it cannot create processing capacity. **Without a gauge, alarm or owner, it hides the delay.** At 20:50, integration engineer Ruben Alves finally checks it. By 21:00 the backlog is four hours; 20:00 orders reach the warehouse after midnight.

The illustrative reconstruction assumes roughly four calls per order against a warehouse limit of 30 per minute. Ruben's June 2027 review also finds about 1,900 platform-side throttles amid traffic from nine apps and connectors (Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation). Warehouse picking capacity itself remains available.

Figure 1: *On Black Friday 2026 the queue did its job and hid the problem: orders piled up between the storefront and the warehouse with no gauge, no alarm and no owner, until the warehouse noticed it had too little work.*

The two-hour sell-out

The sequence diagram shows the two hours that cost €96,000. Each arrow is a scheduled copy, and each copy is late by design.

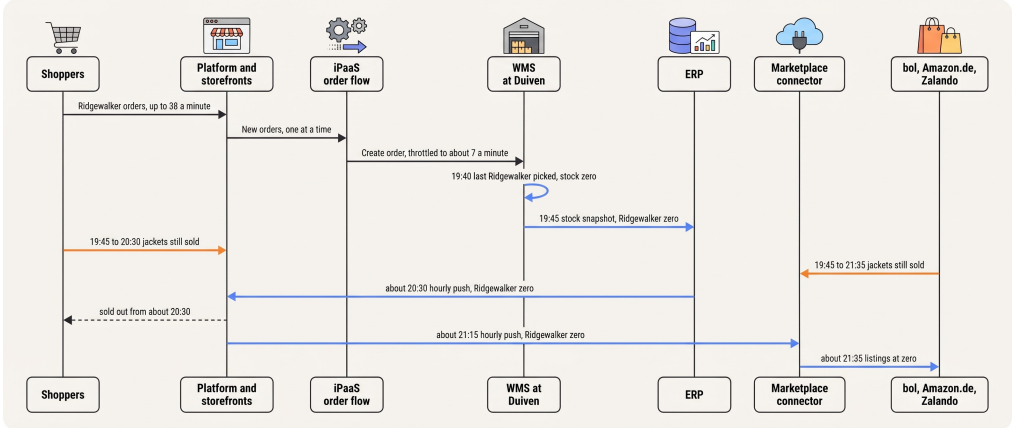


Figure 2: *The sell-out at 19:40 reached the ERP in five minutes, the storefronts in about fifty and the marketplaces in about two hours; in that window the channels sold 1,340 units the warehouse did not have, because stock traveled by three scheduled copies rather than by the event that emptied the shelf.*

The two-hour window was not a fault. It was the chain's design: 15 minutes plus up to an hour plus up to an hour. On every other day of the year the window was harmless, because nothing sold out inside it. Is It in Stock? Inventory, Availability and the Promise You Can Keep explains why the number copied was also the wrong number (physical stock, not available-to-promise) and why the three apps that could write inventory made it worse; this journey only needs the clock.

Three channels sold against the same zero. Of the 1,340 oversold units, about 790 were storefront orders and 550 marketplace orders *(illustrative split from Is It in Stock? Inventory, Availability and the Promise You Can Keep).* The marketplace share was smaller and cost more, because a seller who cancels a confirmed marketplace order is measured for it: about 160 of the 1,120 affected orders were marketplace orders, 140 of them canceled by Kestrel & Pine, and bol issued a strike, its formal warning for a week below its cancellation standard *(illustrative, from Selling on Someone Else's Platform: Marketplaces and Channel Integration).*

What the team could and could not do

By 20:50 four people knew something was wrong, and the table of what they could reach is the most instructive part of the evening.

What someone wanted to do	Was it possible?	What happened
Stop selling the Ridgewalker on the storefronts	Yes, by setting quantities to zero by hand in the platform, one variant at a time (24 variants)	Milan was preparing to when the 20:30 push did it
Stop selling it on the marketplaces	Only by editing listings in the connector by hand; no "pause SKU" switch	Done at 21:05, minutes before the hourly push would have
Speed up the order flow	Only with the WMS vendor's help (raise the limit) or a rewrite to batch calls	Ticket at 21:30; answered 22:40; limit raised and flow switched to batch import at about 00:00
Tell customers their delivery would be late	Only by a mass e-mail from the marketing tool, with no list of which orders were affected	Sent Saturday at 10:00 to everyone who ordered after 17:00 Friday

What someone wanted to do	Was it possible?	What happened
Change "delivered tomorrow" on the Dutch storefront	Yes, by editing the theme text; it was a sentence, not a computed promise	Nobody thought of it until Saturday; the page promised next day until 23:30
Switch off recommendations, search facets, the newsletter pop-up	Not needed; the platform held	—
See orders per minute against the warehouse's intake	No such view existed	Reconstructed on Monday from two exports

The first and third rows say the most. The sell-out could have been stopped on the storefronts at 19:50 by hand, if anyone had known to; the order flow could not have been fixed by anyone at Kestrel & Pine before the vendor woke up. The 2027 plan is largely the table's "No" and "Only" cells turned into switches, gauges and contracts.

Saturday to Monday: Finding Out What Was Sold

Saturday morning the queue was empty and the damage was still invisible. The platform showed 11,500 paid orders. The WMS showed 11,500 orders received, about 1,100 of them with at least one Ridgewalker line it could not allocate. Nothing connected the two lists to the shoppers waiting for a jacket. Support found the oversell the way support always finds it: from the first "my order says processing" tickets.

The clean-up took three people most of the weekend and finance the week after. Daniel's team produced the shortage list from the WMS. Milan matched it to orders in the platform and in the connector by hand, by SKU, because the two systems had no shared order identifier for marketplace orders. Sanne de Wit's team handled the money. The outcome, in the ledger's totals with an illustrative breakdown:

Outcome	Orders	Units	Treatment
Canceled	760	900	Refund by the original method; a €15 goodwill voucher; marketplace cancellations filed as seller cancellations
Delayed	360	440	Held for a supplier delivery due Tuesday 1 December; shipped that week with the voucher

Outcome	Orders	Units	Treatment
Total affected	**1,120**	**1,340**	

The **€96,000** the ledger records for the incident breaks down, illustratively, into lost gross margin on the canceled units (about €65,700: 900 units at roughly €73 of margin each at the promotion price), the goodwill vouchers (€16,800: 1,120 at €15), marketplace penalties and the cost of the strike (about €6,500), and support overtime and re-shipping (about €7,000). The roughly 8,000 orders that missed next-day delivery *(assumption, from Pick, Pack, Ship: Warehouse Management and Fulfillment)* are not in that figure; their cost was a week of tickets and an unmeasured amount of trust.

Three reconciliations ran by hand that weekend that should have been jobs: platform orders against WMS orders (to find the stuck and the duplicated), WMS shortages against orders (to find the oversold), and refunds issued against cancellations (to make sure each canceled order was refunded once; the Double Refund of January 2027 shows what happens when that one is skipped, in The Way Back: Returns, Exchanges, Refunds and Reverse Logistics). Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation turns each into a reconciliation card.

The post-mortem, written in December, found five things: no forecast; no capacity figure for any link; a queue with no gauge; stock copied on a schedule rather than sent on change; and no one with the authority to flip anything. It also recorded Milan's first fix, in place by mid-December: the connector publishes the platform's figure minus ten units per variant to each marketplace, and zero below ten *(illustrative, from Selling on Someone Else's Platform: Marketplaces and Channel Integration)*, a buffer that buys minutes on a Black Friday evening and costs a few unsold units on every other day. Noor Visser, who arrived in January 2027, built the Waypoint proposal on that post-mortem (Meet Kestrel & Pine).

The 2027 Plan: The Same Evening, Rehearsed

Black Friday 2027 falls on **Friday 26 November**; the peak week is 22–28 November and Cyber Monday is the 29th. The plan below was approved with Waypoint in March 2027 and is owned by Noor as incident commander. Every figure is fictional; the dated public practices it borrows from are cited in Black Friday Every Year: Peak Load, Resilience and Recovery.

The forecast and the capacity table

The forecast is **60 orders a minute** in the peak hour, about one and a half times 2026, a convention Shopify's 2025 readiness program also used (roughly 150% of the previous

year's load; Shopify Engineering, November 2025). Every link is tested at **80**, and the slowest link sets the number the waiting room and the promise are tuned to.

Link	2026 capacity *(illustrative)*	2027 target	Proof
Storefront and checkout (the platform)	Held at 38 a minute; buyer traffic is not rate-limited	Vendor's responsibility; status page watched	Vendor's own program; no merchant test
Payment authorization (the PSP)	Held	200 a minute confirmed with the PSP in August, with notice of the flash-sale evening as its documentation asks	PSP confirmation in writing
Order into the OMS (platform webhook to OMS)	Not applicable in 2026	150 a minute measured	Load test, September
OMS to WMS (fulfillment orders in batches within the WMS's limit)	About 7 a minute, one at a time	120 a minute tested; alarm at 10 minutes of lag	Load test with marked orders, September and October
WMS stock event to availability service to channels	15 min plus hourly plus hourly	Under 1 minute end to end; 300 events a minute tested	Load test, October
Marketplace stock push (connector to marketplaces)	Hourly	Within 5 minutes, per marketplace rate limit	Connector vendor's stated rate; observed in October
Picking at Duiven (Friday evening and Saturday)	Never the limit in 2026	9,000 parcels a day with the peak roster *(illustrative)*	Daniel's staffing plan; the delivery promise is set from it

Link	2026 capacity *(illustrative)*	2027 target	Proof
ERP import and invoicing	Hourly	Daily batch, overnight, off the peak path (W5)	—

The 2027 path uses a shared **order management system (OMS)**, introduced for marketplaces on **15 September** and direct consumer orders on **13 October**. It sends warehouse work in batches sized to the API limit. The order number identifies each request, so repeating it cannot create a second pick list: **idempotency**.

The queue now shows its oldest unsent order and alarms at ten minutes. Failed messages enter a **dead-letter queue**, with Ruben responsible for recovery; the team practiced replay in October. Wholesale moves on **8 December**, after the peak. Changes freeze from **15 November to 5 December** (Moving Without Burning the Bridge: Replatforming and Migration).

The stock loop is replaced for the channels that oversold. The WMS emits a **stock event** on every pick, receipt and adjustment; the **availability service** (a module of the OMS, Waypoint W1) recomputes **available-to-promise** per channel (physical stock minus what is already promised, minus a per-channel buffer of three units per SKU for marketplaces and one for the storefronts) and publishes the change to the platform and the connector within a minute. The go/no-go for using it in the peak was **15 October**. Had it failed, the fallback was the 2026 chain with the ten-unit marketplace buffer and a person pausing listings by hand; that fallback was rehearsed too.

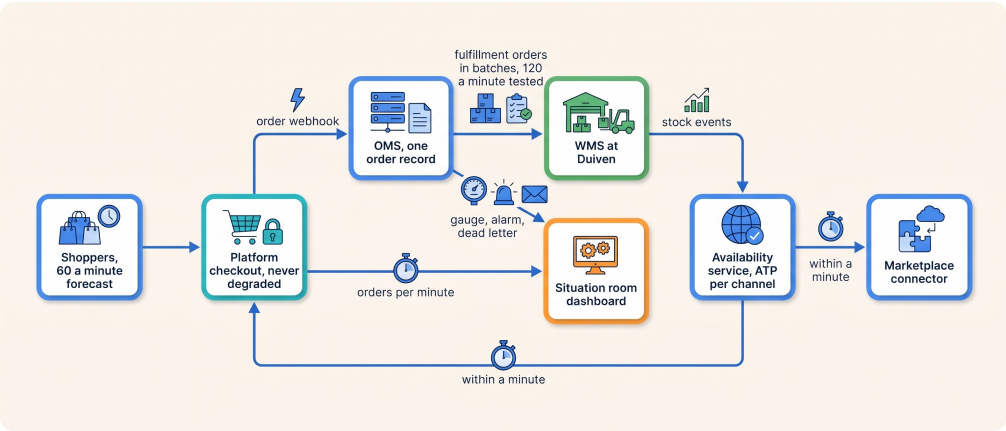


Figure 3: *The 2027 chain: orders flow through the OMS to the WMS in batches with a gauged queue, stock returns as events through the availability service within a minute, and both feed one dashboard in orders per minute; the platform's checkout is the one*

link that is never switched off.

The switches

The **degradation ladder** (switching off non-essential features in a planned order so the essential ones keep working) has eight rungs. Each is a feature switch that exists today, was flipped once in the October rehearsal, and has one owner who may flip it without a meeting. The triggers are the plan's; the thresholds are illustrative.

#	Switch	Owner	Trigger	What it does
1	Recommendations and "recently viewed" off	Milan	Storefront p95 response over 2 s, or a platform status notice	Removes the personalized calls from every page
2	Search facets reduced to size, color and price; two sort options	Milan	Same as 1	Cuts the search service's load per query
3	Stock badges from the buffered copy	Ruben	Availability event lag over 2 minutes	Shows "low stock" under 10 units; never "in stock" from a stale figure
4	Marketplace buffer raised to 25 units for hero SKUs; listings paused below it	Milan	Lag over 2 minutes, or orders for a SKU exceeding its ATP	Protects the channel that is measured for cancellations
5	Delivery promise widened: "tomorrow" becomes "within three working days"	Daniel	Oldest unsent order over 10 minutes, or the Friday pick plan over 80% consumed	The page tells the truth the warehouse can keep

#	Switch	Owner	Trigger	What it does
6	Non-essential checkout options off (gift wrap, gift message); a single payment method off if the PSP reports it degraded	Sanne's team with Milan	Checkout error rate over 1%, or a PSP notice	Shortens the checkout; removes a failing method
7	Waiting room at the CDN	Ruben with the agency	Storefront error rate over 1% or the platform's notice; admission rate set to 75% of the slowest tested link	Converts the spike into a known inflow
8	New-account creation and the newsletter pop-up off	Milan	Only on Noor's call	The last load removed before the money path is touched
—	**Never**: checkout, payment authorization, order creation in the OMS, the order path to the WMS	—	—	Queues grow instead

Rung 5 is the one 2026 lacked most. The Dutch storefront's "ordered before 22:00, delivered tomorrow" was a sentence in a theme file. Since May 2027 the first promise service of Getting It There: Carriers, Labels, Tracking and the Delivery Promise computes the promise from a transit table plus the WMS's cut-off and backlog, so a growing queue already widens it on its own; rung 5 is the override on top, with an owner and a trigger, and for peak week the service is set to "within three working days" from the start, because Daniel's pick plan, not the theme, decides what can be promised.

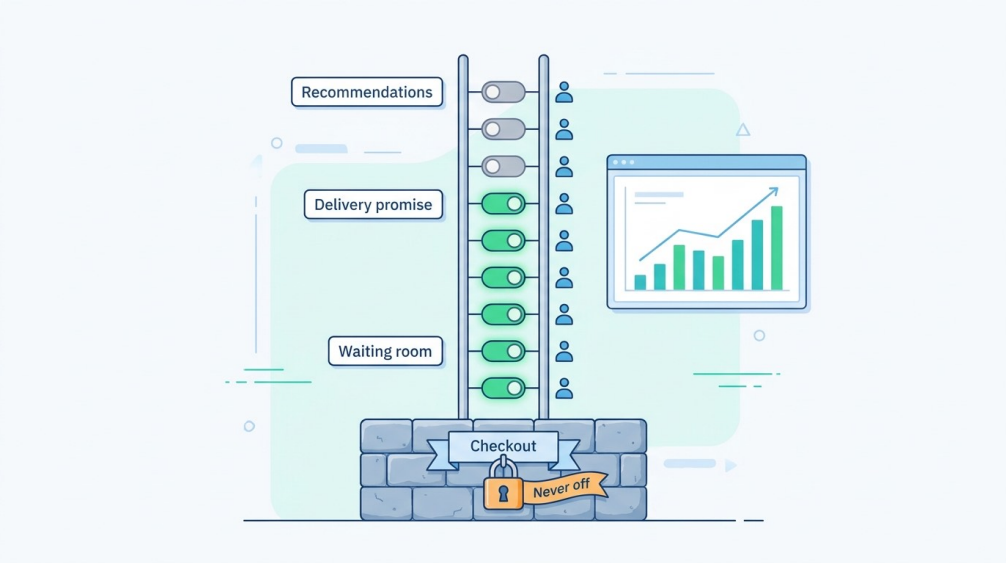


Figure 4: *The 2027 plan's degradation ladder has eight rungs, each a rehearsed switch with one owner; checkout, payment and the order path sit below it and are never switched off.*

The runbook

The runbook is a document of four pages, written in September, rehearsed on 15 November and kept open on the situation-room screen. Its parts:

Roles. Noor is incident commander and decides; nobody else flips rung 8. Ruben owns the OMS-to-WMS path, the availability service and the dead-letter queue. Milan owns the storefronts, the apps and the marketplace listings. Daniel's shift lead owns the warehouse and the delivery promise. Sanne's team owns payment methods and refunds. An agency engineer is on call for the theme and the CDN. The vendors' emergency contacts (platform, OMS, WMS, PSP, connector) are on the first page with ticket priority codes and the hours they are staffed; the WMS vendor's Friday-night gap of 2026 is closed by a contract clause agreed in June.

The dashboard. One screen, refreshed every minute, in business units: orders per minute per channel against the same hour last week; the age of the oldest order not yet in the WMS; stock event lag; dead-letter depth; checkout error rate; PSP authorization rate and decline rate against normal; the WMS's and the PSP's rate-limit headroom. The missing-event alarm fires when orders per minute fall to zero while storefront sessions stay high.

Cadence. The situation room runs on Friday from 16:00 to midnight, with a check-in every 30 minutes until 19:00 and every 15 minutes from 19:00 to 22:00. Each check-in reads the dashboard aloud against the thresholds in the switch table. Saturday and Cyber Monday run on call with the same dashboard.

Communication. Three templates, written in advance: a customer e-mail for a delayed delivery with the order numbers filled from the OMS, not from "everyone who ordered after 17:00"; a note to each marketplace's seller support when listings are paused; and a one-line status to the company every hour. Mara is told at the first switch, not the last.

The morning after. At 08:00 on Saturday the reconciliation report compares orders in the platform, the OMS and the WMS by order number (stuck, duplicated, missing); stock per SKU in the WMS against the availability service; payments captured in the PSP against orders in the OMS; and refunds issued against cancellations. Each difference has a named owner and a fix: replay, manual correction or write-off. Hanna Berg's team produces it from the data warehouse, under the one definition of an order that Waypoint W7 adopted.

The rehearsal

Two load tests and a game day preceded the freeze. In September the team placed marked test orders at 60 a minute through the whole path, platform to OMS to WMS, with the WMS configured to skip them and finance to exclude them, the practice Zalando described for its order-flow simulators in 2020. In October the test ran at 80, then 120 on the OMS-to-WMS link alone, and found the first real defect: the batch size that kept the WMS inside its rate limit at 80 orders a minute exceeded it at 120, and the retries that followed doubled the load, the thundering-herd effect that backoff with jitter exists to prevent. The fix was a smaller batch and a jittered retry, and the test was repeated.

The game day in October rehearsed three failures: the WMS unreachable for thirty minutes (the queue grew to about 1,800 orders, the gauge and the alarm worked, the delivery-promise switch was flipped at minute 11, and replay drained the queue in 22 minutes); the availability service stopped (the stock-badge switch and the marketplace-buffer switch were flipped, and the fallback chain ran for an hour); and the PSP declining one payment method (the method was switched off within eight minutes). Each exercise either proved a runbook page or rewrote it.

When It Goes Wrong in 2027

The plan does not promise a quiet evening. It promises that the failures it expects leave known traces and have known fixes.

If, on 26 November 2027	The plan says
The OMS-to-WMS queue grows past ten minutes	The alarm fires; rung 5 widens the promise; the queue is allowed to grow; replay after the cause is found; no order is lost
The availability service lags past two minutes	Rungs 3 and 4; listings of hero SKUs pause below the raised buffer; the fallback chain is one switch away
The WMS vendor throttles again	The batch size is already inside the limit; the June contract names a Friday-night contact; the queue absorbs the rest
A retry creates a duplicate fulfillment order	It cannot: the order number is the idempotency key; the Saturday report would show it if it did
The PSP degrades	Rung 6 removes the failing method; every captured payment is matched to an order on Saturday
The platform itself has an incident	The vendor's problem to fix and the merchant's to communicate; the waiting room and the status page are the levers; the queues behind the platform are empty, which is the one good thing about it
The forecast was wrong by two	The chain was tested at 80 and the OMS-to-WMS link at 120; above that, the ladder runs from the top and the promise widens

What the plan deliberately leaves slow is the ERP: its daily batch absorbs the week's volume overnight, and if it falls behind, invoices are late, which is the one delay the chain can afford. The 2028 peak will be the first with the promise offered through an API and, if the 2028 evaluation says so, the platform's own POS in the stores.

What to Take Away

Established.

- A queue between a platform and its warehouse system is the right design under peak load, and it is only safe with a gauge, an alarm, an owner and a replay procedure (AWS Builders' Library 2019; Amazon SQS dead-letter documentation; Zalando's 2015 practice of pausing non-essential processing, reported 2020).

- Stock copied on a schedule oversells in proportion to the sales rate and the copy interval; a fifteen-minute plus hourly plus hourly chain is harmless on an ordinary day and a two-hour blind spot at 38 orders a minute.
- A load test of the merchant's own order path, with marked orders that fulfillment skips, is the only test that finds the merchant's bottleneck; the platform vendor's tests find the vendor's (Zalando 2019 and 2020, Shopify 2023 and 2025).

Depends on context.

- The switch-off ladder's order depends on what the business can afford to lose: a brand whose peak is one hero product pauses marketplace listings early; a marketplace-first seller, measured on cancellation rate, protects those listings and widens the storefront's promise first.
- A static but honest peak-week delivery promise is the right interim for a merchant without a computed promise; a merchant with one should let the warehouse's plan set the page automatically and keep the switch as the override.
- Running the OMS cutover before the peak was right for Kestrel & Pine because the parallel run finished in September and the link was load-tested in October; a merchant whose cutover slips into November should run the old path and keep the new one dark until December.

The author's synthesis.

- The lesson of the 2026 evening is not that the systems were too slow; it is that nobody could see them and nobody could touch them. Most of the 2027 plan is visibility and switches, which cost far less than the OMS and would have saved most of the €96,000 on their own.
- A peak plan should be written as a journey, hour by hour, with a name in every row; a plan written as a list of systems leaves the 20:50 moment, when someone finally looks, to chance.

Questions to Ask

1. At what time of day last year did our order flow first fall behind, and who would have seen it?
2. How long does a sell-out at the warehouse take to reach each channel, by the clock, and who tested it?
3. Which of our switches have been flipped in a rehearsal, and who may flip each one without a meeting?
4. Can we change the delivery promise on the page in one minute, and who decides?
5. What are the vendors' Friday-night contacts, and what does the contract say about response time?

6. Which marked test orders have we sent through the whole path at this year's forecast, and what did the warehouse do with them?
7. Which three reconciliations run on Saturday morning, by which key, and who owns each difference?
8. Who is in the room on the evening, and what does the first row of the runbook say?

The journeys end here. Part X asks the questions Kestrel & Pine's 2027 answered in one particular way (buy or build, stay or replatform, how to organize fourteen people around a dozen systems) as questions every merchant faces, and when each answer changes. Continue with Part X — EVOLUTION: Decisions That Change With Scale.

Sources and Evidence

Company, product and investor sources

- How we prepare Shopify for BFCM, Shopify Engineering, 2025 — forecasting at roughly 150% of the previous year's load, scale tests from April to October, Game Days on the money paths.
- Scale and performance testing at Shopify, Shopify Engineering, 2023 — testing with cache bypass so that origins are measured.
- Faster checkout for customers during flash sales, Shopify Changelog, 2018 — payment processing separated from order creation under high volume (eight years old).
- Webhooks, Shopify developer documentation, n.d., accessed 2026-11 — delivery not guaranteed; reconciliation jobs recommended.
- How Zalando prepares for Cyber Week, Zalando Engineering, 2020 — order-flow simulators that fulfillment skips, pausing non-essential asynchronous processing, the Situation Room (six years old).
- End-to-end load testing Zalando's production website, Zalando Engineering, 2019 — production load tests of recorded shopper journeys (seven years old).
- Rate limits, Stripe documentation, n.d., accessed 2026-11 — the advice to tell the PSP about a flash sale in advance, and backoff with jitter.
- Cloudflare Waiting Room, Cloudflare, 2021 — what a waiting room is and how its admission rate is set.

Standards, regulation and official documentation

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- Amazon SQS dead-letter queues, AWS documentation, n.d., accessed 2026-11 — dead-letter queues and redrive.
- Monitoring Distributed Systems, Google SRE book, 2017 — the four golden signals behind the situation-room dashboard.

Research and practitioner sources

- Feature Toggles (aka Feature Flags), Pete Hodgson, martinfowler.com, 2017 — ops toggles as the mechanism of the switch-off ladder.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part X — EVOLUTION: Decisions That Change With Scale

KEY POINTS:

- **At a growing merchant the problems live in the joins**. Build what customers choose you for, buy what every merchant needs, own the joins, and price every option over three years.
- **A stage is a shape, and a migration crosses between stages one seam at a time**. Old and new run side by side, with the old owner kept operable through one peak.
- **Ownership is what makes any of it hold**. A named owner for every system, integration and app, one writer per fact, and AI agents treated as integrations with no write scope on stock, prices or orders.

A merchant adds a store, wholesale channel, marketplace, warehouse or second brand. Its existing systems still run, but they may now need to share stock or coordinate orders in ways their original design never required. Leaving those assumptions unchanged can create overselling and manual work; replacing everything can exceed the team's capacity. This part examines how to choose and sequence the changes that address the new demands.

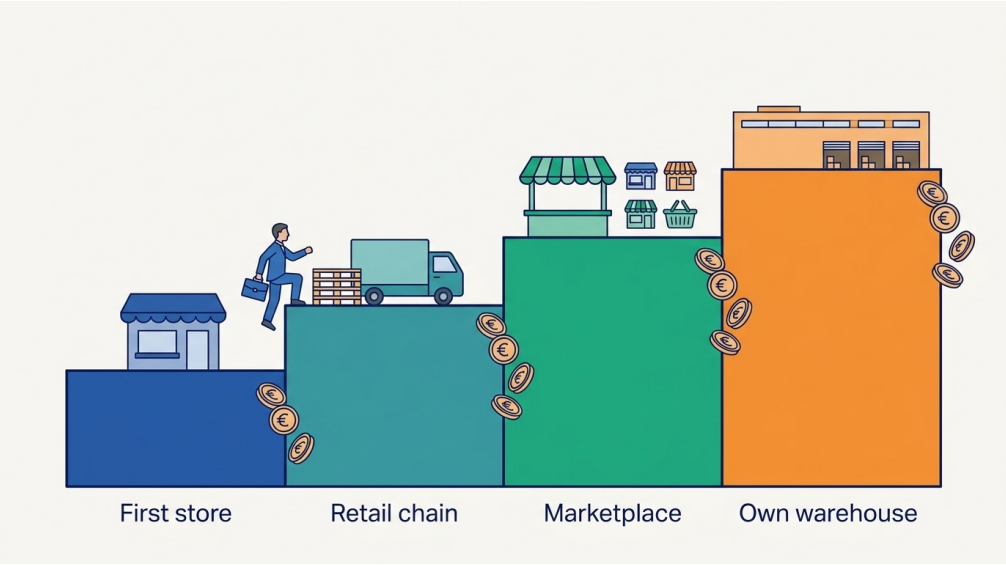

Figure 1: *A merchant grows in jumps, not smoothly: each jump adds a place where stock sits or a channel where it sells, and the money is lost at the jumps rather than between them.*

At Kestrel & Pine, the fictional outdoor brand, a jacket order crosses systems introduced between 2016 and 2025. Their purchase dates explain today's responsibilities and gaps. A new decision must account for that history, the team available to operate it and the customers who still need orders fulfilled during change.

The company faces five questions in 2027: whether to rebuild, how to introduce new order and product systems, which growth stage to design for, who owns the integrations, and whether to add shopping agents. Each requires an explicit choice about ownership, timing and capacity. The chapters use **Waypoint**, Kestrel & Pine's improvement program, to make those choices concrete.

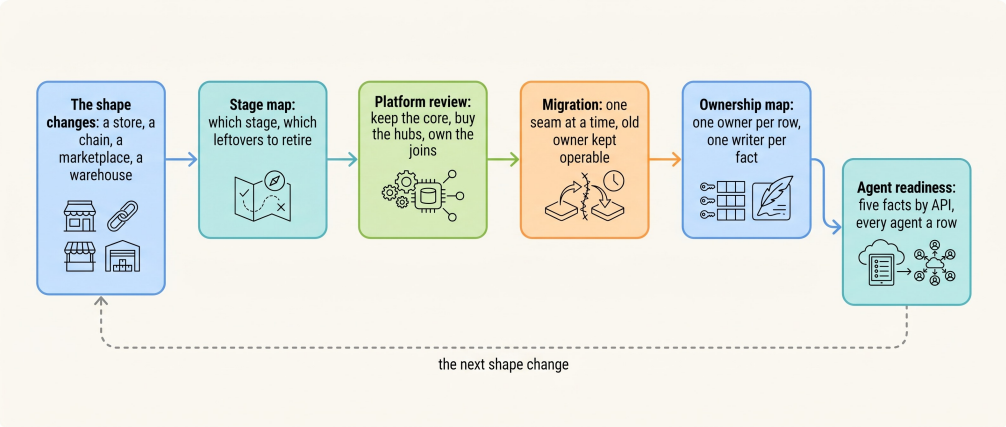

Figure 2: *The five decisions of Part X as one loop: a change of shape is read as a stage, answered by a platform review, carried out as a migration, held in place by an ownership map, and extended to the newest writer, until the shape changes again.*

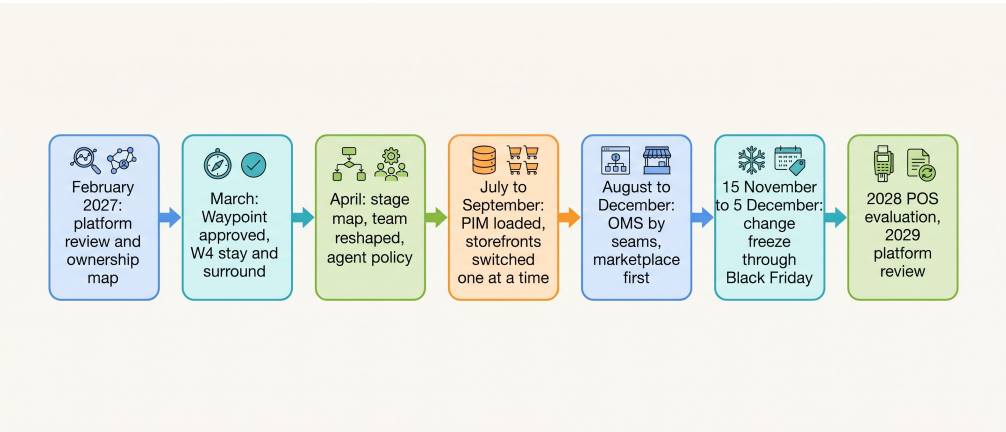

Figure 3: *Kestrel & Pine's 2027 as this part sees it: the decisions are made in the first quarter, carried out seam by seam from July, frozen around the peak, and dated for review rather than treated as final.*

Chapters in this part

Build, Buy or Assemble: Choosing and Changing Platforms compares **bundled products, separate services and custom software by their full cost**. The platform review includes people, integration, migration and exit over three years. It leads to keeping the platform and adding missing capabilities, with a 2029 review.

Moving Without Burning the Bridge: Replatforming and Migration plans **limited transitions with comparison and recovery at every step**. Product publication moves

gradually in 2027. Order channels move on separate dates, with a freeze around Black Friday so migration and peak risk do not compound.

What Changes at Each Scale: From First Store to Group identifies **the business changes that invalidate an earlier design**. Channels, stock locations and brands matter alongside revenue. The stage map separates current coordination problems from possible future group-wide needs.

Who Builds and Who Owns: Teams, Vendors, Agencies and Governance assigns **business and technical responsibility for systems, integrations and apps**. The January inventory finds thirteen of 34 integrations unowned. The chapter addresses supplier contracts, internal ownership and practices a small team can maintain.

Assistants and Agents: AI in Commerce and What Is Real in 2026 compares **AI that recommends an action with AI authorized to perform it**. Both need reliable product and stock answers; actions also need permissions and accountability. A dated review of emerging protocols leads to improving the merchant's data and interfaces before building a shopping agent.

Begin with Build, Buy or Assemble: Choosing and Changing Platforms.

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine is fictional; recommendations and labeled assumptions are the author's judgement.

41. Build, Buy or Assemble: Choosing and Changing Platforms

IN THIS CHAPTER, YOU WILL: Learn to choose between a suite, a hosted platform with apps, a composable assembly and custom build, and to price each one honestly, including the cost of leaving.

KEY POINTS:

- **The choice is not one choice.** A merchant covers the commerce capability map with four kinds of parts (a suite, a hosted platform plus apps, assembled services, custom code), and the real decision is which parts to buy, which to assemble and what, if anything, to build. The answer changes with business model, scale stage and team.
- **The subscription is the smallest line.** Total cost of ownership is dominated by payment fees, agency and partner work, the people who own integrations after launch, and the cost of leaving. A three-year comparison that prices only licenses is wrong by a multiple.
- **Stay and surround is a legitimate architecture, not a failure to decide.** Kestrel & Pine's 2027 review kept its hosted platform and added

an order hub, a product hub and an availability service beside it, because nothing in its checkout differentiates it and everything in its joins does.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Lynchpin** — a platform decision that leaves one vendor, one agency or one connector holding a capability customers depend on, with no exit plan.
- **The Giant's Boots** — a composable or custom architecture copied from a company a hundred times larger, without its team, its traffic or its reasons.
- **The Forked Platform** — a bought platform customized so deeply that it can no longer be upgraded, which turns "buy" into "build" without anyone deciding it.

A merchant has late stock updates, missing orders and a proposal for a new commerce platform. The replacement may help, but the same problems can survive if their causes sit in warehouse feeds or connectors that stay in place. Identifying which work is failing gives the merchant a basis for choosing what to keep, buy together, connect as separate products or build.

Then compare what each option fixes against its cost over several years. Subscription, payment fees, apps, agency work, integration maintenance and eventual exit all belong in that comparison. A lower subscription can require more engineering work, leaving a small team unable to operate the proposed system. This chapter puts those ongoing obligations beside the business problems each option solves.

Kestrel & Pine, the fictional outdoor brand, conducts that review in early 2027. It chooses **stay and surround**: keep the hosted platform and add missing capabilities. That becomes W4 of **Waypoint**, the company's improvement program.

Where we are in the order's journey. A jacket purchase depends on checkout, warehouse, payment, returns and finance systems. This decision assigns those jobs to products and teams, and determines the cost of changing them later.

The 2027 Platform Review at Kestrel & Pine

In February 2027, CEO Mara Lindqvist asks Head of Engineering Noor Visser for a recommendation. Stale stock caused about **€96,000** in Black Friday costs. Separate incidents left orders with conflicting statuses, refunds issued twice and prices inconsistent across channels. Two suppliers cite the oversell as a reason to rebuild.

A consultancy proposes a **composable rebuild**: a separate commerce engine, storefront, order and product management, search and integration services, over eighteen months.

The agency proposes a **headless storefront**, an independent application using the existing platform's APIs, for about **€240,000 over nine months**. It would leave the team with another runtime to operate. Delivering the Storefront: Themes, Headless, Rendering and Performance follows Noor's evaluation of that proposal.

Noor added a third option of her own to the comparison, so that the review would not be a choice between two pitches: an **enterprise suite**, one vendor's bundle of storefront, B2B portal, order management and promotions, priced on a share of online sales and implemented by a partner over about twelve months. Nobody had pitched it; she priced it because a suite is the usual alternative a board asks about.

Noor asks three questions. **What failed?** Checkout held at 38 orders per minute; stock propagation, order ownership and product data did not. **What differentiates the business?** Its products, brand, stores, wholesale relationships and delivery service. **What does each option cost over three years, including people and exit?** The comparison appears below.

The March 2027 decision keeps the hosted platform and adds an **order management system (OMS)** with availability management and a **product information management system (PIM)**. The platform's store tills are evaluated in 2028; the broader choice returns for review in 2029.

Four Ways to Cover the Map

The Map of a Commerce Platform: Capabilities and Layers lays out everything a commerce platform has to do: storefront, catalog, pricing, promotions, search, inventory, cart and checkout, payments, orders, tax and invoicing, warehouse, shipping, returns, customer service, stores, marketplaces, B2B, EDI, ERP, data. No vendor covers all of it well. The question is how you cover it. There are four basic shapes, and most real landscapes mix them.

Shape	What you buy	Who runs it	Where the integration burden lands	Typical pricing	Examples (dated in Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026)
Suite	One vendor's bundle of many capabilities designed to work together	The vendor's cloud, implemented by a partner	Inside the vendor's data model; the joins to the ERP, WMS and carriers remain yours	Negotiated, often linked to GMV or order volume	SAP Commerce Cloud, Salesforce Commerce Cloud, Adobe Commerce
Hosted platform plus apps	A multi-tenant storefront, cart, checkout and admin, extended through an app store	The vendor	Pushed to apps and webhooks; the merchant owns what the apps do	Subscription tier plus payment and transaction fees	Shopify, Commerce.com (BigCommerce), Lightspeed eCom

Shape	What you buy	Who runs it	Where the integration burden lands	Typical pricing	Examples (dated in Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026)
Composable assembly	API-only services (catalog, cart, checkout, search, content) from one or several vendors; the merchant builds or buys the storefront	Vendor clouds; merchant-built front end	On the merchant: it assembles and owns every join	Subscription with GMV or order caps	commercetools, Elastic Path, Medusa, Saleor
Custom build	Nothing; the merchant writes the capability	The merchant	Entirely on the merchant	Salaries and hosting	Large retailers' in-house platforms

Three terms carry this table. **SaaS** (software as a service) is software rented by the month and run by the vendor for many customers at once. **GMV** (gross merchandise value) is the total value of orders through a platform before refunds; several vendors price on it. **Composable** means buying commerce as separate services and assembling them rather than taking one suite; the 2020 acronym **MACH** (microservices, API-first, cloud-native, headless) named the style, though the alliance behind it had reframed its principles as Open, Composable and Connected by 2026 (Case: commercetools, Shopware and the Composable Movement).

Two mixed shapes matter for a merchant of Kestrel & Pine's size. The first is **ERP-**

centered commerce, where the ERP (enterprise resource planning: the system that keeps the books, purchasing and stock valuation) is the hub and the web shop is a front end; Odoo, JTL and PlentyONE sell this, and Microsoft documented in September 2026 that its Shopify connector is preinstalled for new Business Central sign-ups (The Back Office: ERP Integration and Who Is the Hub). The second is the **specialist group**: several specialist products under one owner, pre-integrated with one another. Katapull, a Dutch group formed on 31 March 2025 from five companies and grown to eight labels by its acquisition of the warehouse software Picqer on 7 October 2026, is the public example; its own pages claim "4,500+ customers" and "140+ marketplaces" (vendor figures, accessed November 2026). For a Growing or Omnichannel merchant such a group reduces the joins it has to build, at the price of concentrating several layers under one owner.

The fifth shape is the one most merchants actually have, and the one Kestrel & Pine chose deliberately: **a hosted platform surrounded by specialist layers**. The platform keeps storefront, cart and checkout; an OMS, a PIM, a WMS, a returns portal and a channel connector sit beside it; an integration platform joins them. The composable vendors themselves moved toward this shape in 2025–2026: commercetools launched AgenticLift on 21 January 2026 as a way to "layer agentic commerce capabilities on top of your existing commerce solution", and Adobe's documentation of 21 October 2026 describes Commerce Optimizer as a service that "can work alongside an existing Adobe Commerce or third-party commerce backend". When the champions of replacement start selling layers, the layer is the market.

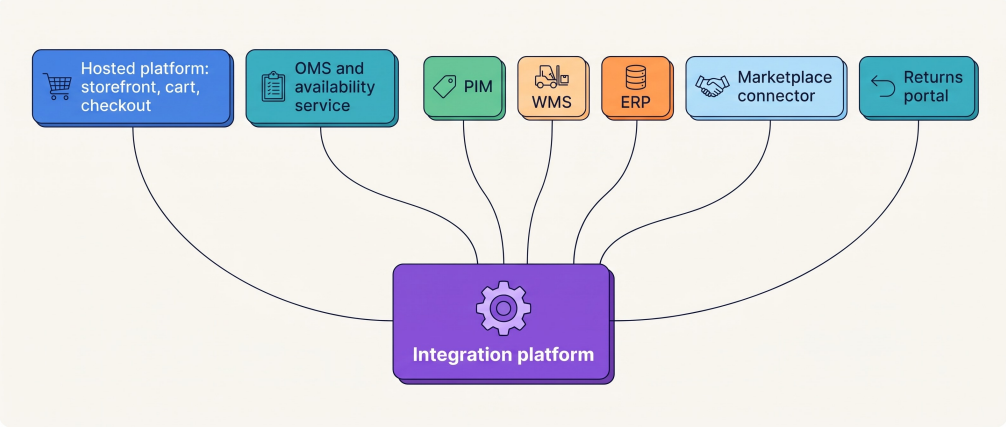

Figure 1: *Stay and surround: the hosted platform keeps the shopper-facing core, specialist systems own the facts it cannot own well, and the integration platform joins them; the merchant's job is the joins, not the core.*

What to Build: Differentiation Against Commodity

The oldest rule in this field is also the most ignored: **build what customers choose you for and nobody sells**; **buy what every merchant needs**; **assemble when the joins between systems are themselves the thing you must own**. In plain words, ask of every capability: if we did this better than anyone, would a customer notice? If not, it is a commodity, and somebody rents it cheaper than you can build it.

For Kestrel & Pine the test was short. Shoppers choose it for jackets, for the brand and for the stores. They do not choose it for its cart. A rebuilt checkout would be a large cost to reproduce a thing the platform already does at 38 orders a minute. What a shopper *does* notice is whether the jacket shown as available is actually there, whether the order's status is the same in the e-mail, the app and the store, and whether the exchange arrives in three days. Those are the joins: availability, order state, product data. That is why W1, W2 and W3 buy an availability service, an OMS and a PIM rather than a new storefront, and why W5 makes integration standards the one thing the team builds itself.

Capability at Kestrel & Pine	Differentiates?	Decision in 2027	Reasoning
Storefront, cart, checkout	No	Keep the hosted platform	Commodity at this scale; held on Black Friday
Available-to-promise per channel	Yes, to the customer	Buy as a module of the OMS; own the rules	The promise is the product; the computation is commodity
One order record across channels	Yes, to the customer and to support	Buy an OMS	A solved category; the ownership decision is the hard part
Product content in one place	Yes, to every channel and to agent readiness	Buy a PIM	A solved category
Integration contracts and reconciliation	Yes, invisibly	Build the standards; run them on the bought iPaaS	Nobody sells your ownership rules
Delivery promise computed from stock and cut-offs	Yes	Compute in the OMS from WMS cut-offs, later	Today a static promise on the page

Figure 2: *Shoppers do not choose Kestrel & Pine for its cart, so checkout is bought; they do notice whether the jacket shown as available is there, so the joins are what it owns.*

The two traps are symmetrical. **Building a commodity** is the Giant's Boots: a merchant copies the architecture of a company with a thousand engineers and spends its fourteen on rendering product pages. **Customizing a bought core until it is a build** is the Forked Platform: every vendor release becomes a project, and the merchant has the costs of building with none of the control (Apps, Plugins and Functions: Extensibility and Ecosystem Governance owns the app side of this).

Total Cost of Ownership: The Lines the Quote Leaves Out

Total cost of ownership (TCO) is everything a choice costs over its life, not what the contract says per month. Shopify's own replatforming guide (published 18 March 2026, vendor content) lists migration costs ("data migration, platform licensing, agency or partner fees, staff retraining, and downtime during transition"), ongoing costs ("hosting, security patches, maintenance, custom development, performance testing") and hidden costs ("peak-traffic surcharges, custom checkout development, additional feature licenses, content management overhead, and app or integration fees not in the original quote"). That is a good list, and it is a vendor's list: it omits the two lines a merchant pays most for. **The people who own the integrations after the agency leaves**, and **the exit**: re-exporting data, redirecting URLs, re-integrating everything that pointed at the old system.

The pricing models themselves shape the cost line (Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026 has the dated figures). A flat tier is cheap to start and forces upgrades on product or variant limits. A subscription with GMV thresholds grows with sales: Commerce.com's Scale plan charges 0.9% above $33,333 of monthly GMV, Shopware requires a paid plan above €1 million of GMV in a rolling year and prices "based on your gross merchandise value", Saleor Cloud charges 0.8% above $200,000 a month (all pricing pages accessed November 2026 and subject to change).

An enterprise contract may add "a variable platform fee based on your revenue and business model", as Shopify's Plus documentation puts it.

Payment fees are the platform's largest revenue line at Shopify and Lightspeed (Lightspeed reported transaction revenue of $815 million against subscription revenue of $371 million for the year to March 2026), and a merchant who brings its own payment provider pays a transaction fee of 0.2% to 2% on Shopify's tiers.

Pricing encodes the vendor's architecture bet: GMV pricing assumes the vendor owns checkout; per-order pricing (warehouse and channel tools) assumes it sees every order; per-request pricing (search) assumes it sits in the request path. Peak days cost money on all three.

The lines, and who usually forgets them

TCO line	What it is	Who forgets it
Subscription or license	The monthly fee per tier or contract	Nobody; it is the only line everyone prices
GMV share and transaction fees	A percentage of sales, or a fee for using another PSP	The merchant, until sales double
Payment processing	PSP fees per transaction; separate from the platform's cut	Rarely forgotten, often assumed to be the platform's problem
Apps and specialist layers	Each app or service with its own fee, often per order or per request	The merchant: 41 apps at Kestrel & Pine, 6 unused
Agency or implementation partner	Build, then retainer	Priced for the build, not for the years after

TCO line	What it is	Who forgets it
Integration	Building and running every join; the iPaaS, the connectors, the scripts	Almost always: it is "included" in the build and then owned by nobody
People	The engineers and application specialists who own the systems and the joins	Treated as a fixed overhead rather than a cost of the option
Migration	Data, URLs, integrations, retraining, and the revenue at risk during cutover (Moving Without Burning the Bridge: Replatforming and Migration)	Underpriced by the vendor, who does not pay it
Exit	What it costs to leave this option in five years	Nearly always; it is nobody's line item until it is everyone's

Kestrel & Pine's three options over three years

The table prices the three options of the February 2027 review (the consultancy's rebuild, the suite Noor costed herself, and stay and surround) over 2027–2029. Every figure is a Kestrel & Pine illustration, built from the ledger's prices where they exist (the platform at €2,300 a month; the agency at €14,000 a month; technology spend of about €2.6 million in 2026) and from labeled assumptions where they do not. Costs common to all three options (the ERP at €3,900, the WMS at €2,100, the marketplace connector at €900, the returns portal at €650 plus €0.95 a return, the iPaaS at €1,200 a month, payment processing, and the existing team) are left out, because they do not change the comparison. The table shows **incremental cost over today's run-rate**, in thousands of euros.

Line (€ thousands, 2027–2029, illustrative)	**A. Composable rebuild**	**B. Enterprise suite**	**C. Stay and surround (W4)**
Platform and new service subscriptions	480 *(assumption: commerce engine with a GMV cap, OMS, PIM, search; after go-live)*	1,350 *(assumption: 1.5% of about €30M online and marketplace sales; the trade describes suite pricing as 1–3% of GMV, which no suite vendor states publicly)*	190 *(assumption: OMS with availability module €4,500 and PIM €1,800 a month from mid-2027; the platform's €2,300 continues unchanged)*
One-time build, implementation and migration	1,200 *(assumption: storefront and B2B portal rebuilt, 26,000 variants and order history migrated, 34 integrations rewired)*	900 *(assumption: partner implementation, data migration, integrations)*	580 *(OMS €320, PIM €110, integration standards and reconciliation €150; inside the €1.1M Waypoint budget)*
Agency or partner retainer, incremental	216 *(assumption: retainer rises from €14,000 to €20,000 a month, because a headless storefront needs continuous development)*	144 *(assumption: an implementation partner's retainer of €18,000 a month replaces the agency's €14,000)*	0 *(the agency's €14,000 continues)*
Added people	765 *(assumption: three engineers for storefront and platform at €85,000 loaded)*	255 *(assumption: one engineer)*	510 *(two integration engineers, the rest of the Waypoint budget)*

Line (€ thousands, 2027–2029, illustrative)	A. Composable rebuild	B. Enterprise suite	C. Stay and surround (W4)
Revenue at risk during cutover	540 *(assumption: an 8% fall in online sales for one quarter, the Marks & Spencer figure of 2014, on €27M of D2C revenue)*	540 *(same assumption)*	about 50 *(phased introductions beside a running platform; each step reversible)*
Three-year incremental total	**about 3,200**	**about 3,190**	**about 1,330**
Exit cost at the 2029 review	High: a merchant-built storefront and every join are bespoke	High: GMV contract, suite data model, partner dependency	Moderate: the platform's export plus the OMS and PIM, each with its own export, remain

Three things stand out, and they generalize beyond the fictional numbers. **The subscription is the smallest line in every column** except the suite's, where GMV pricing turns it into the largest. **People and the agency are the largest lines** in the rebuild, and they persist after the build, which is why an eighteen-month project is a seven-year commitment. **Revenue at risk is a cost**, and the only option that avoids it is the one that never cuts over.

Today's €2.6 million makes the same point from the other side. The subscriptions in the ledger add up to roughly €200,000 a year and the agency to €168,000; the rest is people, hardware, hosting and smaller tools. A platform decision that argues about €2,300 a month is arguing about a few percent of the spend.

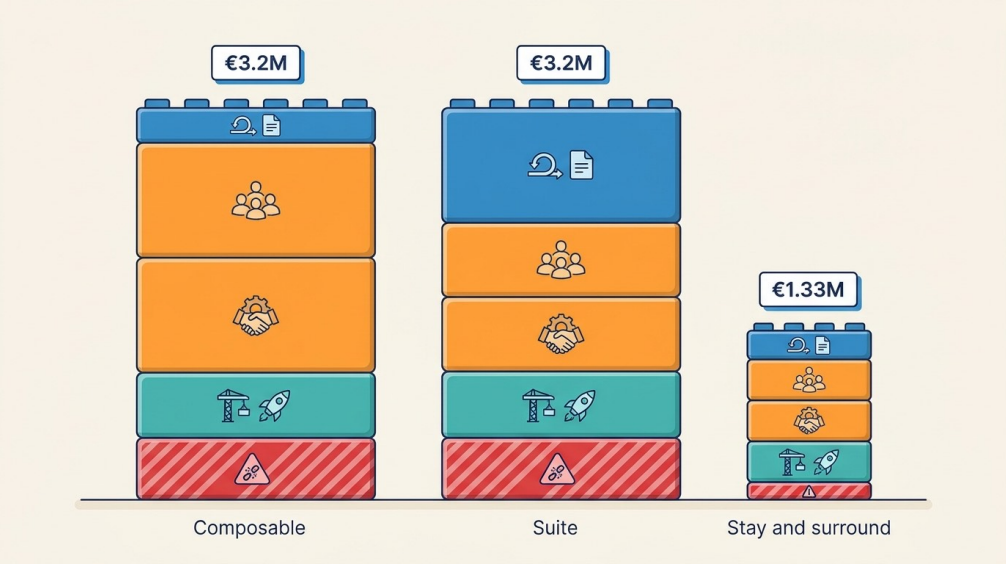

Figure 3: *Over three years the composable rebuild and the enterprise suite each add about €3.2 million against about €1.33 million for stay and surround; the gap is people, agency and revenue at risk, not subscriptions.*

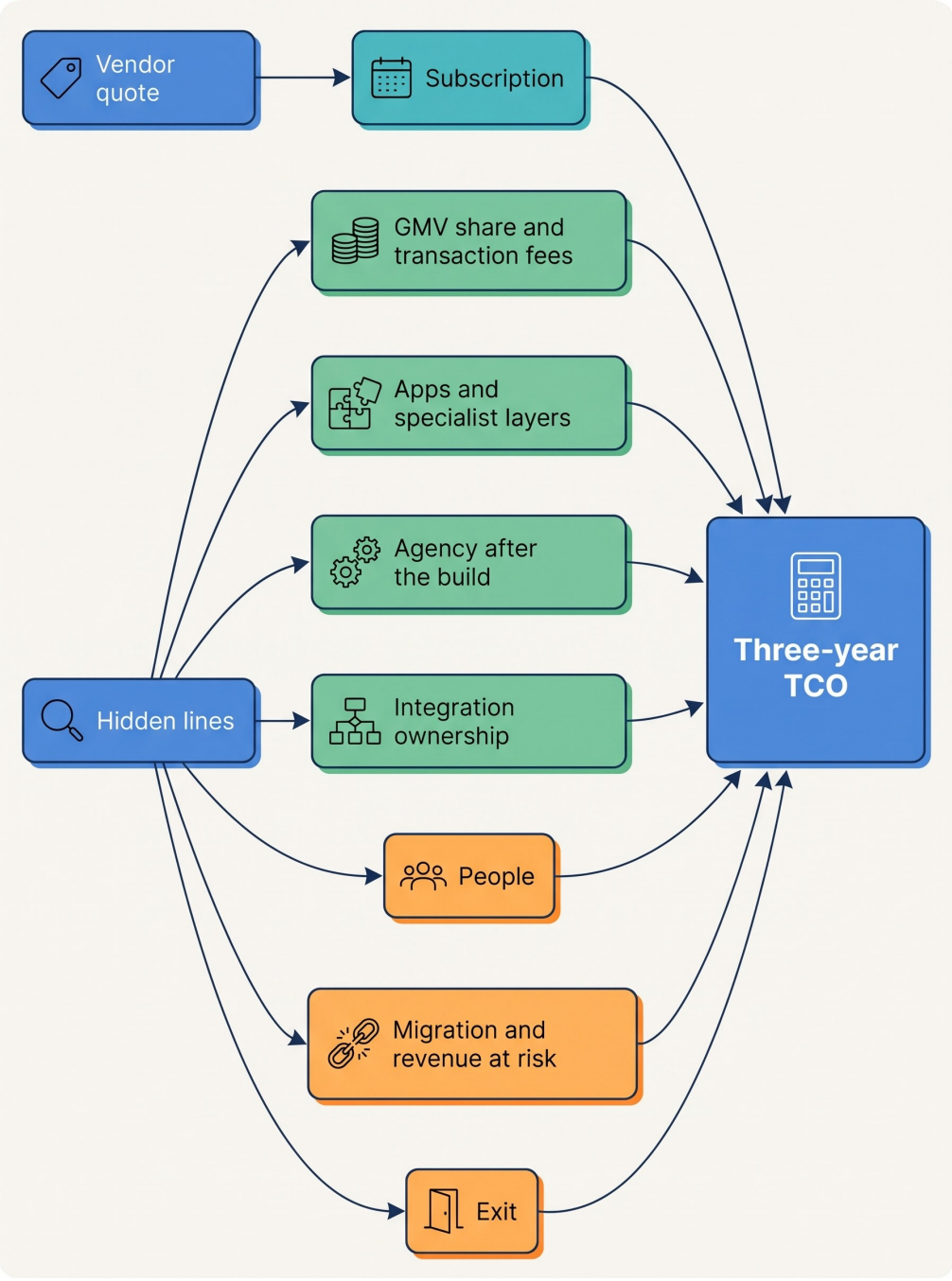

Figure 4: *Total cost of ownership: the quote prices one line; seven others, led by people, agency and exit, decide the total.*

Lock-In and the Cost of Leaving

Vendor lock-in is the condition in which leaving a vendor costs more than staying, whatever the vendor does. It is not a moral failing of vendors; it is the natural result of data, URLs, customizations and integrations accumulating in one place. **Exit cost** is the price of leaving, and it is designed in at purchase or never.

What the platform promises matters here. Shopify's standard Terms of Service (updated 20 October 2026) provide the services "on an 'as is' and 'as available' basis without any warranty", state that Shopify "does not warrant that the Services will be uninterrupted, timely, secure, or error-free", and reserve "the right to modify the Services or any part thereof for any reason, without notice and at any time". Enterprise agreements may add service levels; none is public. Read your own contract, not the marketing uptime.

Regulation now reaches part of this. The EU **Data Act** (Regulation (EU) 2023/2854, applicable from 12 September 2025) requires providers of "data processing services" to enable customers to switch to another provider or to their own infrastructure, to set out switching rights in the written contract, to allow at least 30 days to retrieve data, and, from 12 January 2027, to charge no switching fees at all. Its definition of a data processing service is broad enough to cover SaaS on its face, but whether a hosted commerce platform must therefore offer free switching from 2027 is an open legal question with no guidance or case found as of November 2026; services custom-built for one customer are exempt. The rules point at SaaS platforms; the first cases will tell (Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation).

Whatever the law settles, the exit questions are the same, and you ask them before signing:

- **Data**. Can products, variants, prices, customers, orders, gift-card balances and store credit be exported in full, in a documented format, by you, without a ticket? Password hashes usually cannot move; plan for a reset.
- **URLs**. Do you control the domain and can you emit redirects from every old product and category URL? Marks & Spencer's 2014 relaunch is the cautionary case in Moving Without Burning the Bridge: Replatforming and Migration.
- **Customizations**. Who owns the theme code, the functions, the scripts? If the agency does, the agency is part of the exit.
- **Apps**. Each app holds data (subscriptions, loyalty points, reviews) with its own export, or none.
- **Integrations**. Every connector that points at the platform must be re-pointed; the integration inventory (APIs, Events, Files and Middleware: How Commerce Systems Talk) is the exit's work list.
- **Contract**. Term, notice, GMV floor, what happens to the fee if you move half your traffic away during a strangler migration.

The Lynchpin is lock-in by another name: a capability customers depend on, running through one vendor, one connector, one agency or one person, with no exit plan. Kestrel & Pine had three in January 2027: the EDI consultant, the agency's scripts, and the ERP partner's platform-to-ERP connector. None was a platform decision; all were consequences of one.

Decision Criteria

Five criteria decide more than any feature list. They are the first page of the platform decision card in the Toolkit: Cards and Templates for Commerce Architecture Decisions.

Criterion	What to ask	What it pushes toward
Business model	D2C only, or wholesale, stores and marketplaces too? (Who Sells What to Whom: Business Models and the Economics That Shape Architecture)	A single channel fits a hosted platform; several channels with several stock locations need an OMS whatever the core
Scale stage	Starting, Growing, Omnichannel or Group? (What Changes at Each Scale: From First Store to Group)	Starting and Growing: hosted or ERP-centered; Omnichannel: surround the core; Group: suite versus composable becomes a real question
Team	How many engineers, and do they own integrations today?	A composable assembly needs a team that owns every join; fourteen people with an agency do not have one
Differentiation	What would a customer notice if you did it better than anyone?	Build only that; buy the rest
Time and risk	When is the next peak, and what is the cost of a failed cutover?	Phased introductions over big-bang replacements

Trade-Offs

	Suite	**Hosted platform plus apps**	**Composable assembly**	**Custom build**	**Specialist group**
Time to live	9–18 months with a partner *(industry practice)*	Weeks to months	12–24 months *(industry practice)*	Years	Months per label
Integration burden on the merchant	The joins to ERP, WMS, carriers	The joins, plus governance of apps	All of it	All of it	Reduced inside the group; the joins to everything else remain
Upgrade burden	Vendor, if the merchant has not forked the core	Vendor	Per service, continuous	Merchant	Vendor per label
Flexibility	High within the suite's model	Bounded by the platform's extension points	Very high	Unlimited	Bounded per label
Cost shape	GMV-linked, large partner line	Subscription plus payment share plus apps	Subscriptions with GMV caps, large people line	People	Several subscriptions
Exit	Hard: data model and partner	Moderate: export plus repointing apps and integrations	Hard: everything bespoke	Hardest	Per label, but one owner across layers

	Suite	Hosted platform plus apps	Composable assembly	Custom build	Specialist group
Fits when	Group stage with several brands and a platform team; B2B-heavy models	Starting through Omnichannel when checkout is not the differentiator	Group stage with a team that owns joins, or a vendor	The platform is the product	Growing and Omnichannel merchants in the group's market

My recommendation follows the requirements, so it comes with conditions. **If checkout and storefront are commodities for you, keep a hosted platform and surround it**, buying an OMS, a PIM and a WMS when the business model demands them. **If you have several brands, a platform team and a B2B-heavy model, evaluate a suite and a composable assembly against each other on the people line, not the feature list. Build only a capability that customers choose you for and nobody sells**, and even then, build it beside the bought core. The condition that changes all of this is the team: an assembly is a commitment to own every join forever, and a merchant that cannot name the owner of its integrations today should not sign up to own thirty more.

Across Scale Stages: When It Fits and When It Limits

Stage	Typical choice	What becomes necessary	Warning signs the choice is outgrown
1 — Starting	A hosted platform or a website builder with commerce; a 3PL; the platform's payments	Nothing beyond the platform's export and your own domain	Variant or product limits on the tier; a second channel keyed in by hand

Stage	Typical choice	What becomes necessary	Warning signs the choice is outgrown
2 — Growing	Hosted platform or ERP-centered package; an agency; first marketplace tool, accounting or ERP link, 3PL feed	The integration inventory; a named owner for each connector; the exit questions asked at every new contract	Apps writing stock; the agency owns every script; pricing tier thresholds crossed (platform GMV, PSP volume, per-order quotas) at the same time
3 — Omnichannel	A hosted platform or suite surrounded by OMS, PIM, WMS, returns and channel tools; an iPaaS; in-house integration engineers	Ownership per fact (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record); the hubs bought, the joins owned; a TCO with the people and exit lines	Replatforming proposed to fix problems that live in the joins; the same incident recurring across channels
4 — Group	Suite or composable assembly with a platform team; shared services decided brand by brand	The multi-brand platform decision (What Changes at Each Scale: From First Store to Group); vendor-scale concerns such as tenancy and extensibility	One stack forced on brands before the benefit is measured, or brands kept apart that should share

The judgement across the stages: **the right time to replatform is almost never the time the problems appear, because the problems appear in the joins**. Kestrel & Pine's stage-3 incidents (stock, orders, prices, products) were all ownership problems,

and ownership is decided by contracts and hubs, not by the storefront. A platform change becomes rational at stage 4, when several brands or a vendor-scale concern (tenancy, extensibility, a team that can own an assembly) makes the core itself the limit.

In Practice

The vendors' own turn, 2025–2026. The clearest public evidence that layers beat replacement for mid-sized merchants comes from the vendors who sell replacement. commercetools, the founding MACH vendor, cut about 10% of staff on 26 February 2025 after missing growth targets, then launched AgenticLift on 21 January 2026 as a layer for "your existing commerce solution". Adobe sells Commerce Optimizer beside a kept backend (documentation of 21 October 2026). Commerce.com reported on 6 August 2026 that revenue was up 0.1% and that B2C replatforming "remains softer than historically". The MACH Alliance's own survey of 600 technology decision-makers (18 February 2026) found 37% citing integration complexity as a primary concern, a finding against the surveyor's interest and the cost line the composable promise omitted.

Vendors retreat from what they do not do well. Shopify completed the sale of the majority of Shopify Logistics to Flexport on 6 June 2023 for a 13% equity stake, more than three years ago but still the point: expect your platform vendor to be a payments company first and to push logistics, returns and EDI to partners, which keeps those layers' contracts in your hands at every stage above Starting. Lightspeed's strategic review, closed on 6 February 2025, named North American retail and European hospitality as its growth engines; read a vendor's strategic review as a signal about which products will be funded.

Reading a vendor claim. Shopify's enterprise page (accessed November 2026) claims a "33% lower average total cost of ownership" for a surveyed sample of organizations with GMV from $50 million to $500 million. It may be true for the sample; it says nothing about your people line or your exit. Ask for the unit, the date, the definition and the baseline before comparing any two vendors (Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026).

A build decision from the public record. On 7 August 2009 Target announced it would "build and manage its own online platform", taking back e-commerce technology, fulfillment and guest services from Amazon, which had run Target.com since 2001. The platform launched in 2011 and crashed on the Missoni collection launch of 13 September 2011; the cause was never confirmed publicly, and the sources are fifteen years old. Moving Without Burning the Bridge: Replatforming and Migration takes the migration side; the build-or-buy lesson is narrower: a retailer of Target's size had reasons to own its platform (a competitor ran it), and the cost of the decision was paid not in the build but in the first peak.

Noor records W4 as an **architecture decision record (ADR)** using Michael Nygard's November 2011 format: context, decision, status and consequences. Keeping the reasons

beside the decision lets a later team see whether its assumptions still hold and what would justify changing course.

The context is a platform that held while integrations failed, and a fourteen-person team with an agency. The decision keeps checkout and adds order, availability and product management. The consequences remain explicit: transaction fees and extension limits continue; the merchant owns the integrations; the agency focuses on the storefront. Separate authoritative records and exports preserve options for a later move.

The 2029 review can compare those assumptions with what happened. It does not have to reconstruct why the decision was made.

When It Goes Wrong

The platform decision fails in slow ways, and the detection is organizational before it is technical.

- **The Lynchpin surfaces at renewal**. The agency owns the theme, the scripts and the knowledge; the renewal arrives with a price. *Detect:* no engineer at the merchant can deploy the storefront. *Recover:* ownership of code and credentials in the contract; a second party able to deploy; the ownership map from Who Builds and Who Owns: Teams, Vendors, Agencies and Governance.
- **The Forked Platform turns buy into build**. An upgrade is skipped because a customization would break; two years later the platform is three versions behind and the vendor's support has lapsed. *Detect:* the last upgrade date, and the list of core files touched. *Recover:* move customizations to supported extension points, one per release; if that is impossible, the real decision is whether to rebuild on the current version or leave.
- **The Giant's Boots are bought with a brochure**. A composable assembly is signed because a far larger retailer has one. *Detect:* the proposal names no owner for the joins after go-live. *Recover:* price the people line; if the merchant cannot staff it, do not sign.
- **The pricing cliff**. Growth crosses a GMV threshold, a per-order quota and a PSP volume tier in the same year; the subscription that was the smallest line doubles. *Detect:* the thresholds written next to the forecast. *Recover:* negotiate at the stage change, not after the invoice.
- **The vendor changes direction**. A strategic review, an acquisition, a retired product (Zalando's Connected Retail technology will be discontinued by 30 June 2027, announced April 2026, with fourteen months' notice; Magento 1's end of life on 30 June 2020 forced merchants to move or run unsupported software). *Detect:* read the vendor's investor and press pages twice a year. *Recover:* the exit questions above, answered before they are urgent.

- **The quote was right and the TCO was wrong.** Integration is "included" in the build and then owned by nobody; incidents appear in the joins a year later. *Detect:* the integration inventory has owners missing. *Recover:* W5: every integration has an owner, a contract, idempotent handling and a reconciliation job.

What to Take Away

Established.

- The visible subscription is a minority of total cost; vendors' own guides (Shopify, March 2026) list migration, ongoing and hidden costs, and pricing models with GMV thresholds and transaction fees are documented on vendor pages as of November 2026.
- The composable vendors and suites themselves sell layers beside an existing core as of 2026 (commercetools AgenticLift, January 2026; Adobe Commerce Optimizer, documented October 2026), and the MACH Alliance's own 2026 survey names integration complexity as the top concern.
- The EU Data Act (Regulation (EU) 2023/2854) applies from 12 September 2025 and prohibits switching charges from 12 January 2027; whether it covers hosted commerce platforms is unsettled.
- Decision records (Nygard, 2011) are the lightweight way to keep a platform decision reviewable.

Depends on context.

- Stay and surround is right when checkout and storefront are commodities for the merchant and the problems live in the joins; it is wrong when the core itself limits the business (several brands, vendor-scale tenancy or extensibility needs, a B2B model the platform cannot express).
- A composable assembly fits a team that owns every join and a business whose differentiation lives in the assembly; it is the Giant's Boots for a fourteen-person team with an agency.
- GMV-based pricing aligns the vendor with growth and raises exit cost; it favors high-margin D2C and punishes large, low-margin B2B orders, which is why B2B terms are negotiated separately.
- The specialist-group model reduces joins for merchants in its market and concentrates several layers under one owner; whether that is a benefit depends on the exit terms per label.

The author's synthesis.

- I price every platform option on three lines first, people, agency and exit, and only then on the subscription, because the first three decide the total and the last one decides the meeting.
- I treat "build" as a decision about ownership rather than code: the thing worth building at a mid-sized merchant is the rules for who owns which fact and how the copies are reconciled, which no vendor sells.

Questions to Ask

1. What exactly is broken, and does it live in the core or in the joins between systems?
2. Which capability would a customer notice if we did it better than anyone? Is anyone proposing to build something else?
3. What does each option cost over three years including agency, people, migration, revenue at risk and exit?
4. Who owns each integration the day after go-live, by name?
5. Which pricing thresholds (GMV tier, per-order quota, PSP volume) will growth cross in the next two years, and what happens to the fee?
6. Can we export products, orders, customers and gift-card balances ourselves, in a documented format, today?
7. Who owns the theme code, the scripts and the credentials, and could a second party deploy them?
8. What does the vendor's contract actually promise about availability, deprecation notice and modification without notice?
9. Is there a decision record, and does it name the date and the conditions under which the decision is revisited?

The decision to stay is only half a decision. Kestrel & Pine still has to introduce an OMS and a PIM beside a running platform without creating the very duplicates and gaps the hubs exist to remove. How to move systems without a single cutover date is the subject of Moving Without Burning the Bridge: Replatforming and Migration.

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

42. Moving Without Burning the Bridge: Replatforming and Migration

IN THIS CHAPTER, YOU WILL: Learn to plan a platform or hub migration as a sequence of reversible steps rather than one cutover date, with the data, integrations and rollback each handled on purpose.

KEY POINTS:

- **The documented failures were not feature failures.** Target's 2011 in-house platform went down at its first peak (the cause was never confirmed), Finish Line reverted in 2012 after seventeen days, Marks & Spencer lost 8.1% of online sales in the quarter after its 2014 relaunch. Load, performance, timing and customer habit broke them, not missing features, and one recovered only because the old platform was still running.
- **Migrate along seams, with a bridge that stays up.** The strangler approach moves one channel, one capability or one slice of traffic at

a time behind a router or a copy of the flow, runs old and new side by side with a reconciliation that proves they agree, and keeps the old system operable until the new one has met real load.
- **Data migration is five migrations**. Products (identifiers must not change), customers (consent moves, passwords do not), orders (history is read, not re-created), gift cards and credit (liabilities that must balance to the cent) and URLs (every old address redirected) each need their own plan and check.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Burned Bridge** — everything moves to a new platform on one date, with no way back; this chapter exists to prevent it.
- **The First Snowfall** — a migrated platform, or a newly introduced hub, meets real load for the first time on the busiest day of the year.
- **The Spaghetti Junction** — thirty-four integrations, each with its own owner or none, that all have to be re-pointed when the system in the middle changes.

During a move to a new order system, both the old and new systems may be able to send work to the warehouse. Let both send the same order and two parcels can leave; stop the old one too early and orders can be missed. A migration changes who is allowed to act while the shop keeps trading. Each step needs a clear transfer of responsibility and a way to recover if that transfer fails.

Identify the fact whose owner changes and every integration that reads or writes it. Decide how the new system will be populated, how its answers will be compared, when it may act and what happens if the transition fails. Smaller, reversible steps expose those questions before the whole business depends on the new path.

Kestrel & Pine, the fictional outdoor brand, keeps its commerce platform after a March 2027 review (Build, Buy or Assemble: Choosing and Changing Platforms). It still introduces **product information management (PIM)** in Q3 and an **order management system (OMS)** by Q4. Those migrations must preserve roughly 800 daily online orders.

Where we are in the order's journey. A jacket order can be copied to a new OMS for comparison before that system is allowed to send warehouse work. Product names, sizes and colors can similarly be prepared in a PIM before publication switches. The running shop needs one clear owner at each stage.

Setting the Scene: Two Hubs Beside a Running Shop

In May 2027, a first warehouse-to-platform stock comparison finds differences for **2.3% of active variants**, many caused by three stock-writing apps. Head of Engineering Noor

Visser treats that as a warning: loading a new hub from existing records will also load their disagreements unless somebody resolves them.

The proposed migration plans name dates, not failure behavior. The agency would load historical orders over a weekend, redirect order export Monday and switch the warehouse feed Tuesday. The ERP partner would export the catalog, import it into the PIM and redirect storefronts. Neither explains how to detect an order sent twice or not sent at all.

Noor's plan, which the rest of this chapter follows, had three rules. **Every step is reversible for at least one peak**. **Old and new run side by side with a job that proves they agree before new takes over**. **The smallest channel moves first**. The marketplace channel, 43,000 orders a year, about 120 a day, was the first to be routed through the OMS; the D2C storefronts with 293,000 a year came after; the EDI and manual wholesale orders last, in December, with the stores deferred to the 2028 POS evaluation. Black Friday 2027 sat in a change freeze in the middle of it.

Reasons to Replatform, and Reasons That Are Not

A platform change is justified when the core itself limits the business and no layer beside it can fix that. It is not justified when the problems live in the joins, which, as Build, Buy or Assemble: Choosing and Changing Platforms argues, is where most Omnichannel-stage problems live.

Reason given	Is it a reason?	What to check first
"Stock is wrong across channels"	No: an ownership and availability problem (Is It in Stock? Inventory, Availability and the Promise You Can Keep)	Who is the system of record for stock, and how do the others learn about changes
"Support can't find an order"	No: an order-hub problem (The Order Record: Lifecycle, State and the Order Management System)	Whether an OMS beside the platform solves it
"The site was slow on Black Friday"	Maybe: find the bottleneck	Usually the integrations, not the platform (Black Friday Every Year: Peak Load, Resilience and Recovery)

Reason given	Is it a reason?	What to check first
"The vendor is ending the product"	Yes, on the vendor's date	The notice period; the export; Magento 1's end of life on 30 June 2020 forced merchants to move or run unsupported, non-compliant software
"We have customized the core until it cannot upgrade"	Yes, but the choice is rebuild-on-current-version versus leave	The Forked Platform (Apps, Plugins and Functions: Extensibility and Ecosystem Governance)
"The platform cannot express our B2B model"	Yes, if a B2B layer beside it cannot either	Price lists, contract prices, punchout, approvals (Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals)
"A competitor runs our platform"	Yes, strategically	Target, 2009: see below
"The agency says so"	No, by itself	Who is paid for the build, and who owns the joins after it (Who Builds and Who Owns: Teams, Vendors, Agencies and Governance)

Three Documented Failures, and What They Share

The public record of replatforming is thin, because merchants do not publish post-mortems. Three cases are documented well enough to use. All three are older than a decade, so they describe replaced systems; the lesson is in the shape, not the software.

Target, 2009–2011. Amazon had run Target.com since 2001. On 7 August 2009 Target announced it would "build and manage its own online platform", taking over its "e-commerce technology platform, fulfillment and guest services operations", to launch before the 2011 holiday season.

The new site launched in August 2011. On 13 September 2011, weeks later, the launch of the Missoni for Target collection crashed Target.com; Target's own banner read "we are

suddenly extremely popular", and the site was down most of the day. A second outage of about two and a half hours followed on 25 October 2011, for which Target gave no reason.

Press at the time disagreed on whether the in-house platform or the Missoni demand was the cause, and Target never confirmed it, so the verified chain is the decision, the launch and the two outages, not the root cause. The shape is the First Snowfall on a new platform: the first real peak was the first real test.

Finish Line, 2012. The American footwear retailer switched to new e-commerce site technology on 19 November 2012, the week before Thanksgiving. Seventeen days later it reverted to its former platform, "which it had kept operable", according to Retail Dive's report of 8 January 2013. The switch cost about $3 million in quarterly sales, "despite a 25% increase in online sales for the quarter". Finish Line is the one case in this list with a happy ending, and the reason is in one clause: the old platform had been kept operable. The bridge was not burned.

Marks & Spencer, 2014. M&S launched a new website in early 2014. Its first-quarter trading statement of 8 July 2014 reported that M&S.com sales fell 8.1% in the quarter; the company had warned in May that the new site would take four to six months to "settle in", and said it had since seen "a gradual improvement in sales performance". Retail Week's commentary of 10 July 2014 called it "a timely reminder of the risks involved in major replatforming programmes". The shape here is different: nothing crashed. Customers had to re-register, the search engines had to re-learn the site, and habit broke. A migration can succeed technically and still cost a quarter.

The one that avoided it. Hobbycraft, a British crafts retailer, delayed its website relaunch past Christmas 2012 because it was "too risky to launch the website in peak trading" (Retail Week, 19 February 2013). That is a sourced example of the simplest rule in this chapter: never cut over into a peak.

Case	Date	What failed	What it cost	Was there a way back?
Target (US)	Launch August 2011; outages 13 September and 25 October 2011	Capacity at the first peak (cause not confirmed publicly)	A day of lost sales at a high-profile launch; a second outage; the Target.com president left weeks later	No public evidence of one

Case	Date	What failed	What it cost	Was there a way back?
Finish Line (US)	Switched 19 November 2012; reverted 17 days later	Performance of the new site in the holiday run-up	About $3 million of quarterly sales	**Yes**: the former platform had been kept operable
Marks & Spencer (UK)	Relaunch early 2014; results reported 8 July 2014	Customer habit, search visibility, a four-to-six-month "settling in"	M&S.com sales down 8.1% in the quarter	Not reported; the company chose to settle in rather than revert
Hobbycraft (UK)	Relaunch deferred past Christmas 2012	Nothing: the launch was moved off the peak	A delay	Not needed

What the cases share is the pattern this book calls **The Burned Bridge**: everything moves on one date, into or near a peak, with the new system meeting real load for the first time in public, and with no rehearsed way back. Finish Line is the Burned Bridge without the burning, and it recovered. The rest of this chapter is the engineering of that clause, "kept operable", generalized.

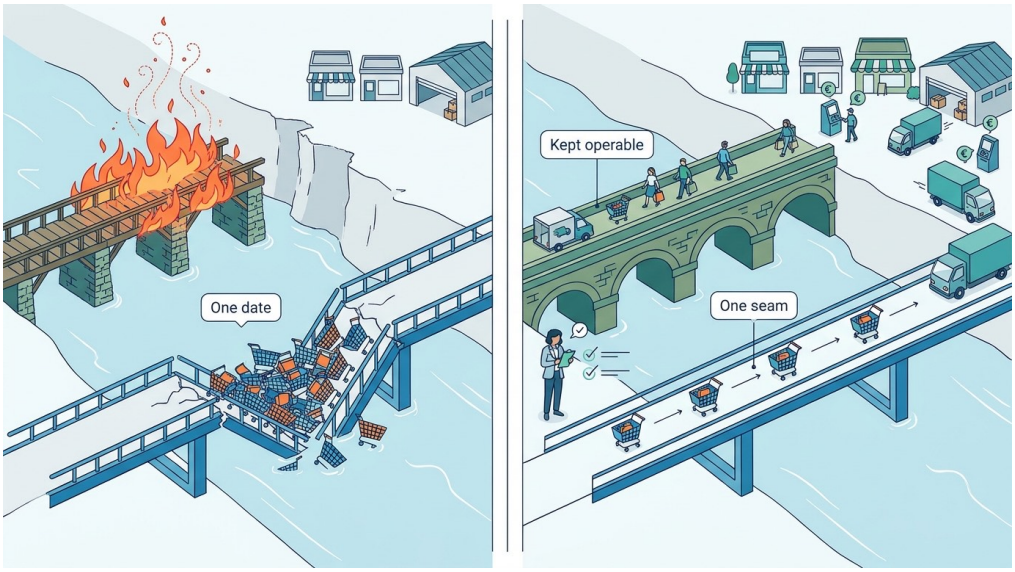

Figure 1: *Finish Line could turn back in 2012 because its old platform had been kept operable: a migration that keeps the old bridge standing until the new one has carried real load always has a way back.*

The Strangler Approach for Commerce

Martin Fowler's note on the **strangler fig** (first written in 2004, restated 22 August 2024) gives the alternative in one sentence: a gradual approach to modernizing a serious system, because plans to replace one wholesale tend to "go down in flames", replacement takes long, users cannot wait for features, and existing behavior is hard to pin down and partly unwanted. The name is from a fig that grows around a host tree until the tree is gone. In plain words: **put something in front of the old system, route one piece of its work to the new system, check, route the next piece, and switch the old system off only when nothing is routed to it any more**.

Thoughtworks' *Patterns of Legacy Displacement* (20 July 2021, revised 5 March 2024) names the moves. **Transitional architecture**: temporary components built only to make the migration safe, and removed afterwards. **Event interception**: capture the changes flowing into the old system and send a copy to the new one. **Divert the flow**: route a slice of traffic or a business flow to the new system. **Legacy mimic**: make the new system look like the old one to the integrations that still expect it. **Dark launching**: run the new path without showing its result. **Canary release**: a small share of traffic first.

The authors warn that defining "as is" functionality "was a huge effort" and that the alternatives to seam-based decomposition "all too often result in Feature Parity and Big Bang releases". **Feature parity**, rebuilding everything the old system did before

switching, is the plan that produces the Burned Bridge, because it postpones every risk to one date.

In commerce the seams are visible and there are three kinds.

- **By channel**. Move the B2B portal first, or the German storefront, or the marketplace orders. Channels are natural slices because each has its own entry point and its own volume, and a merchant can choose the smallest.
- **By capability**. Move search first, or returns, or product content, or availability. This is the stay-and-surround program: each hub is a capability moved from the platform (or from spreadsheets and the ERP) to a new owner.
- **By traffic slice**. A share of sessions, a region, a customer segment. This is the canary, and it needs a router in front of the storefront.

The public example of all three is Zalando's rebuild of its Fashion Store from the monolith it called "Jimmy", described in its engineering posts of 16 October 2015 and 5 October 2016. A team put a Go-based router, Skipper, in front of the shop and migrated the page fragment by fragment: header and footer first, then product pages and wish lists, each served by a new team behind the router while the monolith served the rest. The evidence is ten years old and describes a system that has changed since, but the shape is the strangler in production at a large retailer. The same company's wind-down of its Connected Retail technology, announced 23 April 2026 for discontinuation by 30 June 2027, shows the other end: a channel retired with fourteen months' notice rather than switched off.

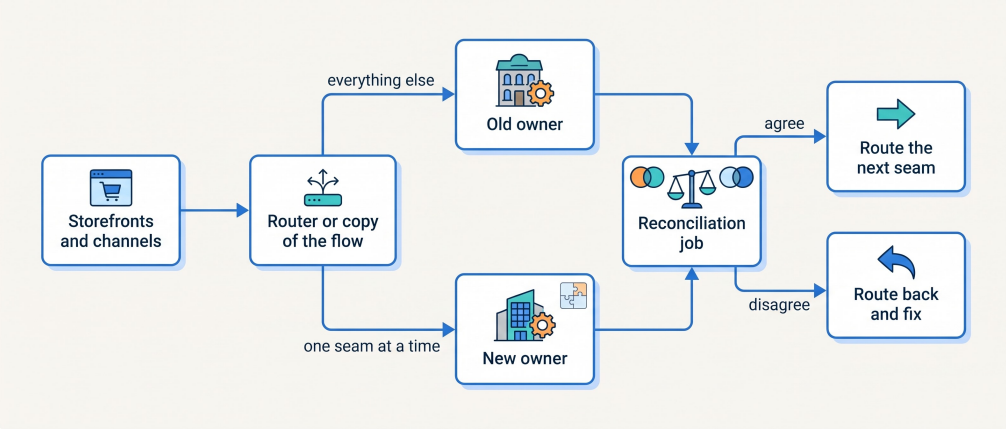

Figure 2: *The strangler in commerce: a router or a copied flow sends one seam at a time to the new owner, a reconciliation job proves old and new agree, and disagreement routes back instead of forward.*

How the Pieces Fit During a Migration

A migration is a period in which **two systems hold the same fact and only one of them is allowed to be right**. The design question is which one, for which fact, for how long, and how the others find out. The requirements change by phase.

Phase	Who owns the fact	Freshness	Consistency check	Latency allowed	What must stay available
Shadow (new system receives a copy; nobody reads it)	Old	Copy within minutes *(assumption)*	Daily reconciliation of counts and samples	Any	Everything on the old path, untouched
Parallel run (both compute; old result is used)	Old	Both within the same minute	Every record compared; differences listed by cause	Any for the new path	Old path; new path may fail silently
Divert one seam (new result used for one channel or capability)	New, for that seam only	Live	Reconciliation per seam; the old path still receives the copy	Production latency for that seam	The old path can take the seam back within an hour *(assumption)*

Phase	Who owns the fact	Freshness	Consistency check	Latency allowed	What must stay available
Divert all	New	Live	Reconciliation continues for one peak	Production	The old system stays operable, read-only or idle, until the peak has passed
Decommission	New	—	Final reconciliation; archive	—	Exports from the old system kept

Two mechanisms carry these phases, and they are easy to confuse.

A parallel run means both systems process the same input and their results are compared, while only one result is acted on. It is the safest way to learn whether the new system agrees with the old one, and it is cheap when the input can be copied (an order created, a product changed). Its weakness is that it finds disagreements, not their causes; the reconciliation job must say *which* records differ and *why* (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation).

Dual writes mean one actor writes the same change to both systems. They sound like a parallel run and are the opposite in risk: two writers to two stores, with no transaction across them, produce exactly the partial failures this book spends Part VI on. If the write to the new system fails and the write to the old one succeeds, the systems diverge and nobody is told. Kestrel & Pine already lived this in March 2026, when a pre-order app and the WMS feed both wrote stock in the platform and left two weeks of phantom reservations on 600 variants. **Prefer one writer and a copied flow** (event interception: the change goes to the owner, and an event carries a copy to the other) over two writers, and where dual writes are unavoidable make them idempotent (a repeated write has the same effect as one) and reconciled (APIs, Events, Files and Middleware: How Commerce Systems Talk).

Data Migration Is Five Migrations

A platform's data is not one export. Each kind of data has its own owner, its own identifiers, its own legal constraints and its own way of going wrong. Plan them

separately.

Data	What moves	The trap	The check
Products and variants	3,400 products, 26,000 variants, attributes, media, translations, categories	Identifiers that change: a SKU or GTIN re-keyed in the new system breaks every integration, feed and barcode (What Are We Selling? Products, Variants and the Catalog)	Every variant in the new system carries the same SKU and GTIN; counts per status match; a sample of fifty variants compared field by field
Prices and promotions	Price lists per market and customer group, active promotions, effective dates	A price-list sync that fails silently, as it did for nine days in October 2026 (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax)	Every active SKU's consumer and wholesale price equal in old and new; a report of differences before go-live
Customers	Accounts, addresses, consent, company accounts and price lists for B2B	Passwords are stored as one-way hashes and usually cannot move; consent must move with its proof; a deletion request mid-migration must reach both systems (Whose Data Is It? Privacy, Consent and Retention)	A password reset flow tested before go-live; consent flags counted; the identity map kept (One Customer, One Product: Identity and Master Data Across Systems)

Data	What moves	The trap	The check
Orders	History for support and returns; open orders in flight	Re-creating history fires every side effect again: notifications, stock reservations, invoices. Orders are read, not replayed (The Order Record: Lifecycle, State and the Order Management System)	History imported read-only with its original numbers; open orders listed and tracked by hand through the cutover window
Gift cards, store credit, loyalty points	Balances, which are liabilities on the balance sheet (Discounts, Coupons and Loyalty: Promotions Without Chaos)	A balance lost is money owed to a customer with no record; a balance duplicated is money given away twice	Sum of balances in old equals sum in new to the cent, signed off by finance
URLs and search presence	Every product, category and content address	A redirect map that is incomplete loses rankings and customers who have bookmarked a page; the M&S quarter	Every old URL returns a redirect to its new equivalent; crawl the old sitemap after go-live

Data	What moves	The trap	The check
Integrations and apps	The 34 flows that read or write the old owner	Each one points at the old endpoint until somebody changes it: the Deprecated Endpoint of February 2027 in reverse (Promises Between Systems: API Contracts, Webhooks and Versioning)	The integration inventory with an owner and a cutover task per flow; a legacy mimic where a flow cannot change in time

Figure 3: *A platform's data is not one export: products, customers, orders, gift cards and URLs each move separately, with their own trap and their own check.*

The order history deserves one more sentence. It is tempting to "migrate the orders" by creating them in the new system. Every order created in a commerce system fires something: a confirmation e-mail, a reservation, a fulfillment request, an invoice. Importing 293,000 D2C orders as new orders is 293,000 e-mails. **History is imported as history**, flagged, read-only, with its original numbers and dates, or it is left in the old system behind a read-only link for support.

Cutover and Rollback

A **cutover** is the moment a seam's traffic is routed to the new owner. Even in a strangler migration there are cutovers; they are just small. Each one needs the same five things.

1. **A window and a freeze.** A time with low volume (not a Friday evening in November), a change freeze around it, and a stop on competing changes to the integrations.
2. **Go and no-go criteria written down beforehand.** The reconciliation shows fewer than an agreed number of differences; the parallel run has lasted an agreed period; the rollback has been rehearsed.
3. **A rehearsed rollback.** Not a plan: a rehearsal, timed, in which the seam is routed back and the records created in the new system during the window are reconciled into the old one. Finish Line's "kept operable" is the minimum; a rehearsal is the standard.
4. **Continuity of identifiers.** The order number family does not restart; the last order number in the old system and the first in the new are recorded; a SKU is the same SKU.
5. **A watch.** Orders per minute, failed messages, reconciliation differences and support tickets watched for the first hours and the first peak (Black Friday Every Year: Peak Load, Resilience and Recovery).

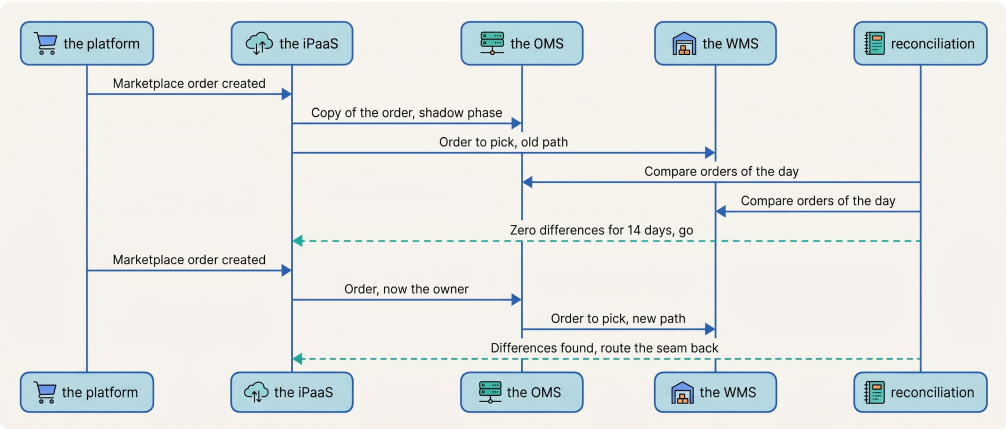

Figure 4: *One seam's cutover at Kestrel & Pine: marketplace orders are copied to the OMS in shadow, reconciled against the WMS for two weeks, then routed through the OMS; a difference routes the seam back rather than forward.*

The rollback has a limit that must be stated: **the longer the new system owns a seam, the more records exist only there.** After a week of the OMS owning marketplace

orders, routing back means reconciling a week of orders into the platform's view. The parallel run narrows that gap, because the old path still receives the copy, but it does not close it. That is why the old system stays operable through one peak and not forever: the window for a cheap rollback closes on its own.

Migrating the Integrations

The system in the middle changes; the thirty-four flows around it do not know. At Kestrel & Pine fourteen of the thirty-four touch orders, stock or product data and have to be re-pointed when the OMS and the PIM arrive (an illustrative count). Three rules, all from Waypoint W5, make that a list of tasks rather than a season of incidents.

- **The inventory first.** Every integration listed with its owner, its direction, its fields, its frequency and the system of record it reads or writes (APIs, Events, Files and Middleware: How Commerce Systems Talk). A flow with no owner gets one before the migration starts, or it gets switched off.
- **A legacy mimic where a flow cannot change in time.** If the agency's script that reads orders from the platform cannot be rewritten before the OMS takes over, the OMS (or the iPaaS) exposes the same shape at the old address for one more season. The mimic is transitional architecture: it has a removal date.
- **Deprecation with notice, in both directions.** The merchant retires its own old endpoints with a published date, as the marketplace connector did in February 2027, and this time the agency script is on the list (Promises Between Systems: API Contracts, Webhooks and Versioning).

Across Scale Stages: When It Fits and When It Limits

Stage	What a migration usually is	What becomes necessary	Warning signs
1 — Starting	Moving from a website builder to a hosted platform; changing 3PL	The platform's export and import; the domain in the merchant's name; a redirect map	The old URLs die; the customer list moves without its consent proof

Stage	What a migration usually is	What becomes necessary	Warning signs
2 — Growing	A first ERP or accounting link; a marketplace tool; a new PSP	A cutover window outside the peak; the integration inventory, because the first connectors are the ones that break	A cutover planned for November; connectors re-pointed by an agency with no owner afterwards
3 — Omnichannel	Hubs introduced beside the platform: OMS, PIM, WMS, availability; a channel re-routed	Strangler seams by channel and capability; parallel runs with reconciliation; the old owner kept operable through one peak; data migrated as five migrations	One weekend planned for an OMS; dual writes to stock; order history "migrated" by re-creating orders
4 — Group	Replatforming a brand; consolidating brands onto one stack; an acquisition's systems	Brand-by-brand migration order (What Changes at Each Scale: From First Store to Group); a platform team that owns the router and the transitional architecture; feature parity resisted per brand	Every brand moved on one date; one brand's customizations rebuilt for all

The judgement: **the migration technique does not change with scale, but the thing being migrated does**. A Starting-stage merchant moves a shop; an Omnichannel merchant moves ownership of a fact; a Group moves brands. The strangler, the parallel run and the kept-operable old system apply at every stage; what grows is the number of seams and the number of integrations each seam drags along.

In Practice: The OMS and PIM Introductions at Kestrel & Pine

The figures below are Kestrel & Pine illustrations; the dates fit the ledger's Waypoint timeline (PIM in Q3 2027, OMS by Q4 2027).

The PIM migration, July–September 2027. The first load combines platform, ERP, spreadsheet and marketplace data. It exposes **1,900 variants with conflicting attributes**. Catalog owner Teun Baas's team resolves those differences before publication, retaining every SKU and GTIN identifier.

The German storefront becomes the first consumer in August. The PIM publishes on a schedule while the platform retains a fallback copy; a nightly job compares them. Dutch and Belgian storefronts and marketplace listings follow in September, then spreadsheets become read-only.

At each step, recovery means stopping publication and keeping the platform's last copy. A catalog correction does not require a single weekend switch for every channel.

The OMS, August to December 2027. The phases follow the table in the previous sections.

Phase	Dates	What happened	Reversible by
Shadow	August	Every order from every channel copied to the OMS through the iPaaS; nobody reads it; the WMS feed unchanged	Switching the copy off

Phase	Dates	What happened	Reversible by
Parallel run	1–14 September	The OMS computes routing and status for every order; the reconciliation compares its view with the platform's, the connector's and the WMS's each night; differences listed by cause (a timing gap, a status mapping, a true error)	Nothing to reverse
Marketplace seam	15 September	Marketplace orders (about 120 a day) routed through the OMS to the WMS; the old path kept as the copy; OMS order numbers continue the platform's sequence, with the marketplace's own order id kept as a reference (The Order Record: Lifecycle, State and the Order Management System)	Routing the seam back; a rehearsed one-hour procedure

Phase	Dates	What happened	Reversible by
D2C seam	13 October	Storefront orders routed through the OMS to the WMS; the platform keeps checkout and the storefront order history, as W2 says	Routing the seam back within the day
Change freeze	15 November – 5 December	Nothing changes; the OMS meets Black Friday week (26 November 2027) with the old path still receiving copies and the WMS integration able to take the old path back	The freeze itself
Wholesale seam	8 December	EDI orders from Bergzicht and manual wholesale orders created in the OMS; the ERP receives them from there for invoicing	Routing back to the ERP's order entry
Decommission the old order paths	January 2028	The platform-to-WMS and connector-to-WMS flows retired; exports archived; the stores deferred to the 2028 POS evaluation	Not needed; the peak has passed

The reconciliation's first fortnight is the part worth reading twice. The parallel run

found differences on about 1.4% of orders in its first week (illustrative), nearly all from two causes: a status mapping (the WMS's "packed" had no equivalent in the platform) and a timing gap (orders created in the last minutes before the nightly comparison). The one true error was an order the connector had delivered twice to the iPaaS, which the old path had fulfilled twice in the past and the OMS, with idempotent handling, created once. **The migration found an existing defect before it caused a new one**, which is the parallel run's quiet payoff.

The order history. The 2026 orders were imported into the OMS as history: original numbers, dates and statuses, flagged as imported, with no side effects fired. The platform kept its own order history for the storefront's account pages, as W2 requires. Support looked in one place from October.

When It Goes Wrong

- **Duplicates across the seam**. During the window, an order is created on the old path and its copy on the new, and both reach the WMS. *Detect:* the reconciliation's "in both" list; a customer with two parcels. *Recover:* idempotency on the WMS inbound keyed on the order number; the duplicate canceled before picking; the cause (a retry at the router) fixed before the next seam.
- **The gap**. An order created in the minutes between the last copy and the cutover reaches neither system. *Detect:* the count of orders in the window against the platform's own count. *Recover:* a replay from the platform's order export for the window; one more reason the old owner must stay readable.
- **Stale copies downstream**. The search index, a cache or the data warehouse still reads the old owner and shows the old product name for a day. *Detect:* the consumers listed in the inventory with their refresh cadence. *Recover:* every consumer re-pointed and refreshed as a cutover task, not an afterthought (Finding the Product: Search, Navigation and Merchandising, Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics).
- **The rollback that cannot roll back**. A week of orders exists only in the new system when a defect appears. *Detect:* the rollback rehearsal's timing against the records created since. *Recover:* reconcile forward rather than back if the gap is large: fix the defect in the new system and keep going; a rollback after a week is often the riskier move.
- **The vendor outage mid-cutover**. The new hub's SaaS has an incident during the window. *Detect:* the watch. *Recover:* the seam routed back; the window rescheduled; the status page's history read before the next one.
- **The redirect map with holes**. Old product URLs return "not found"; rankings fall; the M&S quarter. *Detect:* crawl the old sitemap after go-live; watch zero-result searches and organic traffic by landing page. *Recover:* redirects added in order of traffic; the old sitemap kept live until the crawl is clean.

- **Gift cards that do not balance**. The sum of balances differs by a few hundred euros after the import. *Detect:* the finance sign-off, before go-live. *Recover:* the import re-run from the source; no go-live until the sums match to the cent.
- **The agency script still pointing at the old endpoint**. The Deprecated Endpoint in reverse: two days of stale stock to a marketplace because one flow was not on the list. *Detect:* the integration inventory, complete, with every flow's cutover task ticked. *Recover:* the legacy mimic at the old address for one more season while the script is rewritten.

What to Take Away

Established.

- The documented replatforming failures (Target, 2011; Finish Line, 2012; Marks & Spencer, 2014) broke on capacity, performance, timing and customer habit, not on features, and the one documented recovery depended on the old platform having been kept operable.
- The strangler fig (Fowler, 2004, restated 2024) and the patterns of legacy displacement (Thoughtworks, 2021, revised 2024) are the established alternative to the big bang: seams, transitional architecture, event interception, diverted flows, and resistance to feature parity. Zalando's 2015–2016 fragment-by-fragment shop rebuild is the public commerce example.
- Order history is imported as history, not re-created as orders; gift-card and credit balances are liabilities that must balance to the cent; identifiers (SKU, GTIN, order numbers) do not change in a migration.

Depends on context.

- A parallel run is worth its cost when the input can be copied cheaply (orders, product changes) and the reconciliation can explain differences; for a storefront rendering migration, a traffic canary behind a router is the better seam.
- How long the old system stays operable depends on the next peak: through one peak for an order or availability hub; until the crawl is clean for a storefront; until the final reconciliation for a PIM.
- Which seam moves first depends on volume and blast radius: the smallest channel for an order hub, the smallest storefront for a product hub, a region or a segment for a storefront.
- At the Group stage, brand order and feature-parity pressure dominate; at the Starting stage, the URL map and the customer list are most of the risk.

The author's synthesis.

- I plan every migration backwards from the first peak the new owner must survive, and I do not let a seam move unless the previous one has been reconciled to zero known causes and the route back has been rehearsed with a stopwatch.
- I treat the reconciliation job as the migration's main deliverable rather than its test: it is the thing that stays in production afterwards, and it is what makes the next migration cheaper than this one.

Questions to Ask

1. What is the seam, and what is the smallest one we could move first?
2. Is there a way back after the cutover, how long does it take, and has anyone rehearsed it?
3. What does the old system do while the new one owns the seam: receive a copy, sit idle, or get switched off?
4. Which reconciliation proves that old and new agree, and does it explain differences by cause?
5. Are there two writers for any fact during the migration, and if so, why?
6. How are orders imported: as history with their original numbers, or re-created with their side effects?
7. Do the gift-card and store-credit balances sum to the same number in both systems, and who signed that off?
8. Does every old URL redirect, and who crawled the old sitemap afterwards?
9. Which of the thirty-four integrations reads or writes the thing that is moving, who owns each, and which need a mimic?
10. When is the next peak, and what is frozen around it?

Migrations are the moments when a merchant moves from one scale stage to the next and feels it. The next chapter steps back to the stages themselves: which decisions change at each one, what the signals of a stage ending are, and how Kestrel & Pine's own path from a web shop to an omnichannel brand maps onto them. Continue with What Changes at Each Scale: From First Store to Group.

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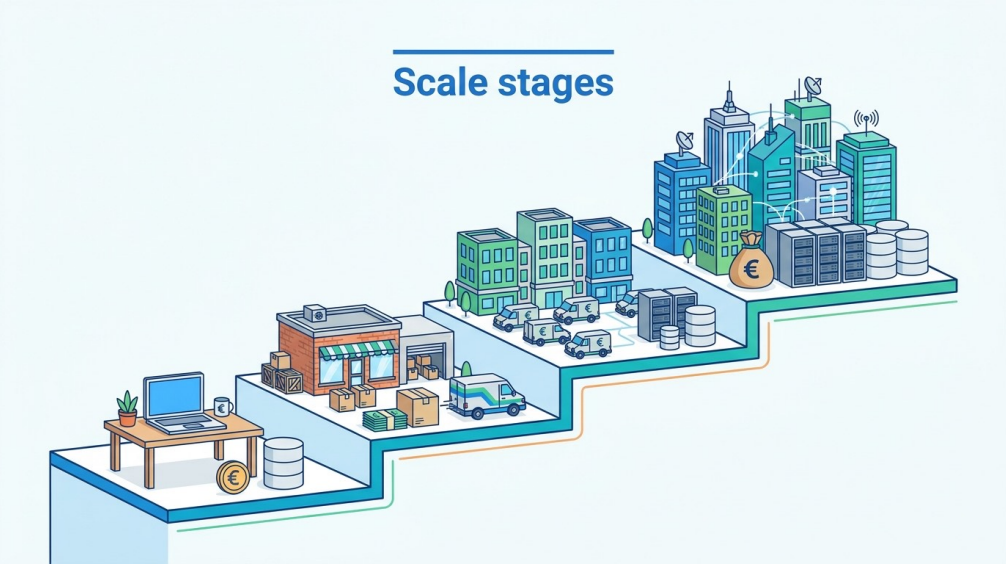
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

43. What Changes at Each Scale: From First Store to Group



IN THIS CHAPTER, YOU WILL: Read one map of the four scale stages, what each demands in systems, teams and decisions, and the signals that a stage is ending.

KEY POINTS:

- **A stage is a shape, not a revenue number.** The stage changes when the business adds a channel or a stock location that the current systems were not built to keep in step; revenue bands are only rough markers. Kestrel & Pine became Omnichannel in 2023, when stores, wholesale, marketplaces and an own warehouse ran on one stock at the same time.
- **Each stage leaves systems behind, bought for the stage before.** Most incidents at the Omnichannel stage are Growing-stage designs (hourly syncs, agency scripts, the finance system as the hub) running under Omnichannel load. The decision at each stage is which leftovers to retire, not which new tool to add.

- **The signals arrive before the architecture does.** Vendor fee cliffs and plan thresholds, a second stock location, a retail chain asking for orders by EDI (electronic data interchange), a connector nobody owns, the first "which number is right" meeting and the first peak-day incident each mark a transition months before it is named.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Giant's Boots** — an architecture copied from a Group-stage company, or from a trend, into a Growing-stage business that cannot staff it.
- **The Spaghetti Junction** — the Growing stage's one-off connectors, each built for one new channel, that nobody retires when the Omnichannel stage arrives.
- **One Size Fits None** — the Group stage's question answered before it is measured: several brands or countries forced onto one stack, or kept apart when they should share.

A merchant opens its first store while continuing to sell online. The servers may have ample capacity, but the business now holds stock in two places and may want either channel to fulfill an order. Systems designed for one warehouse must coordinate new decisions. Similar changes arrive with a first wholesale customer, marketplace or second brand, often before raw transaction volume becomes a problem.

This book uses four **scale stages**: **Starting**, a simple sales operation; **Growing**, with new channels and integrations; **Omnichannel**, coordinating stores, wholesale and online sales; and **Group**, with several brands or business units. The map below brings their recurring decisions together. Use it to recognize transitions, not to copy a larger company's system list.

The stages combine common practice, dated vendor evidence and the fictional example. They are not validated revenue thresholds: the research found no public survey establishing system counts or team sizes by revenue.

Where we are in the order's journey. Eva buys a jacket at Kestrel & Pine, the fictional outdoor brand. Her order crosses products bought at different stages: platform in 2016, marketplace connector in 2021, warehouse and finance systems in 2023, returns portal in 2025. Each still carries assumptions from when it was introduced.

Setting the Scene at Kestrel & Pine: The Stage Map

In April 2027, CEO Mara Lindqvist asks Head of Engineering Noor Visser which planned fixes suit the company now. **Waypoint**, their improvement program, addresses the 2026 failures. A stage map should also show what would make its decisions worth revisiting.

Noor places Kestrel & Pine at Omnichannel since 2023 and maps its thirteen systems to their origins. A second brand, fourth country or revenue near €200 million would prompt a Group-stage review; none is planned for 2027–2028.

Waypoint therefore adds an **order management system (OMS)**, **product information management system (PIM)** and availability service for current coordination problems. It defers group-wide choices to the 2029 review.

The map had one uncomfortable row. Of the ten incidents of 2026 and early 2027, Noor could attribute eight to a system or integration designed for the Growing stage (2019–2022) and still running in 2026: the hourly stock sync built when marketplace orders were a few a day; agency scripts written for one channel at a time; the ERP (the enterprise resource planning system that keeps the books, purchasing and stock valuation) as the hub for stock it did not physically hold; the POS (the store till system) synced nightly because, in 2019, there was one store next to head office. The Black Friday Oversell (1,340 units, 1,120 orders, about €96,000) was the most expensive. **The company had moved stage; the architecture had not noticed.** That row is this chapter's argument.

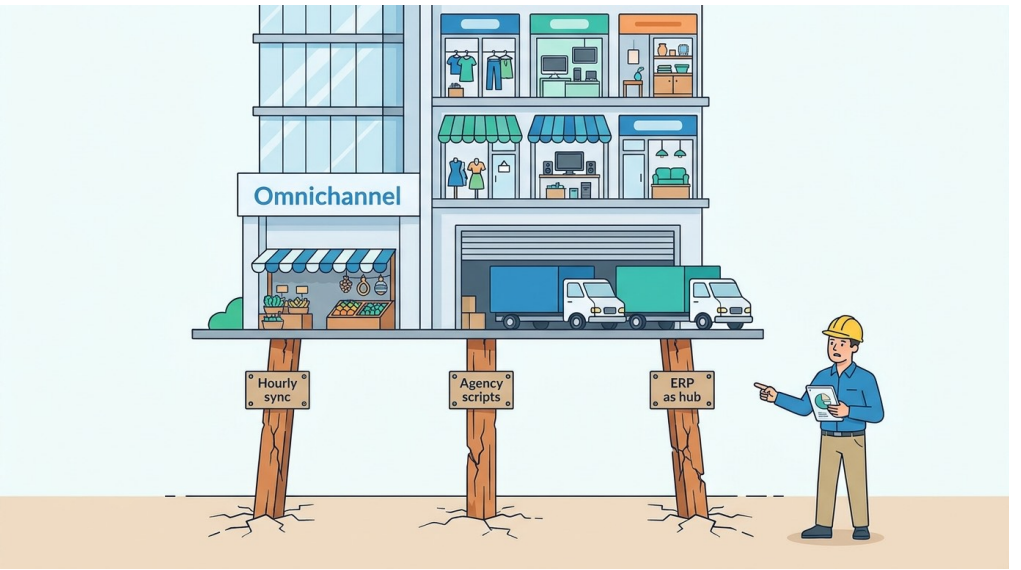


Figure 1: *Eight of Kestrel & Pine's ten incidents traced to Growing-stage designs still carrying Omnichannel load: the company had moved stage; the architecture had not noticed.*

What a Scale Stage Is, and Is Not

A **scale stage** is a shape of business defined by how many channels sell and how many locations hold stock, because those two numbers decide how many systems must agree

about the same facts. A **channel** is any place a customer can buy: a web shop, a store, a wholesale portal, a marketplace, an EDI link (electronic data interchange: orders and invoices exchanged as structured messages between companies' systems). A **stock location** is any place stock physically sits and can be sold from: a back room, a 3PL (a third-party logistics company that stores and ships on the merchant's behalf), an own warehouse, a store, a marketplace's fulfillment center. One channel and one location need no integration. Four channels and four locations need every system to learn, quickly and reliably, what every other system has sold or moved.

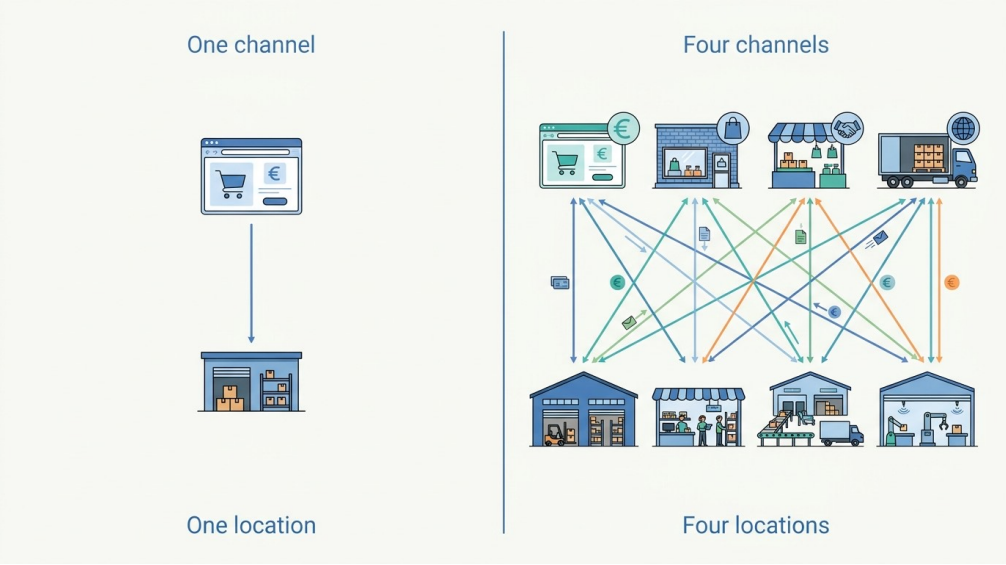


Figure 2: *One channel and one stock location need no integration; four channels and four locations need every system to learn, quickly and reliably, what every other has sold or moved.*

The ledger defines the stages like this, with revenue bands as rough markers:

Stage	Typical size	Shape	Kestrel & Pine
1 — Starting	under about €2M revenue	One channel, a hosted platform, founder-run operations, a 3PL or a back room	2016–2018

Stage	Typical size	Shape	Kestrel & Pine
2 — Growing	about €2M–€20M	Two or three channels, first store or wholesale, first integrations (ERP or accounting, 3PL, marketplace), a small team and an agency	2019–2022
3 — Omnichannel	about €20M–€200M	Stores, wholesale, marketplaces and several countries; own warehouse; a dozen or more systems; the joins become the problem; OMS, PIM and integration standards become necessary	2023– (now)
4 — Group	over about €200M, or several brands or business units	Multiple brands, countries or business units; platform teams; shared services vs autonomy; acquisitions; vendor-scale concerns (tenancy, extensibility)	not yet

The vendors' own price lists confirm that shape and volume, not headcount, are what the market tiers on. Shopify's euro pricing page (accessed November 2026) runs from Basic at €21 a month paid yearly to Plus from €2,100 a month on one- or three-year terms, and its fee for using a payment provider other than its own falls from 2% of each order on Basic to 0.2% on Plus.

Shopware's pricing page (accessed November 2026) says its plans are "based on your

gross merchandise value (GMV) and further individual factors", with a paid plan required above €1 million GMV in a rolling year. BigCommerce's plan page (accessed November 2026) assigns tiers by trailing-twelve-month GMV and auto-upgrades merchants at $2 million.

GMV is gross merchandise value, the total value of orders through a platform before refunds and before the vendor's cut. The numbers change often; the mechanism does not: a vendor's tiers are a map of which decisions it expects you to make at which size.

What a stage is **not**: a target, a maturity score or a verdict. A merchant can stay at the Growing stage profitably forever; the stage only tells you which problems are now yours.

The Map: Four Stages Side by Side

This is the chapter's centerpiece. Each row is a question the other chapters answer in depth; the cells say what is typical and what becomes necessary at each stage. Everything in it is industry practice, dated vendor evidence or Kestrel & Pine; where a cell makes a judgement, it is mine. The abbreviations are the book's: a **WMS** (warehouse management system) runs the warehouse; a **PSP** (payment service provider) takes the money; an **iPaaS** (integration platform as a service) is a hosted tool for building flows between systems; an **SSCC** is the serial shipping container code on a pallet label; **B2B2C** is selling through a partner's consumer channel.

	1 — Starting	2 — Growing	3 — Omnichannel	4 — Group
Shape	One channel, one stock location, founder-run	Two or three channels, a first store or wholesale line, a 3PL or small warehouse	Stores, wholesale, marketplaces, several countries, an own warehouse; a dozen or more systems	Several brands, countries or business units; acquisitions; platform teams

	1 — Starting	2 — Growing	3 — Omnichannel	4 — Group
Channels (Who Sells What to Whom: Business Models and the Economics That Shape Architecture)	Web shop; perhaps one marketplace	Web shop plus a store, a marketplace or wholesale by e-mail	Four or more, including EDI with a retail chain and marketplace-fulfilled stock	Channels per brand and per country; marketplace *operating* or B2B2C may appear
Storefront and checkout (Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026, Delivering the Storefront: Themes, Headless, Rendering and Performance)	Hosted SaaS platform on a monthly plan; a theme; public apps	Same platform, higher tier; an agency customizes the theme; first custom app	Enterprise tier or contract; multi-market in one tenant or several; headless considered and usually declined	Shared front end or per brand; suite, composable or platform-plus-layers becomes a real choice
Where the order lives (The Order Record: Lifecycle, State and the Order Management System)	In the platform	In the platform, with copies in the ERP and the 3PL	Several order records per channel until an **OMS** is introduced	OMS as the hub for all brands, or one per business unit

	1 — Starting	2 — Growing	3 — Omnichannel	4 — Group
Where stock truth lives (Is It in Stock? Inventory, Availability and the Promise You Can Keep)	The platform's stock count	The ERP or the 3PL feed, copied to the platform on a schedule	A **WMS** for the warehouse, the POS for stores, an external location for marketplace fulfillment; an availability service computes what each channel may sell	Stock per brand and region; shared availability rules; allocation between business units
Product data (What Are We Selling? Products, Variants and the Catalog)	The platform's catalog	Platform plus ERP item master plus spreadsheets	A **PIM** as the system of record; platform, connector and marketplaces consume it	PIM per brand or shared, with a shared attribute model
Fulfillment (Pick, Pack, Ship: Warehouse Management and Fulfillment)	Back room or 3PL	3PL or a small warehouse run from the ERP's stock module	Own warehouse on a WMS; pallets and SSCC labels for wholesale; marketplace fulfillment programs	Several warehouses, a network, logistics as a shared service

	1 — Starting	2 — Growing	3 — Omnichannel	4 — Group
Finance and ERP (The Back Office: ERP Integration and Who Is the Hub)	Accounting package; the platform exports	A mid-market ERP, often as the integration hub	The ERP kept for finance, purchasing and valuation only; the hub role moves to the OMS and the integration layer	Group ERP with per-entity books; intercompany flows
Integration style (APIs, Events, Files and Middleware: How Commerce Systems Talk)	None; the platform's own connectors	Point-to-point connectors, agency scripts, vendor connectors; the first iPaaS flows	Events for stock, orders and prices; an owned integration layer; contracts, idempotency and reconciliation per flow	A platform team runs the integration layer as an internal product; API guidelines; shared event bus
Who builds and runs (Who Builds and Who Owns: Teams, Vendors, Agencies and Governance)	The founder and the platform's support; an agency for the theme	The agency builds; a first engineer; nobody owns integrations after launch	A 10–20-person team that owns the joins; the agency as a contributor; named owner per integration and app	Stream-aligned teams per brand or journey; a platform team; vendor management as a function

	1 — Starting	2 — Growing	3 — Omnichannel	4 — Group
Pricing and fees you meet	Monthly plan; blended card rates	Plan upgrades; GMV thresholds; the third-party gateway fee; per-order quotas on tools	Negotiated contracts (platform, PSP, WMS, OMS); interchange-plus pricing; several thresholds crossed at once	Group contracts; volume commitments; exit and switching terms
Peak day (Black Friday Every Year: Peak Load, Resilience and Recovery)	The platform's problem, not yours	The first integration bottleneck: the order export, the stock sync	A peak plan with orders-per-minute capacity per system, a switch-off order and a reconciliation owner	Shared peak tooling and runbooks across brands
Decisions that become necessary	Which platform; which PSP; which 3PL	Who owns stock and price truth; the first connector's contract; one tenant or many	OMS, PIM, availability service; integration standards; stay-and-surround or replatform; one definition of an order	Share or separate per brand; platform team or not; acquisition integration playbook; tenancy and residency
Failure patterns that strike	The Lynchpin (one agency, one person)	The Spaghetti Junction; The Price Nobody Can Explain; The First Snowfall	The Phantom Stock; Schrödinger's Order; The Boomerang; The Familiar Stranger; The Pulled Rug; The Secret Agent	One Size Fits None; The Giant's Boots; The Burned Bridge

	1 — Starting	2 — Growing	3 — Omnichannel	4 — Group
Signals the stage is ending	A second channel or a first store; the 3PL cannot say stock per location	A retailer asks for EDI; a second stock location; a fee cliff; the first "which number is right" meeting	A second brand or business unit; an acquisition; a fourth country with its own books; the platform's tenancy or rate limits bind	(The stage has no end; its question is what to share)
Kestrel & Pine	2016–2018: platform, PSP, 3PL in Tiel, agency from 2017	2019–2022: POS, Belgium, connector and iPaaS, GTINs, Germany, Ruben Alves	2023–2027: Duiven on a WMS, ERP, enterprise tier, stores, EDI, returns portal, Peppol; Waypoint	Not yet; signals named in the April 2027 stage map

Three things to notice. **The order record and the stock record move in the same direction**: from "the platform knows" through "the ERP knows, mostly" to "a hub knows, and everything else is a copy." **The integration style changes faster than the systems do**: a merchant can keep its platform for all four stages and still rebuild the joins twice. And **the failure patterns are stage-specific**: the Truth patterns of Out of Stock, Out of Sync: Sixteen Ways Commerce Platforms Go Wrong cluster in the Omnichannel column because that is where the same fact first lives in four systems.

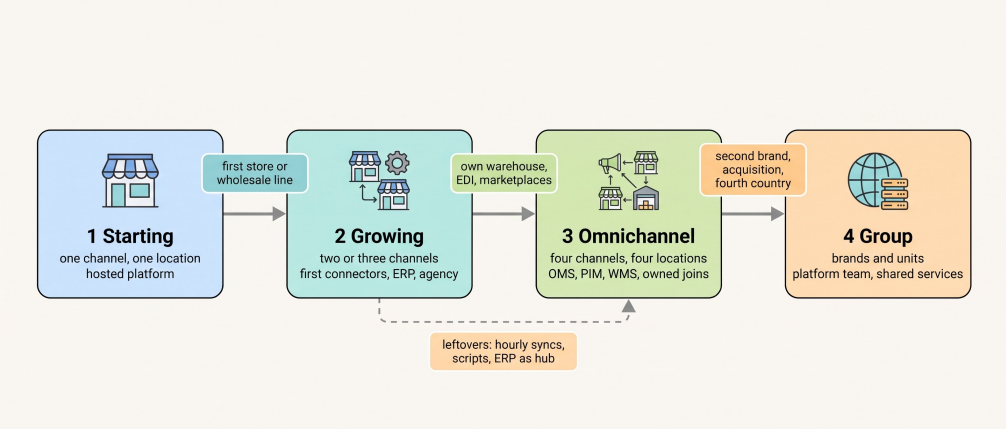


Figure 3: *Each stage is entered by adding a channel or a stock location, and each carries the previous stage's systems with it; the dotted arrow is where most Omnichannel incidents come from.*

How Fresh Must a Fact Be? Requirements by Stage

The reason the joins must be rebuilt is that the same fact needs to travel faster and more reliably at each stage. The table below states the requirement for the three facts that cause most incidents. The latencies are illustrative targets, not benchmarks *(assumption)*; the Kestrel & Pine figures are from the ledger.

Fact	Starting	Growing	Omnichannel	Group
Stock a channel may sell	The platform's own count; freshness is instant because there is one writer	A scheduled copy (hourly is common) from the ERP or 3PL; tolerable while one channel dominates	Seconds to a few minutes, by events, because four channels sell the same unit; Kestrel & Pine's two-hour chain oversold 1,340 units	As Omnichannel, plus allocation rules between brands and regions

Fact	Starting	Growing	Omnichannel	Group
Order status	One record; the customer and support read the same screen	Copies in the ERP and the 3PL; support checks two systems	One state model in an OMS; status events out to the storefront, service tool and market-places within minutes; the Three-Status Order (11 tickets, six days) is the failure	One state model across brands, with per-unit order number families
Price	Set in the platform	Price lists in the platform and the ERP; a sync that can fail silently (the Price Drift: €179 for nine days, about €8,200 lost)	One price source with effective dates; channels consume; parity rules for market-places	Price books per brand and market with shared governance

The point is not the numbers. It is that **a requirement met for free at the Starting stage (one writer, one reader) becomes an engineering problem at the Omnichannel stage (four writers, four readers), and the Growing stage's scheduled copies are the bridge that collapses between them.** Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation owns the mechanics; Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record owns the ownership decision that must come first.

Stage by Stage

Starting: one channel, rented everything

At the Starting stage the architecture is a purchase, not a design. A hosted SaaS platform (software as a service: rented by the month and run by the vendor) provides storefront,

cart, checkout, stock count and order record in one tenant; a payment service provider (PSP) takes the money; a 3PL or a back room ships. Shopify's annual report for 2025, filed in February 2026, says most of its merchants subscribe to its Basic and Grow plans while the majority of its GMV comes from Plus and enterprise merchants: the Starting stage is where most merchants are and most volume is not.

What becomes necessary here is small: a platform whose export you could leave with, a PSP whose fees you understand, an app inventory with each app's permissions written down from day one (Apps, Plugins and Functions: Extensibility and Ecosystem Governance), and the founder, not the agency, holding the vendor logins. The failure pattern that strikes is **The Lynchpin**: one agency or one person through whom everything runs. Kestrel & Pine retained its agency in 2017; ten years later it still holds the theme, the apps and several scripts, at €14,000 a month.

The signal that the stage is ending is a second channel or a first store. The day the business has stock in two places that both sell, it has an integration problem, whether or not it has an integration.

Growing: the first joins, built one at a time

The Growing stage adds a store on a point-of-sale system (POS, the till software that records store sales), a marketplace through a connector, a wholesale line run by e-mail and keyed into an ERP (enterprise resource planning: the finance, purchasing and inventory-valuation system), and often an ERP-centered model where the ERP becomes the hub that every other system syncs with. Microsoft's documentation of 22 September 2026 says its Shopify connector is preinstalled for all new Business Central sign-ups, the clearest public signal that "a hosted storefront with the ERP as hub" is the mainstream Growing-stage shape in at least one vendor channel.

Each join is built when the channel arrives, by whoever is available: the agency, the ERP partner, the connector vendor, a contractor. Each is correct for the volume of the day. Kestrel & Pine's marketplace connector arrived in 2021 with an hourly stock sync, when marketplace orders were a few a day *(illustrative)*; by 2026 the marketplaces took 43,000 orders a year and the hourly sync was one link in a two-hour chain nobody had decided was the architecture. **The Spaghetti Junction is not built; it accretes.**

This is also the stage where fees change character. A merchant that brings its own PSP to a hosted platform pays the platform's third-party gateway fee (2% of each order on Shopify's Basic plan, falling by tier); operational tools price per order (Picqer, a Dutch WMS, lists €379 a month including 3,000 orders, then €0.28 an order, accessed November 2026); and open-source platforms stop being free above Shopware's €1 million GMV line. A merchant moving from Growing to Omnichannel typically crosses several thresholds at once, which is why price lists are a transition signal, not just a cost.

The signals that the stage is ending are concrete. A retail chain asks for EDI (Eurostat's enterprise statistics for 2024, extracted February 2026, show EDI-type sales at 11.07%

of EU enterprise turnover against 8.39% for web sales; by value, European business-to-business electronic ordering is still mostly EDI, so a wholesale line meets EDI before it meets a B2B web shop). A second stock location appears that must be sold from, which vendor guidance converges on as the trigger for an OMS. The first meeting happens in which two reports show two order counts and nobody can say which is right. And the first peak day arrives on which an integration, not the platform, is what fails (**The First Snowfall**).

Omnichannel: the joins become the architecture

At the Omnichannel stage the merchant runs three or four business models on one stock: direct-to-consumer web shops in several countries, stores, wholesale by portal and by EDI, and marketplaces with some stock held by the marketplace itself. Target's annual report for 2025, filed in March 2026, states that its stores fulfilled 97.6% of its merchandise sales, including shipment from stores and store pickup; for a retailer with stores, omnichannel is the business, and an order management system is what makes store stock sellable online. Target does not name its OMS; the mechanism is what matters.

What becomes necessary here is the set of decisions Waypoint makes: a **system of record** per fact (the one system whose value is the truth when copies disagree) for stock, orders and product content (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record); an OMS as the order hub (The Order Record: Lifecycle, State and the Order Management System); a PIM for product content, which Akeneo's guidance (accessed November 2026) says is signaled by content scattered across teams, manual channel launches and inconsistent storefront data (What Are We Selling? Products, Variants and the Catalog); an availability service per channel (Is It in Stock? Inventory, Availability and the Promise You Can Keep); and integration standards with a named owner, a contract, idempotent handling and a reconciliation job per flow (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation). The ERP is demoted from hub to one system among many (The Back Office: ERP Integration and Who Is the Hub).

The temptation at this stage is **The Giant's Boots**. The composable and microservices vocabulary is everywhere, and the public examples are companies with hundreds of engineers. Fowler's "MonolithFirst" note of June 2015 argued that even experienced architects "have great difficulty getting boundaries right at the beginning"; Shopify's engineering post of February 2019 explained why a company with over a thousand developers chose a modular monolith over microservices. The MACH Alliance's own survey of 600 decision-makers, published in February 2026, reports that 37% cite integration complexity as a primary concern, a figure against the interest of the alliance that commissioned it. For a 14-person team the honest reading is: adopt composable per capability (an OMS here, a PIM there) around a kept core, and own the joins. Monolith,

Modular Monolith, Microservices, Composable: What the Evidence Says and Build, Buy or Assemble: Choosing and Changing Platforms work this through.

The signals that the Omnichannel stage is ending are a second brand or business unit, an acquisition, a fourth country with separate books, or the platform's tenancy binding (one tenant's rate limits shared by every app and connector; Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation). Shopify's enterprise page (accessed November 2026) cites organizations with GMV of $50 million to $500 million and more as its enterprise band, the nearest public hint of where a vendor sees the top of this stage.

Group: what to share and what to keep apart

The Group stage is defined by plurality: several brands, business units or countries, each of which could run its own stack. The question is no longer "which systems" but "**which systems are shared, and who owns the shared ones?**"

VF Corporation's annual report for the fiscal year to March 2026 describes a direct-to-consumer business of brand-operated stores, brand e-commerce sites and concession locations across brands such as The North Face and Timberland, at 44% of revenue: one group, many brand storefronts, and a shared question about what sits behind them.

Zalando's 2025 annual report, published March 2026, shows what an acquisition does first: consolidating ABOUT YOU changed the group's reported figures (the partner-business share of GMV fell to 32.1% from 34.2% by dilution) before any system was touched. An acquisition changes the metrics on day one and the architecture over years.

The failure pattern is **One Size Fits None**: forcing brands onto one stack without measuring the benefit, or duplicating work they could share. My default is to share expensive capabilities with common requirements, such as identity, payment contracts, integration tooling, data definitions, peak tooling and security. A platform team can maintain those once for the group. Brands should retain control of storefronts, content, promotions and assortment when those differences matter to their customers. Sharing works only while its common rules leave room for those needs.

The public cautions against too much autonomy are Zalando's and Uber's. Henning Jacobs' 2020 retrospective on a decade at Zalando (more than 260 engineering teams by 2020) reports that team autonomy over technology produced similar services "written in six languages", which "hurts productivity at scale"; Uber's 2020 account reports about 2,200 microservices needing domains, gateways and layers before a root-cause investigation stopped running through fifty services owned by twelve teams.

Both sources are six years old and describe companies far larger than any merchant; the lesson transfers: **autonomy without shared standards produces the Spaghetti Junction at group scale**.

The Group stage's remaining questions are vendor-scale ones: tenancy (one tenant per brand or a tenant of tenants), extensibility (a shared app catalog and an upgrade calendar

aligned to vendors' support clocks), data residency, and exit terms. The EU Data Act (Regulation 2023/2854) prohibits switching charges for data processing services from 12 January 2027; whether hosted commerce platforms fall under it is an open legal question as of November 2026, but a group negotiating several contracts should read the switching chapter before signing (Build, Buy or Assemble: Choosing and Changing Platforms).

Multi-Brand and Multi-Country: The Group's Decision Table

Because the Group stage's question is "what to share", it deserves its own table. The cells are my recommendations with their conditions.

Capability	Share across brands or units when…	Keep separate when…
Storefront and theme	Brands are positioned alike and share a design system	Brand identity is the product; customers must not see the group
Checkout and PSP contract	Always share the contract and the fraud rules; volume pricing rewards it	Legal entities or countries require separate merchant accounts (share the provider, not the account)
Catalog and PIM	Attribute model and data-quality rules; a shared PIM with brand workspaces	Assortments never overlap and teams are in different countries with different seasons
Stock and availability	Warehouses serve several brands; allocation rules must be one	Brands have separate logistics and no cross-selling
Order hub (OMS)	One state model, one service tool; customers buy across brands	Business units have incompatible order types (B2B contract orders vs D2C) with no shared fulfillment
Integration layer and data definitions	Always; this is the platform team's product	Never, in my judgement; separate definitions of "order" per brand is the Group-stage Schrödinger's Order

Capability	Share across brands or units when…	Keep separate when…
Peak tooling, security, compliance	Always	Never

Two conditions govern the table. If the group acquires rather than builds, every acquired brand arrives with its own stack and the question becomes sequencing, which is a migration (Moving Without Burning the Bridge: Replatforming and Migration). And if the group operates a marketplace or a partner program, it has become a platform vendor to other sellers, and the vendor-side concerns of Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation apply to it directly.

The Transition Signals

A transition is easier to see than to name. Every signal below was visible at Kestrel & Pine before the incident that priced it.

Signal	What it usually means	Pattern it precedes	Kestrel & Pine
A vendor fee cliff or plan threshold is crossed	The vendor expects a different kind of customer; read its tier as a map	—	Enterprise tier in 2023
A second stock location that must be sold from	An availability computation is needed, not a copy	The Phantom Stock	Duiven 2023; stores selling online stock from 2026
A retail chain asks for EDI	Order capture moves into the buyer's systems; chargebacks become possible	The Pulled Rug (contract), The Lynchpin (one consultant)	Bergzicht 2025; the ASN Chargeback 2026
Two reports, two order counts	Three definitions of the same fact; a data-definition owner is needed	Schrödinger's Order (in the data)	The three definitions of "order"

Signal	What it usually means	Pattern it precedes	Kestrel & Pine
A connector nobody can name an owner for	The Growing stage's build-it-when-it-arrives has left orphans	The Spaghetti Junction; The Pulled Rug	The Deprecated Endpoint, February 2027
Support opens several systems to answer one question	No order hub	Schrödinger's Order	The Three-Status Order, 11 tickets
An app or script writes a fact another system owns	No ownership rule per fact	The Secret Agent	The App That Wrote Stock, March 2026
The first peak on which an integration fails	The platform scales; the joins do not	The First Snowfall	Black Friday 2026, four-hour WMS queue
A second brand, an acquisition, a fourth country	The Group stage's share-or-separate question has arrived	One Size Fits None	Not yet; on the stage map

The rule I use: **two signals in one year mean the stage has changed, whatever the revenue says**. Kestrel & Pine had six in 2026.

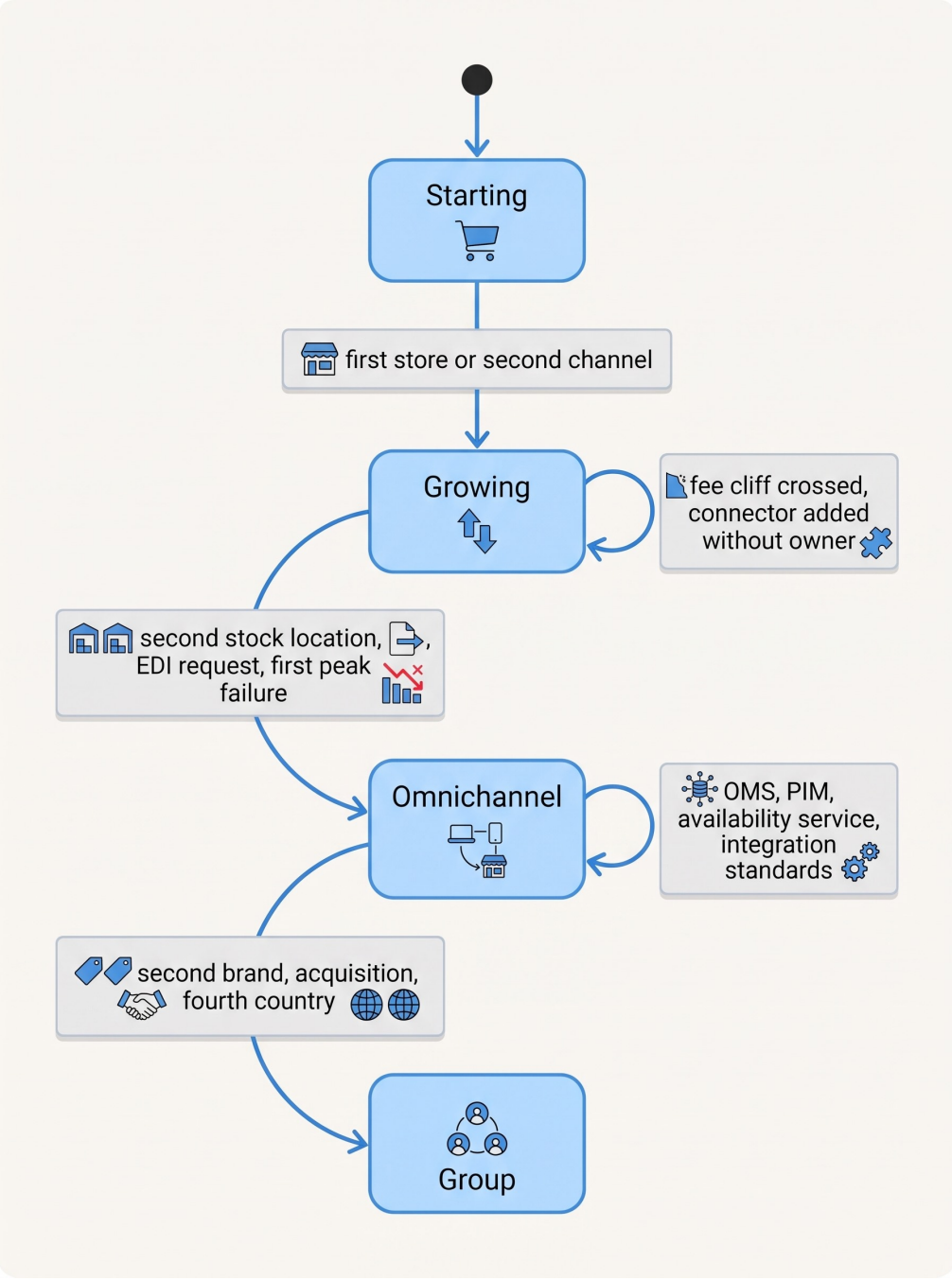


Figure 4: *The transitions are triggered by shape changes, not revenue; the self-loops are where a stage does its work, and where a merchant can stay for years.*

In Practice: Kestrel & Pine's Path, Priced

The stage map is more persuasive with the costs attached. The figures below are Kestrel & Pine's, from the ledger, with one illustrative calculation.

2016–2018, Starting. One Dutch web shop on a hosted platform, one PSP, a 3PL in Tiel, an agency from 2017. The architecture was the vendor's. Nothing from this stage caused a 2026 incident, except the habit it formed: the agency owns the storefront.

2019–2022, Growing. The Arnhem store on a cloud POS from a different vendor (2019), synced nightly. Belgium (2020). The Utrecht store, the marketplace connector with its hourly sync, the iPaaS and GTINs on every own-brand item (2021). Germany, the investor and Ruben Alves as lead integration engineer (2022). Every join was built when its channel arrived. Eight of the ten incidents descend from designs of this era; the clearest are the Store Stock Mirage (37 items promised that Utrecht had sold), the Deprecated Endpoint (an unowned agency script) and, in its cadence, the Black Friday Oversell.

2023–2026, Omnichannel. Duiven on its own WMS, the ERP and the platform's enterprise tier (2023); three more stores and the data warehouse (2023–24); Antwerp, EDI with Bergzicht and the returns portal (2025); Peppol e-invoicing and the Utrecht click-and-collect pilot (2026). The company now had thirteen systems, four order records, four copies of stock and three definitions of "order", and technology spend of about €2.6 million, 5.4% of revenue, high against the 2–4% that secondary sources suggest for store-heavy retailers (no primary benchmark could be verified; a judgement). The incidents of 2026 and January 2027 cost, in round figures, €96,000 (Oversell), €19,400 refunded twice (most recovered), €8,200 of margin (Price Drift) and €14,200 (ASN Chargeback): about €138,000 of direct cost, before support time and goodwill.

The fee example. The fee cliff that matters most to Kestrel & Pine is not the platform subscription (€2,300 a month on the enterprise tier) but payments. On €27.0 million of online D2C revenue, each half a percentage point of blended payment cost is about €135,000 a year *(illustrative arithmetic on ledger figures)*, more than the direct cost of all four priced incidents. That is why Taking the Money: Payment Providers, Authorization, Capture and Refunds and Build, Buy or Assemble: Choosing and Changing Platforms treat the PSP contract as an architecture decision, and why a merchant that never renegotiated its Starting-stage card rate pays a transition tax without knowing it.

2027, the stage map. Waypoint's €1.1 million over two years buys the Omnichannel stage's hubs and standards, retires the Growing stage's chain, and explicitly defers the Group stage. That is the whole use of the stages: not to predict revenue, but to sort decisions by the shape they belong to.

When It Goes Wrong: Too Early, Too Late, Sideways

A stage mismatch fails in three directions, and each has a detection and a recovery.

Too early (The Giant's Boots). A Growing-stage merchant adopts an API-only composable core, or a microservices rebuild, on the strength of case studies from companies with platform teams. Detection: the integration layer is the largest line of engineering effort and nobody can say who is on call for it. Recovery: consolidate the unit of deployment, as Segment did in 2018 when it folded more than 140 services back into one (its own account, now eight years old); keep the capabilities that earn their place and return the rest to the platform.

Too late (The Phantom Stock, Schrödinger's Order). An Omnichannel-stage merchant runs Growing-stage copies: scheduled syncs, the ERP as the stock hub, order records per channel. Detection is cheap: count the systems that hold stock, count the writers, and run one reconciliation between the warehouse and the platform. Kestrel & Pine's first weekly reconciliation in May 2027 found 2.3% of active variants with a different quantity, most from the apps that wrote stock. Recovery is Waypoint: decide the system of record per fact, add the hubs, convert the chain to events, and keep the reconciliation job forever (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation).

Sideways (One Size Fits None). A group forces brands onto one stack by decree, or lets each keep everything, without measuring what sharing would save or cost. Detection: a shared-platform program whose business case is "consolidation" with no per-capability benefit, or as many definitions of "customer" as brands. Recovery: the decision table above, applied per capability, with the shared capabilities run by a team that treats the brands as its customers.

In every direction the recovery runs through the same two tools: an entity–owner table (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record) and a reconciliation job per copy. The stage decides how many rows those tools have, not whether you need them.

What to Take Away

Established.

- Vendors tier on volume and shape, not headcount: Shopify's plans and third-party gateway fee (2% to 0.2% by tier), Shopware's €1 million GMV fair-use line and BigCommerce's GMV-assigned plans (all accessed November 2026) make a merchant's growth visible as fee cliffs and plan thresholds.
- By value, European business-to-business electronic ordering is still mostly EDI (11.07% of EU enterprise turnover against 8.39% for web sales, Eurostat 2024 data),

so a wholesale line meets EDI before it meets a B2B web shop, usually at the Growing-to-Omnichannel transition.
- For a retailer with stores, omnichannel fulfillment is the business, not an add-on (Target's stores fulfilled 97.6% of merchandise sales in 2025), and an order hub is what makes store stock sellable online.
- Autonomy without shared standards is costly at group scale: Zalando's 2020 retrospective (services in six languages) and Uber's 2020 domain-oriented redesign (about 2,200 services) are the dated public evidence.

Depends on context.

- Whether an OMS, a PIM or a WMS is needed depends on channel and location counts, not order volume: an OMS when a second stock location must be sold from; a PIM when the catalog is published to a third channel in a third format; a WMS when physical operations outgrow the ERP's stock module. A single-channel merchant with high volume may need none of them.
- The revenue bands are markers. A €15 million wholesaler with one warehouse is a simpler shape than a €6 million brand with stores, marketplaces and a 3PL; design for the shape.
- The ERP as integration hub is right at the Growing stage and becomes the bottleneck at the Omnichannel stage, when stores, marketplaces and a WMS need stock faster than the ERP's batch cadence.
- At the Group stage, share what the customer cannot see and what is expensive to duplicate; keep brand-visible capabilities separate until a measured benefit says otherwise.

The author's synthesis.

- Most Omnichannel-stage incidents are Growing-stage designs running under Omnichannel load; the first decision at each stage is which leftovers to retire, because the leftover, not the missing tool, is what fails.
- Two transition signals in one year mean the stage has changed, whatever the revenue says; a stage map that lists the signals is worth more to a board than a roadmap of tools.

Questions to Ask

1. Which stage is the business in by shape (channels and stock locations), and which stage was each of its systems bought for?
2. Which vendor thresholds (plan tier, GMV cap, per-order quota, PSP volume pricing) will the business cross in the next eighteen months, and what does each vendor expect of a customer on the other side?

3. How many systems hold stock today, how many write it, and how often is the warehouse's number compared with the storefront's?
4. Is there one order record, or one per channel? How many systems does support open to answer "where is my order"?
5. Which integrations were built when a channel arrived, by whom, and who owns each one now?
6. Which transition signals from this chapter's table appeared in the last twelve months, and which incident followed each?
7. If a second brand, an acquisition or a fourth country arrived next year, which capabilities would be shared on day one, and who would run them?
8. Which decisions in the current program are for the stage the business is in, which for the stage it was in, and which copy a company much larger, and why?

The stages sort the decisions; people make them. Who owns each system, integration and app, how a 14-person team works with an agency and a dozen vendors, and what governance fits a merchant rather than a software company is the subject of Who Builds and Who Owns: Teams, Vendors, Agencies and Governance.

Sources and Evidence

Company, product and investor sources

- Shopify pricing (EUR), Shopify, n.d., accessed 2026-11 — plan prices from Basic to Plus and the third-party transaction fee by tier.
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- VF Corporation Form 10-K, fiscal 2026, SEC filing, 2026 — a multi-brand group's direct-to-consumer definition and share.

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- Top 3 signs you need a PIM, Akeneo (guest post), n.d., accessed 2026-11 — vendor guidance on PIM triggers; vendor content.
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- Introducing Domain-Oriented Microservice Architecture, Uber Engineering, 2020 — about 2,200 services and the cost at the far end of scale; older than three years.
- Goodbye Microservices, Twilio Segment (Alexandra Noonan), 2018 — consolidating 140+ services into one; older than three years.
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Standards, regulation and official documentation

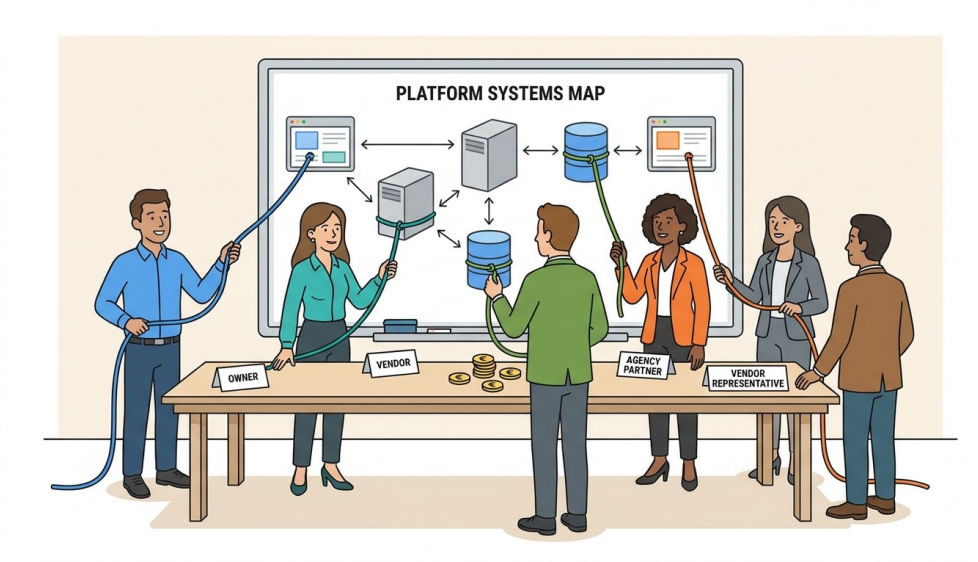
- E-commerce statistics (enterprises), Eurostat, 2026 (2024 reference year) — EDI-type sales 11.07% and web sales 8.39% of EU enterprise turnover.
- Regulation (EU) 2023/2854 (Data Act), EU Publications Office, 2023 — switching charges prohibited from 12 January 2027; application from 12 September 2025.

Research and practitioner sources

- MonolithFirst, Martin Fowler, 2015 — the difficulty of getting boundaries right early; older than three years.
- IT Spending and Staffing Benchmarks 2025/2026, Chapter 8: Retail, Avasant, 2025 — the retail IT-spend benchmark exists but is paywalled; the 2–4% range in the text is judgement.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

44. Who Builds and Who Owns: Teams, Vendors, Agencies and Governance



IN THIS CHAPTER, YOU WILL: Learn who should own each system, integration and app, how to shape a small team around journeys, and how to work with agencies and vendors.

KEY POINTS:

- **Ownership follows money unless it is designed.** An agency is paid to build and a vendor to keep you subscribed; neither is paid to own a connector after launch. The ownership map, one row per system, integration and app with a named owner, is the merchant's first governance artifact, and most incidents at Kestrel & Pine trace to a blank cell in it.
- **Shape teams around journeys, not vendors.** A team per system reproduces the integration problem in the organization chart. For ten to twenty people the honest shape is one or two stream-aligned teams

by journey (buy; fulfill and return), one pair that owns the integration layer as an internal product, and vendors and agencies as services with explicit contracts.
- **Govern lightly, but govern the facts.** Decision records for ownership, vendor and app decisions; one writer per fact; a contract card per integration; a quarterly app review; API versions pinned and recorded. Heavier process will not be followed by fourteen people, and lighter process lets the Secret Agent back in.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Lynchpin** — a capability customers depend on runs through one agency, one consultant or one engineer, with no exit plan.
- **The Secret Agent** — an app, script or integration changes stock, prices or money with no named owner and no record of what it may touch.
- **The Spaghetti Junction** — connectors built by whoever was available when a channel arrived, each with its own owner or none, and no inventory.

An order appears in checkout but never reaches the warehouse. The platform vendor, connector partner and agency may each confirm that their own component is working. The merchant still needs someone to trace the whole path, identify the failed handoff and restore the order. Supplier contracts do not necessarily assign that responsibility, so it must be owned explicitly.

The owner needs authority and information to follow a flow across supplier boundaries. This chapter explains who defines the flow, watches it, changes it and recovers it, so an incident does not become a series of referrals between vendors. It covers team structure, agencies, contracts and engineering practices a small team can sustain on rented software.

The setting is a merchant with roughly a dozen engineers, an agency and several vendors. Its job is to coordinate those suppliers and integrations. Platform vendors organize much larger teams to build the products themselves; Part XI — PRACTICE: How Others Built It examines those different constraints.

Where we are in the order's journey. Eva's jacket purchase and exchange at Kestrel & Pine, the fictional outdoor brand, cross several systems between 13 and 22 April 2027. For each handoff, someone must own the contract, monitoring, changes and recovery. "Probably the agency" is an unresolved responsibility.

Setting the Scene at Kestrel & Pine: The Ownership Map

When Noor Visser becomes Head of Engineering in January 2027, Kestrel & Pine has thirteen systems, 34 integrations and 41 apps. On **8 February**, a marketplace connector retires an API version. A 2021 agency script stops sending stock to bol, but nobody acts until seller support escalates canceled orders two days later.

The agency says the script falls outside its retainer. Ecommerce Manager Milan Petrov did not know it existed. This is the **Deprecated Endpoint** incident: a published change meets an unowned dependency.

Noor maps each system, integration and app to a business owner, technical owner, supplier, renewal date, API version and write permissions. The hosted integration tool, or **integration platform as a service (iPaaS)**, belongs to Ruben Alves's team. The planned **order management system (OMS)** and **product information management system (PIM)** need owners too.

January's inventory found **13 of 34 integrations unowned**. Eight receive owners; **five remain unassigned in April 2027**, the date of the table below.

What	Count	Owned by the integration team	Owned by the agency	Owned by a vendor or partner	No named owner
Integrations	34	14 (flows in the iPaaS, Ruben Alves' team)	9 (scripts, mostly from 2019–2022)	6 (vendor connectors, including the ERP partner's platform ↔ ERP connector)	**5**
Installed apps	41	—	22 installed by the agency	—	19 installed by various people; 6 unused; **3 write inventory**

What	Count	Owned by the integration team	Owned by the agency	Owned by a vendor or partner	No named owner
Systems	13	Business owner known for all; technical owner known for 8	—	—	5 systems with no technical owner inside the company

The five integrations still unowned in April included the bol stock script, a price-list export to the German storefront (the one behind the **Price Drift** of October 2026: €179 instead of €199 for nine days, about €8,200 of margin) and the pre-order app's inventory write (the **App That Wrote Stock**, March 2026). Three of the ten incidents in the ledger sat in the blank cells of one spreadsheet. That is the whole case for the map: **the incidents were not caused by bad code; they were caused by code with no owner**.

What Owning Means

Before team shapes, a definition, because "owner" is used loosely. To **own** a system, integration or app is to carry five duties: to **decide** its contract (what it does, which fields, which direction, how often, which system is the source of truth); to **change** it, or approve changes; to **watch** it, with a monitor and a reconciliation that says when it is wrong; to **fix** it, which means being the one called at 20:40 on Black Friday; and to **retire** it when it is no longer needed or when the vendor retires what it depends on. An agency can do the second and, under contract, the fourth; it will rarely do the first, the third or the fifth, because it is not paid to.

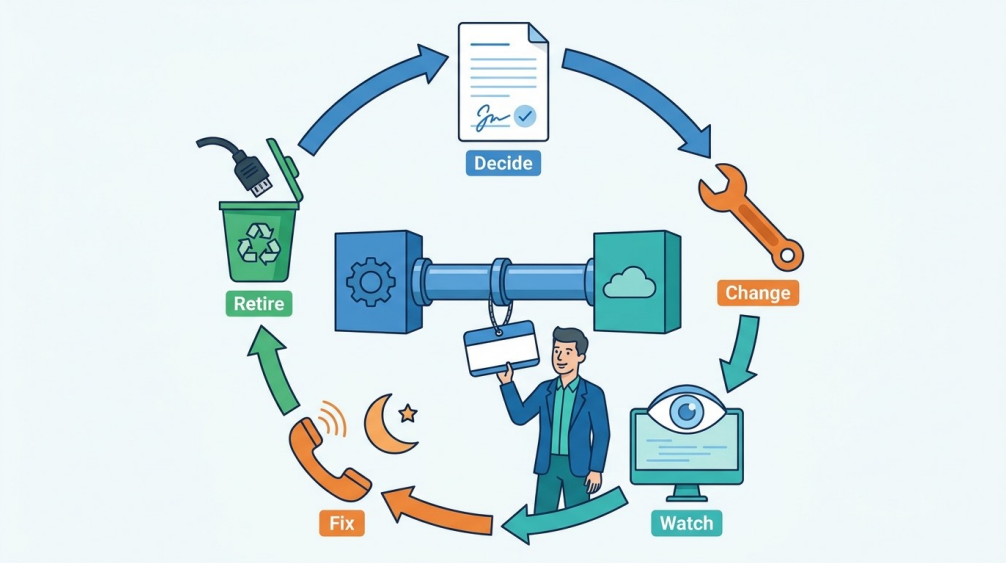


Figure 1: *To own a system, integration or app is to carry five duties: decide its contract, change it, watch it, fix it and retire it.*

Ownership follows money unless it is designed. The vendor-side economics are public. Shopify's partner page (undated, accessed November 2026) pays a partner that builds or transfers a store 20% of the monthly platform fee plus 0.1% of eligible GMV (gross merchandise value: the order value that passes through the platform) for four years, and a one-time $2,500 for a Plus or enterprise referral; partners are tiered Silver to Diamond on attributed GMV and attributed shops, and the older tiers retire on 1 January 2027. An agency on such a program is paid by the platform for keeping the merchant on it and by the merchant for building. Nothing in that arrangement pays for owning a connector after launch, for a reconciliation job, or for an exit. This is not a criticism of agencies; it is the reason the merchant must design ownership rather than inherit it.

A few terms the rest of the chapter uses. A **system of record** is the one system whose value is the truth when copies disagree (Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record). An **integration contract** is a short written agreement for one flow: fields, direction, frequency, source of truth, what happens on a repeat or a failure (APIs, Events, Files and Middleware: How Commerce Systems Talk). An **ADR** (architecture decision record) is a one-page note of a decision, its context and its consequences. A **sandbox** is a copy of a service where nothing real happens. An **API version** is a dated edition of a vendor's interface that the vendor promises to keep working for a stated period. A **stream-aligned team** owns one flow of work end to end; a **platform team** provides internal services to other teams.

The Ownership Map

The map is a table, kept where the integration inventory of APIs, Events, Files and Middleware: How Commerce Systems Talk is kept, and reviewed quarterly. Its columns:

Column	What goes in it	Why
Item	System, integration or app, by role name	One row per thing that can fail
Business owner	The manager accountable for the outcome (Milan for the storefront, Sanne de Wit for the ERP, Daniel Okafor for the WMS)	Decides priorities and accepts risk
Technical owner	The named engineer, team or partner who decides, changes, watches, fixes and retires	The person called
Builder and vendor	Who built it and who runs it	Separates "who wrote it" from "who answers for it"
What it may write	The facts it changes (stock, price, order status, refunds) and in which system	The Secret Agent check: one writer per fact
Contract	Renewal date, notice period, exit terms, support hours	Vendor management lives here
Depends on	API versions pinned, platform release tested against, vendor deprecation calendar	The Pulled Rug check
Watched by	The monitor and the reconciliation job, and who reads them	Ownership without watching is a name on a list

Two rules make the map more than documentation. **Every row has a technical owner inside the company**, even when the builder is a partner; the owner may be the person who manages the partner, but the row is never blank. And **no fact has two writers**: if two rows claim to write stock in the platform, one of them is wrong, and the map says which. At Kestrel & Pine the second rule removed the pre-order app's inventory

permission and became Waypoint decision W1's organizational half.

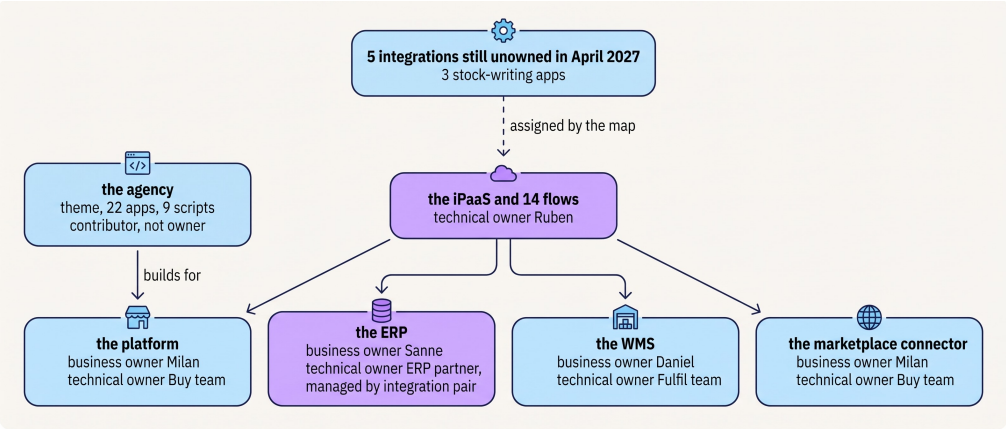


Figure 2: *Kestrel & Pine's ownership map in outline: every system has a business owner and a technical owner inside the company, the agency is a contributor, and the five integrations still unowned in April and the three stock-writing apps are assigned to the integration pair until retired or re-homed.*

Team Shapes for a Commerce Merchant

Conway's 1968 paper observed that organizations "are constrained to produce designs which are copies of the communication structures of these organizations." In a merchant this works in reverse: a landscape of a dozen vendors pushes the organization into one team, or one person, per vendor, and the handoffs between systems become handoffs between people who report to different managers. **The organization copies the vendor landscape, and the vendor landscape's gaps become the organization's.**

Team Topologies (Skelton and Pais, 2019) gives the vocabulary that fits a small team. Four team types: **stream-aligned** (aligned to a flow of work from a segment of the business domain), **platform** (an internal product that accelerates stream-aligned teams), **enabling** (helps a team overcome obstacles and detects missing capabilities) and **complicated-subsystem** (where deep technical expertise is needed). Three interaction modes: collaboration, X-as-a-service, facilitation. One premise: teams can only handle so much complexity before breaking down. The vocabulary was written for software companies; the mapping to a merchant is mine.

The options

By system (one team or person per vendor). The default a vendor landscape produces: the platform person, the ERP person, the WMS person, the agency for the storefront.

Cheap to set up, matches support contracts, and reproduces every integration gap as an organizational gap. Nobody owns the order's journey.

By journey (stream-aligned). One team owns "buy" (storefront, catalog publication, checkout, marketplaces listings, promotions) and one owns "fulfill and return" (order hub, warehouse integration, shipping, returns, EDI, the electronic exchange of orders and invoices with trading partners). Each team touches several vendors; each handoff inside a journey has one owner. The integrations between the two journeys (the order leaving checkout for the OMS) need a contract between the teams.

By layer (front end, integration, data). A storefront team, an integration team, a data team. Fits the skills most merchants hire (front-end, back-end, data) and makes the integration layer a product with an owner. The risk is that the integration team becomes the bottleneck for every change and nobody owns a journey end to end.

Agency-led. The agency owns the storefront and the apps; the merchant owns the accounts and a backlog. Right at the Starting stage, workable at the Growing stage, and the origin of most Omnichannel-stage incidents when the merchant has not taken the joins in-house.

	By system	By journey	By layer	Agency-led
Who owns the order's journey	Nobody	A team	Nobody; the layers do	The agency, implicitly
Who owns the integration layer	The vendor connector or nobody	Must be assigned separately	The integration team	The agency or nobody
Fits team size	Any	8 and up (two teams of 3–5 plus a pair)	10 and up	1–5
Handoffs become	Tickets between people	Contracts between two teams	Queues into the integration team	Change requests
Main risk	The Spaghetti Junction in the org chart	Both teams need the same integration pair	The bottleneck	The Lynchpin

	By system	By journey	By layer	Agency-led
When it fits	Starting; a Group's shared-service vendor manage-ment	Omnichannel	Omnichannel with a strong integration lead	Starting, early Growing

My recommendation for ten to twenty people combines the second and third: one or two stream-aligned teams by journey, one pair (not a team) that owns the integration layer, the ownership map and the reconciliation jobs as an internal product, the data engineers as a second small platform function under the data lead, and the vendors and agency as X-as-a-service providers with written contracts. The pair, not a team, because at this size a full integration team becomes the bottleneck the by-layer option warns of. The condition under which this does not hold: fewer than eight engineers, where one journey team plus one integration engineer is all the structure the people can carry.

Two public cautions bound the autonomy a merchant gives its teams. Henning Jacobs' retrospective of March 2020 on a decade at Zalando reports that team autonomy over technology choice produced similar services "written in six languages", which "hurts productivity at scale", and that a standard stack would have been "a major win".

Uber's July 2020 account of its domain-oriented architecture describes about 2,200 microservices in which a root-cause investigation could run through fifty services owned by twelve teams.

Both are six years old and describe organizations a hundred times larger than any merchant; the lesson scales down without change: **journey teams need a shared integration standard, or each will build its own connector to the same ERP**. Werner Vogels' 2006 line "you build it, you run it", relayed by AWS's blog at the time, is the other half: the team that changes a flow is the team paged when it fails.

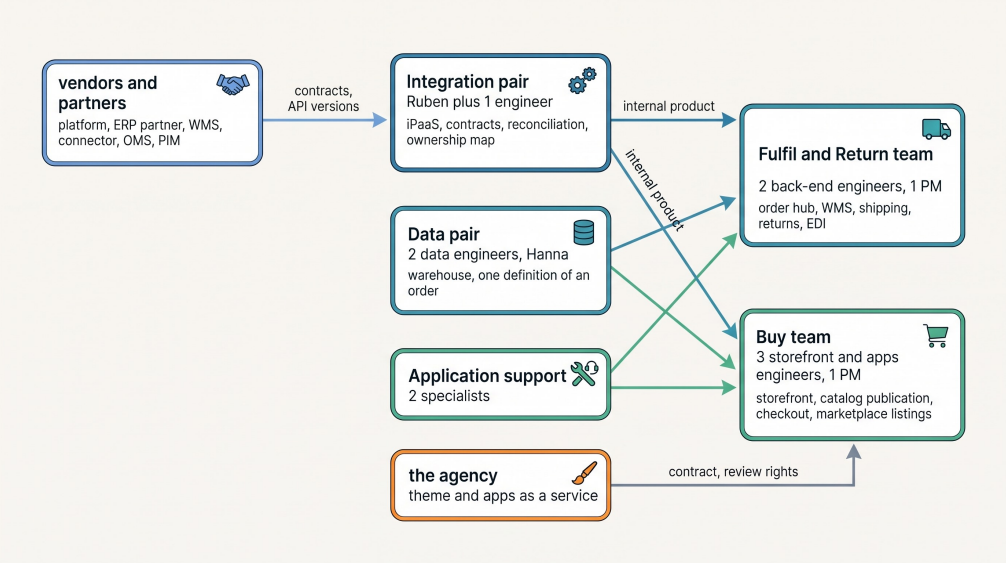


Figure 3: *Kestrel & Pine's 2027 team shape: two journey teams, an integration pair that serves both as an internal product, a data pair, shared application support, and the agency and vendors as services under contract.*

Working With Agencies and Implementation Partners

An agency or implementation partner builds what the merchant cannot staff: the theme, the first apps, a replatforming, an ERP implementation. Hobbycraft's 2022 move to Salesforce Commerce Cloud, reported by a trade site in June 2022, was built by an agency after a competitive pitch, using the agency's accelerator and a stack of six third-party services; the article reports the build, not the outcome, which is typical of agency-sourced evidence. The vendor-side mechanisms for agency work are structural: Shopify's development stores exist so a partner can build a store and transfer it to the merchant, with real transactions disabled until transfer (undated, accessed November 2026); Salesforce's cartridge path layers partner-built code over the vendor's base.

The rules I use, stated as judgement because no dated study of merchant–agency practice in the Netherlands, Belgium or Germany could be found:

- **The merchant owns the repository, the vendor accounts, the API credentials, the ownership map and the decision records.** The agency commits to the merchant's repository and holds review rights, not ownership. If the agency holds the platform login, the merchant has a Lynchpin.
- **Every custom module, app or script the agency builds is a row in the map** with its scopes (what it may read and write), its pinned API version, the platform

release it was tested against, and an exit plan: who could maintain it if the agency left.

- **The statement of work names the API versions and platform releases the deliverable is tested against**, and a deprecation clause: when a vendor announces a retirement, who updates what by when.
- **Ownership after launch is written down before launch.** Either the agency owns the row under a support contract with a named engineer, or the merchant's team does, from the day of go-live. "We'll see" means nobody.
- **Separate building from keeping.** Pay the agency for the first; staff the second.

Kestrel & Pine's agency has been retained since 2017 at €14,000 a month for the theme, apps and small integrations. The 2027 statement of work, after the map, lists each of its nine scripts with an owner and a pinned version; three scripts that write to stock or prices moved in-house to the integration pair, and the rest stay with the agency under a named engineer *(illustrative).* The ERP partner's platform↔ERP connector, built in 2023, gets the same treatment: the partner maintains it; Ruben's pair owns the contract, the monitor and the reconciliation.

Vendor Management: What the Contract Must Say

A merchant on rented software is governed by its vendors' terms more than by its own policies. Shopify's Terms of Service, last updated 20 October 2026, provide the services "as is" and "as available", do not warrant that they "will be uninterrupted, timely, secure, or error-free", and reserve the right to modify the services "for any reason, without notice and at any time". Plus and enterprise agreements may add service levels; none was available to verify, so ask for the actual agreement rather than a marketing uptime figure.

What a merchant can govern is its own dependence. The **vendor card**, one per vendor, sits beside the ownership map:

Field	What to record	Dated example
Term and notice	Contract length, renewal, notice period	Shopify Plus is sold on one- or three-year terms (pricing page, accessed 2026-11)

Field	What to record	Dated example
Versioning and deprecation	How the vendor versions its API and how long versions live	Shopify releases an API version quarterly, supports each for at least twelve months, and "falls forward" retired versions to the oldest supported one (developer documentation, accessed 2026-11); Adobe Commerce gives each release three years of standard support and will stop maintaining cloud environments on unsupported versions from 1 June 2027 (lifecycle policy, accessed 2026-11); WooCommerce states that backward compatibility between releases "is not guaranteed"
Rate limits and quotas	What the tenant may call, shared by every app and connector	Shopify's GraphQL Admin API bucket refills at a plan-dependent rate, from 100 to 2,000 points a second (accessed 2026-11); every app on the store draws from it (Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation)

Field	What to record	Dated example
Export and exit	What data can be exported, in what form, at what cost	The EU Data Act (Regulation 2023/2854) prohibits switching charges for data processing services from 12 January 2027; whether hosted commerce platforms fall under it is an open legal question as of November 2026
Extension retirement	Which mechanisms the vendor has retired and on what notice	Shopify Scripts and the Script Editor were retired on 30 June 2026; scripts no longer run on any store (help documentation, accessed 2026-11)
Service levels in your units	Not the vendor's uptime but your checkout success rate, order-to-WMS latency, stock-sync lag	Google's SRE book (2017) defines an SLO as a target for a measured service level and an SLA as the contract with consequences; the vendor's SLA covers its uptime, and the merchant's integration SLOs are nobody else's
Residency and tenancy	Which region each SaaS system was provisioned in	commercetools' regions are isolated and moving is a migration, not a setting (documentation, accessed 2026-11)

The versioning row deserves emphasis because it is the Pulled Rug's calendar. Kestrel & Pine's Deprecated Endpoint was a published date that nobody had put on a calendar. The vendor card turns each vendor's deprecation policy into a dated list, and the ownership map says who updates which row by when. Promises Between Systems: API Contracts, Webhooks and Versioning owns the mechanics.

Governance That Fourteen People Will Follow

Governance in a merchant has one purpose: to keep the ownership decisions true after the meeting ends. Heavy process (architecture boards, review gates, templates that take an afternoon) will not survive a team of fourteen with a peak season. Light process can, if it is specific.

Decision records. Michael Nygard's note of November 2011 defined the architecture decision record: a short text file with title, context (the forces at play, "described as neutral facts, often in tension"), decision ("We will…"), status (proposed, accepted, deprecated, superseded) and consequences. Without one, a newcomer can only "blindly accept the decision" or "blindly change it". In a merchant the decisions worth a record are the ownership ones (which system is the system of record for stock, price, product, order), the vendor ones (why this OMS, with what exit) and the app ones (which apps may write stock). Kestrel & Pine's Waypoint decisions W1–W8 are ADR-shaped; Noor numbered them as the first eight records *(illustrative)*, and the 2029 review is the date on which W4 is re-opened.

Standing rules. Rules that need no meeting, because they are checkable from the map:

Rule	Check	Pattern it prevents
One writer per fact	The map's "what it may write" column has no duplicates	The Secret Agent, The Phantom Stock
Every integration has an owner, a contract card, idempotent handling and a reconciliation job (Waypoint W5)	No blank cells; each row names its reconciliation	The Spaghetti Junction, The Message in a Bottle
Every app's scopes are recorded and reviewed quarterly; unused apps are removed	41 apps, 6 unused, 3 writing stock becomes 34 apps by July 2027, none unused and none writing stock	The Secret Agent
Every integration pins an API version, recorded with the vendor's retirement date	The vendor card's calendar has no past-due rows	The Pulled Rug

Rule	Check	Pattern it prevents
No change to stock, price or order flows in the four weeks before peak without the peak owner's approval	A dated freeze in the calendar	The First Snowfall

A lightweight review. One 45-minute **architecture hour** a week, open to both journey teams, the integration pair and the data pair, with a standing agenda: new rows in the map, proposed ADRs, deprecation dates in the next ninety days, and one incident replayed against the map. No slides. The output is edits to the map and to ADRs. At Kestrel & Pine this hour replaced three vendor-specific status meetings *(illustrative)*.

The Product Manager in a Merchant

A merchant that buys everything still needs its own product managers, for a reason of asymmetry. The vendor's product manager optimizes the platform's product: GMV through it, tenancy, the app ecosystem. The merchant's product manager's product is **the order's journey**: the share of visits that become orders, the average order value, the share of orders delivered when promised, the return rate and the cost of a return, the time from "where is my order?" to an answer. Those are the metrics of Who Sells What to Whom: Business Models and the Economics That Shape Architecture, and no vendor optimizes them for a particular merchant. Kestrel & Pine's two product managers sit one per journey team; their backlogs are written in those metrics, and a vendor's roadmap is an input, not a plan.

Engineering Practices on Rented Software

Environments and sandboxes. On SaaS the merchant never gets a full copy of production. Shopify's development stores are for development and testing only, cannot process real transactions and are capped at 250 per organization (accessed November 2026); Stripe's sandboxes are isolated test environments where "card networks and payment providers don't process payments" (accessed November 2026); Adobe Commerce Cloud Pro provides integration, staging and production environments, with staging on dedicated hardware matching production (accessed November 2026).

The gap is the chain: no vendor offers a sandbox of platform plus payment provider (PSP) plus ERP plus WMS together. Zalando's 2020 account of its Cyber Week preparation describes order-flow simulators that place marked test orders which fulfillment skips, exercising inventory and payments in production.

A small team can copy the pattern at small scale: a test customer, a test SKU, an order flag the WMS and the ERP recognize, and a weekly run through the whole chain.

Release management. The release calendar is set by the vendors: quarterly API versions on hosted platforms, yearly release lines on suites, connector updates whenever a marketplace or carrier changes its interface. The cheapest control is to pin the version in every connector and record it in the map.

Feature flags. Pete Hodgson's taxonomy of October 2017 (release, experiment, ops and permissioning toggles) gives the team a vocabulary: release toggles to ship unfinished checkout changes dark, ops toggles as the peak-day switch-off list of Black Friday Every Year: Peak Load, Resilience and Recovery, permissioning toggles for staff-only or B2B-only features.

AI coding tools. The dated evidence is mixed and worth stating plainly. DORA's 2024 report (October 2024) found more than 75% of respondents relying on AI for at least one daily task, with a 25% increase in adoption associated with small gains in documentation and code quality but an estimated 1.5% lower delivery throughput and 7.2% lower delivery stability. DORA's 2025 report calls AI "an amplifier, magnifying an organization's existing strengths and weaknesses".

METR's randomized trial of July 2025, with sixteen experienced developers on 246 tasks in mature repositories, measured a 19% *increase* in completion time with early-2025 tools, while the developers believed they had been 20% faster. Shopify's CEO wrote to staff in April 2025 that "reflexive AI usage" is a baseline expectation and that teams must show why AI cannot do a task before asking for headcount.

For a commerce team the stability finding is the one that matters: AI tools raise output on exactly the code that touches stock and prices (connectors, theme code, app glue), so tests against sandboxes, small batches, review and the one-writer rule are preconditions, not afterthoughts. AI shifts the bottleneck from writing a connector to owning it.

Across Scale Stages: When It Fits and When It Limits

	Starting	Growing	Omnichannel	Group
Typical shape	Founder plus the platform's support; an agency for the theme	The agency builds; a first engineer; the ERP partner	10–20 people: journey teams, an integration pair, a data function	Stream-aligned teams per brand or journey; a platform team; vendor management as a function
Who owns integrations	The platform's own connectors; nobody else needed	The agency or the partner, then nobody after launch	The integration pair, by map	The platform team, as an internal product with API guidelines
What becomes necessary	The founder holds every account; an app list with scopes	The first engineer owns the map, the repository and the API pins; a staging store with marked test orders	Named owner per row; ADRs; the architecture hour; quarterly app review; a peak owner	Shared standards across teams; an acquisition integration playbook; group vendor contracts
Warning signs of outgrowing it	A second channel arrives and the agency builds the connector with no owner	Support tickets route to the agency; an incident reveals a script nobody knew	Both journey teams build their own connector to the same ERP; the integration pair is the bottleneck for every change	Teams choose their own stacks and build the same service twice (Zalando's six languages)
Kestrel & Pine	2016–2018	2019–2022: Ruben Alves from 2022	2027 shape above	Deferred to the 2029 review

The judgement: the team shape should change one stage *before* the system landscape does, because the owner has to exist before the connector is built. Kestrel & Pine hired its lead integration engineer in 2022, a year after the connector and the iPaaS arrived; the thirteen integrations without a named owner in January 2027 are the cost of that year.

In Practice: Kestrel & Pine, Before and After

Before (January 2027). Thirteen systems; 34 integrations of which thirteen had no named owner; 41 apps of which three wrote inventory; the agency holding the theme, 22 apps and nine scripts on a €14,000 monthly retainer; the ERP partner holding the platform↔ERP connector; one engineer (Ruben) owning the iPaaS and, informally, everything nobody else owned; technology spend of about €2.6 million a year. Support answered "where is my order?" from five tabs. Nobody had a calendar of vendor deprecation dates.

After (from April 2027). The ownership map with no blank cells. Two journey teams (Buy: three storefront and apps engineers and a product manager, with Milan as business owner; Fulfil and Return: two back-end engineers and a product manager, with Daniel and Lea as business owners), the integration pair (Ruben and one back-end engineer) owning the iPaaS, the contracts, the reconciliation jobs and the map, the data pair under Hanna Berg, and two application support specialists serving both journeys. ADRs KP-001 to KP-008 are Waypoint W1–W8. The two integration engineers Waypoint adds from mid-2027 join one journey team each, so the integration pair stays a pair. A vendor card per vendor with a deprecation calendar. The agency under a statement of work that lists each script with an owner and a pinned version. The architecture hour on Tuesdays.

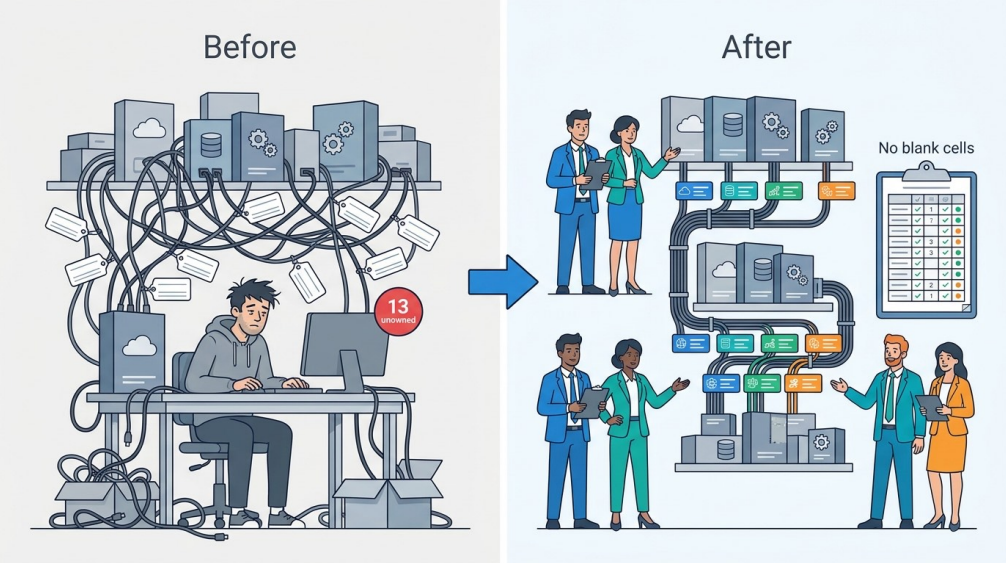


Figure 4: *Before, 13 of 34 integrations had no named owner and one engineer held whatever nobody else owned; after, the ownership map has no blank cells.*

The Deprecated Endpoint, replayed. With the map, the bol stock script has a row: technical owner the integration pair, pinned version recorded, the connector's retirement date on the vendor card ninety days ahead, and a monitor on stock-sync lag to bol. The script is updated in the week the retirement is announced, or retired in favor of the connector's native stock flow. The two stale days do not happen. The cost of the control is one row and one calendar entry.

When It Goes Wrong

The Lynchpin leaves. The one EDI consultant who built Bergzicht's connection, or the one engineer who knows the iPaaS, resigns. Detection: the map shows one name on many rows. Recovery: pair every critical row with a second name, document the contract card, and treat a single-name row as a risk register entry with a date.

The agency contract ends. On exit the agency holds the theme repository, the app credentials and the scripts. Detection: the merchant cannot deploy a theme change without the agency. Recovery: repository and credentials transfer as a condition of every statement of work, from the first one; a handover run through the map, row by row.

The vendor retires the mechanism. Shopify's retirement of Scripts on 30 June 2026 is the dated example of a platform ending an extension mechanism on a published date; Adobe's June 2027 enforcement of supported versions on its cloud is another. Detection: the vendor card's calendar. Recovery: migration to the replacement mechanism

(Functions, in Shopify's case) planned a year ahead, owned by the row's owner (Apps, Plugins and Functions: Extensibility and Ecosystem Governance).

The Secret Agent returns. A new app, installed by a marketer to run a pre-order campaign, asks for an inventory-write scope and gets it. Detection: the quarterly app review, or better, an install rule that routes any write scope on stock, price or orders through the integration pair. Recovery: revoke the scope, reconcile the stock the app touched (the May 2027 reconciliation found 2.3% of active variants different, most from such apps), and record the decision.

Two teams, two connectors. Both journey teams integrate with the ERP, each its own way. Detection: two rows in the map with the same target and different contracts. Recovery: the integration pair owns one flow and the other team consumes it; this is what the pair is for.

What to Take Away

Established.

- Organizations produce designs that copy their communication structures (Conway, 1968); a merchant organized one team per vendor reproduces its integration gaps in the organization chart.
- Vendor partner programs pay agencies to build and to keep merchants on the platform (Shopify: 20% of the platform fee plus 0.1% of GMV for four years, accessed 2026-11), not to own integrations after launch; ownership must be designed by the merchant.
- Decision records with context, decision, status and consequences (Nygard, 2011) are the lightest durable form of architecture governance and fit a team of any size.
- Autonomy without shared standards is costly at scale (Zalando's 2020 "six languages"; Uber's 2020 fifty-service investigations), and the lesson holds for two journey teams sharing one ERP.

Depends on context.

- Journey teams plus an integration pair fit ten to twenty people at the Omnichannel stage; below eight engineers, one journey team and one integration owner is all the structure the people can carry; at the Group stage the pair becomes a platform team.
- Agency-led engineering is right at the Starting stage and workable at the Growing stage, provided the merchant holds the accounts, the repository and the map from the first contract.

- How much to formalize vendor management depends on the vendors' versioning policies: quarterly dated API versions with twelve-month support (Shopify) need a calendar; three-year release lines (Adobe Commerce) need an upgrade plan; "not guaranteed" compatibility (WooCommerce) needs a test suite.
- AI coding tools raise output where the evidence says stability falls (DORA 2024); their value in a commerce team depends on the tests, review and scope governance already in place.

The author's synthesis.

- The ownership map is the merchant's first architecture artifact and the only one that prevents incidents on its own: at Kestrel & Pine three of ten incidents sat in its blank cells.
- Change the team shape one stage before the system landscape changes; the owner must exist before the connector does.

Questions to Ask

1. Is there a map with one row per system, integration and app, and does every row name a technical owner inside the company?
2. Which facts (stock, price, order status, refunds) have more than one writer, and which row says so?
3. Who holds the platform, PSP, ERP and connector logins, the repository and the API credentials: the merchant or the agency?
4. Which vendor deprecation dates fall in the next twelve months, and who updates which integration by when?
5. Does the agency's statement of work name the API versions and releases the deliverable is tested against, and who owns each deliverable after go-live?
6. Does any team own the order's journey end to end, or only systems?
7. Is there a written decision for which system is the system of record for stock, orders and product content, and when it will be reviewed?
8. How does a change to a stock, price or order flow reach production: through which review, which sandbox, which flag, and under which freeze before peak?
9. Which rows in the map carry one name only, and what happens if that person leaves?

Ownership answers "who"; the last chapter of this part asks what the next kind of actor, an AI agent that discovers, compares and buys, will need from the systems and the owners described here. Continue with Assistants and Agents: AI in Commerce and What Is Real in 2026.

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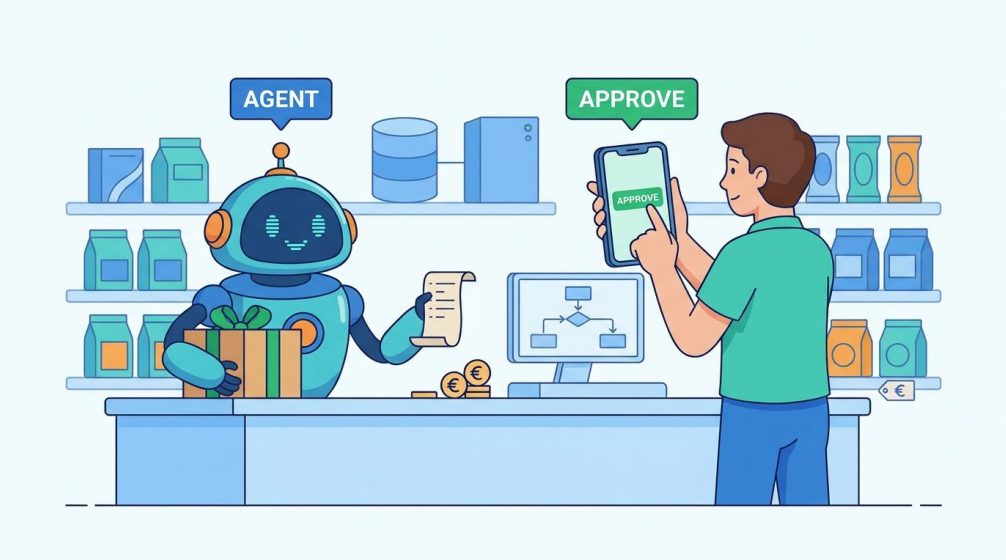
Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

45. Assistants and Agents: AI in Commerce and What Is Real in 2026



IN THIS CHAPTER, YOU WILL: Separate what AI already does in commerce from what agents are promised to do, read the protocol landscape as of November 2026, and build the readiness list a merchant needs before any agent may buy.

KEY POINTS:

- **Recommending a purchase and making it require different permissions.** AI can suggest products, draft descriptions or flag suspicious payments. Software authorized to place an order, issue a refund or change stock also needs limits, an identity and someone accountable for its actions.
- **Agentic commerce is real, early, and not yet in Europe.** Between September 2025 and November 2026 the flagship in-chat checkout launched and was pulled back, three protocols (ACP, UCP, AP2) were published with no settled governance, and every live consumer checkout program is limited to the United States, Canada or Australia.

European payment rules require the shopper to authenticate the exact cart.
- **Agent readiness is a data and API program, not an AI program.** Product data from one owner, availability and a delivery promise by API, order status by API, structured policies and an identity for every agent make a merchant ready for any protocol. Kestrel & Pine's Waypoint decision W8 buys exactly that in 2027, and no shopper-facing agent.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Secret Agent** — an AI agent or admin assistant that changes orders, stock, prices or money without its own identity, a named owner and a person's approval is this pattern's newest form.
- **The Phantom Stock** — an agent channel is the harshest consumer of availability, because a stale copy becomes a promise a third party makes for you.
- **The Giant's Boots** — building your own shopping agent, or implementing three competing protocols in the storefront, copies the vendors' problem into a fourteen-person team.

A shop can use AI to **recommend a jacket** or authorize it to **place an order for that jacket**. A shopper can ignore a recommendation. Placing the order can reserve stock and initiate a payment. Permissions and spending limits bound what the software may commit to; an accountable owner must be able to investigate and correct a mistaken action.

This chapter compares those two uses of AI: **helping someone decide and carrying out the decision**. Both need reliable data. If the stock count is two hours old, an assistant may recommend a sold-out jacket; an agent may order it. The merchant still has to explain the cancellation to the customer. Before adding either, establish where product, price and stock answers come from, and who may change them.

This chapter explains **agentic commerce**, software acting in a shopping or merchant workflow, and the interfaces and controls a merchant needs to support it. Deeper agent design appears in Tool Ecosystems, MCP and the Tool Broker, Side Effects: When Agents Touch the World and Agent Identity, Delegation and Least Privilege.

Where we are in the order's journey. A shopper might ask an assistant for a waterproof jacket under €200, delivered by Thursday. Authorized software might then place the order or request an exchange later. Each step needs current product facts, a clear permission and a confirmed result.

Setting the Scene at Kestrel & Pine: The Agent Test

Kestrel & Pine, the fictional outdoor brand, faces three requests in February 2027. Its agency offers a shopping assistant for **€60,000 to build and €2,500 monthly to run**, both illustrative assumptions. A board observer asks CEO Mara Lindqvist whether shoppers can find the brand in ChatGPT. The platform also enables an AI channel with checkout limited to US shoppers, an assumption modeled on the documented programs below.

Head of Engineering Noor Visser tests the underlying data before judging the proposals. She asks five product questions about the Ridgewalker women's M in moss at two moments: the Black Friday oversell at 20:00 on 27 November 2026 and Eva's order at 21:14 on 13 April 2027.

The existing platform, **warehouse management system (WMS)** and returns portal must supply the answers. An **integration platform as a service (iPaaS)** carries data between them. **Waypoint**, the improvement program, adds **product information management (PIM)** for content and an **order management system (OMS)** for shared orders. The question is whether those records can support a reliable action.

What an agent asks	Which system answers today	True on 27 Nov 2026, 20:00?	True on 13 Apr 2027, 21:14?
What is it?	The platform's product record; the waterproof rating only in a spreadsheet	Partly: no rating, three copies of the description	Same
Is it in stock?	The platform's stock copy, up to two hours behind the WMS	**No**: sold out at 19:40; the copies said yes until about 20:30 (storefronts) and 21:35 (marketplaces)	Yes, by luck
When will it arrive?	A sentence in the theme: "ordered before 22:00, delivered tomorrow"	**No**: the WMS queue was four hours behind	**No**: the 21:00 wave had gone; delivered Thursday, not Wednesday
What does it cost delivered, can I return it?	Price in the platform; policies in page text	Price yes; policies not machine-readable	Same

What an agent asks	Which system answers today	True on 27 Nov 2026, 20:00?	True on 13 Apr 2027, 21:14?
Where is my order, can I exchange it?	Platform, WMS via the iPaaS, returns portal: no single answer	The Three-Status Order was two weeks away	Mostly, for one parcel

Two of five answers were wrong on the Oversell night, one on an ordinary evening, and none came from a system that owned the fact. **An agent would not have fixed any of this; it would have broadcast it.** Noor's memo ended with one line, which became Waypoint decision **W8**: agent readiness, not agents. The agency's assistant was declined; the vendor's AI channel stayed on for discovery only.

Two Kinds of AI, and Why the Difference Is Architectural

In plain words: A **model** is software trained on data to produce an output from an input: a ranking, a sentence, a score. A **large language model** (LLM) is a model trained on text that can read and write natural language, which is what makes chat assistants possible. An **assistant** is an LLM wrapped in a product that helps a person. An **agent** is an assistant that has been given tools and permission to use them: it can call an API that creates a cart, places an order or changes a stock figure. **Agentic commerce** is the business term for agents that discover, compare and buy on a shopper's behalf, inside a chat product or a marketplace app.

The architectural question is **which actions the software is allowed to carry out**, whatever the vendor calls it. Drafting a description differs from publishing it; suggesting a refund differs from sending the money. Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record assigns authority over those facts, and Who Builds and Who Owns: Teams, Vendors, Agencies and Governance assigns accountable owners. Use the same permission checks as other integrations so an agent cannot bypass a stock or payment limit. Record its actions so the owner can explain and correct a mistaken commitment.

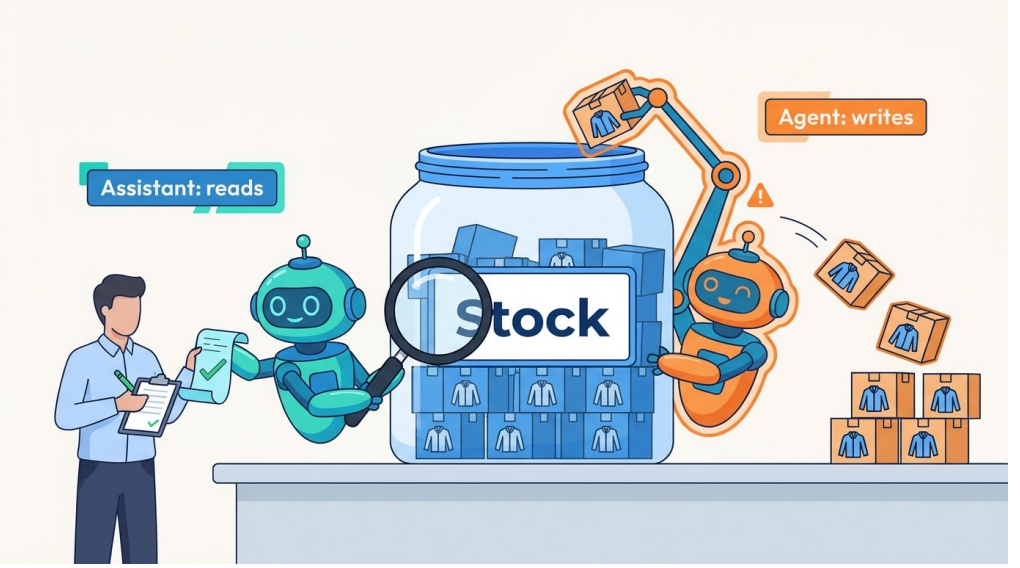


Figure 1: *The assistant uses the stock count to suggest a purchase. The agent can place the order and change the stock count, so it needs permission to do so.*

What AI Already Does in Commerce

Six uses are established. Each is worth having at the right stage, and each has a public record thinner than the marketing.

Search and recommendations. Amazon's item-to-item collaborative filtering (recommending from what other shoppers bought together) was described in a 2003 paper; its 2019 history records that a deep-learning model first lost to the old method and a plain bestseller list: **a strong baseline is hard to beat**. Hosted "AI search" adds vector search, which matches meaning so that "waterproof jacket" finds "rain shell"; every published lift is a customer case without a control (Finding the Product: Search, Navigation and Merchandising). Zalando's assistant, a ChatGPT-based beta in April 2023, had "more than 22 million unique users" in 2026 by its Q3 2026 release (3 November 2026).

Content. Generated listings are common (Amazon reported more than 100,000 sellers using its tools in September 2023). The architectural point is where generated text lands: **in the system of record for product content, with a review state, or the channels drift apart** (What Are We Selling? Products, Variants and the Catalog).

Service bots. Klarna said on 27 February 2024 that its assistant handled 2.3 million conversations in its first month, two thirds of its chats, the work of 700 agents. In May 2025 its CEO told Bloomberg it was rehiring humans because the bots produced "lower quality" output: measure resolution, not **deflection** (the share of contacts a bot absorbs). **A bot is only as good as the APIs behind it**: order status, shipment events, return

eligibility and refund state, the same APIs an external agent needs, and the ones Kestrel & Pine's "five tabs" could not provide (Where Is My Order? Notifications, Service and the Customer Record).

Fraud. Scoring belongs in the PSP (the payment service provider that takes the shopper's money) or a specialist; Stripe's Radar page claims a "32% average reduction in fraud" with no accuracy or false-positive rate published. Agents break the premise of bot defense, that a non-human session is hostile (Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers).

Forecasting. Deep demand forecasting (Amazon's DeepAR paper, 2017) produces a purchase proposal a buyer approves, and it needs clean sales and stock-out history, which a merchant with six writers of stock does not have.

AI traffic. Adobe Analytics reported in January 2026 that visits to US retail sites referred by generative-AI products rose 693% year on year in the 2025 holiday season, growth from an unpublished, small base.

What the six have in common: each reads the merchant's data, and each writes either a derived copy that goes stale or a proposal a person approves. None of them owns a fact.

Assistants in the Admin: The Secret Agent Wears a Badge

The second place AI arrived is the back office. Shopify announced Sidekick in July 2023 and said on 21 May 2025 that it could run reports, create segments, filter orders and set up discount codes "with the merchant's approval"; its help page (accessed November 2026) describes it editing orders and presenting changes for review before applying them, and its Fall 2026 edition claims daily sessions up 4.2 times year on year, a vendor figure. Shopware (June 2026) and Salesforce (July 2026) followed with their own admin agents.

Three things are the same across all of them: **the assistant acts through the platform's own APIs and permission model**; **every vendor describes a draft-and-review step**; and **the assistant's reach ends where the platform's data ends** (Sidekick cannot see Kestrel & Pine's WMS unless an app exposes it). The review step is a vendor default today and a setting someone can switch off; a merchandiser who lets an assistant "clean up" inventory has recreated the App That Wrote Stock. One writer per fact, and an assistant is not it (Human Oversight and Bounded Autonomy).

Agentic Commerce: Fourteen Months, Told in Order

On **16 September 2025** Google announced the **Agent Payments Protocol** (AP2) with more than 60 organizations, built on signed **mandates**: an intent mandate (what the

user asked for, with price limits) and a cart mandate (an unchangeable record of the exact items and price). On **29 September 2025** Stripe and OpenAI launched **Instant Checkout** in ChatGPT: US users could buy from US Etsy sellers inside the chat, under the **Agentic Commerce Protocol** (ACP), with a payment token scoped to one merchant and one cart.

On **11 January 2026** Google and Shopify announced the **Universal Commerce Protocol** (UCP), co-developed with Etsy, Wayfair, Target and Walmart, with checkout in Google's AI Mode and Gemini app for US shoppers.

On **6 March 2026** OpenAI said it was "evolving how we approach commerce in ChatGPT", and on 12 March that Instant Checkout "is moving to Apps"; ChatGPT became a discovery channel with checkout on the merchant's own storefront. On **3 November 2026** Zalando announced Zalando Extended, with its own agent planned to check out at a pilot partner in 2027.

In fourteen months the flagship in-chat checkout launched and retreated, three protocols appeared, Shopify replaced its own agent interface (Storefront MCP, May 2025) with UCP endpoints, and no party published a transaction count. **This is the most volatile part of the book, and the reason the chapter is dated to the day.**

Two shapes exist as of November 2026, and the difference decides what a merchant needs.

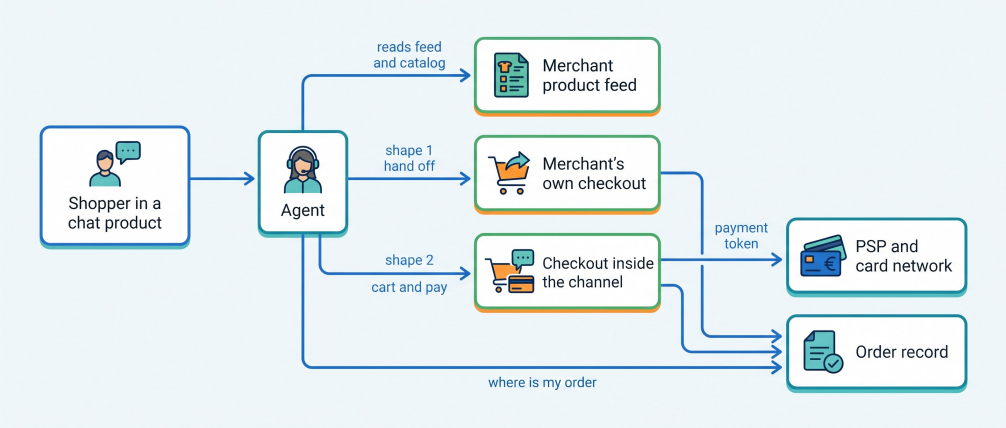


Figure 2: *Discovery with hand-off needs only a correct feed and a stable product URL; checkout inside the channel needs cart, checkout, order and payment-token APIs and an identity for the agent.*

Discovery with hand-off. The agent finds and shows the product; the shopper pays on the merchant's checkout, in an in-app browser or a new tab. The merchant needs a correct, fresh **product feed** (a machine-readable export of the catalog: identifier, title, description, URL, image, brand, price, availability, shipping, returns) and nothing else new. This is where a European merchant can be in 2026.

Checkout inside the channel. The agent builds the cart and pays, with the merchant remaining the **seller of record** (the party that legally sells, collects VAT and owes the withdrawal right). Google's AI Mode and Gemini, Copilot and Meta's Muse work this way, for shoppers in the United States, with Canada and Australia announced. The merchant needs cart, checkout, order and payment APIs, plus a way to know which agent is calling.

The Protocols in Plain Words

A **commerce protocol** is a published contract (feed, cart, checkout, order, payment token) that lets any agent talk to any merchant without a bespoke integration, as EDI lets any retailer send any supplier an order (Documents Between Companies: EDI, Punchout and E-Invoicing). Four names recur, in layers.

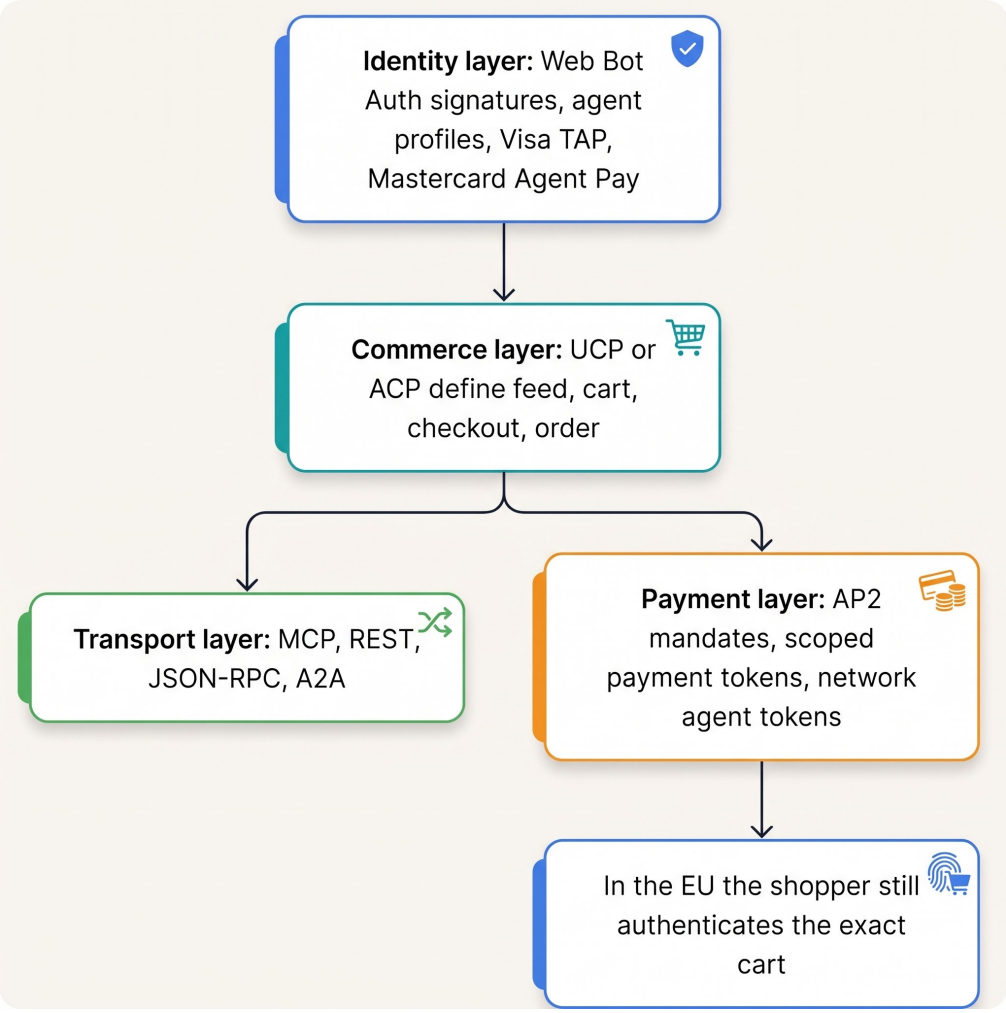


Figure 3: *MCP carries the calls, UCP or ACP say what the calls mean, AP2 and the card networks carry the money, and a signature says who is calling; a European checkout adds the shopper's own authentication at the end.*

MCP (Model Context Protocol) is the transport: how an agent discovers and calls tools such as "search catalog" or "create cart". It says nothing about what a cart is (Tool Ecosystems, MCP and the Tool Broker).

ACP is OpenAI and Stripe's specification: feed, checkout, delegated payment, and since April 2026 cart, orders and authentication. Its repository calls it "in beta", governed by the two founders; its feed upload "currently targets the US".

UCP is the Google-led consortium standard: catalog, cart, checkout, identity linking and order, over REST or JSON-RPC with MCP "built in" and AP2 for payment. Its governance council is Google, Shopify and Stripe; its repository shows no version tags. Google said on 16 September 2026 that UCP "enables direct checkout for hundreds of thousands of

brands and retailers", which counts enabled merchants, not transacting ones.

AP2 is the payment layer under UCP: signed mandates that give a "non-repudiable audit trail" of what the shopper authorized. Its repository holds samples and demos.

Agent identity is the layer the others assume. Cloudflare proposed **Web Bot Auth** in May 2025: an agent signs each HTTP request and publishes its keys at a well-known address, because "user agent headers alone are easily spoofed". It is now an IETF working-group draft (29 October 2026) and underpins Visa's Trusted Agent Protocol and Mastercard's Agent Pay; Shopify's agent profiles work the same way, where "higher trust tiers unlock broader access, including direct checkout completion". Amazon, which sued Perplexity in November 2025 over its shopping agent, insists third-party agents "operate openly" (Case: Amazon, From Monolith to Services to the Everything Store; the identity model is in Agent Identity, Delegation and Least Privilege).

Status as of 7 November 2026

Every row is volatile; re-check the primary source before relying on it.

Protocol or program (owner)	Launched	Status on 7 November 2026	Buyer markets
Instant Checkout (OpenAI, Stripe)	29 Sep 2025	Pulled back March 2026; "moving to Apps"; discovery with hand-off	US
ACP (OpenAI and Stripe as "Founding Maintainers"; Apache 2.0)	29 Sep 2025	Beta; latest spec 17 Apr 2026	Feed "currently targets the US"
AP2 (Google with 60+ organizations)	16 Sep 2025	Specification and samples; used inside UCP	Payment layer
UCP (council: Google, Shopify, Stripe, two seats open)	11 Jan 2026	Shopping spec "latest", no version tags; "hundreds of thousands" of merchants enabled	US; Canada and Australia "early next year"; UK later; EU not mentioned

Protocol or program (owner)	Launched	Status on 7 November 2026	Buyer markets
Shopify Agentic Storefronts	10 Dec 2025	Live; ChatGPT discovery only; direct checkout in Google, Copilot, Meta; Storefront MCP replaced by UCP endpoints; checkout needs signed requests	ChatGPT: must sell to US; Google: US, Canada, Australia
Visa Intelligent Commerce and TAP; Mastercard Agent Pay	Apr and Oct 2025	EU pilots with the shopper authenticating each purchase (Visa July 2026; Mastercard June 2026)	Pilot
Worldline, ING, Mastercard	2 Jun 2026	One production pilot transaction (Netherlands)	Pilot
Zalando Extended	3 Nov 2026	Partner products in the app (Westwing pilot); agent checkout planned 2027	Zalando markets
Web Bot Auth (IETF working group)	May 2025; draft -01 29 Oct 2026	Internet-Draft; deployed by Cloudflare, Visa, Mastercard	n/a

Read the table for its shape: **no protocol is governed by a standards body, no checkout program has published volumes, and the only live European transactions are pilots with the shopper present.**

Europe: The Shopper Still Authenticates the Cart

Strong customer authentication. Under PSD2's technical standards (Delegated Regulation 2018/389, in force since September 2019), a remote electronic payment needs

SCA: two of knowledge, possession and inherence (something you know, have, are), **dynamically linked** to one amount and one payee: "any change to the amount or the payee results in the invalidation of the authentication code". None of the exemptions was written for a delegated purchase whose amount is unknown when the shopper delegates.

That is why the European pilots authenticate at the moment of purchase: Worldline and ING's June 2026 transaction completed "only after the consumer explicitly approved it", and Visa's July 2026 pilot authenticated every purchase with a **passkey** (a device-bound credential, usually unlocked by a fingerprint or face, that counts as possession plus inherence).

A March 2026 law-firm note (Osborne Clarke) is plain: agent payments "remain subject to PSD2 and the RTS", with "no special 'extra-regulatory' regime"; the PSD3 texts read for this book do not address agent-initiated payments.

The consequence for architecture is small and strict. **The cart the agent builds must be identical to the cart the shopper authenticates**: price, delivery promise and tax final before the mandate is signed, the no-drift rule of From Cart to Confirmation: Checkout Orchestration. Unattended delegated purchases in the EU are unproven in public evidence as of November 2026.

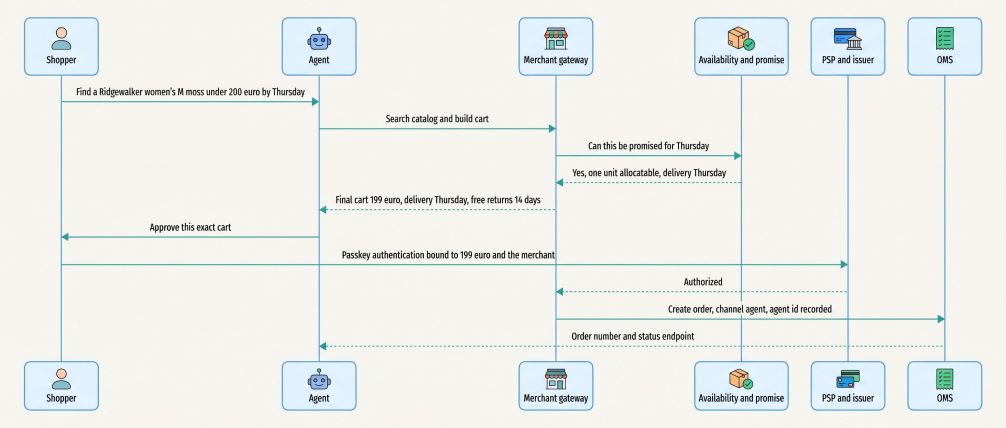


Figure 4: *A European agent purchase as the 2026 pilots shaped it: the agent builds the cart, the merchant finalizes price and promise, the shopper authenticates that exact cart, and the order carries the agent's identity for service and disputes.*

The AI Act. Regulation 2024/1689 applies generally from 2 August 2026. Its Article 50 is the live obligation for commerce: an AI system that interacts with people must tell them so "unless this is obvious", and generated content must be marked as artificial in a machine-readable way. Search and recommendations are not high-risk uses (Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation).

Consumer law. A CERRE issue paper of July 2026 argues that EU consumer law

"implicitly assumes" a human makes the purchase decision; every protocol treats the agent as the consumer's instrument, a working assumption, not settled law.

For a Dutch, Belgian or German merchant in November 2026, no live consumer channel lets an agent complete a checkout: ChatGPT has no in-chat checkout anywhere; Google's covers the US, Canada and Australia; Copilot, Meta, Perplexity and Amazon are US-centered; the card-network pilots authenticate the shopper each time; Zalando's agent checkout is planned for 2027. The realistic 2026–2027 work is feed quality, discovery presence, and payment-side readiness through the PSP and the card networks.

What a Merchant Must Have Before an Agent Can Buy

Agent readiness means the five facts an agent asks for are correct, owned by one system and served by API, work most of this book has already described.

Fact the agent needs	**Owner after Waypoint**	**Freshness**	**Consistency**	**Latency *(assumptions)***	**Availability**
Product: title, attributes, GTIN (the barcode number), images, policies	The PIM (W3), published to platform, connector and feed	Hours; feed rebuilt at every publication	One version everywhere; a review state for generated text	Lookup under about 300 ms	Feed served from a cache, never from the PIM
Availability per channel	The availability service (W1), from WMS stock events	Minutes (Shopify lets agents cache price and availability for at most 15 minutes)	Available-to-promise with a buffer, not a copy of on-hand stock	Under about 200 ms	Degrade to "check at checkout", never to a stale yes

Fact the agent needs	Owner after Waypoint	Freshness	Consistency	Latency *(assumptions)*	Availability
Delivery promise	The promise service (planned 2028)	Per request	The same answer to storefront, marketplaces and agent	Under about 300 ms	Fall back to a conservative lane table
Price and policy: price, VAT, shipping, returns	The platform's price list (What Does It Cost? Price Lists, Contract Prices, Currencies and Tax); policies as structured fields	Minutes; ACP asks for a feed update "when a sale starts or ends"	Checkout price equals feed price	Feed	Feed
Order status, returns, refunds	The OMS (W2), with the returns portal	Seconds after a shipment event	One state model for every channel	Under about 500 ms	Read model served from the OMS, not the WMS

Three observations carry the design. **The feed is most of the work, and you may already have it.** The schema.org `Product` vocabulary, Google Merchant Center's specification, OpenAI's ACP feed (nine required fields) and Shopify's catalog ask for the same fields: identifier, GTIN, title, description, URL, image, brand, price, availability, shipping, returns. The new requirement is **freshness for price and availability in minutes, not nightly. Availability and the promise are the hard part, and they are not AI work.** An agent cannot negotiate; it reads a figure and either shows the product or does not, so an agent channel is the harshest consumer of the available-to-promise of Is It in Stock? Inventory, Availability and the Promise You Can Keep and the computed promise of Getting It There: Carriers, Labels, Tracking and the Delivery Promise. **Order**

status is the same API three times, for the service bot, the order page and the external agent; build it once, on the OMS.

To the list add **an identity and a policy for every agent**: verified agents distinguished from scrapers at the CDN (content delivery network, the edge layer in front of the storefront) or platform level, an allowed set of actions per agent (read the catalog; build a cart; complete checkout only at a trust tier the merchant chose), and no agent writing stock, prices or orders directly. And add **an "agentic" marker on the order** so service, fraud review and consumer-law disclosure can treat the channel on its own terms.

The Choices and Their Trade-offs

Four realistic postures exist, and they are cumulative: a correct feed; the platform's agent channel, where the vendor carries the protocol churn but sells the platform's view of availability; an agent gateway with adapters over the merchant's own services, so a protocol change is a mapping change; and your own shopping agent, which competes with the largest technology companies for attention with none of their reach. If OpenAI found owning the checkout hard enough to retreat, a fourteen-person team should not volunteer.

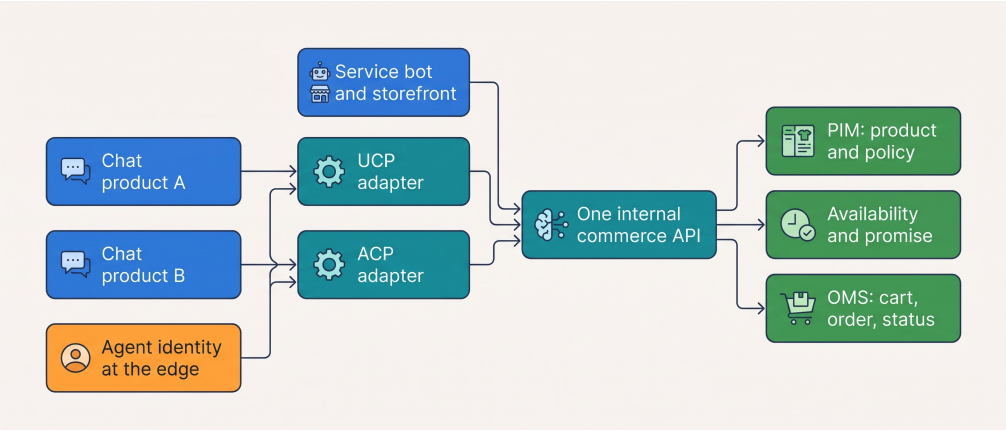


Figure 5: *The low-regret design: adapters per channel over one internal commerce API, so a protocol that changes or disappears costs a mapping, not a rebuild, and every channel reads the same availability and promise.*

Posture	What you build	What you depend on	When it fits	When it limits
Correct feed only	One feed from the product owner, refreshed in minutes	Feed consumers' rules	Every stage; the floor	No order inside a channel
Platform's agent channel	Nothing; switch on, review terms	The vendor's channels, countries, exclusions and availability copy	Starting and Growing; Omnichannel while the platform's copy is good enough	Sells stale availability if the platform is not the stock owner; no EU checkout today
Agent gateway over own services	Adapters per protocol over one internal API; identity at the edge	Your OMS, PIM, availability and promise being right	Omnichannel with Waypoint-grade services; Group	You carry protocol churn; worthless over copies
Own shopping agent	An LLM product with tools and a checkout	Model vendors, every fact above, and shoppers choosing you	Rarely; perhaps a Group with a destination app	The Giant's Boots

I would hold a merchant at the first two postures until a protocol has stable governance, a European checkout program with published volumes exists, and the merchant's own availability and order truth are better than the platform's copy; revisit in mid-2027. **What not to build yet**: a consumer-facing shopping agent; three protocols natively in the storefront; agent authentication from scratch; any agent or assistant that writes stock, prices or orders directly; any promise through an agent that the storefront cannot keep.

Across Scale Stages: When It Fits and When It Limits

Stage	Typical AI use	What becomes necessary	Warning signs
Starting (under about €2M)	The platform's built-in search and assistant; one feed to Google Merchant Center; the PSP's fraud scoring	A correct feed with GTINs and fresh price and stock; multi-factor authentication on the admin; the AI channel for discovery only	Product text edited in three places; the assistant given write permissions "to save time"
Growing (€2M–€20M)	A hosted search service; reviewed generated descriptions; the platform's agent channel; a bot on the platform's inbox	Feeds from one product owner, refreshed in minutes; structured return and shipping policies	The feed publishes the ERP's hourly stock copy; a repricing app with no floor; the bot cannot read shipment events
Omnichannel (€20M–€200M; Kestrel & Pine)	AI search shared by storefront and store staff; a bot in front of the OMS and returns portal; forecasting fed by stock events; admin assistants with review	Availability and the computed promise by API; order status by API on the OMS; an agent policy (identity, allowed actions, no writes); an "agentic" order attribute; AI Act disclosure on every bot	Agent and storefront show different availability; agent orders canceled for stock the warehouse never had; "auto-apply" switched on by a team

Stage	Typical AI use	What becomes necessary	Warning signs
Group (over about €200M or several brands)	A shared discovery platform with per-brand indexes; one order-status contract; an agent gateway per brand on one implementation	Per-brand feeds, identities and markets; one payment and disclosure policy per market; a key inventory for every agent credential	Brands implementing protocols separately; agent orders unattributable to a brand; one credential shared across brands

The readiness list does not change with scale; who assembles it does: the platform at the Starting stage, the merchant across PIM, OMS, WMS and PSP at the Omnichannel stage, a shared service at the Group stage.

In Practice: Waypoint's W8 at Kestrel & Pine

W8 is written as prerequisites with owners and dates, and a sentence about what will not be built. The figures are Kestrel & Pine illustrations.

Prerequisite	Owner	Delivered by	How an agent sees it
Product data from one owner, including the attributes shoppers search for (the Ridgewalker's 20,000 mm waterproof rating lived only in a spreadsheet)	Teun Baas, through the PIM (W3)	Q3 2027: the DE storefront in August, NL/BE and the connector in September	One feed per market from the PIM, with structured shipping and return policies

Prerequisite	Owner	Delivered by	How an agent sees it
Availability per channel from WMS stock events, with buffers	Daniel Okafor and Ruben Alves, through the availability service (W1)	Q4 2027	An availability endpoint; feed availability refreshed every 10 minutes *(assumption)*
A computed delivery promise	Noor Visser; the May 2027 promise service (transit table plus WMS cut-off and backlog), the promise API in 2028	2028	The same date the storefront shows
Order status, returns and refunds by API	The Fulfil and Return team, on the OMS (W2)	Q4 2027, after D2C orders route through the OMS on 13 October	One customer-scoped endpoint, used first by the service tool and the order page
An agent policy	Milan Petrov (business), Ruben Alves (technical)	April 2027	Verified agents may read the catalog; checkout only on the storefront in 2027; no agent or assistant holds a write scope on stock, prices or orders; every agent credential on the ownership map
AI disclosure	The Buy team	April 2027 (Article 50 has applied since 2 August 2026)	The service chat says it is automated; generated images are marked

The policy line cost nothing and mattered most: W8 extends the ownership map of Who Builds and Who Owns: Teams, Vendors, Agencies and Governance to agents by a single rule: **an agent is an integration**, with a row, an owner, a contract, a pinned API version and a reconciliation. The agency's admin-assistant trial in June 2027 ran under that rule, with auto-apply disabled and every change reviewed; it drafted 1,400 product descriptions for the autumn season, about 60% accepted unchanged, and could not touch stock *(illustrative)*. What W8 does **not** do is also recorded: no storefront shopping agent, no in-channel checkout and no custom protocol work in 2027; the gateway posture is the 2028 plan if a European checkout program exists by then. Mara's answer to the board observer: "We will be the merchant whose answers are right when the agents arrive."

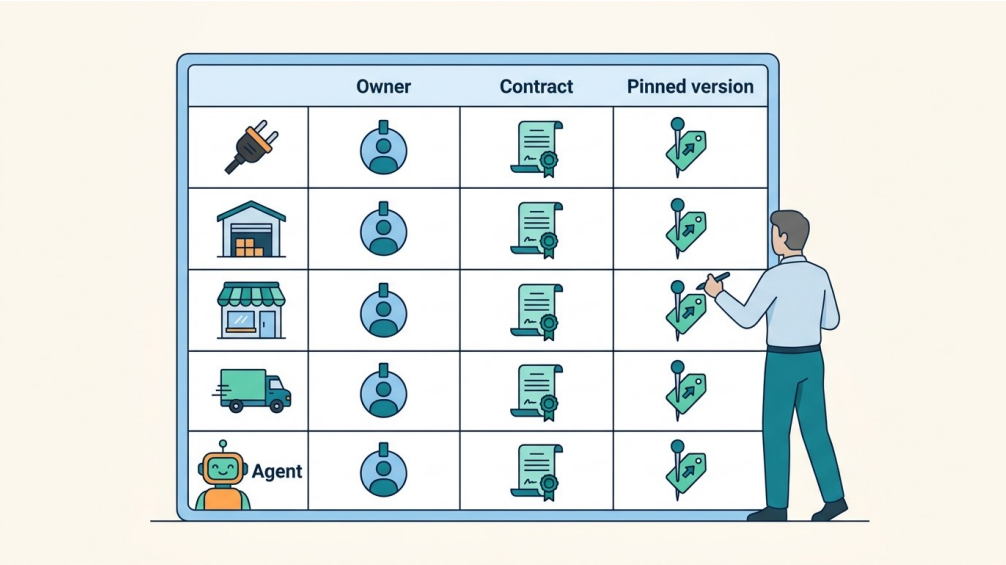


Figure 6: *Waypoint's W8 rule: an agent is an integration, with a row on the ownership map, an owner, a contract, a pinned API version and a reconciliation.*

When It Goes Wrong

The assistant with a write scope. A merchandiser enables auto-apply on an admin assistant to "fix negative stock" before a campaign; the assistant sets quantities in the platform while the WMS feed also sets them. This is the App That Wrote Stock, and the detection is the same: the weekly WMS-to-platform reconciliation (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation) and a rule that no assistant credential holds an inventory write scope. Recovery: revoke, overwrite from the system of record, add the credential to the ownership map.

The stale feed. The feed is rebuilt nightly from the platform's stock copy; a product

sells out at 19:40 and every agent shows it in stock until morning. Detection: a feed-freshness monitor (the age of the last price and stock update per market) and rising agent cancellations. Recovery: availability in the feed from the availability service, refreshed in minutes, and "check at checkout" when the service is down.

Duplicate orders from agent retries. An agent's checkout call times out, the agent retries, two orders exist. The remedy is an **idempotency key** (a unique identifier sent with the request, so a repeat has the same effect as the first) on every cart and checkout call, and a daily match of agent orders to PSP authorizations.

Cart drift before authentication. The agent shows €199 with Thursday delivery; by the time the shopper authenticates, the promotion has ended, and under dynamic linking the authentication is void. Remedy: hold price, promise and tax for the authentication window, re-quote rather than silently charge, and store the mandate reference and agent identity with the order.

The protocol pulled back. A channel you built against disappears or changes its terms within a year, as Instant Checkout did in six months and Shopify's Storefront MCP within twelve: the Pulled Rug of Promises Between Systems: API Contracts, Webhooks and Versioning at protocol scale. Remedy: adapters over one internal API and a contract card per channel.

What to Take Away

Established.

- AI in commerce has two kinds: assistants that read and propose (Amazon 2003 and 2019, Zalando 2023 and 2026, Klarna 2024 and 2025) and agents that act. The second adds a writer, and every writer needs an owner.
- As of 7 November 2026, agentic commerce is three protocols without standards-body governance, one flagship checkout pulled back within six months, and no published transaction volumes; every live consumer checkout is limited to the United States, Canada or Australia.
- In the EU, strong customer authentication and dynamic linking apply to agent payments without a special regime (Osborne Clarke, March 2026); the 2026 pilots all authenticate the shopper on the exact cart; the AI Act's Article 50 requires bots to say they are AI from 2 August 2026.
- Every feed specification asks for the same dozen fields; agent identity is converging on signed HTTP requests with a published key directory (Web Bot Auth, an IETF draft).

Depends on context.

- Switching on the platform's agent channel depends on whether the platform's availability copy is the stock owner's truth: for a merchant whose platform holds stock, yes, now; for one whose stock lives in a WMS and syncs hourly, for discovery only until the availability service exists.
- Building an agent gateway depends on owning better availability, promise and order truth than the platform, and on an EU checkout program existing; before both it is machinery with no traffic.
- How much an admin assistant may apply without review depends on the fact: generated descriptions with a review state, freely; discounts and segments, with approval; stock, prices and refunds, never.

The author's synthesis.

- Agent readiness is the same program as fixing the Phantom Stock and Schrödinger's Order: one owner per fact, availability and promise computed rather than copied, order status from one record. A merchant that does that work is ready for whichever protocol survives; one that chases protocols without it broadcasts its drift.
- Treat every agent as an integration: a row on the ownership map, an identity, a contract, a pinned version, a reconciliation; the Secret Agent is a governance failure first.
- For a European merchant in 2027 the honest plan is feed, discovery, payment-side readiness through the PSP, and a review in mid-2027; anything more is the Giant's Boots, worn early.

Questions to Ask

1. Which system owns each of the five facts an agent asks for (product, availability, promise, price and policy, order status), and can each be read by API from that system today?
2. How old is the stock figure in our product feed when an agent reads it, and what does the agent see when the availability service is down?
3. Which agent channels has our platform switched on by default, under what terms, for which countries, and which of our products are excluded?
4. Does any assistant, app or agent hold a write scope on stock, prices or orders, and is each agent credential on the ownership map with an owner, a contract and a deprecation date?
5. If an agent's checkout call is retried, do we create one order or two?
6. Is the cart the shopper authenticates identical to the cart the agent built, and how long do we hold price and availability for the authentication window?
7. Do our bots say they are AI, and are generated product images and text marked?

8. Which vendor has published a transaction volume for its agent channel, and when will we re-check the protocol table?

Part X ends here, with the newest kind of actor asking the oldest question in the book: who may make which promise, and which system has to keep it. Part XI tests the argument against the public record of real companies, starting with the platform that serves millions of merchants on one codebase. Continue with Part XI — PRACTICE: How Others Built It.

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Research and practitioner sources

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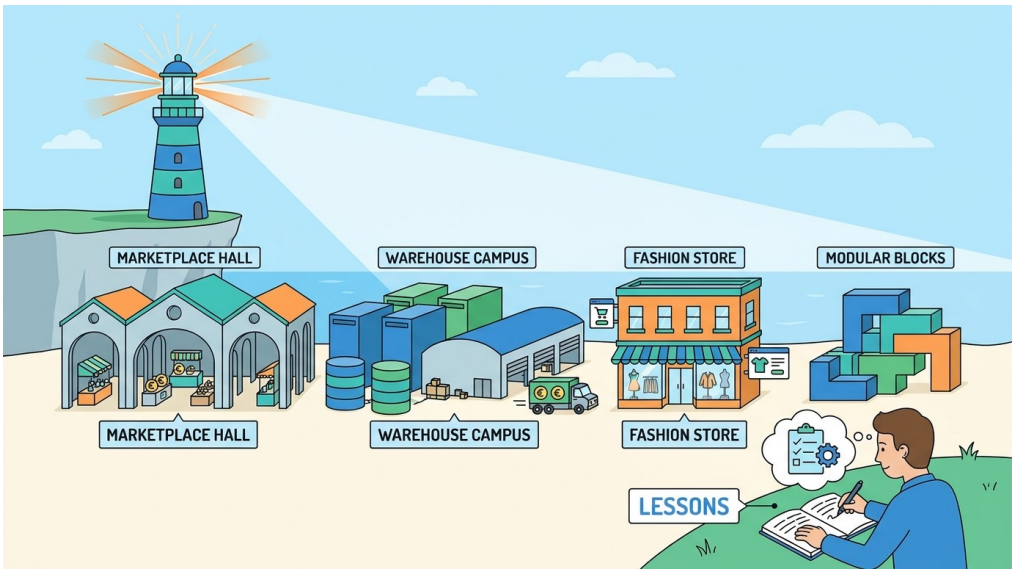
- DeepAR, Salinas, Flunkert and Gasthaus, 2017, and Deep learning based forecasting: a case study from the online fashion industry, Kunz et al., 2023 — forecasting methods, not current systems.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Part XI — PRACTICE: How Others Built It



KEY POINTS:

- **The documented record is narrower than the folklore, and more useful**. In the companies' own dated words, the slogans shrink to rules a small team can use.
- **Every answer was sized to its company**. In a fourteen-person team, Amazon's, Zalando's and the composable vendors' shapes are copies without their reasons.
- **What transfers is discipline, not machinery**. Dated API versions, reconciled integrations and one owner per fact transfer; pods and a hundred services do not.

Shopify operates a platform for merchants; Amazon and Zalando also sell, host sellers and fulfill orders. Composable vendors sell separate services that merchants assemble. Their engineering decisions address different businesses and responsibilities. Comparing those conditions helps you judge whether a documented practice would solve your problem or give your team more systems to operate without a corresponding benefit.

The cases use public filings, papers, engineering posts and documentation researched on **7 November 2026**. Claims carry dates. Older posts may describe replaced systems; unsupported internal explanations remain labeled **INFERENCE, NOT DOCUMENTED**. Each case ends with what the evidence could not establish.

The merchant applying those lessons is **Kestrel & Pine**, **the fictional outdoor brand**. Its fourteen technology staff cannot reproduce a large platform organization. The useful question is which practice helps its order flow, and what operating burden that practice adds.

Figure 1: *The Giant's Boots: each case is an architecture copied by merchants a hundred times smaller, so each chapter names what only the giant needs.*

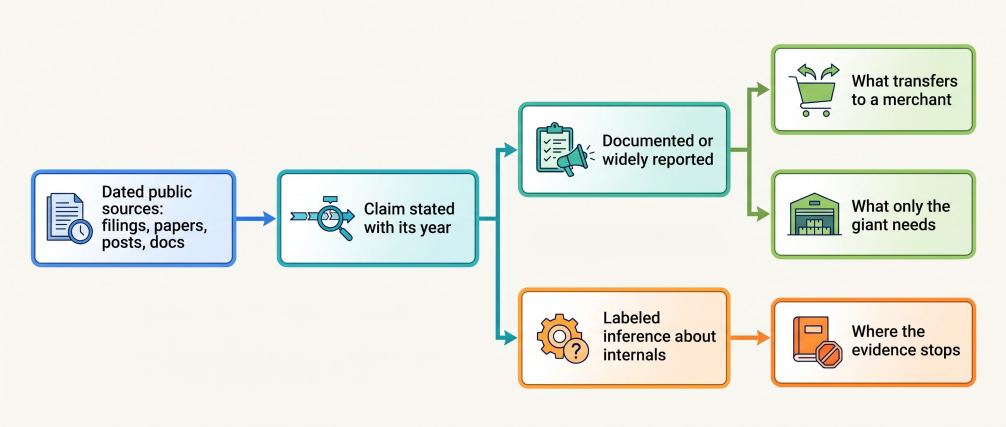


Figure 2: *How each case is read: every claim is dated and typed, documented claims are split into lessons that transfer and machinery that does not, and inferences lead to a list of what the record does not show.*

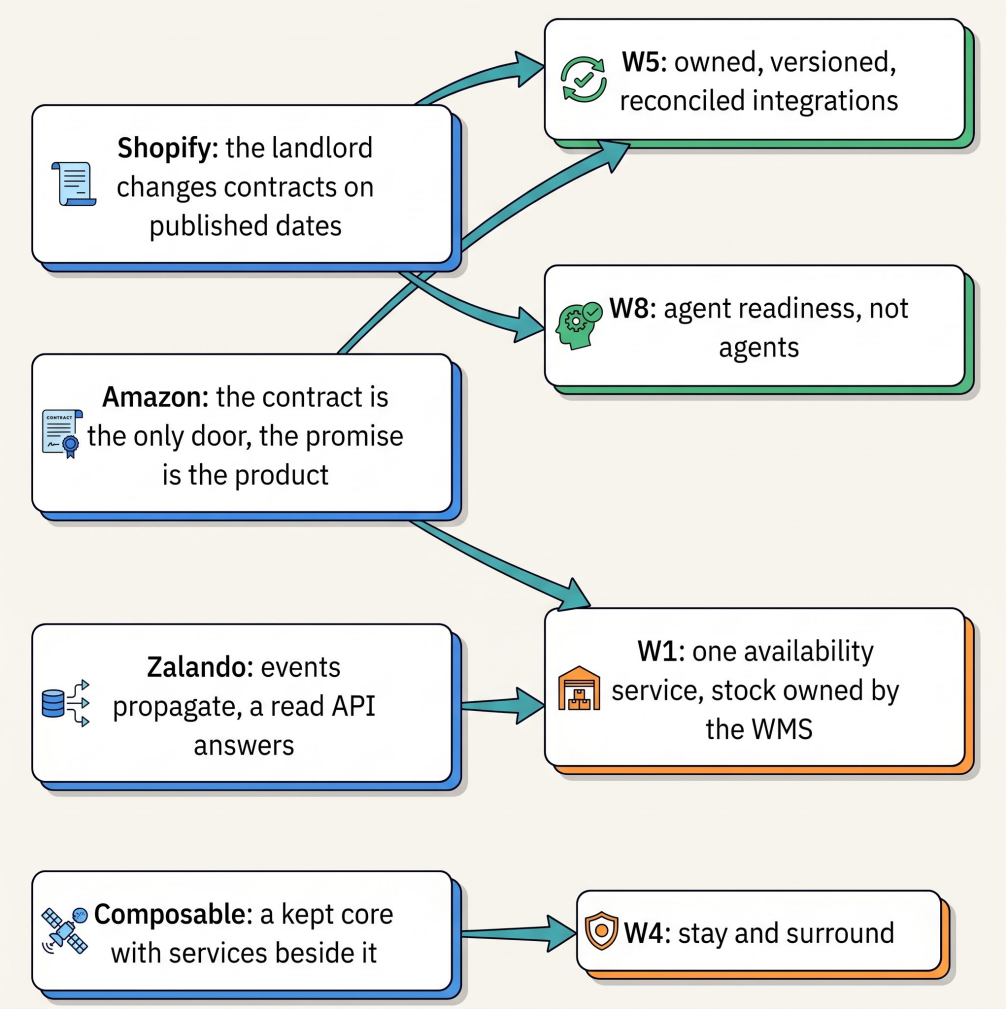


Figure 3: *What Kestrel & Pine takes from the four cases: each dated lesson confirms a Waypoint decision already made, and none changes its architecture.*

Chapters in this part

Case: Shopify, the Modular Monolith That Serves Millions of Merchants examines **modularity, tenant isolation, checkout extensions and integration contracts**. It separates older internal-architecture accounts from current product behavior. A merchant can adopt version tracking and reconciliation without copying the vendor's infrastructure.

Case: Amazon, From Monolith to Services to the Everything Store reads **dated accounts of services, cart design and fulfillment**. The merchant lessons concern reliable contracts, marketplace-held stock as a separate location and delivery promises computed from operations. The chapter states where evidence about today's internals ends.

Case: Zalando, Microservices, Platform Strategy and the Return to Discipline follows **team autonomy, shared API standards and later corrections**. A 2025 price-and-stock example shows why direct queries can replace a delayed event stream for a particular need. The lesson depends on that need, rather than a universal preference for either mechanism.

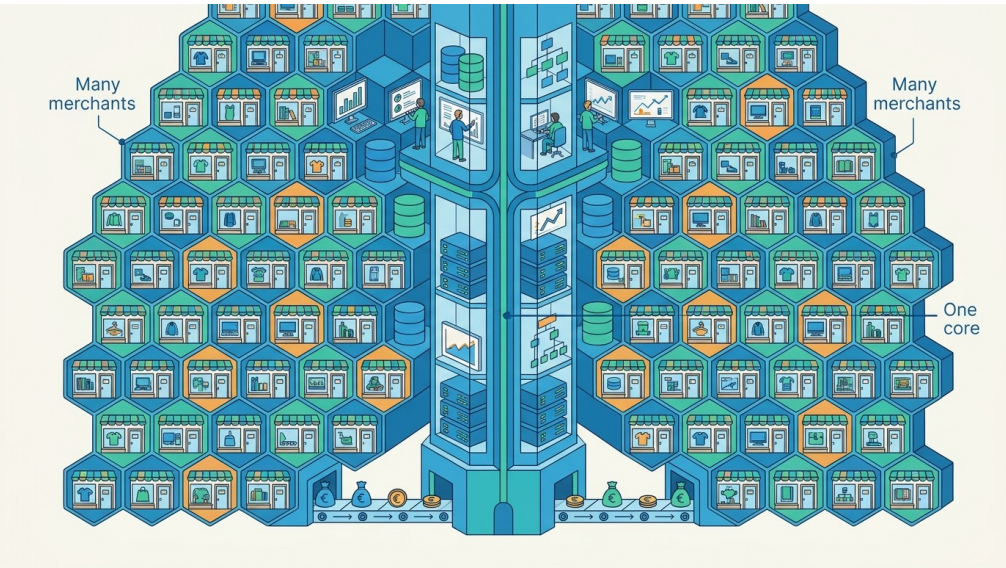
Case: commercetools, Shopware and the Composable Movement tracks **how vendors' promises and offerings changed**. Separate services can be added beside an existing platform as well as used in a rebuild. Kestrel & Pine's selective additions provide a test of the integration costs and ownership involved.

Begin with Case: Shopify, the Modular Monolith That Serves Millions of Merchants.

Sources and Evidence

Every public claim in this part is sourced and dated in the chapter that makes it, under that chapter's Sources and Evidence. Kestrel & Pine is fictional; recommendations, labeled inferences and labeled assumptions are the author's judgement.

46. Case: Shopify, the Modular Monolith That Serves Millions of Merchants



IN THIS CHAPTER, YOU WILL: Read Shopify's own dated accounts of its business and architecture, see what each supports and what it does not, and decide which lessons transfer to a merchant who rents a platform rather than builds one.

KEY POINTS:

- **Shopify is economically a payments business with a subscription as the entry ticket, and its architecture follows the money.** In FY2025 three quarters of its $11.6 billion revenue moved with merchandise volume and two thirds of that volume ran through Shopify Payments, which Shopify resells from licensed processors. Checkout is the most protected path because it is the revenue path.
- **The monolith stayed, and the boundary lessons are the honest part.** Shopify's 2019 to 2024 posts record one codebase of 2.8 million lines

kept as one deployment unit, a tool that gates new dependencies, and the admission that domain boundaries drawn on paper "do not reflect the way Shopify's code actually functions". The shop, not the service, is the unit of isolation, failure and recovery.
- **The platform changes under everyone at once, on published dates.** Free-form checkout customization was withdrawn over three and a half years and replaced by sandboxed extension points; API versions last at least twelve months; the agent interface was replaced within a year. A merchant on any hosted platform inherits that cadence and must budget for it.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Giant's Boots** — pods, shard moves and a custom observability stack are scale-specific machinery that a merchant mistakes for architecture to copy.
- **The Pulled Rug** — checkout.liquid, Shopify Scripts, the REST Admin API and the Storefront MCP endpoints were all withdrawn on published dates; the merchant who did not read the changelog found out on the day.
- **The Secret Agent** — the scope model, the writer problem among 21,000 apps and the trust tiers for AI agents are Shopify's answer to automation that changes facts without an owner.

A merchant considering Shopify needs to understand which work the platform takes on and which work its own team will still operate. Shopify illustrates the kind of hosted, shared platform used by Kestrel & Pine, the fictional merchant in this book. Its published accounts of a **monolith**, a single deployable application, help explain the vendor's engineering choices. The useful question is what those choices make easier or harder for the merchant's integrations and operations.

Keep business figures separate from architecture history. The financial sources include the FY2025 annual filing and the quarter ended September 2026. The major architecture accounts describe pods in 2018, components in 2019 and shard balancing in 2021. A February 2024 retrospective revisits the component work.

Those posts describe decisions made under earlier constraints. They do not establish today's complete internal design, so the chapter dates each account and identifies where later evidence is missing. This lets you use the documented reasoning without assuming that a historical mechanism is still implemented unchanged.

Where we are in the order's journey. A jacket purchase needs product and stock records, pages, checkout, payment and messages to the warehouse. Shopify documents counterparts for each. Related chapters explain the concepts: Is It in Stock? Inventory,

Availability and the Promise You Can Keep, Delivering the Storefront: Themes, Headless, Rendering and Performance, From Cart to Confirmation: Checkout Orchestration, Taking the Money: Payment Providers, Authorization, Capture and Refunds, Promises Between Systems: API Contracts, Webhooks and Versioning and Apps, Plugins and Functions: Extensibility and Ecosystem Governance.

How to Read This Case

Company claims are distinguished as **documented**, **widely reported** through secondary sources, or explicitly labeled **inferences**. Older architecture evidence may describe a replaced system; its age remains part of the explanation.

The research cutoff is **7 November 2026**. Unverified areas include a precise merchant count, checkout-queue design, methodology behind conversion claims, Sidekick's general-availability date and first-party confirmation of a vendor's database account. The final evidence section lists those limits. This is a platform-neutral case study, not a product recommendation.

The Business: Where the Money Comes From

Read the financial measures with Shopify's definitions. **Gross merchandise volume (GMV)** counts facilitated orders net of refunds, including shipping, duties and taxes; it is not Shopify's revenue. A **merchant** means a unique shop paying a subscription, so one brand can count several times. **Monthly recurring revenue (MRR)** is the subscription run-rate at period end. These definitions come from the FY2025 annual filing.

Metric	FY2025	FY2024	Q3 2026
GMV	$378.4 billion (+29%)	$292.3 billion	$119.1 billion (+29%)
Revenue	$11.56 billion (+30%)	$8.88 billion	$3.74 billion (+32%)
Subscription solutions	$2.8 billion (24%)	$2.4 billion	$836 million
Merchant solutions	$8.8 billion (76%)	$6.5 billion	$2.9 billion
Shopify Payments penetration (share of GMV)	65.6%	61.9%	68%
MRR	$205 million	$178 million	$230 million

Metric	FY2025	FY2024	Q3 2026
Merchants	"millions", 175+ countries; 44% US, 31% EMEA	—	—
Apps in the App Store	"more than 21,000" (31 December 2025)	—	—
Employees	about 7,600	—	—

Sources: the FY2025 10-K (filed 11 February 2026) and the Q3 2026 8-K and 10-Q (3 November 2026).

Two revenue lines carry the story. **Subscription solutions** are plans, platform fees, point-of-sale subscriptions, apps, themes and domains. **Merchant solutions** are Shopify Payments processing fees, currency conversion, lending, shipping labels, hardware and advertising, which the 10-K says are "directionally correlated with the level of GMV". Three quarters of revenue therefore moves with what merchants sell, and two thirds of what they sell is paid through Shopify Payments. The 10-K adds the plan mix ("most merchants subscribe to our Basic and Grow plans", while "the majority of our GMV has been generated from merchants subscribing to our Shopify Plus plan and enterprise offerings") and a concentration: "we currently rely on two suppliers to provide the technology we offer through Shopify Payments", Stripe and PayPal, on agreements that renew every twelve months.

> **INFERENCE, NOT DOCUMENTED:** Read together, these facts make Shopify economically a payments-and-platform business whose subscription is the entry ticket. Its architecture priorities follow: the checkout and the payment flow are the most protected paths because they are the revenue paths, and every product that raises Payments penetration (Shop Pay, point of sale, B2B, agent channels paying with Shop Pay) has a direct revenue line. The plan mix means one "single branch" codebase must serve millions of small shops that need safe defaults and a few thousand large ones that need extension points, which is why advanced checkout customization, custom Functions and the richer B2B features are gated to the Plus plan.

The 10-K also describes the technology in one paragraph: "a multi-tenant cloud-based system", with Ruby on Rails as "the technology that powers much of the Shopify platform", run as "'single branch' software, which means that all of our merchants use the latest version of Shopify", on "third-party cloud computing services, such as Google Cloud Platform". **Multi-tenant** means many merchants share one running system and one database fleet, isolated by data rather than by separate installations (Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation). **Single branch** means there is no version a merchant can stay on; every change ships to everyone.

The Monolith That Stayed: 2019 to 2024

A **monolith** is one deployable application that holds all of the shop's logic; a **modular monolith** keeps that single deployment but divides the code into named modules with enforced boundaries. Shopify's account of choosing the second over splitting into services runs from 2019 to 2024, and it is more candid than the slogan.

In February 2019 Shopify described its codebase as "one of the largest Ruby on Rails codebases in existence", worked on for over a decade by more than a thousand developers, and said that around 2016 "the downsides of the monolith were outweighing the benefits". It rejected microservices explicitly, listing the costs: pipelines and infrastructure per service, "not always having the data you need when you need it", network latency and reduced reliability, and coordinated deploys for large refactors. Instead, a small team labeled about 6,000 classes by business domain (orders, shipping, inventory, billing) and moved them in one automated pull request, intending each component to have "a clean, dedicated interface" and "exclusive ownership of its associated data". The payoff claimed was concrete: the isolation work let Shopify replace a legacy tax engine. **(Evidence age: seven years.)**

By September 2020 the core had "over 2.8 million lines of Ruby code", 37 components and an Architecture Guild of about 400 members; "every file has a stewardship team assigned". The same post is honest about what had not worked: "every component depended on over half of all the other components", with "lots of circular dependencies". What did pay off was ownership: exceptions were triaged automatically to the owning component, and framework upgrades could be distributed across owning teams. **(Evidence age: six years.)**

Packwerk, open-sourced the same month, is the tool that survived: it checks references between packages and records existing violations in a list so that teams "stop new ones without fixing everything at once".

The February 2024 retrospective, the latest first-party architecture account, is the most useful document in the record. Privacy checks "were removed from Packwerk with the release of version 3.0" because they pushed teams into premature API design. Several common ways of loading code are "untracked and invisible to the tool". Isolating one base package with zero dependencies "took many months of hard work".

The rate at which the tool found problems "vastly outpaced" teams' capacity to fix them; "we have discussed removing Packwerk from our monolith", and it stayed only because it "has provided value in holding the line against new dependencies at the base layer".

And the sentence that matters most: "domains and the boundaries between them do not reflect the way Shopify's code actually functions in practice", so components were demoted to "top-level organizational tools for grouping one or more packages".

INFERENCE, NOT DOCUMENTED: The honest reading of 2019 to 2024

> is not "Shopify modularized its monolith". It is that a company with a thousand developers spent seven years learning that domain boundaries drawn on an organization chart do not match runtime coupling, got durable value from file ownership and from one cheap gate against new base-layer dependencies, and kept one deployment unit throughout. Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says draws the general lesson; what this case adds is that even the best-resourced practitioner of the modular monolith could not make enforced domain boundaries stick, and said so.

The one documented case of carving a hot path out of the monolith is the **Storefront Renderer**, the server-side application that renders a merchant's theme into HTML. Its rewrite began in January 2019 because storefront code "has much stricter performance requirements" than, say, payment processing. It "always reads from dedicated read replicas" behind three cache layers (a query cache, an in-memory cache of the most frequent queries, a full-page cache); the August 2020 post reported server responses "4x to 6x faster" and a rollout in which a verifier compared legacy and new responses on live traffic, with circuit breakers falling back to the old path. **(Evidence age: six years.)** The split was made for latency, behind a verifier, not for team autonomy.

The Shop as the Unit of Isolation: Pods and Moves

A **shard** is one of several databases that each hold a slice of the data; the **sharding key** is the field that decides which shard a row lives in. Shopify's 2018 post describes the reason and the consequence. By 2015 the database "could no longer be scaled by buying a bigger server", so Shopify sharded by shop. Sharding reduced resilience: shared code running across all shards let one failed shard take an action down platform-wide. In 2016 Shopify reorganized into **pods**: "A pod consists of a set of shops that live on a fully isolated set of datastores." Web servers and job workers are shared, but "all shared resources can only ever communicate to a single pod at a time"; a routing layer assigns each request to one pod, and a "Pod Mover" could move a pod to its recovery data center "in a minute". **(Evidence age: eight years; the pre-cloud era.)**

The 2021 post on shard balancing shows the mechanism that persisted. "A pod is an isolated instance of Shopify" with one MySQL shard plus its caches; "we can attach a `shop_id` to all shop-owned tables and have it serve as a sharding key". A routing table mapping domains to pods lives in a separate, unsharded database. Moving a shop has three phases: Ghostferry copies rows and replays the change log while the store stays live; a cutover takes a brief writer lock, and fails if the lock cannot be acquired in time; then the routing table is updated. "The only opportunity for downtime is during the cutover." Placement uses measured load, not shop counts: before rebalancing the least- and most-used shards differed by "almost four times", and simulation brought the spread "closer to two times". **(Evidence age: five years.)**

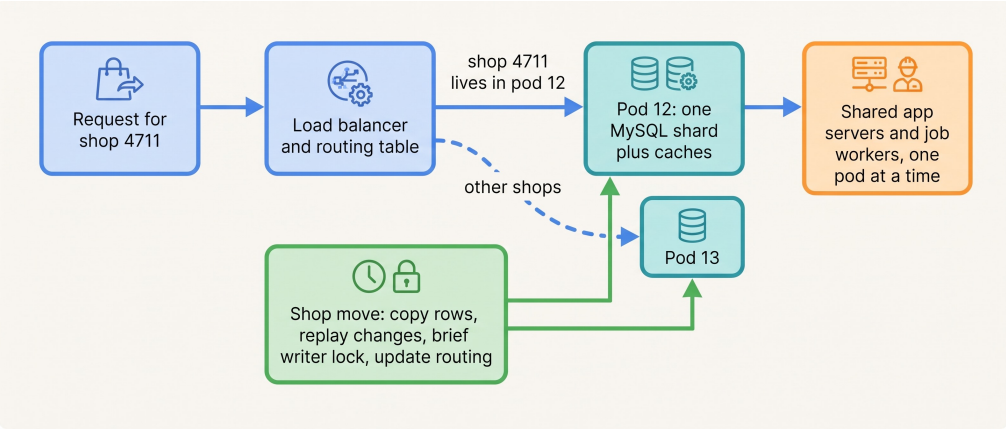


Figure 1: *The shop is the sharding key and the pod is the unit of isolation: a request is routed to one pod, a failing pod is contained, and a shop is moved as a whole with downtime only during a brief cutover (Shopify, 2018 and 2021).*

Two later datastore facts are often misquoted. In January 2024 Shopify described scaling the **Shop app**'s backend with Vitess and said plainly that "Shop app has different properties than the rest of Shopify"; the post does not say the core platform uses Vitess. In February 2026 a database vendor's blog, recapping a Shopify engineer's talk at the vendor's conference, said Shopify runs one of the largest clusters of that vendor's database; no Shopify page confirms it.

> **INFERENCE, NOT DOCUMENTED:** The pod is Shopify's unit of isolation, failure and recovery, and the shop is the sharding key throughout the record: a noisy shop is contained to its pod, and a shop is moved as a whole. Whether the pod topology survived the move to Google Cloud unchanged is not documented; the 2019 backup post and the 2021 balancing post both describe shards on Google Cloud, so pods and the cloud coexisted at least to 2021. The platform-wide scale is reported only for peak weekends: 14.8 trillion database queries and 1.75 trillion writes over the 2025 Black Friday weekend.

For a merchant none of this is machinery to copy. It is a question to ask any SaaS vendor, at the Growing stage already: **what is your unit of isolation, and can you move or restore my tenant alone?** Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation works that question out.

Storefront Delivery

Shopify offers three rendering models side by side. The default is a **theme** (files in Shopify's template language, Liquid) rendered by the Storefront Renderer; at Unite in

June 2021, Online Store 2.0 made themes "fully modular" with sections on every page and let apps add blocks the merchant places in the theme editor. The second is **Hydrogen on Oxygen**: Hydrogen, introduced in 2022 as "our modern React-based framework", is described in the 2026 documentation as "Shopify's headless toolkit", supplying clients for the Storefront and Customer Account APIs and nothing for the admin or the checkout; Oxygen hosts it on Shopify's edge. The third is any framework against the **Storefront API**, which "is available only in GraphQL", has three access modes (tokenless with a complexity cap, a public token, a private server-side token), and promises that "requests from real buyers aren't subject to a fixed request-per-minute limit" while bots are limited "most strictly when unsigned".

The performance gates are specific. Themes need "a minimum average Lighthouse performance score of 60" for Theme Store acceptance; public apps "shouldn't reduce Lighthouse performance scores by more than 10 points"; the admin's performance report warns that "some features can slow down your site", naming theme code and apps.

> **INFERENCE, NOT DOCUMENTED:** Under all three rendering models the cart, the checkout and the Storefront API are the shared substrate, which is why a headless storefront on Shopify still hands off to Shopify's checkout. The "ten Lighthouse points per app" rule is also a tell: apps are the usual cause of slow stores, and the platform polices them at submission because it cannot police them after installation. Delivering the Storefront: Themes, Headless, Rendering and Performance treats the headless decision; for Kestrel & Pine, with a themed storefront and 41 apps, the app gate is the more relevant fact.

Checkout: Freedom Withdrawn on a Schedule

Checkout is where Shopify's "single branch" rule and its revenue model meet, and the history is a timeline of withdrawn freedoms with published dates.

Date	Event
29 June 2021	Unite: first checkout extension (post-purchase) in beta; Scripts rebuilt on WebAssembly promised for Plus
9 February 2023	Winter '23: checkout extensibility "now available"
13 February 2023	Changelog: `checkout.liquid` customizations "will no longer work for in-checkout pages starting August 13, 2024"; non-upgraded checkouts "will be automatically upgraded to an un-customized, out-of-the-box Shopify Checkout"

Date	Event
13 August 2024	`checkout.liquid` stops working for the information, shipping and payment pages (Plus)
28 August 2025	`checkout.liquid` and additional scripts sunset for the Thank you and Order status pages (Plus)
15 April 2026	"Editing and publishing new Shopify Scripts will no longer be possible"
30 June 2026	"All Shopify Scripts will cease to execute entirely"
26 August 2026	Script tags on the Thank you and Order status pages sunset for non-Plus stores

Sources: Shopify's changelog (2023, 2026), the Liquid reference and Help Center (accessed November 2026), the Editions pages (2023).

What replaced the freedom is **checkout extensibility**: a fixed set of slots where apps may render user-interface extensions, a branding API, web pixels for tracking, and **Shopify Functions**, which "allow developers to customize the backend logic of Shopify" in any language that compiles to WebAssembly, invoked by the platform during the purchase flow, never by URL. Functions run in checkout order (cart transform, discount, fulfillment constraints, order routing, delivery and payment customization, validation) under hard limits: a 256 KiB module, 10 MiB of memory, and for carts up to 200 lines about 11 million instructions. Public apps with Functions work on any plan; custom Functions and "advanced checkout customization features are available only on the Shopify Plus plan".

Shop Pay is the other half of the story: "an accelerated checkout option" that stores a buyer's details, "included as part of Shopify Payments". Its conversion claims ("as much as 50% better conversion compared to guest checkout") rest on a 2023 study Shopify has not published; they are vendor claims. The documented reach figure is from Shopify's own peak reporting: 32% of orders over the 2025 Black Friday weekend.

> **INFERENCE, NOT DOCUMENTED:** The arc is free-form template and script access to checkout, withdrawn over three and a half years with dated deadlines, replaced by sandboxed extension points with hard budgets. Shopify traded merchant freedom for an upgrade-safe checkout contained within the card-data security standard's scope (PCI DSS) that it can change under everyone at once, and the sandbox limits are what let it run merchant logic inside its own transaction at peak. For a merchant this is the reference case for "the platform will withdraw an extension point: plan for it", worked in From Cart to Confirmation: Checkout Orchestration and Apps, Plugins and Functions: Extensibility and Ecosystem Governance. The Forked Platform pattern is what Shopify made impossible by design; the Pulled Rug is what it did instead, with notice.

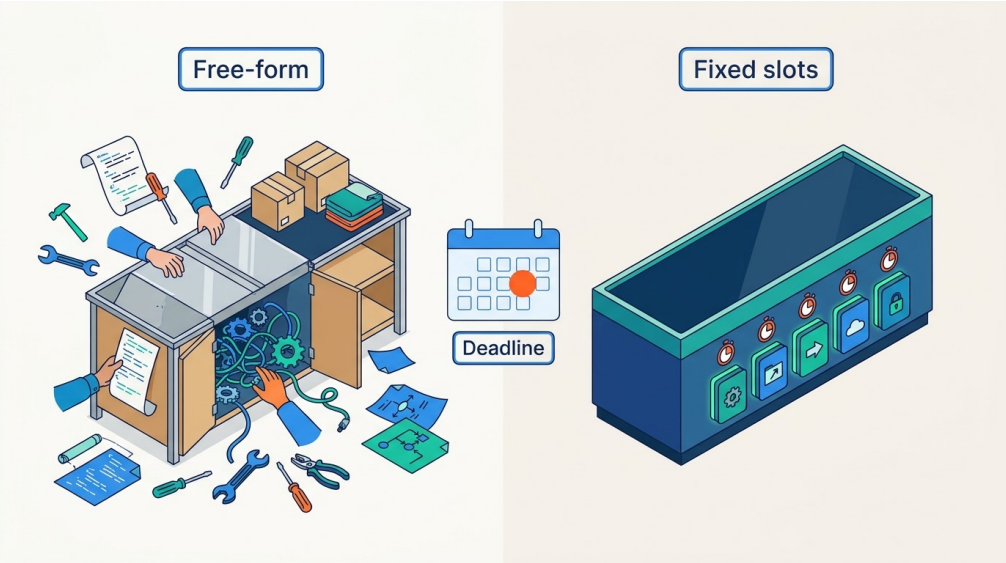


Figure 2: *Over three and a half years and on dated deadlines, free-form checkout access was withdrawn and replaced by sandboxed extension slots with hard budgets.*

Payments: Orchestration, Not a License

Shopify Payments is the clearest public example of a platform that earns from payments without being a payment institution. Its European terms (accessed November 2026) name Shopify International Limited in Ireland as the contracting entity and state: "None of the Shopify entities listed above are licensed as a bank, payment institution, or money services business." The licensed **PSP** (payment service provider, the company that actually moves the money) is a separate party under separate terms; the processor list names, for the Netherlands, Belgium and Germany alike, Stripe Payments Europe, PayPal (Europe) and Adyen N.V., without saying which handles which transaction. Shopify Payments is available in 41 countries, including the three Kestrel & Pine sells in.

> **INFERENCE, NOT DOCUMENTED:** Shopify owns the checkout, the buyer wallet and the merchant contract, and resells licensed processing from Stripe, PayPal and Adyen; "Shopify Payments" is commerce orchestration over a white-labeled processor stack, which is why the 10-K reports two suppliers as a concentration risk. Taking the Money: Payment Providers, Authorization, Capture and Refunds owns the mechanics and hands the ledger side to the sibling book's Payments for Finance Architects: Eleven Reasons the Numbers Disagree; what this case adds is the business shape: whoever

owns the checkout owns the payment revenue, whether or not they hold the license.

Inventory States, Briefly

Shopify's inventory model is a documented worked example of the distinction Is It in Stock? Inventory, Availability and the Promise You Can Keep owns. Stock is held per location in exclusive states: *available* ("the inventory that a merchant can sell"), *committed* ("set aside" by placed but unfulfilled orders), *reserved*, *damaged*, *safety_stock* ("to help guard against overselling"), *quality_control* and *on_hand* ("the total number of units that are physically at a location", the sum of the others), with *incoming* outside on-hand. The states arrived in steps: committed and incoming before February 2023, safety stock, damaged and quality control in August 2023, and on 5 August 2026 holds for draft orders and transfers moved from reserved to committed. Shopify holds no per-customer allocation for B2B and computes no cross-location available-to-promise; those belong to an order management system (OMS), which is why Kestrel & Pine's W1 buys an availability service beside the platform.

The Ecosystem: Apps, Scopes, Webhooks and Versions

The App Store is both a business model and a governance problem. **Business**: "more than 21,000 apps" at the end of 2025; developers pay 0% on the first $1,000,000 of lifetime app revenue and 15% above it. **Governance**: an app must "immediately authorize using OAuth before any other steps occur" (the standard login-and-consent flow); apps "using APIs that will be deprecated within 90 days can't be submitted"; every app must implement three privacy compliance webhooks and return `401` on a bad signature, or "your app will be rejected". **Scopes** control "which store data your app can read and write", declared by the app and approved by the merchant at install: `read_inventory` and `write_inventory`, `read_orders` (covering "orders created within the last 60 days"), with `read_all_orders` and protected customer data needing separate approval.

Webhooks are the integration contract, and Shopify's own words set the baseline every integrator should assume: "Webhook delivery isn't always guaranteed"; "Shopify doesn't guarantee ordering within a topic, or across different topics for the same resource"; "ignore duplicate deliveries using `X-Shopify-Webhook-Id`"; "use reconciliation jobs to periodically fetch data from Shopify so that your app stays consistent". Delivery allows five seconds; "if Shopify receives no response or an error, it retries 8 times over the next 4 hours"; and "after 8 consecutive failures, the subscription is

automatically deleted". Each delivery carries a signature (an HMAC, a keyed hash the receiver can verify). In 2021 the platform sent 13 billion webhooks.

API versioning is dated and guaranteed: "Shopify releases a new API version every three months", each "supported for a minimum of 12 months", and "if your app targets an inaccessible version, Shopify falls forward and responds using the oldest accessible stable version". The REST Admin API "is a legacy API as of October 1, 2024". Rate limits are per app per store and scale with the plan: the GraphQL Admin API allows 100 points a second on standard plans, 1,000 on Plus and 2,000 for enterprise, with no single query over 1,000 points, and "Shopify might temporarily reduce API rate limits to protect platform stability".

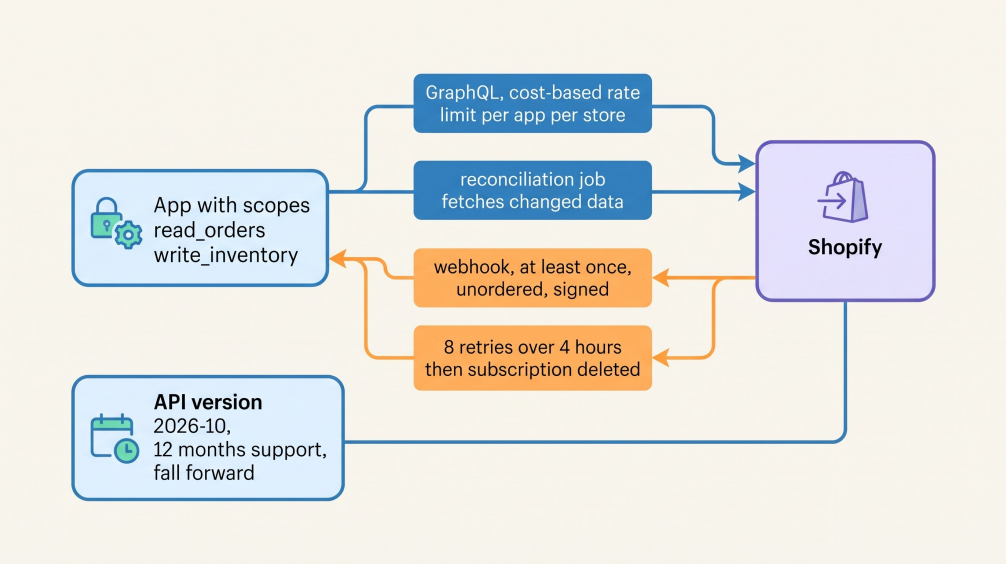


Figure 3: *Shopify's documented integration contract: scoped access, cost-based rate limits, at-least-once unordered webhooks that stop after eight consecutive failures, a reconciliation job the consumer must run, and dated API versions with a guaranteed window.*

INFERENCE, NOT DOCUMENTED: Three rules follow. Webhooks are at best at-least-once and unordered, so every consumer needs idempotent handling and a periodic reconciliation, which is Waypoint W5 in Shopify's own words. The "subscription deleted after eight consecutive failures" rule is the one that silently breaks an integration after an outage on the consumer's side; a stock-sync connector that was down for five hours on a Friday night has no subscription on Saturday morning. And rate limits are a tenancy tool: an integration's throughput on a peak day is bounded by plan tier and by how many apps share the store, so a polling connector can starve

another app. Promises Between Systems: API Contracts, Webhooks and Versioning generalizes the versioning rule; Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation the rate limits.

Peak Days With Definitions

Shopify reports its Black Friday weekend every year, and the figures are useful only with their definitions.

Metric	BFCM 2023	BFCM 2024	BFCM 2025
Sales (GMV, unaudited)	$9.3 billion (+24%)	$11.5 billion (+24%)	$14.6 billion (+27%)
Peak sales per minute	$4.2 million	$4.6 million	$5.1 million (all at 12:01 p.m. EST on Black Friday)
Peak load	967,000 requests per second	284 million edge requests per minute; 80 million+ app-server requests per minute	489 million edge requests per minute; 117 million+ app-server; 31.8 million API
Database over the weekend	—	10.5 trillion queries, 1.17 trillion writes	14.8 trillion queries, 1.75 trillion writes
Shop Pay	sales +60%	sales +58%	sales +39%; 32% of orders
Merchants with their best day ever	55,000+	67,000+	94,900+

Sources: Shopify's announcements of 28 November 2023, 2 December 2024 and 2 December 2025. Every year carries the caveat that the data are "unaudited and subject to adjustment" and that prior years "may not be directly comparable" because methodology changes. The 2023 load figure is per second and does not say edge or app server; no year reports an order count.

How Shopify prepares is documented twice. In 2023: a "full-scale approach, where we scale the systems to their expected BFCM profile", with scripted browse, buy and flash-sale flows that "bypass the CDN for a portion of the traffic" (the content delivery network, the cache layer in front of the platform) so the origins are exercised, and an

abort switch that stops every test at the first sign of degradation. In 2025: preparation "started in March with capacity planning", game days on "critical journeys" (checkout, payment processing, order creation, fulfillment, search) with injected network faults, and five scale tests from April to October reaching "146 million requests per minute and 80,000+ checkouts per minute" by the fourth; in early tests "core operations threw errors and checkout queues backed up", and authenticated checkout "exposed rate-limit paths that anonymous browsing never touches". The checkout queue itself is documented only in a 2018 changelog: during flash sales "processing payments at checkout happens separately from creating orders", so the buyer finishes at once while the order record, and the confirmation e-mail, follow after a delay. **(Evidence age: eight years for the queue; the Storefront API reference still documents a per-minute checkout throttle.)**

> **INFERENCE, NOT DOCUMENTED:** Two ratios matter more than the absolutes. Edge requests are about four times app-server requests (489 million against 117 million in 2025), so roughly three quarters of peak traffic is absorbed by the content delivery network and caches before it reaches application code. And the peak minute ($5.1 million) against the weekend average (about $2.7 million a minute over roughly 5,400 minutes) is a platform-level peak-to-average ratio of only about two, because millions of merchants' peaks do not coincide. A single merchant sees its own 41-times spike, and so do its connectors, its ERP (the finance and back-office system) and its WMS. The transferable practices are the shape of the program, scaled down: test at forecast load with caches bypassed, rehearse failure on the money paths, decouple payment from order creation so the money step stays fast while the record catches up. Black Friday Every Year: Peak Load, Resilience and Recovery and Journey 4: Black Friday, Hour by Hour apply them.

AI and the Agent Surface: Rebuilt Within a Year

Shopify's AI story has a merchant-facing side and an agent-facing side, and both moved fast. **Sidekick**, the assistant in the admin, was announced in July 2023; on 21 May 2025 Shopify said it could run reports, create segments, filter orders and set up discounts "with the merchant's approval", and the help page (accessed November 2026) describes it editing orders and presenting changes for review. No page states its general-availability date; the 4.2-times growth in daily sessions claimed in the Fall 2026 edition is a vendor figure.

The agent-facing side is the clearest dated example of protocol churn in the book. In May 2025 Shopify built a **Storefront MCP** server on the Model Context Protocol, the tool-calling transport that lets an AI agent discover and call operations such as "search catalog" (Tool Ecosystems, MCP and the Tool Broker). On 10 December 2025 **Agentic**

Storefronts became "available to every merchant", syndicating products to AI chat products through a **Catalog**. On 11 January 2026 Shopify co-announced the Universal Commerce Protocol with Google and an **Agentic plan** for brands not on Shopify, at "no monthly fee" and card rates "from 1.9% + €0.25". By November 2026 the developer documentation states that the Storefront MCP catalog and cart tools "were removed" and that UCP "now provides these capabilities" at a new endpoint; every agent request must carry a profile hosted at a well-known URL, Checkout MCP "requires authenticated or signed requests", and "higher trust tiers unlock broader access, including direct checkout completion". The Catalog documentation lets agents cache price and availability for at most fifteen minutes, labels some fields as AI-inferred, prohibits bulk harvesting, and warns that "endpoint URLs might change".

The geography is the fact a European merchant needs. The ChatGPT channel requires that "your store must sell to customers in the United States", with purchases completed on the merchant's own checkout since OpenAI's March 2026 retreat from in-chat checkout. The Google channel offers direct checkout only to customers in the United States, Canada and Australia, excludes subscriptions, bundles and B2B-only products, and does not fire the merchant's analytics pixels. **No EU buyer market is in scope on any agentic channel page read at the cut-off.** Assistants and Agents: AI in Commerce and What Is Real in 2026 owns the subject and the readiness list; what the case adds is a platform's own record of replacing its agent interface within twelve months, which is the strongest argument in the book for adapters over one internal API rather than channel-specific builds.

Shopify's use of AI in its own engineering is self-reported: a chief-executive memo of April 2025 making "reflexive AI usage" a baseline expectation; an agent that "only works in the open" in Slack and co-authors "one in eight merged pull requests" by Shopify's own count (May 2026); a rebuilt native Shop app in twelve weeks, gated by tests, "two adversarial code reviews, and human approval" (September 2026). Who Builds and Who Owns: Teams, Vendors, Agencies and Governance treats these as a prompt for a merchant's own baseline, with the metrics as the vendor's.

What Transfers to a Merchant, and What Does Not

The table is the author's synthesis; each row names the dated evidence behind it.

Shopify practice	Evidence	Transfers?	How
The shop as the unit of isolation, move and recovery	Pods 2018; balancing 2021	As a question to vendors; as a rule for a Group's own services	Ask every SaaS vendor; enforce tenant identifiers in shared services
Pods, Ghostferry, a custom observability platform	2018–2026	No	Scale-specific machinery; use managed services
Gate new boundary violations in code; do not chase zero	Packwerk 2020 and 2024	Yes, for in-house code	A boundary check in the build pipeline with modest targets
Carve out a hot path only on measured latency, behind a verifier with fallback	Storefront Renderer 2020	Yes, the rollout pattern	Shadow-compare before any cutover
Dated API versions with a guaranteed window; deprecate with dates and enforce	Versioning docs; checkout timeline 2023–2026	Yes, as a maintenance budget for every connector	Inventory connectors by API version and support end
Webhooks as at-least-once, unordered, signed, reconciled	Webhook docs	Yes, everywhere	Idempotent consumers and a reconciliation job per flow (W5)
Sandboxed extension points instead of in-process scripts	Functions docs; Scripts sunset 2026	As a governance idea	Rules in bounded places; never in a forked core
Rate limits per app per store, lowered under stress	Rate-limit docs	As a peak-day constraint	Know the budget before Black Friday; one writer per fact

Shopify practice	Evidence	Transfers?	How
Full-scale load tests with cache bypass; game days on money paths	2023 and 2025 posts	Yes, scaled down	Test connectors, ERP and WMS at forecast load

Only Shopify needs the rest: pods and shards balanced by simulation, a renderer deployed at the edge worldwide, an observability platform ingesting 100 million events a second at peak (a vendor's account of a 2026 talk), a readiness program that starts in March. A merchant who copies that list wears the Giant's Boots; a merchant who copies the disciplines in the table, and reads the vendor's changelog as carefully as Shopify's integrators must, gets the useful part for free.

What Kestrel & Pine takes from the case. Four things, each already in Waypoint. From the webhook contract: every integration idempotent, with a reconciliation job, because the platform itself says delivery is not guaranteed (W5). From the versioning rule: every connector and app pinned to a dated API version with its support end on the ownership map, after the Deprecated Endpoint cost two days of stale stock on bol (W5, Who Builds and Who Owns: Teams, Vendors, Agencies and Governance). From the checkout timeline: nothing customized in the checkout that the platform could withdraw, and a rule that custom behavior goes to the lowest rung that works (the app governance rule of Apps, Plugins and Functions: Extensibility and Ecosystem Governance). From the agent surface: readiness through product data, availability and order status by API, and no channel-specific build until a protocol has held still for a year (W8). What it does not take is the architecture: it rents the pods, the renderer and the checkout queue, which is the point of W4, stay and surround.

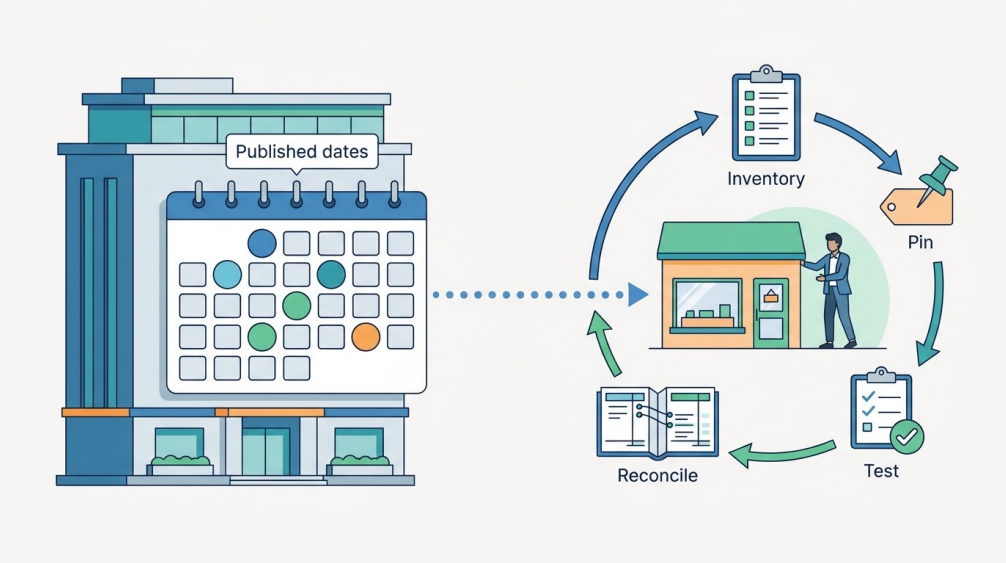


Figure 4: *The landlord changes the platform under everyone on dates it publishes; the merchant's job is to inventory, pin, test and reconcile against that cadence.*

Where the Evidence Stops

Be clear about what this chapter could not establish.

- **Today's topology**. The pod and shard counts, and whether pods survived the Google Cloud migration unchanged, are undocumented after 2021. The component count and Packwerk coverage after February 2024, and deploy frequency after the 2023 figure of "approximately every 30 minutes", are unknown.
- **The checkout queue**. Only a 2018 changelog and a one-line reference to a per-minute throttle exist; community descriptions of a "checkout queue" are not cited here.
- **Merchant and app counts**. Shopify says "millions" of merchants and "more than 21,000" apps; third-party trackers disagree widely and are not used.
- **Shop Pay**. The "150M+ shoppers" figure is reach, not active users; the conversion claims rest on an unpublished 2023 study.
- **Sidekick's general-availability date**, the Agentic plan's country eligibility, the Catalog API's commercial terms (its reference page returned 404), and whether any EU buyer market has a date for agent checkout.
- **The new database**. A vendor's 2026 blog is the only source; Shopify has not published it, and which system it backs is unknown.
- **Scripts' earlier deadline**. An August 2025 date and its extension were seen only in search snippets; the changelog posts opened state only the 2026 dates.

What to Take Away

Established.

- Shopify's revenue is three quarters merchant solutions, correlated with GMV, and two thirds of GMV runs through Shopify Payments, which is orchestrated over licensed processors (10-K FY2025; payment terms, 2026).
- The core platform is one Rails codebase, multi-tenant and single-branch, sharded by shop into isolated pods, with a modularity tool kept as a gate on new base-layer dependencies rather than as enforced domain boundaries (Shopify 2018, 2020, 2021, 2024; 10-K FY2025).
- The integration contract is explicit: at-least-once unordered webhooks that stop after eight consecutive failures, reconciliation jobs expected of the consumer, dated API versions supported for at least twelve months, rate limits per app per store (Shopify developer documentation, 2026).
- Extension points are withdrawn on published dates: `checkout.liquid` announced February 2023, enforced August 2024 and August 2025), Shopify Scripts (June 2026), the REST Admin API (legacy since October 2024), the Storefront MCP catalog and cart tools (replaced by UCP endpoints in 2026).

Depends on context.

- Whether the modular-monolith lesson applies depends on owning code: a merchant on a hosted platform has nothing to modularize; a vendor or a Group with in-house services should gate new dependencies and not chase zero violations.
- Whether to run a Shopify-style readiness calendar depends on where the merchant's peak breaks: for Kestrel & Pine the platform held and the connectors failed, so the calendar belongs to the integrations, not the storefront.
- Whether the Plus plan's extension points are worth their price depends on needing in-checkout logic at all; a merchant that keeps custom behavior out of checkout avoids both the fee and the next deprecation.
- Whether the agent channels are worth switching on depends on the buyer geography, which excludes the EU at the cut-off; for a European merchant they are discovery and nothing more in 2026.

The author's synthesis.

- The useful Shopify is not the architecture but the landlord's behavior: it changes the platform under everyone on dates it publishes, and the merchant's job is to inventory, pin, test and reconcile against that cadence.

- Shopify's own record undermines the slogan made from it. It did not modularize its monolith into clean domains; it learned which gate was worth keeping, and that honesty is the lesson to copy.
- A platform that rebuilt its agent interface within a year is the best argument I know for a merchant to put nothing channel-specific below an adapter.

Questions to Ask

1. Which of our platform's extension points have a published sunset date, and which of our apps and scripts depend on them?
2. Which API version does each of our connectors target, when does that version leave support, and who owns the upgrade?
3. What does our platform promise about webhook delivery, ordering and retries, and which of our flows runs a reconciliation job to cover the gap?
4. What is our platform's unit of isolation, and can the vendor move or restore our tenant alone?
5. What rate-limit budget does each app and connector have on our plan, and which two of them write the same fact?
6. Which of the vendor's figures are audited, which are defined (GMV, merchants, penetration), and which are claims from unpublished studies?
7. Which checkout behavior have we customized that the platform could withdraw, and what is the un-customized fallback?
8. For every agent channel the platform has switched on by default, which countries, which products and whose checkout?

Shopify is the landlord's record; the next case is the retailer that built the everything store and now sells its machinery to others. Continue with Case: Amazon, From Monolith to Services to the Everything Store.

Sources and Evidence

Company, product and investor sources

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- Shopify says purchases are coming inside ChatGPT as OpenAI retreats on Instant Checkout, Modern Retail, 2026 — the March 2026 change to the ChatGPT channel.

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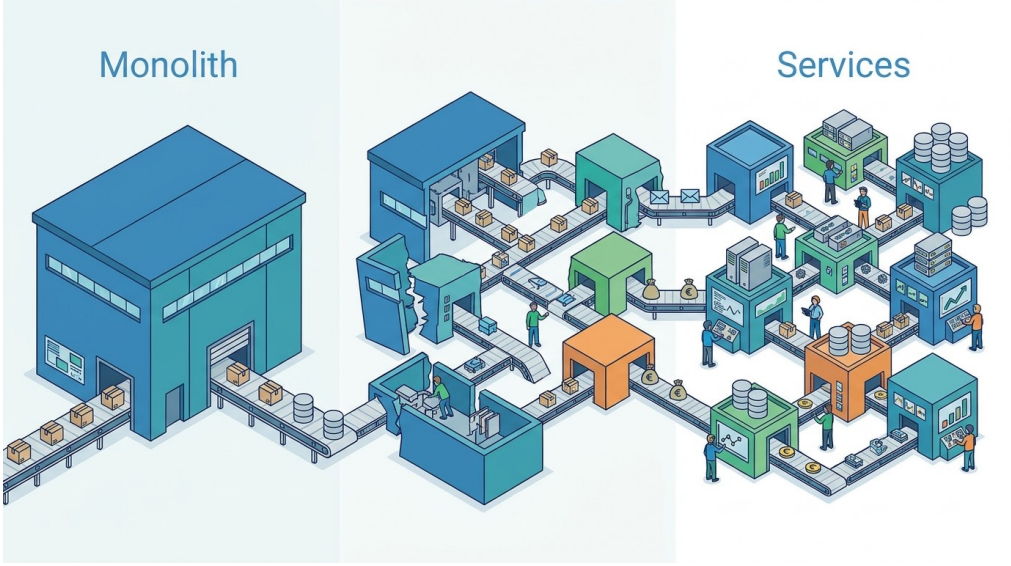
Research and practitioner sources

- None beyond the first-party and trade sources above; the chapter deliberately relies on Shopify's own dated record.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

47. Case: Amazon, From Monolith to Services to the Everything Store



IN THIS CHAPTER, YOU WILL: Read Amazon's own dated accounts of its architecture and business, separate what they document from what people assume, and decide which lessons a merchant can use.

KEY POINTS:

- **Amazon split its shop into services because it hit a wall, not because services were fashionable.** Its 2006 account dates the scaling wall to 2001; by 2007 one page touched 100 to 150 services. The transferable rule is the one that made the split survivable: every team exposes its data only through a service interface, and nobody reads another team's database.
- **The shopping cart is an availability problem, and the delivery promise is the product.** Amazon's 2007 Dynamo paper encodes a business decision in a data store: an add-to-cart must never be refused, so conflicts are merged later. Its 2024 to 2026 letters put the investment into the network that keeps the promise shown at checkout.

- **A retailer became a platform by opening its operations, not its storefront**. Prime (2005) and Fulfillment by Amazon (2006) took third-party sellers from 3% of physical sales in 1999 to 58% in 2018 and over 60% since. The same machinery is now exported to other shops and, since 2025, to AI agents, on Amazon's terms.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Giant's Boots** — Amazon's service decomposition is the architecture most often copied without its reasons, which were a scaling wall and hundreds of teams.
- **The Spaghetti Junction** — the rule Amazon is remembered for, no direct reads of another team's data store, is the cure for a landscape where every system reaches into every other.
- **The Secret Agent** — Amazon's 2025 position on third-party shopping agents (identify yourself, respect the provider's choice) is the clearest public statement of the rule every merchant will need for automation on its own site.

A merchant can sell through Amazon, use its fulfillment services and compete with its own retail offers. Each relationship changes what the merchant controls: the sales rules, the location of stock or the customer relationship. Kestrel & Pine, the fictional outdoor brand, lists jackets on Amazon.de and sends some German marketplace stock to outside fulfillment programs. Understanding the separate roles is necessary to decide which records and commitments its own systems must still track.

Amazon's architecture history is also widely cited. This chapter follows its dated papers, engineering accounts, shareholder letters and annual filings. They explain particular choices about services, carts and fulfillment. **They do not reveal the complete design of today's order, pricing or delivery-promise systems.** Read each account for the problem it actually addresses.

The central architecture evidence comes from **2006–2007** and should be read as history. Financial figures come from 2025–2026 filings; AI features from 2024–2026 announcements. Dates distinguish historical mechanisms from more recent business and product claims.

Where we are in the order's journey. Three topics affect a jacket purchase: adding to a cart, calculating a delivery promise and fulfilling an order from marketplace-held stock. From Cart to Confirmation: Checkout Orchestration, Getting It There: Carriers, Labels, Tracking and the Delivery Promise and Selling on Someone Else's Platform: Marketplaces and Channel Integration explain those decisions in the merchant's setting.

How to Read a Case Study in This Book

Claims are marked as **documented**, **widely reported** or **inferred**. Inferences include their reasoning. Secondary coverage remains identified when a primary page could not be retrieved. Introduction: Why Commerce Architecture Matters explains the book's evidence approach.

Research closed on 7 **November 2026**. Unreachable primary pages included Werner Vogels' 2006 ACM Queue interview, the 2004 profile associated with two-pizza teams, the 2005 Prime announcement and a February 2026 delivery release. The chapter identifies where digests or syndicated reporting substitute for those pages.

2001: The Front End Hits a Wall

Start with vocabulary. A **monolith** is one deployable application that contains all of the shop's logic (catalog, cart, checkout, orders) and usually one shared database. A **service-oriented architecture**, or SOA, splits the shop into many separately deployed **services**, each owning one function and talking to the others over the network. In plain words: one big program with one database, versus many small programs that call each other.

Amazon's own account of its move from the first to the second is Werner Vogels' 2006 interview in *ACM Queue*, "A Conversation with Werner Vogels". As digested by High Scalability in September 2007, it describes the early shop as a "two-tier monolith" and dates the turning point: "In 2001 it became clear that the front-end application couldn't scale anymore." Amazon moved "to a fully-distributed, decentralized, services platform" that "grew into hundreds of services and a number of application servers that aggregate the information from the services". A single page request touched "between 100-150 services".

Amazon's own 2007 paper on Dynamo, presented at the SOSP conference in October 2007, opens with the same statement in its own words: Amazon "uses a highly decentralized, loosely coupled, service oriented architecture consisting of hundreds of services", and those services have "stringent latency requirements which are in general measured at the 99.9th percentile of the distribution".

That last clause carries the engineering cost of the decision. **Tail latency** at the 99.9th percentile is the response time that only one request in a thousand exceeds. A page assembled from 100 services is only as fast as its slowest call, so Amazon measured every service on its worst thousandth, not its average. The split bought independent teams and independent scaling; it cost a discipline about time budgets that a monolith never needed.

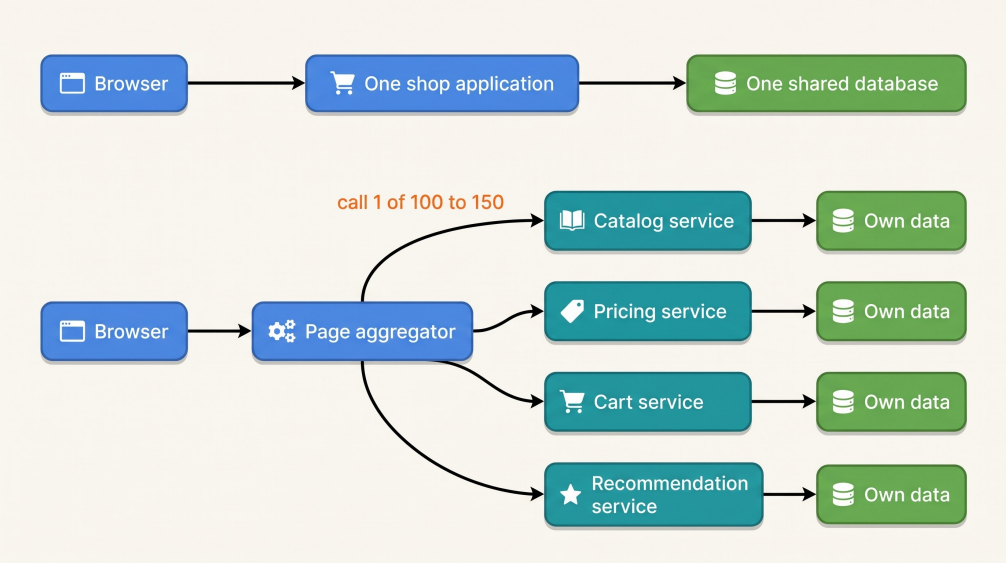


Figure 1: *The two shapes in Amazon's 2006 account: one application on one database until 2001, then a page assembled by an aggregator from hundreds of services that each own their data, where the slowest of 100 to 150 calls sets the page's speed.*

The sequence to take from this is **monolith**, **scaling wall**, **services**, in that order. The record does not say that Amazon decomposed early, or that services were a goal; it says the front end stopped scaling and the decomposition followed. Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says uses this and Zalando's 2015 move as the two dated examples of when a split is justified. The name retellings give the monolith does not appear in any page that could be verified, so it is not used here.

> **INFERENCE, NOT DOCUMENTED**: Amazon's 2001 wall was a wall of both load and people: hundreds of engineers changing one codebase. A merchant at Kestrel & Pine's scale, 293,000 online orders a year with fourteen people in technology, is nowhere near either wall, and nothing in Amazon's record argues for decomposing a shop before it hits one. The lesson transfers at a different altitude: the moment a merchant's product page is assembled from several *systems* (a PIM for content, the ERP for price, the availability service for stock, a reviews app), the merchant inherits Amazon's aggregation problem in miniature, including the tail-latency discipline. That is the Omnichannel stage, and it is where Kestrel & Pine is now.

The Cart That Never Says No

The Dynamo paper is the one place where Amazon explains a system in enough detail to learn from it, and the system it chose to explain is the shopping cart.

The motivation is stated as a business requirement. Customers "should be able to view and add items to their shopping cart even if disks are failing, network routes are flapping, or data centers are being destroyed by tornados". Dynamo targets "an 'always writeable' data store", meaning one "that is highly available for writes". The reasoning is commercial: a refused add-to-cart is a lost sale, while a cart that is briefly out of date is harmless. So the cart "must allow customers to add and remove items from their shopping cart even amidst network and server failures", which "forces us to push the complexity of conflict resolution to the reads".

Two more terms. **Eventual consistency** means that different copies of a piece of data may disagree for a short time and converge later. **Conflict resolution** is what happens when two copies disagree and the system has to produce one answer. Dynamo's rule for the cart is explicit: "an 'Add to Cart' operation can never be forgotten or rejected". If the latest version of the cart is unavailable, a change made to an older version "is still meaningful and should be preserved" but "shouldn't supersede the currently unavailable state of the cart". When versions diverge, they are merged: an added item is never lost, and the paper acknowledges the price, which is that a deleted item can resurface.

What the Dynamo paper documents (2007)	What it does not say
The cart service accepts writes during failures and merges conflicting versions on read	That Amazon runs Dynamo today (the 2007 post says it was internal; the 2012 post says teams moved to managed services)
In 2006–2007 the cart service "served tens of millions requests that resulted in well over 3 million checkouts in a single day"	Anything about the order, pricing or checkout systems behind the cart
Services are measured at the 99.9th percentile	What the budgets were
A business rule (never lose an add) is encoded in a data structure's merge rule	That the same rule is right for orders, payments or stock

The 2012 sequel is as instructive as the paper. Announcing the managed database service DynamoDB in January 2012, Vogels wrote that Dynamo was "originally built to serve the needs of the shopping cart" but "did nothing to reduce the operational complexity of running large database systems"; Amazon's engineers "strongly preferred simplicity to fine-grained control" and chose managed services instead. A **managed service** is a database run by a provider so the team using it does not operate servers, replication or failover. In plain words: Amazon's own teams would not adopt a correct system that was hard to run. The 2022 USENIX paper on DynamoDB gives the scale of the second generation: during the 66-hour Prime Day event of 2021, calls to DynamoDB peaked "at

89.2 million requests per second" at single-digit-millisecond latency.

> **INFERENCE, NOT DOCUMENTED:** The cart is the one place in a commerce platform where availability beats consistency by design, and the Dynamo merge rule is the book's central idea in miniature: a decision about which promise a system may make (the cart promises to remember every add) encoded where it cannot be forgotten. The opposite rule applies one step later. A checkout that charges twice or an inventory that promises the last Ridgewalker to two customers is the Black Friday Oversell, and those systems need the consistency the cart gives up (Is It in Stock? Inventory, Availability and the Promise You Can Keep, Taking the Money: Payment Providers, Authorization, Capture and Refunds). For Kestrel & Pine the practical reading is short: its hosted platform's cart already behaves this way; a merchant that builds its own storefront should buy a managed, highly available store for the cart rather than reinvent Dynamo; and the 2012 lesson, that operability decides adoption, applies to every system a fourteen-person team is asked to run.

Figure 2: *The cart that never says no: diverging copies are merged so an added item is never lost, at the price that a deleted item can resurface (Dynamo, 2007).*

Teams, Ownership and the Contract as the Only Door

Amazon is as famous for how it organizes engineers as for what they built. Two of the three well-known rules are documented by Amazon; the third is not.

Two-pizza teams. Amazon's own DevOps guidance on the AWS documentation site (undated, accessed November 2026) quotes Jeff Bezos: "We try to create teams that are no larger than can be fed by two pizzas. We call that the two-pizza team rule." It ties the size to ownership: the team should see its service "as an end-to-end venture" rather than treating production "as black boxes where they have no visibility". The phrase is commonly dated to a 2004 Fast Company profile, which could not be fetched for this chapter, so the date is secondary. **You build it, you run it**, the rule that the team that writes a service also operates it in production, appears in the 2007 digest of the Vogels interview alongside the team rule.

The "API mandate". This is the rule most often quoted in architecture reviews, and it exists in public only as one person's recollection. Steve Yegge, a former Amazon engineer, wrote in October 2011 (in a post preserved as a public gist) that "back around 2002 I think, plus or minus a year" Bezos issued a mandate: "All teams will henceforth expose their data and functionality through service interfaces"; "Teams must communicate with each other through these interfaces"; no other communication, "no direct reads of another team's data store"; the technology "doesn't matter"; and "All service interfaces, without exception, must be designed from the ground up to be externalizable". No page from Amazon that the brief could fetch confirms the memo. Treat it as **widely reported**, not as an Amazon document.

What Amazon has confirmed, in its own words, is the outcome. In September 2006 Vogels wrote that Amazon "continues on the path of opening up all of its services for customers and partners to use", and that with Fulfillment by Amazon "we are now also opening up our fulfillment network for everyone to use". Ten years later, in his 2016 retrospective on AWS, he wrote that the lesson of building management APIs first was one "we had already learned from our experiences with Amazon retail".

An **externalizable interface** is an API designed so that it *could* be offered to outside developers, which forces a clean contract even when only internal teams use it. In plain words: build every internal door as if a stranger might walk through it. That test is the useful part of the mandate, whoever wrote it, and Promises Between Systems: API Contracts, Webhooks and Versioning carries it forward with Zalando's written, verifiable guidelines (Case: Zalando, Microservices, Platform Strategy and the Return to Discipline) as the documented version.

> **INFERENCE, NOT DOCUMENTED:** For a merchant, the rule applies to vendors before it applies to teams. Kestrel & Pine's 34 integrations include

agency scripts that read systems directly rather than through contracts, and the ledger's App That Wrote Stock is what happens when a second writer reaches into a system that already has an owner. Waypoint decision W5, a named owner and a contract for every integration, is the merchant-scale form of "no direct reads of another team's data store". At the Omnichannel stage the same rule becomes internal: the availability service of W1 exposes available-to-promise through an API, and nobody reads its tables.

Fulfillment as a Platform

This is the part of Amazon's record that most changes how a merchant should think, because it is about the business model, and the business model is documented in filings.

Three terms first. In the **first-party** model (1P), Amazon buys stock and is the seller. In the **third-party** model (3P), an independent seller sells on Amazon's site and Amazon takes fees. The **seller of record** is the legal party that sells to the consumer and carries the obligations: the invoice, the value-added tax, the consumer's right to return. **Fulfillment by Amazon** (FBA) is the service in which the seller sends stock to Amazon's warehouses and Amazon picks, packs, ships and handles returns.

The dates are clean. Amazon Prime launched in February 2005 at $79 a year for unlimited two-day shipping (widely reported; the press release was unreachable). Fulfillment by Amazon launched as a beta on 19 September 2006 with a press release promising small and medium-sized businesses "access to Amazon's order fulfillment, customer service, customer shipping offers, and underlying website technology"; third-party items "Fulfilled by Amazon" could qualify for Prime shipping. The two were designed to meet: a seller's item in Amazon's warehouse could carry Amazon's promise.

Bezos' 2018 shareholder letter, published in April 2019, gives the result as a series. Third-party sellers' share of physical gross merchandise sales:

Year	1999	2002	2005	2010	2012	2015	2018
Third-party share	3%	17%	28%	34%	42%	51%	58%

The letter's own comment: "Third-party sellers are kicking our first party butt. Badly." It credits two investments, "Fulfillment by Amazon and the Prime membership program", with having "meaningfully improved the customer experience of buying from independent sellers". Amazon's 2022 and 2024 Small Business Empowerment Reports (published May 2023 and May 2025) say "more than 60%" of sales in Amazon's store come from independent sellers; the 2024 report adds that those sellers "have shipped more than 80 billion items through Fulfillment by Amazon since its 2006 launch" and that Multi-Channel Fulfillment helped "more than 300,000 sellers" ship orders from their

other sales channels in the past year. Independent trackers put the unit share at 60 to 62% through 2024 to 2026.

The FY2025 Form 10-K, filed 6 February 2026, defines the model in one sentence: "We offer programs that enable sellers to sell their products in our stores, and fulfill orders using our services. We are not the seller of record in these transactions." Amazon earns "fixed fees, a percentage of sales, per-unit activity fees, interest, or some combination thereof", and designs its stores "to enable hundreds of millions of unique products to be sold by us and by third parties". The same filing gives the scale: net sales of **$716.9 billion** in 2025 (North America $426.3 billion, International $161.9 billion, AWS $128.7 billion), fulfillment expense of **$109.1 billion**, and approximately **1,576,000** full- and part-time employees at 31 December 2025. It does not state a count of fulfillment centers; it reports square footage.

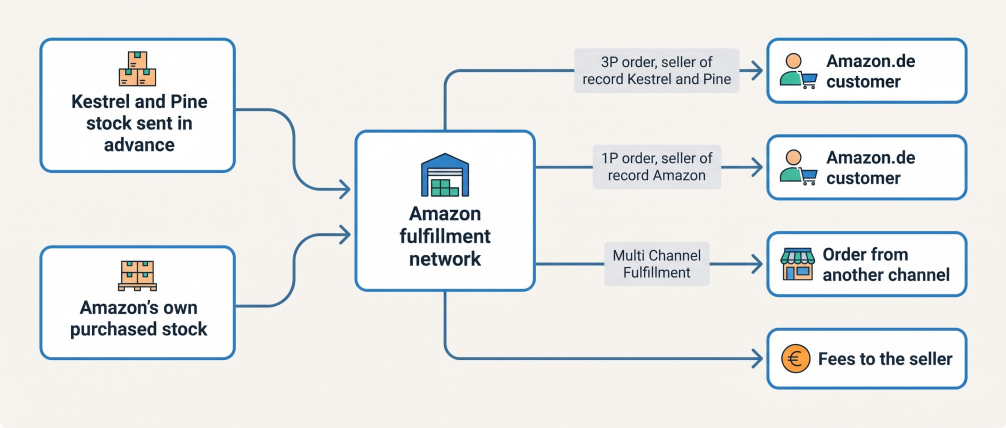


Figure 3: *One fulfillment network serves three flows with different sellers of record: Amazon's own stock, a third-party seller's stock under FBA, and orders from channels outside Amazon under Multi-Channel Fulfillment; the stock pool is shared, the legal and financial flows are not.*

INFERENCE, NOT DOCUMENTED: Amazon does not publish how FBA inventory, 1P inventory and Multi-Channel Fulfillment are kept apart inside its systems, so the diagram shows the documented flows, not the implementation. What the record does establish is that a retailer became a platform *through two operational services*, the promise and the warehouse, rather than through any storefront feature. The architectural consequence of running 1P and 3P side by side is one inventory and fulfillment capability serving two seller-of-record models with different invoices, tax treatment, returns obligations and ledgers. That is the split any merchant faces the day it fulfills a marketplace order from the pool that also serves its own shop, and it is why

Selling on Someone Else's Platform: Marketplaces and Channel Integration

> treats a marketplace order as a different *kind* of order, not a different source of the same one.

For Kestrel & Pine the relationship runs the other way: it is the third-party seller. The stock it sends to Amazon's program (part of the roughly 40% of German marketplace units that marketplace programs ship) is, in Waypoint decision W1, "a tracked external location": stock Kestrel & Pine owns but does not control, counted separately from Duiven, and sold only on the channel that holds it.

The Promise Is the Product

If the 2006–2007 sources are about the shop's internals, the 2024–2026 sources are about one thing: delivery speed. Amazon's recent letters and releases describe the shape of the network, and they describe nothing about the software that computes the promise.

Jassy's 2024 shareholder letter (published April 2025) attributes faster delivery "primarily to our regionalization redesign of our fulfillment network", to new placement algorithms and to same-day fulfillment centers, and confirms the starting scope of Prime: "Prime two-day delivery began with one million products and now covers over 300 million items, with tens of millions available in one day or better."

The 2025 letter (released 9 April 2026) reports over one million robots in fulfillment centers, "more than 85 Same Day Fulfillment Centers" that carry the top 90,000 SKUs across the United States, and over $4 billion committed to a rural delivery network.

The 3 February 2026 press release (syndicated; the primary page was unreachable) reports over 13 billion items delivered same or next day worldwide in 2025, over 8 billion of them in the United States, and a 70% rise in items delivered in under a day. These figures will be out of date within a year; they are quoted for their shape, not their size.

Three terms. **Regionalization** means holding stock in the region where it will be ordered, so most orders ship from nearby. A **placement algorithm** decides which warehouse should hold which SKU in what quantity. A **delivery promise** is the date or hour shown at checkout, which becomes a commitment the network has to keep. The 10-K's framing of Prime is commercial and candid: shipping offers and Prime benefits are "effective worldwide marketing tools" that Amazon intends "to continue offering them indefinitely".

What the promise needs	Amazon's documented answer (2024–2026)	What a merchant at Kestrel & Pine's scale does
Stock near the customer	Regionalized network; 85+ same-day centers with a curated 90,000-SKU set	One warehouse at Duiven; a static "ordered before 22:00, delivered tomorrow" rule for the Netherlands
Deciding where stock sits	Placement algorithms (named, not described)	Buyers decide by season; marketplace stock sent to Germany in advance
Capacity to keep the promise	Own delivery network, rural investment, robots	PostNL, bpost, DHL and DPD through the shipping service; the cut-off is the carrier's
Measuring the promise	Items delivered same or next day, published yearly	Not measured in 2026 *(ledger: the promise is static)*

INFERENCE, NOT DOCUMENTED: Amazon's public account makes the promise, not the catalog, the product, and the investment goes into the physical network that keeps it. The transferable idea is not the network. It is that **the promise engine, the logic that decides which date to show, is a first-class system** with inputs from stock, the network and carrier capacity, and that a promise is a commitment the fulfillment side has to be able to keep on the busiest hour of the year. Kestrel & Pine's "ordered before 22:00" rule is a promise with no engine behind it: it was true on an ordinary Tuesday and false on the Friday of Black Friday week, when the WMS queue backed up for four hours. Getting It There: Carriers, Labels, Tracking and the Delivery Promise owns the promise; Journey 4: Black Friday, Hour by Hour shows what happens when the promise and the capacity part company. Only Amazon's scale justifies 85 single-purpose same-day centers. The idea that transfers is the *curated subset*: a small set of SKUs for which a fast promise is made and kept, rather than one promise for everything.

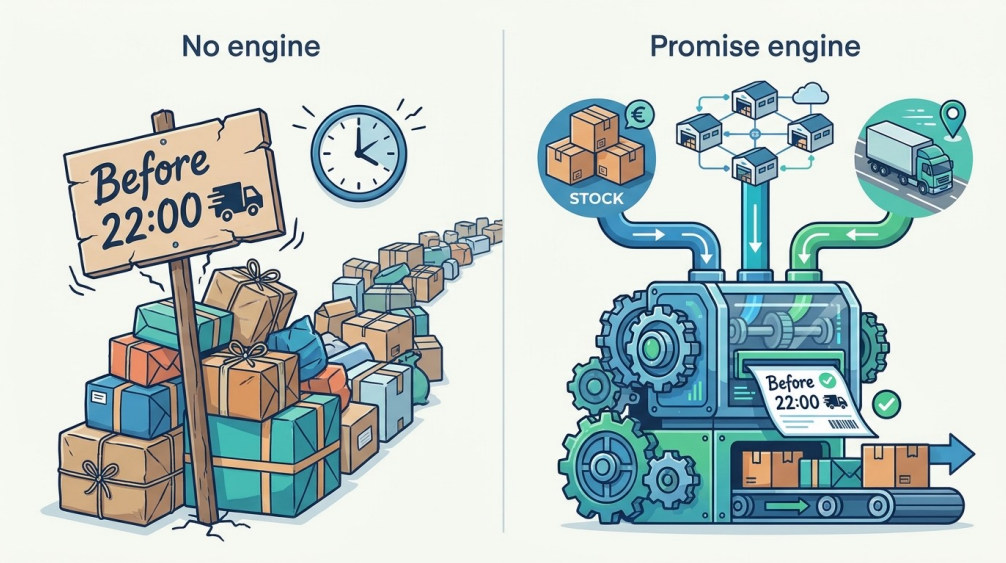


Figure 4: *Kestrel & Pine's "ordered before 22:00" rule is a promise with no engine behind it; the transferable idea is a promise engine fed by stock, the network and carrier capacity.*

Other Shops, Other Agents: 2022 to 2026

Amazon's most recent moves push the same machinery outward, into other merchants' storefronts and into software that shops on a customer's behalf. Everything in this section changes quickly; product names and fees in particular should be checked against the dated pages before being relied on.

Buy with Prime (launched 2022, widely reported) lets a merchant offer Prime delivery and Amazon's checkout on the merchant's own site, with Amazon handling "the payment, processing, storage, picking, packing, and shipping" plus customer service and returns; the product page (accessed November 2026) cites a 16% average lift in revenue per visitor from tests across 167 merchants between July 2023 and June 2024. **Shop Direct** (launched 2025, feeds added 2026) puts products from other stores into Amazon's search results: "over 100 million products from more than 400,000 merchants", synced through existing **product feeds**, structured files of catalog, price and stock, from Feedonomics, Salsify or CEDCommerce.

Rufus, Amazon's generative AI shopping assistant, launched as a beta in the US mobile app in February 2024 (per coverage); Amazon's page records that on 13 May 2026 it was renamed **Alexa for Shopping**. Amazon reports that it "helped over 300 million customers in 2025". The 2026 version adds actions: an automatic purchase when an item reaches a target price (a standing instruction that "stay[s] active for six months or until you cancel", with a 24-hour cancellation window), reorders and scheduled recurring purchases.

Buy for Me (beta, April 2025) goes further. For items Amazon does not sell, the app shows "Shop brand sites directly"; the customer confirms address, taxes, shipping and payment on Amazon's checkout page, and "Amazon completes the purchase on the brand's site". Amazon passes "encrypted name, address, and payment details" to the brand and "cannot see customers' previous or separate orders from other brands' sites"; the brand "handles delivery, returns, exchanges, and customer service" and chooses whether to participate. The catalog grew "from 65K products at launch to over half a million" and later to "tens of millions". Trade coverage of April 2025 reported that Amazon charged brands no commission at the time.

Agentic commerce is the term for software that does not only recommend but acts, placing an order on the customer's behalf. Assistants and Agents: AI in Commerce and What Is Real in 2026 owns it. What this case adds is Amazon's position on agents it does not control.

In its statement on Perplexity's Comet browser, Amazon wrote that "third-party applications that offer to make purchases on behalf of customers from other businesses should operate openly" and "respect service provider decisions whether or not to participate", that such apps "have the same obligations" as food-delivery apps or online travel agencies, and that it had "repeatedly" asked Perplexity to remove Amazon from Comet, citing a "significantly degraded shopping and customer service experience".

The cease-and-desist letter is dated 31 October 2025; Retail Dive reported on 6 November 2025 that Amazon had sued in Northern California and that Perplexity called the suit a "threat to user choice". The case's status was not checked.

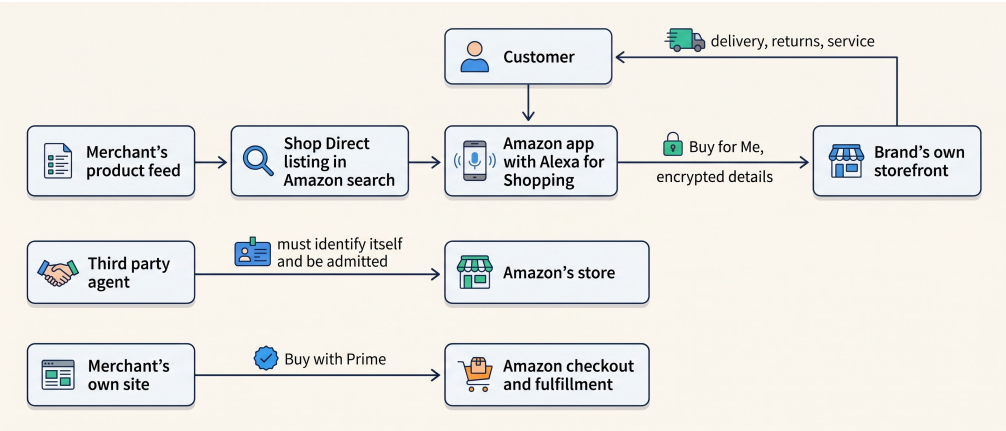


Figure 5: *Three documented directions: Amazon exports its checkout and fulfillment to merchants' sites (Buy with Prime), imports merchants' catalogs into its own surface (Shop Direct, Buy for Me), and admits outside agents only on its terms; the merchant's checkout and warehouse become modules behind someone else's demand.*

> **INFERENCE, NOT DOCUMENTED:** Two patterns are visible. First, Amazon exports its promise-keeping machinery to other storefronts and imports other storefronts' catalogs into its own, so that a merchant's checkout and fulfillment become interchangeable modules behind a dominant demand surface. Second, Amazon's rule for outside agents, identify yourself and respect the provider's choice to participate, is the rule every merchant will need for bots on its own site: a documented, authenticated channel for automation (feeds, APIs, a checkout protocol) and a way to refuse automation that will not identify itself. That is the Secret Agent pattern from the other side. For Kestrel & Pine it is Waypoint decision W8 in practice: accurate product data, availability and order status by API first, and no shopper-facing agent of its own in 2027.

A Nuance: When One Process Beat a Distributed Design

One Amazon post from 2023 is quoted as the reversal of everything above, and the source supports a much narrower claim.

In March 2023 the Prime Video team responsible for audio and video quality monitoring described replacing a **serverless** design, in which small functions run on demand and a managed workflow engine (AWS Step Functions) calls each step and stores intermediate results in object storage, with a single application process. The reason, as summarized by InfoQ on 3 May 2023 (the original post now redirects to a generic page): "running the infrastructure at a high scale was very expensive", the design "supported only around 5% of the expected load", and moving the data in memory instead of through storage cut operational costs by 90%. The team's own framing was that the "distributed approach wasn't bringing a lot of benefits in our specific use case".

That is a cost-and-coupling argument about one internal pipeline, not evidence about the retail site's architecture. The rule it supports, which Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says develops, is that **a boundary should be sized to the data that crosses it**: when most of the work is moving bytes between steps, the steps belong in one process. Zalando reached a similar conclusion about a stream-based integration in 2025 (Case: Zalando, Microservices, Platform Strategy and the Return to Discipline).

What a Merchant Can Learn, and What Only Amazon Can Do

The table is the author's synthesis; each row names the dated evidence it rests on.

Lesson	Evidence	Transfers to a merchant?
Split along ownership lines, and make the contract the only door	2006 interview digest; 2011 recollection of the mandate	Yes, first as a rule for vendor integrations (no direct reads of another system's data), then for internal services
The cart is an availability problem, not a consistency problem	Dynamo, 2007	Yes; buy, do not build, the highly available store; keep the consistency for stock and payment
Operability decides adoption	DynamoDB, 2012	Yes, at every stage; a fourteen-person team cannot run what Amazon's teams refused to run
Fulfillment opened to others turns a retailer into a platform	FBA 2006; 2018 letter series	Partly: the Group stage; an Omnichannel merchant starts by fulfilling marketplace orders from one pool
The promise is the product; shape the network to it	2024 and 2025 letters; February 2026 release	The idea, yes; the network only at Amazon's scale; the curated fast subset transfers
Agents must identify themselves and respect participation	Perplexity statement, 2025	Yes, as a policy and an authenticated channel
Size a boundary to the data flow across it	Prime Video, 2023	Yes

Only Amazon can do the rest: 85-plus single-purpose same-day centers and a million robots; a 300-million-item two-day catalog; a demand surface that 400,000 merchants

feed; a database lineage from Dynamo to DynamoDB. The Giant's Boots pattern is a merchant mistaking the second list for the first.

What Kestrel & Pine takes from the case. Three things, each already a Waypoint decision. From the API rule: every integration gets an owner and a contract, and no script reads another system's data directly (W5). From FBA: the stock in Amazon's program is a tracked external location with its own count, sold only on the channel that holds it, never pooled with Duiven in the availability calculation (W1). From the promise: "ordered before 22:00, delivered tomorrow" needs an engine behind it before Black Friday 2027, fed by the WMS's real capacity, or it will be withdrawn on the day it matters most (Black Friday Every Year: Peak Load, Resilience and Recovery). What it does not take is the architecture: a hosted platform with an OMS, a PIM and an availability service beside it is the right shape for a €48 million merchant (Build, Buy or Assemble: Choosing and Changing Platforms).

Where the Evidence Stops

Be clear about what this chapter could not establish.

- **Today's internals**. No public, dated description exists of Amazon's current order, pricing, promise or inventory systems. The 2006–2007 architecture is history; the Dynamo paper describes a system Amazon said in 2012 its teams had moved away from. Anything you read about "how Amazon's order system works today" is inference.
- **The API mandate** rests on one first-hand recollection from 2011, never confirmed by Amazon in any page fetched. Its content is consistent with Amazon's documented behavior, which is the most that can be said.
- **The monolith's name**, the 2004 dating of the two-pizza rule, the 2005 Prime release and the February 2026 delivery record are widely reported but were not fetched from primary pages. Fulfillment center counts are not in the 10-K, which reports floor area instead.
- **The agent products** (Alexa for Shopping, Buy for Me, Shop Direct) are described as of November 2026; their names, fees and scope have already changed once in 2026 and will again. The Perplexity litigation's outcome was not checked.
- **Costs**. Amazon publishes fulfillment expense in aggregate and nothing about the unit economics of FBA, Prime or same-day delivery.

What to Take Away

Established.

- Amazon's move from a monolith to hundreds of services followed a scaling wall it dates to 2001, and by 2007 a page touched 100 to 150 services measured at the 99.9th percentile (2006 interview digest; Dynamo paper, 2007). Decomposition followed the wall; it did not precede it.
- The shopping cart was built as an "always writeable" store that merges conflicts on read because a refused add-to-cart is a lost sale (Dynamo, 2007); Amazon's own teams then preferred managed services to a system that was hard to operate (DynamoDB, 2012).
- Amazon became a marketplace through Prime (2005) and Fulfillment by Amazon (2006): third-party sellers went from 3% of physical sales in 1999 to 58% in 2018 and "more than 60%" since (2018 letter; 2022 and 2024 reports). Amazon is "not the seller of record" in those sales (10-K, 2026).
- Amazon's stated rule for outside shopping agents is to operate openly and respect a provider's choice to participate (2025 statement).

Depends on context.

- "No direct reads of another team's data store" is right for a merchant as a rule about vendors and scripts at every stage; it becomes a rule about internal services only once there are internal services to govern, which for most merchants is the Omnichannel stage or later.
- Availability-over-consistency is right for the cart and wrong for stock, payment and orders. A merchant on a hosted platform inherits the right behavior; one building a storefront must choose it deliberately and buy the store rather than build it.
- Fulfilling other sellers' orders from your own pool is a Group-stage move for most merchants; the Omnichannel-stage version is fulfilling your own marketplace orders from one pool with one promise.
- A computed delivery promise becomes necessary with a second stock location or a cut-off that varies by day; a static promise serves one warehouse on ordinary days and fails on peak days.

The author's synthesis.

- The useful Amazon is the narrow one. I would strike "Amazon did microservices" from every design review and replace it with the four documented rules: the contract is the only door, the cart never says no, operability decides adoption, and the promise is the product. Those fit a fourteen-person team; the service count does not.
- The most important Amazon decision for a merchant is not architectural: Amazon opened its *operations* to others and let the storefront follow, which is why a merchant's own warehouse, promise and returns handling are the capabilities worth owning well (Pick, Pack, Ship: Warehouse Management and Fulfillment).

Questions to Ask

1. Which of our integrations read another system's data directly instead of through a documented contract, and who owns each one?
2. If our cart store were unavailable for a minute on Black Friday, would the customer see an error or a slightly stale cart? Which do we want?
3. Where is consistency non-negotiable for us (stock, payment, the order record), and does each of those systems have one writer?
4. Could our team run the systems we are proposing to build, on the busiest night of the year, with the people we have?
5. Which stock do we own but not control (a marketplace program, a 3PL, a store), and is it counted as a separate location in our availability calculation?
6. What is our delivery promise, which system computes it, and what does it know about warehouse capacity and carrier cut-offs today?
7. For which small set of SKUs could we make a faster promise and keep it, instead of one promise for everything?
8. What is our policy for automated buyers and agents on our own site, and do we have an authenticated channel for the ones we want?

Amazon's record ends where its publications end, and the next case has a fuller one. Zalando, the European fashion platform on which Kestrel & Pine also sells, has published its architecture decisions on its own engineering blog for ten years, including the ones it later reversed. Continue with Case: Zalando, Microservices, Platform Strategy and the Return to Discipline.

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

48. Case: Zalando, Microservices, Platform Strategy and the Return to Discipline

IN THIS CHAPTER, YOU WILL: Follow Zalando's own dated accounts from the 2015 microservices move through the corrections that followed, and take from them what a merchant can use about modularity, API contracts and becoming a platform.

KEY POINTS:

- **Zalando split its shop for team autonomy, not for load, and said so.** In 2015 a shared codebase and slow deployments blocked 85-plus teams; the fix was organizational first (small teams, an AWS account each, a paved road called STUPS) and architectural second. The service count followed the team count.
- **The durable export is a written API standard, not the service count.** Zalando's guidelines, open since 2016 and still maintained in 2026, say what a merchant should demand of every vendor: extend compatibly

instead of versioning, never break consumers, and sunset an API only with client consent, a date, headers and usage monitoring.
- **Autonomy was followed by a decade of discipline**. Alignment over raw autonomy (2018), fewer languages (2020), one developer portal (2023) and, in 2025, a synchronous read API replacing a price-and-stock event stream that fell 30 minutes behind at Cyber Week and bred divergent copies of product data. Events propagate change; they are not the system of record.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Giant's Boots** — Zalando's microservices move is copied for its shape; the record shows it was made for 85-plus teams and corrected for a decade afterward.
- **The Pulled Rug** — Zalando's deprecation rules (consent, a sunset date, headers, usage monitoring) are the written cure for the vendor API change that breaks an integration before anyone is told.
- **The Message in a Bottle** — Zalando's 2025 post documents what happens when consumers rebuild their own copy of price and stock from a stream they cannot keep up with: delays and divergent copies.

A merchant listing products on Zalando depends on a platform that serves shoppers, connects sellers and offers fulfillment services. Those jobs create coordination problems within Zalando as well as across the merchant's integrations. The Berlin fashion platform reported **€17.6 billion in gross merchandise volume**, **62 million active customers** and **278.6 million orders** for 2025 (March 2026 fact sheet). Its engineering choices address that scale; merchants such as Kestrel & Pine, the fictional brand used here, need to understand which lessons apply to their smaller operation.

From 2015, Zalando described a move to independently deployed services and autonomous teams. Later posts addressed shared standards, fewer technologies and common developer tools. A 2025 account even replaced one event-based integration with direct API calls. The case is useful because it exposes the costs of earlier decisions as well as their rationale.

Read the **2015–2017 accounts as history**, alongside newer business filings and 2023–2026 AI material. Each claim is dated; inferred internals remain labeled. "Return to discipline" is this chapter's interpretation of successive posts, not a program Zalando announced. Case: Amazon, From Monolith to Services to the Everything Store applies the same evidence discipline to another company.

Where we are in the order's journey. A merchant listing a jacket on Zalando needs correct price and stock, an imported order and a maintained API contract. Selling on Someone Else's Platform: Marketplaces and Channel Integration, Is It in Stock? Inventory, Availability and the Promise You Can Keep and Promises Between

Systems: API Contracts, Webhooks and Versioning explain those responsibilities from the merchant's side.

2015: Jimmy, Radical Agility and the Split

A **monolith** is one deployed application with a shared codebase. **Microservices** split work into separately deployed programs that communicate over a network. An **autonomous team** owns its services in production and has freedom over tools and process. The practical question is how much independence helps teams deliver without making coordination harder.

Zalando's monolith, **Jimmy**, blocked teams through slow startup and complicated deployments, according to its October 2015 post. The stated problem was release speed and shared code, rather than traffic alone.

In May 2015, seven engineers and a designer began rebuilding the Fashion Store, then fifteen country sites serving over sixteen million active customers. Teams would own programming, testing, deployment and operations. A router called **Skipper**, later open-sourced, directed requests between the old and new parts.

The organizational side was called **Radical Agility**, introduced in March 2015 and built, by trade accounts of 2016, on autonomy, mastery and purpose, with teams of three to seven people owning a topic end to end. Zalando's own July 2015 post has Eric Bowman, then VP Engineering, at the AWS Summit in Berlin: "We think of teams as a unit of trust"; "we've tried to create the context where people can make mistakes and it's not fatal". By February 2016 a Zalando talk at the MicroXchange conference covered microservices across "our 85+ engineering teams"; by December 2016 InfoQ reported "about 110 teams", more than 1,400 people in Tech across six hubs, and "each team has its own AWS account".

That last fact needed a platform. **STUPS** was Zalando's answer: an **internal platform-as-a-service**, meaning one supported way for every team to deploy to the cloud. Development began in late 2014 and it was available for production to every team in May 2015. InfoQ's April 2016 account describes it as "a convenient and audit-compliant Platform-as-a-Service" on AWS: "Our employees get nearly administrative access to their AWS accounts with almost no restrictions", with near-real-time audit reporting because "Auditors need to know that we're complying with the regulations" (card-industry, financial-reporting and German rules). In plain words: Zalando gave every team its own cloud account and built the tooling that made that auditable.

The shop itself was rebuilt as **fragments**, pieces of a page owned by different teams and deployed separately, assembled behind the router. One year in, in October 2016, the effort was renamed Project Mosaic: header and footer fragments were live, product-page and wish-list fragments were going live, and the author described a shift "from

prototyping to performance optimization" and toward "stable components that other teams can depend on".

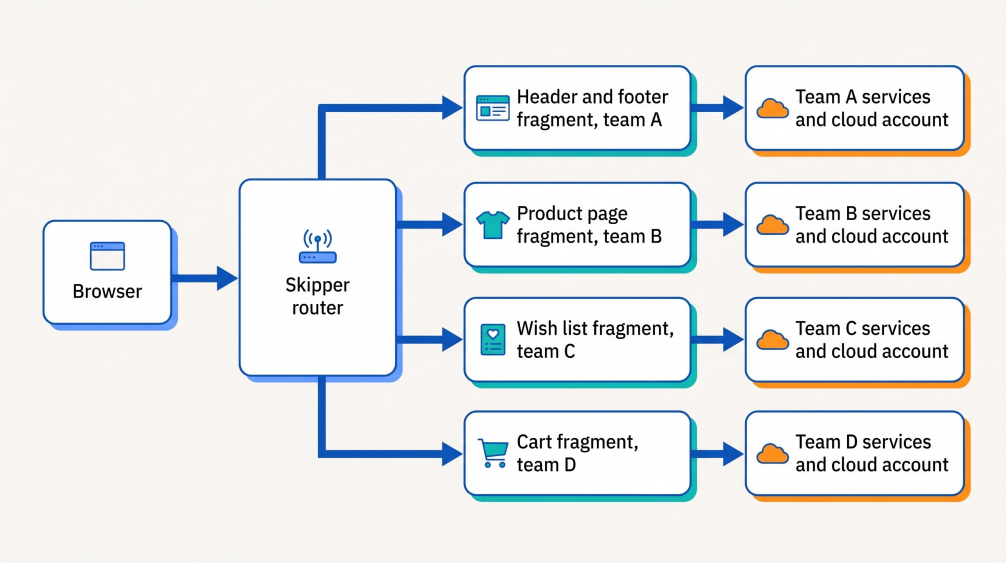

Figure 1: *The 2015–2016 shape as Zalando described it: a router in front, a page assembled from fragments each owned and deployed by one team, and each team on its own cloud account behind a shared, audited platform.*

A smaller post from April 2016 is the most honest about the first months: a team in Dortmund cut its domain into two services, then into eight after modeling the boundaries properly ("Define your Microservices properly and know your boundaries"), and replaced a failed upfront plan with weekly goals.

> **INFERENCE, NOT DOCUMENTED:** Zalando's documented sequence mirrors Amazon's with a thirteen-year lag and a different stated motive. Amazon's 2006 account names a scaling wall; Zalando's 2015 posts name team autonomy. The fix was organizational first (teams, accounts, a paved road) and architectural second, which is why the service count tracked the team count: 85-plus teams in early 2016, about 110 by December, and by 2020 "4,000+ applications" for "1,200+ developers". A merchant with fourteen people in technology, like Kestrel & Pine, is below the team count at which any of this applies. What transfers is the *order of operations*: decide who owns what before deciding how to split what, which is the argument of Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record and Who Builds and Who Owns: Teams, Vendors, Agencies and Governance.

The Cost of Distributed Data, Stated Early

The clearest early statement of what the split would cost came not from the shop teams but from business intelligence. In September 2015 a Zalando post observed that the monolith's "consolidation of data in a single place makes business intelligence fairly simple", whereas "with each service owning its data, data is spread over a significantly larger number of cloud-based microservices"; article and order data alone were spread across eight PostgreSQL databases. Zalando's response was to deploy Kafka, a message log, on AWS "to make our event data available for downstream systems", and to plan a data lake.

Two terms. An **event** is a message sent the moment something happens (an order placed, a price changed) so other systems can react. An **event bus** is the shared channel on which services publish events and others subscribe.

Zalando's event bus was **Nakadi**, started in October 2015: in its own README, "a distributed event bus that implements a RESTful API abstraction on top of Kafka-like queues", with a schema registry so that producers register an event type with a **schema** (the agreed shape of the message) before publishing, and every event is validated against it.

By May 2017 a Zalando post on "third-generation microservices" described the intended style: services "process incoming events and data from other services", and data streaming becomes "a central part of the service fabric".

> **INFERENCE, NOT DOCUMENTED:** This is the earliest honest statement in the record of the rule Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation is built on: once each system owns its own data, there is no single place to query, so analytics and reconciliation need a stream of what happened. Kestrel & Pine meets this problem without a single microservice: its thirteen systems hold orders in four places and product data in four, and its data warehouse has three definitions of an order (Waypoint decision W7 picks one). An Omnichannel merchant meets the *data* problem of distribution long before the *team* problem, and should take the event-stream lesson without the service count.

The API Guild and the Guidelines

The piece of Zalando's 2015 work that is still alive, still public and still maintained in 2026 is its API standard. It began as the **API Guild** in March 2015: an informal group of about 20 people who set standards and review APIs as "a sounding board and review resource, not an approval board, to avoid bottlenecks", spending about 10% of their week on it. The September 2015 post explains why it mattered to the business: Zalando's "open platform

strategy relies on public APIs used by external partners", so "We cannot afford to break our APIs", and "Ideally, all Zalando APIs should look like the same author created them".

The principle is **API First**: the interface is designed and reviewed, as an OpenAPI document (a machine-readable description of an API), before any code is written. By December 2016 the guidelines were open-sourced; Zalando described API definitions as "a source truth, part of a contract between a service provider and consumers". The current guidelines (undated page, accessed November 2026) use the MUST, SHOULD and MAY of internet standards and say teams "must define APIs first, before coding its implementation, using OpenAPI as specification language" and should "Treat your API as product and act like a product owner".

The rules that matter most to a merchant are the ones about change. In plain words, a **compatible extension** adds to an API without breaking anyone who already uses it; **versioning** publishes a second, different API beside the first; **deprecation** announces that an API will stop; a **sunset** is the date it does.

Rule (Zalando guideline number)	What it says, in plain words	What Kestrel & Pine should demand of a vendor
#106 MUST not break backward compatibility	"Change APIs, but keep all consumers running"	Additive changes only inside a version
#107 SHOULD prefer compatible extensions; #113 SHOULD avoid versioning	Add fields rather than publishing a new version	A changelog of additions, not surprise versions
#108 MUST prepare clients to accept compatible extensions	Clients ignore fields they do not know (the **tolerant reader**)	Its own integrations must not fail on a new field
#114 MUST use media type versioning; #115 MUST not use URL versioning	When a version is unavoidable, put it in the content type, not the address	A stated versioning scheme, whichever it is
#185 MUST obtain approval of clients before API shut down; #186 MUST collect external partner consent on deprecation time span	No sunset until "all clients have given their consent on a sunset date"	A deprecation notice with a date agreed in advance

Rule (Zalando guideline number)	What it says, in plain words	What Kestrel & Pine should demand of a vendor
#187 MUST reflect deprecation in API specifications; #189 SHOULD add Deprecation and Sunset headers	`deprecated: true`, the sunset date and migration guidance in the spec; headers on every response	Machine-readable deprecation, so a script can warn before it breaks
#188 MUST monitor usage of deprecated API scheduled for sunset	The provider watches who is still calling	A warning to the merchant while it is still calling
#191 MUST not start using deprecated APIs	A client never builds on something already marked for removal	The agency checks before building

The companion to the guidelines is the **Tech Radar**, created in 2016 "to inspire and support teams to pick the best technologies for new projects". A Technologist Guild of about 25 senior people assigns technologies to rings by vote: ADOPT ("high confidence in to serve our purpose, also in large scale"), TRIAL, ASSESS and HOLD ("not recommended to be used for new projects"). In 2016 "Teams must use the Tech Radar as one input source for their technology decisions"; that wording changes in 2020.

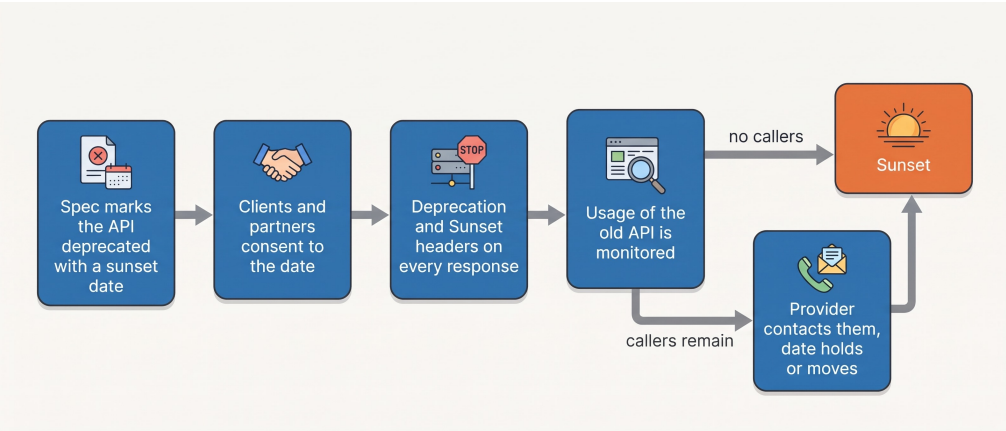

Figure 2: *Deprecation as Zalando's guidelines prescribe it: announce in the specification, obtain consent on a date, signal in headers, monitor who still calls, and only then sunset; every step is a chance to avoid the Pulled Rug.*

Kestrel & Pine learned the value of these rules the expensive way. In February 2027 its marketplace connector retired an API version on a published date; the agency script

that read it had not been updated, and stock to bol was stale for two days (the ledger's Deprecated Endpoint). Every row in the table above describes a step that would have caught it: a machine-readable deprecation, a header the script could have logged, usage monitoring that would have shown the script still calling. Promises Between Systems: API Contracts, Webhooks and Versioning turns the guidelines into the book's contract rules.

The Return to Discipline, 2018 to 2026

No single Zalando post declares that the autonomy era ended. The posts that follow narrow a choice each time, and read in order they tell one story.

Year	What Zalando published	What narrowed
2018	Bowman, by then leading a 500-person group, told InfoQ in September: "Our biggest challenge was that we treated autonomy like an input to teams"; unaligned teams were like "air molecules in a balloon: moving fast but not running together as a group"; autonomy became "an aspiration and a reward, not a gift"; "a set of principles and systems scales better than a set of prescriptions". Teams were embedded across the company with shared objectives and company-wide "Global Projects".	Autonomy without alignment
2018	The agile coaches: "we value principles more than frameworks"; six principles replace a single prescribed method (August).	One mandated process, replaced by principles

Year	What Zalando published	What narrowed
2020	With "1,200+ developers" and "4,000+ applications", Zalando "redefined the ring semantics" of the radar, wrote Technology Selection Principles, gave a Principal Engineering community the decision, required a sponsoring Engineering Head for TRIAL technologies, and stopped new applications in Clojure, Haskell and Rust in favor of Kotlin and TypeScript, because "A broad freedom of choice leads to increased difficulties in supporting software long-term" (July).	Free choice of language
2023	**Sunrise**, an internal developer portal built on Backstage since 2021, replaced "more than 100 disconnected interfaces" with one portal holding "over 40k registered entities (applications, teams, and users)" and 27 integrated tools; old tools were shut down (August).	Many tools, replaced by one paved road
2023	On 21 February the co-CEOs announced the removal of "several hundred overhead roles", about 5% of roughly 17,000 employees, sparing logistics, customer care and outlets (Retail Gazette).	Overhead
2025	The event-driven price and stock pipeline "became a bottleneck, introducing delays and scaling challenges"; it was replaced by a synchronous Product Read API (March).	A stream as the only source of current state

Year	What Zalando published	What narrowed
2026	Connected Retail, the ship-from-store program launched in 2018, is folded into the standard partner program; its technology "will be discontinued by June 30, 2027" (FashionUnited, April).	A separate channel technology

Three terms. A **paved road**, or internal developer platform, is one supported way to build, deploy and find services so teams do not each solve the same problem. A **technology radar with teeth** is a list of allowed technologies teams must follow rather than consult. A **synchronous read API** is one that answers "what is the price now" when asked, instead of making the caller rebuild its own copy from a stream of changes.

The evidence does not support "microservices failed at Zalando". It supports something more specific: **autonomy produced too many technologies**, **too many tools and**, **on one critical path**, **an integration that could not keep up**, and each was fixed by narrowing a choice. Zalando today reports more than 17,000 employees, around 3,000 in technology and more than 7,000 in fulfillment centers (annual report 2025, March 2026).

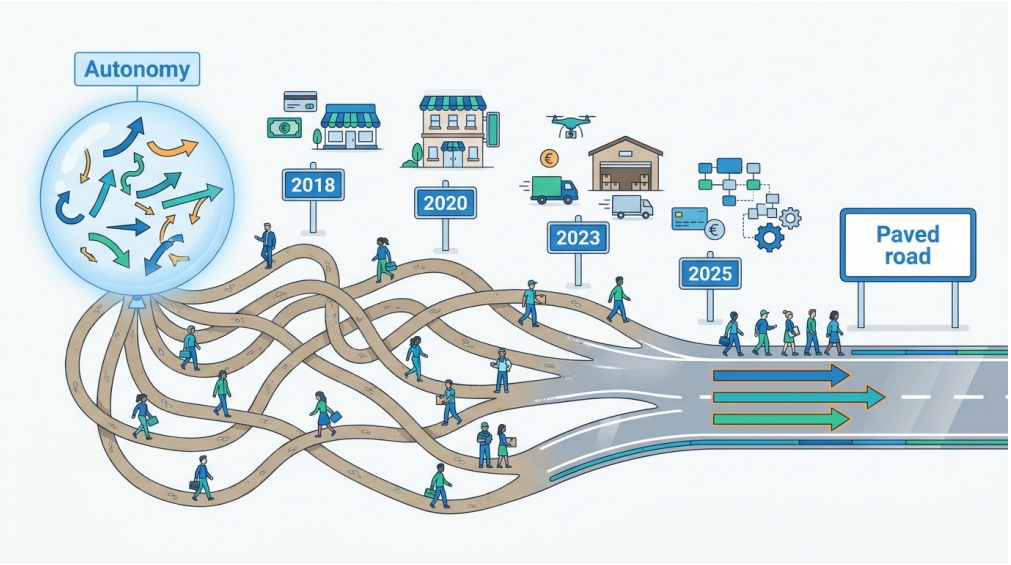

Figure 3: *Zalando's return to discipline narrowed one choice at a time: alignment in 2018, fewer languages in 2020, one developer portal in 2023 and a read API in 2025.*

2025: When the Event Stream Fell Behind

The 2025 post deserves its own section because it is the most useful single document in the case for a merchant, and because it is the Black Friday Oversell told by a platform.

Zalando's price and stock updates were published as events. Consumers, the systems that needed current prices and stock, subscribed to the stream and maintained their own copies. The March 2025 post "From Event-Driven to API" reports what happened at scale: "over 90% of each payload unchanged" was being reprocessed, because an event carried the whole record even when one field had changed; during Cyber Week events "could be delayed by up to 30 minutes"; and teams that could not consume the streams fast enough used other sources, producing "divergent versions of product data". The pipeline was replaced by a synchronous, low-latency **Product Read API** (PRAPI): callers ask for the current price and stock and get an answer now.

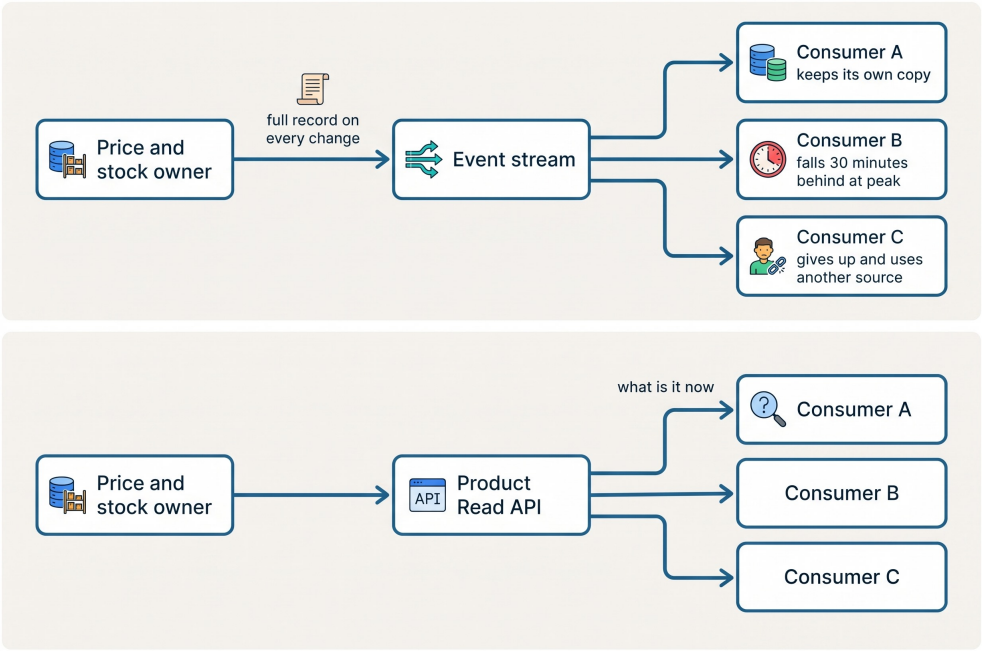

Figure 4: *Before 2025, every consumer rebuilt its own copy of price and stock from a stream of full records and the copies diverged under load; after, consumers ask the owner for the current value, and the stream is no longer the only way to know the truth.*

In plain words: an event is the right way to tell everyone that something changed. It is the wrong way to be the only place the current value can be learned, because then every consumer is a copy, every copy can lag, and on the busiest day the lag is largest. Kestrel & Pine's stock sync chain, WMS to ERP every 15 minutes, ERP to platform hourly,

platform to marketplace connector hourly, is a slower version of the same design: a chain of copies, each one stale by the time it is read, which is how the Ridgewalker kept selling for two hours after it sold out at 19:40 on 27 November 2026.

Figure 5: *Kestrel & Pine's stock sync is a chain of copies, WMS to ERP every 15 minutes, then hourly to the platform and hourly to the connector, which is how the Ridgewalker kept selling for two hours after it sold out.*

> **INFERENCE, NOT DOCUMENTED**: Zalando does not describe PRAPI's internals, its caching or how events and the API now coexist, so the diagram shows the documented before and after, not the implementation. The rule I draw is the one Waypoint decision W1 encodes: keep a **system of record** for stock and price, an **availability service** that answers "can I promise this unit on this channel now" through an API, and use events to propagate change and to feed reconciliation, not as the only source of truth. Events for "what happened"; a read API for "what is it now"; a reconciliation job for "do the copies still agree" (Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation). The 30-minute figure is Zalando's; the two-hour figure is Kestrel & Pine's; the shape of the failure is the same.

The Platform Pieces: What Survived

Zalando open-sourced much of what it built in 2015, and the status of those projects in November 2026 is itself evidence.

Project	What it is, in Zalando's words	Status on 7 November 2026
STUPS (for example Senza)	audit-compliant platform-as-a-service on AWS for autonomous teams	Senza archived; last change March 2023
Skipper	"An HTTP router and reverse proxy for service composition"; used "as shop frontend http router with 350000 routes" and as the ingress controller "in more than 100 production clusters"	Active; changed the day before the check
Patroni	"a Python template for building PostgreSQL high availability (HA) clusters", introduced on Zalando's blog in February 2016, now under its own organization	Active
Nakadi	the event bus described above	Archived June 2024: "no longer developed in the open as we've not seen usage outside of Zalando for many years"
RESTful API and Event Guidelines	"A model set of guidelines for RESTful APIs and Events, created by Zalando"	Active; changed September 2026
Tech Radar	the radar page and its generator	Active; changed October 2026
Zalenium	container-based test grid	Archived September 2021: "no new maintainers were found"

INFERENCE, NOT DOCUMENTED: The pieces that survived are the ones with a life outside Zalando: a router, a database failover template and a written standard. The pieces tied to the 2015 account-per-team model (STUPS) or used only internally (Nakadi) were archived. Read the table as a warning against building platform tooling only one company needs.

Archiving Nakadi says nothing about whether Zalando still uses an event bus internally; the README says only that nobody outside did.

A Retailer Becoming a Platform

Zalando's business story runs parallel to its architecture story, and the two meet at the warehouse.

Three terms. In the **partner program**, a brand sells on Zalando as the seller of record (the legal seller who issues the invoice and owes the consumer rights) and Zalando takes a commission, as opposed to Zalando's **wholesale** model in which it buys the stock and sells it itself. **Ship-from-store** sells a physical store's shelf stock online and ships it from the store. **Logistics as a service** offers a retailer's warehouses, carriers and software to other sellers, for orders on and off its own platform.

Year	Move	Documented figures
2011	**Partner Program** launches: brands sell as seller of record through the zDirect portal	June 2021: "more than 600 partners, representing over 2,000 brands"; about a quarter of GMV; a target of 40% by 2023 and half in the long term (targets now dated)
2018	**Connected Retail** launches: stores' stock sold on Zalando, collected by carriers (DHL in Germany, PostNL in the Netherlands), returns back to the store	March 2020: "more than 1,500" stores in Germany and the Netherlands, commissions waived for April–May 2020; January 2021: "more than 2,600"; February 2021: "over 3,000 active stores" in nine markets
2010s–2021	**Zalando Fulfillment Solutions** (ZFS) and Shipping Solutions: Zalando's warehouses and carriers for partners	June 2021: access to 16 European markets, "more than 2,300 market expansions" in 18 months

Year	Move	Documented figures
2022–2023	Multi-channel fulfillment pilot (2022), rebranded **ZEOS** Fulfillment in October 2023: one stock pool for Zalando, own-shop and other-platform orders, open to sellers "whether or not they sell through Zalando"	End of 2025: 25 markets, "over 600 partners selling across 18 sales channels (including Zalando), covering around 85% of the European marketplace volume" (annual report)
2025	**ABOUT YOU** acquired: intention announced 11 December 2024, completed 11 July 2025 at 91.45%, squeeze-out registered 6 November 2025 at €6.50 a share; B2C runs two brands on shared logistics and payment infrastructure	Targeted group synergies "around €100 million per year"
2025	B2B segment defined as ZEOS Fulfillment (including ZFS and multi-channel fulfillment), the commerce platform **SCAYLE** and the integration product **Tradebyte**	FY2025 B2B revenue €1,091.3 million (+14.6%), adjusted EBIT €53.9 million; 330 online shops on SCAYLE; a ZEOS app for Shopify's European merchants
2026	Connected Retail folded into the partner program; same standard fees for all partners from September 2026; technology discontinued by 30 June 2027	"We are not saying goodbye to brick and mortar stores; we are simply updating the technology they use to connect with us"

The FY2025 fact sheet (12 March 2026) gives the scale around this: GMV €17,560.2 million (+14.7%), revenue €12,346.1 million (+16.8%), adjusted EBIT €590.7 million, 278.6 million orders at 4.8 orders per customer and an average basket of €62.8. Reported growth includes ABOUT YOU; Zalando's guidance for 2026 is pro-forma growth of 5 to 10%, the honest measure of the core business. The partner share of group GMV fell to 32.1% from 34.2% because ABOUT YOU "currently operates with a lower partner business share". Zalando's December 2025 strategy note counts "over 60" live merchants on its fulfillment service; the annual report's 600 partners span all B2B services.

> **INFERENCE, NOT DOCUMENTED:** Zalando's platform story is Amazon's in European form and on a European timetable: marketplace (2011), then ship-from-store (2018), then fulfillment for partners, then fulfillment for

anyone on any channel (2023), then the shop software itself (SCAYLE, in the group from 2025). The architectural consequence is the one Case: Amazon, From Monolith to Services to the Everything Store drew: **one stock pool serving several seller-of-record models and several demand channels**, which is the Omnichannel-stage problem (own site, stores, wholesale, marketplaces) at a much larger scale. Two cautions are in the record. Connected Retail shows that a channel built on its own technology gets consolidated once the general program can do the job. And the partner share falling after the ABOUT YOU merger shows that an acquisition imports the acquired company's model mix, not only its revenue (What Changes at Each Scale: From First Store to Group).

AI at Zalando: Discovery First, Then Identity

On 19 April 2023 Zalando announced a beta "fashion assistant powered by ChatGPT" for selected customers in four countries, letting them describe what they want in their own words and refine the results in conversation. The annual report for 2025 says the Zalando Assistant "was used by over 6 million customers" that year, and that Zalando sees "the emerging channel of agentic commerce as a great opportunity" while noting "the rise of AI agents as digital gatekeepers".

The engineering side came in September 2026. Zalando described an internal agentic platform on Kubernetes and open-sourced an **Agentic Identity Broker** that "lets agents act on behalf of users without ever holding provider tokens".

Two terms: **delegated access** means a user allows an agent to act with a subset of the user's permissions ("Delegated access is the intersection of both the user and agent permissions"), and **token exchange** means the broker swaps the agent's identity for a short-lived credential scoped to one action, so the agent never holds the user's real credentials.

The post adds that consent "should name a business capability … instead of presenting a raw scope string", and that stronger approval is planned for high-risk transactions because users rubber-stamp approvals.

INFERENCE, NOT DOCUMENTED: The 2023 assistant is a search feature: finding products in natural language. The 2026 broker is the infrastructure answer to the question Amazon answered with a policy and a lawsuit: who is this agent, for whom does it act, and what may it do. Read the two cases together. Amazon states the rule for *external* agents; Zalando publishes the identity model for *internal* ones. For Kestrel & Pine the broker is a preview of what Waypoint decision W8 will require before any agent may place or change an order: an identity for the agent, a named person it acts for, and a scope a human can read. Assistants and Agents: AI in Commerce and What

Is Real in 2026 owns the subject; the identity model is also treated in the sibling book at Agent Identity, Delegation and Least Privilege.

What Zalando Teaches

The table is the author's synthesis; each row names its evidence.

Lesson	Evidence	Transfers to a merchant?
Decide ownership before decomposition; the service count follows the team count	2015 posts; 110 teams (2016); 4,000+ applications (2020)	The order of operations, yes; the count, only at the Group stage
A written API standard outlives the architecture it was written for	Guild (2015); guidelines active in 2026	Yes, as what to demand of vendors first and what to write for internal APIs later
Deprecate only with consent, a date, headers and monitoring	Guidelines #185–#191	Yes; it is the cure for the Pulled Rug
Distributed data needs an event stream for analytics and reconciliation	2015 BI post; Nakadi	Yes, at the Omnichannel stage, without the service count
An event stream must not be the only source of current state	2025 Product Read API post	Yes; a read API on the system of record, events for propagation
Autonomy needs alignment, a bounded technology set and one paved road	2018, 2020, 2023 posts	At the Group stage; a small team gets this for free from one platform
Build platform tooling only one company needs and it will be archived	STUPS, Nakadi, Zalenium status	Yes, at every stage
A retailer becomes a platform through logistics, then software; consolidation follows	2011–2026 timeline	The pool-serving-many-channels idea, yes; the business, only at scale

Kestrel & Pine takes two practices from the case. Waypoint, its improvement

program, requires integration owners to track API versions and retirement notices under **W5**. That addresses the February 2027 script failure that left marketplace stock stale for two days.

Under **W1**, stock events feed an availability service that answers current availability through an API, with periodic reconciliation. That avoids reproducing the Black Friday chain of delayed copies. Neither lesson requires copying Zalando's thousands of applications or large engineering organization into a fourteen-person team.

Where the Evidence Stops

- **The current architecture**. Zalando has not published its service inventory, its order or pricing architecture, the internals of ZEOS or how PRAPI and its event streams coexist today. The 2015–2017 posts describe systems that have been replaced.
- "**Radical Agility**" **ended** nowhere in the public record. No post announces it, and no source describes a 2019 reorganization under that name; the return to discipline is this chapter's reading of a sequence of posts and one leadership change (Bowman left in 2019, by professional-profile sources). The widely reported "expanded too much" wording from the February 2023 announcement was not in the page fetched. The 2016 trade accounts of team size (three to seven) are secondary; Zalando's own posts give "85+" teams.
- **Open-source status** is as of 7 November 2026 and will change; the Tech Radar page shows no last-updated date.
- **Business figures** for 2025 include ABOUT YOU for part of the year; the 2026 guidance of 5–10% pro-forma growth is the comparable number. The "over 60" fulfillment merchants and the "over 600" B2B partners are read here as different scopes; Zalando has not reconciled them in public.
- **The Connected Retail wind-down** and the September 2026 fee change are reported by trade press from Zalando's partner communication; the primary page was not fetched.

What to Take Away

Established.

- Zalando rebuilt its shop as microservices owned by autonomous teams from 2015 because a shared codebase and slow deployments blocked its 85-plus teams, and it built an audited platform (STUPS) so each team could have its own cloud account (Zalando posts, 2015–2016; InfoQ, 2016).

- Its API guidelines, open since 2016 and maintained in 2026, require API-first design in OpenAPI, forbid breaking consumers, prefer compatible extension to versioning, and require client consent, a sunset date, headers and usage monitoring before shutdown.
- Zalando narrowed its choices over the following decade, each in its own posts: alignment over raw autonomy (2018), a bounded language set (2020), one developer portal (2023) and a read API replacing a price-and-stock event stream that lagged up to 30 minutes at Cyber Week (2025).
- Zalando became a platform through logistics: partner program (2011), ship-from-store (2018), fulfillment for partners and then for any channel (ZEOS, 2023), commerce software (SCAYLE, 2025); 32.1% of its 2025 GMV came from partners (annual report, 2026).

Depends on context.

- Team-sized modularity is right when there are enough teams to need it, roughly the Group stage; below that, one platform with named owners for each capability gives the ownership without the operating cost.
- An event bus is necessary once data is spread across systems that must be analyzed and reconciled, which for a merchant is the Omnichannel stage; it is sufficient as a source of current state only while consumers can keep up, which the 2025 post shows they cannot on the busiest week.
- The compatibility and deprecation rules apply at every stage, but in different directions: a Starting or Growing merchant demands them of vendors; an Omnichannel merchant writes them into its own integration contracts; a Group-stage platform owes them to external partners.
- Building a separate technology for one channel (as Connected Retail was) is right while the general program cannot serve it, and becomes a consolidation project once it can; plan the exit when you build it.

The author's synthesis.

- Zalando's most valuable export is not an architecture but a habit: writing down the rules for change, and publishing the corrections. I would adopt the compatibility and deprecation chapters of its guidelines as the baseline for every vendor contract a merchant signs.
- The 2025 post is the book's thesis in a platform's own words: a copy is not a system of record, and the busiest day is when the copies drift furthest. Give every fact an owner with a read API, and let events carry the news.

Questions to Ask

1. Which of our vendors publish a deprecation policy with a sunset date, consent and headers, and which retire versions on a date they choose alone?
2. Do our own scripts and connectors log Deprecation and Sunset headers, and would anyone see the warning?
3. For price and stock, which system can be asked "what is it now", and which systems only hold a copy rebuilt from events?
4. How far behind are our copies on an ordinary day, and how far on the peak hour? Has anyone measured it?
5. If we tolerate unknown fields in every payload we receive, would a vendor adding a field break anything of ours?
6. Do we have an API standard our own team and our agency follow, even a short one, and who reviews against it?
7. Which channel runs on technology built only for that channel, and what is the exit if the general program can serve it later?
8. Before an agent may act on an order, can we name the agent, the person it acts for and the capability it was granted, in words a human can read?

The two cases so far are large companies that wrote their own commerce software. The next asks what happened when vendors tried to sell that shape to everyone else: the composable movement, its promise, its costs and its 2025–2026 turn toward working beside the systems merchants already have. Continue with Case: commercetools, Shopware and the Composable Movement.

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Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

49. Case: commercetools, Shopware and the Composable Movement

Composable

IN THIS CHAPTER, YOU WILL: Test the composable promise against the dated public record of its leading vendors and its own alliance, and learn when assembling commerce from parts is right.

KEY POINTS:

- **The movement's own flagship turned to layers.** commercetools cut staff in February 2025 after missing growth targets, launched AgenticLift in January 2026 "on top of your existing commerce solution", and in July 2026 began selling its order engine and catalog as standalone modules "without replacing their existing platforms".
- **The mid-market leader went the other way.** Shopware, first among the 1,000 largest German shops in the 2026 EHI study, launched Nexus (an orchestration layer that replaces middleware) and Shopware Payments on PayPal's rails in June 2026: integration and payments pulled into the platform, not decomposed out of it.

- **The alliance rewrote its principles, and its own survey names the cost**. Between 2023 and 2026 the MACH Alliance's stated principles went from the four-letter acronym to five principles to three, with microservices and headless gone; its February 2026 report found 37% of respondents naming integration complexity as a primary concern. Composable survives as a per-capability practice around a kept core.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Giant's Boots** — composable was defined by vendors whose reference customers were global enterprises with product teams; the same architecture proposed to a €48 million brand is a copy without its reasons.
- **The Burned Bridge** — the composable pitch of 2020–2023 was a replatforming with one cutover; the vendors' 2026 products are the opposite, layers added beside a kept core.
- **The Spaghetti Junction** — every capability bought separately is a join to own; the alliance's own survey says integration complexity is its members' primary concern.

A merchant wants to replace its search service without replacing checkout and the catalog as well. Buying separate services seems to offer that freedom, but each still depends on shared product identifiers, prices and stock updates. Replacement is practical only if those connections can be changed at an acceptable cost. **Composable commerce** is the approach of assembling separate commerce services, and that cost is central to evaluating it.

In June 2020, four vendors formed the **MACH Alliance** around independently deployed services (microservices), interfaces designed for other programs (API-first), cloud operation and a separate storefront (headless). By December 2023, it had more than 100 members. This case follows dated accounts from commercetools, Shopware, Adobe and the alliance, including later services attached to existing platforms. Their changes help show when assembly solves a merchant's problem and when integration work outweighs the flexibility.

The services may cover catalog, cart, checkout, search or content and come from one or several vendors. **Headless** is a narrower choice: separating the storefront from the commerce engine through an **application programming interface (API)**. A merchant can change that separation without committing to a separate product for every commerce function.

Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says and Build, Buy or Assemble: Choosing and Changing Platforms explain the design decisions; Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026 maps the product categories. Here we examine what vendors said and sold over time.

Where we are in the order's journey. Eva buys a jacket from Kestrel & Pine, the fictional outdoor brand. Its storefront, cart, checkout and order record can come from one product or several. The composable decision determines who must connect them and operate those connections (Follow the Order: What Happens Between "Buy" and "Delivered").

What Composable Promised, 2020–2023

EPAM's **24 June 2020** announcement identifies the founding members as commercetools, Contentstack, EPAM and Valtech. It spells out MACH as **Microservices**, **API-first**, **Cloud-Native and Headless**. This is historical evidence of the original proposition; later definitions should be read against their own dates.

The original proposition combined four choices: small independently deployed services, capabilities exposed through documented APIs, vendor-operated cloud software, and a separately built or purchased storefront. Merchants were promised easier component replacement and less dependence on a suite's release cycle.

The alliance's **13 December 2023** announcement reported over 100 members and cited a survey of 500 technology leaders: 85% said their MACH share had increased in the prior year. Its applicant totals were internally inconsistent. These are the alliance's own adoption claims, with definitions that must stay attached; they do not establish whether a merchant reduced costs or improved sales by adopting the approach.

What the 2020 promise says	What it does not say
Capabilities should be reachable through APIs and replaceable one at a time	Who builds and runs the storefront that no longer ships with the engine
A merchant should not be locked into one suite's release cycle	Who owns the joins between the services, and what that costs each year
The vendor runs and scales the services	That the merchant's own team must run everything the vendor does not: storefront, integration, identity, content
Best-of-breed beats one suite	For which size of merchant, with which team, the difference pays for itself

The right-hand column is the subject of the rest of this chapter. The promise was true as a description of how large enterprises with product-engineering teams could build; it said nothing about the merchant with fourteen technology staff.

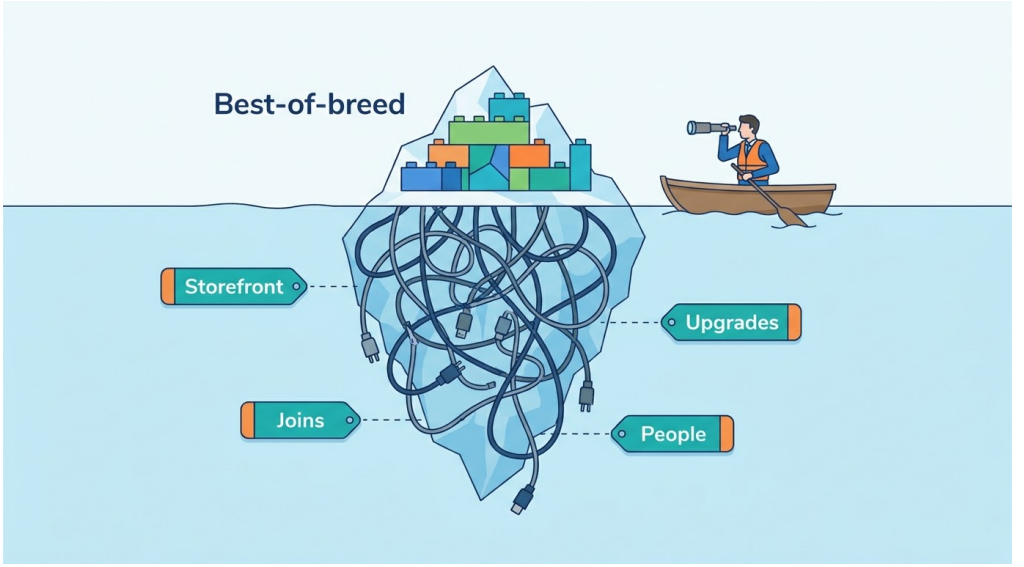

Figure 1: *The 2020 promise showed the services; the storefront, the joins, the upgrades and the people who own them sat below the waterline, assigned to nobody.*

commercetools: From Flagship to Layers

commercetools was founded in Munich in 2006, acquired by the German retail group REWE in 2015, and funded in 2019 ($145 million from Insight Partners at a $300 million valuation) and 2021 ($140 million led by Accel at $1.9 billion), as TechCrunch summarized in February 2025. It is the composable flagship: founding member of the alliance, a Leader in Gartner's Magic Quadrant for Digital Commerce every year since 2020, and, as of its 22 April 2026 release, past a "$100B GMV run-rate" with more than 600 million orders a year across 36 countries. (**GMV**, gross merchandise volume, is the value of orders through the platform, not the vendor's revenue; a **run-rate** annualizes a recent period and is not a full-year figure.)

The turn is in its releases.

Date	Event (source)	What changed
10 July 2024	Andrew Burton, formerly of Rapid7, became CEO; founder Dirk Hoerig moved to Chief Innovation Officer; the release cited 2023 GMV of over $30 billion, up 45% (commercetools release)	Operator management for a growth-stage vendor
26 February 2025	About 10% of staff cut; CEO Burton's memo said the company had not met its "aggressive revenue growth targets" over several quarters, targets that "were never reset for macroeconomic uncertainty"; the CRO and CFO left the executive team (TechCrunch)	The pure composable pitch no longer carried the growth plan
21 January 2026	AgenticLift launched as the "first standalone agentic offering" (**agentic commerce** is the 2025–2026 term for selling through AI assistants that browse catalogs and buy on a shopper's behalf), letting companies "layer agentic commerce capabilities on top of your existing commerce solution", including homegrown or legacy systems (commercetools release)	A product for merchants who keep a non-composable core

Date	Event (source)	What changed
22 April 2026	Q1 2026 results signed by a new CEO, Doug McNary: $100B run-rate, more than $7 billion of GMV from new enterprise customers in the quarter, same-store sales up 27% (commercetools release)	Growth claimed on new logos and same-store volume
9 June 2026	"Autonomous Commerce" announced as "the new category for the AI era", with Sphere, "an enterprise commerce platform", and a preview of MosAIc, an operations tool (commercetools release)	The vocabulary moves from composable to autonomous
15 July 2026	Core Commerce ("enterprise-grade cart and order management") and Product Catalog sold as standalone modules that integrate "with their existing technology stack", so enterprises can adopt capabilities "without replacing their existing platforms" (commercetools release)	The flagship sells itself by the piece, beside a kept backend
22 October 2026	A Leader in Gartner's 2026 Magic Quadrant for Digital Commerce Platforms (report dated 19 October 2026) "for the seventh consecutive year" (commercetools release)	Analyst standing intact through the turn

Three things stand out. First, **the vendor's own products now describe the architecture this book recommends for Omnichannel-stage merchants**: a kept core with layers added beside it. AgenticLift and the July 2026 modules are built for stacks that are not composable.

Second, **the sales vocabulary moved on**. The June 2026 release does not sell "compos-

able"; it sells "Autonomous Commerce", a category the vendor says it is introducing. A newcomer should notice that the category changed while the product's shape (API services the merchant assembles or attaches) did not.

Third, **the numbers need their units**. The April 2026 release says "500+ enterprises"; the July 2026 boilerplate says "more than 600 global brands and enterprises", "over $110 billion in annualized GMV" and "over 700 million orders". The April figures were $100 billion and 600 million orders. Either the business grew about 10% in three months or the units differ; the releases do not say which, and neither should you.

> **INFERENCE, NOT DOCUMENTED:** No commercetools release explains why it moved to standalone modules. The sequence (missed growth targets in 2025, then layers and modules in 2026) is consistent with a vendor finding that full replatforming onto an API-only core sells to fewer merchants than its 2020–2023 pitch assumed, and that Commerce.com's August 2026 remark about B2C replatforming being "softer than historically" describes the same market. That reading is mine.

The CEO sequence also has a gap in the public record: founder Dirk Hoerig to Andrew Burton (July 2024) is documented; Doug McNary is the CEO quoted from April 2026; the handover between them was not found in any release fetched for this book.

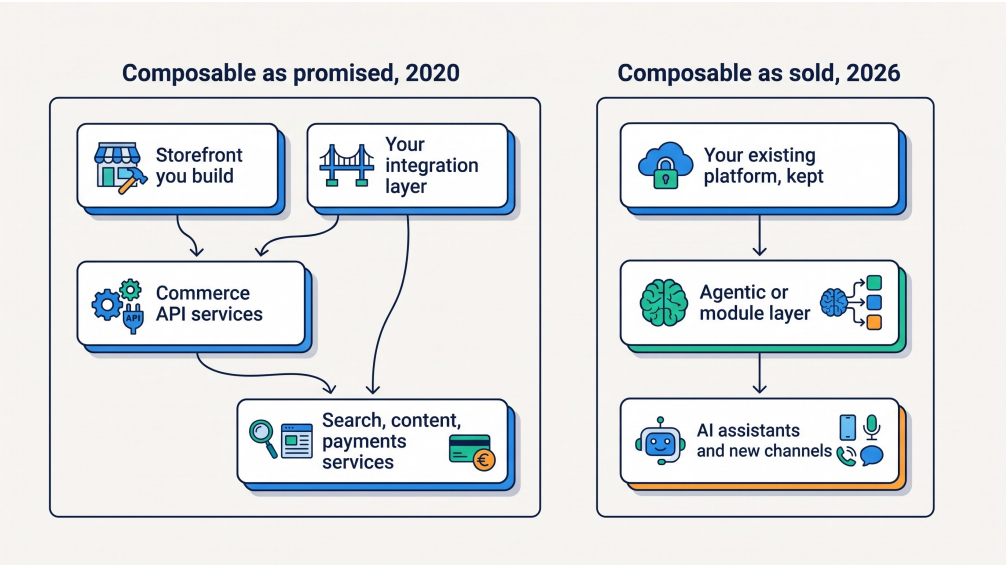

Figure 2: *The 2020 promise asked the merchant to build the storefront and own every join; the 2026 products from the same vendors attach a layer to a platform the merchant keeps.*

Shopware: The Mid-Market Path

Shopware, from Schöppingen in Germany, is the counter-case: an open-source platform whose answer to the composable movement was to pull more into the platform, not less.

Its finances and ownership are documented by its own releases. It was "100% bootstrapped" until February 2022, when Carlyle and PayPal invested USD 100 million, with platform GMV approaching USD 20 billion in 2021 (that release is more than three years old). In late 2025 PayPal raised its stake "from about 11% to more than 40%" by taking over Carlyle's shares, as Shopware's July 2026 boilerplate states and the trade press reported on 23 October 2025 alongside a claim of "more than 50,000 online sellers". On 11 August 2026 Shopware announced that Joachim Weber would become Co-CEO on 17 August and sole CEO on 1 January 2027, with founder Sebastian Hamann becoming Chief Experience Officer and founder Stefan Hamann "Chief Agentic Officer"; the founders keep a majority of shares.

Its market position is specific. The EHI Retail Institute's 2026 report on the 1,000 largest German B2C shops found 12.5% on Shopware, first place for the fifth year (a platform was identified for 60.2% of those shops). W3Techs, which counts websites worldwide, puts Shopware at 0.2% of sites with a known e-commerce technology; both are true, and the difference is the population counted. Analysts placed it as a Visionary in Gartner's 2025 and 2026 Magic Quadrants for Digital Commerce and a Strong Performer, "best fit for B2B organizations" (B2B: business-to-business, selling to other companies), in Forrester's Commerce Solutions Wave of Q3 2026.

The pricing is the first clue to its path. The Community Edition is free; the paid plans (Rise "from €600 a month", Evolve "from €2,400 a month", Beyond on request, November 2026) are "based on your gross merchandise value", and a fair-usage policy requires a paid plan once the shops linked to one account exceed €1 million of GMV in a rolling year. An open-source platform priced on GMV is a platform that wants to sit in the path of every order, which is the opposite of a service you can swap.

The second clue came on 10 June 2026 at Shopware Community Day in Cologne. Shopware announced **Nexus**, "a native, event-driven orchestration layer" with reusable connectors and AI-assisted mapping, which it positions as replacing middleware, and **Shopware Payments**, powered by PayPal's infrastructure, live in Germany and Austria with the EU and US to follow.

The release cited vendor figures that merchants average more than 15 disconnected tools and spend up to 52% of implementation effort wiring them together, and claimed Nexus cuts integration costs by 40%; those are vendor claims without a published method. Nexus is metered: €10 per extra 1,000 executions above the plan's quota (pricing page, November 2026).

In plain words, the **iPaaS** (integration platform as a service: the hosted tool that moves data between systems) that a composable merchant buys separately became a Shopware

product, and so did the **PSP** (payment service provider: the company that takes the shopper's payment and settles the money).

> **INFERENCE, NOT DOCUMENTED:** Shopware has not said that Nexus and Payments are a response to the composable movement. Read together with the GMV-based pricing and PayPal's stake, they describe a platform that earns on orders and payments and wants the integration layer inside its own metering. That is the same economic shape as a hosted SaaS platform (Case: Shopify, the Modular Monolith That Serves Millions of Merchants), reached from the open-source side; the inference is mine.

What Shopware shows a newcomer is that **"open source" and "composable" are different axes**. Shopware's core is open; its commercial path is to be the hub. A merchant on Shopware at the Omnichannel stage can still buy an OMS or a PIM beside it, but the platform's own direction is to absorb the joins, and the fair-usage line means the hub is paid for once the merchant grows.

Adobe, the Forks and Commerce.com: Composable as a Deployment Option

Two more vendors show the word being absorbed rather than adopted.

Adobe Commerce (the former Magento Commerce, renamed in April 2021 after Adobe's 2018 acquisition of Magento for $1.68 billion) now lists three deployment options on its information center (updated 21 October 2026): on-premises, "on cloud infrastructure" (platform-as-a-service: Adobe runs the hosting, the merchant owns custom code) and "as a Cloud Service" (fully managed SaaS with continuous updates and a headless storefront that Adobe hosts). Beside them, **Adobe Commerce Optimizer** is a SaaS for storefront, catalog and merchandising that "can work alongside an existing Adobe Commerce or third-party commerce backend"; the two SaaS products "don't currently integrate" with each other. Partner reports place both launches in June 2025, after a March 2025 announcement; no Adobe press release was found. Here "composable" has become a layer a suite vendor sells beside any backend, including a competitor's.

The forks tell the open-source side. Community leaders organized in 2021 "concerned about the future of Magento Open Source"; the **Mage-OS** Association was incorporated in Poland in 2022 with a founding board of eight directors and describes its distribution as "independent, community-driven" and "highly compatible" with Magento 2 extensions. Mage-OS 4.0.0 shipped on 30 October 2026, based on Magento 2.4.9; 3.6.0, on 14 October 2026, was a security port of Adobe's October patch. The association "has no shareholders or investors" and lists 216-plus financial contributors and 82-plus professional members

(site accessed November 2026). A merchant on Mage-OS has the most freedom of any in this chapter and the longest patch chain: Adobe, then the distribution, then the agency, then the merchant.

Commerce.com is the SaaS side of the same story. BigCommerce Holdings renamed itself Commerce.com on 31 July 2025 and, in Digital Commerce 360's account, described the change as moving "away from an all-in-one platform toward an open, composable system that connects to external AI tools"; its CEO said "this is not a cosmetic change". The engine, BigCommerce, stayed. Its results for the quarter ended 30 June 2026, filed with the SEC on 6 August 2026, showed revenue of $84.5 million, up 0.1%, subscription revenue down 1%, and management saying B2C (business-to-consumer) replatforming "remains softer than historically". A vendor adopted the composable vocabulary and, a year later, reported that the market for moving platforms was weak.

The Alliance Rewrites Its Principles

The clearest evidence of the turn is the alliance's own wording, which changed three times.

Date	The principles as stated (source)
June 2020 and December 2023	"Microservices, API-first, Cloud-Native, and Headless" (EPAM, 2020-06-24); "Microservices based, API-first, Cloud-native SaaS and Headless" (MACH Alliance, 2023-12-13)
5 August 2025	Five guiding principles, Composable, Connected, Incremental, Open and Autonomous, announced with a move to "end-user leadership" by September 2026 (an appointed end-user president, two end-user ambassadors on the board); the acronym is not restated (MACH Alliance via Silicon UK, 2025-08-05)

Date	The principles as stated (source)
November 2026	Three principles: "Open is about what the system is made of — and whether humans and agents can see it, trust it, and verify it"; "Composable means capabilities you can assemble, replace, and evolve independently"; "Connected means designed to be optimized in an ecosystem — not in isolation"; the alliance "certifies ISVs, system integrators, and enablers against these principles" (machalliance.org/mach-principles, accessed 2026-11)

Microservices and headless, two of the four letters, are no longer principles. "Incremental", which appeared in August 2025, is the word that most contradicts the 2020–2023 pitch, because incremental adoption around a kept core is what the vendors now sell. The 2026 page gives no reason for the change. My reading, labeled as such: **the acronym outlived its expansion**. The parts of the promise that survive (open, composable, connected) are properties a merchant can ask of any platform, including a hosted SaaS one with good APIs; the parts that were dropped were prescriptions about how the vendor's own software should be built, which were never a merchant's concern.

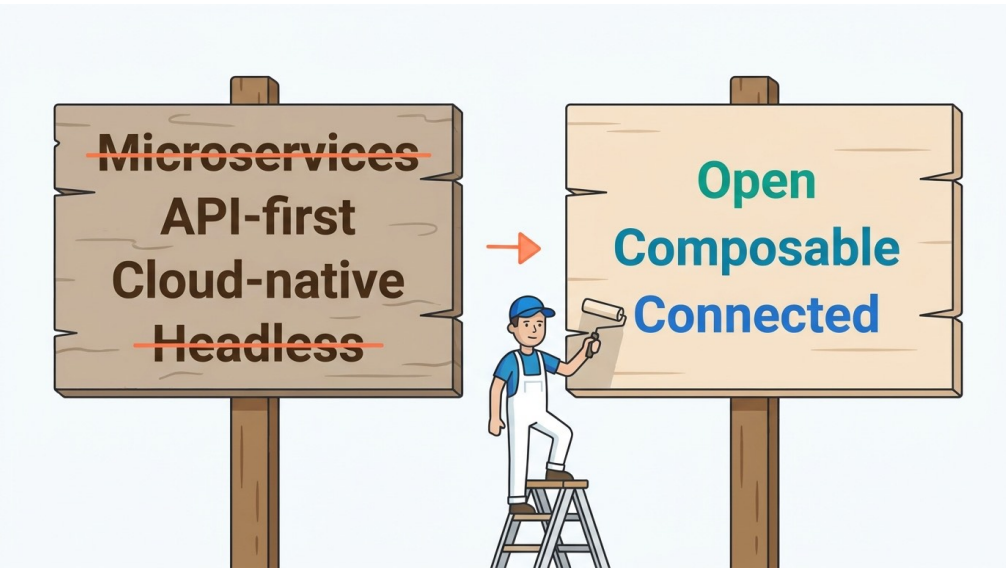

Figure 3: *Between 2020 and November 2026 the alliance went from four words to five principles to three: microservices and headless were dropped, and open, composable and*

connected remained.

What the Movement's Own Data Say

The MACH Alliance's Enterprise Technology Report, published 18 February 2026 from a survey of 600 enterprise technology decision-makers in seven markets, reports that 78% of organizations with fully scaled MACH technology see clear return on investment from AI versus 13% in early planning, that 98% of mature composable adopters can support AI at scale versus 33% of early-stage ones, and that 89% say standards and certifications for AI in composable environments are missing. It also reports that **37% cite integration complexity as a primary concern** and 45% flag data privacy and security.

Read it as what it is: a survey commissioned by a vendor alliance, answered by a population that includes its own adopters, with "clear ROI" undefined. The usable finding is the one against interest. Among the people most committed to composable, more than a third name integration complexity as a primary concern. That is the cost line the 2020 promise left out, and the reason the "Spaghetti Junction" pattern is the one most composable merchants meet first.

Shopware's June 2026 figures (more than 15 disconnected tools per merchant; up to 52% of implementation effort spent wiring them) point the same way and carry the same caveat: a vendor selling an integration product has reasons to count integration cost high. No independent, published study with a stated method was found that measures the total cost of a composable assembly against a hosted platform with layers; where this chapter gives costs, they are the author's judgment or Kestrel & Pine illustrations.

What Composable Actually Costs

A composable assembly has the same bill as any platform, plus lines that the brochure assigns to nobody. The table is my synthesis; the amounts are not from any vendor.

Cost line	In a hosted platform with layers	In a composable assembly	Who pays it
The storefront	Included: a theme, customized by an agency	Built or bought separately, then kept current with every framework and browser change	Your storefront team or agency, every year

Cost line	In a hosted platform with layers	In a composable assembly	Who pays it
The joins	Apps and connectors, each with an owner (34 at Kestrel & Pine)	Every pair of services, plus the integration layer that runs them; the alliance's 37%	Your integration team, every day
Upgrades	The vendor upgrades the core for everyone; you upgrade your apps	The vendors upgrade their services; you upgrade everything you built against them, including the storefront	You, on the vendors' schedules
Identity, content, search	Often in the platform or one app each	Separate products, each with a contract, an API version and a renewal date	Your vendor manager
Licenses	Subscription plus a share of sales	Subscription with GMV caps per service (Saleor Cloud: 0.8% above $200,000 a month; commercetools: not public)	The merchant, with more lines to add up
People	Integration engineers and app owners	Product engineers, platform engineers, integration engineers, and someone to own each service	Headcount that must exist before the first sale

Cost line	In a hosted platform with layers	In a composable assembly	Who pays it
Exit	Export data, rebuild theme and apps	Every join is yours to rebuild; the storefront moves with you, which is the real benefit	Paid once, if ever

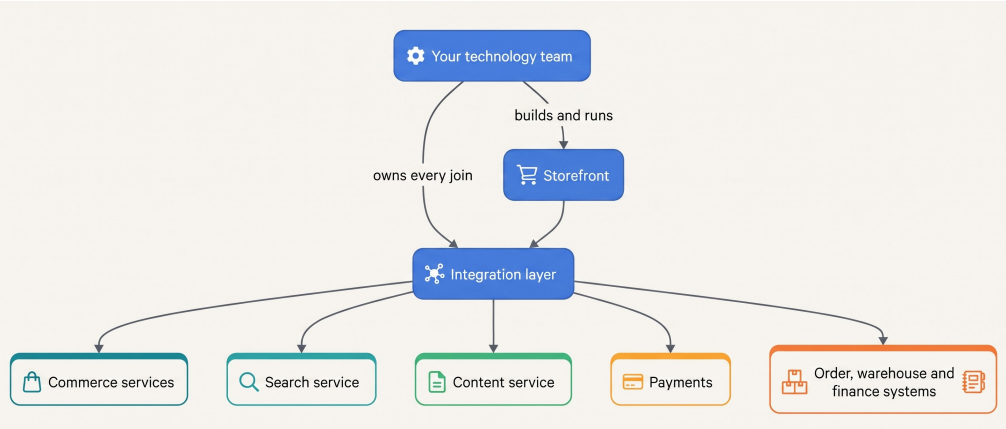

Figure 4: *In a composable assembly the merchant's own team sits at the center, owning the storefront and every join; nothing in the diagram is run by a vendor except the boxes at the edge.*

The storefront line deserves a worked illustration. Kestrel & Pine, the fictional outdoor brand this book follows (Meet Kestrel & Pine), has three storefront and app engineers out of fourteen technology staff and pays an agency €14,000 a month for its theme, apps and small integrations. A built storefront for three countries, with search, content and checkout integrations, is a product in its own right; assume (an illustration, not a ledger fact) that it needs at least the three engineers full time plus the agency, before any of the 34 existing joins are rebuilt against new APIs. The assembly would not have fewer joins than today; it would have more, and all of them would be Kestrel & Pine's. That is what the fourth column of the table means.

When Composable Is Right

The record does not say composable is wrong. It says it was oversold to the wrong merchants. The conditions under which it is right are specific.

Situation	Composable fits when	It limits when
Scale stage	Group: several brands or countries with a platform team and product teams per journey	Starting and Growing: there is no team to own the joins; Omnichannel: only per capability, around a kept core
Differentiation	The storefront or the checkout is where the business competes, and a built one earns its keep	The storefront is a channel like any other and a theme would do
Team shape	Product-engineering organization: people who build and run services and storefronts	Integration organization: people who own connectors and contracts (Kestrel & Pine's shape)
Vendor count	One owner per service and a vendor manager; contracts with versioned APIs and dated deprecations	Each new service adds a renewal, an API version and an owner nobody has named
Reason to move	A capability the current platform cannot own (a global catalog, an OMS, a B2B engine) is adopted as a service beside it	The reason is an incident that an availability service or an OMS would fix without replatforming

The per-capability path is the one the vendors themselves now sell and the one an Omnichannel merchant can absorb: keep the platform, add a PIM, an OMS, a search service or an agentic layer as separate services with their own owners, and treat each as a composable step. Kestrel & Pine's Waypoint program is that path under another name. The full assembly is a Group-stage choice, and even there the evidence from commercetools' July 2026 modules is that enterprises want to adopt it "without replacing their existing platforms".

What Kestrel & Pine Takes From It

Kestrel & Pine's February 2027 consultancy proposal asked fourteen technology staff to operate a separate engine, storefront, search, content system and integration layer. Its

illustrative three-year cost was about **€3.2 million** (Build, Buy or Assemble: Choosing and Changing Platforms).

Waypoint, the improvement program, keeps the hosted platform while buying **order management (OMS)**, availability and **product information management (PIM)** beside it. The warehouse remains authoritative for physical stock, and each integration gets an owner and contract. That selective assembly resembles the later vendor offerings described here. The company will revisit its needs at the 2029 review.

Where the Evidence Stops

- **commercetools' reasons**. No release explains the turn to layers and modules; the link between the February 2025 cuts and the 2026 product line is my inference. The CEO handover from Andrew Burton to Doug McNary is undated in the fetched record, and the April and July 2026 figures ("500+" versus "more than 600"; $100 billion versus $110 billion) are not reconciled by the vendor.
- **Shopware's strategy**. Shopware has not stated that Nexus and Payments respond to composable, nor published a method behind its "40% lower integration costs" claim; its current merchant count ("more than 50,000" in 2025 press) is not on a Shopware page fetched here; PayPal's exact stake is reported as "more than 40%" or "about 41%".
- **Adobe's dates**. The general-availability date of Adobe Commerce as a Cloud Service and Commerce Optimizer rests on partner reports (June 2025); no Adobe press release was found.
- **The alliance's reasons**. The MACH Alliance's principles page gives no reason for dropping microservices and headless; the 2023 release's applicant figures are internally inconsistent; the 2026 survey's method and sample are the alliance's.
- **Costs**. No independent study with a stated method compares the total cost of a composable assembly with a hosted platform plus layers; every cost figure in this chapter is a labeled judgment or a Kestrel & Pine illustration.
- **Analyst placements**. Gartner's 2026 vendor count was not in the fetched release; the placements of Shopify, Salesforce, SAP and Adobe in the 2025–2026 reports were not found.

What to Take Away

Established.

- The composable promise was defined by vendors in June 2020 as microservices, API-first, cloud-native and headless (EPAM release), and by November 2026 the alliance's principles no longer include microservices or headless (machalliance.org, accessed 2026-11).

- The flagship vendor now sells layers and modules for existing stacks: AgenticLift (21 January 2026) and standalone Core Commerce and Product Catalog "without replacing their existing platforms" (15 July 2026), after a February 2025 staff cut tied to missed growth targets (TechCrunch).
- Shopware, first among the 1,000 largest German B2C shops (EHI 2026), moved integration and payments into the platform with Nexus and Shopware Payments (10 June 2026) and prices its plans on GMV (pricing page, November 2026).
- The alliance's own February 2026 survey found 37% of respondents naming integration complexity as a primary concern.

Depends on context.

- A full composable assembly is right for a Group-stage merchant with product and platform teams and a storefront that is itself a competitive product; it is the Giant's Boots for a merchant below that stage or with an integration-shaped team.
- Adopting one capability as a separate service (PIM, OMS, search, an agentic layer) beside a kept platform is right at the Omnichannel stage whenever the platform cannot own that capability well; it limits when the merchant cannot name an owner for the new join.
- Open source and composable are independent choices: Shopware is open and hub-shaped; commercetools is proprietary and service-shaped. Choose on who owns upgrades and joins, not on the license.

The author's synthesis.

- The composable movement was right about properties (open APIs, replaceable capabilities, clear ownership) and wrong about prescriptions (rebuild as services, build your own storefront), because the prescriptions were the vendors' reference architecture, not the merchant's need.
- The 2026 products of commercetools, Adobe and Salesforce are the movement's own correction: the kept core with layers beside it is composable as it should have been sold, and it is the shape most merchants should buy.

Questions to Ask

1. When a vendor or agency says "composable", which of the four 2020 words do they mean, and which of the alliance's three 2026 principles?
2. Who will build and run the storefront that an API-only engine does not ship, and what will that cost each year after launch?
3. For each service in the proposed assembly, who owns the join to the others, its API version and its renewal date?

4. Which capability does our current platform fail to own well, and can it be adopted as one service beside the platform rather than by replacing the platform?
5. Which of the vendor's reference customers is within a factor of ten of our revenue and team size?
6. What does the vendor's own 2026 product line sell to merchants who keep their existing platform, and why would we not buy that instead?
7. What unit, date and definition stand behind every GMV, customer and order figure in the deck, and are the figures in two releases consistent?

The four cases of Part XI end here: a modular monolith that serves millions of merchants, a retailer that became a platform, a microservices pioneer that returned to discipline, and a movement that corrected itself. The reference pages begin with Domain Coverage: From Three Model Families to Forty-Nine Chapters, which maps the research models behind this book to the chapters that cover them.

Sources and Evidence

Company, product and investor sources

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- commercetools, a pioneer in headless commerce, lays off dozens of staff, TechCrunch, 2025 — the February 2025 cut, the CEO memo, the funding history.
- commercetools launches AgenticLift, commercetools, 2026 — the "on top of your existing commerce solution" layer.
- commercetools reports strong Q1 2026, commercetools, 2026 — the $100B run-rate, orders, countries, same-store sales, the CEO.
- commercetools introduces "Autonomous Commerce" category and unveils Sphere, commercetools, 2026 — the June 2026 category and platform naming.
- commercetools launches modular commerce offerings, commercetools, 2026 — standalone Core Commerce and Product Catalog, "without replacing their existing platforms", the boilerplate figures.
- commercetools named a Leader in the 2026 Gartner Magic Quadrant for Digital Commerce Platforms, commercetools, 2026 — the seventh consecutive placement, report dated 19 October 2026.
- Shopware raises USD 100 million from Carlyle and PayPal, Shopware, 2022 — the bootstrapped history and 2021 GMV (more than three years old).
- PayPal increases stake in Shopware, Ecommerce News Europe, 2025 — the stake rising to about 41% and the merchant count.
- Shopware named a Strong Performer in Forrester's 2026 commerce evaluation, Shopware, 2026 — the Forrester placement and the boilerplate on PayPal's stake.

- Shopware Community Day 2026, Shopware, 2026 — Nexus, Shopware Payments, the integration-cost claims.
- Shopware launches Shopware Payments in partnership with PayPal, Shopware, 2026 — transaction pricing and onboarding through PayPal.
- New management structure at shopware AG, Shopware, 2026 — the CEO succession and founder roles.
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- Shopware a Visionary in the 2025 Gartner Magic Quadrant, Shopware, 2025 — report date and 19-vendor count.
- Shopware pricing, Shopware, n.d., accessed 2026-11 — GMV-based plans, the fair-usage threshold, Nexus metering.
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- Magento, Wikipedia, 2026 — the 2018 acquisition and the 2021 rename.
- Mage-OS home, about and founding announcement, Mage-OS Association, 2022–2026 — the distribution, the association, releases 3.6.0 and 4.0.0.
- BigCommerce rebrands as Commerce.com, Digital Commerce 360, 2025 — the rename and the "open, composable system" framing.
- Commerce.com Q2 2026 results, exhibit 99.1, Commerce.com via SEC EDGAR, 2026 — revenue growth of 0.1% and the "softer than historically" remark.
- Saleor pricing, Saleor, n.d., accessed 2026-11 — GMV-capped subscription for an API-only platform.

Standards, regulation and official documentation

- EPAM joins the newly formed MACH Alliance as a founding member, EPAM Systems, 2020 — the founding members and the acronym's expansion (more than three years old; the founding document).
- MACH Alliance surpasses 100 members, MACH Alliance, 2023 — the 2023 formulation, the survey and the applicant figures.
- MACH Alliance advances end-user leadership model, MACH Alliance via Silicon UK, 2025 — the five principles and the governance change.
- MACH principles and MACH explained, MACH Alliance, n.d., accessed 2026-11 — the three principles as stated in 2026.
- MACH Alliance Enterprise Technology Report, MACH Alliance, 2026 — the survey figures, including the 37% integration-complexity finding.

Research and practitioner sources

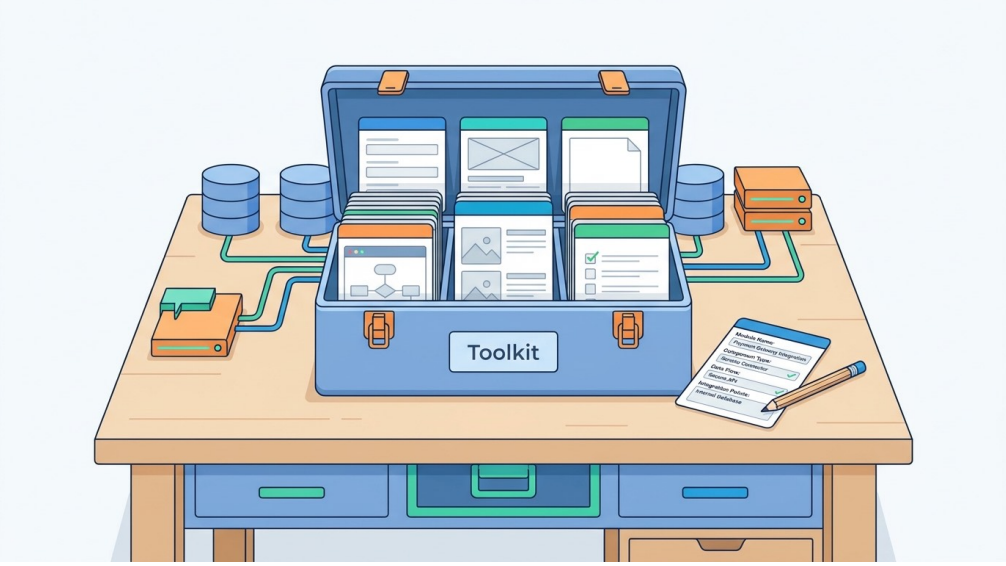
- Usage statistics of e-commerce technologies, W3Techs, 2026 — Shopware's share of websites, contrasted with the EHI population.
- Kibo named a Leader in The Forrester Wave: Commerce Solutions, Q3 2026, Kibo Commerce, 2026 — the ten-vendor count of the wave in which Shopware is a Strong Performer.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.

Reference

The rest of this reference is online: Domain Coverage: From Three Model Families to Forty-Nine Chapters, Glossary: Plain-Language Definitions and Bibliography: Every Source the Book Cites.

Toolkit: Cards and Templates for Commerce Architecture Decisions



IN THIS SECTION, YOU WILL: Find the card that fits the question on your desk, copy its fields, and see it filled in once for Kestrel & Pine, with the chapter that explains every field.

KEY POINTS:

- **Every card names an owner, a system of record and a date.** Those three fields are what turn a diagram into a decision: who is called when it breaks, which system wins when two disagree, and when the card must be looked at again.
- **The cards point at each other.** The capability map says which boxes exist, the entity–owner table says who owns each fact, the integration inventory lists every flow that moves a fact, and each flow has a contract card and a reconciliation card. A decision record changes a row in one of them; nothing else should.
- **Most fields were paid for by an incident.** The writer column came from the App That Wrote Stock, the sunset date from the Deprecated

> Endpoint, the pallet-label source from a shipment-notice mismatch, the summary rule from the Three-Status Order. Write "unknown" rather than guess.

A design discussion can end with agreement about a new system while leaving its data owner or recovery process undecided. These sixteen cards make those gaps visible by recording the business fact, responsible owner, systems involved and expected result. They range from a capability map to a migration plan and decision record. Each includes instructions to start; its linked chapter supplies the reasoning and sources.

Worked examples use **Kestrel & Pine, a fictional outdoor brand** (Meet Kestrel & Pine). Replace its names, dates and figures with your own. Keep the fields that expose a decision or responsibility; adapt the rest to your setting.

Three conventions run through all sixteen cards.

- **Keep cards with the work they describe**. Store them in a shared repository where the people changing the systems can find and update them; a detached slide deck is easy to leave behind.
- **Link related cards by identifier**. A decision record names the ownership rows it changes, and a contract card names its integration inventory row, so a reviewer can trace the effects of a change.
- **Investigate consequential blanks**. A missing owner leaves nobody responsible for recovery; a missing retirement date can leave an obsolete integration running. Use the gap to assign the next piece of work.

Find the Template for the Question on Your Desk

The question in front of you	Tool	Chapter that teaches it
What must the business be able to do, and which system does each job?	Tool 1: Capability map	The Map of a Commerce Platform: Capabilities and Layers
Two systems disagree about a price, a stock level or an order. Which one is right?	Tool 2: Entity–owner table	Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record

The question in front of you	Tool	Chapter that teaches it
How many flows connect our systems, and who owns each one?	Tool 3: Integration inventory	APIs, Events, Files and Middleware: How Commerce Systems Talk
A flow is about to go live, or a vendor has announced a new API version	Tool 4: Integration contract card	Promises Between Systems: API Contracts, Webhooks and Versioning
How do we know a flow actually worked?	Tool 5: Reconciliation card	Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation
What may each channel sell, and where does that number come from?	Tool 6: Availability (ATP) card	Is It in Stock? Inventory, Availability and the Promise You Can Keep
Support sees three different statuses for one order	Tool 7: Order state card	The Order Record: Lifecycle, State and the Order Management System
What delivery date may the checkout promise?	Tool 8: Delivery promise card	Getting It There: Carriers, Labels, Tracking and the Delivery Promise
How much stock should each marketplace be told about?	Tool 9: Channel allocation card	Selling on Someone Else's Platform: Marketplaces and Channel Integration
A trading partner orders by EDI or deducts money for a bad despatch advice	Tool 10: EDI partner card	Documents Between Companies: EDI, Punchout and E-Invoicing
Someone wants to install an app, or an app writes data it should not	Tool 11: App governance card	Apps, Plugins and Functions: Extensibility and Ecosystem Governance
Black Friday is coming	Tool 12: Peak plan and degradation ladder	Black Friday Every Year: Peak Load, Resilience and Recovery

The question in front of you	Tool	Chapter that teaches it
A law changes what a field, an invoice or a checkout must do	Tool 13: Regulatory calendar	Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation
Should we build, buy or assemble a capability, or replatform?	Tool 14: Build, buy or assemble decision card	Build, Buy or Assemble: Choosing and Changing Platforms
A system of record is moving from one system to another	Tool 15: Migration plan card	Moving Without Burning the Bridge: Replatforming and Migration
A decision changes who owns a fact, which vendor we depend on or which app may write	Tool 16: Architecture decision record	Who Builds and Who Owns: Teams, Vendors, Agencies and Governance

Two fuller records live in their chapters: the system and vendor ownership map in Who Builds and Who Owns: Teams, Vendors, Agencies and Governance, and shared order, revenue and return definitions in Counting the Same Thing Everywhere: Data Platforms, Definitions and Analytics.

How the Records Connect

Most changes touch several cards; they meet at the entity–owner table.

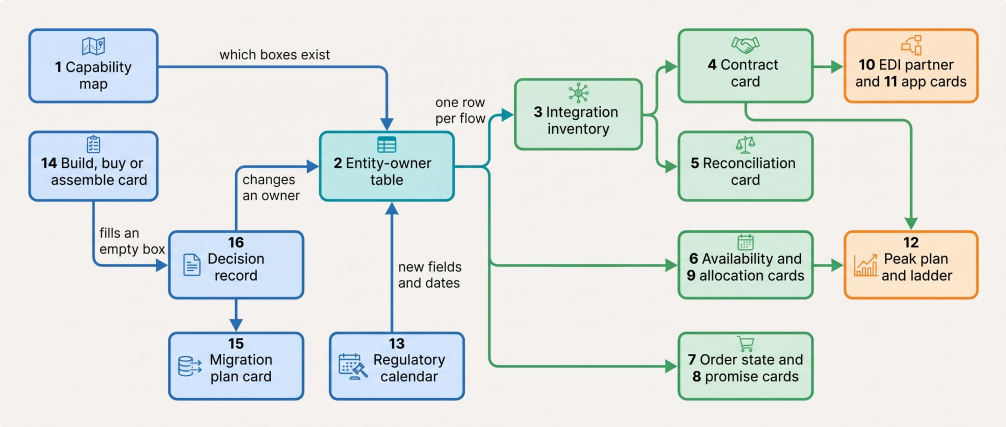


Figure 1: *The capability map says which boxes exist, decision records change who owns each fact, and every flow, partner, app, peak plan and migration is written against the entity–owner table, so a change to one row tells you which other cards to update.*

In plain words: start at the entity–owner table. A fact with no row, or a row with no owner, is the problem.

Tool 1: Capability Map

Purpose. List the jobs the business must perform, independently of today's systems. Then place each system against those jobs. Empty boxes expose missing capability; overlapping boxes prompt a decision about responsibility.

The template. One row per capability, grouped in six layers: experience, offer, transaction, operations, partners and back office, foundations.

Field	What to record
Layer and capability	A business ability as a verb phrase ("hold every order in one state model"), never a system, team or vendor
The fact it is the natural home for	The business fact this capability should own; mark computed facts such as availability and the delivery promise
Systems today	Every system that does any part of the job; two or more is an overlap, none is an empty box

Field	What to record
Writers	Every system or person that changes the fact
Status and evidence	Covered, overlap or empty, with the incident or number behind it
Owner	One named person accountable for the capability

How to use it. Draw it before reading any vendor's diagram; review it yearly and after any new channel or country.

Kestrel & Pine example (fictional; Noor Visser's first Waypoint working document, February 2027):

Field	Inventory and availability
Fact it is the home for	Available-to-promise per channel, derived from physical stock
Systems today	Stock on hand in the WMS, the ERP, the platform and the POS; **nobody computes availability**
Writers	Six writers of Duiven stock: the WMS feed, manual ERP entry, manual platform entry and three apps; the POS writes store stock separately
Status and evidence	**Empty box** for availability, overlap for stock; the Black Friday Oversell, 1,340 units oversold on 27 November 2026
Owner	Daniel Okafor for stock; Ruben Alves for the availability service once W1 delivers it

The whole map had about two dozen capabilities with thirteen systems placed on them and four empty boxes: product information, order management, availability and customer identity.

Used in: The Map of a Commerce Platform: Capabilities and Layers, Build, Buy or Assemble: Choosing and Changing Platforms, What Changes at Each Scale: From First Store to Group, Domain Coverage: From Three Model Families to Forty-Nine Chapters.

Tool 2: Entity–Owner Table

Purpose. Name the **system of record**, the authoritative source for each business fact. Record who may change it, where copies live and how disagreements are detected. The table should let a new colleague answer which value to trust.

The template. One row per fact; stock and customers usually need more than one row.

Field	What to record
Fact	A business fact with a lifecycle: product content, consumer price, physical stock per location, order, payment, return, invoice, customer
System of record	The one system whose value wins; per location where the fact is physical
Who may write	People, scans or processes allowed to change it in that system, and only there
Copies live in	Every system that holds a copy
How copies learn	Event on change, scheduled publication or daily batch, chosen by what a stale copy costs
Disagreement found by	The reconciliation that compares record and copy (Tool 5), with its cadence

How to use it. Write it before buying any hub. Then run a **write audit** each quarter: list every credential, app scope and job that can change each fact, and compare it with the "who may write" cell.

Kestrel & Pine example (fictional; the review of early March 2027, adopted by W1–W3 and the April 2027 price decision):

Fact	System of record	Who may write	How copies learn	Disagreement found by
Physical stock, Duiven	The WMS	Scans, counts and receipts in the WMS only	Stock events (from Q4 2027)	WMS-to-platform reconciliation, weekly; daily in November
Consumer price	The platform's market price lists (NL, BE, DE)	Pricing owner, with effective dates	Price events	Daily price reconciliation across all copies, from May 2027

Fact	System of record	Who may write	How copies learn	Disagreement found by
Order	The OMS (W2, Q4 2027)	Order creation from channels; state changes from events	Order events	Nightly order reconciliation

Used in: Who Owns the Truth: Boundaries, Bounded Contexts and Systems of Record, Is It in Stock? Inventory, Availability and the Promise You Can Keep, What Does It Cost? Price Lists, Contract Prices, Currencies and Tax, What Are We Selling? Products, Variants and the Catalog, One Customer, One Product: Identity and Master Data Across Systems, The Back Office: ERP Integration and Who Is the Hub.

Tool 3: Integration Inventory

Purpose. The integration inventory is one list of every flow that moves data between two systems, with its owner, style and failure handling. It makes **integration debt** (flows whose owner, contract version, failure alert or reconciliation are unknown) visible as a list of rows rather than a feeling.

The template. One row per flow.

Column	What goes in it
Source → target, and what moves	"WMS → ERP: stock on hand per SKU" (a SKU is one sellable variant)
Style and trigger	File, shared database, direct API call or message; scheduled, on event, on request
System of record	Which side wins, from Tool 2
Owner	A named person, not a team or a vendor
Contract version and sunset date	The API version or file format pinned, and when the vendor retires it
Volume and expected rate	Per day, and per minute at peak

Column	What goes in it
Failure handling	Retries, dead-letter queue (a holding queue for messages that failed every retry), alert
Reconciliation	Card reference (Tool 5), or "none"

How to use it. The rule I would adopt anywhere: **no flow goes live without a row, and no row without an owner.** Review it quarterly; a flow whose owner has left is an incident, not a note.

Kestrel & Pine example (fictional; Noor's inventory of February 2027): 34 integrations, only 3 with a reconciliation job. One row:

Column	Value
Source → target	WMS → ERP: stock on hand per SKU
Style and trigger	Scheduled, every 15 minutes, through the iPaaS
System of record	The WMS (W1)
Owner	Ruben Alves
Failure handling	No dead-letter queue
Reconciliation	None; the first stock reconciliation ran in May 2027

Used in: APIs, Events, Files and Middleware: How Commerce Systems Talk, Who Builds and Who Owns: Teams, Vendors, Agencies and Governance, Moving Without Burning the Bridge: Replatforming and Migration, The Back Office: ERP Integration and Who Is the Hub.

Tool 4: Integration Contract Card

Purpose. A contract card is a short written agreement for one flow: what moves, in which direction, how often, which system wins, what happens on a repeat or a failure, and which version of the other side's interface it depends on. It gives every inventory row (Tool 3) a person, a version, a date, an alarm and a check.

The template.

Field	What to record
Flow and owners	Inventory row; source, target, what moves; business and technical owner
Style	Webhook, scheduled push, event, batch, synchronous call
Contract	The interface pinned (API version, message type, file format) and where its changelog is read
Sunset of the pinned version	The vendor's retirement date, and a calendar entry three months ahead
System of record	From Tool 2
Idempotency	The key that makes a repeated message harmless (order number, return id, event id)
Failure handling	Alert time, paging rule, dead-letter queue and its owner; "never exits silently"
Reconciliation	Reference to the reconciliation card (Tool 5)
Expected rate	Volume per run or per minute, and the vendor's rate limit it must fit

How to use it. Write the card before the flow goes live. **Idempotent** handling means a message received twice has the same effect as once; if the card cannot name the key, the first retry will create a duplicate.

Kestrel & Pine example (fictional; the stock push to bol, rewritten by Ruben Alves after the Deprecated Endpoint, in use from March 2027):

Field	Value
Flow	Stock levels, platform to marketplace connector, bol listings
Owner	Milan Petrov (business), Ruben Alves (technical)
Style	Scheduled API push, hourly, until the availability service's events replace it in Q4 2027 (W1)
Contract	The connector's stock API, pinned to its current release; changelog read monthly from the integrations role mailbox
Sunset	Recorded in the inventory; calendar entry three months ahead, assigned to the owner
Idempotency	Last write wins per SKU and location; each push carries its snapshot time so an older push cannot overwrite a newer one

Field	Value
Failure handling	Alert within fifteen minutes; two failed runs in a row page the owner
Reconciliation	Daily comparison of the connector's stock per SKU with the platform's
Expected rate	About 17,000 variants per run, in batches within the connector's limit

It replaced a 2021 agency script whose API version was retired on 8 February 2027, leaving bol with stale stock for two days.

Used in: Promises Between Systems: API Contracts, Webhooks and Versioning, APIs, Events, Files and Middleware: How Commerce Systems Talk, Who Builds and Who Owns: Teams, Vendors, Agencies and Governance, Discounts, Coupons and Loyalty: Promotions Without Chaos, The Way Back: Returns, Exchanges, Refunds and Reverse Logistics.

Tool 5: Reconciliation Card

Purpose. Reconciliation is a job that compares two systems' records on a schedule and reports or repairs the differences. Retries, keys and events reduce disagreement; only reconciliation proves there is none, so **no flow that creates, moves or pays should go live without a card**.

The template.

Field	What to record
What two sets are compared	The record and the copy, and the scope ("all active variants")
Matching key	SKU and location, order number, order number and amount, payout line id
Tolerance	Zero, or a stated threshold, and why
Cadence	How often, and how it changes in peak months
Who gets the exceptions	A named person or team
The fix	Which side is corrected (always the copy, from the system of record) and how the cause is hunted

How to use it. Let the system of record decide the direction of the fix: the copy is overwritten, then the writer that caused the difference is found and removed. A zero tolerance is a choice; for stock it is zero because one wrong variant at peak is one oversell.

Kestrel & Pine example (fictional; the stock reconciliation from May 2027):

Field	Value
Compared	WMS on hand minus reservations vs platform available, per active variant
Matching key	SKU and location
Tolerance	Zero per variant
Cadence	Weekly, Sunday 03:00; daily in November
Exceptions to	Ruben Alves's team and Milan Petrov
The fix	Overwrite the platform from the WMS (W1); find and remove the writer
First result	**2.3%** of active variants, about 390 of 17,000: about 240 from the three apps that wrote stock, about 90 in-flight returns, about 40 manual edits in the platform admin, about 20 unexplained

Used in: Built to Disagree and Recover: Events, Eventual Consistency and Reconciliation, Taking the Money: Payment Providers, Authorization, Capture and Refunds, Tax, Invoices and the Books: From Order to Ledger, Black Friday Every Year: Peak Load, Resilience and Recovery, Moving Without Burning the Bridge: Replatforming and Migration.

Tool 6: Availability (ATP) Card

Purpose. **Available-to-promise** (ATP) is the number a channel may still sell: on hand, minus units committed to placed orders, minus reservations, minus a deliberate buffer. The card records, per channel, who supplies each term of that sum and what happens when the computation is down.

The template. One card per channel, or one table with a column per channel.

Field	What to record
Channel	Storefront NL, BE, DE; each marketplace; B2B portal; click-and-collect per store
Locations it may sell from	Warehouse, stores, external locations
On hand from	The counting system per location (WMS, POS, marketplace report)
Committed from	The order-taking system (platform, connector, later the OMS)

Field	What to record
Buffer	Units held back per SKU for this channel, and who sets it
Publication	Event within seconds, or a schedule, and the measured lag
Checkout rule	Synchronous check and atomic commit at order creation, or not
If the computation is down	The degraded answer: cached value, wider buffer, "call the store"
Reconciliation	Tool 5 reference

How to use it. The warehouse owns on hand; the order-taking system owns committed; **nobody owns "available" by hand**, because it is a computation. If a person or an app can type a number into it, the card has a second writer.

Kestrel & Pine example (fictional; Ridgewalker shell jacket, women's M, moss, NL storefront, 6 April 2027, before W1 is live):

Field	Value
Location	Duiven
On hand	36, from the WMS
Committed	12 to Bergzicht Outdoor's EDI reorder
Buffer	1 unit for the storefronts (3 for marketplaces)
Available for NL	**23**
Publication today	The hourly chain (WMS → ERP every 15 minutes → platform hourly → connector hourly); events to the availability service from Q4 2027

The same 23 is what Eva Janssen's storefront showed on 13 April 2027.

Used in: Is It in Stock? Inventory, Availability and the Promise You Can Keep, From Cart to Confirmation: Checkout Orchestration, Stores Are Systems Too: Point of Sale and Omnichannel Fulfillment, Journey 1: Eva's Jacket, From Search to Doorstep, Journey 3: A Chain Reorders by EDI, From Order to Invoice.

Tool 7: Order State Card

Purpose. An order is not one state machine but several: the order's lifecycle, the money, the goods on each shipment, and each return. The card writes down which machines exist, where each lives, which system moves it, and how the one status a customer sees is computed from them, so that three systems cannot each publish a different truth.

The template.

Field	What to record
Order number family	One sequence across channels; the channel's own id kept as a reference
Machines	Lifecycle, money, goods (per fulfillment order), return; their states
Who moves each machine	The order system, the PSP's events, the warehouse and carrier, the returns portal and warehouse
Customer-facing summary rule	How "Shipped" or "Delivered" is computed; for a split order, the least-advanced shipment
Split rule	When to split a shipment and when to wait; the message the customer gets
Invoice trigger	Per shipment or per order; one finance decision, recorded once

How to use it. "Fulfilled" is not "delivered" unless the carrier's events are fed back, and a two-parcel order is delivered only when both parcels are.

Kestrel & Pine example (fictional; order #1052877, December 2026, as the OMS handles it under W2):

Field	Value
Order	Trailhead fleece €119 and merino socks €18, €137, paid in full
What went wrong	The fleece shipped 7 December and was delivered 8 December; the socks shipped 14 December. Platform *Delivered*, WMS *In picking*, ERP *Not invoiced*; 11 tickets
Machines under W2	Separate money, goods (per fulfillment order) and return machines in the OMS
Summary rule	Least-advanced shipment: "Partially delivered: 1 of 2 parcels"
Split rule	Two-line D2C order under €150: wait up to one working day for replenishment, then split and emit "ships in two parcels"
Invoice trigger	A "shipment to invoice" event per shipment, idempotent on the shipment id, or one invoice at completion if Finance prefers

Used in: The Order Record: Lifecycle, State and the Order Management System, Where Is My Order? Notifications, Service and the Customer Record, Tax, Invoices and the Books: From Order to Ledger, The Way Back: Returns, Exchanges, Refunds and Reverse Logistics, Journey 2: The Jacket Comes Back.

Tool 8: Delivery Promise Card

Purpose. The delivery promise is the date the checkout shows. A static sentence ("ordered before 22:00, delivered tomorrow") is a rule with no data behind it; a computed promise is an estimate from the cut-off, the backlog, the stock location and the carrier's measured transit. The card records the inputs and how the promise is checked against what actually happened.

The template. One row per lane (warehouse, carrier service, destination area).

Field	What to record
Lane	Ship-from location, carrier and service, destination area
Cut-off	The last pick wave or carrier collection that still ships today
Backlog input	The age of the oldest order not yet in the warehouse, and the threshold that moves the ship date
Transit	Working days for the lane
Margin	Extra time for the lane's late tail, from measured performance
Measured	Promised versus delivered per lane, and how often it is reviewed
Owner	The person who owns the warehouse's plan, not the theme

How to use it. Start with a transit table and the warehouse's real cut-off, and measure promised against delivered weekly before adding any model.

Kestrel & Pine example (fictional; Eva's order #1072455, Tuesday 13 April 2027, 21:14):

Field	Value
Lane	Duiven → Nijmegen, PostNL standard home delivery
Cut-off	Last pick wave 21:00; the theme's "before 22:00" dated from the Tiel 3PL era
Backlog	None that evening
Transit	One working day; reliable lane, small margin
Static promise	Wednesday 14 April, a day early
Computed promise	**Thursday 15 April**, the day the parcel arrived
Plan	First promise service May 2027, measured weekly per lane; promise API under W8 in 2028
Owner	Daniel Okafor

Used in: Getting It There: Carriers, Labels, Tracking and the Delivery Promise, Pick, Pack, Ship: Warehouse Management and Fulfillment, Journey 1: Eva's Jacket, From Search to Doorstep, Journey 4: Black Friday, Hour by Hour, Assistants and Agents: AI in Commerce and What Is Real in 2026.

Tool 9: Channel Allocation Card

Purpose. The stock figure sent to a marketplace is the quantity that marketplace may sell without asking again. **Channel allocation** is the decision of how much of your available stock each channel may promise. The card records that policy per channel, with the channel's scorecard and API calendar.

The template. One card per marketplace or sales channel.

Field	What to record
Channel and seller account	Marketplace, countries, account owner
Fulfillment model	Seller-fulfilled, marketplace-fulfilled, or both
External locations	Stock held by the marketplace's program, tracked as its own location
Allocation policy	Shared pool, fixed quantity, buffer, percentage split, or computed ATP per channel
Buffer or share	Units or percentage, and when it changes (peak, hero SKUs)
Scorecard thresholds	The channel's published cancellation, late-shipment and tracking limits

Field	What to record
Interface and sunset	API version used and its retirement date (Tool 4)

How to use it. A buffer is an honest interim while stock moves on a schedule; computed ATP per channel is the target once stock moves by events.

Kestrel & Pine example (fictional; marketplaces through the connector, 2027):

Field	Value
Channels	bol (NL/BE), Amazon.de, Zalando (DE, NL); €3.0M in 2026, owner Milan Petrov
External location	About 40% of marketplace units for Germany ship from the marketplace's fulfillment program
Policy from December 2026	The platform's figure minus a **ten-unit buffer** per variant; zero below ten
Policy from Q4 2027 (W1)	Computed ATP per channel from the availability service, buffer 3 units per SKU
Peak rule 2027	Buffer raised to 25 units for hero SKUs when stock events lag (Tool 12, rung 4)
Order path	Every marketplace order created in the OMS from 15 September 2027

Used in: Selling on Someone Else's Platform: Marketplaces and Channel Integration, Is It in Stock? Inventory, Availability and the Promise You Can Keep, Journey 4: Black Friday, Hour by Hour, What Does It Cost? Price Lists, Contract Prices, Currencies and Tax.

Tool 10: EDI Partner Card

Purpose. EDI (electronic data interchange) is the exchange of orders, despatch advices and invoices between companies' systems as structured documents. Every trading partner publishes its own guideline and enforces it with deductions, so the card holds, per partner, everything needed to keep the messages right, in the merchant's own repository rather than in one consultant's head.

The template.

Field	What to record
Partner and owners	Name, locations, annual value; business owner; internal technical owner who receives partner notices
GLNs	Global Location Numbers for the partner's legal entity and each delivery point
Transport	VAN (a value-added network run by an EDI provider), AS2, web portal, Peppol
Messages and versions	ORDERS, ORDRSP, DESADV (the despatch advice, also called ASN), INVOIC, RECADV; standard and version
Guideline version	The partner's implementation guideline in force, and its next change
Mapping version	Where the mapping lives, who maintains it, an export in your repository
Source of each document	Which system produces the order, the SSCC pallet labels and the DESADV
Compliance rules	Deduction terms, on-time-in-full rules
Open disputes	Amount, deliveries, evidence

How to use it. The one rule the card must enforce: **each document comes from the system that knows its truth.** An SSCC (Serial Shipping Container Code, the number on a pallet label) must come from the system that builds the pallet, so the DESADV must come from the pack confirmation.

Kestrel & Pine example (fictional; Bergzicht Outdoor under W6):

Field	Value
Partner	Bergzicht Outdoor: 38 shops, distribution centre in Zwolle; €2.1M a year; business owner Lea Moreau
GLNs	Recorded per entity and the Zwolle DC; the DESADV is addressed to the GLN the order came from
Transport and messages	The EDI provider's VAN; EDIFACT ORDERS, DESADV, INVOIC; Bergzicht sends no RECADV
Source of each document	Order into the ERP (the OMS from 8 December 2027); SSCC labels from the WMS at pack time; DESADV from the WMS pack confirmation
Terms	Credit limit €450,000 held in the ERP; 30-day payment

Field	Value
Open disputes	€14,200 deducted in October 2026 for nine Q3 2026 deliveries whose SSCC labels did not match the DESADV
Mapping	Built in 2025 by one external consultant; export now in Kestrel & Pine's repository

Used in: Documents Between Companies: EDI, Punchout and E-Invoicing, Selling to Businesses: Accounts, Contract Prices, Quotes and Approvals, Journey 3: A Chain Reorders by EDI, From Order to Invoice, Pick, Pack, Ship: Warehouse Management and Fulfillment.

Tool 11: App Governance Card

Purpose. An app installed in a hosted platform can read and often write the merchant's data through the scopes (permissions) it was granted. The card names an owner per app, records what it may touch and on which API version, and enforces **one writer per fact**, the rule whose absence produced the App That Wrote Stock.

The template. One card per installed app.

Field	What to record
App, vendor, owner	Public or custom; installed when and by whom; a named business owner
Purpose	One sentence a new colleague would understand
Scopes granted	Each read and write scope
Scopes revoked and why	With date and the decision record (Tool 16)
Facts it writes	Checked against Tool 2's writer column
API version and support end	Pinned version and the vendor's end date
Review date	The next quarterly uninstall review
Exit plan	What replaces it, and when

How to use it. Put custom behavior on the lowest rung that works: configuration, then the platform's own functions and workflows, then a public app, then a custom app with an owner. In the quarterly review, remove any app with no owner, no use in the quarter, or a twin.

Kestrel & Pine example (fictional; the pre-order app, from the April 2027 governance

rule):

Field	Value
App, owner	Pre-order app, public, installed 2025; owner Milan Petrov
Purpose	Take pre-orders for next season's Ridgewalker colors
Scopes granted	read_products, read_inventory, write_orders (pre-order tags)
Scopes revoked	write_inventory, June 2027: the WMS feed is the only writer of stock (W1)
API version	2027-01; supported at least to January 2028 *(illustrative)*
Review date	October 2027
Exit plan	Pre-orders handled by the availability service and the OMS from Q4 2027; app removed after the spring 2028 season

Used in: Apps, Plugins and Functions: Extensibility and Ecosystem Governance, Delivering the Storefront: Themes, Headless, Rendering and Performance, Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers, Assistants and Agents: AI in Commerce and What Is Real in 2026.

Tool 12: Peak Plan and Degradation Ladder

Purpose. Measure the order path in orders per minute and test each link. Define a **degradation ladder**, the sequence of optional work to pause under pressure. Each action needs an owner and trigger so checkout, payment and order processing retain capacity.

The template.

Plan field	What to record
Peak dates and owner	The peak week, the peak day, the incident commander
Forecast	Orders per minute in the peak hour, from last year's peak and this year's promotions; owned by whoever owns the promotion calendar
Capacity per link	Tested orders per minute for checkout, order export, warehouse intake, stock loop, PSP authorizations

Plan field	What to record
Load tests and game day	Dates, rates, marked test orders, failures rehearsed
Freeze	From and to; what it covers
Gauges and alarms	Queue age, dead-letter depth, event lag, decline rate against normal
Morning-after reconciliation	Time, what is compared, who produces it

The ladder is a table with one row per rung: the switch, its one owner (who may flip it without a meeting), the measured trigger and the date it was last flipped in a test, ending with a line for what is **never** switched off.

How to use it. A switch never flipped in a test is a hope; the order is set in a calm month by the people who own the revenue.

Kestrel & Pine example (fictional; the 2027 plan approved with Waypoint in March 2027, Noor Visser as incident commander):

Plan field	Value
Peak	Week of 22–28 November 2027; Black Friday 26 November
Forecast and tests	60 orders a minute (about 1.5 times 2026's 38); every link tested at 80; the OMS-to-WMS link at 120 with marked orders the WMS skips
Go/no-go	Availability service, 15 October 2027; fallback the old chain with the ten-unit marketplace buffer
Freeze	15 November–5 December 2027
Situation room	Friday 16:00–24:00
Reconciliation	08:00 Saturday 27 November, by Hanna Berg's team

Three of the eight rungs:

#	Switch	Owner	Trigger
3	Stock badges from the buffered copy	Ruben Alves	Availability event lag over 2 minutes
5	Delivery promise widened to "within three working days"	Daniel Okafor	Oldest unsent order over 10 minutes, or the Friday pick plan over 80% consumed

#	Switch	Owner	Trigger
8	New-account creation and the newsletter pop-up off	Milan Petrov	Only on Noor's call
—	**Never**: checkout, payment authorization, order creation in the OMS, the order path to the WMS	—	—

Used in: Black Friday Every Year: Peak Load, Resilience and Recovery, Journey 4: Black Friday, Hour by Hour, Many Merchants, One Platform: Multi-Tenancy, Pods and Isolation, Protecting the Shop: Card Data, Account Takeover, Bots and Skimmers.

Tool 13: Regulatory Calendar

Purpose. Translate an applicable rule into required data or behavior, a responsible system, a test, an owner and a deadline. Record scope as well as dates; a rule may apply differently by country, product or company size.

The template. One row per rule, sorted by date.

Column	What to record
Date	Application date from the official source; several dates where the rule phases in
Rule	Official name and number, with a link
Scope check	Why it applies to you, or why it does not (size, country, role)
What it requires	In data terms: which field, clock or document
System of record	Where that field or clock lives (Tool 2)
Test	How you know you comply, and how often it runs
Owner	The person who owns the fact, often not an engineer
Status	*Maintain* (in force), *prepare* (date ahead), *watch* (pending)

How to use it. Plan to the scope and schedule to the date. Review the *watch* rows quarterly; this is not legal advice.

Kestrel & Pine example (fictional rows and owners; the rules and dates are documented in Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform

Regulation):

Date	Rule	What it requires	Owner	2027 status
13 Dec 2024	GPSR, Regulation (EU) 2023/988, Art. 19	Manufacturer, responsible person, identification and warnings on every offer	Teun Baas	*Maintain*: the PIM blocks publication without the fields from Q3 2027; about 850 third-party variants to clean up
1 Jan 2026	Belgian e-invoicing	Structured invoices over Peppol to Belgian business customers	Sanne de Wit	*Maintain*: delivered-status report; 210 accounts

Used in: Rules Turned Into Requirements: Consumer, Payment, Invoicing and Platform Regulation, Documents Between Companies: EDI, Punchout and E-Invoicing, Whose Data Is It? Privacy, Consent and Retention, What Are We Selling? Products, Variants and the Catalog.

Tool 14: Build, Buy or Assemble Decision Card

Purpose. A merchant covers the capability map with four kinds of parts: a suite, a hosted platform plus apps, assembled specialist services, and custom code. The card compares the realistic options for one capability or one platform decision on five criteria and on three-year total cost, including the lines a vendor's quote leaves out.

The template. Page one is the criteria.

Criterion	What to record per option
Business model	Which channels and stock locations the option must serve
Scale stage	Starting, Growing, Omnichannel or Group, and the signals of the next stage

Criterion	What to record per option
Team	Engineers needed to own it, and whether you have them
Differentiation	What a customer would notice if you did it better than anyone; build only that
Time and risk	The next peak; the cost of a failed cutover; whether each step is reversible

Page two prices each option over three years: subscriptions and any share of gross merchandise value (GMV); build, implementation and migration; the agency after go-live; people added; revenue at risk during cutover; and the exit cost at the next review.

How to use it. Price every option as incremental cost over today's run rate and label every figure that is not a quote as an assumption.

Kestrel & Pine example (fictional; the February 2027 platform review, incremental cost 2027–2029, illustrative):

	Composable rebuild	Enterprise suite	Stay and surround (W4)
Three-year incremental total	about €3.2M, 18 months	about €3.19M, 12 months	about €1.33M
Revenue at risk	€540k *(assumption)*	€540k *(assumption)*	about €50k
Exit at the 2029 review	High	High	Moderate
Decision	—	—	**Chosen**: OMS with availability module (€4,500/month, €320k to implement), PIM (€1,800/month, €110k), integration standards €150k; recorded as KP-004

Used in: Build, Buy or Assemble: Choosing and Changing Platforms, Suites, SaaS, Open Source and Composable: The Platform Landscape in 2026, Case: commercetools,

Shopware and the Composable Movement, What Changes at Each Scale: From First Store to Group.

Tool 15: Migration Plan Card

Purpose. Plan each change of authoritative system as a limited step: one channel, capability or traffic slice. Specify how old and new results will be compared, who may write during the transition and how to recover if the new path fails.

The template. One card per seam; the data rows once per migration.

Field	What to record
Seam	By channel, by capability or by traffic slice
Phases and dates	Shadow, parallel run, divert this seam, divert all, decommission
Owner of the fact per phase	Old or new
Reconciliation	Tool 5 card; what agrees for how long before going forward
Go/no-go criteria	Written before the date
Rollback	How the seam is routed back, how long it takes, when it was rehearsed
Identifier continuity	Order number family, SKU, GTIN unchanged; last old and first new number recorded
Freeze and peak	Change freeze around the peak; the old path kept operable through one peak
Data migrations	Each data kind with its own check: order history read-only, gift-card balances to the cent

How to use it. Prefer one writer and a copied flow over dual writes (one actor writing both systems).

Kestrel & Pine example (fictional; the OMS introduction under W2, 2027):

Phase	Dates	Reversible by
Shadow copy of all orders	August	Switching the copy off
Parallel run	1–14 September; differences on about 1.4% of orders in week one, mostly status mapping and timing	Nothing to reverse
Marketplace seam	15 September	A rehearsed one-hour route back
D2C seam	13 October	Routing back within the day
Change freeze	15 November–5 December, around Black Friday (26 November)	The freeze itself
Wholesale seam (EDI and manual)	8 December	Routing back to the ERP's order entry
Decommission old order paths	January 2028; 2026 orders imported as read-only history	Not needed

About 14 of the 34 integrations were re-pointed.

Used in: Moving Without Burning the Bridge: Replatforming and Migration, The Order Record: Lifecycle, State and the Order Management System, What Are We Selling? Products, Variants and the Catalog, Case: Zalando, Microservices, Platform Strategy and the Return to Discipline.

Tool 16: Architecture Decision Record

Purpose. An architecture decision record (ADR) is a one-page note of a decision, the facts that forced it, the options considered and what it changes, kept in the repository so a newcomer can understand a decision instead of blindly accepting or reversing it. Michael Nygard defined the form in 2011; this version adds ownership, vendors and a review date.

The template.

```
# KP-NNN: <Decision in a few words>

Status:       proposed | accepted | deprecated | superseded by KP-NNN
Date:         YYYY-MM-DD
Deciders:     <names>; business owner: <name>

## Context       The incident or finding, as neutral facts with dates and numbers.
## Drivers       What a good answer must achieve; the failure pattern it prevents.
## Options       At least two real options, plus "do nothing".
## Decision      "We will ...", with scope, owner and dates.
## Ownership     Rows of the entity-owner table (Tool 2) whose owner or writer changes.
## Consequences  Better / worse / new work, and who carries it.
## Vendors       Contracts, API versions and exit terms this decision depends on.
## Confirmation  How conformance is checked: reconciliation cards, write audits, gauges.
## Review        A date, or the trigger that reopens it.
```

How to use it. Write one for every decision that changes who owns a fact, which vendor you depend on, or which app may write. Keep it to a page, and supersede rather than edit an accepted record.

Kestrel & Pine example (fictional; Waypoint decision W4, numbered KP-004):

```
# KP-004: Stay on the SaaS platform and surround it

Status:       accepted
Date:         March 2027
Deciders:     Mara Lindqvist, Noor Visser; business owner: Mara Lindqvist

## Context       Two February 2027 replatforming pitches cited the Black Friday
                 Oversell (about €96,000). The platform held; the chain behind it
                 failed. Four empty boxes on the capability map.
## Options       Composable rebuild (about €3.2M); enterprise suite (about €3.19M);
                 stay and surround (about €1.33M); do nothing.
## Decision      We will stay on the SaaS platform. OMS, PIM and availability are added
                 beside it (W1-W3); the platform's own POS is evaluated in 2028.
## Ownership     Order -> OMS (Q4 2027); product content -> PIM (Q3 2027);
                 availability per channel -> availability service (Q4 2027).
## Consequences  Better: no cutover; each step reversible. Worse: more systems to
                 integrate. New work: integration standards (W5), two engineers.
## Confirmation  Order, stock and price reconciliation cards; the 2027 peak plan.
## Review        The 2029 review.
```

KP-001 to KP-008 are Waypoint decisions W1 to W8.

Used in: Who Builds and Who Owns: Teams, Vendors, Agencies and Governance, Build, Buy or Assemble: Choosing and Changing Platforms, Monolith, Modular Monolith, Microservices, Composable: What the Evidence Says, Delivering the Storefront: Themes, Headless, Rendering and Performance.

Keeping the Cards True

Cards decay unless three habits keep them current.

- **Review on a calendar**. The capability map yearly; the entity–owner table with every decision record; the integration inventory and app cards quarterly; contract cards at every vendor version; partner cards at every guideline change; the peak plan from March; the regulatory calendar's *watch* rows each quarter.
- **Let incidents add fields**. When a review asks a question no card answers, add a field.
- **Keep the set small**. A field filled by habit and read by nobody should go.

The terms these cards use, from ATP and SSCC to idempotency and system of record, are defined in plain language in Glossary: Plain-Language Definitions, and Bibliography: Every Source the Book Cites lists the sources behind them.

Sources and Evidence

The templates are the author's, derived from the chapters linked in each tool, which cite their sources in full. The external forms and texts they build on:

Company, product and investor sources

- Payout reconciliation report, Stripe, n.d., accessed 2026-11 — a vendor's itemized reconciliation, the finance-side model for Tool 5.
- Webhooks, Shopify developer documentation, n.d., accessed 2026-11 — delivery not guaranteed and reconciliation jobs recommended, the reason Tool 4 always points to Tool 5.

Standards, regulation and official documentation

- GS1 General Specifications, Release 26.0, GS1, 2026 — GTIN, GLN and SSCC as used in Tool 10.
- Regulation (EU) 2023/988 (General Product Safety), European Union, 2023 — the Art. 19 row of Tool 13.

- Service Level Objectives and Monitoring Distributed Systems, Google, *Site Reliability Engineering*, 2017 — measured objectives and the gauges of Tool 12.

Research and practitioner sources

- Documenting Architecture Decisions, Michael Nygard, 2011 — the record anatomy (title, context, decision, status, consequences) that Tool 16 extends; older than three years and still the standard reference.
- Feature Toggles (aka Feature Flags), Pete Hodgson, martinfowler.com, 2017 — ops toggles and kill switches as the mechanism of the degradation ladder in Tool 12.
- Strangler Fig Application, Martin Fowler, 2024 (restating a 2004 note), and Patterns of Legacy Displacement, Ian Cartwright, Rob Horn and James Lewis, 2021, revised 2024 — seams, parallel runs and transitional architecture in Tool 15.

Kestrel & Pine is fictional; architectural recommendations, labeled inferences and labeled assumptions are the author's judgement.